S. Palani

Automatic Control Systems

With MATLAB

Second Edition





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ISBN 978-3-030-93444-6 ISBN 978-3-030-93445-3 (eBook)
https://doi.org/10.1007/978-3-030-93445-3

Jointly published with ANE Books India
The print edition is not for sale in South Asia (India, Pakistan, Sri Lanka, Bangladesh, Nepal and Bhutan) and Africa. Customers from South Asia and Africa can please order the print book from: ANE Books Pvt.Ltd.
ISBN of the Co-Publisher's edition: 978-9-383-65627-1

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This book is
dedicated to My Beloved Teachers

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Preface

Automatic Control Engineering is an exciting field of study which gave significant contributions to the development of latest technologies in diversified applications which include the very sophisticated aircraft and ship control, control of communication satellites, control of boilers and nuclear reactors, complex power system network, war weapons etc. Automatic control theory is divided into two major topics namely classical control theory and modern control theory. The application of classical control theory is restricted to single-input and single-output linear time invariant continuous time systems. Here for the transfer function model of the system, the analysis and design are carried out in time domain as well as in frequency domain. There are several limitations in this method. The limitations of classical control theory were overcome by modern control theory which was introduced in the yearly 1960s. The method uses state space model and can deal with multi-input and multi-output system. Many sophisticated design problems are carried out in this method. Nonlinear, multivariable, adaptive and robust control theory comes under modern control theory.

Originally automatic control theory was applied to linear continuous time system analysis and design. In recent time, computers and intentionally introduced into the system to monitor and control all physical variables. Now the system becomes discrete time system. In all technical universities in India, automatic control theory course is taught for the undergraduate, postgraduate and doctoral level students of Electrical and Electronics Engineering, Electronics and Communication Engineering, Electronics and Instrumental Engineering, Mechanical Engineering, Computer Science and Engineering, Aeronautical Engineering and Chemical Engineering. To write books to the requirements of students of individual branch is really a tough task. Therefore, it is decided to write a common book which will cater to the need of students of all these engineering disciplines of undergraduate level of Indian universities. I have used all my experiences I gained while teaching this course to the students of undergraduate, postgraduate and doctor degree for the past four and a half decades to bring out a good book on this subject.

The book presents a comprehensive coverage of linear system theory which is intended for undergraduate courses in control system engineering in all technical

universities in India. The concept of each topic is well discussed with a full length presentation of numerical examples. Each example is unique in its own way and it is graded in sequential manner. The book highlights and provides with easy methods of getting solution. Select numerical problems worked out bring a test the capability of the student's knowledge and understanding of the basic concepts of the theory. The organization of the book is as follows:

1. Control systems modelling and their representation
2. Time response analysis
3. Frequency response analysis
4. Stability analysis of linear control system
5. Root locus method for analysis
6. Design of compensators and PID controllers
7. State space modelling and analysis

In Chap. 1, representation of system in open loop and closed loop is discussed and different terminologies associated with control theory are defined. The concept of transfer function is explained and system modelling using force-voltage and force-current analogies is discussed. Physical systems are represented in block diagrams and signal flow graph and their reduction technique explained.

In control system study, the ultimate objective is to study of nature of variation of different variables in the system with respect to time. In Chap. 2, transient and steady state responses of first and second order systems are discussed and expressions for the time domain specifications of step input are derived. The performance evaluation of PID controllers when connected in cascade with prototype second order system is also discussed using the results obtained from MATLAB simulation.

Even though performance of many real time control systems is hedged based on the time response of the system, we get very valuable information when the system is tested with sinusoidal input. The test study when the frequency is varied over a wide frequency range is called frequency response. In Chap. 3, we discuss the frequency response using the polar plot, Bod plot and Nichols plot. Expressions for the frequency domain specifications are derived and the use of constant M and constant N circles and Nichols chart explained.

When a control system is designed, it should be stable. If the system is not stable, it is no use and causes severe damages not only to the system itself but also to the environment and human life. In Chap. 4, the concept of linear system stability and different ways of defining stability are discussed. The general stability conditions for linear system stability are derived. The stability tests are carried out using Routh-Hurwitz and Nyquist stability criteria.

It is well established that for the closed loop system to be stable, all the roots of the characteristic equation should be in the left half of the s-plane. In Chap. 5, the root locus method which gives the path of the characteristic roots traced out in the s-plane as the gain of the open loop transfer function is varied is discussed. Root loci are drawn for different types of transfer functions. From root locus plots determination of closed loop system absolute stability, transient response, relative stability, phase margin and gain margin is explained.

Control system design engineer has used his ingenuity to design the system to give optimum performance. Still if the performance obtained by an individual is not optimum, a compensator is inserted into a system in order to compensate for the performance deficiency. In Chap. 6, the design methodologies of lead, lag, lag-lead compensators, and PI, PD and PID controllers in the time domain as well as in the frequency domain are discussed.

System representation in state space model has many advantages compared to transfer function model representation. Recent development in modern control theory has been possible with the state space model only. Here a system is represented as a set of n first order differential/difference equation where physical variables are defined as the state variables. In Chap. 7, continuous and discrete time system modelling in state space has been derived and the solution of the state question obtained. Further, the concepts of controllability and observability are explained with numerical examples.

I take this opportunity to thank **Shri Sunil Sexana**, Managing Director, Ane Books Pvt. Ltd. India for coming forward to publish the book with the same enthusiasm as he showed while publishing my other books earlier on. I would like to express my thanks to **Shri A. Rathinam**, General Manager (South), Ane Books Pvt. Ltd. who took the initiatives to publish the book in a short period of time. I am thankful to my wife **Dr. S. Manimegalai M.B.B.S., M.D.**, who inspires me whenever I write a book. I also thank **Mr. V. Ashok** who has done a wonderful job to key a voluminous book like this with atmost patience and care. Useful suggestions and constructive criticisms are always welcome from the readers.

Tiruchirapalli, India Dr. S. Palani

Acknowledgements

The author would like to express his gratitude to the members of the teaching faculty and student community for extending their patronage to the first edition of the book *Automatic Control Systems*.

Few errors noticed here and there in the first edition have been corrected in the present edition. A very exhaustive treatment of automatic control theory, a large number of variety of memorial problems worked out and the inclusion of a large number of short and long questions and answers in the exercise increased the volume of the previous edition. To limit the number of pages in the present edition, some of the contents such as PID Controller design and MATLAB programs for the design of compensators and controllers have been shortened. Further, a few numerical problems worked out in the text have been shifted to the long type questions as exercise. I hope that the reduced size of the book will be more comfortable for the students to handle it. Since the first edition was exhaustively informative and contained a variety of numerical examples, the author did not make any attempt to include new examples to explain the theory.

The author would like to express his sincere thanks to Shri. Sunil Sexana, Managing Director, ANE Books Pvt. Ltd., for his encouragement given to the authors. I would also like to thank Shri. A. Rathinam, General Manager (South), ANE Books Pvt. Ltd., for taking active steps in promoting the sales of the book. We also thank Mr. V. Ashok for doing nice work in bringing the book in the present form.

Dr. S. Palani

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About the Author



Dr. S. Palani obtained his B.E. degree in Electrical Engineering in the year 1966 from University of Madras, M.Tech. in Control Systems Engineering from IIT, Kharagpur in 1968 and Ph.D. in Control Systems Engineering from University of Madras in 1982 with academic excellence. He has a wide teaching experience of over four decades. He started his teaching career in the year 1968 at the erstwhile Regional Engineering College (Now National Institute of Technology), Tiruchirappalli in the department of EEE and occupied various positions. As Professor and Head, he took the initiatives to start the Instrumentation and Control Engineering Department. After a meritorious service of more than three decades in REC, Tiruchirappalli, he joined Sudharsan Engineering College, Pudukkottai as the Founder Principal. He established various departments with massive infrastructure. Since 2006, he is the Director in the same college.

Dr. S. Palani has published more than hundred research papers in reputed international journals and has won many cash awards. Under his guidance and supervision many research scholars have been awarded Ph.D. He has carried out several research projects funded by Government of India and other agencies. As the theme leader of the Indo - UK, REC Project on energy, he has visited many Universities and industries in United Kingdom. He is the author of the popular books titled Automatic Control Systems, Signals and Systems, Basic System Analysis, Digital Signal Processing (ECE), Digital Signal Processing (EEE), and Digital Signal Processing (CSE).

Chapter 1
Control Systems Modelling and Their Representation



Learning Objectives

After completing this chapter, you should be able to:

- Represent the system in open loop and closed loop forms.
- Define various terminologies related to control system.
- Write the dynamic equations of mechanical, electrical and electromechanical systems and derive their transfer functions.
- Model mechanical systems by Force–Voltage and Force–Current analogies.
- Derive the transfer function of commonly used control system components.
- Represent the system in block diagram and to study block diagram reduction technique.
- Represent the system by signal flow graph and to study the application of Mason's gain formula.

1.1 Introduction

Automatic control has played a vital role in the advancement of science and engineering. It finds wide applications in space vehicle systems, missile guidance systems, robotic systems, anti-aircraft gun control and radar systems. It also finds applications in manufacturing and industrial process control, autopilot systems, aerospace industries, design of cars, trucks and war weapons, and in industrial operations such as controlling pressure, temperature, humidity, viscosity, flow etc. The use of digital computers as the integral part of modern control system has enhanced its applications in recent times.

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S. Palani, *Automatic Control Systems*,
https://doi.org/10.1007/978-3-030-93445-3_1

Thus, control engineering is not limited to any particular engineering but it is applicable to various fields of engineering such as chemical, aeronautical, mechanical, electrical, civil and environmental engineering. Without the advances in control system design, space travel would be impossible, music system would not be so effective, industrial production would be very low and the applications of computers would be severely limited. The application of control theory and technology is so widespread today that one can hardly find a working engineering system without some form of control circuitry or control procedure in it.

In today's highly competitive global economy control engineers have to produce the best quality products at the lowest cost. For the industrial control engineers, improving the product quality and minimizing the cost have been the major tasks. They look at the control systems, at the latest advances in sensors, actuators, control methods and computer systems and recommend how plant performance can be improved for a much lower capital outlay without rebuilding the existing plant which requires huge amounts of capital expenditure. In recent years modern health services have undergone a radical change due to advances in control, sensors, actuators and information technology.

Kidney dialysis, machines, pacemakers, artificial hearts, automation procedures in surgeries, anaesthesia procedures and radiation treatment all use control methods. Control engineering also plays an important role in many other areas of industries such as the processes in pharmaceutical plants, steel works and automobile engine management system. An aircraft control system is a vital control engineering application. Much of our space technology would not be where it is now without control techniques.

Wind turbine systems are probably the most widely used renewable energy method for generating electric power. To connect the wind generator to the grid, the speed of rotation is maintained by the grid frequency. The design and implementation problems call for control, mechanical and electrical engineering skills. However, it is the control engineer who will integrate the various aspects to produce a working system. Many of the most advanced control and instrumentation systems are found in the aerospace and space industries. Almost all the basic maneuvers performed by an aircraft, such as landing, take-off and directional control are automated and implemented by computer systems. A fully integrated flight control system maximizes aircraft performances while ensuring crew and passenger safety. Such a system reduces the burden of decision making and ease the task faced by the human crew.

Due to the environmental disturbances, like wind, wave and sea current, a ferry travelling in a sea experience roll which is unpleasant and unsafe for the travel. Different types of control systems are designed to compensate for the effects of these disturbances and make the sea transport more safe and comfortable.

From the many examples described above, we find that control engineering is an important feature in many aspects of our life. Thus, control is everywhere on which our society depends. In the life style of today, we rely heavily on technology and its applications. Control engineers are needed to analyse these technological systems and provide control systems without which modern society would simply be not possible.

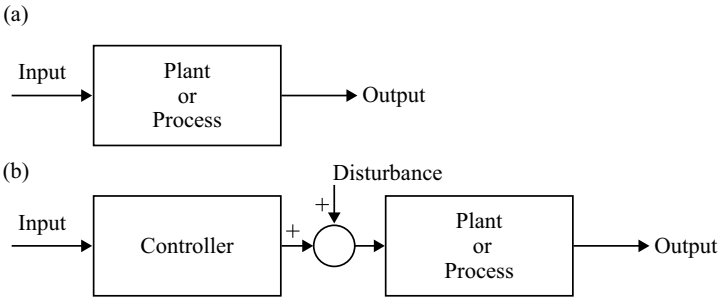


Fig. 1.1 **a** A plant to be controlled and **b** A plant with a controller

1.2 Basic Concepts and Terminologies

As mentioned earlier, there are many control systems that we meet in our daily life. Any portion of a system that we want to control is called a plant or process. The input signal which is applied to the system interacts with the plant and produces the output. For the given plant, to get the desired output, which is our objective we should give appropriate input signal. This is shown in Fig. 1.1a. The control system designer job is to ensure that the plant operates as required. When disturbances occur in the system a controller is inserted in between the plant and the input. The controller output signals produce desired output from the plant. This is represented in Fig. 1.1b. In the analysis and design of control system, various components which are interconnected are represented in block diagram form as shown in Fig. 1.1. Arrows indicate the direction of signal flow. When complex systems are studied, it becomes easier to understand when they are represented in block diagrams. When we talk about control systems, we come across several terminologies and some of them are very commonly used. The definitions of these terminologies are given below.

1.2.1 System

A system is defined as a set of interconnected objects with a definite relationship between the objects and attributes. It is also defined as the interconnection of elements and devices for a desired purpose.

1.2.2 Control System

Control systems are the systems that implement certain objectives. The interconnection of connection of components forming a system configuration that provides the desired output is also called control system.

1.2.3 Variables

Variables are the attributes that are present at different parts of the system.

1.2.4 Input

Input of a system is an attribute of a system which is varied directly and independently by the operator.

1.2.5 Output

Output of a system is an attribute that varies as a result of variation in the input. It cannot be independently varied by the operator.

1.2.6 System Parameters

Parameters are the properties of the given system which is inherently present in the system.

1.2.7 Plant or Process

The device which is under control is called plant or process.

1.2.8 Automation

The control of a process by automatic means is called automation.

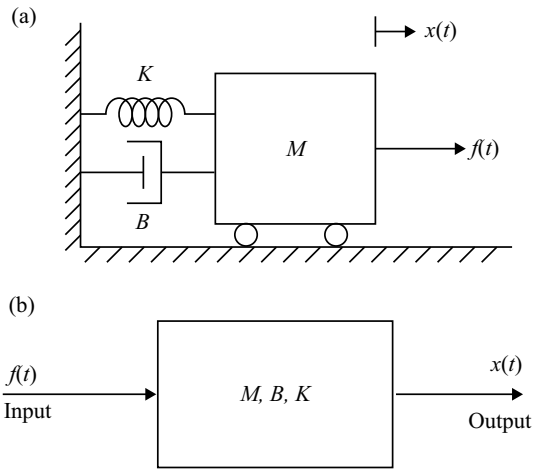


Fig. 1.2 **a** Mechanical system and **b** Block diagram representation of mechanical system

1.2.9 Servomechanism

The control system which follows the frequently changing reference signal is called servomechanism. It is usually referred to system whose output is mechanical position or velocity.

1.2.10 Regulator System

If the reference or set point signal is kept at a steady value for a long period, and the output remains at the prescribed set level rejecting disturbance signals, it is called a regulator system. The above terminologies are explained below.

Consider the mechanical system consisting of mass M, dash-pot B and spring K. $f(t)$ is the force applied and $x(t)$ is the displacement of mass. The elements M, B and K are interconnected and form as a translational mechanical system. $f(t)$ is the input variable (attribute) and $x(t)$ is the output variable (attribute). The following dynamic equation is written for the above mechanical system:

$$M\frac{d^2x(t)}{dt^2} + B\frac{dx(t)}{dt} + Kx(t) = f(t) \tag{1.1}$$

$f(t)$ is the input attribute and it can be independently varied by the operator as the way he wants. However, the output attribute $x(t)$ can be varied by varying the input $f(t)$. M, B, and K are the objects (parameters). These objects are the inherent properties of the system and are fixed at a particular operating condition. Since there is a definite relationship between these objects and the attribute $x(t)$ it is called a system. When

the input is applied, opposing forces are generated by these objects. The mass M generates the opposing force $M\frac{d^2x(t)}{dt^2}$, the dash-pot generates $B\frac{dx(t)}{dt}$ and the spring generates $Kx(t)$. Hence, the interconnection of M, B and C is called a system according to the definition. Further, if the objective is to change the displacement $x(t)$, it can be achieved by changing the input $f(t)$. Hence, it is a control system.

1.3 Concept of Feedback-Open Loop and Closed Loop Systems

1.3.1 Open Loop System

A simple D.C. motor speed control system is shown in Fig. 1.3a. The speed ω of the motor depends on the motor armature voltage V_a. The armature voltage is adjustable with a help of a potentiometer. If the load on the motor increases, the speed ω drops. To maintain the shaft speed constant, the armature voltage V_a had to be increased. If the load on the motor decreases, the shaft speed increases. Now the armature voltage V_a has also to be decreased accordingly. Thus, adjusting the potentiometer setting according to the load variations requires manual operation which may require constant vigil on the part of the human operator which may sometime be not accurate. This type of the system is called **open loop system**. In the above system V_a is the input and it causes control action. Here ω, the speed is the output. When there is a change in the output, it does not have any influence on the control action. Thus, the open loop system is defined as follows.

An open loop system is defined as the system in which the control action (actuating signal) is independent of the output or the desired result.

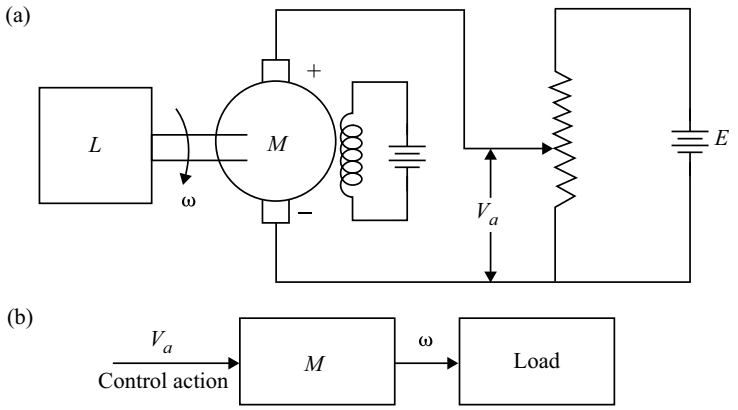


Fig. 1.3 **a** D.C. motor speed control system and **b** Block diagram representation of speed control system

1.3.2 Closed Loop or Feedback System

Now consider D.C. motor speed control system shown in Fig. 1.4a. Here, an additional component, namely a D.C. tachogenerator T is connected to the motor and load shaft. The tachogenerator generates voltage proportional to speed. The output of this tachogenerator V_T is compared with the reference voltage V_a. The difference of these voltages is $V_a - V_T$, as per the connection shown in Fig. 1.4a. Now we say the output is fedback and compared with the reference input. The system now becomes a **feedback or closed loop system**. For the connection shown in Fig. 1.4a, the feedback is **negative**. If the polarity of the tachogenerator is reversed the feedback is **positive**. The working principle of the feedback is explained below.

With the feedback loop kept open and on no load condition, the reference input V_a is adjusted using the potentiometer such that motor runs at desired speed ω. When the load is connected to the machine shaft, the speed ω decreases. Since V_T is proportional to ω, this voltage also decreases. $V_a - V_T$ which is applied to the armature increases which ultimately increases the shaft speed and brought to the desired value. Now suppose, a portion of the load is removed. This will increase the

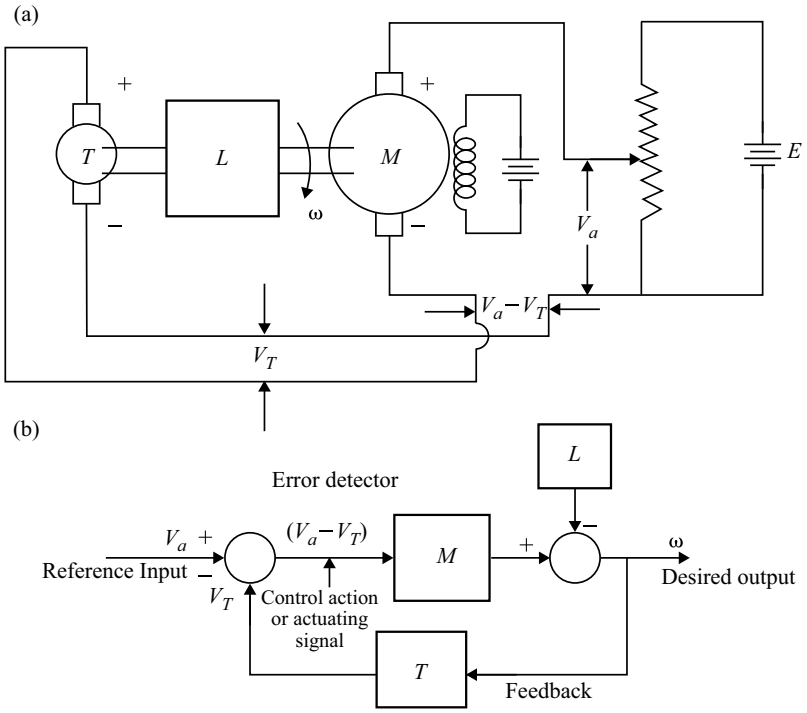


Fig. 1.4 **a** Closed loop D.C. motor speed control system and **b** Block diagram of closed loop D.C. motor speed control system

shaft speed ω which is undesirable when constant speed operation is desired. This increase in ω increases V_T. The control action $(V_a - V_T)$ decreases which decreases the shaft speed ω. Thus, when there is disturbance in the load side, it is rejected automatically by the feedback connection and the desired output is restored may be with marginal acceptable deviation. The control action $(V_a - V_T)$ is also called by the name **Actuating Signal or Error Signal**.

On the other hand, consider the system with a positive feedback. Now the actuating signal becomes $(V_a + V_T)$. When the load is connected to the motor shaft, ω decreases and also V_T. The control action decreases and ω decreases further. This is a cumulative process and finally the speed becomes zero. Suppose, while running, a part of the load is removed. Now the speed, ω increases which results in increase of $(V_a + V_T)$. As a result ω also increases in a cumulative manner and may reach a dangerously high speed causing heavy damage to the entire system. Now we say, the system is in **unstable** mode of operation. This is because of positive feedback. For all practical purposes, only negative feedback is employed and only in selective cases such as design of oscillators positive feedback is used. Now we give below the definition of closed loop system and then give the advantages and disadvantages of open loop and closed loop systems.

A closed loop system is defined as a system in which the measured output is compared with the reference input. Here the control action depends upon the changes in the output.

1.4 Comparison of Open Loop and Closed Loop Systems

1.4.1 Open Loop Systems

1.4.1.1 Advantages

1. Design and construction are simple and economical.
2. Almost all physically realizable open loop system are stable.

1.4.1.2 Disadvantages

1. They are less accurate and due to manual operation sometimes become unreliable.
2. They do not compensate for the effect of internal and external disturbances which result in corrupted output and degrade the performance of the system.

1.4.2 Closed Loop System

1.4.2.1 Advantages

1. Closed loop systems have much greater accuracy than the open loop systems.
2. Due to feedback, they do not require manual supervision. Further, they are fast acting in removing the error.
3. They reject internal and external disturbances and minimize their effects.
4. Closed loop systems have higher bandwidth. This property gives better transient response in the time domain. The range of frequencies over which the system responds is increased.
5. The sensitivity of the system for parameter variation, noise, disturbances and environmental changes are made small by increasing the gain of the feedback loop.
6. If the open loop system is stable by proper choice of feedback gain the system can be made stable.

1.4.2.2 Disadvantages

1. Because of feedback, the overall gain of the system is reduced.
2. It requires more components compared to open loop system. This increases the cost.
3. Since more components are connected the system becomes more complex. System design and analysis become more difficult.
4. In general closed loop system is less stable. But by proper design, it can be well stabilized with faster response and higher accuracy. In view of these important properties, closed loop systems are invariably used and not the open loop system.

1.4.3 Effects of Feedback

The effects of feedback can be summarized as given below.

1. Negative feedback reduces the overall gain of the system while positive feedback increases it. Further, in one frequency range, the overall gain is increased while in another frequency range, the overall gain is reduced for the given feedback.
2. Feedback, if properly designed will improve the stability. Improper design will make the system unstable.
3. The sensitivity function can be made small by increasing the feedback gain. This will remove the harmful effect of parameter variation.
4. Feedback reduces the effect of noise and disturbances.
5. Feedback increases the bandwidth of the system. This increases the fastness of time response of the system in the time domain and increases the frequency range of operation in the frequency domain.

1.5 Examples of Closed Loop System

1.5.1 Automobile Steering Control System

A closed loop automobile steering system controlled by a human operator is shown in Fig. 1.5. Here θ_i is the desired direction of the automobile. The actual direction of the car is judged by the naked eyes of the driver. The eyes compare the difference between the desired direction and the actual direction of the vehicle and this message is sent to the brain of the driver. The arms of the driver steers the wheel in such a direction such that $(\theta_i - \theta_0)$, the error is zero or minimum. Thus, the above system is an example of a closed loop system with man-made controllers.

1.5.2 Regulator System

Consider the voltage regulator system shown in Fig. 1.6 where a D.C. generator run by a prime mover supplies electrical load. G is the gain of the D.C. generator which is expressed in volts per field ampere at a particular speed condition. The block diagram of the system is shown in Fig. 1.6b. Here the reference voltage is V_r and is applied to the field winding of the generator. The system operates in open loop mode since there is no feedback provided. Under no load condition, the reference voltage V_r is adjusted such that the induced voltage V_g in the generator is the same as the load rated voltage V_t. If the generator is loaded, the current through the armature of the generator I_a also increases. At this stage the following equation applies:

$$V_g - I_a R_a = V_t \tag{1.2}$$

As the load current (same as armature current) increases, $I_a R_a$ drop increases which results in decrease of load terminal voltage V_t, which is undesirable. To solve the problem, the reference voltage is adjusted (increased) using the potential divider. The flux in the field winding of the D.C. generator is thus increased and also emf V_g. Thus, the $I_a R_a$ drop is compensated. Suppose, while operating on load, a portion of it is switched off. This results in decrease of I_a which as per Eq. (1.2) increases the

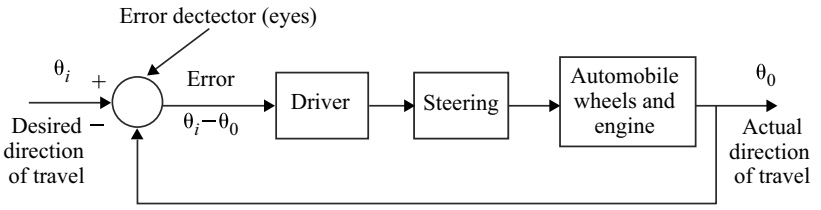


Fig. 1.5 A closed loop automobile steering system

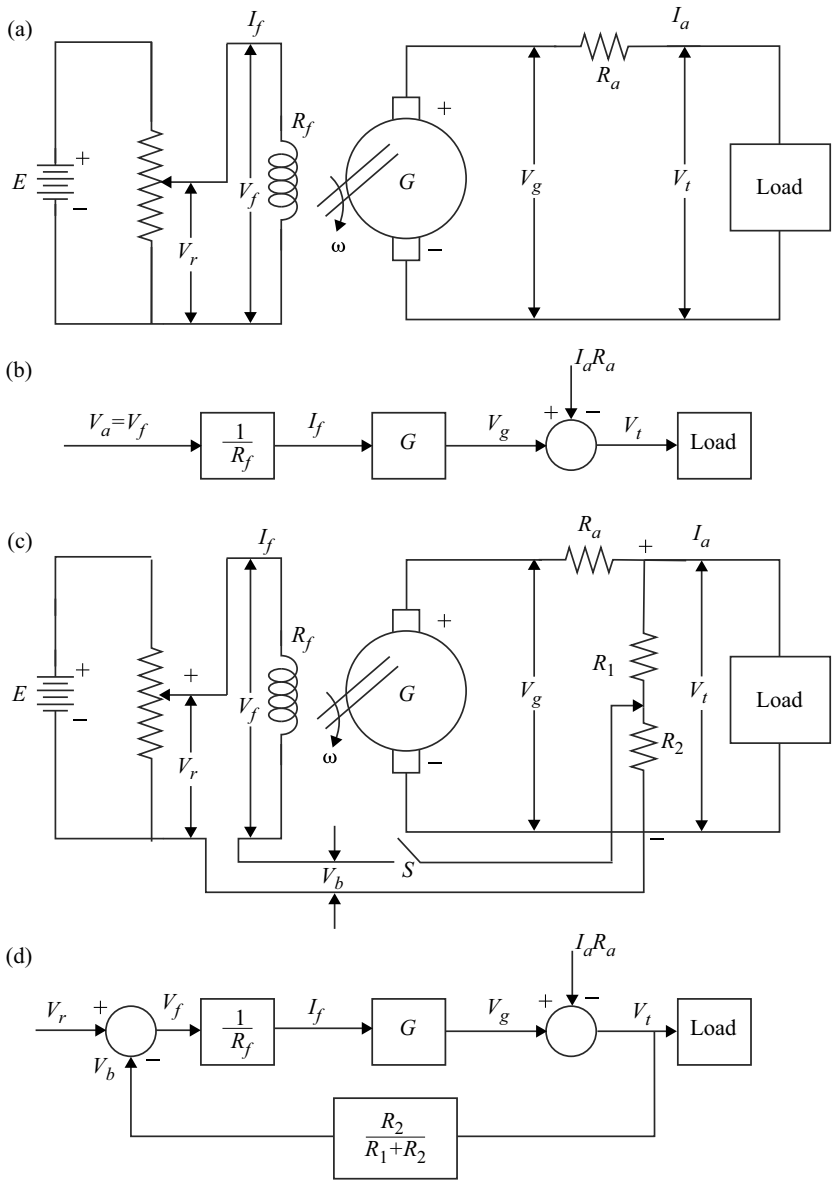


Fig. 1.6 Voltage regulator system. **a** Schematic diagram of open loop regulator system; **b** Block diagram of open loop regulator system; **c** Schematic diagram of closed loop regulator system and **d** Block diagram of closed loop regulator system

load terminal voltage. If this condition is allowed to continue, due to over voltage the load will be damaged.

Now, according to the changes in the load current, the reference voltage V_r is decreased such that, the load terminal voltage is kept at the required rated voltage. Thus, all the time close inspection of the variation of the load is necessitated and the potentiometer setting has to be adjusted manually. Thus, the accuracy and fastness of maintaining the voltage regulation depend upon the manual skill, which on many occasions may not be so accurate.

Now consider the voltage regulator system where feedback is provided which is shown in Fig. 1.6c and its block diagram in Fig. 1.6(d). A portion of the signal V_t is taken and compared with the reference input V_r. They are related by the following equation

$$\begin{aligned} V_b &= \frac{R_2}{(R_1 + R_2)} V_t \\ V_f &= (V_r - V_b) \end{aligned} \tag{1.3}$$

The above feedback is negative since the negative of the feedback voltage is connected to the negative of the reference voltage while their positive points are connected when comparison is done. The system becomes positive feedback if the polarities of the feedback voltage are reversed. The operation of the negative feedback voltage regulator system is explained below.

To start with, under no load condition the feedback loop is broken by using the switch S. The reference voltage V_r is adjusted until the load terminal voltage is at its rated value. Now the feedback loop is closed using the switch S and the load is connected to the main generator. When there is no disturbance, the load terminal voltage will be retained at the prescribed set level. Suppose more load is connected to the generator. This causes increase in the load current and hence the armature current. Since R_1 and R_2 are very high value resistors, the current drawn in that branch is negligible and for all practical purposes the load current and generator current are same.

When the armature current increases when the load in increased, the armature voltage drop $I_a R_a$ also increases. This results in drop in V_t which is not desirable. However, the drop reflects in V_b also. Now, $V_f = (V_r - V_b)$ increases which increases the excitation of the generator and hence the induced emf V_g. Thus, the fall in V_t is automatically corrected without any manual supervision. On the other hand, if some load is dropped, V_t and hence V_b increases and V_f decreases. Thus, V_g is decreased and hence V_t is retained at the prescribed set level. Thus, the system becomes automatically a controlled one.

1.5.3 Liquid Level System

Consider the open loop liquid level system shown in Fig. 1.7a. The liquid level in the tank is adjusted by manual operation of the inlet valve attached to the supply pipe. Through exit pipe, the water is consumed. If the consumption is more than the supply, the tank becomes empty. On the other hand if the consumption is less than the supply, the water overflows. Thus, according to the consumption rate, the inlet valve has to be adjusted manually so that a desired level of water is maintained in the supply. This is an example of open loop regulator system.

Now, consider the same system shown in Fig. 1.7b with a float and a pivot mechanism incorporated and made as a closed loop system. Let us assume that, initially, the water in the tank is in the prescribed set level. If water is consumed through exit pipe, the water level falls and the float comes down. Due to the pivot action, the arm of the rod connected to float comes down while the arm connected to the control valve goes up. This allows more opening for the inflow and fills the water tank until it reaches the set level. If the liquid level goes up, the arm *A* of the pivot goes up and arm *B* goes down. This pushes the control valve down and narrows the gap of incoming flow of the liquid. Once, the level of the liquid in the tank reaches the desired value, the control valve is completely closed and the inflow of the liquid in the tank is stopped. At this level, pivot arm is horizontal. This is the mechanism used in toilet bowel flushing system.

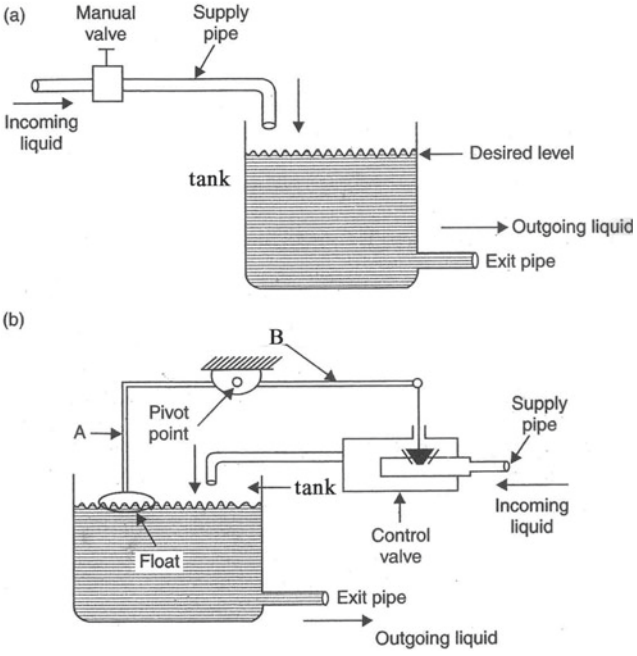


Fig. 1.7 Liquid level system. **a** Open loop system and **b** Closed loop system

1.5.4 Temperature Control System

A simple closed loop temperature control system is shown in Fig. 1.8a and its block diagram in Fig. 1.8b. A process industry requires a constant temperature θ of material being mixed in a tank be held constant. The input is 'r' a voltage and is calibrated in terms of the desired temperature. The actual temperature of the process in the tank is 'θ' and is measured by means of a thermocouple immersed in the tank. The voltage produced in the thermocouple is amplified to produce a voltage $b = K_b\theta$. This voltage is fedback and compared with the reference voltage r. The error signal (actuating signal) $e = (r - b)$ and being a weak signal it is amplified by an amplifier to get the output voltage $e_i = K_a e$. This voltage energizes the solenoid whose current $i_s = (e_i/R)$. The solenoid produces the force $f_s = K_s i_s$ and pulls the mass M which includes the mass of the boiler valve, spring K and viscous friction damping B. This activates the boiler valve which lets the steam out. The position of the valve is denoted as x. The valve position in turn controls the flow q of the hot steam from the boiler into the tank. The temperature θ of the tank is related as $\theta = K_c q$.

The block diagram of the temperature controlled process is shown in Fig. 1.8b. It is desired to keep the temperature θ of the tank constant by feedback. When the temperature falls due to disturbance or load, it is sensed by the thermocouple and its output is now reduced than what it was and b is reduced. The actuating signal $e = r - b$ is increased which increases the excitation current in the solenoid. The force f_s generated by the solenoid is increases the displacement x of the boiler valve opening. Now more steam is let into the tank which increases the temperature θ of the process in the tank. The reverse process happens when θ is increased. This time the actuating signal is reduced which in turn reduces the pull generated by the solenoid

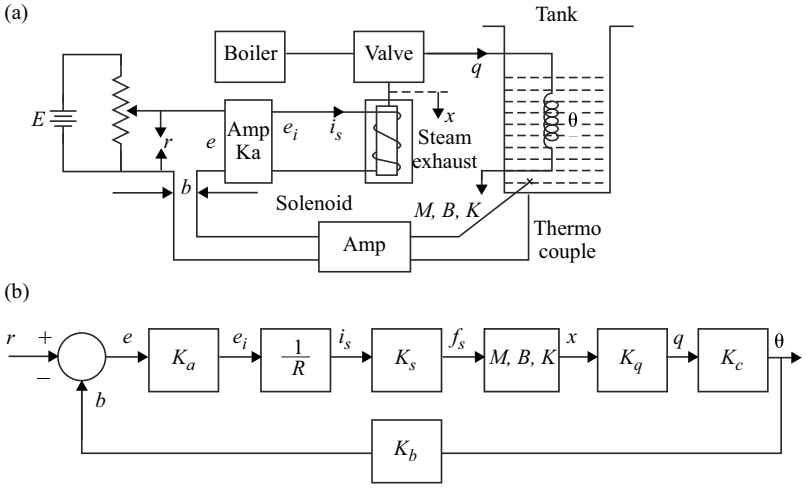


Fig. 1.8 Temperature control system. **a** Process and **b** Block diagram

and hence x. This restricts the flow of steam into the tank which ultimately reduces its temperature to the prescribed set value which is performed automatically.

1.5.5 Position Control System (Servomechanism)

A position control of an anti-aircraft gun system is shown in Fig. 1.9a and its block diagram in Fig. 1.9b. θ_i is the angular position of the input shaft and θ_0 the angular position of the output shaft. These shafts are connected to the movable arms of the potentiometer which sends out signal according to the following equation:

$$V_i = K_p(\theta_i - \theta_0) \tag{1.4}$$

where K_p is expressed in volts per radian error. This signal is weak and it is amplified by an amplifier with a gain A whose output is v_0. This voltage provides excitation to the field winding of a D.C. motor whose electromagnetic torque is given by

$$T_d = K_T I_f \tag{1.5}$$

This electromagnetic torque is used to drive the load which is the anti-aircraft gun. The gun is connected to a worm-gear arrangement. To increase the load torque, the output shaft of the motor and the worm gear are connected with gears of N teeth on the motor shaft and N_L teeth on the load side where $N_L > N$. The output arm of the potentiometer is connected to the motor shaft through gear N_c. The input

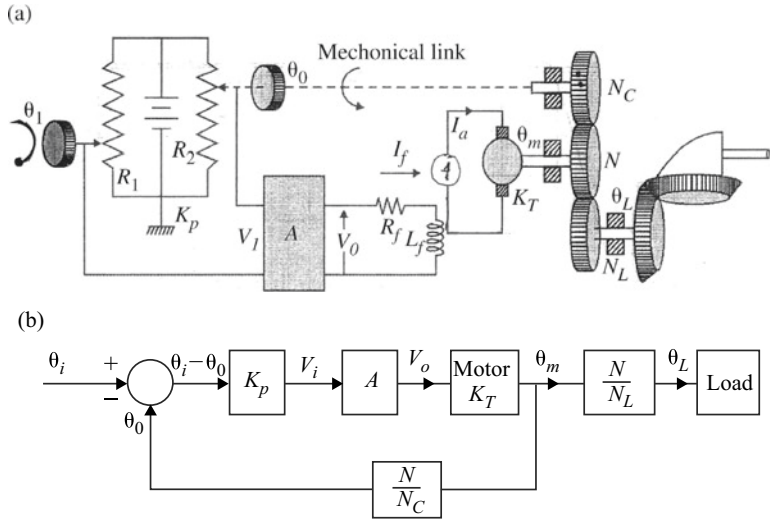


Fig. 1.9 Position control of anti-aircraft gun system. **a** Schematic diagram and **b** Block diagram

arm of the potentiometer is controlled by the radar signal which is converted into angular position. Suppose, the radar signal makes the input arm rotate through 30° in the clockwise direction. An error voltage V_i proportional to 30° appears and is amplified by the amplifier. The excitation current in the field winding of the D.C. motor interacts with the current carrying conductors of the D.C. motor armature and produces electromechanical torque which drives the anti-aircraft gun. The objective is to make the load also move through 30° in the same clockwise direction. Once $\theta_L = 30°$, the difference of $(\theta_i - \theta_0) = 0$ and the motor stops its activation. Similar operation is carried out for anti-clock wise rotation. Since the output is mechanical positioning the system is also called by the name **servomechanism**.

1.5.6 Computer Controlled System

Computer controlled system have now become very popular and intentionally introduced to control multi-variables in the plant. For large scale systems and systems with more complexity, the digital computer is successfully used as a controller. The computer can also perform supervisory functions such as scheduling many required applications in addition to providing closed loop control. The block diagram of a typical computer control system to control plant variables is shown in Fig. 1.10. The input $r(t)$, output $c(t)$ and error $e(t)$ are all continuous signals. The error signal $e(t)$ is converted into a sequence of numbers $m(kT)$ by means of analog to digital (A/D) converter where T is the sampling time. The A/D converter samples $e(t)$ into $e(kt)$. This process is called discritization. The discritized signal is converted into $m(kT)$ and applied to the digital computer. The output of the digital computer appears as $u(kT)$. Since the plant requires continuous time signal $u(t)$, a digital to analog converter (D/A) is used between the digital computer and the plant. In the computer hardware, there is a clock which sends a pulse every T seconds to A/D and D/A converters. The A/D converter sends a number to the digital computer. The A/D converter, digital computer, D/A converter and the clock are all represented by a sampling switch followed by a data extrapolator (hold circuit). Appropriate software is developed to modify the control law in the digital computer. If the output deviates from the desired one, an error appears as $e(t)$, and this error is processed in the digital computer which sends out the signals which correct the deviating variables.

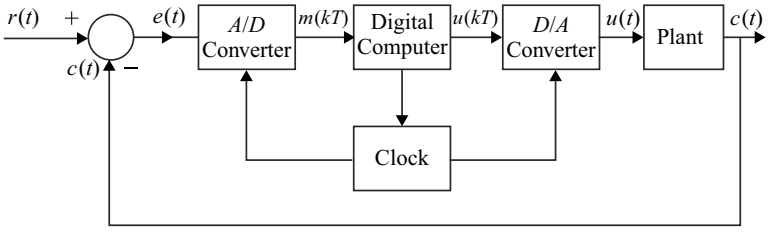


Fig. 1.10 Block diagram of a computer controlled system

1.6 System Classification

For convenience of description and mathematical analysis, systems are classified according to the nature of inputs, number of inputs/outputs inherent characteristics, design solution components used etc. Systems are broadly classified as given below:

1. Linear and non-linear.
2. Time varying and time invariant.
3. Continuous time system and discrete time system.
4. Single input–single output (SISO) and multi-input–multi-output (MIMO).
5. Deterministic and stochastic system.
6. Lumped parameter and distributed parameter system.
7. Optimal/adaptive/fuzzy type/robust control systems.

1.7 The Concept of Transfer Function

In Sect. 1.3, the dynamic behaviour of a simple mechanical system was represented by the differential Eq. (1.1) relating the input and output of the system. However, it is not a so satisfying representation from system perspective. The time domain representation of Eq. (1.1) is not a convenient way when subsystems are connected in cascade. Some analysis are easier to perform in the frequency domain. Converting the time domain model to frequency domain model of a linear time invariant continuous time system is done using Laplace transform. This transformation converts differential equation into algebraic expressions which are easier to manipulate. The Laplace transform converts functions with a real dependent variable such as time into functions with a complex dependent variable such as frequency which is often represented by s which is a complex variable and expressed as $s = \sigma + j\omega$. The transfer function is defined as follows.

The transfer function of a linear time invariant system is defined as the ratio of the Laplace transform of the output variable to the Laplace transform of the input variable with all initial conditions assumed to be zero.

The transfer function is not applicable (defined) to non-linear and time varying systems. Since the transfer function gives the description relating the input-output of the system, the information about the internal structure and its behaviour is missing which is the major disadvantage of this model. Now taking Laplace transform on both sides of the Eq. (1.1) we get

$$\begin{aligned} (Ms^2 + Bs + K)X(s) &= F(s) \\ \frac{X(s)}{F(s)} &= G(s) \\ \frac{X(s)}{F(s)} &= \frac{1}{(Ms^2 + Bs + K)} \end{aligned} \tag{1.6}$$

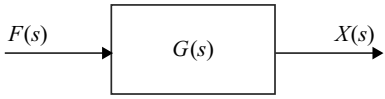


Fig. 1.11 Block diagram representation of transfer function

The transfer function $G(s)$ of system described by Eq. (1.1) is given in Eq. (1.6) and is represented in the block diagram as shown in Fig. 1.11.

From Fig. 1.11, it is to be noted that the transfer function is independent of the input. Further, it is not a function of the real variable, time or any other independent variable. While transfer function of a continuous data system is expressed as the function of the complex variable s together with the system parameter, the transfer function of a discrete time system (described by difference equation) is the function of z.

In Eq. (1.6), the denominator is a polynomial in s. System transfer function may have polynomial in the numerator also. If the order of the numerator polynomial is less than that of the denominator polynomial the transfer function is said to be **strictly proper**. Transfer function whose order of numerator polynomials is the same as that of the denominator polynomial, is called **proper** transfer function. If the order of the numerator polynomial is greater than that of the denominator, such a transfer function is called **improper**. If the denominator polynomial is set to zero, the equation so obtained is called **characteristic equation**. Transfer function is also defined for multi-variable system which has multiple inputs and outputs. Hence, the relationship is between an input–output pair when all other inputs are set to zero. The total effect on any output due to all the inputs acting simultaneously is obtained by adding up the outputs due to each input acting alone. The principle of superposition is applied since the system considered is linear.

Consider the following transfer function whose numerator and denominator are the polynomials in s.

$$G(s) = \frac{b_m s^m + b_{m-1} s^{m-1} + \cdots + b_0}{a_n s^n + a_{n-1} s^{n-1} + \cdots + a_0} \tag{1.7}$$

In Eq. (1.7) if $n > m$, the transfer function is said to be strictly proper, if $n = m$, the transfer function is proper and if $n < m$, the transfer function is improper. Further if the denominator polynomial is equated to zero, we get

$$a_n s^n + a_{n-1} s^{n-1} + \cdots + a_0 = 0 \tag{1.8}$$

Equation (1.8) is called the characteristic equation. In Eq. (1.7) if the coefficient $a_n = 1$, the denominator polynomial is called **monic polynomial**. Equation (1.8) is called as characteristic equation because it decides the stability of the system. Further, the transient and steady state responses, to a large extend depends upon the characteristic equation.

A Historical Note: Pierre-Simon Laplace (1749–1827)


Laplace (1749-1827)

Pierre-Simon, Marquis de Laplace was born on 23rd March, 1749 in France to Pierre Laplace and Marie-Ancesochan, who were farm laborers. Having his school education in his native village normondy, he was sent to university of Caen for higher studies in theology. Laplace did not study theology but was sent to Paris to work under D'Alembert by his enthusiastic teachers who awoke his zeal for mathematics. D'Alembert a mathematic genius received Laplace very poorly and wanted to get rid of him. He gave him a very thick mathematics book and asked him to come back when he had read it. Laplace came back over night. D'Alembert was very impressed by the way Laplace answered all his queries which were difficult to answer and recommended him for a teaching place in a military college. For the next seventeen years from 1771–1787, he devoted his time for the original research work in astronomy.

Laplace was elected as the associated member in the French academy of science at the age of 24. During the period 1799–1825 he published his works in celestial mechanics in five volumes. He formulated Laplace's equations, Laplace transform and the Laplacian differential operator which are widely used now in college level mathematics. One of the biggest contributions of Laplace is the development of nebular hypothesis of the origin of the solar system and developed the theory of black holes and the development of gravitational collapse. He is remembered as one of the greatest scientists of all time and sometimes referred to as a French Newton. He became a Count of the French empire in 1806 and was named a Marquis in 1817. Laplace contemporaries include the great mathematical wizards like Lagrange, Legendre, Monge, Lacorse, Fourier and French ruler Napoleon Bonaparte who himself was great scientist. Laplace married Marie-Charlotte in his late thirties and the couple had a daughter, Sophie and a son, Charles-Emile. The all time great scientist died on 5 March, 1827 in Paris at the age of 77.

1.7.1 Poles and Zeros of the Transfer Function

Consider the transfer function given in Eq. (1.7). The numerator and denominator polynomials can be written in factors as given below:

$$G(s) = \frac{(s+z_1)(s+z_2)\cdots(s+z_m)}{(s+p_1)(s+p_2)\cdots(s+p_n)} \tag{1.9}$$

In Eq. (1.9) the factors in the numerator are called zeros of the transfer function and the factors in the denominator are called poles of the transfer function. The poles and zeros can be located in the complex s-plane at $s = -p_1$, $s = -p_2$ etc., and the zeros at $s = -z_1$, $s = -z_2$ etc. The poles locations are marked with a cross $\times$ and the zero locations are marked with a small circle $\bigcirc$. It is also noticed in Eq. (1.9), at $s = -p_1$ or $s = -p_2$ etc., the T.F. (transfer function) becomes infinitive and at $s = -z_1$ or $s = -z_2$ etc. the T.F. becomes zero. $p_1, p_2, \ldots, p_n$ and $z_1, z_2, \ldots, z_n$ may be real or complex. Thus, the factors of the denominator of the transfer function which are called zeros are defined as "**the points in the complex s-plane at which the transfer function becomes zero**". Similarly the factors of the denominator of the transfer function which are called poles are defined as "**the points in the complex s-plane at which the transfer function becomes infinitive**".

1.7.2 Complex s-Plane and the Locations of Poles and Zeros

The complex variable $s = \sigma + j\omega$ and the s-plane is shown in Fig. 1.12. The horizontal axis represents the positive and negative real parts and the vertical axis represents positive and negative imaginary parts. It contains four quadrants as shown in Fig. 1.12. The infinite space complex s-plane is divided by the vertical imaginary as Left Half Plane (LHP) and Right Half Plane (RHP). The importance of this division plays significant role in the system stability and the transient response studies which will be dealt with in the chapters to follow.

Now let us consider the location of the poles and zeros in the complex s-plane. Consider the following transfer function:

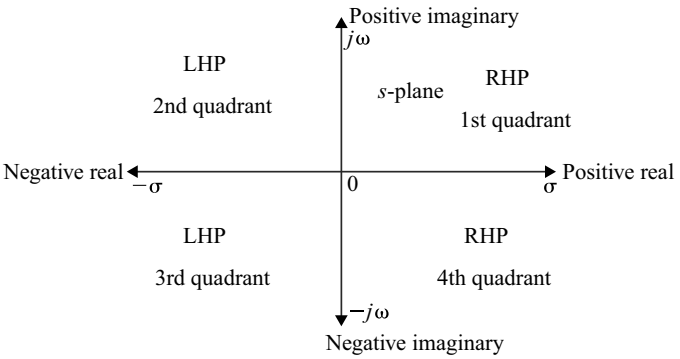


Fig. 1.12 Complex s-plane diagram

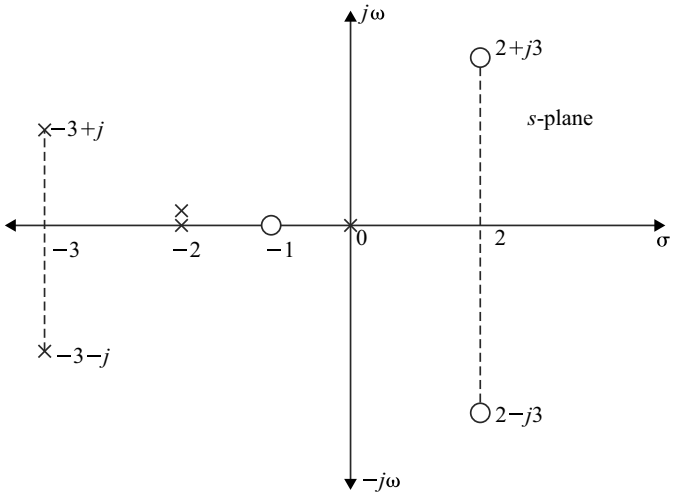


Fig. 1.13 Pole-zero diagram of Eq. (1.10)

$$G(s) = \frac{10(s-2+j3)(s-2-j3)(s+1)}{s(s+2)^2(s+3+j)(s+3-j)} \tag{1.10}$$

In Eq. (1.10) there are three zeros and five poles. Thus, the order of the numerator polynomial is three and that of the denominator is five. In all practically realizable system the order of the denominator polynomial is higher than that of the numerator polynomial. Hence, system Eq. (1.10) is strictly proper. The characteristic equation is $s(s+2)^2(s+3+j)(s+3-j) = 0$. System Eq. (1.10) is called fifth order system. The order of the system is defined as follows.

The order of the system is defined as the highest order of the denominator polynomial or the numerator polynomial whichever is greater.

Now we place the poles and zeros of Eq. (1.10) in the s-plane as shown in Fig. 1.13.

The poles and zeros are located in the s-plane by equating the pole factors and the zero factors to zero. For example consider the zero $(s+1)$. When this is equated to zero we get $s = -1$. At negative real with one magnitude, this zero is located with the marking 0. It is to be noted that the complex poles and zeros which will occur only in pair are also located in the same way. For example the complex pole $s+3+j=0$ or $s=-3-j$ is located in LHP in the third quadrant and its compliment $(s+3-j)$ which is the mirror image is located in LHP in the second quadrant. Also note that there are two poles at $s=-2$ and they are called repeated poles marked twice at $s=-2$. Further, the pole $s=0$, which is called a pure integrator is located at the origin in the s-plane. By the number of poles located at the origin, the systems are designated by its Type. Since there is one pole located at the origin of the s-plane, system Eq. (1.10) is called Type one system. If no pole is located at the origin the system is designated as Type zero and so on. Hence, the Type of the system is defined as follows.

The Type of the system is defined by the number which represents the presence of pure integrators in the transfer function. This is equivalent to saying the number of poles of the T.F. located at the origin of the s-plane.

The number of zeros located at the origin does not define the Type of the system.

1.7.3 Sinusoidal, Minimum Phase, Non-minimum Phase and All Pass Transfer Functions

1.7.3.1 Sinusoidal Transfer Function

Consider system shown in Fig. 1.14a where $R(s)$ is the Laplace transform of input variable $r(t)$ and $C(s)$ is the Laplace transform of the output variable. $G(s)$ is the transfer function of the system which is related by the following equation:

$$\frac{C(s)}{R(s)} = G(s) \tag{1.11}$$

If the input $r(t) = A \sin \omega t$, a sinusoid, for a linear system, that the output is also a sinusoid with a phase shift of ϕ. This is obtained after the input signal is processed by the system. For Fig. 1.14b, the following equation is obtained:

$$\frac{C(j\omega)}{R(j\omega)} = G(j\omega) \tag{1.12}$$

Equation (1.12) is obtained from Eq. (1.11) simply replacing s by $j\omega$.

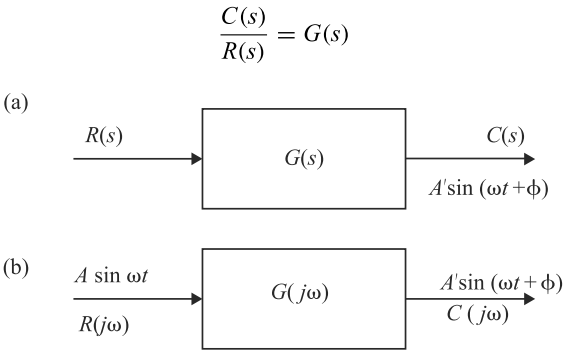


Fig. 1.14 Representation of sinusoidal transfer function

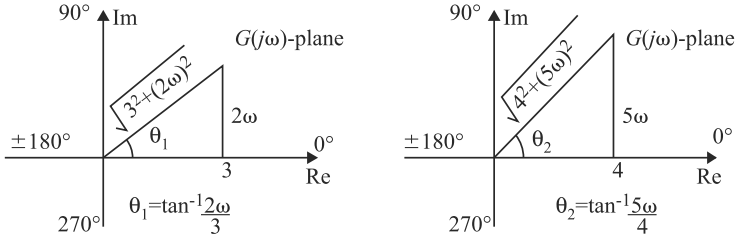


Fig. 1.15 Representation of complex minimum phase function

From transfer function $G(s)$ sinusoidal T.F $G(j\omega)$ is obtained replacing s by $j\omega$. Sinusoidal T.F. is very useful in the study of frequency response method of analysis which will be discussed later in detail.

1.7.3.2 Minimum Phase Transfer Function

Consider the following T.F.

$$G(s) = \frac{10(2s+3)}{(5s+4)}$$

The sinusoidal T.F. is obtained replacing s by $j\omega$.

$$G(j\omega) = \frac{10(j2\omega+3)}{(j5\omega+4)}$$

In the complex plane a complex function is represented with its real and imaginary parts as shown in Fig. 1.15.

From Fig. 1.15, the magnitude and phase angle of $G(j\omega)$ are obtained as,

$$G(j\omega) = \frac{10\sqrt{(9+4\omega^2)}}{\sqrt{(16+25\omega^2)}}\angle\theta_1 - \theta_2 \tag{1.13}$$

Form Fig. 1.15, it is evident that θ_1 and θ_2 when measured from positive real axis (reference axis) for any particular ω, they give minimum phase shift. This is possible if the poles and zeros are in the LHP of the s-plane. Hence, the following definition follows.

A minimum phase transfer function is defined as the transfer function whose poles and zeros are located in the left half s-plane.

Here the zero is located at $s = (-3/2)$ and pole at $s = (-4/5)$.

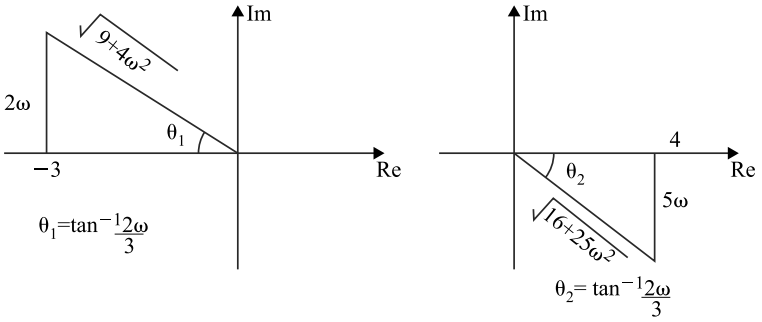


Fig. 1.16 Representation of non-minimum phase transfer function

1.7.3.3 Non-minimum Phase Transfer Function

Consider the following transfer function:

$$G(s) = \frac{(2s-3)}{(-5s+4)}$$

The sinusoidal transfer function is given by

$$\begin{aligned} G(j\omega) &= \frac{(j2\omega-3)}{(-j5\omega+4)} \\ &= \frac{(4\omega^2+9)}{\sqrt{25\omega^2+16}} \frac{\angle 180^\circ - \theta_1}{\angle 360^\circ - \theta_2} \end{aligned}$$

The phase angle contribution of the poles and zero of the given T.F. is not minimum. Maximum value of θ_1 and θ_2 is 90°. However, the zero contributes more than 90° and the pole contributes more than 270°. Hence, they are called non-minimum phase T.F. The pole and zero are located in RHP of the s-plane. Hence, the non-minimum phase transfer is defined as follows.

The non-minimum phase transfer function is defined as the transfer function which has atleast a pole or a zero located in the RHP of the s-plane.

In the above T.F. the zero is located at $s = \frac{3}{2}$ and the pole is located at $s = \frac{4}{5}$. which are points on the positive real in RHP.

1.7.3.4 All Pass Transfer Function

Consider, for example the following transfer functions:

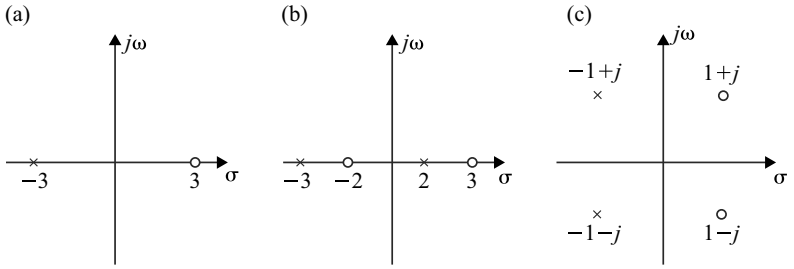


Fig. 1.17 Representation of all pass transfer function

$$\text{(a)} \quad G(s) = \frac{(s-3)}{(s+3)}$$

$$\text{(b)} \quad G(s) = \frac{(s+2)(s-3)}{(s-2)(s+3)}$$

$$\text{(c)} \quad G(s) = \frac{(s^2-2s+2)}{(s^2+2s+2)}$$

The pole-zero locations of the above transfer functions are shown in Fig. 1.17a–c respectively. Here, corresponding to a pole, a zero is symmetrically placed with respect to the $j\omega$ axis. Consider the following T.F.

$$\begin{aligned} G(s) &= \frac{(s-3)}{(s+3)} \\ G(j\omega) &= \frac{(j\omega-3)}{(j\omega+3)} \\ &= \sqrt{\frac{(\omega^2+9)}{(\omega^2+9)}}\frac{\angle 180^\circ - \tan^{-1}\omega/3}{\angle \tan^{-1}\omega/3} \\ &= 1\angle 180^\circ - 2\tan^{-1}\omega/3 \end{aligned}$$

Form the above expression, it is evident that the input signal is processed by the system and the output is sent with a gain of 1. This means that the system passes the signal without amplification or atteneuation. However, there is a phase shift which varies from 180° to 0° for $0 \leq \omega \leq \infty$. Hence, the all pass transfer function is defined as follows.

An all pass transfer function is defined as the T.F. which has poles and zeros symmetrically placed with respect to the $j\omega$ axis. Such systems do not alter the amplitude of the input signal.

1.8 System Representation in Simple Block Diagram

Systems are often represented in a simple block diagram as shown in Fig. 1.18a where the symbols have the following meaning:

$R(s)$ = Laplace transform of the input variable $r(t)$.
$C(s)$ = Laplace transform of the output variable $c(t)$.
$E(s)$ = Laplace transform of the error signal $e(t)$.
$G(s)$ = Forward path or open loop transfer function.
$H(s)$ = Feedback path transfer function.
$GH(s)$ = Loop transfer function.

From Fig. 1.18a, the following equations are written:

$$\begin{aligned} E(s) &= R(s) - CH(s) \\ EG(s) &= C(s) \\ [R(s) - CH(s)]G(s) &= C(s) \end{aligned}$$

Re-arranging the terms, we get

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + GH(s)} \tag{1.14}$$

Equation (1.14) is called the closed loop transfer function of the system. For positive feedback it can be shown as

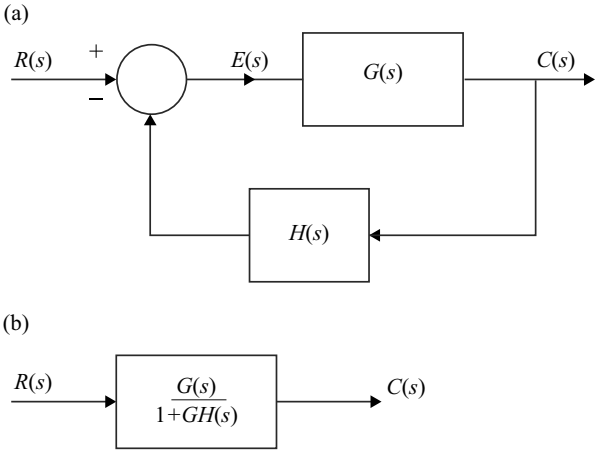


Fig. 1.18 System representation in simple block diagram

Table 1.1 Voltage-current and impedance relationship

Component	Voltage-current	Current-voltage	Impedance $Z(s) = \frac{V(s)}{I(s)}$
Capacitor, C	$v(t) = \frac{1}{C}\int i(t)dt$	$i(t) = C\frac{dv(t)}{dt}$	$\frac{1}{Cs}$
Inductor, L	$v(t) = L\frac{di(t)}{dt}$	$i(t) = \frac{1}{L}\int v(t)dt$	Ls
Resistor, R	$v(t) = i(t)R$	$i(t) = \frac{1}{R}v(t)$	R

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 - GH(s)} \tag{1.15}$$

Unless otherwise, it is mentioned, the system is always given negative feedback. From Eq. (1.14), the characteristic equation is written as,

$$F(s) = 1 + GH(s) = 0 \tag{1.16}$$

For example, if $G(s) = [10/s(s+2)]$ and $H(s) = (s+5)$ the following closed loop T.F. is obtained.

$$\frac{C(s)}{R(s)} = \frac{10}{s^2 + 12s + 50}$$

The characteristic equation is written as

$$s^2 + 12s + 50 = 0.$$

1.9 Transfer Function Model of Electrical Network

In this section, we determine the transfer function of electric circuits from the mathematical model. The electric circuit consists of three passive linear components namely resistor, inductor and capacitor. Mathematical equations are written connecting the input and output following Kirchhoff's laws. Differential equations are written either for the sum of the voltages around loops or sum of currents at nodes and then equated to zero. By taking Laplace transforms of the differential equations, the transfer function is obtained. Alternatively, the transfer function can also be obtained by transfer method. Table 1.1 gives the components and their relationship between voltage and current under zero initial conditions and also the impedance relationship.

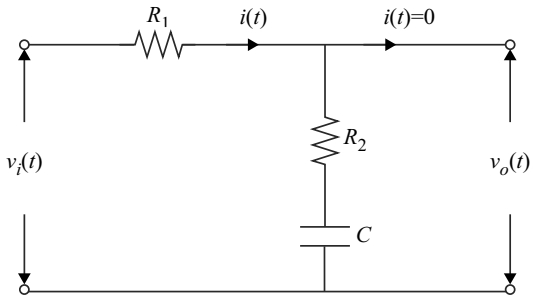


Fig. 1.19 Electrical network for Example 1.1

Example 1.1

Determine the T.F. $\frac{V_0}{V_i}(s)$ for the electrical network shown in Fig. 1.19.

Solution For the circuit shown in Fig. 1.19, the following Kirchhoff's loop equation is written

$$v_i(t) = (R_1 + R_2)i(t) + \frac{1}{C}\int i(t)dt$$
$$v_0(t) = R_2 i(t) + \frac{1}{C}\int i(t)dt$$

Taking Laplace transform on both sides of the above equations we get

$$V_i(s) = (R_1 + R_2)I(s) + \frac{1}{Cs}I(s)$$
$$= \frac{[(R_1 + R_2)Cs + 1]}{Cs}I(s)$$
$$V_0(s) = \left[R_2 + \frac{1}{Cs}\right]I(s)$$
$$= \frac{[(R_2 Cs + 1)]}{Cs}I(s)$$

Dividing the above equations one by the other we get

$$\boxed{\frac{V_0}{V_i}(s) = \frac{(1 + R_2 Cs)}{(1 + (R_1 + R_2)Cs)}}$$

It is to be noted, the time function is written with lower case letters and the s functions are written with upper case letters.

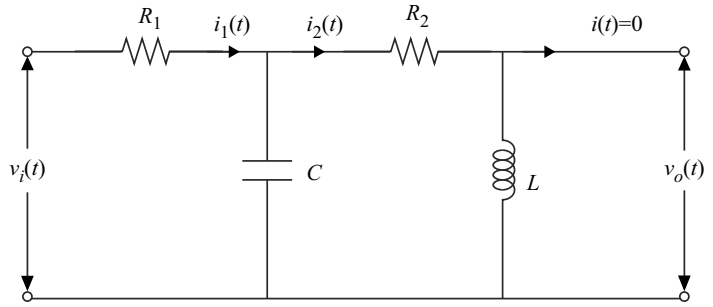


Fig. 1.20 Electrical network for Example 1.2

Example 1.2

Find the T.F. of the network shown in Fig. 1.20.

Solution For the $(v_i(t) - R_1 - C)$ loop we write the following equation.

$$v_i(t) = R_1 i_1(t) + \frac{1}{C}\int (i_1(t) - i_2(t))dt$$

Taking Laplace transform on both sides we get

$$V_i(s) = \left(R_1 + \frac{1}{Cs}\right) I_1(s) - \frac{1}{Cs} I_2(s) \tag{1.17}$$

For the loop consisting of $R_2 - L - C$, the following equation is written

$$R_2 i_2(t) + L\frac{di_2(t)}{dt} + \frac{1}{C}\int (i_2(t) - i_1(t))dt = 0$$

Taking Laplace transform on both sides we get

$$\left(R_2 + Ls + \frac{1}{Cs}\right) I_2(s) = \frac{1}{Cs} I_1(s)$$

Solving for $I_1(s)$ we get

$$I_1(s) = (R_2 Cs + LCs^2 + 1) I_2(s) \tag{1.18}$$

Substituting for $I_1(s)$ in (1.17) we get the following equation

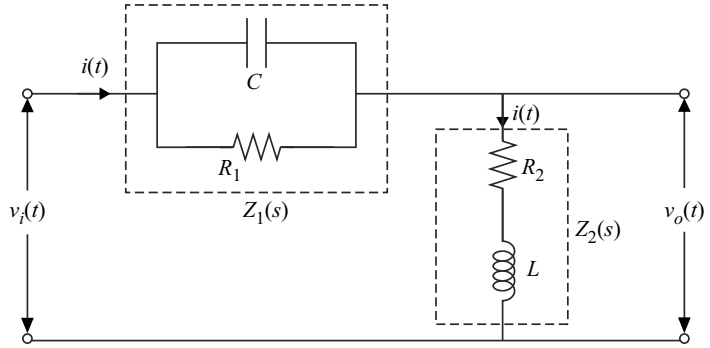


Fig. 1.21 Electrical network for Example 1.3

$$V_i(s) = \frac{[(R_2Cs + LCs^2 + 1)(1 + R_1Cs) - 1]}{Cs} I_2(s)$$

$$v_0(t) = L\frac{di_2(t)}{dt}$$

$$V_0(s) = L\,s\,I_2(s)$$

$$= \frac{Ls\,V_i(s)Cs}{[(R_2Cs + LCs^2 + 1)(1 + R_1Cs) - 1]}$$

$$\boxed{\frac{V_0}{V_i}(s) = \frac{Ls}{[LR_1Cs^2 + (R_1R_2C + L)s + (R_1 + R_2)]}}$$

The problem of this nature can be solved easily using signal flow graph which is explained later.

Example 1.3

Find the T.F. of the electrical network shown in Fig. 1.21 using impedance transform method.

Solution Let $Z_1(s)$ be the impedance function for $R_1 - C$ parallel combination and $Z_2(s)$ for $R_2 - L$ series combination.

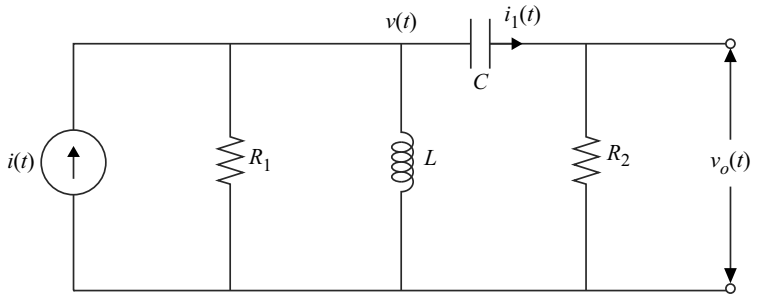


Fig. 1.22 Electrical network for Example 1.4

$$
\begin{aligned}
Z_1(s) &= \frac{\frac{1}{Cs}R_1}{\frac{1}{Cs}+R_1} \\
&= \frac{R_1}{1+R_1Cs} \\
Z_2(s) &= R_2 + Ls \\
V_i(s) &= I(s)[Z_1(s)+Z_2(s)] \\
I(s) &= \frac{V_i(s)}{Z_1(s)+Z_2(s)} \\
V_0(s) &= I(s)Z_2(s) \\
\frac{V_0(s)}{V_i(s)} &= \frac{Z_2(s)}{Z_1(s)+Z_2(s)} \\
&= \frac{R_2+Ls}{\frac{R_1}{(1+R_1Cs)}+R_2+Ls} \\
\frac{V_0}{V_i}(s) &= \frac{(R_2+Ls)(1+R_1Cs)}{R_1+(R_2+Ls)(1+R_1Cs)}
\end{aligned}
$$

$$
\boxed{\frac{V_0}{V_i}(s) = \frac{R_1LCs^2+(R_1R_2C+L)s+R_2}{R_1LCs^2+(R_1R_2C+L)s+R_1+R_2}}
$$

Example 1.4

For the electrical network shown in Fig. 1.22, find the transfer function $\frac{V_0}{I}(s)$ by impedance transform method.

Solution For the electrical network shown in Fig. 1.22 the following equation in terms of Laplace transform variable is written at node $V(s)$.

$$I(s) = \frac{V(s)}{R_1} + \frac{V(s)}{Ls} + (V(s) - V_0(s)Cs) \tag{1.19}$$

$$\begin{aligned} V_0(s) &= I_1(s)R_2 \\ &= (V(s) - V_0(s)CsR_2 \\ V_0(s)(1 + R_2Cs) &= V(s)R_2Cs \end{aligned}$$

$$V(s) = \frac{(1 + R_2Cs)V_0(s)}{R_2Cs} \tag{1.20}$$

Substituting Eq. (1.20) in (1.19) for $V(s)$ we get

$$\begin{aligned} I(s) &= V(s)\left(\frac{1}{R_1} + \frac{1}{Ls} + Cs\right) - V_0Cs \\ &= V_0(s)\left[\frac{1 + R_2Cs}{R_2Cs}\left(\frac{1}{R_1} + \frac{1}{Ls} + Cs\right) - Cs\right] \\ &= V_0(s)\left[\frac{(1 + R_2Cs)(Ls + R_1 + R_1LCs^2)}{R_1R_2CLs^2} - Cs\right] \end{aligned}$$

$$\boxed{\frac{V_0}{I}(s) = \frac{(R_1R_2LCs^2)}{[(R_1 + R_2)LCs^2 + (R_1R_2C + L)s + R_1]}}$$

1.10 Transfer Function Model of Mechanical Systems

Mechanical systems, like electrical networks, have three passive linear components. They are classified as mechanical translational and mechanical rotational systems. In the translational mechanical system, linear displacement, linear velocity and linear acceleration are the variables of motion while force is the input. The mass, spring and dash-pot which offers viscous friction or damping are the three passive elements. The spring and the mass are the energy storage elements while viscous damper dissipates the energy. A lever a matching device rested on a pivot, transmits energy from one part of the system to another so that the force, speed and displacement may be altered and maximum power transfer obtained.

In the rotational mechanical system, the angular displacement, angular velocity and angular acceleration are the variables of motion and torque is the input applied to the system. Like the translational system, there are three elements in rotational system also. While the inertia and the spring store the energy, the viscous damper (dash-pot) dissipates energy. Like a lever in the translational system, gear arrangements are used as a matching device in rotational system which is used to alter torque, angular speed and displacement. Thus, energy is transmitted from one part of the system to another part and maximum power transfer is obtained. Transfer function model of

these mechanical system is obtained in the same way used for the electrical network. This is discussed below.

1.11 Transfer Function Model for Mechanical Translational System

The basic elements of the translational systems are mass, spring, dash-pot and lever. Newton law of motion is applied according to which the algebraic sum of the forces acting at a point is zero. This law, as applied to the above elements is discussed below.

The units of the above elements and the associated variables are given below:

1. Mass, M = kilogram (kg).
2. Viscous friction coefficient, B = N/m/s.
3. Spring constant, K = N/m.
4. Force, $f(t)$ = Newton (N).
5. Displacement, $x(t)$ = metre (m).
6. Velocity, $v(t)$ = m/s.
7. Acceleration, $a(t)$ = m/s^2.

In Fig. 1.23, the components of mechanical translational system and their free-body diagrams are represented. Form the free-body diagrams the following equations are written as per Newton's law of motion.

For mass M,

$$M\frac{d^2x(t)}{dt^2} = f(t)$$

For dash-pot B,

$$B\frac{dx(t)}{dt} = f(t)$$

For spring K,

$$Kx(t) = f(t)$$

For lever

$$\begin{aligned} f_a(t)l_1 &= f_b(t)l_2 \\ \frac{x}{l_1} &= \frac{y}{l_2} \end{aligned} \tag{1.21}$$

Now consider the dash-pot B whose one end is neither connected to any mass nor to any fixed reference frame. Force $f(t)$ is applied to the other end as shown in Fig. 1.24. Its free-body diagram is also shown. At the point where the force $f(t)$ is applied, the following equation is written

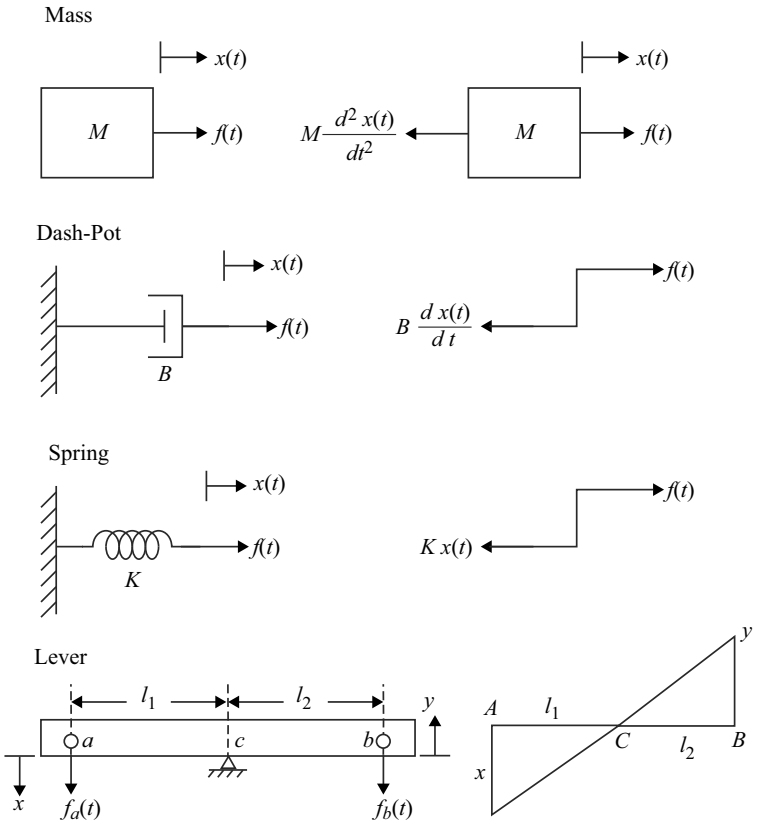


Fig. 1.23 Forces acting on mechanical translational system components and their free-body diagram

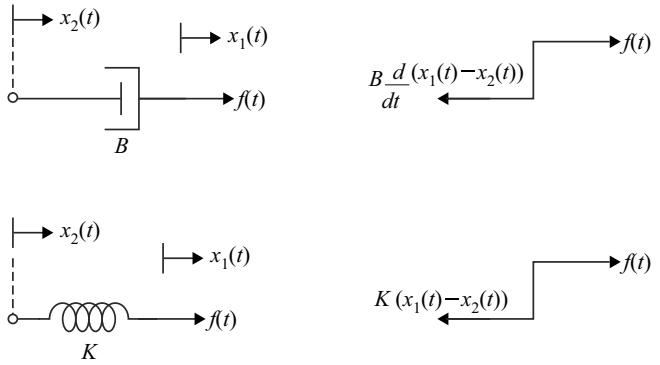


Fig. 1.24 Dash-pot and spring whose ends are not connected to the fixed frame

$$B\frac{d}{dt}(x_1(t) - x_2(t)) = f(t) \tag{1.22}$$

Here $dx_1(t)/dt$ is treated as the final velocity and $\frac{dx_2(t)}{dt}$ as the initial velocity. At the point of application of the applied force, the opposing force generated by the dash-pot is proportional to **the final velocity minus initial velocity**.

Similarly consider the spring whose one end is neither connected to any mass nor to the reference frame. At the point where the force is applied from Fig. 1.24, the following equation is written

$$K(x_1(t) - x_2(t)) = f(t) \tag{1.23}$$

$$K(\text{Final displacement-initial displacement}) = \text{Applied force}$$

In the above two cases, the final velocity and the final displacement are taken at the point of application of the force $f(t)$. The initial velocity and initial displacement are taken from the other end of the dash-pot and spring respectively. If any end of these elements is connect6ed to the mass, that end moves with the displacement and velocity of that particular mass. **It is to be noted that each mass has to be given individual displacement**. If there are n masses in the system, there should be n displacements to be identified in addition to other displacements depending upon the connections made by the spring, dash-pot and levers. Then only complete description is possible. The following step by step procedure is followed while writing equation of motion for translational mechanical system.

Step by Step Procedure

1. For each mass give one independent displacement x. If there are n masses connected in the system, there will be n degree of freedom and their displacements are identified as $x_1, x_2 \ldots\ldots x_n$.
2. Identify the ends of the spring or dash-pot which are neither connected to any mass nor to any reference frame. These ends move with additional degree of freedom and hence with different displacement. Include the lever arrangement connected if any.
3. Draw the free-body diagram for each degree of freedom and mark the various forces acting.
4. Write down the equation for each free-body diagram using D'Alembert's principle according to which Σ forces $= 0$. (The algebraic sum of the forces acting at a point is zero).
5. Take Laplace transform for equations written in step 4. By solving simultaneous equations, retain only the Laplace transform of the output variable and Laplace transform of the input variable and eliminate intermittent variables.
6. The transfer function of the given system is obtained by the ratio of the Laplace transform of output variable to the Laplace transform of the input variable.

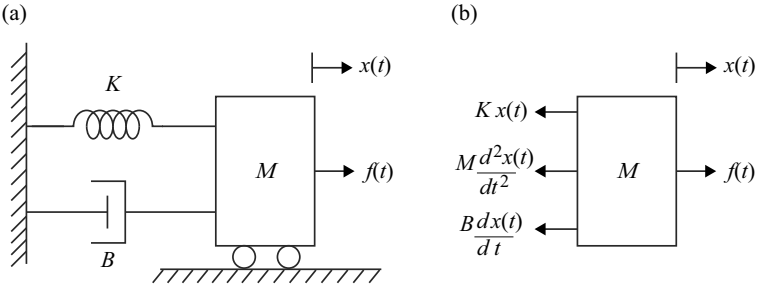


Fig. 1.25 **a** Translational mechanical system and **b** Free-body diagram

The following examples illustrate the above procedure.

Example 1.5

Consider the mechanical system shown in Fig. 1.25a. The applied force $f(t)$ is the input and the displacement $x(t)$ of mass M is the output variable. Obtain the T.F.

Solution

1. The mass M is given a displacement $x(t)$ which acts in the direction of the applied force $f(t)$. The opposing force generated is $Md^2x(t)/dt^2$ and acts opposite to the applied force.
2. One end of the spring K is connected to the mass M. This end moves with a displacement $x(t)$. Its other end is connected to the reference frame. The opposing force generated by the spring is $Kx(t)$ and acts opposite to the applied force.
3. The behavior of the dash-pot is similar to that of the spring except that the opposing force generated is equal to $B\frac{dx(t)}{dt}$.
4. The free body diagram is shown in Fig. 1.25b and various forces acting in the system are marked. Using D'Alembert's principle, the sum of the forces acting towards the left is equated to the sum of the forces acting towards the right. Thus, the following equation is obtained:

$$M\frac{d^2x(t)}{dt^2} + B\frac{dx(t)}{dt} + Kx(t) = f(t)$$

5. Taking Laplace transform on both sides of the above equation, the following equation is obtained:

$$[Ms^2 + Bs + K]X(s) = F(s)$$

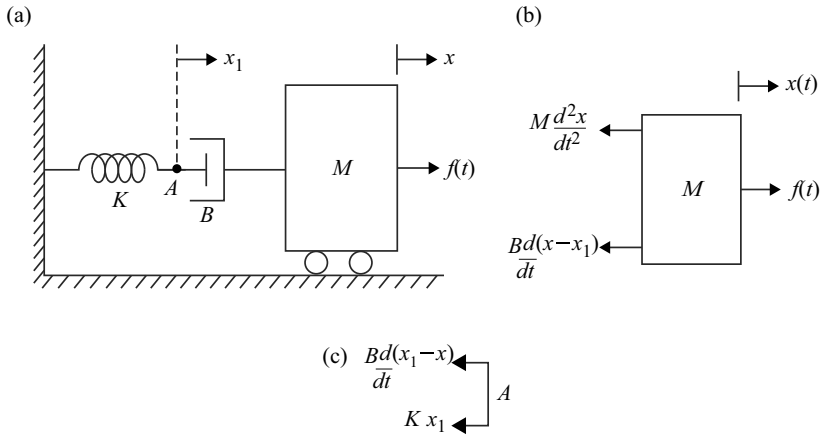


Fig. 1.26 **a** Mechanical translational system **b** and **c** Free-body diagrams

$$\boxed{\frac{X}{F}(s) = \frac{1}{Ms^2 + Bs + K}}$$

The system has only a single degree of freedom. The notation $x(t)$, for simplicity is written as x hereafter.

Example 1.6

Consider the mechanical system shown in Fig. 1.26a, where all the three elements namely mass, damper and spring are connected in cascade. Determine the T.F. with the displacement of mass M as the output variable and force $f(t)$ as the input variable.

Solution

1. The mass M is given independent displacement x. One end of the dash-pot is connected to the mass M and the other and to the spring K. Hence, one end of the dash-pot is neither connected to the mass nor to any reference frame. Hence, this point is marked as A and let it move with a displacement x_1. Thus, the system has two degrees of freedom, one for the mass and another at point A.
2. When we apply D'Alembert's principle at mass M, x is taken as the final displacement and x_1 as the initial displacement. However, at point A, the final displacement is taken as x_1 and initial displacement as x. The free-body diagrams are shown in Fig. 1.26a, b. From these free-body diagrams, the following equations are written

$$M\frac{d^2x}{dt^2} + B\frac{d}{dt}(x - x_1) = f(t)$$

Taking Laplace transform on both sides, we get

$$(Ms^2 + Bs)X(s) - BsX_1(s) = F(s) \tag{1.24}$$

For Fig. 1.26b the following equation is written

$$B\frac{d}{dt}(x_1 - x) + Kx_1 = 0$$

Taking Laplace transform on both sides we get

$$(Bs + K)X_1(s) = BsX(s) \tag{1.25}$$

Substituting for $X_1(s)$ from (1.25) in (1.24) we get

$$(Ms^2 + Bs)X(s) - \frac{B^2s^2}{(Bs + K)}X(s) = F(s)$$
$$\frac{[(Ms^2 + Bs)(Bs + K) - B^2s^2]}{(Bs + K)}X(s) = F(s)$$

$$\boxed{\frac{X}{F}(s) = \frac{(Bs + K)}{s(MBs^2 + MKs + BK)}}$$

Example 1.7

For the mechanical translational systems shown in Fig. 1.27a, b find the transfer function relating the displacement of the mass M and the applied force $f(t)$.

Solution Consider the mechanical system shown in Fig. 1.27a. Its free-body diagram is shown in Fig. 1.27c. From free-body diagram, the following equations are written

$$M\frac{d^2x}{dt^2} + B\frac{dx}{dt} + K_2x + K_1(x - x_1) = 0$$

Taking Laplace transform on both sides we get

$$[Ms^2 + Bs + K_2]X(s) = K_1[X_1(s) - X(s)] \tag{1.26}$$

But

$$K_1(x_1 - x) = f(t)$$
$$K_1[X_1(s) - X(s)] = F(s) \tag{1.27}$$

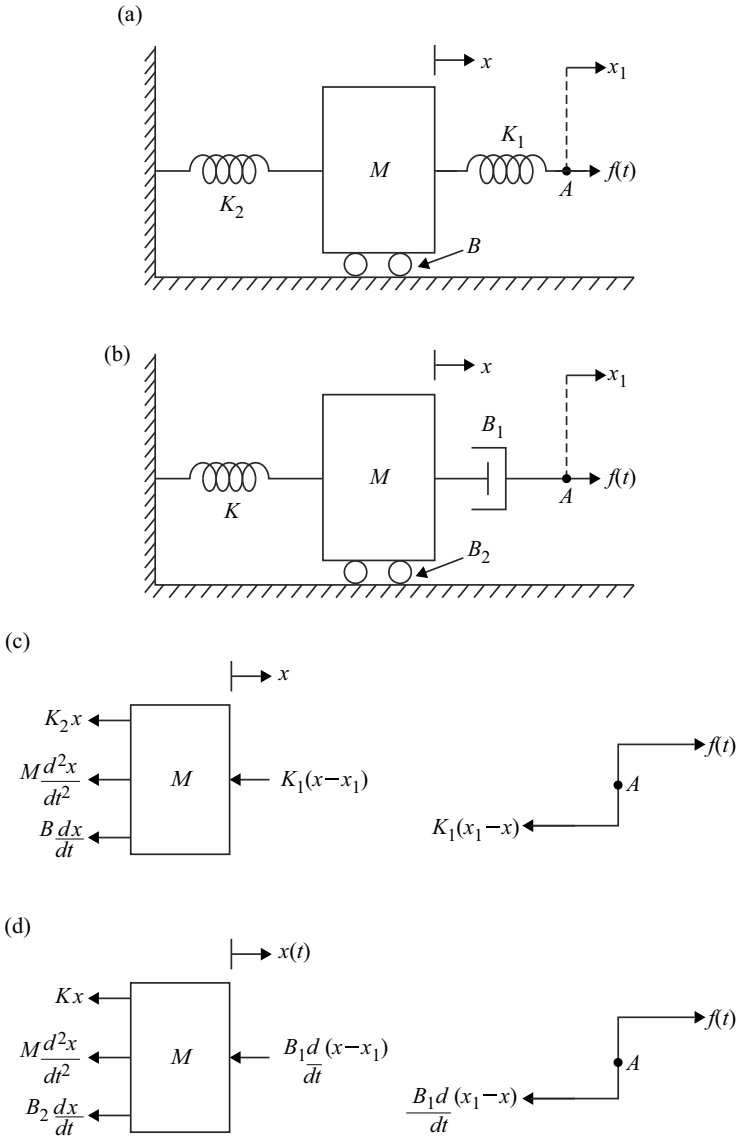


Fig. 1.27 Mechanical systems for Example 1.7. **c** Free-body diagrams of Fig. 1.27a, d Free-body diagram of Fig. 1.27b

Substituting Eq. (1.27) in (1.26) we get

$$\boxed{\frac{X}{F}(s) = \frac{1}{(Ms^2 + Bs + K_2)}}$$



Note: When the force $f(t)$ is applied to the spring K_1, it is transformed and two displacements x_1 and x must be assigned to the end points of the spring. Further it is to be noted that the above T.F. determined is independent of K_1. Here K_1 just transforms $f(t)$ from one point to another point. However, if $\frac{X_1}{F}(s)$ is determined it will contain all system elements including K_1.

Now consider the mechanical system shown in Fig. 1.27b. The free-body diagram of this is shown in Fig. 1.27d. From the free-body diagram, the following equations are written.

$$M\frac{d^2x}{dt^2} + B_2\frac{dx}{dt} + Kx + B_1\frac{d}{dt}(x - x_1) = 0$$

Taking Laplace transform on both sides we get

$$[Ms^2 + B_2s + K]X(s) = B_1s[X_1(s) - X(s)] \tag{1.28}$$

Also,

$$\begin{aligned} B_1\frac{d}{dt}(x_1 - x) &= f(t) \\ B_1s[X_1(s) - X(s)] &= F(s) \end{aligned} \tag{1.29}$$

Substituting Eq. (1.29) in Eq. (1.28) we get

$$\boxed{\frac{X(s)}{F(s)} = \frac{1}{(Ms^2 + B_2s + K)}}$$

Example 1.8

Consider the mechanical system shown in Fig. 1.28a. Determine the T.F. relating the displacement of mass M_1 and the force $f(t)$.

Solution The free-body diagram for mass M_1 and mass M_2 of Fig. 1.28a are shown in Fig. 1.28b, c respectively. For Fig. 1.28b the following equation is written.

$$M_1\frac{d^2x_1}{dt^2} + B_1\frac{dx_1}{dt} + B\frac{d}{dt}(x_1 - x_2) + K(x_1 - x_2) = 0$$

Taking Laplace on both sides, we get

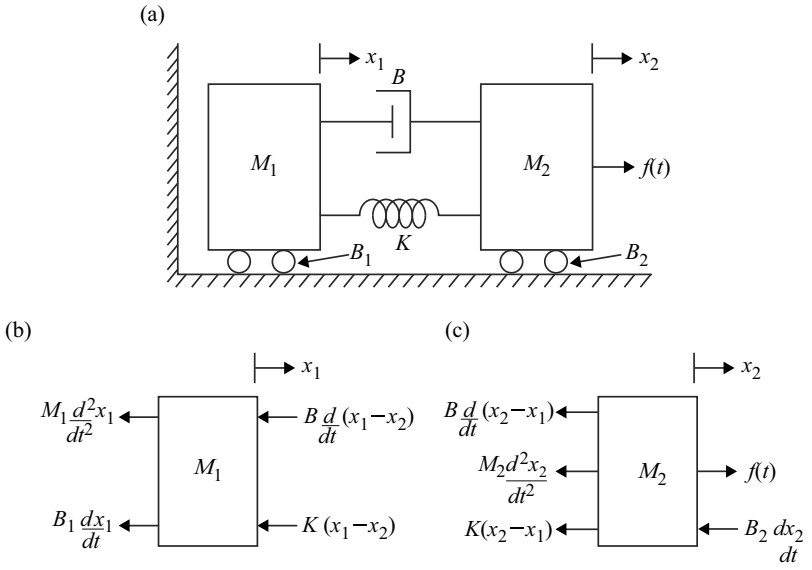


Fig. 1.28 **a** Mechanical System for Example 1.8. **b** and **c** Free-body diagram of mechanical system of Fig. 1.28a

$$[M_1 s^2 + (B_1 + B)s + K]X_1(s) = [Bs + K]X_2(s) \tag{1.30}$$

From the free-body diagram of mass M_2, the following equation is written.

$$M_2 \frac{d^2x_2}{dt^2} + B_2 \frac{dx_2}{dt} + B \frac{d}{dt}(x_2 - x_1) + K(x_2 - x_1) = f(t)$$

Taking Laplace transform on both sides, we get,

$$[M_2 s^2 + (B_2 + B)s + K]X_2(s) - [Bs + K]X_1(s) = F(s) \tag{1.31}$$

Substituting for $X_2(s)$ from Eq. (1.30) in (1.31) we get

$$F(s) = \left[\left\{[M_2 s^2 + (B_2 + B)s + K]\frac{[M_1 s^2 + (B_1 + B)s + K]}{(Bs + K)}\right\} - (Bs + K)\right]X_1(s)$$

$$\frac{X_1(s)}{F(s)} = \frac{Bs + K}{[[M_2 s^2 + (B_2 + B)s + K][M_1 s^2 + (B_1 + B)s + K] - (Bs + K)^2]}$$

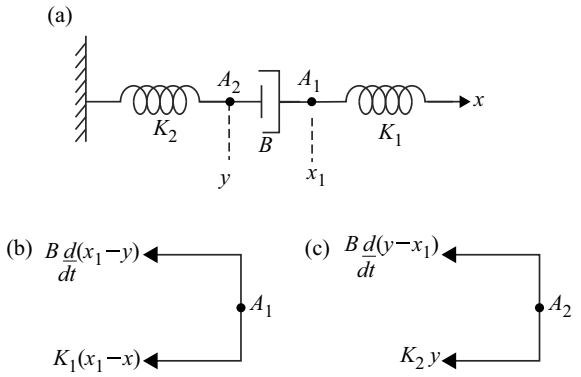


Fig. 1.29 **a** Mechanical system for Example 1.9. **b** Free-body diagrams of Fig. 1.29a

$$\frac{X_1}{F(s)} = (Bs+K) \Big/ s[M_1M_2s^3 + \{M_2(B_1+B) + M_1(B_2+B)\}s^2 + \{K(M_1+M_2) + B_1B_2 + B(B_1+B_2)\}s + K(B_1+B_2)]$$

Example 1.9

Consider the mechanical system shown in Fig. 1.29a. Find the T.F. $\frac{Y}{X}(s)$.

Solution

For Fig. 1.29a, x is the input and output is taken at point A_2 where the displacement is y. The end point of the spring K_1, at A_1 moves with the displacement x_1. The free-body diagrams are shown in Fig. 1.29b, c. From Fig. 1.29b the following equations are written:

$$B\frac{d}{dt}(x_1 - y) + K_1(x_1 - x) = 0$$

Taking Laplace transform on both sides we get

$$(Bs + K_1)X_1(s) = BsY(s) + K_1X(s) \tag{1.32}$$

From Fig. 1.29c, the following equation is written

$$B\frac{d}{dt}(y - x_1) + K_2y = 0$$

Taking Laplace transform on both sides we get

$$(Bs + K_2)Y(s) = BsX_1(s) \tag{1.33}$$

Substituting for $X_1(s)$ in Eq. (1.32) we get

$$\frac{(Bs + K_1)(Bs + K_2)}{Bs} Y(s) - BsY(s) = K_1 X(s)$$

$$[B^2 s^2 + Bs(K_1 + K_2) + K_1 K_2 - B^2 s^2] Y(s) = sBK_1 X(s)$$

$$\boxed{\frac{Y(s)}{X(s)} = \frac{BK_1 s}{[B(K_1 + K_2)s + K_1 K_2]}}$$

Example 1.10

Consider the mechanical system with lever arrangement which is shown in Fig. 1.30a. Find the T.F $\frac{Y}{F}(s)$.

Solution

The free-body diagrams of Fig. 1.30a are shown in Fig. 1.30b, c. Form Fig. 1.30b, the net force acting downwards on the lever at point a is,

$$f_a = f(t) - \left(M_1 \frac{d^2x}{dt^2} + B_1 \frac{dx}{dt} + K_1 x \right) \tag{1.34}$$

f_b is the force acting downwards at point c which opposes motion. It is given as

$$f_b = M_2 \frac{d^2y}{dt^2} + B_2 \frac{dy}{dt} + K_2 y \tag{1.35}$$

These forces and the displacement of the lever are shown in Fig. 1.30d.

Now taking moment at the pivot point b, the following equation is obtained.

$$f_a l_1 - f_b l_2 = 0$$

$$f_a l_1 = f_b l_2$$

$$\left[f(t) - \left(M_1 \frac{d^2x}{dt^2} + B_1 \frac{dx}{dt} + K_1 x \right) \right] l_1 = \left(M_2 \frac{d^2y}{dt^2} + B_2 \frac{dy}{dt} + K_2 y \right) l_2$$

Taking Laplace transform on both sides we get

$$F(s) - (M_1 s^2 + B_1 s + K_1) X(s) = \frac{l_2}{l_1} (M_2 s^2 + B_2 s + K_2) Y(s) \tag{1.36}$$

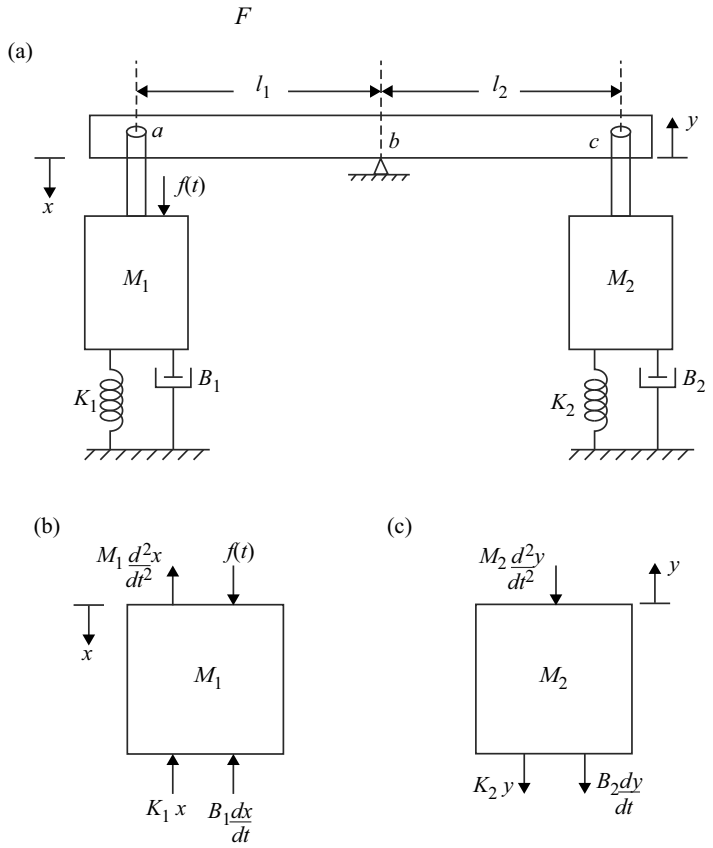


Fig. 1.30 **a** Mechanical system with lever arrangement. **b** Free-body diagrams of Fig. 1.30a. **c** Forces acting on the lever

From Fig. 1.30e we write,

$$\frac{l_1}{x} = \frac{l_2}{y}$$
$$x = \frac{l_1}{l_2} y \tag{1.37}$$

Substituting for x from Eqs. (1.37) in (1.36) we get

$$F(s) - (M_1 s^2 + B_1 s + K_1)\frac{l_1}{l_2} Y(s) = \frac{l_2}{l_1}(M_2 s^2 + B_2 s + K_2) Y(s)$$
$$F(s) = \left[(M_1 s^2 + B_1 s + K_1)\frac{l_1}{l_2} + (M_2 s^2 + B_2 s + K_2)\frac{l_2}{l_1} \right] Y(s)$$

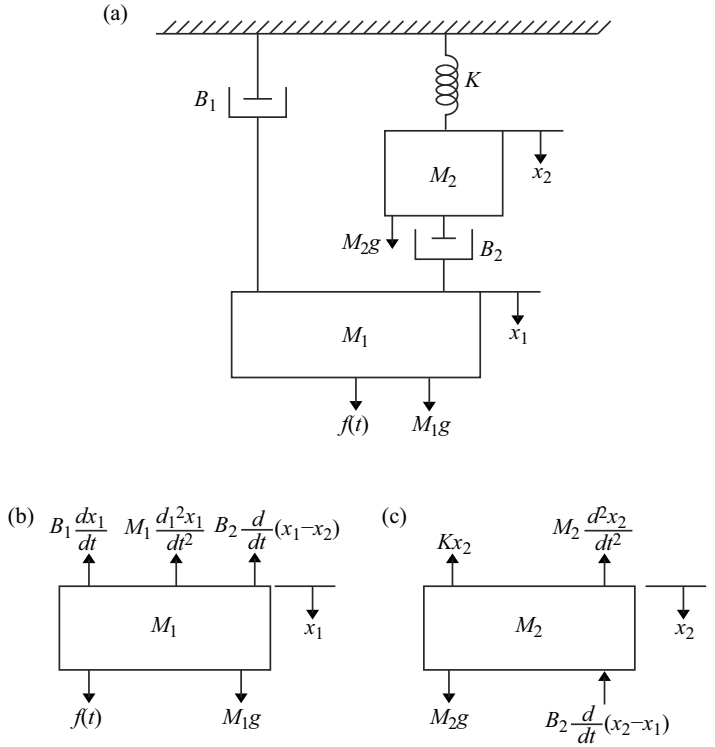


Fig. 1.31 **a** Mechanical system for Example 1.11. **b** Free-body diagrams of Fig. 1.31a

$$\frac{Y(s)}{F(s)} = \frac{l_1 l_2}{l_1^2(M_1 s^2 + B_1 s + K_1) + l_2^2(M_2 s^2 + B_2 s + K_2)}$$

Example 1.11

Consider the mechanical system shown in Fig. 1.31a. Obtain the T.F. connecting the displacement of mass M_2 and the applied force $f(t)$.

(*Anna University, May, 2010*)

Solution

The free-body diagrams of Fig. 1.31a are shown in Fig. 1.31b, c. It is to be noted that the gravitational forces $M_1 g$ and $M_2 g$ are acting on mass M_1 and mass M_2 respectively. **If the system is initially at the equilibrium, the gravitational forces acting in the system are balanced by preloading of the spring and hence g is set to zero.** Hence, the following equations are written for Fig. 1.31b, c.

$$M_1\frac{d^2x_1}{dt^2}+B_1\frac{dx_1}{dt}+B_2\frac{d}{dt}(x_1-x_2)=f(t)$$

Taking Laplace transform on both sides we get

$$[M_1s^2+(B_1+B_2)s]X_1(s)-B_2sX_2(s)=F(s) \tag{1.38}$$

$$M_2\frac{d^2x_2}{dt^2}+B_2\frac{d}{dt}(x_2-x_1)+Kx_2=0$$

$$[M_2s^2+B_2s+K]X_2(s)=B_2sX_1(s)$$

$$X_1(s)=\frac{(M_2s^2+B_2s+K)}{B_2s}X_2(s) \tag{1.39}$$

Substituting Eq. (1.39) in (1.38) we get

$$F(s)=\left[\frac{[M_1s^2+(B_1+B_2)s][M_2s^2+B_2s+K]}{B_2s}-B_2s\right]X_2(s)$$

$$\boxed{\frac{X_2}{F}(s)=\frac{B_2}{[M_1M_2s^3+\{M_1B_2+M_2(B_1+B_2)\}s^2+(M_1K+B_1B_2)s+K(B_1+B_2)]}}$$

1.12 Transfer Function Model for Mechanical Rotational System

The behaviour of mechanical rotational system resembles with that of translational system. The basic difference between the two is that rotational system has angular displacement while translational system has linear displacement. Further, instead of force, torque is applied in the rotational system. Mass is replaced by inertia. The basic elements of the rotational system consists of inertia, rotational dash-pot and spring. Instead of lever, gear arrangements are used as matching device in rotational system which is used to alter torque, angular speed, angular displacement and transfer maximum power. The dynamic behaviour of these elements due to the torque applied are explained below in Fig. 1.32.

In the mechanical rotational system energy is stored in inertia J and spring K. Energy is dissipated in the dash-pot B. Power transfer is made through gear system. The units of the above elements and the associated variables are given below:

1. Inertia, J = Kg-m^2.
2. Viscous friction coefficient, B = N-m/rad/s.
3. Torsional spring, K = N-m/rad.
4. Torque, $T(t)$ = N-m.
5. Angular displacement, $\theta(t)$ = radians (rad).

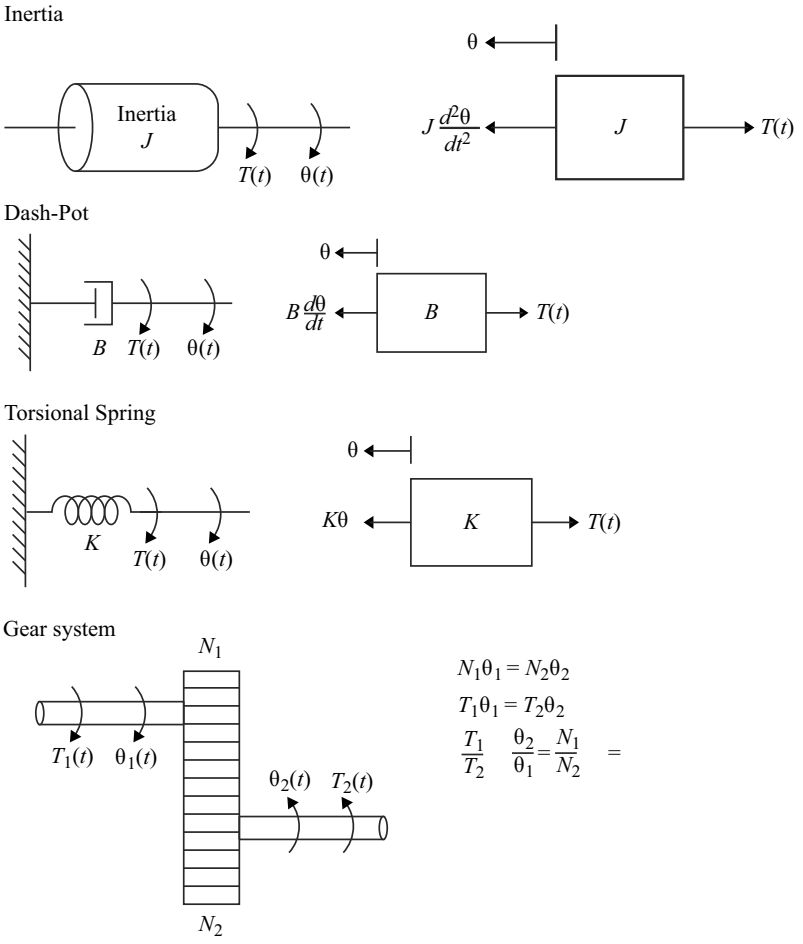


Fig. 1.32 Torque acting on mechanical rotation system components and their free-body diagrams

6. Angular velocity, $\omega(t) =$ rad/s.
7. Angular acceleration, $a(t) =$ rad./s^2.

Example 1.12

For the mechanical rotational system shown in Fig. 1.33a, find $\theta_1/F(s)$.

Solution The Free-body diagrams for Fig. 1.33a are shown in Fig. 1.33b, c. For Fig. 1.33b, the following equation is written

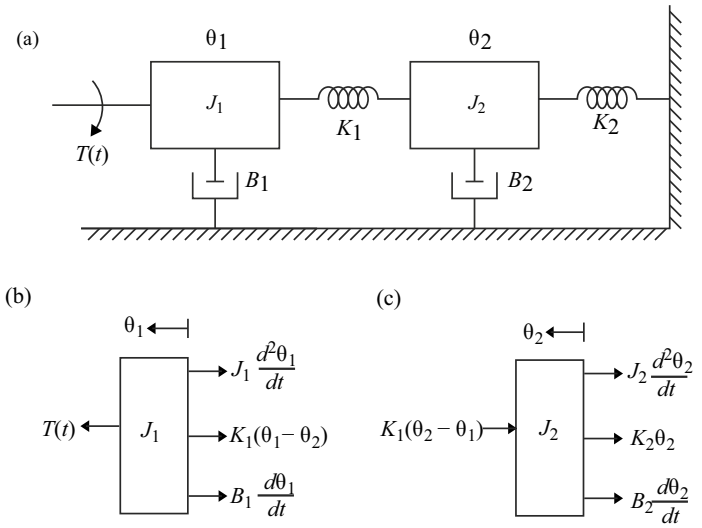


Fig. 1.33 **a** Mechanical rotational system for Example 1.12. **b** Free-body diagram of Fig. 1.33a

$$J_1\frac{d^2\theta_1}{dt^2} + B_1\frac{d\theta_1}{dt} + K_1(\theta_1 - \theta_2) = T(t)$$

Taking Laplace transform on both sides we get,

$$(J_1s^2 + B_1s + K_1)\theta_1(s) - K_1\theta_2(s) = T(s) \tag{1.40}$$

For Fig. 1.33c the following equation is written:

$$J_2\frac{d^2\theta_2}{dt^2} + B_2\frac{d\theta_2}{dt} + K_2\theta_2 + K_1(\theta_2 - \theta_1) = 0$$

Taking Laplace transform on both sides we get

$$[J_2s^2 + B_2s + (K_2 + K_1)]\theta_2(s) = K_1\theta_1(s) \tag{1.41}$$

Substituting for $\theta_2(s)$ from Eq. (1.41) in (1.40) we get

$$(J_1s^2 + B_1s + K_1)\theta_1(s) - \frac{K_1^2\theta_1(s)}{(J_2s^2 + B_2s + K_1 + K_2)} = T(s)$$

$$\frac{[(J_1s^2 + B_1s + K_1)(J_2s^2 + B_2s + K_1 + K_2) - K_1^2]}{(J_2s^2 + B_2s + K_1 + K_2)}\theta_1(s) = T(s)$$

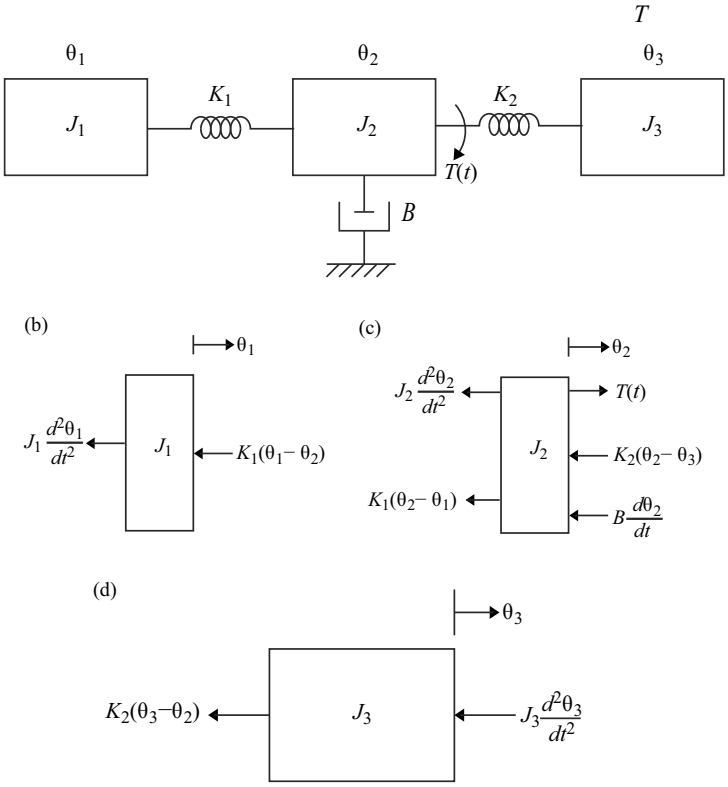


Fig. 1.34 **a** Mechanical translational system for Example 1.13. **b** Free-body diagrams of Fig. 1.34a

$$\frac{\theta_1}{T}(s) = (J_2 s^2 + B_2 s + K_1 + K_2) \Big/ \Big[J_1 J_2 s^4 + (J_1 B_2 + J_2 B_1) s^3 + (J_1 (K_1 + K_2) + B_1 B_2 + J_2 K_1) s^2 + (K_1 + K_2 + K_1 B_2) s + K_1 K_2 \Big]$$

Example 1.13

For the mechanical rotational system shown in Fig. 1.34a, find $\frac{\theta_2}{T}(s)$.

Solution

The free-body diagrams of Fig. 1.34a are shown in Fig. 1.34b–d. For Fig. 1.34b the following equation is written:

$$J_1 \frac{d^2\theta_1}{dt^2} + K_1(\theta_1 - \theta_2) = 0$$

Taking Laplace transform on both sides we get

$$(J_1 s^2 + K_1)\theta_1(s) = K_1\theta_2(s) \tag{1.42}$$

For Fig. 1.34c, following equation is written

$$J_2 \frac{d^2\theta_2}{dt^2} + B\frac{d\theta_2}{dt} + K_1(\theta_2 - \theta_1) + K_2(\theta_2 - \theta_3) = T(t)$$

Taking Laplace transform on both sides we get

$$[J_2 s^2 + Bs + (K_1 + K_2)]\theta_2(s) - K_1\theta_1(s) - K_2\theta_3(s) = T(s) \tag{1.43}$$

For Fig. 1.34d, the following equation is written:

$$J_3 \frac{d^2\theta_3}{dt^2} + K_2(\theta_3 - \theta_2) = 0$$

Taking Laplace transform on both sides we get

$$[J_3 s^2 + K_2]\theta_3(s) = K_2\theta_2(s) \tag{1.44}$$

Substituting for $\theta_1(s)$ and $\theta_2(s)$ from Eqs. (1.42) and (1.44) respectively in (1.43) we get the following equation:

$$\left[J_2 s^2 + Bs + (K_1 + K_2) - \frac{K_1^2}{(J_1 s^2 + K_1)} - \frac{K_2}{(J_3 s^2 + K_2)}\right]\theta_2(s) = T(s)$$

On simplifying the above equation we get

$$\frac{\theta_2}{T}(s) = (J_1 s^2 + K_1)(J_3 s^2 + K_2) \Big/ s\Big[J_1 J_2 J_3 s^5 + J_1 J_3 B s^4 + (J_1 J_2 K_2 + J_2 J_3 K_1 + J_1 J_3 (K_1 + K_2))s^3 + B(J_1 K_2 + J_3 K_1)s^2 + (K_1 K_2 (J_1 + J_2 + J_3))s + B_1 K_1 K_2\Big]$$

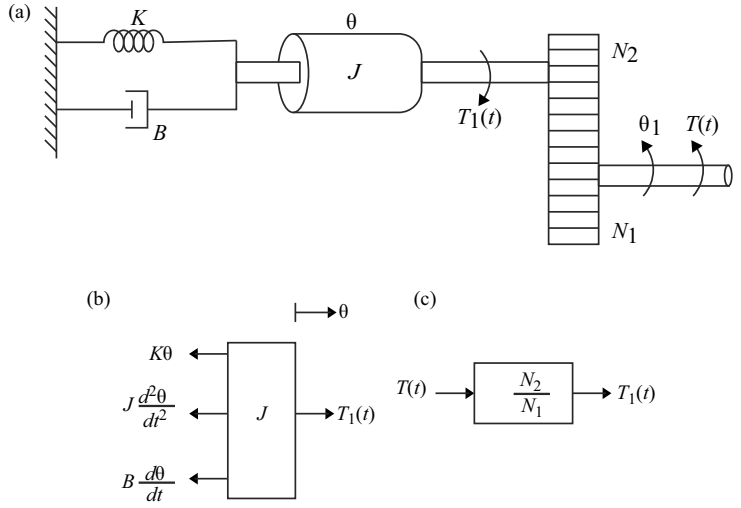


Fig. 1.35 **a** Mechanical system with gear arrangement for Example 1.14. **b** Free-body diagrams of Fig. 1.35a

Example 1.14

Consider the mechanical rotational translational system shown in Fig. 1.35a. Find the T.F. relating the angular displacement of inertia J and the applied torque $T(t)$ and angular displacement of θ_1 and $T(t)$.

Solution The free-body diagrams for Fig. 1.35a are shown in Fig. 1.35b, c.

$$\left[J\frac{d^2\theta}{dt^2} + B\frac{d\theta}{dt} + K\theta\right] = T_1(t)$$

Taking Laplace transform on both sides we get

$$[Js^2 + Bs + K]\theta(s) = T_1(s)$$

But $T_1 N_1 = T N_2$

$$T_1 = \frac{N_2}{N_1}T$$

Substituting this in the above equation we get

$$\boxed{\frac{N_1}{N_2}(Js^2 + Bs + K)\theta(s) = T(s)}$$

$$\boxed{\frac{\theta}{T}(s) = \left(\frac{N_2}{N_1}\right)\frac{1}{(Js^2 + Bs + K)}}$$

Also we have

$$N_1\theta_1 = N_2\theta$$
$$\theta = \frac{N_1}{N_2}\theta_1$$

substituting this in the above equation we get

$$\boxed{\frac{\theta_1}{T}(s) = \left(\frac{N_2}{N_1}\right)^2\frac{1}{(Js^2 + B + K)}}$$

Example 1.15

Consider the mechanical rotational system shown in Fig. 1.36a. Find the T.F. relating the position θ_4 of J_2 and the input torque $T(t)$.

Solution

The free-body diagrams of the given mechanical system are shown in Fig. 1.36b–d. For Fig. 1.36b, the following equation is written:

$$B\frac{d}{dt}(\theta_1 - \theta_2) + K_1(\theta_1 - \theta_2) = T(t)$$

Taking Laplace transform on both sides we get

$$(Bs + K_1)\theta_1(s) - (Bs + K_1)\theta_2(s) = T(s) \tag{1.45}$$

For Fig. 1.36c the following equation is written:

$$J_1\frac{d^2\theta_2}{dt^2} + B\frac{d}{dt}(\theta_2 - \theta_1) + K_1(\theta_2 - \theta_1) + K_2(\theta_2 - \theta_3) = 0$$
$$J_1\frac{d^2\theta_2}{dt^2} + K_2(\theta_2 - \theta_3) = T(t)$$

Taking Laplace transform on both sides we get

$$(J_1s^2 + K_2)\theta_2(s) - K_2\theta_3(s) = T(s) \tag{1.46}$$

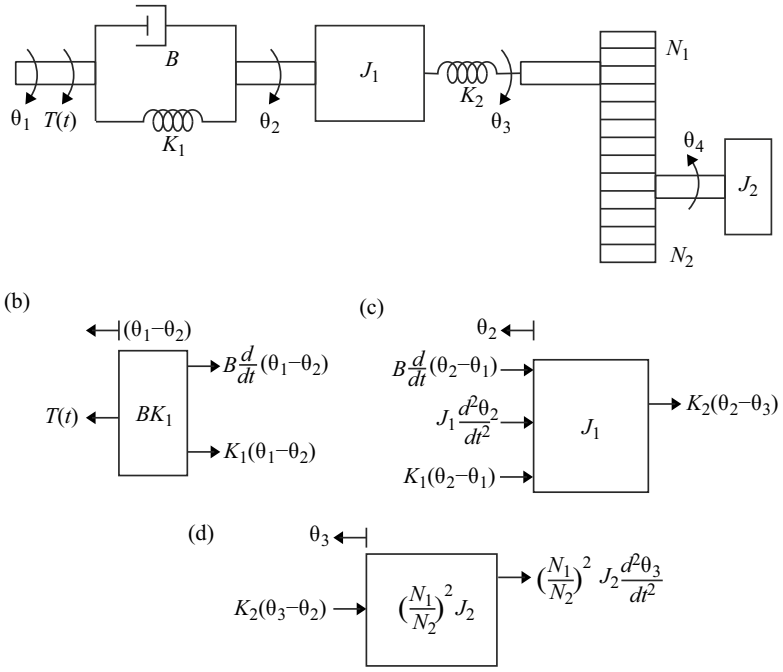


Fig. 1.36 **a** Mechanical rotational system for Example 1.15. **b** Free-body diagrams of Fig. 1.35a

For Fig. 1.36d, the following equation is written:

$$\left(\frac{N_1}{N_2}\right)^2 J_2 \frac{d^2\theta_3}{dt^2} + K_2(\theta_3 - \theta_2) = 0$$

Taking Laplace transform on both sides of the above equation we get

$$\left[\left(\frac{N_1}{N_2}\right)^2 J_2 s^2 + K_2\right]\theta_3 = K_2\theta_2 \tag{1.47}$$

Substituting for θ_2 from Eq. (1.47) in (1.46) we get

$$\left[(J_1 s^2 + K_2)\left(\left(\frac{N_1}{N_2}\right)^2 J_2 s^2 + K_2\right)\frac{1}{K_2} - K_2\right]\theta_3(s) = T(s)$$

$$\left[\left(\frac{N_1}{N_2}\right)^2 J_1 J_2 s^4 + J_1 K_2 s^2 + K_2\left(\frac{N_1}{N_2}\right)^2 J_2 s^2 + K_2^2 - K_2^2\right]\frac{\theta_3(s)}{K_2} = T(s)$$

$$\frac{\theta_3}{T}(s) = \frac{K_2}{\left(\frac{N_1}{N_2}\right)^2 J_1 J_2 s^4 + K_2\left(J_1 + \left(\frac{N_1}{N_2}\right)^2 J_2\right) s^2}$$

But

$$N_1\theta_3 = N_2\theta_4$$
$$\theta_3 = \frac{N_2}{N_1}\theta_4$$
$$J_e = \left(\frac{N_1}{N_2}\right)^2 J_2$$

$$\boxed{\frac{\theta_4}{T}(s) = \frac{K\frac{N_1}{N_2}}{s^2[J_1 J_e s^2 + K_2(J_1 + J_e)]}}$$

1.13 Transfer Function Model of Electro-Mechanical Systems

Real world electromechanical problems require the ability to design magnetic and electric components to produce needed forces and torques, such as motor and solenoid design. Application examples include actuators, alternators and generators, electromagnetic brakes and clutches loudspeakers, magnetic bearings, micro-electro mechanical system (MEMS), motors, relays and contactors, sensors and solenoids. These devices are very extensively used in defence, manufacturing, communication, commercial air travel, energy products etc. The transfer functions of some of the commonly used electromechanical devices are derived in this section.

Example 1.16

Consider the solenoid shown in Fig. 1.37. The force of attraction generated is $K_f i(t)$ Newtons. The back emf constant of the solenoid coil is K_b volts/m/s. Derive the T.F. relating the displacement $x(t)$ of the M of the solenoid core and the applied voltage $e(t)$.

Solution

The electrical circuit of the solenoid coil is shown in Fig. 1.37b. For Fig. 1.37b the following equations are written:

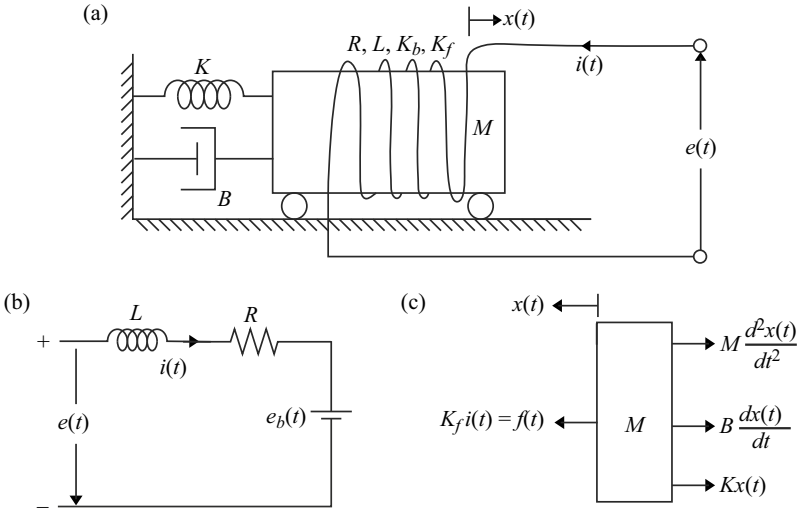


Fig. 1.37 **a** Solenoid coil for Example 1.16. **b** Equivalent electrical circuit and mechanical free-body diagram of Fig. 1.37a

$$\begin{aligned} e(t) &= L\frac{di(t)}{dt} + Ri(t) + e_b(t) \\ &= L\frac{di(t)}{dt} + Ri(t) + K_b\frac{dx(t)}{dt} \end{aligned}$$

Taking Laplace transform on both sides of the above equation we get

$$E(s) = [Ls + R]I(s) + K_b s X(s) \tag{1.48}$$

For Fig. 1.37c, the following equations are written:

$$M\frac{d^2x(t)}{dt^2} + B\frac{dx(t)}{dt} + Kx(t) = K_f i(t)$$

Taking Laplace transform on both sides of the above equation we get

$$[Ms^2 + Bs + K]X(s) = K_f I(s) \tag{1.49}$$

Substituting Eq. (1.49) in (1.48) for $I(s)$ we get

$$E(s) = \left[(Ls + R)\frac{(Ms^2 + Bs + K)}{K_f} + K_b s\right] X(s)$$

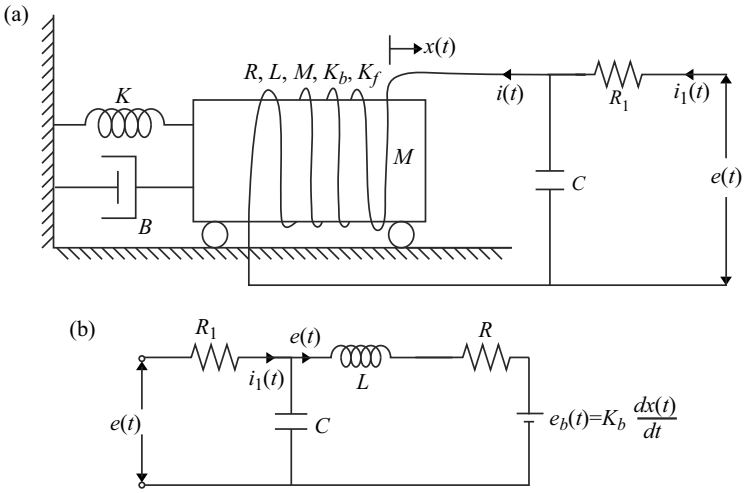


Fig. 1.38 **a** Solenoid coil for Example 1.17. **b** Electrical circuit part of Fig. 1.13a

$$\boxed{\frac{X(s)}{E(s)} = \frac{K_f}{[(Ls+R)(Ms^2+Bs+K)+K_bK_fs]}}$$

Example 1.17

Consider the solenoid coil shown in Fig. 1.38a. The coil is connected to the voltage source via an R-C filter as shown in Fig. 1.38a. Other parameters remain the same as in Example 1.16. Derive the T.F. relating $\frac{X}{E}(s)$.

Solution

By independence transform method, the following equations are written for Fig. 1.38b:

$$E(s) = \left[R_1 + \frac{1}{Cs}\right] I_1(s) - \frac{1}{Cs} I(s)$$

$$I_1(s) = \frac{I(s)}{(1+R_1Cs)} + \frac{1}{(1+R_1Cs)} CsE(s) \tag{1.50}$$

$$\left[Ls + R + \frac{1}{Cs}\right] I(s) - \frac{1}{Cs} I_1(s) + K_bsX(s) = 0$$

$$(LCs^2 + RCs + 1)I(s) - I_1(s) + K_bCs^2X(s) = 0 \tag{1.51}$$

Substituting Eq. (1.50) for $I_1(s)$ in (1.51) we get

$$[(LCs^2 + RCs + 1)(1 + R_1Cs) - 1]I(s) - CsE(s) + K_bCs^2(1 + R_1Cs)X(s) = 0 \tag{1.52}$$

For the mechanical system, the following equation written

$$K_f I(s) = [Ms^2 + Bs + K]X(s) \tag{1.53}$$

Substituting for $I(s)$ from Eq. (1.53) in (1.52) we get

$$CsE(s) = \frac{(Ms^2 + Bs + K)}{K_f}[(LCs^2 + RCs + 1)(1 + R_1Cs) - 1] + K_bCs^2(1 + R_1Cs)]X(s)$$

$$\boxed{\frac{X}{E}(s) = K_f \Big/ \Big[(Ms^2 + Bs + K)(LCR_1s^2 + (R_1RC + L)s + (R_1 + R) + K_fK_bs(1 + R_1Cs))\Big]} \tag{1.54}$$

Example 1.18

The solenoid shown in Fig. 1.39a produces a magnetic force proportional to the current in the coil with proportionality constant being Ki N/ampere. The coil has resistance R and inductance L. Determine the transfer function $\frac{X_2}{E}(s)$. Assume the back emf constant of the solenoid coil as K_b volts/meter.

(*Anna University, December, 2004*)

Solution

The portion of electrical circuit for Fig. 1.39a is shown in Fig. 1.39b. For Fig. 1.39b the following equations is written:

$$e(t) = L\frac{di}{dt} + Ri + K_b\frac{dx_1}{dt}$$

Taking Laplace transform on both sides of the above equation we get

$$E(s) = (R + Ls)I(s) + K_bsX_1(s)$$
$$I(s) = \frac{E(s) - K_bsX_1(s)}{(R + Ls)} \tag{1.55}$$

The electromechanical force developed by the solenoid coil is,

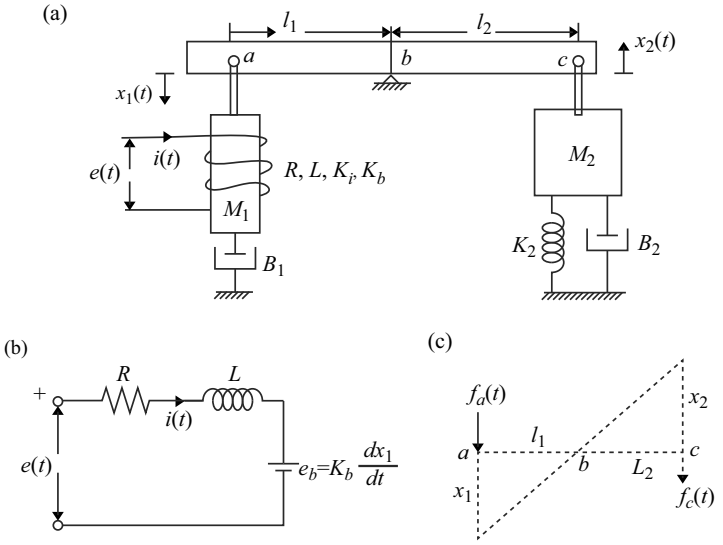


Fig. 1.39 **a** Electromechanical system for Example 1.18. **b** Electrical circuit and free-body diagram of mechanical system for Example 1.18

$$f(t) = K_i i(t)$$
$$F(s) = K_i I(s)$$

Substituting Eq. (1.55) in the above equation we get

$$F(s) = Ki \frac{[E(s) - K_b s X_1(s)]}{(R + Ls)} \tag{1.56}$$

The force acting at point a of the lever is

$$f_a(t) = f(t) - \left(M_1 \frac{d^2 x_1}{dt^2} + B_1 \frac{dx_1}{dt} \right)$$

Taking Laplace transform on both sides we get

$$F_a(s) = F(s) - (M_1 s + B_1) s X_1(s) \tag{1.57}$$

The force acting at point c on the lever is,

$$f_e(t) = M_2 \frac{d^2 x_2}{dt^2} + B_2 \frac{dx_2}{dt} + K_2 x_2$$

Taking Laplace transform on both sides we get

$$F_c(s) = [M_2 s^2 + B_2 s + K_2] X_2(s) \tag{1.58}$$

From Fig. 1.39c we get

$$\frac{x_1}{l_1} = \frac{x_2}{l_2}$$

The Laplace transformed equation is

$$X_1(s) = \frac{l_1}{l_2} X_2(s)$$

Substituting the above equation in (1.57) we get

$$\begin{aligned} F_a(s) &= F(s) - (M_1 s + B_1) s \frac{l_1}{l_2} X_2(s) \\ F(s) &= \frac{K_i E(s)}{R + Ls} - K_b K_i \frac{l_1}{l_2} \frac{X_2(s)}{R + Ls} \end{aligned} \tag{1.59}$$

Taking moment at point b on the pivot in Fig. 1.39c we get

$$\begin{aligned} F_a(s) l_1 &= F_c(s) l_2 \\ F_a(s) &= F_c(s) \frac{l_2}{l_1} \end{aligned}$$

Substituting for $F_a(s)$ in Eq. (1.59) we get

$$\begin{aligned} &\left[K_i \frac{E(s)}{(R + Ls)} - K_b K_i \frac{l_1}{l_2} \frac{l}{(R + Ls)} X_2(s) - (M_1 s + B_1) s \frac{l_1}{l_2} X_2(s) \right] \\ &= (M_2 s^2 + B_2 s + K) X_2(s) \frac{l_2}{l_1} \end{aligned}$$

Rearranging the like terms we get

$$\begin{aligned} K_i \frac{E(s)}{(R + Ls)} &= \left[K_b K_i \frac{l_1}{l_2} \frac{1}{(R + Ls)} + (M_1 s + B_1) s \frac{l_1}{l_2} + \frac{l_2}{l_1} (M_2 s^2 + B_2 s + K) \right] X_2(s) \\ \frac{X_2}{E}(s) &= \frac{K_i}{(R + Ls) \left\{ (M_2 s^2 + B_2 s + K) + (M_1 s + B_1) s \frac{l_1}{l_2} + K_b K_i \frac{l_1}{l_2} \frac{1}{(R + Ls)} \right\}} \end{aligned}$$

$$\boxed{\frac{X_2}{E}(s) = \frac{K_i}{(R + Ls) \left[(M_2 s^2 + B_2 s + K) \frac{l_2}{l_1} + (M_1 s + B_1) s \frac{l_1}{l_2} + \frac{l_1}{l_2} K_b K_i \frac{1}{(R + Ls)} \right]}}$$

1.14 Transfer Function Model of D.C. Motor

The D.C. motor is a power actuator device that delivers to the load. It converts direct current electrical energy into rotational mechanical energy. The torque generated in the rotor shaft is used to drive an external load connected to the shafts. The operation of D.C. motor is based on the principle that if a current carrying conductor is placed in a magnetic field, there will be a force exerted on the conductor. The reverse process, namely if a conductor is moved through a magnetic field there will be a voltage induced in the conductor. This principle is made use of in generators. The D.C. motor consists of a rotating cylinder called armature in which copper conductors are placed in the slots made in the armature. The armature is placed in the magnetic field between stationary magnetic poles, called the field. D.C. motors have traditionally been used for variable speed applications. There are two methods of controlling the speed of a D.C. motor.

1. The armature current is kept constant and the torque is controlled via the field voltage. Such a control is called field controlled D.C. motor.
2. The field is connected to a constant voltage source to produce constant field flux and the torque is controlled via the armature voltage. Such a control is called armature controlled D.C. motor.

In both the cases the electromagnetic torque produced is proportional to the armature current and the field magnetic flux. The D.C. motors have the following advantages:

1. Torque developed is very high.
2. A wide range of speed controllability.
3. Easily portable.
4. Good torque-speed characteristics.
5. Adaptability to various types of control methods.

The control applications include the following:

1. Machine tools
2. Servo valve actuators
3. Robatic manipulators
4. Tape transport mechanisms
5. Disk drives.

1.14.1 Transfer Function of Field Controlled D.C. Motor (No Load)

The field controlled D.C. motor is shown in Fig. 1.40. The symbols and notations have the following meanings:

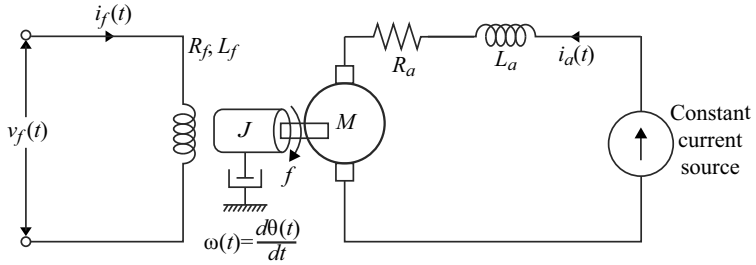


Fig. 1.40 Field controlled D.C. motor

1. J = Moment of inertia of the motor in Kg-m^2.
2. f = Friction coefficients N-m/rad/s.
3. R_f, L_f = Field coil resistance and inductance respectively.
4. R_a, L_a = Armature coil resistance and inductance respectively.
5. $i_a(t)$ = Armature current
6. $v_f(t)$ = Voltage applied across the field coil to produce flux.
7. $\omega(t) = \frac{d\theta(t)}{dt}$ = Motor shaft speed in rad/s.
8. $\theta(t)$ = Motor shaft position in rad.

If the field is unsaturated the air-gap flux of the motor is proportional to the field current $i_f(t)$ and is given by

$$\Phi = K_f i_f(t) \tag{1.60}$$

The electromagnetic torque developed by the motor which is proportional to the field flux and the armature current is given by

$$\begin{aligned} T_d(t) &= K_a \Phi i_a(t) \\ &= K_a K_f i_f(t) i_a(t) \end{aligned} \tag{1.61}$$

From Eq. (1.61) it is evident that any one of $i_f(t)$ or $i_a(t)$ must be maintained constant while the other current becomes the input so that Eq. (1.61) is linear. If $i_a(t)$ is kept constant the motor is field controlled one and Eq. (1.61) becomes

$$T_d(t) = K_t i_f(t) \tag{1.62}$$

where K_t is expressed in N-m field ampere. If $i_f(t)$ is kept constant, the motor is armature controlled one and Eq. (1.61) becomes

$$T_d(t) = K_t i_a(t) \tag{1.63}$$

where K_t is expressed in N-m armature ampere. The developed torque is used to drive the system having the total inertia J and to overcome the friction f. This torque is expressed as

$$T_L(t) = J\frac{d^2\theta(t)}{dt^2} + f\frac{d\theta(t)}{dt} \tag{1.64}$$

Equating the developed torque $T_d(t)$ to the torque used $T_L(t)$ we get, for a field controlled D.C. motor

$$K_t i_f(t) = J\frac{d^2\theta(t)}{dt^2} + \frac{f\, d\theta(t)}{dt}$$

Taking Laplace transform on both sides we get

$$K_t I_f(s) = (Js + f)s\theta(s) \tag{1.65}$$

From Fig. 1.40 we get

$$I_f(s) = \frac{V_f(s)}{R_f + sL_f} \tag{1.66}$$

Substituting Eq. (1.66) in (1.65) we get

$$\frac{\theta(s)}{V_f(s)} = \frac{K_t}{s(R_f + sL_f)(Js + f)}$$

$$\boxed{\frac{\theta(s)}{V_f(s)} = \frac{K}{s(1 + sT_e)(1 + sT_m)}} \tag{1.67}$$

where

$$K = \frac{K_t}{R_f f}$$

$$T_e = \frac{L_f}{R_f} = \text{Electrical time constant}$$

$$T_m = \frac{J}{f} = \text{Mechanical time constant}$$

The block diagram representation of field controlled D.C. motor is shown in Fig. 1.41.

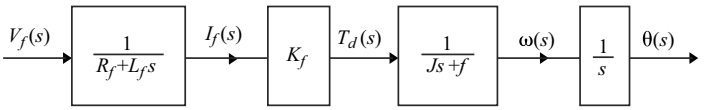


Fig. 1.41 Block diagram of field controlled D.C. motor

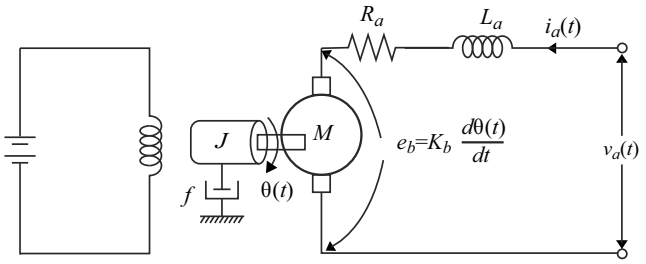


Fig. 1.42 Armature controlled D.C. motor

1.14.2 Transfer Function of Armature Controlled D.C. Motor

For the armature controlled D.C. motor, the output variable is angular displacement $\theta(t)$ and the input variable is the applied voltage $v_a(t)$. The ratio of the Laplace transform of these variables which is defined as the T.F is obtained as follows. The electromagnetic torque developed by the motor is given by Eq. (1.63) while Eq. (1.64) gives the torque used to drive the combined inertia and friction connected to the motor shaft. Equating these two equations we get

$$K_t i_a(t) = J\frac{d^2\theta(t)}{dt^2} + f\frac{d\theta(t)}{dt}$$

Taking Laplace transform on both sides of the above equation we get

$$K_t I_a(s) = (Js + f)s\theta(s) \tag{1.68}$$

The following equation for the armature circuit is written

$$\begin{aligned} v_a(t) &= L_a\frac{di_a(t)}{dt} + R_a i_a(t) + e_b(t) \\ &= L_a\frac{di_a(t)}{dt} + R_a i_a(s) + K_b\frac{d\theta(t)}{dt} \end{aligned}$$

Taking Laplace transform on both sides of the above equation we get

$$V_a(s) = (R_a + L_a s)I_a(s) + K_b s\theta(s) \tag{1.69}$$

Substituting Eq. (1.68) in (1.69) for $I_a(s)$ we get

$$V_a(s) = \left[(R_a + L_a s)\frac{(Js + f)}{K_t} + K_b\right]s\theta(s)$$

$$\frac{\theta}{V_a}(s) = \frac{K_t}{s[(R_a + L_a s)(Js + f) + K_b K_t]}$$

$$\boxed{\frac{\theta}{V_a}(s) = \frac{K_t}{s[(1 + sT_e)(1 + sT_m)R_a f + K_b K_t]}} \tag{1.70}$$

where

$$T_e = \frac{L_a}{R_a} = \text{Electrical time constant}$$
$$T_m = \frac{J}{f} = \text{Mechanical time constant}$$

To Prove $K_b = K_t$

When the motor operates at steady-state and if we neglect the losses in the rotor resistances, the following equations are written. The power input to the rotor is

$$\begin{aligned} P_r &= e_b i_a \\ &= K_b \omega i_a \end{aligned}$$

The power delivered to the shaft

$$\begin{aligned} P_s &= \text{Torque} \times \text{speed} \\ &= T_d \omega \\ &= K_t i_a \omega \end{aligned}$$

Since $P_r = P_s$

$$K_b \omega i_a = K_t \omega i_a$$

$$\boxed{K_b = K_t} \tag{1.71}$$

The block diagram of armature controlled D.C. motor is shown in Fig. 1.43.

1.14.3 Transfer Function of D.C. Generator (On No Load)

The schematic diagram of a D.C. generator run by a constant speed prime mover is shown in Fig. 1.44a. Due to the cutting of the flux produced by the electromagnet, emf is induced in the armature which has constant K_g expressed in volts per field ampere

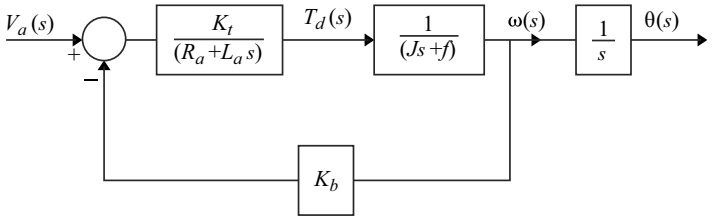


Fig. 1.43 Block diagram of armature controlled D.C. motor

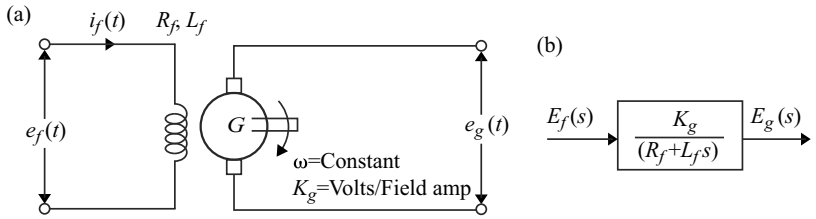


Fig. 1.44 Field excited D.C. generator and its T.F

when speed is constant. Under this condition the following equation is written:

$$e_g(t) = K_g i_f(t)$$

$$E_g(s) = K_g I_f(s) = \frac{K_g E_f(s)}{R_f + L_f s}$$

$$\boxed{\frac{E_g}{E_f}(s) = \frac{K}{(1 + sT_f)}} \tag{1.72}$$

where

$$K = \frac{K_g}{R_f}$$

$$T_f = \frac{L_f}{R_f} = \text{Electrical time constant.}$$

D.C. Tachogenerator

Instead of electromagnet, if a permanent magnet is used in Fig. 1.44a, the generator becomes a D.C. tachogenerator. Now, ω the speed is the input variable and $e_g(t)$ is the output variable. If K_T is the constant of the tachogenerator expressed in V per rad. per s; the T.F. is derived as

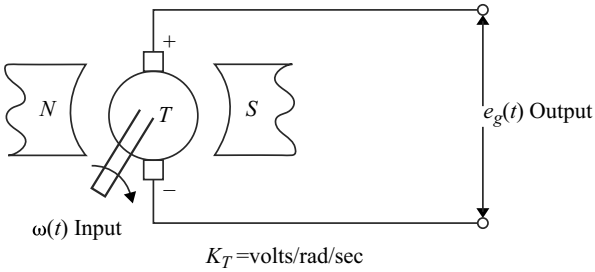


Fig. 1.45 D.C. Tachogenerator

$$\boxed{\frac{E_g}{\omega}(s) = K_T} \tag{1.73}$$

The tachogogenerator is shown in Fig. 1.45.

1.14.4 Performance Comparison of Armature and Field Controlled D.C. Motors

The performance comparison of armature and field controlled D.C. motors are given in Table 1.2.

1.15 Two Phase A.C. Servomotor

The two phase servomotors are basically, two phase, reversible induction motors modified for servo operations. These motors are used in applications which require rapid accurate response characteristics. Therefore, they are designed with small diameter rotor with high resistance rotor coils. The small diameter provides faster response and the high resistance in the rotor provides linear speed torque characteristics with a negative slope which provides positive damping and hence the stability.

The schematic diagram of a two phase servomotor is shown in Fig. 1.46a. It consists of a stator and a rotor. In the rotor slots heavy copper bars of high resistance are inserted and their ends are shorted by end rings. The stator consists of a pair of winding on the inner periphery of the stator with an angle space of 90° with reference to their axes. One winding is called the reference winding while the other is called the control winding. The voltages applied to these windings also differ in phase by 90°. Usually, the reference winding is excited by the voltage $v_r(t)$ from a single phase A.C. supply. The control voltage $v_c(t)$ after sending through a 90° phase shifter is applied to the control winding. The direction of rotation of the rotor depends on the phase relationship of these two voltages. These two phase voltages

Table 1.2 Comparison of armature and field controlled D.C. motors

	Armature controlled D.C. motor	Field controlled D.C. motor
1.	Electrical time constant of the motor is very small and hence the time response is faster	Since the field coil inductance is very high the electrical time constant is high and the time response is sluggish
2.	Efficiency is high	Efficiency is not so good
3.	Due to back emf developed there is inherent feedback inside the system which increases the damping. This avoids unwanted oscillations in the system variables due to disturbances	There is no additional damping here and the oscillations are high when the system is disturbed
4.	Power amplifiers are required while the D.C. motors are used. Design of power amplifiers which are incorporated in the armature circuit is little combursone and the cost is high	Power amplifies are used in the field circuit where the current and power requirements are low. Designing a power amplifier does not pose any problem and the cost is also less
5.	Armature controlled D.C. motor requires constant voltage source in the field circuit which can be easily obtained	Field controlled D.C. motor requires constant current source in the armature circuit which is difficult to get

establish a rotating magnetic field of synchronous speed N_s and sweeps over the rotor conductors. Voltages are induced in these stationary conductors. Since they are short circuited a heavy current is circulated in the rotor conductors. These conductors which are carrying current when situated in a magnetic field experience a force which moves them in the same direction as that of the rotating magnetic field with a speed slightly less than the synchronous speed. The relative speed of the rotor and that of the notating magnetic is thus reduced and rotates at steady speed $\omega(t)$.

The dynamic behavior of the two phase servomotor is decided if the transfer function relating $\omega(t) = d\theta(t)/dt$ where $\theta(t)$, the angular position of the rotor is known. This can be obtained from the torque speed characteristic of the motor. The

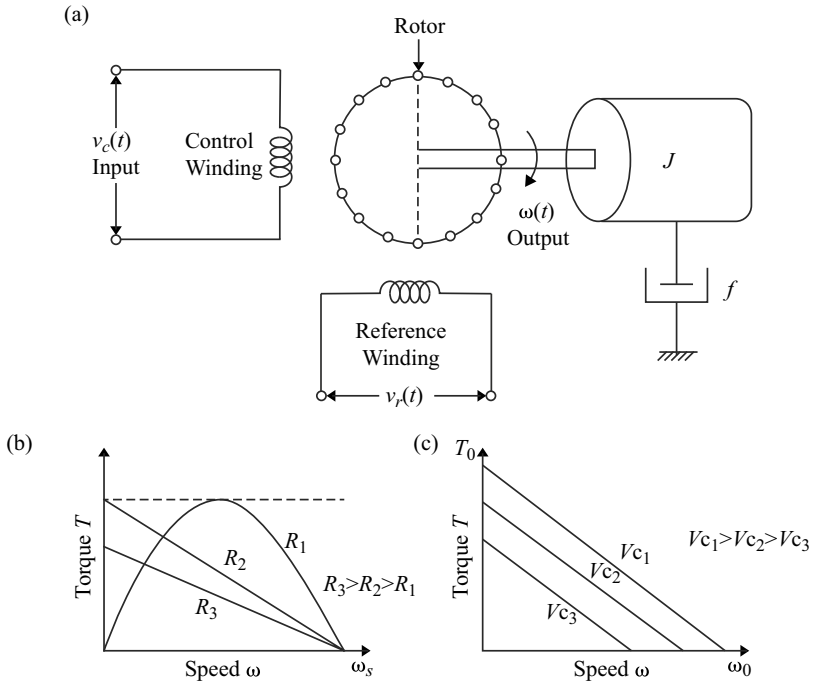


Fig. 1.46 **a** Two phase A.C. servomotor; **b** Torque-speed characteristic of ordinary induction motor and **c** Torque-speed characteristic of two phase A.C. servomotor

torque speed characteristic of the ordinary two phase induction motor is shown in Fig. 1.46b. If these characteristic curves have positive slope, while starting, it can be proved that it will result in unstable operation. By intentionally introducing, high resistance rotor conductors, the torque-speed characteristic can be linearized with a negative slope which ensures stability to the motor but at the cost of efficiency. In control applications, stability and fastness of response of the motor are given priority over efficiency. The following parameters and variables are defined as given below:

$$K = \frac{\text{Block rotor torque at the rated voltage } v_c}{\text{Rated control phase voltage}} \text{ in N-m/V}$$

$$m = \frac{-T_0}{\omega_0} = \text{Slope of the torque} - \text{Speed curve}$$

$$= \frac{\text{Rotor stand still torque}}{\text{No load speed}} \text{ N-m/rad/s}$$

The torque T_d developed by the motor is given by

$$\begin{aligned} T_d &= \text{Static torque} + \text{Dynamic torque} \\ &= K v_c(t) + m\frac{d\theta}{dt} \end{aligned}$$

This torque is used to accelerate the motor and overcome friction and any load connected.

$$T_L = J\frac{d^2\theta}{dt^2} + f\frac{d\theta}{dt}$$

where J is combined inertia connected to the motor shaft in kg-m^2 and f is combined friction connected to the motor shaft in N-m/rad/s.

$$T_d = T_L$$

$$K v_c(t) + m\frac{d\theta}{dt} = J\frac{d^2\theta}{dt^2} + f\frac{d\theta}{dt}$$

Taking Laplace transform on both sides we get,

$$K V_c(s) = s[Js + (f - m)]\theta(s)$$

$$\boxed{\frac{\theta}{V_c}(s) = \frac{K}{s[Js + (f - m)]}} \tag{1.74}$$

$$\frac{\theta}{V_c}(s) = \frac{K_m}{s[1 + sT_m]}$$

where $K_m = K/(f - m)$ is motor gain constant and T_m is mechanical time constant $T_m = J/(f - m)$.

Since m is negative, in Eq. (1.74), the poles are in the left half of the s-plane and the motor is stable with added damping. In case of ordinary induction motor m is positive and $m \gg f$. Now one pole of Eq. (1.74) is in the right half of s-plane and hence the motor **dances** and **jumps** when the supply is initially given until its slope of the torque-speed curve becomes negative. **That is why ordinary two phase induction motors are unsuitable for servo applications**.

A.C. Technogenerator

In Fig. 1.46a if mechanical power is given as the input to the shaft, the machine will give out voltage $v_c(t)$ in the control winding whose magnitude is proportional to the speed $\omega(t)$ and its frequency is the same as that of the reference voltage frequency. The T.F of the A.C. tachogenerator is given as

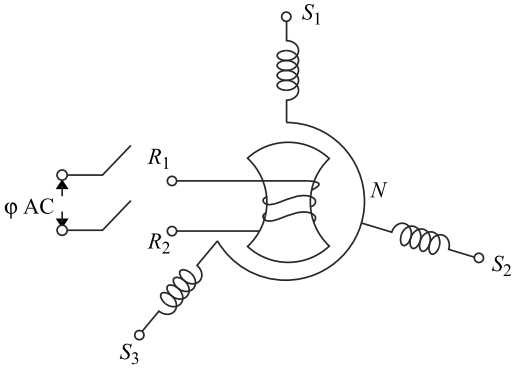


Fig. 1.47 Synchro generator

$$\boxed{\frac{V_c}{\omega}(s) = K_g} \tag{1.75}$$

where K_g is expressed in V/rad./s. For purpose of feedback in A.C. control system which requires device which will generate A.C. voltage proportional to speed but at constant frequency, A.C tachogenarators are used.

1.16 A Pair of Synchros

A synchro is a type of electrical transformer that is used for measuring the angle of a rotating machine such as antenna platform. In its general construction, it is much like a three phase alternator. In servo applications it is mainly used as an error detector. It consists of a pair of synchro which are named as synchro transmitter and synchro receiver. The synchro transmitter is also called as synchro generator (SG) and synchro receiver or control transformer (CT). Their basic construction and principles of operation are explained below.

1.16.1 Synchro Generator

A synchro transmitter consists of a stator and a rotor. In the stator, a Y connected three phase winding is provided. It resembles that of a three phase synchronous generator. The synchro generator has a salient pole rotor over which a coil is wound. Single phase excitation is given to the rotor coil through slip rings R_1R_2. The flux produced by this electro magnet links the stator windings. The rotor shaft is normally connected to any external rotation which is the input to the system. When the rotor rotates, emf's are induced in the stator windings S_{1N}, S_{2N} and S_{3N}. Eventhough

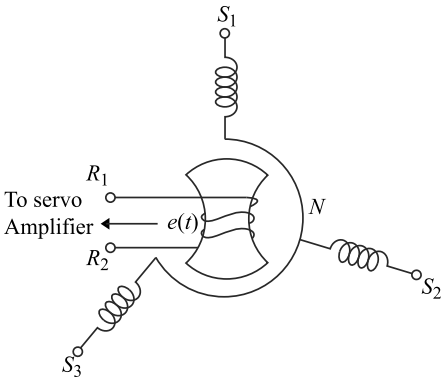


Fig. 1.48 Schematic diagram of synchro control transformer

the construction of the stator winding resembles that of a three phase synchronous generator, the voltages induced in these coils are only single phase voltages.

1.16.2 Control Transformer

The schematic diagram of control transformer is shown in Fig. 1.48. The construction is similar to that of a synchro generator except that the rotor is a bell shaped cylindrical one. This special construction is meant for minimizing the change in rotor impedance when it is rotated. This is necessary because the output of the rotor is connected to the servo-amplifier.

For most of the servo applications, the synchro pair acts as an error detector. The stator coils of SG and CT are connected with the corresponding phase windings. (S_1 to S_1, S_2 to S_2 and S_3 to S_3). Single phase supply is given to the rotor of SG, which produces magnetic flux. When SG rotor is rotated, there will be emf induced in the stator coils of SG. These voltages are now applied to the CT stator windings. These voltages again set up their own flux and induce a voltage in the CT rotor coil. The flux pattern in the air gap of CT rotor is same as that of the SG rotor air gap.

1.17 Electrical Analogy

Mechanical translational and rotational systems could be represented and studied by their equivalent electrical circuits which are more easily constructed than the corresponding models. Similarity between the equations resulting from Kirchhoff's laws for electrical systems and equations of motion of mechanical systems obtained using D'Alembert's principle exist and hence analogous electric circuits can be just constructed by inspection. The variables of the electric circuit so obtained behave exactly as the analogous variables of the mechanical systems. **Conversion of mechani-**

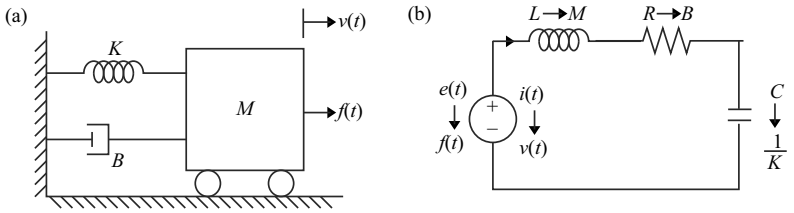


Fig. 1.49 **a** Mechanical translational system and **b** Electrical analogous circuit

cal systems to electrical network is done before writing the describing equations. However, the one to one correspondence between the equations describing the dynamic behaviour of both these systems can be verified by writing the equations corresponding to these two systems. An electric circuit that is analogous to a mechanical system is called electric circuit analog. The equation of motion of a mechanical system when compared to mesh equation of electric circuit it is called a **series analog** and when compared to nodal equations it is called **parallel analog**. Series analog is called **Force-Voltage (Torque-Voltage in rotational system) analog**. Parallel analog is called **Force-Current (Torque-Current in rotational system) analog.**

1.17.1 Force–Voltage Analogy

Consider the mechanical system shown in Fig. 1.49a. As discussed before, the following equation of motion is written

$$M\frac{dv(t)}{dt} + Bv(t) + \frac{1}{K}\int v(t)dt = f(t) \tag{1.76}$$

For the electric circuit of Fig. 1.49b the following mesh equation is written

$$L\frac{di(t)}{dt} + Ri(t) + \frac{1}{C}\int i(dt) = e(t) \tag{1.77}$$

The electrical circuit of Fig. 1.49b is said to be analogous to the mechanical system of Fig. 1.49a with the following identities:

1. Inductor L is analogous to mass M.
2. Resistor R is analogous to viscous friction B.
3. Capacitor C is analogous to reciprocal of spring K.
4. Voltage $e(t)$ is analogous to the applied force $f(t)$.
5. Mesh-current $i(t)$ is analogous to velocity $v(t)$.

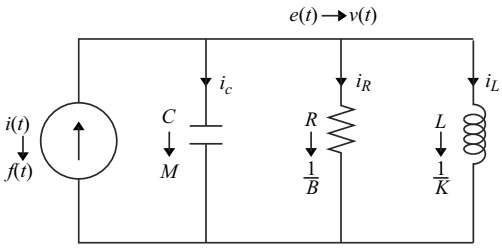


Fig. 1.50 Force-current electric analogous circuit

The above analogy also holds good for mechanical rotational system. Here $e(t)$ is analogous to the torque applied and the mesh current $i(t)$ is analogous to angular velocity $\omega(t)$. Other things remain the same with appropriate equivalent units.

1.17.2 Force–Current Analogy

Consider the electric circuit shown in Fig. 1.50, where the current source $i(t)$ and the circuit elements R, L and C are connected to the voltage node $e(t)$. At node $e(t)$, the sum of the current is equal to zero and hence the following equation is written:

$$i_c + i_R + i_L = i(t)$$
$$C\frac{de(t)}{dt} + \frac{e(t)}{R} + \frac{1}{L}\int e(t)dt = i(t) \tag{1.78}$$

Now compare Eq. (1.78) with Eq. (1.76). It is found C is analogous to the mass M, $1/R$ is analogous to viscous friction B and $1/L$ is analogous to the spring K. The current $i(t)$ is analogous to the applied force $f(t)$ and the nodal voltage $e(t)$ is analogous to the linear velocity $v(t)$. The same analogy can also be extended to the mechanical translational system. The summary of analogy of mechanical and electrical systems are given in Table 1.3.

Example 1.19

Consider the mechanical system shown in Fig. 1.51a. Draw the electrical circuits based on both (a) Force-voltage and (b) Force-current analogies. Write the differential equations governing the dynamic behaviour of these systems.

(*Anna University, May, 2011*)

Solution Force-Voltage Analogy

1. Since $f(t)$ is applied to mass M_1, the voltage source $e(t)$ in Fig. 1.51b is connected to the series connection of $L_1 \rightarrow M_1$ and $R_1 \rightarrow B_1$ which move with velocity v_1. The current flow in this series circuit is therefore $i_1 \rightarrow v_1$.

Table 1.3 Analogy of mechanical and Electrical systems

S. no.	Items	Mechanical system		Electrical system	
		Translational system	Rotational system	Loop system (Force-voltage)	Nodal system (Force-current)
1.	Independent variable	Force $f(t)$	Torque $T(t)$	Voltage $e(t)$	Current $i(t)$
2.	Dependent variable	Linear velocity $v(t)$	Angular velocity $\omega(t)$	Current $i(t)$	Voltage $e(t)$
3.	Power dissipation component	Linear dash-pot	Rotational dash-pot B	Resistance	Conductance G
		where $f(t) = Bv(t)$	where $T = B\omega$	where $e(t) = i(t)R$	where $i(t) = Ge(t)$
4.	Energy storing element	Mass M	Inertia J	Inductance L	Capacitance C
		where $f(t) = M\frac{dv}{dt}$	where $T(t) = J\frac{d\omega}{dt}$	where $e(t) = L\frac{di}{dt}$	where $i(t) = C\frac{de(t)}{dt}$
5.	Energy storing element	Spring K	Spring K	Capacitance C	Inductance L
		where $f(t) = K\int vdt$	where $T(t) = K\int \omega dt$	where $e(t) = \frac{1}{C}\int idt$	where $I = \frac{1}{L}\int e(t)dt$
6.	Physical laws	D'Alembert's principle	D'Alembert's principle	Kirchhoff's voltage law	Kirchhoff's current law
		$\sum f = 0$	$\sum T = 0$	$\sum e(t) = 0$	$\sum i(t) = 0$
7.	Changing the level of the independent variable	Lever $\frac{f_1}{f_2} = \frac{I_2}{I_1}$	Gear $\frac{T_1}{T_2} = \frac{N_1}{N_2}$	Voltage transformer $\frac{e_1}{e_2} = \frac{N_1}{N_2}$	Current transformer $\frac{i_1}{i_2} = \frac{N_2}{N_1}$

2. The mass M_2, spring K_2 and dash-pot B_2 move with a velocity v_2. So in the electrical circuit a branch is created where the current flow is $i_2 \rightarrow v_2$. In this branch $L_2 \rightarrow M_2$, $R_2 \rightarrow B_2$ and $C_2 \rightarrow (1/K_2)$ are connected in series.
3. The inductance K_1 and dash-pot B_{12} move with a velocity difference $(v_1 - v_2)$. Hence, in the electrical circuit a branch in which $(i_1 - i_2) \rightarrow (v_1 - v_2)$ flows is created. In this branch $C_1 \rightarrow (1/K_1)$ and $R_{12} \rightarrow B_{12}$ are connected in series. This completes the force-voltage analogous circuit. This circuit equation are verified with that of the mechanical system.

From Fig. 1.51b, for the loop consisting of L_1, R_1, C_1 and R_{12} the following equation is written:

$$L_1\frac{dt_1}{dt} + i_1R_1 + \frac{1}{C_1}\int (i_1 - i_2)dt + R_{12}(i_1 - i_2) = e(t) \tag{1.79}$$

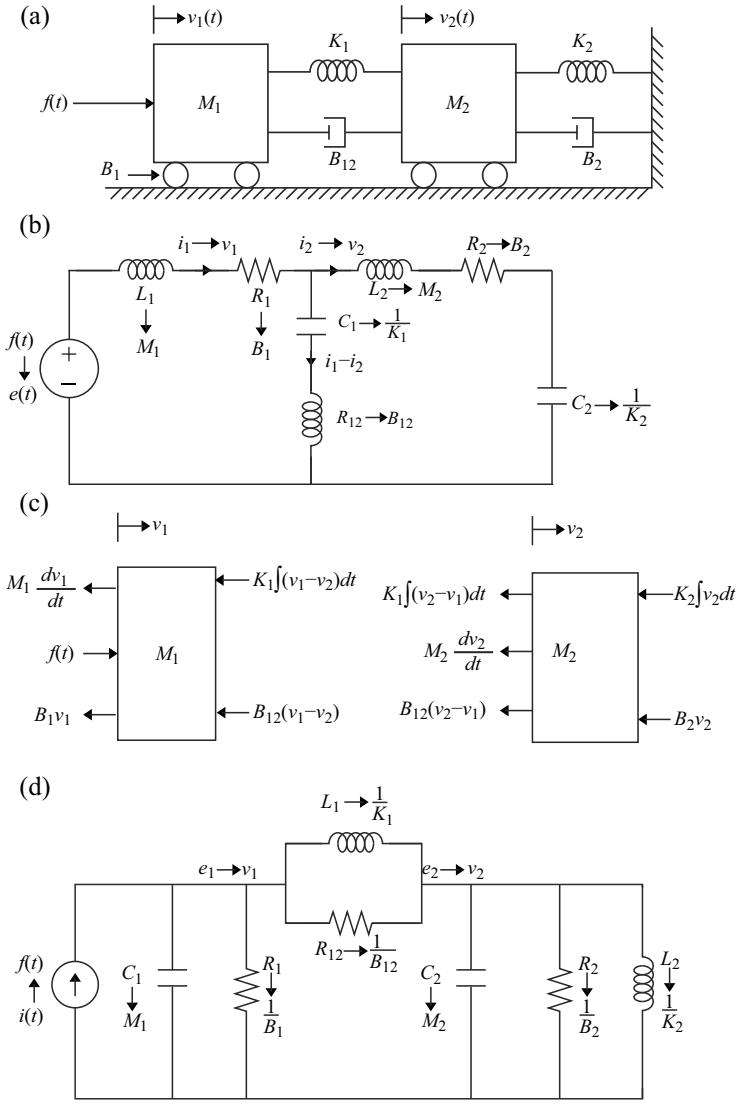


Fig. 1.51 **a** Mechanical translational system for Example 1.19. **b** Force-voltages analogy of example. **c** Free-body diagram of Fig. 1.51a. **d** Force -current analogous circuit of Fig. 1.51a

For the loop consisting of L_2, R_2, C_2, R_{12} and C_1, the following equation is written:

$$L_2\frac{di_2}{dt} + i_2R_2 + R_{12}(i_1 - i_2) + \frac{1}{C_1}\int (i_1 - i_2)dt + \frac{1}{C_2}\int i_2 dt = 0 \tag{1.80}$$

The free-body diagram of the given mechanical system given in Fig. 1.51a is shown in Fig. 1.51c. For Fig. 1.51c the following equations are written:

$$M_1\frac{dv_1}{dt} + B_1v_1 + B_{12}(v_1 - v_2) + K_1\int (v_1 - v_2)dt = f(t) \tag{1.81}$$

$$M_2\frac{dv_2}{dt} + B_2v_2 + B_{12}(v_2 - v_1) + K_1\int (v_2 - v_1)dt + K_2\int v_2 dt = 0 \tag{1.82}$$

From Table 1.3, Eq. (1.79) is analogous to Eqs. (1.81) and (1.80) is analogous too Eq. (1.82). Hence, force-voltage analogy is verified.

Force-Current Analogy

1. In Fig. 1.51a, mass M_1 and dash-pot B_1 move with a velocity v_1. The force is applied to mass M_1. Hence, the one end of current sources $i(t) \rightarrow f(t)$, $C_1 \rightarrow M_1$ and $R_1 \rightarrow (1/B_1)$ are connected to the node $e_1 \rightarrow v_1$. Their other ends are connected to the common point.
2. Mass M_2, dash-pot B_2 and spring K_2 move with velocity v_2. Hence, the one end of $C_2 \rightarrow M_2$, $R_2 \rightarrow (1/B_2)$ and $L_2 \rightarrow (1/K_2)$ are connected to the mode $e_2 \rightarrow v_2$. Their other ends are connected to the common points.
3. The dash-pot B_{12} and spring K_1 move with a velocity difference $(v_1 - v_2)$. Therefore, $R_{12} \rightarrow (1/B_{12})$ and $L_1 \rightarrow (1/K_1)$ are connected in parallel and in between the nodes $e_1 \rightarrow v_1$ and $e_2 \rightarrow v_2$. This completes the force-current analogy. This is verified by writing down the electric circuit equations.

For the node e_1, the following equation is written:

$$C_1\frac{di_1}{dt} + \frac{1}{R_1}e_1 + \frac{1}{L_1}\int (e_1 - e_2)dt + \frac{1}{R_{12}}(e_1 - e_2) = i(t) \tag{1.83}$$

For the node e_2, the following equation is written:

$$C_2\frac{de_2}{dt} + \frac{1}{R_2}e_2 + \frac{1}{L_2}\int e_2 dt + \frac{1}{R_{12}}(e_2 - e_1) + \frac{1}{L_2}\int (e_2 - e_1)dt = 0 \tag{1.84}$$

From Table 1.3, it is evident that Eq. (1.83) is analogous to Eqs. (1.81) and (1.84) is analogous to Eq. (1.80). Thus, force-current analogy is verified.

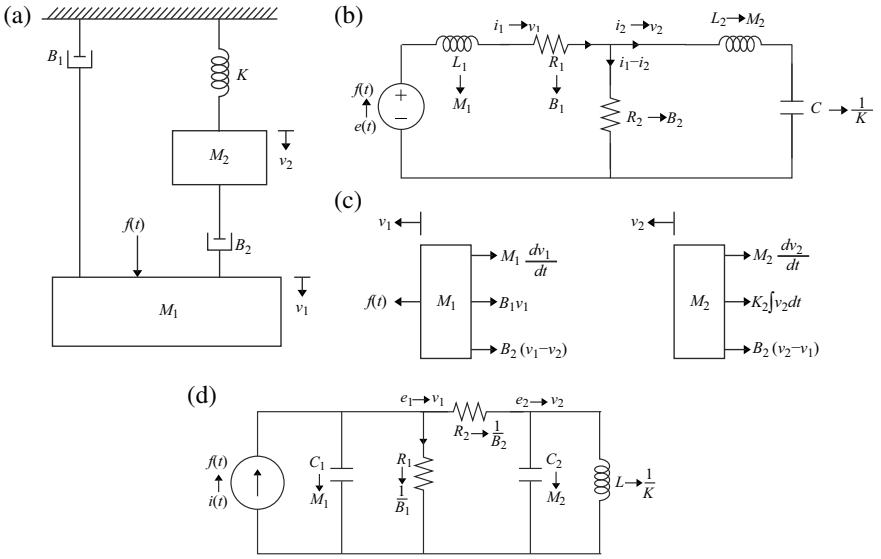


Fig. 1.52 **a** Mechanical system for Example 1.20. **b** Force-voltage analogous circuit of Fig. 1.52a. **c** Free-body diagram of Fig. 1.52a. **d** Force-current analogy of Fig. 1.52a

Example 1.20

Draw the force voltage and force-current analogous circuits for the mechanical translational system given in Fig. 1.52a.

(*Anna University, May, 2010*)

Solution Force-Voltage Analogy

1. Force $f(t)$ is applied to mass M_1 which moves with velocity v_1. Dash-pot B_1 is connected to mass M_1 and it also moves with the same velocity. A voltage source $e(t) \rightarrow f(t)$ is created and in series with $e(t)$, $L_1 \rightarrow M_1$ and $R_1 \rightarrow B_1$ are connected in series as shown in Fig. 1.52b.
2. Mass M_2 moves with velocity v_2 and also the spring K. A loop wherein the current flow is i_2 is created and the series combination of $L_2 \rightarrow M_2$ and $C \rightarrow (1/K)$ is connected.
3. The velocity difference across B_2 is $(v_1 - v_2)$. A branch current $(i_1 - i_2)$ is opened from i_1 and i_2 and $R_2 \rightarrow (1/B_2)$ is connected in that branch. The force voltage analogy is now completed and is shown in Fig. 1.52b.
4. The free-body diagram of the mechanical system is shown in Fig. 1.52c.

For Fig. 1.52b the following equations are written:

$$L_1\frac{di_1}{dt} + R_1 i_1 + R_2(i_1 - i_2) = e(t) \tag{1.85}$$

$$L_2\frac{di_2}{dt} + \frac{1}{C}\int i_2 dt + R_2(i_2 - i_1) = 0 \tag{1.86}$$

For Fig. 1.52c, the following equations are written:

$$M_1\frac{dv_1}{dt} + B_1 v_1 + B_2(v_1 - v_2) = f(t) \tag{1.87}$$

$$M_2\frac{dv_2}{dt} + K\int v_2 dt + B_2(v_2 - v_1) = 0 \tag{1.88}$$

Equation (1.85) is analogous to Eqs. (1.87) and (1.86) to (1.89) as seen from Table 1.3. Hence, force-voltage analogy is verified.

Force-Current Analogy

1. Mass M_1 and dash-pot B_1 move with velocity v_1 and the force is applied to mass M_1. A node $e_1 \rightarrow v_1$ is created and to this node $e(t) \rightarrow f(t)$, $C_1 \rightarrow M_1$ and $R_1 \rightarrow (1/B_1)$ are connected as shown in Fig. 1.52d.
2. Mass M_2 and spring K move with velocity v_2. A node $e_2 \rightarrow v_2$ is created and to this node $C_2 \rightarrow M_2$ and $L \rightarrow (1/K)$ are connected as shown in Fig. 1.52d.
3. The dash-pot B_2 move with a velocity difference of $(v_1 - v_2)$. Hence, $R_2 \rightarrow (1/B_2)$ is connected in between the nodes e_1 and e_2. This completes the force–current analogy.

For Fig. 1.52d the following equation is written at node e_1

$$C_1\frac{de_1}{dt} + \frac{1}{R_1}e_1 + \frac{1}{R_2}(e_1 - e_2) = i(t) \tag{1.89}$$

$$C_2\frac{de_2}{dt} + \frac{1}{L}\int e_2 dt + \frac{1}{R_2}(e_2 - e_1) = 0 \tag{1.90}$$

Equation (1.89) is analogous to Eqs. (1.87) and (1.90) is analogous to Eq. (1.89) and hence force-current analogy is verified.

Example 1.21

Consider the mechanical system shown in Fig. 1.53a. Draw the force-voltage and force-current analogous circuits.

Solution Force-Voltage Analogy

1. In Fig. 1.53a the junction point A of dash-pot B and spring K is neither connected to any mass nor to any reference frame. Hence, that point is assumed to move with a velocity v_3.

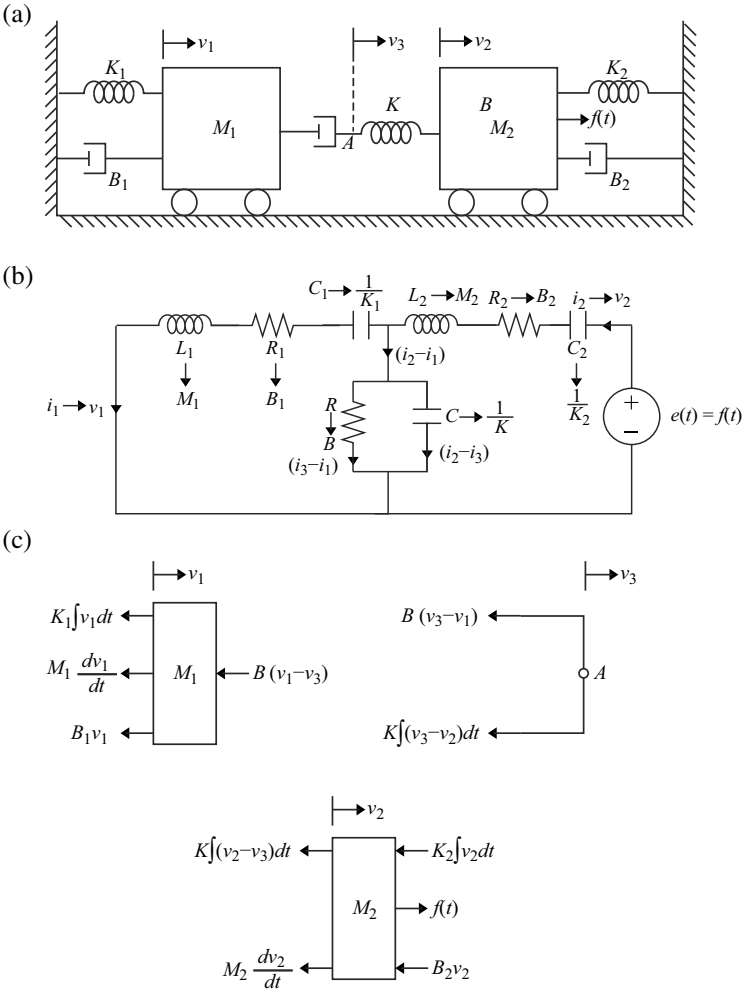


Fig. 1.53 **a** Mechanical system for Example 1.21. **b** Force-voltage analogy of Fig. 1.53a. **c** Free-body diagram of Fig. 1.53a. **d** Force-current analogy of Fig. 1.53a

2. Force $f(t)$ is applied to mass M_2. The spring K_2, dash-pot B_2 and mass M_2 all move with velocity v_2. The voltage source $e(t) \rightarrow f(t)$, $L_2 \rightarrow M_2$, $R_2 \rightarrow B_2$ and $C_2 \rightarrow (1/K_2)$ are all connected in series and the current flow in this branch is $i_2 \rightarrow v_2$.
3. Mass M_1, dash-pot B_1 and spring K_1 all move with velocity v_1. A branch current i_1 is opened and in this branch $L_1 \rightarrow M_1$, $R_1 \rightarrow B_1$ and $C_1 \rightarrow (1/K_1)$ are connected in series.
4. The velocity difference across the cascade connected B and K is $(v_1 - v_3)$. From i_1 and i_2 branches, $(i_2 - i_1)$ branch is created. The velocity difference across K

is $(v_2 - v_3)$. From $(i_2 - i_1)$ branch, a new branch with $(i_2 - i_3)$ current is created and $C \rightarrow (1/K)$ is connected in that branch. Similarly, the velocity difference across B is $(v_3 - v_1)$. The branch current $(i_3 - i_1)$ is created from $(i_2 - i_1)$ and in this branch $R \rightarrow B$ is connected. The series connected mechanical elements B and K in Fig. 1.53a become parallel connected R and C in Fig. 1.53b. This completes the force-voltage analogy. The free-body diagram of Fig. 1.53a is represented in Fig. 1.53b. For Fig. 1.53a the following equation is written:

$$L_1 \frac{di_1}{dt} + R_1 i_1 + \frac{1}{C_1} \int i_1 dt + R(i_1 - i_3) = 0 \tag{1.91}$$

$$L_2 \frac{di_2}{dt} + R_2 i_2 + \frac{1}{C_2} \int i_2 dt + \frac{1}{C} \int (i_2 - i_3) dt = e(t) \tag{1.92}$$

$$R(i_3 - i_1) + \frac{1}{C} (i_3 - i_2) dt = 0 \tag{1.93}$$

From free-body diagram of Fig. 1.53c the following equations are written:

$$M_1 \frac{dv_1}{dt} + B_1 v_1 + K_1 \int v_1 dt + B(v_1 - v_3) = 0 \tag{1.94}$$

$$M_2 \frac{dv_2}{dt} + B_2 v_2 + K_2 \int v_2 dt + K \int (v_2 - v_3) dt = f(t) \tag{1.95}$$

$$B(v_3 - v_1) + K \int (v_3 - v_2) dt = 0 \tag{1.96}$$

Equations (1.91), (1.94), (1.92), (1.95), (1.93) and (1.96) are analogous as seen from Table 1.3. Hence, force-voltage analogy is proved.

Force-Current Analogy

1. Corresponding to the velocities v_1, v_2 and v_3 in the mechanical system, nodal voltage points e_1, e_2 and e_3 are created for force-current analogy as shown in Fig. 1.53d.
2. Mass M_1, dash-pot B_1 and spring K_1 all move with velocity v_1 and therefore the one ends of $C_1 \rightarrow M_1$, $R_1 \rightarrow (1/B_1)$ and $L_1 \rightarrow (1/K_1)$ are connected to the node e_1 while their other ends are connected to the common point.
3. The force $f(t)$ is applied to mass M_2. Mass M_2, spring K_2 and the dash-pot B_2 all move with velocity v_2 and therefore the one ends of $i(t) \rightarrow f(t)$, $C_2 \rightarrow M_2$, $R_2 \rightarrow (1/B_2)$ and $L_2 \rightarrow (1/K_2)$ are connected to the node e_2 and their other ends are connected to the common point.
4. The velocity difference across B is $(v_1 - v_3)$ and therefore $R \rightarrow B$ is connected between e_1 and e_3 nodes. Similarly the velocity difference across K is $(v_2 - v_3)$. The inductance $L \rightarrow (1/K)$ is connected between the nodes e_2 and e_3. This completes the force-current analogy. The following equations are written for Fig. 1.53d and verified with those obtained for the given mechanical system shown in Fig. 1.53a.

At node e_1, the following equation for the algebraic sum of the current entering and leaving is written.

$$C_1\frac{de_1}{dt}+\frac{1}{R_1}e_1+\frac{1}{L_1}\int e_1dt+\frac{1}{R}(e_1-e_3)=0 \tag{1.97}$$

At node e_2, the following equation is written:

$$C_2\frac{de_2}{dt}+\frac{1}{R_2}e_2+\frac{1}{L_2}\int e_2dt+\frac{1}{L}\int(e_2-e_3)dt=i(t) \tag{1.98}$$

At node e_3, the following equation is written:

$$\frac{1}{R}(e_3-e_1)+\frac{1}{L}\int(e_3-e_2)dt=0 \tag{1.99}$$

Equations (1.94), (1.97), (1.95), (1.99), (1.91) and (1.99) are as per Table 1.3 are analogous. Hence, force-current analogy is verified.

Example 1.22

For the rotational mechanical system shown in Fig. 1.54a draw the torque-voltage and torque-current analogous circuits.

Solution

Torque-Voltage Analogy

1. In Fig. 1.54, torque $T(t)$ is applied to inertia J_1. Inertia J_1 and damper B_1 experience the same angular velocity ω_1. Therefore, the voltage source $e(t) \rightarrow T(t)$, inductor $L_1 \rightarrow J_1$ and resistor R_1 are all connected in series and the current flow in this branch is $i_1 \rightarrow \omega_1$.
2. A branch with i_2 current is created. J_2 and B_2 move with a velocity ω_2. Therefore, in the branch where $i_2 \rightarrow \omega_2$ flows, the series combination of $L_2 \rightarrow J_2$ and $R_2 \rightarrow B_2$ are connected.
3. From branches where i_1 and i_2 flow, a third branch $(i_1 - i_2) \rightarrow (\omega_1 - \omega_2)$ is created. The velocity difference across the spring K and dash-pot B_{12} is $(\omega_1 - \omega_2)$. In the branch where the current flow is $(i_1 - i_2)$. $R_{12} \rightarrow B_{12}$ and $C = 1/K$ are connected in series. This completes the torque-voltage analogous circuit. The following equations are written for Fig. 1.54b.

$$L_1\frac{di_1}{dt}+R_1i_1+R_{12}(i_1-i_2)+\frac{1}{C}\int(i_1-i_2)dt=e(t) \tag{1.100}$$

$$L_2\frac{di_2}{dt}+R_2i_2+R_{12}(i_2-i_1)+\frac{1}{C}\int(i_2-i_1)dt=0 \tag{1.101}$$

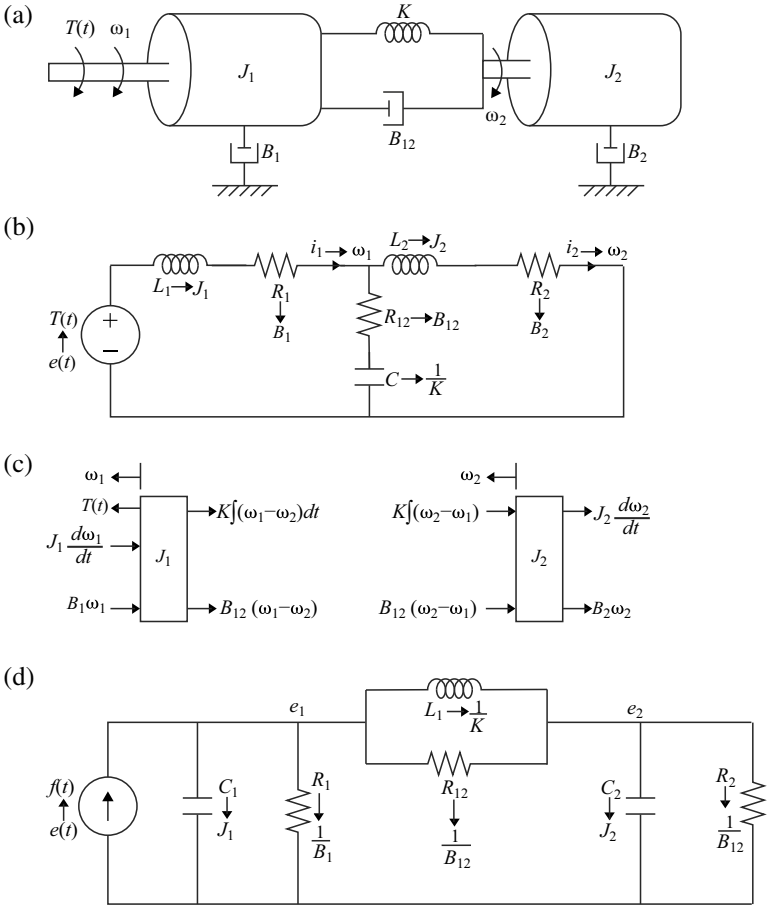


Fig. 1.54 **a** Mechanical rotational system for Example 1.22. **b** Torque-force analogous circuit of Fig. 1.54a. **c** Free-body diagram of Fig. 1.54a. **d** Torque-current analogous circuit of Fig. 1.54a

For the free-body diagram of Fig. 1.54c, the following equations are written:

$$J_1\frac{d\omega_1}{dt} + B_1\omega_1 + B_{12}(\omega_1 - \omega_2) + K\int(\omega_1 - \omega_2)dt = T(t) \quad (1.102)$$

$$J_2\frac{d\omega_2}{dt} + B_2\omega_2 + B_{12}(\omega_2 - \omega_1) + K\int(\omega_2 - \omega_1)dt = 0 \quad (1.103)$$

Equations (1.100), (1.102), (1.101) and (1.103) are analogous as seen from Table 1.3. Hence, the torque-voltage analogy is verified.

Torque-Current Analogy

The following equations are written for Fig. 1.54d. At node e_1 we equate the current entering to the current leaving as given below:

$$C_1\frac{de_1}{dt}+\frac{e_1}{R_1}+\frac{1}{R_{12}}(e_1-e_2)+\frac{1}{L}\int(e_1-e_2)dt=i(t) \tag{1.104}$$

At node e_2 we write,

$$C_2\frac{de_2}{dt}+\frac{1}{R_2}e_2+\frac{1}{R_{12}}(e_2-e_1)+\frac{1}{L}\int(e_2-e_1)dt=0 \tag{1.105}$$

Equations (1.102), (1.104), (1.103) and (1.105) are seen to be analogous as per Table 1.3. Hence, torque-current analogy is verified.

Example 1.23

Consider the mechanical system shown in Fig. 1.55a. Draw the torque-voltage and torque-current analogous circuits.

Solution

Torque-Voltage Analogy

1. In the mechanical system of Fig. 1.55a there are four angular displacements ω_1, ω_2, ω_3 and ω_4. Corresponding to these displacements, branch currents i_1, i_2, i_3 and i_4 are created. Torque $T(t)$ is applied across the spring K_1 and dash-pot B_1. A voltage source $e(t) \rightarrow T(t)$ is created. From this source $i_1 \rightarrow \omega_1$ flows. From i_1 branch i_2 branch and $(i_1 - i_2)$ branch are created.
2. The velocity difference across K_1 and B_1 is $(\omega_1 - \omega_2)$ and therefore $R_1 \rightarrow B_1$ and $C_1 \rightarrow (1/K_1)$ are connected in series with the branch where the current flow is $(i_1 - i_2)$. J_1 moves with angular velocity ω_2. $L_1 \rightarrow J_1$ is connected in the branch where the current flow is i_2.
3. From i_2 branch, two more branches are created, one branch where i_3 current flows and another branch where $(i_2 - i_3)$ flows. Spring K_2 moves with angular speed difference of $(\omega_2 - \omega_3)$ and therefore, $C_2 \rightarrow (1/K_2)$ is connected where the current flow is $(i_2 - i_3) \rightarrow (\omega_2 - \omega_3)$. The current $i_3 \rightarrow \omega_3$ which corresponds to primary gear. i_3 is connected to the primary of the voltage transformer whose turns ratio is $N_2 : N_1$.
4. The secondary gear moves with angular speed ω_4. Hence, in the secondary of the transformer i_4 current flows.
5. Inertia J_2 and dash-pot B_2 rotate with the angular speed ω_4 and therefore $L_2 \rightarrow J_2$ and $R_2 \rightarrow B_2$ are connected in series in the secondary of the transformer. This completes the torque-voltage analogy.

The following equations for mechanical and electrical systems are written and the analogy verified. The free-body diagram for Fig. 1.55a is shown in Fig. 1.55c. For

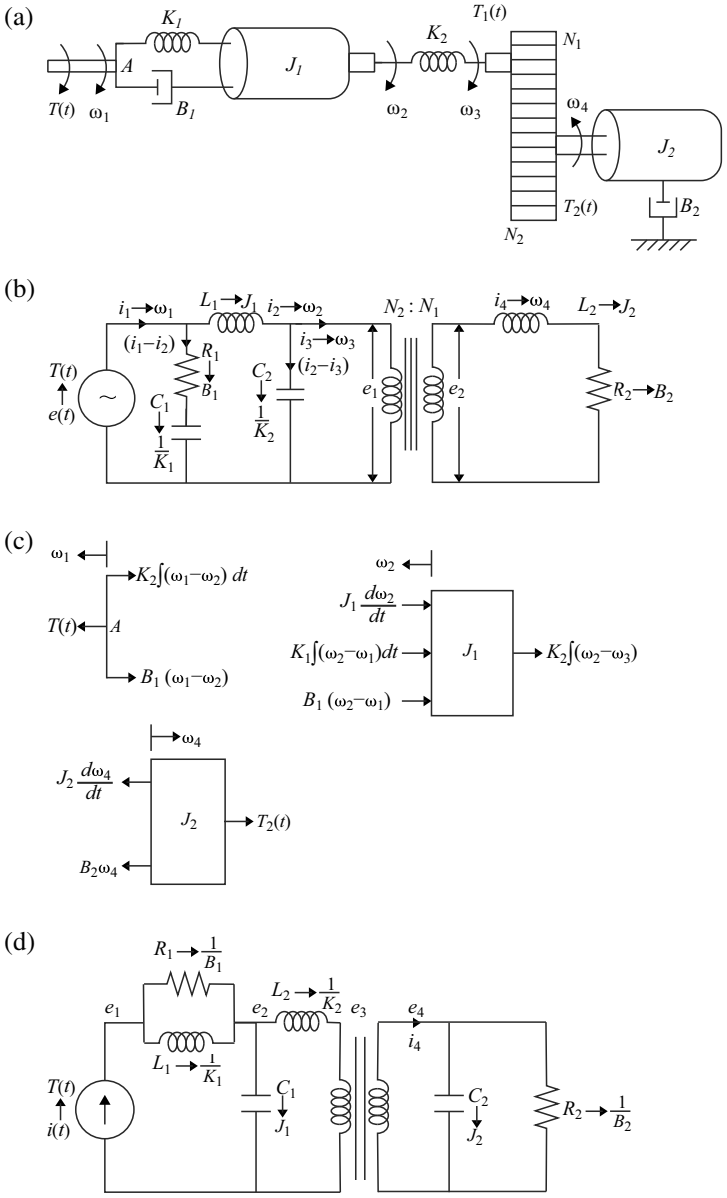


Fig. 1.55 **a** Mechanical rotational system for Example 1.23. **b** Torque-voltage analogy of Fig. 1.55a. **c** Free-body diagram of Fig. 1.55a. **d** Torque-current analogous circuit for Fig. 1.55a

Fig. 1.55b, the following equations are written:

$$R_1(i_1 - i_2) + \frac{1}{C_1}\int (i_1 - i_2)dt = e(t) \quad (1.106)$$

$$L_1\frac{di_2}{dt} + \frac{1}{C_2}\int (i_2 - i_3)dt + \frac{1}{C_1}\int (i_2 - i_1)dt + R_1(i_2 - i_1) = 0 \quad (1.107)$$

$$L_2\frac{di_4}{dt} + R_2 i_4 = e_2 \quad (1.108)$$

$$\frac{T_1}{T_2} = \frac{N_1}{N_2} = \frac{\omega_4}{\omega_3}$$

$$N_1\omega_3 = N_2\omega_4 \quad \text{or} \quad \frac{N_1}{N_2} = \frac{\omega_4}{\omega_3} = \frac{e_2}{e_1}$$

Hence, the turns ratio of the transformer is $N_2 : N_1$. For Fig. 1.55c the following equations are written:

$$B_1(\omega_1 - \omega_2) + K_1\int (\omega_1 - \omega_2)dt = T(t) \quad (1.109)$$

$$J_1\frac{d\omega_2}{dt} + K_2\int (\omega_2 - \omega_3)dt + K_1\int (\omega_2 - \omega_1)dt + B_1(\omega_2 - \omega_1) = 0 \quad (1.110)$$

$$J_2\frac{d\omega_4}{dt} + B_2\omega_4 = T_2(t) \quad (1.111)$$

Equations (1.106), (1.109), (1.107), (1.110), (1.108) and (1.111) are analogous as per Table 1.3. Hence, torque-voltage analogy is verified.

Torque-Current Analogy

Consider Fig. 1.55d. The following equations are written. At node e_1 we get,

$$\frac{1}{R_1}(e_1 - e_2) + \frac{1}{L_1}\int (e_1 - e_2)dt = i(t) \quad (1.112)$$

At node e_2 we get,

$$C_1\frac{de_2}{dt} + \frac{1}{L_2}\int (e_2 - e_3)dt + \frac{1}{L_1}\int (e_2 - e_1)dt + \frac{1}{R_1}(e_2 - e_1) = 0 \quad (1.113)$$

At e_4 node we get,

$$C_2\frac{de_4}{dt} + \frac{1}{R_2}e_4 = i_4 \quad (1.114)$$

Equations (1.106), (1.112), (1.107), (1.113), (1.108) and (1.114) are analogous as per Table 1.3. Hence, torque-current analogy is verified. It is to be noted that the current transformer will have a turns ratio $N_1 : N_2$.

1.18 Block Diagram Reduction Technique

A simple system is described as the operation which transforms one input signal to produce one output signal as shown in Fig. 1.56. The system in general has the cause and effect relationship. The block diagram represents the cause and effect relationship or simply the functional relationship between the input signal and output signal. The input signal $r(t)$ can be altered or manipulated to achieve a desired system behaviour. The output signal $c(t)$ is a variable which is to be controlled in some desired manner. The dynamic systems are very often represented by a set of differential equations. The application of Laplace transform to these set of differential equations results in a set of linear algebraic equations. Thus, a functional relationship between the output variable and input variable is established which is defined as the transfer function.

The cause and effect relationship of the transfer function can be represented by diagrammatic means as in Fig. 1.56 which is called the block diagram. Block diagrams consist of unidirectional operational blocks that represent the transfer function of the variables of interest. It is a pictorial representation of individual components and of the flow of the signal. In the block diagram representation while the blocks represent the transfer function, the direction of the signal flow is represented by arrow. The signal can only pass in the direction of the arrows. The dimensions of the output signal from the block are the dimensions of the input signals multiplied by the dimension of the transfer function $G(s)$.

In Fig. 1.56 individual system is represented by a single block with its input and output. In practice, many such systems are interconnected. Such multiple subsystems can be represented with a single transfer function. The technique to reduce the interconnected subsystem to a single block diagram as in Fig. 1.56 is known as block diagram reduction technique and is discussed below.

Advantages of Block Diagram Representation

The following are the advantages of block diagram representation:

1. It is easy to form the overall block diagram for the entire system by merely connecting the blocks of the components according to the signal flow.
2. It is easy to evaluate the contribution of each component to the overall performance of the system.

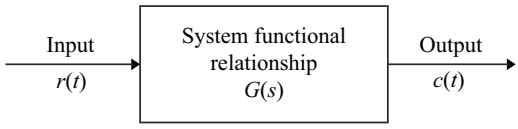


Fig. 1.56 System representation in block diagram

3. In block diagram, the functional operation of the system can be visualized more readily by examining the individual blocks rather than examining the physical system itself.

Disadvantage of Block Diagram Representation
The following are the disadvantages of block diagram representation:

1. The main source of energy flow in the system is not shown in block diagram.
2. For a particular system, different block diagram is drawn. It is not unique.
3. It is difficult to identify the nature of physical elements connected in the system. The second order electrical or mechanical or hydraulic systems are all represented by the same block diagram.
4. Important functions are concealed within the walls of block diagram.
5. Block diagram representation does not take into account the effect of interaction between blocks.
6. Passive symmetrical pi networks are reciprocal since the input and output are inter-changeable. However, in an amplifier the input and output are not inter-changeable. This property is not considered in block diagram representation.

1.18.1 Component Parts of Block Diagram

When multiple subsystems are interconnected new elements such as junction and pick off points are added to the system which are shown in Fig. 1.57.

Few topologies for interconnecting subsystems are available so that a single transfer function, using these topologies are obtained. These topologies include the following:

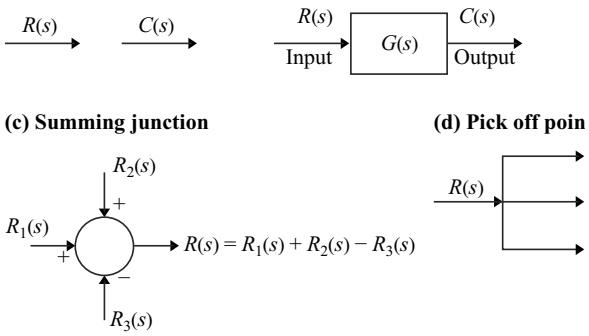


Fig. 1.57 Components of block diagram

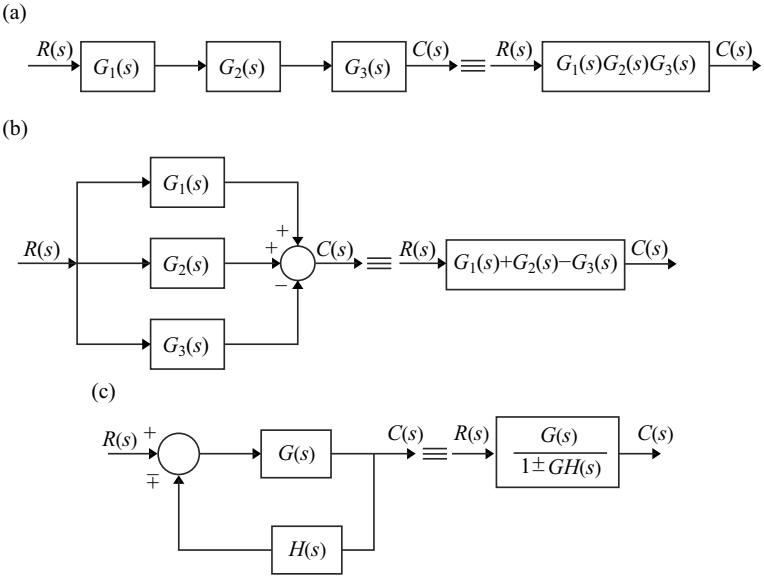


Fig. 1.58 Block diagram topology. **a** Cascade connection, **b** Parallel connection and **c** Feedback connection

1. Cascade form
2. Parallel form
3. Feedback form

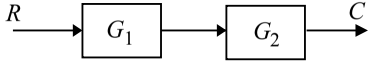
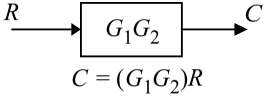
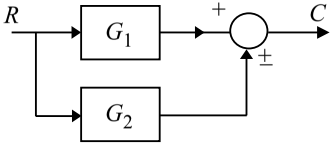
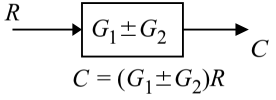
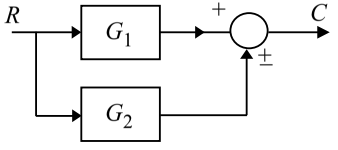
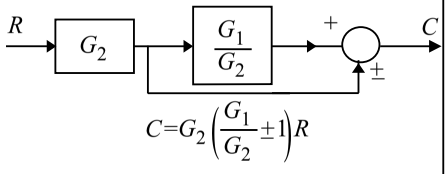
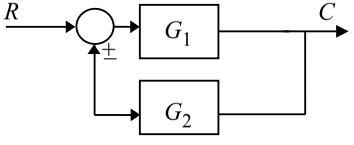
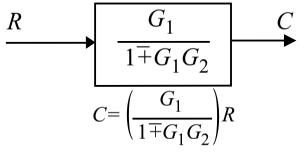
These connections and their equivalence are shown in Fig. 1.58.

1.18.2 Block Diagram Reduction Rule

The reduction of block diagram to a single block representation can be derived by simple algebraic manipulation of the equations representing the blocks. These diagram transformations are known as block diagram reduction rules. These rules are given in Table 1.4. System analysis by the method of block diagram reduction technique gives a better understanding of the contribution of each component than is possible to obtain by the manipulation of equations.

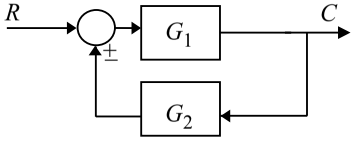
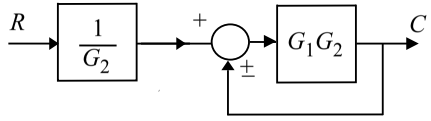
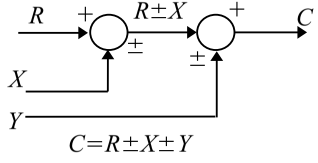
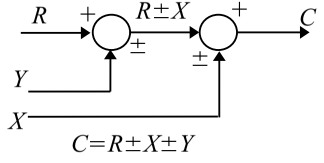
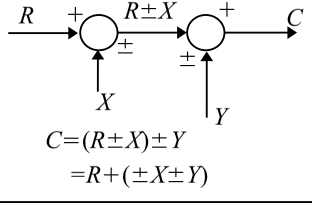
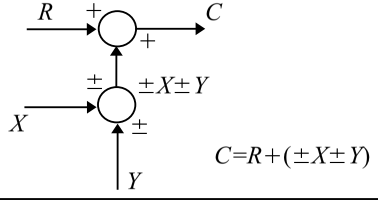
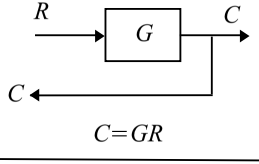
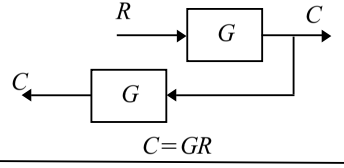
The following examples illustrate the method of block diagram reduction. Table 1.4 is used whereever necessary.

Table 1.4 Block diagram reduction rules

Transformation	Original Block Diagram	Equivalent Block Diagram
1. Combining blocks in Cascade		
2. Combining blocks in parallel or eliminating a forward loop		
3. Removing a block from a forward path		
4. Eliminating a feedback loop		

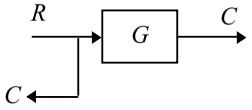
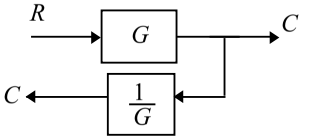
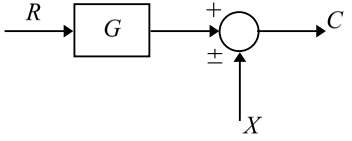
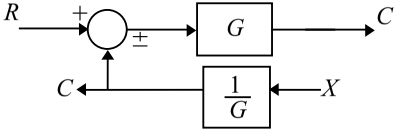
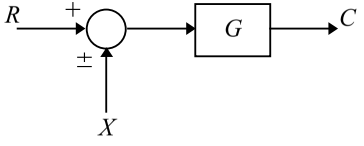
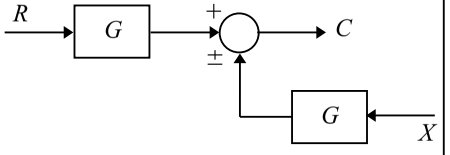
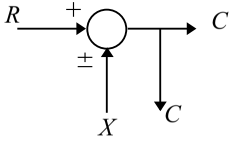
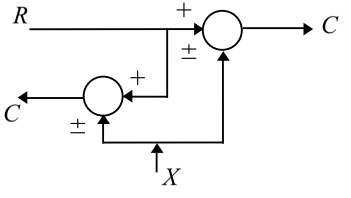
(continued)

Table 1.4 (continued)

5. Removing a block from a feedback loop		
6. Rearranging summing points		
7. Rearranging summing points		
8. Moving a take-off point ahead of a block		

(continued)

Table 1.4 (continued)

9. Moving a take-off point beyond a block	 $C=GR$	 $C=GR$
10. Moving a summing point ahead of a block	 $C=RG\pm X$	 $C=G(R\pm(1/G))X=RG\pm X$
11. Moving a summing point behind a block	 $C=RG\pm XG=G(R\pm X)$	 $C=RG\pm XG=G(R\pm X)$
12. Moving a take-off point ahead of a summing point	 $C=R\pm X$	

(continued)

Table 1.4 (continued)

13. Moving a take-off point beyond a summing point		
14. Moving a take-off point behind a summation		

(continued)

Example 1.24

Reduce the block diagram shown in figure and obtain its closed loop transfer function $\frac{C}{R}(s)$.

(*Anna University, May, 2009*)

Solution The successive reductions of blocks using Table 1.4 are explained below.

1. In Fig. 1.59b G_2 and H_2 blocks are connected in feedback mode and their equivalence is $G_2/1 + G_2H_2$ (Rule 4). The summers S_1 and S_2 are interchanged using

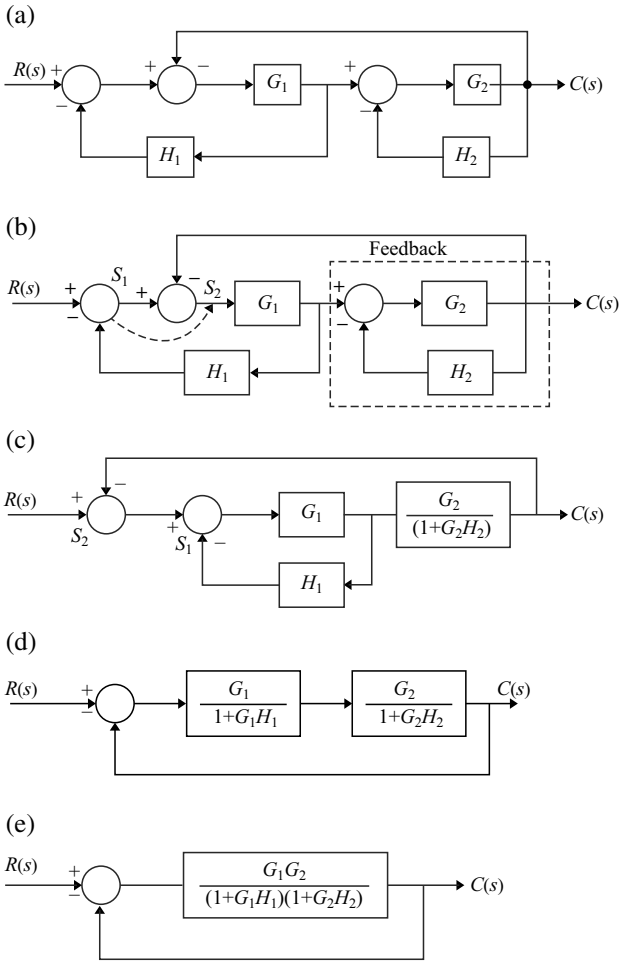


Fig. 1.59 **a** Block diagram for Example 1.24. **c** Reduced block diagram. **d** Reduced block diagram (**e**)

Rule 6 so that the blocks G_1 and H_1 come under feedback mode. The reduced block diagram is shown in Fig. 1.59c.

2. G_1 and H_1 blocks are in feedback mode and they are transformed as $G_1/1 + G_1H_1$. The reduced block diagram is shown in Fig. 1.59d.
3. The two blocks in the forward path are in cascade and they are transformed as

$$\frac{G_1G_2}{(1 + G_1H_1)(1 + G_2H_2)}$$

and is shown in Fig. 1.59e.

4. Figure 1.59e is in feedback mode and is reduced using Rule 4 with the following result.

$$\boxed{\frac{C}{R}(s) = \frac{G_1G_2}{(1 + G_1H_1)(1 + G_2H_2) + G_1G_2}}$$

Example 1.25

Using block diagram reduction technique find the closed loop T.F. of the system whose block diagram is given in Fig. 1.60a.

(*Anna University, December, 2010*)

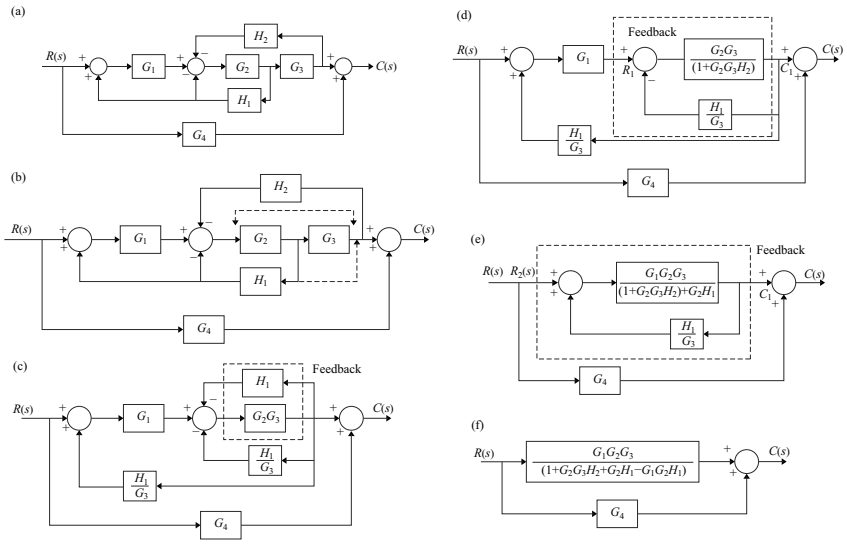


Fig. 1.60 **a** Block diagram for Example 1.25

Solution

1. In Fig. 1.60a, using Rule 9, the take off point of H_1 block is moved beyond G_3 block. This enables G_2 and G_3 blocks in cascade. The reduced block diagram is shown in Fig. 1.60c.
2. The blocks G_2G_3 and H_2 are in feedback mode and they are reduced using Rule 4 as
$$\frac{G_2G_3}{(1+G_2G_3H_2)}.$$
The reduced block diagram is shown in Fig. 1.60d.
3. The blocks inside the dotted line are in feedback mode. Using Rule 4, they are reduced as,
$$\begin{aligned}\frac{C_1}{R_1} &= \frac{G_2G_3/(1+G_2G_3H_2)}{1+(G_2G_3H_1/(1+G_2G_3H_2)G_3)} \\ &= \frac{G_2G_3}{(1+G_2G_3H_2)+G_2H_1}\end{aligned}$$
The reduced block diagram is shown in Fig. 1.60e. Coupled with cascaded block G_1.
4. The blocks inside the dotted line are in positive feedback mode and they are reduced using Rule 4 as,
$$\frac{C_1}{R_2} = \frac{\frac{G_1G_2G_3}{(1+G_2G_3H_2+G_2H_1)}}{1-\frac{G_1G_2G_3}{(1+G_2G_3H_2+G_2H_1)}\frac{H_1}{G_3}} = \frac{G_1G_2G_3}{1+G_2G_3H_2+G_2H_1-G_1G_2H_1}$$
The reduced block diagram is shown in Fig. 1.60f.
5. The two blocks shown in Fig. 1.60f are in parallel connection. Rule 2 is used to reduce these blocks and thus we get,
$$\frac{C(s)}{R(s)} = \frac{G_1G_2G_3}{(1+G_2G_3H_2+G_2H_1-G_1G_2H_1)} + G_4$$

$$\boxed{\frac{C(s)}{R(s)} = \frac{G_1G_2G_3+G_4(1+G_2G_3H_2+G_2H_1-G_1G_2H_1)}{(1+G_2G_3H_2+G_2H_1-G_1G_2H_1)}}$$

Example 1.26

The block diagram of a closed loop system is shown in Fig. 1.61a. Using block diagram reduction technique determine the closed loop T.F.

(*Anna University, December, 2009*)

Solution

1. The blocks G_2 and H_2 are in negative feedback mode and using Rule 4, their equivalence is $G_2/(1 + G_2H_2)$. The reduced block diagram is shown in Fig. 1.61c.
2. The block H_1 in the feedback path is shifted after the summer S_2. Rule 11 is used and the modified block diagram is shown in Fig. 1.61d.
3. From Fig. 1.61d, between $R(s)$ and R_1 variable, a parallel connection exists and this part is written as

$$\frac{R_1}{R} = (1 + G_1)$$

Between R_1 and $C(s)$, the two blocks are connected in negative feedback mode and using Rule 4 we get,

$$\frac{C}{R_1} = \frac{G_2/(1 + G_2H_2)}{1 + (G_1G_2H_1/(1 + G_2H_2))} = \frac{G_2}{(1 + G_2H_2 + G_1G_2H_1)}$$

This is represented in Fig. 1.61e.

4. From Fig. 1.61e the final input/output ratio is obtained as,

$$\boxed{\frac{C(s)}{R(s)} = \frac{G_2(1 + G_1)}{(1 + G_2H_2 + G_1G_2H_1)}}$$

Example 1.27

For the block diagram shown in Fig. 1.62a obtain the overall transfer function $\frac{C(s)}{R(s)}$.

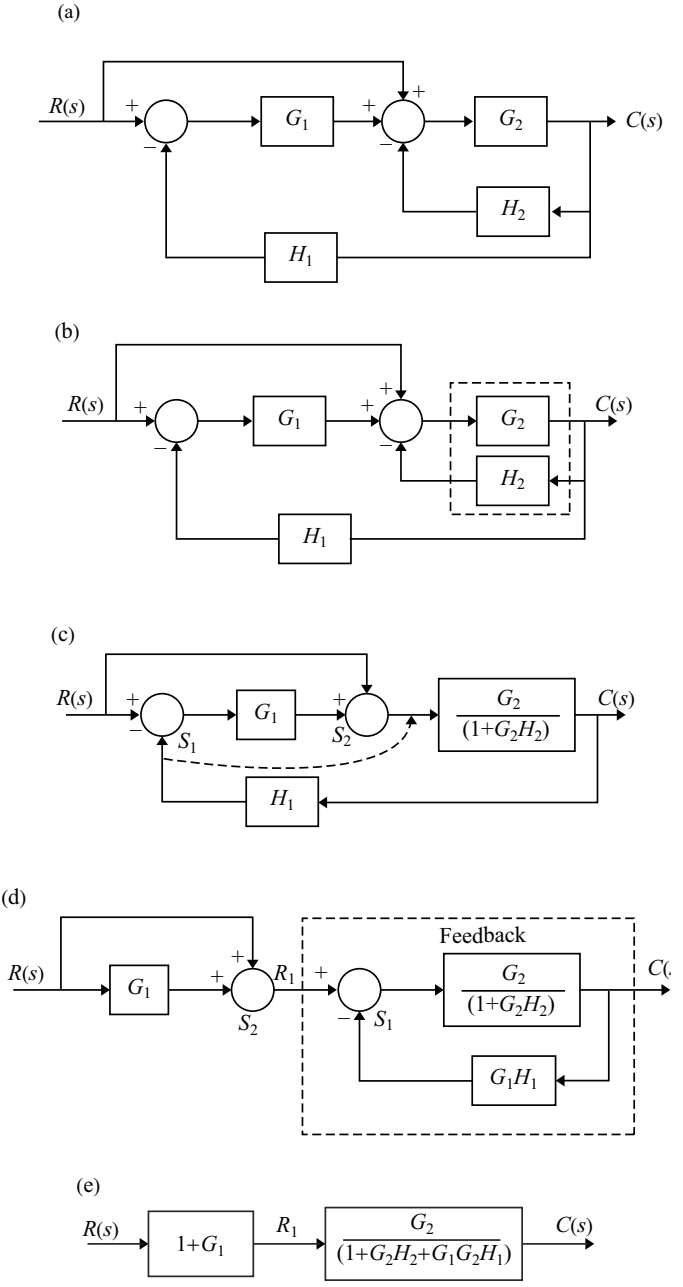


Fig. 1.61 Block diagram for Example 1.26

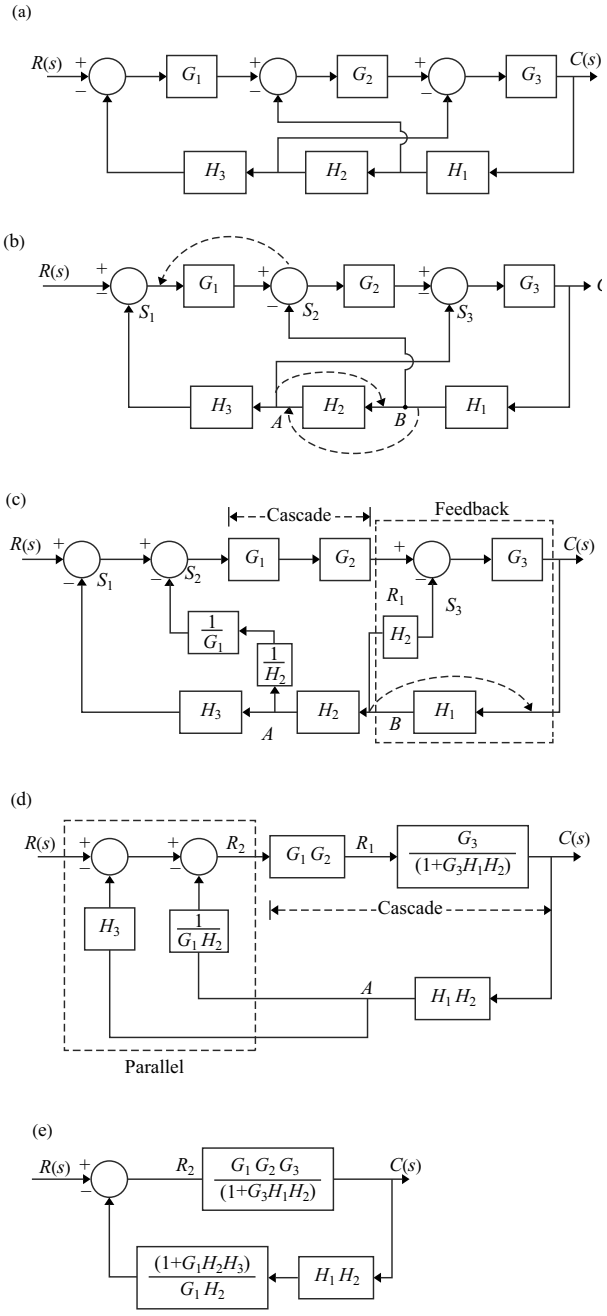


Fig. 1.62 Block diagram for Example 1.27

Solution

1. The summer S_2 is shifted before block G_1. Rule 10 is applied. Take off point A is shifted to take off point B and *vice-versa*. Rules 8 and 9 are respectively applied. The modified block diagram is shown in Fig. 1.62c.
2. From Fig. 1.62c, for the negative feedback indicated,

$$\frac{C(s)}{R_1} = \frac{G_3}{(1 + G_3H_1H_2)}$$

The blocks G_1 and G_2 are in cascade and can be combined as G_1G_2 using Rule 1. The block H_2 is shifted to $C(s)$ using Rule 8. The reduced block diagram is shown in Fig. 1.62d.

3. The blocks H_3 and $1/G_1H_2$ are in parallel and their combined block is

$$H_3 + \frac{1}{G_1H_2} = \frac{(1 + G_1H_2H_3)}{G_1H_2}$$

The cascaded blocks are combined as,

$$\frac{C(s)}{R_2} = \frac{G_1G_2G_3}{(1 + G_3H_1H_2)}$$

The modified block is shown in Fig. 1.62e.

4. The forward path T.F. of Fig. 1.62e is,

$$G = \frac{G_1G_2G_3}{(1 + G_3H_1H_2)}$$

The feedback loop T.F. is,

$$H = \frac{(1 + G_1H_2H_3)H_1}{G_1}$$

Using Rule 4 we get,

$$\frac{C(s)}{R(s)} = \frac{G}{1 + GH} = \frac{\frac{G_1G_2G_3}{(1+G_3H_1H_2)}}{1 + \frac{G_1G_2G_3}{(1+G_3H_1H_2)}\frac{(1+G_1H_2H_3)}{G_1}H_1}$$

$$\boxed{\frac{C(s)}{R(s)} = \frac{G_1G_2G_3}{(1 + G_3H_1H_2) + G_2G_3H_1(1 + G_1H_2H_3)}}$$

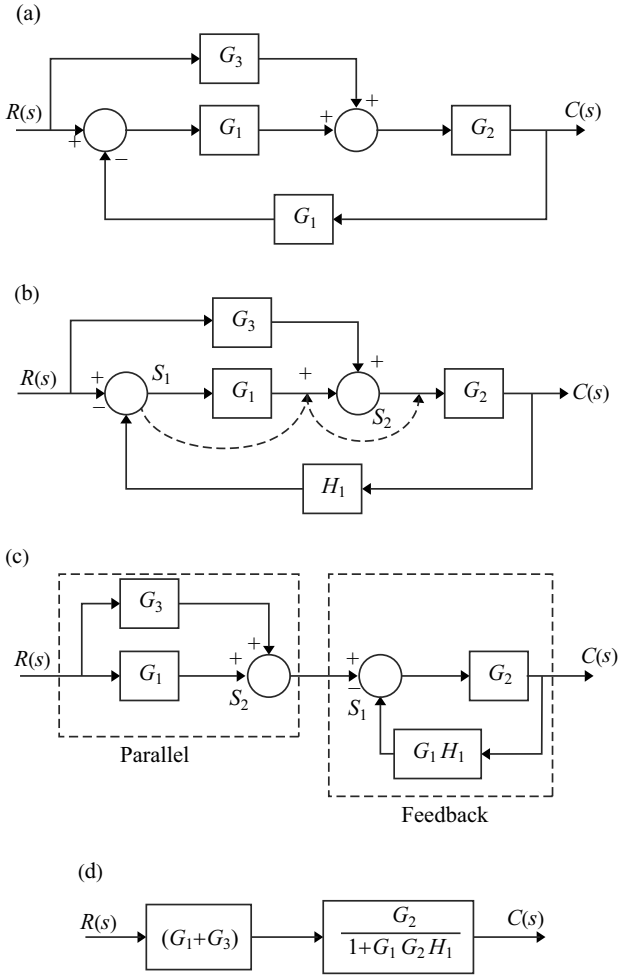


Fig. 1.63 Block diagram for Example 1.28

Example 1.28

Consider the block diagram shown in Fig. 1.63a. Determine the overall T.F. using block diagram reduction technique.

Solution

1. In Fig. 1.63a the summer S_1 is shifted after the block G_1 using Rule 11. Now the block H_1 is multiplied by G_1. Further the summer S_1 and S_2 are interchanged using Rule 6. The reduced block diagram is redrawn as shown in Fig. 1.63c.

2. The parallel connection in Fig. 1.63c is reduced, using Rule 2 as $G_1 + G_3$. The feedback connection is reduced, using Rule 4 as $G_2/1 + G_1G_2H_1$. The reduced block diagram is shown in Fig. 1.63d.
3. In Fig. 1.63d the two blocks are in cascade. Using Rule 1, the overall T.F. is obtained as,

$$\boxed{\frac{C(s)}{R(s)} = \frac{G_2(G_1 + G_3)}{(1 + G_1G_2H_1)}}$$

Example 1.29

For the block diagram shown in Fig. 1.64a, determine the overall T.F. by block diagram reduction technique.

Solution

1. Three points are shifted in the given block diagram. The block G_3 is shifted behind the block G_1 using Rule 8. Block G_3 gain is changed as G_3/G_1. Summer S_3 is shifted behind the block G_2 using Rule 11. This makes the H_2 block as G_2H_2. Further summer S_3 and S_4 are interchanged using Rule 7. The input to summer S_5 when H_2 block shifted is H_1H_2 block. The resultant block diagram is shown in Fig. 1.64c.
2. In Fig. 1.64c, the blocks G_1 and H_1 are in feedback mode and is converted to $G_1/1 + G_1H_1$ as per Rule 4. The parallel blocks shown in Fig. 1.64c is combined as

$$\frac{G_3}{G_1} + G_2 = \frac{(G_3 + G_2G_1)}{G_1}$$

   as per Rule 2. The feedback loop 2, is converted as $1/1 + G_2H_2$. The summer S_1 and S_5 are combined together with a summer S_1 with the input from $C(s)$ as $(1 + H_1H_2)$. The complete reduced block diagram is shown in Fig. 1.64d.
3. From Fig. 1.64d three blocks are in cascade. Their equivalence is,

$$G = \frac{(G_1G_2 + G_3)}{(1 + G_1H_1)(1 + G_2H_2)}$$

   The feedback loop gain is $H = (1 + H_1H_2)$. From Fig. 1.64d using feedback Rule 4, we get,

$$\frac{C(s)}{R(s)} = \frac{G}{1 + GH}$$

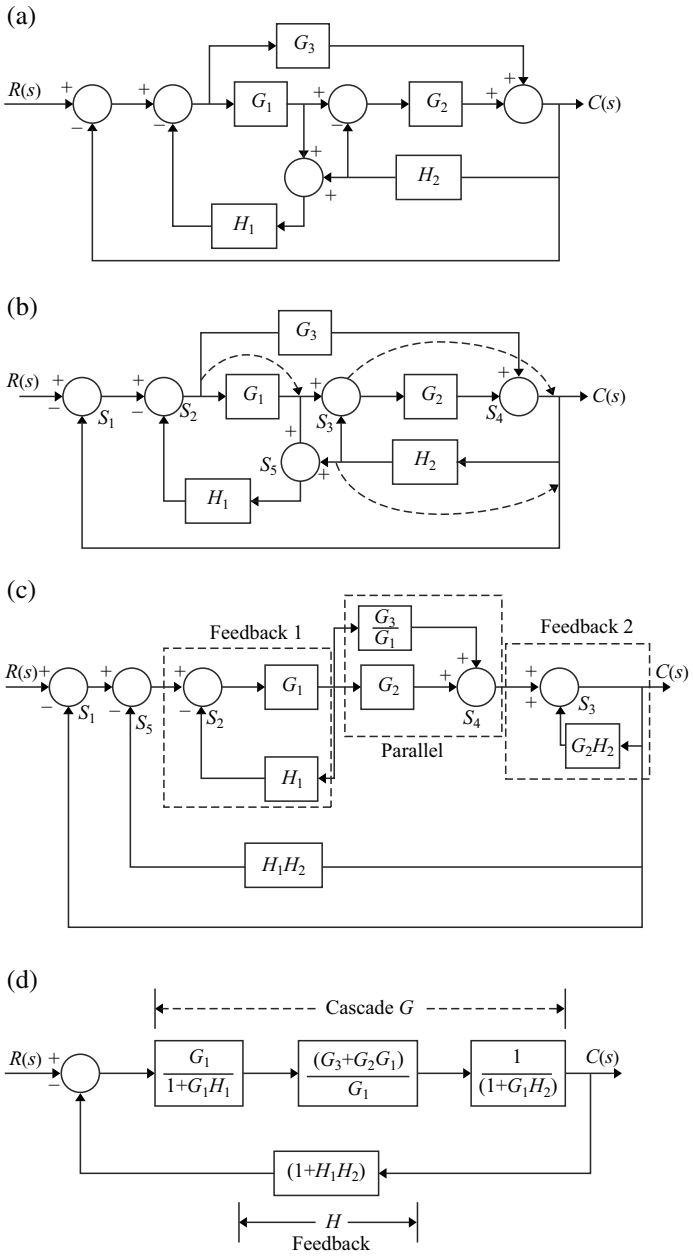


Fig. 1.64 Block diagram for Example 1.29

$$\frac{C(s)}{R(s)} = \frac{(G_1G_2 + G_3)}{[(1 + G_1H_1)(1 + G_2H_2) + (G_1G_2 + G_3)(1 + H_1H_2)]}$$

1.18.3 Block Diagram Reduction for Multiple Inputs

Systems which are given more than one input simultaneously are called multiple input system. If the output taken is more than one, such systems are called multiple output systems. When the input is more than one and output is also more than one such systems are called multiple input and multiple output systems or simply multi-input, multi-output (MIMO) systems. The block diagram reduction is done by following the steps given below:

1. Except one input, assume all the outputs are absent and put them zero.
2. For the particular output and the input considered in step 1, reduce the block diagram using rules given in Table 1.4.
3. Repeat steps one and two for the remaining inputs but for a particular output.
4. Algebraically add the responses (outputs) obtained in steps 1–3 which is the output of the system when all the inputs act simultaneously.
5. Steps 1 to 4 are repeated for some other output.

The following examples illustrate the above procedure.

Example 1.30

For the block diagram shown in Fig. 1.65a determine $C(s)$ for the inputs $R(s)$ and $D(s)$ when they act simultaneously.

Solution

1. $R(s)$ alone is present and $D(s) = 0$. The block diagram is shown in Fig. 1.65b. The G_1 and G_2 are in cascade and the whole block diagram is in negative feedback mode. Let $C_1(s)$ be the output when $R(s)$ alone is present. From Fig. 1.65b, the output is easily obtained as,

$$C_1(s) = R(s)\frac{G_1G_2}{(1 + G_1G_2H_1)}$$

2. $D(s)$ alone is present and the corresponding output is $C_2(s) \cdot R(s) = 0$. For Fig. 1.65c the forward path T.F is G_2 and the feedback path T.F. is G_1H_1 which is negative feedback mode. Using Rule 4, we derive the output $C_2(s)$ as,

$$C_2(s) = D(s)\frac{G_2}{(1 + G_1G_2H_1)}$$

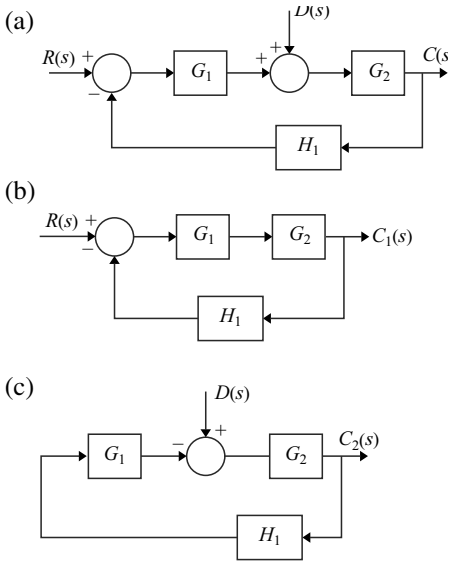


Fig. 1.65 Block diagram for Example 1.30

3. When $R(s)$ and $D(s)$ are simultaneously present, using superposition theorem we get,

$$C(s) = C_1(s) + C_2(s)$$

$$\boxed{C(s) = \frac{[R(s)G_1G_2 + D(s)G_2]}{[1 + G_1G_2H_1]}}$$

Note: The denominator for $C_1(s)$ and $C_2(s)$ is same which is called the determinant of the block diagram.

Example 1.31

For the block diagram shown in Fig. 1.66a determine $C(s)$ when the input $R(s)$ and disturbance $D(s)$ are simultaneously present.

Solution

1. $D(s) = 0$. The output $C_1(s)$ due to $R(s)$ is shown in Fig. 1.66b. The take off point of block H_1 is shifted after the block G_3. Applying Rule 9, H_1 block is modified as H_1/G_3 and is shown in Fig. 1.66c.

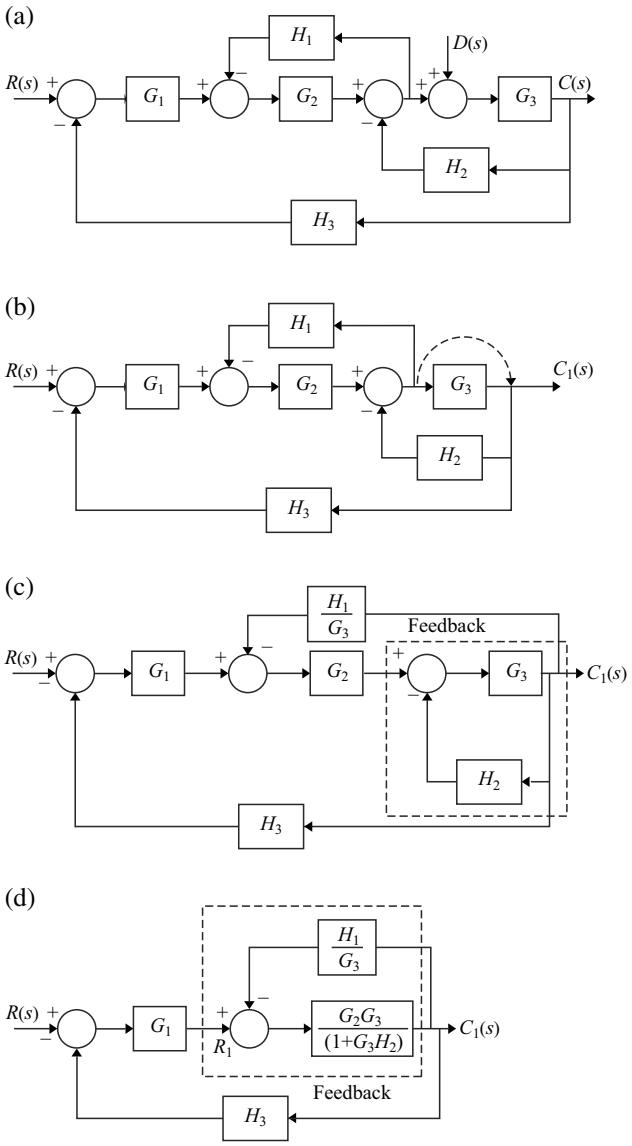


Fig. 1.66 Block diagram for Example 1.31

2. From Fig. 1.66c the equivalence of feedback blocks is obtained as $G_3/(1 + G_3H_2)$ and this is in cascade with G_2. The reduced block diagram is shown in Fig. 1.66d.
3. The feedback mode between R_1 and $C_1(s)$ is reduced as,

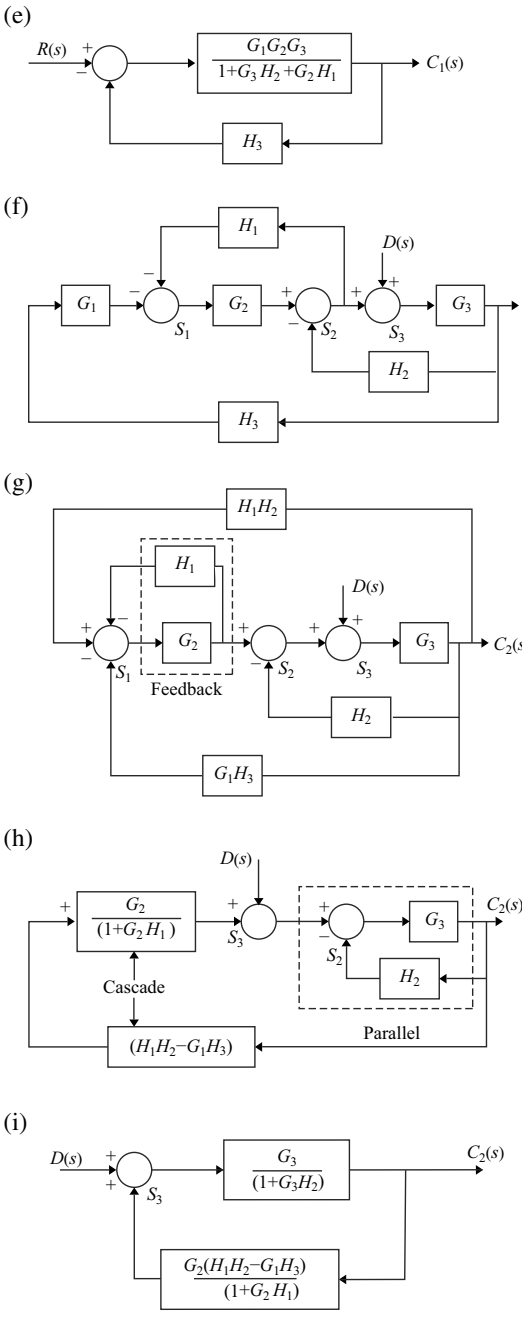


Fig. 1.66 (continued)

$$\frac{C_1(s)}{R_1} = \frac{\dfrac{G_2G_3}{(1+G_3H_2)}}{1+\dfrac{G_2G_3}{(1+G_3H_2)}\dfrac{H_1}{G_3}} = \frac{G_2G_3}{[(1+G_3H_2)+G_2H_1]}$$

The reduced block diagram is shown in Fig. 1.66e after cascading it with the G_1 series block.

4. Figure 1.66e is in feedback mode. Using Rule 4, $C_1(s)$ is obtained as,

$$C_1(s) = \frac{R(s)\dfrac{G_1G_2G_3}{(1+G_3H_2+G_2H_1)}}{\left[1+\dfrac{G_1G_2G_3H_3}{(1+G_3H_2+G_2H_1)}\right]}$$

$$\boxed{C_1(s) = \frac{R(s)G_1G_2G_3}{(1+G_3H_2+G_2H_1+G_1G_2G_3H_3)}}$$

5. Assume $R(s) = 0$ and $D(s)$ alone is present. Now the output is $C_2(s)$. The block diagram is shown in Fig. 1.66f.
6. Shifting the take off point of the H_1 block ahead of the summer S_2, using Rule 12, the block diagram is represented as shown in Fig. 1.66g.
7. The blocks G_1H_3 and H_1H_2 are in parallel and their equivalence is $(H_1H_2 - G_1H_3)$. The blocks G_2 and H_1 are in negative feedback mode and their equivalence is $G_2/(1+G_2H_1)$. Incorporating these changes and interchanging summers S_2 and S_3, the block diagram is represented as shown in Fig. 1.66h.
8. The cascade and feedback connections are converted using Rules 1 and 4 respectively. The reduced block diagram is shown in Fig. 1.66h.
9. Figure 1.66i is in positive feedback mode and using Rule 4, the following expression is obtained for $C_2(s)$.

$$C_2(s) = \frac{D(s)\frac{G_3}{(1+G_3H_2)}}{\left[1-\frac{G_3}{(1+G_3H_2)}\frac{G_2(H_1H_2-G_1H_3)}{(1+G_2H_1)}\right]}$$

$$\boxed{C(s) = \frac{D(s)G_3(1+G_2H_1)}{[1+G_3H_2+G_2H_1+G_1G_2G_3H_3]}}$$

10. $C(s) = C_1(s) + C_2(s)$

$$C(s) = \frac{[R(s)G_1G_2G_3 + D(s)G_3(1 + G_2H_1)]}{[1 + G_3H_2 + G_2H_1 + G_1G_2G_3H_3]}$$

Example 1.32

For the block diagram shown in Fig. 1.67a determine the outputs C_1 and C_2 due to the inputs R_1 and R_2.

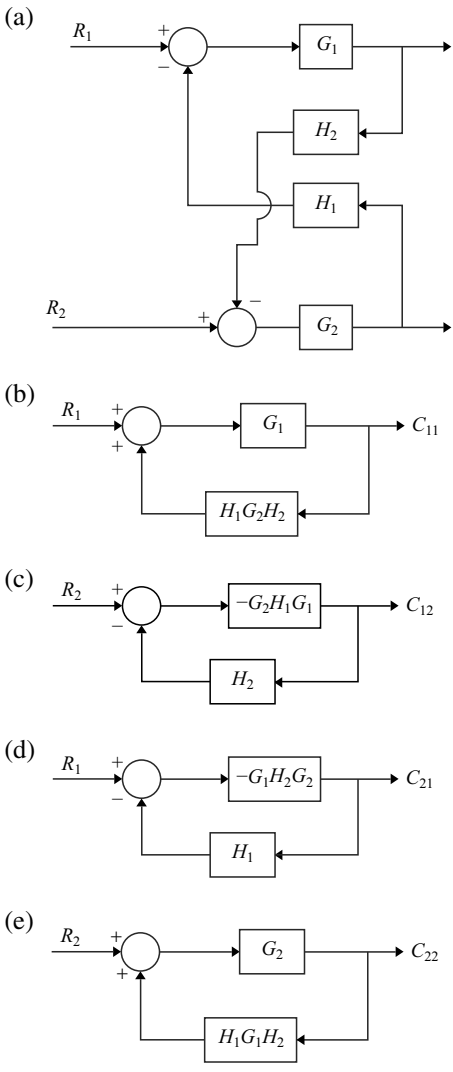


Fig. 1.67 Block diagram for Example 1.32

Solution

1. **Let $R_2 = 0$.**

   Let the output C_1 due to R_1 be C_{11}. The block diagram is shown in Fig. 1.67b. From Fig. 1.67b, the output C_{11} is obtained using Rule 4 for positive feedback mode as

$$C_{11} = \frac{R_1 G_1}{(1 - G_1 G_2 H_1 H_2)}$$

2. **Let R_1 be zero.**

   Let output C_1 be C_{12}. The block diagram for the above condition is shown in Fig. 1.67c.
   Figure 1.67c is in negative feedback mode and using Rule 4 we get

$$C_{12} = \frac{R_2(-G_2 H_1 G_1)}{(1 - G_1 G_2 H_1 H_2)}$$

   The output C_1 when R_1 and R_2 are simultaneously applied we get

$$C_1 = C_{11} + C_{12}$$

$$\boxed{C_1 = \frac{(R_1 G_1 - R_2 G_1 G_2 H_1)}{(1 - G_1 G_2 H_1 H_2)}}$$

3. **Let R_1 be present and R_2 be zero.**

   Let the output C_2 due to R_1 be C_{21}. For this condition the block diagram is shown in Fig. 1.67d.
   Figure 1.67d is in feedback mode and using Rule 4, the following equation for C_{21} is obtained.

$$C_{21} = \frac{-R_1 G_1 G_2 H_2}{(1 - G_1 G_2 H_1 H_2)}$$

4. **Finally, R_1 is assumed to be zero.**

   Let the output C_2 due to R_2 be C_{22}. The block diagram for this condition is shown in Fig. 1.67e.
   From Fig. 1.67e, using Rule 4 for a positive feedback connection, the following equation is obtained for C_{22}

$$C_{22} = \frac{R_2 G_2}{(1 - G_1 G_2 H_1 H_2)}$$

5. The total output C_2 due to R_1 and R_2 when they are simultaneously applied is

$$C_2 = C_{21} + C_{22}$$

$$\boxed{C_2 = \frac{-R_1 G_1 G_2 H_2 + G_2 R_2}{(1 - G_1 G_2 H_1 H_2)}}$$



Note: The denominators in all the output equations are the same which is the determinant of the block diagram.

1.18.4 Signal Flow Graph

Signal flow graph like block diagram is a graphical model representation of feedback control system. It is a pictorial representation of simultaneous equations describing the system. It has a distinct advantage over block diagram in that it is easier to draw and manipulate. It primarily consists of nodes and branches to represent these simultaneous equations. While nodes represent the variables, the branch which is connected between the nodes represents the gain or transmission function.

Representation of basic signal flow graph is shown in Fig. 1.68. x_1 and x_2 are the variables and the line drawn between x_1 and x_2 is the branch. G is the gain or the transmission function. The arrow indicates the direction of signal flow.

1.18.5 Signal Flow Graph Algebra

1. The Addition Rule

Like the rules used in block diagram reduction we have few rules in signal flow graph (SFG) manipulation also. Consider Fig. 1.69, where the variables x_1, x_2, x_3 and x_4

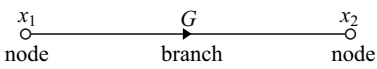


Fig. 1.68 Basic signal flow graph

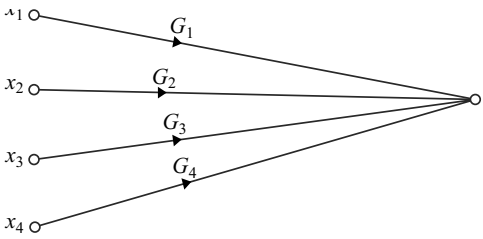


Fig. 1.69 Illustration of addition rule of signal flow graph

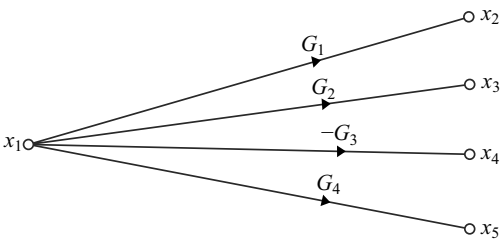


Fig. 1.70 Illustration of transmission rule of SFG

are directed towards the nodes x_5 with their respective branch gain. For Fig. 1.69, the following equation is written:

$$G_1x_1 + G_2x_2 + G_3x_3 + G_4x_4 = x_5 \tag{1.115}$$

Equation (1.115) illustrates the addition rule of SFG. According to the addition rule, the value of the variable represented by the node is equal to the sum of all signals entering that node.

2. The Transmission Rule

According to the transmission rule, the value of the variable designated by a node is transmitted on every branch leaving that node. In Fig. 1.70, x_1 is the designated node and it transmits signals to other nodes x_2, x_3, x_4 and x_5. For Fig. 1.70, the following equations are written:

$$\begin{aligned} x_2 &= G_1x_1 \\ x_3 &= G_2x_1 \\ x_4 &= -G_3x_1 \\ x_5 &= G_4x_1 \end{aligned} \tag{1.116}$$

3. Multiplication Rule Consider Fig. 1.71 in which the nodes x_1 to x_5 are connected in cascade. The cascaded connection of these four branches with branch gain G_1, G_2, G_3 and G_4 can be replaced by a single branch that has the branch gain which is the product of the old one.

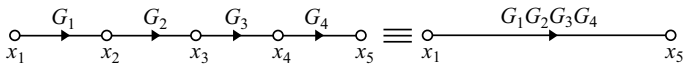


Fig. 1.71 Illustration of multiplication rule of SFG

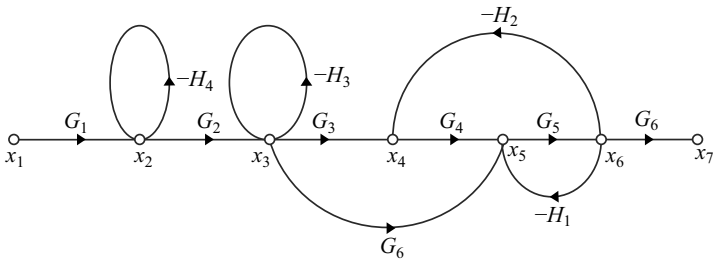


Fig. 1.72 Illustration of terminologies of SFG

1.18.6 Definitions of SFG Terminologies

The following terminologies which are commonly used in SFG are defined as follows:

1. **Node**: A node represents the variable and are represented by a dot 0. x_1, x_2, x_3, x_4, x_5, x_6 and x_7 in Fig. 1.72 are the nodes.
2. **Source**: A source or input node has only outgoing branches. x_1 node is the source node.
3. **Sink**: A sink or output node has only incoming branches. x_7 in Fig. 1.72 is a sink node.
4. **Branch**: Branch represents the transmission of signal from one node to the other node and is represented by a line with an arrow and the transmittance. The arrow represents the direction of transmission between nodes. In Fig. 1.72, $x_1 - x_2$ is a branch with a branch gain G_1. Similarly $x_2 - x_3$ is a branch with a branch gain G_2 and so on.
5. **Mixed nodes**: Nodes which do not belong to either the input node or the output node are called mixed nodes.
6. **Path**: A path is a continuous unidirectional succession of branches along which no node is passed more than once. In Fig. 1.72, $x_1$to x_2 is a path, $x_1 - x_2 - x_3 - x_4 - x_5 - x_6 - x_7$ is a path, and $x_5 - x_6 - x_5$ is a path. There are several such paths in Fig. 1.72.
7. **Forward path**: A forward path is a path from the input node to the output node. In Fig. 1.72, $x_1 - x_2 - x_3 - x_4 - x_5 - x_6 - x_7$ is a forward path. Similarly $x_1 - x_2 - x_3 - x_5 - x_6 - x_7$ is a forward path.
8. **Path gain**: Path gain is the product of the gains of all the branches encountered in a path. In Fig. 1.72 for the path $x_1 - x_2 - x_3$ the path gain is $G_1G_2G_3$.
9. **Loop**: Loop is a closed path which originates from a node and terminates at the same node without touching a node more than once. In Fig. 1.72, $x_4 - x_5 - x_6 - x_4$ is a loop. Similarly $x_5 - x_6 - x_5$ is a loop.
10. **Loop gain**: Loop gain is the product gain of the branches forming a loop. In Fig. 1.72, for the loop $x_4 - x_5 - x_6 - x_4$, the loop gain is $-G_4G_5H_2$. For the loop $x_5 - x_6 - x_5$, the loop gain is $-G_5H_1$.

11. **Self loop**: A self loop is a feedback loop consisting of a single branch. It starts and terminates at the same node. In Fig. 1.72, $x_2 - x_2$ is a self loop with a loop gain $-H_4$. Similarly, $x_3 - x_3$ is a self loop with a loop gain $-H_3$.
12. **Non-touching loops**: Loops which do not have a common node are said to be non-touching. In Fig. 1.72 there are four feedback loops

$$\begin{aligned} L_1 &= x_4 - x_5 - x_6 - x_4 \quad (-G_4G_5H_2 \text{ loop gain}) \\ L_2 &= x_5 - x_6 - x_5 \quad (-G_5H_1 \text{ loop gain}) \\ L_3 &= x_3 - x_3 \quad (-H_3 \text{ loop gain}) \\ L_4 &= x_2 - x_2 \quad (-H_4 \text{ loop gain}) \end{aligned}$$

L_1 and L_4 are non-touching loops L_4 and L_2 are non-touching and so on. When two loops are not touching, they are called two, non-touching loops. In Fig. 1.72, L_1L_3, L_1L_4, L_2L_3, L_2L_4 and L_3L_4 are two non-touching loops. The loops $L_1L_3L_4$ and $L_2L_3L_4$ are called three non-touching loops.

1.18.6.1 Application of Mason's Gain Formula for SFG

To find the overall gain between input node and output node it is possible to write down the input-output relationship from SFG and obtain the same. However, the procedure is somewhat tedious and cumbersome. But this can be accomplished using Mason's gain formula which is explained below. Using this formula, sometimes, the results can be obtained just by inspection of SFG. Because of the simplicity of the formula, SFG has been very extensively used for the system analysis. The Mason's general gain formula for any signal flow graph is

$$A = \frac{\sum_{m=1}^{n} T_m \Delta_m}{\Delta} \tag{1.117}$$

where

A is the total gain between the input node and output node.
T_m is the gain of the mth forward path.

$$\Delta = 1 - \sum_n P_{n1} + \sum_n P_{n2} - \sum_n P_{n3} + \cdots$$

where

P_{nr} = gain product of 'n' possible combinations of 'r' non-touching loops (or)

$\Delta = 1 -$ (sum of individual loop gains)

$+$(sum of gain product of all possible combinations of two non-touching loops)

$-$(sum of gain product of all possible combinations of three non-touching loops) $+ \cdots$

$=$ Determinant of SFG.

Δ_m = The value of Δ for that part of all graph with the feedback loops touching the m_{th} forward path being broken $=$ co-factor of m_{th} forward path.

For simplicity, the individual loop gains in Eq. (1.117) are denoted by L_1, L_2, L_3 etc. The forward path gains are denoted by T_1, T_2, T_3 etc. The following methods illustrate the application of Mason's gain formula to solve SFG.

Example 1.33

For the signal flow graph shown in Fig. 1.73a using Mason's gain formula, determine $C(s)/R(s)$.

Solution

The forward path of Fig. 1.73a is shown in Fig. 1.73b. There is only one forward path. Equation (1.117) for this problem becomes

$$\frac{C(s)}{R(s)} = \frac{T_1 \Delta_1}{\Delta}$$

From Fig. 1.73b,

$$T_1 = G$$

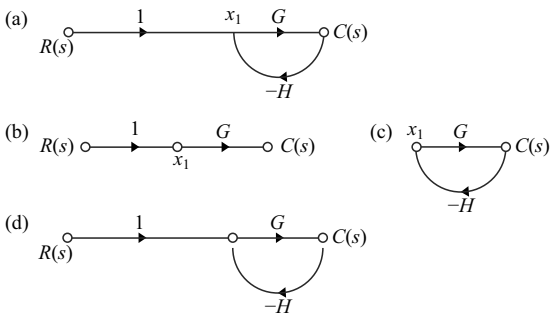


Fig. 1.73 **a** SFG for Example 1.33. **b** Forward path; **c** Feedback loop and **d** Feedback loop touching the forward path when removed it is broken

There is only one feedback loop namely $x_1 - C(s) - x_1$ as shown in Fig. 1.73c. Feedback loop gain L_1 is

$$L_1 = -GH$$

To calculate Δ_1, the co-factor of the graph, the feedback loop touching the forward path T_1 at two points when broken is shown in Fig. 1.73d. This means there is no feedback when Δ_1 is calculated for the forward path T_1.

From Eq. (1.117),

$$\begin{aligned}\Delta &= 1 - L_1 = 1 - (\text{sum of individual loop gain}) \\ &= 1 + GH \\ \Delta_1 &= 1\end{aligned}$$

$$\boxed{\frac{C(s)}{R(s)} = \frac{G}{1+GH}}$$

Example 1.34

The SFG of a system is shown in Fig. 1.74a. Determine $\dfrac{C(s)}{R(s)}$.

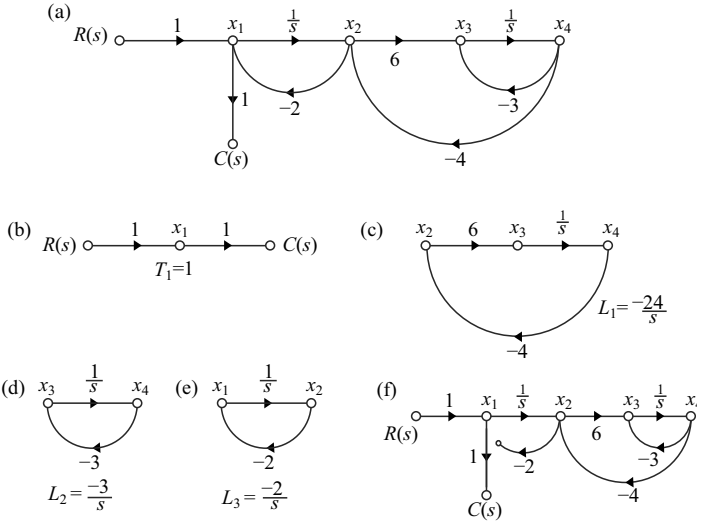


Fig. 1.74 **a** SFG for Example 1.34. **b** Forward path; **c** Feedback loop 1; **d** Feedback loop 2; **e** Feedback loop 3; and **f** feedback touching the forward path when broken

Solution

1. There is only one forward path for the SFG which is shown in Fig. 1.74b. The forward path gain $T_1 = 1$.
2. There are three feedback loops as shown in Fig. 1.79c–e. The feedback loop gains are $L_1 = -24/s$ and $L_2 = -3/s$ and $L_3 = \frac{-2}{s}$. However, L_1 and L_2 feedback loops are touching each other at x_3 and x_4 nodes. The feedback loops L_2 and L_3 are not touching each other.
3. The feedback loops have connections at node x_2 and when broken at x_1, where it is touching the forward path they are shown in Fig. 1.74f.
4. The overall T.F. for a single forward path of SFG is

$$\begin{aligned} A &= \frac{T_1\Delta_1}{\Delta} \\ \Delta &= 1 - (L_1 + L_2 + L_3) + L_2L_3 \\ &= 1 + \left(\frac{24}{s} + \frac{3}{s} + \frac{2}{s}\right) + \frac{6}{s^2} \\ &= \frac{(s^2 + 29s + 6)}{s^2} \end{aligned}$$

5. Δ_1 is calculated from Fig. 1.74f. Δ for T_1 when the feedback loops touching the forward path T_1 gives Δ_1.

$$\begin{aligned} \Delta_1 &= 1 - (L_1 + L_2) \\ &= 1 + \frac{24}{s} + \frac{3}{s} \\ &= \frac{(s + 27)}{s} \end{aligned}$$

6. The overall gain is given by

$$\begin{aligned} \frac{C(s)}{R(s)} &= \frac{T_1\Delta_1}{\Delta} \\ &= \frac{(s + 27)}{s}\left(\frac{s^2}{(s^2 + 29s + 6)}\right) \end{aligned}$$

$$\boxed{\frac{C(s)}{R(s)} = \frac{s(s + 27)}{(s^2 + 29s + 6)}}$$

Example 1.35

Find $C(s)/R(s)$ by using Manson's gain formula for the SFG shown in Fig. 1.75a.

(*Anna University, May, 2009*)

Solution

1. There are two forward paths T_1 and T_2 as shown in Fig. 1.75b, c respectively. The overall gain is therefore given by

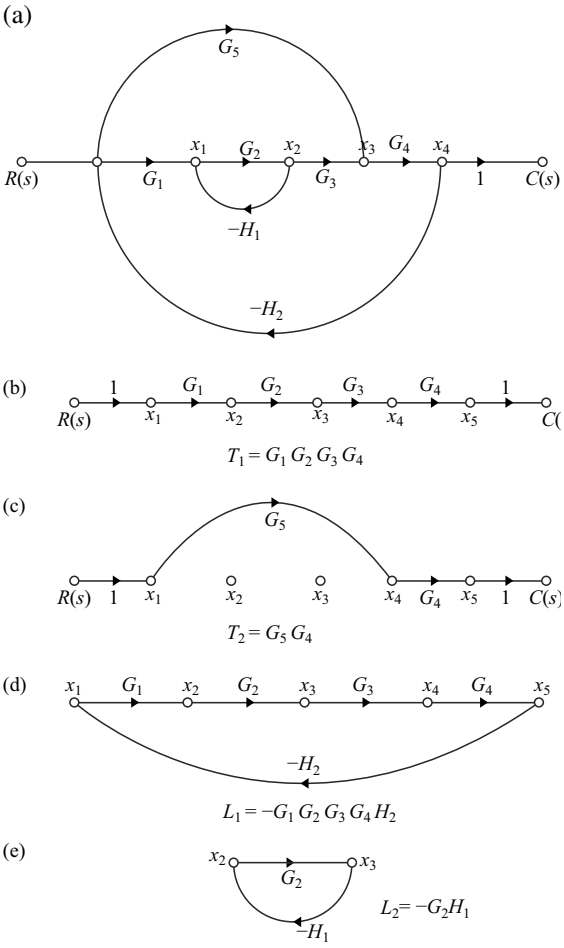


Fig. 1.75 **a** SFG for Example 1.35. **b** Forward path T_1; **c** Forward path T_2; **d** Feedback loop L_1 and **e** Feedback loop L_2. **f** Feedback loop L_3; **g** Two non-touching loops; **h** Feedback loops touching the forward path T_1, when broken; and **i** Feedback loops touching the forward path T_2, when broken

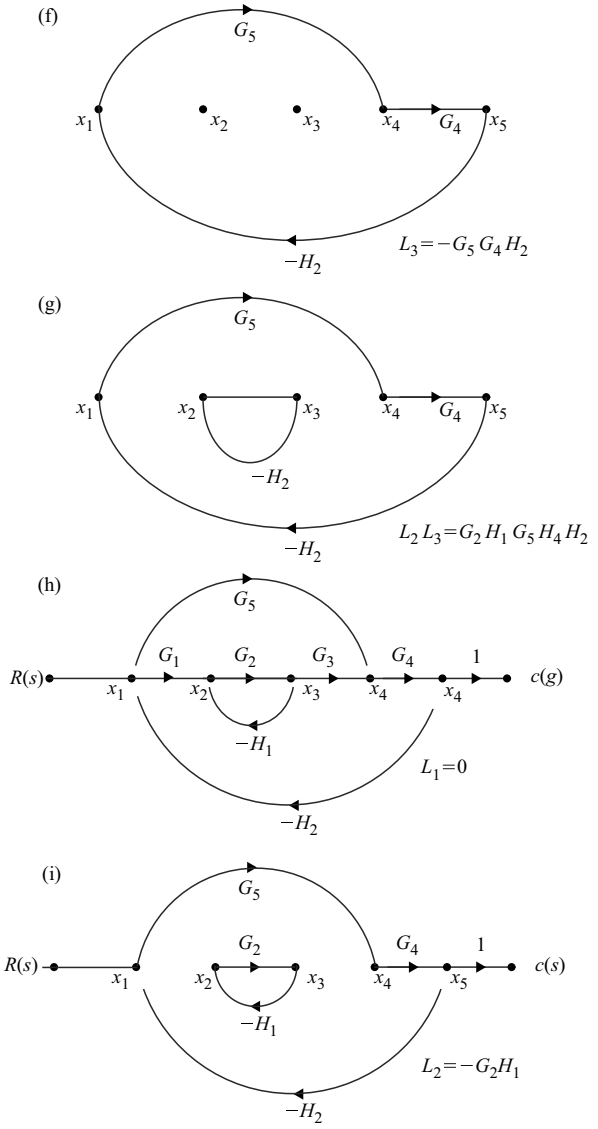


Fig. 1.75 (continued)

$$\frac{C(s)}{R(s)} = \frac{T_1\Delta_1 + T_2\Delta_2}{\Delta}$$

2. From Fig. 1.75b,

$$T_1 = G_1G_2G_3G_4$$

From Fig. 1.75c,

$$T_2 = G_5G_4$$

3. There are three feedback loops for the given SFG. They are shown in Fig. 1.75d–f. The feedback loop gains are

$$L_1 = -G_1G_2G_3G_4H_2$$
$$L_2 = -G_2H_1$$
$$L_3 = -G_5G_4H_2$$

While loops 1 and two are touching each other, loops 2 and three are non-touching. Hence, the gain product of these non-touching loops are $G_2H_1G_5G_4H_2$

$$\begin{aligned}\Delta &= 1 - (L_1 + L_2 + L_3) + (L_2L_3)\\ &= 1 + G_1G_2G_3G_4H_2 + G_2H_1 + G_5G_4H_2 + G_2H_1G_5G_4H_2\end{aligned}$$

4. Fig. 1.75h represents the feedback loop touching the forward path T_1 when it is broken. From this figure there is no feedback loop. Hence, $\Delta_1 = 1$.
5. Fig. 1.75i represents the feedback loop touching the forward path T_2 when it is broken. There is only one feedback loop and its loop gain is $L_2 = -G_2H_1$.

$$\begin{aligned}\Delta_2 &= 1 - (L_2)\\ &= 1 + G_2H_1\end{aligned}$$

6.

$$\frac{C(s)}{R(s)} = \frac{T_1\Delta_1 + T_2\Delta_2}{\Delta}$$

$$\boxed{\frac{C(s)}{R(s)} = \frac{G_1G_2G_3G_4 + G_5G_4(1 + G_2H_1)}{[1 + G_1G_2G_3G_4H_2 + G_2H_1 + G_5G_4H_2 + G_2H_1G_5G_4H_2]}}$$



Note: Just by inspection of SFG, it is possible to identity the forward paths and feedback loops. Hence, the detailed representation of these figures are avoided in future explanation.

Example 1.36

For the system represented by the following equations find the T.F. $\frac{X}{U}(s)$ by signal flow graph technique:

$$\begin{aligned} x &= x_1 + \beta_3 u \\ \dot{x}_1 &= -a_1 x_1 + x_2 + \beta_2 u \\ \dot{x}_2 &= -a_2 x_1 + \beta_1 u \end{aligned}$$

(*Anna University, December, 2009*)

Solution

$$x = x_1 + \beta_3 u \quad \text{or} \quad X(s) = X_1(s) + \beta_3 U(s) \tag{a}$$

$$\dot{x}_1 = -a_1 x_1 + x_2 + B_2 u$$

Taking Laplace transform on both sides we get

$$(s + a_1) X_1(s) = X_2(s) + \beta_2 U(s)$$

Solving for $X_1(s)$ we get

$$X_1(s) = \frac{1}{(s + a_1)} X_2(s) + \frac{\beta_2}{s + a_1} U(s) \tag{b}$$

$$\dot{x}_2(t) = -a_2 x_1 + B_1 u$$

Taking Laplace transform on both sides we get

$$X_2(s) = -\frac{a_2}{s} X_1(s) + \frac{\beta_1}{s} U(s) \tag{c}$$

Equations (a), (b) and (c) are represented in Fig. 1.76. From Fig. 1.76, the following forward paths are identified:

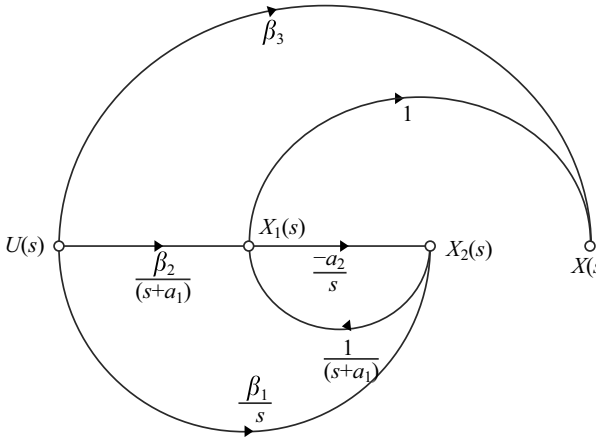


Fig. 1.76 SFG for Example 1.36

	Forward path	Forward path gain
1.	$U - X$	$T_1 = \beta_3$
2.	$U - X_1 - X$	$T_2 = \frac{\beta_2}{s + a_1}$
3	$U - X_2 - X_1 - X$	$T_3 = \frac{\beta_1}{s(s + a_1)}$

The following feedback loops are identified:

	Feedback loop	**Loop gain**
1	$X_1 - X_2 - X_1$	$L_1 = \frac{-a_2}{s(s + a_1)}$

$$\begin{aligned}\Delta &= 1 - (L_1) \\ &= 1 + \frac{a_2}{s(s + a_1)} \\ &= \frac{(s^2 + a_1 s + a_2)}{s(s + a_1)}\end{aligned}$$

For the forward path one, the feedback loop $X_1 - X_2 - X_1$ is not touching the forward path T_1 when broken.

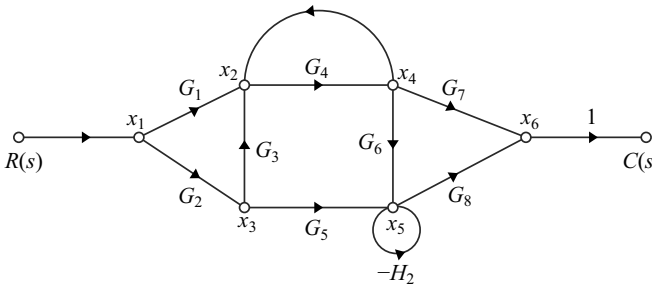


Fig. 1.77 SFG for Example 1.37

$$\begin{aligned}\Delta_1 &= 1 - L_1 \\ &= 1 + \frac{a_2}{s(s+a_1)} \\ &= \frac{(s^2 + a_1 s + a_2)}{s(s+a_1)}\end{aligned}$$

For the forward paths T_2 and T_3, the feedback loops are touching these forward paths and when broken from the forward path, there is no feedback loop.

Hence, $\Delta_2 = 1$ and $\Delta_3 = 1$

$$\begin{aligned}\frac{X(s)}{U(s)} &= T_1\Delta_1 + T_2\Delta_2 + T_3\Delta_3 \\ &= \frac{\frac{\beta_3(s^2+a_1s+a_2)}{s(s+a_1)} + \frac{\beta_2}{(s+a_1)} + \frac{\beta_1}{s(s+a_1)}}{\frac{(s^2+a_1s+a_2)}{s(s+a_1)}}\end{aligned}$$

$$\boxed{\frac{X}{U}(s) = \frac{\beta_3(s^2 + a_1 s + a_2) + \beta_1 + \beta_2 s}{(s^2 + a_1 s + a_2)}}$$

Example 1.37

For the SFG shown in Fig. 1.77 determine $\dfrac{C(s)}{R(s)}$ using Mason's gain formula.

Solution

	Forward path	Forward path gain
1.	$R(s) - x_1 - x_2 - x_4 - x_6 - C(s)$	$T_1 = G_1G_4G_7$
2.	$R(s) - x_1 - x_2 - x_4 - x_5 - x_6 - C(s)$	$T_2 = G_1G_4G_6G_8$
3.	$R(s) - x_3 - x_2 - x_4 - x_6 - C(s)$	$T_3 = G_2G_3G_4G_7$
4.	$R(s) - x_1 - x_3 - x_2 - x_4 - x_5 - x_6 - C(s)$	$T_4 = G_2G_3G_4G_6G_8$
5	$R(s) - x_1 - x_3 - x_5 - x_6 - C(s)$	$T_5 = G_2G_5G_8$
	Feedback loop	Feedback loop gain
1.	$x_2 - x_4 - x_2$	$L_1 = -G_4H_1$
2	$x_5 - x_5$	$L_2 = -H_2$

$$\frac{C(s)}{R(s)} = \frac{T_1\Delta_1 + T_2\Delta_2 + T_3\Delta_3 + T_4\Delta_4 + T_5\Delta_5}{\Delta}$$

$$\Delta = 1 - (L_1 + L_2) + L_1L_2$$

$$= 1 + G_4H_1 + H_2 + G_4H_1H_2$$

For forward path T_1, L_1, is broken and L_2 alone is not touching

$$\Delta_1 = 1 + H_2.$$

For forward path T_2, both L_1 and L_2 are touching and hence when removed from the forward path, they are broken. Hence,

$$\Delta_2 = 1.$$

For forward path T_3, L_1 is touching L_2 is not touching. When the touching portion of L_1 is removed, it is broken. L_2 is the only non-touching loop. Hence,

$$\Delta_3 = 1 + H_2.$$

For the forward path T_4, both L_1 and L_2 are touching the forward path and when they are removed from the forward path, they are broken. Hence, there are no feedback loops.

$$\Delta_4 = 1$$

For the forward path T_5, loop L_2 is touching and loop L_1 is not touching. Thus, L_2 is broken when removed.

$$\Delta_5 = 1 + G_4H_1$$

$$\frac{C(s)}{R(s)} = G_1G_4G_7(1 + H_2) + G_1G_4G_6G_8 + G_2G_3G_4G_7(1 + H_2) + G_2G_3G_4G_6G_8 + G_2G_5G_8(1 + G_4H_1)$$

Example 1.38

For the SFG shown in Fig. 1.78a determine the outputs due to the inputs R_1 and R_2.

Solution The multi-input-multi-output problems are solved in similar way as we followed for block diagram reduction. When the input R_1 is applied the input $R_2 = 0$ and the output is C_{11}. This is shown in Fig. 1.78b.

For Fig. 1.78b,

$$\begin{aligned} T_1 &= R_1 - x_1 - x_2 - C_{11} \\ &= G_1 \\ T_2 &= R_1 - x_1 - x_3 - x_4 - x_2 - C_{11} \\ &= G_3 G_2 G_4 \\ L_1 &= -G_1 H_1; \\ L_2 &= -G_2 H_2; \\ L_3 &= -G_3 G_2 G_4 H_1 \\ \Delta &= 1 - (L_1 + L_2 + L_3) + L_1 L_2 \\ &= 1 + G_1 H_1 + G_2 H_2 + G_2 G_3 G_4 H_1 + G_1 G_2 H_1 H_2 \end{aligned}$$

For T_1, L_1 and L_3 are broken and L_2 is not broken

$$\Delta_1 = 1 + G_2 H_2$$

For T_2, L_1, L_2 and L_3 are broken $\Delta_2 = 1$

$$\begin{aligned} C_{11} &= \frac{T_1 \Delta_1 + T_2 \Delta_2}{\Delta} \\ \therefore \quad C_{11} &= \frac{R_1[G_1(1 + G_2 H_2) + G_2 G_3 G_4]}{[1 + G_1 H_1 + G_2 H_2 + G_1 G_2 H_1 H_2]} \end{aligned}$$

When R_2 is applied, $R_1 = 0$ and the output at C_1 is C_{12}. For Fig. 1.78c we get

$$\begin{aligned} T_1 &= R_2 - x_3 - x_4 - x_2 - C_{12} \\ &= G_2 G_4 \end{aligned}$$

Feedback loops L_1 and L_2 and L_3 are in touch with T_1 and when removed they are broken. Hence, $\Delta_1 = 1$

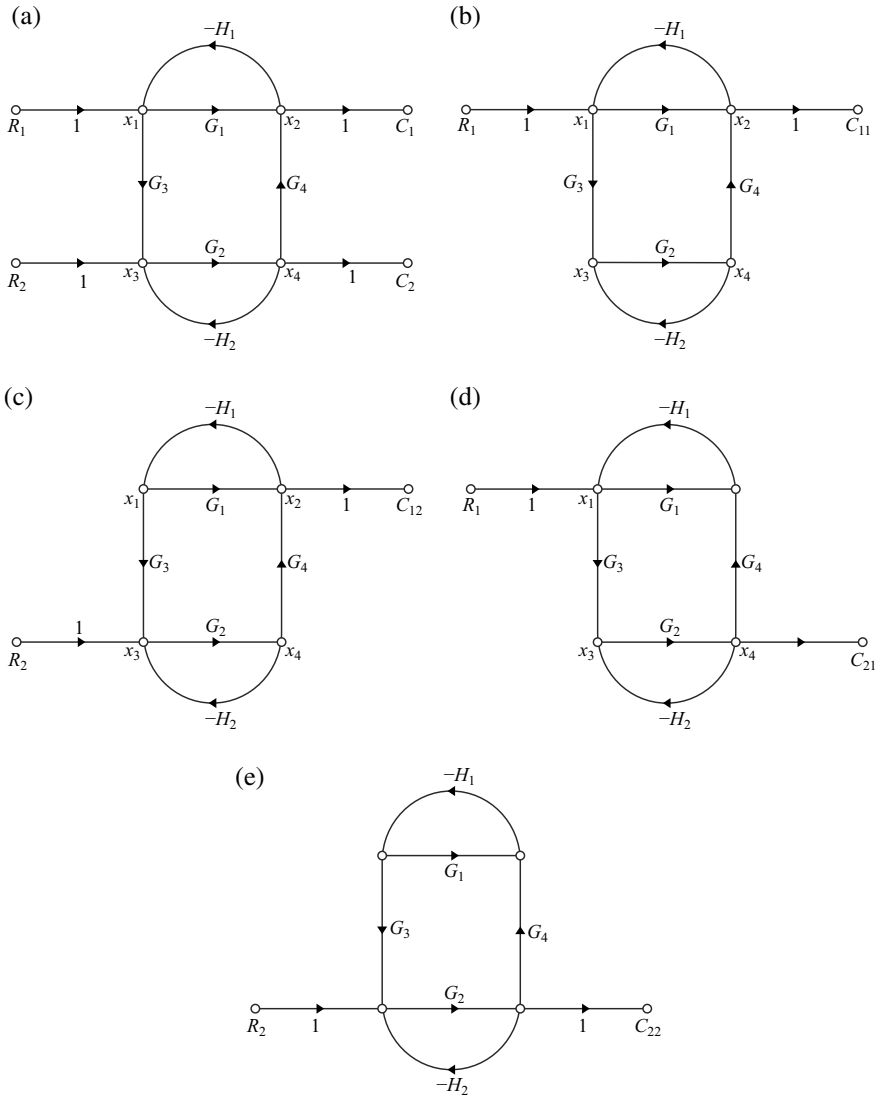


Fig. 1.78 **a** SFG for Example 1.38

$$C_{12} = \frac{T_1 \Delta_1 R_2}{\Delta}$$
$$= \frac{R_2 G_2 G_4}{[1 + G_1 H_1 + G_2 H_2 + G_2 G_3 G_4 H_1 + G_1 G_2 H_1 H_2]}$$
$$C_1 = C_{11} + C_{12}$$

$$\boxed{C_1 = \frac{R_1[G_1(1 + G_2 H_2) + G_2 G_3 G_4] + R_2 G_2 G_4}{[1 + G_1 H_1 + G_2 H_2 + G_2 G_3 G_4 H_1 + G_1 G_2 H_1 H_2]}}$$



Note: Δ is same for the given SFG.

Now let us calculate C_2 due to R_1 and R_2. First we assume R_1 is applied and $R_2 = 0$. The output now is C_{21}. SFG is shown in Fig. 1.78d. There is only on forward path

$$T_1 = R_1 - x_1 - x_3 - x_4 - C_{21}$$
$$T_1 = G_3 G_2$$

The feedback loops touching T_1 when removed are broken. Hence, $\Delta_1 = 1$

$$C_{21} = \frac{R_1 T_1 \Delta_1}{\Delta}$$
$$= \frac{R_1 G_2 G_3}{[1 + G_1 H_1 + G_2 H_2 + G_2 G_3 G_4 H_1 + G_1 G_2 H_1 H_2]}$$

Now R_2 is applied and R_1=0. The output is C_{22}. The SFG is shown in Fig. 1.78e. From Fig. 1.78e

$$T_1 = G_2$$

The feedback loop L_1 is not touching the forward path. Hence,

$$L_1 = -G_1 H_1$$
$$\Delta_1 = 1 - (L_1)$$
$$= 1 + G_1 H_1$$
$$C_{22} = R_2 \frac{T_1 \Delta_1}{\Delta}$$
$$C_{22} = \frac{R_2 G_2 (1 + G_1 H_1)}{\Delta}$$

when R_1 and R_2 are simultaneously applied,

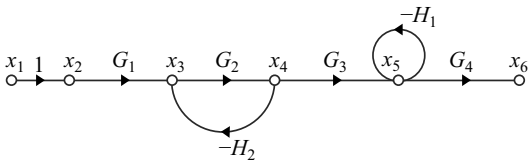


Fig. 1.79 SFG between output node and non-input node

$$\begin{aligned} C_2 &= C_{21} + C_{22} \\ &= R_1 G_2 G_3 + R_2 G_2 (1 + G_1 H_1) \end{aligned}$$

$$\boxed{C_2 = \frac{R_1 G_2 G_3 + R_2 G_2 (1 + G_1 H_1)}{[1 + G_1 H_1 + G_2 H_2 + G_2 G_3 G_4 H_1 + G_1 G_2 H_1 H_2]}}$$

1.18.7 Application of the Gain Formula Between Output Nodes and Non-input Nodes

Mason's gain formula was applied to find the overall gain between the input and output nodes. However, it is also applicable to find the relation between an output node variable and a non-input variable.

Consider the SFG shown in Fig. 1.79. The output node is x_6. We find the ratio x_6/x_2 where x_2 is a non-input node. x_6/x_2 can be written in the following form:

$$\begin{aligned} \frac{x_6}{x_2} &= \frac{x_6}{x_1}\frac{x_1}{x_2} \\ &= \frac{x_6}{x_1}\frac{1}{x_2/x_1} \\ \frac{x_6}{x_1} &= \frac{\sum_{m=1}^{n} T_{ma}\Delta_{ma}}{\Delta} \\ \frac{x_2}{x_1} &= \frac{\sum_{m=1}^{n} T_{mb}\Delta_{mb}}{\Delta} \end{aligned}$$

$$\boxed{\frac{x_6}{x_2} = \frac{\sum_{m=1}^{n} T_{ma}\Delta_{ma}}{\sum_{m=1}^{n} T_{mb}\Delta_{mb}}} \tag{1.118}$$

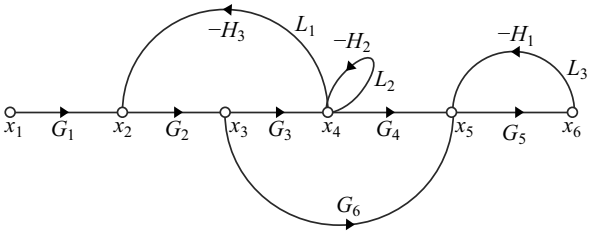


Fig. 1.80 SFG for Example 1.39

Equation (1.118) suggests that there is no necessity to determine Δ, the determinant of SFG in the above case. The following example illustrates this:

Example 1.39

For the figure shown in Fig. 1.80. Find x_6/x_3.

Solution For Fig. 1.80, between x_1 and x_6 nodes,

$$T_{a1} = G_1G_2G_3G_4G_5$$
$$T_{a2} = G_1G_2G_6G_5$$
$$\frac{x_6}{x_1} = \frac{T_{a1}\Delta_{a1} + T_{a2}\Delta_{a2}}{\Delta}$$

There are three feedback loops. They are all touching the forward path T_{a1} and when they are all removed they are all broken. Hence, $\Delta_{a1} = 1$.

Now consider the forward path T_{a2}, the feedback loop with gain $-H_2$ is not torching T_{a2} and it is not broken when removed from T_{a2}. However, the other two feedback loops are in touch with T_{a2} and when removed they are broken. Therefore, $\Delta_{a2} = (1 + H_2)$

$$\frac{x_6}{x_1} = \frac{T_{a1} + T_{a2}(1 + H_2)}{\Delta}$$
$$= \frac{G_1G_2G_5[G_3G_4 + G_6(1 + H_2)]}{\Delta}$$

Now consider the forward path $x_1 - x_2 - x_3$. This is the only forward path between the nodes x_1 and x_3. The forward path gain is

$$T_{b1} = G_1G_2$$
$$\frac{x_3}{x_1} = \frac{T_{b1}\Delta_{b1}}{\Delta}$$

The feedback loops are

$$L_1 = -G_2G_3H_3$$
$$L_2 = -H_2$$
$$L_3 = -G_5H_1$$

The loops L_2 and L_3 are not touching the forward path T_{b1}. Further they are touching themselves and form two non-touching loops. However, loop L_1 is in touch with T_{b1} and when removed it is broken.

$$\begin{aligned}\Delta_{b1} &= 1 - (L_2 + L_3) + L_2L_3 \\ &= 1 + H_2 + G_5H_1 + G_5H_1H_2 \\ \frac{x_3}{x_1} &= \frac{T_{b1}\Delta_{b1}}{\Delta} \\ &= \frac{G_1G_2[1 + H_2 + G_5H_1 + G_5H_1H_2]}{\Delta} \\ \frac{x_6}{x_3} &= \frac{x_6}{x_1}\frac{1}{x_3/x_1} \\ &= \frac{G_1G_2G_5[G_3G_4 + G_6(1 + H_2)]}{G_1G_2[1 + H_2 + G_5H_1 + G_5H_1H_2]}\end{aligned}$$

$$\boxed{\frac{x_6}{x_3} = \frac{G_5[G_3G_4 + G_6(1 + H_2)]}{[1 + H_2 + G_5H_1 + G_5H_1H_2]}}$$

1.18.8 Applications of the Gain Formula to Block Diagrams

Mason's gain formula can be directly applied to block diagram and the input–output gain can be easily determined. However, it becomes easier if an equivalent SFG is drawn for the block diagram before applying the gain formula.

1. The input and the output in the block diagram are identified and are designated with input node and output node respectively in SFG.
2. All take off points in the block diagram are designated with nodes in SFG.
3. The output of the summers in the block diagram are identified and they are also designed with nodes in SFG.

Consider the block diagram shown in Fig. 1.81a. The input and output are $R(s)$ and $C(s)$ respectively and they are represented as input nodes and output nodes in SFG as shown in Fig. 1.81b. There is a take off point after the block G and a node should be created to represent this. But this point is already represented by the output node and hence it coincides with output node. After the summer, a variable e is created and a node is identified for e. The following equations hold good both for block diagram as well as SFG.

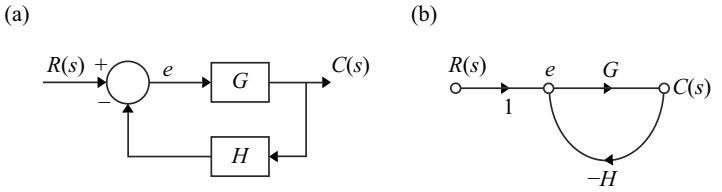


Fig. 1.81 **a** Block diagram and **b** Equivalent SFG

$$e = R(s) - HC(s)$$
$$C(s) = eG$$

The following examples illustrate the above procedure.

Example 1.40

For the block diagram shown in Fig. 1.82a draw the SFG and using Mason's gain formula determine C/R.

(*Anna University, May, 2009*)

Solution

In Fig. 1.82a the output of the summers and the take off points are identified and marked as x_1, x_2 and x_3. The SFG with these nodes and the input and output nodes is drawn and shown in Fig. 1.82b. The following forward paths and feedback loops are identified with their gains.

	Forward path	Forward path gain
1.	$R - x_1 - x_2 - C$	$T_1 = -G_1G_2$
2.	$R - x_1 - x_2 - C$	$T_2 = G_1G_3$
3.	$R - x_3 - x_1 - x_2 - C$	$T_3 = -G_4H_2G_1G_2$
4.	$R - x_3 - x_1 - x_2 - C$	$T_4 = G_4H_2G_1G_3$
	Feedback loop	Loop gain
1.	$x_1 - x_2 - C - x_3 - x_1$	$L_1 = G_1G_2H_1H_2$
2	$x_1 - x_2 - C - x_3 - x_1$	$L_2 = -G_1G_3H_1H_2$

The two feedback loops are touching with each other and therefore there are no two non-touching loops

$$\Delta = 1 - (L_1 + L_2)$$
$$= 1 + G_1H_1H_2(G_3 - G_2)$$

These feedback loops L_1 and L_2 are touching all the four forward paths T_1, T_2, T_3 and T_4. When they are removed from the forward path, they are broken. Thus, $\Delta_1 = 1$; $\Delta_2 = 1$; $\Delta_3 = 1$ and $\Delta_4 = 1$

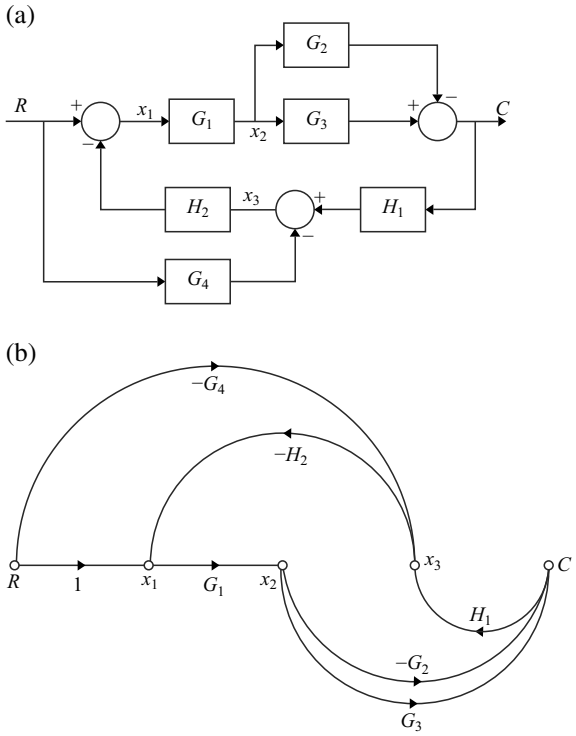


Fig. 1.82 **a** Block diagram for Example 1.40. **b** SFC of Fig. 1.82a

$$\frac{C}{R} = \frac{T_1\Delta_1 + T_2\Delta_2 + T_3\Delta_3 + T_4\Delta_4}{\Delta}$$
$$= \frac{-G_1G_2 + G_1G_3 - G_4H_2G_1G_2 + G_4H_2G_1G_3}{1 + G_1H_1H_2(G_3 - G_2)}$$

$$\boxed{\frac{C}{R} = \frac{G_1(G_3 - G_2)(1 + G_4H_2)}{[1 + G_1H_1H_2(G_3 - G_2)]}}$$

Example 1.41

Reduce the block diagram using reduction rules and obtain $C(s)/R(s)$ for Fig. 1.83a. Verify by SFG.

(*Anna University, May, 2010*)

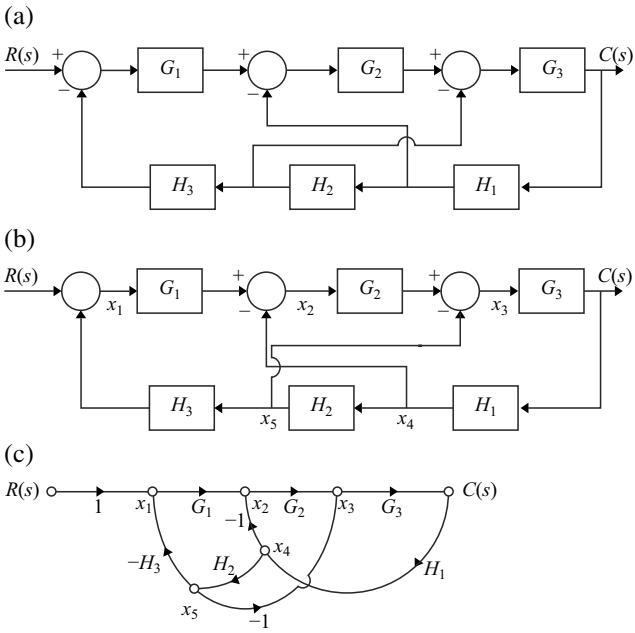


Fig. 1.83 Block diagram for Example 1.41

Solution The block diagram is solved in Example 1.27. To solve the problem by SFG, the output of the summers and the take off points are assigned variables as given below in Fig. 1.83b.

The SFG, with the variables marked in Fig. 1.83b is shown below in Fig. 1.83c.

The forward path $T_1 = R(s) - x_1 - x_2 - x_3 - C(s) = G_1G_2G_3$

	Feedback loop	Loop gain
1.	$x_1 - x_2 - x_3 - C - x_4 - x_5 - x_1$	$L_1 = -G_1G_2G_3H_1H_2H_3$
2.	$x_2 - x_3 - C - x_4 - x_2$	$L_2 = -G_2G_3H_1$
3	$x_3 - C - x_4 - x_5 - x_3$	$L_3 = -G_3H_1H_2$

All the feedback loops are touching with each other. Hence, there is no two non-touching loops

$$\Delta = 1 - (L_1 + L_2 + L_3)$$

$$\Delta = 1 + G_1G_2G_3H_1H_2H_3 + G_2G_3H_1 + G_3H_1H_2$$

All the feedback loops are touching the forward path T_1. When they are removed from the forward path all the three feedback paths are broken. Hence, $\Delta_1 = 1$

$$\frac{C(s)}{R(s)} = \frac{T_1 \Delta_1}{\Delta}$$

$$\boxed{\frac{C(s)}{R(s)} = \frac{G_1 G_2 G_3}{[1 + G_1 G_2 G_3 H_1 H_2 H_3 + G_2 G_3 H_1 + G_3 H_1 H_2]}}$$

Example 1.42

Consider the block diagram shown in Fig. 1.84a. Find C/R using block diagram reduction method. Verify the result using SFG.

Solution Block Diagram Reduction Method

The given block diagram is redrawn and shown in Fig. 1.84b. The outputs of the summers and the take off points are identifies as x_1, x_2, x_3 and x_4. The take off point x_4 is shifted ahead of the block G_2. The modified block diagram is shown in Fig. 1.84c (Rule 8).

The take off point at x_3 is shifted ahead of the summer S_2. Rule 12 is followed (Fig. 1.84d).

Now the summer S_2 is shifted behind the block G_2 using Rule 11. This is shown in Fig. 1.84e.

Interchanging the summers S_2 and S_3, the block diagram is reduced as shown in Fig. 1.84f.

By shifting the take off point ahead of block H_1 and combining the parallel connection the block diagram is reduced as shown in Fig. 1.84g.

Converting the feedback connection and cascading it with $(G_2 + G_3)$ block, the block diagram is reduced as shown in Fig. 1.84h.

The above block diagram is redrawn as shown in Fig. 1.84i.

The feedback is reduced and cascaded. This is shown in Fig. 1.84j.

The above block diagram is reduced using feedback rule.

$$\boxed{\frac{C}{R} = \frac{G_1(G_2 + G_3)}{(1 + G_2 H_1 + G_1 G_2 H_2 - G_1 G_2 G_3 H_1 H_2)}}$$

Signal Flow Graph Method

For the block diagram shown in Fig. 1.84b, the SFG is drawn as shown in Fig. 1.84k. From Fig. 1.84k,

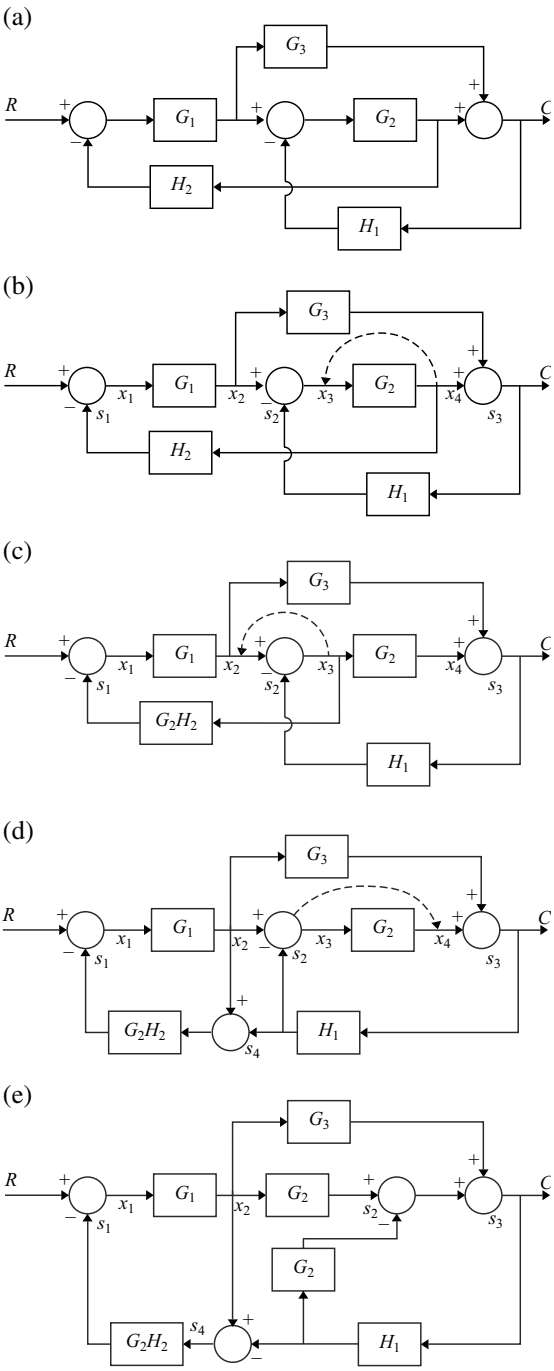
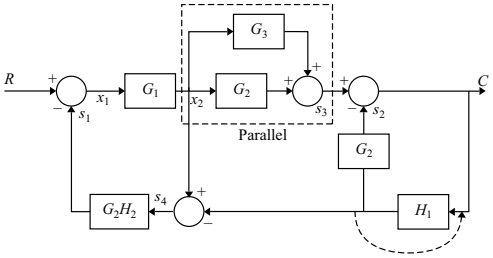


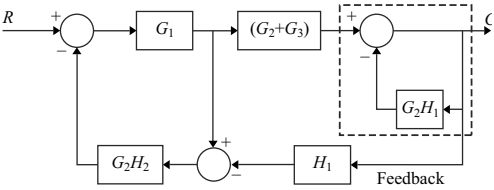
Fig. 1.84 Block diagram for Example 1.42

Fig. 1.84 (continued)

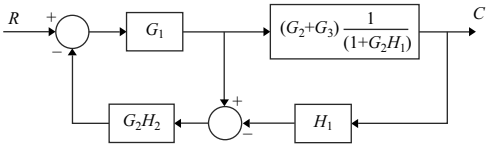
(f)



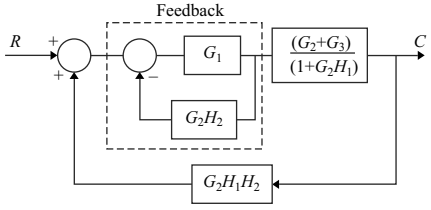
(g)



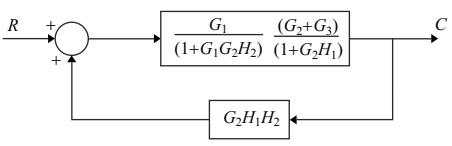
(h)



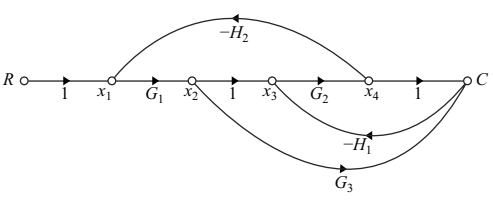
(i)



(j)



(k)



$$\begin{aligned}
T_1 &= R - x_1 - x_2 - x_3 - x_4 - C \\
&= G_1 G_2 \\
T_2 &= R - x_1 - x_2 - C \\
&= G_1 G_3 \\
L_1 &= x_1 - x_2 - x_3 - x_4 - x_1 \\
&= -G_1 G_2 H_2 \\
L_2 &= x_3 - x_4 - C - x_3 \\
&= -G_2 H_1 \\
L_3 &= x_1 - x_2 - C - x_3 - x_4 - x_1 \\
&= G_1 G_3 H_1 G_2 H_2
\end{aligned}$$

All the three loops are touching with each other. Hence, there is no two non-touching loops. Further, all the three feedback loops are touching both the forward paths T_1 and T_2 and the portion touching the forward paths when removed, the loops are broken.

Hence, $\Delta_1 = 1$ and $\Delta_2 = 1$

$$\begin{aligned}
\Delta &= 1 - (L_1 + L_2 + L_3) \\
&= 1 + G_1 G_2 H_2 + G_2 H_1 - G_1 G_2 G_3 H_1 H_2 \\
\frac{C}{R} &= \frac{T_1 \Delta_1 + T_2 \Delta_2}{\Delta} \\
&= \frac{G_1 G_2 + G_1 G_3}{\Delta}
\end{aligned}$$

$$\boxed{\frac{C}{R} = \frac{G_1(G_2 + G_3)}{(1 + G_1 G_2 H_2 + G_2 H_1 - G_1 G_2 G_3 H_1 H_2)}}$$

The same result is obtained using block diagram reduction method with much more tediousness. However, SFG method is much simpler.

1.18.9 Signal Flow Graph for the Electrical Network

The signal flow graph can be easily extended to determine the transfer function of electrical network and other system that are commonly used. In electrical networks, the mesh and nodal equations are written using Kirchhoff's laws. The variables include the mesh currents and nodal voltages and these variables form the nodes in SFG in addition to nodes due to input and output variables. By applying Mason's

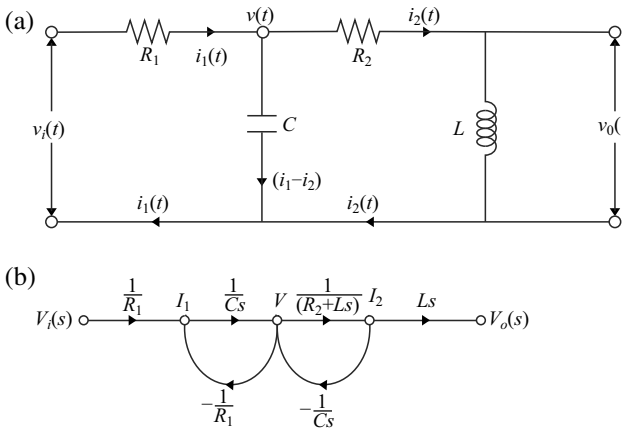


Fig. 1.85 Electrical network for Example 1.43

gain formula, the T.F. of the network is obtained. The following examples illustrate the above procedure.

Example 1.43

For the electrical network shown in Fig. 1.85a determine the T.F.

(*Anna University, May, 2010*)

Solution

The following transform equations for Fig. 1.85a are written

$$V_i(s) = R_1 I_1(s) + V(s)$$

$$I_1(s) = \frac{V_i(s)}{R_1} - \frac{1}{R_1} V(s) \tag{a}$$

$$V(s) = \frac{1}{Cs}[I_1(s) - I_2(s)] \tag{b}$$

Also,

$$V(s) = R_2 I_2(s) + Ls I_2(s)$$

$$I_2(s) = \frac{1}{(R_2 + Ls)} V(s) \tag{c}$$

$$V_0(s) = LsI_2(s) \tag{d}$$

In the above equations, $V_i(s)$ is assigned the input mode and $V_0(s)$, the output node. $I_1(s)$, $V(s)$ and $I_2(s)$ are assigned intermittent nodes. Equations (a), (b), (c) and (d) are represented in SFG as shown in Fig. 1.85b. The following forward paths and feedback loops are identified in Fig. 1.85b.

$$\text{Forward path, } T_1 = \frac{L}{R_1C(R_2 + Ls)}$$

$$\text{Feedback loop, } L_1 = I_1 - V - I_1 = -\frac{1}{R_1Cs}$$

$$\text{Feedback loop, } L_2 = V - I_2 - V = \frac{-1}{Cs(R_2 + Ls)}$$

L_1 and L_2 are touching with each other and therefore there is no two non-touching loops.

$$\begin{aligned}\Delta &= 1 - (L_1 + L_2)\\ &= 1 + \frac{1}{R_1Cs} + \frac{1}{Cs(R_2 + Ls)}\\ &= \frac{LR_1Cs^2 + (R_1R_2C + L)s + (R_1 + R_2)}{R_1C(R_2 + Ls)s}\end{aligned}$$

Since both the feedback loops $L_1$and L_2 are touching the forward path T_1 and when their touching part is removed, both the feedback loops are broken. Hence, for the forward path T_1, there is no feedback loops and hence $\Delta_1 = 1$.

$$\frac{V_0}{V_i} = \frac{T_1\Delta_1}{\Delta} = \frac{L}{R_1C(R_2 + Ls)\Delta}$$

$$\boxed{\frac{V_0}{V_i}(s) = \frac{Ls}{LR_1Cs^2 + (R_1R_2C + L)s + (R_1 + R_2)}}$$

Example 1.44

For the electrical network shown in Fig. 1.86a draw the SFG and using Mason's gain formula determine V_4/V_1.

Solution For Fig. 1.86a, the following equations are written:

$$I_1 = \frac{V_1}{R_1} - \frac{V_2}{R_1} \tag{a}$$

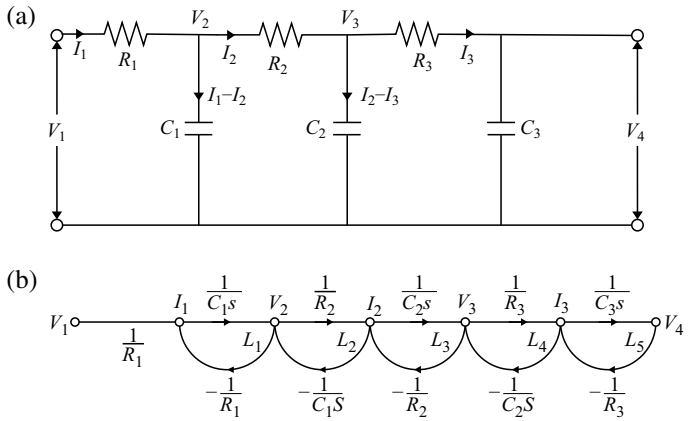


Fig. 1.86 **a** Electrical network for Example 1.44. **b** SFG for Fig. 1.86a

$$V_2 = \frac{I_1}{C_1 s} - \frac{I_2}{C_1 s} \tag{b}$$

$$I_2 = \frac{V_2}{R_2} - \frac{V_3}{R_2} \tag{c}$$

$$V_3 = \frac{I_2}{C_2 s} - \frac{I_3}{C_2 s} \tag{d}$$

$$I_3 = \frac{V_3}{R_3} - \frac{V_4}{R_3} \tag{e}$$

$$V_4 = \frac{I_3}{C_3 s} \tag{f}$$

There are seven variables including the input and output variables and hence seven nodes are created in SFG. The SFG is drawn which satisfies equations (a) to (f) derived above and shown in Fig. 1.86b.

For Fig. 1.86b there is only one forward path with a gain,

$$T_1 = \frac{1}{R_1 R_2 R_3 C_1 C_2 C_3 s^3}$$

There are five feedback paths. The feedback loops and the loop gains are given below:

	Feedback loop	Loop gain
1.	$I_1 - V_2 - I_1$	$L_1 = -\dfrac{1}{R_1 C_1 s}$
2.	$V_2 - I_2 - V_2$	$L_2 = -\dfrac{1}{R_2 C_1 s}$
3.	$I_2 - V_3 - I_2$	$L_3 = -\dfrac{1}{R_2 C_2 s}$
4.	$V_3 - I_3 - V_3$	$L_4 = -\dfrac{1}{R_3 C_2 s}$
5	$I_3 - V_4 - I_3$	$L_5 = -\dfrac{1}{R_3 C_3 s}$

The following pair of loops form two non-touching loops:

(a) $L_1 L_3$ (d) $L_2 L_4$
(b) $L_1 L_4$ (e) $L_2 L_5$
(c) $L_1 L_5$ (f) $L_3 L_5$

There is only one combination of three non-touching loops namely $L_1 L_3 L_5$.

$$\Delta = 1 - (L_1 + L_2 + L_3 + L_4 + L_5) + (L_1 L_3 + L_1 L_4 + L_1 L_5 + L_2 L_4 + L_2 L_5 + L_3 L_5) - L_1 L_3 L_5$$

$$\begin{aligned}\Delta &= 1 + \frac{1}{R_1 C_1 s} + \frac{1}{R_2 C_1 s} + \frac{1}{R_2 C_2 s} + \frac{1}{R_3 C_2 s} + \frac{1}{R_3 C_3 s} + \frac{1}{R_2 C_1 R_3 C_3 s^2} \\ &\quad + \frac{1}{R_2 C_2 R_3 C_3 s^2} + \frac{1}{R_1 R_2 C_1 C_2 s^2} + \frac{1}{R_1 C_1 R_3 C_2 s^2} + \frac{1}{R_1 C_1 R_3 C_3 s^2} \\ &\quad + \frac{1}{R_2 C_1 R_3 C_2 s^2} + \frac{1}{R_1 R_2 R_3 C_1 C_2 C_3 s^3} \\ &= \left(\left[R_1 R_2 R_3 C_1 C_2 C_3 s^3 + (R_2 R_3 C_2 C_3 + R_1 R_3 C_2 C_3 \right.\right. \\ &\quad + R_1 R_3 C_1 C_3 + R_1 R_2 C_1 C_2 + R_1 R_2 C_1 C_3) s^2 + (R_3 C_3 \\ &\quad \left.\left. + R_2 C_3 + R_2 C_2 + R_1 C_3 + R_1 C_2 + R_1 C_1) s + 1 \right] \right) \Big/ R_1 R_2 R_3 C_1 C_2 C_3 s^3\end{aligned}$$

When the feedback loops touching the forward path T_1 are removed all the five feedback loops are broken. Hence, $\Delta_1 = 1$.

$$\frac{V_4}{V_1} = \frac{T_1 \Delta_1}{\Delta}$$

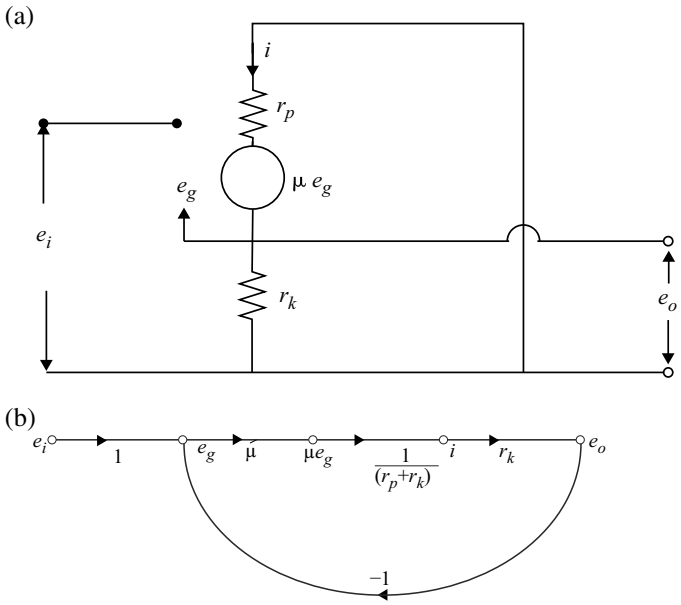


Fig. 1.87 **a** Cathode follower for Example 1.45. **b** SFG for Fig. 1.87a

$$\frac{V_4}{V_1} = 1 \Big/ \Big[R_1R_2R_3C_1C_2C_3s^3 + (R_2R_3C_2C_3 + R_1R_3C_2C_3 + R_1R_3C_1C_3 + R_1R_2C_1C_2 + R_1R_2C_1C_3)s^2 + (R_3C_3 + R_2C_3 + R_2C_2 + R_1C_3 + R_1C_2 + R_1C_1)s + 1\Big]$$

Example 1.45

Determine the SFG and T.F. of the cathode follower circuit shown in Fig. 1.87a.

(*Anna University, December, 2006*)

Solution For Fig. 1.87a, the following equations are written:

$$e_i = e_g + e_o$$

$$e_g = e_i - e_o \tag{a}$$

$$i = \frac{\mu e_g}{r_p r_k} \tag{b}$$

$$e_o = i\, r_k \tag{c}$$

The variables are e_i, e_g, μe_g, i and e_o and they form the nodes in the SFG. Which is shown in Fig. 1.87b.

There is only one forward path. The forward path gain

$$T_1 = \frac{\mu r_k}{r_p + r_k}$$

There is only one feedback loop. The loop gain is

$$L_1 = -\frac{\mu r_k}{r_p + r_k}$$

When the feedback loop touching the forward path is removed it is broken. There is no feedback loop for the forward path T_1. Hence, $\Delta_1 = 1$.

$$\begin{aligned}\Delta &= 1 - L_1 \\ &= 1 + \frac{\mu r_k}{(r_p + r_k)} \\ &= \frac{r_p + r_k(1+\mu)}{(r_p + r_k)} \\ \frac{e_o}{e_i} &= \frac{T_1 \Delta_1}{\Delta}\end{aligned}$$

$$\boxed{\frac{e_o}{e_i} = \frac{\mu r_k}{r_p + r_k(1+\mu)}}$$

Example 1.46

For the electrical network shown in Fig. 1.88a draw the SFG and determine V_3/V_1 using Mason's gain formula.

(*Anna University, April, 2004*)

Solution Assume that the current in the branch where R_4 is connected is I_2. The current from the voltage source V_1 is I_1 (given) and that of the current source is αI_1 (given) where α is any constant. Using Kirchhoff's law the current through R_2 is $(\alpha I_1 - I_2)$ and that of R_3 is $(I_1 - I_2)$. The following loop and nodal equations are written for Fig. 1.88a.

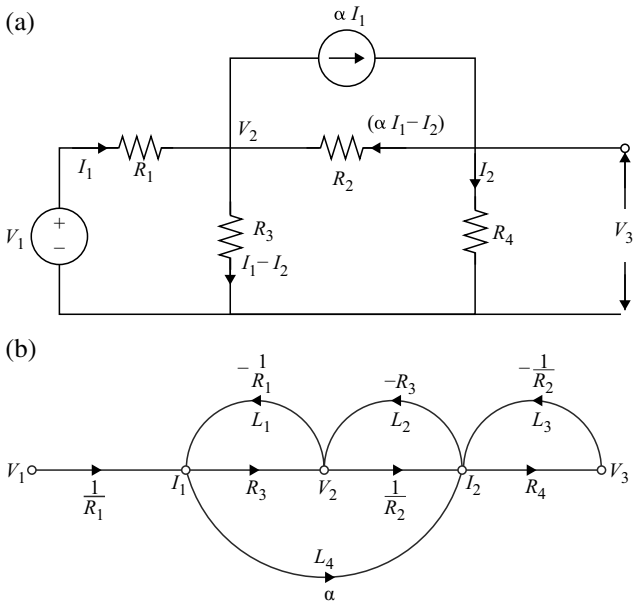


Fig. 1.88 **a** Electrical network for Example 1.46. **b** SFG for Fig. 1.88a

$$I_1 = \frac{V_1}{R_1} - \frac{V_2}{R_1} \tag{a}$$

$$V_2 = I_1 R_3 - I_2 R_3 \tag{b}$$

$$\alpha I_1 - I_2 = \frac{V_3}{R_2} - \frac{V_2}{R_2}$$

$$I_2 = \alpha I_1 + \frac{V_2}{R_2} - \frac{V_3}{R_2} \tag{c}$$

$$V_3 = I_2 R_4 \tag{d}$$

V_1, I_1, V_2, I_2 and V_3 are represented as nodal variables in SFG. Equations (a), (b), (c) and (d) are used to construct SFG which is shown in Fig. 1.88b.

The forward path and the path gain, and the feedback loop and loop gains are given in the following tables:

	Forward path	Forward path gain
1.	$V_1 - I_1 - V_2 - I_2 - V_3$	$T_1 = \dfrac{R_3 R_4}{R_1 R_2}$
2.	$V_1 - I_1 - I_2 - V_3$	$T_2 = \alpha \dfrac{R_4}{R_1}$
	Feedback loop	Loop gain
1.	$I_1 - V_2 - I_1$	$L_1 = -\dfrac{R_3}{R_1}$
2.	$V_2 - I_2 - V_2$	$L_2 = -\dfrac{R_3}{R_2}$
3.	$I_2 - V_3 - I_2$	$L_3 = -\dfrac{R_4}{R_2}$
4	$I_1 - I_2 - V_2 - I_1$	$L_4 = \alpha \dfrac{R_3}{R_1}$

L_1 and L_3 are the only two non-touching loops. Their gain product is given by,

$$
\begin{aligned}
L_1 L_3 &= \frac{R_3 R_4}{R_1 R_2} \\
\Delta &= 1 - (L_1 + L_2 + L_3 + L_4) + L_1 L_3 \\
&= 1 + \frac{R_3}{R_1} + \frac{R_3}{R_2} + \frac{R_4}{R_2} - \alpha \frac{R_3}{R_1} + \frac{R_3 R_4}{R_1 R_2} \\
&= \frac{R_1 R_2 + R_2 R_3 + R_3 R_1 + R_1 R_4 - \alpha R_2 R_3 + R_3 R_4}{R_1 R_2}
\end{aligned}
$$

All the feedback loops touching the forward paths T_1 and T_2 when removed are broken. Hence, $\Delta_1 = 1$ and $\Delta_2 = 1$.

$$
\begin{aligned}
\frac{V_3}{V_1} &= \frac{T_1 \Delta_1 + T_2 \Delta_2}{\Delta} \\
&= \frac{(R_3 R_4 / R_1 R_2) + \alpha (R_4 / R_1)}{\Delta}
\end{aligned}
$$

$$
\boxed{\frac{V_3}{V_1} = \frac{R_3 R_4 + \alpha R_2 R_4}{[R_1 R_2 + R_2 R_3 + R_3 R_1 + R_1 R_4 + R_3 R_4 - \alpha R_2 R_3]}}
$$

Example 1.47

For the electrical network shown in Fig. 1.89a, draw the SFG and determine $\dfrac{V_3}{V_1}(s)$, using Mason's gain formula.

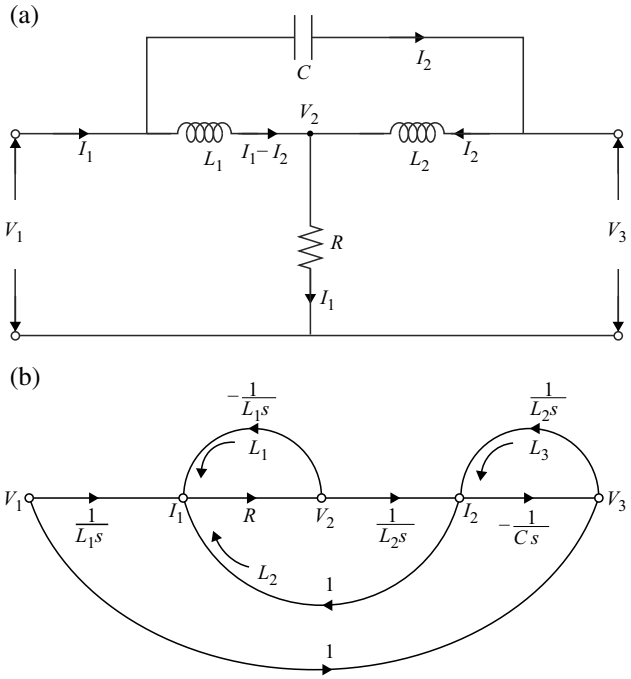


Fig. 1.89 **a** Electrical network for Example 1.47. **b** SFG for Fig. 1.89a

Solution The Laplace transformed network equations for Fig. 1.89a are written as given below.

$$V_1 - V_2 = L_1 s(I_1 - I_2) \tag{a}$$

$$I_1 = \frac{V_1}{L_1 s} - \frac{V_2}{L_1 s} + I_2$$

$$V_2 = I_1 R \tag{b}$$

$$V_1 - V_3 = \frac{1}{Cs} I_2 \tag{c}$$

$$V_3 = V_1 - \frac{1}{Cs} I_2$$

$$V_3 - V_2 = L_2 s I_2 \tag{d}$$

$$I_2 = \frac{V_3}{L_2 s} - \frac{V_2}{L_2 s}$$

The variables V_1, I_1, V_2, I_2 and V_3 form the nodes in SFG. Using equations (a), (b), (c) and (d) SFG is drawn as shown in Fig. 1.89b.

From Fig. 1.89b, the forward paths and their gains, the feedback loops and the loop gains are determined as given below:

	Forward path	Forward path gain
1.	$V_1 - I_1 - V_2 - I_2 - V_3$	$T_1 = \dfrac{R}{L_1 L_2 C s^3}$
2	$V_1 - V_3$	$T_2 = 1$

	Feedback loop	**Loop gain**
1.	$I_1 - V_2 - I_1$	$L_1 = -\dfrac{R}{L_1 s}$
2.	$I_1 - V_2 - I_2 - I_1$	$L_2 = -\dfrac{R}{L_2 s}$
3	$I_2 - V_3 - I_2$	$L_3 = -\dfrac{1}{L_2 C s^2}$

Feedback loops L_1 and L_3 are not touching with each other. Hence, there are two non-touching loops.

$$\begin{aligned}\Delta &= 1 - (L_1 + L_2 + L_3) + L_1 L_3 \\ &= 1 + \frac{R}{L_1 s} + \frac{R}{L_2 s} + \frac{1}{L_2 C s^2} + \frac{R}{L_1 L_2 C s^3} \\ &= \frac{L_1 L_2 C s^3 + R L_2 C s^2 + R L_1 C s^2 + L_1 s + R}{L_1 L_2 C s^3}\end{aligned}$$

All the three feedback loops are touching the forward path T_1 and when removed from the touching part, they are broken. Hence, $\Delta_1 = 1$. The feedback loops L_1 and L_2 are not touching the forward path T_2. However, loop L_3 is touching the forward path T_2 and when the touching portion is removed, this feedback loop is broken. Feedback loops L_1 and L_2 are touching with each other. Hence,

$$\begin{aligned}\Delta_2 &= 1 - (L_1 + L_2) \\ &= 1 + \frac{R}{L_1 s} + \frac{R}{L_2 s} \\ &= \frac{L_1 L_2 s^2 + R(L_1 + L_2)s}{L_1 L_2 s^2} \\ \frac{V_3}{V_1} &= \frac{T_1 \Delta_1 + T_2 \Delta_2}{\Delta} \\ \frac{V_3}{V_1} &= \frac{\dfrac{R}{L_1 L_2 C s^3} + \dfrac{L_1 L_2 s + R(L_1 + L_2)}{L_1 L_2 s}}{\dfrac{L_1 L_2 C s^3 + R L_2 C s^2 + R L_1 C s^2 + L_1 s + R}{L_1 L_2 C s^3}}\end{aligned}$$

$$\boxed{\frac{V_3}{V_1} = \frac{[L_1L_2s + R(L_1 + L_2)]Cs^2 + R}{[L_1L_2Cs^3 + (L_1 + L_2)RCs^2 + L_1s + R]}}$$

SUMMARY

1. Control systems are generally classified as open loop and closed loop systems. Each one has its own advantages and disadvantages. However, because of fastness of response and high accuracy feedback (or closed loop) systems are invariably used in almost all applications.
2. To analyse the dynamic behaviour of a system, systems are modelled in the frequency and time domains. Transfer function model is used in the frequency domain while state space model is used in the time domain. Transfer function model is described in this chapter.
3. Transfer function model, using Laplace transform, is obtained for mechanical, electrical and electromechanical systems.
4. The transfer function of mechanical systems are obtained by writing the dynamic equations using D'Alembert's principle. It is also possible to convert the given mechanical systems to its electrical equivalent and obtain the T.F from the equations written by following Kirchhoff's laws. This is called electrical analog. Force-voltage and force-current analogies are available and one can follow any of them. Since there exists a one to one relationship between the components and variables in the mechanical and electrical systems, the results obtained from one system can be easily transferred to the other system.
5. Components are interconnected to form a system. Each component is represented by a block with its functional relationship kept inside the box. The input/ output relationship of the entire system is obtained by reducing the block diagram to a single block. This is known as block diagram reduction technique. This is obtained using a set of rules.
6. Block diagram reduction method is somewhat tedious and as the complexity of the system increases, it becomes more cumbersome. However, systems can be represented by a graphical model which is known as signal flow graph model. Using Mason's gain formula, the input-output relationship can be easily obtained. Even the block diagram can be converted into its equivalent SFG and by using Mason's gain formula, the over all gain between any two variables can be obtained.

EXERCISE

Short Answer Type Questions

1. **Define a system.**
   A system is defined as the interconnections of objects which have a definite relationship between objects and attributes (variables).
2. **What is a control system?**
   A system, which is formed by interconnection of components when it provides a desired response is called a control system.
3. **What is an open loop system?**
   An open loop system is one in which the control action is independent of the output.
4. **What is a closed loop system?**
   A closed loop system is one in which the control action depends upon the output.
5. **What is control action?**
   Actuating signal also otherwise called error signal is called control action which actuates the power device directly or sometimes indirectly. The actuating signal is the difference of the reference input and part of the output or full output.
6. **What are the advantages of open loop system?**
   Open loop systems are simple to design. Component requirements are less and therefore less costly. Almost all physically realizable open loop systems are stable.
7. **What are the drawbacks of open loop systems?**
   Open loop systems are manually controlled and hence accuracy is less or it depends upon the ingenuity of the human operator. The time response of the system is poor (sluggish operation). Its operation requires close manual inspection during any internal and external disturbances.
8. **What are the advantages of closed loop systems?**
   Closed loop systems are very fast acting and more accurate. It rejects internal and external disturbances automatically without human involvement. Since the bandwidth is high, the range of frequency over which the system responds is high. If the open loop system is unstable, it is possible to make the closed loop system stable by proper feedback.
9. **What are the drawbacks of closed loop systems?**
   Closed loop systems tempt to oscillate. System complexity is increased by addition of components. This increases the cost and reduces the stability limit. The over all gain of the system is reduced.
10. **What is an input?**
    Input is an attribute of the system which can be varied independently.
11. **What is an output?**
    Output is an attribute which can be varied by varying the input.
12. **What is servomechanism? Give examples.**
    Servomechanism is a branch of engineering study if the output is mechanical

position or velocity. Examples include control of position of anti-aircraft gun, radar antenna, rudders in ships etc.

13. What is a regulator system? Give examples.
A regulator system is a system which maintains the system output at a prescribed set level. Temperature, pressure, level control of a process, voltage control of an electrical generator, speed control of electric motors, oil engines etc. are examples of regulator system.

14. Why positive feedback is not used in closed loop system?
For positive feedback systems, the characteristic equation is $1 - GH(s) = 0$ which will yield poles in the right half s-plane. This makes the closed loop system unstable. That is why positive feedback is not employed in closed loop system. However, positive feedback may be employed in the design of oscillators which has sustained oscillations.

15. What is characteristic equation and why is it so called?
The denominator of a transfer function when equated to zero is called the characteristic equation. Since the roots of the characteristic equation determine the character of the transient response and the stability character it is called characteristic equation.

16. Define transfer function of a system. Does it exit for non-linear systems?
Transfer function is defined as the ratio of the Laplace transform of output variable to the Laplace transform of input variable with all initial conditions being zero. Laplace transform does not exist for non-linear system.

17. What do you understand by the T.F being strictly proper?
A transfer function is said to be strictly proper if the order of the denominator polynomial is greater than the order of the numerator polynomial. For example,

$$G(s) = \frac{10(s^2 + 2s + 4)}{(s^3 + 5s^2 + 7s + 8)}$$

is called a strictly proper T.F.

18. What is a proper T.F?
A T.F where numerator and denominator polynomials have the same order is called a proper T.F. For example,

$$G(s) = \frac{10(s^2 + 2s + 4)}{(5s^2 + 7s + 8)}$$

is called a proper transfer function.

19. What is an improper T.F?
A transfer function whose order of the numerator polynomial is greater than the order of the denominator polynomial is called improper T.F. For example,

$$G(s) = \frac{10(s^2 + 2s + 4)}{(2s + 5)}$$

is called an improper T.F.

20. How T.F is defined in terms of impulse response of a system?
The transfer function of a linear system is defined as the inverse Laplace transform of the impulse response of the system with all the initial conditions set to zero.

21. What is a minimum phase T.F? Give example.
A transfer function which has all the poles and zeros placed in the left of the s-plane is called a minimum phase T.F. For example,

$$G(s) = \frac{10(s+2)}{(s+3)(s+9)}$$

is a minimum phase T.F.

22. What is a non-minimum phase T.F?
A T.F which has atleast one pole or a zeros placed in the right half s-plane is called non-minimum phase T.F. For example,

$$G(s) = \frac{(s-2)}{s(s+2)}$$

is a non-minimum phase T.F. A zero is present in the right half s-plane.

23. What is an all pass T.F.?
A T.F. which has its poles and zeros present symmetrically with respect to the imaginary axis is called all pass T.F. For example

$$G(s) = \frac{(s+10)}{(s-10)}$$

is an all pass T.F.

24. Define zero of a T.F.?
The points at which the T.F. becomes zero in the s-plane are called zeros of the T.F. They are also the roots of the numerator polynomial of the T.F.

25. Define pole of a T.F.?
The points at which the T.F. becomes infinity in the s-plane are called poles of the T.F. They are also the roots of the denominator polynomial of the T.F.

26. What is the use of a lever in mechanical system?
Lever is used as a matching device in a mechanical translational system to attain maximum power transfer. It transmits translational motion and forces from one point to another point in the system.

27. What is the use of a gear in mechanical system?
A gear is a matching device in a mechanical rotational system to attain maximum power transfer. It transmits rotational motion and torque from one point to another point in the system.

28. What are the energy storing elements in the mechanical system?
In the translational system, kinetic energy is stored in mass and potential energy

is stored in the spring. In the rotational system, kinetic energy is stored in inertia and potential energy is stored in the spring.

29. Distinguish between ordinary D.C motor and D.C servomotor.

For fastness of response, the time constant of a D.C servomotor should be small. To achieve this, for the given friction the inertia of the servomotor is made small so that the mechanical time constant is small. To reduce the electrical time constant, the inductance of the field is made small if it is field controlled motor and the inductance of armature coil is small if it is armature controlled motor. Further rotor inertia should be low.

30. What are the advantages of armature controlled D.C motor?

The electrical time constant of the armature controlled D.C motor is small and hence its time response is very fast. The back emf developed in the armature circuits provides additional damping which eliminates unwanted oscillations during transients.

31. What are the advantages of field controlled D.C motor?

In field controlled D.C motor, the power requirement is less and there is no need for any power amplifier and it requires voltage amplifier only which is not difficult to design.

32. What are the disadvantages of armature controlled D.C motor?

Armature controlled D.C motor requires power amplifier which is somewhat difficult to design and costly too.

33. What are the disadvantages of field controlled D.C motor?

The electrical time constant of field circuit of the D.C motor is high and hence the time response is very sluggish. Further it requires a constant current source in the armature circuit which is some what difficult to get.

34. What is the necessacity of negative slope torque-speed characteristic of a two phase A.C servomotor?

The negative torque-speed characteristic of a two phase A.C servomotor provides positive damping to the motor which ensures stable operation. In other words, the roots of the characteristic equation in the T.F of the motor will lie in the left half of the s-plane.

35. Why an ordinary two phase induction motor is not useful for control application?

The torque-speed characteristic of an ordinary two phase induction motor has positive slope at low speed which provides negative damping to the motor operation and is unstable. Hence, ordinary two phase induction motors are not suitable for control application.

36. What change is made in the design of two phase servomotor from that of the ordinary two phase A.C motor?

The rotor resistance of the two phase A.C motor is considerably increased. This makes the torque-speed characteristic to have negative slope for entire speed of operation without losing stability but at the cost of efficiency. For control applications stability is more important than efficiency.

37. Can two phase A.C servomotor be operated as A.C technogenerator?

Similar to D.C motor being converted to D.C technogenerator, two phase A.C

servomotor can be used as A.C technogenerator where output is proportional to the speed at constant frequency which is the frequency of the excitation voltage of the reference field winding.

38. What is a pair of synchros?
A pair of synchros is used as an error detector in servo mechanism. It consists of a synchro generator and a control transformer.

39. What is the constructional difference between synchro generator and control transformer?
The synchro generator and control transformer have identical construction of stators where three phase winding is provided. However, there is a major difference in the construction of the rotors. While the rotor of the synchro generator is a salient pole one, the rotor of the control transformer is dump bell shaped cylindrical type.

40. Why the rotor of control transformer is cylindrically shaped?
The CT rotor is cylindrically shaped so that the air gap flux is uniformly distributed to the servo amplifier and hence the change in rotor impedance with the rotor position is minimized.

41. In the stators of S.G and C.T three phase A.C windings are provided. What is the nature of induced voltage in the stator windings?
Even though the stators of S.G and C.T are provided with three phase A.C windings, the induced voltages in the windings are single phase voltages.

42. What is a block diagram?
The pictorial representation of a system is called a block diagram. System inputs and outputs are represented by drawing a single line towards the block and away from the block while the block represents the functional relationship between input and output.

43. What are the basic connections of block diagram?
Series (cascade), parallel and feedback connections of block diagram are the basic connections.

44. What are the basic operations performed in block diagram reduction?
The basic operations performed in block diagram reduction include shifting the blocks before and after the summer, moving a take off point before and after a block and before and after summers, interchanging of summers and reducing blocks in parallel, cascade and feedback.

45. State the block diagram simplification rule for removing the feedback.
If $G(s)$ is the T.F of the forward path and $H(s)$ is that of the feedback path, than the overall T.F is simplified for negative feedback as

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + GH(s)}$$

The sign of $GH(s)$ will be negative for positive feedback.

46. What are the advantages of block diagram representation?
By connecting the blocks of individual components, the block diagram for the

overall system can be easily formed. Further, it is easy to visualize the actual system by examining the block diagram.

47. What are the disadvantages of block diagram representation?

The disadvantages of block diagram are that it is not unique for a given system. The main source of energy flow in the system is not shown in the block diagram. Some important functions are often concealed inside the four walls of the blocks. From the block diagram it is difficult to understand what type of system the block diagram represents since no information about the construction is known. Further, block diagram is drawn under the assumption that what is inside the block is not affected by what is outside of the block which is not true.

48. What is the signal flow graph?

Signal flow graph is a graphical method of signal modelling in which the nodes represent the different variables in the system. The nodes are connected by branches whose gain give functional relationship between any two nodes.

49. Explain Mason's gain formula for signal flow graphs.

According to Mason's gain formula, the total gain between the output variable and the input variable is given by:

$$A = \frac{\Sigma_{m=1}^{n} T_m \Delta_m}{\Delta}$$

where A is total gain and T_m is gain of m_{th} forward path.

$$\Delta = 1 - \sum_n P_{n1} + \sum_n P_{n2} - \sum_n P_{n3} + \cdots$$
$$= \text{Determinant of the graph}$$

P_{nr} is the gain product of n possible combination of r non-touching loops. Δ_m is Δ for mth forward path with the feedback loops touching that forward path when removed is equal to cofactor of the graph.

50. What are input and output and mixed nodes of SFG?

A node which has only outgoing branches is called input while node which has only incoming branches are called output node or sink nodes. A node which is neither an input node nor an output node is called a mixed node.

51. What is path and what is a forward path of SFG?

The traversal of branches connected by nodes is called a path. A path which originates from the input node and terminates at the output node is called a forward path.

52. What is path gain?

The product of the branch gains while traversing the path is called the path gain.

53. What is a feedback loop and loop gain?
If the path which originates from a node and ends at the same node without encountering the same node more than once is known as feedback loop. The product of all the branch gains of the feedback loop is called loop gain.

54. What is a self loop?
The path starting from a node if it ends in the same node without touching any other node is called self node.

55. What is non-touching loop?
Two feedback loops are said to be non-touching if they do not have any common node.

56. What are basic rules of SFG?
The basic rules of SFG include the following. A signal always flows along the branch in the direction of arrow. Its strength gets multiplied by the branch gain. The value of the signal at any node is given by the sum of all the signals entering that node.

57. How block diagram is converted into SFG?
In addition to input and output variables in the block diagram, the outputs of the summers and also the take off points are identified with variables. These variables become nodes is SFG. These nodes are connected by branches which are obtained by inspection of block diagram.

Long Answer Type Questions

1. Find the T.F. of the circuit shown in Fig. 1.90. The initial conditions are zero.

$$\frac{V_0}{V_i}(s) = \frac{10^6}{s^2 + 10^6 s + 10^6}$$

2. For the translational mechanical system shown in Fig. 1.91, determine $\frac{X}{F}(s)$.

$$\frac{X(s)}{F(s)} = \frac{(B_1 s + K_1)}{[MB_1 s^3 + (MK_1 + B_1 B_2)s^2 + (B_1 K_1 + B_2 K_1 + B_1 K_2)s + K_1 K_2]}$$

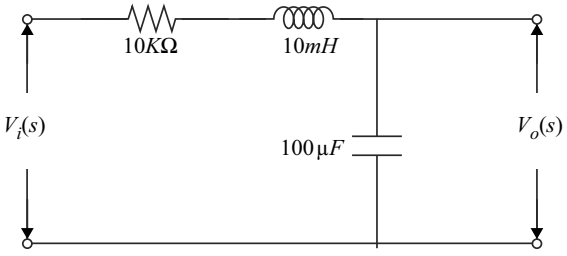


Fig. 1.90 Electrical network for Question 1

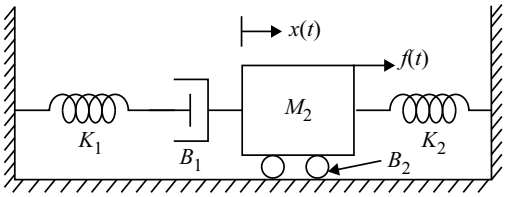


Fig. 1.91 Mechanical system for Question 2

3. Consider the electro mechanical system shown in Fig. 1.92. The force of attraction of the solenoid coil is directly proportional to the current in the coil and the proportionality constant is K_f N/amp. Determine the T.F. $\frac{X}{E}(s)$. Neglect back emf effect of the solenoid coil. The coil has a resistance R and inductance L.

$$\frac{X}{E}(s) = \frac{K_f(Bs + K)}{s(R + Ls)[MBs^2 + MKs + BK]}$$

4. For the block diagram shown in Fig. 1.93 find C/R using block diagram reduction method.

$$\boxed{\frac{C}{R} = \frac{G_1G_2G_3G_4}{[1 + G_2G_3H_2 + G_3G_4H_1 + G_1G_2G_3G_4H_3]}}$$

5. For the block diagram shown in Fig. 1.94, find C/R using block diagram reduction method.

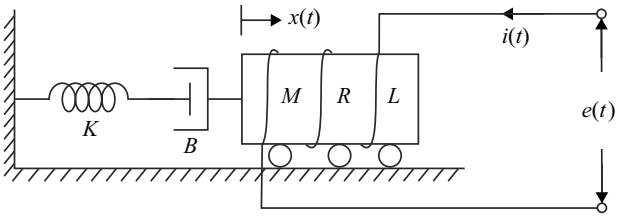


Fig. 1.92 Electromechanical system for Question 3

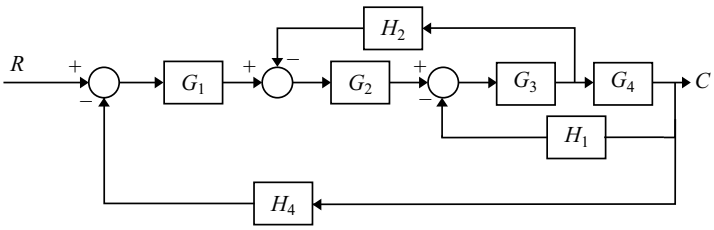


Fig. 1.93 Block diagram for Question 4

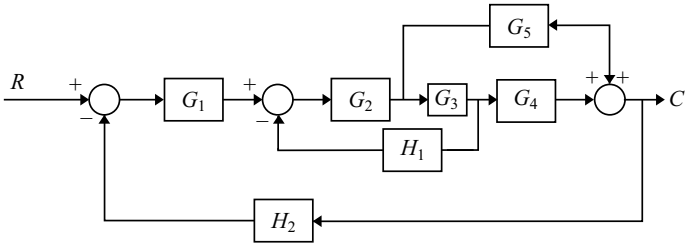
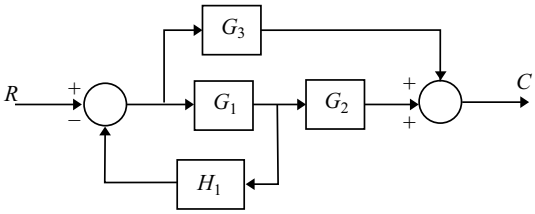


Fig. 1.94 Block diagram for Question 5

Fig. 1.95 Block diagram for Question 6



$$\boxed{\frac{C}{R} = \frac{G_1G_2(G_3G_4 + G_5)}{[1 + G_2G_3H_1 + G_1G_2G_3G_4H_2 + G_1G_2G_5H_2]}}$$

6. For the block diagram shown in Fig. 1.95, obtain C/R using block diagram reduction method.

$$\boxed{\frac{C}{R} = \frac{G_1G_2 + G_3}{1 + G_1H_1}}$$

7. For the block diagram shown in Fig. 1.96, determine C/R using block diagram reduction method.

$$\boxed{\frac{C}{R} = \frac{G_1G_2(G_3 + G_4)}{[1 + G_1G_2H_1 + G_2G_3H_2 + G_2G_4H_2 + G_1G_2(G_3 + G_4)]}}$$

8. For the block diagram shown in Fig. 1.97, determine C/R using block diagram reduction method.

$$\boxed{\frac{C}{R} = \frac{G_2[G_1 + G_3(1 + G_1H_1)]}{[1 + G_1(H_1 + G_2)]}}$$

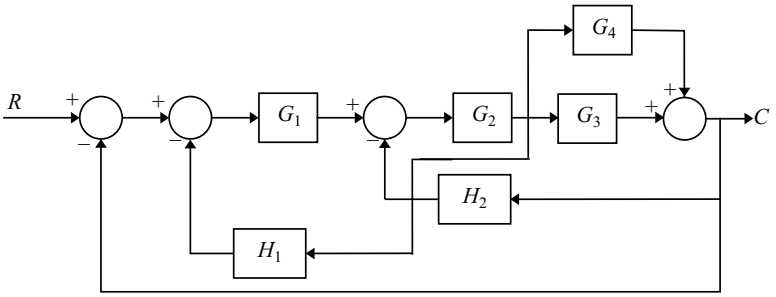
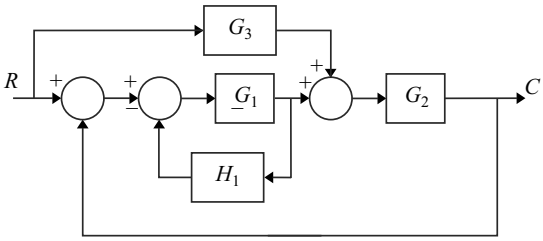


Fig. 1.96 Block diagram for Question 7

Fig. 1.97 Block diagram for Question 8



9. For the block diagram shown in Fig. 1.98, determine C when the inputs R and D are simultaneously present.

$$\boxed{C = \frac{[R(G_1 + G_2)G_3G_4] + D[G_4(1 + G_3H_1)]}{[1 + G_3H_1 + (G_1 + G_2)G_3G_4H_2]}}$$

10. For the block diagram shown in Fig. 1.99a, draw the equivalent SFG and using Mason's gain formula determine C/R.

$$T_1 = G_1G_2G_3; \quad T_2 = G_1G_4;$$
$$L_1 = -G_4H_2; \quad L_2 = -G_1G_2H_1;$$
$$L_3 = -G_2G_3H_2; \quad L_4 = -G_1G_2G_3; \quad L_5 = -G_1G_4;$$
$$\Delta_1 = 1; \quad \Delta_2 = 1; \quad \Delta = 1 - (L_1 + L_2 + L_3 + L_4 + L_5)$$

$$\boxed{\frac{C}{R} = \frac{(G_1G_2G_3 + G_1G_4)}{[1 + G_2G_3H_2 + G_1G_2G_3 + G_4H_2 + G_1G_2H_1 + G_1G_4]}}$$

11. Using Mason's gain formula, obtain the overall T.F. of the SFG shown in Fig. 1.100.

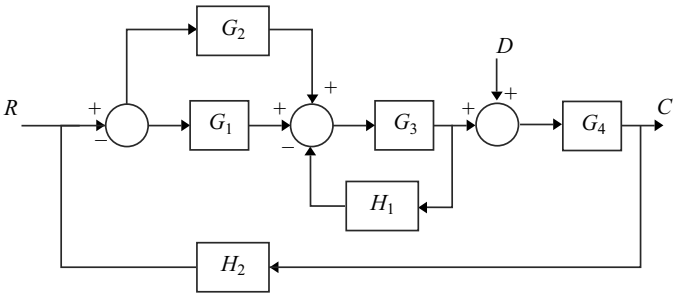


Fig. 1.98 Block diagram for Question 9

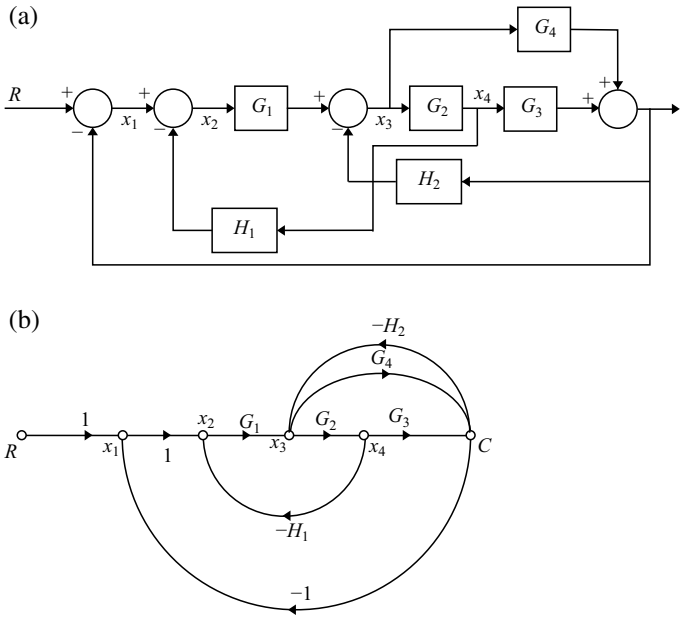


Fig. 1.99 **a** Block diagram for Question 10. **b** SFG for Fig. 1.99a

$$T_1 = G_1G_2G_3G_4G_5G_8; \quad T_2 = G_1G_6G_4G_5G_8; \quad T_3 = G_1G_2G_7;$$
$$L_1 = -G_4G_5H_1; \quad L_2 = -G_2G_3G_4G_5G_8H_2;$$

$$L_3 = -G_6G_4G_5G_8H_2; \quad L_4 = -G_2G_7H_2;$$
$$\Delta_1 = 1; \quad \Delta_2 = 1; \quad \Delta_3 = 1 + G_5G_4H_1;$$
$$\Delta = 1 - (L_1 + L_2 + L_3 + L_4) + L_1L_4$$

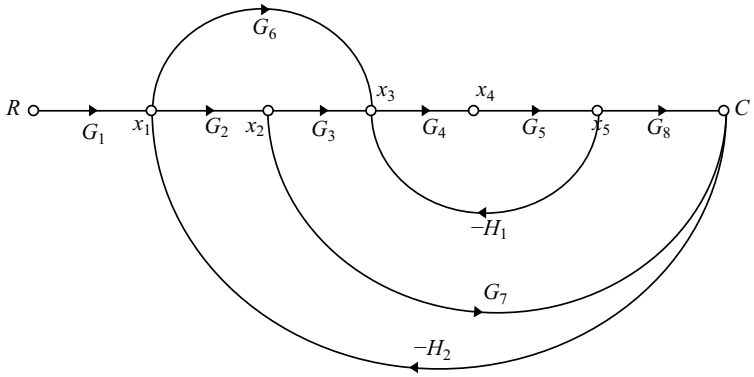


Fig. 1.100 SFG for Question 11

$$\frac{C}{R} = \frac{G_1G_2G_3G_4G_5G_8 + G_1G_6G_4G_5G_8 + G_1G_2G_7(1 + G_4G_5H_1)}{[1+G_4G_5H_1+G_2G_4G_5G_8H_2+G_6G_4G_5G_8H_2+G_2G_7H_2+G_2G_4G_5G_7H_1H_2]}$$

12. For the SFG shown in Fig. 1.101, evaluate C/R using Mason's gain formula.

$$\begin{aligned}
T_1 &= G_1G_2G_3G_4; \qquad T_2 = G_1G_2G_5;\\
L_1 &= -G_2H_4; \qquad L_2 = -G_2G_3H_2;\\
L_3 &= -H_1; \qquad L_4 = -G_2G_3G_4H_3;\\
\Delta_1 &= 1; \qquad \Delta_2 = (1 + H_1);\\
\Delta &= 1 - (L_1 + L_2 + L_3 + L_4) + (L_1L_3)
\end{aligned}$$

$$\frac{C}{R} = \frac{G_1G_2G_3G_4 + G_1G_2G_5(1 + H_1)}{[1 + G_2H_4 + G_2G_3H_2 + H_1 + G_2G_3G_4H_3 + G_2H_1H_4]}$$

13. Using Mason's gain formula obtain the overall T.F. of the SFG given in Fig. 1.102.

$$\begin{aligned}
T_1 &= G_1G_2G_3G_4; \qquad T_2 = G_1G_2G_5G_4;\\
L_1 &= -G_4H_2; \qquad L_2 = -G_1G_2G_3G_4;\\
L_3 &= -G_1G_2G_5G_4; \qquad L_4 = G_2H_1;\\
\Delta_1 &= 1; \qquad \Delta_2 = 1;\\
\Delta &= 1 - (L_1 + L_2 + L_3 + L_4) + L_1L_4
\end{aligned}$$

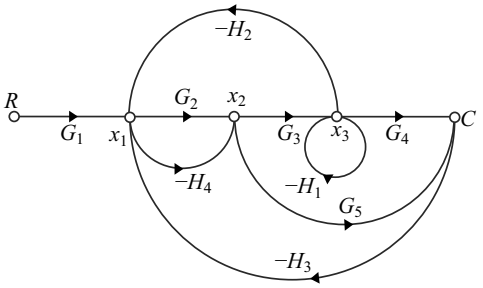


Fig. 1.101 SFG for Question 12

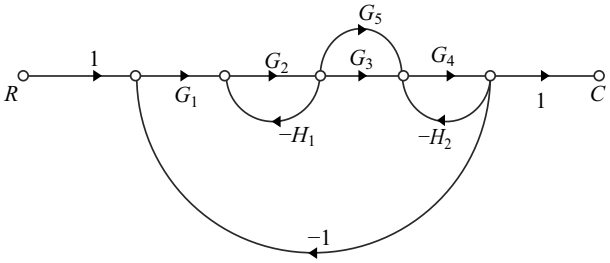


Fig. 1.102 SFG for Question 13

$$\boxed{\frac{C}{R} = \frac{G_1G_2G_4(G_3 + G_5)}{[1 + G_4H_2 + G_1G_2G_4(G_3 + G_5) - G_2H_1(1 + G_4H_2)]}}$$

14. For the SFG shown in Fig. 1.103 obtain the values of C/R.

$$\begin{aligned}
&T_1 = 400; \qquad T_2 = 200;\\
&L_1 = -4; \qquad L_2 = -4; \qquad L_3 = -1;\\
&\Delta_1 = 1 + 1 = 2; \qquad \Delta_2 = 1 + 4 = 5;\\
&\Delta = 1 - (L_1 + L_2 + L_3) + L_1L_3 + L_2L_3;\\
&\Delta = 1 + 4 + 4 + 1 + 4 + 4 = 18
\end{aligned}$$

$$\boxed{\frac{C}{R} = \frac{400 \times 2 + 200 \times 5}{18} = 100}$$

15. For the electrical network shown in Fig. 1.104a, draw the SFG and using Mason's gain formula determine $\frac{V_2}{V_1}(s)$.

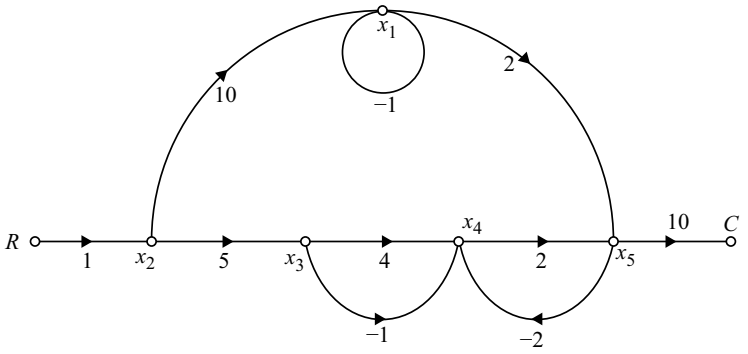


Fig. 1.103 SFG for Question 14

$$T_1 = \frac{1}{R_1 R_2 C_1 C_2 s^2}; \qquad T_2 = 1;$$
$$L_1 = -\frac{1}{R_1 C_2 s}; \qquad L_2 = -\frac{1}{R_2 C_2 s}; \qquad L_3 = -\frac{1}{R_2 C_1 s};$$
$$\Delta = 1 - (L_1 + L_2 + L_3) + L_1 L_3; \qquad \Delta_1 = 1;$$
$$\Delta_2 = 1 + L_1 + L_2$$

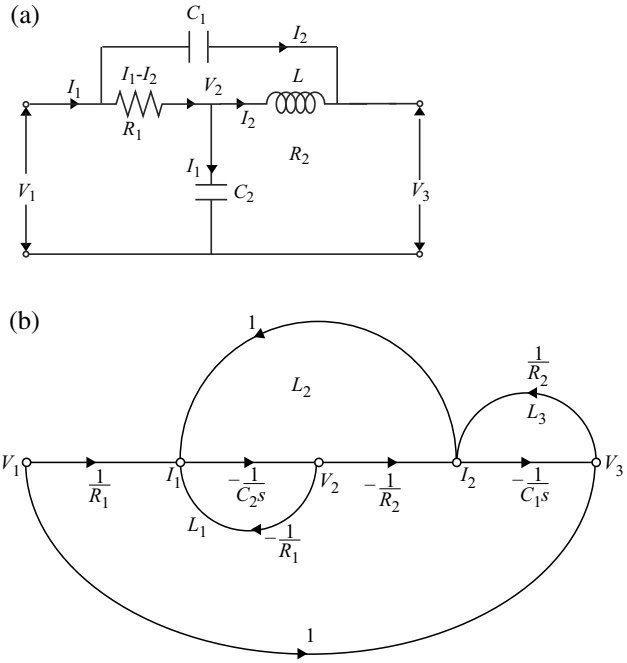


Fig. 1.104 **a** Electrical network for Question 15. **b** SFG for Fig. 1.104a

$$\boxed{\frac{V_3}{V_1} = \frac{[R_1R_2C_1C_2s^2 + (R_1 + R_2)C_1s + 1]}{[R_1R_2C_1C_2s^2 + (R_1C_1 + R_2C_1 + R_1C_2)s + 1]}}$$

Chapter 2
Time Response Analysis



Learning Objectives

After completing this chapter, you should be able to:

- Model first and second order systems.
- Find the time response of first order systems for impulse and step inputs.
- Derive expressions for time domain specifications of first order system.
- Derive the transfer function of second order system in terms of damping ratio and natural frequency of oscillation.
- Classify the second order systems as under damped, critically damped and over damped systems.
- Find the impulse response of a second order system for different damping ratios.
- Find the step response of second order system for different damping ratios.
- Derive expressions for time domain specifications of second order system for step input.
- Find the steady state error using static and dynamic error constants for unity and non-unity feedback systems.
- Find the dynamic behaviour of the system with PID controllers.
- Plot the dynamic response of the system using MATLAB.

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S. Palani, *Automatic Control Systems*,
https://doi.org/10.1007/978-3-030-93445-3_2

2.1 Introduction

When the control system is excited by the input $r(t)$, the output $c(t)$ which is expressed as a function of time t is known as time response of the system. When time t tends to infinity, the output response reaches the steady state. Such a response is called the steady state response. The dynamic behaviour of continuous time system is described by differential equation. The functional relationship between the output and the input of a linear time invariant system is described by transfer function which is defined as the ratio of the Laplace transform of the output variable to the Laplace transform of the input variable with all initial conditions being zero. The system function of linear continuous time system is expressed in terms of poles and zeros. The values of s at which the transfer function becomes infinity are called poles of Linear Time Invariant Continuous system (LTIC). The poles are also known as the factors of the denominator polynomial. Similarly the values of s at which the transfer function becomes zero are called zeros of the transfer function of the systems. They are also the factors of the numerator polynomial of the transfer function. For system performance analysis the transfer function is represented in the form of a block. A complex control system when inter-connected by numerous blocks can be ultimately reduced to a single block with the system required output and the input. This is done by what is known as block diagram reduction technique as described in Chap. 1.

The Laplace transform of the output variable when inverse transformed gives time response of the system. In this chapter, time response of first and second order systems is obtained for impulse and step inputs. Further, expressions for the time domain specifications of these systems are derived in terms of system parameters. The steady state errors for typical inputs such as step, ramp and parabolic are derived in terms of error constants. Expressions for the steady state error are also determined for inputs other than the above inputs. Steady state error expressed as a function of time is derived in terms of generalized error constants. To obtain desired steady state and transient response, PID controllers are introduced and connected in cascade with the plant. For performance evaluation, a prototype second order system is considered. The study includes P, I, D, PI, PD, ID, and PID controllers. The transient response of the system with the above controllers is obtained by simulation study in the digital computer using MATLAB program. Extensive numerical examples are worked out to explain different concepts in the study of transient and steady state responses of the system with and without controllers. The simulation study is carried out in digital computer using MATLAB program.

2.2 First Order Continuous Time System

2.2.1 System Modelling

Consider the following first order differential equation with input $r(t)$ and output $c(t)$.

$$\frac{dc(t)}{dt} + ac(t) = Kr(t) \tag{2.1}$$

Taking Laplace transform on both sides, we get

$$\begin{aligned}(s+a)C(s) &= KR(s)\\ \frac{C(s)}{R(s)} &= G(s)\\ &= \frac{K}{(s+a)}\end{aligned} \tag{2.2}$$

Equation (2.2) gives the transfer function of a first order system. This is represented in block diagram and is shown in Fig. 2.1.

2.2.2 Time Response of First Order System

2.2.2.1 Impulse Response of a First Order System

The impulse input is defined as

$$\delta(t) = \begin{cases}1, & t = 0\\ 0, & \text{otherwise}\end{cases}$$

Here $r(t) = \delta(t)$

$$R(s) = 1$$

From Eq. (2.2), for impulse input the output is written as

$$C(s) = \frac{K}{(s+a)} \tag{2.3}$$

The impulse response of a first order system is obtained by taking inverse Laplace transform. Thus

$$c(t) = Ke^{-at}$$

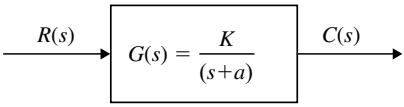


Fig. 2.1 Block diagram representation of first order system

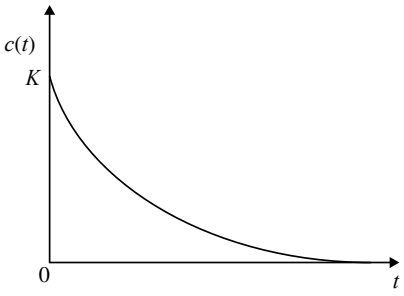


Fig. 2.2 Impulse response of a first order system

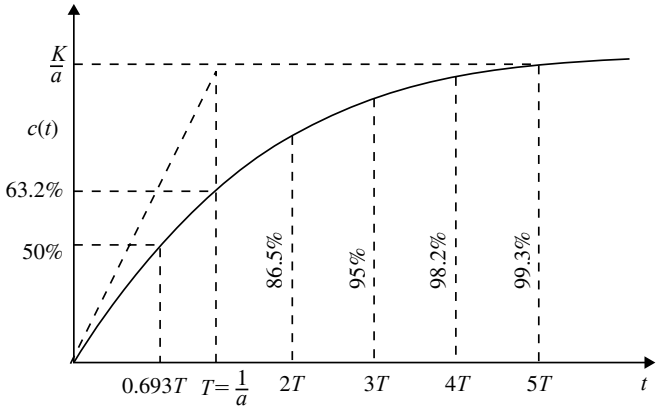


Fig. 2.3 Step response of first order system

The impulse response curve is plotted and is shown in Fig. 2.2.

2.2.2.2 Step Response of First Order System

The unit step input is defined as follows:

$$r(t) = \begin{cases} u(t), \\ 1, & t \geq 0 \\ 0, & t \leq 0 \end{cases}$$

The Laplace transform of unit step input is

$$R(s) = \frac{1}{s}$$

Substituting the above in Eq. (2.2), we get

$$C(s) = \frac{K}{s(s+a)} \tag{2.4}$$

The above equation is put into partial fraction as given below:

$$\begin{aligned} C(s) &= \frac{A_1}{s} + \frac{A_2}{s+a} \\ &= \frac{K}{a}\left[\frac{1}{s} - \frac{1}{s+a}\right] \end{aligned}$$

Taking inverse Laplace transform, we get

$$\boxed{c(t) = \frac{K}{a}[1 - e^{-at}]} \tag{2.5}$$

The transient response curve of Eq. (2.5) is shown in Fig. 2.3.

Note: The time response for ramp input $r(t) = t$ can be obtained by integrating Eq. (2.5) and is given below.

$$\boxed{c(t) = \frac{K}{a^2}\left[-1 + at + e^{-at}\right]} \tag{2.6}$$

2.2.3 Time Domain Specifications

The following time domain specifications are defined for a first order system:

(a) Time constant T.
(b) Rise time t_r.
(c) Time delay t_d.
(d) Settling time t_s.

2.2.3.1 Time Constant

The time constant T is defined as the time taken for the step response to reach 63.2% of its final value for the first time. From Fig. 2.3, the time constant T is obtained as

$$\boxed{T = \frac{1}{a}} \tag{2.7}$$

It is to be noted here that the tangential line of the exponential curve has the slope at $t = 0$ as K. Thus, if the time constant T is small, the response is fast and *vice versa*.

The above equation can be analytically derived as follows. From Eq. (2.5), 6.3.2% output is $0.632(K/a)$ and $t = T$.

$$0.632\frac{K}{a} = \frac{K}{a}[1 - e^{-at}]$$
$$e^{-at} = 0.368$$

Taking $\log_e$ on both sides and putting $t = T$, we get

$$-aT = -1.0$$

$$\boxed{T = \frac{1}{a}}$$

2.2.3.2 Rise Time t_r

The rise time t_r is defined as the time taken for the response to go from 10% to 90% of its final value KT. From Eq. (2.5), at $t = T_1$, let the output be 10% and $t = T_2$ the output be 90%. Thus,

$$0.1\frac{K}{a} = \frac{K}{a}[1 - e^{-aT_1}]$$
$$0.9\frac{K}{a} = \frac{K}{a}[1 - e^{-aT_2}]$$

Dividing the second equation by the first equation, we get

$$9 = \frac{(1 - e^{-aT_2})}{(1 - e^{-aT_1})}$$

Taking $\log_e$ on both sides, we get

$$2.2 = a(T_2 - T_1)$$

Substituting $t_r = (T_2 - T_1)$, the rise time is obtained as

$$\boxed{t_r = 2.2T} \tag{2.8}$$

2.2.3.3 Time Delay t_d

Time delay t_d is defined as the time taken for the response $c(t)$ to reach 50% of its final value for the first time. From Eq. (2.5), for 50% output, we may write the following equation with $t = t_d$.

$$0.5 = 1 - e^{-(t_d/T)}$$
$$e^{-(t_d/T)} = 0.5$$

Taking $\log_e$ on both sides, we get

$$\boxed{t_d = 0.693T} \tag{2.9}$$

2.2.3.4 Settling Time t_s

Settling time t_s is defined as the time taken for the response $c(t)$ to reach and stay within 2% of its final value for the first time.

From Eq. (2.5) for 98% output we may write the following equation with $t = t_s$.

$$0.98 = 1 - e^{\frac{-t_s}{T}}$$

Taking $\log_e$ on both sides, we get

$$\boxed{t_s = 3.91T} \tag{2.10}$$

For 5% error tolerance

$$\boxed{t_s = 3T}$$

For 7% error tolerance

$$\boxed{t_s = 2.66T}$$

It is to be noted that error tolerance may be given as 5, 7%, *etc.*, and the corresponding settling time is determined. Unless otherwise the error tolerance is specified it is always taken as 2% error.

Summary of Time Domain Specifications of First Order System

1	Time constant	$T = \frac{1}{a}$
2	Rise time	$t_r = 2.2T$
3	Time delay	$t_d = 0.693T$
4	Settling time	$t_s = 3.91T$

Example 2.1

Consider the following T.F. of a certain first order system:

$$G(s) = \frac{10}{(s+10)}$$

Derive an expression for the response of the system for $r(t) = 5u(t)$. Find the time constant, settling time, time delay and rise time.

Solution

$$\frac{C(s)}{R(s)} = G(s) = \frac{10}{(s+10)}$$

$$R(s) = \frac{5}{s}$$

$$C(s) = \frac{50}{s(s+10)}$$

$$= 5\left[\frac{1}{s} - \frac{1}{s+10}\right]$$

Taking inverse Laplace transform, we get

$$\boxed{c(t) = 5[1 - e^{-10t}]}$$

$$\text{Time constant } T = \frac{1}{10} = 0.1\,\text{s}$$

$$\text{Settling time } t_s = 3.91T = 0.391\,\text{s}$$

$$\text{Rise time } t_r = 2.2T = 0.22\,\text{s}$$

$$\text{Time delay } t_d = 0.693T = 0.0693\,\text{s}$$

$$\boxed{\begin{aligned} T &= 0.1\,\text{s} \\ t_s &= 0.391\,\text{s} \\ t_r &= 0.22\,\text{s} \\ t_d &= 0.0693\,\text{s} \end{aligned}}$$

Example 2.2

A glass bulb thermometer reads 98.2% of its final value of temperature 1 min after immersing it in hot water. Determine the time constant, rise time, time delay and settling time for 5% error tolerance.

Solution The mercury thermometer is a first order system. From Eq. (2.5), for 98.2% output, the time taken is 60 s.

$$\begin{aligned} \frac{K}{a}\left[1 - e^{\frac{-60}{T}}\right] &= 0.982\frac{K}{a} \\ 4T &= 60\,\text{s} \\ T &= 15\,\text{s} \end{aligned}$$

Rise time

$$\begin{aligned} t_r &= 2.2T \\ &= 2.2 \times 15 \\ &= 33\,\text{s} \end{aligned}$$

Time delay

$$\begin{aligned} t_d &= 0.693T \\ &= 0.693 \times 15 \\ &= 10.4\,\text{s} \end{aligned}$$

For 5% error tolerance, the settling time is

$$\begin{aligned} t_s &= 3T \\ &= 3 \times 15 \\ &= 45\,\text{s} \end{aligned}$$

$$\boxed{\begin{aligned} T &= 15\,\text{s} \\ t_r &= 33\,\text{s} \\ t_d &= 10.4\,\text{s} \\ t_s &= 45\,\text{s} \end{aligned}}$$

2.3 Second Order System Modelling

For model development of a general second order system, consider the electrical motor represented in block diagram form as shown in Fig. 2.4a. The system parameters are as follows:

$$\begin{aligned} J &= \text{Moment of inertia of motor in Kg.m}^2 \\ B &= \text{Motor frictional coefficient in N-m/rad/s} \\ K &= \text{Error detector constant in N-m/rad error} \end{aligned}$$

Torque developed by the motor is

$$\begin{aligned} T_d(t) &= Ke(t) \\ &= K(r(t) - c(t)) \end{aligned}$$

This torque is to overcome the torque opposed by J and B under no load condition. Thus

$$\begin{aligned} T_u(t) &= J\frac{d^2c(t)}{dt^2} + B\frac{dc(t)}{dt} \\ T_u(t) &= T_d(t) \\ J\frac{d^2c(t)}{dt^2} + B\frac{dc(t)}{dt} &= K(r(t) - c(t)) \end{aligned} \tag{2.11}$$

Taking Laplace transform on both sides and expressing the ratio of the output variable to the input variable we get the following equation:

$$\frac{C(s)}{R(s)} = \frac{K}{(Js^2 + Bs + K)} \tag{2.12}$$

The denominator of Eq. (2.12) is a second degree polynomial in s and therefore Eq. (2.12) describes the dynamics of a second order system. Equation (2.12) can be written as

$$\frac{C(s)}{R(s)} = \frac{K/J}{(s^2 + \frac{B}{J}s + \frac{K}{J})} \tag{2.13}$$

Equation (2.13) can be written in a generalized form as given below:

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{(s^2 + 2\zeta\omega_n s + \omega_n^2)} \tag{2.14}$$

where

$$\omega_n = \sqrt{\frac{K}{J}} = \text{Natural frequency of oscillation in rad/s.}$$
$$\zeta = \frac{B}{2\sqrt{KJ}} = \text{Damping ratio and } \zeta\omega_n = \text{Damping factor}$$

Equation (2.14) is called prototype second order system equation. Thus, natural frequency of oscillation and damping ratio are the two parameters of a generalized second order system. The systems may be electrical, mechanical, thermal, hydraulic, biological or in any form. Equation (2.14) is represented in block diagram form as shown in Figs. 2.4b and c which are in the form of open loop and closed loop forms respectively. Equation (2.14) is also called standard equation for a second order system.

The natural frequency ω_n is defined as the frequency of oscillation of a second order system without damping

If the damping is provided to the system, the system time response contains damped oscillations with exponential decay. Now consider Eq. (2.14) which can be written as follows:

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{(s + a_1)(s + a_2)} \tag{2.15}$$

where a_1 and a_2 are the pole locations of Eq. (2.14) and they are expressed in terms of ζ and ω_n. For $0 \leq \zeta \leq 1$.

$$a_1 = -\zeta\omega_n + j\omega_n\sqrt{1 - \zeta^2}$$
$$a_2 = -\zeta\omega_n - j\omega_n\sqrt{1 - \zeta^2} \tag{2.16}$$

In the above case the system is said to be under damped and the pole locations are as shown in Fig. 2.5.

From Fig. 2.5, $\omega_d = \omega_n\sqrt{1 - \zeta^2}$ is called damped frequency of oscillation. For $\zeta = 1$, the system is said to be critically damped and the pole locations are shown in Fig. 2.6. which are repeated poles at $s = -\omega_n$.

If $\zeta > 1$, the system is said to be over damped. In this case the poles are at,

$$a_1 = -\zeta\omega_n + \omega_n\sqrt{\zeta^2 - 1}$$
$$a_2 = -\zeta\omega_n - \omega_n\sqrt{\zeta^2 - 1} \tag{2.17}$$

The pole locations are shown in Fig. 2.7.

2.4 Time Response of a Second Order System

2.4.1 Impulse Response

Consider Equation (2.14),

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{(s^2 + 2\zeta\omega_n s + \omega_n^2)}$$

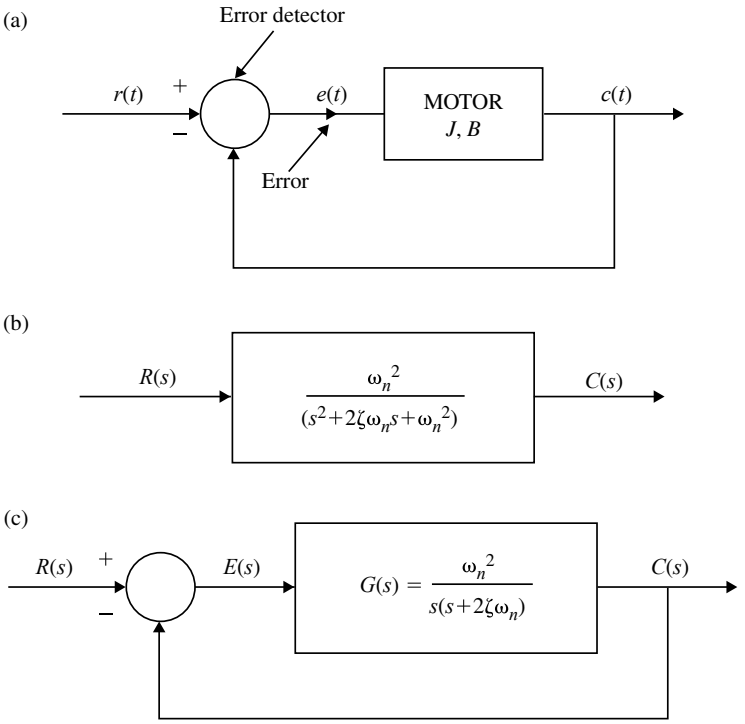


Fig. 2.4 **a** Block diagram representation of a second order systems. Second order system representation. **b** open loop form, **c** closed loop form

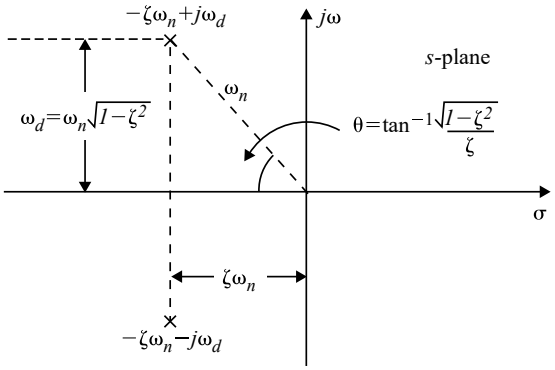


Fig. 2.5 Pole location in the s-plane for complex conjugate poles for $0 \leq \zeta \leq 1$ (under damped)

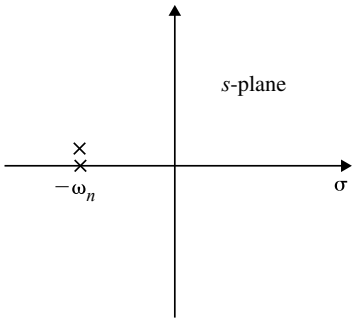


Fig. 2.6 Pole location of a critically damped system ($\zeta = 1$)

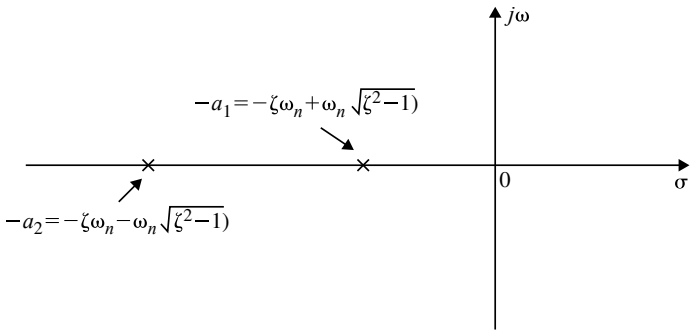


Fig. 2.7 Pole location of an over damped system ($\zeta > 1$)

For an impulse input,

$$R(s) = 1$$

The above equation is written as

$$C(s) = \frac{\omega_n^2}{(s^2 + 2\zeta\omega_n s + \omega_n^2)} \tag{2.18}$$

The impulse response for the following cases are determined:

1. Under damped case ($\zeta < 1$).
2. Over damped case ($\zeta > 1$).
3. Critically damped case ($\zeta = 1$).

2.4.1.1 Under Damped Case ($\zeta < 1$)

For $\zeta < 1$, the second degree denominator polynomial of Eq. (2.18) is written as follows:

$$C(s) = \frac{\omega_n^2}{(s + \zeta\omega_n + j\omega_d)(s + \zeta\omega_n - j\omega_d)}$$

where

$$\omega_d = \omega_n\sqrt{1 - \zeta^2} = \text{Damped frequency of oscillation}$$

Putting the above equation into partial fraction we get,

$$\begin{aligned} C(s) &= \frac{A_1}{s + \zeta\omega_n + j\omega_d} + \frac{A_2}{s + \zeta\omega_n - j\omega_d} \\ &= \frac{A_1(s + \zeta\omega_n - j\omega_d) + A_2(s + \zeta\omega_n + j\omega_d)}{(s^2 + 2\zeta\omega_n s + \omega_n^2)} \\ \omega_n^2 &= A_1(s + \zeta\omega_n - j\omega_d) + A_2(s + \zeta\omega_n + j\omega_d) \end{aligned}$$

Put

$$\begin{aligned} s &= -\zeta\omega_n + j\omega_d \\ \omega_n^2 &= A_2(2j\omega_d) \\ A_2 &= \frac{\omega_n}{j2\sqrt{1 - \zeta^2}} \\ A_1 &= \text{Conjugate of } A_2 \\ &= \frac{-\omega_n}{j2\sqrt{1 - \zeta^2}} \\ C(s) &= \frac{\omega_n}{j2\sqrt{1 - \zeta^2}}\left[\frac{-1}{(s + \zeta\omega_n + j\omega_d)} + \frac{1}{(s + \zeta\omega_n - j\omega_d)}\right] \end{aligned}$$

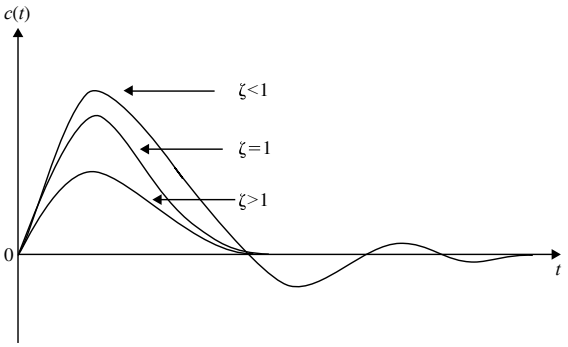


Fig. 2.8 Impulse response curves of a second order system

Taking inverse Laplace transform we get,

$$c(t) = \frac{\omega_n}{j2\sqrt{1-\zeta^2}}\left[-e^{-(\zeta\omega_n + j\omega_d)t} + e^{-(\zeta\omega_n - j\omega_d)t}\right]$$
$$= \frac{\omega_n}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t}\left[\frac{e^{j\omega_d t} - e^{-j\omega_d t}}{2j}\right]$$

$$\boxed{c(t) = \frac{\omega_n}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin \omega_n \sqrt{1-\zeta^2}\, t} \tag{2.19}$$

2.4.1.2 Critically Damped Case ($\zeta = 1$)

Equation (2.18) for $\zeta = 1$ is written as

$$C(s) = \frac{\omega_n^2}{s^2 + 2\omega_n s + \omega_n^2}$$
$$= \frac{\omega_n^2}{(s+\omega_n)^2}$$

Taking inverse Laplace transform we get,

$$\boxed{c(t) = \omega_n^2 t e^{-\omega_n t}} \tag{2.20}$$

2.4.1.3 Over Damped Case ($\zeta > 1$)

Equation (2.18) for $\zeta > 1$ is written as

$$C(s) = \frac{\omega_n^2}{(s + \zeta\omega_n + \omega_n\sqrt{\zeta^2 - 1})(s + \zeta\omega_n - \omega_n\sqrt{\zeta^2 - 1})}$$
$$= \frac{A_1}{(s + \zeta\omega_n + \omega_n\sqrt{\zeta^2 - 1})} + \frac{A_2}{(s + \zeta\omega_n - \omega_n\sqrt{\zeta^2 - 1})}$$
$$\omega_n^2 = A_1(s + \zeta\omega_n - \omega_n\sqrt{\zeta^2 - 1}) + A_2(s + \zeta\omega_n + \omega_n\sqrt{\zeta^2 - 1})$$

Put $s = -\zeta\omega_n - \omega_n\sqrt{\zeta^2 - 1}$

$$\omega_n^2 = A_1(-2\omega_n\sqrt{\zeta^2 - 1})$$
$$A_1 = \frac{-\omega_n}{2\sqrt{\zeta^2 - 1}}$$

Put $s = -\zeta\omega_n + \omega_n\sqrt{\zeta^2 - 1}$

$$\omega_n^2 = A_2(2\omega_n\sqrt{\zeta^2 - 1})$$
$$A_2 = \frac{\omega_n}{2\sqrt{\zeta^2 - 1}}$$

$$C(s) = \frac{\omega_n}{2\sqrt{\zeta^2 - 1}}\left[\frac{-1}{(s + \zeta\omega_n + \omega_n\sqrt{\zeta^2 - 1})} + \frac{1}{(s + \zeta\omega_n - \omega_n\sqrt{\zeta^2 - 1})}\right]$$

Taking inverse Laplace transform we get,

$$\boxed{c(t) = \frac{\omega_n}{2\sqrt{\zeta^2 - 1}} e^{-\zeta\omega_n t}\left[-e^{-\omega_n\sqrt{\zeta^2-1}t} + e^{\omega_n\sqrt{\zeta^2-1}t}\right]} \tag{2.21}$$

The time response curves of Eqs. (2.19), (2.20) and (2.21) are shown in Fig. 2.8.

2.4.1.4 Importance of Impulse Response

1. If the area under the impulse response curve is finite, then the system is said to be Bounded Input, Bounded Output (BIBO) stable.
2. From impulse response, by taking inverse Laplace transform the system transfer function is obtained.
3. If impulse response is known, step response can be obtained by integrating it.

2.4.1.5 Importance of Step Response

Step response of a system is important for the following reasons:

1. It is easy to generate step signal and test the system in the laboratory.
2. The step signal is sufficiently drastic and if satisfactory step response is obtained, then the system is likely to give satisfactory performance for other types of inputs.
3. From step response impulse response can be obtained by differentiating it and useful information may be derived. Similarly from step response, ramp response can be obtained by integrating it.
4. The application of step input is equivalent to the application of numerous sinusoidal signals with a wide range of frequencies.

2.4.2 Step Response of a Second Order System

Consider Eq. (2.14).

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{(s^2 + 2\zeta\omega_n s + \omega_n^2)}$$

For a step input of height R, the Laplace transform

$$R(s) = \frac{R}{s}$$

Substituting this in the above equation we get,

$$C(s) = \frac{R\omega_n^2}{s(s^2 + 2\zeta\omega_n s + \omega_n^2)} \tag{2.22}$$

2.4.2.1 Under Damped Response ($\zeta < 1$)

For $\zeta < 1$, Eq. (2.22) is written in the following form:

$$C(s) = \frac{R\omega_n^2}{s(s + \zeta\omega_n + j\omega_d)(s + \zeta\omega_n - j\omega_d)}$$

$$C(s) = \frac{A_1}{s} + \frac{A_2}{(s + \zeta\omega_n + j\omega_d)} + \frac{A_3}{(s + \zeta\omega_n - j\omega_d)} \tag{2.23}$$

Analytical Method of Determining the Residues A_1, A_2 and A_3

$$R\omega_n^2 = A_1(s^2 + 2\zeta\omega_n s + \omega_n^2) + A_2 s(s + \zeta\omega_n - j\omega_d) + A_3 s(s + \zeta\omega_n + j\omega_d)$$

Putting $s = 0$ in the above equation we get,

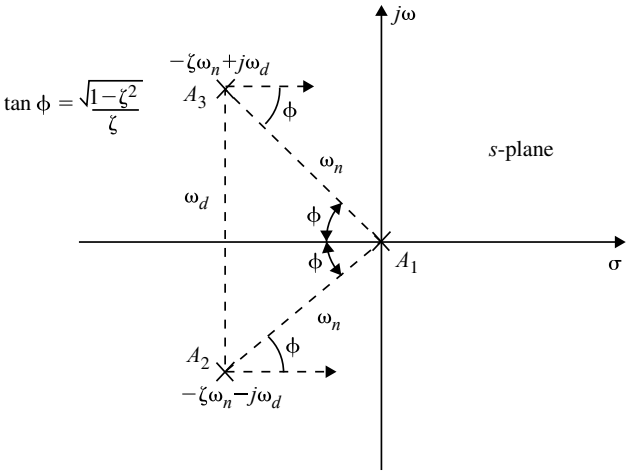


Fig. 2.9 Poles and residue locations of Eq. (2.23)

$$R\omega_n^2 = A_1\omega_n^2$$
$$A_1 = R$$

Putting $s = (-\zeta\omega_n - j\omega_d)$ we get,

$$R\omega_n^2 = A_2(-\zeta\omega_n - j\omega_d)(-j\omega_d - j\omega_d)$$
$$A_2 = \frac{R}{j2\sqrt{1-\zeta^2}(\zeta + j\sqrt{1-\zeta^2})}$$
$$= \frac{R\angle -\phi - \frac{\pi}{2}}{2\sqrt{1-\zeta^2}}$$

where $\tan\phi = \frac{\sqrt{1-\zeta^2}}{\zeta}$

$$A_3 = A_2^* = \frac{R\angle\phi + \frac{\pi}{2}}{2\sqrt{1-\zeta^2}}$$

Graphical Method of Determining the Residues

The residues of Eq. (2.23) can also be determined as explained below. The poles are located along with their residues as shown in Fig. 2.9.

$$A_1 = \frac{\text{Constant} \times \text{Product of directed vector distances from all zeros to point } A_1}{\text{Product of directed vector distances from all poles to point } A_1}$$

$$A_1 = \frac{R\omega_n^2}{\omega_n \angle\phi \; \omega_n \angle -\phi} = R$$

$$A_3 = \frac{R\omega_n^2}{\omega_n \angle\pi - \phi \; 2\omega_d \angle\frac{\pi}{2}} = \frac{R\angle\frac{\pi}{2} + \phi}{2\sqrt{1-\zeta^2}}$$

$$A_2 = A_3^* = \frac{R\angle -(\frac{\pi}{2} + \phi)}{2\sqrt{1-\zeta^2}}$$

The residues determined by analytical method are the same as obtained by graphical method. However, the graphical method is simpler and quicker. Substituting the above residues in Eq. (2.23) we get,

$$C(s) = R\left[1 + \frac{1}{2\sqrt{1-\zeta^2}}\left\{\frac{e^{-j(\frac{\pi}{2}+\phi)}}{(s+\zeta\omega_n + j\omega_d)} + \frac{e^{j(\frac{\pi}{2}+\phi)}}{(s+\zeta\omega_n - j\omega_d)}\right\}\right]$$

Taking inverse Laplace transform we get,

$$\begin{aligned} c(t) &= R\left[1 + \frac{e^{-\zeta\omega_n t}}{2\sqrt{1-\zeta^2}}\left\{e^{-j(\frac{\pi}{2}+\phi+\omega_d t)} + e^{j(\frac{\pi}{2}+\phi+\omega_d t)}\right\}\right] \\ &= R\left[1 + \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}\cos\left(\frac{\pi}{2} + \phi + \omega_d t\right)\right] \end{aligned}$$

$$\boxed{c(t) = R\left[1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}\sin(\omega_n\sqrt{1-\zeta^2}t + \phi)\right]} \tag{2.24}$$

where ϕ is in radians and ω_n = rad/s.

2.4.2.2 Critically Damped Response ($\zeta = 1$)

For $\zeta = 1$,

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = (s+\omega_n)^2$$

Equation (2.14) can be written as

$$C(s) = \frac{R}{s(s+\omega_n)^2}$$
$$= \frac{A_1}{s} + \frac{A_2}{(s+\omega_n)^2} + \frac{A_3}{(s+\omega_n)}$$
$$R = A_1(s+\omega_n)^2 + A_2 s + A_3 s(s+\omega_n)$$

Put $s = 0$

$$A_1 = \frac{R}{\omega_n^2}$$

Put $s = -\omega_n$

$$A_2 = \frac{-R}{\omega_n}$$

Equating the coefficients of s^2 terms we get,

$$0 = A_1 + A_3$$
$$A_3 = \frac{-R}{\omega_n^2}$$
$$C(s) = \frac{R}{\omega_n^2}\left[\frac{1}{s} - \frac{\omega_n}{(s+\omega_n)^2} - \frac{1}{s+\omega_n}\right]$$

Taking inverse Laplace transform we get,

$$\boxed{c(t) = \frac{R}{\omega_n^2}\left[1 - t\omega_n e^{-\omega_n t} - e^{-\omega_n t}\right]} \tag{2.25}$$

2.4.2.3 Over Damped Response ($\zeta > 1$)

For over damped case the time response of a second system for step input can be derived following the method described above. The time response is given below:

$$\boxed{c(t) = R\left[1 + \frac{e^{-\omega_n(\zeta+\sqrt{\zeta^2-1})t}}{2\{\zeta^2 - 1 + \zeta\sqrt{\zeta^2-1}\}} + \frac{e^{-\omega_n(\zeta-\sqrt{\zeta^2-1})t}}{2\{\zeta^2 - 1 - \zeta\sqrt{\zeta^2-1}\}}\right]} \tag{2.26}$$

Using Eqs. (2.24), (2.25) and (2.26), the transient response curves are plotted as shown in Fig. 2.10 for $R = 1$.

The poles locations in the s-plane and the transient response curves are shown in Fig. 2.11a. The effect of roots on damping ratio is shown in Fig. 2.11b.

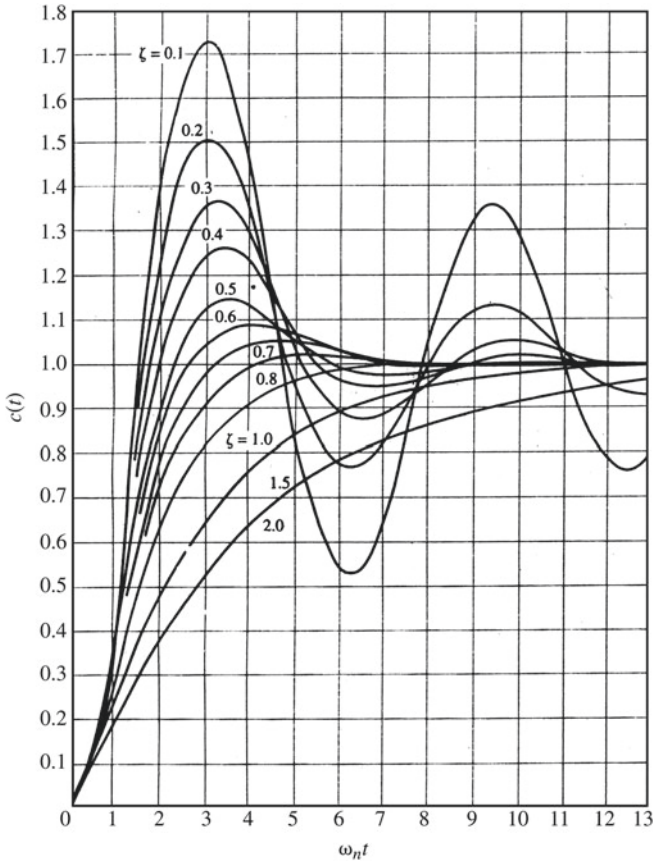


Fig. 2.10 Transient response curves of a second order system

2.4.3 Time Domain Specifications of a Second Order System

The performance of a second order system is measured by the following specifications:

1. Peak overshoot M_p and % peak overshoot $\%\ M_p$.
2. Time at which the peak overshoot occurs peak time t_p.
3. Time constant T.
4. Rise time t_r.
5. Settling time t_s.
6. Time delay t_d.

Expressions for the above specifications are derived in terms of the second order system parameters ζ and ω_n. The transient response curve shown in Fig. 2.12 is followed.

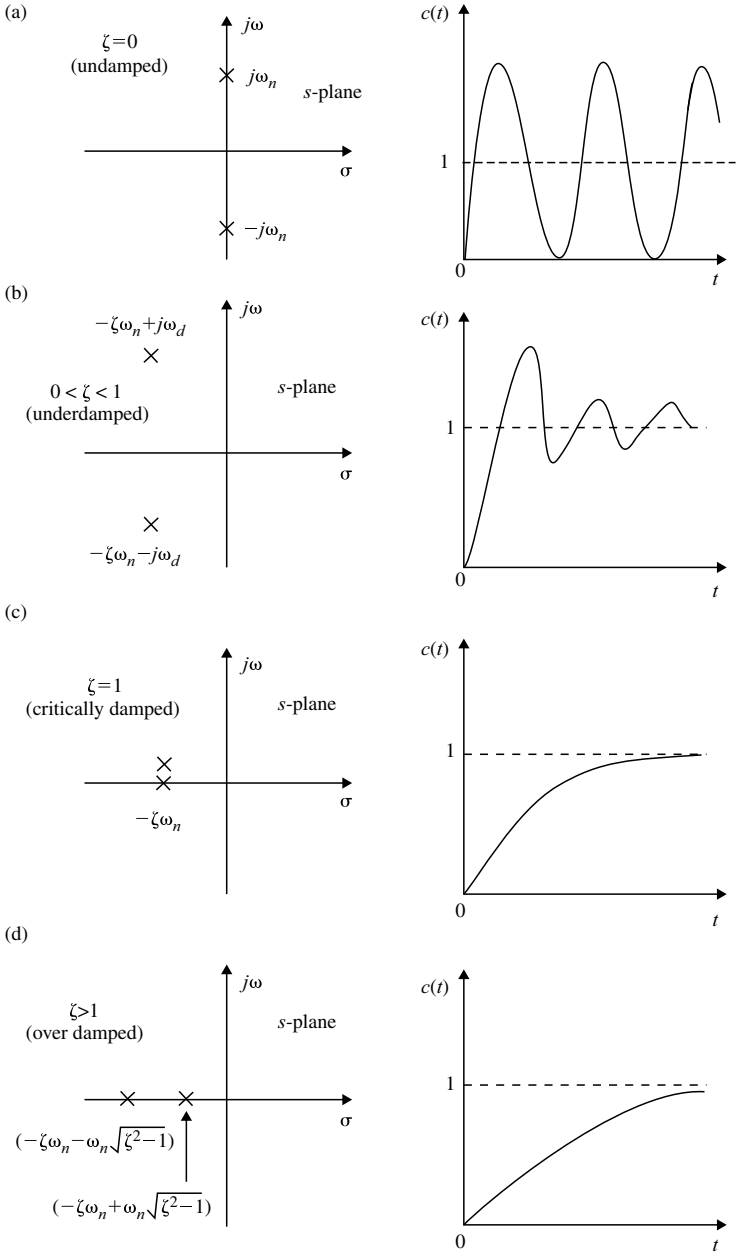


Fig. 2.11 **a** Pole locations and transient response of a second order system. **b** Effect of roots on damping ratio

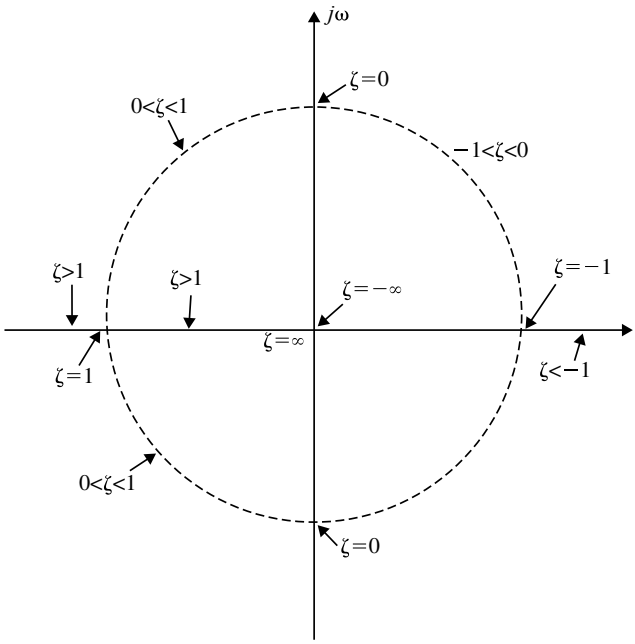


Fig. 2.11 (continued)

2.4.3.1 Peak Over Shoot M_p

Peak overshoot is defined as the amount by which the transient response waveform overshoots the steady value or the final value.

The peak overshoot is denoted by M_P. It is expressed as

$$M_P = C_{\max} - C_{\text{final}}$$

The percentage overshoot is expressed as

$$\%\, M_P = \frac{(C_{\max} - C_{\text{final}})}{C_{\text{final}}} \times 100$$

Peak overshoot can be expressed in terms of the system parameters ζ and ω_n. This occurs for the under damped system response. Consider Eq. (2.24).

$$c(t) = R\left[1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}} \sin(\omega_n\sqrt{1-\zeta^2}t + \phi)\right]$$

$C_{\max}$ is obtained by differentiating $c(t)$ with respect to t which gives t_p, the time at which the maxima occurs and substituting in Eq. (2.24).

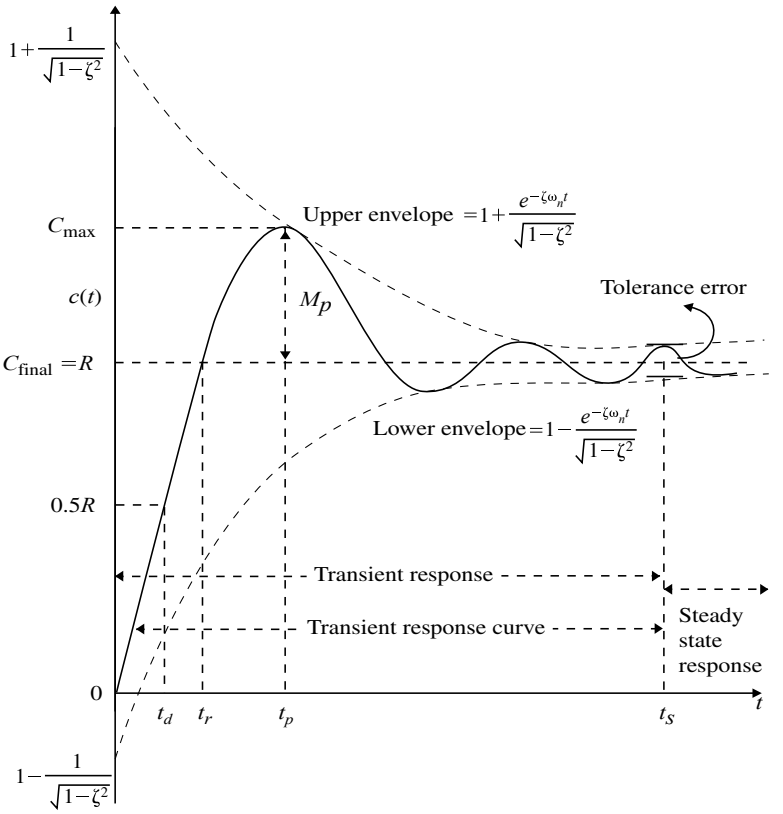


Fig. 2.12 Time domain specifications of a second order system

$$\frac{dc(t)}{dt} = R\left[0 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}\omega_n\sqrt{1-\zeta^2} \times \cos(\omega_n\sqrt{1-\zeta^2}t+\phi) + \frac{e^{-\zeta\omega_n t}(\zeta\omega_n)\sin(\omega_n\sqrt{1-\zeta^2}t+\phi)}{\sqrt{1-\zeta^2}}\right] = 0$$

$$\sqrt{1-\zeta^2}\cos(\omega_n\sqrt{1-\zeta^2}t+\phi) = \zeta\sin(\omega_n\sqrt{1-\zeta^2}t+\phi)$$

$$\tan\phi = \tan(\omega_n\sqrt{1-\zeta^2}t+\phi)$$

$$\omega_n\sqrt{1-\zeta^2}t = n\pi \quad \text{where } n = 0, 1, 2, \ldots$$

For $n = 1$ maxima occurs and $t = t_p$

$$\boxed{t_p = \frac{\pi}{\omega_d}}$$

Substituting $t = t_p$ and $c(t) = C_{\max}$ in Eq. (2.24) we get

$$C_{\max} = R\left[1 - \frac{e^{\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}}}{\sqrt{1-\zeta^2}}\sin(\pi + \phi)\right]$$
$$= R\left[1 + e^{\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}}\right] \quad \left[\because \sin\phi = \sqrt{1-\zeta^2}\right]$$

$$M_P = C_{\max} - C_{\text{final}}$$
$$= R + Re^{\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}} - R$$

$$\boxed{M_P = Re^{\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}}} \tag{2.27}$$

$$\%\, M_P = \frac{M_P}{R} \times 100$$

$$\boxed{\%\, M_P = e^{\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}} \times 100} \tag{2.28}$$

2.4.3.2 Peak Time t_p

The peak time t_p is defined as the time at which the first maximum of transient response waveform occurs. It is expressed as

$$\boxed{t_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}}} \tag{2.29}$$

2.4.3.3 Settling Time t_s

The settling time t_s is defined as the time required for the transient response to reach and stay within the prescribed percentage error. The expression for the settling time of a second order system is derived as follows.

Let

$$\pm e^{-m} \times 100 \quad \text{where } m = 1, 2, 3 \ldots \tag{2.30}$$

be the prescribed error within which the system transient response settles down. Equation (2.28) represents the error and can be written as follows:

$$\%\, M_P = e^{\frac{-\zeta \pi n}{\sqrt{1-\zeta^2}}} \times 100 \tag{2.31}$$

where $n = 0, 1, 2, \ldots$. For odd values of n overshoots occur and for even values of n under-shoots occur. For any error, Eqs. (2.30) and (2.31) are same. Thus,

$$\begin{aligned} e^{-m} &= e^{-\frac{\zeta \pi n}{\sqrt{1-\zeta^2}}} \\ n &= \frac{m\sqrt{1-\zeta^2}}{\zeta \pi} \end{aligned} \tag{2.32}$$

From Eq. (2.29), for any peak value of the response t_p can be written as follows:

$$t_p = \frac{n\pi}{\omega_n \sqrt{1-\zeta^2}} \tag{2.33}$$

For the given error e^{-m}, $t_p = t_s$. Substituting this in Eq. (2.33) and also for n from Eq. (2.32), we get the following expression for the settling time:

$$\begin{aligned} t_s &= \frac{m\sqrt{1-\zeta^2}\pi}{\zeta \pi \omega_n \sqrt{1-\zeta^2}} \\ t_s &= \frac{m}{\zeta \omega_n} \end{aligned}$$

$$\boxed{t_s = mT} \tag{2.34}$$

In Eq. (2.34), $T = \frac{1}{\zeta \omega_n}$ is called the time constant of the system.

Normally the permissible error prescribed for the settling time is 2%. In this case

$$\begin{aligned} e^{-m} &= 0.02 \\ \log_e e^{-m} &= \log_e(0.02) \\ &= -3.91 \\ -m &= -3.91 \end{aligned}$$

$$\boxed{\begin{aligned} m &= 3.91 \\ t_s &= 3.91T \end{aligned}} \tag{2.35}$$

For 5% error, $m = 3$ and

$$\boxed{t_s = 3T}$$

For 7% error, $m = 2.66$

$$\boxed{t_s = 2.66T}$$

2.4.3.4 Time Constant *T*

Time constant is defined as the time taken for the transient response to reach 63.2% of its final value for the first time. From Eq. (2.24) and Fig. 2.12 either from the upper envelope $1 + e^{-\zeta\omega_n t}$ or from lower envelope $1 - e^{-\zeta\omega_n t}$ can be obtained as

$$\boxed{T = \frac{1}{\zeta\omega_n}} \tag{2.36}$$

2.4.3.5 Rise Time t_r

Rise time t_r is defined as the time taken for the transient response to go from 10% to 90% of the final value. Sometimes, the rise time is also defined as the time taken for the transient response to reach the final value for the first time. The expression for the rise time is derived as follows using the second definition stated above. For $c(t) = R$, Eq. (2.24) is written as follows:

$$R = R\left[1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}\sin(\omega_d t + \phi)\right] \tag{2.37}$$

Equation (2.37) gives the solution as

$$\begin{aligned} \sin(\omega_d t + \phi) &= 0 \\ &= \sin n\pi \quad \text{where } n = 0, 1, 2 \ldots \\ \omega_d t + \phi &= n\pi = \pi \quad \text{for } n = 1 \end{aligned}$$

Substituting $t = t_r$ in the above equation we get

$$t_r = \frac{(\pi - \phi)}{\omega_n \sqrt{1 - \zeta^2}} \tag{2.38}$$

In Eq. (2.38) it is to be noted that ϕ is in radians.

2.4.3.6 Time Delay t_d

Time delay is defined as the time taken for the transient response to reach 50% of the final value for the first time. The expression for the time delay is given by the following empirical formula:

$$t_d = \frac{(1 + 0.7\zeta)}{\omega_n} \tag{2.39}$$

2.4.3.7 The Period of Oscillation T_0

The period of oscillation of the transient response curve of a second order system is defined as the time taken for the response to complete one cycle of the damped oscillation and it is given as

$$T_0 = \frac{2\pi}{\omega_d} \tag{2.40}$$

2.4.3.8 The Number of Oscillations Before Settling Time is Reached T_{0s}

The number of oscillations before settling time is reached is given by

$$T_{0s} = \frac{t_s}{T_0} \tag{2.41}$$

Summary of Time Domain Specifications of Second Order System

1	Time constant	$T = \frac{1}{\zeta \omega_n}$
2	Rise time	$t_r = \frac{(\pi - \phi)}{\omega_n \sqrt{1-\zeta^2}}$
3	Time delay	$t_d = \frac{(1+0.7\zeta)}{\omega_n}$
4	Settling time	$t_s = mT$
5	% Peak overshoot	$\%\ M_p = e^{\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}} \times 100$
6	Time at which the peak overshoot occurs	$t_p = \frac{\pi}{\omega_d}$
7	The period of oscillations	$t_0 = \frac{2\pi}{\omega_d}$
8	The number of oscillations before settling	$T_{os} = \frac{t_s}{T_0}$
	Time is reached	

Example 2.3

Obtain the impulse and step responses of the following unity feedback control system with open loop transfer function

$$G(s) = \frac{6}{s(s+5)}$$

(*Anna University, December, 2009*)

Solution The given system is represented in block diagram form as shown in Fig. 2.13. From Fig. 2.13, the following equation is derived.

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)}$$
$$= \frac{6}{s^2+5s+6}$$
$$\frac{C(s)}{R(s)} = \frac{6}{(s+2)(s+3)}$$

For unit step input $R(s) = \frac{1}{s}$.

$$C(s) = \frac{6}{s(s+2)(s+3)}$$
$$= \frac{A_1}{s} + \frac{A_2}{s+2} + \frac{A_3}{s+3}$$
$$6 = A_1(s+2)(s+3) + A_2 s(s+3) + A_3 s(s+2)$$

For $s = 0$

$$A_1 = 1$$

For $s = -2$

$$A_2 = -3$$

For $s = -3$

$$A_3 = +2$$

$$C(s) = \frac{1}{s} - \frac{3}{s+2} + \frac{2}{s+3}$$

Taking inverse Laplace transform we get

$$\boxed{c(t) = 1 - 3e^{-2t} + 2e^{-3t}}$$

The impulse response is obtained by differentiating the step response. If we denote impulse response as $h(t)$, then

$$h(t) = \frac{dc(t)}{dt}$$

$$\boxed{h(t) = 6[e^{-2t} - e^{-3t}]}$$

Alternatively, the impulse response is obtained from first principle.

$$\frac{C(s)}{R(s)} = \frac{6}{(s+2)(s+3)}$$

For an impulse $R(s) = 1$

$$C(s) = H(s) = \frac{6}{(s+2)(s+3)}$$
$$= 6\left[\frac{1}{s+2} - \frac{1}{s+3}\right]$$

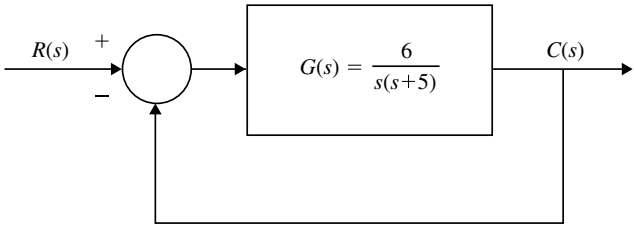


Fig. 2.13 System block diagram for Example 2.3

$$\boxed{h(t) = 6[e^{-2t} - e^{-3t}]}$$

It is to be noted that the poles of the closed loop transfer function lie on the −ve real axis of the s-plane and hence the system is over-damped.

Example 2.4

A certain unity negative feedback control system has an open loop transfer function.

$$G(s) = \frac{10}{s(s+2)}$$

Find the rise time, percentage overshoot, peak time, t_d, T_0, T_{os} and settling time for a step input of 12 units.

(*Anna University, December, 2009*)

Solution

$$G(s) = \frac{10}{s(s+2)}$$

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)} = \frac{10}{s^2+2s+10}$$

Comparing the above equation with the following standard second order equation.

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$$

$$\omega_n = \sqrt{10}$$

$$2\zeta\omega_n = 2$$

$$\zeta = 0.3162$$

Given $R = 12$ units. Using Eq. (2.27),

$$\text{overshoot } M_p = \text{Re}\frac{-\zeta\pi}{\sqrt{1-\zeta^2}} = 12e^{\frac{-0.3162\pi}{\sqrt{1-0.3162^2}}}$$

$$\boxed{M_p = 4.2 \text{ units}}$$

$$\% \text{ overshoot} = \frac{M_p}{R} \times 100 = \frac{4.2}{12} \times 100$$

$$\boxed{\% M_p = 35\%}$$

Using Eq. (2.29), the peak time is obtained.

$$t_p = \frac{\pi}{\omega_n \sqrt{1-\zeta^2}} = \frac{\pi}{\sqrt{10}\sqrt{(1-0.3162^2)}}$$

$$\boxed{t_p = 1.05 \text{ s}}$$

The rise time is obtained using Eq. (2.38)

$$\begin{aligned} t_r &= \frac{(\pi - \phi)}{\omega_n \sqrt{1-\zeta^2}} \\ \omega_n \sqrt{1-\zeta^2} &= \sqrt{10}\sqrt{1-0.3162^2} = 3 \\ \cos\phi &= \zeta \\ \phi &= \cos^{-1} 0.3162 \\ &= 1.25 \text{ radians} \\ t_r &= \frac{(\pi - 1.25)}{3} \end{aligned}$$

$$\boxed{t_r = 0.63 \text{ s}}$$

The settling time is obtained using Eq. (2.34). Here 2% error tolerance is taken. Hence, $m = 3.91$.

$$\text{Time constant, } T = \frac{1}{\zeta \omega_n} = \frac{1}{0.3162\sqrt{10}}$$

$$\boxed{T = 1 \text{ s}}$$

$$t_s = mT$$

$$\boxed{t_s = 3.91\ \text{s}}$$

The time delay is obtained using Eq. (2.39)

$$t_d = \frac{1 + 0.7\zeta}{\omega_n} = \frac{1 + 0.7 \times 0.3162}{\sqrt{10}} = 0.3862$$

$$\boxed{t_d = 0.3862}$$

The period of oscillation is obtained using Eq. (2.40)

$$T_0 = \frac{2\pi}{\omega_n\sqrt{1-\zeta^2}} = \frac{2\pi}{\sqrt{10}\sqrt{1-(0.3162)^2}} = 2.0943$$

$$\boxed{T_0 = 2.0943}$$

The number of oscillations before the settling time is reached is obtained using Eq. (2.41)

$$T_{0s} = \frac{t_s}{T_0} = \frac{3.91}{2.0943} = 1.867$$

$$\boxed{T_{0s} = 1.867}$$

Answers:

$$\boxed{\begin{array}{l} M_p = 4.2\ \text{units};\quad \%\ M_p = 35\%;\quad t_p = 1.05\ \text{s} \\ t_r = 0.63\ \text{s};\quad T = 1\ \text{s};\quad t_s = 3.91\ \text{s} \\ t_d = 0.3862\ \text{s};\quad T_0 = 2.0943\ \text{s};\quad T_{os} = 1.867\ \text{s} \end{array}}$$

Example 2.5

Figure 2.14 shows a unity feedback system. Calculate ζ and ω_n when $K = 0$. Also calculate K when $\zeta = 0.6$.

(*Anna University, December, 2009*)

Solution The given block diagram is reduced and represented as shown in Fig. 2.15.

From Fig. 2.15, the following equation is written:

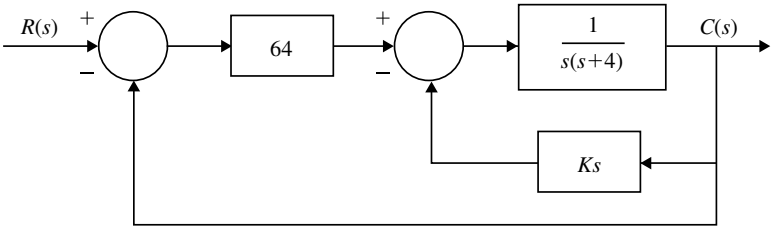
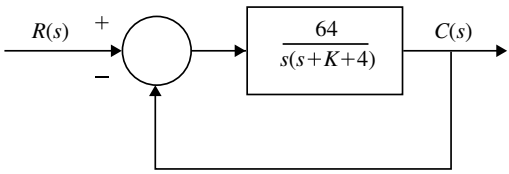


Fig. 2.14 Block diagram of Example 2.5

Fig. 2.15 Reduced block diagram of Fig. 2.14



$$\frac{C(s)}{R(s)} = \frac{64}{s^2 + (K+4)s + 64}$$

Compare this with the following standard second order equation:

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{(s^2 + 2\zeta\omega_n s + \omega_n^2)}$$

For $K = 0$;

$$\begin{aligned} \omega_n &= \sqrt{64} = 8 \\ 2\zeta\omega_n &= 4 \\ \zeta &= 0.25 \end{aligned}$$

For $\zeta = 0.6$;

$$\begin{aligned} 2\zeta\omega_n &= K + 4 \\ K &= 2 \times 0.6 \times 8 - 4 \\ K &= 5.6 \end{aligned}$$

Answers:

$$\boxed{\begin{aligned} \text{For } K = 0; \quad \zeta &= 0.25 \\ \omega_n &= 8 \\ \text{For } \zeta = 0.6; \quad K &= 5.6 \end{aligned}}$$

Example 2.6

The forward path T.Fs of certain unity negative feedback control systems are given below. Determine the damping ratio and natural frequency of oscillation and the nature of damping of each system.

$$\text{(a)} \quad G(s) = \frac{(3s+13)}{(s+3)(s+4)}$$
$$\text{(b)} \quad G(s) = \frac{2(s+2)}{s(s+4)}$$
$$\text{(c)} \quad G(s) = \frac{(s+9)}{s(s+2)}$$

Solution

(a)

$$G(s) = \frac{(3s+13)}{(s+3)(s+4)}$$

The closed loop T.F. of the system is

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)} = \frac{(3s+13)}{(s^2+10s+25)}$$

The characteristic equation of the above system is

$$s^2 + 10s + 25 = 0$$

The characteristic equation of the prototype second order system is

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

Comparing the given characteristic equation with that of the prototype second order system equation, we get

$$\omega_n^2 = 25 \quad \text{or} \quad \omega_n = 5 \text{ rad/s}$$
$$2\zeta\omega_n = 10 \quad \text{or} \quad \zeta = 1$$

Since $\zeta = 1$, the system is critically damped.

(b)

$$G(s) = \frac{2(s+2)}{s(s+4)}$$

The closed loop T.F. of the system is

$$\frac{C(s)}{R(s)} = \frac{2(s+2)/(s(s+4))}{1+[2(s+2)/(s(s+4))]} = \frac{2(s+2)}{s^2+6s+4}$$

The characteristic equation is

$$s^2 + 6s + 4 = 0$$

Comparing the above equation with the standard second order system characteristic equation, we get

$$\omega_n^2 = 4 \quad \text{or} \quad \omega_n = 2 \text{ rad/s}$$
$$2\zeta\omega_n = 6 \quad \text{or} \quad \zeta = 1.5$$

Since $\zeta > 1$, the system is over damped.

(c)

$$G(s) = \frac{(s+9)}{s(s+2)}$$

The closed loop T.F. is

$$\frac{C(s)}{R(s)} = \frac{(s+9)}{s(s+2)+s+9} = \frac{(s+9)}{s^2+3s+9}$$

The characteristic equation is

$$s^2 + 3s + 9 = 0$$

Comparing this with the characteristic equation of a standard second order system, we get

$$\omega_n^2 = 9 \quad \text{or} \quad \omega_n = 3 \text{ rad/s}$$
$$2\zeta\omega_n = 3 \quad \text{or} \quad \zeta = 0.5$$

Since $\zeta < 1$, the system is under damped.

Example 2.7

Consider the following systems with their open loop T.F. Determine the value of K of the unity negative feedback system to give the required damping ratio and natural frequency of oscillation.

$$\text{(a)} \quad G(s) = \frac{(s+K)}{s(s+3)}$$

The required damping ratio $\zeta = 0.5$

$$\text{(b)} \quad G(s) = \frac{(2s+1)}{(3s+4)(s+K)}$$

The required natural frequency of oscillation is 4 rad/s.

Solution

(a)

$$G(s) = \frac{(s+K)}{s(s+3)}$$

The closed loop T.F. is

$$\frac{C(s)}{R(s)} = \frac{(s+K)/s(s+3)}{1+[(s+K)/s(s+3)]} = \frac{(s+K)}{s^2+4s+K}$$

The characteristic equation is

$$s^2 + 4s + K = 0$$

In the above equation

$$\omega_n = \sqrt{K} \quad \text{and} \quad 2\zeta\omega_n = 4$$

Given $\zeta = 0.5, 2\zeta\omega_n = 4$

$$\boxed{K = 16}$$

(b)

$$G(s) = \frac{(2s+1)}{(3s+4)(s+K)}$$

$$\frac{C(s)}{R(s)} = \frac{(2s+1)}{3s^2+(3K+6)s+(4K+1)}$$

The characteristic equation is

$$s^2 + \frac{(3K+6)}{3}s + \frac{(4K+1)}{3} = 0$$

$$\omega_n^2 = \frac{(4K+1)}{3} = 4 \times 4$$

$$\boxed{K = 11.75}$$

$$2\zeta\omega_n = \frac{(3K+6)}{3} = 13.75$$

$$\boxed{\zeta = 1.7188}$$

Example 2.8

A unity feedback control system has the following open loop transfer function:

$$G(s) = \frac{2(s+8)}{s(s+2)}$$

(a) Write the closed loop system T.F. $C(s)/R(s)$ and from this find damping ratio, natural frequency and peak overshoot.
(b) Calculate $c(t)$ for unit step input.
(c) Calculate the steady state error for unit step input and the rise time.

(*Anna University, December, 2010*)

Solution

(a)

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)} = \frac{2(s+8)}{(s^2+4s+16)}$$

The characteristic equation is

$$s^2 + 4s + 16 = 0$$

Comparing this with the standard prototype second order system characteristic equation

$$\begin{aligned} s^2 + 2\zeta\omega_n s + \omega_n^2 &= 0 \\ \omega_n &= \sqrt{16} = 4 \\ 2\zeta\omega_n &= 4 \quad \text{or} \quad \zeta = 0.5 \end{aligned}$$

$$\boxed{\begin{aligned} \omega_n &= 4 \text{ rad/s} \\ \zeta &= 0.5 \end{aligned}}$$

(b) Performance specifications have been derived for the transfer function of the closed loop system which has no zero. However, there is a zero at $s = -8$ for the given closed loop system. To find the peak overshoot, the use of expression Eq. (2.27) will give erroneous result. Hence, it is to be derived from first principle. The characteristic equation is factorised as

$$(s^2 + 4s + 16) = (s + 2 + j3.465)(s + 2 - j3.465)$$

The closed loop T.F. is put into partial fraction as

$$\frac{C(s)}{R(s)} = \frac{2(s+8)}{(s+2+j3.465)(s+2-j3.465)}$$

For unit step input $R(s) = 1/s$

$$C(s) = \frac{A_1}{s} + \frac{A_2}{(s+2+j3.465)} + \frac{A_3}{(s+2-j3.465)}$$

$$2(s+8) = A_1(s^2 + 4s + 16) + A_2 s(s + 2 - j3.465) + A_3 s(s + 2 + j3.465)$$

For $s = 0$,

$$\begin{aligned} 16 &= A_1 \times 16 + 0 + 0 \\ A_1 &= 1 \end{aligned}$$

For $s = -2 - j3.465$,

$$\begin{aligned}
2(-2-j3.465+8) &= A_1 \times 0 + A_2(-2-j3.465)(-2-j3.465+2-j3.465) \\
&\quad + A_3 \times 0 \\
2(6-j3.465) &= A_2(-2-j3.465)(-j6.93) \\
2 \times 6.93\angle -30^\circ &= A_2 4\angle 60^\circ 6.93\angle 90^\circ \\
A_2 &= -0.5 \\
A_3 &= \text{Conjugate of } A_2 \\
A_3 &= -0.5 \\
C(s) &= \frac{1}{s} - 0.5\left[\frac{1}{(s+2+j3.465)} + \frac{1}{(s+2-j3.465)}\right]
\end{aligned}$$

Taking inverse Laplace transform, we get

$$c(t) = 1 - 0.5[e^{-(2+j3.465)t} + e^{-(2-j3.465)t}]$$

$$\boxed{c(t) = 1 - e^{-2t}\cos 3.464t}$$

To find the maximum overshoot the above equation is differentiated with respect to t and equated to zero

$$\begin{aligned}
\frac{dc(t)}{dt} &= e^{-2t}[(-2)\cos 3.465t - 3.465t \sin 3.465t] = 0 \\
\tan 3.465t &= \frac{-2}{3.465} \\
&= \tan\left(n\pi - \frac{\pi}{6}\right)
\end{aligned}$$

The first maximum overshoot occurs for $n = 1$ and correspondingly $t = t_p$.

$$\begin{aligned}
\tan 3.465t_p &= \tan\left(\pi - \frac{\pi}{6}\right) \\
t_p &= \frac{5}{6} \times \frac{\pi}{3.465} = 0.7556 \text{ s}
\end{aligned}$$

Substituting $t = t_p$ in the equation for $c(t)$, we get

$$C_{\max} = 1 - e^{-2\times 0.7556}\cos(3.465 \times 0.7556) = 1.191$$

Peak value

$$\begin{aligned}
M_p &= C_{\max} - 1 \\
&= 1.191 - 1 = 0.191
\end{aligned}$$

$$\boxed{\%M_p = 19.1\%}$$

(c)

$$c(t) = 1 - e^{-2t}\cos(3.465t)$$
$$C(\infty) = 1$$

Steady state error

$$e_{ss} = r(\infty) - C(\infty)$$
$$= 1 - 1$$

$$\boxed{e_{ss} = 0}$$

For the rise time, $c(t) = 1$ or $\cos 3.465\, t_r = \cos \pi/2$

$$\boxed{t_r = 0.4533}$$

Example 2.9

The open loop T.F. of a certain unity negative feedback control system is given by

$$G(s) = \frac{K}{s(T + s)}$$

where K and T are constants and greater than zero.

The system overshoot for unit step input is 60%. It is desired to bring the overshoot to 20% by reducing the gain of the amplifier in the system. Determine by what factor the amplifier gain is to be reduced to meet the above requirement. K_1 corresponds to 60% overshoot and K_2 corresponds to 20% overshoot.

Solution The closed loop T.F. of the given system is

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)}$$
$$= \frac{K}{s^2 + Ts + K}$$

Comparing the characteristic equation with that of the standard second order equation, we get

$$\begin{aligned}\omega_n &= \sqrt{K}\\ 2\zeta\omega_n &= T\\ \zeta &= \frac{T}{2\sqrt{K}}\end{aligned}$$

Let the peak overshoot corresponding to $\zeta = \zeta_1$ be M_{p1} and ζ_2 for M_{p2}. Using Eq. (2.27), we get

$$\begin{aligned}e^{-\frac{\zeta_1\pi}{\sqrt{1-\zeta_1^2}}} &= 0.6\\ e^{-\frac{\zeta_2\pi}{\sqrt{1-\zeta_2^2}}} &= 0.2\end{aligned}$$

Taking $\log_e$ on both sides of the above equations, we get

$$\begin{aligned}\frac{-\zeta_1\pi}{\sqrt{1-\zeta_1^2}} &= -0.5108\\ \zeta_1 &= 0.1605\\ e^{\frac{-\zeta_2\pi}{\sqrt{1-\zeta_2^2}}} &= 0.2\end{aligned}$$

Taking $\log_e$ on both sides, we get

$$\begin{aligned}-\frac{\zeta_2\pi}{\sqrt{1-\zeta_2^2}} &= -1.6094\\ \zeta_2 &= 0.456\\ \zeta_1 &= \frac{T}{2\sqrt{K_1}} = 0.1605\\ \zeta_2 &= \frac{T}{2\sqrt{K_2}} = 0.456\end{aligned}$$

Dividing one by the other and squaring, we get

$$\frac{K_2}{K_1} = \frac{1}{8.07}$$

$$\boxed{K_2 = \frac{1}{8.07}K_1}$$

The amplifier gain has to be reduced by a factor 8.07.

Example 2.10

A certain unity feedback control system when subjected to step input of size 10 gave the response as

$$c(t) = 10 - 30e^{-2t} + 20e^{-3t}$$

Determine the closed loop T.F. Also find the damping ratio and undamped natural frequency of oscillation.

Solution

$$c(t) = 10 - 30e^{-2t} + 20e^{-3t}$$

Taking Laplace transform on both sides of the above equation, we get

$$\begin{aligned} C(s) &= 10\left[\frac{1}{s} - \frac{3}{s+2} + \frac{2}{s+3}\right] \\ &= \frac{10[(s+2)(s+3) - 3s(s+3) + 2s(s+2)]}{s(s+2)(s+3)} \\ &= \frac{10[s^2 + 5s + 6 - 3s^2 - 9s + 2s^2 + 4s]}{s(s+2)(s+3)} \\ &= \frac{10 \times 6}{s(s^2 + 5s + 6)} \end{aligned}$$

For a step size of 10,

$$R(s) = \frac{10}{s}$$

$$C(s) = R(s)\frac{6}{(s^2 + 5s + 6)}$$

$$\boxed{\frac{C(s)}{R(s)} = \frac{6}{(s^2 + 5s + 6)}}$$

The characteristic equation is

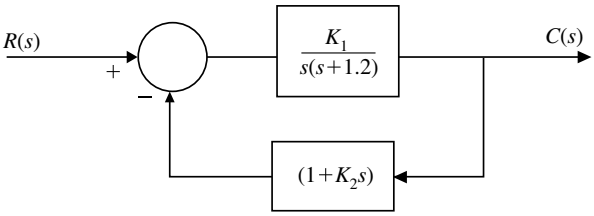


Fig. 2.16 Block diagram for Example 2.11

$$
\begin{aligned}
s^2 + 5s + 6 &= 0 \\
\omega_n &= \sqrt{6} \\
&= 2.45 \text{ rad/s} \\
2\zeta\omega_n &= 5 \\
\zeta &= \frac{5}{2 \times 2.45} = 1.02
\end{aligned}
$$

$$
\boxed{\begin{aligned} \zeta &= 1.02 \\ \omega_n &= 2.45 \end{aligned}}
$$

Example 2.11

The block diagram of a certain control system is shown in Fig. 2.16. Determine the value of K_1 and K_2 so that the maximum overshoot for step response is 25% which occurs at $t = 2$ s. For these values, find the time domain specifications.

Solution From Fig. 2.16, the closed loop T.F. is obtained as

$$
\frac{C(s)}{R(s)} = \frac{K_1/s(s+1.2)}{\left[1 + \frac{K_1(1+K_2 s)}{s(s+1.2)}\right]}
$$

$$
\frac{C(s)}{R(s)} = \frac{K_1}{[s^2 + (K_1K_2 + 1.2)s + K_1]}
$$

The characteristic equation is

$$
\begin{aligned}
s^2 + (K_1K_2 + 1.2)s + K_1 &= 0 \\
\omega_n &= \sqrt{K_1} \\
2\zeta\omega_n &= (K_1K_2 + 1.2)
\end{aligned}
$$

For 25% overshoot, the damping ratio is calculated using Eq. (2.28) as

$$e^{\frac{\zeta\pi}{\sqrt{1-\zeta^2}}} = 0.25$$

Taking $\log_e$ on both sides, we get

$$-\frac{\zeta\pi}{\sqrt{1-\zeta^2}} = -1.3863$$

$$\boxed{\zeta = 0.404}$$

Using Eq. (2.29), we get

$$\frac{\pi}{\omega_n\sqrt{1-\zeta^2}} = t_p$$

$$\omega_n = \frac{\pi}{t_p\sqrt{1-\zeta^2}} = \frac{\pi}{2\sqrt{1-(0.404)^2}} = 1.717$$

$$K_1 = \omega_n^2 = (1.717)^2$$

$$\boxed{K_1 = 2.95}$$

$$(K_1K_2 + 1.2) = 2\zeta\omega_n$$

$$= 2 \times 0.404 \times 1.717$$

$$\boxed{K_2 = 0.0635}$$

$$t_r = \frac{(\pi - \Phi)}{\omega_n\sqrt{1-\zeta^2}} \qquad \text{where } \Phi = \cos^{-1}\zeta = 1.155 \text{ rad}$$

$$= \frac{(\pi - 1.155)}{1.717\sqrt{1-(0.404)^2}}$$

$$\boxed{t_r = 1.265 \text{ s}}$$

$$T = \frac{1}{\zeta\omega_n} = \frac{1}{0.404 \times 1.717}$$

$$\boxed{T = 1.442 \text{ s}}$$

$$t_s = 3.91T = 3.91 \times 1.442$$

$$\boxed{t_s = 5.638 \text{ s}}$$

$$t_d = \frac{(1 + 0.7\zeta)}{\omega_n} = \frac{(1 + 0.7 \times 0.404)}{1.717}$$

$$\boxed{t_d = 0.747}$$

$$T_o = \frac{2\pi}{\omega_d} = \frac{2\pi}{1.717\sqrt{1 - (0.404)^2}}$$

$$\boxed{T_o = 4.0 \text{ s}}$$

$$T_{os} = \frac{t_s}{T_o} = \frac{5.638}{4}$$

$$\boxed{T_{os} = 1.41}$$

$$\boxed{\begin{array}{l} T = 1.442 \text{ s}; \quad t_s = 5.638 \text{ s}; \quad t_d = 0.747 \text{ s} \\ t_r = 1.265 \text{ s}; \quad M_p = 25\%; \quad t_p = 2 \text{ s} \\ T_o = 4.0; \quad T_{os} = 1.41 \end{array}}$$

Example 2.12

The closed loop T.F. of a unity feedback control system is

$$\frac{C(s)}{R(s)} = \frac{20s^2}{(s+1)(s+3)(s+5)}$$

Determine the response of the system when the excitation applied to the input terminal is

$$r(t) = 1 + 2t + \frac{3}{2}t^2$$

(*Anna University, December, 2010*)

Solution

$$r(t) = 1 + 2t + \frac{3}{2}t^2$$

$$R(s) = \frac{1}{s} + \frac{2}{s^2} + \frac{3}{s^3} = \frac{(s^2 + 2s + 3)}{s^3}$$

$$\frac{C(s)}{R(s)} = \frac{20s^2}{(s+1)(s+3)(s+5)}$$

$$C(s) = \frac{20(s^2 + 2s + 3)}{s(s+1)(s+3)(s+5)} = \frac{A_1}{s} + \frac{A_2}{s+1} + \frac{A_3}{s+3} + \frac{A_4}{s+5}$$

$$20(s^2 + 2s + 3) = A_1(s+1)(s+3)(s+5) + A_2 s(s+3)(s+5) + A_3 s(s+1)(s+5) + A_4 s(s+1)(s+3)$$

For $s = 0; \Longrightarrow A_1 = \dfrac{20 \times 3}{3 \times 5} = 4.$

For $s = -1; \Longrightarrow A_2 = \dfrac{20(1 - 2 + 3)}{(-1)(2)(4)} = -5.$

For $s = -3; \Longrightarrow A_3 = \dfrac{20(9 - 6 + 3)}{(-3)(-2)(2)} = 10.$

For $s = -5; \Longrightarrow A_4 = \dfrac{20(25 - 10 + 3)}{(-5)(-4)(-2)} = -9.$

$$C(s) = \frac{4}{s} - \frac{5}{s+1} + \frac{10}{s+3} - \frac{9}{s+5}$$

$$\boxed{c(t) = 4 - 5e^{-t} + 10e^{-3t} - 9e^{-5t}}$$

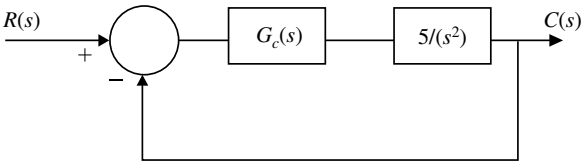


Fig. 2.17 Block diagram for Example 2.13

Example 2.13

A double integrator plant has the transfer function $G(s) = 5/s^2$ with $H(s) = 1$. It is to be compensated to achieve the damping ratio $\zeta = 0.5$ and undamped natural frequency $\omega_n = 5$ rad/s. What is the T.F. of the series compensator?

Solution The system with the series compensator $G_c(s)$ is shown in Fig. 2.17. From Fig. 2.17, we get

$$\frac{C(s)}{R(s)} = \frac{G_c(s)[5/s^2]}{1 + G_c(s)[5/s^2]} = \frac{5G_c(s)}{s^2 + 5G_c(s)}$$

The standard second order system to have $\zeta = 0.5$ and $\omega_n = 5$ rad/s will have the following characteristic equation:

$$s^2 + 5s + 25 = 0$$

Comparing the above two characteristic equations, we get

$$s^2 + 5s + 25 = s^2 + 5G_c(s)$$
$$G_c(s) = (s + 5)$$

$$\boxed{\begin{aligned} G_c(s) &= (s + 5) \\ \frac{C(s)}{R(s)} &= \frac{5(s + 5)}{(s^2 + 5s + 25)} \end{aligned}}$$

Example 2.14

The time response curve of a certain second system for a step input of size 4 is shown in Fig. 2.18. Determine the T.F. of the closed loop system and all time domain specifications.

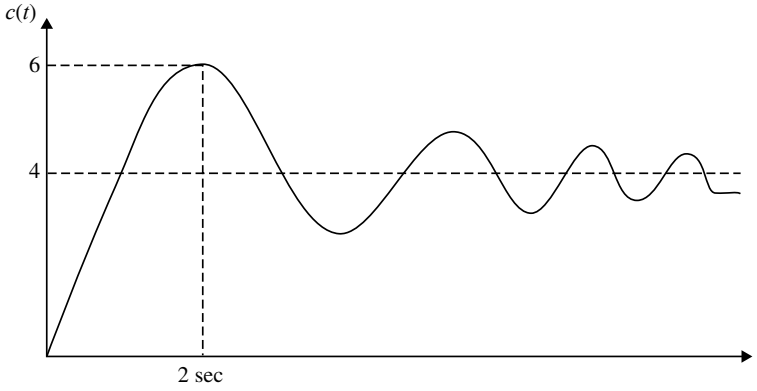


Fig. 2.18 Time response of a second order system for Example 2.14

Solution From Fig. 2.18, % peak overshoot is

$$\%M_p = \frac{(6-4)}{4} \times 100 = 50\%$$

$$e^{\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}} = 0.5$$

Taking l_n on both sides, we get

$$\frac{-\zeta\pi}{\sqrt{1-\zeta^2}} = -0.6931$$

$$\zeta = 0.215$$

From Fig. 2.18, $t_p = 2$ s.

$$t_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}}$$

$$\omega_n = \frac{\pi}{2\sqrt{1-(0.215)^2}} = 1.6084 \text{ rad/s}.$$

Time constant T is given by

$$T = \frac{1}{\zeta\omega_n} = \frac{1}{0.215 \times 1.6084}$$

$$\boxed{T = 2.892 \text{ s}}$$

The rise time t_r is given by

$$t_r = \frac{(\pi - \phi)}{\omega_d}$$

where $\phi = \cos^{-1} 0.215 = 1.354$ rad

$$t_r = \frac{(\pi - 1.354)}{1.6084\sqrt{1 - (0.215)^2}}$$

$$\boxed{t_r = 1.138 \text{ s}}$$

The settling time t_s for 2% tolerance is given by

$$t_s = 3.91T = 3.91 \times 2.892$$

$$\boxed{t_s = 11.3 \text{ s}}$$

The time delay is given by

$$t_d = \frac{1 + 0.7\zeta}{\omega_n} = \frac{1 + 0.7 \times 0.215}{1.6084}$$

$$\boxed{t_d = 0.7153 \text{ s}}$$

The period of oscillation is given by

$$T_o = \frac{2\pi}{\omega_d} = \frac{2\pi}{1.6084\sqrt{1 - (0.215)^2}}$$

$$\boxed{T_o = 4 \text{ s}}$$

The number of oscillation is given by

$$T_{os} = \frac{t_s}{T_o} = \frac{11.3}{4}$$

$$\boxed{T_{os} = 2.825}$$

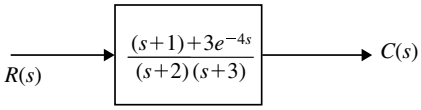


Fig. 2.19 Block diagram of second order system for Example 2.15

$T = 2.892$ s; $t_r = 1.138$ s; $t_s = 11.3$ s
$t_d = 0.7153$ s; $\%M_p = 50\%$; $t_p = 2$ s
$T_o = 4$ s; $T_{os} = 2.825$

Example 2.15

A certain second order system is represented in block diagram shown in Fig. 2.19. Derive an expression for the unit step response of the system.

Solution For the system represented in Fig. 2.19, the T.F. is

$$\frac{C(s)}{R(s)} = \frac{(s+1) + 3e^{-4s}}{(s+2)(s+3)}$$

For unit step input $R(s) = 1/s$ and the above equation is written as

$$\begin{aligned} C(s) &= \frac{[(s+1) + 3e^{-4s}]}{s(s+2)(s+3)} \\ &= \frac{(s+1)}{s(s+2)(s+3)} + \frac{3e^{-4s}}{s(s+2)(s+3)} \\ &= C_1(s) + C_2(s) \end{aligned}$$

where

$$C_1(s) = \frac{(s+1)}{s(s+2)(s+3)}$$
$$C_2(s) = \frac{3e^{-4s}}{s(s+2)(s+3)}$$

Now consider $C_1(s)$ which can be put into partial fraction as given below.

$$C_1(s) = \frac{A_1}{s} + \frac{A_2}{s+2} + \frac{A_3}{s+3}$$
$$(s+1) = A_1(s+2)(s+3) + A_2 s(s+3) + A_3 s(s+2)$$

For $s = 0 \Longrightarrow A_1 = 1/(2 \times 3) = 1/6$

For $s = -2 \Longrightarrow A_2 = (-1)/(-2) = 1/2$
For $s = -3 \Longrightarrow A_3 = (-2)/[(-3)(-1)] = -2/3$

$$C_1(s) = \frac{1}{6s} + \frac{1}{2(s+2)} - \frac{2}{3(s+3)}$$
$$c_1(t) = \left(\frac{1}{6} + \frac{1}{2}e^{-2t} - \frac{2}{3}e^{-3t}\right)u(t)$$

where $u(t)$ indicates that $c(t)$ exits for $t \geq 0$. Now consider $C_2(s)$ without delay e^{-4s} which can be put into partial fraction as given below. This is denoted as $C_3(s)$

$$C_3(s) = \frac{3}{s(s+2)(s+3)}$$
$$= \frac{A_1}{s} + \frac{A_2}{s+2} + \frac{A_3}{s+3}$$
$$3 = A_1(s+2)(s+3) + A_2 s(s+3) + A_3 s(s+2)$$

For $s = 0 \Longrightarrow A_1 = \dfrac{3}{2 \times 3} = \frac{1}{2}$

For $s = -2 \Longrightarrow A_2 = \dfrac{3}{(-2)(1)} = \frac{-3}{2}$

For $s = -3 \Longrightarrow A_3 = \dfrac{3}{(-3)(-1)} = 1$

Thus, $C_2(s)$ without time delay is written as

$$C_3(s) = \frac{1}{2s} - \frac{3}{2(s+2)} + \frac{1}{s+3}$$

Taking inverse Laplace transform we get.

$$c_3(t) = \left(\frac{1}{2} - \frac{3}{2}e^{-2t} + e^{-3t}\right)u(t)$$

The output $c_2(t)$ with delay is written as

$$c_2(t) = c_3(t-4)u(t-4)$$
$$c_2(t) = \left[\frac{1}{2} - \frac{3}{2}e^{-2(t-4)} + e^{-3(t-4)}\right]u(t-4)$$

The symbol $u(t-4)$ indicates that the response starts from $t \geq 4$. For $t < 4$ it is zero. The total response is

$$c(t) = c_1(t) + c_2(t)$$

$$\boxed{c(t) = \left(\frac{1}{6} + \frac{1}{2}e^{-2t} - \frac{2}{3}e^{-3t}\right)u(t) + \left(\frac{1}{2} - \frac{3}{2}e^{-2(t-4)} + e^{-3(t-4)}\right)u(t-4)}$$

Example 2.16

The forward path T.F. of a certain unity negative feedback system is given by

$$G(s) = \frac{K}{s(1+0.1s)}$$

The overshoot to unit step is 10%, and the settling time for 2% error tolerance is 2 s. Determine the value of K.

Solution

$$G(s) = \frac{K}{0.1s^2 + s}$$

$$\frac{C(s)}{R(s)} = \frac{K}{0.1s^2 + s + K} = \frac{10K}{s^2 + 10s + 10K}$$

The characteristic equation is

$$s^2 + 10s + 10K = 0$$

For 10% overshoot,

$$e^{\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}} = 0.1$$

Taking $\log_e$ on both sides, we get

$$\frac{-\zeta\pi}{\sqrt{1-\zeta^2}} = -2.3026$$

$$\zeta = 0.5916$$

The required settling time is $t_s = 2$.

$$t_s = 3.91T$$

$$T = \frac{2}{3.91} = 0.5115$$

$$T = \frac{1}{\zeta\omega_n}$$

$$\omega_n = \frac{1}{\zeta \times T} = \frac{1}{0.5916 \times 0.5115} = 3.3 \text{ rad/s}$$

From the characteristic equation

$$\omega_n = \sqrt{10K}$$
$$3.3 = \sqrt{10K}$$

$$\boxed{K = 1.089}$$

Example 2.17

A certain unity feedback control system has the following open loop T.F.

$$G(s) = \frac{K}{(s + A)(s + 2)}$$

Find the values of K and A so that the damping ratio is 0.707 and the peak time for unit step response is 1.8 s.

Solution

$$G(s) = \frac{K}{s^2 + (A + 2)s + 2A}$$

The closed loop T.F. is

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)} = \frac{K}{s^2 + (A + 2)s + (2A + K)}$$

Given

$$\zeta = 0.707$$
$$t_p = 1.8 \text{ s}$$

But

$$t_p = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}}$$
$$\omega_n = \frac{\pi}{t_p \sqrt{1 - \zeta^2}} = \frac{\pi}{(1.8)\sqrt{1 - (0.707)^2}}$$
$$\omega_n = 2.4683$$

The characteristic equation of the system is

$$
\begin{aligned}
s^2 + (A+2)s + (2A+K) &= 0 \\
2\zeta\omega_n &= A+2 \\
A &= 2\zeta\omega_n - 2 \\
&= 2 \times 0.707 \times 2.4683 - 2 \\
&= 3.49 - 2 \\
A &= 1.49 \\
(2A+K) &= \omega_n^2 \\
K &= \omega_n^2 - 2A \\
&= (2.4683)^2 - 2 \times 1.49 \\
K &= 6.0925 - 2.98 \\
K &= 3.1125
\end{aligned}
$$

$$
\boxed{\begin{aligned} A &= 1.49 \\ K &= 3.1125 \end{aligned}}
$$

Example 2.18

A certain unity negative feedback control system has the following open loop transfer function

$$G(s) = \frac{K}{s^2 + As + K}$$

Determine the values of K and A so that the peak overshoot of the step response is 20% and the rise time is 1.2 s.

Solution For the given system, the closed loop T.F. is

$$
\begin{aligned}
\frac{C(s)}{R(s)} &= \frac{G(s)}{1+G(s)} \\
&= \frac{K}{s^2 + As + 2K}
\end{aligned}
$$

For 20% peak overshoot, the damping ratio is obtained from

$$
\begin{aligned}
\%M_p &= e^{\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}} \times 100 \\
e^{\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}} &= 0.2
\end{aligned}
$$

Taking $\log_e$ on both sides, we get

$$\frac{-\zeta\pi}{\sqrt{1-\zeta^2}} = \log 0.2 = -1.609$$
$$\zeta = 0.456$$

The rise time is given by

$$t_r = \frac{(\pi - \Phi)}{\omega_n\sqrt{1-\zeta^2}}$$

where $\Phi = \cos^{-1} 0.456 = 1.1$ rad

$$\omega_n = \frac{(\pi - 1.1)}{1.2\sqrt{1-(0.456)^2}}$$
$$= 1.912 \text{ rad/s}$$

The characteristic equation of the given system is

$$s^2 + As + 2K = 0$$
$$\omega_n^2 = 2K$$
$$K = \frac{\omega_n^2}{2} = \frac{(1.912)^2}{2}$$

$$\boxed{K = 1.827}$$

$$A = 2\zeta\omega_n$$
$$= 2 \times 0.456 \times 1.912$$
$$A = 1.744$$

$$\boxed{\begin{array}{l} A = 1.744 \\ K = 1.827 \end{array}}$$

Example 2.19

If $r(t)$ is the step input and $c(t)$ is the output of the system described by a differential equation

$$\frac{d^2c(t)}{dt^2} + 4\frac{dc(t)}{dt} + 8c(t) = 8r(t)$$

Determine the undamped natural frequency, damping ratio and damped natural frequency, time for peak overshoot, and settling time.

(*Anna University, May, 2007*)

Solution The given differential equation is

$$\frac{d^2c(t)}{dt^2} + 4\frac{dc(t)}{dt} + 8c(t) = 8r(t)$$

Taking Laplace transform on both sides with zero initial conditions, we get

$$(s^2 + 4s + 8)C(s) = 8R(s)$$
$$\frac{C(s)}{R(s)} = \frac{8}{(s^2 + 4s + 8)}$$

The characteristic equation of the system is

$$s^2 + 4s + 8 = 0$$

Comparing this with the characteristic equation of a standard second order system $(s^2 + 2\zeta\omega_n s + \omega_n^2 = 0)$ we get

$$\omega_n = \sqrt{8} = 2.8284 \text{ rad/s.}$$
$$2\zeta\omega_n = 4$$
$$\zeta = \frac{2}{2.8284} = 0.707$$

The damped frequency of oscillation is given by

$$\omega_d = \omega_n\sqrt{1 - \zeta^2}$$
$$= 2.8284\sqrt{1 - (0.707)^2} = 2 \text{ rad/s.}$$

The time for peak overshoot is given by

$$t_p = \frac{\pi}{\omega_d} = \frac{\pi}{2} \text{ s.}$$

The percentage peak overshoot is given by

$$\%M_p = e^{\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}} \times 100 = e^{\frac{-0.707\pi}{\sqrt{1-(0.707)^2}}} \times 100 = 4.32\%$$

The settling time for 2% error tolerance is given by

$$t_s = \frac{3.91}{\zeta \omega_n}$$

$$t_s = \frac{3.91}{0.707 \times 2.8284} = 1.955 \text{ sec}$$

$$\omega_n = 2.8284 \text{ rad/s}; \quad \zeta = 0.707; \quad \omega_d = 2 \text{ rad/s}$$
$$t_p = \frac{\pi}{2} \text{ s}; \quad M_p = 4.32\%; \quad t_s = 1.955 \text{ s}$$

Example 2.20

A second order system is represented by the T.F.

$$\frac{C(s)}{R(s)} = \frac{1}{(Js^2 + Fs + K)}$$

A step input of 10 N.m is applied to a system and test results are:

1. maximum overshoot = 6%,
2. time at peak overshoot = 1 s. and
3. the steady state value of the output is 0.5 radians.

Determine the value of J, F and K.

(*Anna University, May, 2010*)

Solution

$$\frac{C(s)}{R(s)} = \frac{1}{Js^2 + Fs + K} = \frac{1}{J\left[s^2 + \frac{F}{J}s + \frac{K}{J}\right]}$$

$$R(s) = \frac{10}{s}$$

$$C(s) = \frac{10}{sJ\left[s^2 + \frac{F}{J}s + \frac{K}{J}\right]}$$

Using final value theorem,

$$C(\infty) = \lim_{s \to 0} sC(s) = \lim_{s \to 0} \frac{10}{J\left[s^2 + \frac{F}{J}s + \frac{K}{J}\right]}$$

$$= \frac{10}{K} = 0.5$$

$$K = 20 \text{ N.m/rad.}$$

The percentage overshoot is given by

$$\begin{aligned} e^{\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}} &= 0.06 \\ \frac{-\zeta\pi}{\sqrt{1-\zeta^2}} &= -2.8134 \\ \zeta &= 0.667 \\ t_p &= \frac{\pi}{\omega_n\sqrt{1-\zeta^2}} = 1 \text{ s} \\ \omega_n &= \frac{\pi}{t_p\sqrt{1-\zeta^2}} = \frac{\pi}{\sqrt{1-(0.667)^2}} \\ &= 4.2166 \text{ rad/s}. \end{aligned}$$

The characteristic equation of the given system is

$$s^2 + \frac{F}{J}s + \frac{K}{J} = 0$$

Here $\omega_n = \sqrt{K/J}$ and $2\zeta\omega_n = F/J$

$$\begin{aligned} J &= \frac{K}{\omega_n^2} = \frac{20}{(4.2166)^2} \\ &= 1.125 \text{ Kg-m}^2 \\ F &= 2\zeta\omega_n J \\ &= 2 \times 0.667 \times 4.2166 \times 1.125 \\ &= 6.33 \text{ N.m/rad/s} \end{aligned}$$

$$\boxed{\begin{aligned} J &= 1.125 \text{ Kg-m}^2 \\ F &= 6.33 \text{ N.m/rad/s} \\ K &= 20 \text{ N.m/rad} \end{aligned}}$$

Example 2.21

A certain unity feedback control system has the open loop T.F. given by

$$G(s) = \frac{s+a}{(s+b)^2}$$

The system is to have a natural frequency of oscillation $\omega_n = 11$ rad/s. and the damping factor $\zeta = 0.5$. Find the values of a and b and also the closed loop T.F. Is it possible to find time domain specifications using Eqs. (2.27) to (2.41).

Solution

$$G(s) = \frac{(s+a)}{(s+b)^2}$$

The closed loop T.F. is obtained from

$$\begin{aligned}\frac{C(s)}{R(s)} &= \frac{G(s)}{1+G(s)} \\ &= \frac{(s+a)}{s^2+(1+2b)s+(a+b^2)}\end{aligned}$$

The characteristic equation of the above equation is

$$s^2 + (1+2b)s + (a+b^2) = 0$$

For the given $\zeta = 0.5$ and $\omega_n = 11$ rad/s,

$$\begin{aligned}1+2b &= 2\zeta\omega_n \\ &= 2 \times 0.5 \times 11 \\ b &= 5 \\ a+b^2 &= \omega_n^2 \\ a+25 &= 121 \\ a &= 96\end{aligned}$$

$$\boxed{\begin{aligned}a &= 96 \\ b &= 5 \\ \frac{C(s)}{R(s)} &= \frac{(s+96)}{(s^2+11s+121)}\end{aligned}}$$

Using Eqs. (2.27)–(2.41), the time domain specifications cannot be determined. This is because in the closed loop T.F. there is a zero at $s = -96$, which will give different expressions for the time response and hence different time domain specifications.

Example 2.22

Consider a certain second order system shown in Fig. 2.20.

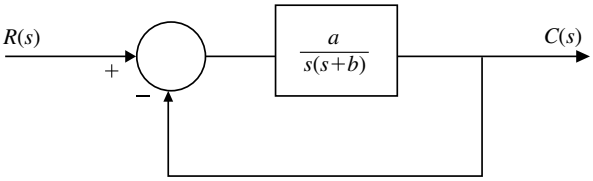


Fig. 2.20 Second order system for Example 2.22

(a) Evaluate the time constant, % peak overshoot, settling time, peak time and the rise time of the closed loop system when subjected to unit step input for the value of $a = 16$ and $b = 4$ and find the closed loop poles.
(b) Evaluate the values of a and b so that the real part of the poles remains constant and the imaginary part is increased 2 times over that of part (a).
(c) Evaluate the values of a and b so that the imaginary part remains constant and the real part is increased 2 times over that of part (a).
(d) Evaluate a and b if the poles are moved along the radial line from the origin by 2 times over that of part (a). In all the above cases find $\%M_p$ and t_r.

Solution From Fig. 2.20 the closed loop T.F. is obtained as

$$T(s) = \frac{C(s)}{R(s)} = \frac{a}{s^2 + bs + a}$$

Substituting $a = 16$ and $b = 4$, we get

$$T(s) = \frac{16}{s^2 + 4s + 16}$$

The T.F. of the prototype second order system is

$$T(s) = \frac{\omega_n^2}{(s^2 + 2\zeta\omega_n s + \omega_n^2)}$$

By comparison we get $\omega_n = 4$; $\zeta = 0.5$

$$\begin{aligned}\omega_d &= \omega_n\sqrt{1 - \zeta^2} \\ &= 4\sqrt{1 - 0.5^2} = 3.464\end{aligned}$$

(a)

$$
\begin{aligned}
T &= \frac{1}{\zeta\omega_n} = \frac{1}{0.5 \times 4} = 0.5 \text{ s} \\
\%M_p &= \frac{-\zeta\pi}{e\sqrt{1-\zeta^2}} \times 100 = \frac{-0.5\pi}{e\sqrt{1-0.5^2}} \times 100 \\
&= 16.3\% \\
t_s &= 3.91T = 3.91 \times 0.5 \\
&= 1.955 \text{ s} \\
t_p &= \frac{\pi}{\omega_d} = \frac{\pi}{3.464} \\
&= 0.907 \text{ s} \\
\phi &= \cos^{-1}\zeta = \cos^{-1} 0.5 \\
&= \frac{\pi}{3} \text{ rad.} \\
t_r &= \frac{(\pi - \phi)}{\omega_d} = \frac{\pi - (\pi/3)}{3.464} \\
&= 0.6046
\end{aligned}
$$

Real part of the pole is

$$\zeta\omega_n = 0.5 \times 4 = 2$$

The imaginary part of the pole is

$$\omega_d = 3.464$$

The poles of the closed loop system is

$$
\begin{aligned}
s_{12} &= -\zeta\omega_n \pm j\omega_d \\
&= -2 \pm j3.464
\end{aligned}
$$

The poles are located as shown in Fig. 2.21.

T	$= 0.5$ s
$\%M_p$	$= 16.3$
t_s	$= 1.955$ s
t_p	$= 0.907$
t_r	$= 0.6046$
$\zeta\omega_n$	$= 2$ (Real part)
ω_d	$= 3.464$ (Imaginary part)

(b) **Real Part Remains Same**

$$\zeta \omega_n = 2$$

Imaginary Part is Increased by 2 Times

$$\omega_d = 2 \times 3.464 = 6.928$$

The pole locations are shown in Fig. 2.22.

$$\omega_n \sqrt{1 - \zeta^2} = 6.928$$
$$\zeta \omega_n = 2$$

Dividing one by the other we get

$$\frac{\sqrt{1 - \zeta^2}}{\zeta} = \frac{6.928}{2} = 3.464$$

Solving for ζ we get

$$\begin{aligned}
\zeta &= 0.2773 \\
\omega_n &= \frac{2}{\zeta} = \frac{2}{0.2773} \\
\omega_n &= 7.211 \\
a &= \omega_n^2 = 7.211^2 = 52 \\
b &= 2\zeta\omega_n \\
b &= 2 \times 0.2773 \times 7.211 = 4 \\
\%M_p &= e^{\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}} \times 100 \\
&= e^{\frac{-0.2773\pi}{\sqrt{1-0.2773^2}}} \times 100 = 40\% \\
\phi &= \cos^{-1}\zeta = \cos^{-1} 0.2773 \\
&= 1.29 \text{ rad} \\
t_r &= \frac{(\pi - 1.29)}{6.928} = 0.2673 \text{ s}
\end{aligned}$$

$$\boxed{\begin{aligned} a &= 52 \\ b &= 4 \\ \%M_p &= 40\% \\ t_r &= 0.2673 \text{ s} \end{aligned}}$$

(c) **Imaginary Part remains the Same.**

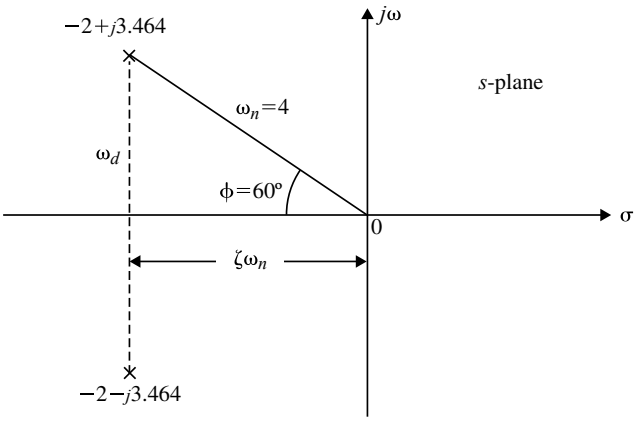


Fig. 2.21 Pole location of a second order system

$$\omega_d = 3.464$$

Real Part is Increased Two Times.

$$\zeta\omega_n = 2 \times 2 = 4$$

The pole–zero location is shown in Fig. 2.23.

$$\omega_n\sqrt{1-\zeta^2} = \omega_d = 3.464$$
$$\zeta\omega_n = 4$$

Dividing one by the other we get

$$\frac{\sqrt{1-\zeta^2}}{\zeta} = \frac{3.464}{4} = 0.866$$

Solving for ζ and ω_n we get

$$\begin{aligned}
\zeta &= 0.756 \\
\omega_n &= 5.29 \\
a &= \omega_n^2 = 5.29^2 = 28 \\
b &= 2\zeta\omega_n \\
&= 2 \times 0.756 \times 5.29 = 8 \\
\%M_P &= e^{\frac{-\zeta\pi}{\sqrt{1-0.756^2}}} \times 100 \\
&= 2.66\%
\end{aligned}$$

$$\begin{aligned}\phi &= \cos^{-1}\zeta = \cos^{-1}0.756\\ &= 0.714 \text{ rad.}\\ t_r &= \frac{(\pi-\phi)}{\omega_d}\\ &= \frac{\pi-0.714}{3.464} = 0.7 \text{ s}\end{aligned}$$

$$\boxed{\begin{aligned}a &= 28\\ b &= 8\\ \%M_P &= 2.66\%\\ t_r &= 0.7 \text{ s}\end{aligned}}$$

(d) **The poles are moved along a radial line originating from the origin by 2 times as in (a).**

$$\omega_n = 4 \times 2 = 8 \text{ rad/s.}$$

ζ remains the same as $\zeta = 0.5$. The pole locations are shown in Fig. 2.24.

$$\begin{aligned}\omega_d &= \omega_n\sqrt{1-\zeta^2}\\ &= 8\sqrt{1-0.25} = 6.928\\ \phi &= \cos^{-1}0.5 = \frac{\pi}{3} \text{ rad.}\\ \zeta\omega_n &= 0.5 \times 8 = 4\\ b &= 2\zeta\omega_n = 2 \times 4 = 8\\ a &= \omega_n^2 = 64\end{aligned}$$

$$\begin{aligned}\%M_p &= e^{\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}} \times 100\\ &= e^{\frac{-0.5\pi}{\sqrt{1-0.25}}} \times 100 = 16.3\%\\ t_r &= \frac{\pi-\phi}{\omega_d}\\ &= \frac{\pi-(\pi/3)}{6.928} = 0.302 \text{ s}\end{aligned}$$

$$\boxed{\begin{aligned}a &= 64\\ b &= 8\\ \%M_p &= 16.3\%\\ t_r &= 0.302 \text{ s}\end{aligned}}$$

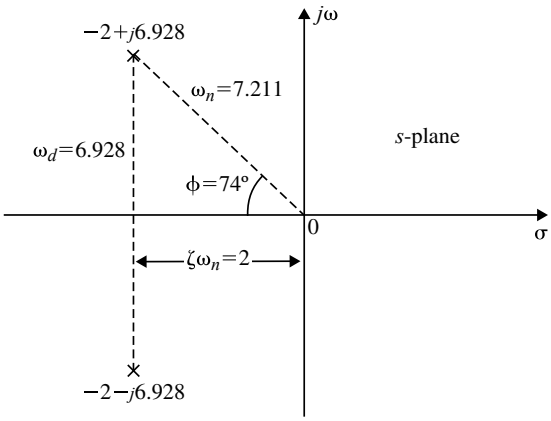


Fig. 2.22 Real part remains constant. Imaginary part is increased by 2 times

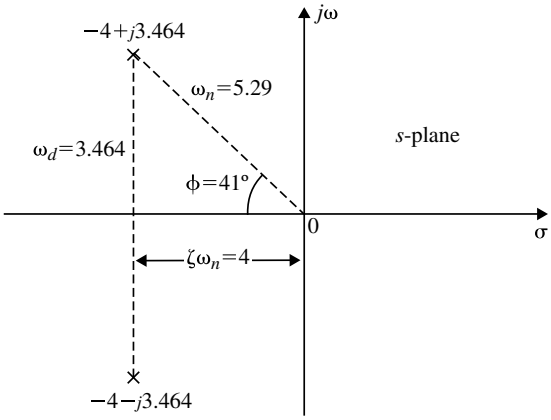


Fig. 2.23 Imaginary part is kept fixed. Real part is increased by 2 times

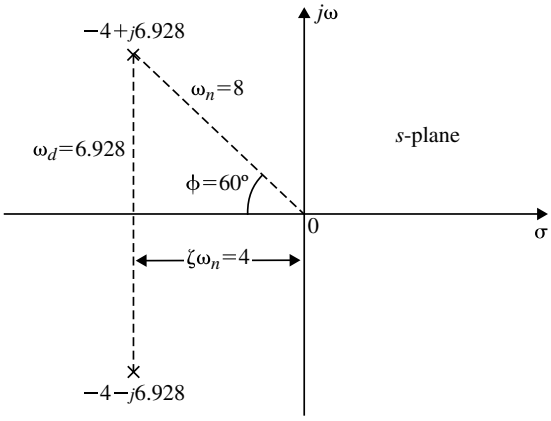


Fig. 2.24 The pole is moved along the radial line by 2 times

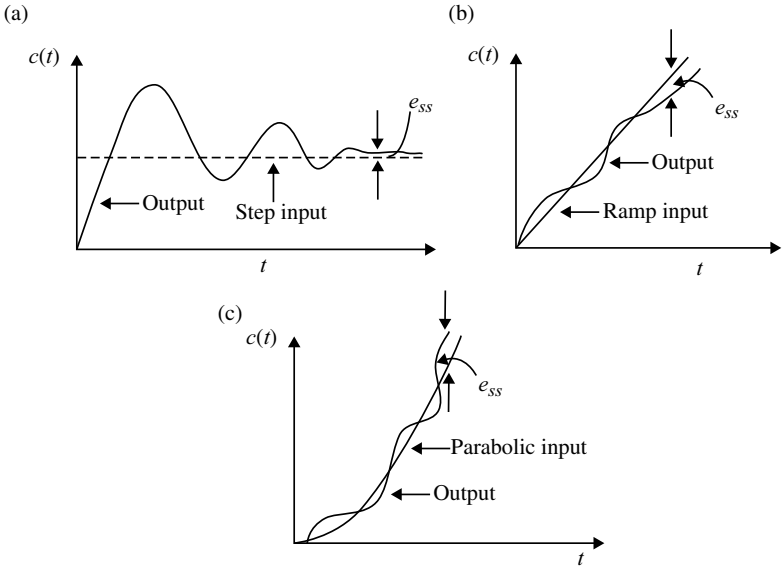


Fig. 2.25 Steady state error for **a** Step input; **b** Velocity input; and **c** Acceleration input

2.5 Steady State Error

Control systems are used to control physical variables. How accurately the system performs is very important. The variables under control should be as close as to the desired value. To be able to measure and predict accuracy in control system, a standard measure of performance is widely used. This standard measure of performance is steady state error. The difference between the input (the desired response) and the output (the actual response) is referred to as the error. This error may be a function of time. This is called transient error. The expression in Eq. (2.27) which is the peak overshoot is the transient error. This error when t tends to ∞ is called the steady state error and is denoted by e_{ss}. The system under test is simulated with standard inputs such as step, ramp and acceleration. The steady state error of certain control system for these signals is shown in Fig. 2.25.

The steady state error in control systems mainly come from non-linearities of some of the components connected in the systems. The non-linearities include the backlash in gear or a motor that will not move unless the input voltage exceeds a threshold. However, here we study about the steady state errors which are due to system configuration and the type of input that is applied.

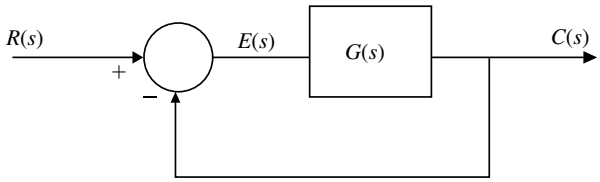


Fig. 2.26 Block diagram showing error function $E(s)$ for a unity feedback system

2.5.1 Type of the System

It can be established that the steady state errors are dependent upon the type of the system. The type of the system is decided by the number of pure integrators that are present in the open loop T.F. $G(s)$. As explained in Chap. 1, consider the following open loop T.F:

$$G(s) = \frac{K(1+sz_1)(1+sz_2)\times\cdots}{s^n(1+sp_1)(1+sp_2)\times\cdots} \tag{2.42}$$

In Eq. (2.42) the value of n in the denominator, or equivalently, the number of pure integrations in the denominator of $G(s)$ gives the Type of the system. Thus, a system with $n = 0$ is a Type 0 system, with $n = 1$, it is Type 1 system and so on. We will establish that the steady state errors are functions of the input and system Type.

2.5.2 Steady State Error Using Static Error Constants

Consider the system with forward path T.F. $G(s)$ connected with unity feedback as shown in Fig. 2.26. In Fig. 2.26, $E(s)$ is the error which is the difference between the input and the output and can be written as

$$\begin{aligned} E(s) &= R(s) - C(s) \\ &= R(s) - E(s)G(s) \\ E(s)[1+G(s)] &= R(s) \end{aligned}$$

$$\boxed{E(s) = \frac{R(s)}{1+G(s)}} \tag{2.43}$$

From Eq. (2.43) it is evident that the error $E(s)$ is a function of system configuration $G(s)$ and the input $R(s)$. By taking inverse Laplace transform of $E(s)$ one can get $e(t)$, the error which is expressed as a function of time t. The steady state error e_{ss} is obtained by substituting $t = \infty$ in $e(t)$. However, this procedure is little cumbersome.

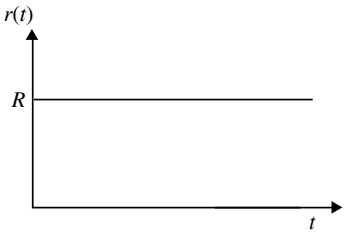


Fig. 2.27 Step input signal $r(t)$

Instead, using Laplace final value theorem the steady state error is obtained from $E(s)$ as

$$e_{ss} = \lim_{s \to 0} sE(s) \tag{2.44}$$

Using Eq. (2.44), the steady state errors are calculated for the step, ramp and parabolic inputs as explained below. It is to be noted that final value theorem as applied to Eq. (2.44) is valid iff the closed loop system is stable and the transfer function $G(s)$ is proper.

2.5.2.1 Steady State Error Due to Step Input-The Position Error Constant

Consider the step input signal shown in Fig. 2.27 which is described as $r(t) = Ru(t)$ or

$$r(t) = \begin{cases} R & t \geq 0 \\ 0 & t < 0 \end{cases} \tag{2.45}$$

The Laplace transform of $r(t)$ is given by

$$R(s) = \frac{R}{s} \tag{2.46}$$

Substituting Eq. (2.46) in (2.43) we get

$$E(s) = \frac{R}{s + sG(s)}$$

The steady state error is obtained by using Eq. (2.44) in the above equation.

$$e_{ss} = \lim_{s \to 0} s \frac{R}{s + sG(s)} = \frac{R}{\lim_{s \to 0} G(s) + 1}$$

$$\boxed{e_{ss} = \frac{R}{1 + K_p}} \tag{2.47}$$

where

$$K_p = \lim_{s \to 0} G(s) \tag{2.48}$$

K_p is called the position error constant. Now consider the system with the following T.F.

$$G(s) = \frac{K(1 + sz_1)(1 + sz_2) \cdots}{s^n(1 + sp_1)(1 + sp_2) \cdots}$$

For $n = 0$, the system is Type 0 system. K_p for Type 0 system is calculated using Eq. (2.48).

$$\begin{aligned} K_p &= \lim_{s \to 0} G(s) \\ &= \lim_{s \to 0} \frac{K(1 + sz_1)(1 + sz_2) \cdots}{(1 + sp_1)(1 + sp_2) \cdots} = K \end{aligned}$$

The steady state error is obtained using Eq. (2.47)

$$e_{ss} = \frac{R}{1 + K} \tag{2.49}$$

For Type 1 system

$$\begin{aligned} K_p &= \lim_{s \to 0} \frac{K(1 + sz_1)(1 + sz_2) \cdots}{s(1 + sp_1)(1 + sp_2) \cdots} = \infty \\ e_{ss} &= \frac{R}{(1 + \infty)} = 0 \end{aligned}$$

For Type 2 and Type 3 systems the steady state error will similarly be zero.

Thus, when step input is applied, for Type 0 system the steady state error is finite and is given by Eq. (2.49). **For systems of Type** 1 **and above the steady state error is zero**.

Example 2.23

Consider the closed loop system with the following forward path T.F.

$$G(s) = \frac{200(s + 2)(s + 5)}{(s + 4)(2s + 6)}$$

A step input of height 12 size is applied. Find the steady state error.

Solution

$$G(s) = \frac{200(s+2)(s+5)}{(s+4)(2s+6)}$$

$$K_p = \lim_{s\to 0} G(s) = \frac{200 \times 2 \times 5}{4 \times 6} = \frac{1000}{12}$$

$$e_{ss} = \frac{R}{1+K_p} = \frac{12}{1+(1000/12)}$$

$$\boxed{e_{ss} = \frac{144}{1012}}$$

Example 2.24

A certain unity negative feedback control system has the following forward path T.F.

$$G(s) = \frac{K}{s^2+2s+10}$$

Determine the value of K to achieve a steady state error $e_{ss} \leq 0.5$ to a step input of size 2.

Solution

$$G(s) = \frac{K}{s^2+2s+10}$$

$$K_p = \lim_{s\to 0} G(s) = \frac{K}{10}$$

$$e_{ss} = \frac{2}{1+K_p} = \frac{20}{10+K}$$

$$0.5 = \frac{20}{10+K}$$

$$\boxed{K = 30}$$

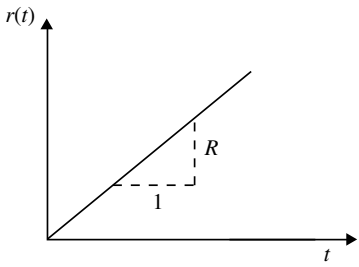


Fig. 2.28 Ramp or velocity input signal

2.5.2.2 Steady State Error Due to Velocity (Ramp) Input-The Velocity Error Constant

Consider the velocity or ramp input signal shown in Fig. 2.28. The slope of the ramp is R. The signal is described as $r(t) = Rtu(t)$ or

$$r(t) = \begin{cases} Rt & t \geq 0 \\ 0 & t < 0 \end{cases}$$

The Laplace transform of the above signal is

$$R(s) = \frac{R}{s^2} \tag{2.50}$$

$$E(s) = \frac{R}{s^2 + s^2 GH(s)}$$

$$e_{ss} = \lim_{s \to 0} sE(s) = \lim_{s \to 0} \frac{R}{sG(s)}$$

$$\boxed{e_{ss} = \frac{R}{K_v}} \tag{2.51}$$

where

$$K_v = \lim_{s \to 0} sG(s) \tag{2.52}$$

K_v is called the velocity error constant and has the unit s^{-1}.

Consider the system with the following T.F.

$$G(s) = \frac{K(1 + sz_1)(1 + sz_2) \cdots}{s^n(1 + sp_1)(1 + sp_2) \cdots}$$

For Type 0 system,

$$K_v = \lim_{s\to 0} \frac{sK(1+sz_1)(1+sz_2)\cdots}{(1+sp_1)(1+sp_2)\cdots} = 0$$
$$e_{ss} = \frac{R}{K_v} = \infty$$

For Type 1 system,

$$K_v = \lim_{s\to 0} \frac{sK(1+sz_1)(1+sz_2)\cdots}{s(1+sp_1)(1+sp_2)\cdots} = K$$
$$e_{ss} = \frac{R}{K}$$

For Type 2 system,

$$K_v = \lim_{s\to 0} \frac{sK(1+sz_1)(1+sz_2)\cdots}{s^2(1+sp_1)(1+sp_2)\cdots}$$
$$= \infty$$
$$e_{ss} = \frac{R}{\infty} = 0$$

Thus for

$$\text{Type 0 system,} \quad e_{ss} = \infty$$
$$\text{Type 1 system,} \quad e_{ss} = \frac{R}{K_v}$$
$$\text{Type 2 system,} \quad e_{ss} = 0$$

For Type 2 and higher Type system $e_{ss} = 0$.

Example 2.25

A negative feedback control system has the following loop T.F.

$$G(s) = \frac{10(2s+3)(s-4)}{s(3s+2)(4s+5)}$$

The system is subjected to ramp input of slope 2 rad/s. Find the steady state error.

Solution

$$G(s) = \frac{10(2s+3)(s-4)}{s(3s+2)(4s+5)}$$
$$K_v = \lim_{s\to 0} sG(s) = \frac{-10\times 3\times 4}{(2)(5)} = -12$$
$$e_{ss} = \frac{R}{K_v} = \frac{-2}{12}$$

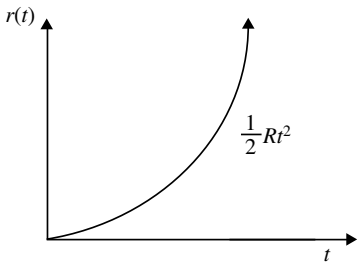


Fig. 2.29 Parabolic or acceleration input signal

$$\boxed{e_{ss} = -\frac{1}{6} \text{ rad.}}$$

The negative sign indicates that the output is higher than the input at steady state.

Example 2.26

Consider a unity negative feedback system with the following open loop T.F.

$$G(s) = \frac{K}{s(s+1)(s+4)}$$

The steady state error $e_{ss} \leq 2$ rad for a velocity input of 2 rad/s. Find K_v and hence K.

Solution

$$G(s) = \frac{K}{s(s+1)(s+4)}$$

$$K_v = \lim_{s\to 0} sG(s) = \lim_{s\to 0} \frac{sK}{s(s+1)(s+4)} = \frac{K}{4}$$

$$e_{ss} = \frac{R}{K_v} = \frac{2\times 4}{K} = \frac{8}{K}$$

$$2 = \frac{8}{K}$$

$$\boxed{K = 4}$$

2.5.2.3 Steady State Error Due to Parabolic Input-The Acceleration Error Constant

Consider the parabolic (acceleration) input signal shown in Fig. 2.29. The signal is described as $r(t) = \frac{1}{2}Rt^2u(t)$ or

$$r(t) = \begin{cases} \frac{1}{2}Rt^2 & t \geq 0 \\ 0 & t < 0 \end{cases}$$

The Laplace transform of the above signal is

$$R(s) = \frac{R}{s^3} \tag{2.53}$$

$$E(s) = \frac{R}{s^3 + s^3G(s)}$$

$$e_{ss} = \lim_{s\to 0} sE(s) = \lim_{s\to 0} \frac{R}{s^2 + s^2G(s)}$$

$$= \frac{R}{\lim_{s\to 0} s^2G(s)}$$

$$\boxed{e_{ss} = \frac{R}{K_a}} \tag{2.54}$$

where

$$K_a = \lim_{s\to 0} s^2G(s) \tag{2.55}$$

K_a is called acceleration or parabolic error constant and has the unit of $\sec^{-2}$. Consider the system with the following T.F.

$$G(s) = \frac{K(1 + sz_1)(1 + sz_2)\cdots}{s^n(1 + sp_1)(1 + sp_2)\cdots}$$

For Type 0 system,

$$K_a = \lim_{s\to 0} s^2G(s)$$

$$= \lim_{s\to 0} \frac{s^2K(1 + sz_1)\cdots}{(1 + sp_1)\cdots} = 0$$

$$e_{ss} = \frac{R}{0} = \infty$$

For Type 1 system,

$$\begin{aligned} K_a &= \lim_{s\to 0} s^2 G(s) \\ &= \lim_{s\to 0} \frac{s^2 K(1+sz_1)\cdots}{s(1+sp_1)\cdots} = 0 \\ e_{ss} &= \infty \end{aligned}$$

For Type 2 system,

$$\begin{aligned} K_a &= \lim_{s\to 0} s^2 G(s) \\ &= \lim_{s\to 0} \frac{s^2 K(1+sz_1)\cdots}{s^2(1+sp_1)\cdots} = K \\ e_{ss} &= \frac{R}{K} \end{aligned}$$

Thus, for parabolic input,

$$\begin{aligned} \text{Type 0 system,} \quad & e_{ss} = \infty \\ \text{Type 1 system,} \quad & e_{ss} = \infty \\ \text{Type 2 system,} \quad & e_{ss} = \frac{R}{K_a} \\ \text{For Type 3 or higher Type system,} \quad & e_{ss} = 0 \end{aligned}$$

Example 2.27

A certain control system has the following T.F.

$$G(s) = \frac{1(s+2)}{s^2(2s+3)(s+5)}$$

The input applied to the system is

$$r(t) = \frac{3}{2}t^2 u(t)$$

Find the steady state error.

Solution Given

$$r(t) = \frac{3}{2}t^2$$

$$R(s) = \frac{3}{s^3}$$

$$K_a = \lim_{s\to 0} s^2 G(s) = \lim_{s\to 0} \frac{s^2(s+2)}{s^2(2s+3)(s+5)} = \frac{2}{15}$$

$$e_{ss} = \frac{R}{K_a}$$

where $R = 3$

$$e_{ss} = \frac{3 \times 15}{2}$$

$$\boxed{e_{ss} = \frac{45}{2}}$$

By using the static error constants K_p, K_v and K_a, the steady state error can be obtained as described above. When these signals are simultaneously applied to the system, the steady state error can be derived for the individual signals and using superposition theorem, the total steady state error can be determined. Further, for any other signals, in a similar way, the steady state error can be obtained from $R(s)$ and $G(s)$ using final value theorem. It is to be noted here that the procedure described above is valid under the following conditions:

(1) The method is applicable to system configuration shown in Fig. 2.26 where the feedback is unity. For non-unity feedback system it is not valid.
(2) The error analysis is determined using final value theorem. Hence, it is important to check whether $sE(s)$ has any pole on the $j\omega$ axis or in the right half s-plane. In such cases, the system is unstable and the use of final value theorem is not valid.
(3) The error constants are defined with respect to the forward path T.F. $G(s)$ with unity feedback. If the system is non-unity feedback, then it is necessary to convert it to an equivalent system whose forward path T.F. is $G_e(s)$ and its corresponding feedback path has unity feedback. For this error constants are found out.

The summary of error analysis showing the error relations among the error constants, the Types of systems and the inputs are given in Table 2.1.

Example 2.28

A unity feedback system is characterised by the open loop T.F. as

$$G(s) = \frac{1}{s(1+0.5s)(1+0.2s)}$$

Table 2.1 Summary of the steady state errors due to step, ramp and parabolic inputs for unity feedback systems

Type of system	Error constants			Steady state error		
	K_p	K_v	K_a	Step input	Ramp input	Parabolic input
				$\left(\frac{R}{1+K_p}\right)$	$\left(\frac{R}{K_v}\right)$	$\left(\frac{R}{K_a}\right)$
0	K	0	0	$\frac{R}{1+K}$	∞	∞
1	∞	K	0	0	$\frac{R}{K}$	∞
2	∞	∞	K	0	0	$\frac{R}{K}$
3	∞	∞	∞	0	0	0

Determine the steady state error for unit step, unit ramp and unit acceleration.

Solution

$$G(s) = \frac{1}{s(1+0.5s)(1+0.2s)}$$

$$K_p = \lim_{s\to 0} G(s) = \lim_{s\to 0} \frac{1}{s(1+0.5s)(1+0.2s)}$$

$$K_p = \infty$$

$$K_v = \lim_{s\to 0} sG(s) = \lim_{s\to 0} \frac{s}{s(1+0.5s)(1+0.2s)} = 1$$

$$K_a = \lim_{s\to 0} s^2 G(s) = \lim_{s\to 0} \frac{s^2}{s(1+0.5s)(1+0.2s)} = 0$$

The steady state error for unit step input is

$$e_{ss} = \frac{1}{1+K_p} = \frac{1}{1+\infty} = 0$$

The steady state error for unit velocity input is

$$e_{ss} = \frac{1}{K_v} = \frac{1}{1} = 1$$

The steady state error for unit acceleration input is

$$e_{ss} = \frac{1}{K_a} = \frac{1}{0} = \infty$$

The steady state error when all the above three inputs are simultaneously applied is given as

$$e_{ss} = 0 + 1 + \infty = \infty$$

$$\boxed{e_{ss} = \infty}$$

Example 2.29

Find the steady state error for a system with open loop T.F.

$$G(s) = \frac{10(s+2)(s+3)}{s(s+1)(s+5)(s+4)}$$
$$H(s) = 1$$

when the input is $r(t) = 3 + t + t^2$.

Solution

$$K_p = \lim_{s\to 0} \frac{10(s+2)(s+3)}{s(s+1)(s+5)(s+4)} = \infty$$
$$K_v = \lim_{s\to 0} \frac{s10(s+2)(s+3)}{s(s+1)(s+5)(s+4)} = 3$$
$$K_a = \lim_{s\to 0} \frac{s^2 10(s+2)(s+3)}{s(s+1)(s+5)(s+4)} = 0$$
$$e_{ss} = \frac{3}{1+K_p} + \frac{1}{K_v} + \frac{2}{K_a}$$
$$= \frac{3}{1+\infty} + \frac{1}{3} + \frac{2}{0}$$
$$= 0 + \frac{1}{3} + \infty$$

$$\boxed{e_{ss} = \infty}$$

Example 2.30

Consider a unity feedback system with a closed loop T.F.

$$\frac{C(s)}{R(s)} = \frac{(Ks+B)}{(s^2+As+B)}$$

Determine the open loop T.F. $G(s)$. Show that the steady state error with unit ramp input is given by $e_{ss} = (A - K)/B$.

(*Anna University, December, 2009*)

Solution

$$\begin{aligned}
\frac{C(s)}{R(s)} &= T(s) \\
&= \frac{G(s)}{1 + G(s)} \\
T(s) + TG(s) &= G(s) \\
G(s) &= \frac{T(s)}{1 - T(s)} \\
&= \frac{\dfrac{(Ks + B)}{s^2 + As + B}}{1 - \dfrac{(Ks + B)}{(s^2 + As + B)}} \\
&= \frac{(Ks + B)}{s^2 + As + B - Ks - B} \\
&= \frac{Ks + B}{s(s + A - K)} \\
K_v &= \lim_{s \to 0} sG(s) \\
&= \lim_{s \to 0} \frac{s(Ks + B)}{s(s + A - K)} \\
&= \frac{B}{A - K}
\end{aligned}$$

The steady state error for unit ramp input is

$$e_{ss} = \frac{1}{K_v}$$

$$\boxed{e_{ss} = \frac{(A - K)}{B}}$$

Example 2.31

For a unity feedback system having open loop T.F. as

$$G(s) = \frac{10(s + 2)}{s^2(s^2 + 2s + 12)}$$

determine the Type of the system, error constant, and the steady state error for parabolic input.

Solution

$$G(s) = \frac{10(s+2)}{s^2(s^2+2s+12)}$$

Since there are two pure integrators (number of poles at the origin is two), the system is Type 2. For Type 2 system, $K_p = \infty$ and $K_v = \infty$. For parabolic input,

$$K_a = \lim_{s\to 0} \frac{s^2 10(s+2)}{s^2(s^2+2s+12)} = \frac{5}{3}$$
$$e_{ss} = \frac{R}{K_a} = \frac{3}{5}$$

Type of the system = 2

$$\boxed{\begin{array}{cc} K_p = \infty; & K_v = \infty \\ K_a = \dfrac{5}{3}; & e_{ss} = \dfrac{3}{5} \end{array}}$$

Example 2.32

A certain unity feedback system when subjected to unit ramp input gives the following output response with zero initial conditions.

$$c(t) = t - 1 + e^{-t}\cos t$$

Find out the steady state error. Determine the forward path T.F. $G(s)$ of the system.

Solution

$$c(t) = t - 1 + e^{-t}\cos t$$
$$r(t) = t$$
$$e(t) = r(t) - c(t) = t - t + 1 - e^{-t}\cos t$$
$$e_{ss} = \lim_{t\to\infty} e(t)$$

$$\boxed{e_{ss} = 1}$$

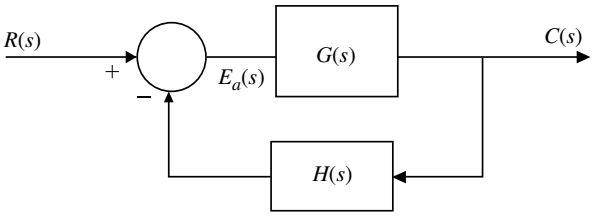


Fig. 2.30 Block diagram of non-unity feedback system

$$c(t) = t - 1 + e^{-t}\cos t$$

Taking Laplace transform on both sides we get

$$\begin{aligned} C(s) &= \frac{1}{s^2} - \frac{1}{s} + \frac{(s+1)}{(s+1)^2+1} \\ &= \frac{(s^2+2s+2) - s(s^2+2s+2) + s^2(s+1)}{s^2(s^2+2s+2)} \\ &= \frac{2}{s^2(s^2+2s+2)} \end{aligned}$$

For unit ramp input $R(s) = 1/s^2$

$$\frac{C(s)}{R(s)} = T(s) = \frac{2s^2}{s^2(s^2+2s+2)} = \frac{2}{(s^2+2s+2)}$$

$$G(s) = \frac{T(s)}{1-T(s)} = \frac{\frac{2}{(s^2+2s+2)}}{1-\frac{2}{s^2+2s+2}}$$

$$\boxed{G(s) = \frac{2}{s(s+2)}}$$

2.5.3 *Steady State Error for Non-unity Feedback Systems*

Consider the non-unity feedback control system shown in Fig. 2.30 where $G(s)$ is the forward path T.F. and $H(s)$ is the T.F. in the feedback path. Here the error is not the difference between the input and the output. The signal at the output of the summer is denoted as $E_a(s)$ which is the actuating signal which is different from the error signal $E(s)$ which was discussed in the previous section to determine the steady state error. Error signal is different from actuating signal and therefore $E_a(s)$

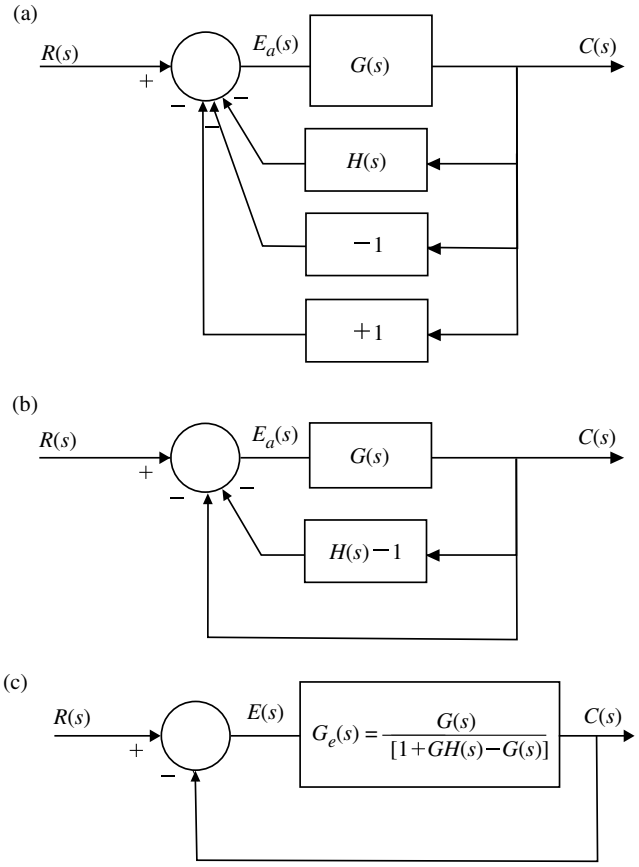


Fig. 2.31 Conversion of non-unity feedback system to unity feedback system

cannot be used in place of $E(s)$ in estimating the steady state error. Figure 2.30 is reduced to its equivalent as shown in Fig. 2.31.

The successive block diagram reduction shows that the equivalent forward path T.F. $G_e(s)$ as

$$G_e(s) = \frac{G(s)}{[1 + GH(s) - G(s)]} \tag{2.56}$$

For a non-unity feedback control system, Eq. (2.56) is used to determine the error constants and the steady state error. In this case the error is

$$\boxed{E(s) = \frac{R(s)}{1 + G_e(s)}}$$

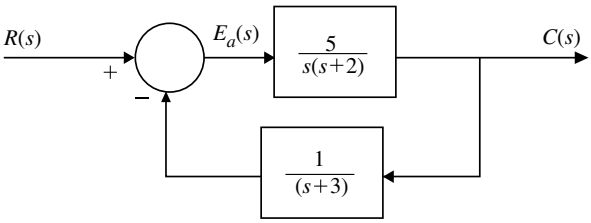


Fig. 2.32 Block diagram for Example 33

Substituting for $G_e(s)$ in the above equation we get

$$E(s) = \frac{R(s)}{1 + \frac{G(s)}{[1+GH(s)-G(s)]}}$$

$$\boxed{E(s) = \frac{R(s)[1 + GH(s) - G(s)]}{1 + GH(s)}} \tag{2.57}$$

Alternatively if the closed loop T.F. $T(s) = C(s)/R(s)$ is known, $G_e(s)$ is expressed as

$$\boxed{\begin{aligned} G_e(s) &= \frac{T(s)}{1 - T(s)} \\ E(s) &= R(s)[1 - T(s)] \end{aligned}} \tag{2.58}$$

Using Eq. (2.58) the static error constants can be calculated and the steady state error determined. For any type of signal, the steady state error is calculated using Eq. (2.58) for the non-unity feedback system. Here the error is defined as

$$\boxed{e(t) = r(t) - c(t)} \tag{2.59}$$

Calculation of steady state error using

$$E(s) = \frac{R(s)}{1 + GH(s)}$$

will give erroneous result. **This gives steady state actuating signal**.

Example 2.33

For the system shown in Fig. 2.32 find the system Type and steady state error for step size of 2 units. Also determine the steady state value of the actuating signal $e_a(\infty)$. Assume input and output values are same.

Solution The equivalent forward path T.F. is obtained using Eq. (2.56).

$$G(s) = \frac{5}{s(s+2)}$$

$$H(s) = \frac{1}{(s+3)}$$

$$G_e(s) = \frac{G(s)}{1 + G(s)H(s) - G(s)} = \frac{\dfrac{5}{s(s+2)}}{1 + \dfrac{5}{s(s+2)(s+3)} - \dfrac{5}{s(s+2)}}$$

$$= \frac{5(s+3)}{(s^3 + 5s^2 + s - 10)}$$

There is no pole of $G_e(s)$ in the origin. Hence, the system is Type 0 system and not Type 1 as seen at the first sight of Fig. 2.32. By applying appropriate stability conditions, it can be proved that the system is stable. Hence, final value theorem is applied to find the error constant.

$$K_p = \lim_{s \to 0} G_e(s) = \lim_{s \to 0} \frac{5(s+3)}{s^3 + 5s^2 + s - 10} = -1.5$$

$$e_{ss} = \frac{R}{1 + K_p} = \frac{2}{1 - 1.5} = -4$$

The negative value of e_{ss} implies that the output is larger than the input. The error constant for the actuating signal is

$$K_{pa} = \lim_{s \to 0} GH(s) = \lim_{s \to 0} \frac{5}{s(s+2)(s+3)} = \infty$$

The steady state actuating signal is

$$e_{ass} = \frac{R}{1 + K_{pa}} = \frac{2}{1 + \infty} = 0$$

Type 0 system

$$\boxed{\begin{aligned} e_{ss} &= -4 \\ e_{ass} &= 0 \end{aligned}}$$

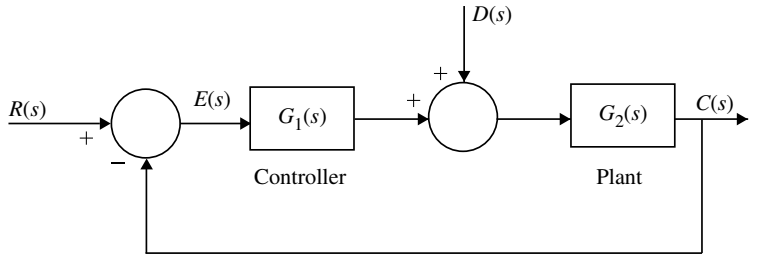


Fig. 2.33 Feedback system with disturbance

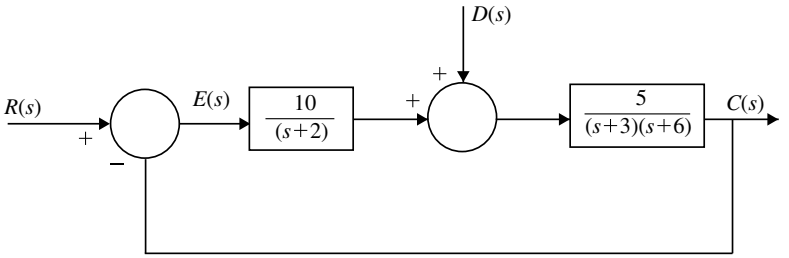


Fig. 2.34 Block diagram for Example 2.34

2.5.4 Steady State Error for Disturbances

Consider the system shown in Fig. 2.33 where $G_1(s)$ and $G_2(s)$ are controller and plant transfer functions respectively. The disturbance $D(s)$ occurs between the plant and the controller. From Fig. 2.33, the following equations are written:

$$C(s) = E(s)G_1(s)G_2(s) + D(s)G_2(s)$$

But $C(s) = R(s) - E(s)$

$$\begin{aligned} R(s) - E(s) &= E(s)G_1(s)G_2(s) + D(s)G_2(s) \\ E(s) &= \frac{R(s)}{[1 + G_1G_2(s)]} - \frac{D(s)G_2(s)}{[1 + G_1G_2(s)]} \end{aligned} \tag{2.60}$$

Example 2.34

Consider the system shown in Fig. 2.34. Find the steady state error due to the input $r(t) = 3u(t)$ and the step disturbance $d(t) = 2u(t)$.

Solution

$$G_1(s) = \frac{10}{(s+2)}$$
$$G_2(s) = \frac{5}{(s+3)(s+6)}$$

where

$$R(s) = \frac{3}{s}; \qquad D(s) = \frac{2}{s}$$

Equation (2.60) can be written as

$$E(s) = E_1(s) + E_2(s)$$

where

$$E_1(s) = \frac{R(s)}{1 + G_1G_2(s)}$$
$$E_2(s) = -\frac{D(s)G_2(s)}{1 + G_1G_2(s)}$$
$$E_1(s) = \frac{3}{s\left[1 + \dfrac{50}{(s+2)(s+3)(s+6)}\right]}$$
$$= \frac{3(s+2)(s+3)(s+6)}{s(s^3 + 11s^2 + 36s + 86)}$$
$$e_{ss1} = \lim_{s\to 0} sE_1(s)$$
$$= \frac{3 \times 2 \times 3 \times 6}{86} = 1.26$$
$$E_2(s) = \frac{-2 \times 5(s+2)}{s[s^3 + 11s^2 + 36s + 86]}$$
$$e_{ss2} = \lim_{s\to 0} sE_2(s)$$
$$= \frac{-2 \times 5 \times 2}{86}$$
$$= -0.23$$
$$e_{ss} = e_{ss1} + e_{ss2}$$
$$= 1.26 - 0.23$$

$$\boxed{e_{ss} = 1.03}$$

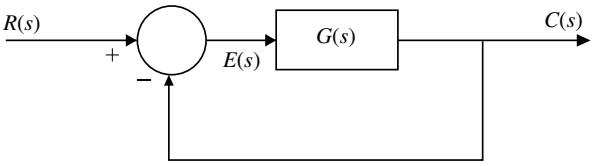


Fig. 2.35 System configuration to determine the generalized error coefficients

2.5.5 The Generalized Error Constants

The steady state errors estimated in Sect. 2.5.3 using static error constants have the following limitations:

1. The steady state errors are estimated using static error constants K_p, K_v and K_a when the input signals applied are step, velocity and parabolic. Such constants are not available for other type of signals. The steady state errors, in these cases to be estimated from $E(s)$ by applying final value theorem, require system stability as the pre requisite.
2. The steady state error estimated using static error constants gives the value of e_{ss} as finite or zero or infinite. No information is known about the accumulation of error with respect to time.

To overcome the above two limitations, generalized error constants or dynamic error constants are used. As stated in Sect. 2.5.3, the error is a function of the input and the system configuration $G(s)$ and $H(s)$. A set of coefficients are derived which are constants but depends upon $G(s)$ and $H(s)$. However, the input is treated as a function of time. The inputs and their derivatives together with the generalized error constants give the error as a function of time and at any time the error can be estimated including $t = \infty$ which gives the steady state error e_{ss}. For establishing dynamic error coefficients the system configuration shown in Fig. 2.35 is used.

From Fig. 2.35, the error $E(s)$ is given as

$$E(s) = \frac{R(s)}{(1 + G(s))} \tag{2.61}$$

$$= R(s)F(s) \tag{2.62}$$

where

$$F(s) = \frac{1}{1 + G(s)} \tag{2.63}$$

From the Laplace transform property of convolution, the following equation is written:

$$e(t) = r(t) * f(t) \tag{2.64}$$

where $e(t)$, $r(t)$ and $f(t)$ are the inverse Laplace transform of $E(s)$, $R(s)$ and $F(s)$ respectively.

Equation (2.64) can be written as

$$e(t) = \int_0^t f(\tau) r(t-\tau) d\tau \tag{2.65}$$

But $r(t-\tau)$ can be expanded in Taylor's series as

$$r(t-\tau) = r(t) - \tau \dot{r}(t) + \frac{\tau^2}{\angle 2} \ddot{r}(t) - \frac{\tau^3}{\angle 3} \dddot{r}(t) + \cdots \tag{2.66}$$

where $\dot{r}(t)$, $\ddot{r}(t)$ *etc.*, are the first, second derivatives of $r(t)$.

Substituting Equation (2.66) in Eq. (2.65), we get

$$\begin{aligned} e(t) &= \int_0^t f(\tau) \left[r(t) - \tau \dot{r}(t) + \frac{\tau^2}{\angle 2} \ddot{r}(t) \cdots \right] d\tau \\ &= \int_0^t f(\tau) r(t) d\tau - \int_o^t \tau \dot{r}(t) f(\tau) d\tau + \cdots \end{aligned}$$

In the above integrations $r(t)$, $\dot{r}(t)$, $\ddot{r}(t)$ *etc.* are constants when integration is done with respect to τ and they are taken out. Hence

$$e(t) = r(t) \int_0^t f(\tau) d\tau - \dot{r}(t) \int_0^t \tau f(\tau) d\tau + \frac{\ddot{r}(t)}{\angle 2} \int_0^t \tau^2 f(\tau) d\tau + \tag{2.67}$$

$$= \left[C_0 r(t) + C_1 \dot{r}(t) + \frac{C_2}{\angle 2} \ddot{r}(t) + \cdots + \frac{C_n}{\angle n} n \dot{r} \right] \tag{2.68}$$

where

$$\begin{aligned} C_0 &= \int_0^t f(\tau) d\tau \\ C_1 &= -\int_0^t f(\tau) d\tau \\ C_2 &= \int_0^t \tau^2 f(\tau) d\tau \\ C_n &= (-1)^n \int_0^t \tau^n f(\tau) d\tau \end{aligned} \tag{2.69}$$

The constants $C_0, C_1, C_2, \ldots, C_n$ in Eq. (2.68) are called generalized or dynamic error coefficients and are defined in Eq. (2.69). However, determination of these constants using the integrals given in Eq. (2.69) is somewhat tedious. Hence, the procedure is simplified as follows. Since we are interested in the steady state error, the upper limit of integration in Eq. (2.69) becomes $t = \infty$. Thus, for example,

$$C_0 = \int_0^\infty f(\tau)d\tau$$

Now consider the Laplace transform of $F(s)$ which is given as

$$F(s) = \int_0^\infty f(\tau)e^{-s\tau}d\tau \tag{2.70}$$

Taking $\lim_{s\to 0}$ on both sides, we get

$$\begin{aligned}\lim_{s\to 0} F(s) &= \lim_{s\to 0}\int_0^\infty f(\tau)e^{-s\tau}d\tau \\ &= \int_0^\infty f(\tau)d\tau \\ &= C_0 \\ C_0 &= \lim_{s\to 0} F(s)\end{aligned}$$

Differentiating Equation (2.70), we get

$$\begin{aligned}\frac{dF(s)}{ds} &= -\int_0^\infty f(\tau)\tau e^{-s\tau}d\tau \\ \lim_{s\to 0}\frac{dF(s)}{ds} &= -\int_0^\infty \tau f(\tau)d\tau \\ &= C_1 \\ C_1 &= \lim_{s\to 0}\frac{dF(s)}{ds}\end{aligned}$$

The procedure is repeated, and we get

$$\begin{aligned}C_2 &= \lim_{s\to 0}\frac{d^2F(s)}{ds^2} \\ C_n &= \lim_{s\to 0}\frac{d^nF(s)}{ds^n}\end{aligned}$$

Thus, the generalized error constants are estimated from

$$\boxed{\begin{aligned}C_0 &= \lim_{s\to 0} F(s) \\ C_1 &= \lim_{s\to 0}\frac{dF(s)}{ds} \\ C_2 &= \lim_{s\to 0}\frac{d^2F(s)}{ds^2} \\ C_n &= \lim_{s\to 0}\frac{d^nF(s)}{ds^n}\end{aligned}} \tag{2.71}$$

where

$$F(s) = \frac{1}{1 + G(s)}$$

Example 2.35

The forward path T.F. of a certain unity negative feedback control system is given as

$$G(s) = \frac{5}{(s + 5)}$$

Calculate the generalized error co-efficients and the steady state error for the input $r(t) = 6 + 5t$.

(*Anna University, May, 2007*)

Solution

$$\begin{aligned} r(t) &= 6 + 5t \\ \dot{r}(t) &= 5 \\ \ddot{r}(t) &= 0 \end{aligned}$$

For the above input, since $\ddot{r}(t)$ is zero, the error series is

$$\begin{aligned} e(t) &= C_0 r(t) + C_1 \dot{r}(t) \\ F(s) &= \frac{1}{1 + G(s)} \\ &= \frac{1}{1 + \frac{5}{s+5}} \\ &= \frac{(s + 5)}{(s + 10)} \\ C_0 &= \lim_{s \to 0} F(s) \\ &= \lim_{s \to 0} \frac{(s + 5)}{(s + 10)} \\ &= 0.5 \\ \frac{dF(s)}{ds} &= \frac{(s + 10) - (s + 5)}{(s + 10)^2} \\ \frac{dF(s)}{ds} &= \frac{5}{(s + 10)^2} \\ C_1 &= \lim_{s \to 0} \frac{dF(s)}{ds} \end{aligned}$$

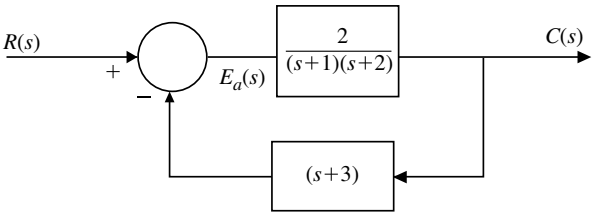


Fig. 2.36 Block diagram for Example 36

$$
\begin{aligned}
&= \lim_{s\to 0} \frac{5}{(s+10)^2} \\
C_1 &= \frac{1}{20} \\
e(t) &= 0.5(6+5t) + \frac{5}{20} \\
&= \frac{13}{4} + 2.5t \\
e_{ss} &= \lim_{t\to\infty} e(t) \\
&= \lim_{t\to\infty} \frac{13}{4} + 2.5t \\
&= \infty
\end{aligned}
$$

$$\boxed{e_{ss} = \infty}$$

Example 2.36

Consider the system shown in Fig. 2.36. The forward path T.F.

$$G(s) = \frac{2}{(s+1)(s+2)}$$

The feedback path T.F. $H(s) = (s+3)$. The input applied is $r(t) = 2 + 3t$. Determine the generalized error constants and hence the steady state error $e(t) = r(t) - c(t)$.

Solution

$$
\begin{aligned}
G(s) &= \frac{2}{(s+1)(s+2)} \\
H(s) &= (s+3)
\end{aligned}
$$

The system is not a unity feedback system. Hence, using Eq. (2.56), the equivalent forward path T.F. is obtained as

$$G_e(s) = \frac{G(s)}{[1 + GH(s) - G(s)]}$$

$$= \frac{\dfrac{2}{(s+1)(s+2)}}{\left[1 + \dfrac{2(s+3)}{(s+1)(s+2)} - \dfrac{2}{(s+1)(s+2)}\right]}$$

$$G_{eq} = \frac{2}{s^2 + 5s + 6}$$

The block diagram representation of the equivalence of Fig. 2.36 is shown in Fig. 2.37.

From Fig. 2.37,

$$F(s) = \frac{1}{1 + G_e(s)}$$

$$= \frac{2}{1 + \dfrac{2}{s^2 + 5s + 6}}$$

$$= \frac{2(s^2 + 5s + 6)}{(s^2 + 5s + 8)}$$

The input

$$r(t) = 2 + 3t$$

$$\dot{r}(t) = 3$$

$$\ddot{r}(t) = 0$$

$$e(t) = C_0 r(t) + C_1 \dot{r}(t)$$

$$C_0 = \lim_{s \to 0} \frac{2(s^2 + 5s + 6)}{(s^2 + 5s + 8)} = \frac{3}{2}$$

$$\frac{dF(s)}{ds} = \frac{(s^2 + 5s + 8)(4s + 10) - 2(s^2 + 5s + 6)(2s + 5)}{(s^2 + 5s + 8)^2}$$

$$C_1 = \lim_{s \to 0} \frac{dF(s)}{ds}$$

$$= \frac{8 \times 10 - 2 \times 6 \times 5}{64}$$

$$= \frac{80 - 60}{64} = \frac{5}{16}$$

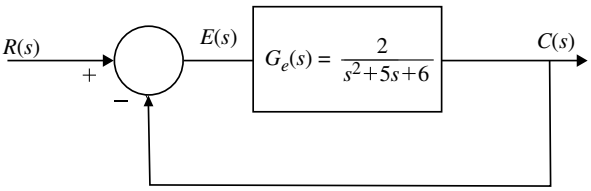


Fig. 2.37 Equivalent block diagram of Fig. 2.36

$$e(t) = \frac{3}{2}(2+3t) + \frac{5}{16} \times 3$$
$$e(t) = 3.9375 + 4.5t$$
$$e_{ss} = \lim_{t \to 0} e(t)$$

$$\boxed{e_{ss} = \infty}$$

Example 2.37

A certain unity negative feedback control system has the following forward path T.F.

$$G(s) = \frac{K(s+a)}{(s+2)(s+b)}$$

The system is subjected to step input of size 5. It is desired to have the closed loop poles at $s = -2 \pm j\sqrt{12}$. Determine the value of K, a and b if the permissible steady state error $e_{ss} = 1$.

Solution

$$G(s) = \frac{K(s+a)}{(s+2)(s+b)}$$
$$K_p = \lim_{s \to 0} G(s) = \frac{Ka}{2b}$$
$$e_{ss} = \frac{R}{1+K_p} = \frac{5}{1+(Ka/2b)}$$
$$1 = \frac{10b}{2b+Ka}$$
$$2b + Ka = 10b$$
$$Ka = 8b \qquad \text{(a)}$$

For the given $G(s)$, the closed loop T.F. is

$$\frac{C(s)}{R(s)} = \frac{\dfrac{K(s+a)}{(s+2)(s+b)}}{1+\dfrac{K(s+a)}{(s+2)(s+b)}} = \frac{K(s+a)}{(s+2)(s+b)+K(s+a)} \tag{b}$$

The characteristics equation of Eq. (b) is

$$s^2 + (2+b+K)s + (2b+Ka) = 0 \tag{c}$$

The desired roots are $s_1 = -2 + j\sqrt{12}$ and $s_2 = -2 - j\sqrt{12}$. The desired characteristics equation is

$$(s+2+j\sqrt{12})(s+2-j\sqrt{12}) = 0$$
$$s^2 + 4s + 16 = 0 \tag{d}$$

Comparing Eqs. (c) and (d), we get

$$(2+b+K) = 4$$
$$b = 2 - K$$

and

$$2b + Ka = 16 \tag{e}$$

Substituting $Ka = 8b$ from Eq. (a), we get

$$2b + 8b = 16$$
$$10b = 16$$
$$b = 1.6$$
$$K = 2 - b = 2 - 1.6 = 0.4$$
$$a = \frac{8b}{K} = \frac{8 \times 1.6}{0.4} = 32$$

$$\boxed{\begin{aligned} K &= 0.4 \\ a &= 32 \\ b &= 1.6 \end{aligned}}$$

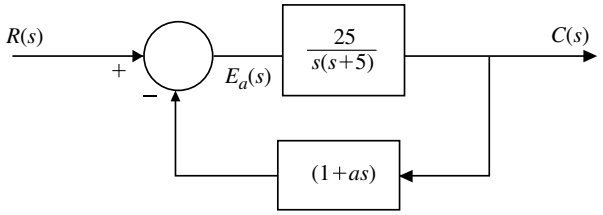


Fig. 2.38 A non-unity feedback system for Example 38

Example 2.38

Consider the non-unity feedback system shown in Fig. 2.38. Determine the value of a so that the peak overshoot for step input is 5%. For this value of a determine the type of the system and the steady state error for a velocity input of slope 4. Verify the results using dynamic error constants.

$$e(t) = r(t) - c(t)$$

Solution

$$\begin{aligned} G(s) &= \frac{25}{s(s+5)} \\ H(s) &= (1+as) \\ \frac{C(s)}{R(s)} &= \frac{G(s)}{1+GH(s)} \\ &= \frac{25/(s(s+5))}{\left[1+\dfrac{25(1+as)}{s(s+5)}\right]} \\ &= \frac{25}{s^2+(5+25a)s+25} \end{aligned}$$

The characteristic equation is

$$s^2 + (5+25a)s + 25 = 0$$

The required overshoot is related as

$$e^{\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}} = 0.05$$

Taking $\log_e$ on both sides, we get

$$\frac{-\zeta\pi}{\sqrt{1-\zeta^2}} = \log_e 0.05 = -3$$

$$\boxed{\zeta = 0.69}$$

From the characteristic equation,

$$\begin{aligned}\omega_n &= \sqrt{25} = 5\\ 2\zeta\omega_n &= 5 + 25a\\ a &= \frac{2\zeta\omega_n - 5}{25}\end{aligned}$$

$$\boxed{a = 0.076}$$

Thus

$$\begin{aligned}G(s) &= \frac{25}{s(s+5)}\\ H(s) &= (1 + 0.076s)\end{aligned}$$

Using Eq. (2.56), we get

$$G_e(s) = \frac{G(s)}{1 + GH(s) - G(s)}$$

$$\begin{aligned}GH(s) &= \frac{25(1+0.076s)}{s(s+5)}\\ G_e(s) &= \frac{\frac{25}{s(s+5)}}{1 + \frac{25(1+0.076s)}{s(s+5)} - \frac{25}{s(s+5)}}\\ &= \frac{25}{s^2 + 5s + 1.9s + 25 - 25}\\ &= \frac{25}{s(s+6.9)}\end{aligned}$$

This is Type 1 system since there is one integrator in the open loop (forward path) T.F. For ramp input,

$$\begin{aligned} K_v &= \lim_{s\to 0} sG_e(s) \\ &= \lim_{s\to 0} \frac{s \times 25}{s(s+6.9)} \\ &= \frac{25}{6.9} \end{aligned}$$

Steady state error for ramp input of slope 4,

$$e_{ss} = \frac{4}{Kv} = \frac{4 \times 6.9}{25}$$

$$\boxed{e_{ss} = 1.1 \text{ rad/s}}$$

Verification using Dynamic Error Constants

$$\begin{aligned} r(t) &= 4t \\ \dot{r}(t) &= 4 \end{aligned}$$

Hence, the dynamic error constants C_0 and C_1 are to be determined.

$$\begin{aligned} F(s) &= \frac{1}{1+G_e(s)} \\ &= \frac{1}{1+\frac{25}{s(s+6.9)}} \\ &= \frac{s(s+6.9)}{s^2+6.9s+25} \\ C_0 &= \lim_{s\to 0} F(s) \\ &= \lim_{s\to 0} \frac{(s^2+6.9s)}{s^2+6.9s+25} = 0 \end{aligned}$$

$$\frac{dF(s)}{ds} = \frac{(s^2+6.9s+25)(2s+6.9) - (s^2+6.9s)(2s+6.9)}{(s^2+6.9s+25)^2}$$

$$\begin{aligned} C_1 &= \lim_{s\to 0} \frac{dF(s)}{ds} \\ &= \frac{25 \times 6.9}{(25)^2} = \frac{6.9}{25} \\ e(t) &= C_1 \dot{r}(t) \\ &= \frac{4 \times 6.9}{25} \end{aligned}$$

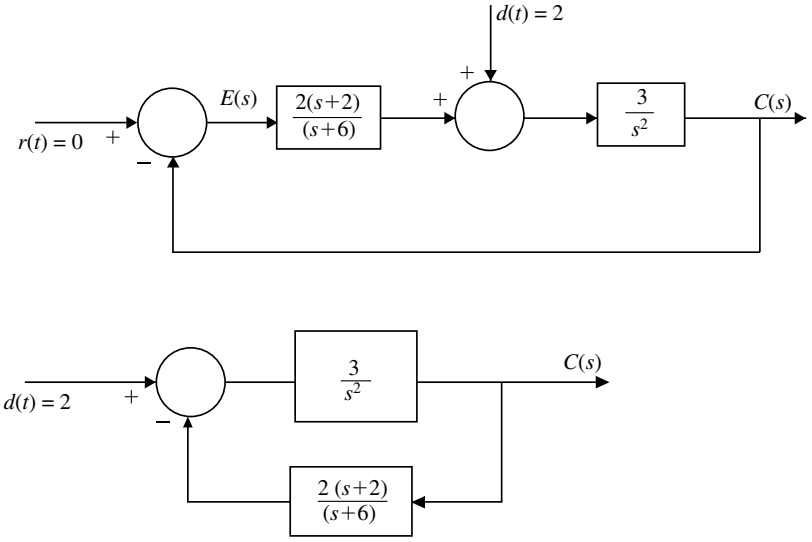


Fig. 2.39 **a** Block diagram for Example 39. **b** Reduced block diagram of Fig. 2.39a

$$\boxed{e_{ss} = 1.1}$$

Thus, the result is verified using dynamic error constants.

Example 2.39

Consider the system shown in Fig. 2.39a. Determine the steady state error for $r(t) = 0$ and $d(t) = 2$.

Solution From Fig. 2.39a when $r(t) = 0$, $E(s) = -C(s)$. The block diagram is redrawn as shown in Fig. 2.39b.

From Fig. 2.39b, $C(s)/D(s)$ can be written as

$$\frac{C(s)}{D(s)} = \frac{3/s^2}{1 + \frac{3}{s^2}\frac{2(s+2)}{(s+6)}} = \frac{3(s+6)}{[s^2(s+6) + 6(s+2)]}$$

$$D(s) = \frac{2}{s}$$

$$C(s) = \frac{2 \times 3(s+6)}{s[s^2(s+6) + 6(s+2)]}$$

The steady value of the output is obtained by using final value theorem

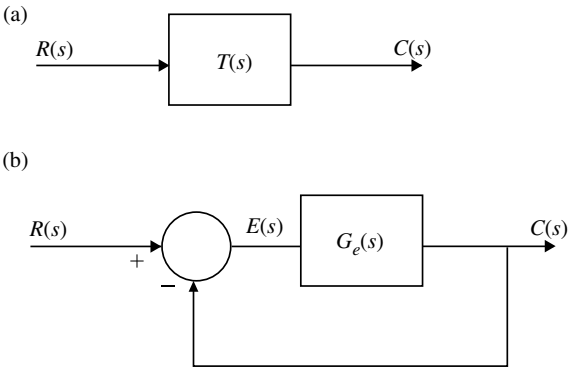


Fig. 2.40 System representation. **a** With closed loop T.F. $T(s)$; and **b** Closed loop system with forward path T.F

$$C_{ss} = \lim_{s\to 0} sC(s) = \lim_{s\to 0} \frac{6s(s+6)}{s[s^2(s+6)+6(s+2)]} = 3$$
$$e_{ss} = -C_{ss}$$

$$\boxed{e_{ss} = -3}$$

2.5.6 *Steady State Error When Closed Loop T.F. is Given*

Consider the system shown in Fig. 2.40a and b. From these figures, the following equations are written:

$$\begin{aligned} \frac{C(s)}{R(s)} &= T(s) \\ C(s) &= R(s)T(s) \\ E(s) &= R(s) - C(s) = R(s) - R(s)T(s) \end{aligned} \tag{2.72}$$

$$\boxed{E(s) = [1 - T(s)]R(s)} \tag{2.73}$$

From Eq. (2.73) steady state error can be determined using final value theorem provided the system is stable. Alternatively $G_e(s)$ from Eq. (2.73) is obtained as

$$\boxed{G_e(s) = \frac{T(s)}{1 - T(s)}} \tag{2.74}$$

Using Eq. (2.74) error constants and hence e_{ss} can be determined.

Example 2.40

The closed loop T.F. of a certain unity feedback control system is given as

$$T(s) = \frac{5}{s^2 + 3s + 5}$$

Find the steady state error for a ramp input of slope 5 rad/s.

Solution

$$T(s) = \frac{5}{s^2 + 3s + 5}$$

$$1 - T(s) = \frac{(s^2 + 3s)}{s^2 + 3s + 5}$$

$$E(s) = [1 - T(s)]R(s) = \frac{5}{s^2}\frac{[s^2 + 3s]}{[s^2 + 3s + 5]}$$

$$e_{ss} = \lim_{s\to 0} sE(s) = \lim_{s\to 0} \frac{5s^2}{s^2}\frac{[s + 3]}{(s^2 + 3s + 5)}$$

$$\boxed{e_{ss} = 3}$$

Alternatively, from Eq. (2.74), $G_e(s)$ determined.

$$G_e(s) = \frac{T(s)}{1 - T(s)} = \frac{5/((s^2 + 3s + 5))}{(s^2 + 3s)/(s^2 + 3s + 5)}$$

$$= \frac{5}{s(s + 3)}$$

$$K_v = \lim_{s\to 0} sG_e(s) = \lim_{s\to 0} \frac{5}{s + 3} = \frac{5}{3}$$

$$e_{ss} = \frac{R}{K_v} = \frac{5}{5} \times 3$$

$$\boxed{e_{ss} = 3}$$

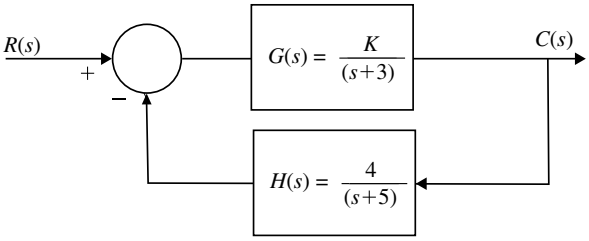


Fig. 2.41 Block diagram for Example 41

Example 2.41

Consider the system shown in Fig. 2.41. Find the value of K so that the steady state error is zero for step input. Find the type of the system. For $K = 15$, find e_{ss} for ramp input. $e(t) = r(t) - c(t)$.

Solution

$$G(s) = \frac{K}{(s+3)}$$

$$H(s) = \frac{4}{(s+5)}$$

$$GH(s) = \frac{4K}{(s+3)(s+5)}$$

$$G_e(s) = \frac{G(s)}{1 + GH(s) - G(s)}$$

$$G_e(s) = \frac{K/s + 3}{1 + \frac{4K}{(s+3)(s+5)} - \frac{K}{s+3}}$$

$$= \frac{K(s+5)}{s^2 + 8s + 15 + 4K - Ks - 5K}$$

For a step input, the position error constant is

$$K_p = \lim_{s \to 0} G_e(s) = \frac{K \times 5}{15 - K}$$

$$e_{ss} = \frac{R}{1 + K_p} = \frac{R}{1 + \frac{5K}{15-K}}$$

$$= \frac{R[15 - K]}{15 + 4K}$$

For e_{ss} to be zero $K = 15$

$$\boxed{K = 15}$$

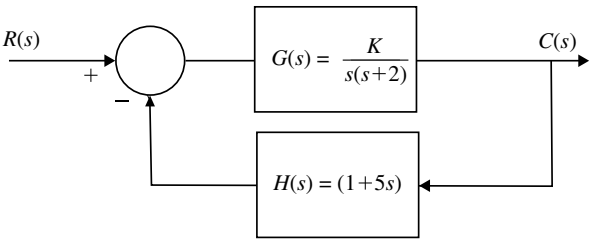


Fig. 2.42 Block diagram for Example 42

For $K = 15$,

$$G_e(s) = \frac{15(s+5)}{s(s-7)}$$

This is Type 1 system,

$$K_v = \lim_{s\to 0} G_e(s) = \frac{-75}{7}$$

$$e_{ss} = \frac{1}{K_v}$$

$$\boxed{e_{ss} = \frac{-7}{75}}$$

Example 2.42

Consider the system shown in Fig. 2.42. Determine the value of K so that the steady state error is 6, for a ramp input of slope 1. Use $e(t) = r(t) - c(t)$.

Solution From Fig. 2.42, the closed loop T.F. is

$$T(s) = \frac{C(s)}{R(s)} = \frac{G(s)}{1 + GH(s)}$$

$$= \frac{\frac{K}{s(s+2)}}{1 + \frac{K(1+5s)}{s(s+2)}}$$

$$= \frac{K}{s(s+2) + 5Ks + K} = \frac{K}{s^2 + (2+5K)s + K}$$

From Eq. (2.73), the error function is written as

$$E(s) = R(s)[1 - T(s)]$$

$$= \frac{1}{s^2}\left[1 - \frac{K}{[s^2 + (2+5K)s + K]}\right]$$

$$E(s) = \frac{1}{s^2}\left[\frac{s^2 + (2+5K)s}{[s^2 + (2+5K)s + K]}\right]$$
$$e_{ss} = \lim_{s\to 0} sE(s)$$
$$e_{ss} = \lim_{s\to 0} \frac{[s + (2+5K)]}{[s^2 + (2+5K)s + K]} = \frac{2+5K}{K} = 6$$
$$2 + 5K = 6K$$

$$\boxed{K = 2}$$

Alternative Method

$$G(s) = \frac{K}{s(s+2)}$$
$$H(s) = (1+5s)$$
$$GH(s) = \frac{K(1+5s)}{s(s+2)}$$
$$G_e(s) = \frac{G(s)}{1 + GH(s) - G(s)}$$
$$= \frac{\frac{K}{s(s+2)}}{1 + \frac{K(1+5s)}{s(s+2)} - \frac{K}{s(s+2)}}$$
$$= \frac{K}{s^2 + 2s + 5Ks}$$
$$K_v = \lim_{s\to 0} sG_e(s)$$
$$= \frac{K}{2+5K}$$
$$e_{ss} = \frac{R}{K_v} \quad \text{where } R = 1$$
$$6 = \frac{2+5K}{K}$$
$$6K - 5K = 2$$

$$\boxed{K = 2}$$

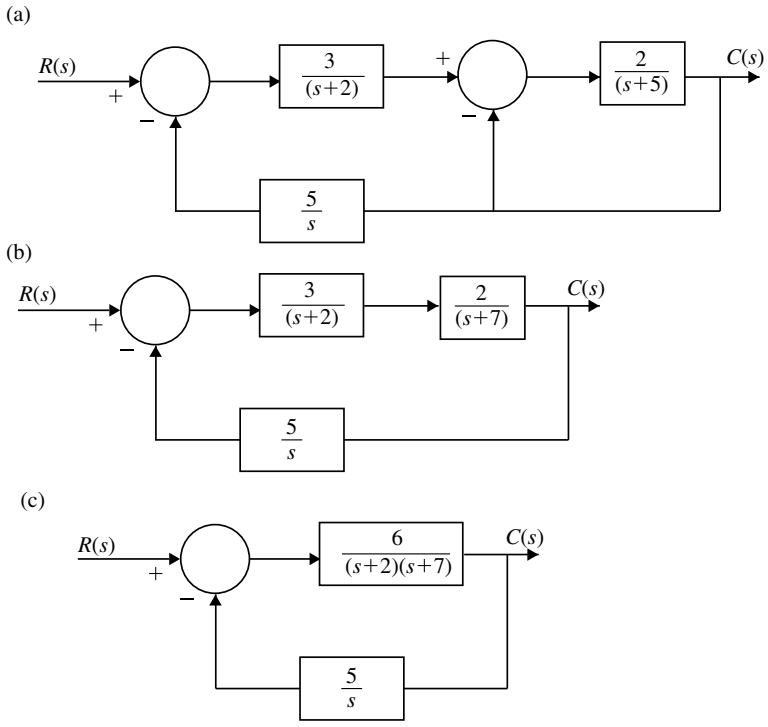


Fig. 2.43 **a** Block diagram for Example 2.43. Reduced block diagram for Fig. 2.43a

Example 2.43

For the system shown in Fig. 2.43a, find the steady state error for unit step input. $e(t) = r(t) - c(t)$.

Solution The given block diagram is shown in Fig. 2.43a, and $T(s) = C(s)/R(s)$ is obtained. The successive block diagram reduction is shown in Fig. 2.43b and c.

From Fig. 2.43c, the closed loop T.F. is obtained as

$$T(s) = \frac{C(s)}{R(s)} = \frac{6/[(s+2)(s+7)]}{1 + [30/(s(s+2)(s+7))]} = \frac{6s}{s^3 + 9s^2 + 14s + 30}$$

$T(s)$ is strictly proper and stable and hence Laplace final value theorem can be applied.

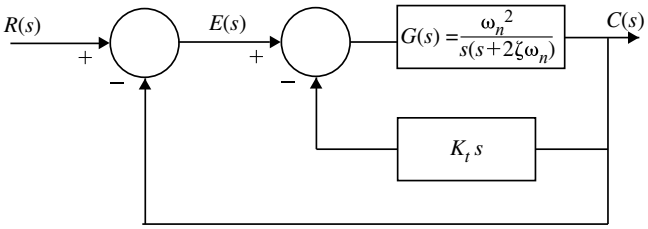


Fig. 2.44 Rate or tachometer feedback controller

$$E(s) = [1 - T(s)]R(s)$$
$$= \left[\frac{s^3 + 9s^2 + 8s + 30}{s^3 + 9s^2 + 14s + 30}\right]\frac{1}{s}$$

$$e_{ss} = \lim_{s \to 0} sE(s)$$

$$\boxed{e_{ss} = 1}$$

2.6 Performance Enhancement By Using Controllers

The transient and steady state responses of a second order system for different types of inputs were discussed in the previous sections. Formulae were derived for transient and steady state performance specifications for typical test signals such as step, velocity and acceleration. However, for a given system, since the damping ratio and natural frequency of oscillations are fixed, the only option to enhance the performance is by adjusting the open loop gain which is called the D.C. gain. By adjusting the D.C. gain, only very little improvement can be obtained in the transient and steady state performance. A viable option is to provide a controller either in the forward path or in the feedback path. Each method has its own advantages and limitations. The commonly used controllers are

1. Rate controller or tachometer feedback controller,
2. PID controller.

These controllers? performances are discussed below when they are connected to a second order system.

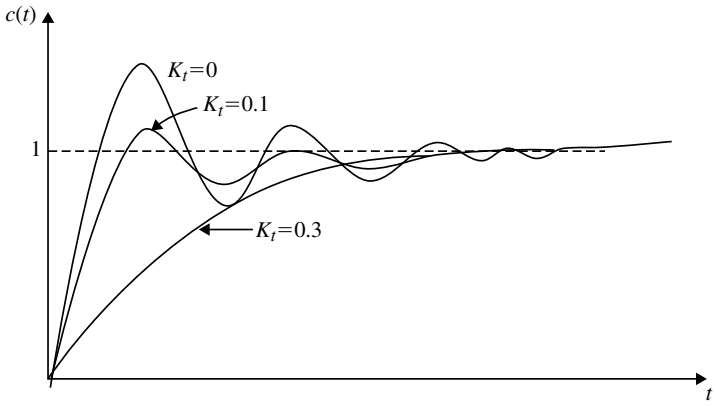


Fig. 2.45 Unit step response curve for different values of error rate damping

2.6.1 Rate Controller or Tachometer Feedback Controller

A second order system whose open loop T.F.

$$G(s) = \frac{\omega_n^2}{s(s + 2\zeta\omega_n)}$$

is taken for the study of transient and steady state performances using controllers. The second order system with rate feedback controller is shown in Fig. 2.44. Here, the derivative of the output signal is fed back and added to the actuating signal in the miner feedback loop. This rate feedback is also called tachometer feedback or velocity feedback. For example, in a mechanical system, if the output is mechanical position, it is differentiated using a tachometer which produces voltage that is proportional to the velocity of the mechanical system. The proportionality constant K_t has the units of volts/rad/s or volts/metre/s.

From Fig. 2.44, the following equation is obtained.

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{[s^2 + (2\zeta\omega_n + K_t\omega_n^2)s + \omega_n^2]} \tag{2.75}$$

The characteristic equation of the closed loop system is

$$s^2 + (2\zeta\omega_n + K_t\omega_n^2)s + \omega_n^2 = 0 \tag{2.76}$$

From Eq. (2.76) the damping ratio of the system is

$$\zeta' = \zeta + \frac{K_t}{2}\omega_n \tag{2.77}$$

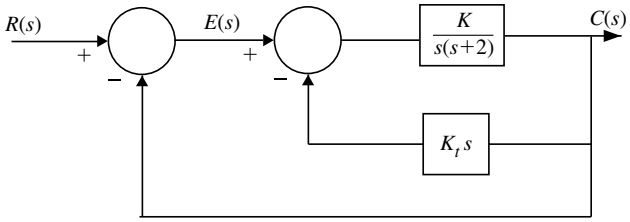


Fig. 2.46 Block diagram for Example 2.44

From Eq. (2.77) it is evident that the effect of the tachometer feedback is to increase the damping of the system by $(K_t/2)\omega_n$. Since ω_n is fixed for the system, by varying K_t the effective damping can be varied for a wide range. Further from Fig. 2.44 the error function is derived as

$$E(s) = \frac{R(s)s[s + 2\zeta\omega_n + K_t\omega_n^2]}{s^2 + (2\zeta\omega_n + K_t\omega_n^2)s + \omega_n^2} \tag{2.78}$$

From Eq. (2.78), the steady state error for step input is zero. However, for the ramp input, the steady state error is

$$e_{ss} = \frac{R}{\omega_n}[2\zeta + K_t\omega_n] \tag{2.79}$$

Without rate feedback the steady error is $2\zeta R/\omega_n$. Hence, by rate feedback the steady state error is increased by RK_t. **Thus, by rate feedback, the effective damping can be improved but it enlarges the steady state error for ramp input**. The time response curves for unit step input with $\zeta = 0.5$ and $\omega_n = 4$ rad/s are shown for $K_t = 0$, 0.1 and 0.3 in Fig. 2.45.

Example 2.44

A certain second order system with rate feedback in the minor loop is shown in Fig. 2.46. The overshoot for a step input of height 2 units should not exceed 0.2. The steady state error for ramp input of slope 4 should be less than 0.5. Find the value of K and K_t. Compare the transient response specifications such as % overshoot, rise time, settling time for step input and e_{ss} for ramp input without rate feedback and with rate feedback.

Solution The required overshoot of the rate feedback system is

$$2e^{\frac{-\zeta_2\pi}{\sqrt{1-\zeta_2^2}}} = 0.2$$

where ζ_2 is the damping ratio of the system with rate feedback.

$$e^{\frac{-\zeta_2\pi}{\sqrt{1-\zeta_2^2}}} = 0.1$$

Taking $\log_e$ on both sides we get

$$\frac{-\zeta_2\pi}{\sqrt{1-\zeta_2^2}} = -2.303$$
$$\zeta_2 = 0.591$$

Let ζ_1 be the damping ratio without rate feedback. Then

$$\frac{C(s)}{R(s)} = \frac{K}{[s^2+2s+K]}$$

From the above equation we get $\omega_n = \sqrt{K}$ and

$$2\zeta_1\omega_n = 2 \quad \text{or} \quad \zeta_1 = \frac{1}{\sqrt{K}}$$

The natural frequency ω_n will remain the same for the system with and without rate feedback. From Eq. (2.79) the steady state error for ramp input of slope 4 is

$$\begin{aligned} e_{ss} &= \frac{4}{\omega_n}[2\zeta_1 + K_t\omega_n] \\ 0.5 &= \frac{4}{\sqrt{K}}\left[\frac{2}{\sqrt{K}} + K_t\sqrt{K}\right] \\ &= \frac{4}{K}[2 + K_tK] \qquad \text{(a)} \\ 0.125K &= [2 + K_tK] \end{aligned}$$

From Eq. (2.77)

$$\begin{aligned} \zeta_2 &= \zeta_1 + \frac{K_t}{2}\omega_n \\ 0.591 &= \frac{1}{\sqrt{K}} + \frac{K_t}{2}\sqrt{K} \\ 2 \times 0.591\sqrt{K} &= 2 + K_tK \\ 1.182\sqrt{K} &= 2 + K_tK \qquad \text{(b)} \end{aligned}$$

From Eq. (a) we get

$$0.125K = 2 + K_t K$$

Substituting in Eq. (b) we get

$$1.182\sqrt{K} = 0.125K$$

$$\boxed{K = 89.42}$$

From Eq. (b) we get after substituting for K,

$$1.182\sqrt{89.42} = 2 + K_t 89.42$$

$$\boxed{K_t = 0.103}$$

$$\zeta_1 = \frac{1}{\sqrt{K}} = \frac{1}{89.42} = 0.106$$

The closed loop T.F. of the rate feedback system from Eq. (2.75) is

$$\begin{aligned}\frac{C(s)}{R(s)} &= \frac{\omega_n^2}{[s^2 + (2\zeta_1\omega_n + K_t\omega_n^2)s + \omega_n^2]} \\ &= \frac{89.42}{[s^2 + (2 \times 0.106 \times 9.456 + 0.103 \times 89.42)s + 89.42]} \\ &= \frac{89.42}{[s^2 + 11.2s + 89.42]}\end{aligned} \tag{c}$$

The closed loop T.F. of the system without rate feedback is

$$\frac{C(s)}{R(s)} = \frac{89.42}{[s^2 + 2s + 89.42]}$$

Time Domain Specifications without Rate Feedback

The steady state error for ramp of slope 4 is

$$
\begin{aligned}
e_{ss1} &= \frac{2\zeta_1 R}{\omega_n} \\
&= \frac{2 \times 0.106 \times 4}{9.456} \\
&= 0.09
\end{aligned}
$$

$$
\text{For step input, overshoot, } Mp_1 = Re^{-\frac{\zeta_1 \pi}{\sqrt{1-\zeta_1^2}}}
$$

$$
\begin{aligned}
Mp_1 &= 2e^{-\frac{0.106\pi}{\sqrt{1-0.106^2}}} \\
&= 1.43
\end{aligned}
$$

The rise time is calculated from

$$
t_{r_1} = \frac{(\pi - \phi)}{\omega_d}
$$

where $\phi = \cos^{-1} \zeta_1 = \cos^{-1} 0.106 = 1.46$ rad

$$
\begin{aligned}
\omega_d &= \omega_n \sqrt{1-\zeta_1^2} \\
&= 9.456\sqrt{1-(0.106)^2} = 9.4 \\
t_{r_1} &= \frac{(\pi - 1.46)}{9.4} = 0.179 \text{ s}
\end{aligned}
$$

The settling time is calculated from (for 2% error tolerance).

$$
\begin{aligned}
t_{s_1} &= \frac{3.91}{\zeta \omega_n} \\
&= \frac{3.91}{0.106 \times 9.456} \\
ts_1 &= 3.9 \text{ s}
\end{aligned}
$$

Time Domain Specifications with Rate Feedback Control

$$
e_{ss2} = 0.5
$$

The peak overshoot,

$$
Mp_2 = 0.2
$$

The rise time is calculated from

$$
\begin{aligned}
t_{r_2} &= \frac{(\pi - \phi)}{\omega_d} \\
\phi &= \cos^{-1} \zeta_2 \\
&= \cos^{-1} 0.591 \\
\phi &= 0.938 \text{ rad} \\
\omega_d &= \omega_n \sqrt{1 - \zeta_2^2} \\
&= 9.456 \sqrt{1 - (0.591)^2} \\
&= 7.628 \\
tr_2 &= \frac{(\pi - 0.938)}{7.628} \\
&= 0.29 \text{ sec}
\end{aligned}
$$

The settling time for 2% error tolerance is given by

$$
\begin{aligned}
t_{s_2} &= \frac{3.91}{\zeta_2 \omega_n} \\
&= \frac{3.91}{0.591 \times 9.456} \\
&= 0.7 \text{ s}
\end{aligned}
$$

The performance specifications obtained above with and without rate feedback are tabulated below.

Performance specifications	Without rate Feedback	With rate Feedback
Steady state error, e_{ss}	0.09	0.5
Overshoot, M_p	1.43	0.2
Rise time, t_r	0.179 s	0.29 s
Settling time, t_s	3.9 s	0.7 s
Damping ratio, ζ	0.106	0.591
Natural frequency of oscillation, ω_n	9.456	9.456

2.6.2 Proportional Plus Integral Plus Derivative (Three Term Controllers) Controllers

A proportional + Integral + Derivative controller (PID controller) is probably the most used feedback control and is widely used in industrial control systems. A PID controller calculates an error value as the difference between a measured process variable and a desired set point. The controller attempts to minimize the error by adjusting the process control inputs. For best performance, the PID parameters used

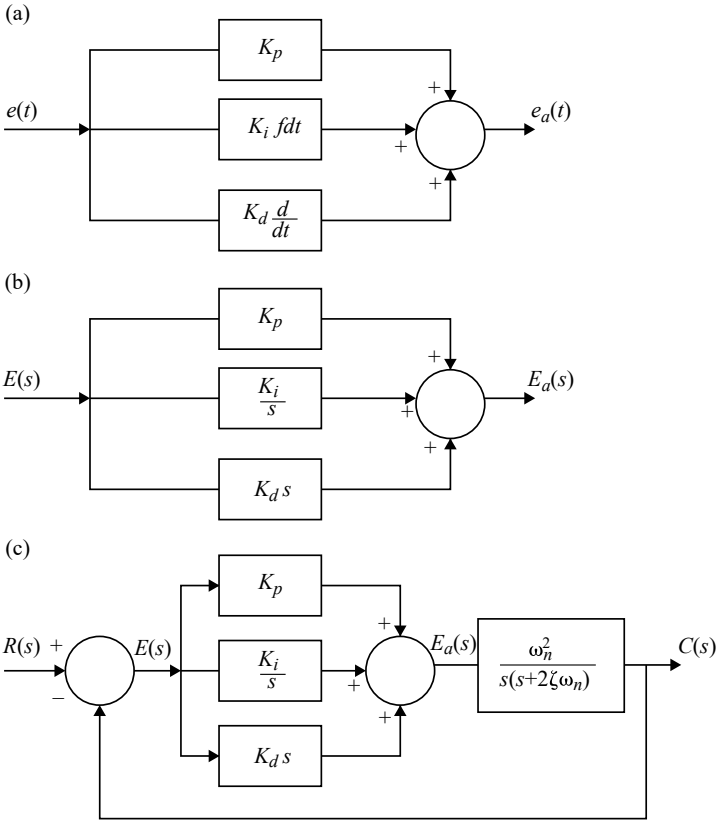


Fig. 2.47 Representation of the PID controller. **a** Time domain representation; **b** Frequency domain representation; and **c** Connected in cascade with the prototype second order system

in the calculation must be tuned according to the nature of the operating conditions of the system.

The PID controller calculation (algorithm) involves three parameters and is accordingly called **three term control: the proportional, the integral and derivative values**, denoted by P, I, and D respectively. The **proportional** value determines the reaction to the current error, the **integral** value determines the reaction based on the sum of recent errors, and the **derivative** value determines the reaction based on the rate at which error has been changing. The weighted sum of these three actions is used to adjust the process *via* a control element. Thus, P depends on the **present** error, I on the accumulation of the **past** error and D is the prediction of **future** errors based on current rate of change.

By tuning the three constants in the PID controller algorithm, the controller can provide the control action designed for specific process requirements. Even though,

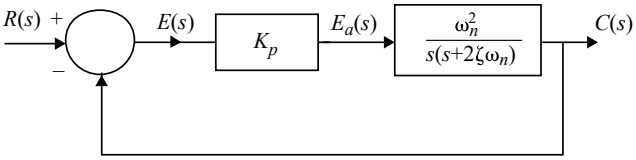


Fig. 2.48 Proportional controller connected in cascade with the second order system

by the use of PID controller, it is possible to achieve some or all of the transient and steady state specifications, it does not guarantee optimal control of the system.

Some applications may require using only one or two modes to provide the appropriate system control. This is achieved by settling the gain of undesired control outputs to zero. A PID controller will be called a PI, PD, P or I controller in the absence of the respective control actions. PI controller are fairly common, since the derivative action is sensitive to noise whereas the absence of the integral value may prevent the system from reaching the target value due to the control action.

While PID controllers are applicable to many control problems, and often perform satisfactorily without even tuning according to the changes in the operating conditions, they can perform poorly in some applications. The fundamental difficulty with PID control is that it is a feedback system with constant parameters and no direct knowledge of the process.

The PID controller is represented in the time domain as shown in Fig. 2.47a. The frequency domain representation is shown in Fig. 2.47b. The error signal $e(t)$ is the input to the controller and its output is $e_a(t)$. The output of the controller is connected to the prototype second order system which is taken up for the performance study of the PID controller. This is shown in Fig. 2.47c. We study below the performance of the individual controllers as well as this combination when connected in series with the standard prototype second order system. For this, the results obtained from the previous sections are referred to.

2.6.2.1 Proportional Controller

The proportional controller connected in cascade with the standard second order system is shown in Fig. 2.48. From Fig. 2.48 the following equations are obtained:

$$\frac{C(s)}{R(s)} = \frac{K_p\omega_n^2}{[s^2 + 2\zeta\omega_n s + K_p\omega_n^2]} \tag{2.80}$$

$$\frac{E(s)}{R(s)} = \frac{s(s + 2\zeta\omega_n)}{[s^2 + 2\zeta\omega_n s + K_p\omega_n^2]} \tag{2.81}$$

The characteristic equation is

$$s^2 + 2\zeta\omega_n s + K_p\omega_n^2 = 0 \tag{2.82}$$

From Eq. (2.82) it is seen that the damping factor of the system with and without the proportional controller remains the same. However, the natural frequency of oscillation of the system with the controller is altered from ω_n to $\sqrt{K_p}\omega_n$. The performance study of the controller with respect to steady state error, peak overshoot, rise time and setting is discussed and compared with and without the controller as given below.

1. **Steady State Error**: The steady state error for the system without controller (system 1) and with the controller (system 2) for step input is zero. However, for ramp input $e_{ss1} = 2\zeta R/\omega_n$ and $e_{ss2} = 2\zeta R/K_p\omega_n$. The suffix one refers to the system without compensator and suffix two refers to system with compensator.
2. **Peak Overshoot**: Since the damping factor is changed by the factor $\zeta' = \zeta/\sqrt{K_p}$ the peak overshoot also changes accordingly.

$$M_{p1} = e^{-\frac{\zeta_1\pi}{\sqrt{1-\zeta_1}}}R$$

$$M_{p2} = e^{-\frac{\frac{-\zeta_1\pi}{\sqrt{K_p}}}{\sqrt{1-\zeta_1^2/K_p}}}R$$

The time at which these overshoot occur differs as given below.

$$t_{p1} = \frac{\pi}{\omega_n\sqrt{1-\zeta_1^2}}$$

$$t_{p2} = \frac{\pi}{\sqrt{K_p}\omega_n\sqrt{1-\zeta_2^2}}$$

Thus, for $K_p > 1$, $t_{p2} < t_{p1}$. The system two has faster transient response.

3. **Rise Time:** The rise time of system 1 is

$$t_{r1} = \frac{(\pi - \phi)}{\omega_n\sqrt{1-\zeta^2}}$$

The rise time of system 2 is

$$t_{r2} = \frac{(\pi - \phi)}{\sqrt{K_p}\omega_n\sqrt{1-\zeta^2}}$$

Thus, $t_{r2} < t_{r1}$ and the system with the controller is faster. Since rise time is inversely proportional to the bandwidth, the bandwidth of the system when controller is incorporated is increased as the proportional controller gain K_p is increased.

4. **Settling Time:** The settling time for system one is

$$t_{s1} = \frac{3.91}{\zeta_1 \omega_{n1}}$$

The settling time for system two is

$$\begin{aligned} t_{s2} &= \frac{3.91}{\zeta_2 \omega_{n2}} \\ &= \frac{3.91}{\sqrt{K_p}\frac{\zeta}{\sqrt{K_p}}\omega_{n1}} = \frac{3.91}{\zeta_1 \omega_{n_1}} \\ t_{s2} &= t_{s1} \end{aligned}$$

The transient response curves of these two systems are shown in Fig. 2.49 for unit step input. The T.F. of the prototype second order system taken for the study is

$$G(s) = \frac{16}{s(s+4)}$$

The natural frequency of the original system is $\omega_n = 4$ rad/s. and $\zeta = 0.5$. The closed loop T.F. of the system without the controller is

$$\frac{C(s)}{R(s)} = \frac{16}{s^2 + 4s + 16}$$

The closed loop T.F. with the proportional controller is

$$\frac{C(s)}{R(s)} = \frac{K_p 16}{s^2 + 4s + 16K_p}$$

The above T.F. is simulated in the digital computer for $K_p = 1, 2, 3$ and 5 For $K_p = 1$, the T.F. is that of the prototype second order system without the controller.

The following MATLAB program is used:

MATLAB Program
% · · · · · · Unit-step response · · · · · ·
num = [0 0 16];
den = [1 4 16];
$t = 0 : 0.05 : 3$;
c = step(num,den,t);
plot(t,c)
title('Unit-Step Response')

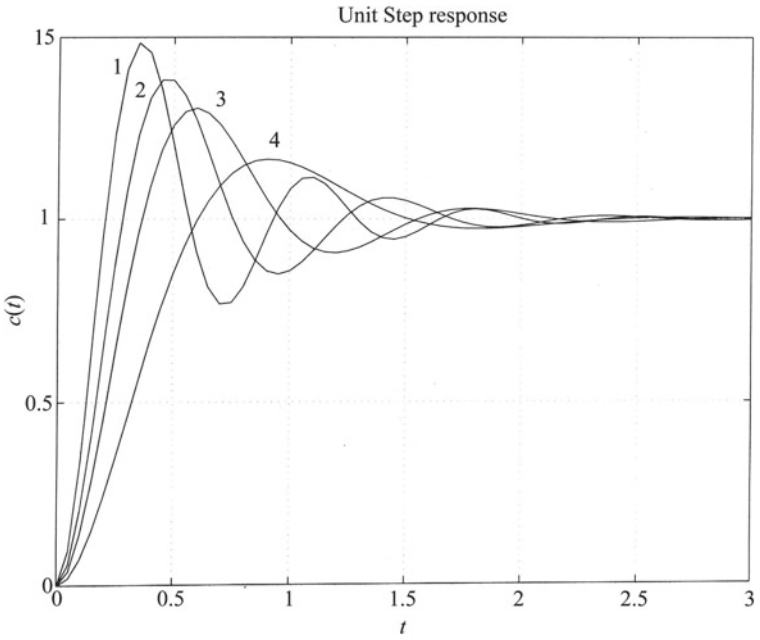


Fig. 2.49 Transient response curves of a second order system for unit step input with proportional controller (1) $K_p = 5$; (2) $K_p = 3$; (3) $K_p = 2$; (4) $K_p = 1$ (without controller)

Figure 2.49 shows the step response curve. From the step responds curve it is observed that curve 4 represents the system without the controller. It has an overshoot of 16.3% and occurs at $t_p = 0.91$ s curves 1, 2 and 3 corresponds to $K_p = 5$; $K_p = 3$ and $K_p = 2$ respectively. For $K_p = 5$ the damping factor $\zeta = 0.2236$; and $\%M_p = 48.6\%$ at $t_p = 0.36$ s, which is much faster than the system without the controller.

Summary of Proportional Controller

1. The system overshoot is changed. As K_p is increased overshoot is also increased.
2. The rise time, settling time and the time at which the peak overshoot occurs are reduced because of fast transient response. However, the settling time is unchanged.
3. The bandwidth is increased.
4. The steady state error for step input is zero and unchanged. However, the steady state error for ramp input is reduced by a factor $1/\sqrt{K_p}$.

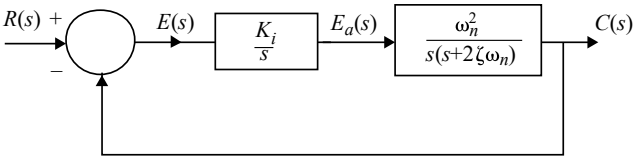


Fig. 2.50 Second order system with integral controller

2.6.3 Integral or Reset Controller

The integral controller connected in the forward path in cascade with a standard second order system is shown in Fig. 2.50. For Fig. 2.50 the following equations are derived.

$$\frac{C(s)}{R(s)} = \frac{K_i\omega_n^2}{[s^3 + 2\zeta\omega_n s^2 + K_i\omega_n^2]} \tag{2.83}$$

The characteristic equation is

$$s^3 + 2\zeta\omega_n s^2 + K_i\omega_n^2 = 0 \tag{2.84}$$

Inspection of the characteristic Eq. (2.84) shows that the coefficient of s is zero and hence the closed loop system will be unstable with a pure integrator connected in cascade with the second order system and hence no further analysis is required. However, its action will be discussed at length when combined with proportional controller. Integral control can be used to control a second order 0 Type system which will be stable. Similarly if an ID controller with the T.F.

$$\begin{aligned} G(s) &= \left[\frac{K_i}{s} + K_d s\right] \\ &= [K_d s^2 + K_i]\frac{1}{s} \end{aligned}$$

when connected in cascade with a prototype second order type 1 system, will have the closed loop T.F. as

$$\frac{C(s)}{R(s)} = \frac{(K_d s^2 + K_i)}{[s^3 + (2\zeta\omega_n + K_d)s^2 + K_i]}$$

From the characteristic equation it is observed that the coefficient of s is zero and hence the system becomes unstable. Therefore, ID controller cannot be used for type 1 and order two system. However, it can be used for zero type order two system.

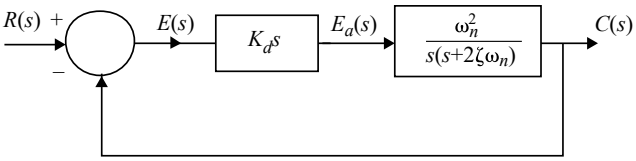


Fig. 2.51 Second order system with derivative controller

2.6.3.1 Derivative or Rate Controller

Consider the second order system connected to the derivative controller in the forward path as shown in Fig. 2.51. The following equations are derived for Fig. 2.51. Here the zero of the controller cancels with the pole of the second order system.

$$\frac{C(s)}{R(s)} = \frac{K_d\omega_n^2}{[s + 2\zeta\omega_n + K_d\omega_n^2]} \tag{2.85}$$

$$\frac{E(s)}{R(s)} = \frac{(s + 2\zeta\omega_n)}{[s + 2\zeta\omega_n + K_d\omega_n^2]} \tag{2.86}$$

The characteristic equation of the system with derivative control in the forward path is given by $[s + 2\zeta\omega_n + K_d\omega_n^2]$.

From Eq. (2.86) it is evident that the given second order system is reduced to a first order system and its pole is placed on the negative real axis in the complex s-plane. The performance study of the controller with respect to steady state error, peak overshoot, rise time and settling time is discussed and compared as given below.

(1) **Steady State Error**: Steady state error for system one for step input is zero and for ramp input $e_{ss1} = 2\zeta R/\omega_n$. However, $e_{ss} = (2\zeta R)/(2\zeta + K_d\omega_n)$ for system with derivative controller for step input. For ramp input $e_{ss} = \infty$.

(2) **Peak Overshoot:** The peak overshoot for step input for system one is as discussed in Sect. 2.6.1. However, for system two, since it is a first order system, the peak overshoot will be zero and the rise time will be $t_{r2} = 2.2T_2$. The time constants of these two systems are

$$T_1 = \frac{1}{\zeta\omega_n}$$

$$T_2 = \frac{1}{2\zeta\omega_n + K_d\omega_n^2}$$

$$= \frac{1}{(2\zeta + K_d\omega_n)\omega_n}$$

$$T_2 < T_1$$

The rise time for system two is

$$t_{r2} = 2.2T_2$$

The settling time for system two is

$$t_{s2} = 3.91T_2$$
$$t_{s2} < T_{s1}$$

For comparison study the following numerical values are chosen. $\omega_n = 4$ rad/s; $\zeta = 0.5$; $K_d = 2$.

$$\begin{aligned} T_1 &= \frac{1}{\zeta\omega_n} \\ &= \frac{1}{4 \times 0.5} = 0.5 \\ T_2 &= \frac{1}{2\zeta\omega_n + K_d\omega_n^2} \\ &= \frac{1}{2 \times 0.5 \times 4 + 2 \times (4)^2} \\ &= 0.0278 \\ Mp_1 &= e^{\frac{-\zeta\pi}{\sqrt{1-\zeta_1^2}}} \times 100 \\ &= e^{\frac{-0.5\times\pi}{4\sqrt{1-0.5^2}}} \times 100 = 16.3 \end{aligned}$$

$$\begin{aligned} t_{r1} &= \frac{\pi - \phi}{\omega_n\sqrt{1-\zeta^2}} \\ &= \frac{\pi - \cos^{-1} 0.5}{4\sqrt{1-0.5^2}} \\ &= \frac{\pi - \frac{\pi}{3}}{4 \times 0.866} \\ t_{r1} &= 0.6 \text{ s} \\ t_{r2} &= 2.2T_2 = 2.2 \times 0.0278 = 0.061 \text{ s} \\ t_{s1} &= 3.91T_1 = 3.91 \times 0.5 = 2.0 \text{ s} \\ t_{s2} &= 3.92T_2 = 3.91 \times 0.0.278 = 0.11 \text{ s} \end{aligned}$$

For unit ramp input,

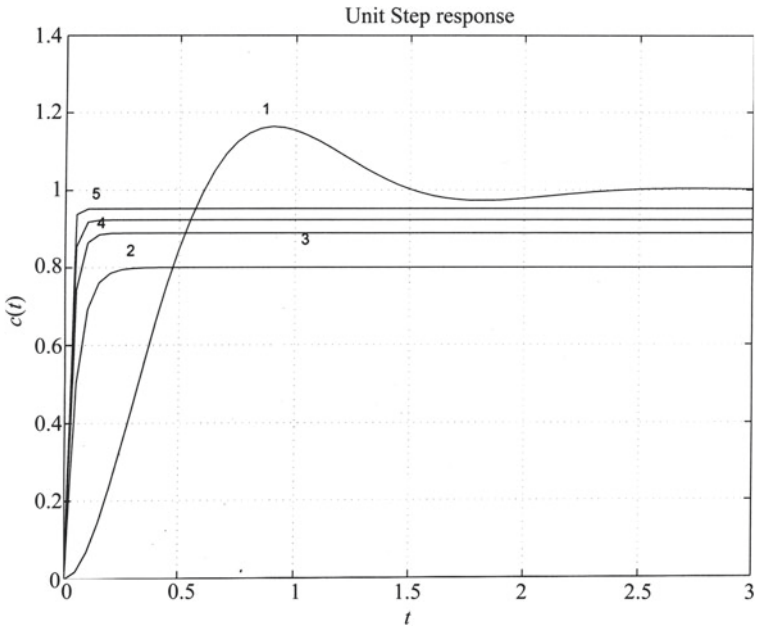


Fig. 2.52 Transient response curves of a second order system for unit step input with derivative controller. (1) $K_d = 1$ (without controller); (2) $K_d = 2$; (3) $K_d = 3$; (4) $K_d = 5$

$$e_{ss1} = \frac{2\zeta}{\omega_n} = \frac{2 \times 0.5}{4} = 0.25$$
$$e_{ss2} = \infty$$

The performance specifications are shown in the table given below:

Performance specifications	Time constant T	Rise time t_r	Settling time t_s	Steady state Error for ramp e_{ss}	Peak Overshoot M_p
System 1 (without controller)	0.5	0.6 s	2.0 s	0.25	16.3%
System 2 (with controller)	0.0278	0.061 s	0.11 s	∞	0

The transient response curves for system 1 (without controller) and system 2 (with derivative controller) are shown in Fig. 2.52 for unit step input for $K_d = 1$, $K_d = 2$, $K_d = 3$ and $K_d = 5$.

Summary of Derivative Controller

The second order system with derivative controller is reduced to a first order system by pole–zero cancellation. Thus, it has no overshoot when subjected to step input. The steady state errors due to step input is finite and for ramp input is zero. The time

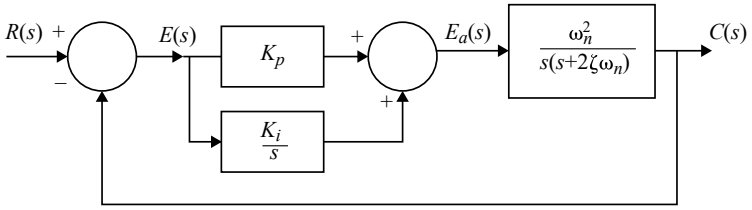


Fig. 2.53 Second order system with PI controller

constant, rise time and settling time are much less for the system with derivative controller.

2.6.4 Proportional Plus Integral (PI) Controller

Figure 2.53 shows the PI controller connected in cascade with a second order system. The control action provided by the controller is proportional to the error as well as the time integral part of the error. Thus, it combines the effects of the proportional controller discussed in Sect. 2.6.1 and integral action discussed in Sect. 2.6.2. The transfer function of the PI controller is

$$\begin{aligned} G_c(s) &= K_p + \frac{K_i}{s} \\ &= \frac{sK_p + K_i}{s} \end{aligned} \tag{2.87}$$

By connecting a PI controller, a pole at $s = 0$ and a zero at $s = -(K_i/K_p)$ are added to the open loop T.F. The equivalent open loop T.F. now becomes

$$G_e(s) = \frac{(K_p s + K_i)}{s} \frac{\omega_n^2}{s(s + 2\zeta\omega_n)} \tag{2.88}$$

The closed loop T.F. is obtained as

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2(K_i + K_p s)}{[s^3 + 2\zeta\omega_n s^2 + K_p\omega_n^2 s + K_i\omega_n^2]} \tag{2.89}$$

From Eqs. (2.88) and (2.89) it is observed that the type of the system is increased by one to a type 2 system. The order of the system is also increased by one to a third order system. The increase in the type and order of the system will have the influence on the steady state error and transient performance respectively. The increase in the type of the system by one improves the steady state error of the original system by order one. Thus, the second order type one system will have zero steady state error

for step input and finite error for ramp input. However, the second order system with PI controller will have zero steady state error for step and ramp inputs irrespective of K_p and K_i. The steady state error for parabolic input is finite which is evident from Eq. (2.90).

$$\frac{E(s)}{R(s)} = \frac{s^2(s + 2\zeta\omega_n)}{[s^3 + 2\zeta\omega_n s^2 + K_p\omega_n^2 s + K_i\omega_n^2]} \tag{2.90}$$

For parabolic input

$$e_{(ss)} = \frac{2\zeta R}{K_i\omega_n} \tag{2.91}$$

As for the stability of the system with PI controller, the order of the system is increased from two to three. While the original second order system is always stable, irrespective of the numerical values of ζ and ω_n ($\zeta > 0$ and $\omega_n > 0$), the system with PI controller may become unstable if the value of K_p and K_i are not judiciously chosen from view point of stability and also transient response specifications.

For retaining stability and improvement in transient response specifications, the values of K_p and K_i are to be properly selected. The PI controller is basically a low pass filter and hence the system with PI controller will have a slower rise time and longer settling time. The values of K_p and K_i are chosen following the procedure given below.

(1) Choose the range of K_p and K_i so that the system with PI controller is stable. Routh–Hurwitz method of stability can be applied.
(2) Select the zero of the PI controller at $s = -(K_i/K_p)$ such that it is relatively closer to the origin and away from the most significant poles of the system.
(3) The values of K_p and K_i should be as small as possible.

Since the order of the system is three and there is a zero in the closed loop T.F; it is not easy to derive expressions for transient response specifications. However, transient response of the original system and the system with PI controller can be simulated and useful inferences can be established.

Again we consider a standard second order system with $\zeta = 0.5$ and $\omega_n = 4$ rad/s. From Eq. (2.89), the characteristic equation is obtained as

$$s^3 + 4s^2 + 16K_p s + 16K_i = 0 \tag{2.92}$$

The system without the controller has the closed loop poles with the real part of $s = -\zeta\omega_n = -2$. Thus, we have to choose $\left|(K_i/K_p)\right| > 2$ or $K_i > 2K_p$. For Eq. (2.92) the following Routh's array is prepared.

s^3	1	$16K_p$
s^2	1	$4K_i$
s^1	$16K_p - 4K_i$	
s^0	K_i	

From Routh's array, to ensure stability of the closed loop system with PI controller $K_i < 4K_p$. Hence, the value of K_i is chosen in the range $2K_p < K_i < 4K_p$.

Time response curves for unit step input are drawn and shown in Figs. 2.54 and 2.55 for the following controller parameters. [$\zeta = 0.5$ and $\omega_n = 4$ rad/s in all cases].

I		II	
1	$K_I = 0$; $K_p = 1$ (without controller)	1	$K_p = 2$
2	$K_i = 0.2$;	2	$K_i = 0.2$
3	$K_i = 0.5$	3	$K_i = 0.5$
4	$K_i = 1$	4	$K_i = 1$

Summary of PI Controller

1. Improves the damping and reduces the maximum overshoot.
2. Increases the rise time.
3. Decreases the bandwidth.
4. Filters out high frequency noise.
5. Improves phase margin and gain margin and hence the relative stability.
6. Increases the type of the system by one and e_{ss} for step and ramp inputs is zero and for parabolic input, it is finite.

2.6.5 Proportional + Derivative (PD) Controller

From Fig. 2.56, the PD controller provides control action $e_a(t)$ which is proportional and time derivative of the error $e(t)$.

$$e_a(t) = K_p e(t) + K_d \frac{de(t)}{dt}$$

Taking Laplace transform on both sides of the above equation we get

$$\begin{aligned} E_a(s) &= [K_p + K_d s]E(s) \\ \frac{E_a}{E}(s) &= G_c(s) = (K_p + K_d s) \end{aligned} \tag{2.93}$$

From Eq. (2.93) it is evident that the PD control is equivalent to adding a zero at $s = -(K_p/K_d)$. Further, it has the characteristic of a high pass filter which increases the

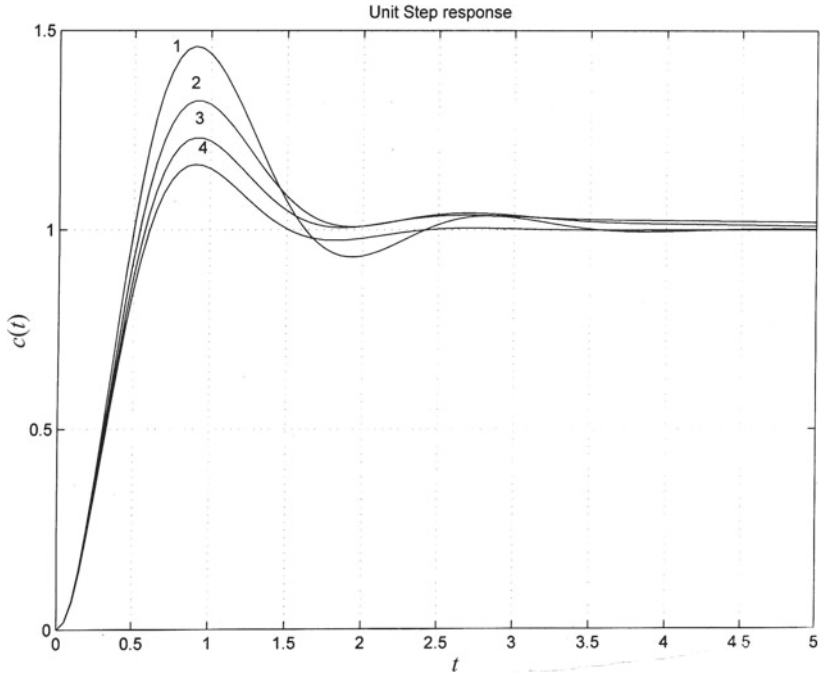


Fig. 2.54 Transient response curves of a second order system for unit step input with proportional plus integral controller with $K_p = 1$. (1) $K_i = 0.2$; (2) $K_i = 0.5$; (3) $K_i = 1.0$; (4) $K_i = 0$ (without controller)

bandwidth and hence reduces the rise time for step input. However, the differentiation characteristic amplifies the noise that enters in the input. The forward path T.F. of the combined PD controller and the second order system is

$$G(s) = \frac{\omega_n^2(K_p + K_d s)}{s(s + 2\zeta\omega_n)}$$

The closed loop T.F is

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2(K_p + K_d s)}{[s^2 + (2\zeta\omega_n + K_d\omega_n^2)s + \omega_n^2 K_p]} \tag{2.94}$$

The error function is given by

$$\frac{E(s)}{R(s)} = \frac{s(s + 2\zeta\omega_n)}{[s^2 + (2\zeta\omega_n + K_d\omega_n^2)s + \omega_n^2 K_p]} \tag{2.95}$$

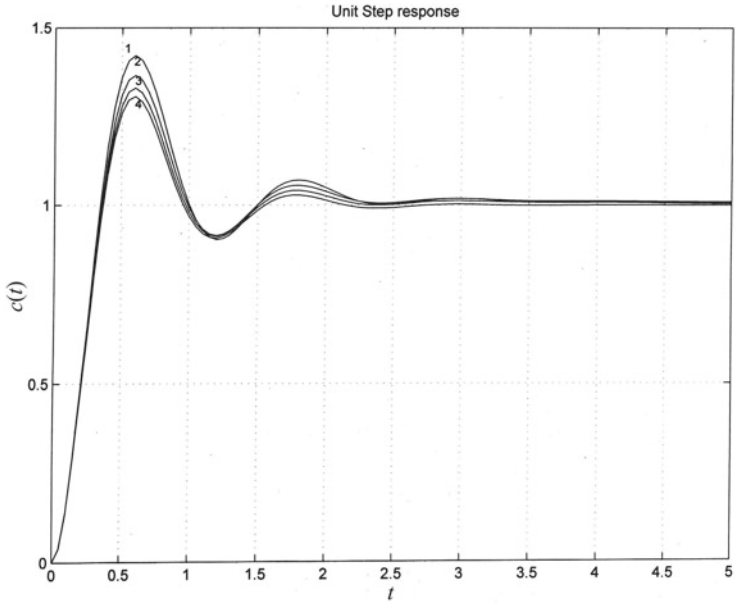


Fig. 2.55 Transient response curves of a second order system for unit step input with proportional plus integral controller with $K_p = 2$. (1) $K_i = 0.2$; (2) $K_i = 0.5$; (3) $K_i = 1.0$; (4) $K_i = 0$ (without controller)

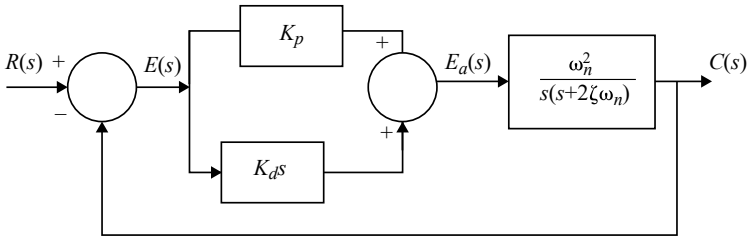


Fig. 2.56 PD controller for a second order system

The steady state error for step input as seen from Eq. (2.95) is zero and for ramp input

$$e_{ss} = \frac{2\zeta R}{\omega_n K_p} \tag{2.96}$$

The characteristic equation of the system is

$$s^2 + (2\zeta\omega_n + K_d\omega_n^2)s + \omega_n^2 K_p = 0 \tag{2.97}$$

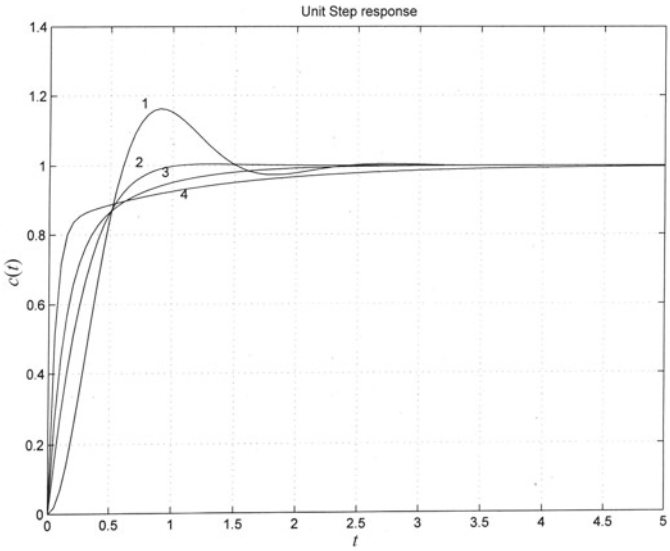


Fig. 2.57 Transient response curves of a second order system for unit step input with proportional plus derivative controller with $K_p = 1$. (1) $K_d = 0$ (without controller); (2) $K_d = 0.2$; (3) $K_d = 0.4$; (4) $K_d = 1$

Because of the introduction of the PD controller, the type and order of the system are not changed. However, the natural frequency and damping factors are changed and are given by

$$\omega_n^1 = \omega_n \sqrt{K_p} \tag{2.98}$$

$$\zeta^1 = \left(\zeta + \frac{K_d}{2}\omega_n\right)\frac{1}{\sqrt{K_p}} \tag{2.99}$$

For the transient response study, the second order system with $\zeta = 0.5$ and $\omega_n = 4$ rad/s is considered. With these parameters, Eq. (2.94) is rewritten as

$$\frac{C(s)}{R(s)} = \frac{16(K_p + K_d s)}{[s^2 + 4(1 + 4K_d)s + 16K_p]} \tag{2.100}$$

If we choose $K_p = 1$, the natural frequency is unaltered and the new damping ratio becomes

$$\zeta' = 0.5 + 2K_d$$

For values of $K_d = 0, 0.2, 0.4$ and 1 and for $K_p = 1$ and $K_p = 2$ transient response curves for unit step input are plotted and are shown in Figs. 2.57 and 2.58 respectively.

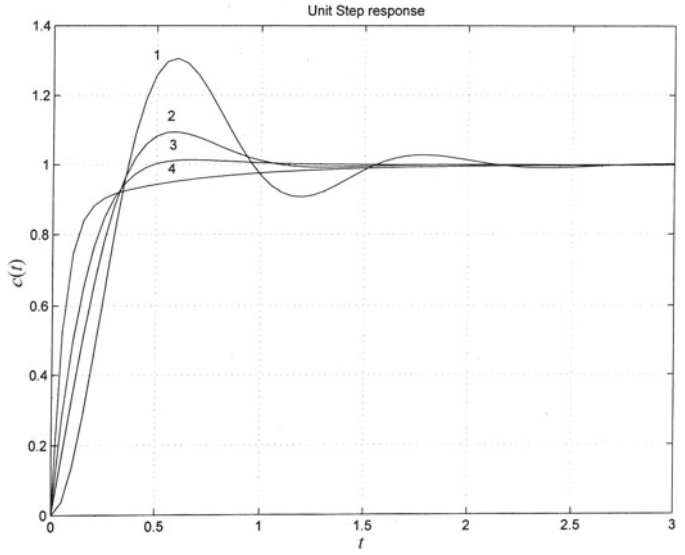


Fig. 2.58 Transient response curves of a second order system for unit step input with proportional plus derivative controller with $K_p = 2$. (1) $K_d = 0$ (without controller); (2) $K_d = 0.2$; (3) $K_d = 0.4$; (4) $K_d = 1$

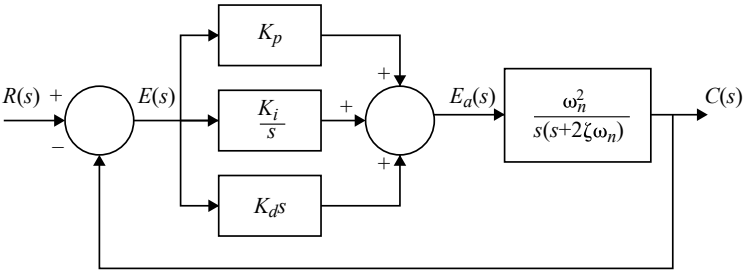


Fig. 2.59 PID controller connected in cascade with a second order system

Summary of PD Controller

1. Improves damping and reduces maximum overshoot.
2. Reduces rise time and settling time.
3. Increases bandwidth.
4. Improves phase margin and gain margin and thus the relative stability.
5. Amplifies noise at higher frequencies.
6. Steady state error for step input is zero and for ramp input it is finite.

2.6.6 Proportional + Integral + Derivative (PID) Controller

From Sects. 2.4 and 2.5 we observed that the PI controller could improve the relative stability and steady state error simultaneously while PD controller could add damping to the system but the steady state response is not affected. Further, the advantages and disadvantages of each of these methods were summarized at the end of the respective section. The best features of each of these controllers are taken to form PID controller. The PID controller connected in cascade with a standard second order system is shown in Fig. 2.59.

The PID controller provides control action $e_a(t)$ which is proportional, time integrated and time derivative of the error $e(t)$. That is

$$e_a(t) = K_p e(t) + K_i \int e(t)dt + K_d \frac{de(t)}{dt}$$

Taking Laplace transform on both sides we get

$$\begin{aligned} E_a(s) &= \left[K_p + \frac{K_i}{s} + K_d s\right] E(s) \\ G_c(s) &= \frac{Ea}{E}(s) \\ &= \left[K_p + \frac{K_i}{s} + K_d s\right] \\ &= \frac{[K_i + K_p s + K_d s^2]}{s} \end{aligned} \tag{2.101}$$

From Eq. (2.101), it is seen that the PID controller has two zeros in the LHP and a pole at $s = 0$.

The forward path T.F. of the entire control system is obtained as

$$G(s) = \frac{(K_i + K_p s + K_d s^2)\omega_n^2}{s^2(s + 2\zeta\omega_n)} \tag{2.102}$$

Equation (2.102) shows that the type of the system is increased by one from one to two. The order of the system is increased by one from two to three. Hence, the following information are available from Eq. (2.102):

1. The steady state errors of the system for step and velocity inputs are zero while it is finite for parabolic input.
2. While the given second order system is always stable, for $\zeta > 0$ and $\omega_n > 0$, stability is not guaranteed for the system when connected with a PID controller.
3. Only judicious selection of controller parameters K_p, K_i and K_d becomes necessary not only for the system stability but also for good transient performance.

From Eq. (2.102), the closed loop T.F. is obtained as

$$\frac{C(s)}{R(s)} = \frac{(K_i + K_p s + K_d s^2)\omega_n^2}{[s^3 + (2\zeta\omega_n + K_d\omega_n^2)s^2 + K_p\omega_n^2 s + K_i\omega_n^2]} \tag{2.103}$$

The error function is derived as

$$\frac{E(s)}{R(s)} = \frac{s^2(s + 2\zeta\omega_n)}{[s^3 + (2\zeta\omega_n + K_d\omega_n^2)s^2 + K_p\omega_n^2 s + K_i\omega_n^2]} \tag{2.104}$$

From Eq. (2.104), while the e_{ss} for step and velocity inputs is zero, it is finite for parabolic input. The e_{ss} for parabolic input is

$$e_{ss} = \frac{2\zeta R}{K_i\omega_n} \tag{2.105}$$

The characteristic equation of the system is given by

$$s^3 + (2\zeta\omega_n + K_d\omega_n^2)s^2 + K_p\omega_n^2 s + K_i\omega_n^2 = 0 \tag{2.106}$$

To ensure the stability of the closed loop system, Routh–Hurwitz stability test is performed.

s^3	1	$K_p\omega_n^2$
s^2	$(2\zeta\omega_n + K_d\omega_n^2)$	$K_i\omega_n^2$
s^1	$(2\zeta\omega_n + K_d\omega_n^2)K_p\omega_n^2 - K_i\omega_n^2$	
s^0	$K_i\omega_n^2$	

For the system to be stable the following condition is to be satisfied which is obtained from sth row

$$(2\zeta + K_d\omega_n)K_p\omega_n > K_i \tag{2.107}$$

For the transient response study of the second order system, with PID controller, as in previous cases, $\zeta = 0.5$ and $\omega_n = 4$ rad/s. are chosen. The PID controller parameters given in Eq. (2.107) become

$$(1 + 4K_d)4K_p > K_i \tag{2.108}$$

The characteristic equation given in Eq. (2.106) is given as

$$s^3 + 4(1 + 4K_d)s^2 + 16K_p s + 16K_i = 0 \tag{2.109}$$

Selection of PID Parameters

Let us consider the PID controller consisting of PD portion connected in series with a PI portion as shown in Fig. 2.60 where it is assumed that the proportional term in PD controller is 1.

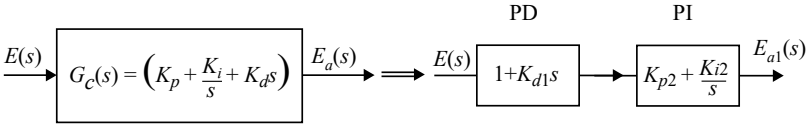


Fig. 2.60 PID controller split into PD and PI controllers

From Fig. 2.60,

$$K_p + \frac{K_i}{s} + K_d s = K_{p2} + K_{d1}K_{p2}s + K_{i2}K_{d1} + \frac{K_{i2}}{s} \tag{2.110}$$

Equating both sides of Eq. (2.110) we get

$$\begin{aligned} K_p &= K_{p2} + K_{d1}K_{i2} \\ K_d &= K_{d1}K_{p2} \\ K_i &= K_{i2} \end{aligned} \tag{2.111}$$

Based on the performance of PD controller alone, choose K_{d1}. Similarly, based on the performance of PI controller alone chose K_{p2} and K_{i2}. Now using (2.111) choose K_p, K_i and K_d. For simulation study, $K_{d1} = 0.4$, $K_{p2} = 0.4$ and $K_{i2} = 0.2$ are chosen. Using (2.111) we get

$$\begin{aligned} K_p &= 0.4 + 0.4 \times 0.2 = 0.48 \\ K_d &= 0.4 \times 0.4 = 0.16 \\ K_i &= 0.2 \\ G_c(s) &= \left[0.48 + \frac{0.2}{s} + 0.16s\right] \\ \frac{C(s)}{R(s)} &= \frac{16[0.2 + 0.48s + 0.16s^2]}{[s^3 + 6.56s^2 + 7.68s + 3.2]} \end{aligned} \tag{2.112}$$

The transient response curve for unit step input is plotted and is shown in Fig. 2.61. Along with this the transient response curves with PD controller whose parameters are $K_p = 1$ and $K_d = 0.4$ and PI controller whose parameters are $K_p = 1$ and $K_i = 0.4$ are also shown for comparison. Note that Eq. (2.108) is satisfied for the PID controller parameters chosen.

Example 2.45

The forward path T.F. of a certain control system is given by

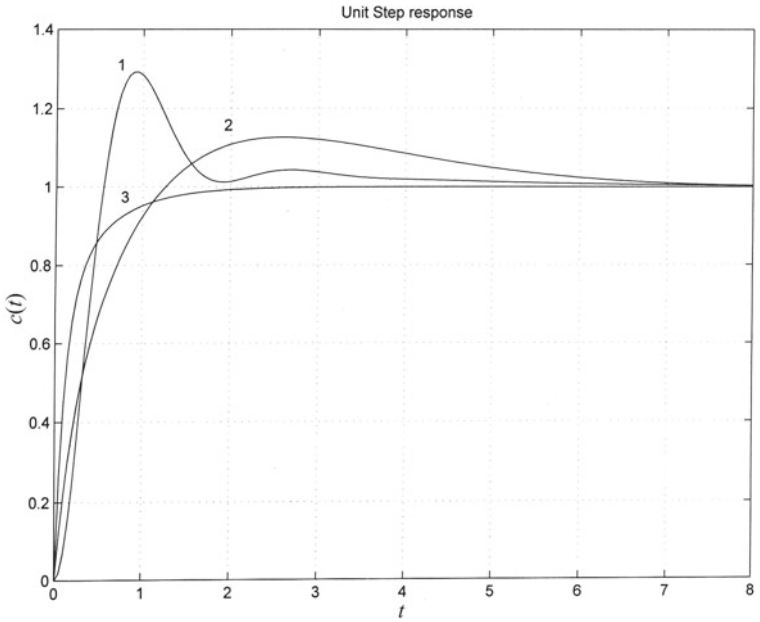


Fig. 2.61 Transient response curves of a second order system for unit step input with (1) PI controller with $K_p = 1$; $K_i = 0.4$; (2) PID controller with $K_p = 0.48$; $K_i = 0.2$; $K_d = 0.16$; (3) PD controller with $K_p = 1$; $K_d = 0.4$

$$G(s) = \frac{4}{s+2}$$

An integral controller with a T.F

$$G_c(s) = \frac{4}{s}$$

is connected in cascade. Determine the damping ratio, natural frequency of the closed loop system. Find the rise time, settling time and % overshoot for unit step input. What is the steady state error for unit ramp input?

Solution The forward path T.F. of the given system is

$$\begin{aligned} G^1(s) &= G(s)G_c(s) \\ &= \frac{16}{s(s+2)} \end{aligned}$$

The closed loop T.F is

$$\frac{C(s)}{R(s)} = \frac{G^1(s)}{1 + G^1(s)} = \frac{16}{(s^2 + 2s + 16)}$$

Comparing this with the standard second order system, we get $\omega_n = \sqrt{16} = 4$. And $2\zeta\omega_n = 2$ or $\zeta = 0.25$.

The rise time t_r is given by

$$t_r = \frac{(\pi - \phi)}{\omega_n\sqrt{1-\zeta^2}} = \frac{\pi - \cos^{-1} 0.25}{4\sqrt{1-(0.25)^2}} = \frac{\pi - 1.318}{4 \times 0.968} = 0.471 \text{ s.}$$

The settling time is given by

$$t_s = \frac{3.91}{\zeta\omega_n} = \frac{3.91}{0.25 \times 4} = 3.91 \text{ s.}$$

The % peak overshoot is given by

$$\%M_p = e^{\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}} \times 100 = e^{\frac{-0.25\pi}{\sqrt{1-(0.25)^2}}} \times 100 = 44.43\%$$

The steady state error for unit ramp is given by

$$e_{ss} = \frac{1}{K_v}$$

where

$$K_v = \lim_{s\to 0} sG^1(s) = \lim_{s\to 0} \frac{s16}{s(s+2)} = 8$$

$$e_{ss} = \frac{1}{8} = 0.125$$

$$\boxed{\begin{aligned} t_r &= 0.471 \text{ s.} \\ t_s &= 3.91 \text{ s.} \\ \%M_p &= 44.43\% \\ e_{ss} &= 0.125 \end{aligned}}$$

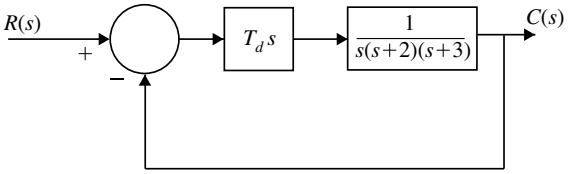


Fig. 2.62 Block diagram for Example 2.46

Example 2.46

Consider a certain third order system shown in Fig. 2.62 connected in cascade with a derivative controller. Determine the value of T_d so that the closed loop system is critically damped. For this value of T_d, find the steady state error for unit step input. Also determine the peak overshoot.

Solution

$$G(s) = \frac{1}{s(s+2)(s+3)}$$
$$G(s) = T_d s$$

The forward path T.F

$$G^1(s) = G(s)G_c(s)$$
$$= \frac{T_d}{(s+2)(s+3)}$$

The closed loop T.F. is given by

$$\frac{C(s)}{R(s)} = \frac{G^1(s)}{1+G^1(s)} = \frac{\frac{T_d}{(s+2)(s+3)}}{\left[1+\frac{T_d}{(s+2)(s+3)}\right]}$$
$$= \frac{T_d}{s^2+5s+(6+T_d)}$$

The characteristic equation is

$$s^2+5s+(6+T_d) = 0$$

Comparing this with the characteristic equation of a standard second order system. We get

$$\omega_n = \sqrt{(6 + T_d)}$$
$$2\zeta\omega_n = 5$$

For the system to be critically damped $\zeta = 1$.

$$\omega_n = \frac{5}{2} = 2.5$$
$$\sqrt{(6 + T_d)} = 2.5$$
$$T_d = 0.25$$
$$G^1(s) = \frac{0.25}{(s+2)(s+3)}$$

The position error constant K_p is

$$K_p = \lim_{s\to 0} G^1(s)$$
$$= \lim_{s\to 0} \frac{0.25}{(s+2)(s+3)} = \frac{0.25}{6}$$
$$e_{ss} = \frac{1}{1+K_p} = \frac{1}{1+\frac{0.25}{6}}$$
$$= \frac{6}{6.25} = 0.96$$

Since the system is critically damped, the overshoot $M_p = 0$.

$$\boxed{\begin{aligned} T_d &= 0.25 \\ e_{ss} &= 0.96 \\ M_P &= 0 \end{aligned}}$$

Example 2.47

A certain unity negative feedback system has the following forward path T.F.

$$G(s) = \frac{36}{s(s+4)}$$

The system is provided with a PD controller in the forward path. Determine the controller parameter K_p and K_d so that the system damping $\zeta = 0.8$ and natural frequency of oscillation $\omega_n = 9$.

Solution For the given system, the characteristic equation is

$$\begin{aligned} 1 + G(s) &= 0 \\ 1 + \frac{36}{s^2 + 4s} &= 0 \\ s^2 + 4s + 36 &= 0 \\ \omega_n &= \sqrt{36} = 6 \\ 2\zeta\omega_n &= 4 \\ \zeta &= \frac{4}{2 \times 6} = 0.333 \end{aligned}$$

The new values of damping factor and natural frequency of oscillation are

$$\begin{aligned} \zeta^1 &= 0.8 \\ \omega_n^1 &= 9 \end{aligned}$$

Using Eq. (2.98)

$$\begin{aligned} \omega_n^1 &= \omega_n \sqrt{K_p} \\ 9 &= 6\sqrt{K_p} \\ K_p &= 2.25 \end{aligned}$$

Using Eq. (2.99)

$$\begin{aligned} \zeta^1 &= \left(\zeta + \frac{K_d}{2}\omega_n\right) \frac{1}{\sqrt{K_p}} \\ 0.8 &= \left(0.333 + K_d \frac{6}{2}\right) \frac{1}{\sqrt{2.25}} \\ K_d &= 0.289 \end{aligned}$$

$$\boxed{\begin{aligned} K_p &= 2.25 \\ K_d &= 0.289 \end{aligned}}$$

Example 2.48

The forward path T.F. of a certain first order system connected with unity feedback is

$$G(s) = \frac{10}{(s + a)}$$

The system is connected in cascade with an integral controller whose T.F. is

$$G_c(s) = \frac{K_i}{s}$$

The permissible steady state error for unit ramp is 0.1. The required $\omega_n = 12$ rad/s. Determine the values of K_i and a.

Solution

$$G(s) = \frac{10}{(s+a)}$$
$$G_c(s) = \frac{K_i}{s}$$
$$G^1(s) = G(s)G_c(s) = \frac{K_i 10}{s(s+a)}$$

For ramp input

$$K_v = \lim_{s\to 0} sG^1(s) = \lim_{s\to 0} \frac{K_i 10}{0+a} = \frac{K_i 10}{a}$$
$$e_{ss} = \frac{1}{K_v} = 0.1$$
$$K_v = 10 \quad \text{or} \quad \frac{10K_i}{a} \quad \text{or} \quad K_i = a.$$

The characteristic equation is

$$1 + G^1(s) = s^2 + as + K_i 10$$

Comparing this with the characteristic equation of a standard second order system equation, we get

$$\omega_n = \sqrt{K_i 10}$$
$$144 = K_i 10$$

$$\boxed{\begin{aligned} K_i &= 14.4 \\ a &= 14.4 \end{aligned}}$$

SUMMARY

1. Time response of a system consists of two parts namely transient response and steady state response. In transient response, the variables are expressed as a

function of time where t is varied in the range $0 < t < \infty$. The steady state response is obtained as time t approaches infinity.

2. Transient response of first and prototype second order systems are obtained for impulse and step inputs.
3. For step input, transient response specifications for first and prototype second order systems are derived in terms of system parameters.
4. Steady state errors for prototype second order system are derived for step, ramp and parabolic inputs when the system is connected with unity feedback. The steady state errors are expressed in terms of static error constants K_p, K_v and K_a.
5. For a unity feedback system, the error can be expressed as a function of time using dynamic error constants. The steady state error is obtained by putting $t = \infty$.
6. For a non-unity feedback system, estimation of steady state error with $GH(s)$ will give erroneous result. The non-unity feedback system has to be converted to an equivalent unity feedback system and for this system only steady state errors are to be determined.
7. To enhance the transient and steady state performances controllers are employed either in the forward or in the feedback paths.
8. The controllers normally used are rate controllers PID controllers and addition of poles and zeros.
9. A prototype second order system is considered for the study of the performance of the controllers.
10. The rate controller is connected in the minor loop of the system. The rate controller increases the damping of the system and suppresses the oscillations in the system.
11. PID controller is the most widely used controller in industrial control systems. This controller is connected in the forward path in cascade of the prototype second order system.
12. The performance enhancement of P, I, D, PI, PD, ID and PID controllers is studied by plotting transient response curves by simulation in digital computer using MATLAB program.
13. For performance enhancement of higher order systems PID controller is the most suited controller. By judicious choice of proportional, integral and derivative term controller parameters, it is possible to obtain good transient response with less overshoot, less rise time and setting time.

EXERCISE

Short Answer Type Questions

1. **What do you understand by time response of a system?**
   The output response, starting at some initial condition expressed as a function of time when the input is applied, is called time response of the system. It is divided into two parts namely transient response and steady state response.

2. **What is transient response?**
   Transient response is the time response which goes from initial state to final state as time becomes large. The variation is expressed as a function of independent variable time t.
3. **What is steady state response?**
   The manner in which the output behaves as time t approaches infinity is called steady state response.
4. **Define impulse, step, ramp and parabolic inputs.**
   (a) Impulse input:

$$r(t) = R \quad t = 0$$
$$r(t) = 0 \quad t \neq 0$$

   (b) Step input:

$$r(t) = \begin{cases} Ru(t) & t \geq 0 \\ 0 & t < 0 \end{cases}$$

   (c) Ramp or velocity input:

$$r(t) = \begin{cases} Rtu(t) & t \geq 0 \\ 0 & t < 0 \end{cases}$$

   (d) Parabolic or acceleration input:

$$r(t) = \begin{cases} \frac{1}{2}Rt^2u(t) & t \geq 0 \\ 0 & t < 0 \end{cases}$$

5. **What are the Laplace transform of signals given in Question 4?**
   (a) $R(s) = R$ (for impulse)
   (b) $R(s) = R/s$ (for step)
   (c) $R(s) = R/s^2$ (for ramp)
   (d) $R(s) = R/s^3$ (for parabolic)
6. **How the impulse, step and ramp signals are related?**

   The relationship between impulse, step and ramp signals are as indicated below.

$$\text{Impulse} \xrightarrow{\text{Integrate}} \text{Step} \xrightarrow{\text{Integrate}} \text{Ramp}$$
$$\text{Ramp} \xrightarrow{\text{Differentiate}} \text{Step} \xrightarrow{\text{Differentiate}} \text{Impulse}$$

7. **How impulse response of a system is important?**

   (a) If the area under the impulse response curve is finite, the system is stable.

(b) By integrating the impulse response, step response can be obtained.
(c) If the impulse response function is known, then by taking the Laplace transform, the T.F. of the system can be obtained.

8. **Why systems are tested with step input?**
It is easy to generate a step function. If the step response is satisfactory, the given system is likely to give satisfactory performance to other commonly used signals. If the step response is known, impulse response is obtained by differentiating it and velocity response is obtained by integrating it.
9. **What are the performance specifications of a first order system?**
The performance specifications of first order system are
(a) Time constant T,
(b) Rise time t_r,
(c) Setting time t_s, and
(d) Time delay t_d.
10. **Define time constant of the first order system. How is it related to the location of the pole?**
Time constant T of a first order system is defined as the time taken for the step response to reach 63.2% of its final value for the first time. It is expressed as $T = 1/a$ where a is the location of the pole on the negative real axis in the s-plane.
11. **Define rise time of a first order system.**
Rise time t_r is defined as the time taken for the step response to go from 10% to 90% of its final value. It is expressed as $t_r = 2.2T$.
12. **Define time delay of a first order system.**
Time delay t_d of a first order system is defined as the time taken for the step response to reach 50% of its final value for the first time. It is expressed as $t_d = 0.693T$.
13. **Define settling time of a first order system.**
The settling time t_s of a first order system is defined as the time taken for the step response to reach and stay within 2% of its final value for the first time. It is expressed as $t_s = 3.91T$. For 5% error tolerance $t_s = 3T$.
14. **What information we get from performance specifications of first order system?**
The time constant of a first order system is inversely proportional to the pole located on the negative real axis of the s-plane. If the pole is moved away from the origin, the time constant is less. The rise time, settling time and the time delay are all directly proportional to the time constant. Hence, the time response is faster. The time response will be slower if the pole is moved towards the origin.
15. **What do you understand by prototype second order system?**
For a unity feedback system, the forward path T.F. and the closed loop T.F are respectively writing as

$$G(s) = \frac{\omega_n^2}{s(s + 2\zeta\omega_n)}$$

$$T(s) = \frac{C(s)}{R(s)}$$

$$= \frac{\omega_n^2}{(s^2 + 2\zeta\omega_n s + \omega_n^2)}$$

where ζ and ω_n are system parameters. The system with the above equations is called prototype second order system.

16. **What are the parameters of a prototype second order system and how they are differently expressed?**
ζ and ω_n are the only two parameters of a prototype second order system where ζ is called the damping ratio and ω_n is the natural frequency of oscillation expressed in rad/s. $\zeta\omega_n$ is called the damping factor and $\omega_d = \omega_n\sqrt{1 - \zeta^2}$ where ω_d is called damped frequency of oscillation in rad/s $\omega_n \geq \omega_d$.
17. **What is characteristic equation? Why is it called so?**
The dominator of the closed loop T.F of a system when equated to zero is called the characteristic equation. The corresponding polynomial is called characteristic polynomial. These polynomials contain the closed loop poles whose locations in the s-plane decide the stability and the time response of the system. Since it characterizes the system response, it is called characteristic equation.
18. **What is the characteristic equation and characteristic polynomial of a prototype second order system?**
The characteristic equation and the characteristic polynomial of a prototype second order system are respectively given below as

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

$$F(s) = s^2 + 2\zeta\omega_n s + \omega_n^2$$

19. **List the time domain specifications of a prototype second order system?**
The time domain specifications of a second order system are
(a) Peak overshoot,
(b) The time at which the peak overshoot occurs,
(c) Rise time,
(d) Settling time,
(e) Time delay,
(f) Time constant,
(g) The period of oscillations, and
(h) The number of oscillations during the settling time.
20. **For what type of input, the time domain specifications mentioned in Question 19 are defined?**
The time domain specifications are defined for the step input.

21. **How many specifications completely determine the time response?**
The percentage overshoot, rise time and settling time completely determine the time response.
22. **How maximum or peak overshoot is defined? What is the percentage maximum overshoot and how it is related to the damping ratio?**
The maximum overshoot is defined as the difference between the maximum value of the output $C_{\max}$ at any instant and the steady value C_{ss}. Thus, $M_p = C_{\max} - C_{ss}$. The percentage maximum overshoot is the amount by which the time response curve overshoots the steady state or final value at the peak time expressed as a percentage of the steady state value.

$$\%M_p = \frac{\text{Maximum overshoot, } M_p}{C_{ss}} \times 100$$

The % peak overshoot of a prototype second order system is

$$\%M_p = \frac{-\zeta\pi}{e\sqrt{1-\zeta^2}} \times 100$$

23. **What pole location characterize: (a) undamped, (b) the over damped, (c) the under damped and (d) critically damped and (e) negatively damped?**

(a) Undamped, $\zeta = 0;\ s_1,\ s_2 = \pm j\omega$

(b) Over damped, $\zeta > 1;\ s_1,\ s_2 = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$

(c) Under damped, $\zeta < 1;\ s_1,\ s_2 = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$

(d) Critically damped, $\zeta = 1;\ s_1,\ s_2 = -\omega_n$

(e) Negatively damped, $\zeta < 0;\ s_1,\ s_2 = \zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$

24. **How can you justify pole–zero cancellation in the T.F.?**
If the ratio $C(s)/R(s)$ is expressed in terms of poles and zeros they can be put into partial fraction. If any residue is negligibly small, that pole can be cancelled with a zero which is very close to that pole. The small value of the residue gives very insignificant contribution to the transient response and thus pole–zero cancellation is justified.
25. **What is an insignificant pole?**
The pole which is far away from the imaginary axis of the s-plane corresponds to fast decaying time response. If the real part of the magnitude of the pole is 5 to 10 times more than the real part of the dominant pole, it is called insignificant pole.
26. **Name any two conditions under which the response generated by a pole can be neglected?**
The response generated by a pole can be neglected if:
(a) Pole–zero cancellation takes place; and
(b) Insignificant poles are omitted.

27. **What are dominant poles?**
Poles that are closer to the imaginary axis of the left half of the s-plane give rise to transient response that will decay very slowly. Such poles are called dominant poles.
28. **In a system the real part of a pole generate what response?**
In a system, the real part of the pole generates exponential decay if it is in LHP and exponential rise if it is in RHP of the s-plane.
29. **In a system, the imaginary part of a pole generate what response?**
In a system, the imaginary part of a pole generates sinusoidal response.
30. **What is the effect of the presence of repeated poles in the imaginary axis?**
The presence of repeated poles in the imaginary axis will make the system unstable with unbounded output for bounded input.
31. **In a system if a pole is moved with a constant imaginary part what will the response have in common?**
If a pole is moved with a constant imaginary part, as shown in Fig. 2.63, the damped frequency of oscillation ω_d is constant. The more negative poles indicate higher value of damping factor $\zeta\omega_n$, which makes the system transient response more heavily damped. The damping ratio ζ is increased which results in less % peak overshoot. The time constant T, rise time t_r and the time at which peak value occurs t_p are shortened.
32. **If a pole is moved with a constant real part what will the response have in common?**
Figure 2.64 shows the movement of complex poles with their real parts remaining constant at $\zeta\omega_n$. For upward movement of the pole, the time constant remains the same. However, the damping ratio decreases and ω_n and ω_d increase. This amounts to high-overshoot. The settling time is unaffected.
33. **If a pole is moved along a radial line extending from the origin what will the response have in common?**
In Fig. 2.65, the movement of complex poles along a radial line extended from the origin is shown. As the poles move away from the origin, the natural frequency of oscillation is increased. The damping ratio ζ remains the same. This amounts to constant peak overshoot. The time constant T, rise time t_r settling time t_s are all decreased. The rate of decay is faster.
34. **Consider the unity feedback system with the following forward path T.F.**

$$\text{(a)} \quad G(s) = \frac{16}{s(s+4)}$$

$$\text{(b)} \quad G(s) = \frac{(s+16)}{s(s+3)}$$

Determine, in each case, the characteristic equation damping factor and natural frequency of oscillation.

$$\text{(a)} \quad \frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)} = \frac{16}{(s^2+4s+16)}$$

The characteristic equation is $s^2 + 4s + 16 = 0$, $\omega_n = 4$; and $\zeta = 0.5$

$$\text{(b)} \quad \frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)} = \frac{(s + 16)}{(s^2 + 4s + 16)}$$

The characteristic equation is $s^2 + 4s + 16 = 0$, $\omega_n = 4$; and $\zeta = 0.5$.

35. **In Question 34, the systems (a) and (b) have the same ζ and ω_n. Will they have the same time domain specifications?**
Eventhough, ζ and ω_n are the same for system (a) and system (b), their time domain specifications will be totally different. The formulae derived in Sect. 2.4.4 are valid for system (a) and they are not valid for system (b). This is due to the presence of a zero at $s = -16$ for system (b).

36. **Consider the closed loop T.F. of the following second order systems. Find ζ, ω_n and the nature of damping.**

$$\text{(a)} \quad T(s) = \frac{(8s + 4)}{s^2 + 5s + 25}$$
$$\text{(b)} \quad T(s) = \frac{(4 - 3s)}{s^2 + 4s + 4}$$
$$\text{(c)} \quad T(s) = \frac{(s - 6)}{s^2 + 10s + 16}$$

(a) $s^2 + 5s + 25 = 0$; $\omega_n = 5$; $\zeta = 0.5$; $\zeta < 1$. The system is under damped.
(b) $s^2 + 4s + 4 = 0$; $\omega_n = 2$; $\zeta = 1$; $\zeta = 1$. The system is critically damped.
(c) $s^2 + 10s + 16 = 0$; $\omega_n = 4$; $\zeta = 1.25$; $\zeta > 1$. The system is over damped.

37. **For a system with unity feedback,**

$$G(s) = \frac{(25 - Ts)}{s(s + 10)}$$

Find the value of T so that the system is undamped. What is the undamped frequency?

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)}$$
$$= \frac{25 - Ts}{s^2 + (10 - T)s + 25}$$

$$10 - T = 0 \text{ or } T = 10$$

The characteristic equation is

$$s^2 + 25 = 0$$
$$\omega_n = 5$$

38. **For the following closed loop T.F find the value of K so that $\zeta = 1$.**

$$T(s) = \frac{(s+2)}{5s^2 + 2s + K}$$
$$K = 0.2$$

39. **For the following closed loop T.F find the value of K so that $\zeta = 0.6$.**

$$T(s) = \frac{(s+0.6)}{[4s^2 + 2.4s + (K+6)]}$$
$$K = 10$$

40. **For the following closed loop T.F find the value of K so that $\omega_d = 2$ rad/s**

$$T(s) = \frac{(Ks+2)}{5s^2 + (2K+5)s + 25}$$
$$K = 2.5$$

41. **For the following closed loop T.F find the value of K so that the undamped natural frequency of oscillation is 8 rad/s**

$$T(s) = \frac{(s+K)}{Ks^2 + 5s + 100}$$
$$K = 1.5625$$

42. **For the following closed loop system T.F. find the % peak overshoot.**

$$T(s) = \frac{125}{5s^2 + 10s + 500}$$
$$\%M_p = 72.94\%$$

43. **For the following closed loop T.F find t_p.**

$$T(s) = \frac{64}{s^2 + 8s + 64}$$
$$t_p = 0.4534 \text{ s}$$

44. **For the following closed loop T.F. find the rise time t_r.**

$$T(s) = \frac{100}{5s^2 + 10s + 100}$$
$$t_r = 0.412 \text{ s}$$

45. **For the following closed loop T.F find the settling time t_s for 5% error tolerance.**

$$T(s) = \frac{60}{8s^2 + 32s + 72}$$
$$t_s = 1.5 \text{ s}$$

46. **Define steady state error.**

Steady state error is the difference between the given input and the output as $t \to \infty$ which amounts to the natural response having decayed to zero.

47. **What are the sources of steady state error?**
The sources of steady state errors are
(a) System configuration,
(b) Type of input applied, and
(c) Non-linear sources such as back lash in gears or a motor that will not move unless the input voltage exceeds the threshold value, saturation in generator *etc.*.

48. **How the steady state error is calculated for a unity feedback system if the forward path T.F is known?**
If the forward path T.F. $G(s)$ is known, then the steady error is calculated from

$$e_{ss} = \lim_{s \to 0} s \frac{R(s)}{1 + G(s)}$$

where $R(s)$ is the Laplace transform of the input.

49. **For a unity feedback system, if the closed loop T.F $T(s)$ is known how the steady state error is calculated?**
For a unity feedback system if the closed loop T.F. $T(s)$ is known, then the steady state error is calculated from

$$e_{ss} = \lim_{s \to 0} sR(s)[1 - T(s)]$$

50. **If $G(s)$ is the forward path T.F and $H(s)$ is the T.F of the feedback path does the following equation give the steady state error? If not, what information we get form this equation.**

$$\lim_{s \to 0} \frac{sR(s)}{[1 + GH(s)]}$$

The above equation does not give the steady state error. It gives the steady state actuating signal.

51. **How do you calculate the steady state error for a non-unity feedback system?**
The steady state error for non-unity feedback system is calculated by converting $G(s)$ and $H(s)$ into $G_e(s)$ which is the equivalent forward path T.F of the unity feedback system.

$$G_e(s) = \frac{G(s)}{1 + G(s)H(s) - G(s)}$$
$$e_{ss} = \lim_{s \to 0} \frac{sR(s)}{1 + G_e(s)}$$

52. **What are static error constants and how they are used to determine the steady state error?**
The position error constant K_p, velocity error constant K_v and acceleration error constant K_a are called static error constants. They are defined as follows:

$$K_p = \lim_{s \to 0} G(s)$$
$$K_v = \lim_{s \to 0} sG(s)$$
$$K_a = \lim_{s \to 0} s^2 G(s)$$

53. **What is the effect of increasing the system gain upon the steady state error?**
By increasing the system gain the steady state error is reduced.
54. **What is the effect of increasing the system gain of a prototype second order system upon the maximum overshoot?**
If the system gain of a prototype second order system is increased, the maximum overshoot is also increased.
55. **Identify the inputs to which the static error constants are related.**
The position error constant K_p is defined when the input is step. The velocity error constant K_v is defined when the input is ramp and the acceleration error constant is defined when the input is parabolic.
56. **What is the disadvantage of using position error constant?**
From the position error constant, the information we get about the steady state error is either it is zero, or finite or infinite. The information about the accumulation of the error is missing.
57. **Define dynamic error constants.**
The dynamic error constants are defined as follows:

$$C_0 = \lim_{s \to 0} F(s)$$
$$C_1 = \lim_{s \to 0} \frac{dF(s)}{ds}$$
$$\vdots \quad \vdots$$
$$C_n = \lim_{s \to 0} \frac{d^n F(s)}{ds^n}$$

where $F(s) = 1/1 + G(s)$.

58. **What is the advantage of dynamic error constants over static error constants?**
The dynamic error constants give error as a function of time. Thus, the accumulation of error with respect to time is estimated.
59. **How many integrations in the forward path should be there in order that the steady state error is zero when the inputs are (a) step; (b) ramp; and (c) parabolic?**
(a) For step input there should be one integration.
(b) For ramp input there should be two integrations.
(c) For parabolic input there should be three integrations.
60. **A certain feedback control system has the following T.F.**

$$G(s) = \frac{16}{s(s+4)}$$
$$H(s) = \frac{2}{(s+1)}$$

What is the type of this system?

$$G_e(s) = \frac{16(s+1)}{s^3 + 6s^2 - 8s + 16}$$

Since there is no integrator in the $G_e(s)$ the system is Type 0 and not Type 1 which is erroneously obtained from

$$GH(s) = \frac{32}{s(s+4)(s+1)}$$

61. **If the following input is applied to the feedback system, what are the dynamic error constants needed to determine the steady state error?**

$$r(t) = t^3$$

The dynamic error constants C_0, C_1, C_2, and C_3 need to be determined to estimate the steady state error.
62. **Consider the system shown in Fig. 2.66. Will the characteristic equation be different for $C(s)/R(s)$ and $C(s)/D(s)$?** The characteristic equation of any system will be same irrespective of the type of input given and the place where it is applied. For the system shown in Fig. 2.66, the characteristic equation is $1 + G_1G_2H(s) = 0$.
63. **What are the effects of rate feedback controller on the system performance?**
By rate feedback controller, the effective damping is increased. However, the steady state error is also increased.

64. **What are the effects of rate feedback controller on M_p, t_r, t_s and ω_n?**
Overshoot is decreased. Rise time is increased. Settling time is decreased. Natural frequency of oscillation remains unchanged.
65. **What is a three term controller?**
The proportional plus the integral plus the derivative controller which has three parameters K_p, K_i and K_d is called three term controller.
66. **What are the effects of proportional controller on the prototype second order system performance specifications?**
As the proportional gain is increased, M_p is increased. t_p is decreased, t_s is unchanged. The bandwidth is increased. The steady state error for ramp input is reduced by a factor $1/\sqrt{K_p}$.
67. **What are the effects of derivative controller on the prototype second order system performance specifications?**
The second order system with derivative controller is reduced to a first order system by pole–zero cancellation. There is no peak overshoot. The time constant, rise time and settling time are less and hence the time response is faster.
68. **What is the effect of I controller on the prototype second order system performance?**
When the integral controller is connected in cascade with a prototype second order system, it becomes unstable. However in general, the I controller permits the rejection of a step disturbance. It accelerates the movement of the process set point and eliminates the residual steady state error.
69. **What are the effects of PI controller on the system performance?**
PI controller improves the damping and reduces the maximum overshoot. Increases the rise time and decreases the bandwidth. Filters out high frequency noise. Increases the type of the system by one and the e_{ss} for step and ramp inputs is zero. For parabolic input e_{ss} is finite.
70. **What are the effects of PD controller when connected with a second order system?**
The PD controller when connected in cascade with a second order system:

    (a) Improves damping and reduces overshoot,
    (b) Reduces t_r and t_s.
    (c) Increases bandwidth.
    (d) Improves phase margin and gain margin and thus increases the relative stability.
    (e) Amplifies noise at higher frequencies.

71. **What are the effects of PID controller on the transient response of higher order system?**
By proper tuning of the PID controller parameters, the stability of the system is improved. The maximum overshoot can be reduced. System rise time and settling time can be reduced. The system bandwidth can be increased.

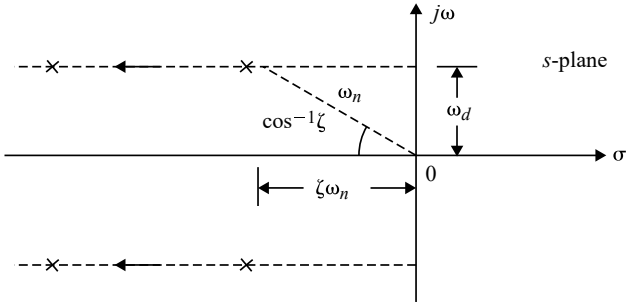


Fig. 2.63 Movement of pole with constant imaginary part

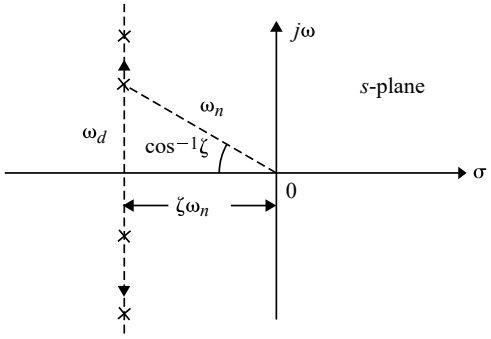


Fig. 2.64 Movement of pole with constant real part

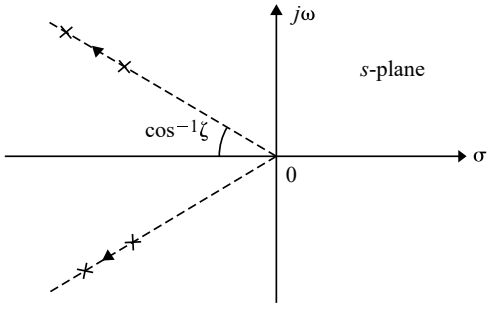


Fig. 2.65 Movement of pole along a radial line from the origin

Long Answer Type Questions

1. For the R-L circuit shown in Fig. 2.67a, the input voltage $v(t) = u(t)$. Drive the expression for the current $i(t)$ and sketch the wave form. Determine the time constant and rise time.

$$c(t) = 0.5(1 - e^{-2t})$$

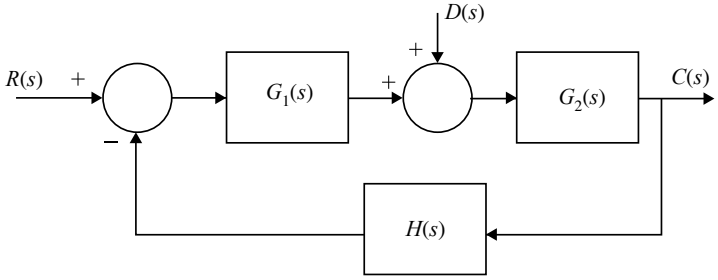


Fig. 2.66 Block diagram for Question 62

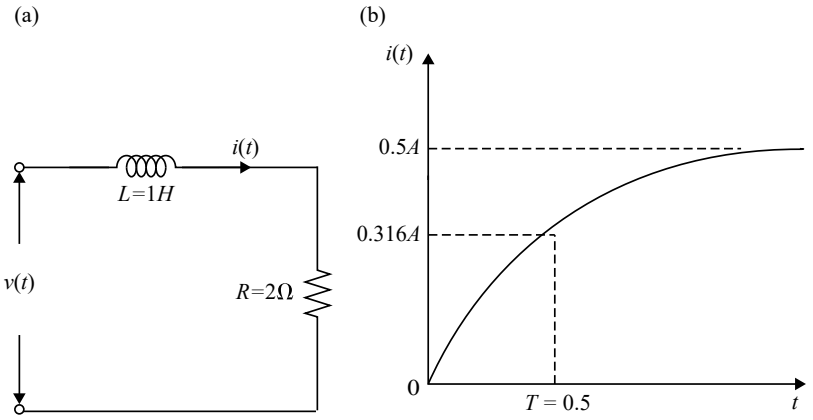
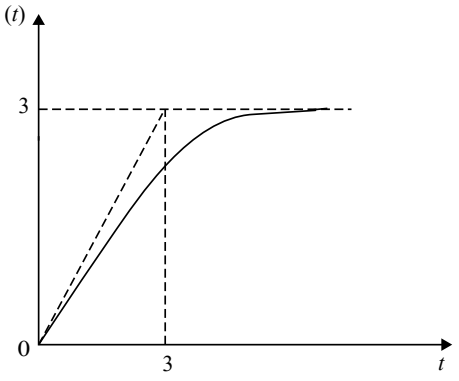


Fig. 2.67 **a** Electric circuit; and **b** current wave form

Fig. 2.68 Time response curve



Time constant $T = 0.5$ sec; rise time $t_r = 1.1$ s.

2. The time response of the first order system for unit step input is shown in Fig. 2.68. The system T.F is $G(s) = K/(s + a)$. Find K and a.

$$a = \frac{1}{3}; \quad K = 1$$

3. Consider the first order system shown in Fig. 2.69. Find the time constant, rise time, time delay and settling time for unit step input for the closed loop system. $T = 0.25$ s; $t_r = 0.55$ s; $t_d = 0.1733$ s; $t_s = 0.98$ s.
4. A certain non-unity feed back control system is represented as shown in Fig. 2.70. Derive an expression for the time response of the system as a function of time for (a) unit impulse; (b) unit step; (c) unit ramp inputs; and (d) also find T, t_r, t_d and t_s and C_{ss} for unit step input.

(a) $c(t) = 0.8e^{-2.4t}$

(b) $c(t) = \frac{1}{3}[1 - e^{-2.4t}]$

(c) $c(t) = \left[\frac{1}{3}t - 0.139 + 0.139e^{-2.4t}\right]$

(d) $T = 0.4167sec$; $t_r = 0.9167$; $t_d = 0.2888$ s; $t_s = 1.63$ s; $C_{ss} = \frac{1}{3}$

5. A certain second order unity feedback system is described by the following forward path T.F.

$$G(s) = \frac{25}{s(s + 6)}$$

Derive an expression for the output response of the system for unit step input. Also find t_r, t_p, T, M_p and t_s.

$$c(t) = 1 - 1.25e^{-3t} \sin(4t + 0.928 \text{ rad}).$$

$t_r = 0.5535$ sec; $t_p = 0.785$ sec; $T = 0.333$ sec; $t_s = 1.3$ sec; $M_p = 9.5\%$.

6. A certain unity feedback control system has the following open loop T.F.

$$G(s) = \frac{6}{s(s + 2)}$$

Find the values for t_r, t_p, t_s, M_p and % M_p for a step input of 10 units.

$t_r = 0.724$ sec; $t_s = 3.91$ sec; $t_p = 1.4$ sec; $M_p = 2.456$ units; $\%M_p = 24.56\%$.

7. For the system represented in block diagram shown in Fig. 2.71 write down the characteristic equation for (a) $K = 9.5$; (b) $K = 11$; and (c) $K = 12$.

(a) $F(s) = s^3 + 6s^2 + 11s + 63$
(b) $F(s) = s^3 + 6s^2 + 11s + 72$
(c) $F(s) = s^3 + 6s^2 + 11s + 78$

8. For a certain unity feedback control system, the forward path T.F. is given by,

$$G(s) = \frac{10(1 + Ks)}{s(s + 2)}$$

Determine the value of K so that the damping ratio is 0.5. For this value of K determine the maximum overshoot, time constant and rise time for unit step input. What is the steady state error for unit ramp input with and without K? Also derive expression for the time response for unit step input for the above cases.

$$K = 0.1162$$

Time domain specifications	T	t_r	t_p	$\%M_p$	e_{ss} for **ramp input**
$K = 0$	1 s	0.63 s	1.05 s	36%	0.2
$K = 0.1162$	0.6325	0.65 s	1.032 s	17.3%	0.2

For $K = 0$; $c(t) = 1 - 1.054e^{-t}\sin(3t + 1.25 \text{ rad})$
For $K = 0.1162$; $c(t) = 1 - 1.02e^{-t}\sin(2.74t + 1.361 \text{ rad})$

9. A unity feedback control system has the following forward path T.F.

$$G(s) = \frac{K}{s(1 + sT)}$$

where K and T are constants and greater than zero. Determine the factor by which K should be multiplied to reduce the overshoot for step input from 85 to 35%.

$$K_2 = 0.02658K_1$$

where K_1 corresponds to 85% overshoot and K_2 corresponds to 35% overshoot. The gain has to be reduced by a factor 0.02658.

10. A certain unity feedback control system has the following forward path T.F.

$$G(s) = \frac{K(s + b)}{(s + a)^2}$$

where K, a and b are constants and greater than zero. The steady state error due to unit step input is to be limited to 0.2 with a damping ratio of 0.4 and undamped frequency of oscillation as 6 rad/s. Find K, a and b.

$$K = 0.328; \quad a = 4.472; \quad b = 15.24$$

11. A certain unity feedback control system has the following forward path T.F.

$$G(s) = \frac{25.4}{s(s^2 + 8s + 17)}$$

Find the dominant poles and the insignificant pole of the closed loop system.

$$\text{Dominant poles, } s_{12} = -1 \pm j1.8$$
$$\text{Insignificant pole, } s_3 = -5.9$$

12. For the rate feedback system shown in Fig. 2.72 it is desired to have the closed loop system to be placed at $s_{12} = -2 \pm j4$.

Find the values of K and K_t. What is the type and order of the system? Find the steady state error for unit ramp input.

(a) $K = 23.34; \quad K_t = 0.0428$
(b) Type 0 and order 2 system
(c) $e_{ss} = 10.1714$

13. A certain unity feedback system has the following forward path T.F.

$$G(s) = \frac{K(s + a)}{s(s + b)}$$

where K, a and b are constants and greater than zero. It is required that the steady state error for unit ramp input is less than 0.1. To improve transient response, the closed loop system poles are to be placed at $s_{12} = -2 \pm j4$. Find the values of K, a and b.

$$K = 2; \quad a = 10; \quad b = 2$$

14. Consider the system shown in Fig. 2.73. (a) Find the value of K so that the damping ratio of the closed system is 0.5. (b) Determine the values of t_r, T and t_s if the system is subjected to unit step input. (c) Derive an expression for the impulse response of the system.

(a) $K = 0.2$;

(b) $t_r = 0.605$ s; $T = 0.5$ s; $t_s = 1.955$ s;

(b) $c(t) = 4.62e^{-2t} \sin 3.464t$

15. A certain unity feedback system has the following forward path T.F.

$$G(s) = \frac{40(s+2)}{s(s+1)(s+4)}$$

Determine the type and order of the system, the static error coefficients and the steady state error for ramp input with magnitude 4.

$K_p = \infty$; $K_v = 20\ \text{s}^{-1}$; $K_a = 0$; $e_{ss} = 0.2$, Type 1 and order 3

16. A certain feedback control system has the following forward and feedback path T.F.

$$G(s) = \frac{20}{(s+1)(s^2+10s+6)}$$
$$H(s) = \frac{5}{(s+3)}$$

Find the steady state error when the system inputs are (a) $r(t) = 5$; (b) $r(t) = 4t$ and (c) $r(t) = \frac{3}{2}t^2$.

$$G_e(s) = \frac{20(s+3)}{s^4+14s^3+49s^2+34s+58}$$

(a) $K_p = \frac{60}{58}$; $e_{ss=2.46}$ (b) $K_v = 0$; $e_{ss} = \infty$ (c) $K_a = 0$; $e_{ss} = \infty$.

17. For a unity feedback system, the forward path T.F. is

$$G(s) = \frac{K}{s^2(s+2)(s+3)}$$

Find the value of K to limit the steady state error to zero when the input to the system is $r(t) = 1 + 10t + 20t^2$.

The steady state error is zero for all values of K since $K_p = \infty$ and also $K_v = \infty$ which corresponds to $r(t) = 1$ and $r(t) = 10t$ respectively.

18. Using the generalized error series calculate the steady state error of a unity feedback system having the open loop T.F. as

$$G(s) = \frac{10}{(s+2)}$$

For the following excitation: (a) $r(t) = 2$; (b) $r(t) = 2t$; (c) $r(t) = \frac{1}{2}t^2$; (d) $r(t) = 1 + 2t + \frac{1}{2}t^2$.

(a) $C_0 = \dfrac{1}{6}; \quad e(t) = \dfrac{1}{3}; \quad e_{ss} = \dfrac{1}{3}$

(b) $C_1 = \dfrac{10}{144}; \quad e(t) = \dfrac{2}{3}t + \dfrac{20}{144}; \quad e_{ss} = \infty$

(c) $C_2 = -2.5; \quad e(t) = \dfrac{1}{12}t^2 + \dfrac{10}{144}t - 1.25; \quad e_{ss} = \infty$

(d) $e(t) = \dfrac{1}{6}\left(1 + 2t + \dfrac{1}{2}t^2\right) + \dfrac{10}{144}(2 + t) - 1.25; \quad e_{ss} = \infty$

19. The open loop T.F. of a unity feedback system is

$$G(s) = \frac{s + 3.15}{s(s + 1.5)(s + 0.5)}$$

Find the type and order of the system. Find the static error constants and the steady state error corresponding to them.

Type 1 order 3 system

$$K_p = \infty; \quad K_v = 4.2; \quad K_a = 0$$

$$e_{ss} = \frac{1}{1 + \infty} + 0.238 + \frac{1}{0} = \infty$$

20. The closed loop T.F. of a unity feedback system is

$$T(s) = \frac{20}{s^2 + 4s + 25}$$

Find the steady state error for unit step input.

$$e_{ss} = 0.2$$

21. Consider the system shown in Fig. 2.74. The steady state error due to a unit ramp input is 0.2 and due to unit step disturbance is -0.5. Determine the values of K_1 and K_2.

$$K_1 = 1.3333; \quad K_2 = 125$$

22. Consider the system shown in Fig. 2.75. What is the type of the system? Find the value of K to get 2% steady state error for unit step input.

Type 0 system; $K = 1.0204$

23. For the system shown in Fig. 2.76 determine the type and order. Identify the input for which the steady state error is finite and calculate the same.

    Type 0 and order 3 system. For step input only the steady state error is finite. $e_{ss} = R$.

24. For the system shown in Fig. 2.77 find K_1 and K_2 so that the percentage error for step input is 10% with a rise time of 0.2 s. What is the steady state error for unit ramp input?

$$K_1 = 18.6; \quad K_2 = 1.411; \quad e_{ss} = 0.0866$$

.

25. For the system shown in Fig. 2.78 find the steady state error for unit step input?

    The system is unstable. Steady state error does not exist.

26. For the system shown in Fig. 2.79, find the value of K so that the steady state error for unit step input is zero. For this value of K what is the steady state error for unit ramp input?

$$K = 7.5; \quad e_{ss} = \frac{1}{45}$$

    for ramp input.

27. For the system shown in Fig. 2.80 find the value of K so that the steady state error for a ramp input is less than 0.1.

$$K = 40$$

28. The forward path T.F. of a unity feedback system is

$$G(s) = \frac{40}{s(0.2s + 1)}$$

    Obtain the generalized error constants. Also find its steady state error when the input is $r(t) = (3 + 4t)t$.

$$C_0 = 0; \quad C_1 = \frac{1}{40}; \quad C_2 = -\frac{1}{400}$$

$$e(t) = \frac{1}{40}(3 + 8t) - \frac{1}{100}; \quad e_{ss} = \infty$$

29. The forward path T.F. of a unity feedback system is

$$G(s) = \frac{10}{(s + 2)}$$

Find the steady state error for a step input of magnitude 5 for the following controllers which are connected in cascade with $G(s)$. (a) PI controller; and (b) PD controller.

$$\text{(a) } e_{ss} = 0; \quad \text{(b) } e_{ss} = \frac{5}{(1 + 5K_p)}$$

where K_p is the proportional term constant of the PD controller.

30. The forward path T.F. of a unity feedback system is

$$G(s) = \frac{5}{s(s^2 + 3s + 5)}$$

Find the steady state error for unit ramp input when the following controllers are connected is cascade with $G(s)$. (a) PI controller (b) PD controller.

$$(a)\ e_{ss} = 0; \quad (b)\ e_{ss} = \frac{1}{K_p}$$

31. A PD controller is connected in cascade with a plant as shown in Fig. 2.81. Find the PD controller parameters K_p and K_d so that the closed loop system damping ratio is 0.5, and the natural frequency of oscillation is 10 rad/s.

$$K_p = 5.25; \quad K_d = 0.5$$

32. Consider the system shown in Fig. 2.82

Choose the value of the proportional controller K_p so that the time constant of the closed loop system is one fourth of its open loop value.

$$K_p = 0.15$$

33. Consider the system shown in Fig. 2.83

(a) Find the values of K_p and K_d so that the damping ratio is 0.8 and undamped natural frequency of oscillation is 12 rad/s for the closed loop system. (b) Find the values of K_P and K_d if the closed loop poles are to be at $s = -4$ and $s = -8$.

$$\text{(a)} \quad K_p = 1.44; \quad K_d = 0.092$$
$$\text{(b)} \quad K_p = 0.32; \quad K_d = 0.02$$

34. The closed loop T.F. of a unity feedback control system is given as

$$T(s) = \frac{K(s + 5)}{[s^3 + 10s^2 + (3K + 5)s + 5K]}$$

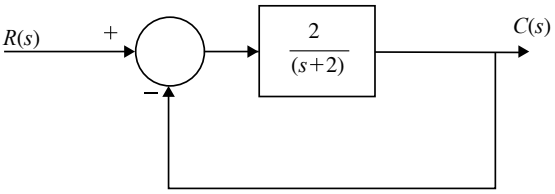


Fig. 2.69 Block diagram for Question 3

Find K so that the steady state error for unit ramp input is 0.9.

$$K = 2$$

35. A certain control system is described by the following forward and feedback path transfer functions.

$$G(s) = \frac{1}{s^2(s+5)}$$
$$H(s) = 3(s+5)$$

The error function is defined as

$$e(t) = \frac{1}{K_h} r(t) - c(t)$$

Find the steady state error for unit ramp input

$$e_{ss} = \frac{1}{75}$$

36. The forward path T.F. of a certain unity feedback system is

$$G(s) = \frac{K}{(s+2)(s+a)}$$

$K_p = 5$ and the required overshoot is less than 16.3%. Find K, a and the steady state error.

The above conditions are satisfied for the following values of K and a.

(a) $K = 74.64$; $a = 7.464$; $e_{ss} = 16.7\%$
(b) $K = 5.35$; $a = 0.535$; $e_{ss} = 16.7\%$

37. The closed loop T.F. of a certain control system is given by

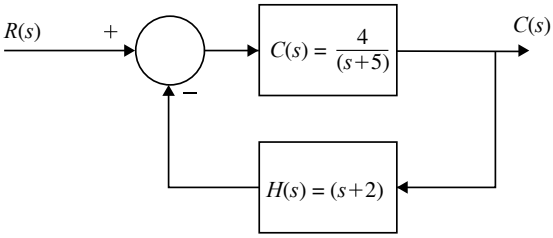


Fig. 2.70 Block diagram for Question 4

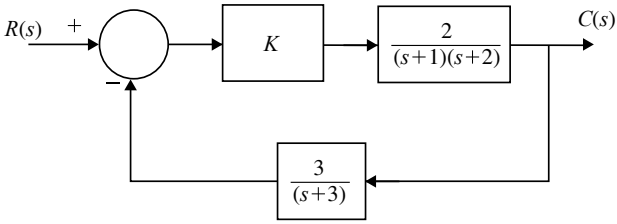


Fig. 2.71 Block diagram for Question 7

Fig. 2.72 Rate feedback system for Question 12

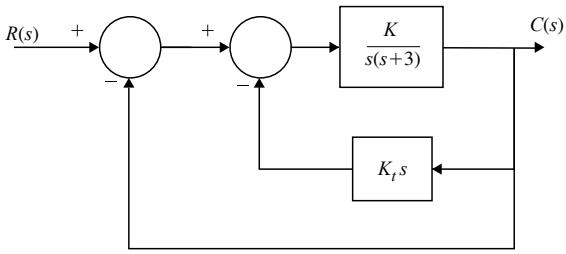
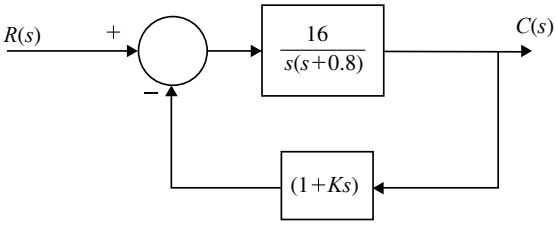


Fig. 2.73 Rate feedback system for Question 14



$$T(s) = \frac{225}{(s^2 + s + 9)(s^2 + 2s + 25)}$$

Determine the damping ratio of the dominant characteristic roots.

(a) $\zeta_1 = \frac{1}{6}$; (b) $\zeta_2 = \frac{1}{5}$

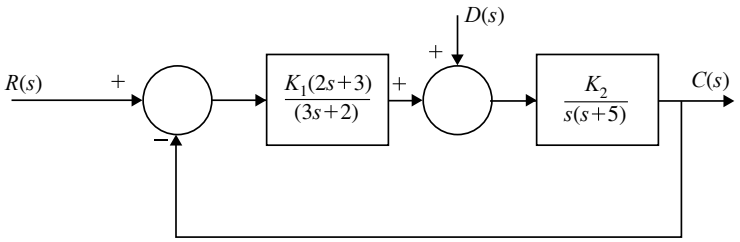


Fig. 2.74 Block diagram for Question 21

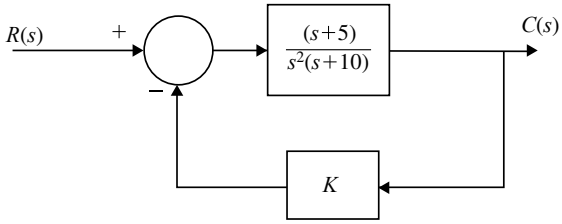


Fig. 2.75 Block diagram for Question 22

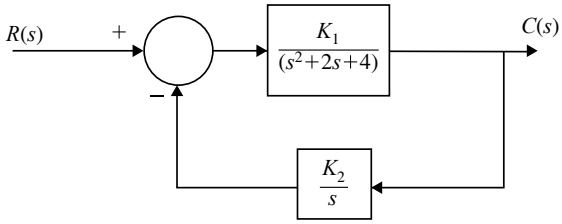


Fig. 2.76 Block diagram for Question 23

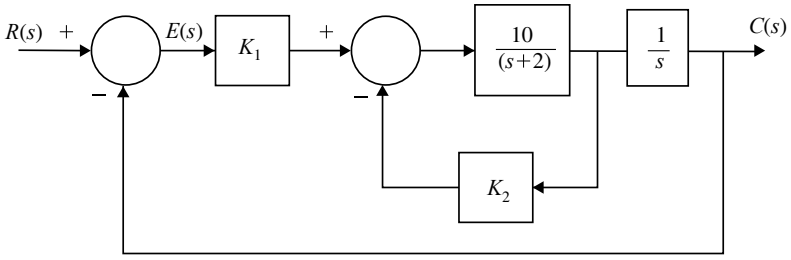


Fig. 2.77 Block diagram for Question 24

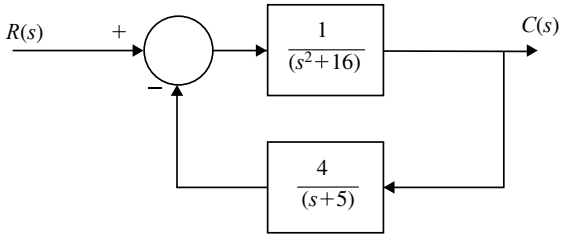


Fig. 2.78 Block diagram for Question 25

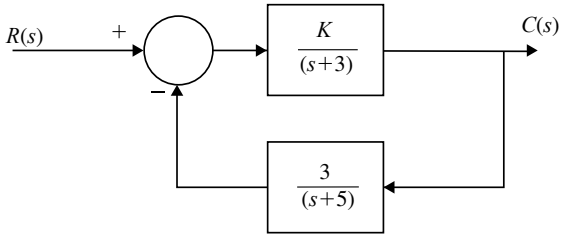


Fig. 2.79 Block diagram for Question 26

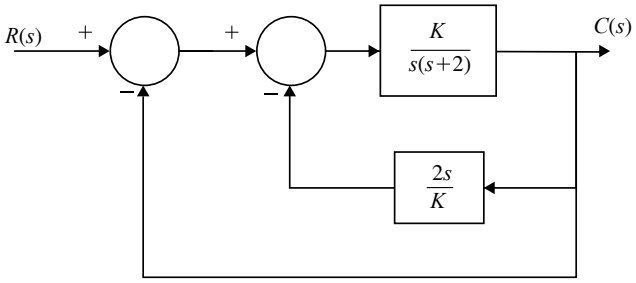


Fig. 2.80 Block diagram for Question 27

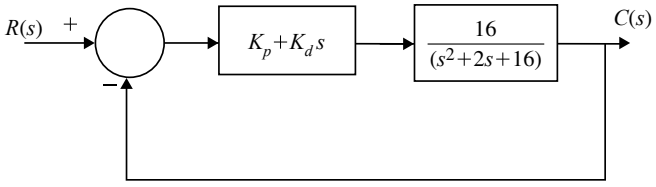


Fig. 2.81 PD Controller block diagram for Question 31

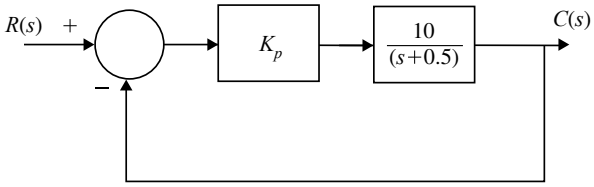


Fig. 2.82 Block diagram for Question 32

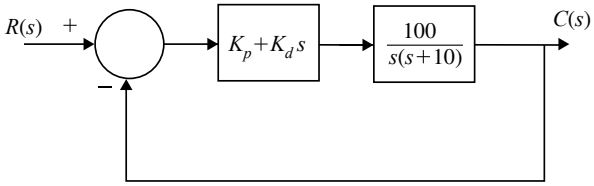


Fig. 2.83 Block diagram for Question 33

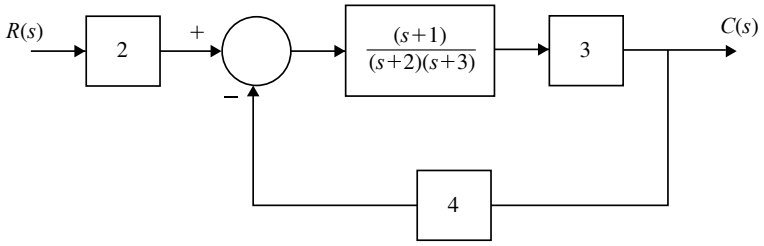


Fig. 2.84 Block diagram for Question 38

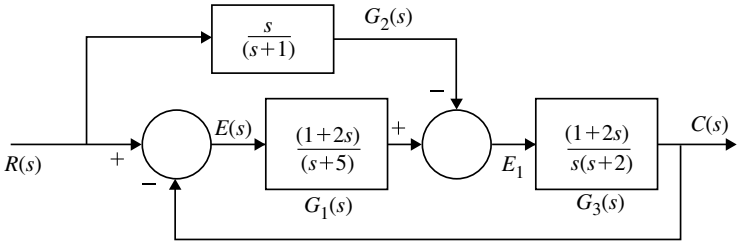


Fig. 2.85 Block diagram for Question 40

38. For the system shown in Fig. 2.84, find the steady state error for unit step input. The error is defined as $e(t) = r(t) - c(t)$.

$$e_{ss} = 0.4$$

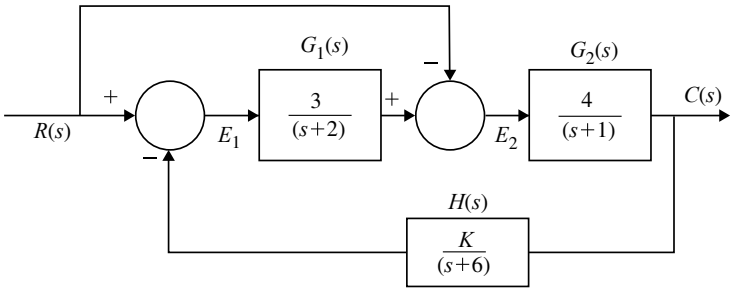


Fig. 2.86 Block diagram for Question 41

39. For the following closed loop T.F. determine the steady state error for unit parabolic input. The error is defined as $e(t) = r(t) - c(t)$

$$T(s) = \frac{(-2s^2 + 3s + 10)}{(s^3 + 5s^2 + 3s + 10)}$$
$$e_{ss} = 0.7$$

40. Consider the system shown in Fig. 2.85. Determine the steady state error for the input $r(t) = 3t$.

$$e_{ss} = 45$$

41. Consider the block diagram shown in Fig. 2.86. Find the value of K so that the steady state error due to step input is zero.

    For the value of K find the steady state error for unit ramp input.

$$e(t) = r(t) - c(t)$$

$$K = 1$$
$$e_{ss} = 1.667$$

Chapter 3
Frequency Response Analysis



Learning Objectives

After completing this chapter, you should be able to

- Understand the concept of frequency response.
- Draw the polar plot for minimum phase, non-minimum phase and all phase transfer functions.
- Draw the Bode plot.
- Determine the transfer function from the Bode plot.
- Establish the correlation between transient response and frequency response.
- Define frequency domain specifications and derive expressions for these specifications.
- Determine the closed loop frequency response from open loop frequency response using constant M and constant N circles and Nichols chart.
- Adjust the gain to satisfy the frequency response specifications.
- Use MATLAB program for the frequency response analysis.

3.1 Introduction

In Chap. 2, the dynamic response of linear time invariant continuous time system was studied when typical test signals such as impulse, step and ramp were applied. The performance of many real-time control systems is judged based on the time response of the system due to the above test signals. However, a simple and useful test signal is a sine wave and we get valuable information when applied to a certain class of

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S. Palani, *Automatic Control Systems*,
https://doi.org/10.1007/978-3-030-93445-3_3

systems such as communication system where most of the signals to be processed are either sinusoidal or composed of sinusoidal signals. When the input applied to a system is a sine wave the output obtained from the system is also a sine wave but with different amplitude and phase. By measuring the output amplitude and phase of a system over a range of frequency of the input sine wave, a different dynamic response is observed and this is called the frequency response of the system which represents the system's response to sinusoidal inputs at varying frequencies.

The frequency response is defined as the plot of magnitude and phase difference between the input and output sinusoids over a range of frequencies.

The frequency response method may be less intuitive than the other methods described in the previous chapter. However, it has the following advantages and disadvantages.

Advantages of Frequency Response Method

1. It is easy to generate sine wave and collect data from the output.
2. Experimental information can be used with ease for design purpose.
3. Raw measurements of the output amplitude and phase of the system are enough to design a suitable feedback control. It *directly* quantifies system performance and *directly* designs control systems.
4. In real-life situations such as modelling the transfer function no intermediate processing of the data is required.
5. It offers the most cost-effective design method especially for systems that are stable in open loop.
6. It is the easiest method to use for designing compensation.
7. The tests involve measurement under steady state conditions which are simpler to analyse compared to measurements of transient responses.
8. From test results it is easy to get the control system order, gain, error constant, resonant frequencies, *etc.*
9. The frequency response method can be very well used for the analysis and design of certain non-linear systems too using describing function method.
10. Since only one frequency at a time is used, the amplitude of the sine wave can be increased or decreased as one wishes.
11. The method is frequency selective and can therefore reject the noise in the output in an excellent way.

Disadvantages of Frequency Response Method

1. It is not often easy to correlate the results of the frequency response characteristics of higher order systems to that of time domain characteristics.
2. Systems like process control operate at low frequencies. It is difficult to generate low-frequency signals and obtain necessary measurements. Measurement may require several hours in such cases.
3. In frequency response method, the analysis is done at one frequency at a time which requires very long test time.
4. If the spacing of the test frequencies is too far apart, it is possible to miss some important frequency response characteristics.

5. The frequency response method as considered by many researches and professors is an old-fashioned method even though for many practical control and signal processing engineers it is a very useful tool.

3.2 Obtaining Steady State Output to Sinusoidal Input

Consider the linear time invariant system whose transfer function is $G(s)$. The system is excited with a sine wave

$$r(t) = R \sin \omega t \tag{3.1}$$

The system representation and input and output wave forms are shown in Fig. 3.1a, b representing the steady state response of the system to a sinusoidal input which can be directly obtained from $G(s)$ by replacing s by $j\omega$.

$$\frac{C}{R}(j\omega) = G(j\omega) \tag{3.2}$$

is called the sinusoidal transfer function, which is the ratio of $C(j\omega)$ to $R(j\omega)$. Thus, by knowing the T.F. $G(s)$, the magnitude characteristic $|G(j\omega)|$ and phase characteristic $\angle G(j\omega)$ completely describe the steady state frequency response of the system. Given the poles and zeros of $G(s)$, the frequency response at any frequency $\omega = \omega_1$ and other frequencies also can be evaluated graphically as illustrated below.

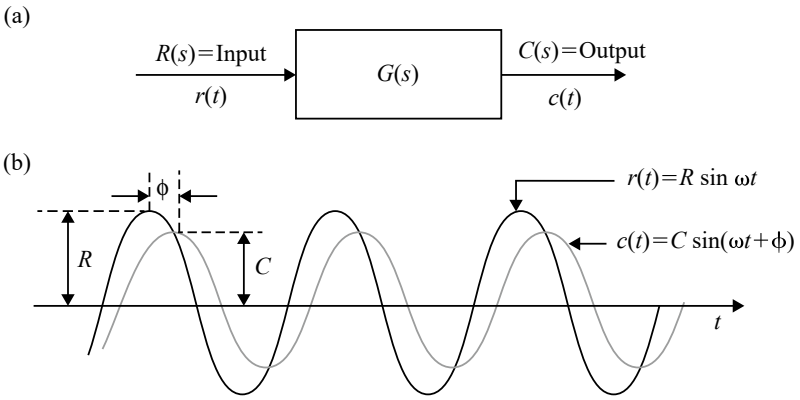


Fig. 3.1 **a** Linear system with sine wave input and sine wave output. **b** Input-output wave forms with their amplitude and phase

Example 3.1

A certain control system has the following transfer function

$$G(s) = \frac{5(s+2)}{(s+1)(s+4)}$$

Find the frequency response at $\omega = 5$ rad/sec.

Solution The poles and zeros of $G(s)$ are located as shown in Fig. 3.2a. Point A which corresponds to $\omega = 5$ in the imaginary axis is identified. Vectors from the poles and zeros of $G(s)$ are drawn to the point A and their magnitude and phase identified. The frequency response at $\omega = 5$ is calculated as

$$\begin{aligned} G(j5) &= \frac{5 \times \text{Product of directed vector distances from all the zeros of } G(s)}{\text{Product of directed vector distances from all the poles of } G(s)} \\ &= \frac{5 \times \sqrt{29}\angle 68.2^\circ}{\sqrt{26}\angle 78.69^\circ \sqrt{41}\angle 51.34^\circ} \\ &= 0.8247\angle -61.83^\circ \end{aligned}$$

The ratio of the output to input magnitude is 0.8247 and the output lags behind the input by an $\angle 61.83^\circ$ at $\omega = 5$ rad/sec. Similar procedure is adopted at other frequencies.

The graphical method described above is somewhat tedious. The frequency response at any frequency $\omega = \omega_1$ can be obtained from the sinusoidal T.F. directly as explained below.

The vectorial representation of $G(j\omega)$ for the sinusoidal T.F. is shown in Fig. 3.2b. From Fig. 3.2b, the magnitude and phase contributed at any frequency in general can be determined. For any required frequency, the response can be obtained by substituting for the given ω

$$\begin{aligned} G(j\omega) &= \frac{5(j\omega+2)}{(j\omega+1)(j\omega+4)} \\ &= \frac{5\sqrt{(\omega^2+4)}\angle \tan^{-1}\frac{\omega}{2}}{\sqrt{(\omega^2+1)(\omega^2+16)}\angle \tan^{-1}\omega + \tan^{-1}\frac{\omega}{4}} \\ &= 0.8242\angle -61.83^\circ \\ |G(j\omega)| &= 0.8247 \\ \angle G(j\omega) &= -61.83^\circ \end{aligned}$$

Obtaining frequency response using Fig. 3.2b is much simpler than using Fig. 3.2a.

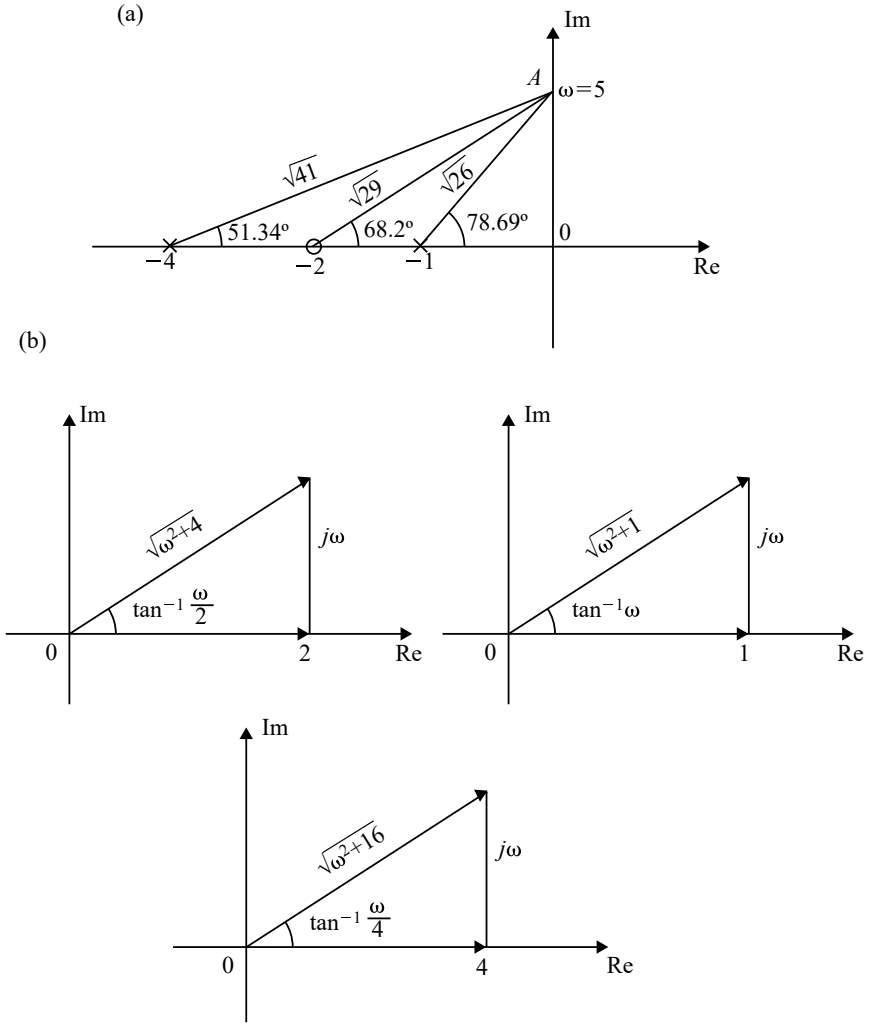


Fig. 3.2 **a** Sketching frequency response from pole–zero plot. **b** Vectorial representation of $G(\,j\omega)$

3.3 Plotting Frequency Response

The frequency response of a system can be plotted in several ways using the sinusoidal transfer function $G(\,j\omega)$. As we vary input frequency over a range, we get a vector of frequency response data in the form of a complex vector. The complex vector can be displayed in a variety of ways as given below.

1. Linear Frequency Response Plots

These plots include the plot magnitude of $G(\,j\omega)$ *versus* ω and phase angle of $G(\,j\omega)$ *versus* ω. The horizontal axis in both plots is frequency ω with linear scaling.

2. Linear Frequency Response Plots *versus* log ω
These plots are similar to the first case but the horizontal axes are log arithmetically scaled (semi-log plots). This allows a larger frequency scale to be used.

3. Polar (Nyquist) Plot
This is the plot of the imaginary *versus* the real part of $G(j\omega)$ drawn in a polar graph with ω as the varying parameter.

4. Bode Plots
It has two separate plots one for the magnitude and another for phase angle of $G(j\omega)$. The magnitude is expressed in decibel ($20\log_{10}|G(j\omega)|$). By this very large magnitude can be accommodated. The horizontal axes are logarithmic by which we can accommodate larger frequency scale.

5. Log Magnitude *versus* Phase Plot (Nichols Plot)
This is the plot of the magnitude of $G(j\omega)$ in dB *versus* the phase in degrees with ω as the varying parameter.

For control system analysis and design, the first two methods are not used. The widely used methods are the polar plot, Bode plot and Nichols plot. These three methods are described below.

3.4 Polar Plot (Nyquist Plot)

The polar or otherwise called Nyquist plot of a sinusoidal transfer function $G(j\omega)$ is the plot of the magnitude of $G(j\omega)$ *versus* the phase angle of $G(j\omega)$ on polar coordinates as ω is varied from zero to infinity. In polar coordinates the gain of the sinusoidal T.F. $G(j\omega)$ is plotted as the radial coordinate, while the phase of $G(j\omega)$ is plotted as the angular coordinate. It has a unique advantage of having the display of both amplitude and phase angle on a single plot over a wide frequency range. It has the property to show whether a closed loop system is stable or unstable. However, the only disadvantage of this method is that it does not clearly indicate the contributions made by each pole and zero. We illustrate below how a polar plot is drawn for the given T.F.

The polar plot is one of the various graphical plots available to represent the sinusoidal T.F. $G(j\omega)$. To represent the frequency response of $G(j\omega)$ in polar plot $G(j\omega)$ should be stable. Frequency response can be represented in polar, Bode and log magnitude *versus* phase plots. However, all these plots need not be frequency response plots.

Consider the T.F. shown in Example 3.1

$$G(j\omega) = \frac{5\sqrt{(\omega^2+4)}}{\sqrt{(\omega^2+1)(\omega^2+16)}} \angle \tan^{-1}\frac{\omega}{2} - \tan^{-1}\omega - \tan^{-1}\frac{\omega}{4}$$

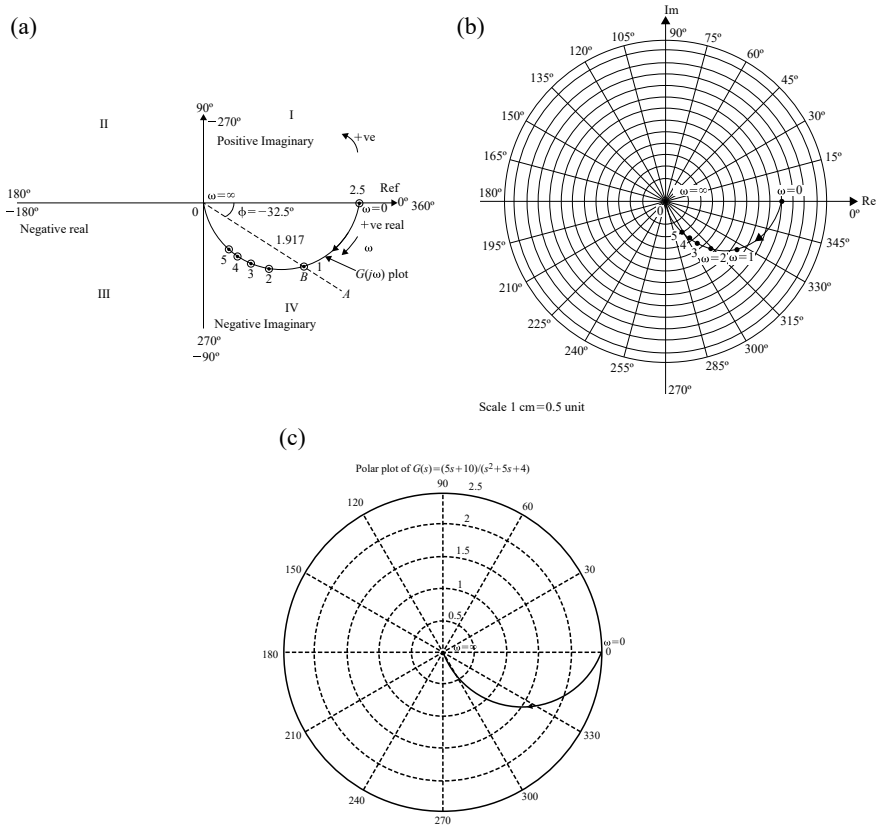


Fig. 3.3 **a** Polar plot for $G(s) = 5(s + 2)/[(s + 1)(s + 4)]$. **b** Frequency response plot in polar graph

For the frequency $0 \leq \omega \leq \infty$, $|G(\,j\omega)|$ and $\angle G(\,j\omega)$ are calculated and tabulated below.

ω	0°	1	2	3	4	5	∞
$\phi = \angle G(\,j\omega)$	0°	−32.5°	−45°	−52.1°	−57.6°	−61.8°	−90°
$\|G(\,j\omega)\|$	2.5	1.917	1.414	1.14	0.959	0.8247	0

The tabulated values of $\phi = \angle G(\,j\omega)$ and $|G(\,j\omega)|$ are transferred to polar graph as follows. The polar coordinates are drawn as shown in Fig. 3.3a. It is divided into four quadrants. The first quadrant lies between 0° and 90°, the second quadrant between 90° and 180°, the third quadrant between 180° and 270° and the fourth quadrant lies between 270° and 0°. The reference line is along the 0° line. Angle is measured positive from the reference line when moved anticlockwise and if moved in the clockwise direction, the angle is measured negative with respect to the reference. Thus, $+90° = -270°; +180° = -180°; +270° = -90°; 0° = 360°$. To transfer the

frequency response say for $\omega = 1$, $\phi = -32.5°$ and $|G(j\omega)| = 1.917$ units, draw the radial line $0A$ from the origin with $\phi = -32.5°$. Choose any scale for $|G(j\omega)|$. Measure $0B = 1.917$ on $0A$. Similar procedure is followed to locate $|G(j\omega)|$ in the polar graph for other values of ω. Join these points by a smooth graph. By the side of B put the value of ω. Put arrow in the $G(j\omega)$ plot pointing in the direction of increasing values of ω. This completes the polar plot which is manually done.

Ready-made polar graph sheets are available. Radial lines have been drawn for each 1° angular space. By choosing appropriate scale factor for the magnitude, the polar plot is drawn in polar graph as shown in Fig. 3.3b.

The polar plot can also be obtained using MATLAB program. The graph can be plotted either in polar radial and angular coordinates or coordinates of real and imaginary axes. The necessary programmers and the plots obtained are shown ahead.

Polar Plot in Radial and Angular Coordinates
The following MATLAB program is written to get this plot. The plot is shown in Fig. 3.3c.

MATLAB Program

```
num=[0 5 10];
den=[1 5 4];
[re, im]-nyquist(num,den,w)
Z = re + li × im
r = abs(z)
theta=angle(z)
polar(theta,r)
grid
title('polar plot of G(s) = (5s + 10)/(s^2 + 5s + 4)')
```

3.4.1 General Shape of Polar Plot and Type and Order of the System

As stated in the previous sections, polar plot plays an important role in the study of linear system stability especially the Nyquist stability analysis. In the stability study, in most of the cases, mere sketching of Nyquist plot is good enough and point by point information may not be always necessary. In these cases, the $|G(j\omega)|$ where it cuts the negative real axis will be enough to assess the stability. The Type and order of the system have the influence in the shape of polar plot and this is discussed below.

Polar Plot of Type 0 and Order 1 System
Consider the following T.F. whose Type is 0 and order is 1

$$G(s) = \frac{K}{(1 + sT)} \tag{3.3}$$

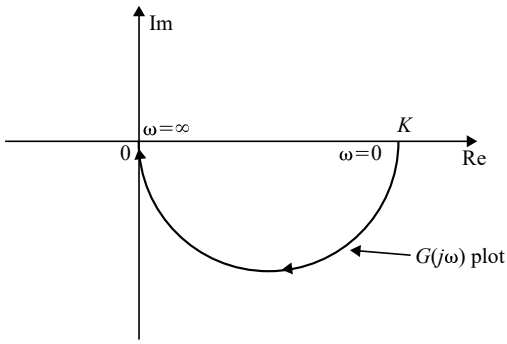


Fig. 3.4 Polar plot of Type 0 order 1 system $G(s) = K/(1 + sT)$

The sinusoidal T.F. is

$$G(j\omega) = \frac{K}{(1 + j\omega T)} = \frac{K\angle \tan^{-1} \omega T}{\sqrt{1 + \omega^2 T^2}}$$
$$|G(j\omega)| = \frac{K}{\sqrt{1 + \omega^2 T^2}}$$
$$\phi = \angle G(j\omega) = \angle - \tan^{-1} \omega T$$

For $\omega = 0$;

$$G(j\omega) = K\angle 0^\circ$$

For $\omega = \infty$;

$$G(j\omega) = 0\angle -90^\circ$$

The polar plot starts from +ve real at K, traverses through the fourth quadrant as ω increases and terminates at the origin as $\omega \to \infty$. The polar plot is shown in Fig. 3.4.

Polar Plot of Type 0 Order 2 System

Consider the following T.F. of a Type 0, order 2 system

$$G(s) = \frac{K}{(1 + sT_1)(1 + sT_2)} \tag{3.4}$$

The sinusoidal T.F. is

$$G(j\omega) = \frac{K}{(1 + j\omega T_1)(1 + j\omega T_2)}$$
$$= \frac{K\angle - \tan^{-1} \omega T_1 - \tan^{-1} \omega T_2}{\sqrt{(1 + \omega^2 T_1^2)(1 + \omega^2 T_2^2)}}$$

For $\omega = 0$,

$$G(j\omega) = K\angle 0^\circ$$

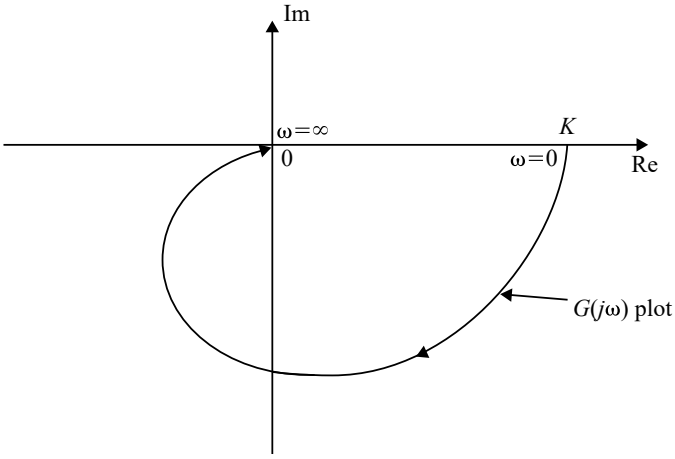


Fig. 3.5 Polar plot of Type 0 order two system $G(s) = K/[(1 + sT_1)(1 + sT_2)]$

For $\omega = \infty$,

$$G(\,j\omega) = 0\angle -180^\circ$$

The polar plot is shown in Fig. 3.5.

The polar plot starts from +ve real at $|G(\,j\omega)| = K$ for $\omega = 0$, traverses through the fourth and third quadrants and terminates at the origin when $\omega = \infty$.

Polar Plot for Type 0 and Order 3 System

Consider the following T.F. of a Type 0 order 3 system

$$G(s) = \frac{K}{(1 + sT_1)(1 + sT_2)(1 + sT_3)} \tag{3.5}$$

$$G(\,j\omega) = \frac{K}{(1 + j\omega T_1)(1 + j\omega T_2)(1 + j\omega T_3)}$$

$$|G(\,j\omega)| = \frac{K}{\sqrt{(1 + \omega^2 T_1^2)(1 + \omega^2 T_2^2)(1 + \omega^2 T_3^2)}}$$

$$\phi = \angle G(\,j\omega) = -\tan^{-1}\omega T_1 - \tan^{-1}\omega T_2 - \tan^{-1}\omega T_3$$

For $\omega = 0$,

$$G(\,j\omega) = K\angle 0^\circ$$

For $\omega = \infty$;

$$G(\,j\omega) = 0\angle -270^\circ$$

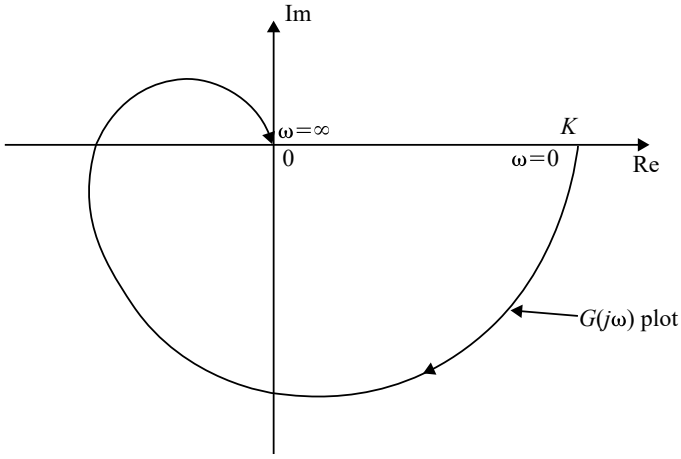


Fig. 3.6 Polar plot of Type 0 order 3 system $G(s) = \dfrac{K}{[(1+sT_1)(1+sT_2)(1+sT_3)]}$

The polar plot is shown in Fig. 3.6. It starts from +ve real at $|G(\,j\omega)| = K$, for $\omega = 0$ traverses through the fourth, third and second quadrants and terminates at the origin when $\omega = \infty$.

Polar Plot for Type 1 Order 2 System
Consider the following T.F. of a Type 1 order 2 system

$$G(s) = \frac{K}{s(1+sT_1)} \tag{3.6}$$

$$G(\,j\omega) = \frac{K}{j\omega(1+j\omega T_1)}$$

$$= \frac{K\angle -90^\circ - \tan^{-1}\omega T_1}{\omega\sqrt{1+\omega^2 T_1^2}}$$

For $\omega = 0$;

$$G(\,j\omega) = \infty\angle -90^\circ$$

For $\omega = \infty$;

$$G(\,j\omega) = 0\angle -180^\circ$$

The polar plot is shown in Fig. 3.7. It starts from ∞ from negative imaginary for $\omega = 0$, traverses through the third quadrant and terminates at the origin when $\omega = \infty$.

Polar Plot for Type 1 Order 3 System
Consider the following T.F. of a Type 1 order 3 system

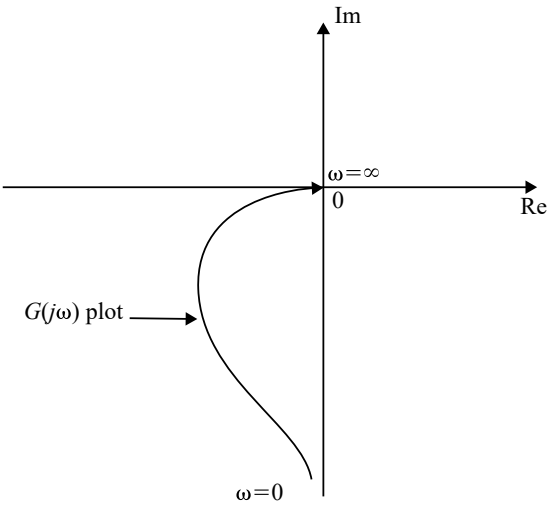


Fig. 3.7 Polar plot of Type 1 order 2 system $G(s) = K/[s(1 + sT_1)]$

$$G(s) = \frac{K}{s(1+sT_1)(1+sT_2)} \tag{3.7}$$

$$G(j\omega) = \frac{K}{j\omega(1+j\omega T_1)(1+j\omega T_2)}$$

$$|G(j\omega)| = \frac{K}{\omega\sqrt{(1+\omega^2T_1^2)(1+\omega^2T_2^2)}}$$

$$\phi = \angle G(j\omega) = -90^\circ - \tan^{-1}\omega T_1 - \tan^{-1}\omega T_2$$

For $\omega = 0$;

$$G(j\omega) = \infty\angle -90^\circ$$

For $\omega = \infty$,

$$G(j\omega) = 0\angle -270^\circ$$

The polar plot is shown in Fig. 3.8. It starts from ∞ from negative imaginary axis, traverses through the third quadrant, cuts the negative real axis, traverses through the second quadrant and terminates at the origin when $\omega = \infty$.

From Eqs. (3.8) to (3.12), the following inferences are drawn:

1. The starting portion of $G(j\omega)$ plot is decided by the type of the system. Type 0 system starts from +ve real. Type 1 system starts from −ve imaginary from infinity. Type 2 system starts from −ve real from infinitive and so on.
2. The finishing portion of the $G(j\omega)$ plot is decided by the order of the system when $\omega = \infty$. First order system terminates in the fourth quadrant with $\phi = -90^\circ$. The second order system terminates in the third quadrant with $\phi = -180^\circ$. The third order system terminates in the second quadrant with $\phi = -270^\circ$.

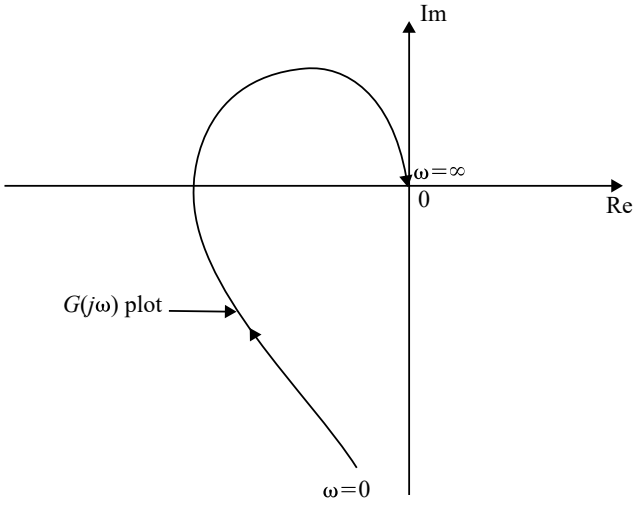


Fig. 3.8 Polar plot of Type 1 order 3 system $G(s) = K/[s(1 + sT_1)(1 + sT_2)]$

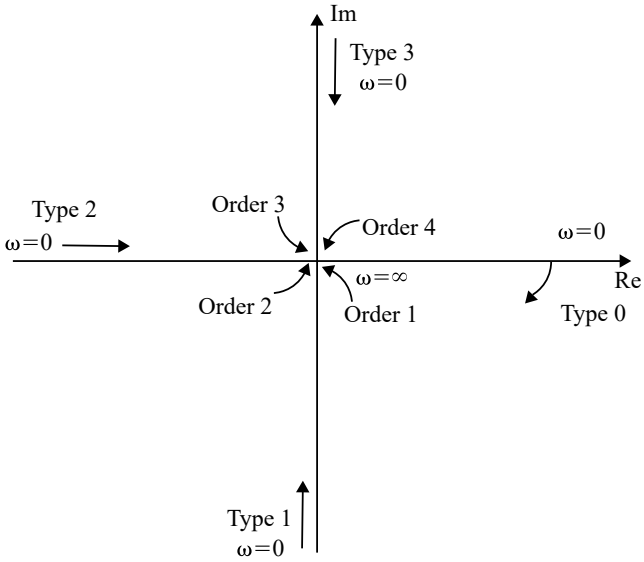


Fig. 3.9 General shape of polar plot at the start and finish

The above inferences are true iff (if and only if) there are no zeros in the T.F. and it is a minimum phase T.F. This is illustrated in Fig. 3.9.

If the T.F. is strictly proper (numerator polynomial is greater than the denominator polynomial), as ω tends to ∞, the polar plot converges to the origin. Further, it is to

be noted that the phase angle of the poles and zeros decide the general shape of the polar plot.

Polar Plot of a T.F. with a Zero
Consider the following T.F. whose Type is 1 and order is three with a zero.

$$\begin{aligned} G(s) &= \frac{K(1+sT_1)}{s(1+sT_2)(1+sT_3)} \\ G(j\omega) &= \frac{K(1+j\omega T_1)}{j\omega(1+j\omega T_2)(1+j\omega T_3)} \\ |G(j\omega)| &= \frac{K\sqrt{(1+\omega^2 T_1^2)}}{\omega\sqrt{(1+\omega^2 T_2^2)(1+\omega^2 T_3^2)}} \\ \angle G(j\omega) &= -90^\circ - \tan^{-1}\omega T_2 - \tan^{-1}\omega T_3 + \tan^{-1}\omega T_1 \end{aligned} \tag{3.8}$$

For $\omega = 0$,

$$G(j\omega) = -90^\circ$$

For $\omega = \infty$,

$$G(j\omega) = 0\angle -180^\circ$$

With the addition of zero, the starting point of the polar plot is not affected. However, as ω increases and tends to ∞, the polar plot traverses through the third quadrant and terminates at the origin as shown in Fig. 3.10a. The finishing portion of the polar plot is like that of a second order system. As ω tends to ∞, the phase contribution of a zero gets nullified by the phase contribution of a pole. In general if n is the degree of the denominator polynomial of $G(j\omega)$ and m is the degree of numerator polynomial, the order of the system is considered as $(n-m)$ for the purpose of deciding the finishing portion of the polar plot as ω tends to ∞. The polar plot of Eq. (3.8) with zero is sketched as shown in Fig. 3.10a and without zero is sketched in Fig. 3.10b. The general shape of finishing portion of the polar plot in terms of $(n-m)$ is shown in Fig. 3.10c.

The summary of starting and finishing portion of polar plot and magnitude and phase angle contribution of poles and zeros are given in Table 3.1.

3.4.2 Polar Plot of Non-minimum Phase Transfer Function

Depending upon the type and order of the system, the general shape of the polar plot can be decided and sketched for a system with a minimum phase T.F. as discussed in Sect. 3.4.1. The inferences derived in Sect. 3.4.1 will not be valid for a system with a non-minimum phase T.F. where at least a pole or a zero lies in RHP. The magnitude of the T.F. of a non-minimum phase T.F. remains the same as that of the minimum

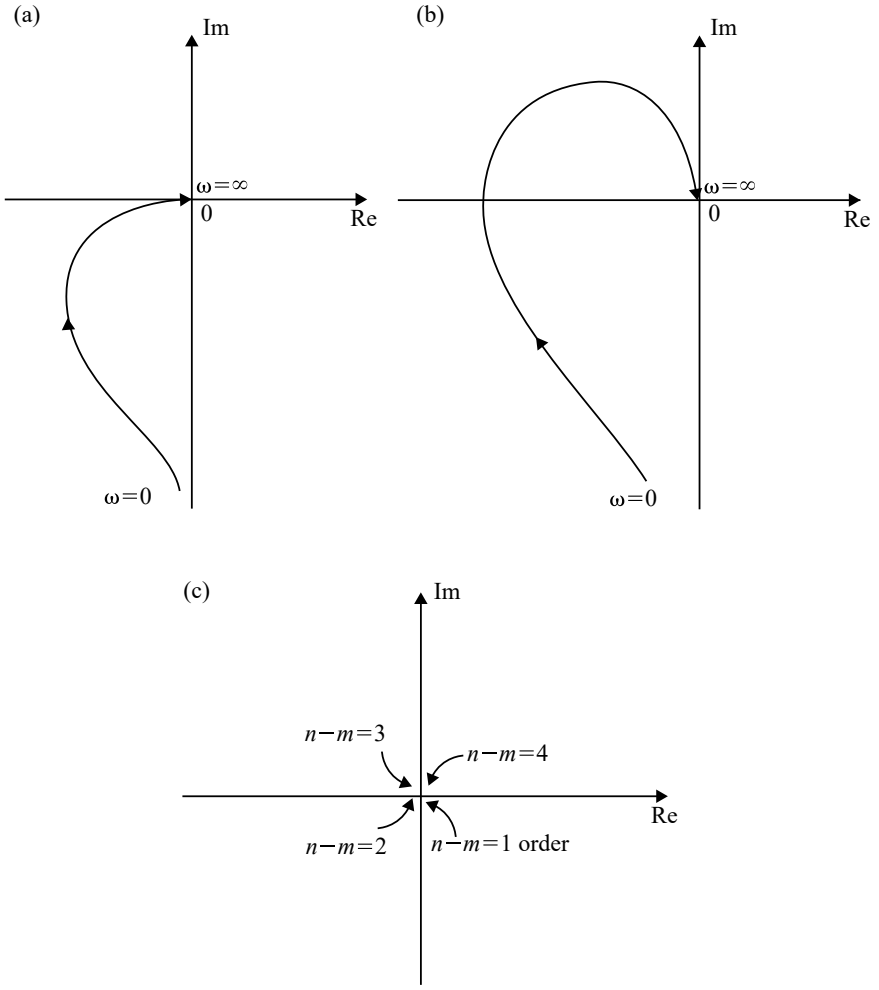


Fig. 3.10 **a** Polar plot of Eq. (3.8). **b** Polar plot of Eq. (3.8) without zero and **c** General shape of $G(j\omega)$ plot as ω tends to ∞ in terms of n and m

phase T.F. The phase angle of the T.F. is totally shifted. In this section, we discuss the sketching of polar plot for non-minimum phase poles and zeros. Consider the following zero of a certain T.F.

$$G(s) = (2 - 3s)$$
$$G(j\omega) = (2 - j3\omega)$$

The zero is located at $s = 2/3$ in RHP +ve real. The real and imaginary parts of $G(j\omega)$ are marked as shown in Fig. 3.11a.

Table 3.1 Magnitude and phase angle contribution of poles and zeros of $G(s)$

$G(s)$	$\frac{1}{s}$	$\frac{1}{s^2}$	$(1+sT)$	$(1+sT)^2$	$\frac{1}{(1+sT)}$	$\frac{1}{(1+sT)^2}$
$\angle G(j\omega)$	$-90°$	$-180°$	$\tan^{-1}\omega T$	$2\tan^{-1}\omega T$	$-\tan^{-1}\omega T$	$-2\tan^{-1}\omega T$
$\lvert G(j\omega)\rvert$	$\frac{1}{\omega}$	$\frac{1}{\omega^2}$	$\sqrt{(1+\omega^2T^2)}$	$(1+\omega^2T^2)$	$\frac{1}{\sqrt{1+\omega^2T^2}}$	$\frac{1}{(1+\omega^2T^2)}$
$\angle G(j\omega)$ $\omega=0$	$-90°$	$-180°$	$0°$	$0°$	$0°$	$0°$
$\angle G(j\omega)$ $\omega=\infty$	$-90°$	$-180°$	$90°$	$180°$	$-90°$	$-180°$
$\lvert G(j\omega)\rvert$ $\omega=0$	∞	∞	1	1	1	1
$\lvert G(j\omega)\rvert$ $\omega=\infty$	0	0	∞	∞	0	0

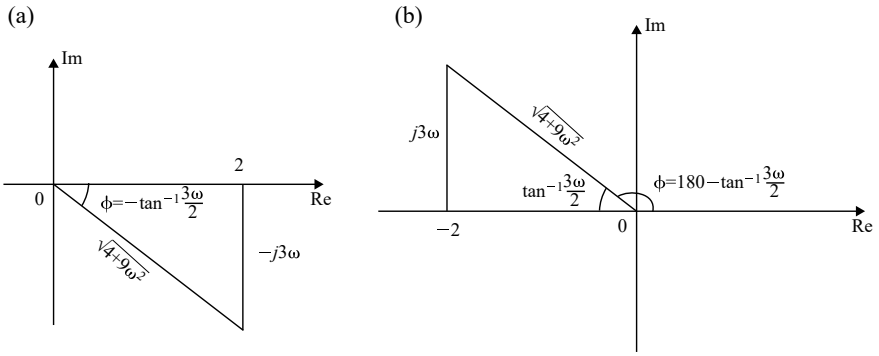


Fig. 3.11 **a** Magnitude and phase representation of non-minimum phase zero. **b** Magnitude and phase representation of a non-minimum zero

From Fig. 3.11a, the magnitude and phase of $G(j\omega)$ are obtained as

$$|G(j\omega)| = \sqrt{4+9\omega^2}$$
$$\angle G(j\omega) = -\tan^{-1}\frac{3\omega}{2}$$

The magnitude of $G(j\omega)$ remains the same as that of the minimum phase T.F. However, the phase angle is $-\tan^{-1}(3\omega/2)$ which alters the polar plot to a completely different quadrant.

Now consider the T.F. with the following zero which has a non-minimum phase

$$G(s) = -2+3s$$
$$G(j\omega) = -2+j3\omega$$

This zero is also located at $s = 2/3$ in RHP +ve real. The magnitude and phase of $G(\,j\omega)$ are determined from Fig. 3.11b.

From Fig. 3.11b, the magnitude and phase of $G(\,j\omega)$ are obtained as

$$|G(\,j\omega)| = \sqrt{4 + 9\omega^2}$$
$$\angle G(\,j\omega) = 180 - \tan^{-1}\frac{3\omega}{2}$$

Here also, the magnitude of $G(\,j\omega)$ remains the same as that of minimum phase zero but the phase is shifted from $\tan^{-1}(3\omega/2)$ to $180 - \tan^{-1}(3\omega/2)$. In a similar way, the phase angle of non-minimum phase poles is determined.

For a system with non-minimum phase poles and zeros, Fig. 3.11a or b is drawn as the case may be and the phase angle is expressed as a function of ω.

Example 3.2

Consider the following T.F.

$$G(s) = \frac{5(s-2)}{(s+1)(s+4)}$$

Sketch the polar plot.

Solution

$$G(s) = \frac{5(s-2)}{(s+1)(s+4)}$$
$$G(\,j\omega) = \frac{5(\,j\omega - 2)}{(\,j\omega + 1)(\,j\omega + 4)}$$
$$G(\,j\omega) = \frac{5\sqrt{(\omega^2 + 4)}}{\sqrt{(\omega^2 + 1)(\omega^2 + 16)}}$$
$$\angle G(\,j\omega) = 180 - \tan^{-1}\frac{\omega}{2} - \tan^{-1}\omega - \tan^{-1}\frac{\omega}{4}$$

For $\omega = 0$,

$$G(\,j\omega) = 2.5\angle 180^\circ$$

For $\omega = \infty$,

$$G(\,j\omega) = 0\angle -90^\circ$$

The polar plot starts from negative real at -2.5, traverses through the second, first and fourth quadrants and terminates at the origin when $\omega = \infty$. The polar plot is sketched as in Fig. 3.12.

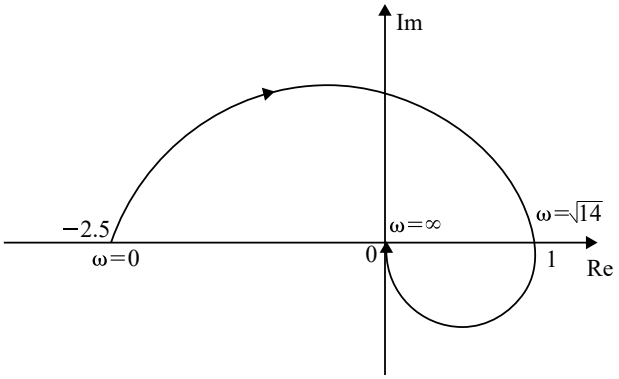


Fig. 3.12 Polar plot for $G(s) = 5(s-2)/[(s+1)(s+4)]$

Example 3.3

Sketch the polar plot for the following T.F.

$$G(s) = \frac{5(s+2)}{(s+1)(-s+4)}$$

Solution

$$G(s) = \frac{5(s+2)}{(s+1)(-s+4)}$$

$$G(j\omega) = \frac{5(j\omega+2)}{(j\omega+1)(-j\omega+4)}$$

$$|G(j\omega)| = \frac{5\sqrt{\omega^2+4}}{\sqrt{(\omega^2+1)(\omega^2+16)}}$$

$$\angle G(j\omega) = \tan^{-1}\frac{\omega}{2} + \tan^{-1}\frac{\omega}{4} - \tan^{-1}\omega$$

The polar plot starts from +ve real from 2.5 for $\omega = 0$ traverses through the fourth quadrant since initially, $\tan^{-1}\frac{\omega}{2} + \tan^{-1}\frac{\omega}{4}$ contributes less phase than $\tan^{-1}\omega$.

When $\angle\tan^{-1}\frac{\omega}{2} + \tan^{-1}\frac{\omega}{4} = \angle\tan^{-1}\omega$, it cuts the +ve real axis at $\omega = \omega_c$ and traverses through the first quadrant. Now $(\tan^{-1}\frac{\omega}{2} + \tan^{-1}\frac{\omega}{4})$ contributes more phase than $\tan^{-1}\omega$. At $\omega = \infty$, the total phase is $+90^\circ$ and the magnitude becomes zero. The frequency ω_c at which the phase angle becomes zero is calculated as follows:

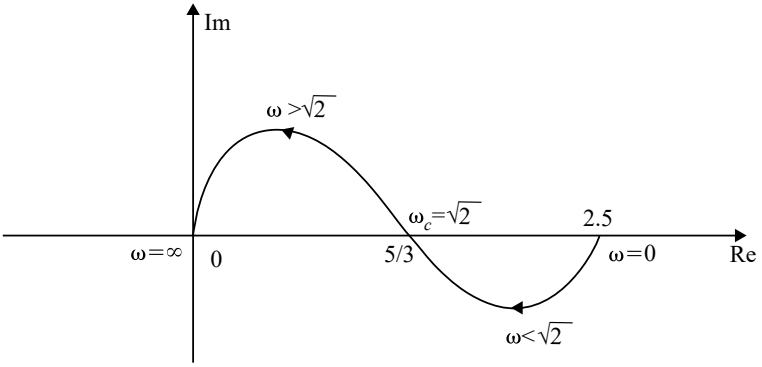


Fig. 3.13 Polar plot for $G(s) = \dfrac{5(s+2)}{(s+1)(-s+4)}$

$$\tan^{-1}\frac{\omega_c}{2} + \tan^{-1}\frac{\omega_c}{4} - \tan^{-1}\omega_c = 0$$
$$\tan^{-1}\frac{\omega_c}{2} + \tan^{-1}\frac{\omega_c}{4} = \tan^{-1}\omega_c$$

Using the formula

$$\tan^{-1}A + \tan^{-1}B = \tan^{-1}\frac{A+B}{1-AB}$$

the above equation is written as

$$\tan^{-1}\frac{\frac{\omega_c}{2}+\frac{\omega_c}{4}}{1-\frac{\omega_c^2}{8}} = \tan^{-1}\omega_c$$
$$\tan^{-1}\frac{6\omega_c}{8-\omega_c^2} = \tan^{-1}\omega_c$$

Taking tan on both sides, we get

$$\frac{6\omega_c}{8-\omega_c^2} = \omega_c$$
$$\omega_c = \sqrt{2}$$

The magnitude at this frequency is $\frac{5}{3}$. The polar plot is sketched as shown in Fig. 3.13.

Example 3.4

Sketch the polar plot for the following T.Fs.

$$\text{(a)} \quad G(s) = \frac{(1-2s)}{(1+2s)}$$

$$\text{(b)} \quad G(s) = \frac{2(s+5)}{(3s+4)}$$

$$\text{(c)} \quad G(s) = \frac{(1-2s)(1+5s)}{(1+2s)(1-5s)}$$

Solution (a)

$$G(s) = \frac{(1-2s)}{(1+2s)}$$

$$G(j\omega) = \frac{(1-j\omega 2)}{(1+j\omega 2)}$$

$$|G(j\omega)| = 1$$

$$\angle G(j\omega) = -2\tan^{-1} 2\omega$$

For $\omega = 0$,

$$G(j\omega) = 1\angle 0^\circ$$

For $\omega = \infty$,

$$G(j\omega) = 1\angle -180^\circ$$

The polar plot is a semi-circle of unit radius. It starts from +ve real with magnitude 1, traverses through fourth and third quadrants and terminates at $(-1, 0)$ on the negative real axis as shown in Fig. 3.14a.

(b)

$$G(s) = \frac{2(s+5)}{(3s+4)}$$

$$G(j\omega) = \frac{2(j\omega+5)}{(3j\omega+4)}$$

$$|G(j\omega)| = \frac{2\sqrt{\omega^2+25}}{\sqrt{9\omega^2+16}}$$

$$\angle G(j\omega) = \tan^{-1}\frac{\omega}{5} - \tan^{-1}\frac{3\omega}{4}$$

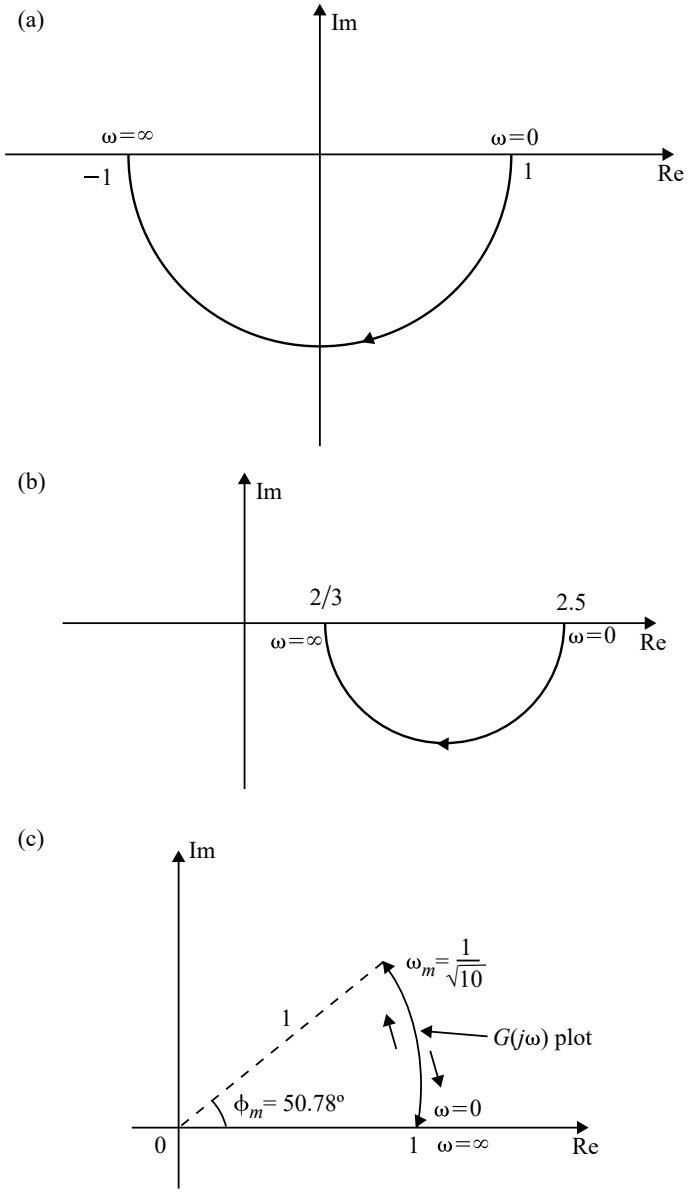


Fig. 3.14 **a** Polar plot for $G(s) = (1 - 2s)/(1 + 2s)$. **b** Polar plot for $G(s) = 2(s + 5)/(3s + 4)$. **c** Polar plot for $G(s) = \dfrac{(1 - 2s)(1 + 5s)}{(1 + 2s)(1 - 5s)}$

For $\omega = 0$,

$$G(j\omega) = 2.5\angle 0^\circ$$

For $\omega = \infty$;

$$G(j\omega) = \frac{2}{3}\angle 0^\circ$$

At any frequency ω, $-\tan^{-1}\frac{3\omega}{4}$ is more dominant than $\tan^{-1}\frac{\omega}{5}$ except at $\omega = 0$ where the phase contribution of each one of them is 0 and at $\omega = \infty$ where the phase contribution of each one of them is 90°. The polar plot starts from +ve real where its magnitude is 2.5, traverses through the fourth quadrant and for $\omega = \infty$, it terminates at 2/3. This is shown in Fig. 3.14b.

(c)

$$\begin{aligned} G(s) &= \frac{(1-2s)(1+5s)}{(1+2s)(1-5s)} \\ G(j\omega) &= \frac{(1-j\omega 2)(1+j\omega 5)}{(1+j\omega 2)(1-j\omega 5)} \\ |G(j\omega)| &= 1 \\ \phi = \angle G(j\omega) &= -2\tan 2\omega + 2\tan^{-1} 5\omega \\ \frac{\phi}{2} &= \tan^{-1} 5\omega - \tan^{-1} 2\omega \end{aligned}$$

$\tan^{-1} 5\omega$, at any frequency ω dominates over $\tan^{-1} 2\omega$ except at $\omega = 0$ they are equal with 0° and at $\omega = \infty$, they are equal at 90°. They polar plot starts ($\omega = 0$) from +ve real at 1, traverses through the first quadrant with a magnitude of 1 at all frequencies reaches the maximum phase ϕ_m at $\omega = \omega_m$, where $0 < \phi < 90^\circ$ and comes back to 0° when $\omega = \infty$. The values of ω_m and ϕ_m are determined as follows

$$\begin{aligned} \frac{\phi}{2} &= \tan^{-1} 5\omega - \tan^{-1} 2\omega \\ &= \tan^{-1}\frac{5\omega - 2\omega}{1 + 10\omega^2} \qquad \left[\tan^{-1} A - \tan^{-1} B = \tan^{-1}\frac{A-B}{1+AB}\right] \\ &= \tan^{-1}\frac{3\omega}{1+10\omega^2} \end{aligned}$$

Let $N = \tan\frac{\phi}{2}$

$$\begin{aligned} \frac{\phi}{2} &= \tan^{-1} N \\ \tan^{-1} N &= \tan^{-1}\frac{3\omega}{1+10\omega^2} \end{aligned}$$

Taking tan on both sides, we get

$$N = \frac{3\omega}{1 + 10\omega^2}$$

Differentiating N with respect to ω and equating $dN/d\omega = 0$ we get ω_m and corresponding phase as ϕ_m.

$$\begin{aligned}
\frac{dN}{d\omega} &= \frac{(1 + 10\omega^2)3 - 3\omega(20\omega)}{(1 + 10\omega^2)^2} \\
&= 0 \\
1 + 10\omega^2 &= 20\omega^2 \\
\omega &= \frac{1}{\sqrt{10}} \\
\omega_m &= \frac{1}{\sqrt{10}}
\end{aligned}$$

At $\omega = \omega_m$, $\phi = \phi_m$

$$\begin{aligned}
\phi_m &= 2[\tan^{-1} 5\omega_m - \tan^{-1} 2\omega_m] \\
&= 2[57.7^\circ - 32.3^\circ] \\
&= 50.78^\circ
\end{aligned}$$

The polar plot is sketched as shown in Fig. 3.14c.

3.4.3 Polar Plot with Transportation Lag (Time Delay)

Consider the transfer function of a first order system

$$G(s) = \frac{K}{(1 + sT_1)}$$

The transfer function of the above system with a time delay (transportation lag) is written as

$$G(s) = \frac{Ke^{-sT}}{(1 + sT_1)}$$

The sinusoidal T.F. is

$$G(j\omega) = \frac{Ke^{-j\omega T}}{(1 + j\omega T_1)}$$
$$|G(j\omega)| = \frac{K}{\sqrt{1 + \omega^2 T_1^2}}$$
$$\angle G(j\omega) = -\tan^{-1}\omega T_1 - \angle \omega T$$

With the introduction of the time delay, the magnitude of the original system is unchanged. However, the phase angle of $G(j\omega)$ is changed by $-\omega T$.

Example 3.5

Consider the following T.F. of a certain control system with time delay. Draw the polar plot and compare this plot with the one obtained without time delay

$$G(s) = \frac{100e^{-0.2s}}{(2 + 3s)(2s + 5)}$$

Solution

$$G(s) = \frac{100e^{-0.2s}}{(2 + 3s)(2s + 5)}$$
$$G(j\omega) = \frac{100e^{-0.2j\omega}}{(2 + j3\omega)(j2\omega + 5)}$$
$$|G(j\omega)| = \frac{100}{\sqrt{(4 + 9\omega^2)(25 + 4\omega^2)}}$$
$$\angle G(j\omega) = -\tan^{-1}\frac{3}{2}\omega - \tan^{-1}\frac{2}{5}\omega - 0.2 \times 57.3\omega$$
$$= -\tan^{-1} 1.5\omega - \tan^{-1} 0.4\omega - 11.46\omega$$

The magnitude and phase of $G(j\omega)$ are tabulated below for different values of ω.

ω	0	1	2	3	4	5	∞
$\lvert G(j\omega)\rvert$	10	5.15	2.47	1.39	0.87	0.6	0
$\angle G(j\omega)$ without delay	0°	−78.1°	−110.3°	−127.66°	−138.5°	−146°	−180°
$\angle 11.46\omega$	0°	−11.46°	−22.92°	−34.38°	−45.84°	−57.3°	0
$\angle G(j\omega)$ with delay	0°	−89.56°	−133.22°	−162.04°	−184.34°	−203.3°	−180°

The polar plots of $G(j\omega)$ with and without delay are shown in Fig. 3.15.

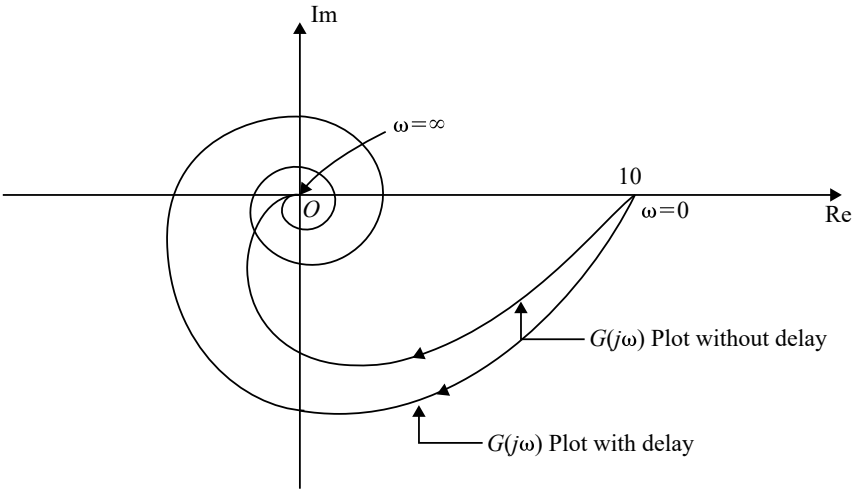


Fig. 3.15 Polar plot with and without delay

3.4.4 Polar Plot of Prototype Second Order System

Consider the following T.F. of a prototype second order system

$$G(s) = \frac{\omega_n^2}{(s^2 + 2\zeta\omega_n s + \omega_n^2)} \tag{3.9}$$

$$G(j\omega) = \frac{\omega_n^2}{-\omega^2 + j2\zeta\omega_n\omega + \omega_n^2}$$

$$= \frac{1}{(1 - \frac{\omega^2}{\omega_n^2}) + j2\zeta\frac{\omega}{\omega_n}}$$

$$|G(j\omega)| = \frac{1}{\sqrt{(1 - \frac{\omega^2}{\omega_n^2})^2 + 4\zeta^2\frac{\omega^2}{\omega_n^2}}} \tag{3.10}$$

$$\angle G(j\omega) = -\tan^{-1}\frac{2\zeta\frac{\omega}{\omega_n}}{\left(1 - \frac{\omega^2}{\omega_n^2}\right)} \tag{3.11}$$

For $\omega = 0$,

$$G(j\omega) = 1\angle 0^\circ$$

For $\omega = \infty$,

$$G(j\omega) = 0\angle -180^\circ$$

At $\omega = \omega_n$,

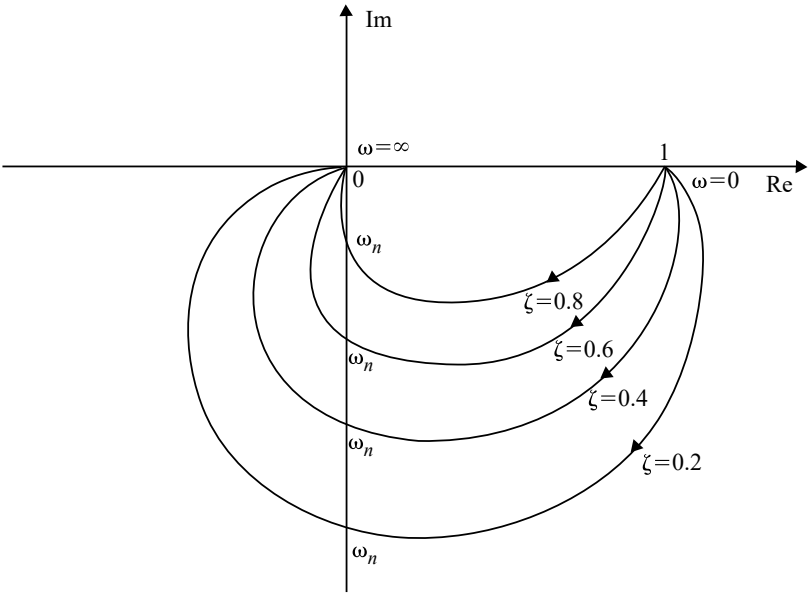


Fig. 3.16 Polar plot of a prototype second order system

$$\angle G(j\omega) = -90^\circ$$

The polar plot starts from +ve real for $\omega = 0$ from 1, traverses through the fourth and third quadrants and ends at the origin when $\omega = \infty$. The phase angle of $G(j\omega)$ is -90° when $\omega = \omega_n$. For $\zeta = 0.2, 0.4, 0.6$ and 0.8, polar plots are sketched and shown in Fig. 3.16. For lower values of ζ, the polar plot is larger in size. As ω is increased, the size shrinks and the polar plot more or less takes the shape of a semi-circle.

3.5 Correlation Between Transient Response and Frequency Response

In Chap. 2, expressions were derived for transient response specifications such as overshoot, rise time, settling time and so on in terms of damping ratio ζ and undamped natural frequency ω_n of a prototype second order system. Similar expressions in the frequency domain can be derived for prototype second order systems. These expressions form the specifications in the frequency domain. Thus, if specifications in the frequency domain are known, the damping ratio and undamped frequency of oscillation can be determined. Once these parameters are known, the time domain specifications are determined. Unfortunately, this correlation exists only for a second order system and it is not valid for a higher order system. However, the higher order

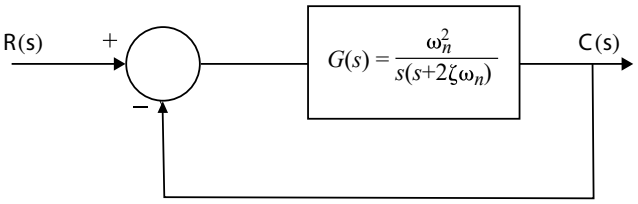


Fig. 3.17 Prototype second order system

systems are reduced to lower second order system and one-to-one correlation can be approximately established.

Consider the following T.F. of a prototype second order system shown in Fig. 3.17

$$\frac{C(s)}{R(s)} = M(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \tag{3.12}$$

The sinusoidal T.F. of Eq. (3.12) is

$$\begin{aligned} M(j\omega) &= \frac{\omega_n^2}{-\omega^2 + j2\zeta\omega\omega_n + \omega_n^2} \\ &= \frac{1}{\left(1 - \frac{\omega^2}{\omega_n^2}\right) + j2\zeta\frac{\omega}{\omega_n}} \end{aligned}$$

Let $u = \omega/\omega_n$

$$\begin{aligned} M(ju) &= \frac{1}{(1-u^2) + j2\zeta u} \\ &= \frac{1\angle -\tan^{-1}\frac{2\zeta u}{(1-u^2)}}{\sqrt{(1-u^2)^2 + 4\zeta^2 u^2}} \end{aligned} \tag{3.13}$$

The maximum value of M is obtained by differentiating $M(ju)$ with respect to u and equating $dM(ju)/du = 0$. The corresponding peak value is called the **resonant peak** denoted by M_r. The frequency at which the resonant peak occurs is called **Resonant Frequency** denoted by ω_r

$$\begin{aligned} \frac{dM(ju)}{du} &= \frac{[2(-2u)(1-u^2) + 8\zeta^2 u]}{2[(1-u^2)^2 + 4\zeta^2 u^2]^{\frac{3}{2}}} = 0 \\ 4u(1-u^2) &= 8\zeta^2 u \\ u &= \sqrt{1-2\zeta^2} \end{aligned}$$

Substituting for u, we get the resonant frequency as

$$\boxed{\omega_r = \omega_n\sqrt{1-2\zeta^2}} \tag{3.14}$$

Substituting $u = \sqrt{1-2\zeta^2}$ in Eq. (3.14), the resonant peak is obtained as

$$M_r = \frac{1}{\sqrt{(1-1+2\zeta^2)^2 + 4\zeta^2(1-2\zeta^2)}}$$

$$\boxed{M_r = \frac{1}{2\zeta\sqrt{1-\zeta^2}}} \tag{3.15}$$

$$\angle M(j\omega) = -\tan^{-1}\frac{2\zeta\frac{\omega}{\omega_n}}{1-\frac{\omega^2}{\omega_n^2}} \tag{3.16}$$

Equation (3.15) is valid for $0 \leq \zeta \leq 0.707$. For $\zeta > 0.707$ there won't be any resonance. For $\zeta > 0.707$, the resonant frequency $\omega_r = 0$ and $M_r = 1$. As seen from Eq. (3.15), the resonant peak is a function of the damping ratio. If the resonant peak is higher, the system is loosely damped. If the resonant peak is smaller, the system is heavily damped. Thus, in the frequency domain specifications, resonant peak and resonant frequency are known, then it is possible to get the damping ratio ζ and undamped frequency ω_n. From this, the time domain specifications can be obtained. Thus, a correlation is obtained between time domain and frequency domain which is true only for a second order system. More frequency response specifications are obtained and discussed in the later part of the chapter. The linear frequency response plots of Eqs. (3.15) and (3.16) are shown in Fig. 3.18a and b, respectively.

3.6 Bandwidth of a Second Order System

The frequency response of a certain second order system is as shown in Fig. 3.19. As established in Sect. 3.5, correlation between the frequency response and time response can be established by another frequency response specification called **BANDWIDTH**. The bandwidth ω_b of the **closed loop frequency response** is defined as the frequency at which the magnitude response curve $|M(j\omega)|$ does not fall below 0.707 of its value at zero frequency. In terms of decibel (dB), it is defined as the frequency at which $20\log_{10}|M(j\omega)|$ does not fall below -3 dB from its value at zero frequency. For a prototype second order system the expression for the bandwidth is derived as given below. From Eq. (3.12) we write

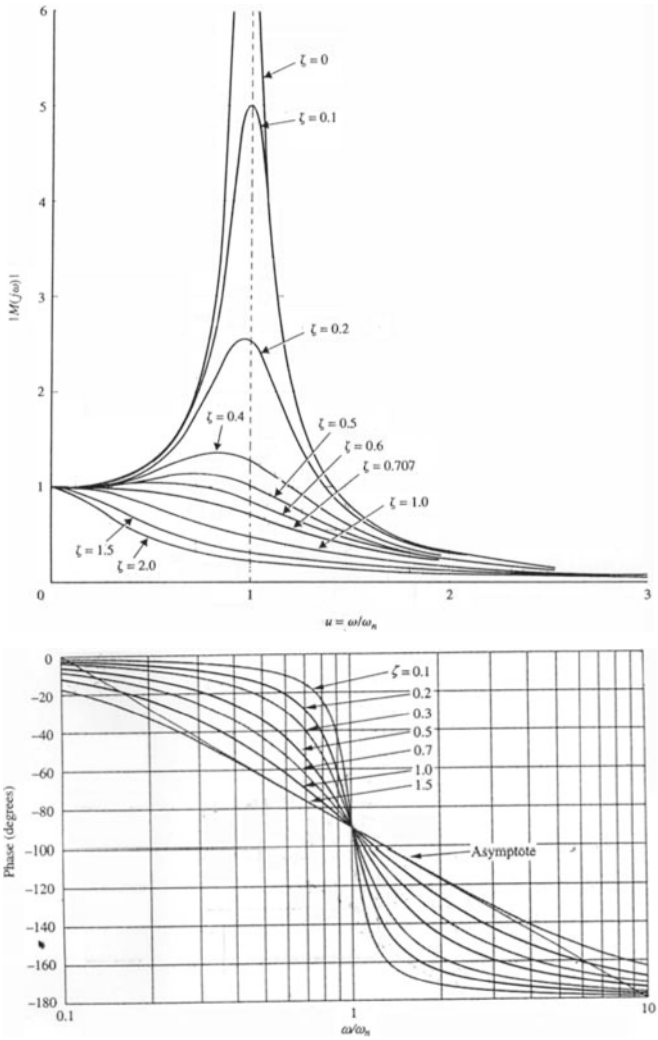


Fig. 3.18 **a** Magnitude *versus* normalized frequency. **b** Phase angle *versus* normalized frequency of a second order closed loop system

$$M(j\omega) = \frac{\omega_n^2}{-\omega^2 + j2\zeta\omega\omega_n + \omega_n^2}$$

$$|M(j\omega)| = \frac{\omega_n^2}{\sqrt{(\omega_n^2 - \omega^2)^2 + 4\zeta^2\omega^2\omega_n^2}}$$

Let ω_b be the frequency at which the D.C. gain (gain at $\omega = 0$) is reduced to 0.707 or $1/\sqrt{2}$. Substituting $\omega = \omega_b$ and $|M(j\omega_b)| = 1/\sqrt{2}$ in the above equation we get

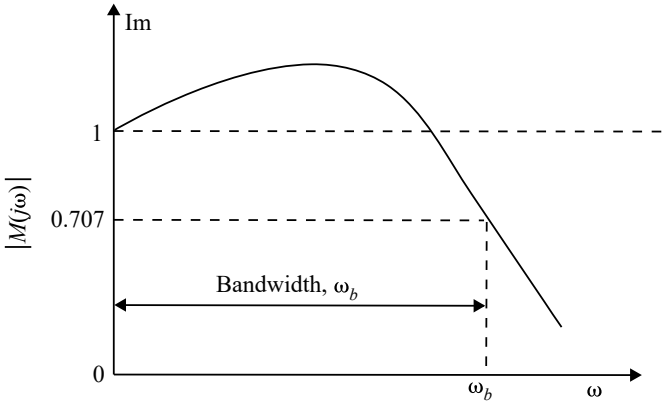


Fig. 3.19 Frequency response and bandwidth of a second order system

$$\frac{1}{\sqrt{2}} = \frac{\omega_n^2}{\sqrt{(\omega_n^2 - \omega_b^2)^2 + 4\zeta^2\omega_b^2\omega_n^2}}$$

Solving the above equation, we get

$$\boxed{\text{Bandwidth} = \omega_b = \omega_n\sqrt{1 - 2\zeta^2 + \sqrt{(4\zeta^4 - 4\zeta^2 + 2)}}} \tag{3.17}$$

ω_b/ω_n *versus* ζ is plotted and shown in Fig. 3.20. The bandwidth plays an important role in the frequency response characteristic. Large bandwidth means the system will pass a wide range of frequency of input signal. In the time domain it amounts to smaller rise time and hence fastness of transient response. Bandwidth is inversely proportional to the rise time. A flat frequency response characteristic gives higher bandwidth which requires high-cost equipment. However, higher bandwidth allows noise to pass through the system. Further, certain systems require low bandwidth and hence high frequency noise is attenuated.

3.6.1 Cut off Frequency and Cut Off Rate

In Fig. 3.19, the frequency ω_b at which the magnitude curve has $1/\sqrt{2}$ times its D.C. value is called the cut off frequency. The slope of the magnitude curve at cut off frequency is called cut off rate. $dM(j\omega)/d\omega$ is called cut off rate.

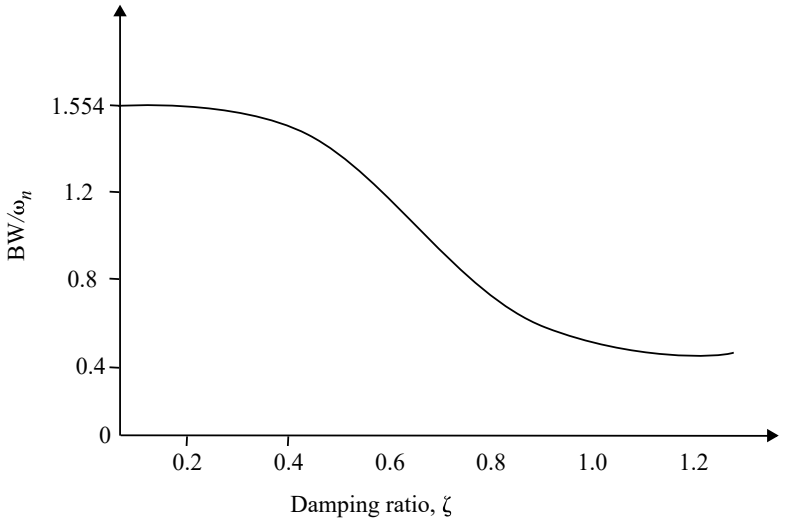


Fig. 3.20 Bandwidth/ω_n *versus* damping ratio of a second order system

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Example 3.6

The open loop T.F.s of certain control systems are given. In each case, determine the resonant peak, resonant frequency and the bandwidth of the closed loop system.

$$\text{(a)} \quad G(s) = \frac{64}{s(s+8)}$$

$$\text{(b)} \quad G(s) = \frac{16}{s(s+8)}$$

Solution

(a)

$$G(s) = \frac{64}{s(s+8)}$$

$$M(s) = \frac{G(s)}{1+G(s)}$$

$$= \frac{64}{(s^2+8s+64)}$$

Comparing the above equation with the standard second order system, we get

$$\frac{\omega_n^2}{(s^2 + 2\zeta\omega_n s + \omega_n^2)} = \frac{64}{s^2 + 8s + 64}$$
$$\omega_n = 8$$
$$\zeta = 0.5$$

Using Eq. (3.15), we get

$$M_r = \frac{1}{2\zeta\sqrt{1-\zeta^2}}$$
$$= \frac{1}{2 \times 0.5\sqrt{1-(0.5)^2}}$$
$$= 1.1547$$

Using Eq. (3.14), we get

$$\omega_r = \omega_n\sqrt{1-2\zeta^2}$$
$$= 8\sqrt{1-2(0.5)^2}$$
$$= 5.6569 \text{ rad/sec}$$

Using Eq. (3.17), we get

$$\omega_b = \omega_n\sqrt{1-2\zeta^2+\sqrt{4\zeta^4-4\zeta^2+2}}$$
$$= 8\sqrt{1-2(0.5)^2+\sqrt{4(0.5)^4-4(0.5)^2+2}}$$
$$= 8\sqrt{1-0.5+1.118}$$
$$= 10.176 \text{ rad/sec}$$

$$\boxed{\begin{aligned} M_r &= 1.1547 \\ \omega_r &= 5.6569 \text{ rad/sec} \\ \omega_b &= 10.176 \text{ rad/sec} \end{aligned}}$$

(b)

$$G(s) = \frac{16}{s(s+8)}$$
$$M(s) = \frac{16}{s^2 + 8s + 16}$$

Comparing this with the standard second order system, we get $\omega_n = 4$ and $\zeta = 1$. Since $\zeta > 0.707$, $M_r = 1$ and $\omega_r = 0$. The bandwidth is calculated using Eq. (3.17)

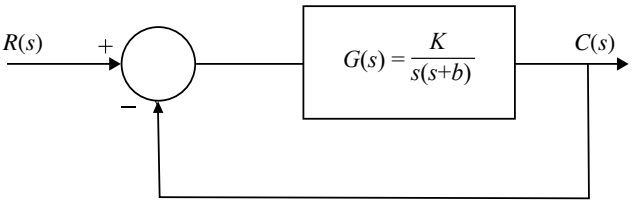


Fig. 3.21 Transfer function, for Example 3.7

$$\omega_b = 4\sqrt{1-2+\sqrt{2}}$$
$$= 2.5737 \text{ rad/sec}$$

$$\boxed{\begin{aligned} M_r &= 1 \\ \omega_r &= 0 \\ \omega_b &= 2.5737 \text{ rad/sec} \end{aligned}}$$

Example 3.7

For the system shown in Fig. 3.21 find the value of K and b to satisfy the following frequency domain specifications. $M_r = 1.2$; $\omega_r = 14$ rad/sec. For these values of K and b, calculate the settling time and bandwidth.

(*Anna University, May, 2011*)

Solution From Eq. (3.15), we have

$$M_r = \frac{1}{2\zeta\sqrt{1-\zeta^2}}$$
$$1.2 = \frac{1}{2\zeta\sqrt{1-\zeta^2}}$$

Squaring and cross multiplying, we get

$$\zeta^2(1-\zeta^2) = 0.1736$$
$$\zeta^4 - \zeta^2 + 0.1736 = 0$$

Solving the above equation, we get

$$\zeta = 0.88 \quad \text{or} \quad \zeta = 0.473$$

Since resonance occurs for $\zeta < 0.707$, the value of $\zeta = 0.473$ alone is considered.

From Eq. (3.14), we have

$$\begin{aligned}\omega_r &= \omega_n\sqrt{1-2\zeta^2}\\ \omega_n &= \frac{\omega_r}{\sqrt{1-2\zeta^2}}\\ &= \frac{14}{\sqrt{1-2\times 0.473^2}} = 18.834\end{aligned}$$

From Fig. 3.21, we get

$$\frac{C(s)}{R(s)} = \frac{K}{(s^2+bs+K)} \tag{a}$$

For $\zeta = 0.473$ and $\omega_n = 18.834$

$$\begin{aligned}\frac{C(s)}{R(s)} &= \frac{\omega_n^2}{(s^2+2\zeta\omega_n s+\omega_n^2)}\\ &= \frac{354.72}{s^2+17.817s+354.72}\end{aligned} \tag{b}$$

Comparing Eqs. (a) and (b), we get

$$\begin{aligned}K &= 354.72\\ b &= 17.817\end{aligned}$$

The settling time for 2% error tolerance is

$$\begin{aligned}t_s &= \frac{3.91}{\zeta\omega_n}\\ &= \frac{3.91}{0.473\times 18.834} = 0.439 \text{ sec}\end{aligned}$$

From Eq. (3.17), the bandwidth is

$$\begin{aligned}\omega_b &= \omega_n\sqrt{1-2\zeta^2+\sqrt{(4\zeta^4-4\zeta^2+2)}}\\ &= 18.834\sqrt{1-2(0.473)^2+\sqrt{4(0.473)^4-4(0.473)^2+2}}\\ &= 18.834\sqrt{1-0.447+1.3}\\ &= 25.673 \text{ rad/sec}\end{aligned}$$

$$\boxed{\begin{aligned} K &= 354.72 \\ b &= 17.817 \\ \omega_b &= 25.673 \end{aligned}}$$

Example 3.8

The magnitude of frequency response of an under-damped second order system is 5 at a frequency of 0 rad/sec. and peaks to $\sqrt{5/3}$ at a frequency of $5\sqrt{2}$ rad/sec. Determine the transfer function.

Solution

$$\begin{aligned} M_r &= \sqrt{\frac{5}{3}} \\ \omega_r &= 5\sqrt{2} \\ M_r &= \frac{1}{2\zeta\sqrt{1-\zeta^2}} = \sqrt{\frac{5}{3}} \end{aligned}$$

Squaring and cross multiplying, we get

$$\begin{aligned} \zeta^2(1-\zeta^2) &= 0.15 \\ \zeta^4 - \zeta^2 + 0.15 &= 0 \end{aligned}$$

Solving the above equation, we get

$$\zeta = 0.429 \quad \text{or} \quad \zeta = 0.9$$

Since $\zeta = 0.9$ is greater than 0.707, only the value of $\zeta = 0.429$ is considered

$$\begin{aligned} \omega_n &= \frac{\omega_r}{\sqrt{1-2\zeta^2}} = \frac{5\sqrt{2}}{\sqrt{1-2\times(0.429)^2}} \\ &= 8.894 \end{aligned}$$

The closed loop T.F. is written as

$$\begin{aligned} M(s) &= \frac{K\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \\ &= \frac{K79.1}{s^2 + 7.639s + 79.1} \\ M(0) &= K \end{aligned}$$

But it is given that $M(0) = 5$ and $K = 5$

$$M(s) = \frac{5 \times 79.1}{s^2 + 7.639s + 79.1}$$

$$\boxed{M(s) = \frac{395.5}{(s^2 + 7.639s + 79.1)}}$$

Example 3.9

A certain second order closed loop system when subjected to step input exhibits a peak overshoot of 16.3% and the rise time is observed as 0.2 sec. Find the values of resonant peak M_r, resonant frequency ω_r and the bandwidth.

Solution The peak overshoot is given by

$$M_p = e^{\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}} = 0.163$$

Taking l_n on both sides of the equation, we get

$$\frac{-\zeta\pi}{\sqrt{1-\zeta^2}} = -1.814$$

Solving the above equation, we get

$$\begin{aligned}
\zeta &= 0.5 \\
\phi &= \cos^{-1}\zeta = \frac{\pi}{3} \text{ rad} \\
t_r &= \frac{(\pi - \phi)}{\omega_n\sqrt{1-\zeta^2}} \\
\omega_n &= \frac{(\pi - \phi)}{t_r\sqrt{1-\zeta^2}} = \frac{\pi - (\pi/3)}{0.2\sqrt{1-(0.5)^2}} \\
&= 12.1 \text{ rad/sec}
\end{aligned}$$

Resonant peak is given by

$$\begin{aligned}
M_r &= \frac{1}{2\zeta\sqrt{1-\zeta^2}} = \frac{1}{2 \times 0.5\sqrt{1-(0.5)^2}} \\
&= 1.155
\end{aligned}$$

The resonant frequency is given by

$$\omega_r = \omega_n\sqrt{1-2\zeta^2} = 12.1\sqrt{1-2(0.5)^2}$$
$$= 8.556 \text{ rad/sec}$$

The bandwidth is given by

$$\omega_b = \omega_n\sqrt{1-2\zeta^2+\sqrt{4\zeta^4-4\zeta^2+2}}$$
$$= 12.1\sqrt{1-2(0.5)^2+\sqrt{4(0.5)^4-4(0.5)^2+2}}$$
$$= 12.1\sqrt{1-0.5+1.25}$$
$$\omega_b = 14.82$$

$$\boxed{\begin{aligned} M_r &= 1.155 \\ \omega_r &= 8.556 \\ \omega_b &= 14.82 \end{aligned}}$$

Example 3.10

A certain unity feedback control system when subjected to sinusoidal input has a resonant peak of 1.2 and the corresponding frequency is 8 rad/sec. Determine the bandwidth of the closed loop system. Also determine the % overshoot, rise time and settling time when subjected to step input.

Solution

$$M_r = \frac{1}{2\zeta\sqrt{1-\zeta^2}}$$
$$4\zeta^2(1-\zeta^2) = \frac{1}{M_r^2} = \frac{1}{(1.2)^2}$$
$$\zeta^2(1-\zeta^2) = 0.1736$$
$$\zeta^4 - \zeta^2 + 0.1736 = 0$$

Solving for ζ, we get

$$\zeta^2 = 0.7764 \quad \text{or} \quad \zeta^2 = 0.2236$$
$$\zeta = 0.882 \quad \text{or} \quad \zeta = 0.473$$

Resonance occurs only for $\zeta < 0.707$. Hence, $\zeta = 0.473$ is considered

$$
\begin{aligned}
\omega_r &= \omega_n\sqrt{1-2\zeta^2} \\
\omega_n &= \frac{\omega_r}{\sqrt{1-2\zeta^2}} = \frac{8}{\sqrt{1-2\times 0.2236}} \\
&= 10.76 \text{ rad/sec} \\
\omega_d &= \omega_n\sqrt{1-\zeta^2} \\
&= 10.76\sqrt{1-0.2236} = 9.481 \\
\phi &= \cos^{-1}\zeta = \cos^{-1} 0.473 \\
&= 61.77^\circ \quad \text{or} \quad 1.078 \text{ rad.} \\
\zeta\omega_n &= 0.473 \times 10.76 = 5.1 \\
\omega_b &= \omega_n\sqrt{1-2\zeta^2+\sqrt{4\zeta^4-4\zeta^2+2}} \\
&= 10.76\sqrt{1-2\times 0.2236+\sqrt{4(0.2236)^2-4(0.2236)+2}} \\
&= 14 \text{ rad/sec}
\end{aligned}
$$

$$
\%M_p = e^{\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}} \times 100 = e^{\frac{-0.473\pi}{\sqrt{1-0.2236}}} \times 100 = 18.52\%
$$

$$
t_r = \frac{(\pi-\phi)}{\omega_d} = \frac{(\pi - 1.078)}{9.481} = 0.2177 \text{ sec}
$$

$$
t_s = \frac{3.91}{\zeta\omega_n} = \frac{3.91}{5.1} = 0.767 \text{ sec}
$$

$$
\boxed{\begin{array}{ll} \omega_b = 14 \text{ rad/sec}; & \%M_p = 18.52\% \\ t_r = 0.2177 \text{ sec}; & t_s = 0.767 \text{ sec} \end{array}}
$$

3.7 Bode Plots

In Sect. 3.4, the display of frequency response in polar graph was explained for the frequency range $0 \le \omega \le \infty$. The polar plot has the main drawback in that information near the origin could not be obtained with clarity since the magnitude and frequency both vary over many powers of 10 and they are got compressed. Further, the method does not clearly indicate the contribution made by each pole and zero in the frequency response plot which will be very useful in the design. To overcome these difficulties, Hendrik Wade Bode, who worked in Bells Telephone Laboratories, U.S.A. in the year 1938, developed his asymptotic phase and magnitude plots what is now popularly known as Bode plots.

These plots are the most widely used means of displaying and communicating frequency response information. Bode plot is very commonly used to communicate

frequency response specifications in stereo equipment such as amplifiers, speakers, microphones and headsets. Further, these plots give useful information for designing a variety of control systems. Stability criteria can be interpreted on Bode plots, and furthermore, there are numerous design techniques available which are based on Bode plots. Bode plot contains two separate plots and they are called magnitude and phase plots. In magnitude plot the gain $|G(j\omega)|$ is expressed in decibel (dB) and displayed in the vertical axis. $20\log_{10}|G(j\omega)|$ is called a decibel which is written in the abbreviated form as dB. The horizontal axis represents frequency on a log scale. In the second plot which is called the phase plot $\angle G(j\omega)$ is plotted in the vertical axis while frequency ω is represented in the horizontal axis which is graduated on a log scale.

Advantages of Bode Plots

1. Bode plots are sketched by approximating the magnitude and phase with straight lines which are easily sketched in Bode plot.
2. Since the horizontal axis is on log scale, it takes care of logarithmic value of frequency ω. Thus, a wide range of frequency study is possible with clarity at all frequencies.
3. The vertical axis represents the magnitude of $G(j\omega)$ which is expressed in logarithmic form. Very large amplitude of $|G(j\omega)|$ can be represented in this way.
4. The magnitude of $G(j\omega)$ is expressed in dB. The multiplications and divisions in $|G(j\omega)|$ become addition and subtraction in Bode plot display which can be easily manipulated. Similarly, phase angles of zeros and poles of $G(j\omega)$ are added and subtracted from each other algebraically.
5. Frequency response specifications such as phase margin, gain margin, phase crossover and gain crossover frequencies are more easily determined from the Bode plot than the polar plot.
6. In the design using Bode plots, the effects of adding controllers and their parameters can be easily visualized than on the polar plot.

Disadvantages of Bode Plot

1. If the Bode plots are manually drawn without the use of computers, the plots become approximate. Unless otherwise, necessary corrections are made in appropriate places in the Bode plots, which is very tedious to accomplish, the results obtained deviate from the actual ones.
2. Absolute and relative stability of the system are obtained from Bode plots provided the system under study is of minimum phase. For non-minimum phase transfer function, the Bode plot cannot be used to assess the absolute and relative stability.

A Historical Note: Hendrik Wade Bode (1905–1982)


Hendrik Wade Bode
(1905-1982)

Hendrik Wade Bode was an American engineer, researcher, inventor, author and scientists of Dutch ancestry. Bode was born in Madian, Wisconsin, USA in the year 1905. His father was a professor of education at the University of Illinois at Urbana. He received his B.A. degree in 1924 at age 19 his M.A. degree in 1926 both in mathematics and his Ph.D. in physics from Columbia University in the year 1935. He joined Bell Laboratories in New York City as a design engineer of electronic filters and equalizers. During World War II, Dr. W.H. Bode joined the armed services of USA. During this period, he developed automatic anti-aircraft gun control system whereby radar information was used to locate the enemy aircraft. The information was feedback to anti-aircraft artillery to shoot down the enemy's aircrafts. He also developed robot weapons which brought down the NaZe V-1 flying bombs, the weapon which terrorized London city during World War II. His research work impacted many other engineering disciplines such as computer, robot and mobile phone. In academic circle, worldwide he is widely known for developing the asymptotic magnitude and phase plots that bear his name, the **Bode Plot**. He was married to Barbara Bode. They had two children: Dr. Katharine Bode Darlington and Mrs. Anne Hathway Bode Aarnes. Hendrik Wade Bode died on 21 June 1982 at the age of 76, at his home in Cambridge, Massachusetts.

3.7.1 Bode Asymptotic Magnitude Plot

Consider the following transfer function

$$G(s) = \frac{K(1+sT_1)(1+sT_2)\cdots(1+sT_m)}{s^k(1+sT_a)(1+sT_b)\cdots(1+sT_n)} \tag{3.18}$$

Equation (3.18) is said to be written in **Time Constant Form**. The sinusoidal T.F. of the above equation is written as

$$G(j\omega) = \frac{K(1+j\omega T_1)(1+j\omega T_2)\cdots(1+j\omega T_m)}{(j\omega)^k(1+j\omega T_a)(1+j\omega T_b)\cdots(1+j\omega T_n)} \tag{3.19}$$

$$|G(j\omega)| = \frac{K\sqrt{(1+\omega^2T_1^2)(1+\omega^2T_2^2)\cdots(1+\omega^2T_m^2)}}{\omega^k(1+\omega^2T_a^2)(1+\omega^2T_b^2)\cdots(1+\omega^2T_n^2)} \tag{3.20}$$

$$\begin{aligned}\angle G(j\omega) = {} & -k\times 90^\circ + \tan^{-1}\omega T_1 + \tan^{-1}\omega T_2 + \cdots + \tan^{-1}\omega T_m \\ & -(\tan^{-1}\omega T_a + \tan^{-1}\omega T_b + \cdots + \tan^{-1}\omega T_n)\end{aligned} \tag{3.21}$$

Bode magnitude plot is drawn for $20\log_{10}|G(\,j\omega)|$ *versus* $\log_{10}\omega$ and phase plot is drawn for $\angle G(\,j\omega)$ *versus* $\log_{10}\omega$. We discuss below drawing of magnitude plot first followed by the phase plot. $20\log|G(\,j\omega)|$ is called a decibel which is written in the abbreviated form as dB. Now taking $\log_{10}$ on both sides of Eq. (3.20) we get (hereafter $\log_{10}$ is simply written as log)

$$\begin{aligned}\text{dB} = 20\log K + 20\log\sqrt{(1+\omega^2T_1^2)} + \cdots + 20\log\sqrt{(1+\omega^2T_m^2)}\\ -20\log\sqrt{(1+\omega^2T_a^2)} - \cdots - 20\log\sqrt{(1+\omega^2T_n^2)}\end{aligned} \tag{3.22}$$

In Eq. (3.22), each term on the right-hand side is plotted separately and added or subtracted accordingly and the total magnitude plot in dB is obtained. The curve is sketched on a semi-log graph. dB is represented in the vertical axis, and frequency ω is represented in the horizontal axis which is marked in logarithmic scale.

In Eq. (3.19), $G(\,j\omega)$ can contain just five simple factors which are given below

1. $G(\,j\omega) = K$ (a constant factor)
2. $G(\,j\omega) = K(\,j\omega)^{\pm p}$ (poles or zeros at the origin of order p)
3. $G(\,j\omega) = (1 + j\omega T)^{\pm q}$ (poles or zeros at $s = -(1/T)$ of order q)
4. $G(\,j\omega) = [\omega_n^2/-\omega^2 + j2\zeta\omega_n\omega + \omega_n^2]^{\pm r}$ (complex poles and zeros of order r)
5. $G(\,j\omega) = e^{-j\omega Td}$ (Time delay)

We discuss below how Bode magnitude plots are sketched for the above five factors.

(1)

$$\begin{aligned}G(s) &= K\\ G(\,j\omega) &= K\\ |G(\,j\omega)| &= K\\ \text{dB} &= 20\log K\end{aligned} \tag{3.23}$$

The magnitude of $G(\,j\omega)$ is independent of frequency and hence dB is constant at all frequency. For $K = 1$, $\text{dB} = 20\log 1 = 0$. For $K > 1$, dB is +ve and for $K < 1$, dB is −ve. The Bode magnitude plot is shown in Fig. 3.22.

(2)

$$\begin{aligned}G(s) &= Ks\\ G(\,j\omega) &= jK\omega\\ |G(\,j\omega)| &= K\omega\\ \text{dB} &= 20\log K\omega\\ \text{At } \omega = \frac{1}{K},&\quad \text{dB} = 20\log 1 = 0\\ \text{At } \omega = \frac{10}{K},&\quad \text{dB} = 20\log 10 = 20\end{aligned} \tag{3.24}$$

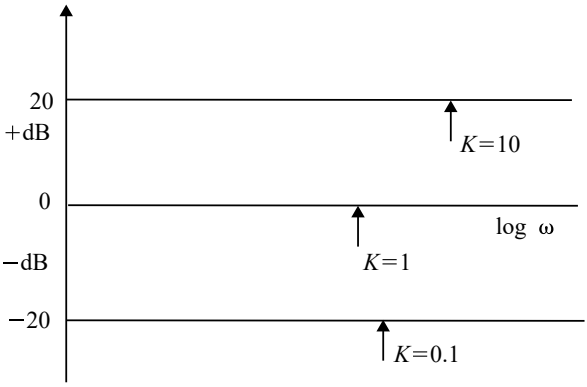


Fig. 3.22 Bode plot of $G(j\omega) = K$

$$\text{At } \omega = \frac{0.1}{K}, \quad \text{dB} = 20 \log 1 = -20$$

This is a straight line. The Bode plot is shown in Fig. 3.23a.
Figure 3.23a is drawn using the following steps:

1. Find ω_l, the frequency at which the magnitude of $G(j\omega) = 1$. In the above case $|G(j\omega)| = K\omega$ and hence $\omega_l = 1/K$. At this frequency the dB is $20 \log 1 = 0$. This corresponds to point A in Fig. 3.23a.
2. Choose $\omega = 10\ \omega_l = 10/K$. At this point, the dB is $20 \log 10K/K = 20$. This corresponds to point B.
3. Join points A and B and extend it as much as you want. The straight line drawn corresponds to Bode plot for $G(j\omega) = jK\omega$.
4. The above straight line is called +20dB/decade line. This is the slope of the straight line. The meaning of **decade** is the frequency range which is multiplied or divided by the factor 10. If the frequency 1 is multiplied by 10, the frequency range from 1 to 10 is a decade. The frequency range from 10 to 100 is a decade. The frequency range from 100 to 1000 is a decade and so on. The frequency range from 1 to 100 is a two decade. The frequency range from 1 to 1000 is a 3 decade and so on.
   The meaning of the slope is to be well understood. For every decade of frequency rise, the increase in dB is 20. Thus, for the frequency range from 1 to 10, the increase in dB is 20, from 10 to 100, the increase in dB is 20 and so on. For the frequency range from 1 to 100, the increase is dB 40. There are two decades in the frequency range from 1 to 100. The straight line simply follows the law $y = mx$, where m is the slope which is determined by the ratio of the vertical height to the base. x is the logarithmic frequency difference. That is $x = \log \omega_2/\omega_1$. y is the increase in the decibel.
5. The straight line in Fig. 3.23a can also be expressed as 6 dB/octave straight line. Octave means, the frequency is multiplied or divided by a factor 2. Thus, at $\omega = 2\omega_l$ in Fig. 3.23a dB $= 20 \log 2\omega_l/\omega_l = 6$. Thus, 20 dB/dec

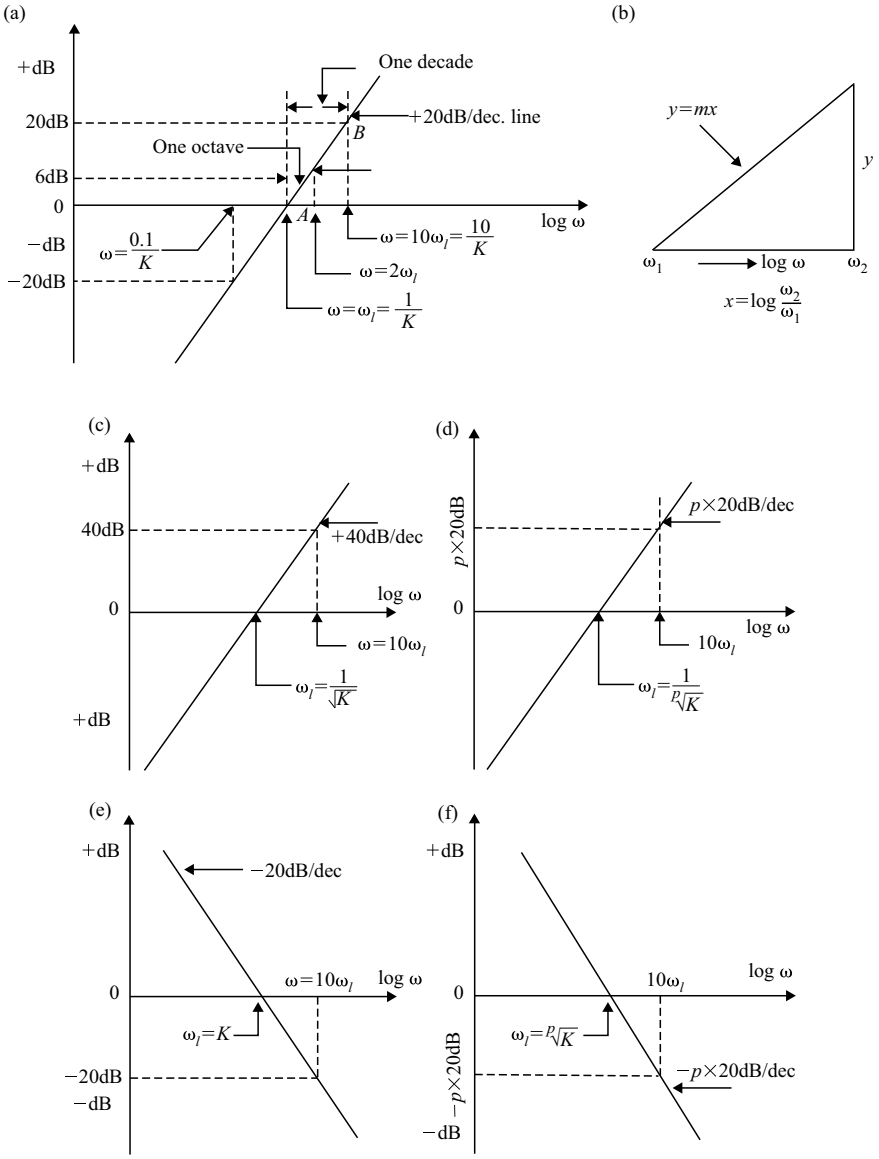


Fig. 3.23 **a** Bode plot of $G(s) = Ks$. **c** Bode plot for $G(s) = Ks^2$, and **d** $G(s) = Ks^m$. **e** Bode plot for $G(s) = \dfrac{K}{s}$, and **f** $G(s) = \dfrac{K}{s^p}$

has the same meaning as 6 dB/octave. In this book, we follow 20 dB/dec. The frequency ω_l is calculated from $K\omega_l = 1$.

Now consider the following T.F.

$$\begin{aligned} G(s) &= Ks^2 \\ |G(j\omega)| &= K\omega^2 \\ K\omega_l^2 &= 1 \\ \omega_l &= \frac{1}{\sqrt{K}} \end{aligned}$$

At $\omega = 10\omega_l$, dB is $20 \log K(10/\sqrt{K})^2 = 40$ dB. Thus, the straight line has a slope of +40 dB/dec slope as shown in Fig. 3.23c. In general for $G(s) = Ks^p$, the Bode plot is a straight line with a slope of $p \times 20$ dB/dec slope and $\omega_l = \frac{1}{\sqrt[p]{K}}$. Now consider the following T.F.

$$\begin{aligned} G(s) &= \frac{K}{s} \\ |G(j\omega)| &= \frac{K}{\omega} \end{aligned}$$

$|G(j\omega)| = 1$ at $\omega = K$ or $\omega_l = K$. At this frequency the dB $= 20 \log 1 = 0$. At $\omega = 10\omega_l = 10K$

$$\text{dB} = 20 \log \frac{K}{10K} = -20$$

Thus, the slope m here is -20 dB/dec and the Bode plot is a straight line. In general, for $G(s) = K/s^p$, $\omega_l = \sqrt[p]{K}$ and the slope is $-p \times 20$ dB/dec. These plots are shown in Fig. 3.23e and f, respectively.

It is to be noted that as the order of integration or differentiation increases, the slope of the straight line plot also increases and plot becomes much steeper.

(3)

$$\begin{aligned} G(s) &= (1 + sT) \\ G(j\omega) &= 1 + j\omega T \\ |G(j\omega)| &= \sqrt{(1 + \omega^2 T^2)} \end{aligned} \tag{3.25}$$

Now consider the T.F. shown above. In the Bode plots drawn for the transfer functions in cases 1 and 2, there was absolutely no approximation incorporated in the Bode plot. However, for quicker sketching of Bode plot, now some approximations are made. These approximations can be improved, if necessary by adding corrections in appropriate places and totally nullified by computer programmed plots using MATLAB. For the T.F. described in case 3 above for $\omega T <<< 1$,

$$|G(j\omega)| = 1$$
$$\text{dB} = 20\log 1 = 0$$

For $\omega T >>> 1$,

$$|G(j\omega)| = \omega T$$

The above plot is similar to case 2. It is a straight line with +20 dB/dec slope and $\omega_l = 1/T$. Thus, an approximate plot is drawn with 0 dB upto $\omega = \omega_l = 1/T$. Another approximate plot is drawn with +20 dB/dec slope for the frequency range $\omega_l \leq \omega < \infty$. These approximate plots are called **Asymptotes**. The points at which these two asymptotes meet in the horizontal axis and the corresponding frequency is called **Corner frequency** ω_c. Thus, ω_l is now replaced by the conventional symbol ω_c when asymptotic plots are drawn.

The corner frequency of a Bode plot is therefore defined as the frequency at which the two asymptotes (approximate plots) meet.

The Bode plot for $G(s) = (1 + sT)$ is shown in Fig. 3.24a.

In the asymptotic plot, by introducing necessary corrections at strategic points, a Bode plot which is reasonably closer to the actual Bode plot can be drawn. These points are chosen at frequencies one octave above the corner frequency and one octave below the corner frequency.

Correction at the Corner Frequency

$$|G(j\omega)| = \sqrt{1 + \omega^2 T^2}$$
$$\text{dB}|_{\omega=\frac{1}{T}} = 20\log\sqrt{2} = 3\,\text{dB}$$

Correction at One Octave Above the Corner Frequency

Actual dB at $\omega = 2/T$ is

$$\text{dB} = 20\log\sqrt{1+4} = 7$$

Asymptotic dB at $\omega = 2/T$ is

$$\text{dB} = 20\log 2 = 6$$

The correction to be made is $(7 - 6) = 1$ dB.

Correction at One Octave Below the Corner Frequency

Actual dB at $\omega = 1/2T$ is

$$\text{dB} = 20\log\sqrt{1 + \frac{1}{4}} = 1$$

Asymptotic dB at $\omega = 1/2T$ is

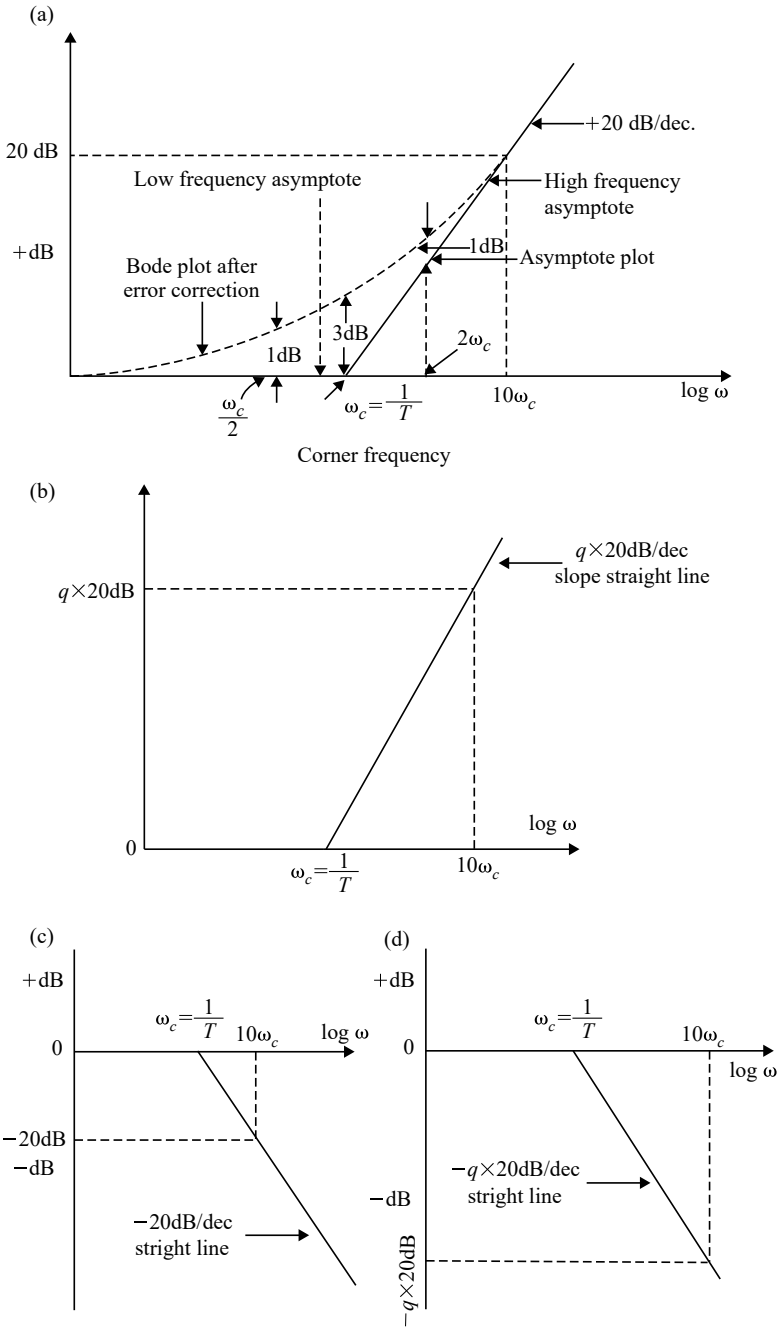


Fig. 3.24 **a** Bode plot for $G(s) = (1 + sT)$. **b** Bode plot for $G(s) = (1 + sT)^q$. Bode plot for **c** $G(s) = \dfrac{1}{(1 + sT)}$, and **d** $G(s) = \dfrac{1}{(1 + sT)^q}$

$$\text{dB} = 0$$

The correction to be made is $1 - 0 = 1\,\text{dB}$.
The complete Bode plot is shown in Fig. 3.24a. Since the Bode plot is very extensively used for the design of control system compensators, where the Bode plot has to be drawn according to the poles and zeros alterations, the asymptotic Bode plot is invariably used. However, the final design is verified by simulation study of actual Bode plot obtained with the help of MATLAB. We discuss below the asymptotic Bode plot of other factors.
Now consider the following T.F.

$$\begin{aligned} G(s) &= (1 + sT)^q \\ G(j\omega) &= (1 + j\omega T)^q \\ |G(j\omega)| &= [\sqrt{(1 + \omega^2 T^2)}]^q \\ \text{dB} &= 10 \times q \log(1 + \omega^2 T^2) \end{aligned}$$

This is the straight line with $\omega_c = 1/T$ and slope $q \times 20\,\text{dB/dec}$ as shown in Fig. 3.34b.
The Bode plot for $G(s) = 1/(1 + sT)$ and $G(s) = 1/(1 + sT)^q$ are the mirror images of Fig. 3.24a and b, respectively, and they are shown in Fig. 3.24c, d. It is to be noted that the corner frequency for $G(s) = (1 + sT)$ and for $G(s) = (1 + sT)^q$ are the same at $\omega = 1/T$.

(4) Complex quadratic factors

$$\begin{aligned} G(s) &= \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \\ G(j\omega) &= \frac{\omega_n^2}{-\omega^2 + j2\zeta\omega\omega_n + \omega_n^2} \\ |G(j\omega)| &= \frac{\omega_n^2}{\sqrt{(\omega_n^2 - \omega^2)^2 + 4\zeta^2\omega^2\omega_n^2}} \\ &= \frac{1}{\sqrt{\left(1 - \frac{\omega^2}{\omega_n^2}\right)^2 + 4\zeta^2\frac{\omega^2}{\omega_n^2}}} \end{aligned} \tag{3.26}$$

For at $\omega <<< \omega_n$,

$$\begin{aligned} |G(j\omega)| &= 1 \\ \text{dB} &= 0 \end{aligned}$$

For $\omega >>> \omega_n$,

$$|G(j\omega)| = \frac{\omega_n^2}{\omega^2}$$
$$\text{dB} = 20\log\frac{\omega_n^2}{\omega^2}$$
$$= -40\text{dB/dec with } \omega_c = \omega_n$$

At $\omega = \omega_n$

$$|G(j\omega)| = \frac{1}{2\zeta}$$
$$\text{dB} = -20\log 2\zeta$$

For $\zeta = 0.5$, dB $= 0$
$\zeta > 0.5$, dB $= -$ve
$\zeta < 0.5$, dB $= +$ve

For the quadratic factor, the Bode asymptotic plot is drawn with -40 dB/dec straight line with corner frequency $\omega_c = \omega_n$. This is represented in Fig. 3.25a. The variation at $\omega = \omega_n$ with respect to ζ is shown in Fig. 3.25b.

(5) Time delay

$$G(s) = e^{-sTd}$$
$$|G(j\omega)| = e^{-j\omega T_d}$$
$$|G(j\omega)| = 1$$

The magnitude plot is a zero dB line at all frequencies.

3.7.2 *Bode Asymptotic Phase Angle Plots*

For the five factors for which the Bode magnitude plot was discussed, we give below the asymptotic phase angle plots (Table 3.2).

(1)

$$G(s) = K$$
$$G(j\omega) = K$$

The phase angle is zero at all frequencies.

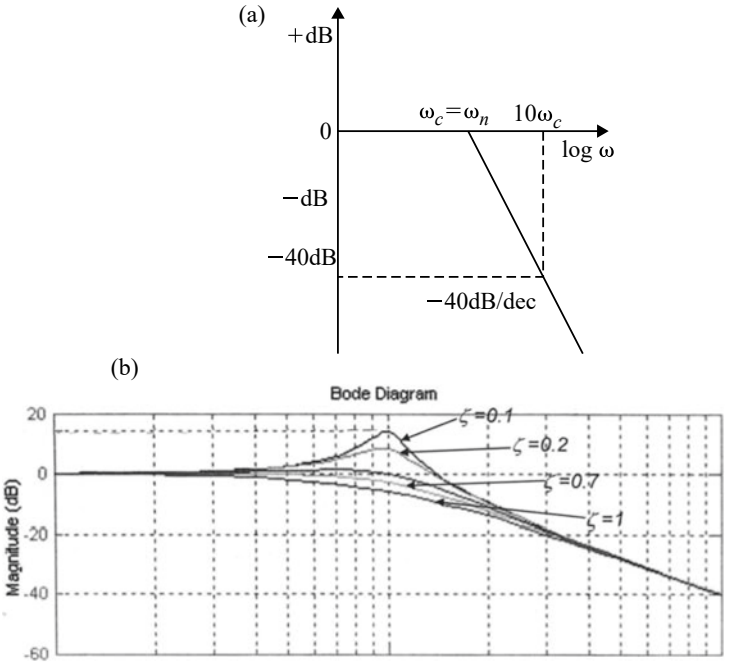


Fig. 3.25 Bode plot for $G(s) = \omega_n^2/s^2 + 2\zeta\omega_n s + \omega_n^2$. **a** Asymptotic and **b** Effect of ζ at ω_c

Table 3.2 Summary of Bode magnitude plot

T.F. $G(s)$	K	$\frac{K}{s}$	$\frac{K}{s^2}$	Ks	Ks^2	$(1+sT)$
Corner frequency, ω_c	–	K	$\sqrt{K}$	$\frac{1}{K}$	$\frac{1}{\sqrt{K}}$	$\frac{1}{T}$
Slope, dB/decade	0	−20	−40	20	40	20
T.F. $G(s)$	$(1+sT)^2$	$\frac{1}{(1+sT)}$	$\frac{1}{(1+sT)^2}$	$\frac{\omega_n^2}{(s^2+2\zeta\omega_n s+\omega_n^2)}$		
Corner frequency, ω_c	$\frac{1}{T}$	$\frac{1}{T}$	$\frac{1}{T}$	ω_n		
Slope, dB/decade	40	−20	−40	−40		

Note Phase plots are normally drawn for the actual phase angle which is calculated at each ω. The asymptotic phase plot is usually avoided

(2)

$$\begin{aligned} G(s) &= Ks \\ G(j\omega) &= jK\omega \\ \angle G(j\omega) &= 90^\circ \\ G(s) &= Ks^2 \\ G(j\omega) &= -K\omega^2 \\ \angle G(j\omega) &= 180^\circ \end{aligned}$$

In general for

$$\begin{aligned} G(s) &= Ks^p \\ \angle G(j\omega) &= \angle p \times 90^\circ \end{aligned}$$

For $G(s) = K/s^p$

$$\angle G(j\omega) = \angle -p \times 90^\circ$$

The phase angle plots are shown in Fig. 3.26.

(3)

$$\begin{aligned} G(s) &= (1 + sT) \\ G(j\omega) &= 1 + j\omega T \\ \angle G(j\omega) &= \tan^{-1} \omega T \end{aligned}$$

The corner frequency is $\omega_c = 1/T$. At ω_c, $\phi = \tan^{-1} 1 = 45^\circ$. At $\omega = 0$, $\phi = 0$; At $\omega = \infty$, $\phi = 90^\circ$. For $G(s) = (1 + sT)^q$ these angles will be multiplied by the factor q. Similarly, for $G(s) = 1/(1 + sT)$, the phase angle is the negative of $(1 + sT)$. The phase angle plots are shown in Fig. 3.27.

(4) Quadratic factor

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$G(j\omega) = \frac{\omega_n^2}{(\omega_n^2 - \omega^2) + j2\zeta\omega\omega_n}$$

$$\phi = -\tan^{-1} \frac{2\zeta\omega\omega_n}{\omega_n^2 - \omega^2}$$

For $\omega = 0$,

$$\phi = 0$$

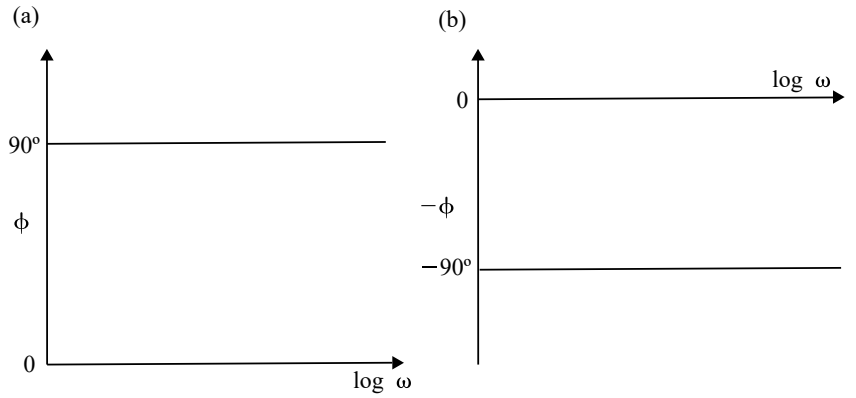


Fig. 3.26 Bode phase plots. **a** $G(s) = Ks$ and **b** $G(s) = \dfrac{K}{s}$

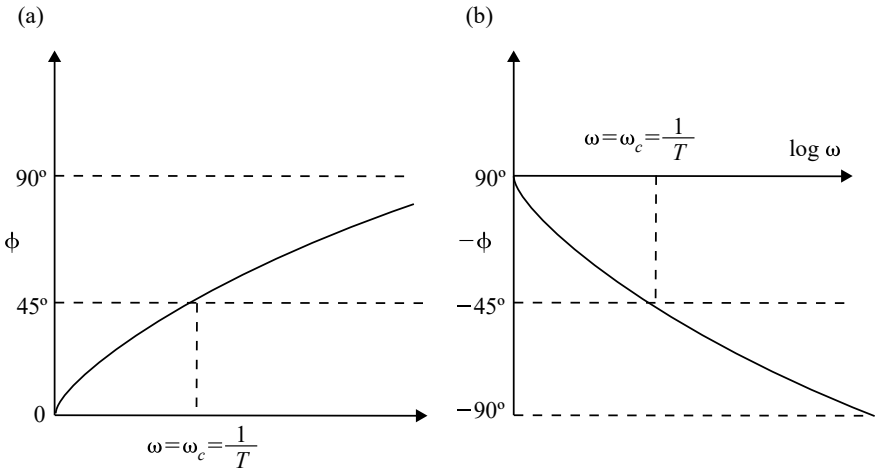


Fig. 3.27 Bode phase angle plots. **a** $G(s) = (1 + sT)$ and **b** $G(s) = \dfrac{1}{(1 + sT)}$

At corner frequency

$$\omega = \omega_n$$
$$\phi = -90^\circ$$

For $\omega \to \infty$

$$\phi \to -180^\circ$$

The phase plots for different values of ζ are shown in Fig. 3.28.

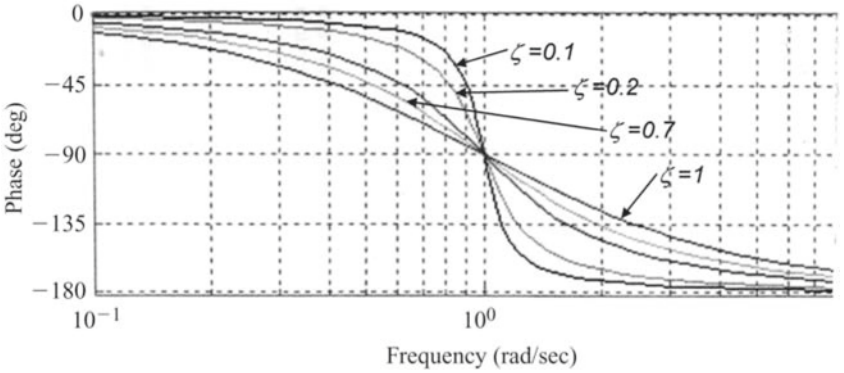


Fig. 3.28 Phase angle plots for $G(s) = \dfrac{\omega_n^2}{(s^2 + 2\zeta\omega_n s + \omega_n^2)}$

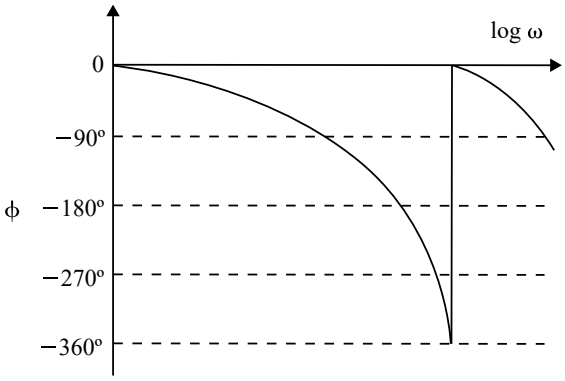


Fig. 3.29 Phase angle plots of $G(s) = e^{-sT_d}$

(5) Time delay

$$\begin{aligned} G(s) &= e^{-sTd} \\ G(j\omega) &= e^{-j\omega T_d} \\ G(j\omega) &= \angle -\omega T_d \text{ in rad} \\ &= \angle -57.3\omega T_d \text{ in degree} \end{aligned}$$

This is a straight line with a slope T_d. The phase angle plot is shown in Fig. 3.29.

3.8 Step by Step Procedure to Draw Bode Magnitude Plot

1. The poles and zeros of the given T.F. is written in the time constant form as given in Eq. (3.18).

2. Identify the low frequency term in the transfer function. This may be K, Ks, Ks^2, K/s, K/s^2, *etc.* The Bode plot starts at low frequency from any of these terms. If it contains pure constant, the low frequency response starts with $20\log K$. Otherwise find ω_l where Ks or K/s, *etc.* magnitude becomes 1 or its decibel is zero.
3. Identify the corner frequencies of each pole and zero and write in the ascending order. Locate these corner frequencies in the $\log\omega$ scale in the horizontal axis. Draw vertical dotted lines passing through these corner frequencies. Also locate ω_l in the $\log\omega$ axis.
4. Start the Bode plot for the low frequency part of the T.F. which will pass through ω_l had there been no other pole or zeros to interfere. (If there is no pure integrator or differentiator, the low frequency response is due to K only and a horizontal line is drawn).
5. When a pole or a zero interferes with the low frequency magnitude response, stop and calculate the dB at that point. If a zero interferes, the slope is changed by $+20$ dB/dec. If two zero interfere the slope is changed by $+40$ dB/dec and so on. On the other hand if a pole interferes, the slope of the straight line is changed by -20 dB/dec. When two poles interfere, the slope is changed by -40 dB/dec and so on. Thus, with the new slope the magnitude plot is extended up to the next corner frequency where poles or zeros interfere. At that point, the Bode plot is to be stopped and the dB is calculated. Depending upon whether the zero interferes or the pole interferes, the new slope of the magnitude plot is decided. The procedure is repeated till the Bode plot crosses past all the corner frequencies and the dB either goes to ∞, 0 finite or $-\infty$ depending upon $G(s)$.

Example 3.11

Consider the following T.F.

$$G(s) = \frac{3000}{(s+2)(1+\frac{s}{10})(s+50)}$$

Sketch the Bode magnitude plot. What is the gain crossover frequency? Also determine the position error constant K_p.

Solution

(1)

$$G(s) = \frac{3000}{(s+2)(1+\frac{s}{10})(s+50)}$$

The given T.F. is written in time constant form as

$$G(s) = \frac{30}{(1 + \frac{s}{2})(1 + (s/10))(1 + (s/50))}$$

The factors of $G(s)$ written in ascending order of corner frequency are

$$G_1(s) = 30$$
$$G_2(s) = \frac{1}{(1 + \frac{1}{2}s)}$$
$$G_3(s) = \frac{1}{(1 + \frac{1}{10}s)}$$
$$G_4(s) = \frac{1}{(1 + \frac{1}{50}s)}$$

The range of corner frequencies and the corresponding slopes of the straight lines are given in the table below. The corner frequencies are $\omega_c = 2$, 10 and 50.

Corner frequency range	$\omega \leq 2$	$2 \leq \omega \leq 10$	$10 \leq \omega \leq 50$	$\omega \geq 50$
Slope in dB/dec	0	−20	−40	−60

(2) The corner frequencies are identified as 2, 10 and 50 and marked in Fig. 3.30a.
(3) The low frequency factor of the T.F. is 30. Taking decibel for this we get

$$\text{dB} = 20 \log 30$$
$$= 29.5$$

A horizontal line ab starting from 29.5 dB is drawn.
(4) At point b, a pole starts and changes the slope from 0 to −20 dB/dec. Line bc is drawn to represent this. The height h in the triangle bgc is obtained as

$$|h| = 20 \log \frac{\text{(higher frequency)}}{\text{(lower frequency)}}$$
$$= 20 \log \frac{10}{2}$$
$$= 14$$

The point c is located at $(29.5 - 14) = 15.5$ dB.
(5) At point c, where the corner frequency is 10, a pole interferes and changes the slope from −20 dB/dec to −40 dB/dec. Straight line cde is drawn. The decibel at e is calculated as in step 4.

$$|h| = 40 \log \frac{50}{10}$$
$$= 28$$

The decibel at $e = 15.5 - 28 = -12.4$.

The frequency where the Bode magnitude plot crosses the 0 dB line is called gain crossover frequency. The gain crossover frequency is calculated as

$$40 \log \frac{\omega_{gc}}{10} = 15.5$$
$$\omega_{gc} = 24.4$$

(6) At point e, another pole interferes and changes the slope from -40 dB/dec to -60 dB/dec. Any frequency $\omega > 50$ is chosen. We choose $\omega = 70$. At this frequency

$$|h| = 60 \log \frac{70}{50}$$
$$= 8.8$$

The point f is chosen with a dB of $-12.4 - 8.8 = -21.2$. The straight line connecting e and f is drawn and extended. This completes the Bode magnitude plot.

(7) The position error constant is calculated from the zero dB slope line. From Fig. 3.30a.

$$20 \log K_p = 29.5$$
$$K_p = 30$$

Figure 3.30a is transferred to semi-log graph and is shown in Fig. 3.30b.

$$\boxed{\begin{aligned} K_p &= 30 \\ \omega_{gc} &= 24.4 \text{ rad/sec} \end{aligned}}$$

How to Draw Bode Plot in Semi-log Graph

1. In semi-log graph in the vertical axis, the decibel is plotted by choosing appropriate scale factor.
2. For the given transfer function, the lower and upper frequencies are determined so that the Bode plot is drawn with clarity at all frequencies. The lower frequency is chosen as 0.01 or 0.1, or 1 or 10, *etc.* depending upon the requirement. The frequency range 0.01 to 0.1 is called one decade. Similarly, 0.1 to 1.0 is another decade.
3. In the decade 0.01 to 0.1, the frequencies are marked for each division as 0.01, 0.02, 0.03, 0.04, 0.05, 0.06, 0.07, 0.08, 0.09 and 0.1. For the next decade the frequencies are marked as 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9 and 1. Like this the frequencies in each decade are marked.
4. Since the frequency axis has already been converted into logarithmic scale, one can choose the division directly for the given ω.

Now we transfer Fig. 3.30a to semi-log sheet and is shown in Fig. 3.30b.

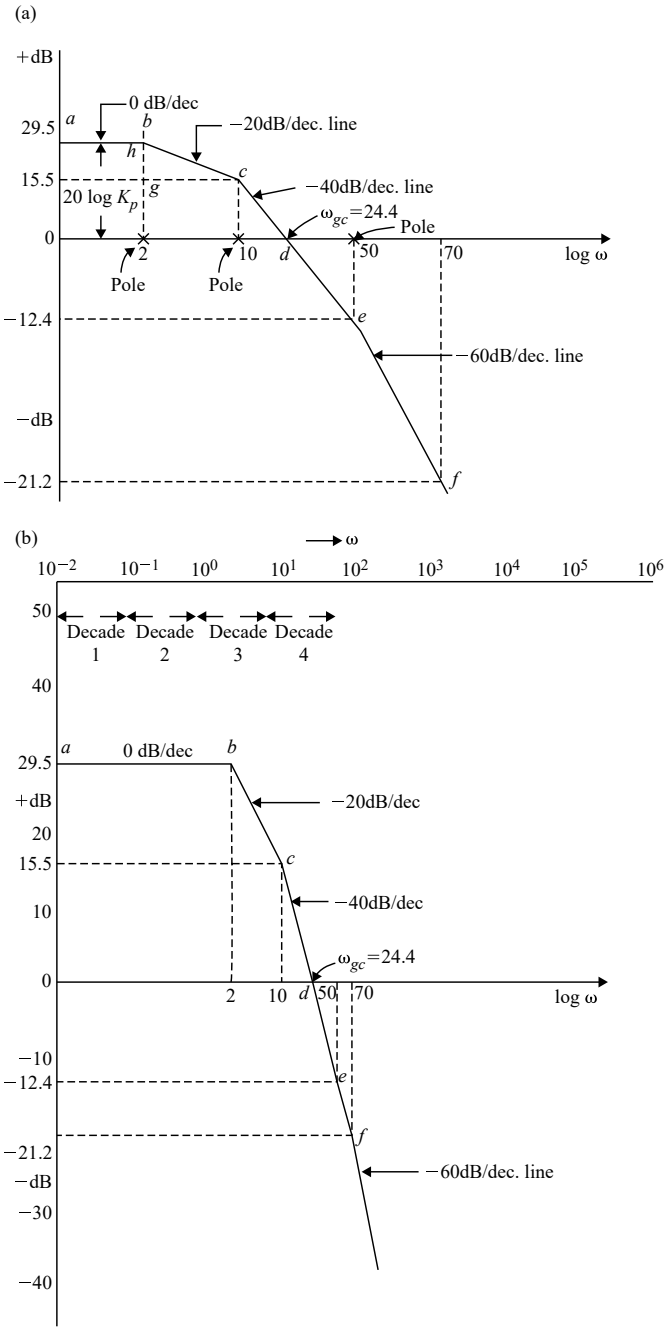


Fig. 3.30 **a** Bode magnitude plot for $G(s) = \dfrac{3000}{(s+2)(1+\frac{s}{10})(s+50)}$. **b** Bode magnitude plot for $G(s) = \dfrac{3000}{(s+2)(1+\frac{s}{10})(s+50)}$ drawn in semi-log graph

Example 3.12

Consider the following transfer function

$$G(s) = \frac{15(1 + \frac{s}{10})}{s(1 + \frac{s}{4})(1 + \frac{s}{20})}$$

Sketch the Bode magnitude plot. Find the velocity error constant. What is the gain crossover frequency?

Solution

(1) The given transfer function is in standard time constant form. The corner frequencies are 4, 10 and 20. The frequency ω_l where $15/\omega = 1$ or dB of $15/\omega$ is 0 is obtained as $\omega_l = 15$. For a type 1 system this is K_v. Hence, $K_v = 15\ \text{sec}^{-1}$. The factors of $G(s)$ in ascending order of corner frequencies are written as given below.

$$G_1(s) = \frac{15}{s}$$
$$G_2(s) = \frac{1}{(1 + \frac{1}{4}s)}$$
$$G_3(s) = \left(1 + \frac{1}{10}s\right)$$
$$G_4(s) = \frac{1}{(1 + \frac{1}{20}s)}$$

The range of corner frequencies and the corresponding slope of the straight lines are given in the following table:

Corner frequency range	$\omega \leq 4$	$4 \leq \omega \leq 10$	$10 \leq \omega \leq 20$	$\omega > 20$
Slope in dB/dec	-20	-40	-20	-40

2. For $\omega \leq 4$, the magnitude contribution comes only from $15/\omega$. Hence

$$\text{dB} = 20 \log \frac{15}{\omega}$$

This is a straight line with slope -20 dB/dec with $\omega_l = 15$. At $\omega = \omega_l = 15$, $\text{dB} = 0$. At $\omega = 4$,

$$\text{dB}|_{\omega=4} = 20 \log \frac{15}{4} = 11.5$$

and point b is located. The points at ω_l and b are joined by a straight line and extended beyond $\omega < 4$. Only line ab is drawn, and the remaining portion is drawn by dotted line as shown in Fig. 3.31.

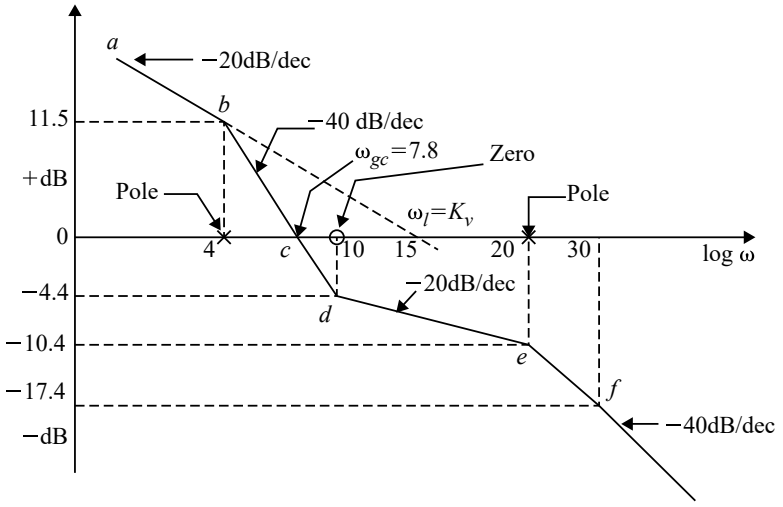


Fig. 3.31 Bode magnitude plot for $G(s) = \frac{15(1+\frac{1}{10s})}{s(1+\frac{s}{4})(1+\frac{s}{20})}$

3. At point b, the pole $1/\left(1+\frac{1}{4}s\right)$ changes the slope from -20 dB/dec to -40 dB/dec and the straight line goes with this slope till $\omega = 10$. At $\omega = 10$, the dB is calculated as

$$\text{dB}|_{\omega=10} = 11.5 - 40\log\frac{10}{4} = -4.4$$

Line bd is drawn. The Bode plot crosses the 0 dB line. The gain crossover frequency is obtained from

$$40\log\frac{\omega_{gc}}{4} = 11.5$$
$$\omega_{gc} = 7.8$$

4. At point d, the zero $\left(1+\frac{1}{10}s\right)$ interferes and changes the slope from -40 dB/dec to -20 dB/dec and goes upto $\omega = 20$. The decibel at $\omega = 20$ is calculated as

$$\text{dB}|_{\omega=20} = -4.4 - 20\log\frac{20}{10} = -10.4$$

Thus, point e is fixed and the straight line de is drawn.

5. At $\omega = 20$, the pole $1/(1+\frac{1}{20}s)$ interferes and changes the slope from -20 dB/dec to -40 dB/dec. Choose any frequency ω which is greater than 20; $\omega = 30$ is chosen. At $\omega = 30$, the dB is calculated as

$$\text{dB} = -10.4 - 40\log\frac{30}{20} = -17.4$$

The point f is fixed at $-17.4\,\text{dB}$ and the straight line ef is drawn and extended beyond $\omega > 30$. This completes the Bode magnitude plot

$$\boxed{\begin{aligned} K_v &= 15\,\text{sec}^{-1} \\ \omega_{gc} &= 7.8\,\text{rad/sec} \end{aligned}}$$

Example 3.13

Consider the following T.F.

$$G(s) = \frac{4(s+5)(s+10)}{s^2(s+20)}$$

Sketch the Bode magnitude plot. Find the acceleration error constant and the gain crossover frequency.

Solution

1.

$$\begin{aligned} G(s) &= \frac{4(s+5)(s+10)}{s^2(s+20)} \\ &= \frac{10\left(1+\frac{1}{5}s\right)\left(1+\frac{1}{10}s\right)}{s^2\left(1+\frac{s}{20}\right)} \end{aligned}$$

The factors of $G(s)$ are written in the ascending order of corner frequency as

$$\begin{aligned} G_1(s) &= \frac{10}{s^2} \\ G_2(s) &= \left(1+\frac{1}{5}s\right) \\ G_3(s) &= \left(1+\frac{1}{10}s\right) \\ G_4(s) &= \frac{1}{\left(1+\frac{1}{20}s\right)} \end{aligned}$$

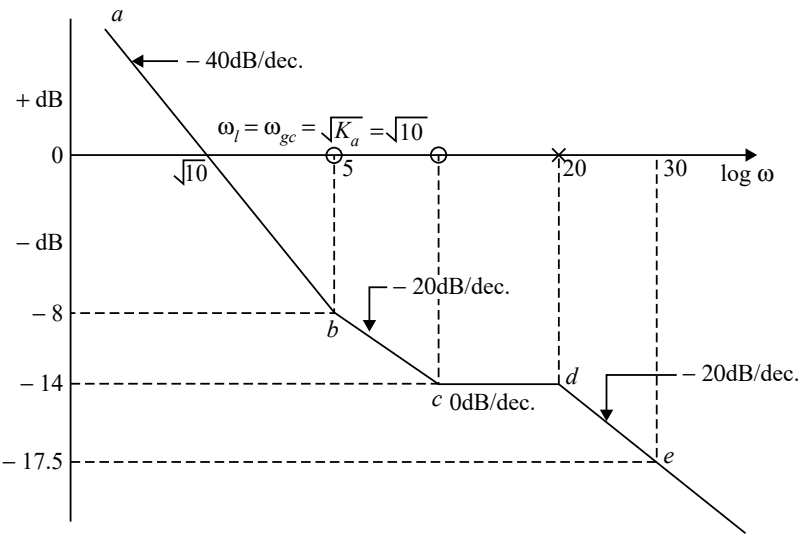


Fig. 3.32 Bode magnitude plot for $G(s) = \dfrac{4(s+5)(s+10)}{s^2(s+20)}$

The range of corner frequencies and the corresponding slopes of the straight line plots are shown in the following table. The corner frequencies are $\omega_c = 5$, 10 and 20 and $\omega_l = \sqrt{10}$.

Corner frequency range	$\omega \leq 5$	$5 \leq \omega \leq 10$	$10 \leq \omega \leq 20$	$\omega \geq 20$
Slope in dB/dec	-40	-20	0	-20

The corner frequencies and ω_l are fixed in the semi-log graph as shown in Fig. 3.32.

2. The low frequency factor is $10/\omega^2$. By equating this to 1 we get $\omega_l = \sqrt{10}$. For $\omega \leq 5$, the magnitude contribution is due to $10/\omega^2$ only. At $\omega = 5$, dB is

$$\begin{aligned} \text{dB}|_{\omega=5} &= 20 \log \frac{10}{\omega^2}\bigg|_{\omega=5} \\ &= 20 \log \frac{10}{25} \\ &= -8 \end{aligned}$$

Thus, point b is fixed. A straight line joining $\omega_l = \sqrt{10}$ and point b is drawn and extended below $\omega < \sqrt{10}$ as shown in Fig. 3.32.

3. At point b, the zero $(s + 5)$ changes the slope from -40 dB/dec to -20 dB/dec. The straight line goes up to $\omega = 10$. At $\omega = 10$, the dB is

$$\begin{aligned} \text{dB}|_{\omega=10} &= -8 - 20\log\frac{10}{5} \\ &= -14 \end{aligned}$$

Thus, point c is fixed. The straight bc is drawn which has a slope of -20 dB/dec.

4. At $\omega = 10$, another zero $(s + 10)$ comes and alters the slope from -20 dB/dec to 0 dB/dec. This is a horizontal line upto $\omega = 20$ without any change in dB. At $\omega = 20$, the point d is fixed. Horizontal straight line cd is drawn.
5. At $\omega = 20$, the pole changes the slope from 0 dB/dec to -20 dB/dec. The Bode plot for $\omega > 20$ is extended as follows. Choose $\omega = 30$. At this frequency, the dB is

$$\begin{aligned} \text{dB}|_{\omega=30} &= -14 - 20\log\frac{30}{20} \\ &= -17.5 \end{aligned}$$

The point e is fixed at -17.5 dB. The straight line joining the points de is extended beyond $\omega > 20$. Thus, the Bode magnitude plot becomes complete.

6. As seen from Fig. 3.32, the dB becomes zero at $\omega = \sqrt{10}$. This is then ω_{gc}. Further, for a type 2 system $\omega_l = \sqrt{K_a}$. Hence, $K_a = 10\,\text{sec}^{-2}$

$$\boxed{\begin{aligned} K_a &= 10\,\text{sec}^{-2} \\ \omega_{gc} &= \sqrt{10}\,\text{rad/sec} \end{aligned}}$$

Example 3.14

Consider the following T.F.

$$G(s) = \frac{10s^2(s+10)}{(s+20)(s+5)(s+30)}$$

Sketch the Bode magnitude plot. What is the gain crossover frequency?

Solution

1.

$$G(s) = \frac{10s^2(s+10)}{(s+5)(s+20)(s+30)}$$

The given T.F. is rewritten in time constant form as

$$G(s) = \frac{1}{30}s^2\frac{(1+\frac{s}{10})}{(1+\frac{s}{5})(1+\frac{s}{20})(1+\frac{s}{30})}$$

The corner frequencies are 5, 10, 20 and 30 with $\omega_l = \sqrt{30}$. The factors of $G(s)$ are written as follows with the ascending order of corner frequencies

$$G_1(s) = \frac{1}{30}s^2$$
$$G_2(s) = \frac{1}{\left(1 + \frac{1}{5}s\right)}$$

$$G_3(s) = \left(1 + \frac{1}{10}s\right)$$
$$G_4(s) = \frac{1}{(1 + \frac{s}{20})}$$
$$G_5(s) = \frac{1}{(1 + \frac{1}{30}s)}$$

Range of corner frequencies	$\omega \leq 5$	$5 \leq \omega \leq 10$	$10 \leq \omega \leq 20$	$20 \leq \omega \leq 30$	$\omega \geq 30$
Slope in dB/dec	+40	+20	+40	+20	0

2. The low frequency response is due to $s^2/30$ or $\omega^2/30$ in sinusoidal T.F. By equating $\omega^2/30 = 1$ we get $\omega_l = \sqrt{30} = 5.5$. For the factor $\omega^2/30$, the Bode plot is a straight line with +40 dB/dec slope and $\omega_l = 5.5$. A pole interferes at $\omega = 5$ and changes the slope. Hence, the dB at $\omega = 5$ is calculated as

$$\begin{aligned} \text{dB}|_{\omega=5} &= 20 \log \frac{\omega^2}{30}\bigg|_{\omega=5} \\ &= 20 \log \frac{25}{30} \\ &= -1.6 \end{aligned}$$

The two points with $\omega = 5$ and dB $= -1.6$ and $\omega = 5.5$ and dB $= 0$ are joined by a straight line. Only the response for $\omega \leq 5$ is sketched. Straight line ab is drawn.

3. At $\omega = 5$ the pole $1/(1 + \frac{1}{5}s)$ changes the slope from +40 dB/dec to +20 dB/dec in the frequency range $5 \leq \omega \leq 10$. At $\omega = 10$, the dB is calculated

$$\begin{aligned} \text{dB}|_{\omega=10} &= -1.6 + 20 \log \frac{10}{5} \\ &= 4.4 \end{aligned}$$

In this frequency range the magnitude plot crosses the 0 dB line. The gain crossover frequency is calculated from

$$20\log\frac{\omega_{gc}}{5} = 1.6$$
$$\omega_{gc} = 6$$

Straight line bc is drawn.

4. At $\omega = 10$, the zero $\left(1 + \frac{1}{10}s\right)$ interferes and changes the slope from +20 dB/dec to +40 dB/dec. A straight line in the frequency range $10 \leq \omega \leq 20$ is to be drawn. The dB at $\omega = 20$ is calculated from

$$\text{dB}|_{\omega=20} = 4.4 + 40\log\frac{20}{10}$$
$$= 16.4$$

The straight line cd is drawn.

5. At $\omega = 20$, the pole $1/(1 + \frac{1}{20}s)$ interferes and changes the slope from +40 dB/dec to +20 dB/dec. A straight line in the frequency range $20 \leq \omega \leq 30$ is to be drawn. The dB at $\omega = 30$ is obtained from

$$\text{dB}|_{\omega=30} = 16.4 + 20\log\frac{30}{20}$$
$$= 20$$

The straight line de is drawn.

6. At $\omega = 30$, the pole $1/(1 + \frac{1}{30}s)$ interferes and changes the slope from +20 dB/dec to 0 dB/dec for $\omega \geq 30$. Hence, the horizontal line ef is drawn and extended beyond $\omega > 30$. The complete Bode magnitude plot is shown in Fig. 3.33

$$\boxed{\omega_{gc} = 6\,\text{rad/sec}}$$

Example 3.15

Consider the following transfer function

$$G(s) = \frac{s(s+10)}{(s^2 + 2s + 9)}$$

Sketch the Bode magnitude plot. Find the gain crossover frequency if any?

Solution

1.

$$G(s) = \frac{s(s+10)}{(s^2 + 2s + 9)}$$

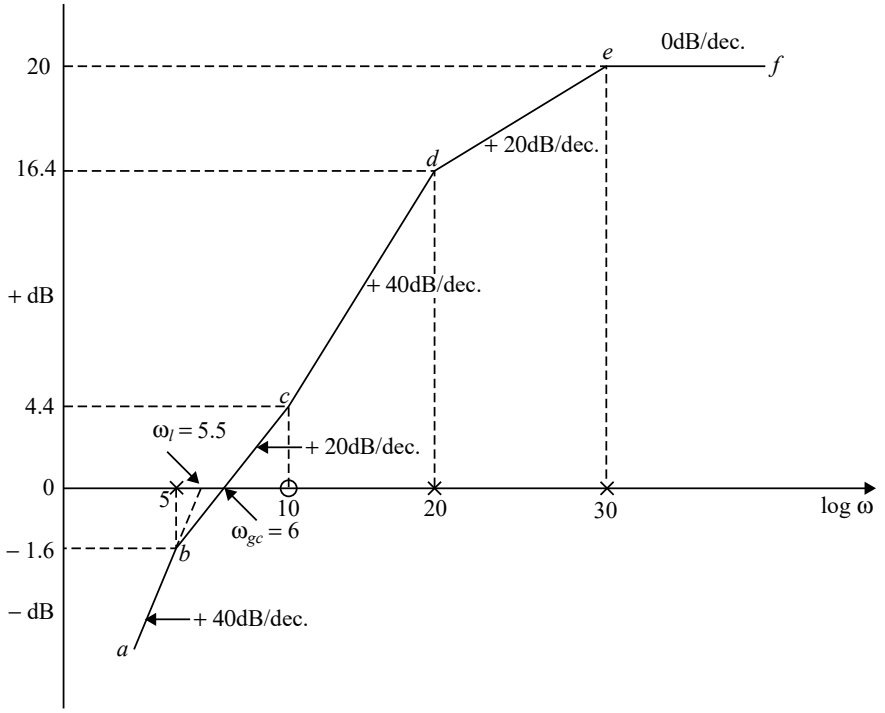


Fig. 3.33 Bode magnitude plot for $G(s) = \dfrac{10s^2(s+10)}{(s+20)(s+5)(s+30)}$

The given T.F. is rewritten in time constant form of poles zeros and the standard prototype second order system as given below.

$$G(s) = \frac{10}{9}\frac{s\left(1+\frac{1}{10}s\right)9}{(s^2+2s+9)}$$

The following factors of $G(s)$ are written in the ascending order of corner frequency

$$G_1(s) = \frac{10}{9}s$$
$$G_2(s) = \frac{9}{s^2+2s+9}$$
$$G_3(s) = \left(1+\frac{1}{10}s\right)$$

The corner frequencies are 3 and 10 with $\omega_l = \frac{9}{10} = 0.9$. The range of corner frequencies and the corresponding slopes of the straight lines are shown in the following table:

Range of corner frequency	$\omega \leq 3$	$3 \leq \omega \leq 10$	$\omega \geq 10$
Slope in dB/dec	+20	−20	0

2. The low frequency response is due to the factor $G_1(j\omega) = \frac{10}{9}\omega$. This is a straight line with $\omega_l = \frac{9}{10}$ for the frequency range $\omega \leq 3$. At $\omega = 3$, the dB is

$$\text{dB}|_{\omega=3} = 20 \log \frac{10}{9} 3 = 10.5$$

Straight line ab is drawn which passes through 0 dB at $\omega_l = 9/10$. Straight line ab is drawn.

3. At $\omega = 3$, the quadratic factor $(9/(s^2 + 2s + 9))$ interferes and changes the slope from +20 dB/dec to −20 dB/dec. The straight line is drawn for the frequency range $3 \leq \omega \leq 10$. The dB at $\omega = 10$ is calculated as

$$\text{dB}|_{\omega=10} = 10.5 - 20 \log \frac{10}{3} = 0$$

The straight line bc is drawn.

4. At $\omega = 10$, the zero $\left(1 + \frac{1}{10}s\right)$ interferes and changes the slope from −20 dB/dec to 0 dB/dec in the frequency range $\omega \geq 10$. A horizontal line cd is drawn and the Bode plot is completed as shown in Fig. 3.34.

Example 3.16

Consider the following T.F.

$$G(s) = \frac{30(s^2 + 4s + 16)}{8(s + 1)(s + 20)}$$

Sketch the Bode magnitude plot. Find the gain crossover frequency.

Solution

1.

$$G(s) = \frac{30(s^2 + 4s + 16)}{8(1 + s)(s + 20)}$$

The T.F. is rewritten in time constant and standard second order forms as given below

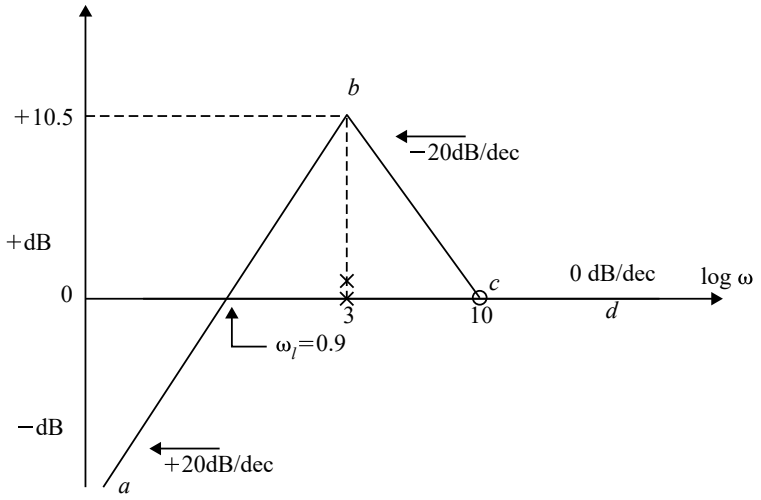


Fig. 3.34 Bode magnitude plot for $G(s) = \dfrac{s(s+10)}{(s^2+2s+9)}$

$$G(s) = \frac{30 \times 16}{8(1+s)\left(1+\frac{1}{20}s\right)20}\frac{(s^2+4s+16)}{16}$$
$$= \frac{3}{(1+s)(1+\frac{1}{20}s)}\frac{(s^2+4s+16)}{16}$$

The factor of $G(s)$ in the ascending order of frequency are written as follows:

$$G_1(s) = 3$$
$$G_2(s) = \frac{1}{1+s}$$
$$G_3(s) = \frac{(s^2+4s+16)}{16}$$
$$G_4(s) = \frac{1}{(1+\frac{1}{20}s)}$$

The corner frequencies are 1, 4 and 20. The corner frequency range and the slope of the Bode magnitude are given in the following table.

Corner frequency range	$\omega \leq 1$	$1 \leq \omega \leq 4$	$4 \leq \omega \leq 20$	$\omega \geq 20$
Slope in dB/dec	0	−20	20	0

2. The low frequency response is due to $G_1(s) = 3$. The magnitude plot is a horizontal line with 0 dB/dec with dB $= 20 \log 3 = 9.5$ upto $\omega \leq 1$. Line ab is drawn.

3. At $\omega = 1$, the pole $1/1 + s$ starts which changes the slope from 0 dB/dec to -20 dB/dec in the frequency range $1 \leq \omega \leq 4$. The dB at $\omega = 4$ is calculated as

$$\text{dB} = 9.5 - 20 \log \frac{4}{1} = -2.5$$

The straight bc is drawn. It cuts 0 dB line at ω_{gc1}. This frequency is calculated from

$$20 \log \frac{\omega_{gc1}}{1} = 9.5$$
$$\omega_{gc1} = 3$$

4. At $\omega = 4$, the quadratic factor $(s^2 + 4s + 16)/16$ starts and changes the slope from -20 dB/dec to $+20$ dB/dec in the frequency range $+4 \leq \omega \leq 20$. The dB at point d is obtained form

$$\text{dB}|_{\omega=4} = -2.5 + 20 \log \frac{20}{4} = 11.5$$

Straight line cd is drawn. This line crossover the 0 dB at ω_{gc2} which is calculated from

$$20 \log \frac{\omega_{gc2}}{4} = 2.5$$
$$\omega_{gc2} = 5.3$$

5. At $\omega = 20$, the pole $1/(1 + \frac{1}{20}s)$ starts which changes the slope from $+20$ dB/dec to 0 dB/dec. Hence, for $\omega \geq 20$, the dB is constant at 11.5. The complete Bode magnitude plot is shown in Fig. 3.35. The T.F. has two gain crossover frequencies at $\omega_{gc1} = 3$ and $\omega_{gc2} = 5.3$.

Example 3.17

Consider the following T.F.

$$G(s) = \frac{100(s + 2)}{s(s + 5)(s + 10)}$$

Sketch the asymptotic graph and hence plot the actual Bode plot. Also sketch the Bode phase plot in semi-log graph.

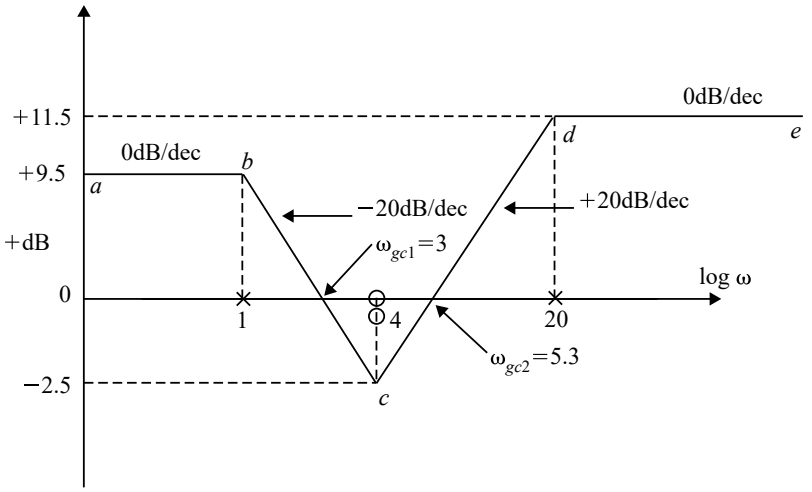


Fig. 3.35 Bode magnitude plot for $G(s) = \dfrac{30(s^2 + 4s + 16)}{8(s + 1)(s + 20)}$

Solution

1.

$$G(s) = \frac{100(s + 2)}{s(s + 5)(s + 10)} = \frac{4(1 + \frac{s}{2})}{s(1 + \frac{s}{5})(1 + \frac{s}{10})}$$

The corner frequencies are 2, 5 and 10 with $\omega_l = 4$. The factors of $G(s)$ are written in the ascending order of corner (break) frequency as follows:

$$G_1(s) = \frac{4}{s}; \qquad G_2(s) = \left(1 + \frac{1}{2}s\right)$$
$$G_3(s) = \frac{1}{(1 + \frac{1}{5}s)}; \qquad G_4(s) = \frac{1}{\left(1 + \frac{1}{10}s\right)}$$

The slopes of the straight lines are given in the following table:

Frequency range	$\omega \leq 2$	$2 \leq \omega \leq 5$	$5 \leq \omega \leq 10$	$\omega \geq 10$
Slope in dB/dec	−20	0	−20	−40

2. The low frequency response is solely due to $4/\omega$ with $\omega_l = 4$ and it is a straight line with −20 dB/dec slope. At $\omega = 2$

$$\begin{aligned}\text{dB}|_{\omega=2} &= +20\log\frac{4}{\omega}\bigg|_{\omega=2}\\ &= +20\log\frac{4}{2}\\ &= +6\end{aligned}$$

Straight line ab is drawn which passes through 6 dB at $\omega = 2$ and 0 dB at $\omega = 4$.

3. At $\omega = 2$, the zero $\left(1+\frac{1}{2}s\right)$ starts changing the resultant slope as 0 dB/dec. Horizontal straight line bc is drawn.
4. At $\omega = 5$, the pole $1/(1+\frac{1}{5}s)$ starts and changes the slope from 0 dB/dec to -20 dB/dec in the frequency range $5 \le \omega \le 10$. At $\omega = 10$, the dB is

$$\text{dB}|_{\omega=10} = 6 - 20\log\frac{10}{5} = 0$$

Straight line cd is drawn.

5. At $\omega = 10$, the pole $1/\left(1+\frac{1}{10}s\right)$ starts making the change in the slope as -40 dB/dec. For $\omega \ge 10$. Any frequency say $\omega = 20$ is chosen. At $\omega = 20$, the dB is

$$\text{dB}|_{\omega=20} = -40\log\frac{20}{10} = -12$$

The straight line de is drawn and extended. The Bode magnitude plot is drawn as shown in Fig. 3.36a.

6. To draw the phase angle plot, the sinusoidal T.F. is

$$G(j\omega) = \frac{100(j\omega+2)}{j\omega(j\omega+5)(j\omega+10)}$$

$$\phi = \angle G(j\omega) = -90 + \tan^{-1}\frac{\omega}{2} - \tan^{-1}\frac{\omega}{10} - \tan^{-1}\frac{\omega}{5}$$

For $\omega = 0$;

$$\phi = -90^\circ$$

For $\omega = \infty$;

$$\phi = -180^\circ$$

In between, few values of ω are chosen and the phase angle is calculated and tabulated below

ω	0	1	2	5	8	12	20	30	∞
ϕ	-90°	-80°	-78°	-94°	-111°	-127°	-145°	-156°	-180°

The data from the above table is transferred to the semi-log sheet where the vertical axis represents the phase angle while the horizontal axis represents the frequency

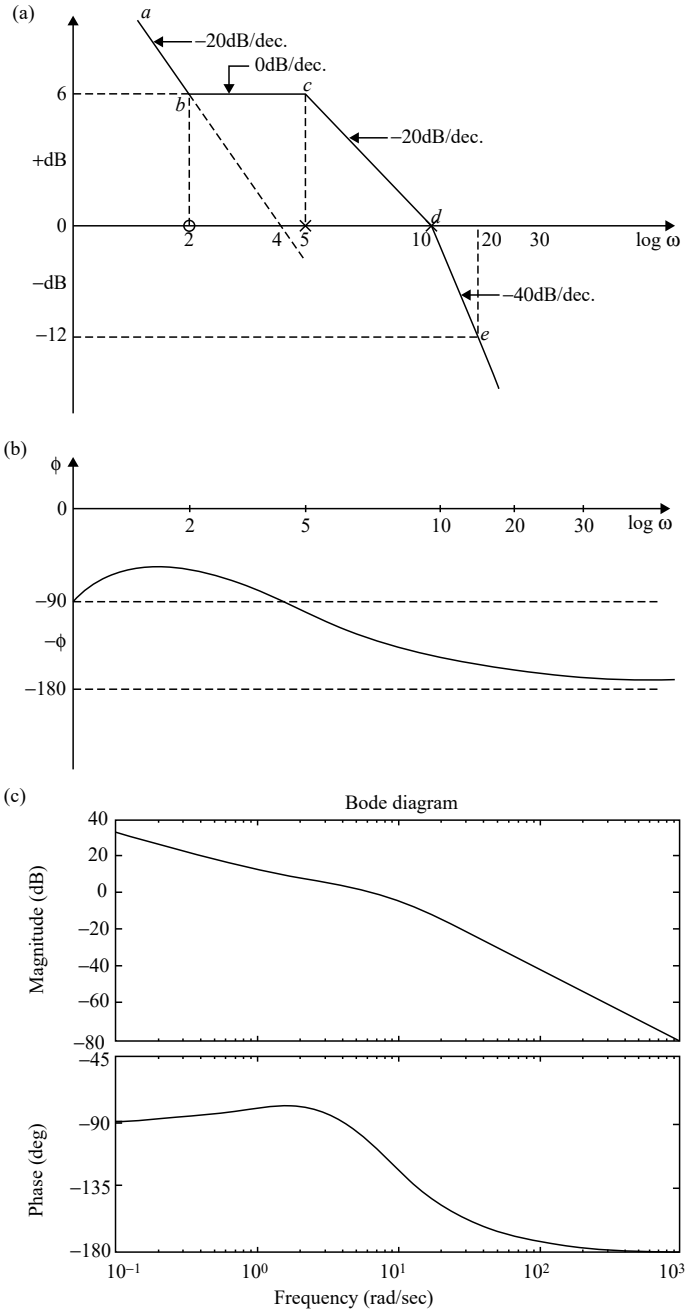


Fig. 3.36 **a** Bode magnitude plot. **b** Bode phase plot. **c** Bode magnitude and phase plots (actual) drawn using MATLAB for the T.F. $G(s) = \dfrac{100(s+2)}{s(s+5)(s+10)}$

in logarithmic scale. The phase angle plot is shown in Fig. 3.36b. Usually both magnitude and phase plots are drawn one below the other to get information easily. The plot of Fig. 3.36 is obtained (actual magnitude and phase angle plots) using the following MATLAB program.

Bode Diagram
Transfer Function

$$\frac{100s + 200}{s^3 + 15s^2 + 50s}$$

Program

```
num=[0 0 100 200];
den=Conv([1 0], Conv([1 5],[1 10]));
sys=tf(num,den)
bode (sys)
grid
```

The actual Bode magnitude and phase plots are shown in Fig. 3.36c. The error between the actual plot and the asymptotic plots is obviously seen. Since the access to MATLAB is easy to get nowadays, control system design problems are invariably carried out with actual Bode plots.

3.8.1 Transfer Function from Bode Plot

In Sect. 3.7, for the given T.F. the Bode magnitude as well as the phase plots were drawn. By the reverse process, that is given the Bode plot, the transfer function can be obtained. If the magnitude plot alone is given, then it is possible to find the T.F. only if the T.F. is of minimum phase. However, if both magnitude and phase plots are given, then it is possible to find the T.F. of non-minimum system also. We discuss below the method to find the minimum phase T.F. when the Bode magnitude plot is given. The step-by-step procedure is given below

1. Identify the low frequency response of the system in the Bode plot. This may correspond to the factors K, Ks, Ks^2, K/s or K/s^2 in the system T.F.
2. Wherever there is a change of slope, identify that with a corner frequency T_1, T_2, T_3, *etc.*.
3. A change of -20 dB/dec slope at $\omega_c = T_1$, there is a pole $1/(1 + \frac{1}{T_1}s)$ and -40 dB/dec slope at $\omega_c = T_2$, there is a pole $1/(1 + \frac{1}{T_2}s)^2$ and so on. Similarly for $+20$ dB/dec change of slope at corner frequency $\omega_c = T_3$, there is a zero $(1 + \frac{1}{T_3}s)$. For $+40$ dB/dec change of slope at corner frequency $\omega_c = T_4$, there are two zeros $(1 + \frac{1}{T_4 s})^2$ and so on.
4. From the Bode plot, information about the decibel values and corner frequencies are obtained and K, T_1, T_2, T_3, T_4 are known and hence the T.F.

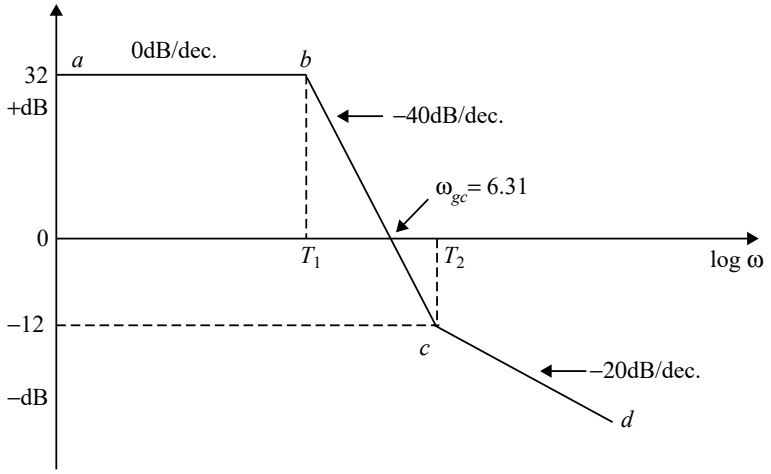


Fig. 3.37 Bode plot, for Example 3.18

Example 3.18

From the Bode magnitude plot shown in Fig. 3.37, determine the T.F.

Solution

1. Straight line ab which has 0 dB/dec. slope represents the low frequency response. The factor for this portion of the T.F. is

$$\begin{aligned} G_1(s) &= K \\ \text{dB} &= 20 \log K = 32 \quad \text{for } \omega \leq T_1 \\ \text{or} \quad K &= 39.8 \\ G_1(s) &= 39.8 \end{aligned}$$

2. There are changes in the slopes at points b and c. At these points, the corner frequencies are identified as T_1 and T_2.
3. At point b, the T.F. $G_2(s) = 1/(1 + \frac{1}{T_1}s)^2$ starts. For this

$$\begin{aligned} 40 \log \frac{\omega_{gc}}{T_1} &= 32 \\ 40 \log \frac{6.31}{T_1} &= 32 \\ T_1 &= 1 \\ G_1(s) &= \frac{1}{(1+s)^2} \end{aligned}$$

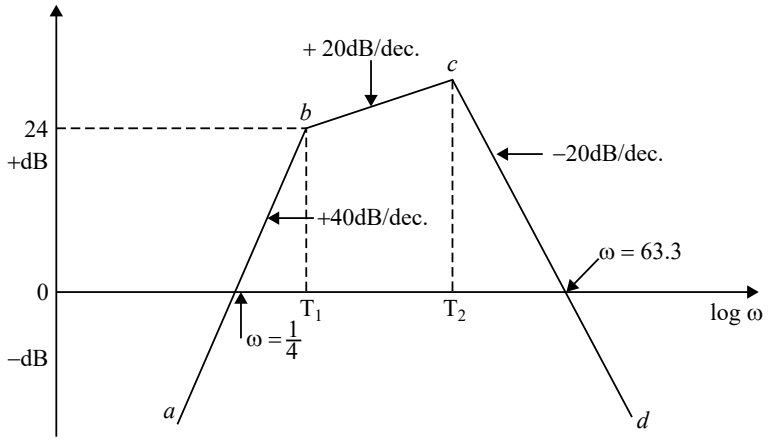


Fig. 3.38 Bode plot, for Example 3.19

4. At point c zero with the T.F. $G_2(s) = (1 + \frac{1}{T_2}s)$ starts. T_2 is determined from

$$-40 \log \frac{T_2}{\omega_{gc}} = -12$$
$$-40 \log \frac{T_2}{6.31} = -12$$
$$T_2 = 12.6$$
$$G_2(s) = (1 + 0.0794s)$$

5.

$$G(s) = G_1(s)G_2(s)G_3(s)$$

$$\boxed{G(s) = \frac{39.8(1 + 0.0794s)}{(1 + s)^2}}$$

Example 3.19

For the Bode plot shown in Fig. 3.38, determine the T.F.

Solution

1. There are two corner frequencies in Fig. 3.38. These frequencies are identified as T_1 and T_2.
2. The low frequency response is contributed by the straight line ab. The T.F. of line ab is

$$G_1(s) = Ks^2$$

3. At the corner frequency T_1 the slope changes from +40 dB/dec to +20 dB/dec. A pole interferes with the following T.F.

$$G_2(s) = \frac{1}{(1 + \frac{1}{T_1}s)}$$

4. At the corner frequency T_2 the slope changes from +20 dB/dec to −20 dB/dec. Hence, two repeated poles start from T_2 as the corner frequency. The T.F. of these poles is

$$G_3(s) = \frac{1}{(1 + \frac{1}{T_2}s)^2}$$

5. Now K, T_1 and T_2 are to be determined. From straight line ab, the decibel contribution is solely due to $G_1(s)$. At $\omega = \frac{1}{4}$

$$\begin{aligned} |G_1(j\omega)|_{\omega=\frac{1}{4}} &= 1 \\ K\omega^2|_{\omega=\frac{1}{4}} &= 1 \\ K &= 16 \\ G_1(s) &= 16s^2 \end{aligned}$$

6. Also for the straight line ab, the following equation is written

$$\begin{aligned} 40 \log 4T_1 &= 24 \\ T_1 &= 1 \\ G_2(s) &= \frac{1}{(1+s)} \end{aligned}$$

7. At point c, the equation for the dB is written as

$$\text{dB}|_{\omega=T_2} = 24 + 20 \log \frac{T_2}{T_1}$$

dB at $\omega = T_2$ can also be written as

$$\text{dB}|_{\omega=T_2} = 20 \log \frac{63.3}{T_2}$$

Equating these equations after substituting $T_1 = 1$, we get

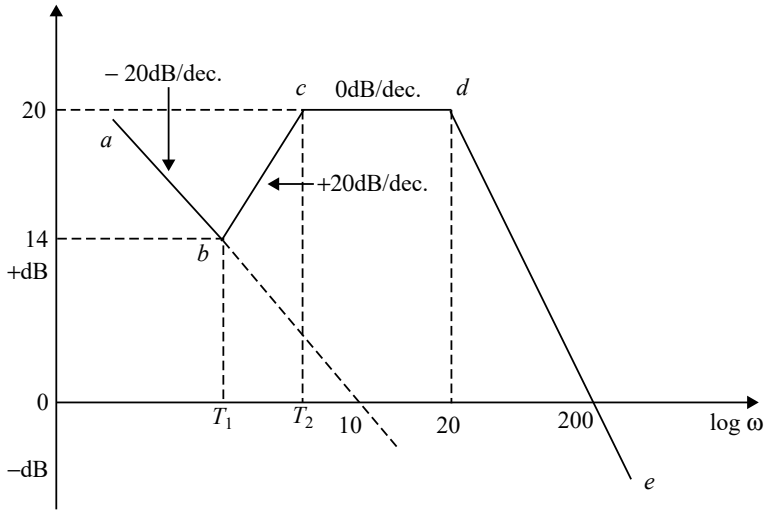


Fig. 3.39 Bode magnitude plot, for Example 3.20

$$24 + 20 \log T_2 = 20 \log \frac{63.3}{T_2}$$

$$20 \log \frac{63.3}{T_2^2} = 24$$

$$T_2 = 2$$

$$G_3(s) = \frac{1}{(1 + 0.5s)^2}$$

8.

$$G(s) = G_1(s)G_2(s)G_3(s)$$

$$\boxed{G(s) = \frac{16s^2}{(1 + s)(1 + 0.5s)^2}}$$

Example 3.20

For the Bode plot shown in Fig. 3.39, determine the T.F.

Solution

1. From Fig. 3.39, the straight line ab forms the low frequency response solely. This transfer function is

$$G_1(s) = \frac{K}{s}$$

2. At points b, c and d there are changes in the slope of the straight lines. At these points, the corner frequencies are identified as T_1, T_2 and 20. The values of K, T_1 and T_2 are to be determined.
3. At corner frequency $\omega = T_1$, the change of slope is from -20 dB/dec to $+20$ dB/dec. There should be two repeated zeros at this corner frequency. Hence, the T.F.s of these zeros are

$$G_2(s) = \left(1 + \frac{1}{T_1}s\right)^2$$

4. At corner frequency $\omega = T_2$, there is a change of slope from $+20$ dB/dec to zero dB/dec. There should be a pole starting from this corner frequency. The T.F. of this pole is

$$G_3(s) = \frac{1}{\left(1 + \frac{1}{T_2}s\right)}$$

5. At corner frequency $\omega = 20$, the 0 dB/dec slope is changed to a negative slope. The frequency range between $\omega = 20$ and $\omega = 200$ is one decade. In this frequency range, the fall in dB is 20 dB. Hence, the slope of the straight line de is -20 dB/dec. Therefore, at corner frequency $\omega = 20$, a pole starts. The T.F. of this straight line is

$$\begin{aligned} G_4(s) &= \frac{1}{(1 + \frac{1}{20}s)} \\ &= \frac{1}{(1 + 0.05s)} \end{aligned}$$

6. The straight line ab has the magnitude

$$\begin{aligned} |G_1(j\omega)| &= \frac{K}{\omega} \\ \frac{K}{\omega}\bigg|_{\omega=10} &= 1 \\ K &= 10 \\ G_1(s) &= \frac{10}{s} \end{aligned}$$

7. At corner frequency $\omega = T_1$, the dB due to $|G_1(j\omega)|$ is 14. Hence, we write the following equation

$$
\begin{aligned}
20\log\frac{K}{\omega}\bigg|_{\omega=T_1} &= 14\\
20\log\frac{10}{T_1} &= 14\\
T_1 &= 2\\
G_2(s) &= \left(1+\frac{1}{T_1}s\right)^2\\
&= (1+0.5s)^2
\end{aligned}
$$

8. For the frequency range $T_1 \le \omega \le T_2$, the following equation is written

$$
\begin{aligned}
20\log\frac{T_2}{T_1} &= 6\\
20\log\frac{T_2}{2} &= 6\\
T_2 &= 4\\
G_3(s) &= \frac{1}{\left(1+\frac{1}{4}s\right)} = \frac{1}{(1+0.25s)}\\
G(s) &= G_1(s)G_2(s)G_3(s)G_4(s)
\end{aligned}
$$

$$
\boxed{G(s) = \frac{10(1+0.5s)^2}{s(1+0.25s)(1+0.05s)}}
$$

Example 3.21

The approximate Bode magnitude plot of a minimum phase system is shown in Fig. 3.40. Determine the T.F. of the system.

Solution

1. The slopes of the low frequency and high frequency responses are given as 0 dB/dec. This means the poles and zeros are equal in number. Further, the corner frequencies where the change of slope occurs are given as 0.1, 10, 100 with a frequency range of 2 decades and one decade. The constant term and the slope of the straight lines are to be determined to get to T.F.
2. For the frequency range $\omega \le 0.1$, the dB is constant with 0 dB/dec. Hence

$$G_1(s) = K$$

But $20\log K = 20$

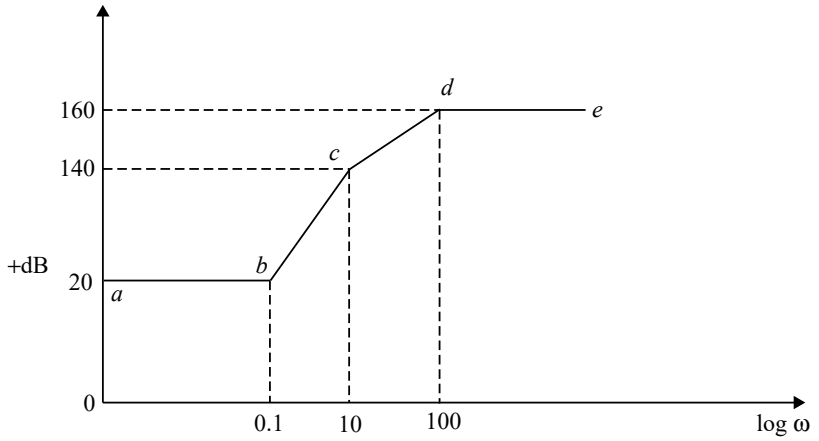


Fig. 3.40 Bode magnitude plot, for Example 3.21

$$K = 10$$

3. For the frequency range $0.1 \leq \omega \leq 10$, there are two decades. The rise in dB is $(140 - 20) = 120$ dB. Hence, the slope of the straight line is +60 dB/dec. Hence, three repeated zeros with a corner frequency of $\omega_c = 0.1$ start. The T.F. of these zeros is

$$G_2(s) = (1 + 10s)^3$$

4. For the frequency range $10 \leq \omega \leq 100$, the change of frequency is one decade and rise in dB is $(160 - 140) = 20$ dB. The slope of the straight line is therefore 20 dB/dec. At $\omega_c = 10$, the slope changes from +60 dB/dec to +20 dB/dec. This is possible if two repeated poles start from $\omega_c = 10$. The T.F. of these poles is

$$G_3(s) = \frac{1}{(1 + 0.1s)^2}$$

5. For the frequency $\omega \geq 100$, the slope of the straight line is 0. This is possible if a pole starts from the corner frequency $\omega_c = 100$. The T.F. of the corresponding pole is

$$G_4(s) = \frac{1}{(1 + 0.01s)}$$

6.

$$G(s) = G_1(s)G_2(s)G_3(s)G_4(s)$$

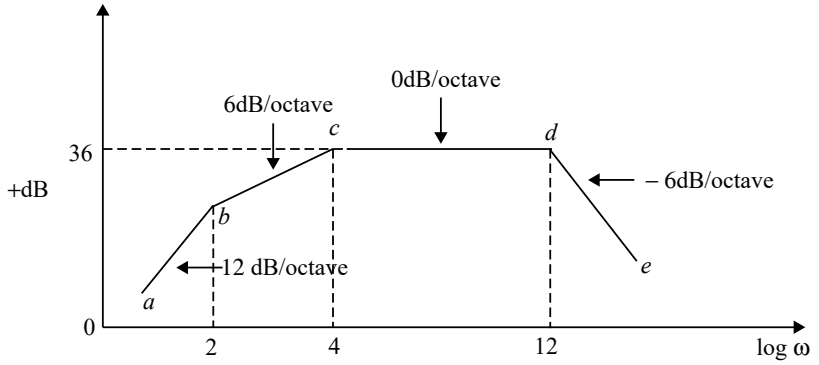


Fig. 3.41 Bode asymptotic magnitude plot, for Example 3.22

$$G(s) = \frac{10(1+10s)^3}{(1+0.1s)^2(1+0.01s)}$$

Example 3.22

Consider the Bode magnitude asymptotic plot of a certain minimum phase T.F. shown in Fig. 3.41. Determine the T.F.

Solution

1. While drawing the Bode plot or getting the T.F. from Bode plot the following points are to be noted:
   (a) 20 dB/dec = 6 dB/octave. It is easier to handle 20 dB/dec. So 6 dB/octave slopes are converted into 20 dB/dec slopes when calculations are made.
   (b) Each pole will contribute −20 dB/dec slope starting at its corner frequency. Similarly, each zero will contribute +20 dB/dec slope starting at its corner frequency. The constant term will have 0 dB/dec slope.
2. For the frequency range $\omega \leq 2$, the low frequency response starts and it is the straight line ab which has a slope of 12 dB/oct or 40 dB/dec. Corresponding to this, the T.F. is

$$G_1(s) = Ks^2$$

3. For the frequency range $2 \leq \omega \leq 4$, which is one octave, the dB is 6 dB/oct or 20 dB/dec. The slope has changed from 40 dB/dec to 20 dB/dec. A pole with a corner frequency $\omega_c = 2$ starts. The transfer function of this pole is

$$G_2(s) = \frac{1}{\left(1 + \frac{1}{2}s\right)} = \frac{1}{(1 + 0.5s)}$$

4. For the frequency range $4 \leq \omega \leq 12$, the slope remains constant. This is possible if a pole with a corner frequency $\omega_c = 4$ starts. The T.F. of this pole is

$$G_3(s) = \frac{1}{\left(1 + \frac{1}{4}s\right)} = \frac{1}{(1 + 0.25s)}$$

5. For the frequency range $\omega \geq 12$, the slope of the straight line is -6 dB/oct or -20 dB/dec. The change from 0dB/dec to -20 dB/dec is by a pole with $\omega_c = 12$. The T.F. of this pole is

$$G_4(s) = \frac{1}{(1 + \frac{1}{12}s)} = \frac{1}{(1 + 0.0833s)}$$

6. For the frequency range $\omega_c = 2$ and $\omega_c = 4$, it is an octave. The slop is 6 dB/oct. The point c is at 36 dB. Hence, point b is at dB $= 36 - 6 = 30$ dB.
7. At $\omega = 2$, the dB contribution is solely by $K\omega^2$. Hence

$$\begin{aligned} \text{dB}|_{\omega=2} &= 20 \log K\omega^2|_{\omega=2} = 30 \\ \log 4K &= 1.5 \\ K &= 7.9 \end{aligned}$$

8.

$$G(s) = G_1(s)G_2(s)G_3(s)G_4(s)$$

$$\boxed{G(s) = \frac{7.9\, s^2}{(1 + 0.5s)(1 + 0.25s)(1 + 0.0833s)}}$$



Note: The order of the numerator polynomial is 2 which will contribute a total slope of 40 dB/dec. The order of the denominator polynomial is 3 which will contribute a total slope of -60 dB/dec. The net slope of these two polynomials is (40 dB/dec$-$60 dB/dec) $= -20$ dB/dec $= -6$ dB/oct as ω tends to ∞ as seen from Fig. 3.41.

3.9 Log Magnitude Versus Phase Plot (Nichols Plot)

The frequency response characteristics of $G(j\omega)$ were displayed in Bode diagrams in two separate parts as magnitude plot and phase plot. In the log magnitude *versus* phase plot which is also known as **Nichols plot** the information about the magnitude and phase of $G(j\omega)$ are combined together and displayed in one graph. The horizontal axis represents the phase of $G(j\omega)$ while the vertical axis represents the magnitude of $G(j\omega)$ expressed in dB. The function $G(j\omega)$ is expressed in its magnitude and phase angle as a function of frequency ω. For various frequencies $20\log|G(j\omega)|$ and $\angle G(j\omega)$ are calculated and plotted. Thus, the frequency response of $G(j\omega)$ is plotted by preparing $20\log|G(j\omega)|$ and $\phi = \angle G(j\omega)$ for different values of ω. Alternatively, if Bode plot is available, it can be easily transferred to a log magnitude-phase plot with ω as the variable parameter.

Advantages of Log Magnitude-Phase Plot

1. Information about the magnitude of $G(j\omega)$ and phase of $G(j\omega)$ along with ω as variable parameter is available in a single graph. The plot is easily drawn with few easy computing which is not so in Bode plot.
2. Relative stability such as phase margin and gain margin can be quickly determined and compensation can be worked out easily.
3. Frequency domain specifications such as resonant peak, resonant frequency, gain crossover and phase crossover frequencies and the bandwidth of the closed loop system can be easily determined. Some of these things are not represented in Bode plot.
4. Closed loop frequency response can be determined from open loop frequency response by superimposing on the Nichols chart.
5. When the loop gain of $G(j\omega)$ is changed, the frequency response plot is simply shifted up and down vertically. Similarly when a constant phase is added, the frequency response plot is simply shifted horizontally without any distortion. By these two important properties, the amount of gain to be adjusted or the amount of phase to be added is determined to achieve the required frequency domain specification.
6. The effect of time delay is to simply add $-57.3T_d^\circ$ and shift the $G(j\omega)$ plot horizontally. Thus, the effect of time delay is easily studied.

Disadvantage of Log-Magnitude-Phase Plot

1. In the Bode plot, the effect of individual poles and zeros are clearly known which will help modify the plot according to the desired response. However, when Bode plot is transferred to log magnitude-phase plot this important information is not carried over.

Example 3.23

Consider the following T.F.

$$G(s) = \frac{20}{s(s+1)(s+10)}$$

Draw (a) Bode asymptotic magnitude and actual phase plots. (b) Log magnitude phase plot.

Solution

(a) **Bode Plot**

$$G(s) = \frac{20}{s(s+1)(s+10)} = \frac{2}{s(1+s)\left(1+\frac{1}{10}s\right)}$$

The corner frequencies are 1 and 10. $\omega_l = 2$. The factors of $G(s)$ are written in the ascending order of the corner frequencies as follows:

$$G_1(s) = \frac{2}{s}$$
$$G_2(s) = \frac{1}{(1+s)}$$
$$G_3(s) = \frac{1}{\left(1+\frac{1}{10}s\right)}$$

The range of corner frequencies and the slope of the corresponding straight lines are given below

Range of corner frequency	$\omega \leq 1$	$1 \leq \omega \leq 10$	$\omega \geq 10$
Slope in dB/dec	-20	-40	-60

$$\phi = \angle G(j\omega) = -90^\circ - \tan^{-1}\omega - \tan^{-1}\frac{\omega}{10}$$

For $0 \leq \omega \leq \infty$, the phase angle ϕ is calculated and shown in the table below

ω	0	0.5	1	1.5	2	3	5	10	∞
ϕ	-90°	-119.5°	-140.7°	-154.8°	-164.7°	-178°	-195°	-220°	-270°

At $\omega = 1$

$$\text{dB}|_{\omega=1} = 20\log\frac{2}{\omega} = 6$$

At $\omega = 10$

$$\text{dB}|_{\omega=10} = 6 - 40 = -34$$

At $\omega = 15$ (say)

$$\text{dB}|_{\omega=15} = -34 - 60\log\frac{15}{10} = -44.6$$

The Bode magnitude and phase angle plots are shown in Fig. 3.42.

(b) **Log Magnitude-Phase Plot**

$$G(s) = \frac{20}{s(s+1)(s+10)}$$

$$|G(j\omega)| = \frac{20}{\omega\sqrt{(\omega^2+1)(\omega^2+100)}}$$

At any ω, the decibel is

$$20\log|G(j\omega)| = 20\log\frac{20}{\omega\sqrt{(1+\omega^2)(100+\omega^2)}}$$

$$\phi = -90^\circ - \tan^{-1}\omega - \tan^{-1}\frac{\omega}{10}$$

dB and phase for different values of ω are calculated and tabulated below

ω	0.5	1	1.5	2	3	5	10
$\text{dB} = 20\log\|G(j\omega)\|$	11.1	3	−2.7	−7.18	−13.9	−23.2	−37.1
$\phi = \angle G(j\omega)$	−119°	−140.7°	−154.8°	−165°	−178°	−195°	−220°

The values of dB and ϕ for different values of ω are transferred to the log magnitude-phase graph as shown in Fig. 3.43 with ω as the varying parameter.

3.10 Frequency Domain Specifications

In Chap. 2, expressions for time domain specifications such as peak overshoot M_p, time at which peak overshoot occurs t_p, rise time t_r, time delay t_d, settling time t_s, *etc.* were derived when the system was subjected to step input for a prototype second order system. For the same second order system, expressions for the resonant peak, resonant frequency and bandwidth were derived in the frequency domain when the

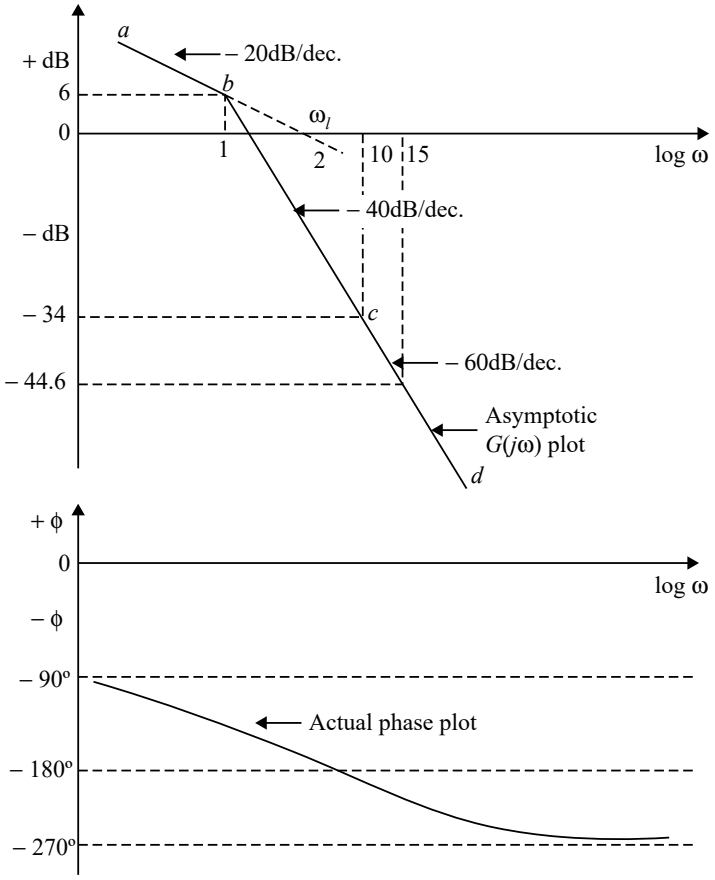


Fig. 3.42 Bode magnitude and phase plots for the T.F. $G(s) = \dfrac{20}{s(s+1)(s+10)}$

input applied is a sinusoidal. These expressions were derived in terms of the second order system parameters damping ratio ζ and undamped natural frequency of oscillation ω_n. Thus, a correlation between time domain specifications and some frequency domain specifications was established. By knowing the specifications in one domain, the specifications in the other domain are obtained. However, for higher order systems such correlation does not exist. Further, the three frequency domain specifications mentioned above are not adequate enough to study the performance of the linear system which need not be a second order system. These specifications are useful to study the system relative stability, fastness of time response, *etc.* They are useful to design the right type of compensators which ensure the desired performance. The following frequency domain specifications are defined and the method to evaluate them for the linear system of any order is discussed. These frequency domain specifications are

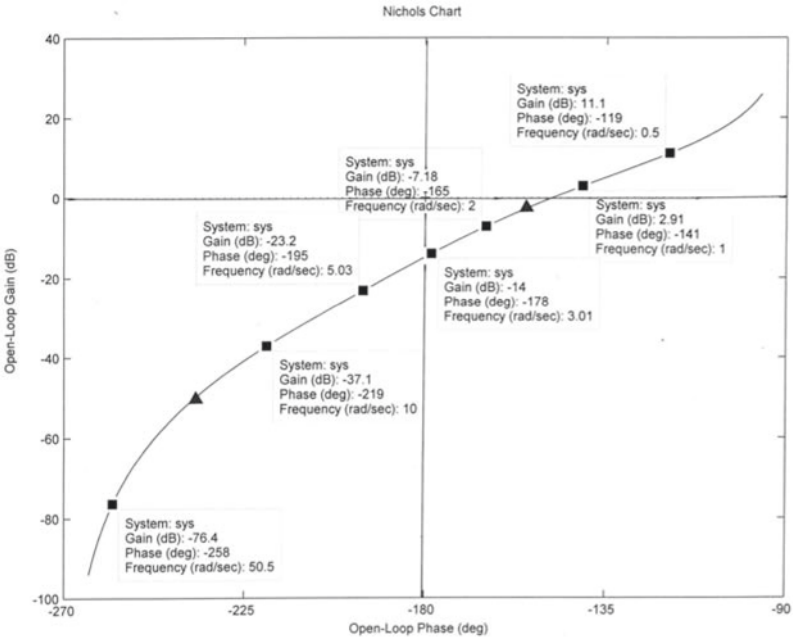


Fig. 3.43 Log magnitude-phase plot for $G(s) = \dfrac{20}{s(s+1)(s+10)}$

1. Resonant peak, M_r
2. Resonant frequency, ω_r
3. Bandwidth (BW), ω_b
4. Phase margin, ϕ_m
5. Gain margin, G_m
6. Phase crossover frequency, ω_{pc}
7. Gain crossover frequency, ω_{gc}

The above specifications are determined using the following frequency response plots:

1. Polar plot (Nyquist plot)
2. Bode plot
3. Log magnitude-phase plot (Nichols plot).

3.10.1 Phase Margin and Gain Margin via Nyquist (Polar) Diagram

For the analysis and design of linear system, system stability is classified as **absolute stability** and **relative stability**. Even though we give exclusive study of the topic on stability in Chap. 4, we take relative stability as performance specification in frequency domain and discuss it below.

The term absolute stability refers to whether the given system is stable or unstable with the answer yes or no. If the answer is yes, how far it is closer to the stability limit. This is called relative stability. Quantitatively the relative stability is measured, in the frequency domain by means of **gain margin (G.M.)** and **phase margin (P.M.)** and these quantities are symbolically represented by G_m and ϕ_m, respectively.

Consider the Nyquist plot of $G(j\omega)$ shown in Fig. 3.44a in polar coordinates for a minimum phase transfer function. From open loop frequency response plot shown in Fig. 3.44a the closed loop system is stable as it does not enclose $(-1, 0)$ point. (Detailed stability study is given in Chap. 4). The frequency response plot where it cuts the negative real axis is called **phase crossover** and the corresponding frequency ω_{pc} is called the **phase crossover frequency**. The phase crossover is denoted by d where d is calculated from

$$d = |G(j\omega)||_{\omega=\omega_{pc}}$$

The gain margin which is the measure of relative stability is defined as

$$\boxed{G_m = \frac{1}{d}} \tag{3.27}$$

The gain margin is expressed in decibel dB as

$$\boxed{G_m = -20 \log d} \tag{3.28}$$

From Fig. 3.44, we give the definition for the following:

1. Phase crossover is defined as the intersection of $G(j\omega)$ plot at the negative real axis.
2. Phase crossover frequency ω_{pc} is defined as the frequency at the phase crossover where the phase angle of $G(j\omega)$ is 180°.
3. (a) Gain margin is defined as the reciprocal of $|G(j\omega)|$ at the phase crossover frequency ω_{pc}.
   (b) Gain margin is defined as the factor (gain) by which the open loop system gain can be increased before the closed loop system becomes unstable.
   (c) Gain margin is defined as the amount of gain in decibel (dB) that can be added to the open loop system before the closed loop system becomes unstable.

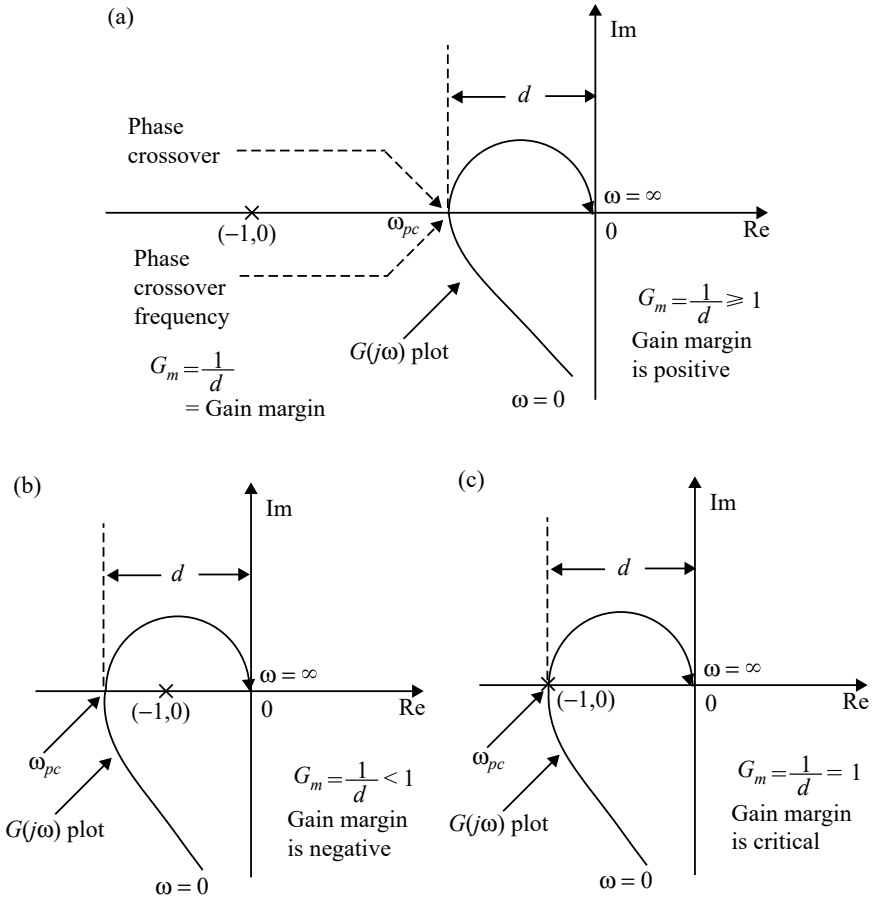


Fig. 3.44 **a** Definition of gain margin in polar coordinates. **b** Negative gain margin and **c** Critical gain margin

From Fig. 3.44, the following observations are made with respect to gain margin:

1. If the $G(j\omega)$ plot does not intersect the negative real axis (system of order two and less)

$$G_m = \infty$$

2. If the $G(j\omega)$ plot passes through $(-1, 0)$ point

$$G_m = 1 \quad \text{or} \quad G_m = 0 \text{ dB}$$

The system is oscillatory now. (Fig. 3.44c).

3. If the $G(j\omega)$ plot intersects the negative real axis between 0 and $(-1, 0)$ points

$$G_m > 1 \quad \text{or} \quad G_m > 0 \text{ dB (Fig. 3.44a)}$$

Here, the G_m in decibel is +ve and the $G_m > 1$. The gain margin is said to be positive. For the closed loop system to be stable the gain margin should be positive.

4. If the $G(j\omega)$ plot has the crossover to the left of $(-1, 0)$ point

$$G_m < 1 \quad \text{or} \quad G_m < 0 \text{ dB (Fig. 3.44b)}$$

Here, the G_m in decibel is negative and $G_m < 1$ and the gain margin is said to be negative. If the gain margin is negative, the system is unstable.
Higher the gain margin, higher is the relative stability. If the gain margin is negative, the closed loop system is unstable. If the gain margin is positive, the closed loop system is stable.

Phase Margin: in Polar (Nyquist) Plot

If the gain margin alone is positive, that does not guarantee the stability of the closed loop system. As stated earlier, another quantitative measurement of relative stability is the phase margin (P.M) denoted by ϕ_m.

Consider Fig. 3.45a where $G(j\omega)$ is plotted in the $G(j\omega)$ plane. The unit circle with the origin of the $G(j\omega)$ plane is drawn. The $(-1, 0)$ is also marked. For a minimum phase T.F. the $G(j\omega)$ plot may have positive gain margin. Due to phase shift near $(-1, 0)$ point the $G(j\omega)$ plot may be very closer but due to low gain at phase crossover, the gain margin may be high. High gain margin implies high relative stability. But low phase margin implies low relative stability. The effect of phase shift is studied below *via* phase margin. We give below a few definitions.

1. **Gain crossover:** The intersection point of $G(j\omega)$ plot on the unit circle ($|G(j\omega)| = 1$) is called gain crossover.
2. **Gain crossover frequency ω_{gc}:** The frequency at the gain crossover is called the gain crossover frequency denoted by ω_{gc}. It is the frequency at which the gain of

$$G(j\omega)|_{\omega=\omega_{gc}} = 1$$

3. **Phase Margin:** Phase margin is defined as the amount of phase angle through which the $G(j\omega)$ plot should be rotated about the origin so that the gain crossover passes through $(-1, 0)$ point. Phase margin is also defined as 180°+ the phase angle of $G(j\omega)$ at unity gain

$$\phi_m = -180^\circ + \angle G(j\omega)|_{\omega=\omega_{gc}} \tag{3.29}$$

It is to be noted that while phase margin is measured at the gain crossover, the gain margin is measured at the phase crossover. In other words to calculate phase margin, gain of $G(j\omega)$ is known which is unity. To calculate gain margin, phase angle of $G(j\omega)$ is known which is 180°.

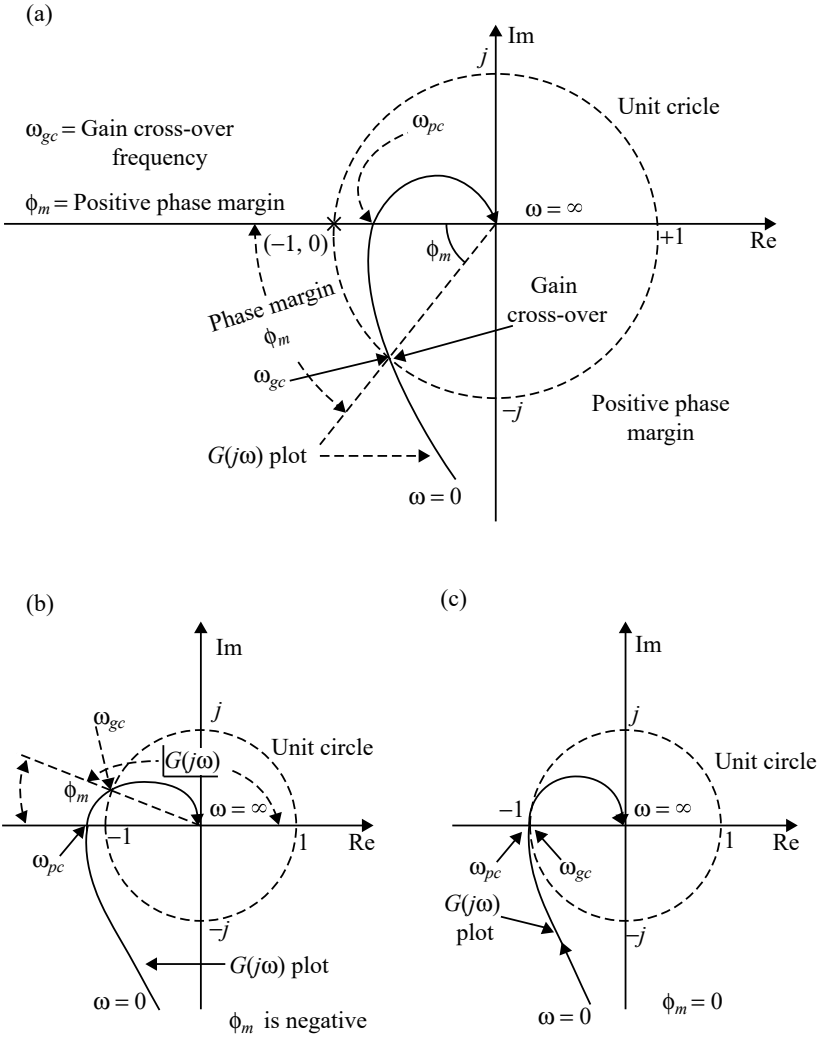


Fig. 3.45 **a** Definition of phase margin in polar coordinates. **b** Negative phase margin in polar coordinate. **c** Zero phase margin in polar coordinate

In Fig. 3.45a the $G(j\omega)$ plot of a third order type 1 system is shown. The $G(j\omega)$ plot cuts the unit circle in the third quadrant. The intersection point is called gain crossover. This point is connected by a radial line. The angle between this radial line and the negative real axis gives the phase margin. If the gain crossover lies in the third quadrant of the complex $G(j\omega)$ plane the **phase margin is positive**. For the system with positive phase margin, an additional phase can be included in the system before the system becomes unstable.

Now consider Fig. 3.45b where the gain crossover is identified in the second quadrant. The radial line from the origin to the gain crossover is drawn. The phase margin is measured between this straight line and the negative real axis. Now the system is said to have negative phase margin. With negative phase margin, the system is unstable.

Consider Fig. 3.45c where the gain crossover is identified with $(-1, 0)$ point. At this point, the phase margin is zero. Further, it has zero dB gain margin. Systems which have zero phase margin and zero dB gain margin are said to be oscillatory. Here $\omega_{gc} = \omega_{pc}$.

Important points to remember with regard to phase margin and gain margin.

1. Gain margin is calculated at phase crossover. To calculate this the phase angle of $G(j\omega)$ is known which is 180°. If $G_m > 1$, the gain margin is positive. If $G_m < 1$, the gain margin is negative. For the system to be stable it is necessary that the gain margin is positive. But this is not sufficient. Further, higher the gain margin, higher is the relative stability.
2. The phase margin is calculated at gain crossover. To calculate this the magnitude of $G(j\omega)$ is known which is one. If $\angle G(j\omega) > 180°$ the phase margin is positive. If $\angle G(j\omega) < 180°$ the phase margin is negative. For the system to be stable it is necessary (but not sufficient) that the phase margin is positive. The higher the phase margin, the higher is the relative stability.
3. For the linear system with minimum phase transfer function, it is necessary and sufficient that both phase margin and gain margin are positive. However, if any one of them is negative, the system is unstable.
4. If the $G(j\omega)$ passes through $(-1, 0)$ point, the gain crossover and phase crossover lie at the same point at $(-1, 0)$. Now the phase crossover and gain crossover frequencies are the same $(\omega_{pc} = \omega_{gc})$. The system is neither stable nor unstable. It is said to be oscillatory.
5. For a stable system $\omega_{pc} > \omega_{gc}$. For an unstable system $\omega_{pc} < \omega_{gc}$.
6. The phase margin which is between 30° and 60° and gain margin which is greater than +6 dB give satisfactory performance of the closed loop system.
7. In the Bode diagram if the slope of log magnitude curve is −40 dB/dec at the gain crossover frequency a phase margin between 30° and 60° is obtained. The slope of −20 dB/dec at the gain crossover frequency is more desirable. However, −60 dB/dec slope magnitude curve at the gain crossover frequency the system is likely to be unstable.
8. The damping ratio ζ of the system is roughly given by the following relation with the phase margin

$$\boxed{\zeta = \frac{\pi}{360}\phi_m} \tag{3.30}$$

The following examples illustrate the method of calculating the phase margin and gain margin using polar plot. Both analytical and graphical methods are described.

Example 3.24

A certain unity feedback system has the following forward path T.F.

$$G(s) = \frac{1}{s(s+1)^2}$$

Find the gain margin and phase margin using Nyquist plot. What are phase and gain crossover frequencies?

(*Anna University, 2007*)

Solution

1.

$$G(s) = \frac{1}{s(s+1)^2}$$

The sinusoidal T.F. is obtained by replacing s by $j\omega$

$$\begin{aligned} G(j\omega) &= \frac{1}{j\omega(j\omega+1)^2} \\ &= \frac{1}{\omega(1+\omega^2)} \angle -90^\circ - 2\tan^{-1}\omega \\ |G(j\omega)| &= \frac{1}{\omega(1+\omega^2)} \\ \angle G(j\omega) &= -90 - 2\tan^{-1}\omega \end{aligned}$$

Here, it is not necessary to compute $|G(j\omega)|$ and $\angle G(j\omega)$ over a range of values of ω. The given T.F. is Type 0 order 3 system. It starts from the 3rd quadrant when $\omega = 0$, crosses over the negative real axis at $\omega = \omega_{pc}$ and terminates at the origin when $\omega = \infty$. Only 2 selective points will be enough to find G_m, ϕ_m, ω_{pc} and ω_{gc}. This is explained in step 2, Fig. 3.46 is referred to.

2. **To find the gain margin:** To find the gain margin, phase angle $\angle G(j\omega) = 180^\circ$. The corresponding frequency is the phase crossover frequency ω_{pc}

$$\begin{aligned} -90^\circ - 2\tan^{-1}\omega_{pc} &= -180^\circ \\ \tan^{-1}\omega_{pc} &= 45^\circ \\ \omega_{pc} &= 1 \text{ rad/sec} \end{aligned}$$

The phase crossover or d is obtained by substituting ω_{pc} in $|G(j\omega)|$

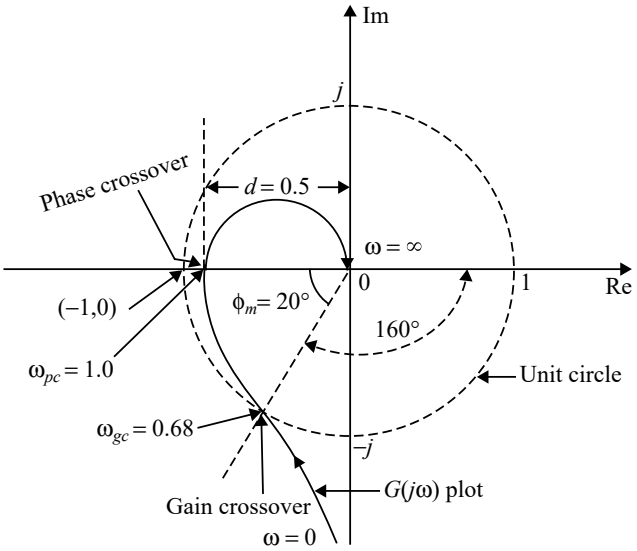


Fig. 3.46 Gain margin and phase margin, for Example 3.24

$$
\begin{aligned}
d &= |G(j\omega)|_{\omega=\omega_{pc}} \\
&= \frac{1}{\omega_{pc}(1+\omega_{pc^2})} \\
&= \frac{1}{1(1+1)} = 0.5 \\
\text{Gain margin, } G_m &= \frac{1}{d} = \frac{1}{0.5} \\
&= 2 \text{ (positive)}
\end{aligned}
$$

3. **To find the phase margin:** To find the phase margin $|G(j\omega| = 1$. The corresponding frequency is the gain crossover frequency. A unit circle with origin as the centre is drawn and the gain crossover is identified. ω_{gc} is calculated from $|G(j\omega)| = 1$

$$\frac{1}{\omega_{gc}(1+\omega_{gc}^2)} = 1$$

Solving the above equation by trial and error, we get $\omega_{gc} = 0.68$. At the gain crossover, the phase angle is

$$
\begin{aligned}
\angle G(j\omega)|_{\omega=\omega_{gc}} &= -90^\circ - 2\tan^{-1}\omega_{gc} \\
&= -90^\circ - 2\tan^{-1} 0.68 = -160^\circ
\end{aligned}
$$

Phase margin is

$$\begin{aligned}\phi_m &= 180^\circ - 160^\circ \\ &= 20^\circ \text{ (positive)}\end{aligned}$$

$$\boxed{\begin{aligned} G_m &= 2 \text{ (positive)} \\ \phi_m &= 20^\circ \text{ (positive)} \\ \omega_{pc} &= 1 \text{ rad/sec} \\ \omega_{gc} &= 0.68 \text{ rad/sec}\end{aligned}}$$

Both phase margin and gain margin are positive and $\omega_{pc} > \omega_{gc}$. **The system is stable**.

Example 3.25

A certain unity feedback control system has the following forward path T.F.

$$G(s) = \frac{Ke^{-0.2s}}{s(s+2)(s+8)}$$

Determine K so that (1) $G_m = 6\,\text{dB}$; (2) $\phi_m = 45^\circ$. Use Nyquist plot in polar coordinate.

(*Anna University, 2009*)

Solution

$$G(s) = \frac{Ke^{-0.2s}}{s(s+2)(s+8)}$$

$$G(j\omega) = \frac{Ke^{-j0.2\omega}}{j\omega(j\omega+2)(j\omega+8)}$$

$$|G(j\omega)| = \frac{K}{\omega\sqrt{(\omega^2+4)(\omega^2+64)}} \tag{a}$$

$$\angle G(j\omega) = -90^\circ - \tan^{-1}\frac{\omega}{2} - \tan^{-1}\frac{\omega}{8} - 0.2 \times 57.3^\circ\omega \tag{b}$$

1. To Find K for the given Gain Margin (Analytical Method)
Method 1

The T.F. of the given system is Type 1 order 3 with a transportation lag. The $G(j\omega)$ is sketched as shown in Fig. 3.47a. The gain margin is given as 6 dB. From this d is calculated as

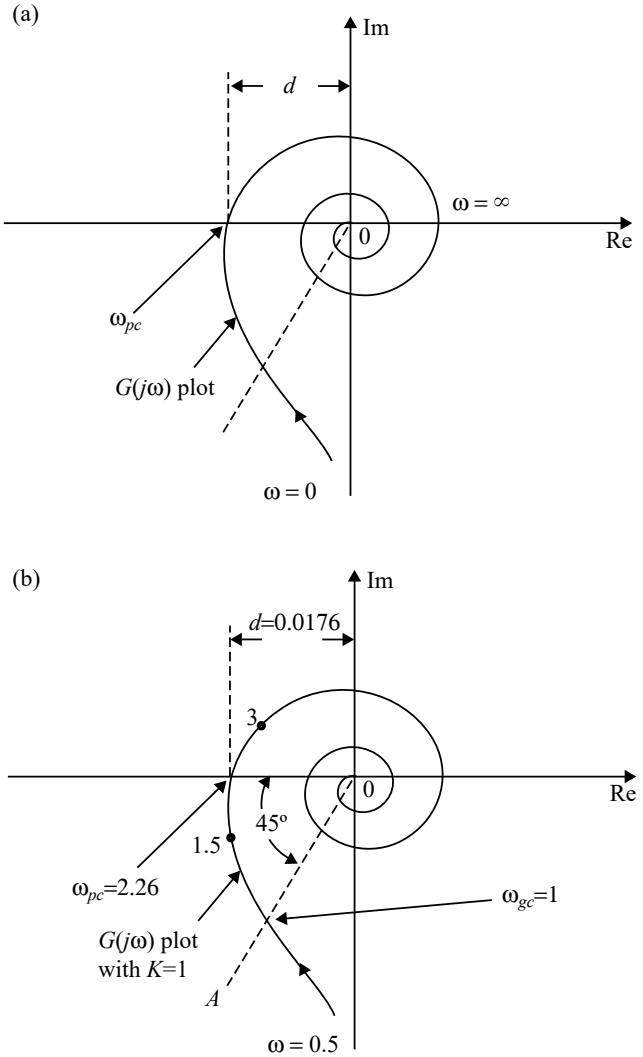


Fig. 3.47 **a** $G(j\omega)$ plot, for Example 3.25. **b** $G(j\omega)$ plot, for Example 3.25 with $K = 1$

$$20 \log \frac{1}{d} = 6$$
$$d = 0.5$$

Thus, the phase crossover is identified as $(-0.5, 0)$. The phase crossover frequency is calculated from the following equation

$$-90^\circ - \tan^{-1}\frac{\omega_{pc}}{2} - \tan^{-1}\frac{\omega_{pc}}{8} - 0.2 \times 57.3\omega_{pc} = 180^\circ$$

$$\tan^{-1}\frac{\omega_{pc}}{2} + \tan^{-1}\frac{\omega_{pc}}{8} = 90^\circ - 0.2 \times 57.3\omega_{pc}$$

$$\tan^{-1}\frac{\frac{\omega_{pc}}{2} + \frac{\omega_{pc}}{8}}{1 - \frac{\omega_{pc}^2}{16}} = 90^\circ - 0.2 \times 57.3\omega_{pc}$$

$$\tan^{-1}\frac{10\omega_{pc}}{16 - \omega_{pc}^2} = 90^\circ - 0.2 \times 57.3\omega_{pc}$$

Taking tan on both sides, we get

$$\frac{10\omega_{pc}}{16 - \omega_{pc}^2} = \tan(90 - 0.2 \times 57.3\omega_{pc})$$

$$= \cot(0.2 \times 57.3\omega_{pc})$$

$$= \frac{1}{\tan(0.2 \times 57.3\omega_{pc})}$$

$$10\omega_{pc}\tan(0.2 \times 57.3\omega_{pc}) = 16 - \omega_{pc}^2$$

$$\omega_{pc}^2 + 10\omega_{pc}\tan(0.2 \times 57.3\omega_{pc}) - 16 = 0$$

Solving for ω_{pc} by trial and error, we get

$$\omega_{pc} = 2.26\,\text{rad/sec}$$

At the phase crossover, $|G(j\omega_{pc})| = 0.5$

$$\frac{K}{\omega_{pc}\sqrt{(\omega_{pc+4}^2)(\omega_{pc}^2 + 64)}} = 0.5$$

$$\frac{K}{2.26\sqrt{9.1 \times 69.1}} = 0.5$$

$$K = 28.35$$

Method 2: To Find K for the given G_m by Graphical Method
Let us assume $K = 1$. For different values of ω, $|G(j\omega)|$ and $\angle G(j\omega)$ are calculated and tabulated as given below

ω	0	0.5	1.0	1.5	2.0	2.26	3	∞
$\lvert G(j\omega)\rvert$	$-\infty$	0.121	0.055	0.033	0.022	0.0176	0.0108	0
$\angle G(j\omega)$	-90°	-113.3°	-135°	-154.7°	-172°	-180°	-201°	0°

The above data in the table is transferred to draw Nyquist plot in the polar coordinates. This is shown in Fig. 3.47b.

From Fig. 3.47b, the values of $d = 0.0176$ is obtained. For the required gain margin of 6 dB, the value of $d = 0.5$. Hence, the value of necessary to achieve this

is

$$K(0.0176) = 0.5$$

$$\boxed{K = 28.4}$$

The above value which is obtained by graphical method is closer to the values as obtained by analytical method.

2. To Find K for the Required Phase Margin

Method 1: Graphical Method
In Fig. 3.47b, with $\phi_m = 45°$ radial line OA is drawn. It cuts the $G(j\omega)$ plot and the corresponding frequency is obtained as 1 rad/sec which is the gain crossover frequency. At gain crossover $|G(j\omega_{gc})| = 1$

$$\frac{K}{\omega_{gc}\sqrt{(\omega_{gc}^2+4)(\omega_{gc}^2+64)}} = 1$$

$$\frac{K}{1\sqrt{5\times 65}} = 1$$

$$\boxed{K = 18}$$

Method 2: Analytical Method
The required phase margin is +45°. Hence

$$\angle G(j\omega)|_{\omega=\omega_{gc}} = -180° + 45° = -135°$$

$$-90° - \tan^{-1}\frac{\omega_{gc}}{2} - \tan^{-1}\frac{\omega_{gc}}{8} - 0.2\times 57.3\omega_{gc} = -135°$$

$$\tan^{-1}\frac{\omega_{gc}}{2} + \tan^{-1}\frac{\omega_{gc}}{8} = 11.46\omega_{gc} + 45°$$

$$\tan^{-1}\frac{\frac{\omega_{gc}}{2}+\frac{\omega_{gc}}{8}}{1-\frac{\omega_{gc}^2}{16}} = 11.46\omega_{gc} + 45°$$

$$\tan^{-1}\frac{10\omega_{gc}}{16-\omega_{gc}^2} = 11.46\omega_{gc} + 45°$$

Solving by trial and error, we get $\omega_{gc} = 1$

$$\frac{K}{\omega_{gc}\sqrt{(\omega_{gc}^2+4)(\omega_{gc}^2+64)}} = 1$$

$$K = 1\sqrt{5\times 65} = 18$$

$$\boxed{K = 18}$$

Example 3.26

The forward path T.F. of a unity feedback control system is

$$G(s) = \frac{(as+1)}{s^2}$$

Find the value of a so that the phase margin is $\pi/4$.

Solution

$$G(s) = \frac{(as+1)}{s^2}$$

$$G(j\omega) = \frac{ja\omega + 1}{(-\omega^2)}$$

$$G(j\omega) = \frac{\sqrt{(1+a^2\omega^2)}}{\omega^2} \angle 180 + \tan^{-1} a\omega$$

$$|G(j\omega)|_{\omega=\omega_{gc}} = 1$$

$$\frac{\sqrt{1+a^2\omega_{gc}^2}}{\omega_{gc}^2} = 1$$

$$a^2\omega_{gc}^2 + 1 = \omega_{gc}^4 \tag{a}$$

$$\angle G(j\omega) = 180 + \tan^{-1} a\omega$$

$$\angle G(j\omega)|_{\omega=\omega_{gc}} = 180^\circ + \frac{\pi}{4}$$

$$\tan^{-1} a\omega_{gc} = \frac{\pi}{4}$$

Taking tan on both sides

$$a\omega_{gc} = 1 \tag{b}$$

Substituting Eq. (b) in Eq. (a), we get

$$\omega_{gc}^4 = 1 + 1 = 2$$

$$\omega_{gc} = 1.1892$$

$$a = \frac{1}{\omega_{gc}} = \frac{1}{1.1892}$$

$$\boxed{a = 0.841}$$

3.10.2 Phase Margin and Gain Margin via Bode Plots

The Bode log magnitude and phase plots of a certain third order minimum phase T.F. are shown in Fig. 3.48. To find the phase margin, the gain crossover is known which has 0 dB and the corresponding frequency is the gain crossover frequency ω_{gc}. From ω_{gc} if we come down straight, the phase angle plot is crossed. The phase margin is obtained as $-180+$ phase angle at ω_{gc} which is 22° in Fig. 3.48a. The phase margin is said to be +ve and we can add more phases before the system becomes unstable. To obtain the gain margin, the phase crossover is identified. This corresponds to 180° phase shift and the frequency at this point is phase crossover frequency ω_{pc}. From ω_{pc} go straight up to the log-magnitude plot and identify the dB at this frequency. This gives the gain margin. In Fig. 3.48a corresponding to ω_{pc}, the dB $= -12$. The gain margin is therefore 12 dB and is positive. In Fig. 3.48a both gain margin and phase margin are positive. Further $\omega_{pc} > \omega_{gc}$. Hence, the closed loop system is stable. On the other hand observe the Bode magnitude and phase plots given in Fig. 3.48b. Here corresponding to ω_{gc}, the phase angle is $-210°$. This is $-30°$ above $-180°$. The system has a phase margin of $-30°$ and is unstable. To make the system stable a phase angle greater than $+30°$ is to be added. Similarly, corresponding to ω_{pc}, the log-magnitude has +ve dB of 18. The gain margin is -18 dB negative and the closed loop system is unstable. To make the system stable, the system gain has to be reduced at least by a factor which corresponds to 18 dB.

Summary of Phase Margin and Gain Margin from Bode Plots

1. For a minimum phase linear system, the gain margin is positive if the magnitude of $G(j\omega)$ at the phase crossover is negative in dB. This means the gain margin is measured below the 0 dB axis. If the gain margin is measured above the 0 dB axis, the gain margin is negative.
2. For a minimum phase linear system, the phase margin is positive if the phase angle of $G(j\omega)$ is greater than $-180°$ at the gain crossover. This means the phase margin is measured positive above the $-180°$ axis. If the phase margin is measured below the $-180°$ axis, the phase margin is negative.
3. For the system to be stable both the phase margin and gain margin should be positive and $\omega_{pc} > \omega_{gc}$.
4. If $\omega_{pc} = \omega_{gc}$, the phase margin is 0° and the gain margin is 0 dB. The system executes sustained oscillations.
5. Increasing or decreasing the system T.F. gain K, the Bode magnitude plot is shifted upwards or downwards while the phase plot is unaltered.

Example 3.27

A certain unity negative feedback system has the following open loop T.F.

$$G(s) = \frac{80}{s(s+2)(s+20)}$$

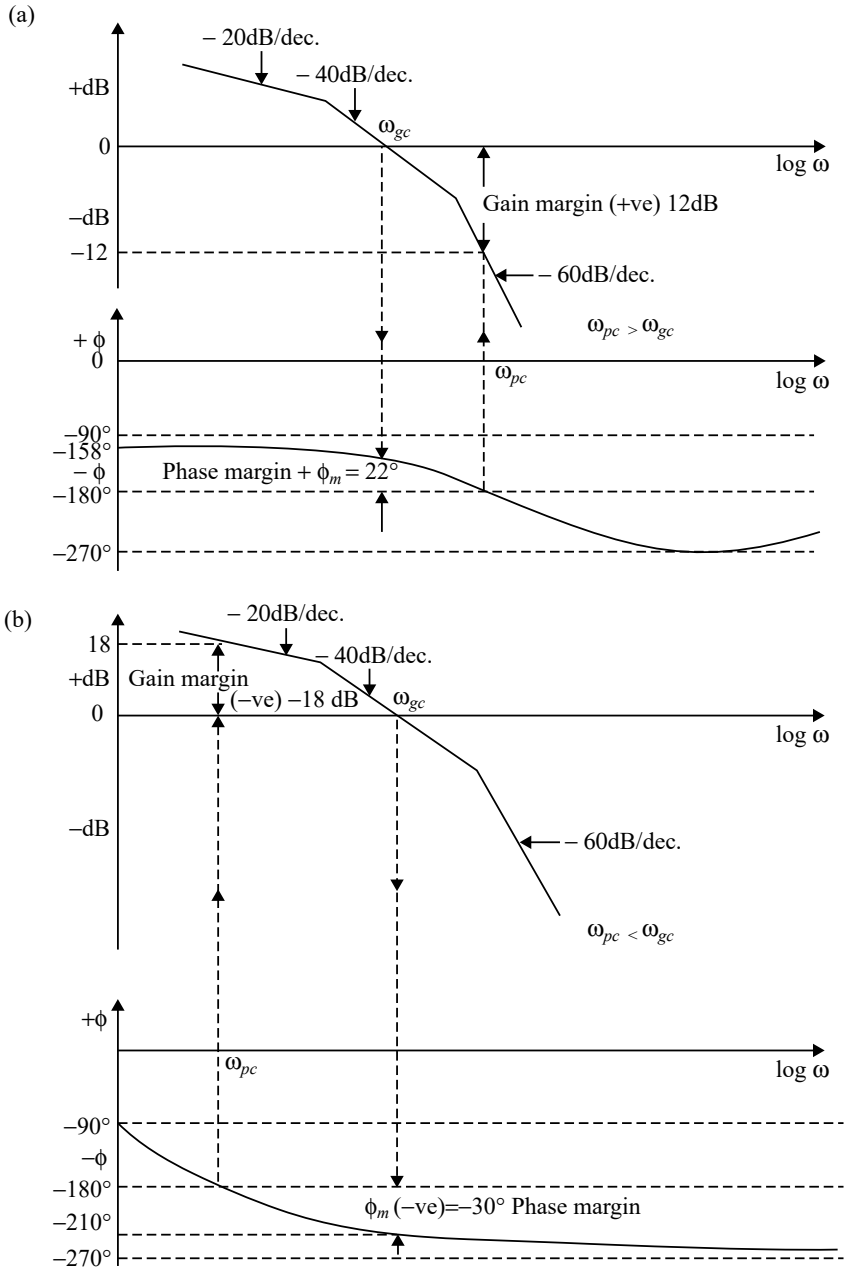


Fig. 3.48 Phase margin and gain margin from Bode plots. **a** Stable system (both ϕ_m and G_m are positive). **b** Unstable system (both ϕ_m and G_m are negative)

Using Bode plot determine the gain margin, phase margin, phase crossover frequency and gain crossover frequency.

Solution

$$G(s) = \frac{80}{s(s+2)(s+20)} = \frac{2}{s(1+\frac{s}{2})(1+\frac{1}{20}s)}$$

The corner frequencies are $\omega_c = 2, 20$ and $\omega_l = 2$

$$G_1(s) = \frac{2}{s}; \quad G_2(s) = \frac{1}{(1+\frac{s}{2})}; \quad G_3(s) = \frac{1}{(1+\frac{1}{20}s)}$$

1. The low frequency T.F. starts with a corner frequency of 2. At $\omega = 1$, $\text{dB}|_{\omega=1} = 20\log\frac{2}{\omega} = 20\log 2 = 6$. At $\omega = 1$, the dB = 6 and $\omega = 2$, dB = 0. A straight line ab is drawn. This represents −20 dB/dec slope.
2. At $\omega = 2$, $G_2(s)$ starts. For the frequency range $2 \le \omega \le 20$ the slope of the straight line is −40 dB/dec. This frequency range is one decade and hence point c is at −40 dB. Straight line bc is drawn.
3. Choose any frequency $\omega > 20$. $G_3(s)$ starts from $\omega = 20$ and the total slope of the straight line now becomes −60 dB/dec. At $\omega = 30$, the dB is

$$\text{dB}|_{\omega=30} = -40 - 60\log\frac{30}{20} = -50.6$$

   Straight line cd is drawn and extended. The Bode magnitude plot is completed and is shown in Fig. 3.49.
4. To draw the phase plot, ω_{gc} and ω_{pc} are necessary. At $\omega = 2$, the dB = 0. Hence, $\omega_{gc} = 2$. The expression for the phase angle is

$$\phi = -90^\circ - \tan^{-1}\frac{\omega}{2} - \tan^{-1}\frac{\omega}{20}$$
$$\phi|_{\omega=2} = -140.7^\circ$$

   Phase margin $\phi_m = 180 - 140.7^\circ = 39.3^\circ$

$$\boxed{\phi_m = +39.3^\circ}$$

5. To find ω_{pc}, the phase angle is known which is 180°

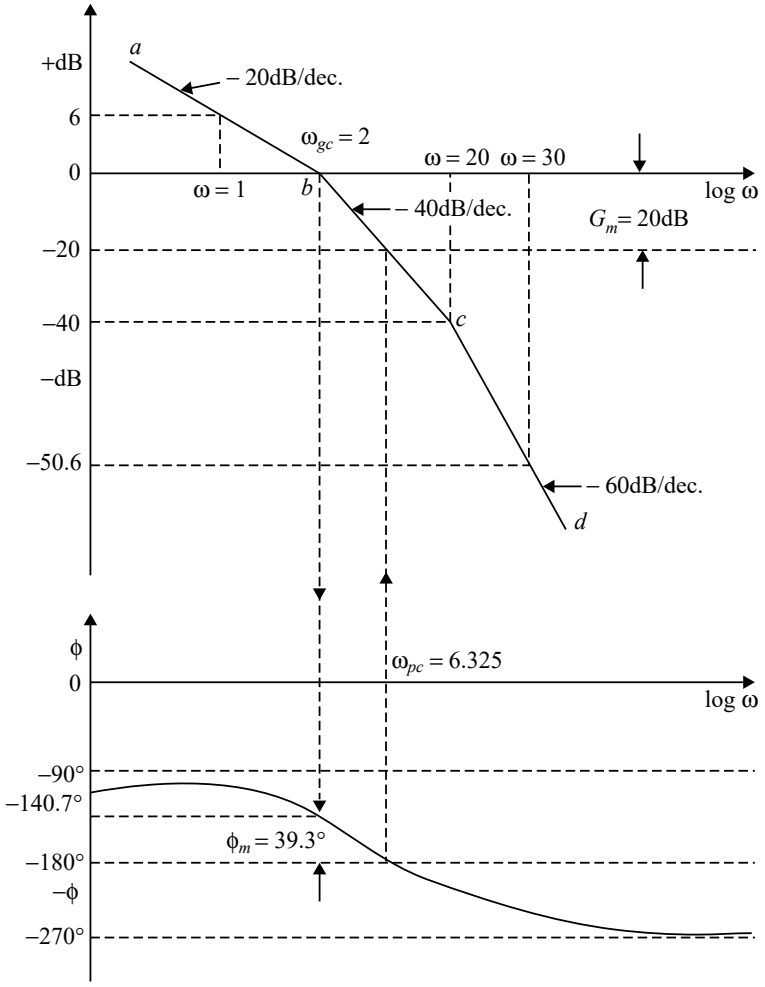


Fig. 3.49 Bode plot, for Example 3.27

$$-90 - \tan^{-1}\frac{\omega_{pc}}{2} - \tan^{-1}\frac{\omega_{pc}}{20} = 180°$$

$$\tan^{-1}\frac{\frac{\omega_{pc}}{2} + \frac{\omega_{pc}}{20}}{1 - \frac{\omega_{pc}^2}{40}} = 90°$$

Taking tan on both sides, we get

$$\tan\left[\tan^{-1}\frac{22\omega_{pc}}{40-\omega_{p_2^2}}\right] = \tan 90^\circ$$
$$= \infty$$
$$\frac{22\omega_{pc}}{40-\omega_{pc}^2} = \infty$$

This is possible if $40 - \omega_{p^2c} = 0$

$$\omega_{pc} = \sqrt{40} = 6.325$$

6. At this frequency, the dB is

$$\text{dB}|_{\omega_{pc}} = -40\log\frac{6.325}{2} = -20$$

$G_m = 20\,\text{dB}$ (+ve); $\quad \phi_m = 39.3^\circ$ (+ve)
$\omega_{pc} = 6.325\,\text{rad/sec}$; $\quad \omega_{gc} = 2\,\text{rad/sec}$
$\omega_{pc} > \omega_{gc}$

The system is stable.

Example 3.28

The open loop T.F. of a certain unity feedback control system is given by

$$G(s) = \frac{K}{s(1+0.5s)(1+0.2s)}$$

Using Bode asymptotic plot find K so that

(a) Gain margin is 6 dB.
(b) Phase margin is 25°.

(*Anna University, December, 2010*)

Solution

$$G(s) = \frac{K}{s(1+0.5s)(1+0.2s)}$$
$$G(j\omega) = \frac{K}{j\omega(1+0.5j\omega)(1+0.2j\omega)}$$
$$|G(j\omega)| = \frac{K}{\omega\sqrt{(1+0.25\omega^2)(1+0.04\omega^2)}}$$

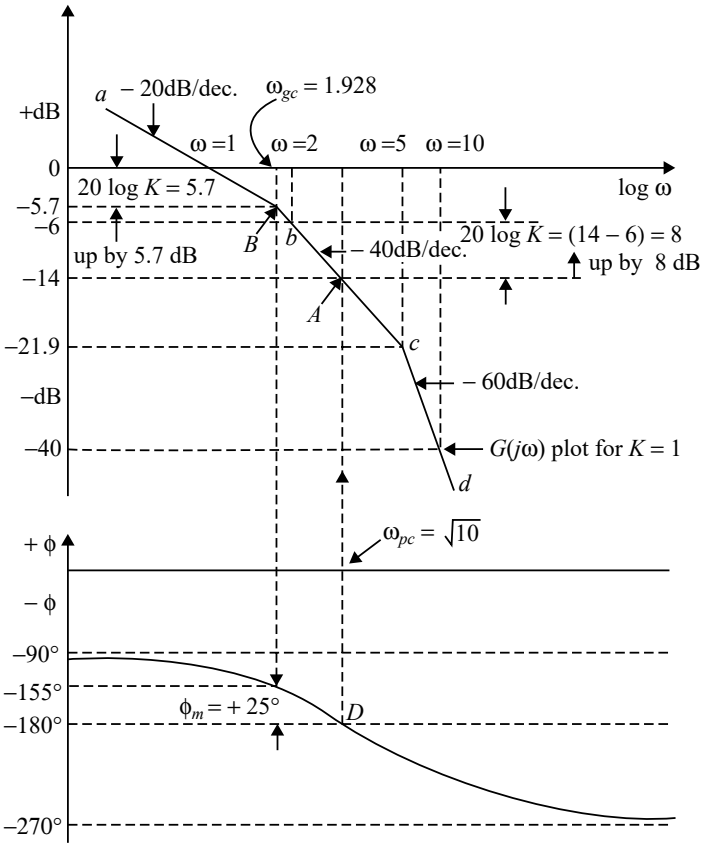


Fig. 3.50 Phase margin and gain margin, for Example 3.28

$$\angle G(j\omega) = -90 - \tan^{-1} 0.5\omega - \tan^{-1} 0.2\omega$$

The corner frequencies are 2 and 5 and $\omega_l = K$. Let us put $K = 1$ and sketch the Bode magnitude plot as shown in Fig. 3.50 as $abcd$. To draw the phase plot, the following table is prepared.

ω	0	1	1.928	$\sqrt{10}$	5	∞
$\angle G(j\omega)$	$-90°$	$-128°$	$-155°$	$-180°$	$-203°$	$-270°$

The phase plot is also sketched in Fig. 3.50.

(a) **To Determine K Value for the Desired $G_m = 6$ dB**
From Fig. 3.50, the phase crossover frequency is obtained as $\omega_{pc} = \sqrt{10}$. This is obtained as follows:

$$-90 - \tan^{-1} 0.5 \times \omega_{pc} - \tan^{-1} 0.2\omega_{pc} = 180^\circ$$
$$\tan^{-1} \frac{0.5\omega_{pc} + 0.2\omega_{pc}}{1 - 0.1\omega_{pc}^2} = 90^\circ$$

Taking tan on both sides, we get

$$\frac{0.7\omega_{pc}}{1 - 0.1\omega_{pc^2}} = \tan 90^\circ = \infty$$

This gives

$$1 - 0.1\omega_{p^2c} = 0$$

or

$$\omega_{pc} = \sqrt{10}$$

This point is identified as point D in phase plot. If we move vertically up the log-magnitude plot is intercepted at point A. At point A, the dB is

$$\text{dB}|_{\omega=\sqrt{10}} = -6 - 40 \log \frac{\sqrt{10}}{2}$$
$$= -14$$

The gain margin occurs at phase crossover. The required gain margin is 6 dB. In other words, the Bode magnitude plot should be shifted vertically upwards so that point A is located at -6 dB. The required upward shift is 8 dB. That is

$$20 \log K = 8$$

$$\boxed{K = 2.51}$$

(b) **To Determine K Value for the Required Phase Margin of $\phi_m = 25^\circ$**
The phase margin is calculated at gain crossover which is calculated as follows:

$$\phi_m = 180^\circ + \angle G(j\omega)|_{\omega=\omega_{gc}}$$
$$25^\circ = 180^\circ + \angle G(j\omega)|_{\omega=\omega_{gc}}$$
$$\angle G(j\omega_{gc}) = -155^\circ$$
$$-90 - \tan^{-1} 0.5\omega_{gc} - \tan^{-1} 0.2\omega_{gc} = -155^\circ$$
$$\tan^{-1} \frac{0.7\omega_{gc}}{1 - 0.1\omega_{gc}^2} = 65^\circ$$
$$0.7\omega_{gc} = (1 - 0.1\omega_{gc}^2)(2.1445)$$
$$0.21445\omega_{gc}^2 + 0.7\omega_{gc} - 2.1445 = 0$$
$$\omega_{gc} = 1.928$$

Corresponding to 1.928, the dB should be zero. For $K = 1$

$$\text{dB}|_{\omega=1.928} = 20 \log 1.928 = 5.7$$

This corresponds to point B in the log-magnitude plot. The point B should be shifted upwards by adding +5.7 dB so that it is located at zero dB. The required gain for the upward shift of 5.7 dB is

$$20 \log K = 5.7$$

$$\boxed{K = 1.928}$$

Example 3.29

Consider the following open loop T.F. of a certain closed loop system

$$G(s) = \frac{20}{s(1 + 0.5s)(1 + 0.05s)}$$

Sketch the Bode asymptotic log-magnitude and phase plots and determine the ϕ_m, G_m, ω_{pc} and ω_{gc}.

Solution

$$G(s) = \frac{20}{s(1 + 0.5s)(1 + 0.05s)}$$

$$G(j\omega) = \frac{20\angle -90^\circ - \tan^{-1} 0.5\omega - \tan^{-1} 0.05\omega}{\omega\sqrt{(1 + 0.25\omega^2)(1 + 0.0025\omega^2)}}$$

The asymptotic Bode log-magnitude plot is sketched as straight line segments $abcd$ shown in Fig. 3.51. The gain crossover frequency is obtained as

$$40 \log \frac{\omega_{gc}}{2} = 20$$

$$\omega_{gc} = 6.32$$

At this frequency, the phase angle of $G(j\omega)$ is

$$\begin{aligned}\phi &= -90 - \tan^{-1} 0.5\omega_{gc} - \tan^{-1} 0.05\omega_{gc} \\ &= -90^\circ - \tan^{-1} 0.5 \times 6.32 - \tan^{-1} 0.05 \times 6.32 \\ &= -180^\circ\end{aligned}$$

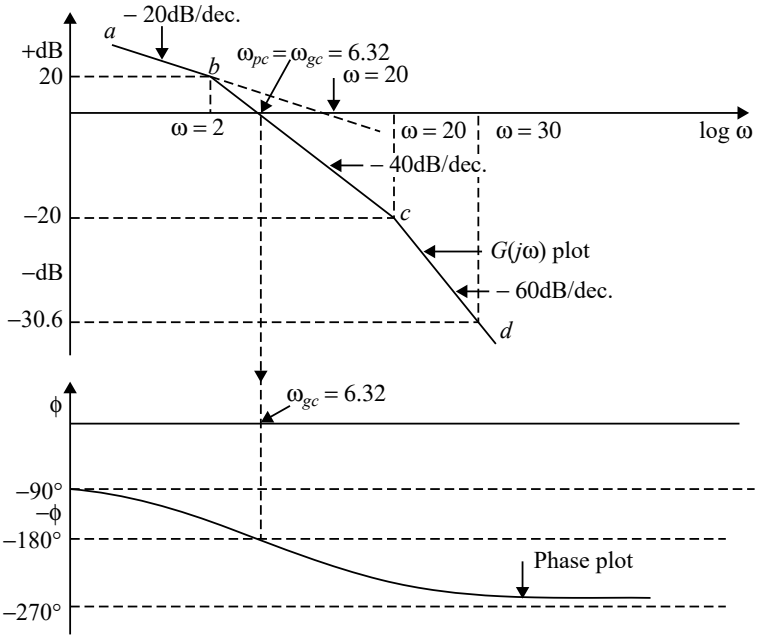


Fig. 3.51 Bode plots, for Example 3.29

Since $\phi = -180^\circ$ at ω_{gc}, the phase margin is 0. To find the gain margin, the phase crossover is determined. For this $\angle G(j\omega) = -180^\circ$. Corresponding to -180° phase, the phase crossover frequency is $\omega_{pc} = 6.32$. Corresponding to this frequency dB $= 0$

$$\boxed{\begin{aligned} G_m &= 0\,\text{dB} \\ \phi_m &= 0^\circ \\ \omega_{gc} &= 6.32\,\text{rad/sec} \\ \omega_{pc} &= 6.32\,\text{rad/sec} \end{aligned}}$$

The system is oscillatory.

3.10.3 Phase Margin and Gain Margin Using MATLAB

Phase margin and gain margin play an important role in the study of relative stability. The examples illustrated above make use of asymptotic Bode magnitude plot and therefore the results obtained are not accurate. Standard MATLAB programs are available to draw the Bode log-magnitude and phase plots very accurately and quickly. The symbols used in the MATLAB program are

G_m = Gain margin
p_m = Phase margin
ω_{cp} = Phase crossover frequency
ω_{cg} = Gain crossover frequency

The following example is solved using MATLAB program to obtain the above frequency domain specifications.

Example 3.30
The open loop T.F. of a certain unity feedback control system is given by

$$G(s) = \frac{10(s+2)}{s(s+6)(s^2+4s+9)}$$

Determine, using MATLAB program, gain margin in dB, phase margin, phase crossover frequency and gain crossover frequency.

Solution

$$G(s) = \frac{10(s+2)}{s(s+6)(s^2+4s+9)} = \frac{(10s+20)}{(s^2+6s+0)(s^2+4s+9)}$$

The MATLAB program is given below

Bode plots
To find Phase margin and Gain margin

num=[0 0 0 10 20];
den=conv([1 6 0],[1 4 9]);
sys=tf(num,den);
w=logspace(−1, 2, 100);
bode (sys,W)
[G_m, p_m, ω_{cp}, ω_{cg}]=margin(sys);
G_m dB = 20* log 10(G_m);
[G_m dB, p_m, ω_{cp}, ω_{cg}]
 	
ans= 22.5146 87.4196 4.3310 0.3768

From Fig. 3.52, the following results are obtained

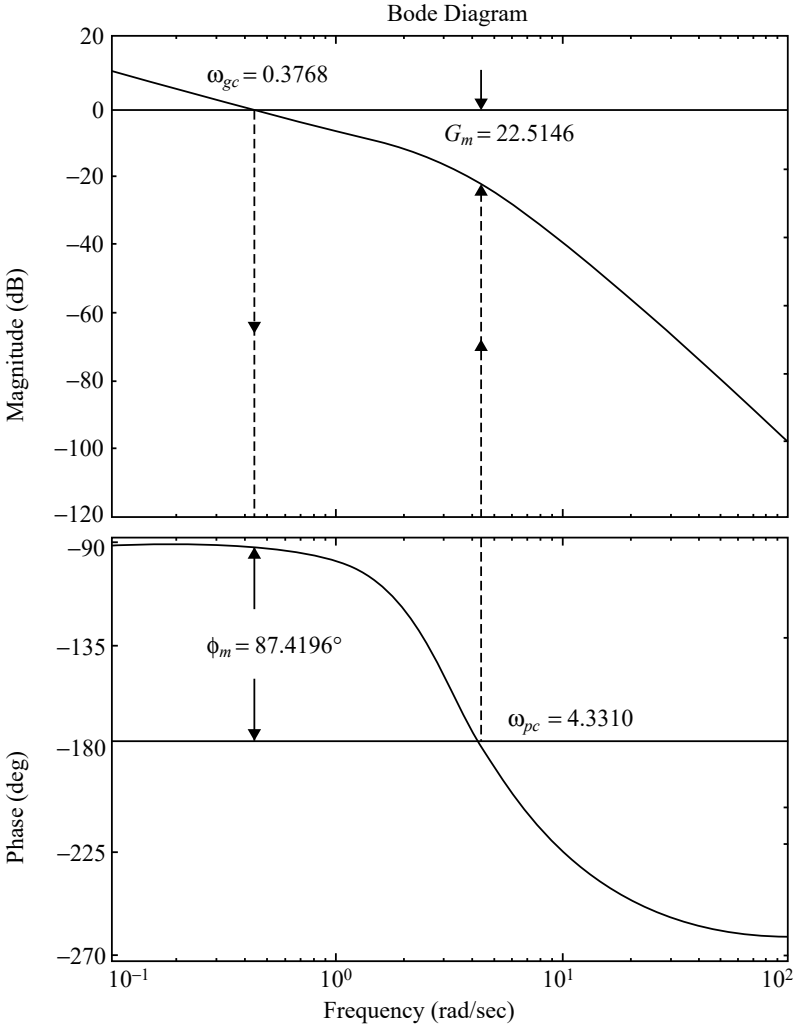


Fig. 3.52 Bode log-magnitude and phase plots obtained by MATLAB program to determine phase margin, gain margin, phase crossover and gain crossover frequencies

$$\boxed{\begin{aligned} G_m &= +22.5146\,\text{dB} \\ \phi_m &= +87.4196^\circ \\ \omega_{gc} &= 0.3768\,\text{rad/sec} \\ \omega_{pc} &= 4.3310\,\text{rad/sec} \end{aligned}}$$

Phase margin and gain margin are +ve. $\omega_{pc} > \omega_{gc}$. The system is stable.

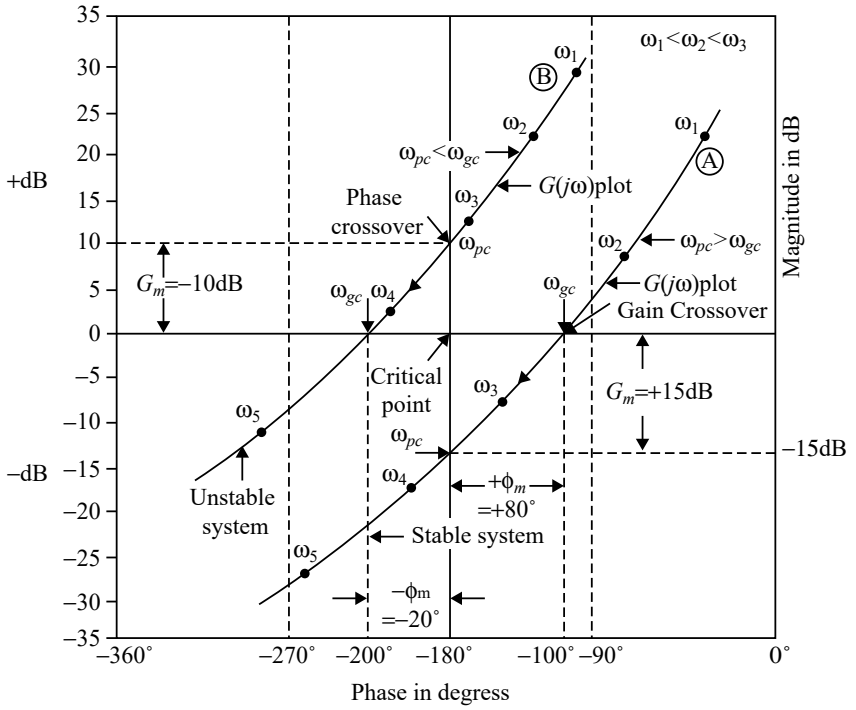


Fig. 3.53 Phase margin and gain margin is log magnitude-phase plot of $G(j\omega)$

3.10.4 Phase Margin and Gain Margin via Log-Magnitude-Phase Plot

The magnitude-phase plot has several advantages over the Bode plots and Nyquist plot as discussed earlier. The relative stability parameters such as the phase margin, gain margin, phase crossover frequency and gain crossover frequency can be easily and quickly determined with high degree of accuracy. The log-magnitude-phase plots of $G(j\omega)$ of two different systems A and B are shown in Fig. 3.53. The following points are to be noted in magnitude-phase plot:

1. The $G(j\omega)$ plot is drawn with frequency as the varying parameter. The arrow in $G(j\omega)$ plot indicates that the value of ω increases in the direction of the arrow.
2. The gain crossover is where the $G(j\omega)$ plot intersects the 0 dB axis. The corresponding frequency is gain crossover frequency ω_{gc}.
3. The phase crossover is where the $G(j\omega)$ plot intersects the $-180°$ axis. The corresponding frequency is phase crossover frequency ω_{pc}.
4. The critical point is the intersection of 0 dB horizontal axis and $-180°$ vertical axis.

5. The phase margin is the horizontal distance in degrees measured from the gain crossover to the critical point. If the gain crossover is to the right of the critical point, the phase margin is measured as positive. If it is to the left of the critical point, the phase margin is measured as negative.
6. The gain margin is the vertical distance in dB measured from the phase crossover to the critical point. If the phase crossover is below 0 dB horizontal axis, it is measured as positive. If the phase crossover is above the 0 dB horizontal axis, it is measured as negative.

It is to be noted that in the horizontal axis the phase of open loop T.F. $\angle G(j\omega)$ is represented. Similarly, in the vertical axis $20\log|G(j\omega)|$ is represented.

Now consider the log magnitude-phase plot of system A shown in Fig. 3.53. The gain crossover is at ω_{gc} and it is to the right of critical point. The phase margin as indicated in Fig. 3.53 is 80° and is positive. For the same system, the phase crossover is at ω_{pc}. It is below the 0 dB horizontal line which is at −15 dB. The gain margin is therefore +15 dB positive. Here $\omega_{pc} > \omega_{gc}$. The system is stable.

Consider the log magnitude-phase plot of system B shown in Fig. 3.53. The gain crossover is at ω_{gc} and it is to the left of the critical point. The phase margin as indicated in Fig. 3.53 is −20° and it is negative. For the same system, the phase crossover is at ω_{pc}. It is above 0 dB horizontal line at +10 dB. The gain margin is −10 dB and is negative. $\omega_{pc} < \omega_{gc}$. Both phase margin and gain margin are negative and the system is unstable.

Example 3.31

A certain closed loop system has the following open loop T.F.

$$G(s) = \frac{0.75(1+0.1s)}{s(1+0.5s)(1+0.2s)}$$

Draw the log-magnitude-phase plot and determine the phase margin, gain margin, phase crossover frequency and gain crossover frequency. Comment on the stability of the system.

Solution

$$G(s) = \frac{0.75(1+0.1s)}{s(1+0.5s)(1+0.2s)}$$

$$G(j\omega) = \frac{0.75(1+0.1j\omega)}{j\omega(1+j0.5\omega)(1+j0.2\omega)}$$

$$|G(j\omega)| = \frac{0.75\sqrt{1+0.01\omega^2}}{\omega\sqrt{(1+0.25\omega^2)(1+0.04\omega^2)}}$$

$$\angle G(j\omega) = \phi = \tan^{-1}0.1\omega - 90 - \tan^{-1}0.5\omega - \tan^{-1}0.2\omega$$

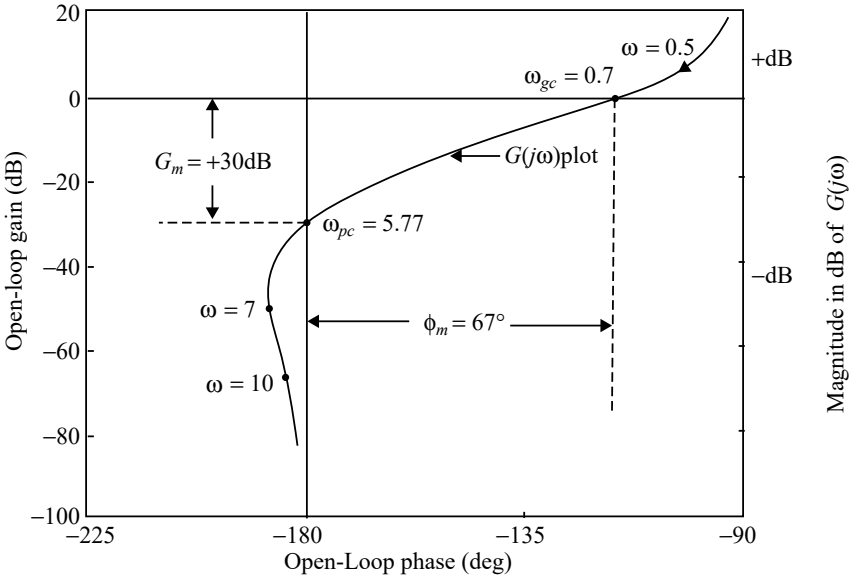


Fig. 3.54 Log magnitude-phase plot, for Example 3.31

For different values of ω in the range $0 \leq \omega \leq \infty$, $|G(j\omega)|$, $20\log|G(j\omega)|$ and ϕ are calculated as shown in the table given below

ω	0	0.5	0.7	5.77	7	10
ϕ	$-90°$	$-107°$	$-113°$	$-180°$	$-183°$	$-187°$
$\lvert G(j\omega)\rvert$	∞	1.45	1	0.032	0.021	0.0092
$\text{dB} = 20\log\lvert G(j\omega)\rvert$	∞	3.23	0	-30	-33.6	-40.6

The data from the above table is transferred to the log magnitude-phase plot, and it is shown in Fig. 3.54. The values of G_m, ϕ_m, ω_{gc} and ω_{pc} are given below.

$$\begin{aligned} G_m &= +30\,\text{dB} \\ \phi_m &= +67° \\ \omega_{gc} &= 0.7\,\text{rad/sec} \\ \omega_{pc} &= 5.77\,\text{rad/sec} \end{aligned}$$

Both phase margin and gain margin are positive and $\omega_{pc} > \omega_{gc}$. Hence, the system is stable.

Example 3.32

The open loop T.F. of a certain feedback control system is

$$G(s) = \frac{Ks^2}{[(1+0.2s)(1+0.02s)]}$$

Determine the value of K for the gain crossover frequency to be 5 rad/sec. Use log magnitude-phase plot.

Solution

$$G(s) = \frac{Ks^2}{(1+0.2s)(1+0.02s)}$$

Let us put $K = 1$

$$G(s) = \frac{250\,s^2}{(s+5)(s+50)}$$

$$|G(j\omega)| = \frac{250\omega^2}{\sqrt{(\omega^2+25)(\omega^2+2500)}}$$

$$\phi = \angle G(j\omega)$$

$$= 180^\circ - \tan^{-1}\frac{\omega}{5} - \tan^{-1}\frac{\omega}{50}$$

The following table is prepared to draw $G(j\omega)$ plot in the log magnitude-phase plot:

ω	1	3	5	7	10
ϕ	-192.5°	-214.4°	-230°	-242.5°	-254.7°
$\lvert G(j\omega)\rvert$	0.98	7.7	17.59	28.2	43.85
$\text{dB} = 20\log\lvert G(j\omega)\rvert$	-0.172	17.7	24.9	29	32.84

The log magnitude-phase plot is shown in Fig. 3.55. For $\omega_{gc} = 5$ rad/sec, the log magnitude plot should be shifted vertically down so that at $\omega_{gc} = 5$, the dB is zero. This means the gain has to be reduced by a factor of 24.9 dB. This is related in terms of K as

$$20\log K = -24.9$$

$$\boxed{K = 0.05688}$$

If the same problem is solved using Bode asymptotic plot, the answer obtained will be slightly different because Bode plot is drawn with some approximation. However, in Bode plot obtained using MATLAB will give the same value.

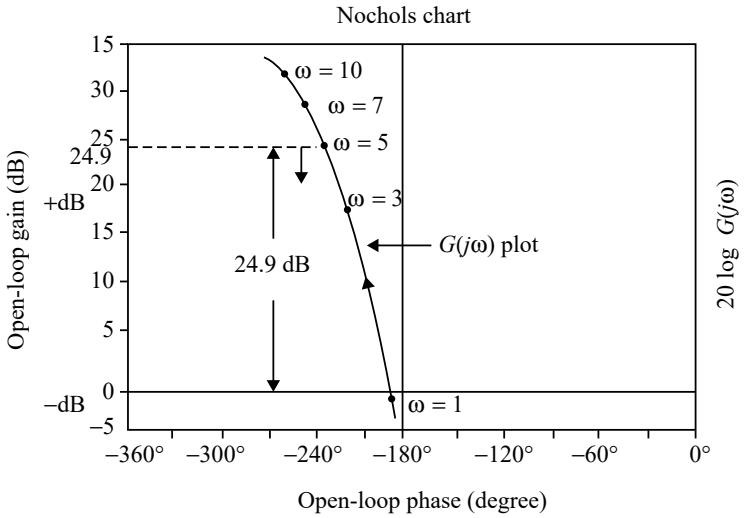


Fig. 3.55 Log magnitude-phase plot, for Example 3.32

Example 3.33

The forward path transfer function of a certain unity feedback control system is given by

$$G(s) = \frac{K}{s(1 + 0.2s)(1 + 0.05s)}$$

Sketch the log-magnitude phase plot and determine the value of K for

(a) Gain margin is 25 dB.
(b) Phase margin is 65°.

Solution

$$G(s) = \frac{K}{s(1 + 0.2s)(1 + 0.05s)}$$

Put $K = 1$

$$G(s) = \frac{1}{s(1 + 0.2s)(1 + 0.05s)}$$

$$= \frac{100}{s(s + 5)(s + 20)}$$

$$G(\,j\omega) = \frac{100}{j\omega(j\omega + 5)(j\omega + 20)}$$

$$|G(j\omega)| = \frac{100}{\omega\sqrt{(\omega^2+25)(\omega^2+400)}}$$

$$\angle G(j\omega) = -90 - \tan^{-1} 0.2\omega - \tan^{-1} 0.05\omega$$

For different values of ω, $|G(j\omega)|$ and $20\log|G(j\omega)|$ and $\angle G(j\omega) = \phi$ are calculated as shown in the table given below

ω	0.5	0.98	4	8	10	15
$\phi = \angle G(j\omega)$	$-97°$	$-104°$	$-140°$	$-170°$	$-180°$	$-198°$
$\lvert G(j\omega)\rvert$	1.99	1	0.1914	0.0615	0.04	0.01687
$\text{dB} = 20\log\lvert G(j\omega)\rvert$	6	0	-14.36	-24.22	-28	-35.5

The data from the table is transferred, and the log magnitude-phase plot is drawn as shown in Fig. 3.56. For $K = 1$, from Fig. 3.56, the phase margin is $+76°$ and gain margin is $+28$ dB. The gain crossover frequency is $\omega_{gc} = 0.98$ rad/sec and the phase crossover frequency is 10 rad/sec

(a) Determination of K for the required gain margin $+25$ dB.
The required gain margin is 25 dB. The actual gain margin is 28 dB. To achieve this the $G(j\omega)$ plot is to be shifted vertically up by 3 dB by increasing the gain. Under this condition

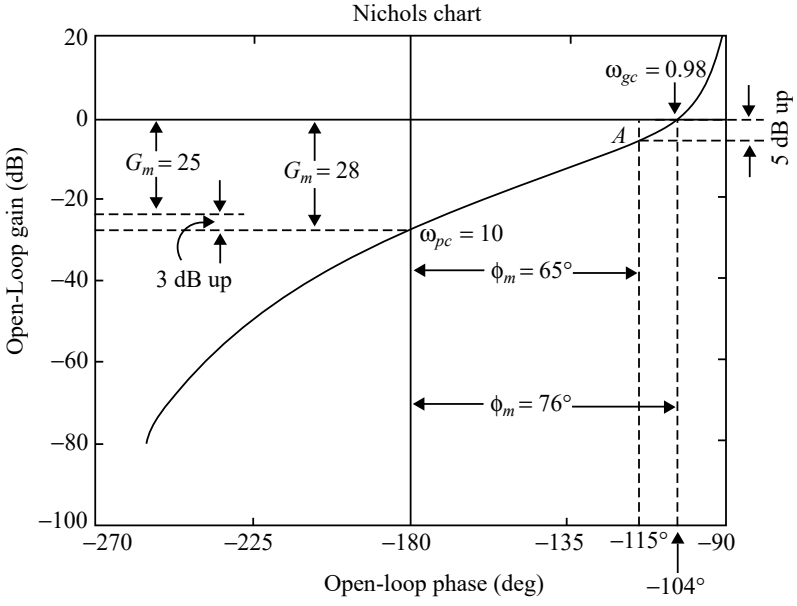


Fig. 3.56 Log magnitude-phase plot, for Example 3.33

$$20 \log K = 3$$

$$\boxed{K = 1.4125}$$

The phase crossover frequency remains the same at $\omega_{pc} = 10\,\text{rad/sec}$

(b) Determination of K for the required phase margin $+65°$.
Corresponding to 65° phase margin point A is identified in the $G(j\omega)$ plot. This point is down by $-5\,\text{dB}$ from 0 dB horizontal axis. To make point A to be on the horizontal axis $G(j\omega)$ plot should be shifted vertically up by $+5\,\text{dB}$ by increasing the gain. Under this condition

$$20 \log K = 5$$

$$\boxed{K = 1.778}$$

By the vertical shift, the gain crossover frequency is changed and the new gain crossover frequency is roughly estimated as $\omega_{gc} = 2\,\text{rad/sec}$

Example 3.34

A certain unity feedback control system has the following open loop T.F. with transportation lag.

$$G(s) = \frac{0.1e^{-Ts}}{s(1 + 0.1s)(1 + 0.5s)}$$

Find the value of T so that the phase margin is $+25°$. Use log magnitude-phase plot.

Solution

$$G(s) = \frac{0.10e^{-Ts}}{s(1 + 0.1s)(1 + 0.5s)}$$

$$G(j\omega) = \frac{0.10e^{-j\omega T}}{j\omega(1 + 0.1j\omega)(1 + 0.5j\omega)}$$

$$= \frac{2(e^{-j\omega T})}{j\omega(j\omega + 10)(j\omega + 2)}$$

$$|G(j\omega)| = \frac{2}{\omega\sqrt{(\omega^2 + 100)(\omega^2 + 4)}}$$

$$\angle G(j\omega) = -90° - \tan^{-1} 0.1\omega - \tan^{-1} 0.5\omega - 57.3T\omega°$$

Let T be zero

$$G(j\omega) = -90^\circ - \tan^{-1} 0.1\omega - \tan^{-1} 0.5\omega$$

ω	0.05	0.1	1	3	4.472	8	10
$\phi = \angle G(j\omega)$	-92°	-93.5°	-122.26°	163°	-180°	-205°	-214°
$\lvert G(j\omega)\rvert$	2	1	0.089	0.0177	0.00833	0.00237	0.00139
$\text{dB} = 20\log \lvert G(j\omega)\rvert$	6	0	-21	-35	-41.6	-52.5	-57

From Fig. 3.57, the following frequency domain specifications are observed.

$$\begin{aligned} G_m &= +41.6\,\text{dB} \\ \phi_m &= +86.5^\circ \\ \omega_{gc} &= 0.1 \\ \omega_{pc} &= 4.472 \end{aligned}$$

The required phase margin is $+25^\circ$. The balance of $-(86.5^\circ - 25^\circ) = -61.5^\circ$ is to be contributed by the transportation large at $\omega = \omega_{gc} = 0.1$

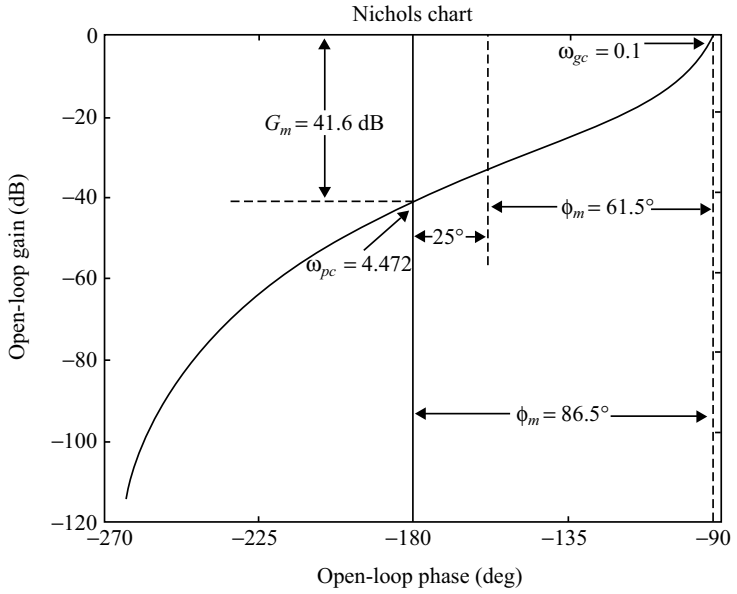


Fig. 3.57 Log magnitude-phase plot, for Example 3.34

$$-57.3\omega_{gc}T = -61.5$$
$$5.73T = 61.5$$
$$\boxed{T = 10.73}$$

It is to be noted that when phase angle is added the $G(j\omega)$ plot is horizontally shifted without affecting the dB of $|G(j\omega)|$. The gain crossover frequency ω_{gc} moves in the same horizontal 0 dB axis hence it is unchanged. However, the phase crossover frequency is changed during horizontal shift.

3.10.4.1 MATLAB Programme to Determine Phase Margin and Gain Margin from Log Magnitude-Phase (Nichols) Plot

MATLAB toolbox is very extensively used in solving control system problems especially in frequency response plots which otherwise involves tedious calculations. The frequency response plots can be quickly and very accurately drawn using MATLAB. The several advantages of log magnitude-phase (Nichols) plot were listed. We illustrate below a few examples to draw exact Nichols plot and determine phase margin, gain margin, phase crossover and gain crossover frequencies.

The same program which was used to determine these parameters from Bode plot can be used except the command bode (sys, W) is replaced by Nichols (sys, W).

Example 3.35

Consider the closed loop system described in Example 3.30 which has the open loop T.F.

$$G(s) = \frac{10(s+2)}{s(s+6)(s^2+4s+9)}$$

Using MATLAB program draw the Nichols plot and determine gain margin, phase margin, phase crossover frequency and gain crossover frequency. Comment on the stability of the closed loop system.

Solution The MATLAB program for the given T.F.

$$G(s) = \frac{10(s+2)}{s(s+6)(s^2+4s+9)}$$

is given below. The program is exactly same as for the Bode diagram except that bode (sys, W) is replaced by Nichols (sys, W). The log magnitude phase plot is shown in Fig. 3.58. From the print out and Fig. 3.58, the following results are obtained

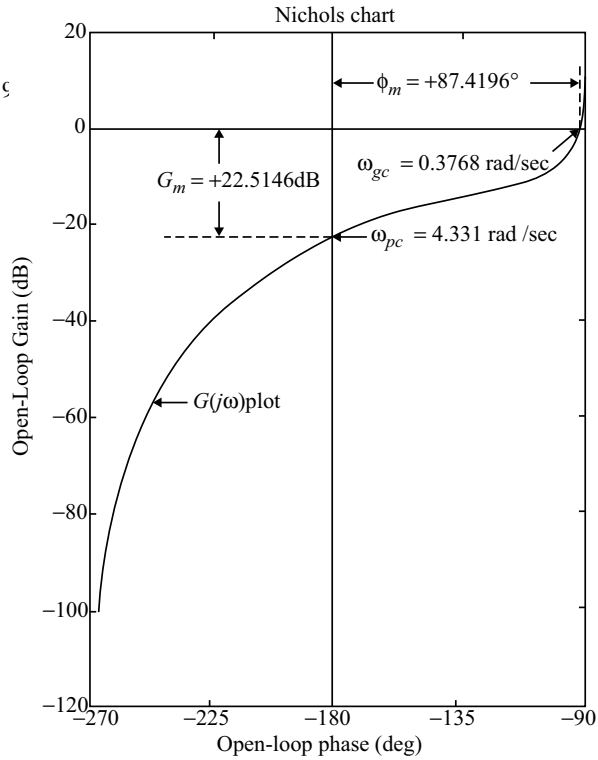


Fig. 3.58 To determine G_m, ϕ_m, ω_{pc} and ω_{gc} from Nichols plot for $G(s) = 10(s+2)/[s(s+6)(s^2+4s+9$

$$\begin{aligned} G_m &= 22.5146\,\text{dB} \\ \phi_m &= 87.4196° \\ \omega_{pc} &= 4.3310\,\text{rad/sec} \\ \omega_{gc} &= 0.3768\,\text{rad/sec} \end{aligned}$$

Since both gain margin and phase margin are +ve and $\omega_{pc} > \omega_{gc}$, the system is STABLE. The same result is obtained using Bode plot in Example 3.30.

MATLAB Program

Nichols plot
To find Phase margin and Gain margin

num=[0 0 0 10 20];
den=conv([1 6 0],[1 4 9]);
sys=tf(num,den);
w=logspace(−1, 2, 100);
Nichols (sys,W)
[G_m, p_m, ω_{cp}, ω_{cg}]=margin(sys);
G_m dB = 20* log 10(G_m);
[G_m dB, p_m, ω_{cp}, ω_{cg}]
 	
ans=
22.5146 87.4196 4.3310 0.3768

Example 3.36

Consider the following open loop T.F. of a certain closed loop system.

$$G(s) = \frac{(s+4)(s+5)}{s(s+1)(s+2)(s+30)}$$

Write a MATLAB program for the frequency response plot of the open loop system in the Nichols plot and determine G_m, ϕ_m, ω_{pc} and ω_{gc}. Determine the stability of the closed loop system. How the above specifications are changed if the gain is changed by a factor of 10.

Solution

$$G(s) = \frac{(s+4)(s+5)}{s(s+1)(s+2)(s+3)} = \frac{(s^2+9s+20)}{(s^2+s+0)(s^2+5s+6)}$$

The MATLAB program is written as given below and the $G(j\omega)$ plot in log magnitude-phase is shown in Fig. 3.59. The following results are given in the print out and marked in Fig. 3.59

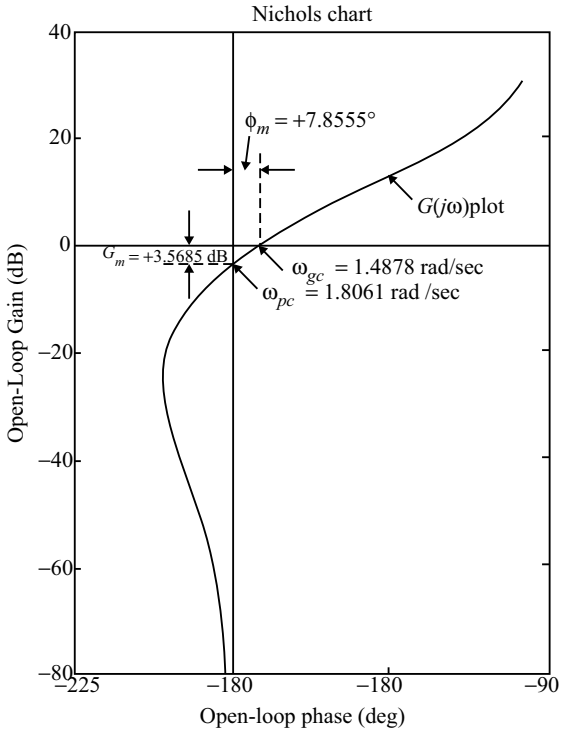


Fig. 3.59 To determine G_m, ϕ_m, ω_{pc} and ω_{gc}. by Nichols plot for $G(s) = \dfrac{(s+4)(s+5)}{s(s+1)(s+2)(s+3)}$

$$\begin{aligned} G_m &= +3.5685\,\text{dB} \\ \phi_m &= +7.8555^\circ \\ \omega_{pc} &= 1.8061\,\text{rad/sec} \\ \omega_{gc} &= 1.4878\,\text{rad/sec} \end{aligned}$$

Both gain margin and phase margin are positive. $\omega_{pc} > \omega_{gc}$. The closed loop system is stable.

MATLAB Program

Nichols plot
To find Phase margin and Gain margin

num=[0 0 1 9 20];
den=conv([1 1 0],[1 5 6]);
sys=tf(num,den);
w=logspace(−1, 2, 100);
Nichols (sys,W)
[G_m, p_m, ω_{cp}, ω_{cg}]=margin(sys);
G_m dB = 20* log 10(G_m);
[G_m dB, p_m, ω_{cp}, ω_{cg}]
ngrid
 	
ans=
3.5685 7.8555 1.8061 1.4878

G_m, ϕ_m, ω_{pc} and ω_{gc} and stability when the system gain is increased

$$G(s) = \frac{10(s+4)(s+5)}{s(s+1)(s+2)(s+3)} = \frac{(10s^2+90s+200)}{(s^2+s+0)(s^2+5s+6)}$$

The MATLAB program is written as given below and the $G(j\omega)$ plot in log magnitude-phase is shown in Fig. 3.60. The following results are given in the print out and marked in Fig. 3.60.

$$\begin{aligned} G_m &= -16.4315\,\text{dB} \\ \phi_m &= -18.8153^\circ \\ \omega_{pc} &= 1.8061\,\text{rad/sec} \\ \omega_{gc} &= 3.9713\,\text{rad/sec} \end{aligned}$$

Both gain margin and phase margin are negative and $\omega_{pc} < \omega_{gc}$. The closed loop system is unstable. An increase in gain of 10 is equivalent to increase in dB by +20 dB. Fig. 3.59 is vertically shifted up by +20 dB. The gain crossover is unaffected and the dB at this point is $-3.5685 + 20 = +16.4315$. Hence, $G_m = -16.4315$ dB as seen is Fig. 3.60.

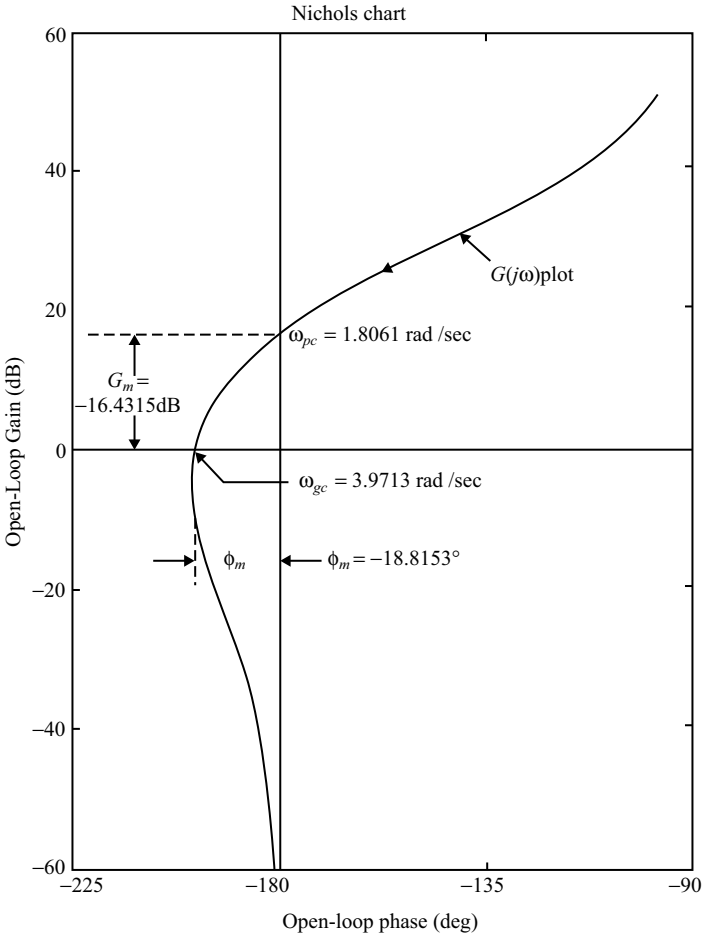


Fig. 3.60 To determine G_m, ϕ_m, ω_{pc} and ω_{gc}. by Nichols plot for $G(s) = \dfrac{10(s+4)(s+5)}{s(s+1)(s+2)(s+3)}$

MATLAB Program

Nichols plot
To find Phase margin and Gain margin

num=[0 0 10 90 200];
den=conv([1 1 0],[1 5 6]);
sys=tf(num,den);
w=logspace(−1, 2, 100);
Nichols (sys,W)
[G_m, p_m, ω_{cp}, ω_{cg}]=margin(sys);
G_m dB = 20* log 10(G_m);
[G_m dB, p_m, ω_{cp}, ω_{cg}]
Warning: The closed-loop system is unstable.
ans= −16.4315 −18.8153 1.8061 3.9713

3.11 Determining Closed Loop Frequency Response from Open Loop Frequency Response (Unity Feedback System)

Consider the block diagram of a certain unity feedback control system with the forward path T.F. as shown in Fig. 3.61a. The closed loop T.F. is obtained as

$$\frac{C(s)}{R(s)} = M(s) = \frac{G(s)}{1 + G(s)} \tag{3.31}$$

The sinusoidal T.F. of Eq. (3.31) is

$$M(j\omega) = \frac{G(j\omega)}{1 + G(j\omega)} \tag{3.32}$$

Instead of using Eq. (3.32) which is complex in nature to evaluate the closed loop frequency response $M(j\omega)$, it can be easily obtained by graphical means. Further earlier methods developed to determine resonant peak M_r, resonant frequency ω_r and the BW are restricted to second order system only. The information that we get from Bode and log magnitude-phase plots is about the phase margin and gain margin which is used to determine the absolute and relative stability. From the graphical method we discuss below, we get information about M_r, ω_r and BW for higher order systems in addition to getting information about the closed loop frequency response which is obtained very easily.

Consider the frequency response $G(j\omega)$ of the open loop system shown in Fig. 3.61b. For any frequency ω_1, point B in $G(j\omega)$ can be located. Draw vectors OB from the origin and AB from $(-1, 0)$ point in the $G(j\omega)$ complex plane.

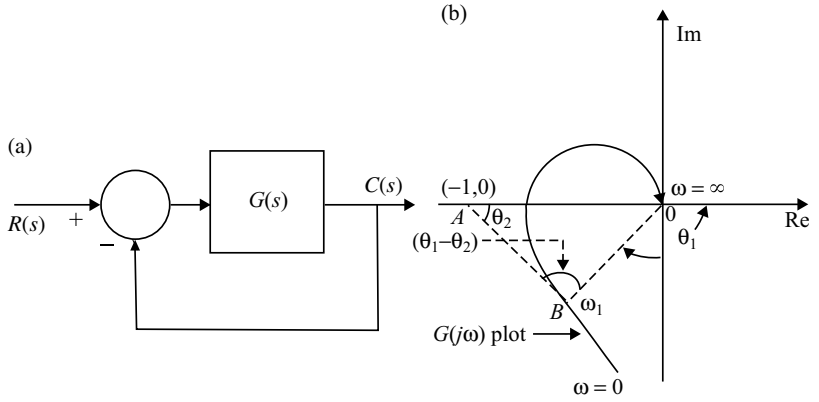


Fig. 3.61 **a** Block diagram of a unity feedback control system. **b** $G(\,j\omega)$ plot in the polar coordinate

The vectors in Fig. 3.61b represent the following

$$\begin{aligned}
\overrightarrow{OA} &= (-1, 0) \\
\overrightarrow{OB} &= G(\,j\omega) \\
\overrightarrow{AB} &= \overrightarrow{OA} + \overrightarrow{AB} = 1 + G(\,j\omega) \\
|M(j\omega)| = \frac{\overrightarrow{OB}}{\overrightarrow{AB}} &= \frac{|G(\,j\omega)|}{|1 + G(\,j\omega)|} \qquad (3.33) \\
\angle M(j\omega) &= \angle\overrightarrow{OB} - \angle\overrightarrow{AB} \\
&= -\theta_1 + \theta_2 \\
&= -(\theta_1 - \theta_2) \\
&= -\angle ABO \qquad (3.34)
\end{aligned}$$

From Eq. (3.33) by calculating the ratio of OB to AB, the closed loop magnitude $M(j\omega)|_{\omega=\omega_1}$ is obtained. Similarly, from Eq. (3.34) the negative value of ABO gives $\angle M(j\omega)|_{\omega=\omega_1}$. The procedure is repeated for other values of ω and for each frequency the magnitude and phase of the closed loop system are obtained. However, the above procedure is still further simplified. A set of circles each indicating a particular value of closed loop magnitude and closed loop phase angle separately are available. These sets of circles are called constant M circles and constant N circles respectively. The $G(\,j\omega)$ plot is simply superimposed on the set of constant M circles. The points of intersections of $G(\,j\omega)$ plot with the constant M circles give the closed loop magnitude at different values of frequency. Similarly, by superimposing $G(\,j\omega)$ plot on the set of constant N circles, the closed loop phase angles are obtained for different values of frequency. Once $|M(j\omega)|$, $\angle M(j\omega)$ and ω are known, closed loop frequency response can be drawn either in polar coordinate, Bode diagram or

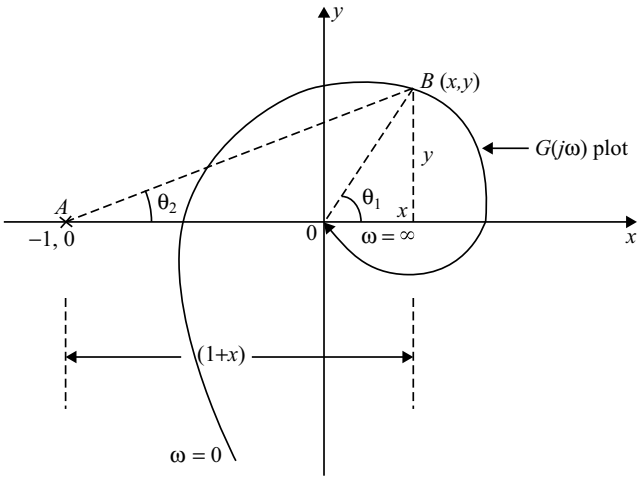


Fig. 3.62 $G(j\omega)$ plot in the complex x-y plane

log magnitude-phase plot. We give below, how the magnitude loci from constant M circles and the phase angle loci from constant N circles are obtained.

3.11.1 Constant M Circles

Consider the $G(j\omega)$ plot drawn in the complex $x - y$ plane as shown in Fig. 3.62. The symbols used in Fig. 3.62 have the same meaning as those used in Fig. 3.61b. From Fig. 3.62, at point B, the closed loop magnitude is obtained from Eq. (3.33) as

$$M(j\omega) = \frac{|\overrightarrow{OB}|}{|\overrightarrow{AB}|}$$

$$M = \frac{\sqrt{x^2 + y^2}}{\sqrt{(1+x)^2 + y^2}}$$

Squaring and cross multiplying, we get

$$M^2[(1+x)^2 + y^2] = x^2 + y^2$$

Rearranging the terms, we get (for $M < 1$)

$$(1 - M^2)x^2 + (1 - M^2)y^2 - 2M^2x = M^2 \tag{3.35}$$

For $M > 1$, the above equation becomes

$$(M^2-1)x^2+(M^2-1)y^2-2M^2x=M^2 \tag{3.36}$$

By dividing Eq. (3.35) by $(1-M^2)$ and adding $\left[\frac{M^2}{(1-M^2)}\right]^2$ on both sides we get

$$x^2+y^2+\frac{2M^2x}{(1-M^2)}+\left[\frac{M^2}{(1-M^2)}\right]^2=\left[\frac{M^2}{(1-M^2)}\right]^2+\frac{M^2}{(1-M^2)}$$
$$\left[x-\frac{M^2}{(1-M^2)}\right]^2+y^2=\left[\frac{M}{(1-M^2)}\right]^2 \tag{3.37}$$

Equation (3.37) represents a circle with centre at centre,

$$x=\frac{M^2}{(1-M^2)};\qquad y=0 \tag{3.38}$$

and

$$\text{radius, } r=\frac{M}{(1-M^2)}\quad \text{for } M<1 \tag{3.39}$$

For $M>1$, from Eq. (3.36), on similar line described above, we get

$$\left[x-\frac{M^2}{(M^2-1)}\right]^2+y^2=\left[\frac{M}{(M^2-1)}\right] \tag{3.40}$$

The centre and radius of the circle for any particular value of M is

$$\text{Center, } x=\frac{M^2}{(M^2-1)};\qquad y=0$$
$$\text{Radius, } r=\frac{M}{(M^2-1)} \tag{3.41}$$

For $M=1$, Eq. (3.35) is written as

$$x=-\frac{1}{2} \tag{3.42}$$

This is the vertical straight line passing through $x=-\frac{1}{2}$. Using Eqs. (3.38), (3.41) and (3.42), for any value of M chosen, the centre and radius using the above equations are drawn. For $M=\infty$, the circle converges to a point at $(-1,0)$. These circles are known as Constant M circles, and they are shown in Fig. 3.63. Ready-made constant M circles charts are also available.

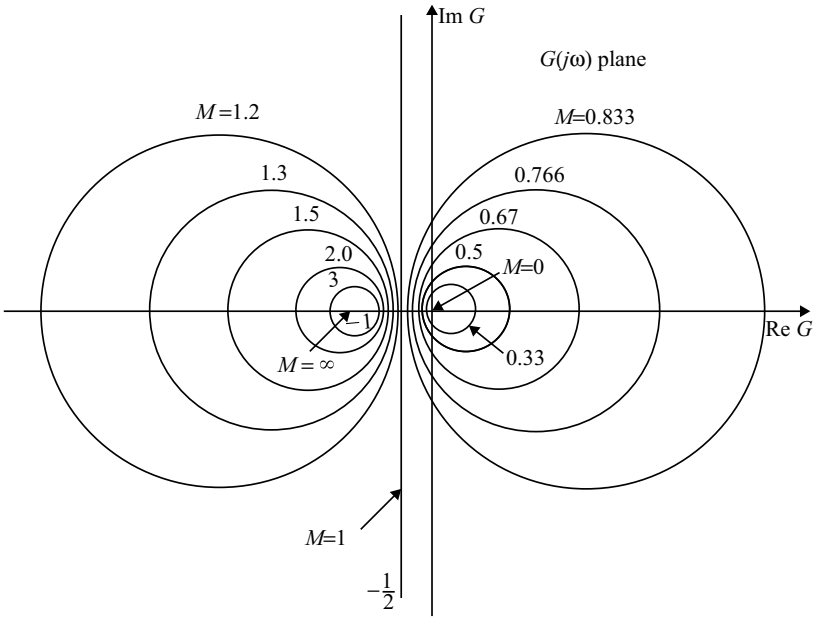


Fig. 3.63 Constant M circles

3.11.2 Constant N Circles

Similar to constant M circles, a set of circles are available to determine the closed loop phase angle from open loop frequency response. These circles are called constant N circles where $\tan \angle M(j\omega) = \tan^{-1} N$. Referring to Fig. 3.62

$$\angle M(j\omega) = \frac{\angle OB}{\angle AB} = (\theta_1 - \theta_2)$$

$\tan\theta_1 = \frac{y}{x}$ and $\tan\theta_2 = \frac{y}{1+x}$. Let

$$\begin{aligned}
\angle M(j\omega) &= \theta \\
\theta &= (\theta_1 - \theta_2) \\
\tan^{-1} N &= \tan^{-1}\frac{y}{x} - \tan^{-1}\frac{y}{1+x} \\
&= \tan^{-1}\frac{\frac{y}{x} - \frac{y}{1+x}}{1 + \frac{y^2}{x(1+x)}} \\
&= \tan^{-1}\frac{y}{x^2 + y^2 + x}
\end{aligned}$$

Taking tan on both sides, we get

$$N = \frac{y}{x^2 + y^2 + x}$$

$$x^2 + x + y^2 - \frac{y}{N} = 0$$

Adding $\left(\frac{1}{4} + \frac{1}{4N^2}\right)$ on both sides, we get

$$x^2 + x + y^2 - \frac{y}{N} + \left(\frac{1}{4} + \frac{1}{4N^2}\right) = \left(\frac{1}{4} + \frac{1}{4N^2}\right)$$

$$\left(x + \frac{1}{2}\right)^2 + \left(y - \frac{1}{2N}\right)^2 = \left[\sqrt{\left(\frac{1}{4} + \frac{1}{4N^2}\right)}\right]^2 \tag{3.43}$$

Equation (3.43) represents the circle with centre at $(-1/2, 1/2N)$ and radius $\sqrt{\frac{1}{4} + \frac{1}{4N^2}}$

$$\text{Center, } x = -\frac{-1}{2} \quad \text{and} \quad y = \frac{1}{2N}$$

$$\text{Radius, } r = \sqrt{\frac{1}{4} + \frac{1}{4N^2}} \tag{3.44}$$

For any chosen value of N, the centre and radius of the circles are obtained and the circle is drawn. Unlike M circles, the constant N circles are not marked with closed loop phase N. Instead N is converted into degrees. For example, for $N = 1$, $x = \frac{-1}{2}$, $y = \frac{1}{2}$; $r = \frac{1}{\sqrt{2}}$. The closed loop phase angle of the circle is not marked with $N = 1$ but $\theta = \tan^{-1} N = \tan^{-1} 1 = 45°$. Ready-made constant phase angle circles are available for use. The constant N circles are shown in Fig. 3.64.

3.11.3 The Nichols Chart

In Sects. 3.11.1 and 3.11.2, obtaining the closed loop frequency response from open loop frequency response using constant M and constant N circles was discussed. The data has to be obtained twice, once from constant M circles and the other time from the constant N circles. The laborious task is very much simplified by combining constant M circles and constant N circles in one chart called Nichols Chart which was introduced by Nathaniel B. Nichols (1914–1997). The Nichols chart is derived from constant M and N circles and represented in log magnitude-phase coordinates.

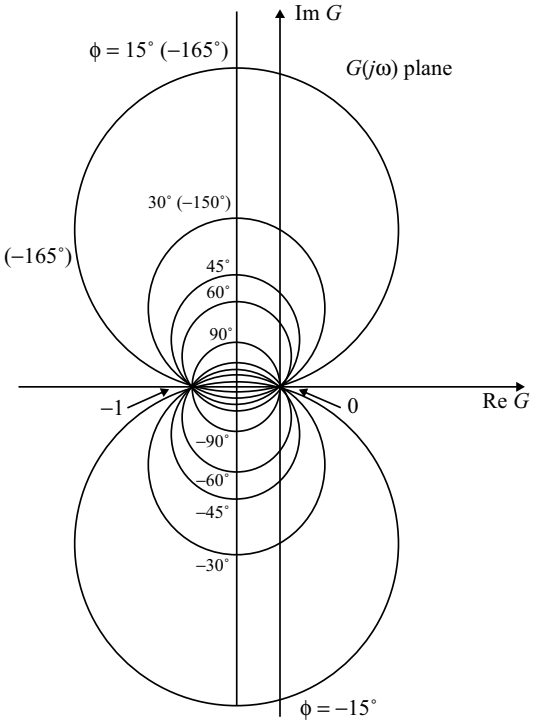


Fig. 3.64 Constant *N* circles

Advantages of Nichols Chart

1. The Nichols chart gives both the gain and phase characteristics of the closed loop system simultaneously at one instant.
2. The phase margin, gain margin, phase crossover and gain crossover, the critical point, the resonant peak, resonant frequency and bandwidth which are the frequency domain specifications can be easily and quickly determined from the open loop frequency response plot $G(j\omega)$.
3. Using Nichols chart, the closed loop frequency response can be easily obtained from the open loop frequency response.
4. The required gain of the system can be obtained for the desired phase margin, gain margin, bandwidth and resonant peak.

A Historical Note: Natheniel B. Nichols (1914–1997)


Nathaniel B. Nichols

N.B. Nichols was born in 1914 in Michigan, USA. He received his B.S. degree from Central Michigan University in 1936 and M.S. degree in physics in 1937 from the University of Michigan. He joined Taylor Instruments in Rochester in 1937 as a process control engineer and worked with John Ziegler. These two intellectuals worked together at Taylor and their job included determining means for tuning PID controllers and successfully developed Ziegler–Nichols tuning parameters which were rough approximations to optimal settings for open loop T.F. to ensure prescribed behaviour in closed loop PID realizations. After working for several years at Taylor, Nichols turned his attention to technical innovations of war weapon design. It was the beginning of World War II then. His innovative ideas included the then emerging field of control of radar and ground anti-aircraft gun and angle tracking systems. Nichols developed a graphical design aid to design this system. This graphical aid now we call as Nichols chart forms a standard chapter in automatic control theory. Nichols contributions to the concepts, design and performance of electric drive systems became very popular worldwide at that time and it was considered several orders of magnitude greater than the introduction of the famous Ward Leonard drives. In 1987, Nichols visited China to participate in technical conference. On that occasion every control engineer in China wanted to meet Dr. Nichols, talk with him and sit next to him for breakfast, lunch and dinner. Everyone wanted to have a personal conversation with this great engineer who had made feedback control theory understandable. His influence on the control field has been continuous since 1942 and for every contribution he was honoured by U.S. government. On April 17, 1997, Nick died after a prolonged illness.

3.11.4 Nichols Chart from Constant M and Constant N Circles

The Nichols chart is a tool for the designer to read off closed loop gain and phase directly from the plot of the open loop frequency response. It is derived from constant M and constant N circles. When constant M circles are transferred to log magnitude-phase plot, then it becomes constant dB loci. Similarly when constant N circles are transferred to log magnitude-phase plot, they become constant phase loci. Consider Fig. 3.65, where a constant M circle is shown. Let us consider the circle with $M = 1.2$. The centre is

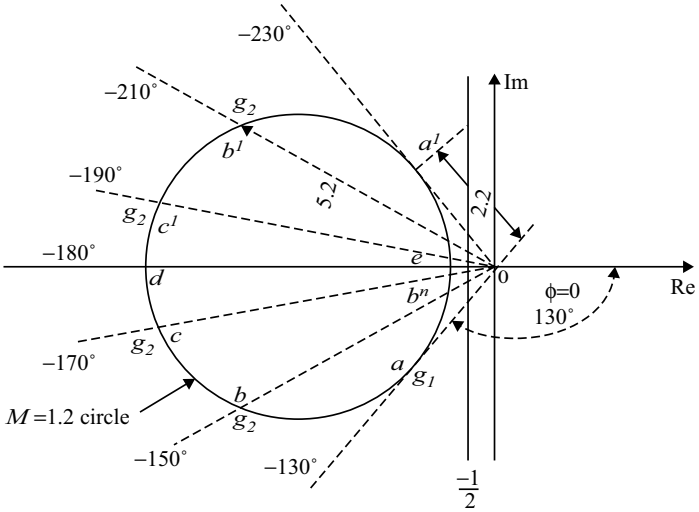


Fig. 3.65 Constant M-circles in $G(j\omega)$ plane

$$c = \frac{M^2}{M^2 - 1} = \frac{1.44}{0.44} = 3.2727$$

and radius

$$r = \frac{M}{M^2 - 1} = \frac{1.2}{1.44 - 1} = 2.7272$$

The corresponding dB to $M = 1.2$ is $20 \log 1.2 = 1.58$ dB.

ϕ	−230°	−210°	−190°	−180°	−170°	−150°	−130°
g_1	2.2	1.0	0.8	0.81	0.8	1.0	2.2
$20 \log g_1$	6.85	0.0	−1.94	−1.95	−1.94	0.0	6.85
g_2	–	5.2	6.2	6.21	6.2	5.2	–
$20 \log g_2$	–	14.32	15.85	15.86	15.85	14.32	–

Radial lines from the origin are drawn to the constant M circle. Consider the line oa^1. The phase angle is −230° and the distance $g_1 = 2.2$ which corresponds to $20 \log 2.2 = 6.85$ dB. Locate the point a^1 in the log magnitude-phase coordinate in Fig. 3.66. Draw another radial line ob^1. The angle this radial line makes with the reference axis is $\phi = -210°$. It intersects the constants M circle at $g_1 = 1$ and $g_2 = 5.2$. Corresponding to g_1, the dB $= 20 \log 1 = 0$ and corresponding to $g_2 = 5.2$, the dB $= 20 \log 5.2 = 14.32$. These points are fixed at b^1 and b^{11} respectively in Fig. 3.66. Thus, for different radial lines, ϕ and dB are calculated from this constant M circle and transferred to log magnitude-phase coordinate. The complete constant dB locus for $M = 1.2$ is shown in Fig. 3.66 as 1.58 dB curve. By similar procedure

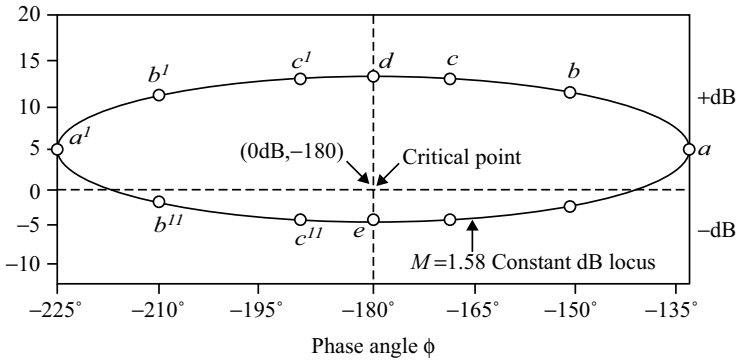


Fig. 3.66 Constant dB locus for $M = 1.2$ or dB $= 1.58$

other constant M circles are transferred to the log magnitude phase coordinate. A set of constant dB lines are now available for use.

Now consider any constant N circle. The same procedure as was done for transferring the constant M circles, constant N circles is transferred to the log magnitude-phase coordinate. These loci are called constant phase angle loci.

The constant dB loci together with constant phase angle loci is called Nichols chart which is shown in Fig. 3.67. Nichols charts are readily available and can be used to solve control system problems very easily and quickly. From Fig. 3.67, it is seen that the Nichols chart is symmetric about the $-180°$ vertical axis. The point $(0\,\text{dB}, -180°)$ is called the critical point. The $G(j\omega)$ plot which is tangential to any one of constant dB lines gives the resonant peak M_r and the corresponding frequency is the resonant frequency ω_r. The bandwidth of the closed loop system is obtained at the intersection point of $G(j\omega)$ plot with $M = -3\,\text{dB}$ locus. If the gain of the open loop system is small, the $G(j\omega)$ will not be tangent to any of the constant dB loci and for such cases the resonant peak does not exist for the closed loop system.

Example 3.37

A certain unity feedback control system has the following open loop transfer function

$$G(s) = \frac{15}{(s+1)(s+2)(s+3)}$$

Draw the $G(j\omega)$ plot in log magnitude-phase plot and determine resonant peak M_r, resonant frequency ω_r and BW of the closed loop system. Also determine the phase margin ϕ_m, gain margin G_m, phase crossover frequency ω_{pc} and gain crossover frequency ω_{gc}.

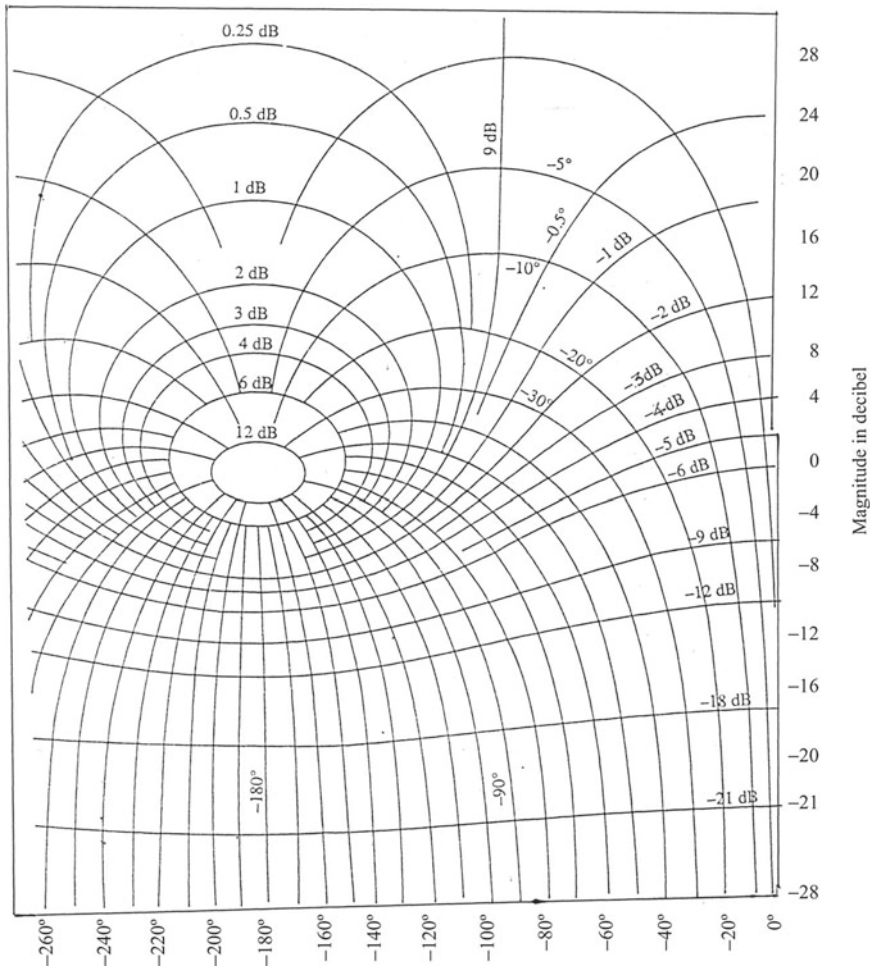


Fig. 3.67 Nichols chart derived from constant M and constant N circles

Solution

$$G(s) = \frac{15}{(s+1)(s+2)(s+3)}$$

$$G(j\omega) = \frac{15}{(j\omega+1)(j\omega+2)(j\omega+3)}$$

$$|G(j\omega)| = \frac{15}{\sqrt{(\omega^2+1)(\omega^2+4)(\omega^2+9)}}$$

$$\phi = \angle G(j\omega) = -\tan^{-1}\omega - \tan^{-1}\frac{\omega}{2} - \tan^{-1}\frac{\omega}{3}$$

$$\text{dB} = 20\log 15 - 20\log\sqrt{(\omega^2+1)} - 20\log\sqrt{(\omega^2+4)} - 20\log\sqrt{(\omega^2+9)}$$

For different values of ω, ϕ and dB are calculated and tabulated below

ω	0.5	1.5	2.5	3.5	11	40
$\angle G(j\omega)$	$-50°$	$-120°$	$-159°$	$-183°$	$-239°$	$-261°$
$\text{dB} = 20\log\lvert G(j\omega)\rvert$	6.6	-0.07	-7.0	-13.1	-39.5	-72.6

The frequency response data in the table is transferred to log magnitude phase Nichols plot and is shown in Fig. 3.68. The following results are obtained from Fig. 3.68.

1. The phase margin and gain crossover frequency are obtained where the $G(j\omega)$ plot intersects 0 dB horizontal axis. This intersection is at point B in Fig. 3.68. $\phi_m = 180° - 119° = 61°$ and $\omega_{gc} = 1.49\,\text{rad/sec}$
2. The gain margin and phase crossover frequency ω_{pc} are obtained where the $G(j\omega)$ plot intersects $-180°$ vertical axis at point D. $G_m = 12\,\text{dB}$ and $\omega_{pc} = 3.32\,\text{rad/sec}$

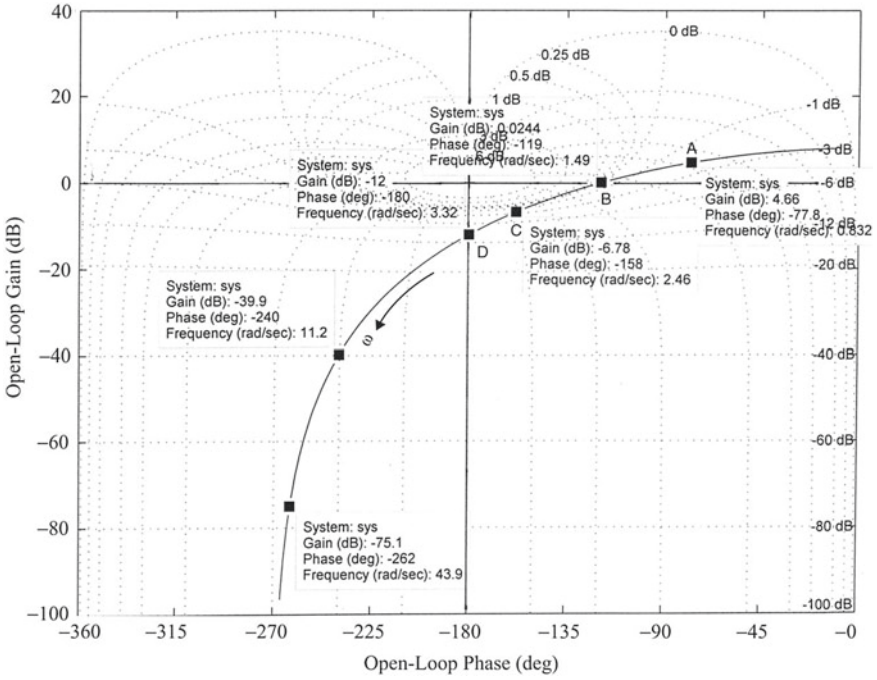


Fig. 3.68 Nichols plot for $G(s) = \frac{15}{(s+1)(s+2)(s+3)}$

3. The bandwidth of the closed loop system is obtained from the intersection of the $G(j\omega)$ plot with -3 dB constant phase angle locus. This intersection is at point C in Fig. 3.68. This dB is different from the dB marked in the vertical axis. The frequency at this point is BW which is denoted by $\omega_b = 2.46$ rad/sec
4. The resonant peak M_r and resonant frequency ω_r are obtained from the intersecting point of $G(j\omega)$ plot with a constant dB locus which is tangential to it. From Fig. 3.68, at point B, the $G(j\omega)$ plot is just tangential to 0 dB constant dB locus. Hence, $M_r = 0$ dB or $M_r = 1$. The corresponding frequency is the resonant frequency $\omega_r = 1.49$ rad/sec

Resonant peak, M_r	$= 1$
Resonant frequency, ω_r	$= 1.49$ rad/sec
Bandwidth, BW ω_b	$= 2.46$ rad/sec
Phase margin, ϕ_m	$= +61^\circ$
Gain crossover frequency, ω_{gc}	$= 1.49$ rad/sec
Gain margin, G_m	$= +12$ dB
Phase crossover frequency, $\omega_p c$	$= 3.32$ rad/sec

$\omega_{pc} > \omega_{gc}$; ϕ_m and G_m are $+ve$. The closed loop system is stable.

Example 3.38

Consider the following open loop T.F. of a certain closed loop system

$$G(s) = \frac{K(1 + 0.1s)}{s(1 + 0.5s)(1 + 0.2s)}$$

Find the value of K so that the resonant peak of the closed loop system is 6 dB.

Solution

$$\begin{aligned} G(s) &= \frac{K(1 + 0.1s)}{s(1 + 0.5s)(1 + 0.2s)} \\ &= \frac{K(s + 10)}{s(s + 2)(s + 5)} \\ G(j\omega) &= \frac{K(j\omega + 10)}{j\omega(j\omega + 2)(j\omega + 5)} \\ |G(j\omega)| &= \frac{K\sqrt{(\omega^2 + 100)}}{\omega\sqrt{(\omega^2 + 4)(\omega^2 + 25)}} \\ \phi &= \tan^{-1} 0.1\omega - 90^\circ - \tan^{-1} 0.5\omega - \tan^{-1} 0.2\omega \end{aligned}$$

Let $K = 1$

$$|G(j\omega)| = \frac{\sqrt{(\omega^2 + 100)}}{\omega\sqrt{(\omega^2 + 4)(\omega^2 + 25)}}$$

For different values of ω, dB $= 20 \log |G(j\omega)|$ and ϕ are calculated and are tabulated below

ω	0.5	1.0	1.5	6	30
ϕ	$-104°$	$-122°$	$-135°$	$-181°$	$-184.5°$
$2\,\text{dB} = 20 \log \|G(j\omega)\|$	5.7	-1.1	-5.73	-28.1	-88

The frequency response plot of $G(j\omega)$ is represented in Nichols chart as shown in Fig. 3.69.

From Fig. 3.69, the $G(j\omega)$ plot is not tangential to 6 dB locus. The $G(j\omega)$ plot is vertically shifted upwards. At point A it is just tangential to 6 dB constant magnitude locus. The amount of the shift of point B which is vertically below to point A is measured as 11 dB +ve. The required gain is calculated from

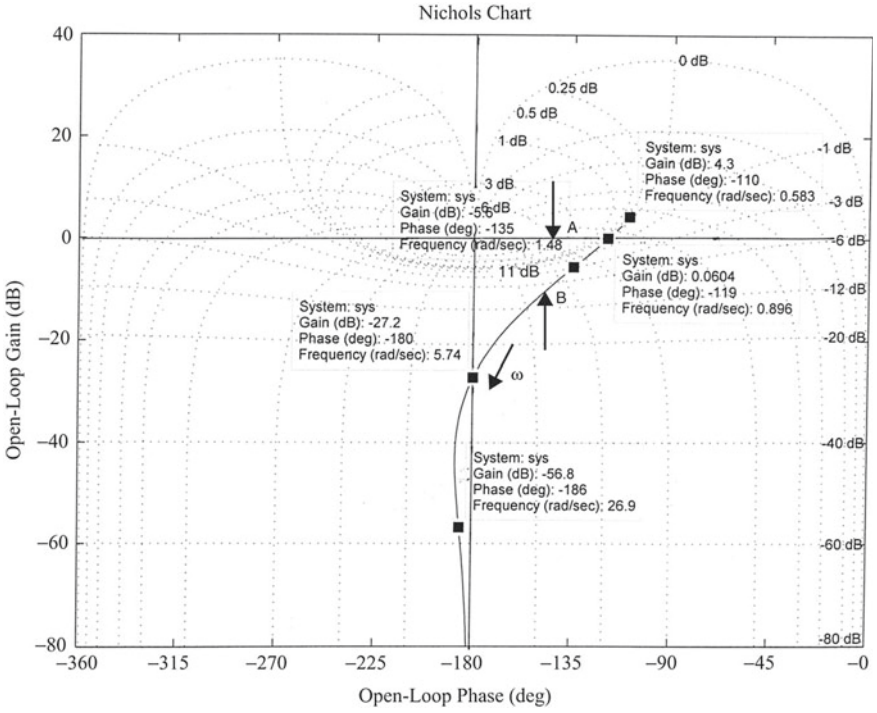


Fig. 3.69 Frequency response in Nichols chart for $G(s) = \dfrac{(1 + 0.1s)}{s(1 + 0.5s)(1 + 0.2s)}$

$$20 \log K = 11$$

$$\boxed{K = 3.55}$$



Note: If the vertical shift is downwards, the dB is measured negative. For example, for a downward shift of 11 dB, the value of K may be calculated as

$$20 \log K = -11$$
$$K = \frac{1}{3.55} = 0.2817$$

Example 3.39

A certain unity feedback control system has the following open loop T.F.

$$G(s) = \frac{10\,K}{(s+1)(s+2)(s+3)}$$

It is desired to have a resonant peak $M_r = 6$ dB. or $M_r = 2$. Determine the value of K. For this value of K, find ϕ_m, G_m, ω_{pc}, ω_{gc}. Is the closed loop system stable?

Solution

$$G(s) = \frac{10\,K}{(s+1)(s+2)(s+3)}$$

Put $K = 1$

$$|G(j\omega)| = \frac{10}{\sqrt{(\omega^2+1)(\omega^2+4)(\omega^2+9)}}$$
$$\angle G(j\omega) = \phi = -\tan^{-1}\omega - \tan^{-1}\frac{\omega}{2} - \tan^{-1}\frac{\omega}{3}$$

For different values of ω, ϕ, $|G(j\omega)|$ and $\text{dB} = 20 \log |G(j\omega)|$ are calculated and tabulated below

ω	1	2	3	4	25
ϕ	$-90°$	$-142°$	$-173°$	$-193°$	$-254°$
$\lvert G(j\omega)\rvert$	1	0.4385	0.2067	0.108	6.33×10^{-4}
$\text{dB} = 20 \log \lvert G(j\omega)\rvert$	0	-7.16	-13.69	-19.3	-64

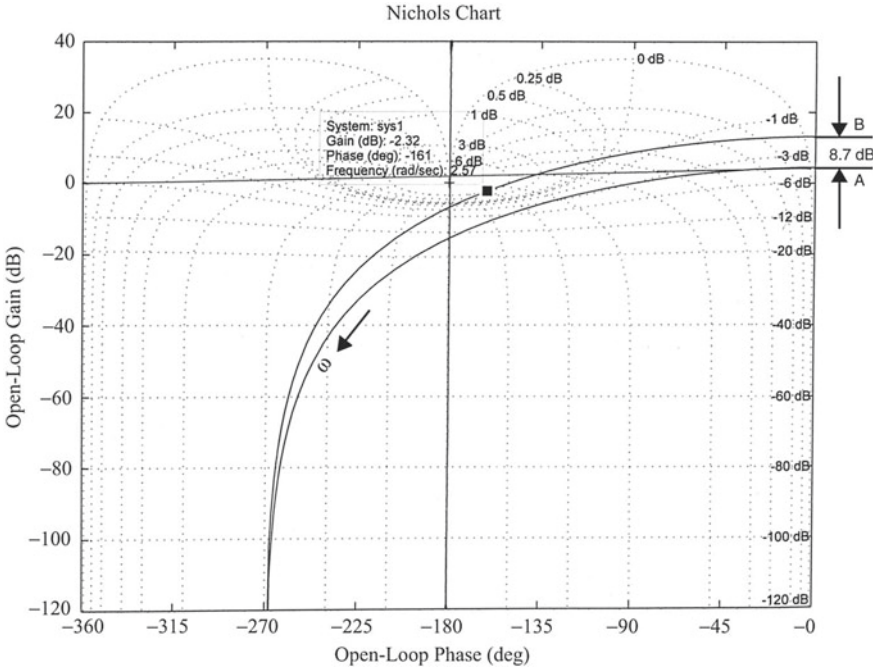


Fig. 3.70 Determine of K for $M_r = 6\text{dB}$ $G(s) = \dfrac{10K}{(s+1)(s+2)(s+3)}$

The above data is transferred to Nichols chart. This is shown in Fig. 3.70 as curve A. The graph A is vertically shifted up until any point in the graph is just tangential to the 6 dB constant dB locus. The curve A when shifted vertically up is drawn as curve B. The amount of vertical shift is observed as 8.7. Since the shift is upwards

$$20 \log K = 8.7$$

$$\boxed{K = 2.72}$$

The new T.F. of the system now becomes

$$\boxed{G(s) = \frac{27.2}{(s+1)(s+2)(s+3)}}$$

The following MATLAB program may be used to get graph A and graph B.

MATLAB Program

Nichols plot

Gain Adjustment for desired M_r

$G(s) = 10K/(s+1)(s+2)(s+3)$

```
num=[0 0 0 10]
den=conv([1 3 2],[1 3]);
sys=tf(num,den);
[Gm, pm, wcp, wcg]=margin(sys);
[Gm dB, pm, wcp, wcg]
num=[0 0 0 27.2]
den1=conv([1 3 2],[1 3]);
sys1=tf(num1,den1);
[Gm1, pm1, wcp1, wcg1]=margin(sys1);
[Gm dB1, pm1, wcp1, wcg1]
Nichols (sys,sys1)

ans=
  15.5630   90.0000   3.3166   1.0000

ans=
  6.8717   29.6873   3.3166   2.2222
```

The answer for G_m, ϕ_m, ω_{pc}, ω_{gc} for $K = 1$ and $K = 2.72$ is given at the end of the program. In both cases, ϕ_m and G_m are +ve and $\omega_{pc} > \omega_{gc}$. Hence, the closed loop system is stable. Further, $M_r = 6$ dB or $M_r = 2$. Resonant frequency is obtained as $\omega_r = 2.5$ rad/sec

---

Example 3.40

Consider the system shown in Example 3.39 where

$$G(s) = \frac{10\,K}{(s+1)(s+2)(s+3)}$$

Find the value of K so that the gain margin is +30 dB. Use Nichols plot.

Solution

$$G(s) = \frac{10\,K}{(s+1)(s+2)(s+3)}$$

Put $K = 1$;

$$|G(j\omega)| = \frac{10}{\sqrt{(\omega^2+1)(\omega^2+4)(\omega^2+9)}}$$
$$\angle G(j\omega) = -\tan^{-1}\omega - \tan^{-1}\frac{\omega}{2} - \tan^{-1}\frac{\omega}{3}$$

The frequency response data prepared in Example 3.39 is used and is plotted in Nichols chart as shown in Fig. 3.71. Graph A represents $G(j\omega)$ plot for $K = 1$. with $\omega_{pc} = 3.32$ and gain $= -15.5$ dB. The curve is shifted down so that it reaches point B where the gain is -30 dB. Hence, the decrease in dB is $-(30 - 15.5) = -14.5$ dB. The required gain is calculated from

$$20\log K = -14.5$$
$$K = 0.188$$

The T.F. of the system becomes

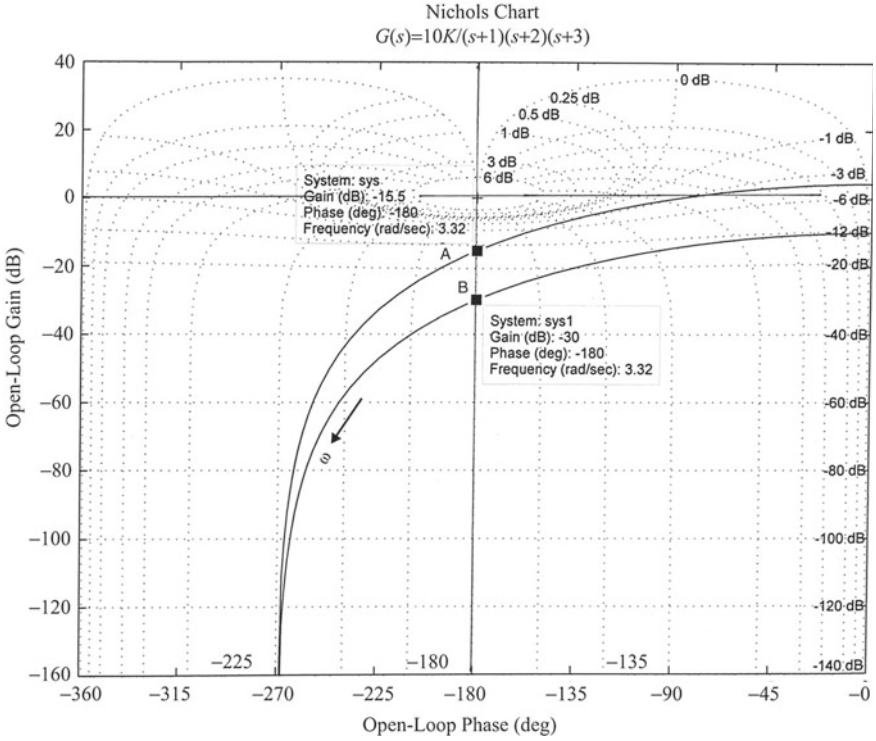


Fig. 3.71 Gain Adjustment for the desired $G_m = 30$ dB; $G(s) = \dfrac{10K}{(s+1)(s+2)(s+3)}$

$$\boxed{G(s) = \frac{1.88}{(s+1)(s+2)(s+3)}}$$

Example 3.41

Consider the Example 3.39 where

$$G(s) = \frac{10\,K}{(s+1)(s+2)(s+3)}$$

Determine the value of K so that phase margin of the system is 45°.

Solution

$$G(s) = \frac{10\,K}{(s+1)(s+2)(s+3)}$$

For $K = 1$, the data prepared in Example 3.39 is used to plot $G(\,j\omega)$ in Nichols chart and is shown in Fig. 3.72 as curve A. The phase margin is now found as 90°. The point A is vertically shifted up. Now the G_m is decreased. ϕ_m is also decreased. ω_{gc} is also decreased. But ω_{pc} is unchanged. The vertical shift is made until the phase margin is 45° as seen from graph B. From point A to point B, the upward shift is $(15.6 - 9.64) = 5.96$. The corresponding value of K is calculated from

$$20 \log K = 5.96$$

$$\boxed{K = 2}$$

The T.F. becomes

$$G(s) = \frac{20}{(s+1)(s+2)(s+3)}$$

$G_m = +9.64$; $\omega_{gc} = 1.82$; $\omega_{pc} = 3.33$.

Example 3.42

Consider the system shown below, where

$$G(s) = \frac{K(s+0.5)}{(s+1)(s+2)(s+3)}$$

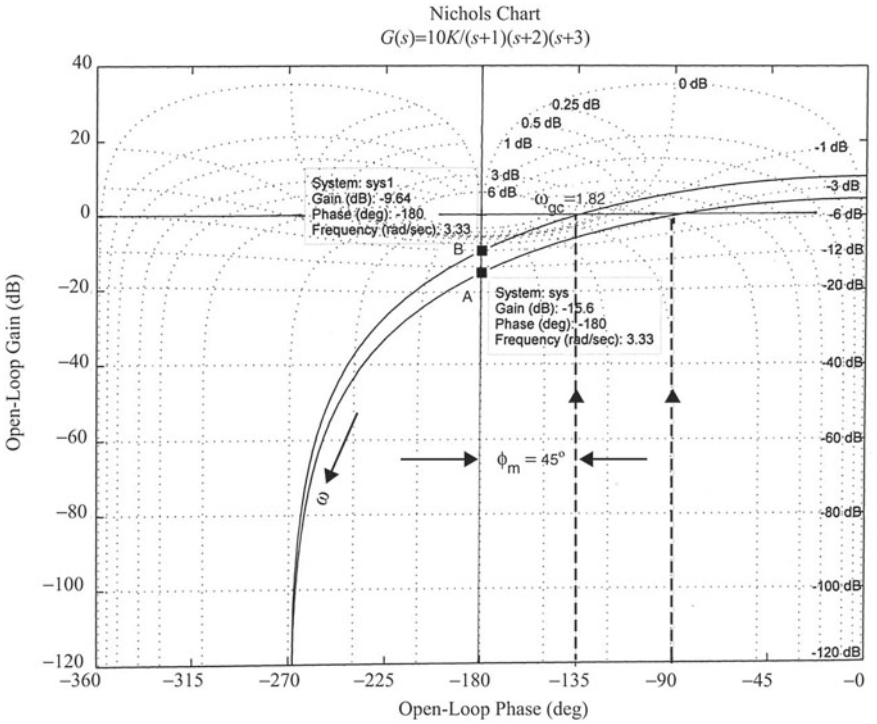


Fig. 3.72 Gain adjustment to get the desired $\phi_m = 45°$; $G(s) = \dfrac{10K}{(s+1)(s+2)(s+3)}$

It is desired to have a BW= 11 rad/sec by adjusting the gain K. Determine the value of K. Under this condition find ϕ_m and G_m.

Solution

$$G(s) = \frac{K(s+0.5)}{(s+1)(s+2)(s+3)}$$

Putting $K = 1$;

$$G(s) = \frac{(s+0.5)}{(s+1)(s+2)(s+3)}$$

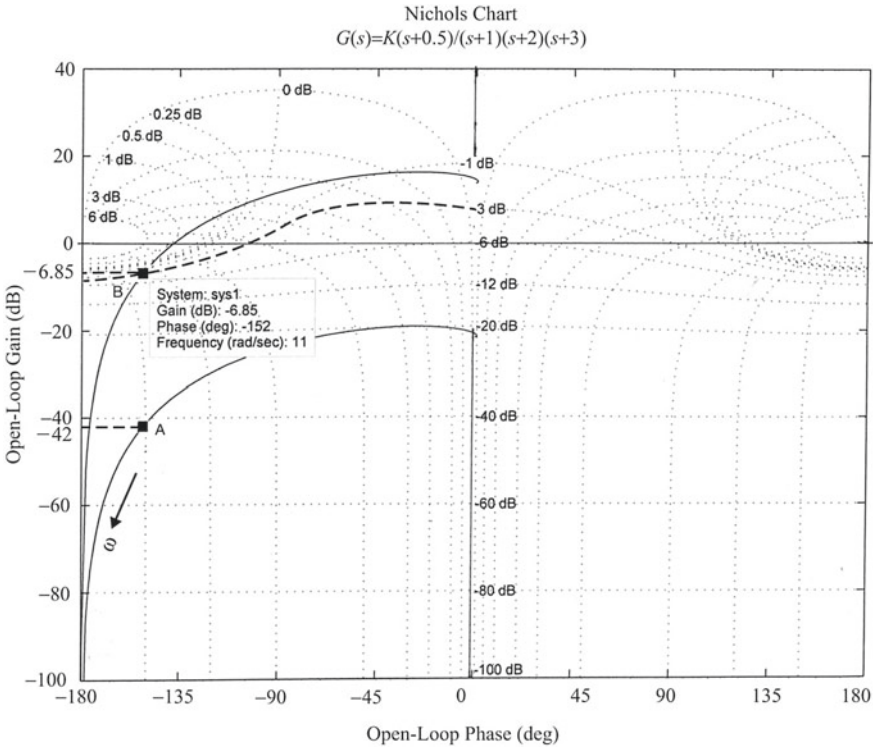


Fig. 3.73 Gain adjustment for the desired BW= 11 rad/sec $G(s) = \dfrac{K(s+0.5)}{(1+s)(s+2)(s+3)}$

ω	1	3	5	8	11	15
$\phi = \angle G(j\omega)$	$-26.6°$	$-92.32°$	$-121.6°$	$-142°$	$-151.8°$	$-159.2°$
$\lvert G(j\omega)\rvert$	0.118	0.062	0.0313	0.01422	7.82×10^{-3}	4.31×10^{-3}
dB	-18.56	-24	-30	-37	-42	-47.3

The data for $G(j\omega)$ plot is prepared and used to plot $G(j\omega)$ in Nichols chart and is shown in Fig. 3.73 as curve A. The curve does not intersect -3 dB constant dB plot. Hence, the curve A is shifted vertically upwards until -3 dB locus is intersected. Further it is shifted such that at the point of intersection the value of ω on $G(j\omega)$ curve is 11 rad/sec. This corresponds to the required BW. At point B of the shifted curve $\omega = 11$. The amount of vertical shift of point A to point B in Fig. 3.73 is obtained as 35.15 dB. The value of K is calculated from

$$20 \log K = 35.15$$

$$\boxed{K = 57}$$

$$G(s) = \frac{57(s+0.5)}{(s+1)(s+2)(s+3)}$$

From Fig. 3.73, we obtain

$$\begin{aligned} \phi_m &= +42.3^\circ \\ G_m &= +\infty \\ \omega_{gc} &= 7.15\,\text{rad/sec} \\ \omega_b &= 11\,\text{rad/sec} \\ K &= 57 \end{aligned}$$

Example 3.43

A certain unity feedback control and system has the following open loop T.F.

$$G(s) = \frac{e^{-10T_d s}}{s(s+2)(s+8)}$$

Determine the value of T_d so that the phase margin is 45°. What is the gain crossover frequency?

Solution

$$G(s) = \frac{e^{-10T_d s}}{s(s+2)(s+8)}$$

Let $T_d = 0$

$$\begin{aligned} G(s) &= \frac{1}{s(s+2)(s+8)} \\ |G(j\omega)| &= \frac{1}{\omega\sqrt{(\omega^2+4)(\omega^2+64)}} \\ \angle G(j\omega) = \phi &= -90^\circ - \tan^{-1}\frac{\omega}{2} - \tan^{-1}\frac{\omega}{8} \end{aligned}$$

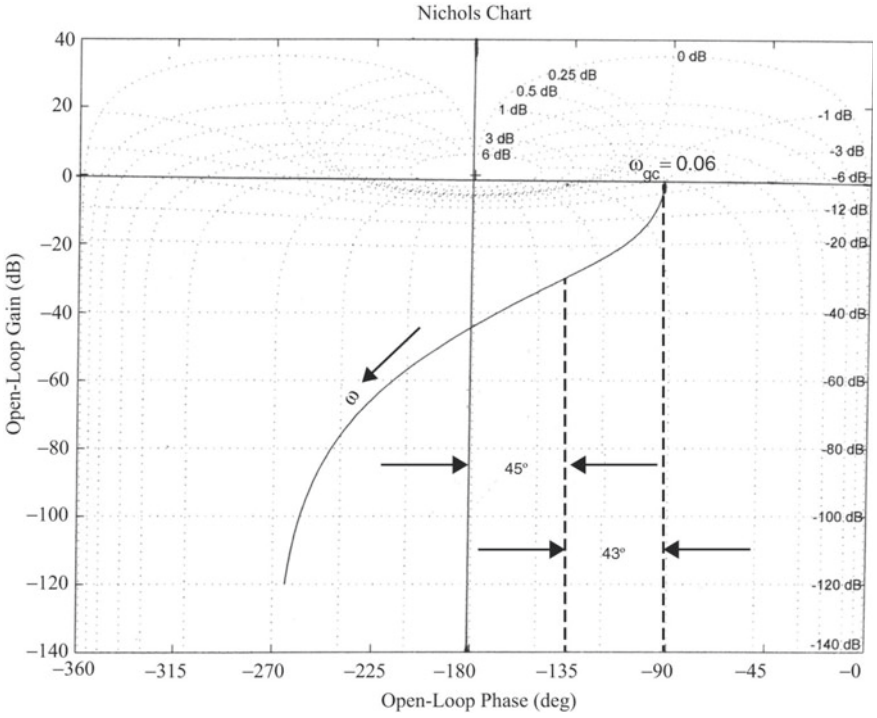


Fig. 3.74 Determination of transportation lag for the required phase margin $\phi_m = 45°$; $G(s) = e^{-10T_d s}/s(s+2)(s+8)$

The following table is prepared.

ω	0.06	0.5	1	2	4	10
$\angle G(j\omega)$	$-92°$	$-108°$	$-124°$	$-149°$	$-180°$	$-220°$
$\lvert G(j\omega)\rvert$	1	0.121	0.0555	0.0214	6.25×10^{-3}	7.66×10^{-4}
dB	0	-18.3	-25.1	-33.4	-44	-62.3

The $G(j\omega)$ plot without time delay is shown in Fig. 3.74. It is noted that $\phi_m = 88°$ at $\omega_{gc} = 0.06$ rad/sec. The required phase margin is 45°. Hence, the $G(j\omega)$ plot is horizontally shifted towards left by an $\angle -43°$ which makes the $\phi_m = 45°$ with time delay. The magnitude contribution by time delay is 0 dB at all frequency. Hence, at $\omega = \omega_{gc} = 0.06$, the phase of $-43°$ is contributed by $e^{-T_d s}$. In other words

$$-10T_d\omega_{gc} \times 57.3 = -43$$
$$10T_d \times 0.06 \times 57.3 = 43$$

$$\boxed{T_d = 1.25 \text{ sec}}$$

SUMMARY

1. The frequency response is the plot of magnitude and phase difference between the input and output sinusoids over a range of frequencies.
2. It has several advantages and a few disadvantages too in comparison with time domain approach.
3. When sinusoidal input is applied to a linear system, the output is also sinusoidal with a different amplitude and a phase difference at steady state.
4. The ratio of the output to input sinusoid is called magnitude or amplitude ratio and the phase difference is called phase angle which are functions of frequency. The magnitude and phase angle of the given linear system is obtained by replacing s by $j\omega$ in the T.F. provided the given linear system is stable.
5. The plot of magnitude ratio and phase angle with frequency as the variable parameter is called frequency response plots.
6. The frequency response can be represented in graph which can be used for the analysis and design. The very extensively used methods include (a) Polar plot, (b) Bode plot and (c) Log magnitude phase (Nichols) plot. Each of these methods has its own merits and demerits.
7. A correlation between time domain and frequency domain can be established for a second order system. Similar to time domain specifications which are expressed in terms of damping ratio and natural frequency of oscillations, in the frequency domain, expressions for resonant peak M_r, resonant frequency ω_r and the bandwidth ω_b are derived. Thus, there exists a one-to-one relationship between time domain and frequency domain specifications. However, such a relationship is valid only for a second order system.
8. In the frequency domain, other important specifications such as gain margin, phase margin, gain crossover frequency and phase crossover frequency are defined which are applicable to higher order systems also.
9. Gain margin and phase margin are the important measures of relative stability of linear system. The higher the phase margin and gain margin are the higher is the degree of stability.
10. For the linear system to be stable, both phase margin and gain margin should be positive. If any one of them is negative then the system is unstable. Further the condition $\omega_{pc} \geq \omega_{gc}$ should be satisfied.
11. Frequency response methods not only give information about the stability of the closed loop system but they also give information about the transient response of the system *via* phase margin, gain margin, bandwidth resonant peak and resonant frequency.
12. Obtaining frequency response of an open loop system by analytical methods as well as by experimental method is easy. However, due to non-availability of closed loop poles, obtaining closed loop frequency response is somewhat difficult if not impossible.

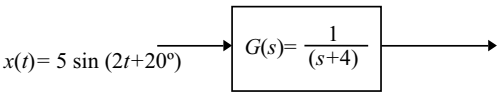
13. Closed loop frequency response can be obtained from open loop frequency response by using graphical aids such as constant M and constant N circles and Nichols chart.
14. With the availability of computers and the MATLAB software today, frequency response plots described in this chapter which are drawn manually can be obtained very quickly and accurately. Side by side the required MATLAB program and the plots obtained are given.

EXERCISE

Short Answer Type Questions

1. **Define frequency response.**
   The frequency response is the steady state response of the system to a sinusoidal input whose frequency is varied over a certain range.
2. **What are the two components of frequency response?**
   The two components of frequency response are the magnitude component and phase component.
3. **What is a sinusoidal transfer function?**
   In the transfer function of a stable system if s is replaced by $j\omega$, then $G(s)$ becomes $G(j\omega)$. $G(j\omega)$ is called the sinusoidal T.F.
4. **How magnitude and phase components of frequency response is obtained for the given T.F.?**
   In the T.F. $G(s)$, if s is replaced by $j\omega$, then the sinusoidal T.F. is obtained as $G(j\omega)$ which is a complex quantity. $|G(j\omega)|$ which is the ratio of the output amplitude to input amplitude gives the magnitude component. $\angle G j\omega$ gives the phase shift of the output from the input and this represents the phase component of the frequency response.
5. **Mention any three major advantage of frequency response method.**
   The advantages of frequency response methods are as follows:
   (a) It is easy to generate sine wave and collect data to draw frequency response which can be used for design purpose.
   (b) The frequency response tests involve measurement under steady state conditions which are simpler to analyse.
   (c) It can be applied to a certain class of non-linear system also.
6. **Mention any two disadvantages of frequency response method.**
   The disadvantages of frequency response methods are as follows:
   (a) It is not often easy to correlate the results of the frequency response characteristics of higher order system (order greater than two) to that of time domain characteristics.
   (b) In frequency response method, the analysis is done at one frequency at a time which requires very long test time.

Fig. 3.75 T.F. for Question 7



7. **For the system shown in Fig. 3.75 find the output.**

$$\begin{aligned} G(j\omega)|_{\omega=2} &= \frac{1}{j2+4} \\ &= \frac{1}{4.472}\angle -63.4^\circ \\ y(t) &= 1.118\sin(2t-43.4^\circ) \end{aligned}$$

8. **What are the magnitude and phase angle components of a prototype second order system at $\omega = \omega_n$?**

$$\text{The magnitude component } |G(j\omega)| = 1/2\zeta$$
$$\text{The phase angle component } \angle G(j\omega) = \angle -90^\circ$$

9. **What are the methods available to represent the frequency response in graph?**
   The methods available to represent the frequency response in graph are as follows:
   (a) Linear frequency response plots *vs.* ω
   (b) Linear frequency response plots *vs.* $\log \omega$
   (c) Polar (Nyquist) plot
   (d) Bode plot
   (e) Log magnitude *vs.* phase (Nichols) plot.
10. **What is a polar (Nyquist) plot?**
    Polar plot is the plot of the imaginary *versus* the real part of the complex sinusoidal T.F. $G(j\omega)$ drawn in a polar graph with ω as the varying parameter.
11. **What are the advantages and disadvantages of polar plot representation of frequency response?**
    The advantage of polar plot representation of frequency response is that both amplitude and phase angle are displayed on a single plot over a wide frequency range. It has the property to show clearly, from the open loop frequency response plot whether the closed loop system is stable or unstable. The famous Nyquist stability criterion has been developed using polar plot only. The only disadvantage of polar plot is that it does not clearly indicate the contributions made by each pole and zero of the T.F.
12. **What part of the T.F. influences the starting portion of a polar plot?**
    The Type of the system decides the starting portion of the polar plot. For example,

the polar plot of a type o system starts from +ve real axis and a Type 1 system starts from $-j\infty$.

13. **What part of the T.F. influences the ending part of a polar plot?**
The difference in the number of poles and zeros of the T.F. decides the ending portion of the polar plot. For example, if there are 4 poles and one zero, the polar plot ends in the second quadrant at the origin with a phase angle of $-(4-1)\pi = -270°$.

14. **For the following T.Fs. determine the magnitude and phase angle of the sinusoidal T.F.**

$$\text{(a)} \quad G(s) = \frac{K(s+10)}{s(s+5)}$$
$$\text{(b)} \quad G(s) = \frac{K(s+10)}{s^2(s+5)}$$
$$\text{(c)} \quad G(s) = \frac{K(s-10)}{s(-s+5)}$$
$$\text{(d)} \quad G(s) = \frac{-K(s+10)}{s^2(s+5)}$$
$$\text{(e)} \quad G(s) = \frac{K(s+10)}{s^2+4s+16)}$$

Ans:

$$\text{(a)} \quad G(j\omega) = \frac{K\sqrt{(\omega^2+100)}}{\omega\sqrt{(\omega^2+25)}} \angle \tan^{-1}\frac{\omega}{10} - 90° - \tan^{-1}\frac{\omega}{5}$$
$$\text{(b)} \quad G(j\omega) = \frac{K\sqrt{(\omega^2+100)}}{\omega^2\sqrt{(\omega^2+25)}} \angle \tan^{-1}\frac{\omega}{10} - 180° - \tan^{-1}\frac{\omega}{5}$$
$$\text{(c)} \quad G(j\omega) = \frac{K\sqrt{(\omega^2+100)}}{\omega\sqrt{(\omega^2+25)}} \angle 90° - \tan^{-1}\frac{\omega}{10} + \tan^{-1}\frac{\omega}{5}$$
$$\text{(d)} \quad G(j\omega) = \frac{K\sqrt{(\omega^2+100)}}{\omega^2\sqrt{(\omega^2+25)}} \angle \tan^{-1}\frac{\omega}{10} - \tan^{-1}\frac{\omega}{5}$$
$$\text{(e)} \quad G(j\omega) = \frac{K\sqrt{(\omega^2+100)}}{(16-\omega^2)^2+16\omega^2} \angle \tan^{-1}\frac{\omega}{10} - \tan^{-1}\frac{4\omega}{(16-\omega^2)}$$

15. **For the following T.F.s Sketch the polar plots.**

$$\text{(a)} \quad G(s) = K(s+10)$$
$$\text{(b)} \quad G(s) = K(s-10)$$
$$\text{(c)} \quad G(s) = K(-s+10)$$
$$\text{(d)} \quad G(s) = \frac{K}{(s+10)}$$

$$\text{(e)} \quad G(s) = \frac{K}{(s-10)}$$

$$\text{(f)} \quad G(s) = \frac{K}{(-s+10)}$$

$$\text{(g)} \quad G(s) = \frac{K(10-s)}{(s-5)}$$

$$\text{(h)} \quad G(s) = \frac{K(10-s)}{s(s-5)}$$

$$\text{(i)} \quad G(s) = \frac{K}{s(s^2+4s+16)}$$

$$\text{(j)} \quad G(s) = \frac{K(s-2)}{(s+2)}$$

16. **What are Bode plots or Bode diagrams?**
The log-magnitude and phase frequency response curves plotted as a function of logarithmic frequency are called Bode plots or Bode diagrams.
17. **Mention any three advantages of Bode plots.**
In Bode magnitude and phase plots, the horizontal axis is on log scale. Since the horizontal axis represents frequency, a wide range of frequency can be included with clarity at all the frequencies. The vertical axis of the magnitude plot represents magnitude of $G(j\omega)$ in dB. This enables to provide large amplitude of $G(j\omega)$ being represented. Multiplication and division in the sinusoidal T.F. become addition and subtraction when magnitude is expressed in dB. This provides clear information about the magnitude contribution of each pole and zero at any frequency range.
18. **What are the disadvantages of Bode plots?**
Bode plots are sketched using straight line approximation which introduces error. The estimation of relative stability using phase margin and gain margin is applicable for minimum phase T.F. and not for non-minimum phase T.F.
19. **What are asymptotes in Bode magnitude plot?**
Bode magnitude plot is sketched by approximating the magnitude with straight line segments. These approximate straight line segments are called the asymptotes.
20. **What is break frequency or corner frequency in Bode plot?**
The break frequency or corner frequency which is denoted by ω_c is the frequency at which the low frequency asymptote and high frequency asymptote meet.
21. **What do you understand by octave and decade?**
If the frequency is multiplied by a factor two, this frequency range is called an octave. If the frequency is multiplied by a factor ten, this frequency range is called a decade. The frequency range 2 to 4 is an octave. The frequency range from 2 to 20 is a decade.
22. **What do you understand by dB/oct and dB/dec?**
The slope of the straight line of the magnitude plot is expressed as dB/oct or

dB/dec. dB/oct represents the rise or fall in dB in the frequency range of one octave. Similarly, dB/dec represents the rise or fall in dB in the frequency range of one decade.

23. **How dB/oct and dB/dec are related?**
    6 dB/oct is same as 20 dB/dec.
24. **Each pole and each zero of a system contributes how much of a slope to the Bode magnitude plot?**
    Each pole contributes -20 dB/dec and each zero contributes $+20$ dB/dec. slope to the slope of Bode magnitude plot.
25. **What is the slope of the magnitude Bode plot at high frequency for a system with four poles and two zeros?**
    The slope of the Bode magnitude plot at high frequency for a system with 4 poles and 2 zeros is -40 dB/dec.
26. **The Bode magnitude plot of a certain system is shown below with frequency in logarithmic scale. Find the rise in dB between ω_1 and ω_2.**
    Between $\omega_1 = 5$ and $\omega_2 = 80$, there are 4 octaves. Hence, the rise in dB between ω_1 and $\omega_2 = 4 \times 12 = 48$ dB. Alternatively, 12 dB/oct $= 40$ dB/dec. Hence, the rise in dB $= 40 \log \frac{80}{5} = 48$ dB.
27. **What is the error introduced at ω_c by asymptotic plot for a pole.**
    The error introduced at the corner frequency ω_c for a pole is -3 dB.
28. **How the error is corrected in Bode magnitude plot for a pole with a corner frequency ω_c.**
    The error is corrected as follows. At corner frequency ω_c, -3 dB is located. At one octave above and below corner frequency -1 dB points are located below the asymptotes. By joining these points by a smooth curve, the error is corrected to the minimum value.
29. **A certain system has the following T.F.**

$$G(s) = K(s+4)^3$$

    What is the corner frequency and how much error is to be added to the magnitude plot?
    The corner frequency is 4 rad/sec. The error is 9 dB at $\omega = 4$ and $+9$ dB is to be added to the asymptotic plot at this frequency.
30. **What is the effect of time delay e^{-sTd} in the Bode diagram?**
    The sinusoidal T.F. of the time delay is

$$e^{-j\omega Td} = 1\angle -57.3\omega Td$$

    The magnitude is 1 and the dB is zero. Hence, the magnitude plot of the system with time delay is not affected. However, a phase of $-57.3\omega Td$ degrees is to be added at each frequency in the phase angle plot of Bode diagram.
31. **What is log magnitude-phase (Nichols) plot?**
    The log magnitude-phase plot of $G(j\omega)$ is a plot of $G(j\omega)$ magnitude in dB

versus its phase in degrees with ω as a parameter on the curve. The Nichols plot can be derived from Bode diagram very easily.

32. **Of the methods namely Nyquist, Bode and Nichols which one is more suitable for stability analysis?**
Nyquist or polar plot is the most suitable one for stability study. The stability analysis of system with non-minimum phase T.F. has to be very carefully analysed using Bode and Nichols plots. However, in Nyquist plot both minimum phase and non-minimum phase T.F.s are treated in a similar way.
33. **Bode diagram is more suitable for what applications?**
Bode plot is an ideal one for the determination of phase margin, gain margin, phase crossover and gain crossover frequencies of minimum phase systems. It gives ω_{pc} and ω_{gc} very explicitly. It is also widely used in the design of compensator where frequent adjustment of poles and zeros of the compensators is required.
34. **Nichols plot is more suitable for what applications?**
Nichols plot is extensively used to plot the closed loop frequency response from open loop frequency response using Nichols chart. It is a very ideal method to apply for the determination of gain for the desired resonant peak and BW of the closed loop system and also to meet the required phase margin and gain margin of the closed loop system.
35. **Define bandwidth.**
The bandwidth BW is the frequency ω_b at which $|M(j\omega)|$ drops to 70.7 per cent of, or 3 dB down from its zero frequency value.
36. **What are frequency domain specifications?**
The frequency domain specification are as follows:

    (a) Resonant peak,
    (b) Resonant frequency,
    (c) Bandwidth,
    (d) Phase margin,
    (e) Gain margin,
    (f) Phase crossover frequency, and
    (g) Gain crossover frequency.

37. **What are constant M circles?**
The constant M circles are the family of circles drawn on the $G(j\omega)$ plane. These circles represent the magnitude of the closed loop system. The centre of the circle is given by $c = \left[0, M^2/(M^2-1)\right]$ and the radius is $r = M/(M^2-1)$.
38. **How the constant M loci are placed in the $G(j\omega)$ plane?**
The constant M circles are symmetrical with respect to $M = 1$ line and the real axis. For $M = 1$, the circle becomes a vertical straight line passing through $(-1/2, 0)$ point. The circles to the left of $M = 1$ correspond to $M > 1$ and circles to the right of $M = 1$ correspond to $M < 1$. At $(-1, 0)$ point $M = \infty$.

39. **What are the applications of the constant M circles?**

    (a) The points of intersection of $G(j\omega)$ plot when superimposed on constant M loci give the closed loop magnitude at the corresponding frequency on the $G(j\omega)$ curve. Thus, if the open loop frequency response is superimposed on the constant M loci, the closed loop magnitudes at different frequencies can be obtained.
    (b) The resonant peak of the closed loop system is obtained where the $G(j\omega)$ plot is tangential to the constant M circle. The corresponding frequency on the $G(j\omega)$ gives the resonant frequency ω_r.
    (c) When the $G(j\omega)$ plot intersects $M = 0.707$ constant M circle, the corresponding frequency on the $G(j\omega)$ curve gives the BW of the closed loop system.

40. **What are constant N circles?**
    The constant N circles are the family of circles drawn on the $G(j\omega)$ plane. These circles represent the phase angle of the closed loop system. The centre of the circle is given by $c = [-1/2, 1/2N]$ and radius $r = \sqrt{(1 + N^2)/4N^2}$, where $N = \tan\phi$. Here ϕ is the closed loop phase angle.

41. **How constant N circles are used to get closed loop phase if open loop frequency response is known?**
    The $G(j\omega)$ plot is super imposed on the loci of constant N circles. The points of intersections of $G(j\omega)$ with these loci give the closed loop phase angle at the corresponding frequency on the $G(j\omega)$ curve.

42. **How closed loop frequency response is obtained if the open loop frequency response is known using constant M and constant N circles?**
    The $G(j\omega)$ plot which represents the open loop frequency response is superimposed on the constant M-loci and constant N loci separately. From the points of intersection of $G(j\omega)$ on constant M loci closed loop magnitude at different frequency values on the $G(j\omega)$ curve are obtained and tabulated. Similarly, for the same frequency values, the points of intersection when $G(j\omega)$ is superimposed on constant N circles are obtained. These values give the closed loop system phase. Knowing closed loop system magnitude and phase for different frequencies, the closed loop frequency response is drawn.

43. **Can we use constant M and constant N circles for non-unity feedback system also?**
    Constant M and constant N circles cannot be used to get the closed loop frequency response of a non-unity feedback system. However, it can be used with small modification.

44. **The $G(j\omega)$ plot of certain closed loop system passes through the left of $(-1, 0)$ point. Is it possible to get closed loop magnitude from constant N circles?**
    For a system with minimum phase T.F. if $G(j\omega)$ passes to the left of $(-1, 0)$ point, the closed loop system becomes unstable and the output becomes unbounded. Under this condition, the constant M circles no longer have any meaning.

45. **What are the disadvantages of using constant M and constant N circles?**
To get the closed loop frequency response from open loop frequency response, the procedure has to be followed twice, once using the constant M loci and at another time using the consent N loci. This is the main disadvantage of using constant M and N loci separately.
46. **What is a Nichols Chart?**
Log magnitude *versus* phase plot is called Nichols plot with frequency as the variable parameter. A set of radial lines drawn from the origin of $G(j\omega)$ plane intersect a constant M circle. The points of intersection of these radial lines are converted into dB. The phase angles of these radial lines are also measured and transferred to the log-magnitude and phase plane. The loci of these circles become constant dB loci. Similar procedure is adopted to the constant N loci and transferred as constant phase angle loci which together form Nichols Chart (*see* Sect. 3.11.4).
47. **What are the advantages of Nichols Chart?**
    (a) The Nichols chart gives both the gain and phase characteristics of the closed loop system simultaneously at one instant.
    (b) The frequency response specifications can be easily and quickly determined from Nichols chart in addition to obtaining closed loop frequency response from open loop frequency response.
48. **What is a critical point in the Nichols chart? Why is it called so?**
(0 dB, $-180°$) point is called the critical point in the Nichols chart. It is called so because, for a minimum phase system, if the $G(j\omega)$ plot passes to the right of it, the closed loop system is stable. On the other hand if the $G(j\omega)$ plot passes through the left of the critical point, the system is unstable.
49. **Define gain margin in the polar coordinate.**
Gain margin is defined as the amount by which the loop gain is increased before the closed loop system becomes unstable. It is the reciprocal of the gain of $G(j\omega)$ plot when its phase angle is $-180°$.
50. **Define phase margin in the polar coordinate?**
Phase margin is defined as the phase angle in degrees through which the $G(j\omega)$ plot be rotated about the origin before the closed loop system becomes unstable. It is $180°+$ phase angle of $G(j\omega)$ when $|G(j\omega)| = 1$.
51. **What is gain crossover and gain crossover frequency in the polar coordinate?**
The gain crossover is defined as a point on the $G(j\omega)$ plane at which $|G(j\omega)| = 1$. The corresponding frequency on the $G(j\omega)$ curve gives the gain crossover frequency denoted by ω_{gc}.
52. **What is phase crossover and phase crossover frequency in the polar coordinate?**
The phase crossover is defined as the point on the $G(j\omega)$ plane at which the plot intersects the negative real axis. The corresponding frequency on the $G(j\omega)$ plot is called phase crossover frequency denoted by ω_{pc}.

53. **For the estimation of gain margin what is known in $G(j\omega)$?**
When gain margin is to be estimated, the phase angle of $G(j\omega)$ is known which is $-180°$. By this ω_{pc} is calculated.
54. **For the estimation of phase margin what is known in $G(j\omega)$?**
When phase margin is to be estimated, the magnitude of $G(j\omega)$ is known which is unity. By this ω_{gc} is calculated.
55. **How relative stability is defined in the frequency domain?**
In the frequency domain, the relative stability is defined by how close the Nyquist plot of $G(j\omega)$ is to the $(-1, 0)$ point.
56. **What are the measures of relative stability in the frequency domain?**
In the frequency domain, relative stability is measured in terms of phase margin and gain margin. For the system to be stable both phase margin and gain margin should be positive. Further $\omega_{pc} > \omega_{gc}$. The higher the phase margin and gain margins are the higher is the relative stability.
57. **If $\omega_{pc} = \omega_{gc}$, what is the status of stability of the closed loop system?**
If $\omega_{pc} = \omega_{gc}$, the $G(j\omega)$ plot passes through the critical point $(-1, 0)$ in the $G(j\omega)$ plane and the system is said to be critically stable or oscillatory.
58. **Under what condition the positive gain margin is not an indication of closed loop system stability?**
For a system with non-minimum phase T.F. positive gain margin may not ensure closed loop system stability.
59. **Define gain margin in Bode plot?**
In Bode plot, the gain margin is defined as the amount of gain in dB that can be added to the loop before the closed loop system becomes unstable.
60. **Define phase margin in Bode plot.**
In Bode plot, the phase margin is defined as the amount of the phase angle that can be added to the loop before the closed loop system becomes unstable.
61. **How to identify positive and negative gain margins in polar plot?**
If the gain margin is greater than one, the gain margin is +ve and $\omega_{pc} > \omega_{gc}$. If the gain margin is less than one, G_m is negative and $\omega_{pc} < \omega_{gc}$. If $G_m = 1$, the system is oscillatory and $\omega_{pc} = \omega_{gc}$.
62. **How do you identify positive and negative phase margin in polar plot?**
If at unity gain of $G(j\omega)$, the phase angle of $G(j\omega)$ is less than $-180°$, the phase margin is positive and now $\omega_{pc} > \omega_{gc}$. If at unity gain of $G(j\omega)$, the phase angle of $G(j\omega)$ is greater than $-180°$, the phase margin is negative and now $\omega_{pc} < \omega_{gc}$. The phase margin is zero when $\omega_{pc} = \omega_{gc}$ and the $G(j\omega)$ plot passes through $(-1, 0)$ point and the closed loop system has oscillatory behaviour.
63. **What is the corner frequency of a prototype second order system in Bode plot? What is the error introduced at ω_c by approximation?**
The natural frequency of oscillation ω_n of the prototype second order system is the corner frequency in the Bode plot. The error introduced at ω_c by the asymptote is given by $-20 \log 2\zeta$.
64. **For a second order system, how the phase margin is related with the damping factor?**
For a second order system, the phase margin is related with damping factor as

$$\zeta = \frac{\pi}{360^\circ}\phi_m$$

65. **What is the effect of bandwidth on transient response?**
A large bandwidth allows high frequency signals more easily to pass through. This corresponds to faster time response in the time domain. If the BW is small the time response will be slow. The rise time and BW are inversely proportional to each other. However, large bandwidth increases noise and component cost.
66. **How static error constants are determined from Bode diagrams?**
From Bode magnitude plot the static error constants are determined. These constants are determined from low frequency response part of Bode plot. The position error constant K_p exits only for a Type 0 system. For Type 0 system the frequency Bode magnitude is $20 \log K = K_p$. The velocity error constant exits for a Type 1 system. This is obtained from ω where $\frac{K}{\omega}$ cuts the frequency axis. That is $K_v = \omega$ where $\omega = K$. The acceleration error constant exits for Type 2 system. This is obtained from ω where K/ω^2 cuts the frequency axis. That is $K_a = \omega^2$ where $\omega = \sqrt{K}$.
67. **How to identify positive and negative gain margins from Bode plots?**
The gain margin is measured positive if the magnitude of the $G(\,j\omega)$ plot at the phase crossover is negative in dB. The gain margin is measured +ve below the 0 dB axis. Here $\omega_{pc} > \omega_{gc}$. The gain margin is measured −ve if it is measured above the 0 dB axis at the phase crossover. Here $\omega_{pc} < \omega_{gc}$ for a minimum phase T.F.
68. **How to identify the positive and negative phase margins from Bode plots?**
The phase margin is measured positive if the phase angle of $G(\,j\omega)$ plot at the gain crossover is greater than -180°. The phase margin is measured +ve above the -180° axis. Here $\omega_{pc} > \omega_{gc}$. The phase margin is measured −ve if it is measured below the -180° axis at the gain crossover. Here $\omega_{pc} < \omega_{gc}$.
69. **What is phase margin and gain margin when $\omega_{pc} = \omega_{gc}$?**
When $\omega_{pc} = \omega_{gc}$, the phase margin is 0° and the gain margin is 0 dB. The system is now oscillatory.
70. **What is the effect of increasing and decreasing the value of open loop system gain on the Bode diagrams?**
When the gain of open loop system is increased, the Bode magnitude plot is vertically shifted up. When the gain of the open loop system is decreased, the Bode magnitude plot is vertically shifted down. In both the cases, the phase plot is unaffected.
71. **What is the effect of incorporating a transportation lag on the Bode diagram? How the relative stability is affected?**
By incorporating a transportation lag in the system, the magnitude plot will not be affected. However, the phase angle plot is shifted downwards very rapidly as ω increases. ω_{pc} is shifted towards left which decreases the gain margin. ω_{gc} is unaffected. Since the phase curve is shifted down, the phase margin is also decreased.

72. **How do you identify positive and negative gain margin in Nichols plot?**
If the dB of $G(j\omega)$ at the critical point (0 dB, $-180°$) is negative, the gain margin has +ve dB. That is the gain margin is +ve at the phase crossover and measured below the 0 dB axis. Here $\omega_{pc} > \omega_{gc}$. The gain margin is measured negative (dB is +ve) at the phase crossover and is above the 0 dB axis. Here $\omega_{pc} < \omega_{gc}$.
73. **How do you identify positive and negative phase margin in Nichols plot?**
In the Nichols plot, the phase margin is measured positive at the gain crossover if it is to the right of the critical point (0 dB, $-180°$). That is the $G(j\omega)$ plot is to the right of $-180°$ vertical axis. Here $\omega_{pc} > \omega_{gc}$. If the gain crossover is to the left of the critical point, the phase margin is negative. Here $\omega_{pc} < \omega_{gc}$.
74. **What will be the nature of Nichols plot when $\omega_{pc} = \omega_{gc}$?**
For $\omega_{pc} = \omega_{gc}$, the $G(j\omega)$ plot will pass through the critical point (0 dB, $-180°$). The system will have oscillatory behaviour.
75. **How do you find BW from Nichols plot?**
The Nichols plot of $G(j\omega)$ is superimposed on the Nichols chart. The point of intersection of $G(j\omega)$ plot with the constant -3 dB locus is found out. The value of ω on the $G(j\omega)$ curve at this intersection point gives the BW of the closed loop system.
76. **How the gain of the system is determined for the desired gain margin using Nichols chart?**
The $G(j\omega)$ plot with $K = 1$ is superimposed on the Nichols chart. ω_{pc} is identified on the $-180°$ vertical axis. The $G(j\omega)$ plot is shifted up or down so that at the phase crossover the required gain margin is obtained. For upward shift the required gain is calculated from $20 \log K = +$ ve dB and for downward shift from $20 \log K = -$ve dB.
77. **How do you calculate the gain of the system for the desired phase margin using Nichols chart?**
The $G(j\omega)$ plot with $K = 1$ is superimposed on the Nichols chart. The $G(j\omega)$ plot is shifted up or down so that at gain crossover the required phase margin is obtained. For upward shift, the gain is calculated from $20 \log K = +$dB and for downward shift, $20 \log K = -$ dB.
78. **Sketch the polar plot for the following T.F. for $0 \leq \omega \leq \infty$.**

$$\text{(a)} \quad G(s) = \frac{10}{(s^2 + 4)}$$

$$\text{(b)} \quad G(s) = 10^{-0.1s}$$

79. **For the following T.F.s. Sketch the Bode magnitude plot and determine K_p, K_v and K_a respectively.**

$$\text{(a)} \quad G(s) = \frac{12}{(s+5)}$$
$$\text{(b)} \quad G(s) = \frac{120}{s(s+5)}$$
$$\text{(c)} \quad G(s) = \frac{200}{s^2(s+5)}$$

80. **What is the damping ratio when the percentage overshoot is 100%? What is the corresponding resonant peak?**
For 100% overshoot, the damping ratio $\zeta = 0$. The corresponding resonant peak $M_r = \infty$.
81. **The log magnitude-phase plot of a certain open loop T.F. gave the following data.**

$$G(j\omega)|_{\omega=5} = -18\,\text{dB}\angle -180^\circ$$
$$G(j\omega)|_{\omega=2} = 0\,\text{dB}\angle -140^\circ$$

Find ϕ_m, G_m, ω_{pc} and ω_{gc} and the stability of the closed loop system.

$$\phi_m = +40^\circ$$
$$G_m = +18\,\text{dB}$$
$$\omega_{pc} = 5\,\text{rad/sec}$$
$$\omega_{gc} = 2\,\text{rad/sec}$$

The closed loop system is stable.
82. **What are the recommended values of phase margin and gain margin of a system for satisfactory performance?**
For satisfactory performance of a system, the recommended value of phase margin is between 30° and 60° and the gain margin is greater than 6 dB.
83. **What is the desirable slope of the log magnitude plot at the gain crossover frequency for the system to be stable?**
−20 dB/dec slope characteristic of the log magnitude plot at the gain crossover frequency is the desirable slope for the system to be stable (Figs. 3.76, 3.77, 3.78 and 3.79).

Long Answer Type Questions

1. **For the following T.F. sketch the polar plot. What is the phase crossover frequency and gain margin?**

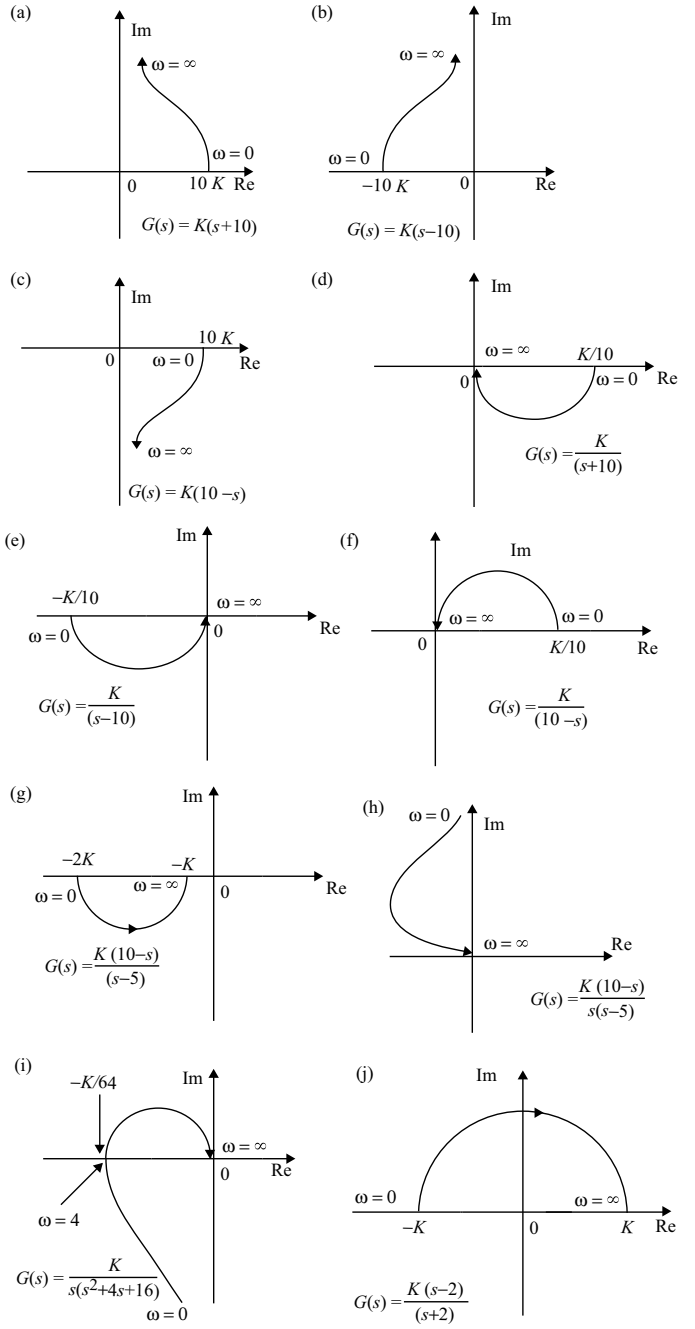


Fig. 3.76 Polar plot for different T.F.s of Question 15

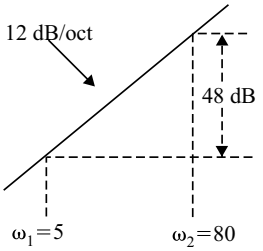


Fig. 3.77 Sample Bode magnitude plot for Question 26

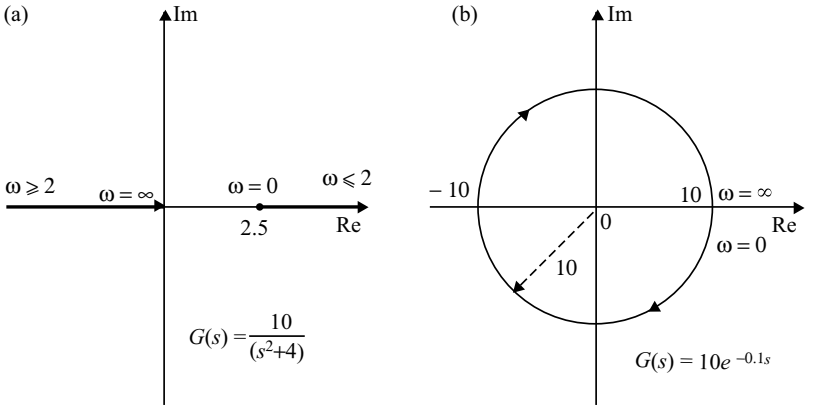


Fig. 3.78 Polar plot of Question 78

$$G(s) = \frac{10}{s(2s+1)(s+1)}$$

Ans: $\omega_{pc} = 0.707\,\text{rad/sec}$; $G_m = -0.15$.

2. **For the following T.F. sketch the polar plot and find phase margin, gain margin, ω_{pc} and ω_{gc}.**

$$G(s) = \frac{20(s+6)(s+8)}{(s-4)(s-7)}$$

Ans: $G_m = -0.0394$; $\omega_{pc} = 6.06$ rad/sec; ϕ_m and ω_{gc} do not exist since unit circle does not intersect the $G(\,j\omega)$ plot. The system is unstable.

3. **For the following T.F. sketch the polar plot and obtain G_m, ϕ_m, ω_{pc} & ω_{gc}.**

$$G(s) = \frac{100}{s(s+2)^2}$$

Ans: $G_m = -0.16$; or $-15.9\,\text{dB}$; $\phi_m = -40.6°$; $\omega_{pc} = 2$ rad/sec; $\omega_{gc} = 4.35$ rad/sec.

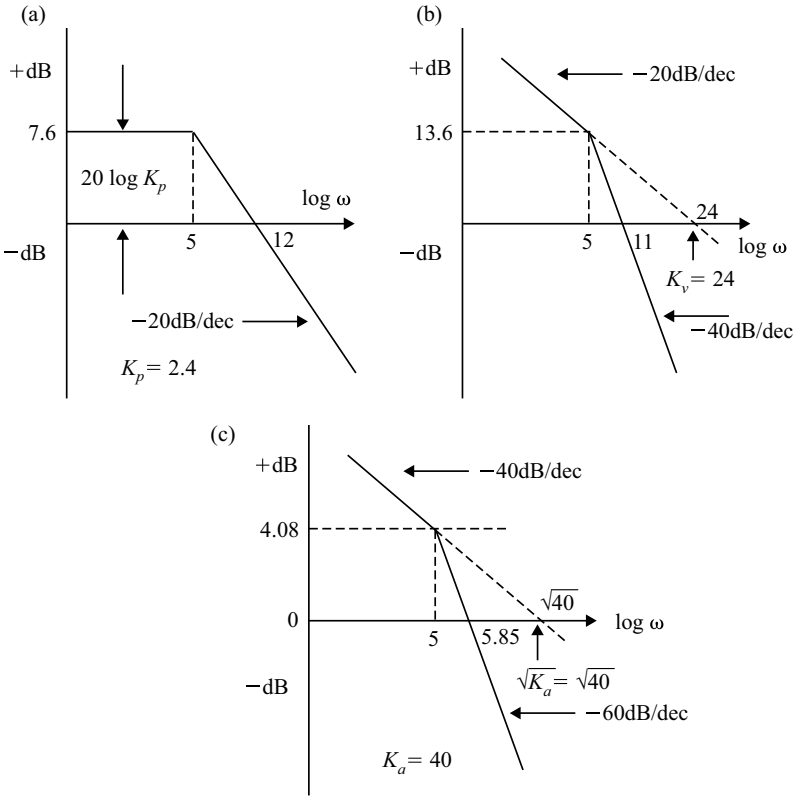


Fig. 3.79 Bode plots of Question 79

4. **For the following T.F. sketch the polar plot and obtain ϕ_m, G_m, ω_{pc} and ω_{gc}.**

$$G(s) = \frac{10s^2}{(s+1)(s+2)(s+3)}$$

Ans: $\phi_m = -90°$; $\omega_{gc} = 1$ rad/sec; $G_m = \infty$; $\omega_{pc} = 0$.

5. **Consider the following open loop T.F. of a certain unity feedback system.**

$$G(s) = \frac{20e^{-0.1T_d s}}{s(s+2)(s+4)}$$

Find ϕ_m, G_m, ω_{pc} and ω_{gc} of the system without transportation lag. With delay included what is the critical value of T_d?
Ans: Without delay: $G_m = +7.6\,\text{dB}$; $\phi_m = +25.64°$; $\omega_{pc} = 2.82$ rad/sec; $\omega_{gc} = 1.73$ rad/sec. Critical value of $T_d = 25.65/(5.73 \times 1.73) = 2.59$ sec.

6. **The forward path T.F. of a certain unity feedback system is given by.**

$$G(s) = \frac{200}{(s+2)^3}$$

By drawing the polar find ϕ_m, G_m, ω_{pc} and ω_{gc}. Is the system stable? If unstable, by what value the gain is to be reduced. With 90% of this value what are the new ϕ_m, G_m, ω_{pc} and ω_{gc}.
Ans: $\phi_m = -30°$; $G_m = -9.84\,\text{dB}$; $\omega_{pc} = 3.46$; $\omega_{gc} = 5.5$. The gain should be < 64 for the system to be stable. For 90% of this value is $K = 57.6$. At this value $G_m = 0.886\,\text{dB}$; $\phi_m = +3.6°$; $\omega_{pc} = 3.46$ rad/sec and $\omega_{gc} = 3.3$ rad/sec. Both ϕ_m and G_m are +ve $\omega_{pc} > \omega_{gc}$ system is stable now.

7. **For the forward path T.F. draw the polar plot and find ϕ_m, G_m, ω_{pc} and ω_{gc}. Is the system stable?**

$$G(s) = \frac{(5s^2 + 2s + 10)}{(2s^2 - 3s - 2)}$$

Ans: $\phi_m = -13.3°$; $G_m = 3.52\,\text{dB}$; $\omega_{pc} = 1.16$; rad/sec; $\omega_{gc} = 1.82$ rad/sec. Since ϕ_m is negative the system is unstable.

8. **For the following T.F. draw the asymptotic Bode magnitude and phase plots. (a) Find ϕ_m, G_m, ω_{pc}, ω_{gc} and stability of the closed loop system for (a) $K = 10$. (b) Find K for $\phi_m = 60°$ and (c) Find K for $G_m = 40\,\text{dB}$.**

$$G(s) = \frac{K}{s(s+1)(s+10)}$$

Ans: (a) $G_m = 20.82\,\text{dB}$; $\phi_m = 47.4°$; $\omega_{pc} = 3.16\,\text{rad/sec}$; $\omega_{pc} = 0.784\,\text{rad/sec}$; closed loop system is stable. (b) $K = 4.47$ and (c) $K = 1.1$ (Results obtained from actual Bode plots).

9. **Consider the following T.F.**

$$G(s) = \frac{12e^{-0.1T_d s}}{s\left(1 + \frac{1}{2}s\right)}$$

Using Bode plot determine T_d so that the phase margin $\phi_m = 45°$.
Ans: $T_d = 0.8132\,\text{sec}$. $(-0.1T_d \times 4.7 \times 57.3 = -21.9)$ (Results obtained from actual Bode plots).

10. **A certain control system has the following T.F.**

$$G(s) = \frac{K e^{-0.2s}}{s(s+2)(s+8)}$$

Using Bode plots determine the value of K if.

(a) $\phi_m = 45^\circ$
(b) $G_m = 6\,\text{dB}$

Ans:

(a) For required $\phi_m = 45^\circ$, the vertical upward shift required is 25 dB. Hence, $20 \log K = 25$ or $K = 17.8$.
(b) For required $G_m = 6\,\text{dB}$, the required vertical upward shift is $(31 - 6) = 25\,\text{dB}$. Hence, $20 \log K = 25$ or $K = 17.8$. Thus, with $K = 17.8$ the required ϕ_m of 45° and the required G_m of 6 dB are achieved. (Results obtained from actual Bode plots).

11. **Consider the following open loop T.F.**

$$G(s) = \frac{80}{s(s+2)(s+20)}$$

Find ϕ_m, G_m, ω_{pc} and ω_{gc} using Bode diagrams. Is the closed loop system stable? By what factor the gain is to be increased so that the gain crossover frequency $\omega_{gc} = 0.2\,\text{rad/sec}$? What is the new ϕ_m?
Ans: $G_m = 20.8\,\text{dB}; \phi_m = 47.4^\circ; \omega_{pc} = 6.32\,\text{rad/sec}; \omega_{gc} = 1.57\,\text{rad/sec}$. Both G_m and ϕ_m are positive. $\omega_{pc} > \omega_{gc}$. The closed loop system is stable. The gain has to be increased by a factor of 0.1 so that ω_{gc} is at 0.2 rad/sec. The new phase margin $\phi_m = 84^\circ$. (Results obtained from actual Bode plots).

12. **Consider the following T.F.**

$$G(s) = \frac{K(1 + 0.2s)}{(1+s)(s^2 + 16s + 80)}$$

For $K = 1000$ draw Bode diagram and find ϕ_m, G_m, ω_{pc} and ω_{gc} and the stability of the closed loop system. What is the value of K to have $\omega_{gc} = 1\,\text{rad/sec}$ and what is the corresponding ϕ_m and G_m?
Ans: For $K = 1000$; $\phi_m = 51.8°$; $G_m = \infty$; $\omega_{gc} = 12.5\,\text{rad/sec}$. For $\omega_{gc} = 1$; $K = 80$; $\phi_m = 135°$. (Results obtained from actual Bode plots).

13. **Consider the following T.F.**

$$G(s) = \frac{K}{s(s+1)(1+0.5s)}$$

Find the value of K so that $\omega_{gc} = 0.5\,\text{rad/sec}$. Use Bode diagrams.
Ans: $K = 0.5762$, and $\phi_{=}50°$. (Results obtained from actual Bode plots).

14. **Consider the following T.F.**

$$G(s) = \frac{K}{s(s+10)}$$

Determine the value of K using Bode diagrams so that $\phi_m = 60°$. What is the gain margin for this value of K?
Ans: For upward shift $20\log K = 36.7$. $K = 68.5$ for $\phi_m = 60°$. $G_m = \infty$ for all values of K. (Results obtained from actual Bode plots).

15. **Consider the following open loop T.F. of a closed loop system.**

$$G(s) = \frac{15}{s(s+1)(s^2+4s+9)}$$

Sketch the log magnitude-phase plot. Determine ϕ_m, G_m, ω_{pc} and ω_{gc}. Also find the BW, ω_r, M_r and the stability of the closed loop system.
Ans: $G_m = 11.4\,\text{dB}$; $\phi_m = 90°$; $\omega_{pc} = 3.6\,\text{rad/sec}$; $\omega_{gc} = 1.34\,\text{rad/sec}$; $\omega_b = 2.23\,\text{rad/sec}$. M_r does not exist; closed loop system is stable.

16. **Using Nichols chart determine the closed loop frequency response of the unity feedback system whose open loop frequency response is** (Figs. 3.80, 3.81, 3.82, 3.83, 3.84, 3.85, 3.86, 3.87, 3.88, 3.89, 3.90, 3.91, 3.92, 3.93, 3.94 and 3.95)

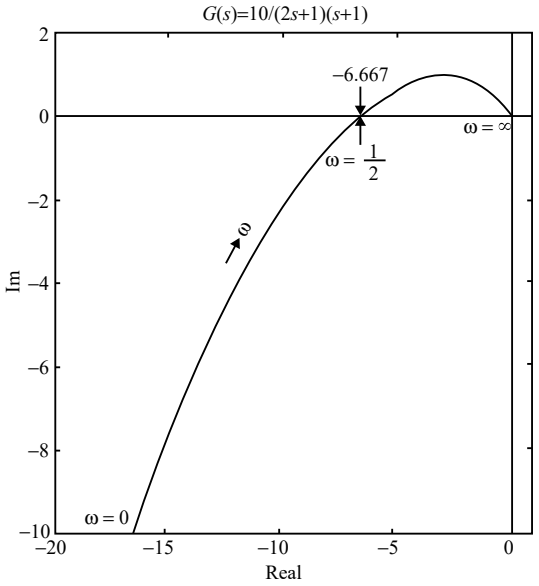


Fig. 3.80 Polar plot for $G(s) = \dfrac{10}{s(2s+1)(s+1)}$

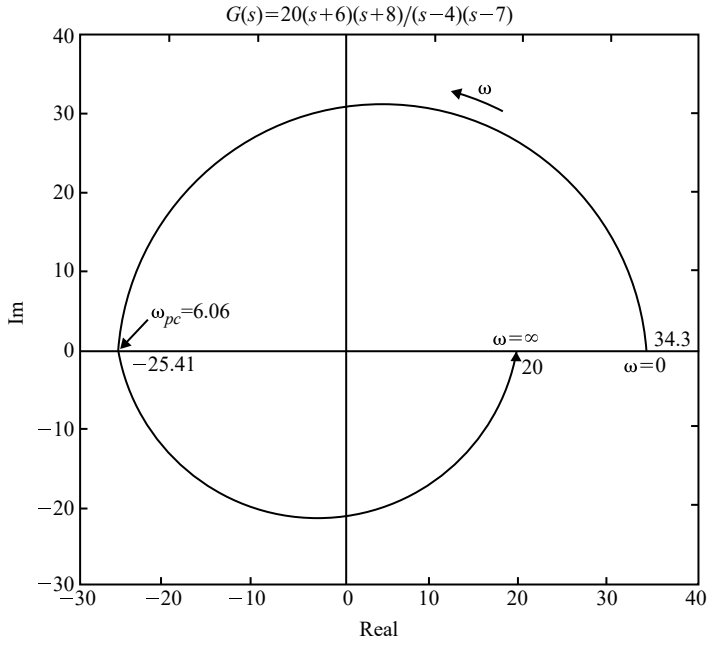


Fig. 3.81 Polar plot for $G(s) = 20(s+6)(s+8)/(s-4)(s-7)$

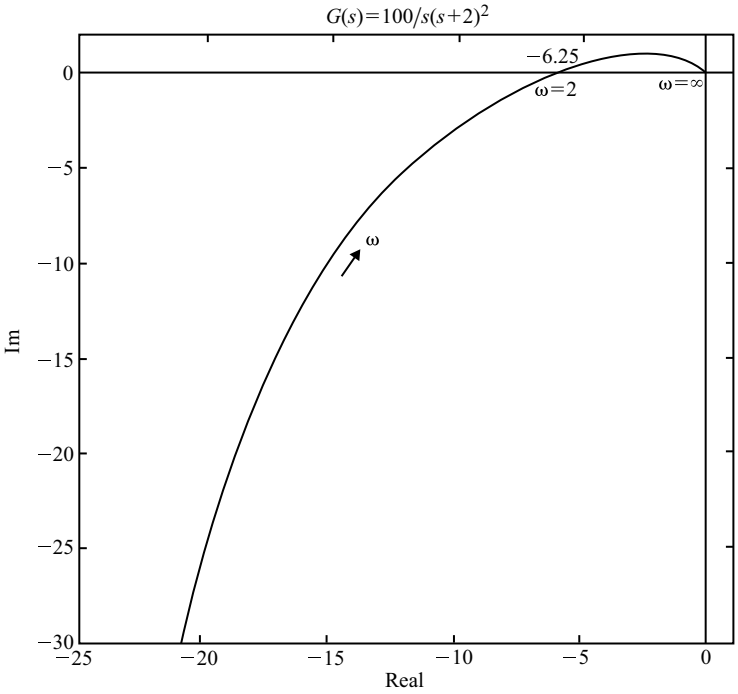


Fig. 3.82 Polar plot for $G(s) = 100/s(s+2)^2$

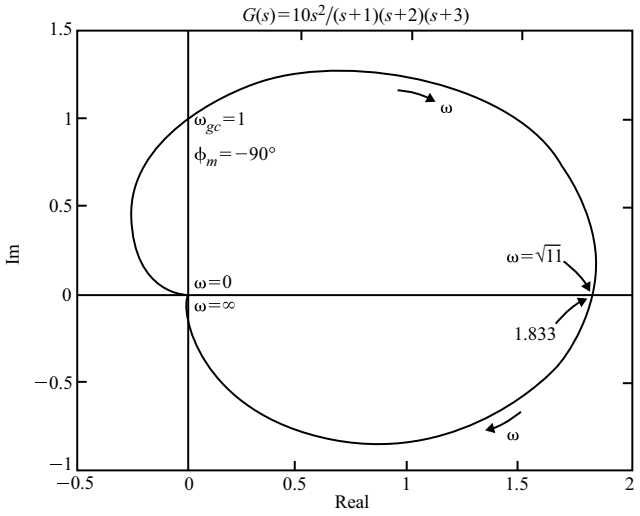


Fig. 3.83 Polar plot for $G(s) = \dfrac{10s^2}{(s+1)(s+2)(s+3)}$

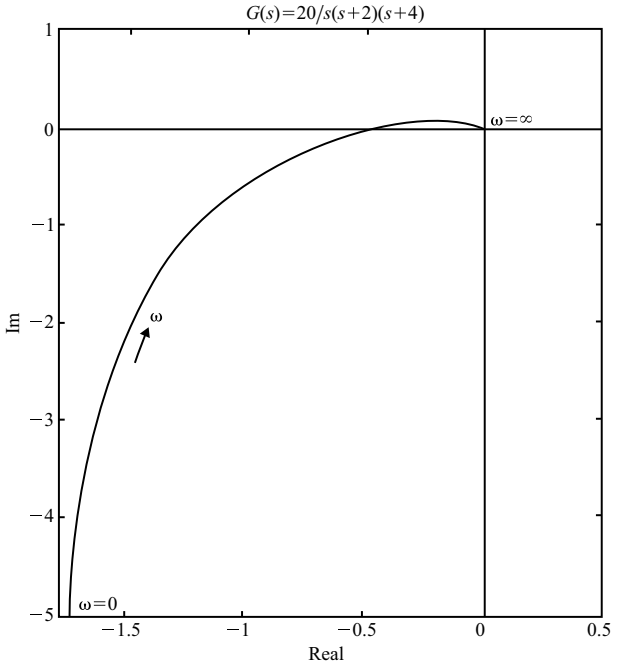


Fig. 3.84 Polar plot for $G(s) = \dfrac{20e^{-0.1T_d s}}{s(s+2)(s+4)}$

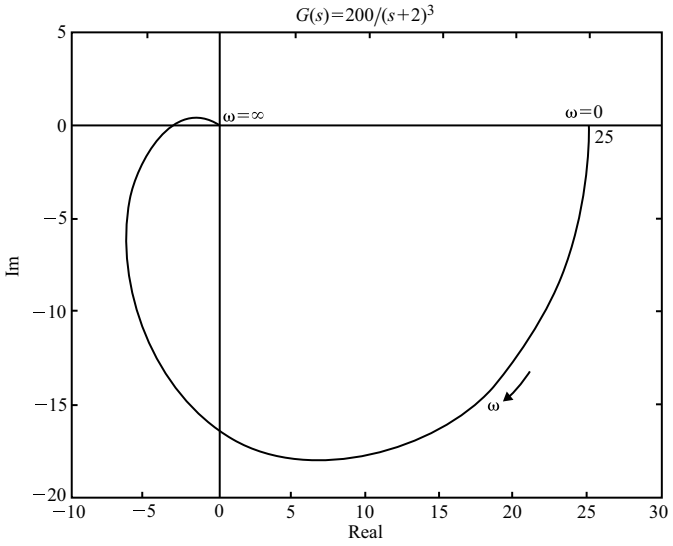


Fig. 3.85 Polar plot for $G(s) = 200/(s+2)^3$

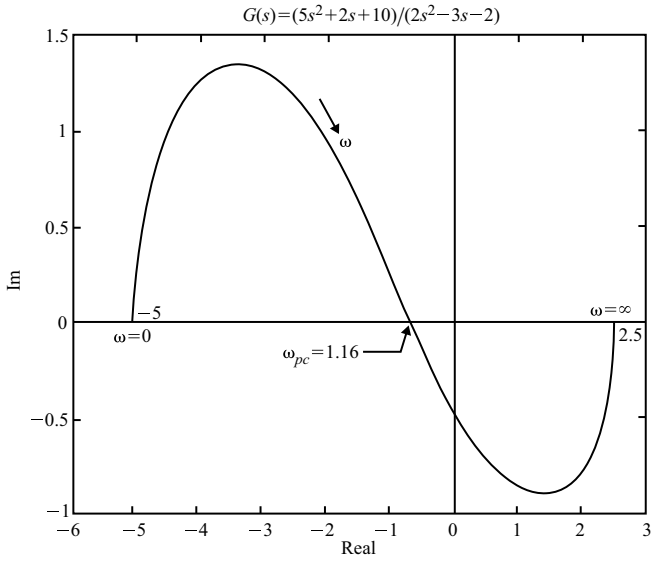


Fig. 3.86 Polar plot for $G(s) = (5s^2 + 2s + 10)/(2s^2 - 3s - 2)$

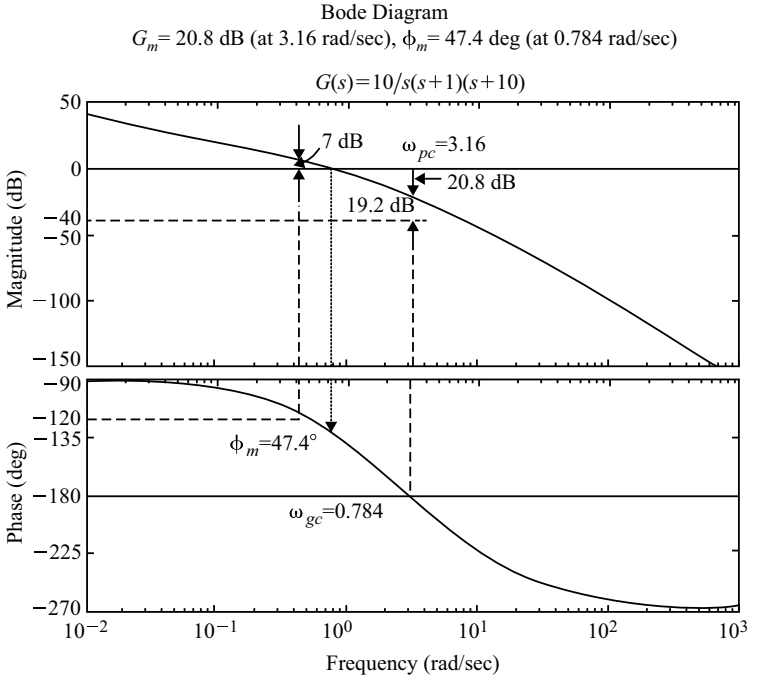


Fig. 3.87 Polar plot for $G(s) = \dfrac{10}{s(s+1)(s+10)}$

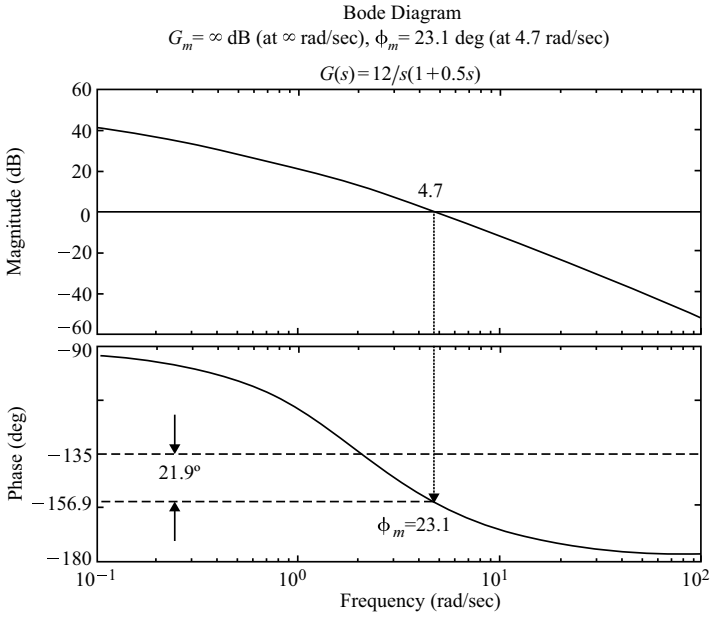


Fig. 3.88 Bode diagram for $G(s) = \dfrac{12}{s(s + 0.5s)}$

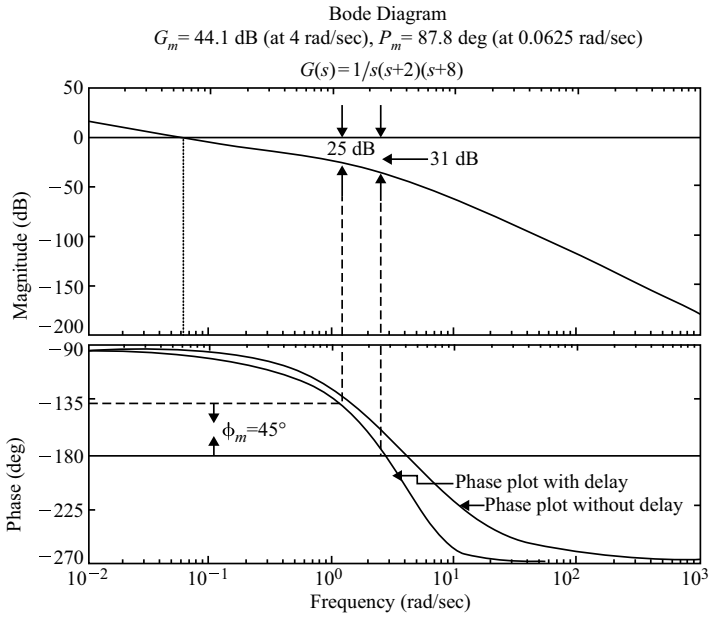


Fig. 3.89 Bode diagram for $G(s) = \dfrac{e^{-0.2s}}{s(s + 2)(s + 8)}$

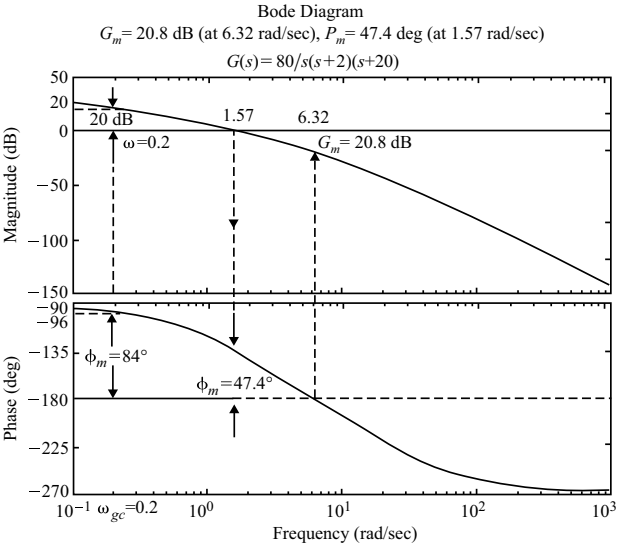


Fig. 3.90 Bode diagram for $G(s) = \dfrac{80}{s(s+2)(s+20)}$

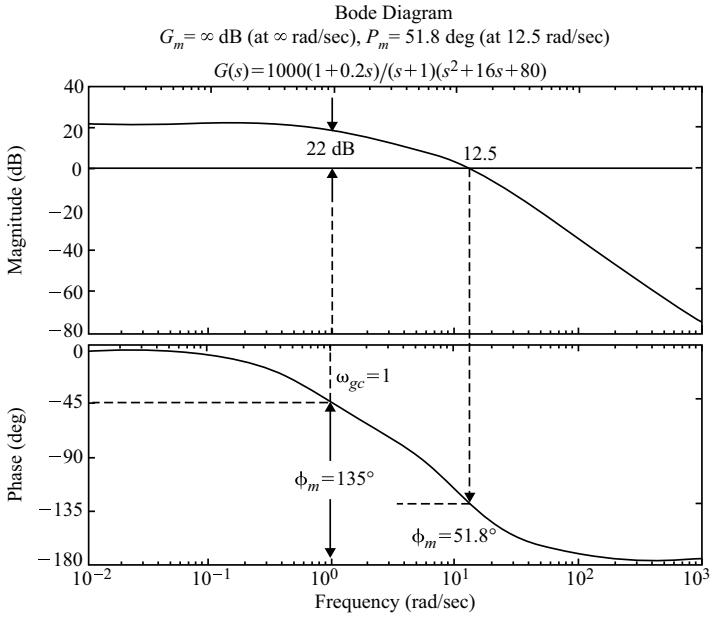


Fig. 3.91 Bode diagram for $G(s) = \dfrac{1000(1+0.2s)}{(s+1)(s^2+16s+80)}$

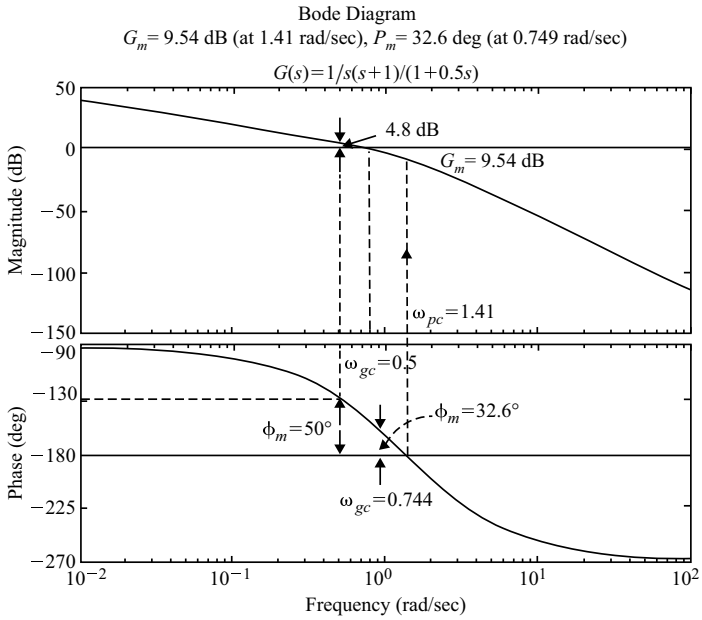


Fig. 3.92 Bode diagram for $G(s) = \dfrac{1}{s(s+1)(1+0.5s)}$

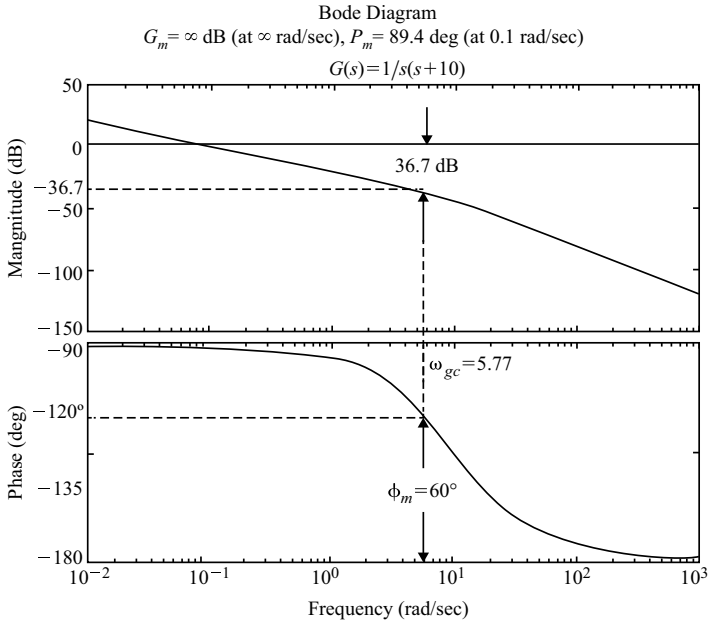


Fig. 3.93 Bode diagram for $G(s) = \dfrac{1}{s(s+10)}$

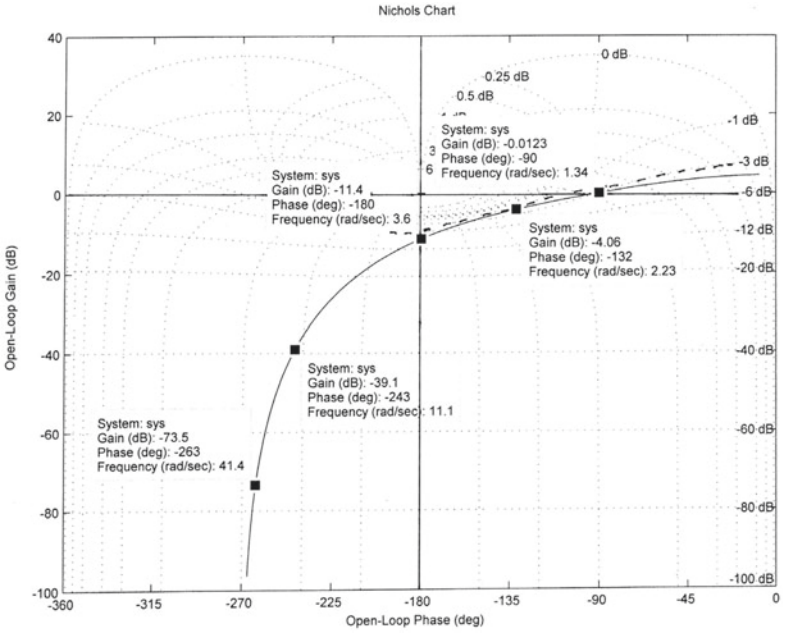


Fig. 3.94 Bode diagram for $G(s) = \dfrac{15}{s(s+1)(s^2+4s+9)}$

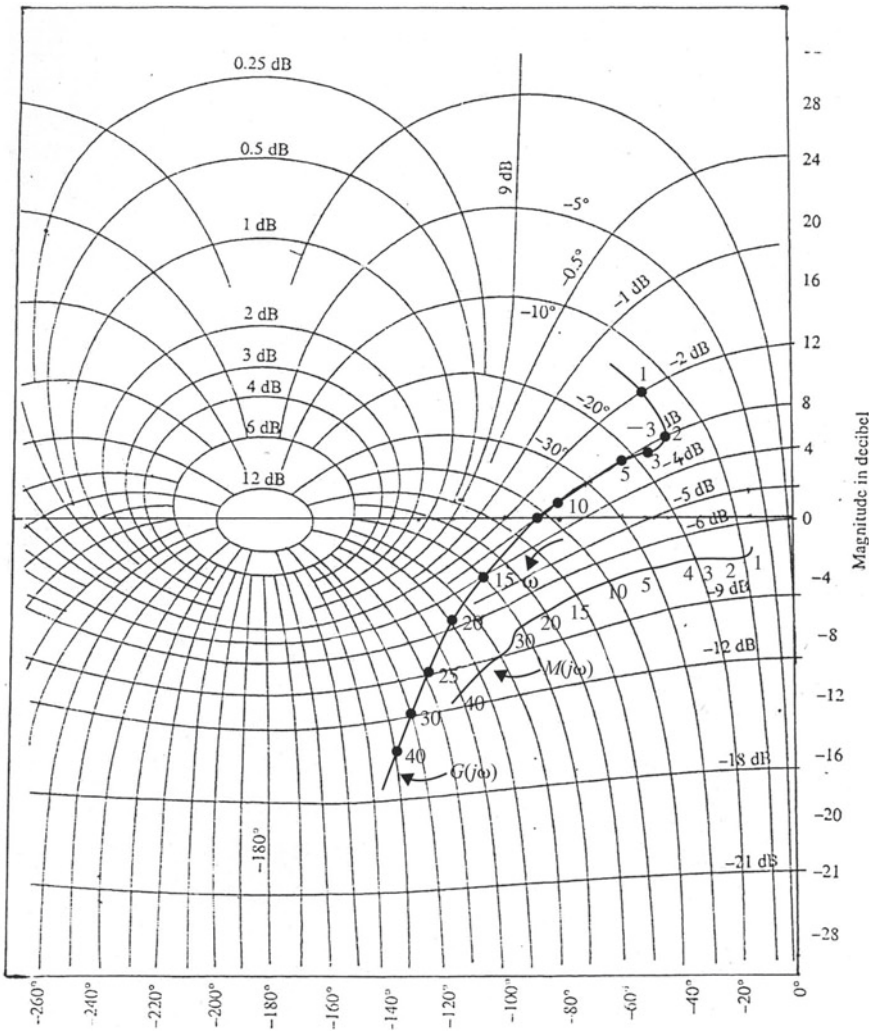


Fig. 3.95 Closed loop frequency response from open loop frequency response in Nichols chart

$$G(s) = \frac{200(s+1)}{s(s+10)^2}$$

$G(\,j\omega)$ Plot:

ω	1	2	3	5	10	15	20	25	30	40
$\angle G(\,j\omega)$	$-56°$	$-49°$	$-52°$	$-64°$	$-96°$	$-117°$	$-130°$	$-138°$	$-145°$	$-154°$
$\lvert G(\,j\omega)\rvert$	2.8	2.15	1.93	1.63	1.0	0.616	0.4	0.276	0.2	0.116
$20\log \lvert G(j\omega)\rvert$ dB	8.9	6.7	5.73	4.25	0	-4.2	-8.0	-11.2	-14	-18

$M(j\omega)$:

ω	1	2	3	5	10	15	20	25	30	40
$\angle M(j\omega)$	$-14°$	$-15°$	$-18°$	$-22°$	$-37°$	$-70°$	$-90°$	$-110°$	$-120°$	$-130°$
$\lvert M(j\omega)\rvert$ dB	-2	-3	-2.8	-3	-3	-4	-6	-9	-12	-14

Chapter 4
Stability Analysis of Linear Control System



Learning Objectives

After completing this chapter, you should be able to:

- Define linear system stability in time and frequency domains.
- Understand different methods available for stability study.
- Establish the general stability conditions for linear system.
- Apply Routh–Hurwitz stability criterion for the linear system.
- Give definitions of Nyquist stability criterion *via* Cauchy's principle of the argument.
- Apply Nyquist stability criterion for the stability analysis of linear system.

4.1 Introduction

In Chaps. 2 and 3, time domain and frequency domain specifications were defined as applied to the linear system. A control system, when designed, has to satisfy some of these specifications required by the individual. In these chapters, one other important specification, namely system stability was not discussed in detail. If this important specification is not satisfied, the system is of no use. The time and frequency domain specifications discussed earlier have no meaning for an unstable system. Physically, for an unstable system, the time response grows without bound and causes severe damages not only to the system itself, but also to the environment and human life. Stability studies have been carried out both in the time domain and frequency domain. Routh–Hurwitz methods, root locus technique proposed by W. R. Evans and Liapunov stability criterion are the time domain approaches developed in the past for the stability studies. While Routh–Hurwitz and root locus methods are applicable

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S. Palani, *Automatic Control Systems*,
https://doi.org/10.1007/978-3-030-93445-3_4

only for a linear system, Liapunov method is applicable for both linear as well as non-linear system. The methods proposed by Nyquist and Popov belong to the frequency domain approach. While Nyquist stability criterion which is based on Cauchy's principle of the argument is applicable only to a linear system, Popov's method can be extended to non-linear systems with certain constraints. In the time domain, the method developed by Routh–Hurwitz and Evan's root locus is extensively used, while Nyquist stability criterion is more popular in the frequency domain. In this chapter, we deal with the application of Routh–Hurwitz stability criterion and Nyquist stability criterion. A separate chapter is devoted to the study of root locus technique. We discuss below the concept of stability and various terminologies associated with it.

4.2 The Concept of Stability

Stable systems are defined as the systems which will return to a stable equilibrium position after some small disturbance. On the other hand, systems which are given small initial disturbance, move off position and do not return to stable equilibrium position are termed as unstable systems. Consider the simple pendulum shown in Fig. 4.1a. When small disturbance is given to the pendulum, it oscillates horizontally and comes to the same vertical equilibrium position, which is stable. Now consider the inverted simple pendulum shown in Fig. 4.1b. When the pendulum is disturbed it falls to the horizontal. This is an example of an unstable system.

There are many definitions for stability depending upon the kind of system. We restrict our definitions to linear time invariant continuous time system. They are given below.

Linear system stability is defined in terms of the system time response. The time response consists of two parts. One is due to the initial condition and the other is the response due to the input. The total response = zero state response + zero input response.

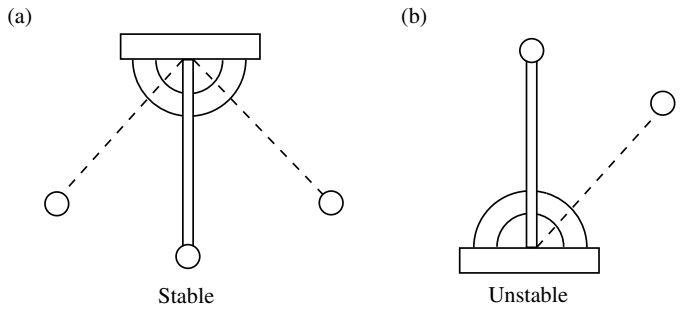


Fig. 4.1 **a** Simple pendulum and **b** Inverted simple pendulum

1. **Zero input response**: The zero input response is due to the initial conditions only. All the inputs are zero.
2. **Zero state response**: The zero state response is the response of the system due to the input only. All the initial conditions are zero.
3. **Zero input stability**: It refers to the stability condition due to zero input response which is due to the initial conditions only and all the inputs are zero. This stability also depends on the roots of the characteristic equation.

A linear time invariant system is said to be zero input stable if the zero input response of the system with finite initial conditions reaches zero as time t tends to infinity.

4.2.1 Asymptotic (Internal) Stability

A linear time invariant system is said to be in its equilibrium state if it remains in a particular state indefinitely in the absence of external input. For a linear time invariant system, zero state is the equilibrium state. When the system is given a small disturbance, the zero initial state is changed to non-zero state (initial conditions). These initial conditions will generate signals consisting of characteristic modes (roots) in the system. Let to itself if every mode in the system created by non-zero initial conditions returns back to zero as time t tends to infinity, the system is said to be asymptotically stable. On the other hand, even if any one of the modes grows with time and does not come back to zero state, the system is said to be asymptotically unstable. Asymptotic stability is also called internal stability or stability in the sense of Liapunov.

4.2.2 Marginal (Neutral) Stability

When a linear time invariant system is disturbed, the equilibrium is shifted from zero state to non-zero state. Some characteristic modes of the system neither decay to zero state nor grow indefinitely but remains constant, while other characteristic modes decay to zero state. These systems are said to be in borderline case. Such systems are said to be marginally stable with neutral equilibrium.

4.2.3 Bounded Input Bounded Output (BIBO) (External) Stability

The bounded input bounded output (BIBO) stability is defined with respect to the total response which is the sum of the zero input and zero state responses.

A linear time invariant system is said to be bounded input bounded output (BIBO) stable if every bounded input yields a bounded output. BIBO stability is also called as external stability.

4.2.4 *Relative Stability*

System stability is classified as *absolute stability* and *relative stability*. Absolute stability refers to the condition whether the given system is stable or not. If the system is stable, the closeness of stability is measured by relative stability. In the frequency domain, as discussed in Chap. 3, the relative stability is measured by *phase margin* and *gain margin*. In the time domain, the relative stability is measured by the closeness of the real part of the characteristic roots to the $j\omega$ axis. If the real part of the characteristic root is closer to the $j\omega$ axis, the relative stability is less and *vise versa*. Consider the following T.F. of a certain system.

$$G(s) = \frac{K(s+1)}{(s+5)(s+2+j3)(s+2-j3)} \tag{4.1}$$

The poles and zero are located in Fig. 4.2. From Fig. 4.2, the relative stability unit of the system is 2 units. This is the nearest distance of the pole of $G(s)$ to the $j\omega$ axis. Zero location does not affect the stability or the relative stability of the system.

4.2.5 *BIBO Stability via Impulse Response Function*

Consider the system with T.F. $G(s)$ represented as in Fig. 4.3. $g(t)$ is the impulse response function, while $r(t)$ is the input and $c(t)$ is the external output. The external output $c(t)$ is obtained using the following convolution integral

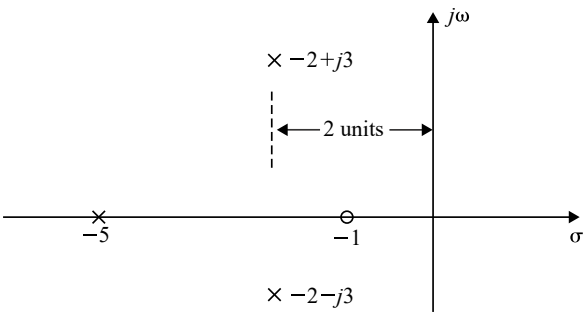


Fig. 4.2 Pole–zero locations of Eq. (4.1)

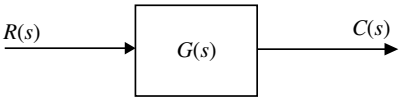


Fig. 4.3 Block diagram representation of a system with T.F.$G(s)$

$$\begin{aligned} c(t) &= g(t) * r(t) \\ &= \int_0^\infty g(\tau) r(t-\tau) d\tau \\ |c(t)| &= \left| \int_0^\infty g(\tau) r(t-\tau) d\tau \right| \end{aligned} \tag{4.2}$$

Equation (4.2) can be rewritten using the following property

$$\left| \int_0^\infty g(\tau) r(t-\tau) d\tau \right| \leq \int_0^\infty |g(\tau)||r(t-\tau)| d\tau \tag{4.3}$$

$$|c(t)| \leq \int_0^\infty |g(\tau)||r(t-\tau)| d\tau \tag{4.4}$$

For the input $r(t)$ being bounded, it can be expressed as $|r(t)| < A$. Now Eq. (4.4) is written as

$$|c(t)| \leq A \int_0^\infty |g(\tau)| d\tau \tag{4.5}$$

For the output $c(t)$ to be bounded, it can be expressed as

$$c(t) \leq B < \infty \tag{4.6}$$

Substituting Equation (4.6) in Eq. (4.5), we get

$$\int_0^\infty |g(\tau) d\tau| \leq \frac{B}{A} < \infty \tag{4.7}$$

Equation (4.7) gives the area of the impulse response curve. For the system to be BIBO stable, the area of the impulse response curve should be finite in the time interval $0 < t < \infty$.

The above requirement is shown in Fig. 4.4a and b.

4.2.6 BIBO Stability and the Characteristic Roots

It is not always possible to test the system with impulse input and measure the output with respect to time to asses the BIBO stability requirement. It can be easily

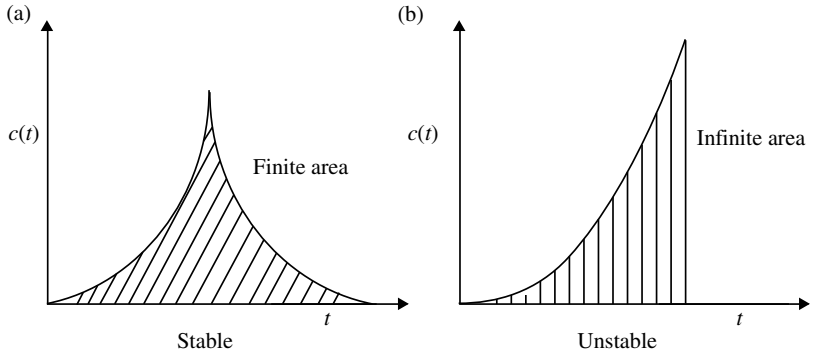


Fig. 4.4 Impulse response curve with **a** Finite area; and **b** Infinite area

determined by knowing the characteristic roots of the system. Consider the system shown in Fig. 4.3. By definition of Laplace transform, $G(s)$ can be written as

$$G(s) = \int_0^\infty g(t)e^{-st}dt \tag{4.8}$$

Making use of the property stated in Eq. (4.5), we get

$$|G(s)| \leq \int_0^\infty |g(t)||e^{-st}|dt \tag{4.9}$$

$$\leq \int_0^\infty |g(t)||e^{-\sigma t}|dt \tag{4.10}$$

where σ is the real part of $s = \sigma + j\omega$ and $|e^{j\omega t}| = 1$. $G(s) = \infty$ at s-poles of $G(s)$. Thus, Eq. (4.10) is written as

$$\infty \leq \int_0^\infty |(g(t))||e^{-\sigma t}|dt \tag{4.11}$$

For one or more number of poles in RHP of the s-plane or in the $j\omega$ axis, $\sigma \geq 0$ and

$$|e^{-\sigma t}| \leq M \quad \text{or} \quad e^{-\sigma t} = 1 \tag{4.12}$$

Substituting Equation (4.12) in Eq. (4.11), we get

$$\int_0^\infty |g(t)|Mdt \geq \infty$$

$$\int_0^\infty g(t)dt \geq \infty \tag{4.13}$$

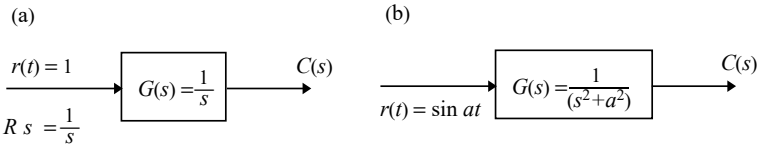


Fig. 4.5 System poles at **a** Origin and **b** $j\omega$-axis

Equation (4.13) implies that the system is BIBO unstable when atleast one pole lies in RHP or on the $j\omega$ axis. It is to be noted that the stability condition (4.13) does not depend upon the input applied. A linear time invariant continuous time system is said to be BIBO stable if all the roots are located in the left half s-plane. A system is unstable if it is not BIBO stable.

Consider the system whose pole is at the origin of the s-plane. The input is the unit step which is bounded. The output

$$c(t) = L^{-1}\frac{1}{s^2} = t$$

The output is unbounded and hence it is BIBO unstable.

Consider the system shown in Fig. 4.5b. The poles of $G(s)$ are located at $s = \pm ja$. The input $r(t) = \sin at$ is bounded, whereas the output

$$c(t) = L^{-1}\frac{1}{(s+a^2)^2} = t\sin at$$

is unbounded.

The above two examples illustrate that for BIBO stability, there should not be even a single root in RHP, and further the characteristic roots should not be at the origin as well as in the $j\omega$ axis. All the characteristic roots should be in LHP for the system to be BIBO stable.



Note: When a pole at the origin is intentionally introduced the system is said to be stable.

4.2.7 Relationship Between BIBO and Asymptotic Stability

The impulse response of a system is nothing but the linear combination of system characteristic modes. If the characteristic root λ is in LHP, the corresponding mode $e^{-\lambda t}$ is absolutely integrable. On the other hand, if the characteristic root is either in RHP or in the imaginary axis, the characteristic mode is not integrable. The first statement implies that if the system is asymptotically stable, then it is BIBO stable. According to the second statement, marginally stable system or asymptotically

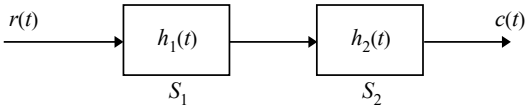


Fig. 4.6 Two systems connected in cascade

unstable system is BIBO unstable. The converse is that BIBO stability does not always imply asymptotic stability.

Consider two systems S_1 and S_2 with impulse response functions $h_1(t) = e^{-2t}u(t)$ and $h_2(t) = e^{-3t}u(t)$ connected in cascade. The characteristic roots are $\lambda_1 = -2$ and $\lambda_2 = -3$ and $\int_0^\infty e^{-2t}dt$ and $\int_0^\infty e^{-3t}dt$ are absolutely integrable. The system is asymptotically stable. From Fig. 4.6, $H_1(s) = 1/(s+2)$ and $H_2(s) = 1/(s+3)$ and

$$H(s) = H_1(s)H_2(s) = \frac{1}{(s+2)(s+3)}$$

The overall impulse response of the system is

$$h(t) = (e^{-2t} - e^{-3t})u(t)$$

The area of the impulse response curve is finite and the system is BIBO stable. The system is asymptotically and BIBO stable.

Now consider Fig. 4.6 which has $h_1(t) = (\delta(t) - 4e^{-2t})$ and $h_2(t) = e^{2t}$. The characteristic root of system S_2 is $\lambda = 2$ and the corresponding characteristic mode $\int_0^\infty e^{+2t}dt$ and is not integrable. Hence, system S_2 is asymptotically unstable with the characteristic root 2 in RHP. However,

$$H_1(s) = \left(1 - \frac{4}{s+2}\right) = \frac{(s-2)}{s+2}$$

and

$$H_2(s) = \frac{1}{s-2}$$

The overall impulse response function is

$$\begin{aligned} H(s) &= H_1(s)H_2(s) \\ &= \frac{(s-2)}{s+2}\frac{1}{s-2} \\ &= \frac{1}{(s+2)} \end{aligned}$$

The output of the system is $c(t) = e^{-2t}u(t)$ which is finite, and therefore, it is BIBO stable. In the above example, BIBO stable system does not imply that it is asymptotically stable. This is due to pole–zero cancellation in the system. With pole–zero cancellation, the system is not controllable or observable. Thus, if the system is both

controllable and observable, BIBO stability for a linear time invariant continuous system guarantees asymptotic stability.

Locations of the characteristic root and zero input response are shown in Fig. 4.7. The characteristic roots location and system stability is shown in Fig. 4.8.

Figure 4.7a shows the characteristic root in LHP only. The system is **asymptotically stable.** For such pole location, the impulse response will be exponentially decaying and hence the system is **BIBO stable.**

Figure 4.7b shows the presence of a single root at the origin and no root in RHP. The system is **marginally stable.** The impulse response curve is a step in nature and the area under the curve is not finite as t tends to ∞. Hence, the system is **BIBO unstable**.

Figure 4.7c shows complex conjugate roots in LHP and no root in RHP and the $j\omega$ axis. Hence, the system is **asymptotically stable**. The impulse response curve is decaying with oscillations. However, the area under the curve is finite and hence the system is BIBO stable.

Figure 4.7d shows the presence of complex conjugate roots in the $j\omega$ axis. The system is **marginally stable**. The impulse response curve has sustained oscillations and hence the area under this curve is not finite. Hence, the system is **BIBO unstable**.

Figure 4.7e shows one characteristic root in RHP. The system is, therefore, **asymptotically unstable**. The corresponding impulse (zero input) response curve is exponentially increasing, and therefore, **BIBO unstable**.

Figure 4.7f shows two characteristic roots placed at the origin of the s-plane. Hence, the system is **asymptotically unstable**. The corresponding impulse response curve is linearly increasing with time and hence it is unbounded. The system is **BIBO unstable**.

Figure 4.7g shows two complex roots are placed in the RHP of the s-plane. Hence, the system is **asymptotically unstable**. The impulse response curve is exponentially increasing with oscillations. Hence, the system is **BIBO unstable**.

Figure 4.7h shows that there are two pairs of repeated roots in the $j\omega$ axis. Hence, the output is **asymptotically unstable**. The impulse response curve is linearly increasing with respect to time with unbounded oscillations. Hence, the system is **BIBO unstable**.

In Fig. 4.8, the stable region, unstable region and the marginally stable region in the s-plane are sketched and shown. Characteristic roots in LHP indicate asymptotic stability, the roots in RHP region indicates instability and the presence of single root in the $j\omega$ axis indicates marginal stability. Presence of repeated roots in the $j\omega$ axis indicates instability.

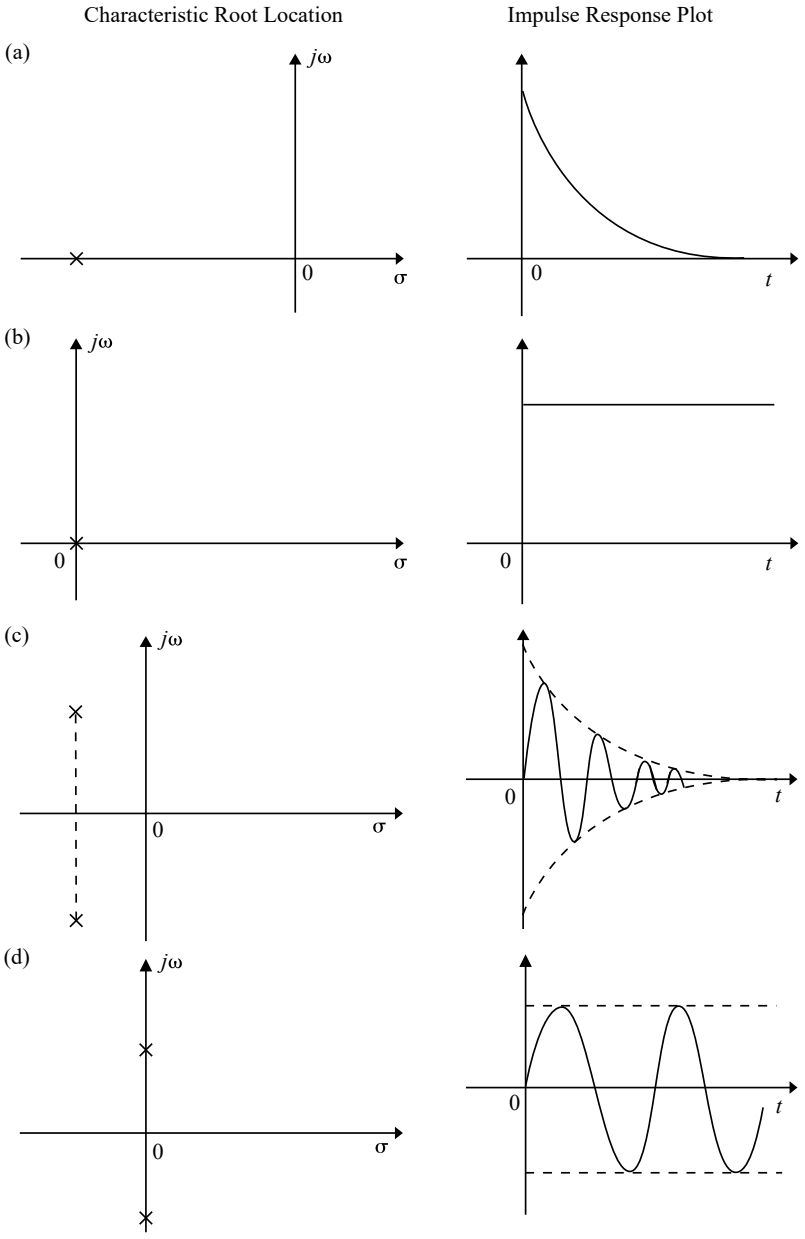


Fig. 4.7 Locations of characteristic roots and corresponding zero input response

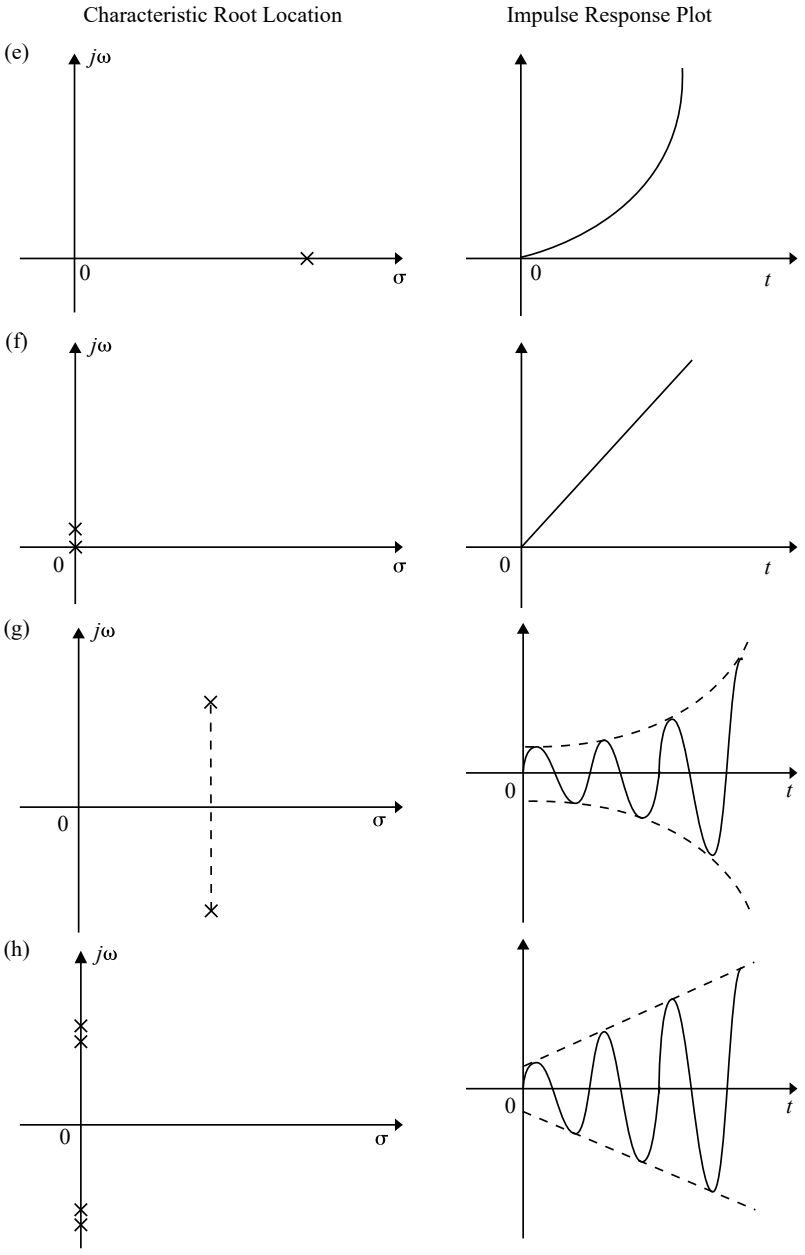


Fig. 4.7 (continued)

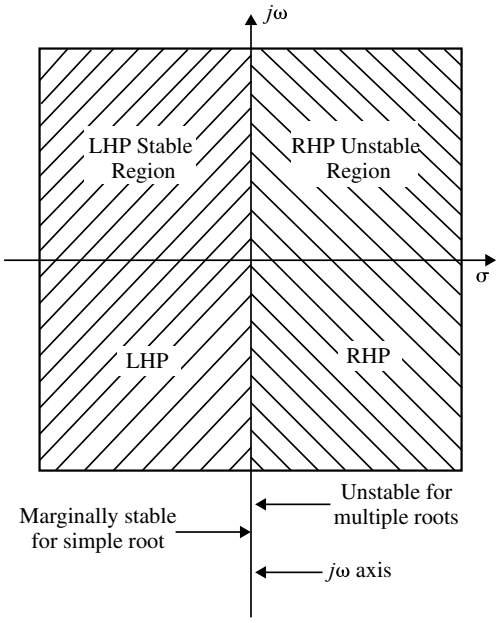


Fig. 4.8 Characteristic root location in the complex s-plane and system stability.

To Summarize

1. A linear time invariant continuous time system is asymptotically stable if all the characteristic roots are in LHP.
2. A linear time invariant continuous time system is unstable either at least one characteristic root is in the RHP or there are repeated roots in the $j\omega$ axis.
3. A linear time invariant continuous time system is marginally stable if there are no roots in the RHP and there are some unrepeated roots in the $j\omega$ axis.
4. An asymptotically stable system is also BIBO stable. An asymptotically unstable system or a marginally stable system is BIBO unstable.
5. BIBO stable system need not be asymptotically stable. This may occur when there is pole–zero cancellation in the system.
6. If the system is both controllable and observable, BIBO stability guarantees asymptotic stability also.

Example 4.1

Investigate the asymptotic and BIBO stability of the following linear time invariant continuous time system with the following characteristic polynomial:

(1) $F(\lambda) = (\lambda + 2)(\lambda + 2 + j3)(\lambda + 2 - j3)$
(2) $F(\lambda) = (\lambda - 2)(\lambda + 2 + j3)(\lambda + 2 - j3)$
(3) $F(\lambda) = (\lambda + 1)(\lambda^2 + 9)$
(4) $F(\lambda) = (\lambda + 1)(\lambda^2 + 9)^2$
(5) $F(\lambda) = \lambda(\lambda + 3)$
(6) $F(\lambda) = \lambda^2(\lambda + 3)$

Solution

(1)

$$F(\lambda) = (\lambda + 2)(\lambda + 2 + j3)(\lambda + 2 - j3)$$

The characteristic roots are

$$\lambda_1 = -2$$
$$\lambda_2 = -2 + j3$$
$$\lambda_3 = -2 - j3$$

All the characteristic roots are in LHP and hence the system is asymptotically stable.
The characteristic modes corresponding to the above characteristic roots are as given below

$$\text{For } \lambda_1, \quad \int_0^\infty e^{-2t} dt$$
$$\text{For } \lambda_2, \quad \int_0^\infty e^{-(2-j3)t} dt$$
$$\text{For } \lambda_3, \quad \int_0^\infty e^{-(2+j3)t} dt$$

The above characteristic modes are absolutely integrable and hence the system is BIBO stable.

Asymptotically and BIBO stable

(2)

$$F(\lambda) = (\lambda - 2)(\lambda + 2 + j3)(\lambda + 2 - j3)$$

The characteristic roots are

$$\lambda_1 = 2$$
$$\lambda_2 = -2 - j3$$
$$\lambda_3 = -2 + j3$$

Since $\lambda_1 = 2$ is in RHP, the system is Asymptotically unstable.
Corresponding to $\lambda_1 = 2$, the characteristic mode is

$$\int_0^\infty e^{2t} dt$$

and it is not integrable and the system is BIBO unstable.

Asymptotically and BIBO unstable

(3)

$$F(\lambda) = (\lambda + 1)(\lambda^2 + 9)$$

The characteristic roots are

$$\lambda_1 = -1$$
$$\lambda_2 = +j3$$
$$\lambda_3 = -j3$$

There are unrepeated roots in the $j\omega$ axis. The system is marginally stable but BIBO unstable.

Marginally stable and BIBO unstable

(4)

$$F(\lambda) = (\lambda + 1)(\lambda^2 + 9)^2$$

The characteristic roots are

$$\lambda_1 = -1$$
$$\lambda_2 = +j3$$
$$\lambda_3 = +j3$$
$$\lambda_4 = -j3$$
$$\lambda_5 = -j3$$

The roots λ_2 and λ_3 and the roots λ_4 and λ_5 are repeated in the $j\omega$ axis. Hence, it is asymptotically and hence BIBO unstable.

Asymptotically and BIBO unstable

(5)

$$F(\lambda) = \lambda(\lambda + 3)$$

The characteristic roots are

$$\lambda_1 = 0$$
$$\lambda_2 = -3$$

Since there is a single root in the $j\omega$ axis, the system is marginally stable but BIBO unstable.

Marginally stable and BIBO unstable

(6)

$$F(\lambda) = \lambda^2(\lambda + 3)$$

The characteristic roots are

$$\lambda_1 = 0$$
$$\lambda_2 = 0$$
$$\lambda_3 = -3$$

Since there are repeated roots in the $j\omega$ axis, the system is asymptotically unstable and hence BIBO unstable.

Asymptotically and BIBO unstable

Example 4.2

A system consists of two subsystems connected in cascade with impulse responses as $h_1(t)$ and $h_2(t)$. Find whether the overall system is asymptotically and BIBO stable (Fig. 4.9).

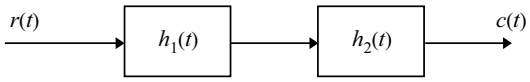


Fig. 4.9 Subsystems connected in cascade for Example 4.2

1. $h_1(t) = (\delta(t) - 2e^{-4t})u(t)$
   $h_2(t) = e^{4t}u(t)$
2. $h_1(t) = (\delta(t) - 8e^{-4t})u(t)$
   $h_2(t) = e^{4t}u(t)$

Solution

1.

$$\begin{aligned} h_1(t) &= (\delta(t) - 2e^{-4t})u(t) \\ H_1(s) &= \left(1 - \frac{2}{(s+4)}\right) \\ &= \frac{(s+2)}{s+4} \\ h_2(t) &= e^{4t}u(t) \\ H_2(s) &= \frac{1}{(s-4)} \end{aligned}$$

The overall impulse response function is

$$\begin{aligned} H(s) &= H_1(s)H_2(s) \\ &= \frac{(s+2)}{(s+4)(s-4)} \end{aligned}$$

The characteristic roots are

$$\begin{aligned} \lambda_1 &= -4 \\ \lambda_2 &= 4 \end{aligned}$$

Since λ_2 is in RHP, the system is asymptotically unstable. When it is asymptotically unstable, it is also BIBO unstable.

Asymptotically and BIBO unstable

2.

$$\begin{aligned} h_1(t) &= (\delta(t) - 8e^{-4t})u(t) \\ h_2(t) &= e^{4t}u(t) \end{aligned}$$

$$H_1(s) = 1 - \frac{8}{(s+4)}$$
$$= \frac{(s-4)}{(s+4)}$$
$$H_2(s) = \frac{1}{(s-4)}$$

The overall impulse response function is

$$H(s) = H_1(s)H_2(s)$$
$$= \frac{(s-4)}{(s+4)(s-4)}$$

The characteristic roots are

$$\lambda_1 = -4 \quad \text{and} \quad \lambda_2 = 4$$

Since λ_2 is in RHP, the system is internally or asymptotically unstable.
If we measure the output $c(t)$, it is due to $h(t) = L^{-1}H(s)$

$$h(t) = L^{-1}\frac{1}{s+4}$$
$$= e^{-4t}u(t)$$

The overall impulse response function is integrable and finite. Hence, the system is BIBO stable or externally stable. This is due to pole–zero cancellation.

Asymptotically or Internally unstable.
BIBO or Externally stable.

4.3 Routh–Hurwitz Criterion

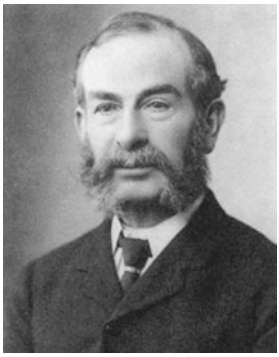
Routh–Hurwitz criterion is a time domain approach to determine the number of roots of the characteristic polynomial with constant real coefficients with respect to the LHP, RHP and the $j\omega$ axis of the s-plane. The method requires two steps. For the given characteristic equation, using the coefficients, the Routh table is prepared. From the Routh table, the information about how many roots of the characteristic polynomial are in the LHP, RHP and the $j\omega$ axis is interpreted. Even though this information can be easily obtained using modern calculators and computers, and also determine the exact location of the closed loop system poles, the power of this method lies in design rather than analysis. The Routh–Hurwitz criterion is applied to determine the following:

1. Absolute stability of the system.
2. Relative stability of the system in the time domain.
3. The gain of the system can be increased before the closed loop system becomes unstable.
4. To factorize the given polynomial in some special cases and locate the roots in the s-plane.
5. To find the number of LHP, RHP and $j\omega$ axis roots.
6. Stability of the system is determined without solving the characteristic equation. It is a straightforward method, and therefore, invariably used in the design of compensators.

Disadvantages of Routh's Method

1. If the system is stable, Routh's criterion does not tell how stable it is.
2. If the system is unstable, the method does not suggest how to improve the stability of the system.
3. In the preparation of Routh's array, when the complete row contains zero coefficients, and there is no sign change in the first column, the hasty conclusion that the system is stable is erroneous. In such cases, careful evaluation of roots in the $j\omega$ axis is very much essential. Presence of simple roots in the $j\omega$ axis indicates that the system is neither stable nor unstable. It is called marginally stable. However, if repeated roots are identified in the $j\omega$ axis, the system is no more marginally stable but it is unstable. Routh–Hurwitz criteria will not reveal this form of instability which is a major disadvantage of this method.
4. Routh–Hurwitz criterion is applicable for the characteristic polynomial whose coefficients are real. For complex coefficient polynomial, it is not applicable. Further, if the characteristic equation is not algebraic, such as containing exponential functions or sinusoidal functions of s, the Routh–Hurwitz criterion cannot be applied.
5. $j\omega$ axis is considered as the stability boundary when the Routh–Hurwitz criterion is applied. The method cannot be applied to any other stability boundaries in a complex plane. Thus, for a discrete time system, where the unit circle is the stability boundary, Routh–Hurwitz criterion cannot be applied. By linear transformation, the unit circle has to be transformed to $j\omega$ axis and then Routh criterion is applied.

A Historical Note: **Edward John Routh (1831–1907)**



Edward John Routh was born on 20 January, 1831 in Quebec, Canada. He was an English mathematician. Routh came to England aged eleven and entered University College, London, in 1847, and chose a career in mathematics. He obtained his B.A.; and M.A.; degrees in mathematics in 1849 and 1853, respectively. On graduation, Routh took up work as a private mathematics teacher in Cambridge for students preparing for Tripos examination. He published a textbook titled "*Dynamics of a System of Rigid Bodies*".

In addition to teaching and writing, he also contributed original research work such as the Routh–Hurwitz theorem. Routh's stability criterion, Routh's array, Routhian, Routh's theorem and Routh's Algorithm. J.J. Thomson and George Darwin are his notable students. Routh married Hilda in 1864, and the couple had five sons and a daughter. One of his sons, John Routh served in the Indian Civil Service. Routh died on 7 June 1907, at the age of 76.

Adolf Hurwitz (1859–1919)



Adolf Hurwitz was a German mathematician. His father, was in the manufacturing business, and his mother died when he was just three years old. He entered the University of Munich in 1877 and then migrated to the University of Berlin. In 1881, he obtained his doctoral degree from the University of Leipzig. In 1884, Hurwitz married Ida Sumuel, the daughter of a professor of medicine. They had three children. Hurwitz spent most of his time in Zurich, Switzerland. He suffered from severe migraines when he was a student in Munich. In 1905, one of his kidneys became diseased and the other one was removed. He died on 18 November 1919, at the age of 60.

4.3.1 Routh–Hurwitz Criterion to Determine Absolute Stability

Routh–Hurwitz stability criterion was developed independently by Routh in 1892 and Hurwitz in 1895. The combined theory suggests the analytical method of finding the number roots of the characteristic polynomial in the LHP, RHP and the $j\omega$-axis without actually solving for them.

Consider the system shown in Fig. 4.10. The closed loop T.F. of the above system is given by

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + GH(s)} \tag{4.14}$$

For a strictly proper T.F., Eq. (4.14) in polynomial form is as given below

$$\frac{C(s)}{R(s)} = \frac{b_0 s^m + b_1 s^{m-1} + \cdots + b_{m-1} s + b_m}{a_0 s^n + a_1 s^{n-1} + \cdots + a_{n-1} s + a_n} \tag{4.15}$$

where $n \geq m$ and $a's$ and $b's$ are constants and real. The characteristic equation of (4.15) is

$$a_0 s^n + a_1 s^{n-1} + \cdots + a_{n-1} s + a_n = 0 \tag{4.16}$$

The roots of the characteristic Eq. (4.16) should all be in the LHP for the system to be stable. For this, the following conditions are necessary before proceeding further to investigate the absolute stability using Routh–Hurwitz stability criterion.

1. The coefficient $a_0, a_1, \ldots, a_{n-1}, a_n$ must be positive.
2. All the coefficients in the characteristic equation must be present. If any one coefficient is zero, there will be a root or roots in the imaginary axis or in the RHP, which means the system will be unstable.

Formation of Routh's Array

If the above necessary conditions are satisfied, we can proceed further to establish the sufficient condition required for the closed loop system to be stable. For this, Routh's array is prepared as explained below.

The coefficients of Eq. (4.16) are arranged in rows and columns with even powers of s in one row and odd power of s in the next row. If the highest degree of the polynomial is even, then the even row is started first followed by the odd row. If the highest degree of the polynomial is odd, the odd row components are started first followed by even power components. By this, the first two rows of the Routh array are formed, from which subsequent rows are formed as explained below.

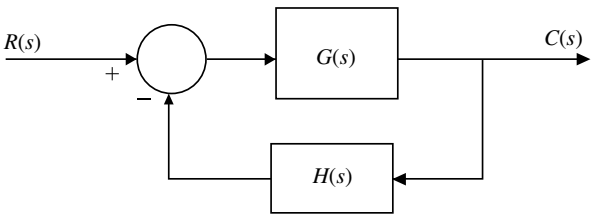


Fig. 4.10 Control system with forward and feedback transfer functions

$$\begin{array}{c|cccc} s^n & a_0 & a_2 & a_4 & \cdots \\ s^{n-1} & a_1 & a_3 & a_5 & \cdots \\ s^{n-2} & b_1 & b_2 & b_3 & \cdots \\ s^{n-3} & c_1 & c_2 & c_3 & \cdots \\ \vdots & \vdots & \vdots & & \\ s^2 & d_1 & d_2 & & \\ s^1 & e_1 & & & \\ s^0 & f_1 & & & \end{array}$$

For an nth degree characteristic polynomial, there will be $(n + 1)$ Routh's rows. The first two rows are prepared from the coefficients of the given characteristic equation. The coefficients of the remaining $(n - 1)$ rows are evaluated as follows:

$$\begin{aligned} b_1 &= \frac{a_1 a_2 - a_0 a_3}{a_1} \\ b_2 &= \frac{a_1 a_4 - a_0 a_5}{a_1} \\ b_3 &= \frac{a_1 a_6 - a_0 a_7}{a_1} \\ &\vdots \end{aligned}$$

$$\begin{aligned} c_1 &= \frac{b_1 a_3 - a_1 b_2}{b_1} \\ c_2 &= \frac{b_1 a_5 - a_1 b_3}{b_1} \\ c_3 &= \frac{b_1 a_7 - a_1 b_4}{b_1} \\ &\vdots \end{aligned} \tag{4.17}$$

The process is continued until the s^0th row is completed. The complete array of coefficients looks like a triangle. After preparing the Routh's array following the procedure described above, the Routh's stability criterion is applied to Eq. (4.16).

The Routh–Hurwitz criterion states that the number of roots of the characteristic equation with positive real parts is equal to the number of changes in sign of the first column of the Routh array.

In assessing the sign changes in the first column array, suitable modification in the calculation procedure should be followed, especially in the following cases:

1. Zero only in the first column.
2. Zeros in one complete row.

3. Zeros in one complete row with repeated roots in the $j\omega$ axis.

If case 3 is not properly evaluated, false indication of stability condition will be arrived at.



Note: For simplifying the numerical calculation, the entire row may be multiplied or divided by a positive number by which the stability condition is unaffected.

Example 4.3

For the following characteristic polynomial, determine whether the closed loop system is stable. Use Routh–Hurwitz stability criterion

$$(1)\quad F(s) = s^5 + 4s^4 - 8s^3 + 11s^2 + 6s + 4$$
$$(2)\quad F(s) = s^6 + 7s^5 + 4s^3 + 2s^2 + 10s + 6$$

Solution

(1)

$$F(s) = s^5 + 4s^4 - 8s^3 + 11s^2 + 6s + 4$$

In the above characteristic polynomial, -8 is one coefficient with negative sign. This indicates that there is at least one root of the characteristic equation in RHP and hence the system is unstable. There is no need to prepare Routh's array.

(2)

$$F(s) = s^6 + 7s^5 + 4s^3 + 2s^2 + 10s + 6$$

In the above characteristic polynomial, the coefficient of s^4 is zero. Hence, all the roots will not be in LHP. Some roots may be in the $j\omega$ axis or in RHP. Hence, the system is not stable.

In the above two cases, the necessary condition for Routh–Hurwitz stability criterion is not satisfied and the system is unstable. There is no need to prepare the Routh's array.

Example 4.4

Determine the stability of the system determined by

$$F(s) = s^4 + 6s^3 + 23s^2 + 40s + 50$$

Solution

$$F(s) = s^4 + 6s^3 + 23s^2 + 40s + 50$$

s^4	1	23	50
s^3	6	40	
s^2	16.33	50	
s^1	21.63		
s^0	50		

Since there is no sign change in the first column of Routh's array, no root is in RHP. Hence, the system is stable.

Example 4.5

Find the number of roots in this equation

$$s^6 + 2s^5 + 8s^4 + 8s^3 + 20s + 10s + 16 = 0$$

with positive real part, zero real part and negative real part.

Solution

$$F(s) = s^6 + 2s^5 + 8s^4 + 8s^3 + 20s^2 + 10s + 16$$

The following Routh's array is formed.

s^6	1	8	20	16
s^5	2	8	10	
s^4	4	15	16	
s^3	0.5	2		
s^2	−1	16		
s^1	10			
s^0	16			

There are two sign changes from s^3th row to s^2th row and from s^2th row to s^1th row as shown in the Routh's table. The polynomial has 2 RHP roots.

To find the LHP roots the polynomial $F(-s)$ is obtained and the Routh's array is formed. Here the number of sign changes will give the number of LHP roots

$$F(-s) = s^6 - 2s^5 + 8s^4 - 8s^3 + 20s^2 - 10s + 16$$

s^6	1	8	20	16
s^5	−2	−8	−10	
s^4	4	15	16	
s^3	−0.5	−2		
s^2	−1	16		
s^1	−10			
s^0	16			

As seen from Routh's array, there are four sign changes in the first column. Hence, there are 4 LHP roots. There is no root in the $j\omega$ axis.

RHP root 1
LHP root 4
$j\omega$ axis root Nil

4.3.2 Routh–Hurwitz Criterion: Special Cases

When the Routh array is prepared, sometimes we will have the first column of any row becoming zero or the entire row becoming zero. When these cases occur, we say the Routh array is broken and it cannot be completed. These are called special cases and they are discussed ahead.

First Column Becoming Zero

When the first column becomes zero, division by the zero becomes infinity and the Routh's array cannot be completed. This phenomenon can be avoided by following any one of the methods described below:

1. By forming the reciprocal of the polynomial.
2. A small value epsilon ε is assigned to replace the zero.
3. The given polynomial is multiplied by a factor $(s + a)$ where $a > 0$ and the Routh's array is formed for the new polynomial.

Method one is computationally easier than the epsilon method. However, when the reciprocal of a polynomial is formed, for the new polynomial also the first column becoming zero may repeat and the method fails. The third method increases the degree of the polynomial by one and an additional row is required in the Routh's array. The special case is illustrated in the following examples.

Example 4.6

Examine the stability by Routh's criterion for

$$s^5 + s^4 + 2s^3 + 2s^2 + 3s + 15 = 0$$

(*Anna University, May, 2009*)

Solution

$$F(s) = s^5 + s^4 + 2s^3 + 2s^2 + 3s + 15$$

The following Routh's array is formed.

s^5	1	2	3
s^4	1	2	15
s^3	0	-12	

The Routh array is broken with the first column of s^3th row becoming zero. This is solved using the method given below.

Method 1: By Forming Reciprocal of the Polynomial

Substitute $s = 1/x$ and form $F(x)$

$$F(x) = \frac{1}{x^5} + \frac{1}{x^4} + \frac{2}{x^3} + \frac{2}{x^2} + \frac{3}{x} + 15 = 0$$

$$15x^5 + 3x^4 + 2x^3 + 2x^2 + x + 1 = 0$$

The Routh array is formed for the above polynomial as given below.

x^5	15	2	1
x^4	3	2	1
x^3	-2	-1	(After dividing by 4)
x^2	0.5	1	
x^1	3		
x^0	1		

There are two sign changes in the first column of Routh's array. The system is unstable.

Method 2: Epsilon Method
The first column of s^3th row becomes zero. Zero is replaced by a small value ε^+ and the Routh's array is completed. After completion, ε is put to zero and the number of sign changes in the first column is counted.

	With ε			With $\varepsilon = 0$		
s^5	1	2	3	1	2	3
s^4	1	2	15	1	2	15
s^3	ε^+	-12		0^+	-12	
s^2	$\dfrac{2\varepsilon + 12}{\varepsilon}$	15		$+\infty$	15	
s^1	$\dfrac{-15\varepsilon^2 - 24\varepsilon - 144}{2\varepsilon + 12}$			-12		
s^0	15			15		

There are two sign changes in the first column of Routh's array and hence there are 2 RHP roots. The system is unstable.

Method 3: Multiplying by a Factor $(s + 1)$
The given polynomial is multiplied by any factor and the new polynomial is obtained. For the new polynomial, Routh's array is formed. Arbitrarily, we choose the factor $(s + 1)$. The new polynomial is

$$\begin{aligned} F(s) &= (s + 1)(s^5 + s^4 + 2s^3 + 2s^2 + 3s + 15) \\ &= s^6 + 2s^5 + 3s^4 + 4s^3 + 5s^2 + 18s + 15 \end{aligned}$$

For the above polynomial, the following Routh's array is formed.

s^6	1	3	5	15
s^5	2	4	18	
s^4	1	−4	15	
s^3	12	−12		
or	1	−1	After dividing by 12	
s^2	−3	15		
s^1	4			
s^0	15			

There are two sign changes in the first column of the Routh's array. There are two RHP roots. The system is unstable.

Example 4.7

Consider the following characteristic polynomial

$$F(s) = 2s^4 + s^3 + 2s^2 + s + 2$$

Find whether the system is stable.

Solution

$$2s^4 + s^3 + 2s^2 + s + 2 = 0$$

The following Routh's array is formed

s^4	2	2	2
s^3	1	1	
s^2	0	2	

The Routh array is broken. We substitute $s = 1/x$ and get the following characteristic polynomial is x

$$2x^4 + x^3 + 2x^2 + x + 2 = 0$$

Both original polynomial and the polynomial of the reciprocal are the same; the method fails. We follow the method by multiplying by the factor $(s + 2)$

$$\begin{aligned} F(s) &= (s + 2)(2s^4 + s^3 + 2s^2 + s + 2) \\ &= 2s^5 + 5s^4 + 4s^3 + 5s^2 + 4s + 4 \end{aligned}$$

The Routh's array is formed below.

s^5	2	4	4
s^4	5	5	4
s^3	2	2.4	
s^2	-1	4	
s^1	10.4		
s^1	4		

There are two sign changes and there are two roots in RHP. The system is unstable.

By Epsilon Method

	With ε			With $\varepsilon = 0$		
s^4	1	1	1 (After dividing by 2)	1	1	1
s^3	1	1		1	1	
s^2	ε	1		0^+	1	
s^1	$\frac{\varepsilon-1}{\varepsilon}$			$-\infty$		
s^0	1			1		

There are 2 sign changes and there are 2RHP roots. The system is unstable.

Routh Array with Entire Row Being Zero

The second special case of Routh array is when the entire row becomes zero which is due to even polynomial which is a factor of the original polynomial. This should be handled in a different way from the case of a zero only in the first column of a row. Let us consider a sixth degree polynomial to demonstrate how to construct and interpret the Routh array when an entire row of zero is present

$$a_0 s^6 + a_1 s^5 + a_2 s^4 + a_3 s^3 + a_4 s^2 + a_5 s + a_6 = 0 \tag{4.18}$$

The following Routh array is formed. We assume that s^3th row has zeros in all the columns.

s^6	a_0	a_2	a_4	a_6	
s^5	a_1	a_3	a_5		
...	.....				
s^4	b_1	b_2	b_3		
s^3	0	0		$A(s) = b_1 s^4 + b_2 s^2 + b_3$	
or	$4b_1$	$2b_2$		$dA(s)/ds = 4b_1 s^3 + 2b_2 s$	
s^2	c_1	c_2			
s^1	d_1				
s^0	e_1				

To complete the Routh's array, we use the following procedure. Draw a horizontal dotted line just above the row which has zeros in the entire row. The equation of the

row which is just below the dotted line is called auxiliary equation. For the example cited above, the auxiliary equation is

$$A(s) = b_1 s^4 + b_2 s^2 + b_3 \tag{4.19}$$

The auxiliary equation is differentiated with respect to s and the following equation is obtained

$$\frac{dA(s)}{ds} = 4b_1 s^3 + 2b_2 s \tag{4.20}$$

The elements of the s^3th row are taken from Eq. (4.20) and the Routh array is completed as shown. Now the interpretation of Routh table is very important. An entire row of zeros in the Routh array is possible when purely even or purely odd polynomial is a factor of the original polynomial.

Equation (4.19) is a factor of the original polynomial. This is an even polynomial. If this polynomial divides the original polynomial, the reminder is also a factor. The even polynomial of Eq. (4.19) has the following properties:

1. The roots of the even polynomial are symmetric about the origin and real.
2. Some of the roots of the even polynomial could be on the $j\omega$-axis as simple roots are **repeated roots**.
3. The roots are quadrantal.

The root locations are shown in Fig. 4.11.

(a) Real and symmetrical about the origin
(a) Simple $j\omega$ axis roots symmetrical about the origin
(c) Repeated $j\omega$ axis roots symmetrical about the origin
(d) Quadrantal and symmetrical roots about the origin.

Routh Array Interpretation

1. Routh array from the row of the even polynomial (in the above example s^4th row) down to the s^0th row is completed and examined.
2. If there are any sign changes, from s^4th row to s^0th row the number of sign changes give the RHP roots of the even polynomial. By symmetry, the same number of roots are in LHP. The remaining roots are in the $j\omega$ axis symmetrically placed with respect to the origin. If there is no sign change, then RHP as well as LHP roots are zero for the even polynomial.
3. From step two, the number of roots in the $j\omega$ axis roots are known. These roots may be simple (Fig. 4.11b) or repeated (Fig. 4.11c). For simple roots, the system is marginally stable. For repeated roots, the system is unstable.
4. For repeated roots in the $j\omega$ axis, the auxiliary Eq. (4.19) is of the form $A(s) = (s^2 + a^2)$.
5. Routh array is now interpreted above the horizontal dotted line to evaluate the remaining roots. The number of sign changes give the number of RHP roots. (The sign change is evaluated from s^6th row to s^5th row).

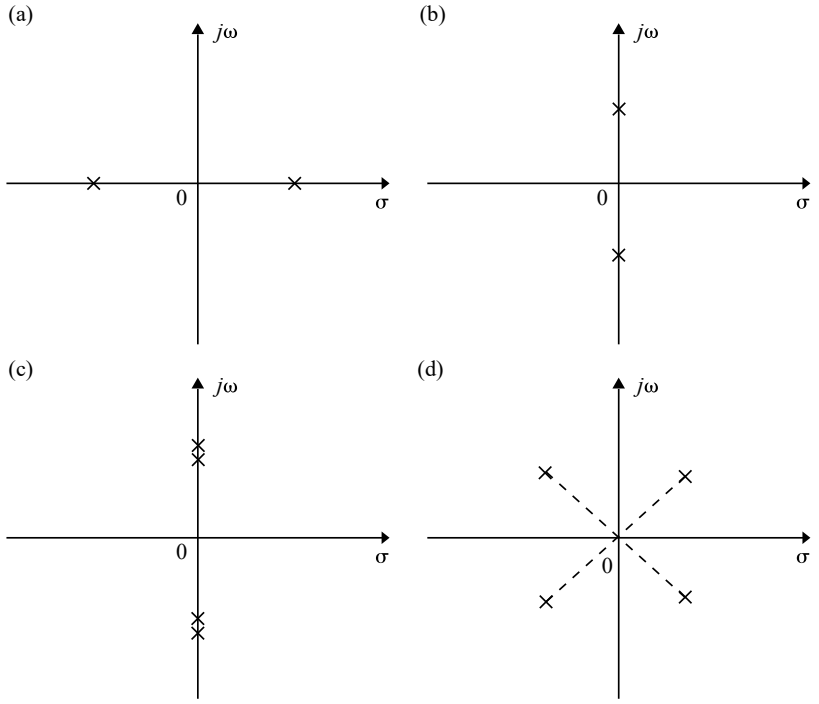


Fig. 4.11 Root locations of an even polynomial

6. The sum of the roots found in steps 2 and 5 together give the LHP, RHP and $j\omega$ axis roots. The following examples illustrate the above procedure.

Example 4.8

For the characteristic polynomial given below find positive real, negative real and imaginary roots. Is the system stable? Factorize the polynomial and locate the roots in the s-plane

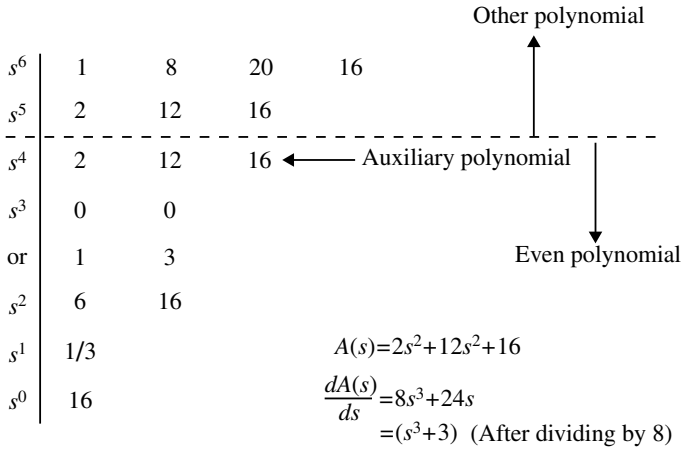
$$s^6 + 2s^5 + 8s^4 + 12s^3 + 20s^2 + 16s + 16 = 0$$

(*Anna University, April, 2010*)

Solution

$$F(s) = s^6 + 2s^5 + 8s^4 + 12s^3 + 20s^2 + 16s + 16$$

The Routh's array is formed as given below



Routh array interpretation

1. The Routh array is first inspected from s^4th row down to s^0th row. It is seen that there is no sign change in the first column. This implies that there is no RHP root of the even polynomial.
2. If there is no RHP root for the even polynomial, by symmetry (*see* Fig. 4.11d), there should not be any LHP root also for the even polynomial.
3. All the 4 roots of the even polynomial are in the $j\omega$ axis.
4. Now we inspect the other polynomial. This starts from s^6th row down to s^5th row. Since there is no sign change, there is no RHP root. There are 2 LHP roots.
5. In step 3, it is evaluated that there are 4 roots in the $j\omega$-axis. In step 4, it is known that there is no RHP root and two roots are in LHP.
6. We divide the characteristic polynomial by the auxiliary polynomial as given below.

$$
\begin{array}{r}
s^2+2s+2 \\
\hline
s^4+6s^2+8 \;\big)\, s^6+2s^5+8s^4+12s^3+20s^2+16s+16 \\
\underline{s^6 \quad +6s^4 \quad +8s^2} \\
2s^5+2s^4+12s^3+12s^2+16s+16 \\
\underline{2s^5 \quad +12s^3 \quad +16s} \\
2s^4+12s^2+16 \\
\underline{2s^4+12s^2+16} \\
0
\end{array}
$$

$$s^6+2s^5+8s^4+12s^3+20s^2+16s+16 = (s^4+6s^2+8)(s^2+2s+2)$$

$$s^2+2s+2 = (s+1+j)(s+1-j)$$

$$
\begin{aligned}
(s^4+6s^2+8) &= (s^2+4)(s^2+2) \\
&= (s+j2)(s-j2)(s+j\sqrt{2})(s-j\sqrt{2}).
\end{aligned}
$$

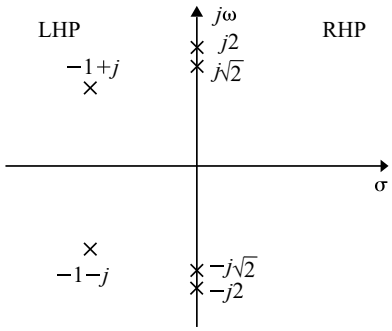


Fig. 4.12 Locations of poles of $F(s)$

The poles of $F(s)$ are located as shown in Fig. 4.12.

From Fig. 4.12, the following conclusions are drawn.

1. There is no pole in RHP.
2. There are 4 poles in the $j\omega$ axis. They are distinct and not repeated. This implies the system is marginally stable.

RHP roots = 0
LHP roots = 2
$j\omega$-axis roots = 4
System is marginally stable

It is erroneous to come to the conclusion that the system is stable since there is no sign change in the first column of Routh's array.

Example 4.9

Consider the following characteristic polynomial

$$s^7 + 9s^6 + 24s^5 + 24s^4 + 24s^3 + 24s^2 + 23s + 15 = 0$$

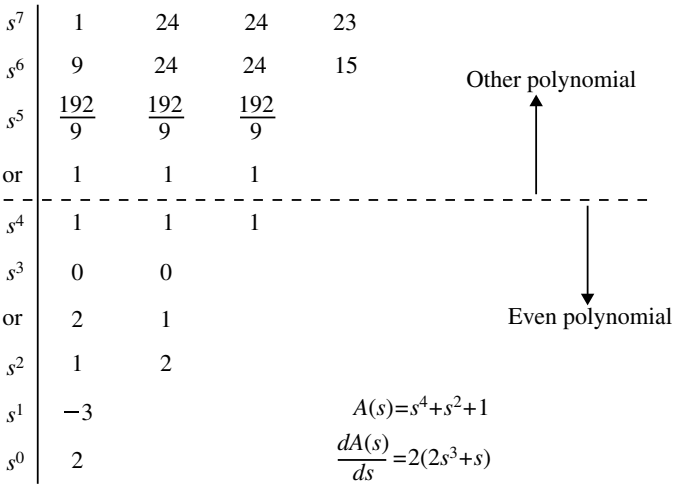
By using Routh–Hurwitz stability criterion, determine the stability of the system. Also, determine the number of roots in RHP, LHP and $j\omega$ axis.

(*Anna University, December, 2011*)

Solution

$$F(s) = s^7 + 9s^6 + 24s^5 + 24s^4 + 24s^3 + 24s^2 + 23s + 15$$

The following Routh's array is formed.



1. The even polynomial starts from s^4th row and comes down to s^0th row. The number of sign changes are two. Hence, there are 2 RHP roots. By symmetry, there will be 2 LHP roots for the even polynomial. There is no $j\omega$ axis roots.
2. By inspection of other polynomial which starts from s^7th row down to s^5th row, there is no sign change. All the three roots are in LHP.
3. The overall root positions are

RHP = 2
LHP = 5
$j\omega$-axis = Nil
System stability = unstable

Example 4.10

Find the number of roots in the equation

$$s^6 + 4s^5 + 3s^4 - 16s^2 - 64s - 48 = 0$$

with positive, negative and zero real parts. Determine the stability of the system.

(*Anna University, December, 2011*)

Solution The characteristic polynomial is

$$F(s) = s^6 + 4s^5 + 3s^4 - 16s^2 - 64s - 48$$

By inspection of $F(s)$, it is found that the system is unstable because of negative sign in the coefficient -16, -64 and -48. The Routh's array is formed below.

s^6	1	3	-16	-48	Other polynomial
s^5	1	0	-16	(After dividing by 4)	↑
s^4	1	0	-16	(After dividing by 3)	↓
s^3	0	0			Even polynomial
or					
s^3	4	0			
s^2	ϵ	-16		$A(s)=s^4-16$	
s^1	$\frac{64}{\epsilon}$			$\frac{dA(s)}{ds}=4s^3$	
s^0	-16				

By inspection of even polynomial of Routh's array, it is found that there is one sign change. Hence, there is 1 RHP root and 1 LHP root and $2j\omega$ axis roots. In the other polynomial, there is no sign change in the first column. Both the roots are in LHP. The roots are distributed as

RHP = 1
LHP = 3
$j\omega$-axis = 2

The system is unstable.

Example 4.11

Consider the following characteristic polynomial

$$F(s) = s^6 + 4s^5 + 11s^4 + 32s^3 + 40s^2 + 64s + 48$$

Using Routh–Hurwitz stability criterion, determine the stability of the system.

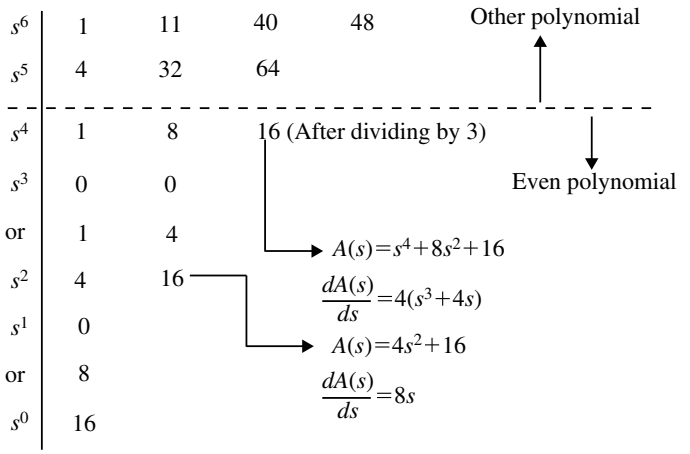
Solution

$$F(s) = s^6 + 4s^5 + 11s^4 + 32s^3 + 40s^2 + 64s + 48$$

The following Routh's array is prepared.

Inspection of first column of Routh's array for even polynomial shows that there are no sign changes. This shows that there is no RHP and LHP roots for the even polynomial. All the 4 roots are in the $j\omega$-axis. Inspection of first column of other polynomial shows that there is no sign change. This shows that the remaining 2 roots are in LHP. To determine the stability of the system, it is necessary to investigate the locations of the $j\omega$-axis poles. For the even polynomial, auxiliary equation is a factor

$$A(s) = (s^4 + 8s^2 + 16) = (s^2 + 4)^2$$



For the above equation, the 4 poles are in the $j\omega$-axis $(s^2 + 4)^2 = 0$

$$s_1 = +j2 \qquad s_3 = -j2$$
$$s_2 = +j2 \qquad s_4 = -j2$$
$$\text{LHP roots} = 2; \qquad \text{RHP roots} = 0; \qquad j\omega - \text{axis} = 4$$

The poles at $s = +j2$ and $s = -j2$ are repeated. The impulse response for these poles will be $t \sin 2t$ which is unbounded. **The system is unstable**.

It is erroneous to conclude that the system is stable since there is no sign change in the first column of the overall Routh array.

System is unstable

4.3.3 *Parameters Estimation via Routh–Hurwitz Stability Criterion*

Any one parameter which is varied to obtain desired system performance can be estimated by applying Routh–Hurwitz stability criterion. This is illustrated in the following example.

Example 4.12

Apply Routh–Hurwitz criterion to determine the value of K for the stability of a control system whose open loop T.F. is given by

$$GH(s) = \frac{5}{s(s^2 + 2Ks + 4 + K)}$$

(*Anna University, December, 2010*)

Solution

$$GH(s) = \frac{5}{s(s^2 + 2Ks + 4 + K)}$$

The characteristic equation is

$$\begin{aligned} F(s) &= 1 + GH(s) = 0 \\ &= s^3 + 2Ks^2 + (4 + K)s + 5 \end{aligned}$$

The following Routh's array is formed.

s^3	1	$(K + 4)$
s^2	$2K$	5
s^1	$\frac{2K^2 + 8K - 5}{2K}$	
s^0	5	

For the system to be stable, there should not be any sign change in the first column of Routh's array. Thus, we get the following conditions:

1. $K > 0$
2. $2K^2 + 8K - 5 > 0$
3. $(K + 4.55)(K - 0.55) > 0$
4. $K > -4.55$
5. $K > 0.55$

If the condition $K > 0.55$ is satisfied, $K > -4.55$ and $K > 0$ conditions are satisfied. Hence, for the system to be stable.

$$\boxed{K > 0.55}$$

Example 4.13

The open loop T.F. of a unity feedback control system is given by

$$G(s) = \frac{K}{(s+2)(s+4)(s^2+6s+25)}$$

By applying the Routh criterion, discuss the stability of the closed loop system as a function of K. Determine the values of K which will cause sustained oscillations in the closed loop system. What are the corresponding oscillation frequencies?

(*Anna University, May, 2009*)

Solution

$$G(s) = \frac{K}{(s+2)(s+4)(s^2+6s+25)}$$

The characteristic equation is

$$1 + G(s) = 0$$
$$(s+2)(s+4)(s^2+6s+25) + K = 0$$
$$F(s) = s^4 + 12s^3 + 69s^2 + 198s + (200 + K)$$

The following Routh's array is formed.

s^4	1	69	$(200+K)$
s^3	12	198	
s^2	52.5	$(200+K)$	
s^1	$7995 - 12K$		
s^0	$200 + K$		

For the system to be stable, the following conditions are to be satisfied:

$$(1) \quad 7995 - 12K > 0$$
$$K < 666.25$$
$$(2) \quad 200 + K > 0$$
$$K > -200$$

The system is stable for the value of K in the following range

$$\boxed{-200 < K < 666.25}$$

At $K = 666.25$ the system oscillates. The oscillating frequency is obtained from the auxiliary equation which is given by

$$52.5s^2 + 200 + K = 0$$

Substituting $s = j\omega$ in the above equation, we get

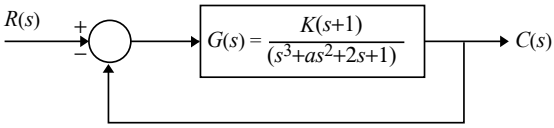


Fig. 4.13 System representation for Example 4.14.

$$-52.5\omega^2 + 200 + 666.25 = 0$$

$$\boxed{\omega = 4.06\,\text{rad/sec}}$$

Example 4.14

A certain unity feedback system is represented in Fig. 4.13. Find the +ve value of K and a so that the system oscillates at a frequency of 2 rad/s.

Solution

$$G(s) = \frac{K(s+1)}{s^3 + as^2 + 2s + 1}$$

The characteristic equation is

$$1 + G(s) = 0$$
$$s^3 + as^2 + (K+2)s + (K+1) = 0$$

The following Routh's array is formed.

s^3	1	$(K+2)$
s^2	a	$(K+1)$
s^1	$a(K+2) - (K+1)$	
s^0	$(K+1)$	

For the system to oscillate

$$a(K+2) = (K+1)$$
$$a = \frac{(K+1)}{(K+2)}$$

Under this condition, the auxiliary equation is

$$as^2 + (K+1) = 0$$

Substituting $s = j\omega$ in the above equation, we get

$$\begin{aligned}\omega^2 &= \frac{(K+1)}{a} \\ &= \frac{(K+1)(K+2)}{(K+1)} \\ &= K+2\end{aligned}$$

For $\omega = 2$ rad/sec

$$\boxed{\begin{aligned} K &= 2 \\ a &= 0.75 \end{aligned}}$$

Example 4.15

Consider the following characteristic polynomial of a certain control system

$$F(s) = s^4 + 2s^3 + 6s^2 + Ks + 10$$

Determine the real value of K so that the system is stable.

Solution

$$F(s) = s^4 + 2s^3 + 6s^2 + Ks + 10$$

The following Routh's array is formed.

s^4	1	6	10
s^3	2	K	
s^2	$6 - 0.5K$	10	
s^1	$\dfrac{K(6-0.5K)-20}{6-0.5K}$		
s^0	10		

For the system to be stable, the following conditions are to be satisfied

1. $6 - 0.5K > 0$
   $K < 12$
2. $K(6 - 0.5K) - 20 > 0$
   $-0.5K^2 + 6K - 20 > 0$

Solving for the quadratic equation for K, we get

$$K = \frac{-6 \pm \sqrt{36 - 40}}{(-1)}$$
$$= 6 \pm j2$$

Since K is complex the inequalities cannot be satisfied for any real value of K. For any value $K < 12$ in condition (1) would not satisfy condition (2). Thus, the system has RHP roots for all values of K and **it is always unstable**.

Example 4.16

The characteristic equation of a feedback control system is

$$s^4 + 3s^3 + 12s^2 + 16s + K = 0$$

Determine the range of gain K so that the system is stable.

Solution

$$s^4 + 3s^3 + 12s^2 + 16s + K = 0$$

For the above characteristic equation, the following Routh's array is formed.

s^4	1	12	K
s^3	3	16	
s^2	20	$3K$	
s^1	$320 - 9K$		
s^0	K		

From the Routh's array, the following conditions are to be satisfied for the system to be stable

(1) $320 - 9K > 0$

$$K < \frac{320}{9}$$

$$K < 35.556$$

(2) $K > 0$

The range of K for the system to be stable is

$$\boxed{0 < K < 35.556}$$

Example 4.17

The open loop T.F. of a certain unity feedback system is given by

$$G(s) = \frac{K(s+10)}{s^3}$$

Determine the value of K so that the closed loop system is stable. If the closed loop system is unstable for all values of K, determine the RHP and LHP roots.

Solution

$$G(s) = \frac{K(s+10)}{s^3}$$

The characteristic equation is given by

$$1 + G(s) = 0$$
$$s^3 + Ks + 10K = 0$$

Since the coefficient of s^2 is missing in the Routh's array, there will be a zero in the first column. The characteristic polynomial is multiplied by the factor $(s+1)$ and the new characteristic equation is obtained as

$$(s+1)(s^3 + Ks + 10K) = 0$$
$$s^4 + s^3 + Ks^2 + 11Ks + 10K = 0$$

The following Routh's array is formed.

s^4	1	K	$10K$
s^3	1	$11K$	
s^2	$-K$	K	
s^1	$11K+1$		
s^0	K		

For all positive values of K, the system is unstable. There are two sign changes in the Routh's array. There are 2 RHP roots and one LHP root for the closed loop system.

4.3.4 *Relative Stability Using Routh–Hurwitz Stability Criterion*

In most of the applications, Routh–Hurwitz stability criterion is applied to know the immediate answer whether the system is stable or not. This information is useful for the system design. Once the system is designed with absolute stability, it can be tested how far it is closer to absolute stability which is termed as relative stability. To find the relative stability, the s-plane axis is shifted to the desired value and Routh–Hurwitz criterion in the new complex s-plane is applied. If the $j\omega$ axis of the s-plane is to be shifted to the left by σ where σ is the desired relative stability factor, then we substitute $s = (\lambda - \sigma)$ in the characteristic equation and form the characteristic polynomial $F(\lambda)$. The Routh–Hurwitz test is carried out for $F(\lambda)$. The number of changes of sign in the first column of the new Routh's array give the number of roots located to the right of the vertical line passing through $s = -\sigma$. The following example illustrates the above procedure.

Example 4.18

Consider the following characteristic polynomial

$$F(s) = s^3 + 12s^2 + 47s + 60$$

Test whether the system has a relative stability of 2 units.

Solution

$$F(s) = s^3 + 12s^2 + 47s + 60$$

The desired relative stability unit is $\sigma = -2$. Substitute $s = (\lambda - 2)$ in the original equation.

$$\begin{aligned} s^3 &= (\lambda - 2)^3 \\ &= \lambda^3 - 6\lambda^2 + 12\lambda - 8 \\ 12s^2 &= 12(\lambda - 2)^2 \\ &= 12\lambda^2 - 48\lambda + 48 \\ 47s &= 47(\lambda - 2) \\ &= 47\lambda - 94 \\ F(\lambda) &= \lambda^3 + 6\lambda^2 + 11\lambda + 6 \end{aligned}$$

The following Routh's array is formed.

$$\begin{array}{c|cc} \lambda^3 & 1 & 11 \\ \lambda^2 & 6 & 6 \\ \lambda^1 & 10 & \\ \lambda^0 & 6 & \end{array}$$

There is no sign change in the first column of Routh's array. This means all the roots of $F(s)$ are to the left of the vertical line passing through $(-2, 0)$ point. The system has a relative stability unit of more than 2.

4.3.5 Routh's Test for System with Transportation Lag

In the process control system, the effect of transportation lag is important. The transfer function of the transportation lag is given as e^{-sT}. The effect of transportation lag cannot be studied by Routh's criterion. e^{-sT} is expressed in the approximated algebraic form. There are few methods available in the literature. We use the following approximation here

$$e^{-sT} = (1 - sT)$$

Example 4.19

A certain unity feedback control system has the following open loop T.F.

$$G(s) = \frac{10e^{-sT}}{(s+1)(s^2+2s+4)}$$

Determine the value of T, so that the closed loop system is stable.

Solution

$$G(s) = \frac{10e^{-sT}}{(s+1)(s^2+2s+4)}$$

The characteristic equation is

$$\begin{aligned} (s+1)(s^2+2s+4) + 10e^{-sT} &= 0 \\ s^3 + 3s^2 + 6s + 4 + 10(1 - sT) &= 0 \\ s^3 + 3s^2 + (6 - 10T)s + 14 &= 0 \end{aligned}$$

The following Routh's array is formed.

$$\begin{array}{c|cc} s^3 & 1 & 6-10T \\ s^2 & 3 & 14 \\ s^3 & 4-30T & \\ s^0 & 1 & \end{array}$$

For the system to be stable $4 - 30T > 0$ or $T < 0.1333$

$$\boxed{T < 0.1333}$$

4.4 Nyquist Stability Criterion

Nyquist stability criterion was developed by H. Nyquist, in 1932, at Bell Telephone Laboratories, USA. It determines the closed loop system stability if the open loop frequency response and open loop poles are known H.S. Black, at Bell Laboratory was developing the negative feedback amplifier for use in telephone repeater circuits using vacuum tubes. When the amplifier became an oscillator, the reason was not understood by any one at that time. Nyquist was asked to study the problem, and as a result, he developed the famous stability test. One morning, going to work on the State Island ferry, before the Verrazano Narrows bridge, Nyquist had an inspiration and wrote his work, literally, on the back of an envelope as he rode. Today millions of control system engineers and students apply the Nyquist stability criterion for the system design. There is hardly any control system textbook that does not include the Nyquist stability criterion for the study. This semi-graphical method which is an alternative to Routh–Hurwitz and root locus methods has the following features.

Advantages of Nyquist Stability Criterion

1. Nyquist stability criterion, like Routh–Hurwitz criterion, gives information about the absolute stability of the closed loop system. In addition, it gives information about the relative stability of the system in terms of phase margin and gain margin. Further, it indicates how to improve the relative stability if required. Furthermore, if the closed loop system is unstable, it indicates how to stabilize the system.
2. Since Nyquist stability criterion is developed in the frequency domain, it is easy to get information about the frequency domain specifications such as resonant peak M_r, resonant frequency ω_r and BW of the closed loop system.
3. Systems with time delay are difficult to handle with accuracy by Routh–Hurwitz criterion and root locus technique. However, Nyquist criterion can be easily extended to linear system with pure time delay.
4. The Nyquist plot is based on the open loop frequency response of $GH(j\omega)$ and the location of open loop poles of $GH(s)$. These two things are very easy to obtain. $GH(j\omega)$ is easily plotted using the MATLAB program nowadays.
5. Even though Nyquist stability criterion is developed from Cauchy's principle of the argument, which requires the knowledge of the theory of complex variable,

interpretation of stability condition from the Nyquist plot is very simple and straightforward.

A Historical Note: Harry Nyquist (1889–1976)



Harry Theodor Nyquist was born on February 7, 1889 in Sweden. His parents had seven children. The family of Nyquist emigrated to the USA in 1907. He received the B.S. and M.S. degrees in electrical engineering, respectively, from North Dakota University and a Ph.D. degree in Physics in 1917 from Yale University. He worked at AT&T's Department of Development and Research from 1917 to 1934 and he continued his service there when it changed its name as Bell Telephone Laboratories in 1934. He retired in the year 1954 on superannuation. Nyquist early theoretical work was on determining the BW requirement for transmitting information. In 1927, Nyquist formulated sampling theorem according to which the number of independent pulses that can be sent through a telegraph channel is limited to twice the bandwidth of the channel.

Nyquist contribution as an engineer at Bell Laboratory was immense. His important works include the study of thermal noise, the stability of feedback amplifiers, telegraphy, facsimile, television and other communication problems. In 1932, he published a classical research paper on the stability of feedback amplifiers. The Nyquist stability criterion can now be found in all textbooks on feedback control theory. Nyquist was honoured for his "Fundamental Contributions to a Quantitative Understanding of Thermal Noise, Data Transmission and Negative Feedback", "theoretical analysis and practical inventions in the field of communications systems", "theories of telegraph transmission", "thermal noise in electric conductors", and many fundamental contributions in Engineering. Nyquist lived in Texas state after his retirement and the great engineer died in Harbinger on April 4, 1976 at the age of 87.

4.4.1 Concepts Used in Nyquist Stability Criterion

Consider the feedback system shown in Fig. 4.14. The closed loop system T.F. is written as

$$\frac{C(s)}{R(s)} = M(s) = \frac{G(s)}{[1 + GH(s)]} \tag{4.21}$$

where $1 + GH(s) = 0$ is the characteristic equation. The characteristic equation $1 + GH(s) = 0$ in Eq. (4.21) determines the stability of the closed loop system.

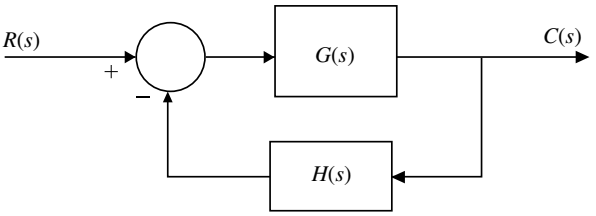


Fig. 4.14 Closed loop system

Nyquist stability criterion tells whether any roots of the characteristic equation are in RHP. Before the derivation of Nyquist stability criterion, the following concepts which are associated with the criterion need to be explained. These concepts include the following:

1. Relationship between the poles of $GH(s)$ and the poles of $1 + GH(s)$.
2. The relationship between the poles of the closed loop T.F. $M(s)$ and the zeros of $1 + GH(s)$.
3. The concept of mapping points.
4. The concept of mapping contour.

4.4.1.1 Relationship Between the Poles of $GH(s)$ and the Poles of $1 + GH(s)$

Consider the following loop T.F.

$$GH(s) = \frac{K(s+z_1)(s+z_2)\ldots(s+z_m)}{(s+p_1)(s+p_2)\ldots(s+p_n)} \tag{4.22}$$

where

$$(s+z_1), (s+z_2)\ldots(s+z_m) = \text{Open loop zeros}$$
$$(s+p_1), (s+p_2)\ldots(s+p_n) = \text{Open loop poles}$$

For simplicity, let us choose $H(s) = 1$ without loss of generality. The closed loop T.F. is written as

$$M(s) = \frac{\dfrac{K(s+z_1)(s+z_2)\cdots(s+z_m)}{(s+p_1)(s+p_2)\cdots(s+p_n)}}{\left[1 + \dfrac{K(s+z_1)(s+z_2)\cdots(s+z_m)}{(s+p_1)(s+p_2)\cdots(s+p_n)}\right]} \tag{4.23}$$

$$= \frac{\dfrac{K(s+z_1)(s+z_2)\cdots(s+z_m)}{(s+p_1)(s+p_2)\cdots(s+p_n)}}{\left[\dfrac{(s+p_1)(s+p_2)\cdots(s+p_n) + K(s+z_1)\cdots(s+z_m)}{(s+p_1)(s+p_2)\cdots(s+p_n)}\right]} \tag{4.24}$$

From Eq. (4.24), it is evident that the poles of $GH(s)$ are the same as the poles of $[1 + GH(s)]$.

$$\boxed{\text{Poles of the open loop T.F. } = \text{ Poles of the characteristic equation}} \tag{4.25}$$

4.4.1.2 Relationship Between the Zeros of the Characteristic Equation and the Closed Loop Poles

Let us rewrite Eq. (4.24) with the zeros of the characteristic equation in factored form as

$$(s + p_1)(s + p_2)\cdots(s + p_n) + K(s + z_1)(s + z_2)\cdots(s + z_m) = (s + Z_1)(s + Z_2)\cdots(s + Z_n)$$

for $n > m$

$$M(s) = \frac{\dfrac{K(s + z_1)(s + z_2)\cdots(s + z_m)}{(s + p_1)(s + p_2)\cdots(s + p_n)}}{\left[\dfrac{(s + Z_1)(s + Z_2)\cdots(s + Z_n)}{(s + p_1)(s + p_2)\cdots(s + p_n)}\right]} \tag{4.26}$$

$$= \frac{K(s + z_1)(s + z_2)\cdots(s + z_m)}{(s + Z_1)(s + Z_2)\cdots(s + Z_n)} \tag{4.27}$$

From Eq. (4.26), $(s + Z_1)(s + Z_2)\cdots(s + Z_n)$ are the zeros of the characteristic equation. From Eq. (4.27), $(s + Z_1)(s + Z_2)\cdots(s + Z_n)$ are the poles of the closed loop T.F.

$$\boxed{\text{Poles of the closed loop T.F. } = \text{ Zeros of the characteristic equation}}$$

For the closed loop system to be stable the closed loop poles should all be in the LHP. The presence of even a single root in RHP will make the system unstable, whereas the presence of $j\omega$ axis roots, which are distinct, will make the closed loop system marginally stable which is BIBO unstable. If the roots in the $j\omega$ axis are repeated, the closed loop system is unstable. To tell the result straightaway, the zeros of the characteristic equation, which are the same as the poles of the closed loop T.F., are not in the form of factors. The Nyquist stability criterion, aims at, without factorizing the zeros of the characteristic equations, finding the number of roots of this equation in RHP, LHP and the $j\omega$ axis. For the closed loop system to be stable, the roots of

$$(s + Z_1)(s + Z_2)\cdots(s + Z_n)$$

should all be in LHP.

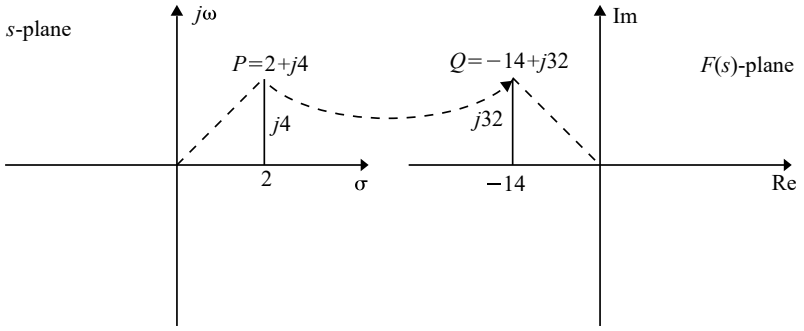


Fig. 4.15 Mapping a point on the s-plane on to $F(s)$-plane

4.4.1.3 The Concepts of Mapping Points

Consider the characteristic equation

$$F(s) = 1 + GH(s) \tag{4.28}$$

If any complex number in the complex s-plane is chosen and substituted in Eq. (4.28), another complex number results in the $F(s)$ plane. This is called mapping a point from s-plane to $F(s)$ plane. s-plane is a complex plane where σ represents the horizontal axis and $j\omega$ represents the vertical axis. $F(s)$ plane is represented by Re as the horizontal axis and Im as the vertical axis.

Consider the following function

$$F(s) = s^2 + 4s - 10 \tag{4.29}$$

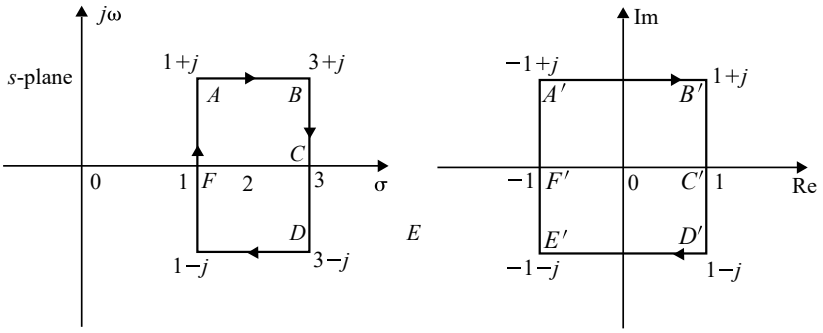
Let us choose any point $s = 2 + j4$ in the complex s-plane. Substituting this value of s in Eq. (4.29), we get

$$\begin{aligned} F(s)|_{s=2+j4} &= (2 + j4)^2 + 4(2 + j4) - 10 \\ &= -14 + j32 \end{aligned}$$

The mapping from s-plane to $F(s)$ plane is shown in Fig. 4.15. The point P in the s-plane is mapped as point Q in the $F(s)$-plane.

4.4.1.4 The Concept of Mapping Contour

The concept of mapping a point from s-plane to $F(s)$ plane can be extended to collection of points. The connection of points in the complex plane is called a contour. Let us consider the following function.



(a) s-plane contour (b) $F(s)$-plane contour

Fig. 4.16 Mapping of s-plane contour to $F(s)$ plane contour (a zero enclosed inside the contour)

$$F(s) = (s - 2)$$

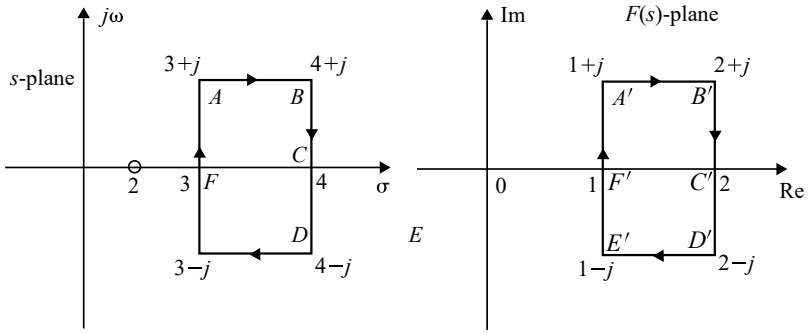
Let us choose a contour in the s-plane as shown in Fig. 4.16a. We choose six points A, B, C, D, E, F in the s-plane contour. The mapping of these points from s-plane to $F(s)$ plane is done as follows. The following table is prepared (Table 4.1).

The s-plane contour and $F(s)$ plane contour are shown in Fig. 4.16. For $F(s) = s - 2$, a zero is placed inside the contour. The contour $ABCDEF$ encloses the zero. When this contour is moved in the clockwise direction, the points are mapped as A', B', C', D', E', F' in the $F(s)$ plane and forms another contour. This contour also moves in **the same clockwise direction** and encircles the origin once. A point within a contour to the right of the traversal of the contour is considered to be the point enclosed by the contour. **The clockwise traversal** of a contour is taken as positive and the point on the right side of the contour is said to be enclosed.

A point is said to be encircled by a closed path if it is found inside the path (to the right) during the traversal. The clockwise encirclement is taken as positive, while encirclement in the counter clockwise is taken as negative.

Table 4.1 Mapping s-plane contour to $F(s)$-plane contour

Point in the s -plane	s	$F(s) = (s - 2)$
A	$1 + j$	$A' = -1 + j$
B	$3 + j$	$B' = 1 + j$
C	$3 + j0$	$C' = 1 + j0$
D	$3 - j$	$D' = 1 - j$
E	$1 - j$	$E' = -1 - j$
F	$1 + j0$	$F' = -1 + j0$



(a) s-plane contour (b) $F(s)$-plane contour

Fig. 4.17 Mapping of s-plane contour to $F(s)$ plane (a zero not enclosed)

Table 4.2 Mapping s-plane contour to $F(s)$-plane contour

Point in the s -plane	s	$F(s) = (s-2)$
A	$3+j$	$A' = 1+j$
B	$4+j$	$B' = 2+j$
C	$4+j0$	$C' = 2+j0$
D	$4-j$	$D' = 2-j$
E	$3-j$	$E' = 1-j$
F	$3+j0$	$F' = 1$

Table 4.3 Mapping s-plane contour to $F(s)$-plane contour

Point in the s -plane	s	$F(s) = 1/(s-2)$
A	$1+j$	$A' = -0.5-0.5j$
B	$3+j$	$B' = 0.5-0.5j$
C	$3+j0$	$C' = 1+j0$
D	$3-j$	$D' = 0.5+j0.5$
E	$1-j$	$E' = -0.5+j0.5$
F	$1+j0$	$F' = -1+j0$

Now let us choose a different contour in the s-plane which does not enclose the zero at $s = 2$. This is shown in Fig. 4.17a. The following table is prepared (Table 4.2).

The s-plane contour which does not enclose a zero when mapped into $F(s)$ plane does not encircle the origin.

Let $F(s) = 1/(s-2)$. Let the contour $ABCDEF$ enclose the pole at $s = 2$ in the s-plane. For the mapping of the points A, B, C, D, E, F from s-plane to $F(s)$-plane, the following table is prepared (Table 4.3).

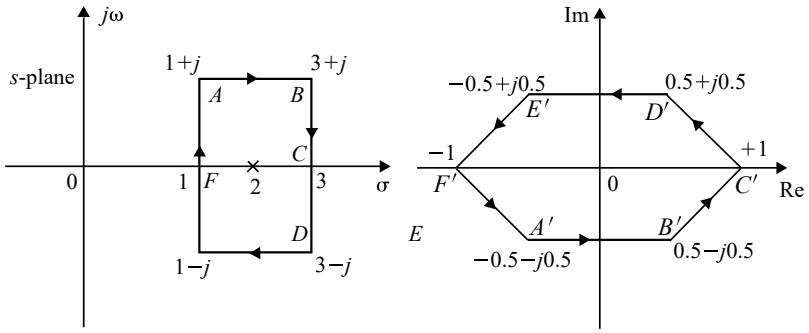


Fig. 4.18 Mapping s-plane contour to $F(s)$ plane contour (a pole enclosed inside the contour)

The s-plane contour when mapped in the $F(s)$ plane is as shown in Fig. 4.18. The $F(s)$ plane contour encircles the origin of $F(s)$ plane in the counter clockwise direction once. From the above discussions, the following conclusions are derived:

1. An s-plane contour with a zero of $F(s)$ enclosed when mapped from s-plane to $F(s)$ plane in the clockwise direction will encircle the origin of the s-plane once in the same clockwise direction. It can be shown that if the s-plane contour encloses two zeros of $F(s)$ when mapped from s-plane to $F(s)$ plane will encircle the origin of the s-plane twice in the same clockwise direction. For each additional zero enclosed inside the s-plane contour, there will be one additional clockwise encirclement of the origin in the same direction.
2. An s-plane contour with no zero or pole enclosed when mapped from s-plane to $F(s)$ plane will not at all encircle origin of the $F(s)$-plane.
3. An s-plane contour with a pole of $F(s)$ enclosed when mapped from s-plane to $F(s)$ plane in the clockwise direction will encircle the origin of the $F(s)$-plane once in the anticlockwise direction. For each additional pole enclosed inside the s-plane contour, there will be one additional encirclement in the $F(s)$-plane and the encirclement is in the counter clockwise direction.
4. In general, if there are Z zeros and P poles enclosed by the s-plane contour, when it is mapped into $F(s)$ plane with the points in the s-plane contour moving in the clockwise direction, there will be $Z - P$ encirclements around the origin of $F(s)$ plane in the same clockwise direction provided $Z > P$. If $P > Z$, there will be $P - Z$ encirclements in the counter clockwise direction.
5. Conversely, if there are say 3 encirclements in the $F(s)$-plane contour, in the clockwise direction, there are Z zeros and P poles inside the s-plane contour which satisfies the condition $Z - P = 3$, where $Z > P$. For counter clockwise encirclement, the condition to be satisfied is $P - Z = 3$, where $P > Z$.

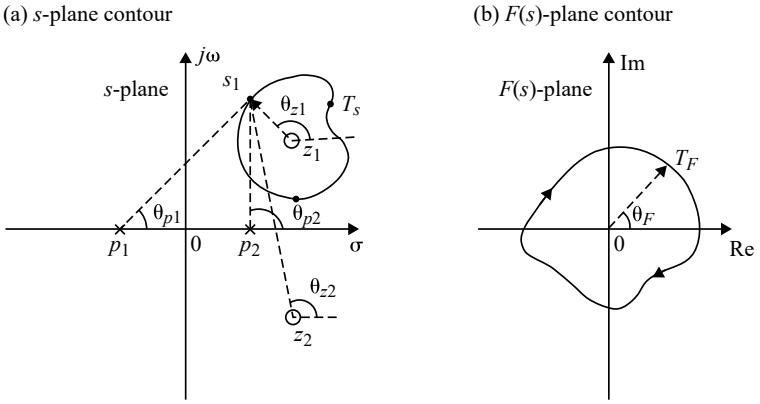


Fig. 4.19 Evaluation of net angle of T_F

4.4.2 The Principle of the Argument—Cauchy's Theorem

The Nyquist stability criterion was developed from the theory of conformal mapping and the principle of the argument which is based on Cauchy's theorem. Cauchy's theorem is based on mapping a function $F(s)$ that has a finite number of poles and zeros within the contour. Consider $F(s)$ in the following form

$$F(s) = \frac{K(s+z_1)(s+z_2)\cdots(s+z_m)}{(s+p_1)(s+p_2)\cdots(s+p_n)} \tag{4.30}$$

For simplicity, we consider two poles and two zeros in Equation (4.30)

$$F(s) = \frac{K(s+z_1)(s+z_2)}{(s+p_1)(s+p_2)} \tag{4.31}$$

The zeros are taken as complex conjugate and poles are real as shown in Fig. 4.19.

Now consider a small contour T_s in the s-plane which has enclosed the zero at $s = z_1$. Take any point s_1 on the contour T_s. At point s_1, Eq. (4.31) is written as

$$F(s_1) = |F(s_1)|\angle\theta_{z1} + \angle\theta_{z2} - \angle\theta_{p1} - \angle\theta_{p2}$$

By mapping, the net angle in the $F(s)$ plane is

$$\theta_F = \theta_{z1} + \theta_{z2} - \theta_{p1} - \theta_{p2}$$

As s_1 traverses along the contour T_s, for a full rotation of 360°, the angles of θ_{z2}, θ_{p1} and θ_{p2} will be zero while θ_{z1} will be 360°. Thus, for one zero enclosed by T_s contour in the s-plane, the contour T_F in the $F(s)$ plane will have one encirclement

around the origin of the $F(s)$-plane. Suppose there are Z zeros and P poles inside the s-plane contour T_s. As s is traversed in the clockwise direction, the net angle of $F(s)$ is $2\pi(Z - P)$. If we denote this as N encirclements in the $F(s)$ plane, the following relation is true.

$$2\pi N = 2\pi(Z - P)$$

$$\boxed{N = Z - P} \tag{4.32}$$

We illustrate below in Table 4.4 the number of poles and zeros inside the s-plane contour and the number of encirclements N made by the $F(s)$ plane contour around the origin of the $F(s)$ plane using Eq. (4.32). As stated earlier, the clockwise encirclement is taken as positive, while counter clockwise encirclement is taken as negative.

4.4.3 The Nyquist Stability Criterion

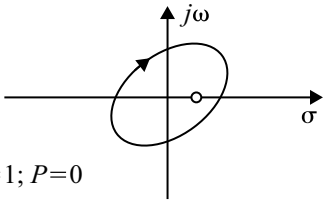
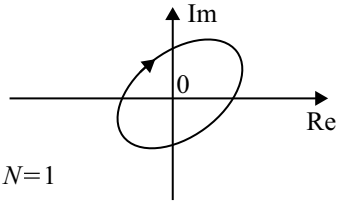
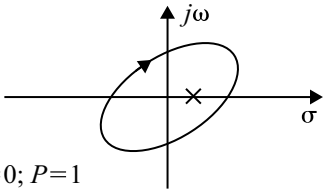
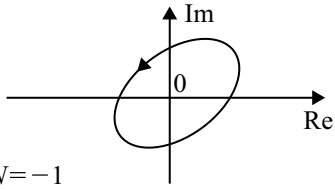
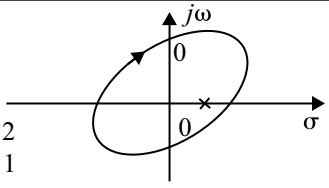
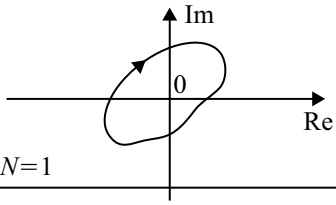
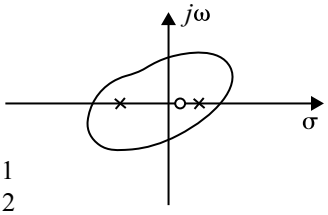
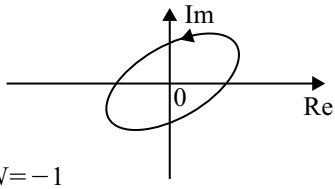
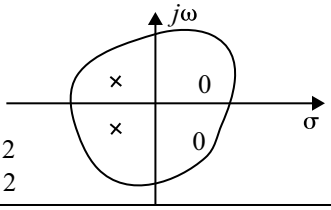
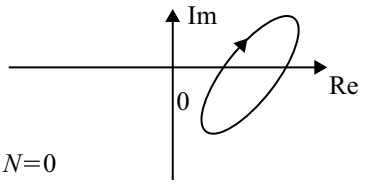
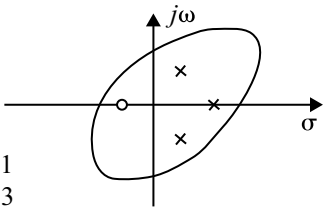
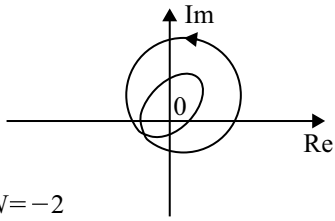
Now consider Eq. (4.26). The closed loop system will be stable if all the roots of the characteristic equation are in LHP. In Eq. (4.26), Z_1, Z_2 *etc.*., should have a negative real part. Now Eq. (4.32) can be applied to Eq. (4.26) to find the number of zeroes of the characteristic equation that are in the RHP. For this, the s-plane contour is formed that will include the entire RHP including the $j\omega$ axis. **However, the contour be such that it does not pass through poles or zeroes of** $F(s)$. Since the poles of open loop T.F. are the poles of the characteristic equation, the number of poles of $F(s)$ that are in RHP is known just by the inspection of $GH(s)$. By conformal mapping, the contour in the s-plane is now mapped to $F(s)$ plane and the number of encirclement N made around the origin is determined. Then using Eq. (4.32) Z, the number of zeroes of the characteristic equation that are in RHP is known. The closed loop system is stable if $Z = 0$. If Z is not zero, then the closed loop system is unstable.

Relationship Between $GH(s)$ Plane and $F(s)$ Plane
Using Eq. (4.32) the stability of the closed loop system is determined by counting the number of encirclements N of $1 + GH(j\omega)$ around the origin of the $F(s)$ plane. As seen from Fig. 4.20, this is equivalent to determine the number of encirclements of $GH(j\omega)$ around $(-1, 0)$ point in the s-plane since plotting of $GH(j\omega)$ is much easier than plotting of $1 + GH(j\omega)$. Hence, the stability condition given by Eq. (4.32) is modified as given below and is still valid. The following symbols with suffix 'r' are used in Eq. (4.32) which are defined as follows:

1. P_r = Number of poles of $GH(s)$ in the right half s-plane.
2. Z_r = Number of zeros of $1 + GH(s)$ in the right half s-plane.
3. N_r = Number of clockwise encirclements of the $(-1, 0)$ point when s-plane Nyquist contour is mapped into GH plane.

Equation (4.32) is rewritten as

Table 4.4 Mapping of s-plane contour to $F(s)$ plane contour

s-plane contour	$F(s)$-plane contour
 $Z=1;\ P=0$	 $N=1$
 $Z=0;\ P=1$	 $N=-1$
 $Z=2$ $P=1$	 $N=1$
 $Z=1$ $P=2$	 $N=-1$
 $Z=2$ $P=2$	 $N=0$
 $Z=1$ $P=3$	 $N=-2$

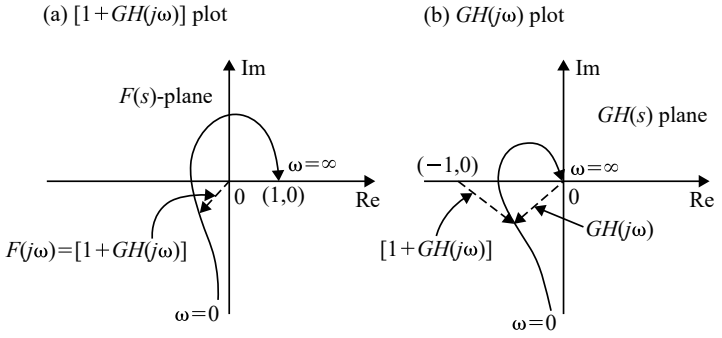


Fig. 4.20 $[1 + GH(j\omega)]$ in the $F(s)$ plane and $GH(s)$ plane

$$\boxed{Z_r = N_r + P_r} \tag{4.33}$$

If no pole of the open loop system is in RHP (open loop system is stable), then

$$\boxed{Z_r = N_r} \tag{4.34}$$

For the closed loop system to be stable $Z_r = 0$. Thus, the problem of determining Z_r is converted to finding N_r. Using Eqs. (4.34) and (4.33), the Nyquist stability criterion is stated as follows.

Open Loop System is Stable $(P_r = 0)$
If the open loop system is stable $(P_r = 0)$, then for the closed loop system also to be stable $(Z_r = 0)$, the s-plane Nyquist contour when mapped into GH-plane should not at all encircle the $(-1, 0)$ point.

Open Loop System is Unstable $(P_r \neq 0)$
If the open loop system is unstable, then for the closed loop system to be stable, the s-plane Nyquist contour when mapped into GH-plane should encircle $(-1, 0)$ point in the GH plane as many times as there are number of open loop poles in RHP and the encirclement should be in the counter clockwise direction of the s-plane contour.

How to Count N_r
The following procedure is followed to count N_r:

1. In the Nyquist diagram, put the arrow in the proper direction for the $GH(j\omega)$ plot, mirror image and infinite semi-circles if any.
2. Identify the $(-1, 0)$ point and draw any radial line from this point, directing preferably in the 3rd quadrant of the GH-plane. This radial line cuts the Nyquist diagram at different points.
3. For each intersection point, draw a curved arrow which is in the same direction as in the Nyquist diagram.

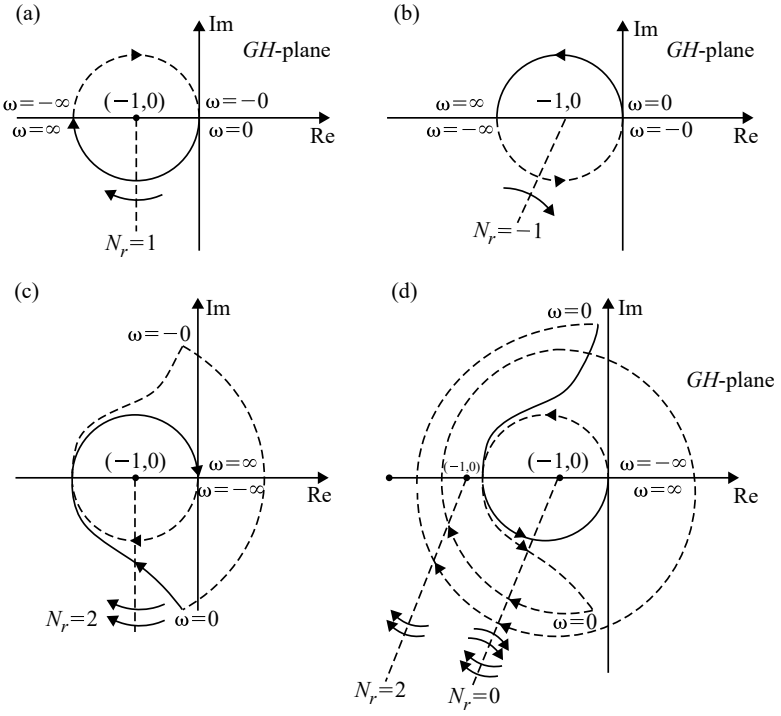


Fig. 4.21 Counting of N_r of Nyquist diagram

4. If the arrow is in the clockwise direction, count $N_r = 1$. If the arrow is in the anticlockwise direction, count $N_r = -1$. Sum up all the $N_r's$ which gives the total N_r for the given Nyquist plot.

Figure 4.21 illustrates the procedure to count N_r for different types of Nyquist plots.

Example 4.20

A certain unity feedback control system has the following open loop T.F. with minimum phase. Determine the stability of the closed loop system using the Nyquist stability criterion.

$$G(s) = \frac{K}{(s + T_1)(s + T_2)}$$

Solution

See Fig. 4.22.

Type 0, **Order** 2 **System**

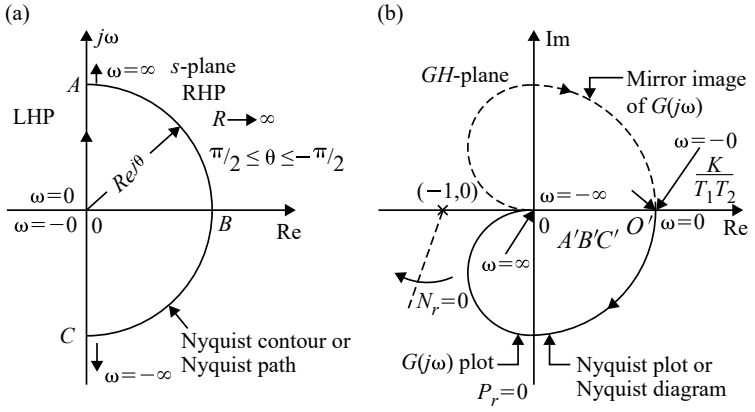


Fig. 4.22 **a** Nyquist contour in the s-plane. **b** Nyquist plot in the GH-plane $G(s) = K/[(s + T_1)(s + T_2)]$

1. Since we are interested in whether there is any zero of $1 + GH(s)$ in the $j\omega$ axis and the RHP of the s-plane, a Nyquist contour as shown in Fig. 4.22a is chosen. This contour covers the entire RHP as R tends to ∞ and varied in the range $\pi/2 \leq \theta \leq -\pi/2$. Further, the contour includes the $j\omega$ axis also to identify whether there is any zero in the $j\omega$ axis as well. The s-plane contour is called Nyquist path.
2. Since the given T.F. is of minimum phase, the poles $s_1 = -T_1$ and $s_2 = -T_2$ are in LHP. There is no RHP poles of $G(s)$. Hence, $P_r = 0$. For the closed loop system to be stable, $Z_r = 0$. Equation (4.34) is applied and N_r should then be zero.
3. To get N_r, the s-plane Nyquist contour is mapped to GH-plane. This is done section by section mapping for the given $G(s)$ as explained below.

$$\text{Section, } OA; \quad s = j\omega; \quad 0 \leq \omega \leq \infty$$

$$G(j\omega) = \frac{K}{(j\omega + T_1)(j\omega + T_2)} = \frac{K\angle -\tan^{-1}\frac{\omega}{T_1} - \tan^{-1}\frac{\omega}{T_2}}{\sqrt{(\omega^2 + T_1^2)(\omega^2 + T_2^2)}}$$

For $\omega = 0$;

$$G(j\omega) = \frac{K\angle 0^\circ}{T_1 T_2}$$

For $\omega = \infty$;

$$G(j\omega) = 0\angle -180^\circ$$

The $G(j\omega)$ plot starts from +ve react at $K/(T_1T_2)$ for $\omega = 0$, traverses through the fourth and third quadrants in the GH-plane and terminates at the origin for $\omega = \infty$. Section OA in the s-plane is mapped into $O'A'$ in the GH-plane as shown in Fig. 4.22b.

4. Section ABC, $s = Re^{j\theta}$ where $R \to \infty$ and $\pi/2 \le \theta \le -\pi/2$.

   For the Section ABC

$$\begin{aligned} G(Re^{j\theta}) &= \lim_{R\to\infty} \frac{K}{(Re^{j\theta} + T_1)(Re^{j\theta} + T_2)} \\ &= 0 \end{aligned}$$

   Section ABC in the s-plane is mapped into the origin of the GH-plane as $A'B'C'$.

5. Section CO, $s = -j\omega$; $\quad -\infty \le \omega \le -0$.

   This is the mirror image of $G(j\omega)$ and is shown as $C'O'$ in dotted line in the GH-plane.

6. The closed Nyquist contour in the s-plane is mapped into another closed contour in the GH-plane. The GH-plane contour is called Nyquist plot or Nyquist diagram. The s-plane contour is known as Nyquist path.

7. In Fig. 4.22b, $(-1, 0)$ is identified and from there a radial dotted line is drawn. This radial line does not intersect the Nyquist plot. Hence, $N_r = 0$. Using Eqs. (4.33) or (4.34), we get

$$\begin{aligned} Z_r &= N_r + P_r \\ &= 0 + 0 \end{aligned}$$

$$\boxed{Z_r = 0}$$

   Since $Z_r = 0$, there is no zero of the characteristic equation (closed loop pole) in RHP and the imaginary axis.

$$\boxed{\text{The closed loop system is stable}}$$



Note: There is no need for numerical values for K, T_1, and T_2 for the above problem.

Example 4.21

Consider the following loop T.F. with minimum phase of a certain closed loop system

$$GH(s) = \frac{K}{s(s+T_1)(s+T_2)}$$

Determine the stability of the closed loop system using Nyquist stability criterion.

(*Anna University, 2005*)

Solution

$$GH(s) = \frac{K}{s(s+T_1)(s+T_2)}$$

1. Type 1, Order 3, System and $P_r = 0$

   The Nyquist contour of Fig. 4.22a is slightly modified as shown in Fig. 4.23a. The given $GH(s)$ has a pole at $s = 0$ and the Nyquist contour of Fig. 4.22a if not modified, it will not satisfy Cauchy's theorem. Hence, a small circle around the origin of s-plane is drawn in the RHP with centre at the origin and radius $r \to 0$ and $-\pi/2 \le \phi \le \pi/2$. Now, section by section mapping is done from s-plane to GH-plane.
2. Section AB; $s = j\omega$; $\quad 0 \le \omega \le \infty$

$$GH(j\omega) = \frac{K}{j\omega(j\omega+T_1)(j\omega+T_2)}$$

$$|GH(j\omega)| = \frac{K}{\omega\sqrt{(\omega^2+T_1^2)(\omega^2+T_2^2)}}$$

$$\angle GH(j\omega) = -90^\circ - \tan^{-1}\frac{\omega}{T_1} - \tan^{-1}\frac{\omega}{T_2}$$

   For $\omega = 0$

$$GH(j\omega) = \infty\angle -90^\circ$$

   For $\omega = \infty$

$$GH(j\omega) = 0\angle -270^\circ$$

   The polar plot starts from $-\infty$ in the $j\omega$ axis for $\omega = 0$, traverses through the third quadrant, cuts the negative real axis for $\omega = \omega_{pc}$, and terminates at the origin of the GH-plane in the second quadrant.
3. The point in the negative real axis at which the $GH(j\omega)$ plot cuts is very important to assess the stability. This is estimated as described below. On the negative real axis, the phase angle is 180° and the corresponding frequency is called the phase crossover frequency which is determined as follows:

$$-90^\circ - \tan^{-1}\frac{\omega}{T_1} - \tan^{-1}\frac{\omega}{T_2} = -180^\circ$$

$$\tan^{-1}\frac{\frac{\omega}{T_1}+\frac{\omega}{T_2}}{1-\frac{\omega^2}{T_1T_2}} = 90^\circ$$

Taking tan on the both sides, we get

$$\frac{\frac{\omega}{T_1}+\frac{\omega}{T_2}}{1-\frac{\omega^2}{T_1T_2}} = \tan 90^\circ = \infty$$

To satisfy the above condition, $\omega = \sqrt{T_1T_2}$ which is ω_{pc}. At ω_{pc}, the magnitude of $GH(j\omega)$ is

$$|GH(j\omega)|_{\omega=\omega_{pc}} = \frac{K}{\sqrt{T_1T_2}\sqrt{(T_1T_2+T_2^2)(T_1T_2+T_1^2)}}$$
$$= \frac{K}{T_1T_2(T_1+T_2)}$$

4. Section BCD; $s = Re^{j\theta}$; $\quad R \to \infty; \pi/2 \le \theta \le -\pi/2$

$$GH(j\omega) = \lim_{R\to\infty}\frac{K}{Re^{j\theta}(Re^{j\theta}+T_1)(Re^{j\theta}T_2)}$$
$$= 0$$

This is plotted into the origin of the GH-plane.

5. Section DE; $s = -j\omega$ $\quad -\infty \le \omega \le -0$

This is the mirror image of section AB. This is drawn in dotted line.

6. Section EFA; $s = re^{j\phi}$; $\quad r \to 0\ -\pi/2 \le \phi \le \pi/2$

$$GH(re^{j\phi}) = \frac{K}{re^{j\phi}(re^{j\phi}+T_1)(re^{j\phi}+T_1)}$$
$$= \infty e^{-j\phi}$$

where $-\pi/2 \le \phi \le \pi/2$. This is one semi-circle of ∞ radius in the clockwise direction. The GH-plane contour becomes automatically closed.

7. The Nyquist diagram shown in Fig. 4.23b is interpreted as follows. Since, the numerical values for K, T_1 and T_2 are not given, if

$$|GH(j\omega)|_{\omega_{pc}} > 1$$

or

$$\frac{K}{T_1T_2(T_1+T_2)} > 1$$

$(-1, 0)$ point is enclosed by the Nyquist plot. As seen in Fig. 4.23b, $N_r = +2$. Since $P_r = 0$

$$\begin{aligned} Z_r &= N_r + P_r \\ &= 2 \end{aligned}$$

There are 2 RHP roots of the characteristic equation in RHP, and the system is unstable.

8. If

$$\frac{K}{T_1 T_2 (T_1 + T_2)} < 1$$

The Nyquist plot does not encircle $(-1, 0)$ point and hence $N_r = 0$. Since $P_r = 0$

$$\begin{aligned} Z_r &= N_r + P_r \\ &= 0 + 0 = 0 \end{aligned}$$

No root of the characteristic equation is in RHP and the imaginary axis and the system is stable

1.	$\frac{K}{T_1 T_2 (T_1 + T_2)} > 1$;	system is unstable
2.	$\frac{K}{T_1 T_2 (T_1 + T_2)} < 1$;	system is stable
3.	$\frac{K}{T_1 T_2 (T_1 + T_2)} = 1$;	system is marginally stable or oscillatory

One important point which is to be noted in the above 2 examples is that one semi-circle of infinite radius is drawn in the clockwise direction for Type 1 system so that the contour in the GH-plane becomes closed.

Example 4.22

Consider the following loop T.F of a certain closed loop system

$$GH(s) = \frac{K}{s^2(s + T)}$$

where K and T are greater than zero. Determine the stability of the closed loop system using Nyquist stability criterion.

Solution

1. Type 0, Order 3 system. $Pr = 0$.

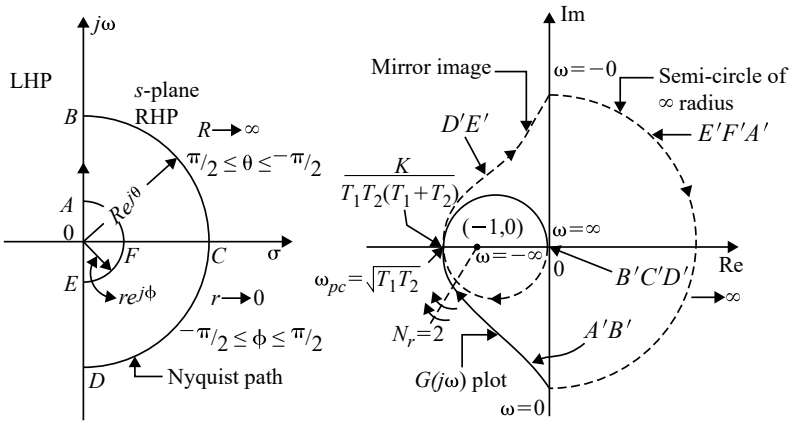


Fig. 4.23 **a** Nyquist path and **b** Nyquist plot for Example 4.21

$$GH(s) = \frac{K}{s^2(s+T)}$$

The Nyquist contour of Fig. 4.23a can be used for this also. For $T > 0$, $P_r = 0$. The section by section mapping of s-plane Nyquist contour into GH-plane is explained below.

2. Section AB; $s = j\omega$; $\quad 0 \le \omega \le \infty$

$$GH(s) = \frac{K}{s^2(s+T)}$$

$$GH(j\omega) = \frac{K}{\omega^2\sqrt{\omega^2+T^2}}\angle -180^\circ - \tan^{-1}\frac{\omega}{T}$$

$$\text{For } \omega = 0, \quad GH(j\omega) = \infty\angle -180^\circ$$

$$\text{For } \omega = \infty, \quad GH(j\omega) = 0\angle -270^\circ$$

The $GH(j\omega)$ plot starts from negative real at $-\infty$ in the second quadrant, traverses through the same quadrant with phase angle of $GH(j\omega)$ reduced as ω increases and ends at the origin of GH-plane with phase angle -270°. This is shown as $A'B'$ in the Nyquist plot.

3. Section BCD; $s = Re^{j\theta}$; $\quad R \to \infty; \pi/2 \le \theta \le -\pi/2$

$$GH(Re^{j\theta}) = \frac{K}{Re^{j2\theta}(Re^{j\theta}+T)} = 0$$

This plots into the origin of GH-plane as $B'C'D'$.

4. Section DE; $s = -j\omega$; $\quad -\infty \le \omega \le -0$

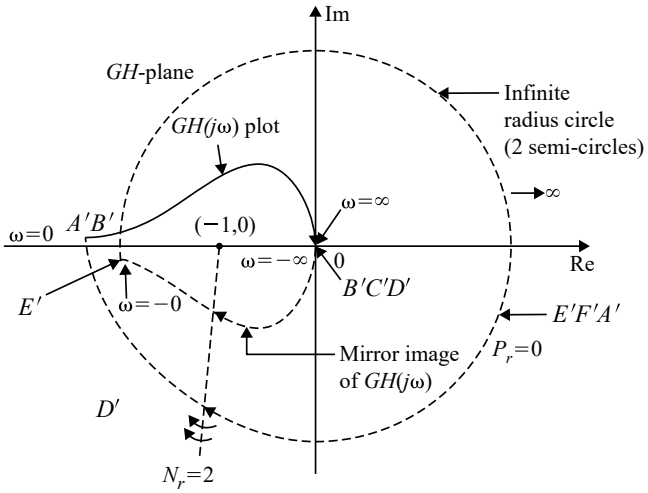


Fig. 4.24 Nyquist plot for Example 4.22

$$GH(-j\omega) = \frac{K}{\omega^2\sqrt{-j\omega + T}}$$

This is the mirror image of AB. This is drawn as $D'E'$.

5. Section EFA; $s = re^{j\phi}$; $\quad r \to 0; -\pi/2 \le \phi \le \pi/2$

$$\begin{aligned} GH(re^{j\phi}) &= \frac{K}{re^{j2\phi}(re^{j2\phi} + T)} \\ &= \infty e^{-j2\phi} \end{aligned}$$

For $-\pi/2 \le \phi \le \pi/2$, section EFA is mapped as a circle (2 semi-circles) of ∞ radius drawn in the clockwise direction.

6. The Nyquist plot gets closed in the GH-plane. Choose point $(-1, 0)$ and draw a radial line to count N_r. As seen from Fig. 4.24, $N_r = 2$. Using Eq. (4.33), Z_r is determined.

$$\begin{aligned} Z_r &= N_r + P_r \\ &= 2 + 0 \\ &= 2 \end{aligned}$$

There are $2RHP$ roots of characteristic equation.

7.

2 RHP roots Closed loop system is unstable for all values of $K > 0$

To Summarize

The general procedure to obtain the closed loop system stability using Nyquist criterion was illustrated by the above three examples. We summarize as follows:

1. From the given T.F. $GH(s)$, just by inspection find P_r, the number of open loop poles in RHP.
2. Draw the $GH(j\omega)$ plot for $0 \leq \omega \leq \infty$. (The traversal of the $GH(j\omega)$ plot is decided by phase angle expression $\angle GH(j\omega)$. By careful assessment of this, one can easily sketch the $GH(j\omega)$ plot.)
3. Draw the mirror image of $GH(j\omega)$ plot by dotted line.
4. For a Type 0 system, the Nyquist plot gets automatically closed. If $GH(s)$ is a Type 1 system, draw one semi-circle in the clockwise direction. For Type 2 system, draw 2 semi-circles in the clockwise direction. Thus, the number of semi-circles to be drawn depends on the Type of the system. Now the Nyquist plot gets closed in the GH-plane.
5. Draw a dotted radial line from $(-1, 0)$ point directing in the 3rd quadrant of the GH-plane and count N_r.
6. Using Eq. (4.33), determine Z_r. For the closed loop system to be stable, $Z_r = 0$. Otherwise, the system is unstable.

In the example we work out below, straightaway we follow the above six steps. Section by section mapping of Nyquist contour from s-plane to GH-plane is avoided as it will be automatically satisfied when one follows the above six steps. However, in certain special cases, the contour mapping will have to be considered. Further, it should be noted that the $GH(j\omega)$ which gives important information about the stability of the closed loop system need not be drawn after calculating the magnitude $|GH(j\omega)|$ and the phase $\angle GH(j\omega)$ for $0 \leq \omega \leq \infty$. In many practical cases, the intersection of $GH(j\omega)$ on the negative real axis alone will suffice to find the stability. In the Nyquist method of finding the stability estimation, the only part which requires some calculation is the plot of $GH(j\omega)$. When this is done judiciously by calculating $|GH(j\omega)|$ at ω_{pc}, the stability study becomes simpler and quicker. This is how many problems in this book are solved.

Example 4.23

Sketch the Nyquist plot for the following open loop T.F. and examine the closed loop system stability

$$GH(s) = \frac{K}{(s+1)(s^2+2s+5)}$$

(*Anna University, December, 2010*)

Solution

1.

$$GH(s) = \frac{K}{(s+1)(s^2+2s+5)}$$

Since it is a minimum phase T.F., $P_r = 0$

$$GH(j\omega) = \frac{K}{(j\omega+1)(j2\omega+(5-\omega^2))}$$

$$|GH(j\omega)| = \frac{K}{\sqrt{(\omega^2+1)[4\omega^2+(5-\omega^2)^2]}}$$

$$\angle GH(j\omega) = -\tan^{-1}\omega - \tan^{-1}\frac{2\omega}{5-\omega^2}$$

For $\omega = 0$

$$GH(j\omega) = \frac{K}{5}\angle 0^\circ$$

For $\omega = \infty$

$$GH(j\omega) = 0\angle -270^\circ$$

Since the given $GH(s)$ is a minimum phase T.F., which is of Type 0 and order three, it starts from +ve real axis at $K/5$ when $\omega = 0$, traverses through the fourth and third quadrants, cuts the negative real axis, passes through the second quadrant and terminates at the origin when $\omega = \infty$. The $GH(j\omega)$ plot for $0 \leq \omega \leq \infty$ is shown in Fig. 4.25.

2. The mirror image of $GH(j\omega)$ is drawn in dotted line. The Nyquist plot gets closed.
3. The phase crossover frequency ω_{pc} and the point of intersection of $GH(j\omega)$ plot with the −ve real axis are determined as follows:

$$-\tan^{-1\omega} - \tan^{-1}\frac{2\omega}{5-\omega^2} = -180^\circ$$

$$180 - \tan^{-1}\omega = \tan^{-1}\frac{2\omega}{5-\omega^2}$$

Taking tan on both sides of the above equation, we get

$$\tan[180 - \tan^{-1}\omega] = \tan\tan^{-1}\frac{2\omega}{5-\omega^2}$$
$$-\omega = \frac{2\omega}{5-\omega^2}$$
$$\omega_{pc} = \sqrt{7}$$

Substituting $\omega = \sqrt{7}$ in $GH(j\omega)$, we get

$$|GH(j\omega)|_{\omega=\sqrt{7}} = \frac{K}{\sqrt{(\omega^2+1)[4\omega^2+(5-\omega^2)^2]}}$$
$$= \frac{K}{\sqrt{8\times(4+28)}}$$
$$= \frac{K}{16}$$

4. For $K/16 > 1$, or $K > 16$, $(-1, 0)$ point is encircled by the Nyquist plot and $N_r = 2$. Now

$$Z_r = N_r + P_r$$
$$= 2 + 0 = 2$$

The closed loop system has two RHP roots and the system is unstable.

5. For $K = 16$, the Nyquist plot passes through the $(-1, 0)$ point. The system is marginally stable or oscillatory.
6. For $K < 16$, the Nyquist plot does not encircle $(-1, 0)$ point. $N_r = 0$ and

$$Z_r = N_r + P_r$$
$$= 0 + 0 = 0$$

The system is stable.

7.

$K < 16$, the system is stable.
$K = 16$, the system is oscillatory.
$K > 16$, the system is unstable.

Example 4.24

The polar diagram of a conditionally stable system for an open loop gain $K = 1$ is shown in Fig. 4.26 in thick lines. The open loop T.F. of the system is known to be stable. Determine the range of K for which the closed loop system is stable.

Solution

1. The $G(j\omega)$ plot drawn in thick line is given. Since the open loop system is stable, the corresponding T.F. has only LHP poles and hence $P_r = 0$. For $\omega = 0$, the $GH(j\omega)$ plot starts from $-$ve $j\omega$ axis, and therefore, it is a Type 1 system.
2. The mirror image is drawn with dotted line for $-\infty \leq \omega \leq -0$. For Type 1 system, one semi-circle of infinite radius is drawn.
3. For $K < 1/8$, the entire Nyquist plot is to the right of $(-1, 0)$ point and $N_r = 0$. Since $P_r = 0$;

$$Z_r = N_r + P_r = 0 + 0 = 0$$

   The closed loop system is stable.
4. From Fig. 4.26, if $(-1, 0)$ point is within loop 1 and loop 3, $N_r = 2$ and

$$Z_r = N_r + P_r = 2 + 0 = 2$$

   There will be 2 RHP roots and the closed loop system will be unstable.
5. If $(-1, 0)$ point is within the loop 2, $N_r = 0$. Since $P_r = 0$ then $Z_r = 0$ and the closed loop system will be stable. If $K > 0.5$ and $K < 5$, the loop 2 will have $(-1, 0)$ point inside it.
6. The closed loop system is stable if

$$\boxed{\begin{array}{ll} (1) & K < \dfrac{1}{8} \quad \text{or} \\ (2) & 0.5 < K < 5 \end{array}}$$

Example 4.25

By Nyquist stability criterion, determine the value of K for the stability of the closed loop system whose open loop T.F. is given by

$$GH(s) = \frac{K(s+4)}{(s+1)(s-1)}$$

(*Anna University, December, 2009*)

Solution

1.

$$GH(s) = \frac{K(s+4)}{(s+1)(s-1)}$$

   The pole $s = 1$ is in RHP and hence $P_r = 1$.

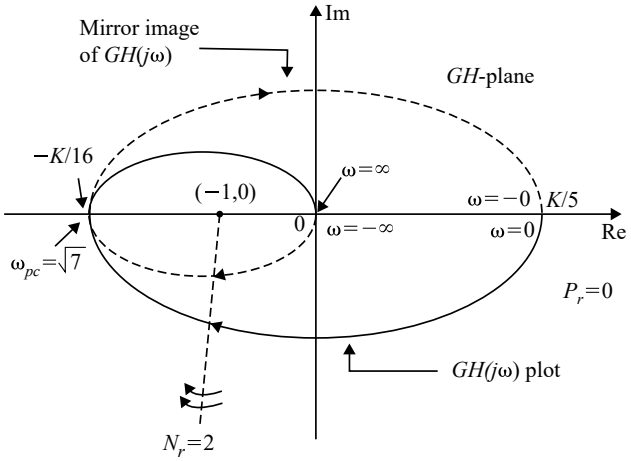


Fig. 4.25 Nyquist plot for Example 4.23

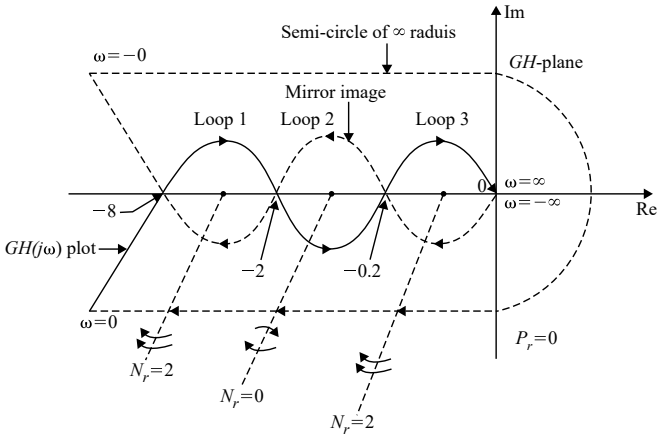


Fig. 4.26 Nyquist plot for Example 4.24

2.

$$GH(j\omega) = \frac{K(j\omega + 4)}{(j\omega + 1)(j\omega - 1)}$$

$$|GH(j\omega)| = \frac{K\sqrt{\omega^2 + 16}}{(\omega^2 + 1)}$$

$$\angle GH(j\omega) = 180^\circ + \tan^{-1}\frac{\omega}{4}$$

For $\omega = 0$,

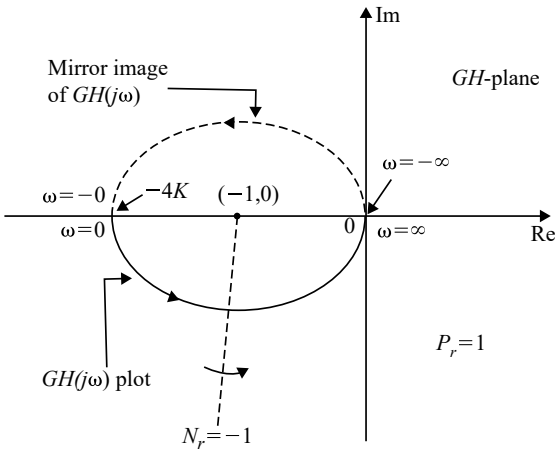


Fig. 4.27 Nyquist plot for Example 4.25

$$GH(j\omega) = 4K\angle 180^\circ$$

For $\omega = \infty$,

$$GH(j\omega) = 0\angle 270^\circ$$

For $0 \leq \omega \leq \infty$, the $G(j\omega)$ plot starts from $-$ve real axis at $(-4K, 0)$ for $\omega = 0$, traverses through the third quadrant and terminates at the origin for $\omega = \infty$.

3. The mirror image is drawn. The Nyquist plot gets closed at $\omega = -0$ and $\omega = 0$. The complete Nyquist plot is shown in Fig. 4.27. From Fig. 4.27, for $K > 0.25$, N_r is counted as -1. Since $P_r = 1$,

$$Z_r = N_r + P_r = -1 + 1 = 0$$

The closed loop system is stable.

4. For $K = 0.25$, the $G(j\omega)$ passes through $(-1, 0)$ point and the system is oscillatory.
5. For $K < 0.25$, the entire Nyquist path will be to the right of the $(-1, 0)$ point and hence $N_r = 0$. In this case

$$Z_r = N_r + P_r = 0 + 1 = 1$$

There will be 1 RHP root and the closed loop system us unstable.

6.

1.	$K > 0.25$ the system is stable.
2.	$K = 0.25$ the system is oscillatory.
3.	$K < 0.25$ the system is unstable.

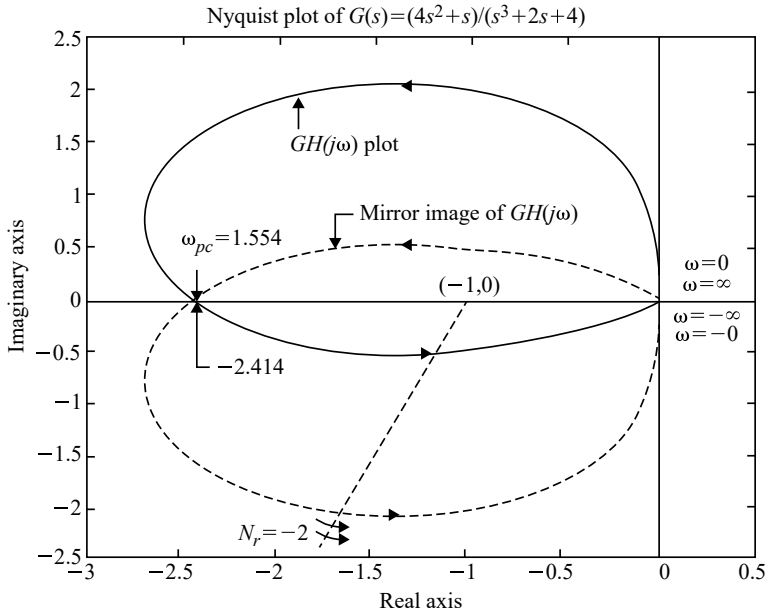


Fig. 4.28 Nyquist plot for Example 4.26

Example 4.26

A certain feedback control system has the following loop T.F

$$GH(s) = \frac{s(1+4s)}{(s^3+2s+4)}$$

Determine the stability of the closed loop system using Nyquist stability criterion. Give MATLAB program.

Solution

1.

$$GH(s) = \frac{s(1+4s)}{(s^3+2s+4)}$$

The system is Type 0 order 3 system

$$GH(j\omega) = \frac{j\omega(1 + j4\omega)}{(4 + j\omega(2 - \omega^2))}$$

$$|GH(j\omega)| = \frac{\omega\sqrt{1 + 16\omega^2}}{\sqrt{16 + \omega^2(2 - \omega^2)^2}}$$

$$\angle GH(j\omega) = 90^\circ + \tan^{-1} 4\omega - \tan^{-1} \frac{\omega(2 - \omega^2)}{4}$$

$$\text{For } \omega = 0, \quad GH(j\omega) = 0\angle 90^\circ$$
$$\text{For } \omega = \infty, \quad GH(j\omega) = 0\angle -90^\circ$$

The $GH(j\omega)$ starts from the origin of GH-plane in the second quadrant, traverses through the second quadrant, cuts the negative real axis, traverses through the third quadrant as ω increases and terminates at the origin when $\omega = \infty$.

2. The mirror image of $GH(j\omega)$ plot is drawn. Since the problem is drawn using the MATLAB program, the mirror image is also drawn in thick line.
3. Since the given system is of Type 0, the Nyquist plot gets closed.
4. The point of intersection of $GH(j\omega)$ plot with the negative real axis for ω_{pc} is obtained as explained below

$$90^\circ + \tan^{-1} 4\omega - \tan^{-1} \frac{\omega(2 - \omega^2)}{4} = 180^\circ$$

$$\tan^{-1} 4\omega = \left(\tan^{-1} \frac{\omega(2 - \omega^2)}{4} + 90^\circ\right)$$

Taking tan on both sides, we get

$$4\omega = -\cot \tan^{-1} \frac{\omega(2 - \omega^2)}{4}$$

$$4\omega = \frac{-1}{\tan \tan^{-1} \frac{\omega(2-\omega^2)}{4}}$$

$$= \frac{-4}{\omega(2 - \omega^2)}$$

$$-4\omega^4 + 8\omega^2 = -4$$

$$4\omega^4 - 8\omega^2 - 4 = 0$$

Substitute $\omega^2 = x$

$$x^2 - 2x - 1 = 0$$
$$x = 2.414$$
$$\omega = \sqrt{x}$$
$$= \sqrt{2.414}$$
$$\omega_{pc} = 1.554$$

Substituting this value in the equation for $|GH(j\omega)|$, we get

$$|GH(j\omega)| = \frac{\omega\sqrt{(1+16\omega^2)}}{\sqrt{16+\omega^2(2-\omega^2)^2}}$$
$$|GH(j\omega)| = \frac{1.554\sqrt{1+16\times 2.414}}{\sqrt{16+2.414(2-2.414)^2}}$$
$$= 2.414$$

The Nyquist plot is shown in Fig. 4.28. From Fig. 4.28, N_r is counted as -2.

5. P_r is found from the open loop T.F. The characteristic equation of the open loop system is $s^3 + 2s + 4 = 0$. The following Routh's array is formed.

				For $\varepsilon = 0$
s^3	1	2	1	2
s^2	ε	4	0^+	4
s^1	$\frac{2\varepsilon-4}{\varepsilon}$		$-\infty$	
s^0	4		4	

There are two sign changes in the first column. Hence, there are two RHP roots of the open loop system. Hence, $P_r = 2$.

6.

$$Z_r = N_r + P_r = -2 + 2 = 0$$
$$Z_r = 0$$

The closed loop system is stable.

7. The MATLAB program is shown below.

MATLAB Program

```
%... To find stability using Nyquist plot...%
num= [0   4   1   0]
den[1   0   2   4]
```

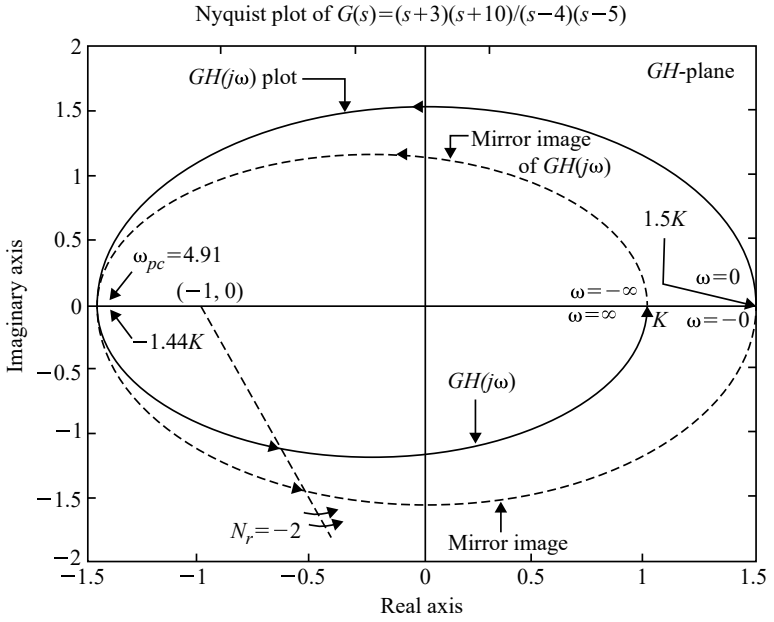


Fig. 4.29 Nyquist plot for Example 4.27

```
nyquist(num,den)
axis auto
title ('Nyquist plot of G(s) = (4s^2 + s)/(s^3 + 2s + 4)')
```

Example 4.27

A certain feedback control system has the following loop T.F

$$GH(s) = \frac{K(s+3)(s+10)}{(s-5)(s-4)}$$

Determine the value of K so that the closed loop system is stable. Use Nyquist stability criterion (Fig. 4.29).

Solution

1.

$$GH(s) = \frac{K(s+3)(s+10)}{(s-5)(s-4)}$$

There are two poles of $GH(s)$ in RHP at $s = 5$ and $s = 4$. Hence, $P_r = 2$.

2.

$$GH(j\omega) = \frac{K(j\omega + 3)(j\omega + 10)}{(j\omega - 5)(j\omega - 4)}$$

$$|GH(j\omega)| = K\frac{\sqrt{(\omega^2 + 9)(\omega^2 + 100)}}{\sqrt{(\omega^2 + 25)(\omega^2 + 16)}}$$
$$\angle GH(j\omega) = \tan^{-1}\frac{\omega}{3} + \tan^{-1}\frac{\omega}{10} + \tan^{-1}\frac{\omega}{5} + \tan^{-1}\frac{\omega}{4}$$

3. For $\omega = 0$

$$GH(j\omega) = 1.5K\angle 0^\circ$$

For $\omega = \infty$

$$GH(j\omega) = K\angle 0^\circ$$

The polar plot starts from +ve real axis at $1.5K$ for $\omega = 0$. As ω increases, the $GH(j\omega)$ plot traverses through the 1st, 2nd, 3rd and 4th quadrants in the GH-plane and terminates on the +ve real axis at K for $\omega = \infty$. The point of intersection of $GH(j\omega)$ plot on the negative real axis is calculated as follows.

4. At $\omega = \omega_{pc}$

$$\tan^{-1}\frac{\omega}{3} + \tan^{-1}\frac{\omega}{10} + \tan^{-1}\frac{\omega}{5} + \tan^{-1}\frac{\omega}{4} = 180^\circ$$
$$\tan^{-1}\frac{\frac{\omega}{3} + \frac{\omega}{10}}{1 - \frac{\omega^2}{30}} + \tan^{-1}\frac{\frac{\omega}{5} + \frac{\omega}{4}}{1 - \frac{\omega^2}{20}} = 180^\circ$$
$$\tan^{-1}\frac{13\omega}{30 - \omega^2} = 180^\circ - \tan^{-1}\frac{9\omega}{20 - \omega^2}$$

Taking tan on both sides

$$\frac{13\omega}{30 - \omega^2} = \tan\left(180 - \tan^{-1}\frac{9\omega}{20 - \omega^2}\right)$$
$$= \frac{-9\omega}{20 - \omega^2}$$
$$13(20 - \omega^2) = -9(30 - \omega^2)$$
$$22\omega^2 = 530$$

$$\omega^2 = 24.1$$
$$\omega = 4.91$$
$$\omega_{pc} = 4.91$$

5.
$$|GH(j\omega)|_{\omega=\omega_{pc}} = \frac{K\sqrt{(24.1+9)(24.1+100)}}{\sqrt{(24.1+25)(24.1+16)}}$$
$$= 1.44K$$

6. The mirror image of $GH(j\omega)$ is drawn.
7. If $1.44K > 1$ or $K > 0.694$, the $(-1, 0)$ point is within the $GH(j\omega)$ loops. A radial line from $(-1, 0)$ point is drawn and N_r is counted as $N_r = -2$. Since $P_r = 2$

$$Z_r = N_r + P_r$$
$$= -2 + 2 = 0$$

The closed loop system is stable.
8. If $K < 0.694$, the Nyquist plot does not encircle $(-1, 0)$ point and hence $N_r = 0$

$$Z_r = N_r + P_r$$
$$= 0 + 2 = 2$$

The closed loop system has 2RHP poles and the system is unstable.
9.

$K > 0.694$ stable.
$K = 0.694$ marginally stable.
$K < 0.694$ unstable.

10. The open loop system which is unstable is made stable by feedback.

Example 4.28

A certain control system has the following loop transfer function

$$GH(s) = \frac{K(s+2)(s+0.5)}{(1+0.2s)(1-5s)}$$

Using Nyquist stability criterion, determine the value of K so that the closed loop system is stable.

Solution

1.

$$GH(s) = \frac{K(s+2)(s+0.5)}{(1+0.2s)(1-5s)}$$

There is pole at $s = 0.2$ in RHP. Hence, $P_r = 1$.

2.

$$GH(j\omega) = \frac{K(j\omega+2)(j\omega+0.5)}{(1+0.2j\omega)(1-j\omega 5)}$$

$$|GH(j\omega)| = \frac{K\sqrt{(\omega^2+4)(\omega^2+0.25)}}{\sqrt{(1+0.04\omega^2)(1+25\omega^2)}}$$

$$\angle GH(j\omega) = \tan^{-1}\frac{\omega}{2} + \tan^{-1}\frac{\omega}{0.5} - \tan^{-1}0.2\omega + \tan^{-1}5\omega$$

3. For $\omega = 0$

$$GH(j\omega) = K\angle 0^\circ$$

For $\omega = \infty$

$$GH(j\omega) = K\angle 180^\circ$$

For $0 \le \omega \le \infty$, the $GH(j\omega)$ starts from +ve real at $|GH(j\omega)| = K$, traverses through the first, second and third quadrants as ω increases and terminates at $-K$ when $\omega = \infty$ as seen from $\angle GH(j\omega)$.

4. The mirror image of $GH(j\omega)$ is drawn as shown in Fig. 4.30 for $-\infty \le \omega \le -0$. The Nyquist plot gets closed.
5. Since $P_r = 1$, the Nyquist plot should encircle $(-1, 0)$ point in the counter clockwise direction once.
6. The phase crossover frequency is calculated as follows. At $\omega = \omega_{pc}$

$$\tan^{-1}0.5\omega + \tan^{-1}2\omega - \tan^{-1}0.2\omega + \tan^{-1}5\omega = 180^\circ$$

$$\tan^{-1}\frac{2.5\omega}{(1-\omega^2)} + \tan^{-1}\frac{4.8\omega}{(1+\omega^2)} = 180^\circ$$

$$\tan^{-1}\frac{2.5\omega}{(1-\omega^2)} = 180^\circ - \tan^{-1}\frac{4.8\omega}{(1+\omega^2)}$$

Taking tan on both sides, we get

$$\frac{2.5\omega}{(1-\omega^2)} = -\frac{4.8\omega}{(1+\omega^2)}$$

Solving the above equation for ω, we get

$$\omega^2 = 3.1739$$
$$\omega_{pc} = 1.7815$$

7. At $\omega = \omega_{pc}$

$$|GH(j\omega)|_{\omega=\omega_{pc}} = \frac{K\sqrt{(3.1739+4)(0.25+3.1739)}}{\sqrt{(1+0.04\times 3.1739)(1+25\times 3.1739)}}$$
$$= 0.521K$$

For loop 1, $N_r = 1$ and $Z_r = N_r + P_r = 1 + 1 = 2$ unstable.

9. For loop 2 to encircle $(-1, 0)$ point in the anticlockwise direction ($N_r = -1$),

$$0.521K > 1$$
$$K > 1.92$$
$$Z_r = N_r + P_r$$
$$= -1 + 1 = 0 \text{ stable}$$

10.

$K > 1.92$ system is stable.
$K = 1.92$ system is oscillatory.
$K < 1.92$ system is unstable.

Example 4.29

A certain feedback control system has the following loop T.F

$$GH(s) = \frac{K(s+2)}{(s^3+2s^2-5)}$$

Determine the value of K, using Nyquist stability criterion so that the closed loop system is stable.

Solution

1.

$$GH(s) = \frac{K(s+2)}{(s^3+2s^2-5)}$$

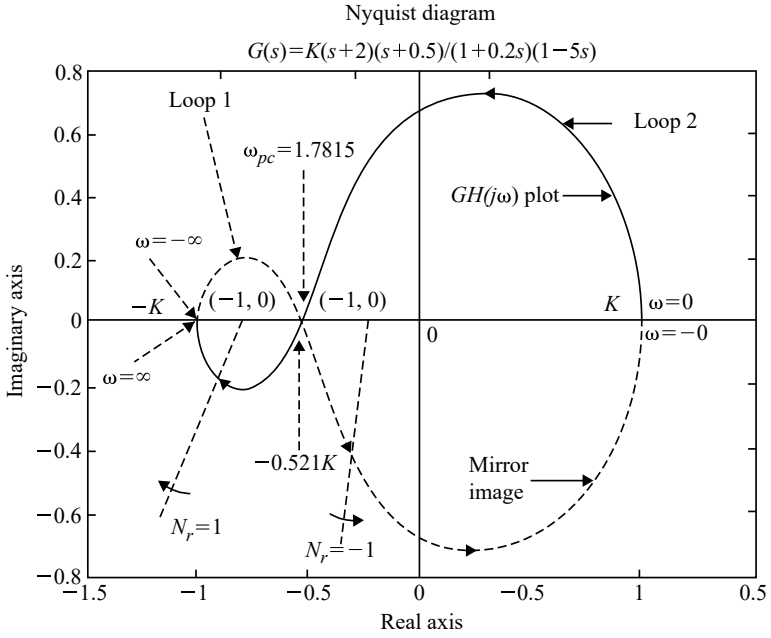


Fig. 4.30 Nyquist plot for Example 4.28

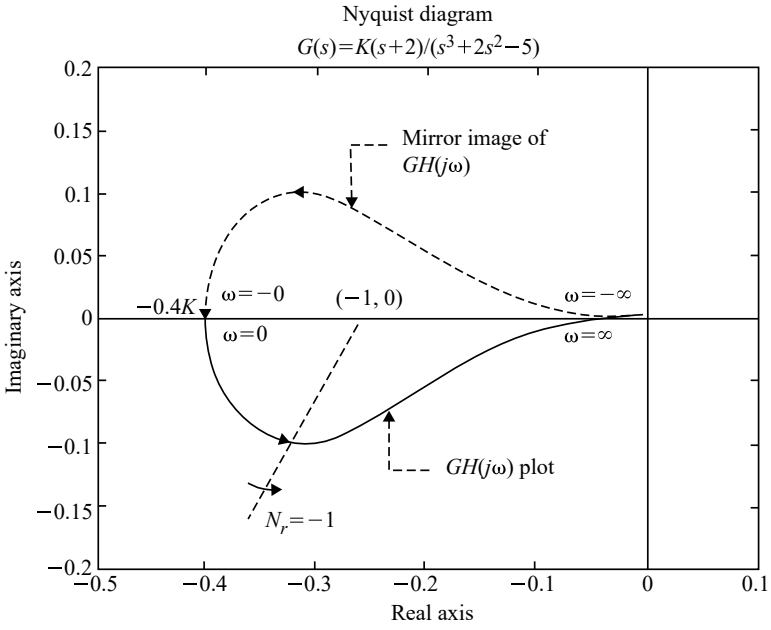


Fig. 4.31 Nyquist plot for Example 4.29

The characteristic equation of the open loop system is

$$s^3 + 2s^2 - 5 = 0$$

The following Routh's array is formed for the above equation.

s^3	1	0
s^2	2	-5
s^1	2.5	
s^0	-5	

There is one sign change in the first column of Routh's array. Hence, $P_r = 1$.

2.

$$GH(j\omega) = \frac{K(j\omega + 2)}{-j\omega^3 - 2\omega^2 - 5}$$

$$|GH(j\omega)| = \frac{K\sqrt{(\omega^2 + 4)}}{\sqrt{\omega^6 + (5 + 2\omega^2)^2}}$$

$$\angle GH(j\omega) = 180^\circ + \tan^{-1}\frac{\omega}{2} - \tan^{-1}\frac{\omega^3}{(5 + 2\omega^2)}$$

3. For $\omega = 0$

$$GH(j\omega) = 0.4K\angle 180^\circ$$

For $\omega = \infty$

$$GH(j\omega) = 0\angle 180^\circ$$

4. For $0 \leq \omega \leq \infty$, the Nyquist plot starts from negative real axis at $-0.4K$, traverses through the third quadrant as ω increases and terminates at the origin of GH-plane as shown in Fig. 4.31.
5. The mirror image of $GH(j\omega)$ is drawn for $-\infty \leq \omega \leq -0$. The Nyquist plot gets closed.
6. Since $P_r = 1$, $N_r = -1$ for Z_r to be zero. The Nyquist plot should encircle $(-1, 0)$ point once in the anticlockwise direction. For this, $0.4K > 1$ or $K > 2.5$.
7.

$K > 2.5$ system is stable.
$K = 2.5$ system is oscillatory.
$K < 2.5$ system is unstable.

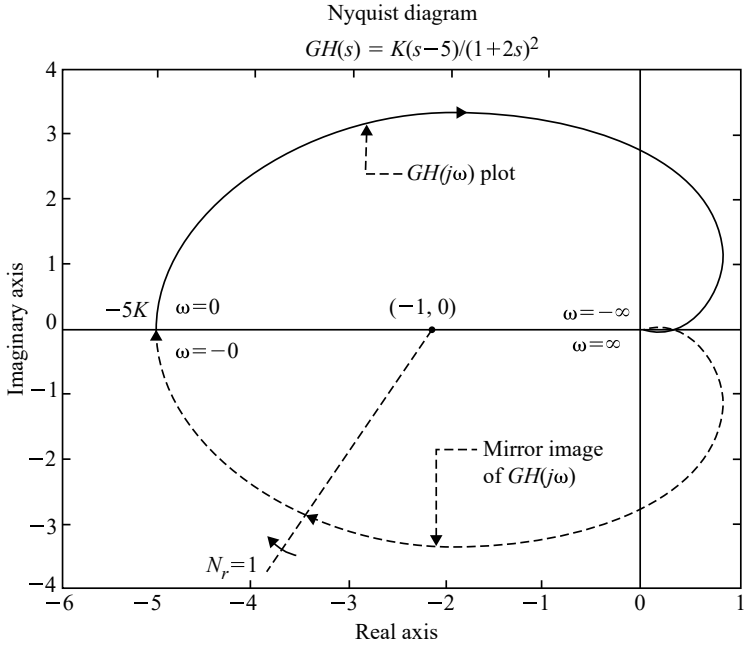


Fig. 4.32 Nyquist plot for Example 4.30

Example 4.30

Consider the following loop T.F. of a certain feedback control system

$$GH(s) = \frac{K(s-5)}{(1+2s)^2}$$

Determine the value of K so that the closed loop system is stable. Use Nyquist stability criterion.

Solution

1. $$GH(s) = \frac{K(s-5)}{(1+2s)^2}$$

   The open loop poles are in LHP and hence $P_r = 0$

2. $$GH(j\omega) = \frac{K(j\omega-5)}{(1+2j\omega)^2}$$

$$|GH(j\omega)| = \frac{K\sqrt{(\omega^2+25)}}{(1+4\omega^2)}$$
$$\angle GH(j\omega) = 180^\circ - \tan^{-1}\frac{\omega}{5} - 2\tan^{-1}2\omega$$

3. For $\omega = 0$

$$GH(j\omega) = 5K\angle 180^\circ$$

For $\omega = \infty$

$$GH(j\omega) = 0\angle -90^\circ$$

4. The Nyquist plot starts from negative real at $-5K$ when $\omega = 0$, traverses through the second and first quadrants as ω increases, cuts the +ve real axis, traverses through the fourth quadrant and terminates at the origin of the GH-plane as shown in Fig. 4.32. The point of intersection of $GH(j\omega)$ plot with the +ve real axis is not exceed.
5. For $-\infty \leq \omega \leq -0$, the mirror image of $GH(j\omega)$ plot is drawn. The Nyquist plot gets closed.
6. Since $P_r = 0$, for the closed loop system to be stable, $N_r = 0$. As seen from Fig. 4.32, the Nyquist plot should not encircle $(-1, 0)$ point. This is possible if $5K < 1$ or $K < 0.2$.
7.

$K < 0.2$ system is stable.
$K = 0.2$ system is oscillatory.
$K > 0.2$ system is unstable.

Example 4.31

A certain closed loop control system has the following open loop T.F

$$GH(s) = \frac{K}{(s+5)^3}$$

Determine the value of K so that the closed loop system is stable. Use Nyquist stability criterion.

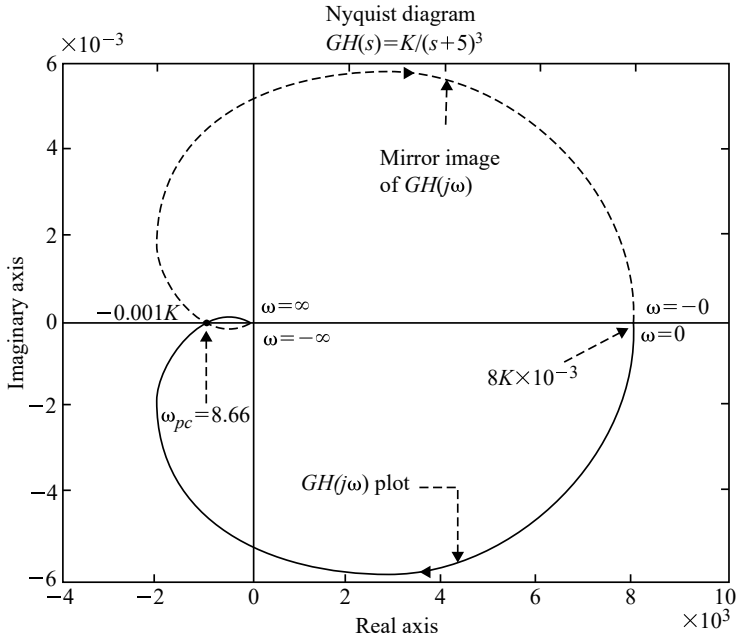


Fig. 4.33 Nyquist plot for Example 4.31

Solution

1.

$$GH(s) = \frac{K}{(s+5)^3}$$

All the three open loop poles are in LHP and $P_r = 0$.

2.

$$GH(j\omega) = \frac{K}{(j\omega+5)^3}$$

$$|GH(j\omega)| = \frac{K}{(\omega^2+25)^{3/2}}$$

$$\angle GH(j\omega) = -3\tan^{-1}\frac{\omega}{5}$$

3. For $\omega = 0$

$$GH(j\omega) = K8\times 10^{-3}\angle 0^\circ$$

For $\omega = \infty$

$$GH(j\omega) = 0\angle -270^\circ$$

4. For $0 \leq \omega \leq \infty$, the Nyquist plot starts from +ve real at $K8 \times 10^{-3}$, traverses through the fourth and third quadrants of the GH-plane, cuts the negative real axis when $\omega = \omega_{pc}$ and terminates at the origin in the second quadrant as shown in Fig. 4.33.
5. For $-\infty \leq \omega \leq -0$, the mirror image of $GH(j\omega)$ is drawn and the Nyquist plot gets closed.
6. The magnitude $|GH(j\omega)|_{\omega=\omega_{pc}}$ is very important which decides the stability of the system and is calculated as follows:

$$\begin{aligned} -3\tan^{-1}\frac{\omega}{5} &= -180^\circ \\ \omega_{pc} &= 8.66 \\ GH(j\omega)|_{\omega=\omega_{pc}} &= \frac{K}{(\omega^2+25)^{3/2}} \\ &= 0.001K \end{aligned}$$

7. For the system to be stable, the Nyquist plot should not encircle $(-1, 0)$ point. In otherwords, $0.001K < 1$ or $K < 1000$.
8.

$K < 1000$ system is stable.
$K = 1000$ system is oscillatory.
$K > 1000$ system is unstable.

Example 4.32

A certain feedback system has the following loop T.F

$$GH(s) = \frac{(1+Ks)}{s(s-4)}$$

Investigate the stability of the closed loop system using Nyquist stability criterion. Determine K, where $K > 0$.

Solution

1.

$$GH(s) = \frac{(1+Ks)}{s(s-4)}$$

There is one pole of $GH(s)$ at $s = 4$ in RHP. $P_r = 1$.

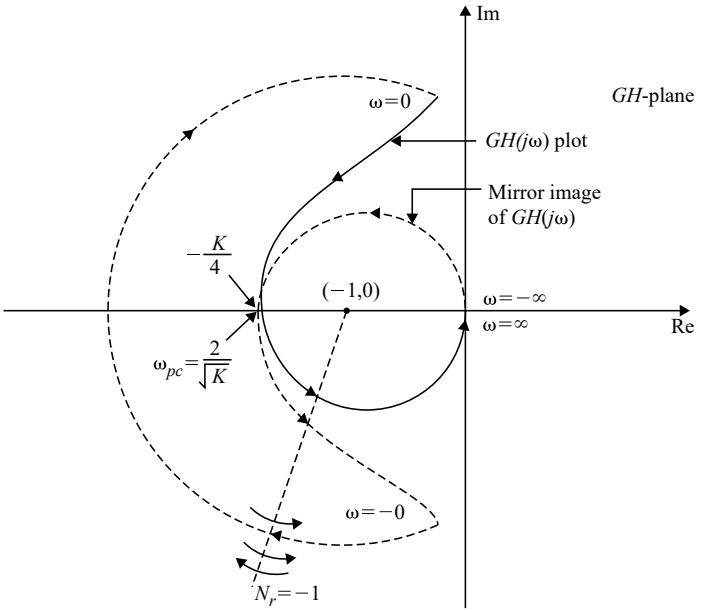


Fig. 4.34 Nyquist plot for Example 4.32

2.

$$GH(j\omega) = \frac{(1 + jK\omega)}{j\omega(j\omega - 4)}$$

$$|GH(j\omega)| = \frac{\sqrt{1 + K^2\omega^2}}{\omega\sqrt{\omega^2 + 16}}$$

$$\angle GH(j\omega) = 90^\circ + \tan^{-1} K\omega + \tan^{-1} \frac{\omega}{4}$$

3. For $\omega = 0$

$$GH(j\omega) = \infty\angle 90^\circ$$

For $\omega = \infty$

$$GH(j\omega) = 0\angle 270^\circ$$

The $GH(j\omega)$ plot starts from +ve imaginary at ∞ when $\omega = 0$, traverses through the second quadrant, cuts the negative real axis when $\omega = \omega_{pc}$, passes through

the third quadrant and terminates at the origin of GH-plane when $\omega = \infty$ as shown in Fig. 4.34.

4. For $-\infty \leq \omega \leq -0$, the mirror image of $GH(j\omega)$ is drawn.
5. Since the Type of the system is 1, a semi-circle of ∞ radius in the clockwise direction is drawn. The Nyquist plot gets closed in the GH-plane.
6. The magnitude of $GH(j\omega_{pc})$ is obtained as follows:

$$\begin{aligned} 90 + \tan^{-1} K\omega + \tan^{-1} \frac{\omega}{4} &= 180^\circ \\ \tan^{-1} K\omega + \tan^{-1} \frac{\omega}{4} &= 90^\circ \\ \tan^{-1} \frac{K\omega + \frac{\omega}{4}}{1 - \frac{K}{4}\omega^2} &= 90^\circ \end{aligned}$$

Taking tan on both sides, we get

$$\frac{K\omega + \frac{\omega}{4}}{1 - \frac{K}{4}\omega^2} = \tan 90^\circ = \infty$$

This is possible if

$$\begin{aligned} 1 - \frac{K}{4}\omega^2 &= 0 \\ \omega^2 &= \frac{4}{K} \\ \omega_{pc} &= \frac{2}{\sqrt{K}} \end{aligned}$$

7.

$$\begin{aligned} |GH(j\omega_{pc})| &= \frac{\sqrt{1 + K^2 \frac{4}{K}}}{\frac{2}{\sqrt{K}}\sqrt{\frac{4}{K} + 16}} \\ &= \frac{K}{2}\frac{\sqrt{(1 + 4K)}}{\sqrt{4 + 16K}} \\ &= \frac{K}{4} \end{aligned}$$

For the system to be stable, N_r should be -1. For this

$$\frac{K}{4} > 1$$

8.

$K > 4$ system is stable.
$K = 4$ system is oscillatory.
$K < 4$ system is unstable.

Example 4.33

A certain feedback control system has the following loop T.F.

$$GH(s) = \frac{(1 + Ks)^2}{s^3}$$

Find the value of K so that the closed loop system is stable. Use Nyquist stability criterion.

Solution

1.

$$GH(s) = \frac{(1 + Ks)^2}{s^3}$$

No pole of $GH(s)$ is in RHP. Hence, $P_r = 0$.

2.

$$GH(j\omega) = \frac{(1 + jK\omega)^2}{(j\omega)^3}$$

$$|GH(j\omega)| = \frac{(1 + K^2\omega^2)}{\omega^3}$$

$$\angle GH(j\omega) = -270^\circ + 2\tan^{-1} K\omega$$

3. For $\omega = 0$

$$GH(j\omega) = \infty\angle -270^\circ$$

For $\omega = \infty$

$$GH(j\omega) = 0\angle -90^\circ$$

4. The $GH(j\omega)$ plot starts from $+j\infty$ when $\omega = 0$, traverses through the second quadrant, cuts the negative real axis passes through the third quadrant as ω increases and terminates at the origin of GH-plane when $\omega = \infty$.
5. For $-\infty \leq \omega \leq -0$, the mirror image of $GH(j\omega)$ is drawn as shown in Fig. 4.35.
6. Since the Type of the open loop system is 3, three semi-circles of infinite radius as shown in Fig. 4.35 is drawn in dotted lines in the clockwise direction.

7. For $\omega = \omega_{pc}$, the $GH(j\omega)$ plot cuts the negative real axis. The point of intersection is calculated as follows. At $\omega = \omega_{pc}$,

$$\begin{aligned} \angle GH(j\omega_{pc}) &= 180^\circ \\ -270^\circ + 2\tan^{-1} K\omega &= 180^\circ \\ 2\tan^{-1} K\omega &= 90^\circ \\ K\omega &= 1 \\ \omega_{pc} &= \frac{1}{K} \end{aligned}$$

8.

$$|GH(j\omega_{pc})| = (1+1)K^3 = 2K^3$$

9. When $(-1, 0)$ point is within loop 2, $N_r = 2$ and since $P_r = 0$, $Z_r = 2$. The system will be unstable. If $(-1, 0)$ point is within loop 1, $N_r = 0$ and $Z_r = 0$ and the system will be stable. So $(-1, 0)$ point should be to the right of $-2K^3$. For this

$$\begin{aligned} 2K^3 &> 1 \\ K &> 0.7937 \end{aligned}$$

10.

$K > 0.7937$ system is stable.
$K = 0.7937$ system is oscillatory.
$K < 0.7937$ system is unstable.

Example 4.34

A certain control system has the following open loop T.F.

$$GH(s) = \frac{9}{(s^2 + 16)}$$

Determine the stability of the closed loop system using Nyquist stability criterion. If the system is oscillatory, find the frequency of oscillation.

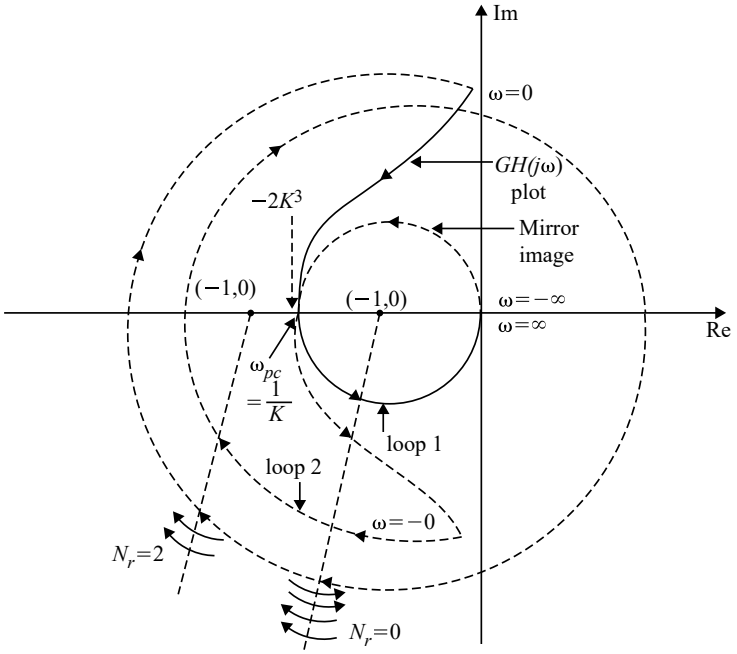


Fig. 4.35 Nyquist plot for Example 4.33

Solution

1.

$$GH(s) = \frac{9}{s^2 + 16}$$

$GH(s)$ poles are at $s_1 = +j4$ and $s_2 = -j4$. No pole is in RHP. Hence, $P_r = 0$. As per Cauchy's theorem, the s-plane contour should not pass through any pole. Hence, these poles are by-passed by a small circle as shown in Fig. 4.36a. The section by section mapping of s-plane contour is discussed below.

2. Section OA, $s = j\omega$; $\quad 0 \le \omega \le 4$

$$GH(j\omega) = \frac{9}{-\omega^2 + 16}$$

$$\text{For } \omega = 0, \quad GH(j\omega) = \frac{9}{16}\angle 0^\circ$$

$$\text{For } \omega = 4, \quad GH(j\omega) = \infty\angle 0^\circ$$

This is shown as $O'A'$ in GH-plane.

3. Section AB, $s = j4 + re^{j\phi}$; $\quad r \to 0; -\pi/2 \le \phi \le \pi/2$

$$\begin{aligned} GH(s) &= \frac{9}{s^2+16} \\ &= \frac{9}{(s+j4)(s-j4)} \\ GH(j4+re^{j\phi}) &= \frac{9}{(j4+re^{j\phi}+j4)(j4+re^{j\phi}-j4)} \\ &= \frac{9}{(j8+re^{j\phi})re^{j\phi}} \end{aligned}$$

As $r \to 0$

$$GH(j4+re^{j\phi}) = \infty e^{-j\phi} \qquad -\pi/2 \le \phi \le \pi/2$$

This represents a semi-circle of infinite radius in the clockwise direction. This is shown as $A'B'$ in the GH-plane.

4. Section BC; $s = j\omega$; $\quad 4 \le \omega \le \infty$

$$\begin{aligned} GH(j\omega) &= \frac{-9}{(\omega^2-16)} \\ \text{For } \omega = 4, \quad GH(j\omega) &= -\infty \\ \text{For } \omega = \infty, \quad GH(j\omega) &= 0\angle 180^\circ \end{aligned}$$

This is plotted as $B'C'$ on the negative real axis in the GH-plane.

5. Section CDE; $s = Re^{j\theta}$; $\quad R \to \infty; \pi/2 \le \theta \le -\pi/2$

$$\begin{aligned} GH(s) &= \frac{9}{s^2+16} \\ GH(Re^{j\theta}) &= \frac{9}{(Re^{j\theta}+16)} \\ &= 0 \quad \text{for } R = \infty \end{aligned}$$

This is plotted into the origin of GH-plane.

6. Section EF; $s = -j\omega$; $\quad -\infty \le \omega \le -4$

This is the mirror image of BC and is shown as $E'F'$ is Fig. 4.36b.

7. Section FG; $s = -j4 + re^{j\phi}$; $\quad r \to 0; -\pi/2 \le \phi \le \pi/2$

This is the mirror image of AB and is shown as $F'G'$.

8. Section GO; $s = j\omega$; $\quad -4 \le \omega \le -0$

This is the mirror image of OA. This is shown as $G'O'$. The Nyquist plot is closed. From Fig. 4.36b, it is observed that the Nyquist plot passes through $(-1, 0)$. Hence, the closed loop system is oscillatory.

In Fig. 4.36a, for clarify, the Nyquist plot in the GH-plane on the real axis is slightly shifted above and below the real axis to differentiate the overlapping part.

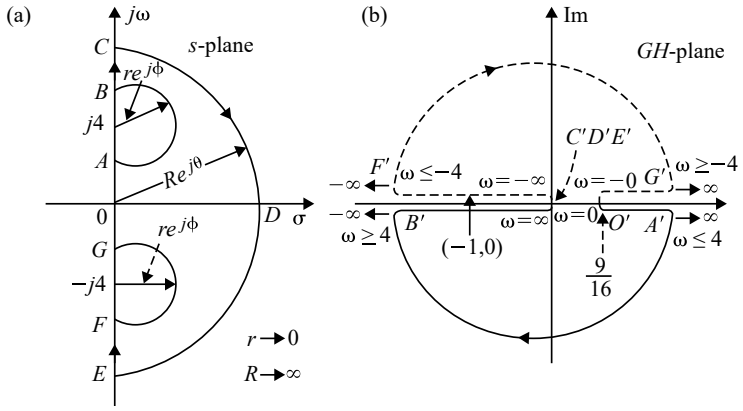


Fig. 4.36 **a** Nyquist contour in the s-plane and **b** Nyquist plot in GH-plane

The system is oscillatory.

9. In Fig. 4.36b, the section $B'C'$ passes, through $(-1, 0)$ point where the frequency range is $4 \leq \omega \leq \infty$ and $s = j\omega$

$$\begin{aligned} GH(j\omega) &= \frac{9}{-\omega^2 + 16} \\ &= -1 \\ \omega^2 - 16 &= 9 \\ \omega &= 5 \end{aligned}$$

The system oscillates with a frequency of

$$\omega = 5\,\text{rad/sec.}$$

Step-by-Step Procedure to Draw Nyquist Plot for Poles of $GH(j\omega)$ on the $j\omega$ Axis

1. Draw $GH(j\omega)$ plot for $0 \leq \omega \leq \omega_1$ where ω_1 is the location of the pole on the $j\omega$ axis.
2. Draw the infinite semi-circle in the clockwise direction.
3. Draw the $GH(j\omega)$ plot for the frequency range $\omega_1 \leq \omega \leq \infty$.
4. Draw the mirror image of step 3.
5. Draw the semi-circle of infinite radius in the clockwise direction.
6. Draw the mirror image of step 1.
7. The Nyquist plot gets closed if the Type of the system is zero. If the Type of the system is one, draw a semi-circle of infinite radius in the clockwise direction, and for Type two system, two semi-circles and so on.

8. From $GH(s)$, find the number of open loop poles in RHP. P_r is known.
9. Draw a radial line from $(-1, 0)$ point and count N_r.
10. Using $Z_r = N_r + P_r$, find Z_r. If $Z_r = 0$, the closed loop system is stable. Otherwise, it is unstable.

Example 4.35

A certain feedback control system has the following loop T.F

$$GH(s) = \frac{(s+9)}{(s^2+9)(s+2)}$$

Using Nyquist stability criterion, determine the stability of the closed loop system.

Solution

1.

$$GH(s) = \frac{(s+9)}{(s^2+9)(s+2)}$$
$$(s^2+9) = (s+j3)(s-j3)$$

Two roots at $j\omega$ axis are at $s = j3$ and $s = -j3$. No pole of $GH(s)$ is in RHP and hence $P_r = 0$. For s-plane Nyquist contour, refer to Fig. 4.36a.

2. For $0 \leq \omega < 3$

$$GH(j\omega) = \frac{(j\omega+9)}{(9-\omega^2)(j\omega+2)}$$
$$|GH(j\omega)| = \frac{\sqrt{(\omega^2+81)}}{(9-\omega^2)\sqrt{\omega^2+4}}$$
$$\angle GH(j\omega) = \tan^{-1}\frac{\omega}{9} - \tan^{-1}\frac{\omega}{2}$$
$$\angle GH(j\omega)|_{\omega=3} = 6^\circ - 26^\circ = -20^\circ$$
$$\text{For } \omega = 0, \quad GH(j\omega) = 0.5\angle 0^\circ$$
$$\text{For } \omega = \infty, \quad GH(j\omega) = \infty\angle -20^\circ$$

The plot $O'A'$ is drawn.

3. For the point $j3$, a semi-circle of infinite radius is drawn as $A'B'$.
4. For $3 \leq \omega \leq \infty$

$$|GH(j\omega)| = \frac{\sqrt{\omega^2+81}}{(\omega^2-9)\sqrt{\omega^2+4)}}$$
$$\angle GH(j\omega) = 180^\circ + \tan^{-1}\frac{\omega}{9} - \tan^{-1}\frac{\omega}{2}$$
For $\omega \geq 3$, $|GH(j\omega)| = \infty$
$$\angle GH(j\omega) = 160^\circ$$
For $\omega = \infty$, $|GH(j\omega)| = 0$
$$\angle GH(j\omega) = 180^\circ$$

The plot $B'C'$ is drawn.

5. For $-\infty < \omega \leq -3$, the mirror image of step 4 is drawn as $E'F'$.
6. For the point $-j3$ the mirror image of step 3 is drawn. This is $F'G'$.
7. For $-0 \leq \omega \leq -\infty$, the mirror image of step 2 is drawn. This is $G'O'$. The complete Nyquist plot is shown in Fig. 4.37.
8. From Fig. 4.37, $N_r = 2$. Since $P_r = 0$, $Z_r = N_r + P_r = 2 + 0 = 2$. There are three RHP roots.

The system is unstable.

Example 4.36

A certain feedback control system has the following loop T.F.

$$GH(s) = \frac{(s+1)}{(s^2+9)(s+2)}$$

By using Nyquist stability criterion, test whether the closed loop system is stable.

Solution

1.

$$GH(s) = \frac{(s+1)}{(s^2+9)(s+2)}$$

There are two poles on the $j\omega$ axis at $s = j3$ and $s = -j3$. There are no poles in RHP. Hence, $P_r = 0$. For s-plane Nyquist contour, refer to Fig. 4.36a.

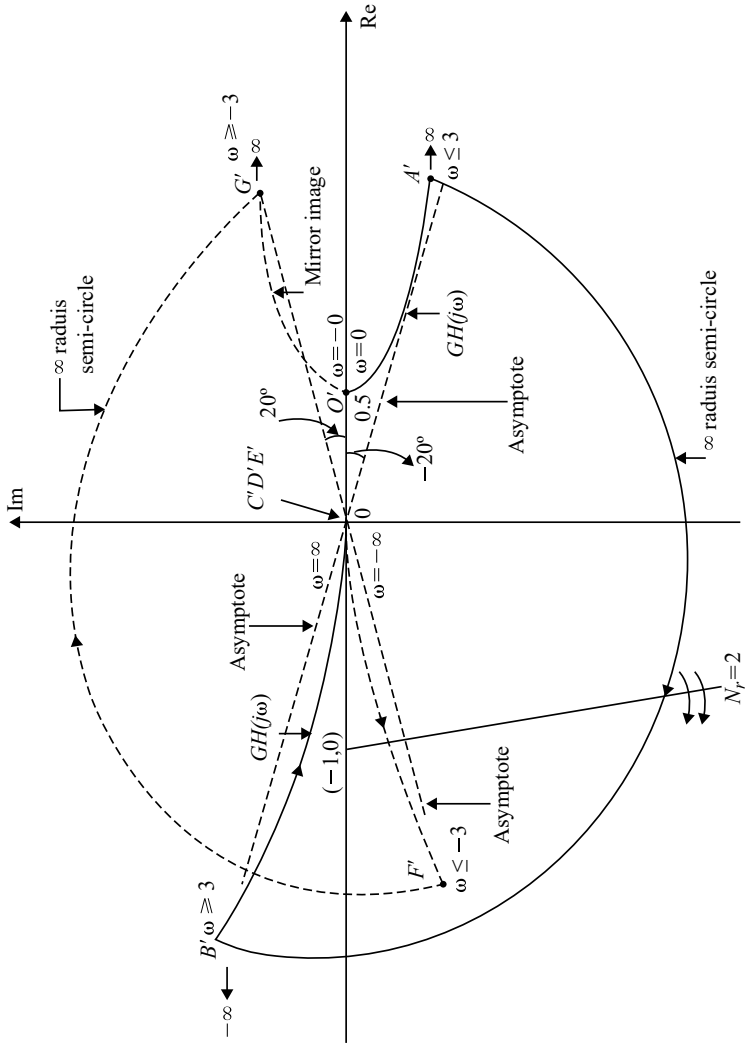


Fig. 4.37 Nyquist plot for Example 4.35

2.

$$GH(j\omega) = \frac{(j\omega + 1)}{(9 - \omega^2)(j\omega + 2)}$$

$$|GH(j\omega)| = \frac{\sqrt{\omega^2 + 1}}{(9 - \omega^2)\sqrt{\omega^2 - 4}}$$

$$\angle GH(j\omega) = \tan^{-1}\omega - \tan^{-1}\frac{\omega}{2}$$

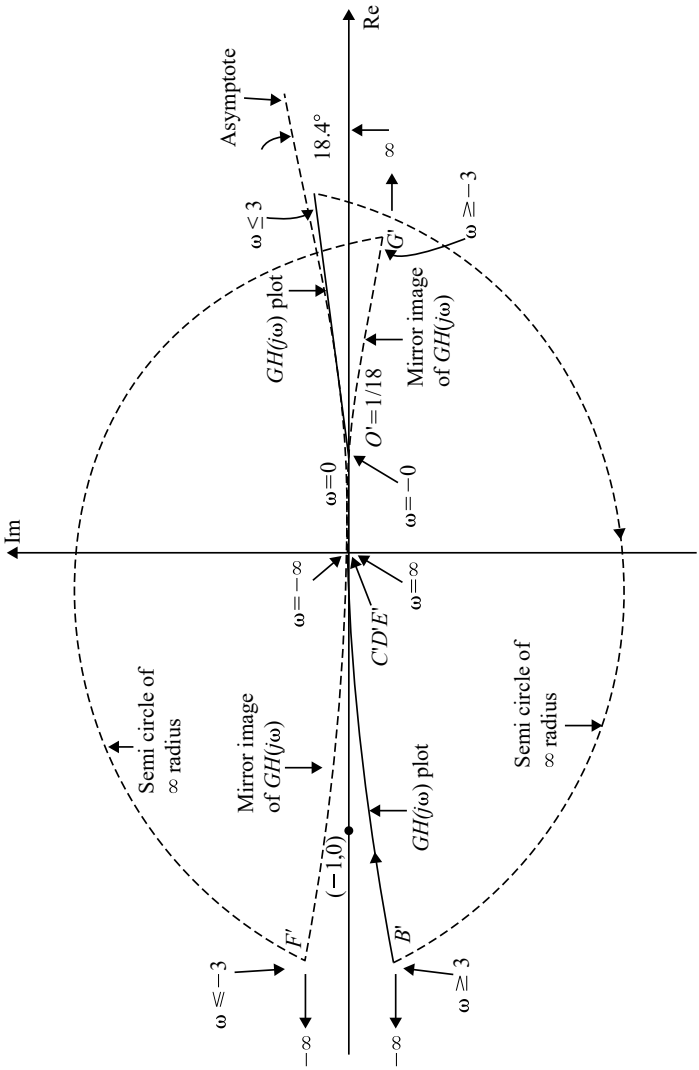


Fig. 4.38 Nyquist plot for Example 4.36

3. For the frequency range $0 \leq \omega \leq 3$

$$\text{For } \omega = 0, \quad GH(j\omega) = \frac{1}{18}\angle 0^\circ$$
$$\text{For } \omega = 3, \quad GH(j\omega) = \infty\angle 45^\circ - 26.6^\circ$$
$$= \infty\angle 18.4^\circ$$

The $GH(j\omega)$ plot is drawn as $O'A'$ as shown in Fig. 4.38.

4. For the frequency $\omega = +j3$, a semi-circle with infinite radius is drawn in the clockwise direction as $A'B'$.
5. For the frequency range $3 \leq \omega \leq \infty$

$$|GH(j\omega)| = \frac{\sqrt{\omega^2+1}}{(\omega^2-9)\sqrt{\omega^2+4}}$$

$$GH(j\omega) = 180^\circ + \tan^{-1}\omega - \tan^{-1}\frac{\omega}{2}$$

$$\text{For } \omega = 3, \quad |GH(j\omega)| = \infty\angle 198.4^\circ$$

$$\text{For } \omega = \infty, \quad GH(j\omega) = 0\angle 180^\circ$$

The $GH(j\omega)$ plot is drawn as $B'C'$ as shown in Fig. 4.38.

6. The ∞ semi-circle BCD in s-plane is mapped into the origin as $B'C'D'$ in the GH-plane.
7. For the frequency range $-\infty \leq \omega \leq -3$, the mirror image of step 5 is drawn and shown as $D'E'$ in the Fig. 4.38.
8. For the frequency $\omega = -3$, a semi-circle of ∞ radius is drawn in the clockwise direction and is shown as $E'F'$. This is also the mirror image of step 4.
9. For the frequency range $-3 \leq \omega \leq -0$, the mirror image of step 3 is drawn. The Nyquist plot in the GH-plane is closed and complete.
10. From $(-1, 0)$ point, a radial line is drawn and N_r is counted. From Fig. 4.38, the Nyquist plot does not encircle $(-1, 0)$ point and hence $N_r = 0$.
11.

$$\begin{aligned} Z_r &= N_r + P_r \\ &= 0 + 0 = 0 \end{aligned}$$

There is no closed loop root in RHP and the system is stable.

Closed loop system is stable.

Example 4.37

Consider the following open loop T.F. of a certain feedback control system

$$GH(s) = \frac{Ks(s+2)}{(s^3+64)}$$

Determine the value of K so that the closed loop system is stable. Use Nyquist stability criterion.

Solution

1.

$$GH(s) = \frac{Ks(s+2)}{(s^3+64)}$$

The characteristic equation of the open loop system is

$$\begin{aligned} s^3 + 64 &= 0 \\ s^3 &= -64 \\ s &= 4\angle(2n+1)\pi/3 \end{aligned}$$

where $n = 0, 1, 2,$

$$\begin{aligned} s_1 &= 4\angle 60^\circ = 4\cos 60^\circ + j4\sin 60^\circ \\ &= 2 + j3.464 \\ s_2 &= 4\angle 180^\circ = -4 \\ s_3 &= 4\angle 300^\circ = 4\cos 300^\circ + j4\sin 300^\circ \\ &= 2 - j3.464 \end{aligned}$$

The poles s_1 and s_3 are in RHP. Hence, $P_r = 2$.

2.

$$\begin{aligned} GH(s) &= \frac{Ks(s+2)}{s^3+64} \\ GH(j\omega) &= \frac{Kj\omega(j\omega+2)}{-j\omega^3+64} \\ |GH(j\omega)| &= \frac{K\omega\sqrt{(\omega^2+4)}}{\sqrt{\omega^6+(64)^2}} \\ \angle GH(j\omega) &= 90^\circ + \tan^{-1}\frac{\omega}{2} + \tan^{-1}\left(\frac{\omega}{4}\right)^3 \end{aligned}$$

3. Frequency range $0 \le \omega \le \infty$

From phase angle expression, it is evident that it varies from 90° to 270°. Therefore, the $GH(j\omega)$ plot lies in the second and 3rd quadrant in the GH-plane. For $\omega = 0$

$$GH(j\omega) = 0\angle 90^\circ$$

For $\omega = \infty$

$$GH(j\omega) = 0\angle 270^\circ$$

The $GH(j\omega)$ plot starts from the origin for $\omega = 0$, traverses through the second quadrant, cuts the negative real axis, passes through the third quadrant and terminates at the origin when $\omega = \infty$.

4. The phase crossover frequency ω_{pc} and the point of intersection of the $GH(j\omega)$ plot on the $-$ve real axis are calculated as follows. For $\omega = \omega_{pc}$

$$90^\circ + \tan^{-1}\frac{\omega}{2} + \tan^{-1}\left(\frac{\omega}{4}\right)^3 = 180^\circ$$

$$\tan^{-1}\left(\frac{\omega}{4}\right)^3 = 90^\circ - \tan^{-1}\frac{\omega}{2}$$

Taking tan on both sides, we get

$$\left(\frac{\omega}{4}\right)^3 = \cot\tan^{-1}\frac{\omega}{2}$$

$$= \frac{1}{\tan\tan^{-1}\frac{\omega}{2}} = \frac{2}{\omega}$$

$$\omega^4 = 128$$

$$\omega = 3.36$$

$$\omega_{pc} = 3.36$$

At ω_{pc} the magnitude is

$$|GH(j\omega_{pc})| = \frac{K3.36\sqrt{(3.36)^2+4}}{\sqrt{(3.36)^6+(64)^2}} = 0.1765K$$

5. The mirror image of $GH(j\omega)$ is drawn for the frequency range $-\infty \leq \omega \leq -0$. This is shown in dotted line in Fig. 4.39.
6. From $GH(s)$, P_r is obtained as 2. Hence, the Nyquist plot should have $N_r = -2$. From Fig. 4.39, this is possible if

$$-0.1765K > -1$$

$$K > 5.66$$

7.

$K > 5.66$ the system is stable.
$K = 5.66$ the system is oscillatory.
$K < 5.66$ the system is unstable.

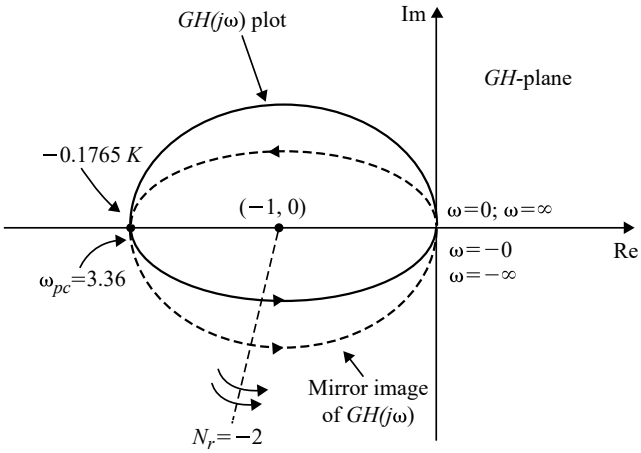


Fig. 4.39 Nyquist plot for Example 4.37

Example 4.38

The loop T.F. of a certain control system is given by

$$G(s) = \frac{K(s+5)(s+2)}{(s^3 - 20)}$$

Using Nyquist stability criterion, determine the value of K so that the closed loop system is stable.

Solution

1.

$$GH(s) = \frac{K(s+5)(s+2)}{(s^3 - 20)}$$

The characteristic equation of the open loop system is

$$s^3 - 20 = 0$$
$$s = \sqrt[3]{20} \quad \angle 2n\pi/3$$

where $n = 0, 1, 2$

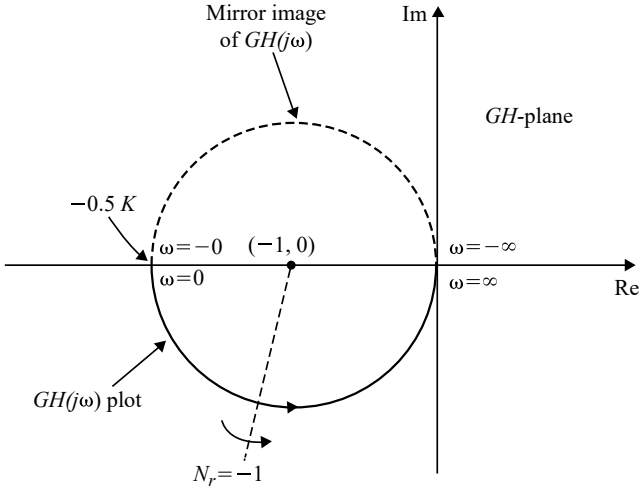


Fig. 4.40 Nyquist plot for Example 4.38

$$s_1 = 2.7144\angle 0^\circ$$
$$s_2 = 2.7144\angle 120^\circ$$
$$s_3 = 2.7144\angle 240^\circ$$

One pole $s_1 = 2.7144\angle 0^\circ$ is in RHP. Hence, $P_r = 1$.

2.

$$GH(j\omega) = \frac{K(j\omega+5)(j\omega+2)}{(-j\omega^3-20)}$$
$$|GH(j\omega)| = \frac{K\sqrt{(\omega^2+25)(\omega^2+4)}}{\sqrt{\omega^6+400}}$$
$$\angle GH(j\omega) = 180^\circ + \tan^{-1}\frac{\omega}{5} + \tan^{-1}\frac{\omega}{2} - \tan^{-1}\frac{\omega^3}{20}$$

3. For $\omega = 0$

$$GH(j\omega) = -0.5K$$

For $\omega = \infty$

$$GH(j\omega) = 0\angle 270^\circ$$

For small values of ω, $\tan^{-1}\frac{\omega}{5}+\tan^{-1}\frac{\omega}{2}$ will contribute more phase than $\tan^{-1}\frac{\omega^3}{20}$ and the phase angle is in the 3rd quadrant of GH-plane. For high value of ω, each of them will contribute 90° which results in total phase 270°.

4. The $GH(j\omega)$ plot starts from −ve real at $0.5K$ for $\omega=0$, traverses through the 3rd quadrant as ω increases and terminates at the origin for $\omega=\infty$ as shown in Fig. 4.40 in thickline.
5. The mirror image of $GH(j\omega)$ for $-\infty\le\omega\le-0$ is drawn in dotted line. The Nyquist plot is closed.
6. Since $P_r=1$, N_r should be -1 for Z_r to be zero. This is possible if $-0.5K$ is to the left of $(-1,0)$ point. For the system to be stable $0.5K>1$ or $K>2$.
7.

$K>2$, the system is stable.
$K=2$, the system is oscillatory.
$K<2$, the system is unstable.

Application of Nyquist stability criterion for system with improper and strictly proper T.F.

Example 4.39

The loop T.F. of a certain feedback control system is

$$GH(s)=\frac{e^{-0.1s}}{s(s+1)(s+2)}$$

Determine whether the closed loop system is stable. Use Nyquist stability criterion. Is the system stable without time delay?

Solution

1.

$$GH(s)=\frac{5e^{-0.1s}}{s(s+1)(s+2)}$$

The system open loop pole in RHP is zero. Thus, $P_r=0$.

2.

$$GH(j\omega)=\frac{5e^{-0.1j\omega}}{j\omega(j\omega+1)(j\omega+2)}$$

$$|GH(j\omega)|=\frac{5}{\omega\sqrt{(\omega^2+1)(\omega^2+4)}}$$

$$\angle GH(j\omega)=-5.73\omega-90^\circ-\tan^{-1}\omega-\tan^{-1}\frac{\omega}{2}$$

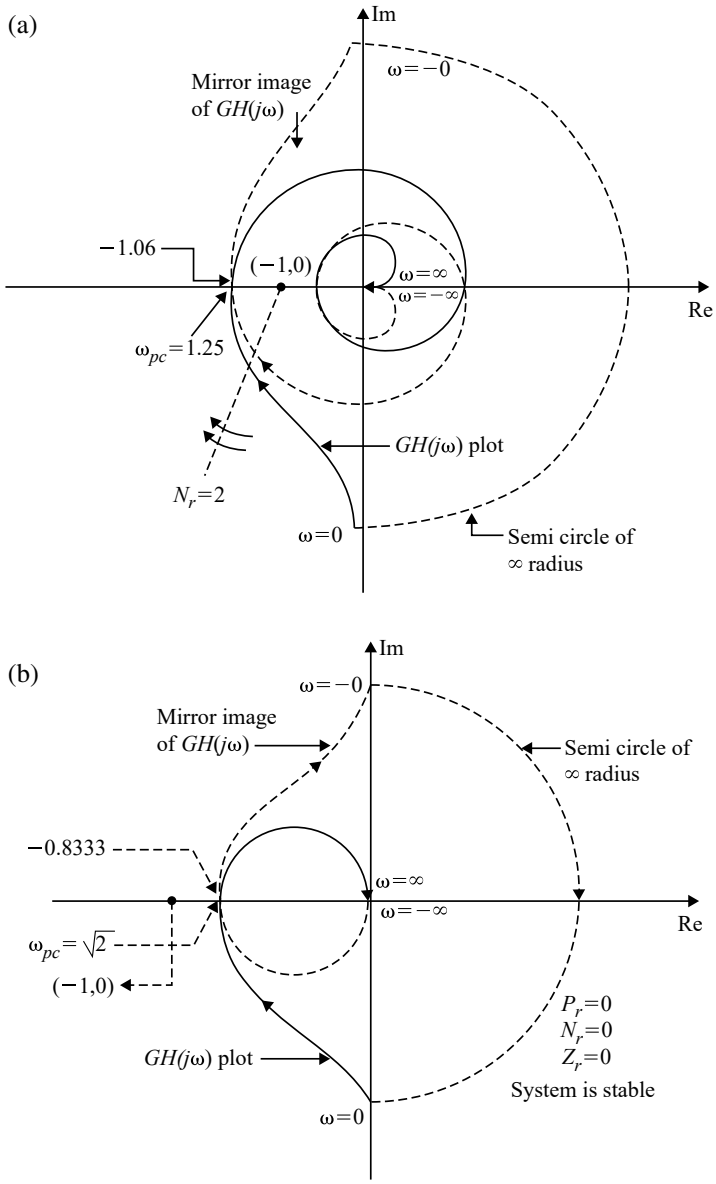


Fig. 4.41 **a** Nyquist plot of system with transportation lag (Example 4.39). **b** Nyquist plot for $GH(s) = \frac{5}{s(s+1)(s+2)}$

3. The transportation lag does not make any change in the magnitude of the system without delay. However, at each frequency, it adds a phase of -5.73ω. Hence, the $GH(j\omega)$ plot of the original system is only phase shifted.
4. Frequency range $0 \leq \omega \leq \infty$. For $\omega = 0$

$$GH(j\omega) = \infty\angle -90^\circ$$

For $\omega = \infty$

$$GH(j\omega) = 0\angle 0^\circ$$

The $GH(j\omega)$ plot starts from $-$ve $j\omega$ axis for $\omega = 0$, traverses through the 3rd quadrant, cuts the $-$ve real axis for $\omega = \omega_{pc}$, traverses through the 2nd and first quadrants as ω increases and terminates at the origin of the GH-plane when $\omega = \infty$. This is shown in thick line in Fig. 4.41a.
5. Frequency range $-\infty \leq \omega \leq -0$

For the above frequency range, the Nyquist plot is the mirror image of $GH(j\omega)$ of step 4.
6. Since the type of the system is one, a semi-circle of infinite radius in the clockwise direction is drawn and the Nyquist plot in the GH-plane gets closed.
7. The phase crossover frequency ω_{pc} and the point of intersection of $GH(j\omega)$ plot with the $-$ve real axis are important to calculate the stability of the closed loop system. They are calculated as follows. For $\omega = \omega_{pc}$, the phase angle is 180°. Thus

$$-5.73\omega - \tan^{-1}\omega - \tan^{-1}\frac{\omega}{2} - 90^\circ = 180^\circ$$
$$\tan^{-1}\omega + \tan^{-1}\frac{\omega}{2} + 5.73\omega = 90^\circ$$

By trial and error, ω is calculated to satisfy the above equation. ω is obtained as 1.25 rad/sec

$$\omega_{pc=1.25}$$

At ω_{pc}, the magnitude of $GH(j\omega)$ is

$$|GH(j\omega_{pc})| = \frac{5}{1.25\sqrt{[(1.25)^2+1][(1.25)^2+4]}}$$
$$= 1.06$$

8. From Fig. 4.41, N_r is calculated as 2

$$Z_r = N_r + P_r$$
$$= 2 + 0$$

There are 2 RHP roots of the characteristic equation of the closed loop system.

The system is unstable.

System Without Time Delay
The T.F. of the system without time delay is

$$GH(s) = \frac{5}{s(s+1)(s+2)}$$
$$GH(j\omega) = \frac{5}{j\omega(j\omega+1)(j\omega+2)}$$
$$|GH(j\omega)| = \frac{5}{\omega\sqrt{(\omega^2+1)(\omega^2+4)}}$$
$$\angle GH(j\omega) = -90^\circ - \tan^{-1}\omega - \tan^{-1} 0.5\omega$$

At phase crossover frequency, $\angle GH(j\omega_{pc}) = 180^\circ$

$$-90 - \tan^{-1}\omega - \tan^{-1} 0.5\omega = 180^\circ$$
$$\tan^{-1}\omega + \tan^{-1} 0.5\omega = 90^\circ$$
$$\tan^{-1}\frac{1.5\omega}{1-0.5\omega^2} = 90^\circ$$

Taking tan on both sides, we get

$$\frac{1.5\omega}{1-0.5\omega^2} = \tan 90^\circ = \infty$$
$$1 - 0.5\omega^2 = 0$$
$$\omega = \omega_{pc} = \sqrt{2}$$

Substituting this value in $|GH(j\omega)|$, we get

$$|GH(j\omega_{pc}) = \frac{5}{\sqrt{2}\times 3\times 6} = 0.833$$

The Nyquist plot is shown in Fig. 4.41b.
From Fig. 4.41b, it is observed that the $GH(j\omega)$ plot does not encircle $(-1, 0)$ point and hence $N_r = 0$. Since $P_r = 0$,

$$Z_r = N_r + P_r$$
$$= 0 + 0 = 0$$

The system without delay is stable.

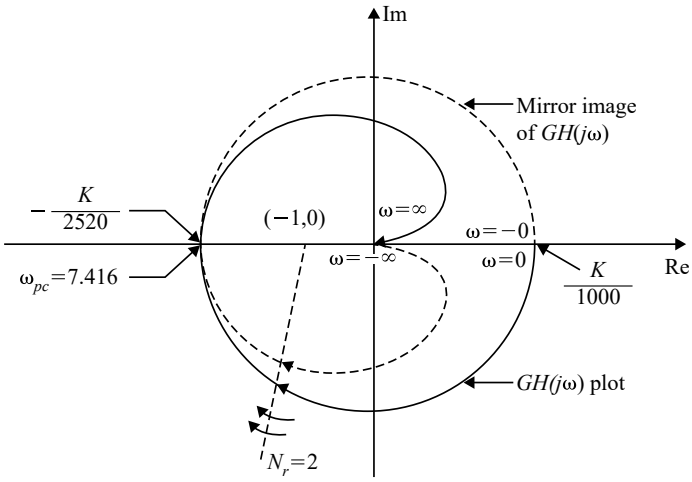


Fig. 4.42 Nyquist plot for Example 4.40

Example 4.40

A certain feedback control system has the following loop T.F

$$GH(s) = \frac{K}{(s^2 + 3s + 10)(s^2 + 3s + 100)}$$

Determine the value of K so that the closed loop system is stable. Use Nyquist stability criterion.

Solution

1.

$$GH(s) = \frac{K}{(s^2 + 3s + 10)(s^2 + 3s + 100)}$$

Since the roots of quadratic equation with +ve coefficients have negative real part, all the four poles of $GH(s)$ are in LHP. Hence, $P_r = 0$.

2.

$$GH(j\omega) = \frac{K}{(-\omega^2 + j3\omega + 10)(-\omega^2 + j3\omega + 100)}$$

$$|GH(j\omega)| = \frac{K}{\sqrt{[(10 - \omega^2)^2 + 9\omega^2][(100 - \omega^2)^2 + 9\omega^2]}}$$

$$\angle GH(j\omega) = -\tan^{-1}\frac{3\omega}{(10 - \omega^2)} - \tan^{-1}\frac{3\omega}{(100 - \omega^2)}$$

3. Frequency range $0 \leq \omega \leq \infty$.

   Since $GH(s)$ is of Type 0 order 4 system, the $GH(j\omega)$ plot starts from +ve real axis at $K/1000$ when $\omega = 0$, traverses through the 4th, 3rd, 2nd and 1st quadrants of GH-plane as ω increases, and terminates at the origin when $\omega = \infty$. This is shown in Fig. 4.42.
4. Frequency range $-\infty \leq \omega \leq -0$.

   The mirror image of $GH(j\omega)$ is drawn in dotted lines. This completes the Nyquist plot.
5. The phase crossover frequency ω_{pc} and the point of intersection of $GH(j\omega)$ on the negative real axis are determined as follows. At the phase crossover frequency ω_{pc}

$$\begin{aligned} \angle GH(j\omega_{pc}) &= 180^\circ \\ -\tan^{-1}\frac{3\omega}{(10-\omega^2)} - \tan^{-1}\frac{3\omega}{(100-\omega^2)} &= 180^\circ \\ \tan^{-1}\frac{3\omega}{(10-\omega^2)} &= 180 - \tan^{-1}\frac{3\omega}{(100-\omega^2)} \end{aligned}$$

   Taking tan on both sides, we get

$$\begin{aligned} \frac{3\omega}{(10-\omega^2)} &= \frac{-3\omega}{(100-\omega^2)} \\ 100-\omega^2 &= \omega^2 - 10 \\ \omega^2 &= 55 \\ \omega &= \omega_{pc} \\ &= 7.416 \\ |GH(j\omega_{pc})| &= \frac{K}{\sqrt{(45^2+9\times 55)(45^2+9\times 55)}} \\ &= \frac{K}{2520} \end{aligned}$$

6. For the system to be stable, N_r should be zero since $P_r = 0$. For this, the following condition is to be satisfied.

$$\frac{K}{2520} < 1$$

7.

$K < 2520$, the system is stable.
$K = 2520$, the system is oscillatory.
$K > 2520$, the system is unstable.

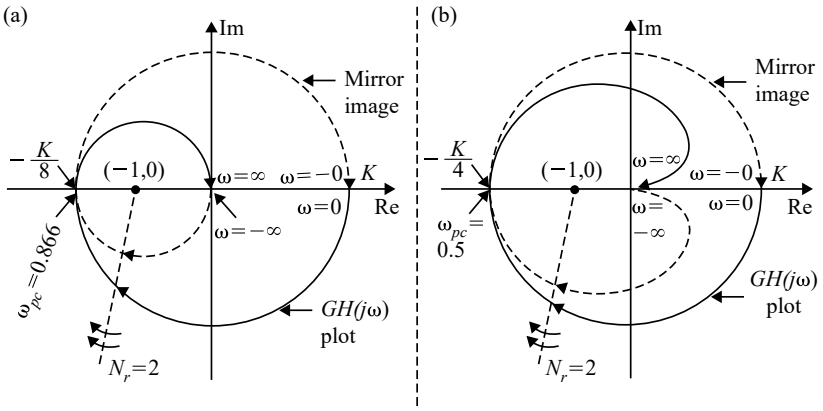


Fig. 4.43 Nyquist plots for Example 4.41. **a** $GH(s) = \frac{K}{(1+2s)^3}$; and **b** $GH(s) = \frac{K}{(1+2s)^4}$

Example 4.41

A certain feedback control system has the following loop T.F. Determine the value of K in each case for the closed loop system to be stable. Use Nyquist stability criterion

$$\text{I.} \quad GH(s) = \frac{K}{(1+2s)^3}$$

$$\text{II.} \quad GH(s) = \frac{K}{(1+2s)^4}$$

Solution

I. 1.

$$GH(s) = \frac{K}{(1+2s)^3}$$

All the 3 poles of $GH(s)$ are at $s = -\frac{1}{2}$ in LHP. No RHP pole of $GH(s)$. Hence, $P_r = 0$.

2.

$$GH(j\omega) = \frac{K}{(1+j2\omega)^3}$$

$$|GH(j\omega)| = \frac{K}{(1+4\omega^2)^{3/2}}$$

$$\angle GH(j\omega) = -3\tan^{-1} 2\omega$$

3. For the frequency range $0 \leq \omega \leq \infty$. For $\omega = 0$

$$GH(j\omega) = K\angle 0^\circ$$

For $\omega = \infty$

$$GH(j\omega) = 0\angle -270^\circ$$

The given $GH(s)$ is a Type 0 and order 3 system with minimum phase. The $GH(j\omega)$ plot starts from +ve real at K for $\omega = 0$, traverses through the 4th and 3rd quadrants, cuts the −ve real axis when $\omega = \omega_{pc}$, passes through the 2nd quadrant and terminates at the origin of GH-plane when $\omega = \infty$. This is shown in thick line in Fig. 4.43a.

4. For the frequency range $-\infty \leq \omega \leq -0$, the Nyquist plot is the mirror image of $GH(j\omega)$ of step 3 and is shown in dotted line in Fig. 4.43a. The Nyquist plot gets closed now.
5. The phase crossover frequency ω_{pc} and the magnitude of $GH(j\omega)$ at this frequency is important to calculate the stability of the closed loop system. This is calculated as explained below.
6. For $\omega = \omega_{pc}$

$$\begin{aligned} \angle GH(j\omega_{pc}) &= -180^\circ \\ -3\tan^{-1} 2\omega &= -180^\circ \\ \tan^{-1} 2\omega &= 60^\circ \end{aligned}$$

Taking tan on both sides, we get

$$\begin{aligned} 2\omega &= \tan 60^\circ = 1.732 \\ \omega = \omega_{pc} &= 0.866 \end{aligned}$$

Substituting this value, the magnitude of $GH(j\omega)$ is calculated

$$|GH(j\omega_{pc})| = \frac{K}{[1 + 4(.866)^2]^{3/2}} = \frac{K}{8}$$

7. If $K/8$ is to the left of $(-1, 0)$ point, $N_r = 2$. Since $P_r = 0$,

$$\begin{aligned} Z_r &= N_r + P_r \\ &= 2 + 0 = 2 \end{aligned}$$

The system will be unstable. For the system to be stable, $K < 8$.

8.

$K < 8$, the system is stable.
$K = 8$, the system is oscillatory.
$K > 8$, the system is unstable.

II. 1.

$$GH(s) = \frac{K}{(1+2s)^4}$$

All the 4 poles of $GH(s)$ are at $s = -1/2$ in LHP. No poles of $GH(s)$ is in RHP. Hence, $P_r = 0$.

2.

$$GH(j\omega) = \frac{K}{(1+j\omega^2)^4}$$
$$|GH(j\omega)| = \frac{K}{(1+4\omega^2)^2}$$
$$\angle GH(j\omega) = -4\tan^{-1} 2\omega$$

3. For the frequency range $0 \leq \omega \leq \infty$. For $\omega = 0$

$$GH(j\omega) = K\angle 0^\circ$$

For $\omega = \infty$

$$GH(j\omega) = 0\angle 360^\circ$$

The given $GH(s)$ is a Type 0 order 4 system T.F. Hence, the $GH(j\omega)$ plot starts from +ve real at K for $\omega = 0$, traverses through the 4th and 3rd quadrants, cuts the negative real axis for $\omega = \omega_{pc}$, traverses through the 2nd and 1st quadrants and terminates at the origin of GH-plane when $\omega = \infty$. This is shown in thick line in Fig. 4.43b.

4. For the frequency range $-\infty \leq \omega \leq -0$
This is the mirror image of $GH(j\omega)$ of step 3. This is drawn is dotted line in Fig. 4.43b. The Nyquist plot now gets closed. The calculation of ω_{pc} and the corresponding magnitude of $GH(j\omega_{pc})$ is important for the stability analysis. They are calculated as follows.

5. At $\omega = \omega_{pc}$

$$\begin{aligned}\angle GH(j\omega_{pc}) &= -180^\circ \\ -4\tan^{-1} 2\omega &= -180^\circ \\ \tan^{-1} 2\omega &= 45^\circ \\ \omega = \omega_{pc} &= 0.5\end{aligned}$$

The magnitude at this frequency is calculated as

$$\begin{aligned}|GH(j\omega_{pc})| &= \frac{K}{(1+4\times 0.25)^2} \\ &= \frac{K}{4}\end{aligned}$$

6. If $\frac{-K}{4}$ is to the left of $(-1, 0)$ point, $N_r = 2$ as seen in Fig. 4.43b and

$$\begin{aligned}Z_r &= N_r + P_r \\ &= 2 + 0 = 2\end{aligned}$$

The closed loop system will be unstable. Hence, $K < 4$ for the system to be stable.

7.

$K < 4$, the system is stable.
$K = 4$, the system is oscillatory.
$K > 4$, the system is unstable.

SUMMARY

1. Stability concept for linear time invariant continuous time system is explained. Zero input response, zero state response, zero input stability, asymptotic (internal) stability, marginal (neutral) stability, bounded input bounded output stability and relative stability are defined.
2. BIBO stability *via* impulse response and the characteristic roots location in the s-plane are explained.
3. If the linear is asymptotically stable, it is also BIBO stable. But all BIBO stable systems need not be asymptotically stable.
4. For BIBO and asymptotic stability, no root of the characteristic equation should be in RHP.
5. Presence of distinct characteristic roots in the $j\omega$ axis implies that the system is marginally stable, but BIBO and asymptotically unstable. If the roots are repeated in the $j\omega$ axis, the system is unstable.

6. Stability analysis is carried out in the time as well as frequency domains. In the time domain, Routh–Hurwitz and root locus methods are extensively used, while Nyquist stability criterion is the most popular method in the frequency domain.
7. Using Routh's stability criterion, absolute and relative stability can be obtained. Further, the method can be used to find the gain that can be increased before the system becomes unstable. Furthermore, in few cases, the characteristic polynomial can be factorized and the root locations in the s-plane can be determined.
8. In the frequency response method, Nyquist stability criterion is applied. The method applies Cauchy's mapping theorem and the principle of the arguments to establish the stability criterion.
9. Nyquist stability analysis has many advantages over Routh–Hurwitz method. By using Routh–Hurwitz method, we find the answer as yes or no regarding the stability of the system. If the system is unstable, it does not suggest how to improve the system stability. However, in the Nyquist stability method, in addition to the information we get from the Routh–Hurwitz method, it suggests methods to improve the system stability if it is unstable.
10. Nyquist stability method can be applied to determine frequency response specifications of closed loop system in addition to stability of the system.
11. Nyquist stability criterion can be applied to system whose open loop T.F. is of minimum phase, non-minimum phase, improper and strictly proper type. It can also be applied to system with transportation lag.

EXERCISE

Short Answer Type Questions

1. **What do you understand by stability of a linear time invariant system? What is instability?**
   When a linear time invariant system is subjected to disturbances the total response of the system is the sum of the forced and zero input (natural) responses. A linear time invariant system is said to be stable if the zero input response approaches zero as time approaches infinity. The system is said to be unstable if the zero input response grows without bound as time approaches infinity.
2. **Does the stability of a linear system depend upon the input?**
   The stability of a linear system is independent of the input. It depends on only the locations of the characteristic roots of the system.
3. **What happens when the system becomes unstable?**
   If the system is unstable, the output grows without bound and causes severe damages not only to the system itself, but also to the environment and human life.

4. **What are the different methods available for the stability study of linear system?**
   Time domain and frequency domain approaches are available for the stability study of linear system. Routh–Hurwitz method, root locus technique and Liapunov methods are the widely used time domain methods while the method proposed by Nyquist is extensively used in the frequency domain.
5. **Define asymptotic stability as applied to the linear system?**
   For a linear system, zero state is the equilibrium state. When the system is given a small disturbance, the zero initial state is changed to non-zero initial state. If the system left to itself returns back to zero state as time tends to infinity, the system is said to be asymptotically stable.
6. **What are the different names given to asymptotic stability?**
   The different names given to asymptotic stability are zero input stability, internal stability and simply stability. It is also called as stability in the sense of Liapunov.
7. **Define marginal stability of a system.**
   A linear system when disturbed, gets its equilibrium state shifted. Some characteristic modes of the system neither decay to zero nor grow indefinitely, but remains constant. Such systems are said to be marginally stable with neutral equilibrium. Marginal stability is also called as neutral stability.
8. **What is bounded input and bounded output (BIBO) stability?**
   A linear time invariant system is said to be BIBO (externally) stable if every bounded input yields a bounded output.
9. **What is the condition that guarantees BIBO stability?**
   For a linear system to be BIBO stable, that all the transfer function poles should be in LHP.
10. **How impulse response of a system is correlated with BIBO stability?**
    A linear time invariant system is BIBO stable if the area under the impulse response curve is finite.
11. **If the system is marginally stable, will it be BIBO stable?**
    A marginally stable system is not BIBO stable since the area under the impulse response curve of a marginally stable system is not finite.
12. **What is the common requirement for the linear system to be asymptotically, marginally and BIBO stable?**
    For the linear system, if the T.F. has all the poles in LHP, the system is said to be asymptotically, marginally and BIBO stable.
13. **What is the relationship between asymptotic stability and BIBO stability?**
    If the linear system is asymptotically stable, then it is also BIBO stable. However, if the linear system is BIBO stable, it does not guarantee asymptotic stability.
14. **Under what condition, BIBO stability and asymptotic stability are equivalent?**
    In the overall T.F., if there is no pole–zero cancellation in RHP, then BIBO stability is equivalent to asymptotic stability. The above condition implies that the given system should be controllable and observable.

15. The T.F. of a linear system has repeated roots in the $j\omega$ axis. Is the system marginally stable or unstable?

If the linear system has repeated roots in the $j\omega$ axis, the output grows indefinitely with respect to time. Hence, the system is unstable.

16. The T.F. of a linear system has a pole at the origin of s-plane. How the stability is termed?

If the pole at the origin is intentionally introduced, the system is said to be stable. Otherwise, it is termed as marginally stable.

17. What is relative stability of a linear system?

If the linear system is stable, it is said to be absolutely stable. How far it is closer to the stability limit is called relative stability.

18. What are the measures of the relative stability?

In the frequency domain, the relative stability is measured in terms of phase margin and gain margin. Higher the phase margin and gain margin, higher the relative stability. In the time domain, it is measured by the root location in LHP. If the nearest root in LHP is placed far away from the $j\omega$ axis, the relative stability is high.

19. What is a conditionally stable system?

A linear system may be stable for a range of values of a system parameter. For other values, the same system will be unstable. Such systems are called conditionally stable systems.

20. Mention any three applications of Routh's–Hurwitz stability criterion?

Routh's–Hurwitz criterion can be applied to determine the absolute stability, relative stability and the system gain that can be increased before the system becomes unstable. It can be used to find out the number of roots of the characteristic polynomial in RHP, LHP and the $j\omega$ axis. In few cases, the exact locations of the characteristic roots in s-plane can be determined.

21. Mention any two disadvantages of Routh–Hurwitz method.

Routh–Hurwitz criterion is applicable only to characteristic polynomial which are real. If the coefficients are complex, the method is not applicable. The $j\omega$ axis is considered as the stability boundary for Routh–Hurwitz criterion. For other boundaries, it is not applicable. When repeated roots are present in the $j\omega$ axis, it cannot be found using Routh–Hurwitz criterion and the erroneous conclusion that the system is marginally stable is derived, whereas the system is actually unstable.

22. What are the necessary conditions to apply Routh–Hurwitz stability condition?

To apply Routh–Hurwitz stability criterion, it is necessary that all the coefficients of the characteristic polynomial are positive real and no coefficient should be zero.

23. State Routh–Hurwitz criterion.
The Routh–Hurwitz criterion states that the number of roots of the characteristic equation with +ve real parts is equal to the number of changes in sign of first column of the Routh array.

24. For simplifying the numerical calculation, in the Routh array what method is followed?
For simplifying the numerical calculations, the entire row may be multiplied or divided by a positive number by which the stability condition is unaffected.

25. What are the difficulties encountered when Routh's array is prepared?
When Routh's array is prepared, the coefficients of complete array becomes zero and the Routh's array is broken. Routh's array is also broken when the first column alone becomes zero.

26. When the complete row becomes zero, how the Routh's array is completed?
When the complete row has zero coefficients in all the columns of that row, the equation of the previous row is considered and is taken as the auxiliary equation. The auxiliary equation is differentiated with respect to s, and the coefficients of this equations are taken for the row which has zeros in all the columns.

27. What are the properties of the auxiliary equation of even polynomial?
The even polynomial of auxiliary equation has the following properties:

1. The roots of the even polynomial are symmetric about the origin and real.
2. They could be on the $j\omega$ axis as simple roots or repeated roots.
3. The roots are quadrantal (complex roots symmetrically placed with respect to the $j\omega$ axis).

28. When the complete row becomes zero in the Routh's array, does it imply the presence of roots in the $j\omega$ axis?
When the coefficients of a complete row becomes zero, the even polynomial may have any one property mentioned in Question 27. If the even polynomial possesses property one or property three, only symmetrically placed real and complex roots exist both in RHP as well as in LHP. In these cases, there won't be any root in the $j\omega$ axis.

29. If the characteristic polynomial has $j\omega$ axis roots, when the Routh's array is formed will it have complete row with zero coefficients?
If the characteristic polynomial has $j\omega$ axis roots, Routh's array will have a complete row with zero coefficients.

30. How stability is interpreted when complete row becomes zero?
The Routh's array is completed as explained in Question 26. The array is divided in to two parts: one below the auxiliary equation and the other above the auxiliary equation. For the rows of even (auxiliary) polynomial, the number of sign changes is determined which gives RHP roots. In this polynomial, equal number of LHP roots exist. The remaining roots are in the $j\omega$ axis. For the other polynomial, which is above the even polynomial, the number of sign changes are determined and the RHP roots found. The remaining roots of this polynomial are in LHP. There won't be any root in the $j\omega$ axis for the other polynomial.

Thus, for the whole characteristic polynomial, LHP, RHP and $j\omega$ axis roots are determined. For the system to be stable, there should not any RHP root.

31. For a system with complete row becoming zero, there is no sign change in the first column of Routh array. Does it mean the system is stable?

No. When the rows of even polynomial are interpreted, it is to be ensured that there is no root being repeated more than once in $j\omega$ axis. If there are repeated roots in the $j\omega$ axis, the system is unstable.

32. How to identify the presence of repeated roots in the $j\omega$ axis?

The auxiliary equation is a factor of the characteristic polynomial. By long division of the characteristic polynomial with the auxiliary polynomial, the exact root locations can be found out not only in the $j\omega$ axis, but also in RHP and LHP. From this, we can interpret whether there exits repeated roots in the $j\omega$ axis.

33. If the first column of Routh array becomes zero, what are the methods available to complete the Routh's array?

When the first column becomes zero, the Routh's array is broken. The Routh's array is completed using any one of the following methods:

1. By forming the reciprocal of the given polynomial.
2. A small value ε (epsilon) is assigned to replace the zero.
3. The given polynomial is multiplied by a factor $(s + a)$ where $a > 0$ and the Routh's array is formed for the new polynomial.

34. If the Routh's array does not have the coefficients of complete row becoming zero, how the number of roots in RHP, LHP and $j\omega$ axis are determined?

For the given characteristic polynomial the Routh's array is formed. The number of sign changes in the first column of Routh's array gives the number of roots in RHP. The characteristic polynomial $F(-s)$ is formed from $F(s)$ and the Routh's array is formed. The number of sign changes in the first column, now gives the number of roots of $F(s)$ in LHP. By subtracting LHP and RHP root number from the order of the polynomial of $F(s)$, the number of roots in the $j\omega$ axis is determined.

35. How to find relative stability from Routh's array?

Suppose the required relative stability unit is $-\sigma$. In the characteristic polynomial $F(s)$, s is replaced as $(s - \sigma)$ and the new characteristic polynomial is formed and the Routh's array is prepared. If there is no sign change in the first column of Routh's array, the system possesses the given relative stability limit.

36. Using Routh–Hurwitz criterion, the critical gain of the system K is found before the system becomes unstable. If K is found to be complex, how the stability is decided?

If the gain K of the system is found to be complex using Routh's stability criterion, the system is unstable.

37. What is the relationship between the poles of open loop T.F. $GH(s)$ and the poles of the characteristic equation $1 + GH(s)$?

The poles of open loop T.F. and the poles of the characteristic equation are one and the same.

38. How the zeros of the characteristic equation $1 + GH(s)$ are related to the poles of closed loop T.F.?
The zeros of the characteristic equation $1 + GH(s)$ are the same as the poles of closed loop T.F.

39. What is the necessary and sufficient condition for the closed loop system to be stable?
The necessary and sufficient condition for the closed loop system to be stable, the poles of the closed loop system (zeros of the characteristic equation) should be placed in LHP of the s-plane.

40. What do you understand by s-plane contour?
The connection of points in the complex s-plane is called s-plane contour.

41. What do you understand by a point being enclosed and being encircled?
A point in the complex plane is considered to be the point enclosed by the contour if it lies to the right of the traversal. A point is said to be encircled by a closed path if it is found inside the path (to the right) during the traversal.

42. There are P poles and Z zeros inside a contour in the s-plane. For clockwise traversal of the s-plane contour, how many encirclements around the origin are made when mapped into $F(s)$ plane in the following cases?

1. $P = 5; Z = 3$
2. $P = 3; Z = 5$
3. $P = 3; Z = 3$

1. Two encirclements in the counter clockwise direction.
2. Two encirclements in the clockwise direction.
3. No encirclement.

43. On what principle, the Nyquist stability criterion is based upon?
Nyquist stability criterion is based upon the conformal mapping and the principle of the argument which are derived by the application of Cauchy's theorem.

44. What is Nyquist stability criterion?
Nyquist stability criterion is stated in two forms:

i **Open loop system is stable**: If the open loop system is stable, then for the closed loop system also to be stable, the s-plane contour when mapped into GH-plane should not at all encircle $(-1, 0)$ point.

ii **The open loop system is unstable**: If the open loop system is unstable then for the closed loop system to be stable, the s-plane Nyquist contour when mapped into GH-plane should encircle $(-1, 0)$ point as many times as there are number of open loop poles in RHP and the encirclement should be in the counter clockwise direction of the s-plane contour.

45. How mathematically Nyquist stability condition is expressed?
Nyquist stability condition is mathematically expressed as follows:

$$Z_r = N_r + P_r$$

where

Z_r= Number of zeros of the characteristic equation located in RHP.
P_r is the number of poles of the open loop T.F. located in RHP.
N_r is the number of counter clockwise encirclements of the $(-1, 0)$ point when the s-plane clockwise Nyquist contour is mapped into GH-plane.

For the closed loop system to be stable, $Z_r = 0$.

46. What is Nyquist contour and what is Nyquist path?
The s-plane closed contour which includes the +ve and −ve $j\omega$ axes and the entire right half s-plane is called Nyquist contour. Such a contour should avoid any pole or zero in the $j\omega$ axis by-passing it with a small circle in RHP to satisfy Cauchy's theorem. This Nyquist contour is called Nyquist path.

47. Why Nyquist contour encloses the entire RHP and not the LHP?
The Nyquist contour encloses the entire RHP in search of whether there is any zero of the characteristic equation $1 + GH(s) = 0$ is present in RHP. As far as stability is concerned, it is immaterial whether there is any zero of $1 + GH(s) = 0$ is in LHP or not. That is why Nyquist contour does not cover LHP.

48. A certain system has the following poles of $GH(s)$ $s = \pm j4$ and $s = 0$. How Nyquist contour bypasses these poles?

1. For $s = j4$, the contour is $s = j4 + re^{j\theta}$ where $r \to 0$ and $-\pi/2 \leq \theta \leq \pi/2$.
2. For $s = -j4$, the contour is $s = -j4 + re^{j\theta}$ where $r \to 0$ and $-\pi/2 \leq \theta \leq \pi/2$.
3. For $s = 0$, the contour is $s = re^{j\theta}$ where $r \to 0$ and $-\pi/2 \leq \theta \leq \pi/2$.

49. What is Nyquist plot or Nyquist diagram? Why is it called a frequency response method?
The s-plane contour when mapped into GH-plane, the contour so obtained is called Nyquist plot or Nyquist diagram. Since the part of the Nyquist diagram is nothing but the polar plot of $GH(j\omega)$, it is a frequency response method.

50. For a stable open loop system, how Nyquist criterion is simplified?
If the open loop system is stable, then for the closed loop system also to be stable, the Nyquist plot should not encircle $(-1, 0)$ point.

51. For an unstable open loop system, how Nyquist criterion is simplified?
If the open loop system is unstable, then for a closed loop system to be stable, the Nyquist plot should encircle $(-1, 0)$ as many times as these are number of open loop poles in RHP and the number of encirclement should be in the counter clockwise direction.

52. How an oscillatory system is identified when Nyquist stability criterion is applied?
If the Nyquist plot passes through (-1.0) point, the closed loop system gives sustained oscillations. Such systems as per Nyquist criterion, are oscillatory systems.

53. What does Nyquist criterion tell us?
By applying Nyquist criterion, we get the information about the number of zeros of the characteristic equation that are in RHP. For the closed loop system to be stable, there should not be any zero of the characteristic equation in RHP.

54. What is the advantage of having the Nyquist plot in the GH-plane rather than $(1 + GH)$-plane?
A part of Nyquist plot requires polar plot of $GH(j\omega)$ or $1 + GH(j\omega)$. Plotting of $GH(j\omega)$ for the Nyquist path is much easier than plotting of $1 + GH(j\omega)$. Sometimes one can experimentally get $GH(j\omega)$. That is why plotting of $GH(j\omega)$ and counting the number of encirclements around $(-1, 0)$ in the GH-plane is preferred rather than plotting $1 + GH(j\omega)$ and counting the encirclements around the origin.

Long Answer Type Questions

1. The characteristic equation of a feedback control system is

$$s^4 + 20s^3 + 15s^2 + 2s + K = 0$$

Determine the range of values of K for the system to be stable. Can the system be marginally stable? if so, find the required value of K and the frequency of sustained oscillations. (*Anna University, May, 2011*)
Ans:

1. For $0 < K < 1.49$ the system is stable.
2. For $K = 1.49$, the system is marginally stable. The frequency of sustained oscillation is $\omega = 0.316$ rad/sec.

2. A certain system has the following characteristic equation

$$F(s) = s^6 + 3s^5 + 4s^4 + 6s^3 + 5s^2 + 3s + 2$$

Examine the stability of the system using Routh–Hurwitz criterion.

(*Anna University, December, 2010*).

Ans: There are 2 LHP roots and 4 roots in $j\omega$ axis. They are repeated twice at $s = +j$ and $s = -j$. The system is unstable even though there is no sign change in the first column of Routh's array.

3. Find the range of values of K so that the system with the characteristic equation

$$s(s^2 + s + 1)(s + 4) + K = 0$$

will be stable using Routh's criterion. (*Anna University, May, 2009*).

Ans: $0 < K < 3.36$

4. For a system with
$$F(s) = s^4 + 22s^3 + 10s^2 + s + K$$
obtain the marginal value of K and the frequency of oscillations at that value of K. (*Anna University, May, 2007*).

   Ans: $K = 0.452$; $\omega = 0.2132$ rad/sec.
5. Test the stability using Routh's criterion for the polynomial
$$s^4 + 2s^3 + 3s^2 + 4s + 5 = 0$$
Determine RHP, LHP and $j\omega$ axis roots.

   Ans: RHP roots =2; LHP roots=2; $j\omega$ axis roots=nil
6. Determine the range of K for the system to be stable if the characteristic equation is
$$s^4 + 3s^3 + 3s^2 + 2s + K = 0$$

   Ans: $K < 1.555$
7. For the characteristic equation
$$s^5 + 2s^4 + 24s^3 + 48s^2 - 25 - 50 = 0$$
Perform complete Routh stability analysis. (*Anna University, May, 2007*)

   Ans: There are 1 RHP root, 2 LHP roots and two $j\omega$ axis roots. The roots in the $j\omega$ axis are distinct and the system is marginally stable.
8. Using Routh's criterion, discuss the stability of the system whose characteristic equation is
$$s^3 + 10s^2 + 50s + 500 = 0$$

   Ans: 1 LHP root; 2 $j\omega$ axis roots. The system is marginally stable. No sign change in the first column of Routh's array.
9. Test their stability of the system using Routh–Hurwitz criterion. Also, comment on the location of roots for the systems whose characteristic equations are

   (a) $s^6 + 3s^5 + 2s^4 + 9s^3 + 5s^2 + 12s + 20 = 0$
   (b) $s^5 + s^4 + 2s^3 + 3s + 5 = 0$

   (*Anna University, December, 2009*).
   Ans: (a) Unstable. RHP roots=2; LHP=4 and (b) Unstable. RHP roots=2; LHP roots =3.

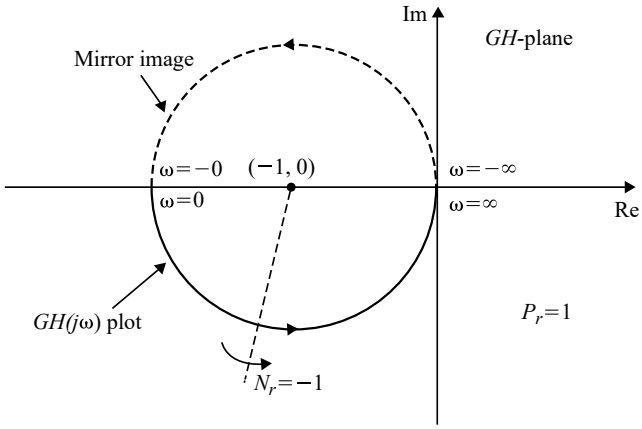


Fig. 4.44 Nyquist diagram for Question 11

10. Consider the following characteristic equation

$$s^4 + 6s^2 - 16 = 0$$

Using Routh–Hurwitz criterion, determine the stability of the system and also the locations of RHP, LHP and $j\omega$ axis roots.
Ans: system is unstable. RHP root=1; LHP root= 1; $j\omega$ axis roots =2. The roots are at $s = \sqrt{2}$, $s = -\sqrt{2}$, $s = +j\sqrt{8}$ and $s = -j\sqrt{8}$.

11. The Nyquist plot of $GH(s)$ is shown in Fig. 4.44. $GH(s)$ has one pole in RHP. Complete the Nyquist diagram and determine the stability of the closed loop system.

Ans: $P_r = 1$; $N_r = -1$; $Z_r = 1 - 1 = 0$. The closed loop system is stable.

12. The loop T.F. of a certain feedback control system is given by

$$GH(s) = \frac{s}{(s + 200)^3}$$

Sketch the Nyquist plot and determine the stability of the closed loop system.

Ans: $P_r = 0$; $N_r = 0$; $Z_r = 0 + 0 = 0$. The closed loop system is stable.

13. The loop T.F. of a certain feedback control system is given as

$$GH(s) = \frac{Ks^2}{(s^3 + 5s + 25)}$$

Using Nyquist criterion, determine the value of K so that the closed loop system is stable.

Ans: For the given system $P_r = 2$; N_r should be -2 for $Z_r = 0$. For this, $K > 5$. For $K > 5$, the closed loop system is stable.

14. The forward path T.F. of a certain unity feedback system is given as

$$G(s) = \frac{K(2s+1)(s+1)}{(5s+1)(s-1)}$$

Using Nyquist stability criterion, determine the value of K so that the closed loop system is stable.

Ans: $P_r = 1$; For $K > 1.33$, $N_r = -1$; $Z_r = 0$. The closed loop system is stable for $K > \frac{4}{3}$.

15. A certain feedback control system has the following loop T.F

$$GH(s) = \frac{K(s+3)(s+2)}{(1+5s)(s-2)}$$

By applying the Nyquist stability criterion, determine the value of K so that the closed loop system is stable.

Ans: $P_r = 1$; N_r should be -1 so that $Z_r = 0$. For this, $K > 1.8$.

16. A certain control system has the following forward path T.F

$$GH(s) = \frac{K(s+3)(s+5)}{(s-2)(s-4)}$$

By applying Nyquist stability criterion, determine the value of K so that the closed loop system is stable.

Ans: For the given system, $P_r = 2$; For Z_r to be zero N_r should be -2. This is possible if $\frac{4}{3}K > 1$ or $K > 0.75$.

17. Consider the following loop T.F. of a certain closed loop system

$$GH(s) = \frac{K}{(1+s)(1+2s)(1+3s)}$$

Draw the Nyquist plot and determine the value of K so that the closed loop system is stable.

Ans: $K < 10$ for the closed loop system to be stable. For the given $GH(s)$, $P_r = 0$. For the closed loop system to be stable $N_r = 0$. For this, at ω_{pc}, $-0.1K$ should be to the right of $(-1, 0)$ point. Hence, $K < 10$.

18. Consider the following loop T.F. of a certain closed loop system

$$GH(s) = \frac{K(6s-1)}{(s^2+2s+2)}$$

Draw the Nyquist plot and determine the value of K so that the closed loop system is stable.

Ans: $P_r = 0$; N_r should be zero for the closed loop system to be stable. $K < 2$ for the system to be stable.

19. A certain feedback control system has the following open loop T.F.

$$GH(s) = \frac{K(s-2)^2}{(s+2)^2(s+4)^2}$$

Draw the Nyquist plot and determine the value of K so the closed loop system is stable.

Ans: $K < 18$ for the closed loop system to be stable. Since $P_r = 0$, N_r should be zero. This implies $K < 18$.

20. A certain closed loop system has the following open loop T.F.

$$GH(s) = \frac{K(s+2)}{(s^3+8s^2-10)}$$

Sketch the Nyquist diagram and determine the value of K so that the closed loop system is stable.

Ans: $P_r = 1$; N_r should be -1 so that $Z_r = 0$. For this, $0.2K > 1$ or $K > 5$ for the closed loop system to be stable.

21. A certain feedback control system has the following loop T.F

$$GH(s) = \frac{K}{(s^2-16)(s+2)}$$

Investigate the stability of the closed loop system, using Nyquist stability criterion.

1. $P_r = 1$; If $K > 32$, $N_r = 1$ and $Z_r = 2$. The system is unstable.
2. $P_r = 1$; If $K < 32$, $N_r = 0$ and $Z_r = 1$. The system is unstable.

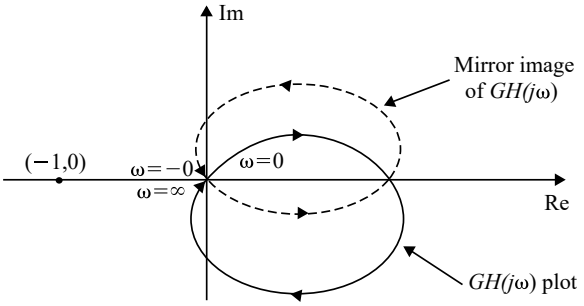


Fig. 4.45 Nyquist diagram for Question 12

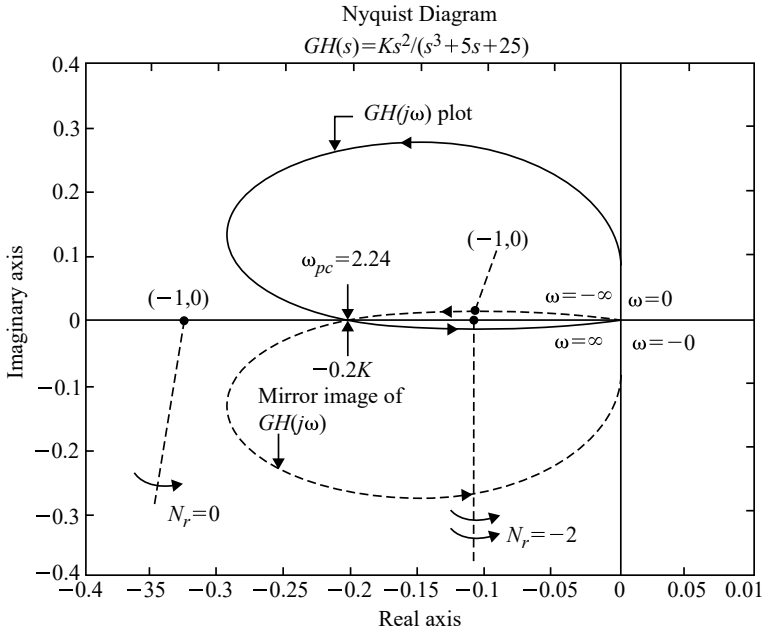


Fig. 4.46 Nyquist diagram for Question 13

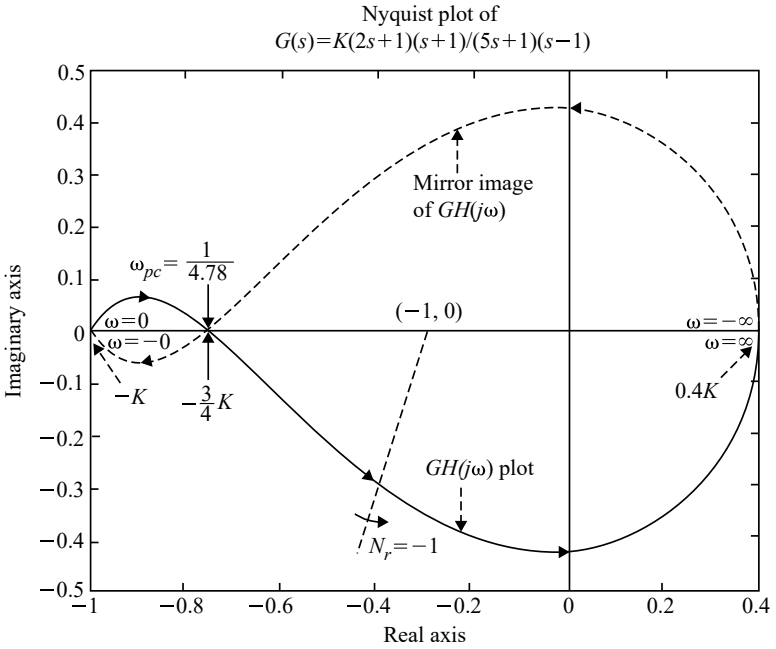


Fig. 4.47 Nyquist diagram for Question 14

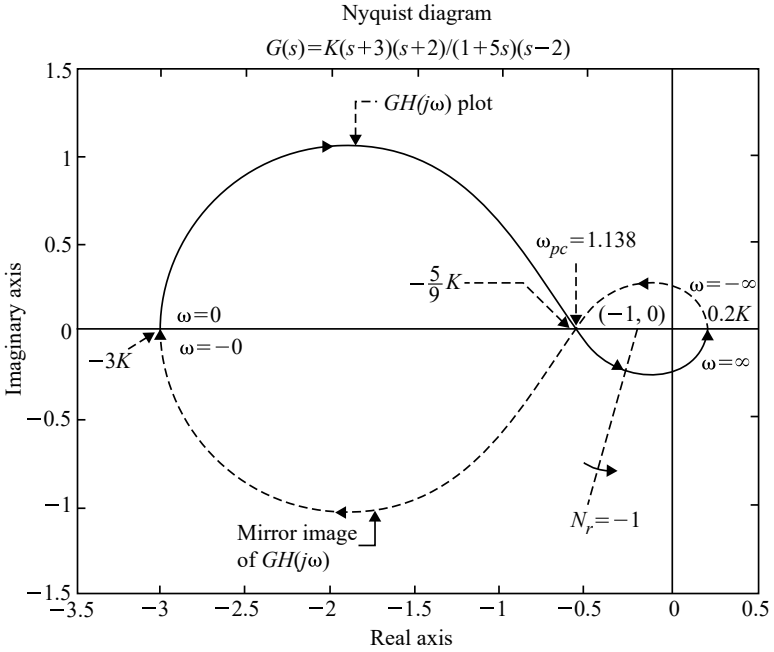


Fig. 4.48 Nyquist diagram for Question 15

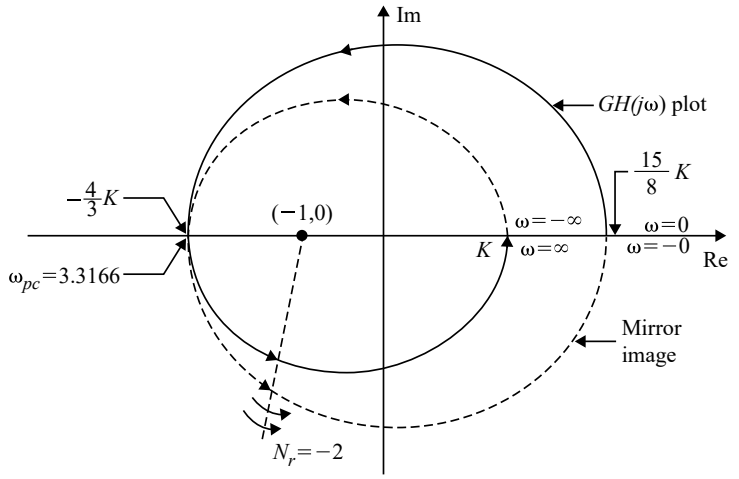


Fig. 4.49 Nyquist diagram for Question 16

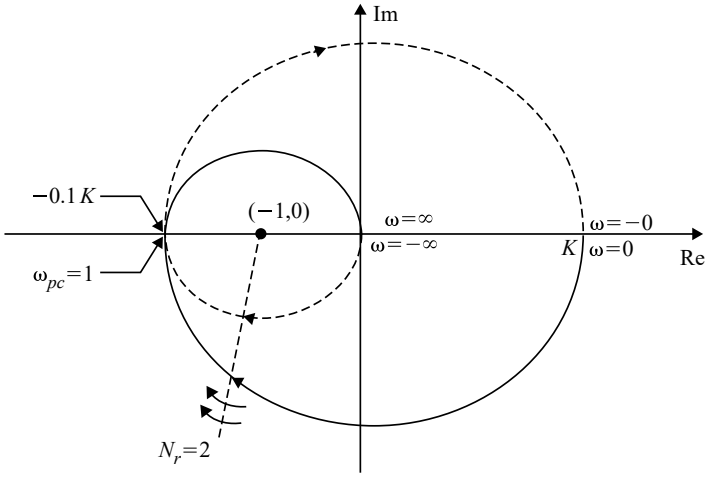


Fig. 4.50 Nyquist diagram for Question 17

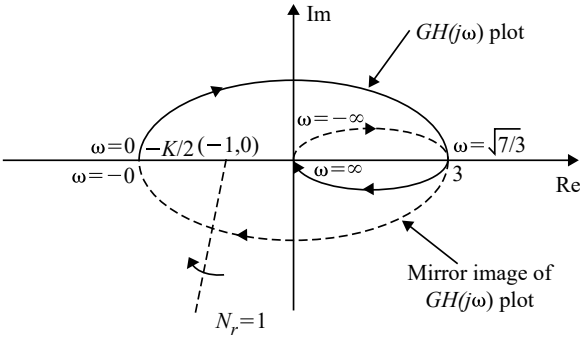


Fig. 4.51 Nyquist diagram for Question 18

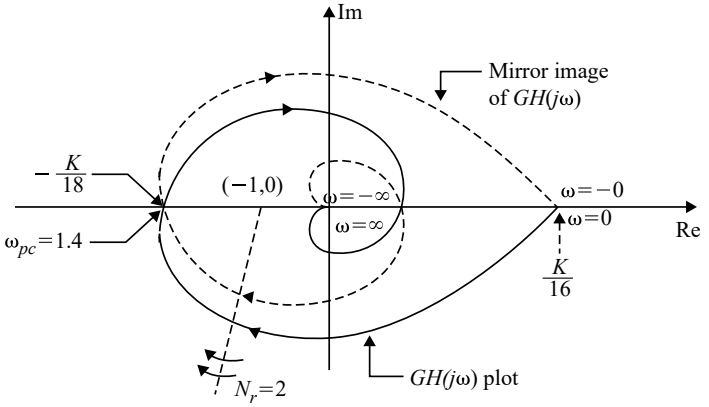


Fig. 4.52 Nyquist diagram for Question 19

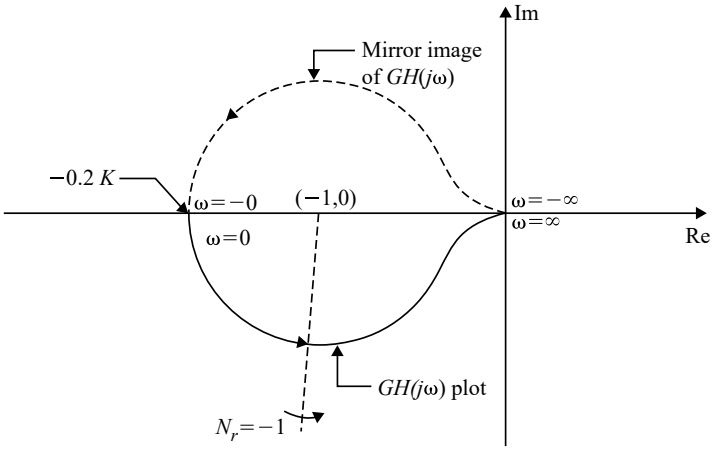


Fig. 4.53 Nyquist diagram for Question 20

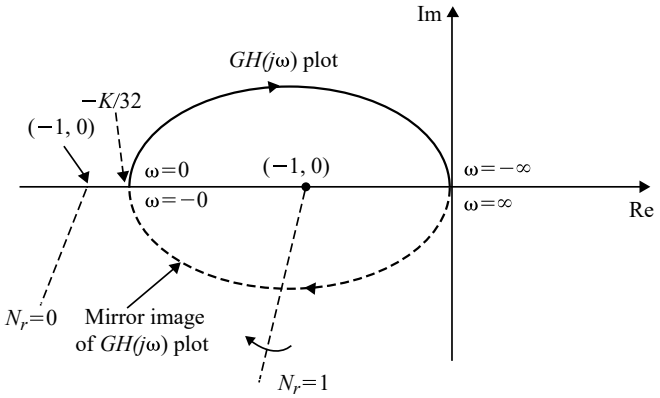


Fig. 4.54 Nyquist diagram for Question 21

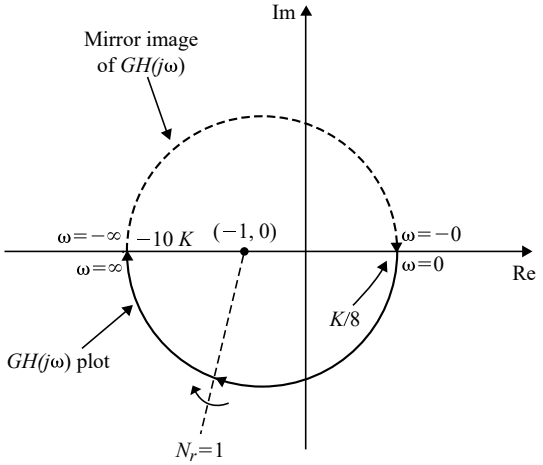


Fig. 4.55 Nyquist plot for Question 22

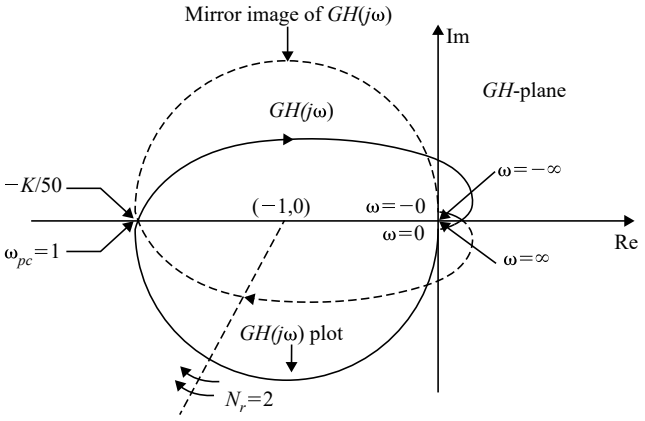


Fig. 4.56 Nyquist plot for Question 23

22. A certain closed loop T.F. has the following T.F. which is strictly proper. Using Nyquist stability criterion determine the value of K so that the closed loop system is stable

$$GH(s) = \frac{K(1 - 10s^2)}{(s + 2)(s + 4)}$$

$K < 0.1$, the system is stable.
$K = 0.1$, the system is oscillatory.
$K > 0.1$, the system is unstable.

23. A certain feedback control system has the following open loop T.F

$$GH(s) = \frac{Ks^3}{(s+2)^2(s+3)^2}$$

Determine the value of K so that the closed loop system is stable (Figs. 4.45, 4.46, 4.47, 4.48, 4.49, 4.50, 4.51, 4.52, 4.53, 4.54, 4.55, 4.56).

$K < 50$, the system is stable.
$K = 50$, the system is oscillatory.
$K > 50$, the system is unstable.

Chapter 5
Root Locus Method for Analysis



Learning Objectives

After completing this chapter, you should be able to

- Understand the root locus concept.
- Understand the magnitude and angle criteria of root locus.
- Understand the step-by-step procedure and the rules to draw the root locus.
- Draw the root locus for variation of system parameters.
- Determine the closed loop roots for the given gain K.
- Determine transient response for the desired damping.
- Determine the phase margin and gain margin from root locus.
- Draw the root locus for the negative value of gain.
- Draw the root locus for the positive feedback system.
- Study the effect of the addition of poles and zeros.

5.1 Introduction

In the previous chapters, stability and relative stability of linear systems were discussed in the time and frequency domains. In the time domain approach, it was seen that the system stability and relative stability can be improved by adjusting the gain of the system. It was further established that a linear time invariant system is said to be absolutely stable if all the roots of the characteristic equation are in LHP. Furthermore that if these poles in the LHP are placed farther from the imaginary

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S. Palani, *Automatic Control Systems*,
https://doi.org/10.1007/978-3-030-93445-3_5

axis of the complex s-plane, the relative stability of the system increases. In the frequency domain, the absolute stability of the system was discussed using the Nyquist stability criterion. Using Bode plots and Nichols chart, the relative stability of the system was discussed. However, these methods do not give information about the exact root locations of the closed loop system which give useful information about time domain specifications. The trajectories of the roots of the characteristic equation when investigated carefully as the gain of the system is varied give all information about absolute and relative stabilities like the other methods discussed in previous chapters. In addition, the method gives the exact root locations of the closed loop system. The method is called the root locus method, and the trajectories of the closed loop poles in the s-plane are called root loci. **The root locus is the path of the characteristic equation traced out in the s-plane as the system parameter is varied.** The method developed by W. R. Evans in 1948 is a very powerful method and extensively used in control engineering analysis and design. The root locus is a way of representing graphical information about system behaviour. It has the virtue of being a good design tool for continuous time systems when we work in the s-plane and discrete time system when we work in the z-plane. The concept of root locus is very simple enough, but the problem is that the closed loop poles can exhibit many strange and wonderful movements in the s-plane when the system is even a little bit complex.

A root locus plot is simply a plot of the s zeros and the s poles on a graph with real and imaginary coordinates. It represents a curve of the location of the poles of a transfer function as some parameter is varied. The locus of the roots of the characteristic equation of the closed loop system as the gain varies from $-\infty$ to ∞ gives the name of the method. Such a plot shows clearly the contribution of each open loop pole or zero to the locations of closed loop poles. General rules to construct the root locus exist and if the designer follows them, sketching of the root loci becomes a simple matter. However, a simple MATLAB program is also available to draw the root locus very accurately, and all necessary information can be easily obtained from the graph so drawn. The information include absolute stability, relative stability, locations of closed loop poles and time and some frequency domain specifications. **Thus, root locus when used has the problem of plenty but without plenty of problems**.

5.1.1 Advantages of Root Locus Method

1. Absolute and relative stabilities can be determined using root locus.
2. For the given system, the gain of the system can be determined for the system to be marginally stable. The corresponding frequency of oscillation can be determined.
3. For a given value of the system gain K, the closed loop pole locations can be determined. Similarly, for the required damping ratio of the dominant closed loop system, the closed loop pole locations can be determined. From this, the

time response of the closed loop system for the given input can be obtained. Further, it is possible to determine the closed loop frequency response also.
4. For the given gain of the system, time domain specifications can be determined.
5. From the root locus, the frequency domain specifications such as phase margin and gain margin can be obtained, which are the measures of relative stability.
6. The root locus shows clearly the contribution of each open loop pole or zero to the locations of the closed loop poles which ultimately decide the absolute and relative stabilities and also give information about the time and frequency domain specifications.
7. Like other methods, a very simple MATLAB program is available to draw root locus which makes the drawing of root loci simpler. That is why root locus is extensively used in the design of compensators in the time domain.

5.2 The Concept of Root Locus

Consider the closed loop system shown in Fig. 5.1. The closed loop T.F. is obtained as

$$\frac{C(s)}{R(s)} = \frac{KG(s)}{1 + KG(s)} \tag{5.1}$$

where K is a real constant and $G(s)$ is a function of s with numerator and denominator polynomials of any order. The characteristic equation of Eq. (5.1) is

$$F(s) = 1 + KG(s) = 0 \tag{5.2}$$

$$KG(s) = -1$$

$$|KG(s)| = 1 \tag{5.3}$$

$$\angle KG(s) = 180^\circ + n \times 360^\circ \tag{5.4}$$

where $n = 0, \pm 1, \pm 2, \pm \ldots$.

The root locus is the path of the roots of the characteristic equation that moves in the complex s-plane as the system parameter varies in the range $-\infty \leq K \leq \infty$.

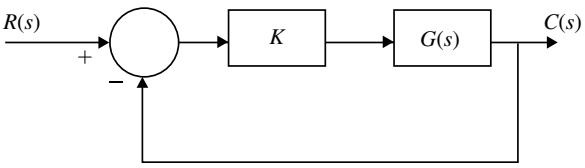


Fig. 5.1 Closed loop system with variable parameter K

Table 5.1 Variation of roots with respect to gain K

K	0	2	4	6	8	∞
s_1	0	-0.586	-2	$-2+j2$	$-2+j4$	$-2+j\infty$
s_2	-4	-3.414	-2	$-2-j2$	$-2-j4$	$-2-j\infty$

At present, we restrict our discussion by allowing K to vary in the range $0 \leq K \leq \infty$. The root locus should always satisfy Eqs. (5.3) and (5.4), where Eq. (5.3) is called magnitude criterion and Eq. (5.4) is called angle criterion. Let us consider the following simple example to trace the root locus as K is varied from 0 to ∞. It is evident from Eqs. (5.4) and (5.3) that angle criterion is used to trace the root locus while magnitude criterion is used to calibrate the root locus in terms of K.

The angle criterion states that for any point s_1 in the s-plane to be a point on the root locus, the sum of the angles of the vectors drawn from all the zeros of $GH(s)$ minus the sum of the angles of the vectors drawn from all the poles of $GH(s)$ to the point s_1 should be an odd multiple of 180°.

According to the magnitude condition, at any point s_1 in the s-plane where root locus exists, the value of K is calculated as

$$K = \frac{\text{Product of the lengths of the vectors drawn from the poles of } GH(s) \text{ to } s_1}{\text{Product of the lengths of the vectors drawn from the zeros of } GH(s) \text{ to } s_1}$$

Now consider the following open loop T.F

$$KG(s) = \frac{K}{s(s+4)}$$
$$F(s) = s^2 + 4s + K = 0 \tag{5.5}$$

Equation (5.5) is solved for the two roots as

$$s_1 = -2 + \sqrt{4-K}$$
$$s_2 = -2 - \sqrt{4-K}$$

For a few values of K, the roots s_1 and s_2 as given by the above equation are calculated and are shown in Table 5.1.

The poles of $GH(s)$ are at $s = 0$ and $s = -4$. From Table 5.1, for $K = 0$, $s_1 = 0$ and $s_2 = -4$. Thus, the root locus starts from the poles of $G(s)$ and the value of K at these points is zero. As K increases, s_1 moves towards the left and s_2 moves towards the right on the negative real axis. For $K = 4$, $s_1 = -2$ and $s_2 = -2$. The root loci originating from the respective poles meet each other at $s = -2$ and break away. For further increases of K, the roots are complex conjugate with real part remaining constant at -2 and the imaginary part increasing. As K tends to ∞, the two root loci also tend to ∞. The root locus plot of Table 5.1 is shown in Fig. 5.2.

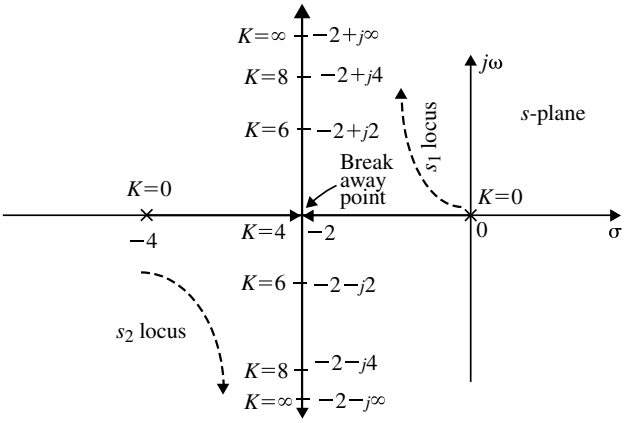


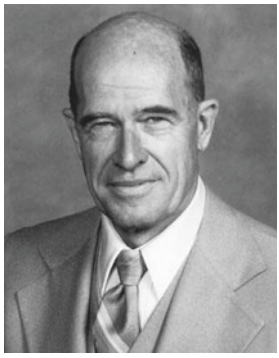
Fig. 5.2 Root locus for $G(s) = \dfrac{K}{s(s+4)}$

From Fig. 5.2, we observe that

1. The root loci start from poles of $G(s)$ where $K = 0$.
2. The root loci terminate at ∞ and $-\infty$ as $K \to \infty$ in the absence of zeros of $G(s)$.
3. There are as many root loci as the order of the system.
4. At any point on the root locus, Eqs. (5.3) and (5.4) are satisfied.

The above procedure is presented to illustrate the concept of root locus. However, for complex systems, where $G(s)$ has many poles and zeros, it is more or less impossible to draw the root locus. Further, using Eqs. (5.3) and (5.4) alone, one cannot draw the root locus. The path of the root is decided by the angle criterion of Eq. (5.4). It is extremely difficult to search for a point in the vast area complex s-plane which will satisfy the angle criterion of root locus. To make the drawing of root locus simpler, a set procedure is available and they are to be followed. The procedure includes the properties of root locus or the rules of root locus. They are discussed below with illustrations.

A Historical Note: Walter R. Evans (1920–1999)



Walter Richard Evans a noted American control theorist was born on 15 January 1920, in St. Louis, Missouri. Young Evan wanted to become an engineer like his father. He received his B.E. Degree in Electrical Engineering from Washington University in St. Louis, Missouri, in 1941 and his M.S. in Electrical Engineering from the University of California, Los Angeles, in 1951. He started his career as a trainee in General Elec-

tric Company in 1944 and worked as an instructor at Washington University from 1946 to 1948. Evans' principal contribution to the automatic control theory was his invention of the Evans Root Locus Method in 1948. Evans' root locus method has widespread application in the design of control systems. Now, root locus is included as a standard chapter in feedback control systems textbooks. Evans had worked as an engineer in several companies including General Electric Rockwell International and Ford Automatic Company. He published a book titled "*Control System Dynamics*" with McGraw Hill in 1954. He was awarded the Prestigious Rufus Oldenburger Medal in 1967 and Richard E. Bellman Control Heritage Award in 1988 in recognition of his outstanding contributions in the fields of automatic control theory. Although a genius, he never looked down on others. His ambition in life was to teach, instruct, be a model and serve. Evans was married to Arline and the couple had four children. The world lost a great technological genius on 10 July 1999.

5.3 Properties of the Root Loci (Rules of the Root Loci)

The construction of root locus manually is made simpler using the properties of the root loci. These properties are also called rules of root loci. Using these rules, certain key points in the s-plane are indentified. These points lie on the root loci. The connection of these points gives approximate or asymptotic root loci. However, by smoothening the asymptotic plots, root loci which are closer to the actual root loci are drawn. These rules are discussed below.

Rule 1: The Starting Points of the Root Loci
Consider the characteristic equation given in Eq. (5.2)

$$\begin{aligned} 1 + KGH(s) &= 0 \\ GH(s) &= -\frac{1}{K} \end{aligned} \tag{5.6}$$

As the magnitude K approaches zero, $|GH(s)| = \infty$. This implies that s must approach the poles of $GH(s)$. Since

$$GH(s) = \frac{(s+z_1)(s+z_2)\cdots(s+z_m)}{(s+p_1)(s+p_2)\cdots(s+p_n)} \tag{5.7}$$

the root loci start from the poles of $GH(s)$ where the value of $K = 0$.

Rule 2: The Ending Points of Root Loci
Starting from the poles of $GH(s)$, as the value of K increases, the root loci move in the s-plane. For $K = \infty$, Eq. (5.6) becomes zero. This is possible if s approaches the zeros of $GH(s)$ or ∞ of the s-plane.

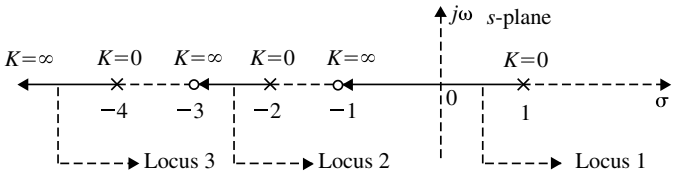


Fig. 5.3 Root locus showing the number of root loci, starting and ending parts and the value of K

The root loci end at the zeros of $GH(s)$ or at ∞ where the value of $K = \infty$. From Fig. 5.1, it can be seen that the two root loci start at $s = 0$ and $s = -4$ where the value of $K = 0$. Since there are no zeros of $GH(s)$, they all end at ∞ where $K = \infty$.

Rule 3: Number of Root Loci of 1+GH(s)
The number of root loci of the characteristic equation $1 + GH(s) = 0$ is equal to the order of the polynomial of Eq. (5.7). Consider the following loop T.F.

$$GH(s) = \frac{K(s+1)(s+3)}{(s-1)(s+2)(s+4)}$$

The numerator polynomial is of order two while the denominator polynomial is of order three. Hence, there should be three root loci as shown in Fig. 5.3.

Rule 4: Symmetry of the Root Loci
The root loci are symmetrical with respect to the real axis of the s-plane. The rule is derived based on the property that for real coefficients of characteristic polynomial, the roots must be real or in complex conjugate form. Consider the following loop T.F

$$GH(s) = \frac{K}{s(s+4)(s+2)}$$

The root locus is shown in Fig. 5.4. Loci 1 and 2 are symmetrical with complex conjugate root loci with respect to the real axis.

Rule 5: Angle of the Asymptote of the Root Loci
If there are P number of poles and Z number of zeros of $GH(s)$, and if $P = Z$, all the P root loci originating from their respective poles will end at the zeros where the value of $K = \infty$. However, if $P > Z$, then $(P - Z)$ loci will end at ∞ where $K = \infty$. The root loci near ∞ in the s-plane are described by the asymptotes of the root loci. They are giving guidance to the actual root locus. The angles of the asymptotes are given by

$$\theta = \frac{(2n+1) \times 180^\circ}{(P - Z)} \tag{5.8}$$

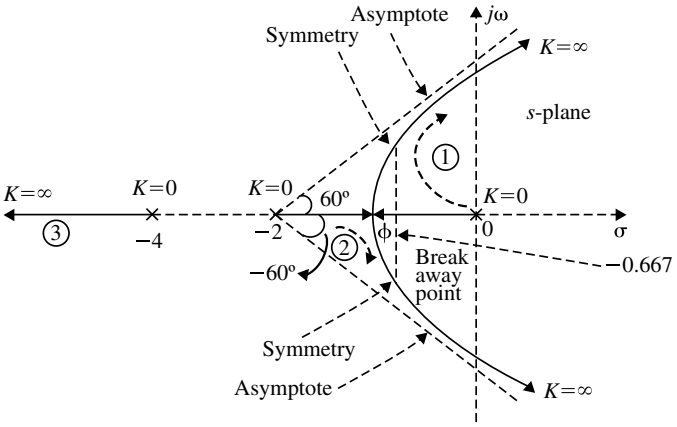


Fig. 5.4 Root locus which illustrates complex symmetry

where

P = Finite number of poles of $GH(s)$.
Z = Finite number of zeros of $GH(s)$.
$n = 0, 1, 2, \ldots, (P - Z - 1)$.

For the example illustrated in Fig. 5.3, the angles of the asymptotes are $\theta = 60°, 180°$ and $-60°$.

Rule 6: Intersect of the Asymptote (Centroid)
The intersect of the asymptotes of the root loci lies on the real axis of the s-plane at

$$\sigma = \frac{\sum \text{ real part of poles of } GH(s) - \sum \text{ real part of zeros of } GH(s)}{(P - Z)} \quad (5.9)$$

The intersect of the asymptotes σ is the centre of gravity of the root loci, and hence it is also called centroid. The value of σ is always a real number. For the example illustrated in Fig. 5.3, the centroid is

$$\sigma = \frac{0 - 2 - 4 - 0}{3}$$
$$= -2$$

Rule 7: Root Loci on the Real Axis
At any point s_1 on the real axis, the angles of the vectors drawn from the complex poles and zeros of $GH(s)$ add up to zero. Only the poles and zeros of $GH(s)$ on the real axis contribute angle to any point s_1 in the real axis. The angle contributions by the poles and zeros of $GH(s)$ to the left of the point s_1 is zero. However, each pole to the right of s_1 will contribute $-180°$, while each zero of $GH(s)$ to the right of

the point s_1 will contribute $+180°$ ($+180°$ and $-180°$ are same). Thus, to satisfy the angle criterion of root locus, given by Eq. (5.4), any point s_1 on the real axis to be a point on the root locus, the total number of poles and zeros of $GH(s)$ to the right of that point should be odd. Three poles and one zero will make a total number 4 which is even. Two poles and one zero will make the total number 3 which is odd.

Consider the root locus drawn in Fig. 5.3. Consider any point s_1 between $s = -1$ and $+1$. To the right of this point, one pole at $s = 1$ exists which is an odd number. Therefore, root locus on the real axis exists between $s = 1$ and $s = -1$. Now consider any point s_1 between $s = -1$ and $s = -2$ on the real axis. To the right of s_1 even number of zeros and poles exists between $s = -1$ and $s = -2$ on the real axis, and there won't be any root locus.

Rule 8: Angle of Departure of the Root Locus

The angle of departure of root locus at a pole of $GH(s)$ is nothing but the angle of the tangent to the locus near the pole. The angle of departure is given by

$$\theta_d = (2n + 1) \times 180° - (\theta_P - \theta_Z) \tag{5.10}$$

where θ_P and θ_Z are the net angle contribution to the concerned pole by all other poles and zeros of $GH(s)$, respectively. This is illustrated in Fig. 5.4 for the root locus drawn for the loop T.F.

$$GH(s) = \frac{K(s + 6)}{(s + 3)(s^2 + 2s + 2)}$$

The root locus plot is shown in Fig. 5.5. The root loci from $s = -1 + j$ and $s = -1 - j$ depart with a curved line. The angle of departure at $s = -1 + j$ is calculated by drawing vectors from poles and zeros of $GH(s)$ as shown in Fig. 5.5. Using Eq. (5.10), we get

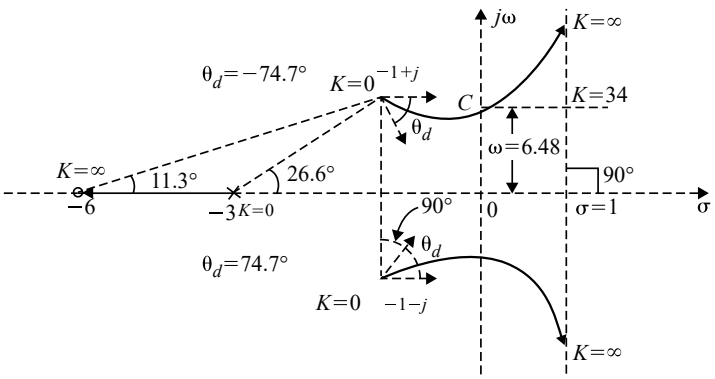


Fig. 5.5 Illustration of angle of departure for $GH(s) = \dfrac{K(s + 6)}{(s + 3)(s^2 + 2s + 2)}$

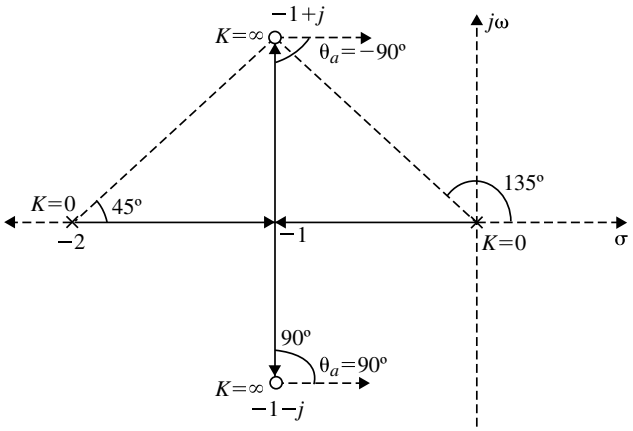


Fig. 5.6 Illustration of angle of arrival for $GH(s) = \dfrac{K(s^2+2s+2)}{s(s+2)}$

$$\begin{aligned}\theta_d &= 180^\circ + (90^\circ + 26.6^\circ - 11.3^\circ)\\ &= 285.3^\circ\\ &= -74.7^\circ\end{aligned}$$

Similarly, for the conjugate pole at $s = -1 - j$, we get

$$\theta_d = 74.7^\circ$$

Rule 9: Angle of Arrival of the Root Locus
The angle of arrival of the root locus at the zero of the root locus is represented by θ_a, and it is the angle of the tangent to the locus near the zero. The angle of arrival is given by

$$\theta_a = (2n+1) \times 180^\circ - (\theta_Z - \theta_P) \tag{5.11}$$

where the symbols θ_Z and θ_P have the same meaning but now drawn to the point of the concerned zero. The angle of arrival is illustrated in Fig. 5.6.

Figure 5.6 shows the root locus for the loop T.F.

$$GH(s) = \frac{K(s^2+2s+2)}{s(s+2)}$$

The two root loci starting from the poles $s = 0$ and $s = -2$ breakaway at $s = -1.0$ and arrive at the zeros at $s_1 = -1 + j1$ and $s_2 = -1 - j1$. The angle of arrival at s_1 is determined using Eq. (5.11)

$$\begin{aligned}\theta_a &= 180^\circ + 45^\circ + 135^\circ - 90^\circ \\ &= -90^\circ\end{aligned}$$

The angle of arrival at s_2 is $\theta_a = 90^\circ$.

Rule 10: Breakaway and Break-in Points or Saddle Points on the Root Locus The point at which more than one root loci meet and then depart is called the breakaway point. At the breakaway point, double roots are present. A root locus may have more than on breakaway point. Breakaway points may be real and complex conjugate. The breakaway point is obtained by satisfying the condition

$$\frac{dK}{ds} = 0 \tag{5.12}$$

in the characteristic equation $1 + GH(s)$ and solving for s. Further, the point s so obtained must also be a point on the root loci for some real value of K. Consider the loop T.F.

$$\begin{aligned}GH(s) &= \frac{K}{s(s+4)(s+2)} \\ 1 + GH(s) &= s^3 + 6s^2 + 8s + K \\ &= 0\end{aligned}$$

Differentiating the above equation with respect to s and equating $dK/ds = 0$, we get

$$\begin{aligned}3s^2 + 12s + 8 + \frac{dK}{ds} &= 0 \\ 3s^2 + 12s + 8 &= 0\end{aligned}$$

$s_1 = -3.33$ and $s_2 = -0.667$. The root locus is shown in Fig. 5.4. The breakaway point $s_1 = -3.33$ is not a point on the root locus, and hence it is not a breakaway point. However, s_2 is a point on the root locus. Hence, the breakaway point is at $s_2 = -0.667$.

When more than one root locus approach a real axis and depart, such a point is called the break-in point. The conditions applied to breakaway points apply to break-in points also.

Rule 11: Angle of Arrival and Departure of Root Loci at the Breakaway Points The angles at which the root loci arrive or depart from a breakaway or break-in point (Fig. 5.7b) depend on the number of locus breakaway or break-in. In Fig. 5.7a, b, two loci meet and breakaway and break-in, respectively. In these cases, the angle of arrival and departure is 90°. However, in Fig. 5.7c there are four root loci that meet at the breakaway point and depart with an angle of 45°. Thus, in general if there are n root loci that meet at the breakaway point, they arrive or depart at $180^\circ/n$.

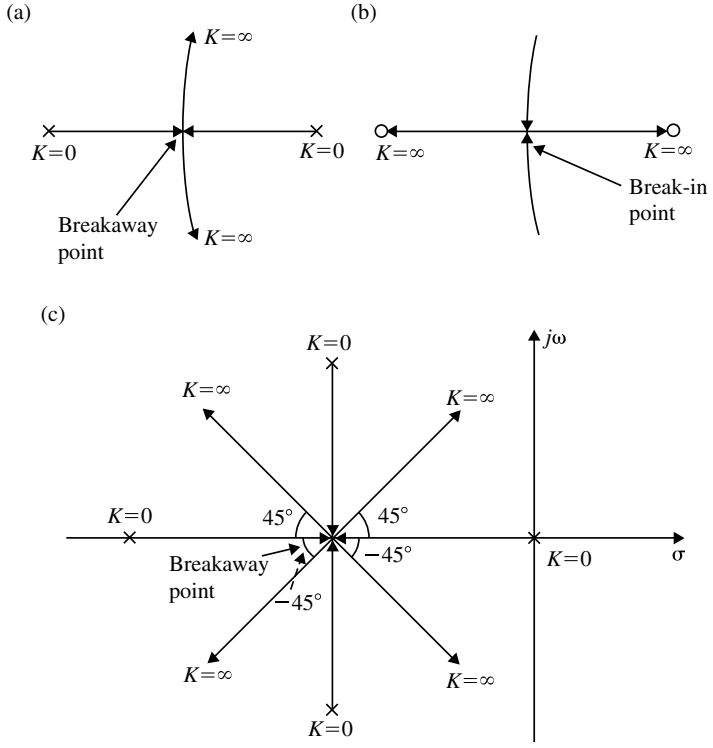


Fig. 5.7 Representation of angle of breakaway points

Rule 12: Intersection of Root Loci with the Imaginary Axis

For some systems, as the parameter K is increased, the root loci cross the imaginary axis. The value of K at this point c is calculated using the Routh–Hurwitz criterion. The system is now marginally stable and has sustained oscillations, and the oscillation frequency can also be determined by solving the auxiliary equation in Routh's array. Consider the following loop T.F.:

$$GH(s) = \frac{K(s+6)}{(s+3)(s^2+2s+2)}$$

The root locus for the above T.F. is shown in Fig. 5.5. The critical value or marginal value of K when the root loci cross from LHP to RHP is calculated by forming Routh's array for the following characteristic equation:

$$1 + \frac{K(s+6)}{(s+3)(s^2+2s+2)} = 0$$
$$(s+3)(s^2+2s+2) + K(s+6) = 0$$
$$s^3 + 5s^2 + (K+8)s + 6(K+1) = 0$$

The following Routh's array is formed.

s^3	1	$(K+8)$
s^2	5	$6(K+1)$
s^1	$-K+34$	
s^0	$(K+1)$	

For the system to be marginally stable, $-K + 34 = 0$ or $K = 34$. The auxiliary equation from Routh's array is written as

$$5s^2 + 6(34+1) = 0$$

In the $j\omega$ axis, $s = j\omega$ and substituting this above equation we get

$$5\omega^2 = 42$$
$$\omega = 6.48$$

This is represented in Fig. 5.5. The properties or the rules of root locus are summarized in Table 5.2.

5.4 Procedure to Construct Root Locus

The following procedure is followed to construct root locus:

1. Express the given open loop T.F. in pole–zero form as

$$GH(s) = \frac{K(s+z_1)(s+z_2)\cdots(s+z_m)}{(s+p_1)(s+p_2)\cdots(s+p_n)}$$

   Locate the poles and zeros of $GH(s)$ on the s-plane.
2. The root loci start from the poles of $GH(s)$ where the value of $K = 0$ and end at the zeros of $GH(s)$ or at ∞ if the order of the numerator polynomial m is less than or equal to the order of the denominator polynomial n. If $m > n$, then $(m - n)$ loci will start from ∞ and end at the zeros of $GH(s)$. The value of K at the zeros is zero.
3. Identify the root loci on the real axis. For any point s_1 on the root locus to be on the real axis, to the right of that point the sum total of poles and zeros should be odd.

Table 5.2 Summary of properties (rules) of root locus

1.	Magnitude criterion	$K =$ (Product of directed distances drawn from all the poles of $GH(s)$ to the point s_1) $\Big/$ Product of directed distances drawn from all the zeros of GH(s) to the point s_1
2.	Angle criterion	(Sum of all the angles of vectors drawn from all the zeros of $GH(s)$ to any point s_1-sum of all the angles of vectors drawn from all the poles of $GH(s)$ to any point s_1) = odd multiple of 180°
3.	The starting points of the root loci	The root loci start from the poles of $GH(s)$ where the value of $K = 0$
4.	The ending points of the root loci	The root loci end at the zeros of $GH(s)$ or at ∞ where the value of $K = \infty$
5.	Number of root loci of $1 + GH(s)$	The number of root loci of the characteristic equation $1 + GH(s) = 0$ is equal to the order of the numerator polynomial or the denominator polynomial of $GH(s)$ whichever is higher
6.	Symmetry of the root loci	The root loci is symmetrical with respect to the real axis of the s-plane
7.	Angle of the asymptote of the root loci	The angles of the asymptotes of the root loci are given by $\theta = \frac{(2n+1)180^\circ}{(P-Z)}$, where $P =$ finite number of poles of $GH(s)$; $Z =$ finite number of zeros of $GH(s)$; $n = 0, 1, 2, \ldots, (P - Z - 1)$
8.	Intersect of the asymptote (Centroid)	$\sigma = \sum$ Real part of poles of $GH(s)$ $- \sum$ Real part of zeros of $GH(s) \Big/ (P - Z)$
9.	Root loci on the real axis	For any point s_1 on the real axis to be a point on the root locus, the total number of poles and zeros of $GH(s)$ to the right of that point should be odd
10	The angle of departure of root locus	$\theta_d = (2n + 1) \times 180^\circ - (\theta_P - \theta_Z)$ where θ_P and θ_Z are the net angle contribution to the concerned pole by all other poles and zeros of $GH(s)$, respectively
11.	The angle of arrival of root locus	$\theta_a = (2n + 1) \times 180^\circ - (\theta_Z - \theta_P)$
12.	Breakaway and break-in points on the root locus	The breakaway and break-in points on the root locus are obtained by satisfying the condition $dK/ds = 0$ in the characteristic equation and solving for s

(continued)

Table 5.2 (continued)

13.	Angle of arrival and departure of root loci at the breakaway or break-in points	If there are n root loci that meet at breakaway or break-in points, they arrive or depart at $180°/n$
14.	Intersection of the root loci with the imaginary axis	The root loci move from LHP to RHP as the value of K increases and when it intersects the imaginary axis, the system becomes marginally stable. The critical value of gain is determined by forming Routh's array. From the auxiliary equation the frequency of oscillation is determined

4. Determine the angles of the asymptotes using

$$\theta = \frac{(2n+1) \times 180°}{(P-Z)}$$

5. Determine the centroid which is given by

$$\sigma = \frac{\sum \text{ real part of poles of } GH(s) - \sum \text{ real part of zeros of } GH(s)}{(P-Z)}$$

Using steps 4 and 5, draw the asymptotes.

6. Find the breakaway and break-in points of the root loci. For this, express the characteristic equation as

$$P(s) + KQ(s) = 0$$

$$K = \frac{-Q(s)}{P(s)}$$

Differentiate K with respect to s and put $dK/ds = 0$. Solve for s which gives the breakaway or break-in points. However, only those points which lie only on the root loci alone are to be taken.

7. Determine the angle of arrival and angle of departure of the root loci using Eqs. (5.10) and (5.11), respectively.
8. Find the intersection of the root loci with the imaginary axis where the system is marginally stable using the Routh–Hurwitz criterion.
9. By joining the points obtained from steps 6 to 8, the root locus is sketched. The root loci coincide with the asymptote as K tends to ∞. However, in the neighbourhood of the origin of the s-plane, root loci are to be checked whether they satisfy the angle criterion. By following the procedure cited above reasonability accurate root loci are drawn. However, if very accurate root loci are needed, MATLAB may be used rather than hand calculation.

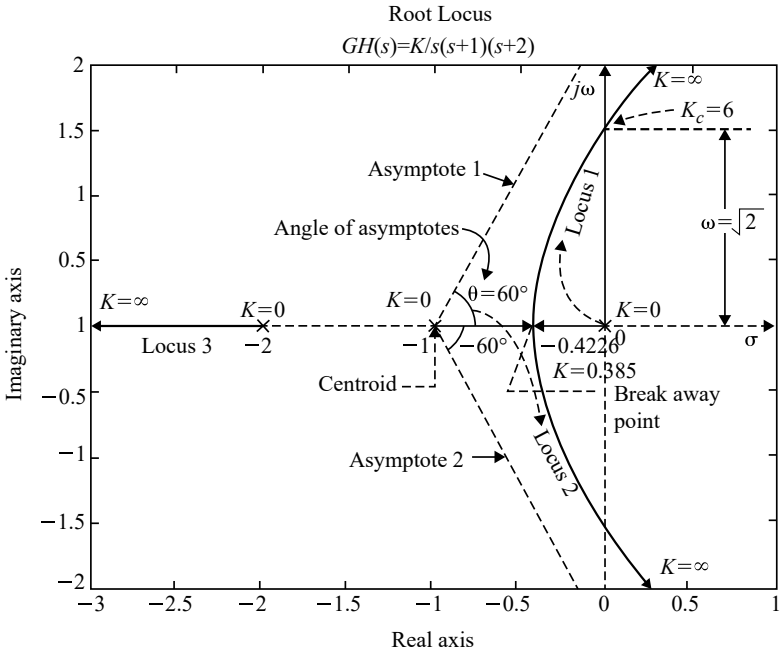


Fig. 5.8 Root locus for Example 5.1

10. After the root loci are drawn, the value of K at some intervals in the root loci are obtained using magnitude condition given by Eq. (5.3).

Example 5.1

A certain negative feedback control system has the following loop T.F.

$$GH(s) = \frac{K}{s(s+1)(s+2)}$$

Draw the root locus for $0 \leq K \leq \infty$.

(*Anna University, November, 2009*)

Solution

1.

$$GH(s) = \frac{K}{s(s+1)(s+2)}$$

$GH(s)$ is in pole–zero form. There are 3 poles at $s = 0$, $s = -1$ and $s = -2$. There are no zeros of $GH(s)$. The poles are located as shown in Fig. 5.8.

2. The order of the numerator polynomial of $GH(s)$ is zero and that of the denominator polynomial is 3. Hence, there should be three separate root loci.
3. The root loci on the real axis are identified between $s = 0$ and $s = -1$ and between $s = -2$ and $s = -\infty$.
4. The root loci start from the poles of $GH(s)$ where $K = 0$ and end at ∞ since there are no zeros of $GH(s)$. At $s = \infty$, the value of $K = \infty$. (**There are ∞ number of poles and zeros at $s = \infty$**).
5. The angle of the asymptote is given by

$$\theta = \frac{(2n+1) \times 180^\circ}{(P - Z)}$$

where $P = 3$ and $Z = 0$; $n = 0, 1, 2$

$$\theta = 60^\circ, 180^\circ \text{ and } 300^\circ$$

6. The centroid (intersection of the asymptote with the real axis) is calculated from

$$\sigma = \frac{\sum \text{real part of poles of } GH(s) - \sum \text{real part of zeros of } GH(s)}{(P - Z)}$$

$$\sigma = \frac{0 - 1 - 2 - 0}{3} = -1$$

The two asymptotes with $\theta = 60^\circ$ and $\theta = 300^\circ$ or -60° are drawn in dashed lines. The third asymptote with $\theta = 180^\circ$ coincides with the negative real axis.

7. The root loci starting from $s = 0$ and $s = -1$ move on the negative real axis between $s = 0$ and $s = -1$, meet each other and breakaway. The breakaway point is calculated as follows. The characteristic equation of the system is

$$1 + GH(s) = 0$$
$$1 + \frac{K}{s(s+1)(s+2)} = 0$$
$$s^3 + 3s^2 + 2s + K = 0$$

Differentiating the above equation with respect to s, we get

$$3s^2 + 6s + 2 + \frac{dK}{ds} = 0$$

Putting $dK/ds = 0$, we get

$$3s^2 + 6s + 2 = 0$$

Solving for s, we get the breakaway points at

$$s = -51.5774 \quad \text{and} \quad s = -0.4226$$

Since at $s = -1.5774$, there is no root locus, the point $s = -0.4226$ alone is considered. At the breakaway point, the gain is calculated as

$$\begin{aligned} K &= 0.4226 \times 1.5774 \times 0.5774 \\ &= 0.385 \end{aligned}$$

8. The critical value of K which is denoted as K_c when the root loci cross from LHP to RHP is determined along with the frequency ω at that point as follows. The characteristic equation of the system is

$$s^3 + 3s^2 + 2s + K = 0$$

The following Routh's array is formed.

s^3	1	2
s^2	3	K
s^1	$\frac{(6-K)}{3}$	
s^0	K	

The system becomes oscillatory when $K = 6$. Hence, the critical value of K is $K_c = 6$. The auxiliary equation is

$$3s^2 + K_c = 0$$

Substituting $s = j\omega$ and $K_c = 6$ in the above equation, we get

$$\begin{aligned} -3\omega^2 + 6 &= 0 \\ \omega &= \sqrt{2} \end{aligned}$$

9. The third root loci which starts from $s = -2$ moves towards the left on the $-$ve real axis and goes to $-\infty$, where $K = \infty$.
10. The asymptotes, the breakaway point and the intersection of the root loci with the imaginary axes give guidance for the movements of the root loci. They are joined by a smooth curve and the complete root locus diagram is shown in Fig. 5.8.

Example 5.2

Consider the following loop T.F. of a certain closed loop control system.

$$GH(s) = \frac{K}{s(s+1)(s+2)(s+3)}$$

Draw the root locus for $0 \leq K \leq \infty$.

(*Anna University, April, 2010*)

Solution

1. $$GH(s) = \frac{K}{s(s+1)(s+2)(s+3)}$$

   There are four poles and no zeros of $GH(s)$. Hence, there should be four root loci.
2. The poles are at $s = 0$; $s = -1$; $s = -2$; and $s = -3$. The root loci start from these four poles where $K = 0$, and they all end at ∞ in the s-plane.
3. On the real axis, root loci exist between $s = 0$ and $s = -1$ and between $s = -2$ and $s = -3$. In these regions, to the right of any point odd number of poles exist.
4. The angles of asymptotes are calculated as

   $$\theta = \frac{(2n+1) \times 180^\circ}{(P - Z)}$$
   $$= \frac{(2n+1) \times 180^\circ}{4}$$

   where $n = 0, 1, 2, 3$

   $$\theta = 45^\circ, 135^\circ, 225^\circ \quad \text{and} \quad 315^\circ$$

5. The centroid is calculated as

   $$\sigma = \frac{\sum \text{real part of poles } GH(s) - \sum \text{real parts of zeros of } GH(s)}{(P - Z)}$$
   $$= \frac{0 - 1 - 2 - 3}{4}$$
   $$= -1.5$$

6. The breakaway points are calculated using the characteristic equation which is given by

$$s(s+1)(s+2)(s+3)+K=0$$
$$s^4+6s^3+11s^2+6s+K=0$$

Differentiating the above equation with respect to s and putting $dK/ds=0$, we get

$$4s^3+18s^2+22s+6=0$$

Solving the above equation for s, we get

$$s_1=-2.5$$
$$s_2=-0.3$$
$$s_3=-1.7$$

At $s=-1.7$, there is no root locus and it is not a breakaway point. Hence, s_1 and s_2 are taken as breakaway points.

7. The intersections of the root loci with the imaginary axis is calculated from Routh's array formed by the characteristic equation

s^4	1	11	K
s^3	6	6	
s^2	10	K	
s^1	$\frac{60-6K}{10}$		
s^0	K		

From Routh's array the critical value of gain $K_c=10$. The auxiliary equation is

$$10s^2+K_c=0$$

Substituting $s=j\omega$ and $K_c=10$, we get $\omega=1$. The value of K_c obtained can be verified using magnitude condition. According to the magnitude condition, at any point on the root locus, the value of K is

$$K=\frac{\text{Product of directed distances drawn from all the poles to that point}}{\text{Product of directed distances drawn from all the zeros to that point.}}$$

Since the root loci intersect the imaginary axis at $\omega=1$,

$$K_c=1\times\sqrt{2}\times\sqrt{5}\sqrt{10}=10$$

The complete root locus is shown in Fig. 5.9.

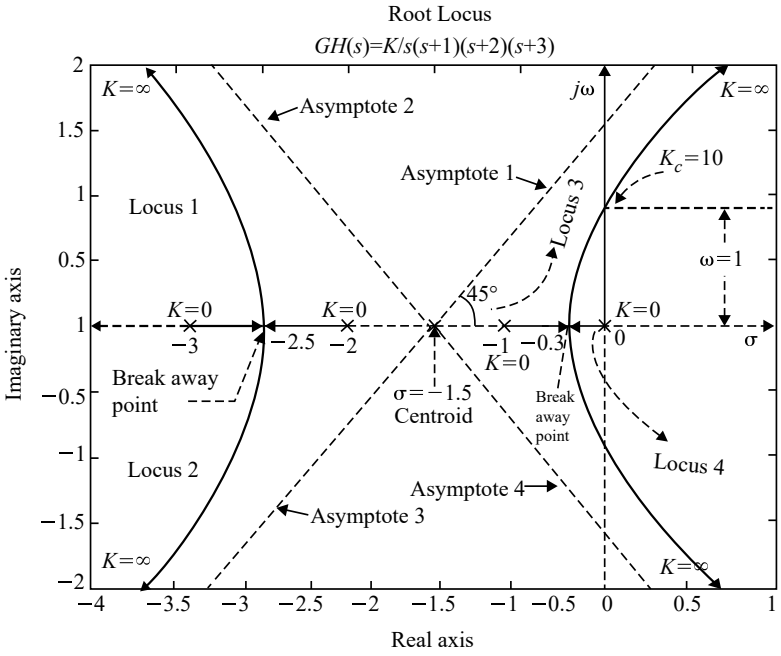


Fig. 5.9 Root locus for Example 5.2

To Determine Closed Loop System Poles for the given Gain K

Example 5.3

A certain closed loop system has the following loop T.F.:

$$GH(s) = \frac{K}{s(s+4)(s^2+4s+20)}$$

Draw the root locus for $0 \leq K \leq \infty$. Identify the following:

1. Breakaway points and the value of K at these points.
2. Critical value of K and the oscillation frequency.
3. The angle of departure and the angle of arrival if any.
4. Find the closed loop system poles for $K = 3000$ and also the closed loop T.F.
5. Is the system stable for this value of K?

(*Anna University, December, 2010*)

Solution

1.

$$GH(s) = \frac{K}{s(s+4)(s^2+4s+20)}$$
$$(s^2+4s+20) = (s+2+j4)(s+2-j4)$$
$$GH(s) = \frac{K}{s(s+2)(s+2+j4)(s+2-j4)}$$

There are four poles and no zero for the T.F. There should be four root loci. The poles of $GH(s)$ are located in the s-plane as shown in Fig. 5.10.

2. The root loci start from the poles $s = 0, s = -2, s = -2 + j4$ and $s = -2 - j4$ where the value of $K = 0$ and end at ∞ where $K = \infty$.
3. The root loci, on the real axis, exist in between $s = 0$ and $s = -4$ where odd number of poles exists to the right of any point in the negative real axis.
4. The angles of the asymptotes are given by

$$\theta = \frac{(2n+1) \times 180^\circ}{(P-Z)}$$
$$= \frac{(2n+1) \times 180^\circ}{4}$$

where $n = 0, 1, 2, 3$

$$\theta = 45^\circ, 135^\circ, 225^\circ \quad \text{and} \quad 315^\circ$$

5. The centroid is given by

$$\sigma = \frac{\sum \text{real part of poles of } GH(s) - \sum \text{real part of zeros of } GH(s)}{(P-Z)}$$
$$= \frac{0-2-2-4}{4}$$
$$= -2$$

6. The breakaway points of the root loci are obtained from the characteristic equation $1 + GH(s) = 0$.

$$s^4 + 8s^3 + 36s^2 + 80s + K = 0$$

Differentiating the above equation with respect to s, we get

$$4s^3 + 24s^2 + 72s + 80 + \frac{dK}{ds} = 0$$

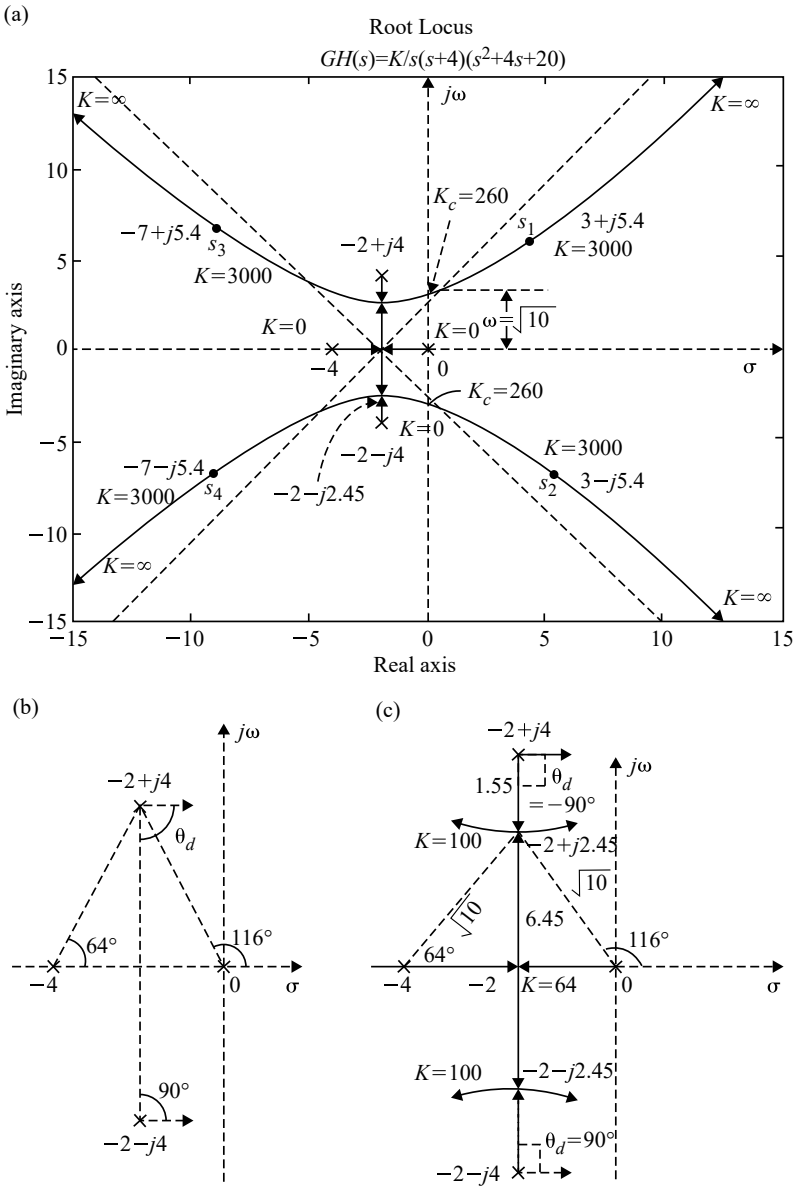


Fig. 5.10 Root locus for Example 5.3

Putting $dK/ds = 0$ and solving for s, we get

$$\begin{aligned} s_1 &= -2 \\ s_2 &= -2 + j2.45 \\ s_3 &= -2 - j2.45 \end{aligned}$$

7. The angle of departure of the root locus from the pole at $s = -2 + j4$ is obtained by drawing vectors from $s = 0$, $s = -4$ and $s = -2 - j4$. Thus, as shown in Fig. 5.10c:

$$\begin{aligned} \theta_d &= 180^\circ - (90^\circ + 116^\circ + 64^\circ) \\ &= -90^\circ \end{aligned}$$

For the conjugate pole at $s = -2 - j4$, we will get the conjugate of the other pole as $\theta_d = +90^\circ$.

8. The intersection of the root loci with the imaginary axis is obtained as follows. The following Routh's array is formed for the characteristic equation.

s^4	1	36	K
s^3	1	10	
s^2	26	K	
s^1	$260 - K$		
s^0	K		

From Routh's array, the critical value of gain is obtained as $K_c = 260$. At this value of K, the root locus crosses from LHP to RHP. The point at which it is critical is determined by taking the auxiliary equation, which is obtained from Routh's array as

$$26s^2 + K_c = 0$$

Substituting $s = j\omega$ and $K_c = 260$, we get

$$\omega = \sqrt{10}$$

9. The root loci are drawn as follows. The root loci starting from $s = 0$ and $s = -4$ meet each other at $s = -2$ and break away and move in the complex s-plane, one vertically upward and another vertically downward. These loci and the loci starting from the poles $s = -2 \pm j4$ meet each other and break away. The breakaway points as determined in step 6 are at $s = -2 \pm j2.45$. These breaking loci move towards the right and towards the left taking guidance from the asymptotes as shown in Fig. 5.10a.
10. To find the roots for $K = 3000$, choose any point s_1 on any one root locus. Draw vectors from all the poles of $GH(s)$ to the point s_1. The value so obtained gives the value of K. **This is obtained by trial and error only**. For one such point $s_1 = 3 + j5.4$, the value of K is calculated as

$$\begin{aligned} K &= 6.1 \times 8.8 \times 10.6 \times 5.3 \\ &= 3000 \end{aligned}$$

For this value of K, for the conjugate root locus, the root is at

$$s_2 = -3 - j5.4$$

By searching in the third and fourth root loci, by trial and error for $K = 3000$, we get

$$\begin{aligned} s_3 &= -7 + j5.4 \\ s_4 &= -7 + j5.4 \end{aligned}$$

11. At the breakaway points, the values of K are calculated by referring to Fig. 5.10c. At the breakaway point $s = -2$

$$\begin{aligned} K &= 2 \times 2 \times 4 \times 4 \\ &= 64 \end{aligned}$$

At the breakaway point $s = -2 + j2.45$,

$$\begin{aligned} K &= \sqrt{10} \times \sqrt{10} \times 6.45 \times 1.55 \\ &= 100 \end{aligned}$$

12. The closed loop transfer function is obtained as

$$\begin{aligned} \frac{C(s)}{R(s)} &= \frac{G(s)}{1 + G(s)} \\ &= \frac{K}{s^4 + 8s^3 + 36s^2 + 8s + K} \end{aligned}$$

For $K = 3000$,

$$\begin{aligned} \frac{C(s)}{R(s)} &= \frac{3000}{[(s + 7 + j5.4)(s + 7 - j5.4)(s - 3 + j5.4)(s - 3 - j5.4)]} \\ &= \frac{3000}{(s^2 + 14s + 78.2)(s^2 - 6s + 38.2)} \end{aligned}$$

13. Since there are 2RHP roots for $K = 3000$, the closed loop system is unstable.

Answers:

1. Breakaway points -2 with $K = 64$ and $-2 \pm j2.48$ with $K = 100$
2. $K_c = 260$ and $\omega = \sqrt{10}$
3. $\theta_d = -90^\circ$ and $+90^\circ$. No angle of arrival
4. Closed loop poles for $K = 3000$
   $s_1 = 3 + j5.4$
   $s_2 = 3 - j5.4$
   $s_3 = -7 + j5.4$
   $s_4 = -7 - j5.4$
5. The closed loop T.F. for $K = 3000$
   $$\frac{C(s)}{R(s)} = \frac{3000}{(s^2 + 14s + 78.2)(s^2 - 6s + 38.2)}$$
6. The closed loop system is unstable

To Get Root Locus by MATLAB Program

Example 5.4

Consider the following loop T.F. of a certain closed loop system

$$GH(s) = \frac{K(s^2 + 2s + 10)}{s(s + 2)(s + 4)}$$

Draw the root locus for $0 \leq K \leq \infty$ using the MATLAB program. From the root locus so obtained, verify the angle of arrival and the breakaway point. At the breakaway point, what is the value of K? For this value of K, determine the closed loop T.F.

Solution

$$GH(s) = \frac{K(s^2 + 2s + 10)}{s(s + 2)(s + 4)}$$

The following MATLAB program is used to plot the root locus.

MATLAB Program
To draw Root Locus

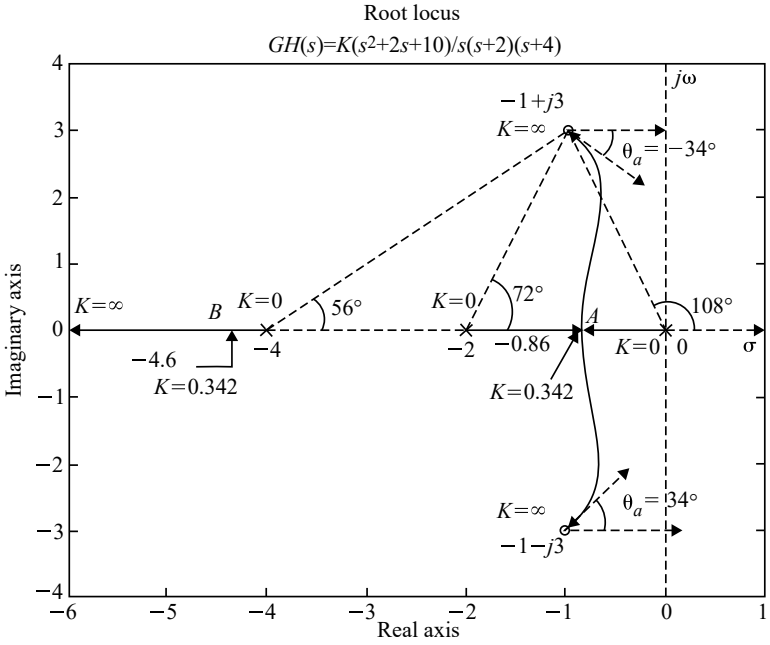


Fig. 5.11 Root locus for Example 5.4

```
clc;
% Let K = 1
num=[0 1 2 10];
den=conv([1 0], conv([1 2], [1 4]);
rlocus(num,den);
title('Root Locus');
xlabel('Real axis');
ylabel('Imaginary axis');
```

1.

$$GH(s) = \frac{K(s^2 + 2s + 10)}{s(s+2)(s+4)}$$

$$(s^2 + 2s + 10) = (s + 1 + j3)(s + 1 - j3)$$

$$GH(s) = \frac{K(s + 1 + j3)(s + 1 - j3)}{s(s+2)(s+4)}$$

The poles and zeros are marked in the s-plane as shown in Fig. 5.11.

2. The root loci on the real axis exist between $s = -4$ and $s = -\infty$ and between $s = 0$ and $s = -2$ as shown in Fig. 5.11. The root locus starting from $s = -4$

moves towards the left and ends at $-\infty$. The root loci starting from $s = 0$ and $s = -2$ meet each other and break away and end at the zeros at $s = -1 + j3$ and $s = -1 - j3$, where the value of $K = \infty$.

3. The breakaway point is obtained from the characteristic equation which is given by

$$(s^3 + 6s^2 + 8s) + K(s^2 + 2s + 10) = 0$$

$$K = \frac{-(s^3 + 6s^2 + 8s)}{(s^2 + 2s + 10)}$$

Differentiating K with respect to s, we get

$$\frac{dK}{ds} = 0$$

$$= s^3 + 8.5s^2 + 30s + 20$$

Solving the above equation, we get

$$s_1 = -0.86$$
$$s_2 = -3.83 + j3$$
$$s_3 = -3.83 - j3$$

4. The angle of arrival of the root locus at the zero at $s = -1 + j3$ is obtained as

$$\theta_a = 180^\circ - (90^\circ - 108^\circ - 72^\circ - 56^\circ)$$
$$= -34^\circ$$

and at $s = -1 - j3$ is obtained as

$$\theta_a = +34^\circ$$

Since there are no root loci at s_2 and s_3, as seen in Fig. 5.11, these points are not the breakaway points. Hence, there is only one breakaway point which is at $s = -0.86$. The gain K at point A marked in Fig. 5.9 is calculated as follows.

$$K\Big|_A = \frac{\text{Product of directed distances drawn from poles of } GH(s) \text{ to point } A}{\text{Product of directed distances drawn from zeros of } GH(s) \text{ to point } A}$$
$$= \frac{0.86 \times 1.14 \times 3.14}{3.01 \times 3.01}$$
$$= 0.342$$

5. There are two closed loop roots at point A where the value of the gain $K = 0.342$. In the third root locus which lies between $s = -4$ and $s = -\infty$, there will be

another root. This is searched by trial and error. This is obtained at point B in Fig. 5.11. The point B is obtained from

$$K\Big|_B = \frac{0.6 \times 4.6 \times 2.6}{4.6 \times 4.6}$$
$$= 0.342$$

The three closed loop poles are at $s = -0.86$, $s = -0.86$ and $s = -4.6$. The closed loop T.F. is obtained from

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + GH(s)}$$
$$= \frac{K(s + 1 + j3)(s + 1 - j3)}{(s^3 + 6s^2 + 8s) + K(s^2 + 2s + 10)}$$

For $K = 0.342$,

$$\frac{C(s)}{R(s)} = \frac{0.342(s + 1 + j3)(s + 1 - j3)}{(s + 0.86)^2(s + 4.6)}$$

$$\boxed{\frac{C(s)}{R(s)} = \frac{0.342(s^2 + 2s + 10)}{(s + 0.86)^2(s + 4.6)}}$$

To Determine the Closed Loop Poles for the given Damping Ratio of Dominant Poles

---

Example 5.5

A certain closed loop system has the following open loop T.F.

$$GH(s) = \frac{K(s + 2)}{s(s + 1)(s + 4)}$$

Draw the root locus for $0 \leq K \leq \infty$. For the damping ratio of $\zeta = 0.707$, determine the value of K and find closed loop poles and the T.F.

Solution

1.

$$GH(s) = \frac{K(s + 2)}{s(s + 1)(s + 4)}$$

There are three poles at $s = 0$, $s = -1$ and $s = -4$ and one zero at $s = -2$. There will be three root loci. They start from the poles of $GH(s)$. One locus ends at the zero and the remaining two end at ∞.

2. The centroid of the root locus is

$$\sigma = \frac{0 - 1 - 4 + 2}{2} = -1.5$$

The angles of the asymptotes are

$$\theta = \frac{(2n + 1) \times 180^\circ}{2} = 90^\circ, 270^\circ$$

3. The characteristic equation is

$$\begin{aligned} s(s + 1)(s + 4) + K(s + 2) &= 0 \\ K &= \frac{-s(s + 1)(s + 4)}{(s + 2)} \\ &= \frac{-(s^3 + 5s^2 + 4s)}{(s + 2)} \end{aligned}$$

Differentiating K with respect to s, we get

$$(s + 2)(3s^2 + 10s + 4) - (s^3 + 5s^2 + 4s) = \frac{dK}{ds}$$

where $dK/ds = 0$.

$$\begin{aligned} s^3 + 5.5s^2 + 10s + 4 &= 0 \\ (s + 0.55)(s + 2.45 + j1.313)(s + 2.45 - j1.313) &= 0 \end{aligned}$$

Since root loci do not break away in the complex s-plane in this particular case, the breakaway point at $s = -0.55$ is taken. At this point, the value of the gain is

$$\begin{aligned} K &= \frac{0.55 \times 0.45 \times 3.45}{1.45} \\ &= 0.59 \end{aligned}$$

4. Smooth curves from the breakaway point are drawn in the complex s-plane, taking guidance from the asymptote. It is checked at a few points that these two loci satisfy the angle criterion of root locus. That is at any point on the root locus, the total sum of the angles contributed by all the zeros minus the total sum of all the angles contributed by the poles should be an odd multiple of 180°. The

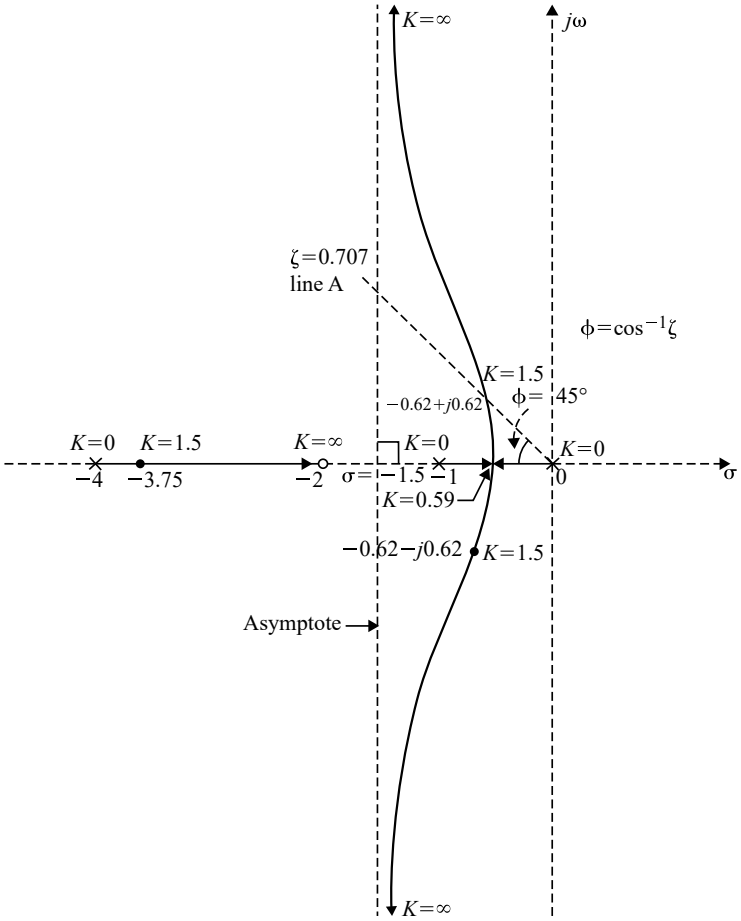


Fig. 5.12 Root locus for $GH(s) = \frac{K(s+2)}{s(s+1)(s+4)}$

third root locus which starts from $s = -4$ moves towards the right and ends at the zero at $s = -2$.

5. Damping factor $\zeta = 0.707$ corresponds to $\phi = \cos^{-1} 0.707 = 45°$. The straight line OA with $\phi = 45°$ is drawn as shown in Fig. 5.12. It cuts the root locus at $s = -0.62 + j0.62$. The value of K at this point is calculated using magnitude criterion as

$$K = \frac{0.87 \times 0.73 \times 3.45}{1.49}$$
$$= 1.5$$

6. In the third root loci, by trial and error it is calculated that at $s = -3.75$, the value of $K = 1.5$.

$$K = \frac{0.25 \times 2.75 \times 3.75}{1.75}$$
$$= 1.5$$

Thus, corresponding to $K = 1.5$, the three closed loop roots are

$$s_1 = -0.62 + j0.62$$
$$s_2 = -0.62 - j0.62$$
$$s_3 = -3.75$$

7. The closed loop T.F. is

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)}$$
$$= \frac{K(s+2)}{s(s+1)(s+4) + K(s+2)}$$

For $K = 1.5$

$$\frac{C(s)}{R(s)} = \frac{1.5(s+2)}{(s+3.75)(s+0.62+j0.62)(s+0.62-j0.62)}$$

$$\boxed{\frac{C(s)}{R(s)} = \frac{1.5(s+2)}{(s+3.75)(s^2+1.24s+0.7688)}}$$

To Find the Transient Response of the System

Example 5.6

Consider the following loop T.F. of a certain unity feedback system

$$GH(s) = \frac{K}{(s+4)(s^2+2s+2)}$$

Draw the root locus for $0 \leq K \leq \infty$. For $K = 20$, find the closed loop system poles and hence determine the transient response for a unit step input.

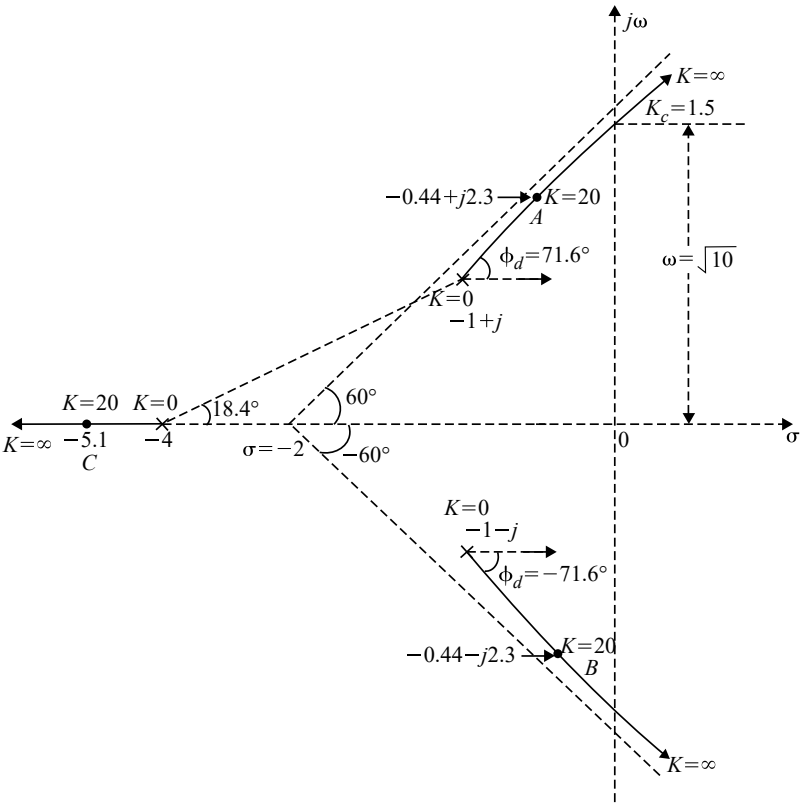


Fig. 5.13 Root locus for $GH(s) = \frac{K}{(s+4)(s^2+2s+2)}$

Solution

1.

$$GH(s) = \frac{K}{(s+4)(s^2+2s+2)}$$
$$= \frac{K}{(s+4)(s+1+j)(s+1-j)}$$

Since the denominator polynomial order is three, there will be three root loci and they start from the poles of $GH(s)$ where the value of $K = 0$ and end at $s = \infty$ where $K = \infty$. The open loop poles are located as shown in Fig. 5.13.

2. The root locus on the real axis exists between $s = -4$ and $s = -\infty$. The other two root loci are complex conjugate.
3. The centroid of the root loci is at

$$\sigma = \frac{-4-1-1}{3}$$
$$= -2$$

The angles of the asymptotes are

$$\theta = \frac{(2n+1) \times 180^\circ}{3}$$
$$= 60^\circ, 180^\circ \text{ and } 300^\circ$$

4. The characteristic equation is

$$1 + GH(s) = 0$$
$$s^3 + 6s^2 + 10s + (K+8) = 0$$

The following Routh's array is formed.

s^3	1	10
s^2	6	$8+K$
s^1	$52-K$	
s^0	$8+K$	

For the system to be critically stable, $K_c = 52$. The auxiliary equation is

$$6s^2 + (8+K) = 0$$

At $s = j\omega$, and $K = K_c = 52$, the above equation is written as

$$6\omega^2 = 60$$
$$\omega = \sqrt{10}$$

5. The breakaway point is obtained from the characteristic equation

$$K = -(s^3 + 6s^2 + 10s + 8)$$
$$\frac{dK}{ds} = 0$$
$$= s^2 + 4s + 3.3$$

Solving the above equation for s, we get

$$s_1 = -2.837$$
$$s_2 = -1.16$$

In these points, there is no root locus. Hence, there is no breakaway point.

6. The angles of departure from the complex poles are obtained as follows. For the pole at $s = -1 + j$,

$$\begin{aligned}\theta_d &= 180^\circ - (\theta_P - \theta_Z) \\ &= 180^\circ - (90^\circ + 18.4^\circ) \\ &= 71.6^\circ\end{aligned}$$

For the pole at $s = -1 - j$, $\theta_d = -71.6^\circ$.

7. From the information obtained above and taking guidance from the asymptotes, the three root loci are drawn. The closed loop roots for the given K are determined from each locus by trial and error using the magnitude condition

$$K = \text{Product of directed vector distances drawn from the poles of } GH(s)$$

Any point A is chosen in one root locus and vectors are drawn from all the poles of $GH(s)$, and K is found

$$\begin{aligned}K &= 4.3 \times 1.4 \times 3.4 \\ &= 20\end{aligned}$$

Point A corresponds to $s = -0.44 + j2.3$. By conjugate property at point B, $s = -0.44 - j2.3$. These are the two closed loop poles. In this locus at point C, by trial and error the value of K is calculated at $s = -5.1$.

$$\begin{aligned}K &= 4.3 \times 4.3 \times 1.1 \\ &= 20\end{aligned}$$

The third root is at $s = -5.1$.
Corresponding to $K = 20$, the three closed loop roots are obtained as $(s + 5.1)(s + 0.44 + j2.3)(s + 0.44 - j2.3)$.

8. The closed loop T.F. is

$$\begin{aligned}\frac{C(s)}{R(s)} &= \frac{G(s)}{1 + G(s)} \\ &= \frac{K}{(s+4)(s^2 + 2s + 2) + K}\end{aligned}$$

For $K = 20$,

$$\frac{C(s)}{R(s)} = \frac{20}{(s + 5.1)(s + 0.44 + j2.3)(s + 0.44 - j2.3)}$$

For unit step input, $R(s) = 1/s$

$$C(s) = \frac{20}{s(s+5.1)(s+0.44+j2.3)(s+0.44-j2.3)}$$

This is put into partial fraction as

$$C(s) = \frac{A_1}{s} + \frac{A_2}{(s+5.1)} + \frac{A_3}{(s+0.44+j2.3)} + \frac{A_4}{(s+0.44-j2.3)}$$
$$20 = A_1(s+5.1)(s^2+0.88s+5.5) + A_2 s(s^2+0.88s+5.5)$$
$$+A_3 s(s+5.1)(s+0.44-j2.3) + A_4 s(s+5.1)(s+0.44+j2.3)$$

For $s = 0$,

$$A_1 = 0.714$$

For $s = -5.1$,

$$A_2 = \frac{20}{(-5.1)((-5.1)^2 - 0.88 \times 5.1 + 5.5)}$$
$$= -0.145$$

For $s = -0.44 - j2.3$,

$$A_3 = \frac{20}{(-.44-j2.3)(-.44-j2.3+5.1)(-j4.6)}$$
$$= \frac{20}{2.38\angle 79^\circ 5.2\angle -26^\circ \times 4.6\angle 90^\circ}$$
$$= 0.35\angle -143^\circ$$
$$A_4 = 0.35\angle 143^\circ$$
$$C(s) = \frac{0.714}{s} - \frac{0.145}{(s+5.1)} + \frac{e^{-j143^\circ}0.35}{s+0.44+j2.3} + \frac{e^{+j143^\circ}0.35}{s+0.44-j2.33}$$

Taking inverse Laplace transform, we get

$$c(t) = 0.714 - 0.145e^{-5.1t} + 0.35e^{-0.44t}[e^{-j143^\circ}e^{-j2.3t} + e^{j143^\circ}e^{j2.3t}]$$
$$= 0.714 - 0.145e^{-5.1t} + 0.7e^{-0.44t}\cos(2.3t + 143^\circ)$$

$$\boxed{c(t) = 0.714 - 0.145e^{-5.1t} - 0.7e^{-0.44t}\sin(2.3t + 0.93\text{rad})}$$

To Find Phase Margin and Gain Margin from Root Locus

To Find Gain Margin G_m

The root locus for a certain loop T.F. is shown in Fig. 5.14. Suppose K_d is the desired gain margin as shown in Fig. 5.14. The gain margin is the amount by which K_d is increased so that the system is not unstable. This is nothing but the ratio of K_c/K_d.

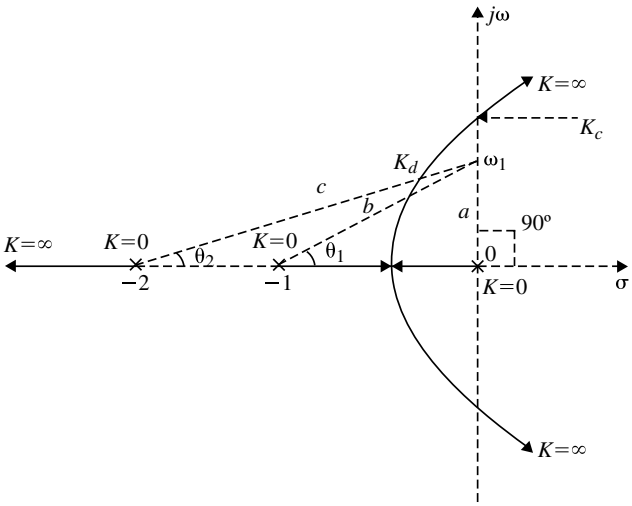


Fig. 5.14 To find phase margin and gain margin from root locus

$$\boxed{G_m = \frac{K_c}{K_d}}$$

To Find Phase Margin

For the given K_d, find ω_1 in the $j\omega$ axis which satisfies the magnitude condition of root locus. From Fig. 5.14, $K_d = abc$ at ω_1. Draw vectors from poles and zeros of $GH(j\omega_1)$ and find sum total of angles drawn from zeros of $GH(s)-$ Sum total of angles drawn from poles of $GH(s)$ to the point ω_1. Let it be ϕ. Then the phase margin is given by

$$\begin{aligned}\phi_m &= 180^\circ + GH(j\omega_1)\\ &= 180^\circ + \phi\end{aligned}$$

In Fig. 5.14, ϕ is given by

$$\phi = -(90^\circ + \theta_1 + \theta_2)$$

Example 5.7

The characteristic equation of a certain unity feedback control system is given by

$$F(s) = s(s+4)(s^2+6s+10) + K$$

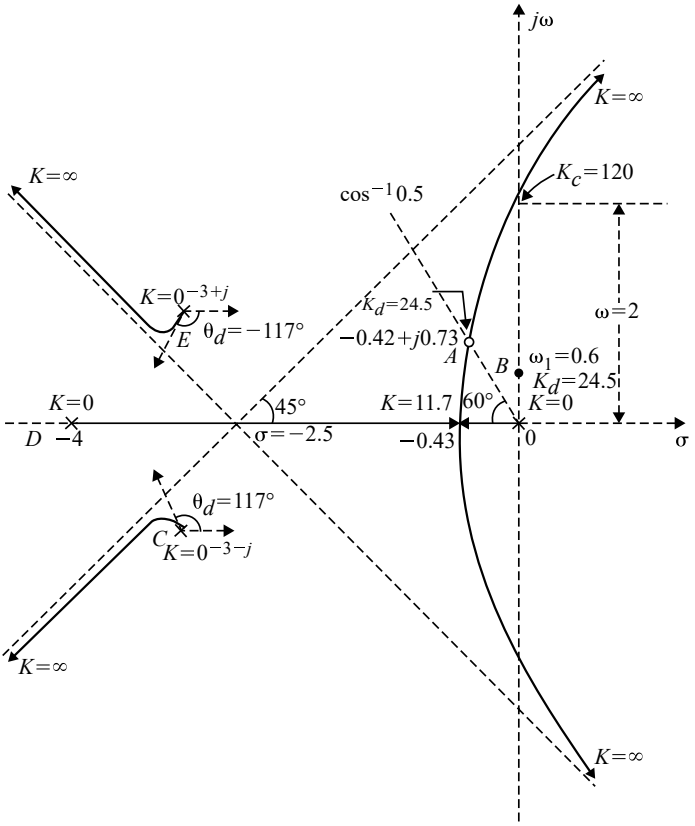


Fig. 5.15 Root locus to determine phase margin and gain margin

Draw the root locus for $0 \leq K \leq \infty$. For the damping ratio $\zeta = 0.5$ for the dominant roots find K. For this value of K, find the phase margin and gain margin.

Solution

1.

$$F(s) = s(s+4)(s^2+6s+10) + K$$

$$G(s) = \frac{K}{s(s+4)(s^2+6s+10)}$$

$$= \frac{K}{s(s+4)(s+3+j)(s+3-j)}$$

The poles are located as shown in Fig. 5.15. There are four poles and there should be four root loci.

2. The centroid of the root loci is

$$\sigma = \frac{0-4-3-3}{4}$$
$$= -2.5$$

The angles of the asymptotes are

$$\theta = \frac{(2n+1) \times 180^\circ}{4}$$

where $n = 0, 1, 2, 3$

$$\theta = 45^\circ, 135^\circ, 225^\circ, 315^\circ$$

3. From the characteristic equation, we get

$$K = -(s^4 + 10s^3 + 34s^2 + 40s)$$
$$\frac{dK}{ds} = 0 = 4s^3 + 30s^2 + 68s + 40$$

Solving the above equation, we get

$$s_1 = -0.43$$
$$s_2 = -3.5 + j1.22$$
$$s_3 = -3.5 - j1.22$$

For the breakaway point $s_1 = -0.43$, root loci exist. By intuition, the other two breakaway points do not have a root locus.

4. For the characteristic equation, the following Routh's array is prepared.

s^4	1	34	K
s^3	1	4	
s^2	30	K	
s^1	$120 - K$		
s^0	K		

The critical value of $K = 120$. The auxiliary equation is

$$30s^2 + K_c = 0$$

Substituting $s = j\omega$ and $K_c = 120$, we get

$$30\omega^2 = 120$$
$$\omega = 2$$

5. For the pole at $s = -3 + j$, the angle of departure is obtained as

$$\begin{aligned}\theta_d &= 180^\circ - (162^\circ + 90^\circ + 45^\circ) \\ &= -117^\circ\end{aligned}$$

For the pole at $s = -3 - j$, the angle of departure is

$$\theta_d = 117^\circ$$

6. Starting from the poles of $G(s)$ where the value of $K = 0$, the root loci from the complex poles move in the direction of the asymptotes with an angle of departure -117° and $+117^\circ$. The root loci, starting from $s = -4$ and $s = 0$, move on the $-$ve real axis, break at $s = -0.43$ and move in the complex s-plane. They cross the $j\omega$ axis with $K_c = 120$ and $\omega = 2$ rad/sec. A smooth graph is drawn.
7. For a damping ratio $\zeta = 0.5$, $\theta = \cos^{-1} 0.5 = 60^\circ$ line as shown in Fig. 5.15 is drawn. It cuts the root locus at point A which is at $s = -0.42 + j0.73$. The value of K at point A is the desired value which is denoted by K_d. At this point using magnitude condition, K_d is calculated as

$$\begin{aligned}K_d &= |\overrightarrow{OA}||\overrightarrow{DA}||\overrightarrow{EA}||\overrightarrow{CA}| \\ &= 0.836 \times 3.67 \times 2.58 \times 3.09 \\ &= 24.5\end{aligned}$$

8. The gain margin is given by

$$\begin{aligned}G_m &= \frac{K_c}{K_d} \\ &= \frac{120}{24.5} = 4.9\end{aligned}$$

9. To determine the phase margin, on the $j\omega$ axis, point B is chosen where the gain is $K_d = 24.5$. At ω_1,

$$\begin{aligned}K_d &= |\overrightarrow{OB}||\overrightarrow{CB}||\overrightarrow{DB}||\overrightarrow{EB}| \\ &= 0.6 \times 3.3 \times 4.1 \times 3 \\ &= 24.5\end{aligned}$$

This is done by trial and error. ω_1 is calculated as 0.6 rad/sec. From $G(s)$

$$\begin{aligned}G(j\omega) &= \frac{24.5}{(j\omega)(j\omega + 4)(-\omega^2 + j6\omega + 10)} \\ \angle G(j\omega) &= -90^\circ - \tan^{-1}\frac{\omega}{4} - \tan^{-1}\frac{6\omega}{10 - \omega^2}\end{aligned}$$

Substituting $\omega = 0.6$,

$$\begin{aligned}\phi = \angle G(j\omega)_{\omega=0.6} &= -90^\circ - \tan^{-1}\frac{0.6}{4} - \tan^{-1}\frac{6\times 0.6}{10-0.36}\\ &= -90^\circ - 8.5^\circ - 20.5^\circ\\ &= -119^\circ\end{aligned}$$

Hence, phase margin

$$\begin{aligned}\phi_m &= 180^\circ - 119^\circ\\ &= 61^\circ\end{aligned}$$

The same result can also be obtained graphically by drawing vectors from all the poles of $G(s)$ to point B, and total sum is subtracted from 180° to get the phase margin

$$\begin{aligned}\phi &= \angle\overrightarrow{OB} + \angle\overrightarrow{CB} + \angle\overrightarrow{DB} + \angle\overrightarrow{EB}\\ &= 90^\circ + 29^\circ + 9^\circ - 9^\circ\\ &= 119^\circ\\ \phi_m &= 180^\circ - 119^\circ\\ &= 61^\circ\end{aligned}$$

10.

Gain margin, $G_m = 24.5$
Phase margin, $\phi_m = 61^\circ$

Example 5.8

Consider the following loop T.F. of a certain unity feedback control system

$$GH(s) = \frac{K(s+3)}{s(s+2)(s^2+4s+13)}$$

(a) Draw the root locus for $0 \le K \le \infty$.
(b) Identify whether there are any repeated roots.
(c) For each of the cases in (b), find the closed loop T.F.

(*Anna University, May, 2010*)

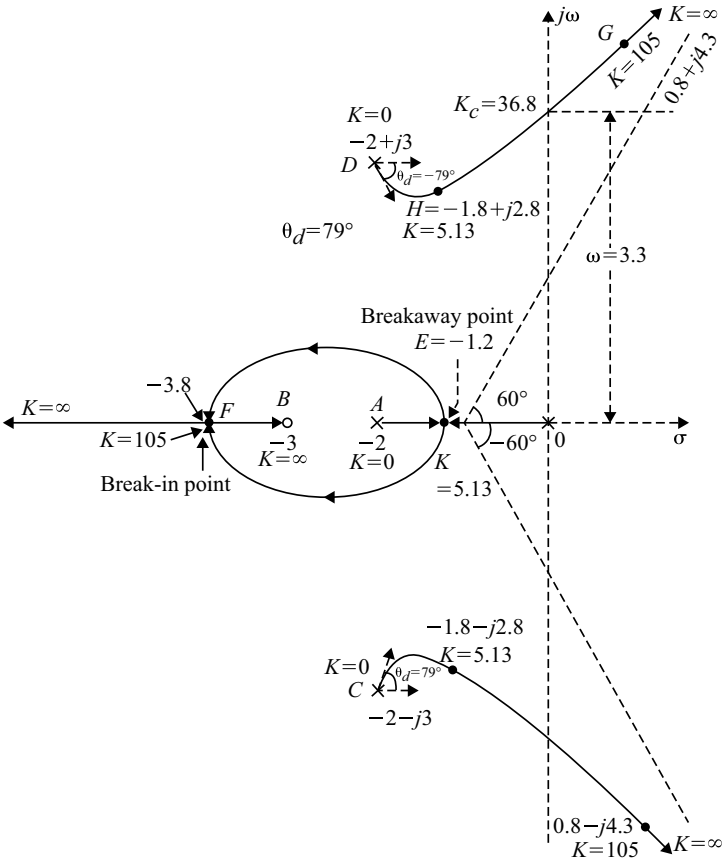


Fig. 5.16 Root locus for $GH(s) = \frac{K(s+3)}{s(s+2)(s^2+4s+13)}$

Solution

1.

$$GH(s) = \frac{K(s+3)}{s(s+2)(s^2+4s+13)}$$
$$= \frac{K(s+3)}{s(s+2)(s+2+j3)(s+2-j3)}$$

The poles and zeros are located as shown in Fig. 5.16.

2. There are 4 poles and 1 zero. Hence, there should be four root loci which originate from the four poles where $K = 0$. One of them ends at the zero $s = -3$ and remaining end at $s = \infty$.
3. The centroid of the asymptote is given by

$$\sigma = \frac{-0-2-2-2+3}{3}$$
$$= -1$$

The angles of the asymptotes are

$$\theta = \frac{(2n+1) \times 180^\circ}{3}$$

where $n = 0, 1, 2$

$$\theta = 60^\circ, 180^\circ, 300^\circ$$

4. The characteristic equation is

$$1 + GH(s) = 0$$
$$1 + \frac{K(s+3)}{s(s+2)(s^2+4s+13)} = 0$$
$$s(s+2)(s^2+4s+13) + K(s+3) = 0$$
$$K = 0 = \frac{-(s^2+2s)(s^2+4s+13)}{(s+3)}$$

$$\frac{dK}{ds} = 0$$
$$= (s+3)(4s^3+18s^2+42s+26)$$
$$-(s^4+6s^3+21s^2+26s) - s^4+8s^3+25s^2+41s+26 = 0$$

Solving by trial and error, we get

$$s_1 = -3.8$$
$$s_2 = -1.2$$
$$s_3 = -1.5 + j1.77$$
$$s_4 = -1.5 - j1.77$$

5. Root loci on the negative real axis exist between $s = 0$ and $s = -2$ and $s = -3$ and $s = -\infty$. The breakaway points s_1 and s_2 are in the root loci on the $-$ve real axis. It can be shown that for s_3 and s_4, angle criterion of root loci will not be satisfied

$$GH(s) = \frac{K(s+3)}{s(s+2)(s+2+j3)(s+2-j3)}$$

At $s = -1.5 + j1.77$, the phase angle of $GH(j\omega)$ is

$$\begin{aligned} GH(j\omega) &= K(-1.5 + j1.77 + 3) \Big/ (-1.5 + j1.77)(-1.5 + j1.77 + 2) \\ &\quad \times (-1.5 + j1.77 + 2 + j3)(-1.5 + j1.77 + 2 - j3) \\ &= \frac{K(1.5 + j1.77)}{(-1.5 + j1.77)(0.5 + j1.77)(0.5 + j4.77)(-0.5 - j1.23)} \\ \angle GH(j\omega) &= \tan^{-1}\frac{1.77}{1.5} + 180^\circ + \tan^{-1}\frac{1.77}{1.5} - \tan^{-1}\frac{1.77}{0.5} \\ &\quad - \tan^{-1}\frac{4.77}{0.5} - \tan^{-1}\frac{1.23}{0.5} + 180^\circ \\ &= 50^\circ + 50^\circ - 74^\circ - 84^\circ - 68^\circ \\ &= -126^\circ \end{aligned}$$

This is not an odd multiple of 180°. So, this point is not a point on the root locus. By conjugate property, $s = -1.5 - j1.77$ is also not a point on the root locus. At point D

$$\begin{aligned} \theta_d &= 180^\circ - (90^\circ + 124^\circ + 90^\circ - 45^\circ) \\ &= -79^\circ \end{aligned}$$

6. There are two closed loop roots at the breakaway point $s = -1.2$ and the value of K at this point is

$$\begin{aligned} K &= \frac{|\overrightarrow{OE}||\overrightarrow{CE}||\overrightarrow{AE}||\overrightarrow{DE}|}{|\overrightarrow{BE}|} \\ &= \frac{1.2 \times 3.1 \times 0.8 \times 3.1}{1.8} \\ &= 5.13 \end{aligned}$$

7. There are two closed loop roots at the breakaway point at $s = -3.8$. The value of K at this point is

$$\begin{aligned} K &= \frac{|\overrightarrow{OF}||\overrightarrow{AF}||\overrightarrow{CF}||\overrightarrow{DF}|}{|\overrightarrow{BF}|} \\ &= \frac{3.8 \times 1.8 \times 3.5 \times 3.5}{0.8} \\ &= 105 \end{aligned}$$

8. The characteristic equation is

$$s^4 + 6s^3 + 21s^2 + (26 + K)s + 3K = 0$$

The following Routh's array is formed.

$$\begin{array}{c|ccc} s^4 & 1 & 21 & 3K \\ s^3 & 6 & 26+K & \\ s^2 & 100-K & 18K & \\ s^1 & -K^2-34K+2600 & & \\ s^0 & K & & \end{array}$$

From Routh's array, the following conditions are derived for stability:

1. $K < 100$
2. $-K^2 - 34K + 2600 > 0$
   $(-K + 36.8)(K + 71.0) > 0$
   $K > -71$ and $K < +36.8$
3. $K > 0$

The condition $K < 36.8$ satisfies the stability condition $K_c = 36.8$. The oscillating frequency is obtained from the auxiliary equation which is given by

$$(100 - 36.8)s^2 + 18 \times 36.8 = 0$$

Putting $s = j\omega$, we get

$$63.2\omega^2 = 18 \times 36.8$$
$$\omega = 3.3$$

The root locus is drawn using the data obtained above.

9. There are two closed loop roots at the breakaway point at $s = -3.8$. The value of K at this point is

$$\begin{aligned} K &= \frac{|\overrightarrow{OF}||\overrightarrow{AF}||\overrightarrow{CF}||\overrightarrow{DF}|}{|\overrightarrow{BF}|} \\ &= \frac{3.8 \times 1.8 \times 3.5 \times 3.5}{0.8} \\ &= 105 \end{aligned}$$

10. Corresponding to $K = 105$, a root is searched at point G by trial and error

$$\begin{aligned} K &= \frac{|\overrightarrow{OG}||\overrightarrow{AG}||\overrightarrow{CG}||\overrightarrow{DG}|}{|\overrightarrow{BG}|} \\ &= \frac{4.4 \times 5.13 \times 7.83 \times 3.09}{5.2} \\ &= 105 \end{aligned}$$

The point is at $s = 0.8 + j4.3$. In the fourth root locus, at the conjugate point for $K = 105$, the root is located at $s = 0.8 - j4.3$. Hence, corresponding to $K = 105$, the four closed loop roots are at

$$\begin{aligned} s_1 &= -3.8 \\ s_2 &= -3.8 \\ s_3 &= 0.8 + j4.3 \\ s_4 &= 0.8 - j4.3 \end{aligned}$$

Since two roots s_3 and s_4 are in RHP, the closed loop system is unstable. The closed loop T.F. is

$$\boxed{\frac{C(s)}{R(s)} = \frac{105(s+3)}{(s+3.8)^2(s^2 - 1.6s + 19.1)}}$$

11. Corresponding to $K = 5.13$, a root is searched at point H by trial and error

$$\begin{aligned} K &= \frac{|\overrightarrow{OH}||\overrightarrow{AH}||\overrightarrow{CH}||\overrightarrow{DH}|}{|\overrightarrow{BH}|} \\ &= \frac{3.33 \times 2.8 \times 5.9 \times 0.2828}{3.05} \\ &= 5.13 \end{aligned}$$

The point H is at $s = -1.8 + j2.8$. The fourth root is at $s = -1.8 - j2.8$ in the fourth root locus. The four closed loop poles are at

$$\begin{aligned} s_1 &= -1.2 \\ s_2 &= -1.2 \\ s_3 &= -1.8 + j2.8 \\ s_4 &= -1.8 - j2.8 \end{aligned}$$

All the four roots are in LHP, and the closed loop system is stable. The closed loop T.F. for $K = 5.13$

$$\frac{C(s)}{R(s)} = \frac{5.13(s+3)}{(s+1.2)^2(s^2 + 3.6s + 11)}$$

Circular Root Locus

Example 5.9

A certain feedback control system has the following loop T.F.

$$GH(s) = \frac{K(s+4)(s+5)}{(s+1)(s+3)}$$

Show that part of root locus is a circle for $0 \leq K \leq \infty$. Draw the actual root locus. Find the repeated roots of the closed loop system and the corresponding value of K and the closed loop T.F. in each case.

(*Anna University, December, 2007*)

Solution

1. $$GH(s) = \frac{K(s+4)(s+5)}{(s+1)(s+3)}$$

   There are two poles at $s = -1$ and $s = -3$ and two zeros at $s = -4$ and $s = -5$. The two root loci originating from the poles where the value of $K = 0$ end at the zeros where the value of $K = \infty$. In such problems, we will not get guidance from the asymptotes. However, theoretically it can be established that part of the root loci is a circle and from that exact root locus can be obtained as explained below.
2. Substituting $s = \sigma + j\omega$ in $GH(s)$, we get

$$GH(\sigma + j\omega) = \frac{K(\sigma + j\omega + 4)(\sigma + j\omega + 5)}{(\sigma + j\omega + 1)(\sigma + j\omega + 3)}$$

$$\angle GH((\sigma + j\omega)) = \tan^{-1}\frac{\omega}{\sigma+4} + \tan^{-1}\frac{\omega}{\sigma+5} - \tan^{-1}\frac{\omega}{\sigma+1} - \tan^{-1}\frac{\omega}{\sigma+3}$$

3. Applying angle criterion to the above equation, we get

$$\tan^{-1}\frac{\omega}{\sigma+4} + \tan^{-1}\frac{\omega}{\sigma+5} = 180^\circ + \tan^{-1}\frac{\omega}{\sigma+1} + \tan^{-1}\frac{\omega}{\sigma+3}$$

$$\tan^{-1}\frac{\frac{\omega}{\sigma+4} + \frac{\omega}{\sigma+5}}{[1 - \frac{\omega^2}{(\sigma+4)(\sigma+5)}]} = 180^\circ + \tan^{-1}\frac{\frac{\omega}{\sigma} + \frac{\omega+1}{\sigma+3}}{[1 - \frac{\omega^2}{(\sigma+1)(\sigma+3)}]}$$

$$\tan^{-1}\frac{\omega(2\sigma+9)}{(\sigma^2+9\sigma+20-\omega^2)} = 180^\circ - \tan^{-1}\frac{\omega(2\sigma+4)}{(\sigma^2+4\sigma+3-\omega^2)}$$

Taking tan on both sides, we get

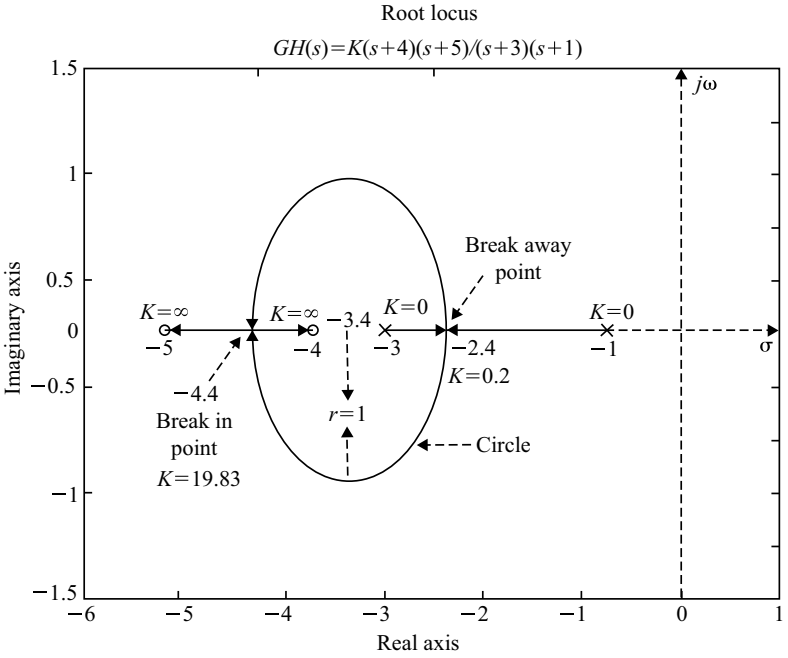


Fig. 5.17 Root locus for Example 5.9

$$\frac{(2\sigma+9)}{\sigma^2+9\sigma+20-\omega^2} = \frac{(2\sigma+4)}{(\sigma^2+4\sigma+3-\omega^2)}$$

$$(2\sigma+9)(\sigma^2+4\sigma+3-\omega^2) = (2\sigma+4)(\sigma^2+9\sigma+20-\omega^2)$$

$$\sigma^2+6.8\sigma+\omega^2+10.6 = 0$$

$$(\sigma+3.4)^2+\omega^2 = (1)^2$$

This is the equation of a circle. The centre of the circle is at $(-3.4, 0)$ and radius $r = 1$.

4. On the real axis, root loci exist between $s = -1$ and $s = -3$ and between $s = -4$ and $s = -5$. The root loci starting from poles $s = -1$ and $s = -3$ break at $s = -2.4$, traverse through the complex s-plane and break—in at $s = -4.4$ and end at $s = -4$ and $s = -5$. The root locus with centre at $s = -3.4$ and radius $r = 1$ is drawn as shown in Fig. 5.17.
5. At the breakaway and break-in points, there are repeated roots. At $s = -2.4$, the gain K is

$$K = \frac{0.6 \times 1.4}{1.6 \times 2.6}$$
$$= 0.2$$

The closed loop T.F. now is

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)} = \frac{\frac{K(s+4)(s+5)}{(s+1)(s+3)}}{1+\frac{K(s+4)(s+5)}{(s+1)(s+3)}}$$

For $K = 0.2$,

$$\frac{C(s)}{R(s)} = \frac{0.2(s+4)(s+5)}{(s+1)(s+3)+0.2(s+4)(s+5)} = \frac{0.2(s+4)(s+5)}{(1.2s^2+5.8s+7)} = \frac{0.2(s+4)(s+5)}{1.2(s^2+4.8s+5.8)}$$

$$\boxed{\frac{C(s)}{R(s)} = \frac{0.167(s+4)(s+5)}{(s+2.4)^2}}$$

6. At the break-in point, the value of K is

$$K = \frac{3.4 \times 1.4}{0.4 \times 0.6} = 19.83$$

The closed loop T.F. at this value of K is

$$\boxed{\frac{C(s)}{R(s)} = \frac{0.952(s+4)(s+5)}{(s+4.4)^2}}$$

Example 5.10

A certain closed loop system has the following open loop T.F.

$$GH(s) = \frac{K(s+5)}{s(s+3)}$$

Draw the root locus for $0 \leq K \leq \infty$. Determine the following:

(a) The breakaway and break-in points.
(b) The minimum value of damping that the system can provide and the corresponding value of K and the closed loop roots.
(c) The repeated roots and the corresponding closed loop T.F.
(d) The range of values of K for which the system is under-damped.
(e) The value of K for critical damping ratio.

Solution

1.

$$GH(s) = \frac{K(s+5)}{s(s+3)}$$

Substitute $s = \sigma + j\omega$.

$$GH(\sigma + j\omega) = \frac{K(\sigma + j\omega + 5)}{(\sigma + j\omega)(\sigma + j\omega + 3)}$$
$$\angle G(\sigma + j\omega) = \tan^{-1}\frac{\omega}{\sigma+5} - \tan^{-1}\frac{\omega}{\sigma} - \tan^{-1}\frac{\omega}{\sigma+3}$$

2. Root locus satisfies angle criterion of odd multiple of 180°

$$\tan^{-1}\frac{\omega}{\sigma+5} - \tan^{-1}\frac{\omega}{\sigma} - \tan^{-1}\frac{\omega}{\sigma+3} = 180°$$
$$\tan^{-1}\frac{\omega}{\sigma+5} - \tan^{-1}\frac{\omega}{\sigma+3} = 180° + \tan^{-1}\frac{\omega}{\sigma}$$
$$\tan^{-1}\frac{\omega(-2)}{(\sigma+5)(\sigma+3)+\omega^2} = 180 + \tan^{-1}\frac{\omega}{\sigma}$$

Taking tan on both sides, we get

$$-\sigma 2 = (\sigma+5)(\sigma+3) + \omega^2$$
$$-2\sigma = \sigma^2 + 8\sigma + 15 + \omega^2$$
$$\sigma^2 + 10\sigma + 15 + \omega^2 = 0$$
$$(\sigma+5)^2 + \omega^2 - 10 = 0$$
$$(\sigma+5)^2 + \omega^2 = (\sqrt{10})^2$$

This is the equation of a circle with centre at $(-5, 0)$ and radius $r = \sqrt{10} = 3.16$.

3. On the negative real axis, the root loci lie in between $s = 0$ and $s = -3$ and $s = -5$ and $s = -\infty$. The complete root locus is shown in Fig. 5.18.
4. The breakaway point is obtained in between $s = 0$ and $s = -3$ as -1.84 and the corresponding value of K at this point is

$$K = \frac{|\overrightarrow{OE}||\overrightarrow{BE}|}{|\overrightarrow{CE}|}$$

$$K = \frac{1.84 \times 1.16}{3.16}$$
$$= 0.675$$

5. The break-in point is obtained in between $s = -5$ and $s = -\infty$ as -8.16. The corresponding value of K at this point is

$$K = \frac{|\overrightarrow{OD}||\overrightarrow{BD}|}{|\overrightarrow{CD}|}$$
$$= \frac{8.16 \times 5.16}{3.16}$$
$$= 13.32$$

6. A radial line from the origin is drawn which is tangential to the circle which is part of the root locus. It is tangential at point A in the circle. The angle θ is measured. This point gives the minimum value of damping ratio that the system can provide. The damping ratio is given by

$$\zeta = \cos\theta$$
$$= \cos 38^\circ$$
$$= 0.788$$

At this point, the value of K is calculated as

$$K = \frac{|\overrightarrow{OA}||\overrightarrow{BA}|}{\overrightarrow{CA}}$$
$$= \frac{3.8 \times 2.3}{3.1}$$
$$= 2.82$$

7. The repeated roots occur at the breakaway and break-in points. From Fig. 5.18, at the breakaway point the repeated roots are at $s = -1.84$. The repeated poles are $(s + 1.84)^2$. The closed loop T.F. corresponding to this point where the value of $K = 0.675$ is

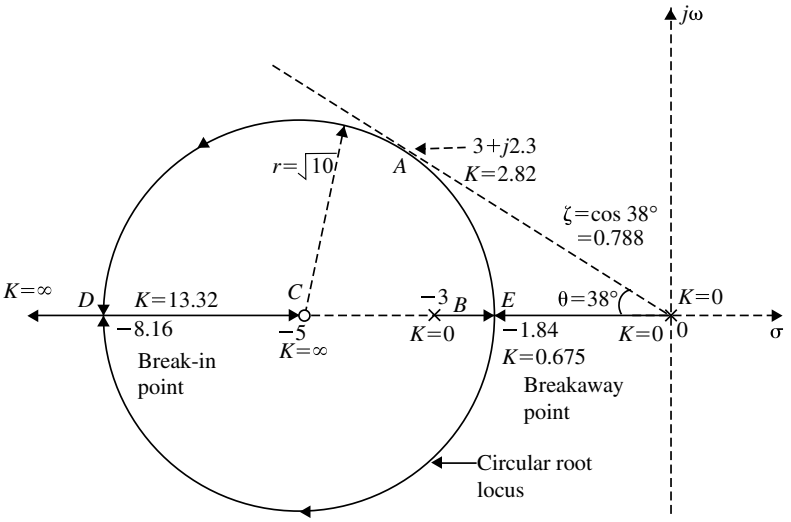


Fig. 5.18 Root Locus for $GH(s) = \frac{K(s+5)}{s(s+3)}$

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)}$$
$$= \frac{\frac{0.675(s+5)}{s(s+3)}}{1+\frac{0.675(s+5)}{s(s+3)}}$$

$$\boxed{\frac{C(s)}{R(s)} = \frac{0.675(s+5)}{(s+1.84)^2}}$$

The break-in point is at $s = -8.16$. The value of K at this point is $K = 13.32$. Corresponding to this point, the repeated roots are $(s + 8.16)^2$. The closed loop T.F. is given by

$$\frac{C(s)}{R(s)} = \frac{\frac{13.32(s+5)}{s(s+3)}}{1+\frac{13.32(s+5)}{s(s+3)}}$$

$$\boxed{\frac{C(s)}{R(s)} = \frac{13.32(s+5)}{(s+8.16)^2}}$$

8. If the root loci are complex, the damping ratio $\zeta < 1$. At the breakaway and break-in points, $\zeta = 1$ and on the real axis $\zeta > 1$. The system is critically damped ($\zeta = 1$) for the value of $K = 0.675$ and $K = 13.32$.
9. (a) Breakaway point $s = -1.84$
   Break-in point $s = -8.16$
   (b) Minimum value of $\zeta = 0.788$
   $K = 2.82$
   (c) At the breakaway point at $(s + 1.84)^2$, poles are repeated

$$\frac{C(s)}{R(s)} = \frac{0.675(s+5)}{(s+1.84)^2}$$

   At the break-in point, the roots are repeated as $(s + 8.16)^2$. The closed loop T.F. is

$$\frac{C(s)}{R(s)} = \frac{13.32(s+5)}{(s+8.16)^2}$$

   (d) The range of values of K for which the system is under-damped is

$$0.675 \leq K \leq 13.32$$

   (e) The values of K for which $\zeta = 1$ is

$$K = 0.675$$
$$K = 13.32$$

Example 5.11

A certain closed loop control system has the following open loop T.F.:

$$GH(s) = \frac{K(s+4)(s+3)}{(s+2)(s-2)}$$

Draw the root locus for $0 \leq K \leq \infty$. Show that part of the root locus is a circle. Determine the value of K for which the system is stable. At this value of K, find the closed loop system roots and hence the closed loop T.F. What are the repeated roots and the corresponding values of K? Also, show that $s = -2.29 + j1.11$ is on the root locus.

(*Anna University, December, 2003*)

Solution

1.

$$GH(s) = \frac{K(s+4)(s+3)}{(s+2)(s-2)}$$

Substitute $s = \sigma + j\omega$

$$GH(\sigma + j\omega) = \frac{K(\sigma + j\omega + 4)(\sigma + j\omega + 3)}{(\sigma + j\omega + 2)(\sigma + j\omega - 2)}$$

$$\angle GH(\sigma + j\omega) = \tan^{-1}\frac{\omega}{\sigma+4} + \tan^{-1}\frac{\omega}{\sigma+3} - \tan^{-1}\frac{\omega}{\sigma+2} - \tan^{-1}\frac{\omega}{\sigma-2}$$

To satisfy the angle criterion of root locus,

$$\tan^{-1}\frac{\omega}{\sigma+4} + \tan^{-1}\frac{\omega}{\sigma+3} - \tan^{-1}\frac{\omega}{\sigma+2} - \tan^{-1}\frac{\omega}{\sigma-2} = 180^\circ$$

$$\tan^{-1}\frac{\omega}{\sigma+4} + \tan^{-1}\frac{\omega}{\sigma+3} = 180^\circ + \tan^{-1}\frac{\omega}{\sigma+2} + \tan^{-1}\frac{\omega}{\sigma-2}$$

$$\tan^{-1}\frac{\frac{\omega}{\sigma+4} + \frac{\omega}{\sigma+3}}{1 - \frac{\omega^2}{(\sigma+4)(\sigma+3)}} = 180^\circ + \tan^{-1}\frac{\frac{\omega}{\sigma+2} + \frac{\omega}{\sigma-2}}{1 - \frac{\omega^2}{(\sigma+2)(\sigma-2)}}$$

$$\tan^{-1}\frac{\omega(2\sigma+7)}{\sigma^2+7\sigma+12-\omega^2} = 180^\circ + \tan^{-1}\frac{\omega 2\sigma}{\sigma^2-4-\omega^2}$$

Taking tan on both sides, we get

$$(2\sigma+7)(\sigma^2-4-\omega^2) = 2\sigma(\sigma^2+7\sigma+12-\omega^2)$$

$$\sigma^2 + 4.58\sigma + \omega^2 + 4 = 0$$

$$(\sigma + 2.29)^2 + \omega^2 = (1.11)^2$$

This is the equation of a circle with centre $(-2.29, 0)$ and radius $r = 1.11$.

2. On the real axis, root loci exist between $s = 2$ and $s = -2$ and between $s = -3$ and $s = -4$ where odd number of poles and zeros exist to the right of any point on the root loci.
3. The root loci originate from $s = 2$ and $s = -2$ where the value of $K = 0$, move on the real axis and breakaway. These loci traverse through the complex s-plane in a circular path, break in on the negative real axis and end at the zeros at $s = -3$ and $s = -4$ where the value of $K = \infty$.
4. The circular path of the root loci is drawn with centre at -2.29 and radius $r = 1.11$. The complete root locus is shown in Fig. 5.19.

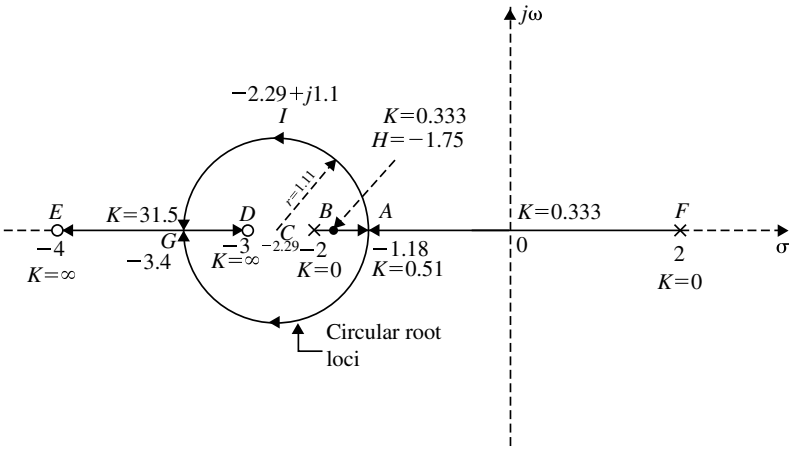


Fig. 5.19 Root locus for $GH(s) = \dfrac{K(s+3)(s+4)}{(s+2)(s-2)}$

5. The root locus crosses the imaginary axis at point $s = 0$. At this point, the value of K is calculated as

$$K = \frac{\overrightarrow{|FO|}\overrightarrow{|BO|}}{\overrightarrow{|DO|}\overrightarrow{|EO|}}$$

$$K = \frac{2.0 \times 2.0}{3.0 \times 4.0} = 0.333$$

For $K > 0.333$, the system is stable. At this point, the value of $s = 0$ and for $K = 0.333$, there is another root. This root is determined by trial and error. Choose any point $H = -1.75$. At this point, the value of K is calculated as

$$K = \frac{\overrightarrow{|FH|}\overrightarrow{|BH|}}{\overrightarrow{|DH|}\overrightarrow{|EH|}}$$

$$= \frac{3.75 \times 0.25}{1.25 \times 2.25} = 0.333$$

Hence, corresponding to $K = 0.333$, the other root is at $s = -1.75$. The closed loop T.F. for this value of K is given by

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)} = \frac{0.333\frac{(s+3)(s+4)}{(s+2)(s-2)}}{1 + \frac{0.333(s+3)(s+4)}{(s+2)(s-2)}}$$

$$\boxed{\frac{C(s)}{R(s)} = \frac{0.25(s+3)(s+4)}{s(s+1.75)}}$$

6. The breakaway point as seen in Fig. 5.19 is at $s = -1.18$. The repeated roots are $(s + 1.18)^2$. The value of K at this point is

$$K = \frac{\overrightarrow{|FA|}\overrightarrow{|BA|}}{\overrightarrow{|DA|}\overrightarrow{|EA|}} = \frac{3.18 \times 0.82}{1.82 \times 2.82}$$

$$\boxed{K = 0.51}$$

Repeated roots occur at break-in point also. This occurs at point G where $s = -3.4$. The value of K is calculated as

$$K = \frac{\overrightarrow{|FG|}\overrightarrow{|BG|}}{\overrightarrow{|DG|}\overrightarrow{|EG|}} = \frac{5.4 \times 1.4}{0.4 \times 0.6}$$

$$\boxed{K = 31.5}$$

7. Let the point I be denoted as $-2.29 + j1.11$. Using angle criterion, at point I

$$\begin{aligned}\angle GH(-2.29 + j1.11) &= \overrightarrow{DI} + \overrightarrow{EI} - \overrightarrow{FI} - \overrightarrow{BI} \\ &= 58^\circ + 34^\circ - 168^\circ - 104^\circ = -180^\circ\end{aligned}$$

This is an odd multiple of 180°. Hence, the point $s = -2.29 + j1.11$ is on the root locus. The above result can be obtained analytically by substituting $s = -2.29 + j1.11$ in $GH(s)$ as explained below

$$\begin{aligned}GH(s)|_{s=-2.29+j1.11} &= \frac{(-2.29 + j1.11 + 3)(-2.29 + j1.11 + 4)}{(-2.29 + j1.11 + 2)(-2.29 + j1.11 - 2)} \\ &= \frac{(0.71 + j1.11)(1.71 + j1.11)}{(-0.29 + j1.11)(-4.29 + j1.11)} \\ \phi &= \tan^{-1}\frac{1.11}{0.71} + \tan^{-1}\frac{1.11}{1.71} + 180^\circ + \tan^{-1}\frac{1.11}{0.29} + 180^\circ \\ &\quad + \tan^{-1}\frac{1.11}{4.29} \\ &= 58^\circ + 33^\circ + 76^\circ + 13^\circ = 180^\circ\end{aligned}$$

The phase angle is an odd multiple of 180°. Hence, the point $-2.29 + j1.11$ is on the root locus.

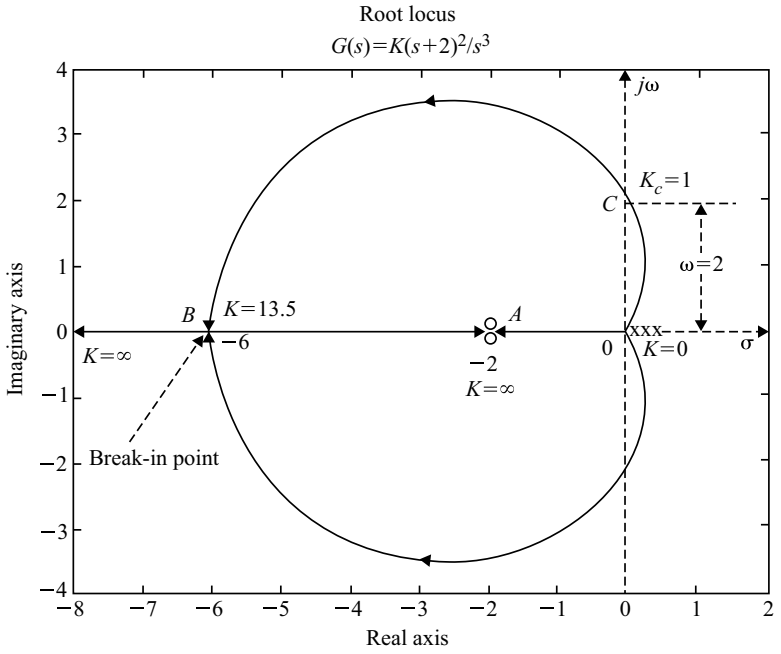


Fig. 5.20 Root locus for Example 5.12

Example 5.12

Consider the following loop T.F.

$$GH(s) = \frac{K(s+2)^2}{s^3}$$

Draw the root locus for $0 \leq K \leq \infty$. Determine the breakaway and break-in points if any and find the corresponding value of K. Determine the range of K for which the system is stable.

Solution

1.

$$GH(s) = \frac{K(s+2)^2}{s^3}$$

There are three poles and two zeros of $GH(s)$. The poles and zeros are located as shown in Fig. 5.20. There should be three root loci.

2. The root loci exist in the entire negative real axis since to any point in the negative real axis, an odd number of poles and zeros exist. There won't be any root loci on the positive real axis.
3. There are three poles at the origin. One such locus lies between $s = 0$ and $s = -2$. The other two loci traverse in the complex s-plane symmetrically with respect to the real axis and again break in on the negative real axis. One of them ends at $s = -2$ where $K = \infty$ and the other ends at $s = -\infty$ where $K = \infty$ as shown in Fig. 5.20.
4. The break-in point is obtained as follows. The characteristic equation is

$$\begin{aligned} 1 + GH(s) &= 0 \\ 1 + \frac{K(s+2)^2}{s^3} &= 0 \\ K &= -\frac{s^3}{(s+2)^2} \\ \frac{dK}{ds} = 0 &= (s+2)^2(3s) - s^3(2)(s+2) \\ 3(s+2) &= 2s \\ s &= -6 \end{aligned}$$

The break-in point is at

$$\boxed{s = -6}$$

At the break-in point, the value of K is calculated as

$$K = \frac{\overrightarrow{|OB|}^3}{\overrightarrow{|AB|}^2} = \frac{6^3}{4^2} = 13.5$$

$$\boxed{K = 13.5}$$

5. At point C, the root locus passes from RHP to LHP. This point can be determined using Routh's criterion. The characteristic equation is

$$\begin{aligned} 1 + GH(s) &= 0 \\ 1 + \frac{K(s+2)^2}{s^3} &= 0 \\ s^3 + Ks^2 + 4Ks + 4K &= 0 \end{aligned}$$

The following Routh's array is formed

$$\begin{array}{c|cc} s^3 & 1 & 4K \\ s^2 & K & 4K \\ s^1 & 4K(K-1) & \\ s^0 & K & \end{array}$$

From Routh's array, it is evident that $K > 1$ for the system to be stable. Hence, $K_c = 1$. The auxiliary equation is

$$Ks^2 + 4K = 0$$

$$s^2 + 4 = 0$$
$$\omega = 2$$

6.

The break-in point is at $s = -6$
The value of K at this point is $K = 13.5$
The system is stable for $K > 1$

Example 5.13

The forward path T.F. of a certain feedback control system is given by

$$GH(s) = \frac{K(s+2)(s+6)}{s^2(s+4)}$$

Draw the root locus for $0 \leq K \leq \infty$. Choose any point P on the root locus which is complex and graphically show that it satisfies the angle criterion. Calculate the value of K at this point using magnitude condition. Also calculate the value of K at the break-in point (Fig. 5.21).

Solution

1.

$$GH(s) = \frac{K(s+2)(s+6)}{s^2(s+4)}$$

There are three poles and two zeros. Hence, there will be three root loci. They start from the poles at $s = 0$ (repeated) and $s = -4$ where the value of $K = 0$.

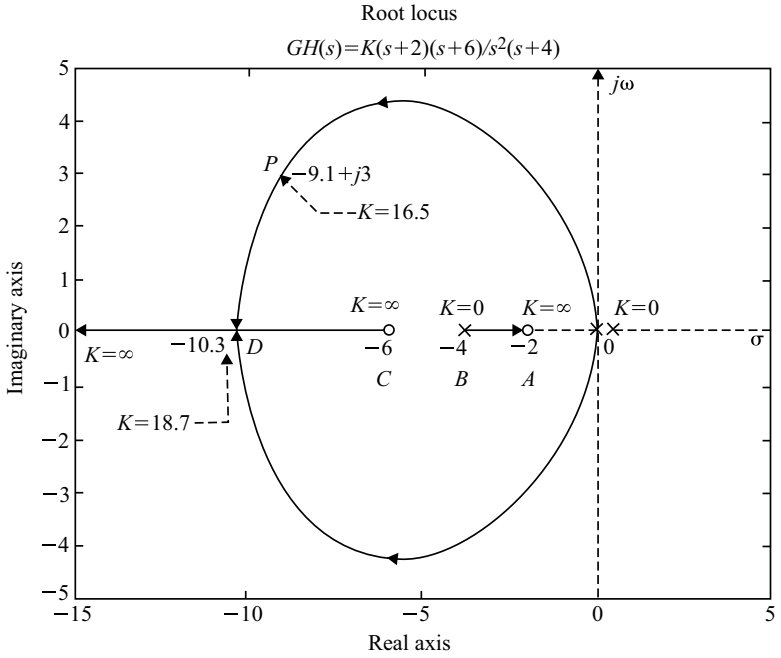


Fig. 5.21 Root locus for Example 5.13

They end at the zero at $s = -2$ and $s = -6$ where $K = \infty$. The third root locus ends at $-\infty$ where the value of $K = \infty$.

2. On the real axis, root loci exist between $s = -4$ and $s = -2$ and between $s = -6$ and $s = -\infty$. The root loci originating from the origin move in the complex s-plane symmetrically with respect to the real axis. They break in on the real axis. One locus ends at $s = -6$ and the other at $s = -\infty$.
3. The break-in point is calculated as follows. The characteristic equation is given by

$$1 + \frac{K(s^2 + 8s + 12)}{(s^3 + 4s^2)} = 0$$

$$K = \frac{-(s^3 + 4s^2)}{(s^2 + 8s + 12)}$$

$$\frac{dK}{ds} = 0 = (s^2 + 8s + 12)(3s^2 + 8s) - (s^3 + 4s^2)(2s + 8)$$

$$s^3 + 16s^2 + 68s + 96 = 0$$

Solving the above equation, we get

$$s_1 = -10.3 \quad \text{and} \quad s_{2,3} = -2.85 \pm j1.08$$

The break-in point occurs at $s = -10.3$. The value of K at this point is obtained from

$$K = \frac{\overrightarrow{|OD|}\overrightarrow{|OD|}\overrightarrow{|BD|}}{\overrightarrow{|AD|}\overrightarrow{|CD|}} = \frac{10.3 \times 10.3 \times 6.3}{8.3 \times 4.3} = 18.7$$

$$\boxed{K = 18.7}$$

4. The point P is chosen at $s = -9.1 + j3$

$$\begin{aligned}\angle GH(-9.1 + j3) &= \angle\overrightarrow{AP} + \angle\overrightarrow{CP} - 2\angle\overrightarrow{OP} - \angle\overrightarrow{BP} \\ &= 158^\circ + 136^\circ - 2 \times 162^\circ - 150^\circ\end{aligned}$$

$$\boxed{\phi = -180^\circ}$$

This is an odd multiple of 180°. Hence, point P is in the root locus. The value of K at point P is calculated as

$$|GH(-9.1 + j3)| = \frac{\overrightarrow{|OP|}^2\overrightarrow{|BP|}}{\overrightarrow{|AP|}\overrightarrow{|CP|}} = \frac{(9.6)^2 \times 5.9}{7.7 \times 4.3}$$

$$\boxed{K = 16.5}$$

Example 5.14

Consider the following loop T.F. of a certain feedback control system:

$$GH(s) = \frac{K(s^2 + 6s + 25)}{s(s + 3)(s^2 + 8s + 25)}$$

Draw the root locus for $0 \leq K \leq \infty$. Find the breakaway and break-in points and the corresponding values of K (Fig. 5.22).

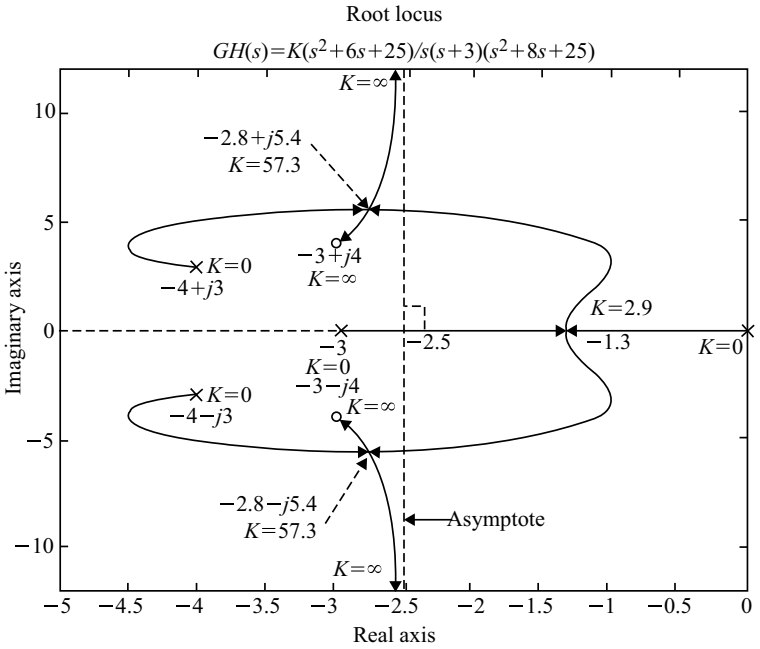


Fig. 5.22 Root locus for Example 5.14

Solution

1.

$$GH(s) = \frac{(s^2 + 6s + 25)}{s(s+3)(s^2 + 8s + 25)}$$
$$s^2 + 6s + 25 = (s + 3 + j4)(s + 3 - j4)$$
$$s^2 + 8s + 25 = (s + 4 + j3)(s + 4 - j3)$$
$$GH(s) = \frac{K(s + 3 + j4)(s + 3 - j4)}{s(s+3)(s + 4 + j3)(s + 4 - j3)}$$

The poles and zeros are located as shown in Fig. 5.22.

2. There are four poles and two zeros. Hence, there should be four separate root loci. Four root loci start from $s = 0, s = -3$ and $s = -4 + j3$ and $s = -4 - j3$. Two of them end at the zeros at $s = -3 + j4$ and $s = -3 - j4$. The remaining two root loci end at ∞.
3. The root loci on the real axis exist between $s = 0$ and $s = -3$ only. The root loci starting from $s = 0$ and $s = -3$ move on the negative real axis and break away in the complex plane symmetrically with respect to the real axis. These root loci and the root loci originating from the other poles $s = -4 + j3$ and

$s = -4 - j3$ meet and break away. Two of them end at the zeros at $s = -3 + j4$ and $s = -3 - j4$. The remaining two root loci go to ∞.

4. The centroid of the root loci

$$\sigma = \frac{0 - 3 - 4 - 4 + 3 + 3}{2}$$
$$= -2.5$$

The angle of the asymptote

$$\theta = \frac{(2n + 1) \times 180^\circ}{2}$$

where $n = 0, 1$

$$\theta = 90^\circ, 270^\circ$$

The asymptotes are drawn as shown in Fig. 5.22.

5. We can expect three breakaway points, one of them on the negative real axis and the remaining two in the complex s-plane. They are calculated as explained below. The characteristic equation of the system is

$$1 + \frac{K(s^2 + 6s + 25)}{s(s + 3)(s^2 + 8s + 25)} = 0$$
$$K = -\frac{(s^2 + 3s)(s^2 + 8s + 25)}{s^2 + 6s + 25}$$
$$= \frac{-(s^4 + 11s^3 + 49s^2 + 75s)}{(s^2 + 6s + 25)}$$
$$\frac{dK}{ds} = 0$$
$$= (s^2 + 6s + 25)(4s^3 + 33s^2 + 98s + 75)$$
$$-(s^4 + 11s^3 + 49s^2 + 75s)(8s + 6)$$
$$0 = s^5 + 14.5s^4 + 116s^3 + 522s^2 + 1225s + 938$$

Solving the above equation is by trial and error, we get

$$s_1 = -1.3$$
$$s_2 = -2.8 \pm j5.4$$
$$s_3 = -3.8 \pm j4.5$$

From the root locus on the real axis, it is seen that $s = -1.3$ is a point on the root locus. Hence, s_1 is a breakaway point. Now we see whether $s = -2.8 \pm j4.5$ are in the root locus which satisfy the angle criterion

$$GH(s) = \frac{K(s+3+j4)(s+3-j4)}{s(s+3)(s+4+j3)(s+4-j3)}$$

Substituting $s = -2.8 + j5.4$, we get

$$\begin{aligned} GH(-2.8+j5.4) &= K(-2.8+j5.4+3+j4)(-2.8+j5.4+3-j4) \\ &\quad \Big/\Big[(-2.8+j5.4)(-2.8+j5.4+3)(-2.8+j5.4+4+j3) \\ &\quad (-2.8+j5.4+4-j3)\Big] \\ &= \frac{K(0.2+j9.4)(0.2+j1.4)}{[(-2.8+j5.4)(0.2+j5.4)(1.2+j8.4)(1.2+j2.4)]} \\ \phi &= \Big[\tan^{-1}\frac{9.4}{0.2} + \tan^{-1}\frac{1.4}{0.2} + 180^\circ + \tan^{-1}\frac{5.4}{2.8} - \tan^{-1}\frac{5.4}{0.2} \\ &\quad - \tan^{-1}\frac{8.4}{1.2} - \tan^{-1}\frac{2.4}{1.2}\Big] \\ &= 180^\circ + 89^\circ + 82^\circ + 63^\circ - 88^\circ - 82^\circ - 64^\circ \\ &= 180^\circ \end{aligned}$$

This is an odd multiple of 180°. Hence, $s = -2.8 \pm j5.4$ are the breakaway points and they lie in the root locus. Now, we see whether $s = -3.8 \pm j4.5$ are in root loci and satisfy angle criterion

$$\begin{aligned} GH(-3.8+j4.5) &= K(-3.8+j4.5+3+j4)(-3.8+j4.5+3-j4) \\ &\quad \Big/\Big[(-3.8+j4.5)(-3.8+j4.5+3) + (-3.8+j4.5+j3) \\ &\quad (-3.8+j4.5+4-j3)\Big] \\ &= \frac{K(-0.8+j8.5)(-0.8+j0.5)}{[(-3.8+j4.5)(-0.8+j4.5)(0.2+j7.5)(0.2+j1.5)]} \\ \phi &= \Big[180^\circ - \tan^{-1}\frac{8.5}{0.8} + 180^\circ - \tan^{-1}\frac{0.5}{0.8} 180^\circ + \tan^{-1}\frac{4.5}{3.8} \\ &\quad +180^\circ + \tan^{-1}\frac{4.5}{0.8} - \tan^{-1}\frac{7.5}{0.2} - \tan^{-1}\frac{1.5}{0.2}\Big] \\ &= -85^\circ - 32^\circ + 50^\circ + 80^\circ - 89^\circ - 82^\circ \\ &= -158^\circ \end{aligned}$$

This is not an odd multiple of 180°. Hence, $s = -3.8 + j4.5$ is not a breakaway point since it is not in the root locus.

6. The values of K at the breakaway points are calculated as given below. At $s = -1.3$

$$K = \frac{1.3 \times 2.7 \times 4 \times 4}{4.4 \times 4.4}$$

$$\boxed{K = 2.9}$$

At $s = -2.8 \pm j5.4$,

$$K = \frac{6.08 \times 5.4 \times 2.7 \times 8.5}{1.4 \times 9.4}$$

$$\boxed{K = 57.3}$$

Example 5.15

A certain feedback control system has the following forward path T.F.:

$$GH(s) = \frac{K(1 + 2s)}{s^2(s + 4.5)}$$

Draw the root locus for $0 \leq K \leq \infty$ and find the repeated roots if any. What is the value of K at this point? For this value of K, determine the closed loop T.F (Fig. 5.23).

Solution

1. $$GH(s) = \frac{K(1 + 2s)}{s^2(s + 4.5)}$$

   The given T.F. is rewritten in the following pole–zero form:

   $$GH(s) = \frac{2K(s + 0.5)}{s^2(s + 4.5)} = \frac{K'(s + 0.5)}{s^2(s + 4.5)}$$

   where $K' = 2K$. The poles and zeros are as shown in Fig. 5.23. The variable parameter is K'.
2. There are three poles and one zero. Hence, there should be three root loci.
3. On the real axis, the root loci exist between $s = -0.5$ and $s = -4.5$. The two root loci originating from the origin move through the complex s-plane, and break in on the negative real axis. One locus ends at the zero at $s = -0.5$ and other two go to ∞.
4. The centroid of the root locus is

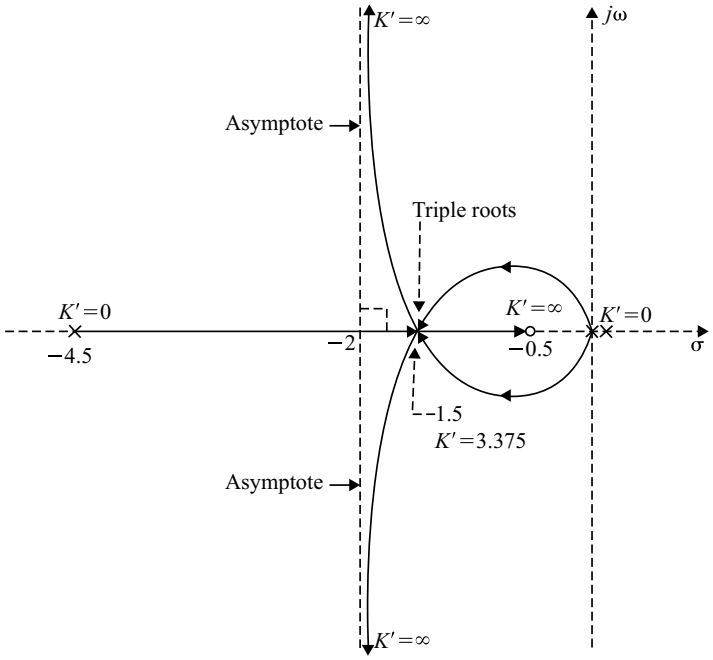


Fig. 5.23 Root locus for Example 5.15

$$\sigma = \frac{-0 - 0 - 4.5 + 0.5}{2}$$
$$= -2$$

The angle of the asymptote is

$$\theta = \frac{(2n+1) \times 180^\circ}{2}$$

where $n = 0, 1$

$$\theta = 90^\circ, 270^\circ$$

5. The characteristic equation is given by

$$1 + GH(s) = 0$$
$$1 + \frac{K'(s+0.5)}{s^2(s+4.5)} = 0$$
$$K' = \frac{-(s^3 + 4.5s^2)}{(s+0.5)}$$

$$\frac{dK'}{ds} = 0 = (s + 0.5)(3s^2 + 9s) - (s^3 + 4.5s^2)$$
$$s^2 + 3s + 2.25 = 0$$
$$(s + 1.5)^2 = 0$$
$$s = -1.5$$

The breakaway point is at $s = -1.5$. At the breakaway point, the value of K' is obtained as

$$K' = \frac{1.5 \times 1.5 \times 3}{1} = 6.75$$
$$K = \frac{6.75}{2} = 3.375$$

$$\boxed{K = 3.375}$$

6. The closed loop roots for $K = 3.375$ are $(s + 1.5)^3$. The closed loop T.F. is

$$\boxed{\frac{C(s)}{R(s)} = \frac{6.75(s + 0.5)}{(s + 1.5)^3}}$$

Example 5.16

Consider the following loop T.F. of a certain closed loop system:

$$GH(s) = \frac{K(s + 2)}{(s^2 + 2s + 2)}$$

Show that part of the root locus is a circle. Determine the break-in point and the value of K at this point. For this value of K, determine the closed loop T.F (Fig. 5.24).

Solution

1.

$$GH(s) = \frac{K(s + 2)}{(s^2 + 2s + 2)} = \frac{K(s + 2)}{(s + 1 + j)(s + 1 - j)}$$

There are two poles and one zero. Hence, there should be two root loci. The part of the root locus is a circle which can be proved as follows.

2. Substitute $s = \sigma + j\omega$ in $GH(s)$

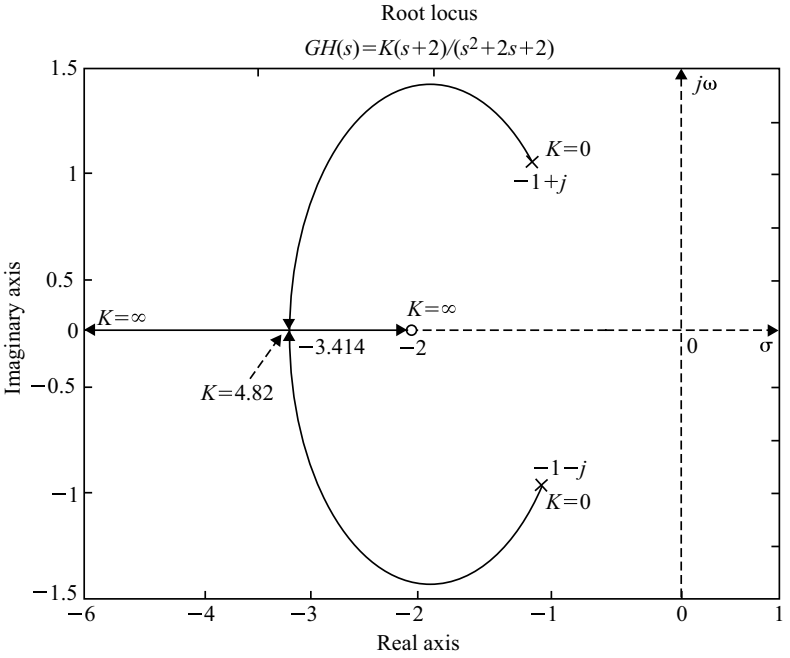


Fig. 5.24 Root locus for Example 5.16

$$GH(\sigma + j\omega) = \frac{K(\sigma + j\omega + 2)}{(\sigma + j\omega)^2 + 2(\sigma + j\omega) + 2}$$
$$= \frac{K(\sigma + j\omega + 2)}{(\sigma^2 + 2\sigma + 2 - \omega^2) + j2\omega(\sigma + 1)}$$
$$\angle GH(\sigma + j\omega) = \tan^{-1}\frac{\omega}{\sigma + 2} - \tan^{-1}\frac{2\omega(\sigma + 1)}{(\sigma^2 + 2\sigma + 2 - \omega^2)}$$

To satisfy the angle criterion of root locus, we get

$$\tan^{-1}\frac{\omega}{\sigma + 2} - \tan^{-1}\frac{2\omega(\sigma + 1)}{(\sigma^2 + 2\sigma + 2 - \omega^2)} = 180^\circ$$
$$\tan^{-1}\frac{\omega}{\sigma + 2} = 180^\circ + \tan^{-1}\frac{2\omega(\sigma+1)}{(\sigma^2+2\sigma+2-\omega^2)}$$

Taking tan on both sides and cross multiplying, we get

$$\sigma^2 + 2\sigma + 2 - \omega^2 = 2\sigma^2 + 6\sigma + 4$$
$$\sigma^2 + 4\sigma + \omega^2 + 2 = 0$$
$$(\sigma + 2)^2 + \omega^2 = 2$$

This is the equation of a circle with centre at $(-2, 0)$ and radius $\sqrt{2}$.

3. The root loci on the negative real axis exists between $s = -2$ and $s = -\infty$. The root loci starting from the complex conjugate poles break in on the $-$ve real axis. One locus goes to the zero at $s = -2$ and the other goes to $-\infty$.
4. The break-in point is calculated as -3.414. The value of K at this point is

$$K = \frac{2.61 \times 2.61}{1.414} = 4.82$$

$$\boxed{K = 4.82}$$

5. At the break-in point, the roots are repeated as $(s + 3.414)^2$. The closed loop T.F. is obtained as

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)}$$
$$= \frac{4.82(s + 2)}{(s^2 + 2s + 2) + 4.82(s + 2)}$$

$$\boxed{\frac{C(s)}{R(s)} = \frac{4.82(s + 2)}{(s + 3.414)^2}}$$

SUMMARY

1. Root loci are the trajectories of the characteristic equation when a system parameter is varied. It is an extensively used technique for the design of a compensator for continuous and discrete time systems. However, in this chapter, it is constructed only for linear time invariant continuous time systems.
2. Magnitude criterion and angle criterion are used to construct root locus. The angle criterion is used to locate the points in the root locus. Using the magnitude criterion, the values of the variable parameter in the root loci are found.
3. Searching of points in the vast s-plane which lie in the root loci trajectories is a very tedious process. However, by using certain properties of the root locus, the job is very much simplified. These properties are also called rules to construct root loci.
4. The properties of the root loci include the following:

   (a) Starting and termination of root loci.

(b) Number of root loci.
(c) Symmetry of the root loci.
(d) The centroid and angles of the asymptotes.
(e) Root loci on the real axis.
(f) Angle of arrival and angle of the departure of root loci.
(g) Intersection of the root loci with the imaginary axis and determination of critical value of the gain.
(h) Breakaway and break-in points.

5. After drawing the root loci, using the properties, exact root loci are obtained by choosing points closer to these loci and verifying whether they satisfy the angle criterion.
6. From root loci, we can obtain the closed loop system poles and hence the closed loop T.F. for any given value of K.
7. When time domain specifications are given, one can identify the point in the dominant root loci. The value of K at this point is determined. For this value of K, in the other root loci, closed loop roots are determined using the magnitude criterion by trial and error.
8. It is possible to find the frequency domain specifications such as phase margin and gain margin.
9. Using root locus, not only can we find the absolute stability of the system but also the relative stability in time and frequency domains.
10. To draw the root locus, for positive feedback, the angle criterion is different. Some of the properties are also different. By following the modified angle criterion and the properties, root loci for a positive feedback system can be drawn.
11. The root locus drawn for $-\infty \leq K \leq 0$ is called complementary root locus. Here, the angle criterion is the same as for the positive feedback system. Most of the properties also resemble those for the positive feedback system.
12. It is not true that one can always use the properties of the root loci and construct them. For certain transfer functions, the properties of root loci will not be of any help to us.
13. For a few T.F.s, a part of root loci will be a circle. For such T.F.s, analytically it can be derived that the root loci have the equation of a circle with centre and radius. Knowing these two, one can quickly construct the exact root locus.
14. For a linear system with transportation lag, root locus can be drawn by expressing the time lag in terms of a single pole and a single zero which is approximate only.
15. A simple MATLAB program is available to draw exact root loci for any complex T.F. We can obtain all required information from the plots so drawn.

EXERCISE

Short Answer Type Questions

1. **What is root locus?**
   Root locus is the trajectory of the closed loop poles as the gain of the system is varied in the range $0 \leq K \leq \infty$.
2. **What is complementary root locus?**
   The root locus drawn for the variation of the system parameter K in the range $-\infty \leq K \leq 0$ is called complementary root locus.
3. **What criteria are used to construct root locus?**
   Magnitude and angle criteria are used to construct root locus.
4. **What criterion decides the shape of the root locus trajectories?**
   Angle criterion decides the shape of the root locus trajectories.
5. **How magnitude criterion is used in the root locus plot?**
   The root loci are drawn for the variation of the gain parameter K. The values of K at different points of the root loci are determined using magnitude criterion. At any point s_1 in the s-plane, the K value is calculated from

$$K = \frac{\text{Product of directed distances drawn from all the poles to the point } s_1}{\text{Product of directed distances drawn from all the zeros to the point } s_1}$$

6. **How is angle criterion used to construct root loci?**
   The angle criterion is used to construct root locus in the cases described as follows::
   (a) **Negative feedback;** $0 \leq K \leq 0$: For any point s_1 in the complex s-plane to be a point on the root locus, the difference of angles drawn from all the zeros and the poles of $GH(s)$ to the point s_1 should be an odd multiple of 180°.
   (b) **Negative feedback;** $-\infty \leq K \leq 0$: As above in (a) but the angle difference should be even multiple of 180° which includes 0° also.
   (c) **Positive feedback;** $0 \leq K \leq \infty$: As in the case of (b).
7. **Are the root loci continuous curves or is any of them broken with discontinuity?**
   Root loci are continuous curves. They are not broken in the middle.
8. **What is the nature of symmetry of root locus in the complex s-plane?**
   Root loci are symmetrical with respect to the real axis of the complex s-plane. They have the complex conjugate property.
9. **What are the important properties of root loci?**
   Some of the important properties of root loci include the starting of the root loci, the ending of the root loci, the number of root loci, the centroid, the angle of intersection of the asymptote, breakaway and break-in points, the points at

which root loci cross the $j\omega$ axis, root loci on the real axis, angle of departure and angle of arrival.

10. **What information do we get from the properties of root loci?**
    The properties of root loci give guidance for the approximate root loci and not the exact root loci.
11. **Given the loop T.F.** $GH(s)$, **how do you determine the total number of root loci?**
    For the given $GH(s)$, the number of root loci is determined from the number of poles or zeros of $GH(s)$ whichever is greater.
12. **What are the starting and ending points of the root loci?**
    The root loci start from poles of $GH(s)$ where the value of $K = 0$ and end at zeros or ∞ where the value of $K = \infty$ if P, the number of poles of $GH(s)$ is greater than or equal to Z the number of zeros of $GH(s)$.
13. **There are three poles and five zeros. From where do the root loci start and where do they end?**
    There will be five root loci. Three of them will start from the three poles and end at three zeros. The remaining two root loci start from ∞ and end at the remaining two zeros.
14. **Do the zeros of the system change their positions as a result of variation of the gain** K?
    If the system gain K is varied, the positions of the zeros of the system in the s-plane are unaffected.
15. **How do you find the existence of root locus on the real axis for** $-\infty \leq K \leq \infty$?
    For $0 \leq K \leq \infty$, on the real axis root loci exist in the region to the right of which an odd number of poles and zeros are present. For $-\infty \leq K \leq 0$, on the real axis root loci exist in the region to the right of which an even number of poles and zeros are present.
16. **What are asymptotes of root loci?**
    The asymptotes of the root loci give guidance to the actual root loci at $s \to \infty$.
17. **How do you get the asymptotes for the given** $GH(s)$?
    The asymptotes are identified by their intersection and the angles with the real axis. The intersect is given by

$$\sigma = \frac{\sum \text{Real part of pole of } GH(s) - \sum \text{Real part of zeros of } GH(s)}{(P - Z)}$$

The angles of the asymptotes are given by

$$\theta = \frac{(2n + 1) \times 180^\circ}{(P - Z)}$$

where $n = 0, 1, 2, \ldots, (P - Z - 1)$.

18. **What is centroid of root locus?**
The intersect σ of the asymptote is the centre of gravity of the root loci. The intersect of the asymptote on the real axis is called the centroid.
19. **How many asymptotes will be there for the given $GH(s)$? Will they intersect at any other point in the s-plane other than the real axis?**
There are as many asymptotes as there are the number of root loci. Since σ the intersect of the asymptotes is a real number, asymptotes intersect the real axis only.
20. **What are breakaway and break-in points of root loci?**
When two or more than two branches of the root loci meet at one point, they depart from that point and reach their destiny. These points are called breakaway or break-in points.
21. **What are the properties of the breakaway and break-in points?**

    (a) For the given $GH(s)$, there may be more than one breakaway or break-in point and they may simultaneously present.
    (b) These points may be complex conjugate in the s-plane.
    (c) At these points, multiple order roots of the characteristic equation are present.

22. **What are the necessary conditions that the breakaway points must satisfy? Will it be sufficient also?**
The necessary condition that the breakaway point must satisfy is that

$$\frac{dK}{ds} = 0 = \frac{d}{ds}[GH(s)]$$

The above condition is not sufficient because the point obtained may not be a point in the root locus for some real value of K. Hence, it is further necessary and sufficient that the value of s so obtained is a point on the root locus.
23. **What are the methods available to determine the point of intersection of the root loci with the $j\omega$ -axis?**

    (a) By using the Routh–Hurwitz stability criterion, the critical value K_c can be determined. Using this value in the auxiliary equation, the point of intersections of the root loci with the $j\omega$ axis can be determined.
    (b) By trial and error, any point ω_1 in the $j\omega$-axis is chosen that satisfies the angle criterion. By using the magnitude criterion, the critical value K_c is obtained.

24. **How is the absolute stability of the closed loop system determined using root locus plot?**
For the given value of K, the roots of the characteristic equation are found in their respective root locus. If all the roots so found are in LHP, the closed loop system is absolutely stable.

25. **How to determine the relative stability from the root locus plot?**
There are two methods to determine the relative stability from the root locus plot.

(a) For the given value of K, determine the roots of the characteristic equations. Of these roots, the roots which are closer to the $j\omega$-axis give the relative stability. If they are away from the $j\omega$-axis and in LHP, the relative stability is high.
(b) By finding the phase margin and gain margin, the relative stability can be determined. The higher the phase margin and gain margin are and positive, the higher will be the relative stability.

26. **How do you determine phase margin and gain margin from root locus plot?**

(a) Let $K = K_d$ be the desired value. Then the gain margin is given by

$$G_m = \frac{K_c}{K_d}$$

where K_c is the critical value of K when the root loci cross the $j\omega$-axis.
(b) Determine any point ω_1 on the $j\omega$-axis that will satisfy the magnitude criterion for the value of $K = K_d$. At this point ω_1, find the phase angle difference from zeros and poles. Let it be ϕ. Then phase margin is given by

$$\phi_m = 180^\circ - \phi$$

27. **When is the centroid not used and why?**
When the difference between the number of poles and the number of zeros $(P - Z)$ is either zero or one, the centroid is not used. For these values, the closed loop poles do not move off the real axis on the way towards infinity.
28. **What is the effect of addition of pole to** $GH(s)$?
By addition of a pole to $GH(s)$, the order of the system is increased by one. Now the root loci bend towards the right and in general the system becomes less stable.
29. **What is the effect of addition of zero to** $GH(s)$?
By addition of zero to $GH(s)$, the root loci bend towards left. In General, the system becomes more stable.
30. **Write down the expression for the angle of departure and angle of arrival.**
The angle of departure is given by

$$\theta_d = 180^\circ - (\theta_P - \theta_Z)$$
$$\theta_a = 180^\circ - (\theta_Z - \theta_P)$$

where θ_P is the sum of angles of vectors drawn from other poles to the pole concerned and θ_Z is the sum of angle of vectors drawn from other zeros to the pole concerned.

31. **Can root locus be drawn for a system with delay? If not, how is it accounted for to draw the root locus?**
Root locus is drawn for rational polynomials of the T.F. When the time delay e^{-Ts} is considered, the polynomial becomes irrational and root locus cannot be applied. However, using **Pade** approximation e^{-Ts} is approximated as

$$e^{-Ts} = -\frac{(s - \frac{2}{T})}{(s + \frac{2}{T})}$$

32. **How do you find the angle of arrival and angle of departure of root loci at the breakaway and break-in points?**
If there are n root loci that meet at breakaway or break-in points, they arrive or depart at $\frac{180^\circ}{n}$.
33. **For the given$K$, how do you find the closed loop poles and the closed loop T.F.?**
If the value of K is given, in each root locus, using magnitude criterion, determine the roots by trial and error. These roots are nothing but the roots of the characteristic equation or the poles of the closed loop system. The closed loop T.F. is determined using the equation

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)}$$

The numerator of $[1 + GH(s)]$ is represented by the poles of the characteristic equation.
34. **For a given damping ratio ζ, how do you find the closed loop poles from root locus?**
For the dominant complex conjugate poles, the damping factor is defined as $\phi = \cos^{-1}\zeta$. Draw a radial line from the origin of the s-plane with an angle $(180^\circ - \phi)$ which cuts the dominant root loci. Using magnitude criterion, find the value of K at this point. This point and the conjugate of this give the closed loop roots. For the remaining root loci, the roots are found, by trial and error for the value of K fixed on the dominant root loci. All these poles represent closed loop poles.
35. **For the following Loop T.F., sketch the root locus for** $0 \leq K \leq \infty$.

(a) $GH(s) = \dfrac{K(s-2)}{(s+2)(s+3)}$

(b) $GH(s) = \dfrac{K(s+4)(s+2)}{(s^2+4)}$

(c) $GH(s) = \dfrac{K(s+2)(s+4)}{(s-2)(s-4)}$

(d) $GH(s) = \dfrac{K}{(s^2+4)}$

$$\text{(e)} \quad GH(s) = \frac{K}{s(s^2+4)}$$

$$\text{(f)} \quad GH(s) = \frac{K}{(s^2-4)}$$

36. **Consider the following loop T.F. of a certain feedback control system.**

$$GH(s) = \frac{K(s+a)}{s(s+b)}$$

Under what condition is part of the root loci a circle? If this condition is not satisfied, what will be the nature of damping available in the system?
For part of the root loci to be circle $a > b$. If $a < b$, the root loci exist only on the negative real axis, and the system is over-damped for all values of K (Figs. 5.25, 5.26, 5.27, 5.28, 5.29, 5.30, 5.31, 5.32, 5.33, 5.34, 5.35, 5.36, 5.37, 5.38, 5.39 and 5.40).

Long Answer Type Questions

1. A certain closed loop unity feedback control system has the following open loop T.F.:

$$G(s) = \frac{K}{s(s+5)(s+10)}$$

Draw the root locus for $0 \leq K \leq \infty$. For $K = 70$, find the closed loop roots. For this value of K, determine the closed loop T.F.
Ans: Closed loop poles are at $s_1 = -11.05$; $s_{2,3} = -2 \pm j1.6$

$$\frac{C(s)}{R(s)} = \frac{70}{(s+11.05)(s^2+4s+6.6)}$$

2. Consider the following open loop T.F. of certain closed loop system.

$$G(s) = \frac{1}{s(1+0.5s)(1+0.2s)}$$

Determine the damping ratio and natural frequency of oscillations of the dominant roots. Find the closed loop T.F. using the root locus method.
Ans: $\zeta = \cos 60° = 0.5$; $\omega_n = 1.5$ rad/s. For $K = 10$, $s_1 = -5.5$; $s_{2,3} = -0.75 \pm j1.3$

$$\frac{C(s)}{R(s)} = \frac{10}{(s+5.5)(s^2+1.5s+2.3)}$$

3. Consider the following open loop T.F. of a certain closed loop system:

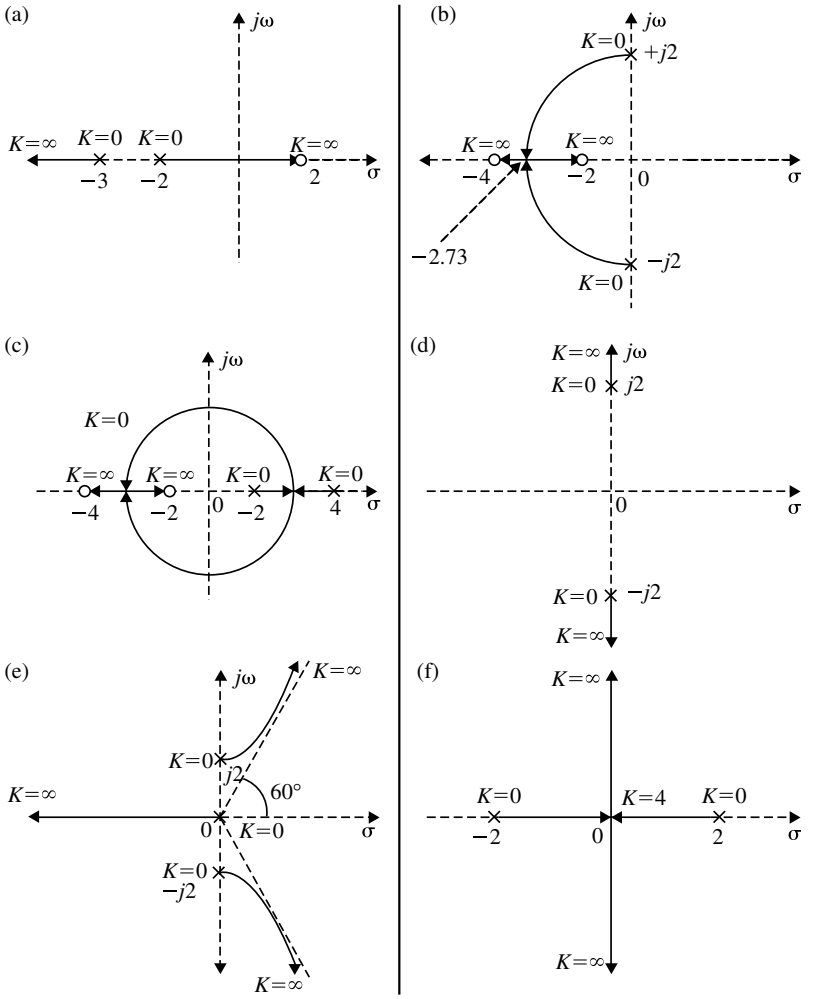


Fig. 5.25 Root locus for Question 35

$$GH(s) = \frac{K(s+9)}{s(s^2+4s+11)}$$

Draw the root locus for $0 \leq K \leq \infty$. Determine the value of K to have a damping ratio of $\zeta = 0.3$ for the dominant roots. For this value of K, determine gain margin. What is the closed loop T.F.?

Ans: $K = 2.26$; Gain margin, $G_m = 2.2$

$$\frac{C(s)}{R(s)} = \frac{2.26(s+9)}{(s+2)(s^2+2s+10)}$$

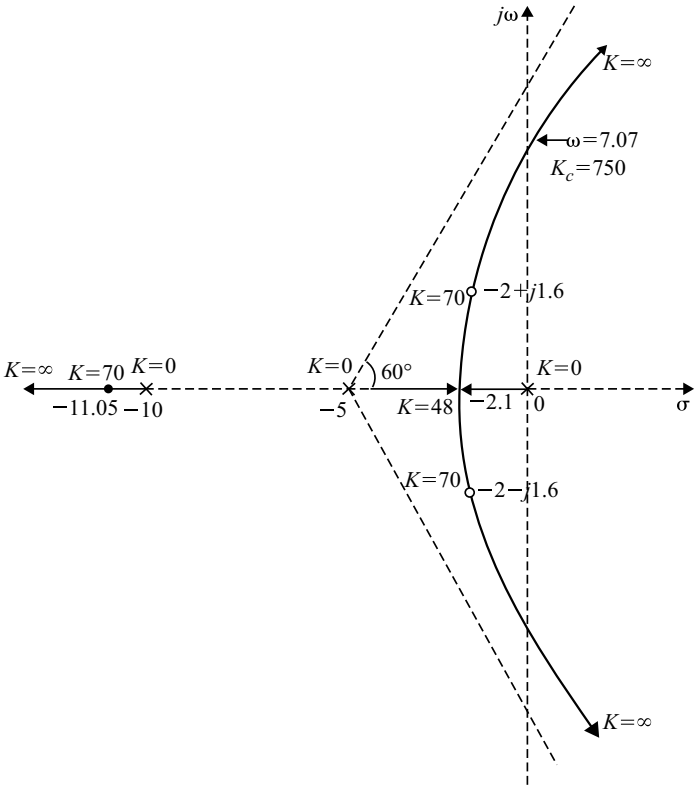


Fig. 5.26 Root locus for $GH(s) = \dfrac{K}{s(s+5)(s+10)}$

4. Consider the following loop T.F. of a certain closed loop system:

$$GH(s) = \frac{K(s+1)}{s(s+4)(s^2+6s+10)}$$

Draw the root locus for $0 \leq K \leq \infty$. For $K = 18$, determine the closed loop system roots. For this value of K, determine the phase margin and gain margin. **Ans:** $K = 18$ and the closed loop system roots are $s_1 = -5.7$; $s_2 = -0.39$; $s_3 = -1.4 + j2.4$; and $s_4 = -1.4 - j2.4$, $\phi_m = 93°$ positive and $G_m = 12$ positive.

5. Consider the following open loop system of a certain unity negative feedback control system:

$$GH(s) = \frac{K(s+2)}{(s+1)^2}$$

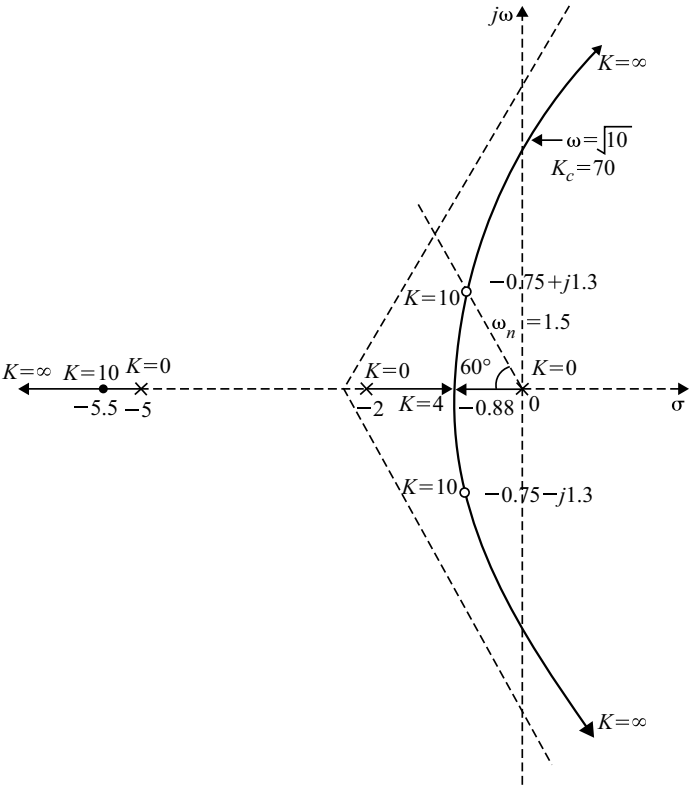


Fig. 5.27 Root locus for $GH(s) = \dfrac{K}{s(s+2)(s+5)}$

Draw the root locus for $0 \leq K \leq \infty$. Show that part of the root locus is a circle. Find the centre and radius of the circle.

Ans: Centre is at $c = (-2, 0)$ and radius $r = 1$. Break-in point is at $s = -3$, where $K = 4$.

6. Consider the following loop T.F. of a certain closed loop system. Sketch the root locus. Identify the value of K for which the system is stable.

$$GH(s) = \frac{K}{s^2(s^2 + 4s + 10)}$$

Ans: For all values of K, where $0 \leq K \leq \infty$, there are two roots in the RHP. The system is always unstable.

7. A certain closed loop control system has the following open loop T.F.:

$$GH(s) = \frac{K(s+2)(s^2+4)}{(s^2 - 16)}$$

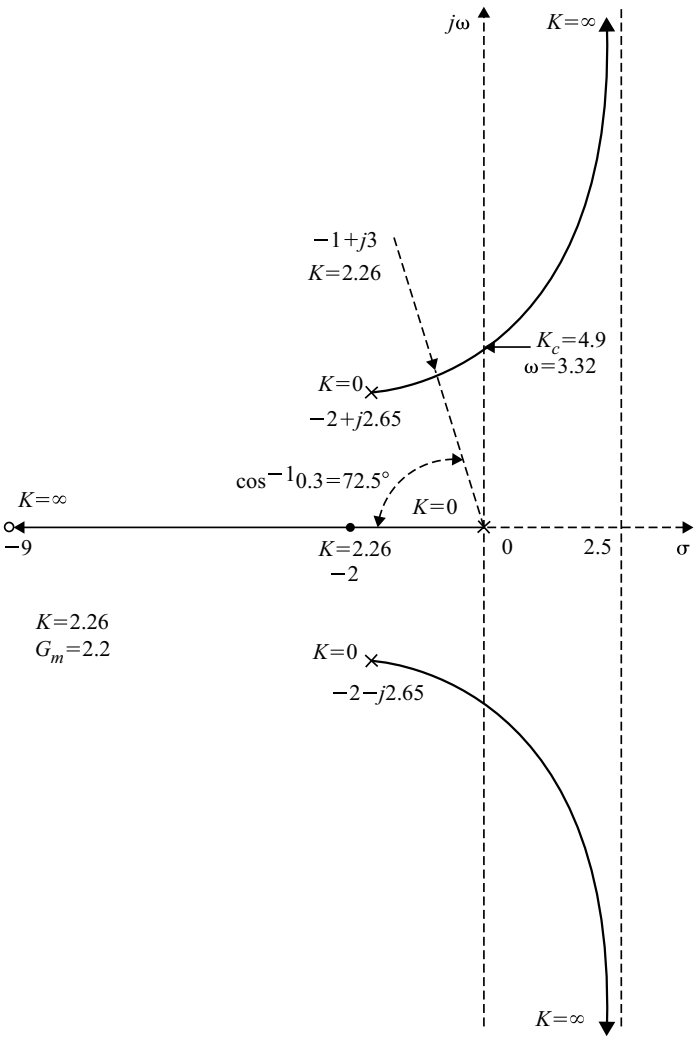


Fig. 5.28 Root locus for $G(s) = \dfrac{K(s+9)}{s(s^2+4s+11)}$

Draw the root for $0 \leq K \leq \infty$. For what value of K is the system stable? For what value of K does the system have repeated poles and what are these pole locations? Is the system stable for this value of K?

Ans:

1. The system is stable for $K > 2$.
2. The breakaway point is at $s = -6.67$.

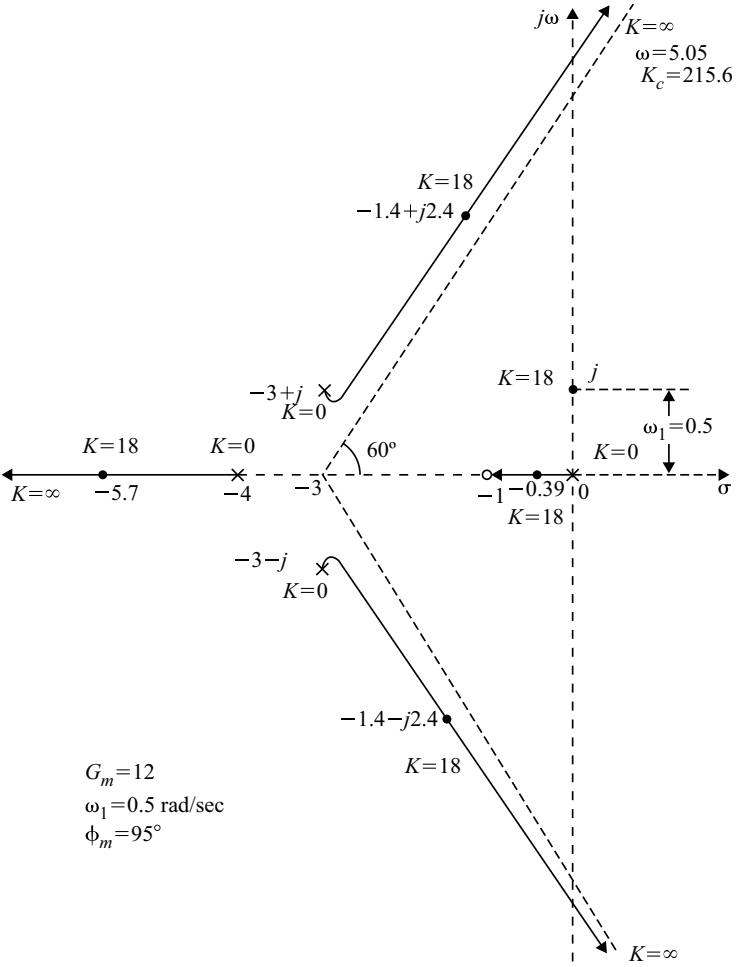


Fig. 5.29 Root locus for $G(s) = \dfrac{K(s+1)}{s(s+4)(s^2+6s+10)}$

3. The repeated poles are $(s+6.67)^2$ and the corresponding value of $K = 0.126$.
4. The system is unstable for this value of K since $K < 2$.

8. A certain control system has the following open loop T.F.:

$$GH(s) = \frac{K(s-2)(s-3)}{s(s+2)}$$

Draw the actual root locus for $0 \leq K \leq \infty$. Show that part of the root locus is a circle. Find the centre and radius of the circle. Find the value of K for which

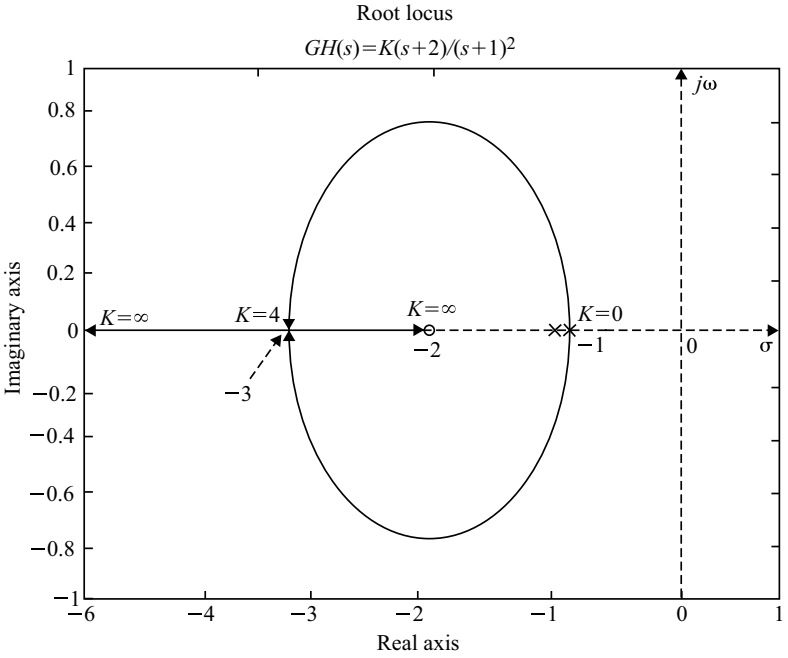


Fig. 5.30 Root locus for Question 5

the damping is zero. For what value of K is the system critically damped? For this value of K, determine the closed loop poles and hence closed loop T.F.
Ans:

1. Centre of the circle, $c = 0.86$.
2. Radius of the circle, $r = 1.55$.
3. The damping is zero for $K = 0.4$.
4. For $K = 0.091$, the system is critically damped.
5. For $K = 0.091$, the closed loop poles are $(s + 0.69)^2$.
6. The closed loop T.F. for the above value of K is

$$\boxed{\frac{C(s)}{R(s)} = \frac{0.0834(s-2)(s-3)}{(s+0.69)^2}} \tag{5.13}$$

9. A certain feedback control system has the following loop T.F.:

$$GH(s) = \frac{K(s^2 - 4s + 5)}{(s+1)(s+2)}$$

Draw the root locus for $0 \leq K \leq \infty$. Find the value of K for which the system is critically stable. Determine the value of K for which the closed loop system

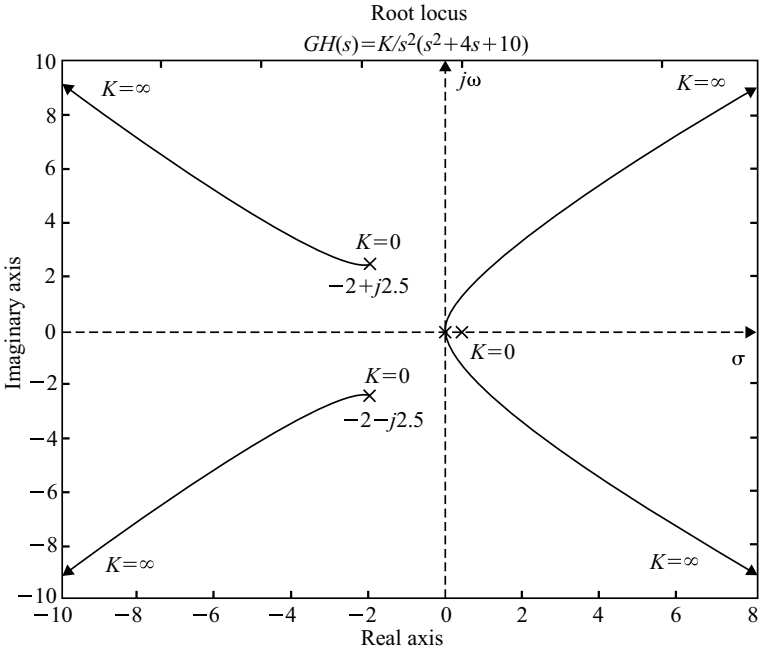


Fig. 5.31 Root locus for $GH(s) = \dfrac{K}{s^2(s^2+4s+10)}$

poles are repeated. What are these repeated roots? Is any part of the root locus circular. If so find the centre and radius.

Ans:

1. The critical value of K is $K_c = 0.75$.
2. At $K = 0.0192$, the root loci break away, and there exist double roots at $s = -1.44$.
3. The root locus is a circle with centre at $s = 0.38$ and radius $r = 1.82$.

10. A certain closed loop system has the following open loop T.F.:

$$GH(s) = \frac{K(s^2 - 4s + 5)}{s(s+1)(s+2)}$$

Draw root locus for $0 \leq K \leq \infty$. Determine the breakaway point and the value of K at that point. What is the critical value of K for the system to be stable? How does the addition of a pole at $s = 0$ affect the stability of the system.

Ans:

1. The breakaway point is at $s = -0.34$ and the corresponding value of $K = 0.058$.

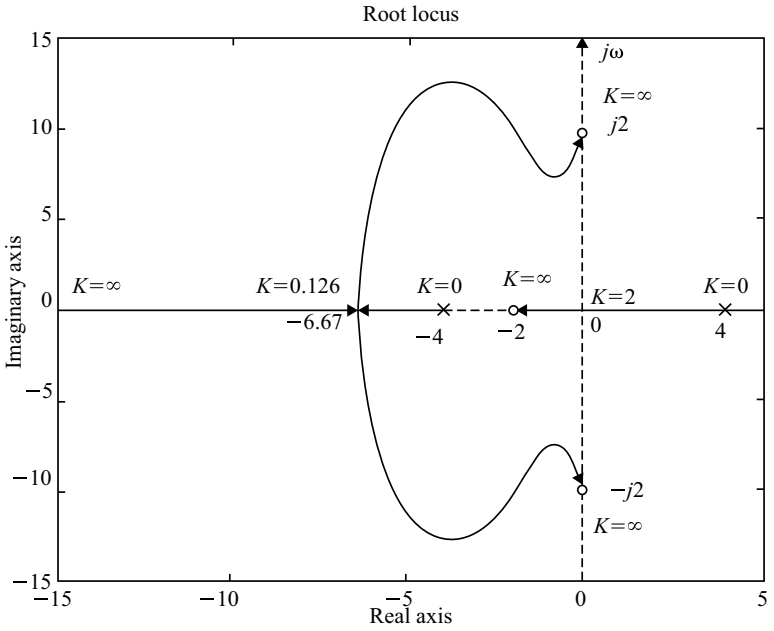


Fig. 5.32 Root locus for $GH(s) = \dfrac{K(s+2)(s^2+4)}{(s^2-16)}$

2. The critical value of K is $K_c = 0.4$ which is 0.75 when a pole at $s = 0$ is not added. Thus, the addition of a pole reduces the stability margin.

11. Consider the following loop T.F. of a certain closed loop system:

$$GH(s) = \frac{K(s-2)(s-3)}{(s+2)(s+3)}$$

Draw the root locus for $0 \leq K \leq \infty$. In this case, the pole at the origin of the system described in Question 8 is shifted to $s = -3$. Compare the critical stability limit and the transient response of these two systems.
Ans:

1. The critical value of stability is increased from $K_c = 0.4$ to $K_c = 1$.
2. For $0.01 < K < 1$, the system is under-damped. However, the system described in Question 8 is under-damped for $0.091 < K < 0.4$. For any given value of gain K, the system described in Question 11 is better damped than the one given in Question 8.

12. Consider the system described in Question 11. The locations of the poles and zeros are interchanged and the loop T.F. is

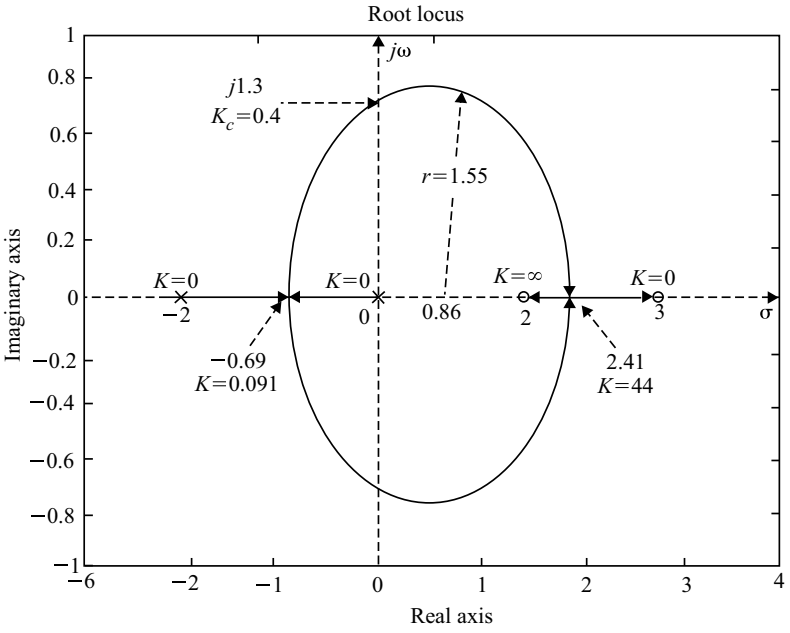


Fig. 5.33 Root locus for $GH(s) = \dfrac{K(s-2)(s-3)}{s(s+2)}$

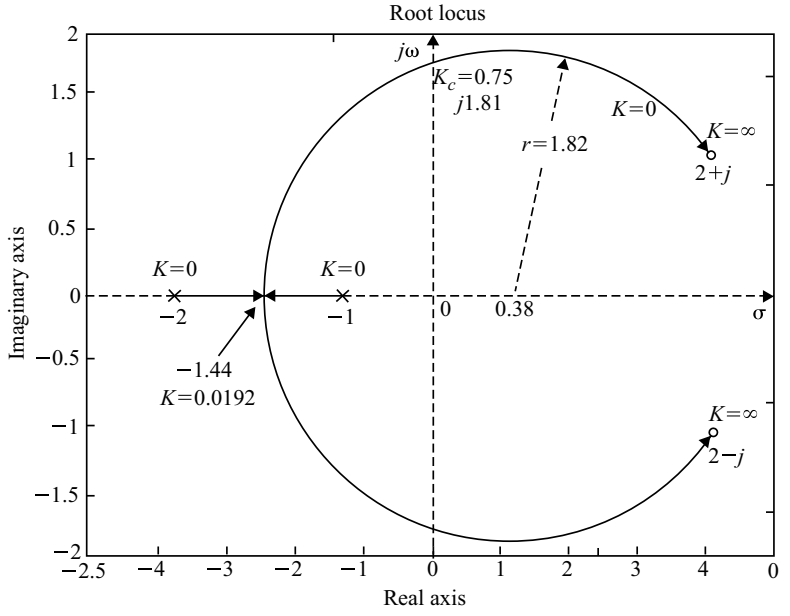


Fig. 5.34 Root locus for $GH(s) = \dfrac{K(s^2-4s+5)}{(s+1)(s+2)}$

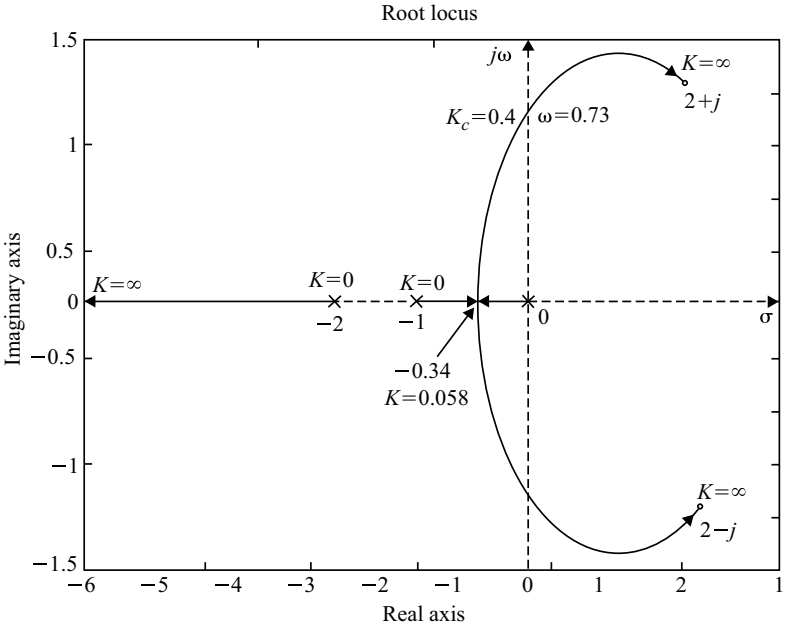


Fig. 5.35 Root locus for $GH(s) = \dfrac{K(s^2 - 4s + 5)}{s(s+1)(s+2)}$

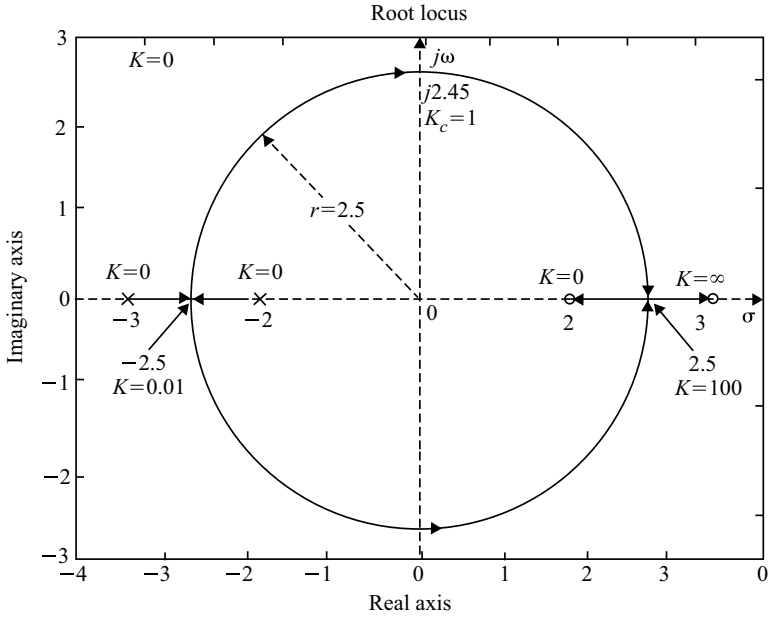


Fig. 5.36 Root locus for $GH(s) = \dfrac{K(s-2)(s-3)}{(s+2)(s+3)}$

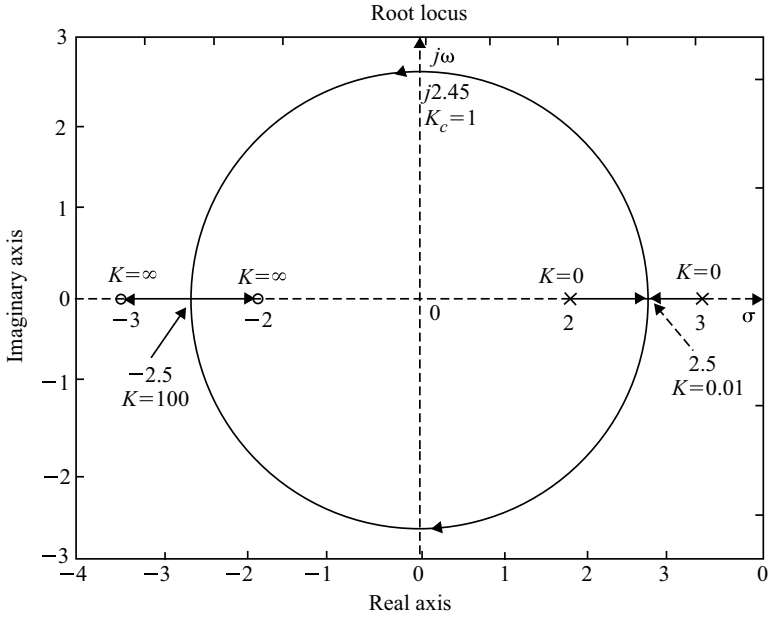


Fig. 5.37 Root locus for $GH(s) = \dfrac{K(s+2)(s+3)}{(s-2)(s-3)}$

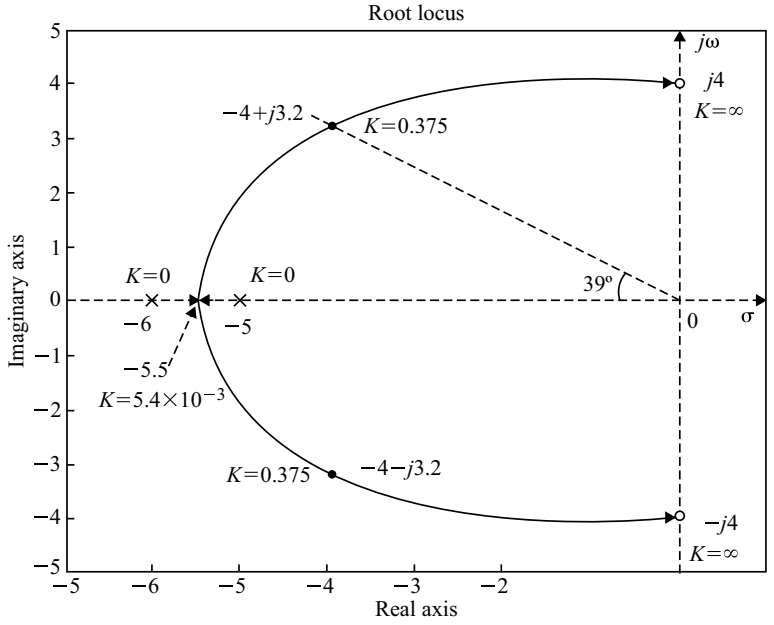


Fig. 5.38 Root locus for $GH(s) = \dfrac{K(s^2+16)}{(s+5)(s+6)}$

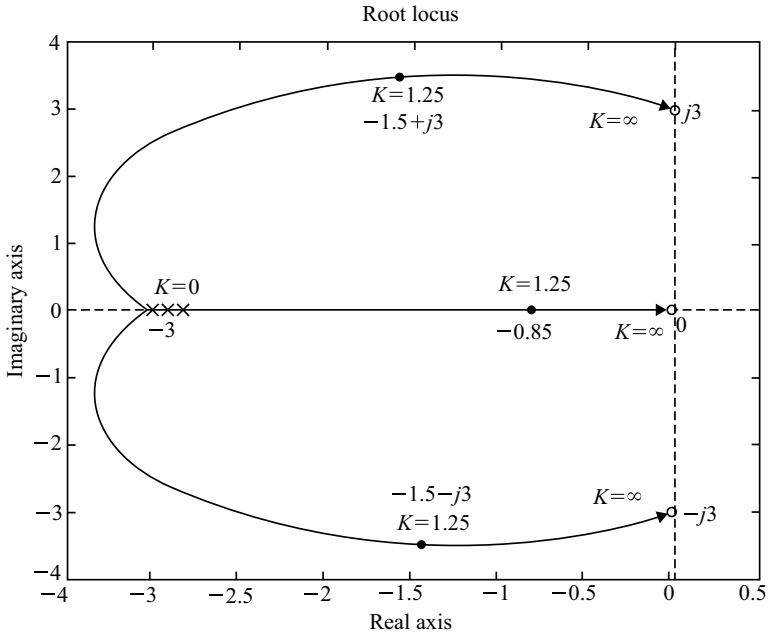


Fig. 5.39 Root locus for $GH(s) = \dfrac{Ks(s^2+9)}{(s+3)^3}$

$$GH(s) = \frac{K(s+2)(s+3)}{(s-2)(s-3)}$$

Draw the root locus for $0 \leq K \leq \infty$. Comment on the stability limit and transient response of these two systems.

Ans:

1. The critical value of K for stability in both the cases is $K_c = 1$. However, for $K > 1$, the system becomes more stable and for $K < 1$, it becomes unstable which is the reverse process of the system in Question 11.
2. As the gain is increased in the range $1 \leq K \leq 100$, the system is underdamped and hence one can choose any value of K in this range which is wider so that the desired damping ratio is achieved.

13. Consider the following loop T.F. of a closed loop system:

$$GH(s) = \frac{K(s^2+16)}{(s+5)(s+6)}$$

Draw the root locus for $0 \leq K \leq \infty$. Find the roots of the characteristic equation for a damping factor of $\zeta = 0.78$.

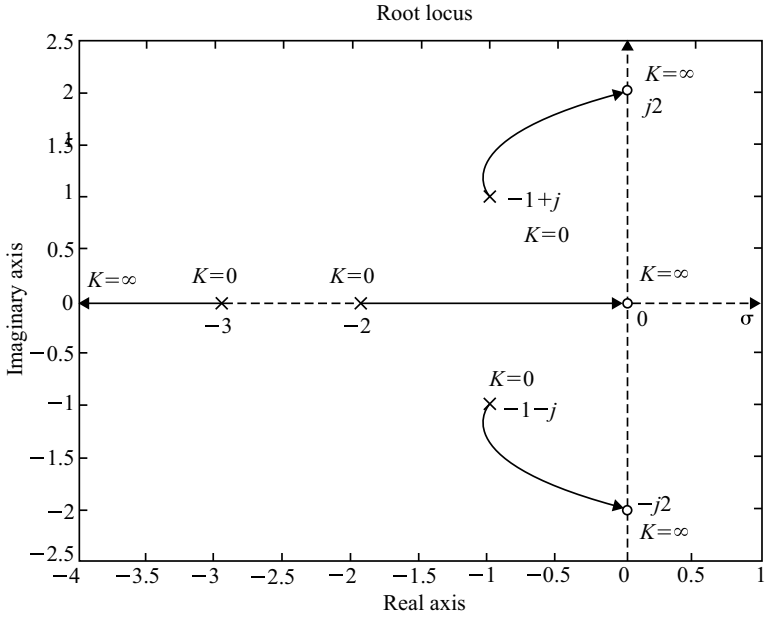


Fig. 5.40 Root locus for $GH(s) = \dfrac{Ks(s^2+4)}{(s+2)(s+3)(s^2+2s+2)}$

Ans: For $\zeta = 0.78$, $K = 0.375$. The roots of the characteristic equations are $s_1 = -4 + j3.2$ and $s = -4 - j3.2$.

14. Consider the following open loop T.F.:

$$GH(s) = \frac{Ks(s^2+9)}{(s+3)^3}$$

Draw the root locus for $0 \le K \le \infty$. For $K = 1.25$, determine the closed loop roots and hence the closed loop T.F.

Ans:

1. For $K = 1.25$, the closed loop roots are $s_1 = -0.85$, $s_2 = -1.5 + j3$ and $s_3 = -1.5 - j3$.
2. The closed loop T.F. is

$$\boxed{\frac{C(s)}{R(s)} = \frac{0.555\,s(s^2+9)}{(s+0.85)(s^2+3s+11.25)}}$$

15. Consider the following open loop T.F. of a certain closed loop system:

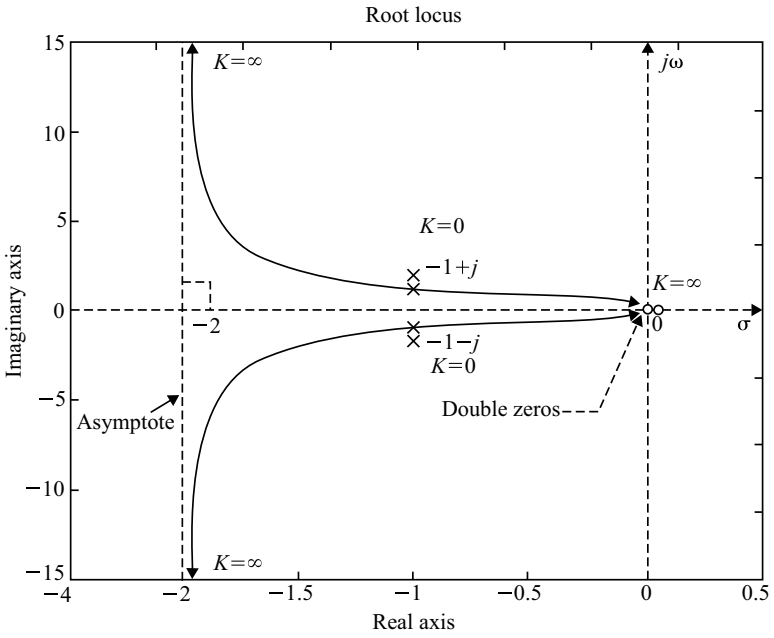


Fig. 5.41 Root locus for $GH(s) = Ks^2/(s^2 + 2s + 2)^2$

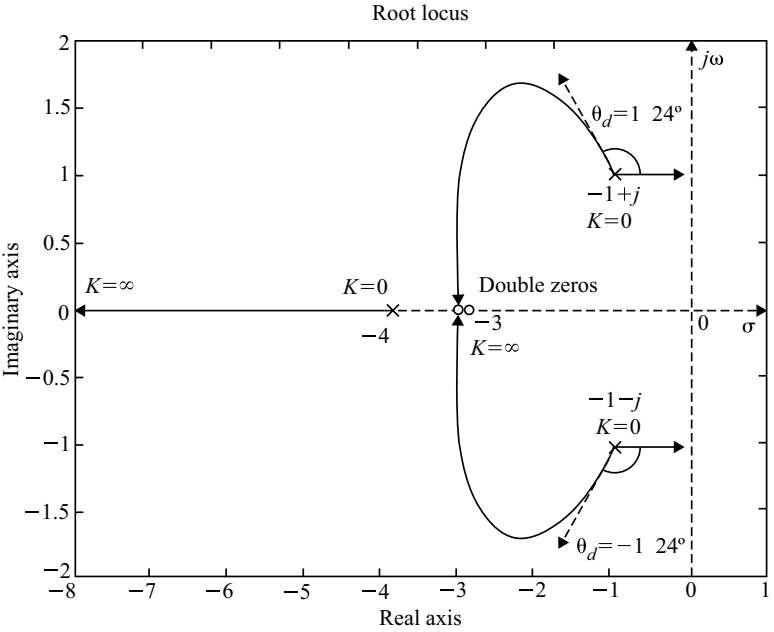


Fig. 5.42 Root locus for $GH(s) = K(s + 3)^2/(s + 4)(s^2 + 2s + 2)$

$$GH(s) = \frac{Ks(s^2+4)}{(s+2)(s+3)(s^2+2s+2)}$$

Draw the root locus for $0 \leq K \leq \infty$.

16. A certain closed loop system has the following open loop T.F.:

$$GH(s) = \frac{Ks^2}{(s^2+2s+2)^2}$$

Draw the root locus for $0 \leq K \leq \infty$ (*see* Fig. 5.41).

17. A certain closed loop control system has the following open loop T.F.:

$$GH(s) = \frac{K(s+3)^2}{(s+4)(s^2+2s+2)}$$

Draw the root locus for $0 \leq K \leq \infty$ (*see* Fig. 5.42).

Chapter 6
Design of Compensators



Learning Objectives

After completing this chapter, you should be able to:

- Understand the performance criteria of compensators.
- Design compensator in the frequency and time domains.
- Design lead compensator in the frequency and time domains.
- Design lag compensator in the frequency and time domains.
- Design lag–lead compensator in the frequency and time domains.

6.1 Introduction

It is the intention of the control engineer to design a system which gives optimum performance. However, very often it happens that the specifications offered by different users vary, and therefore, it becomes necessary to do some adjustments either in the design of the system or insert some external devices to obtain the required performance. The control system design engineer has used his ingenuity to design the system or process to give optimum performance in general. Therefore, the given system has been altered as much as possible to meet the performance specifications of many users. Further alterations of the plant parameters may not give optimum performance in general. Now the option is to insert some devices according to the requirement of the individuals so that the already designed system together with the additional devices inserted gives optimum performance. These additional components or devices which are inserted into the system equalize or compensate the performance deficiency. These compensating devices may include electric mechani-

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S. Palani, *Automatic Control Systems*,
https://doi.org/10.1007/978-3-030-93445-3_6

cal, hydraulic or pneumatic components and they are called compensators. Thus, the process of adjustment or alteration of a control system to get the desired performance specification is called compensation. The additional components or circuits that are inserted into the system in order to compensate for a performance deficiency are called compensators. In this chapter, we discuss the design procedure for electric circuit compensators.

6.2 Performance Criteria for Compensators

A control engineer uses his ingenuity to have a well designed control system that must have less transient and steady state error, good accuracy, good speed of response, good relative stability and good damping so that undue overshoot, etc. can be avoided. The first attempt to achieve the above requirements is by the adjustment of the gain. However, by adjustment of gain alone it is very difficult to achieve satisfactory result. A higher value of the gain improves the steady state behaviour but at the cost of transient response which becomes poor. Thus, the task becomes challenging which requires adding external devices called compensators to fulfil the specification requirements of the individual.

Control system design problem involves two basic steps. The first step involves the choice of the components that should be inserted into the system and the second step is about the location of these devices. Generally, compensators are inserted in the forward path, feedback path or the input path of the feedback loop or the combination of the above. Having chosen the location of the compensator in the control loop, the problem is to design the components that act together to compensate for the performance deficiency of the original system. Care should also be taken to use minimum components since the use of more comments becomes more expensive.

The compensators are located in a feedback system as:

1. Series compensator
2. Parallel compensator
3. Series-parallel compensator

A series or cascade compensator with the T.F. $G_c(s)$ connected in series with the plant whose T.F. is $G(s)$ is shown in Fig. 6.1a. In this scheme, the signal flows from a lower energy level to a higher energy level. Therefore, additional amplifiers are required to increase the gain as well as to provide accessory isolation. It has more number of components and hence it is expensive.

The location of the parallel compensator is shown in Fig. 6.1b. In this scheme, the flow of energy is from higher energy level to lower energy level. Therefore, additional amplifiers are not required in this scheme. However, series compensation is simpler than parallel (feedback) compensation.

The location of the series-parallel compensator is shown in Fig. 6.1c. In this scheme, the advantage of series and parallel schemes are achieved.

In control system design, the transient response specifications are more important and ultimate measures of performance indices. These specifications include: peak

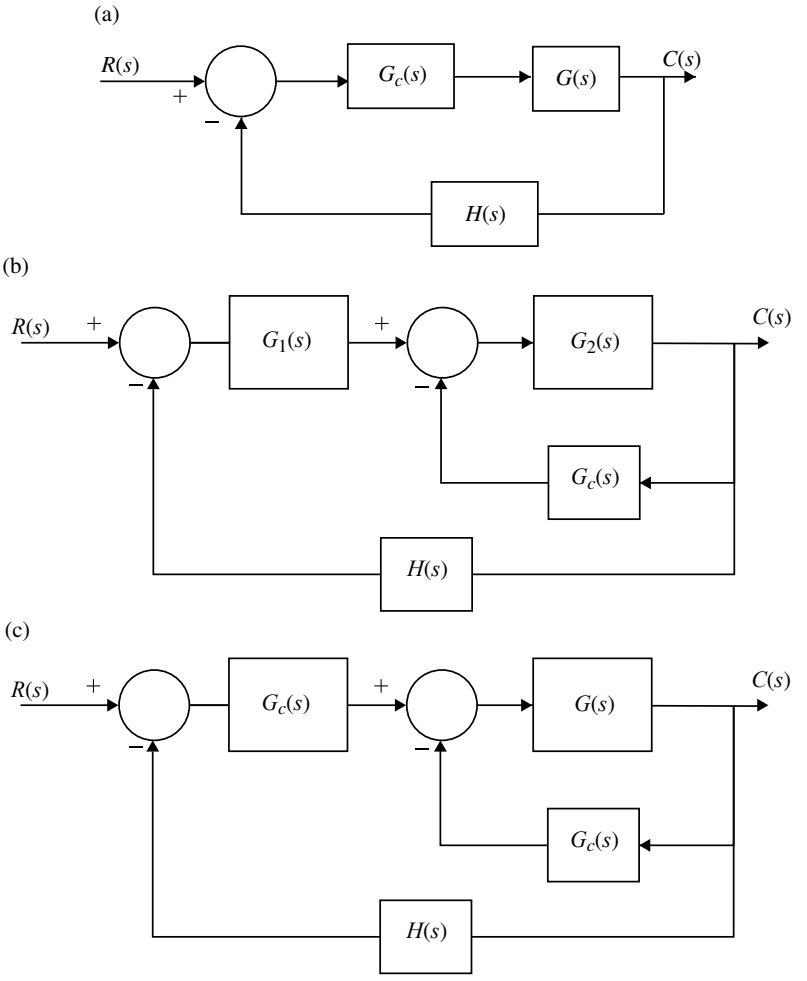


Fig. 6.1 **a** Location of a series compensator; **b** Parallel compensator and **c** Series-parallel compensator

overshoot, rise time, settling time and the steady state error. The root locus method is used to design compensators which satisfy time domain specifications. By this method, the shape of the root locus is altered by the addition of poles and zeros of the compensator so that the required performance specifications are met.

Although a variety of frequency response methods are available for the system analysis, the design of electric network compensator is carried out using either Bode diagram or Nichols chart. The one big advantage of using Bode diagram in preference to Nichols chart is that, while designing a compensator in the frequency domain, if the given specifications are not met, then it is necessary to reshape only that particular part of the Bode diagram which is contributed by the pole or the zero of the compensator.

It is easy to identify this in the Bode diagram, whereas in the Nichols chart, the entire frequency response plot has to be redrawn. Thus, we choose, Bode diagram to design the compensator in the frequency domain.

6.3 Time and Frequency Domain Approaches

We choose the root locus method to design a compensator to meet the time domain specifications. The poles and zeros of the compensator reshape the root locus of the system so that the closed loop poles are placed in locations in the s-plane that give desired transient response. In the frequency domain approach, the Bode diagram is used for the design of the compensator. When the compensator is added to the Bode

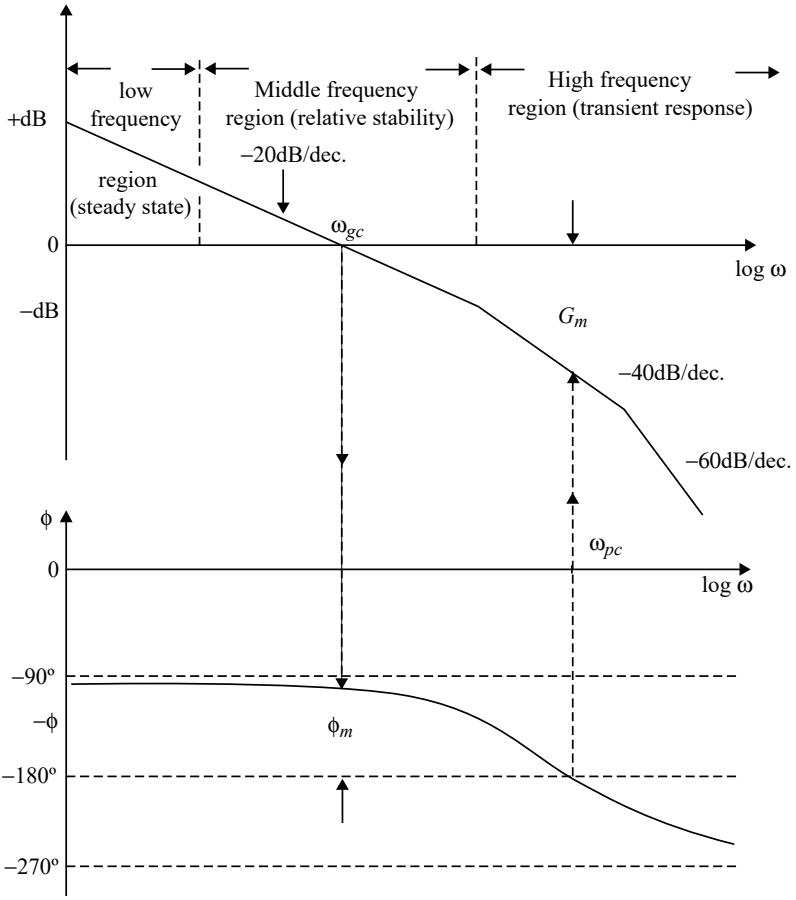


Fig. 6.2 Frequency response characteristic illustrating, low, middle and high frequency regions

plot of the original system, only the portion that corresponds to the poles and zeros of the compensator need to be reshaped in the original Bode diagram. Further, if the open loop gain is varied, the magnitude curve of the Bode plot is shifted up or down without changing the slope of any part of Bode diagram while retaining the original phase curve. In the Bode diagram, the frequency range which is far below the gain crossover frequency is termed as low frequency region. This region indicates the steady state behaviour of the closed loop system. The region nearer to the gain crossover frequency is called medium frequency region which indicates the relative stability of the closed loop system. The region far above the gain crossover frequency is called high frequency region and indicates the complexity of the system. In this region, it is difficult to control the transient response of the system. Therefore, to achieve steady state accuracy, the gain in the low frequency region should be as high as possible. To get satisfactory relative stability, the slope of the log-magnitude curve in the Bode diagram near the gain crossover frequency should be −20 dB/dec and to achieve the desired phase margin, this slope should be extended over a wide frequency range. To improve the transient response and to suppress the effects of noise, the gain should be attenuated in the high frequency region of the Bode diagram. This is illustrated in Fig. 6.2.

6.4 Compensator Design in the Frequency Domain

We describe below the design methodology for series (cascade) compensators in the frequency domain. The most commonly used series compensators are:

1. Lead compensator
2. Lag compensator
3. Lag–lead compensator.

We discuss below, the design of the above compensators in the frequency domain using electric components.

6.4.1 Design of Lead Compensator

A typical lead compensator electric network is shown in Fig. 6.3. The transfer function of the lead compensator is determined as given below. Let Z_1 be the equivalent impedance of the parallel combination of R_1 and C

$$Z_1 = \frac{R_1}{(1 + R_1 C s)}$$

The T.F. of the compensator is

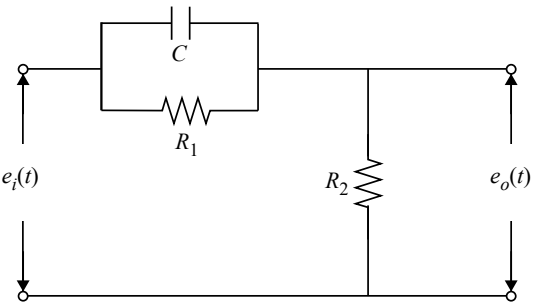


Fig. 6.3 Lead compensator electric network

$$\frac{E_o}{E_i}(s) = \frac{R_2}{R_2 + \frac{R_1}{(1+R_1Cs)}}$$
$$= \frac{R_2(1+R_1Cs)}{[(R_1+R_2)+R_1R_2Cs]}$$

$$\frac{E_o}{E_i}(s) = \frac{R_2}{(R_1+R_2)} \frac{(1+R_1Cs)}{\left[1+\frac{R_2R_1Cs}{(R_1+R_2)}\right]}$$

$$\boxed{\frac{E_o}{E_i}(s) = \alpha \frac{(1+Ts)}{(1+\alpha Ts)}} \tag{6.1}$$

where

$$\alpha = \frac{R_2}{R_1+R_2} < 1$$
$$T = R_1C$$

The sinusoidal T.F. of Equation (6.1) is written as

$$\frac{E_o}{E_i}(j\omega) = \frac{\alpha(1+j\omega T)}{(1+j\alpha\omega T)}$$
$$\frac{E_o}{E_i}(j\omega) = \frac{\alpha\sqrt{(1+\omega^2T^2)}}{\sqrt{(1+\alpha^2\omega^2T^2)}} \angle \tan^{-1}\omega T - \tan^{-1}\alpha\omega T \tag{6.2}$$

From Eq. (6.2), it is evident that the phase contribution of the lead network over the frequency range $0 \leq \omega \leq \infty$ is from 0° to 90°. In other words, the output phase leads the input and that is why it is called a lead network.

The lead network is inserted into the system so that the system performance deficiency is compensated. For this, appropriate values for α and T are to be chosen.

α and T depend on network parameters R_1, R_2 and C. Some useful information is obtained from the frequency response characteristics of the lead network which are given below.

The pole–zero diagram is shown in Fig. 6.4a. For the small value of α, the pole is located far to the left in LHP. The smaller the value of α, the higher is the angle contribution from the lead compensator. However, the lower limit of α is normally restricted to $\alpha = 0.07$ from the view point of physical realization. Further, if $1/\alpha$ becomes high, noise is amplified. For $\alpha = 0.07$, the maximum phase lead contribution is restricted to $\theta_m = 60°$. If the phase lead requirement exceeds 60°, then it is advisable to connect two-phase lead compensators in a cascade.

The polar plot of lead compensator is drawn using Eq. (6.2) and is shown in Fig. 6.4b. From Fig. 6.4b, the following relationships are obtained.

Let ω_m be the frequency at which the phase angle contribution from the lead network is maximum. Let this be denoted by θ_m. From Eq. (6.2), for $\omega = 0$, $|G(j\omega)| = \alpha\angle 0°$. For $\omega = \infty$; $G(j\omega) = 1$. Thus, the polar plot is a semi-circle with centre $(1 + \alpha)/2$ and radius $r = (1 - \alpha)/2$. From the right angle triangle OAB, the value of $G(j\omega)$ at point A is obtained as $\sqrt{\alpha}$. Further

$$\sin\theta_m = \frac{(1-\alpha)/2}{(1+\alpha)/2} = \frac{(1-\alpha)}{(1+\alpha)}$$

$$\boxed{\alpha = \frac{(1-\sin\theta_m)}{(1+\sin\theta_m)}} \tag{6.3}$$

The frequency ω_m at which the maximum phase angle is contributed is obtained as follows:

$$\begin{aligned}\theta_m &= \tan^{-1}\omega_m T - \tan^{-1}\alpha\omega_m T \\ &= \tan^{-1}\frac{\omega_m(1-\alpha)T}{(1+\omega_m^2 T^2\alpha)} \\ \tan\theta_m &= \frac{\omega_m(1-\alpha)T}{(1+\omega_m^2 T^2\alpha)}\end{aligned}$$

From Fig. 6.4b and the triangle OAB, we get

$$\begin{aligned}\tan\theta_m &= \frac{(1-\alpha)}{2\sqrt{\alpha}} \\ \frac{(1-\alpha)}{2\sqrt{\alpha}} &= \frac{\omega_m(1-\alpha)T}{(1+\omega_m^2 T^2\alpha)}\end{aligned}$$

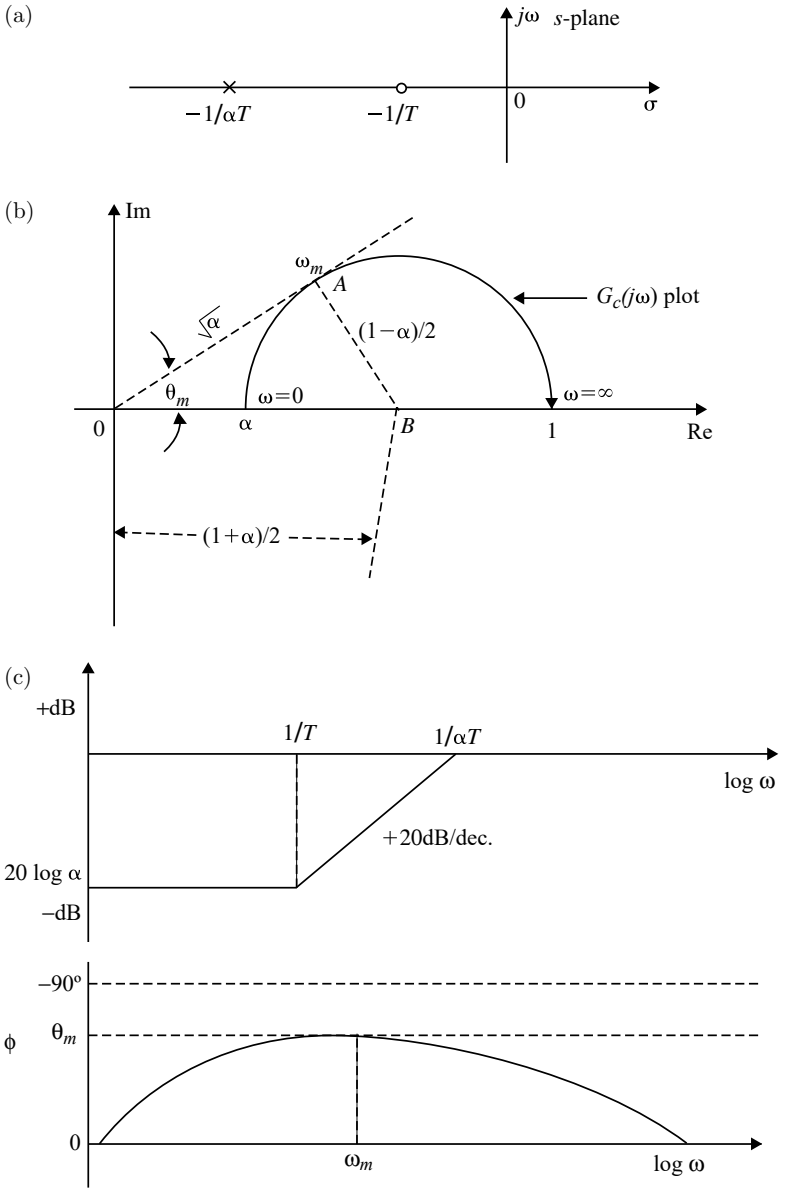


Fig. 6.4 **a** Pole–zero diagram of lead compensator. **b** Polar plot of lead compensator. **c** Bode diagram of lead compensator

Solving the above equation, we get

$$\boxed{\omega_m = \frac{1}{\sqrt{\alpha}T}} \tag{6.4}$$

The Bode diagram for the lead network is drawn using Eq. (6.1). The corner frequencies are $1/T$ and $1/\alpha T$. The log-magnitude *versus* ω and phase angle ϕ *versus* ω are shown in Fig. 6.4c. From the magnitude curve, it is evident that the low frequencies are attenuated and the high frequencies are passed and acts as a high-pass filter. From the phase angle characteristics, it is observed that the maximum phase θ_m contributed by the lead network occurs at $\omega = \omega_m$ which is the geometric mean of the two corner frequencies and is given by Eq. (6.4). The geometric mean is given by

$$\log \omega_m = \frac{1}{2}\left(\log \frac{1}{T} + \log \frac{1}{\alpha T}\right)$$

$$\omega_m = \frac{1}{\sqrt{\alpha}T}$$

The same result is derived from the polar plot shown in Fig. 6.4b.

6.4.2 *Design Procedure for Lead Compensator in the Frequency Domain*

The phase lead compensator is inserted into the system to compensate for the deficiencies in the frequency response specifications such as phase margin, gain margin, resonant peak and steady state error. The primary objective of inserting the phase lead network is to reshape the frequency response of the system by providing phase lead angle which offsets the excessive phase lag that is inherently present in the original system. By doing so, the deficiencies in the frequency response specifications are compensated. The step-by-step procedure is given below:

1. Let $G(s)$ be the T.F. of the uncompensated system and $G_c(s)$ be the T.F. of the lead compensator as given in Eq. (6.1). Let K be the gain of the uncompensated system. Determine the gain K which satisfies the steady state error requirement which is normally given in terms of static error constants.
2. For this value of K obtained in step 1, draw the frequency response plot for $G(j\omega)$ in the Bode diagram and determine the phase margin.
3. If the phase margin determined in step 2, it is not sufficient enough to meet the given phase margin in the specification, determine the amount of deficiency which is to be contributed by the lead compensator. Let it be ϕ.
4. When the lead compensator is inserted into the system, the gain crossover frequency of the composite system (original system + compensator) is shifted to the

right and the phase margin is now decreased. To account for this, add an additional phase angle of roughly about 7° to 12° in ϕ obtained in step 3. This gives θ_m.

5. Using Eq. (6.3), determine the attenuation factor α.
6. At $\omega_m = 1/\sqrt{\alpha}T$, the maximum phase shift θ_m occurs. At this frequency, the magnitude of $G(j\omega)$ is $-20\log(\frac{1}{\sqrt{\alpha}})$. Knowing α, determine ω_m. This frequency is the gain crossover frequency ω_{gc} of the compensated system $G_c(j\omega)G(j\omega)$. Make $\omega_m = \omega_{gc}$.
7. Knowing α from step 5 and $\omega_{gc} = \omega_m$ in step 6, determine T using Eq. (6.4). Thus, α and T, the lead compensator parameters are determined. The zero of the compensator is at $s = -1/T$ and the pole of the compensator is at $s = -1/\alpha T$.
8. For the compensated system, determine whether the given specifications are satisfied. If not, change the locations of the poles and zeros of the compensator and repeat the design steps from 3 to 7.

Example 6.1

The open loop T.F. of an uncompensated system is

$$G(s) = \frac{5}{s(s+2)}$$

Design a lead compensator for the system so that the static velocity error constant $K_v \geq 20\ \text{sec}^{-1}$, the phase margin is at least 55° and the gain margin is at least 12 dB.

(*Anna University, December, 2005*)

Solution

$$G(s) = \frac{5}{s(s+2)}$$

$$K_v = \lim_{s\to 0} sG(s) = \lim_{s\to 0} \frac{s5}{s(s+2)} = 2.5$$

But the required $K_v = 20$. Hence, an amplifier with a gain of 8 is to be connected in cascade with the compensator. Now the T.F. becomes

$$G(s) = \frac{20}{s(1+\frac{s}{2})}$$

(To draw the Bode plot, the T.F. is to be in time constant form). The Bode plot of the uncompensated system is shown in Fig. 6.5a.

The Bode diagram for the uncompensated system is sketched as shown in Fig. 6.5a. From magnitude plot, the gain crossover frequency ω_{gc} is obtained as 6.32. Corresponding to this the phase angle is

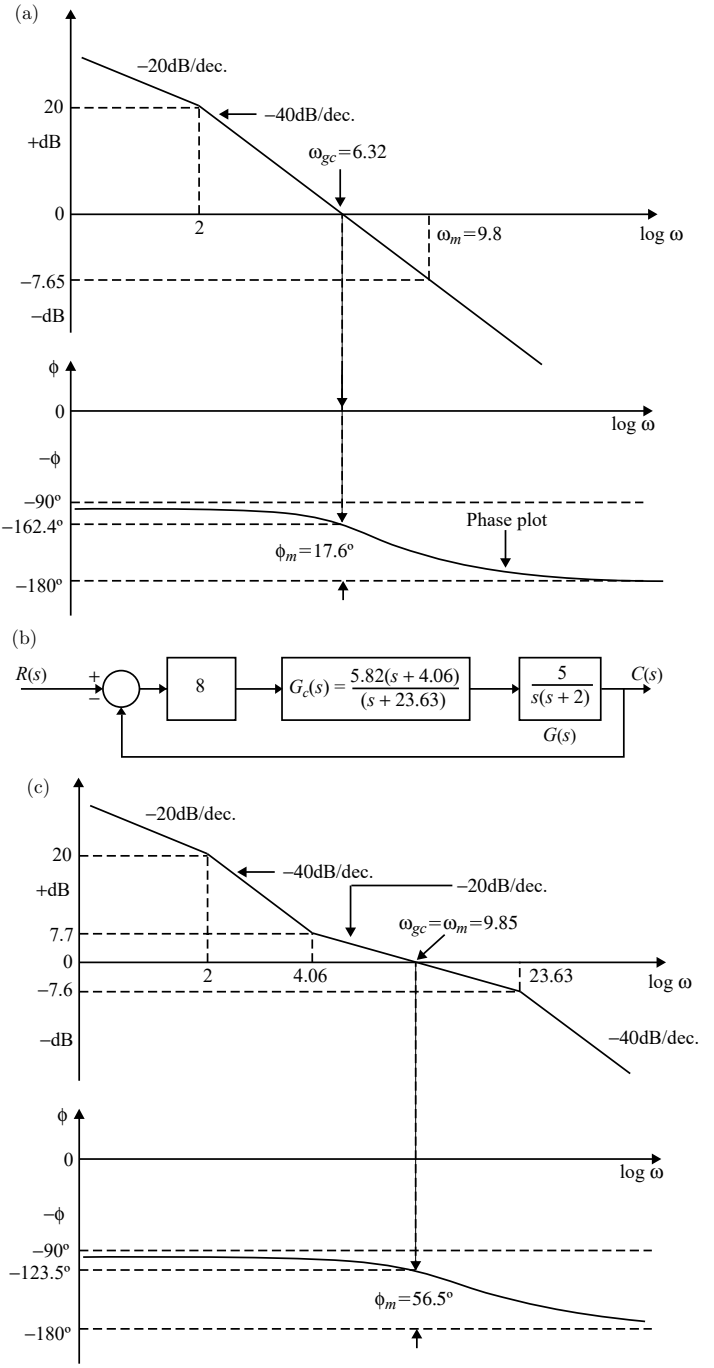


Fig. 6.5 **a** Bode diagram of uncompensated system. **b** Block diagram of the compensated system. **c** Block diagram of the compensated system

$$\phi = -90 - \tan^{-1}\frac{6.32}{2} = -162.4^\circ$$

Hence, the phase margin of the uncompensated system is $180^\circ - 162.4^\circ = 17.6^\circ$. However, the required phase margin is 55°. Hence, the additional phase lead to be provided by the lead compensator is $55^\circ - 17.6^\circ = 37.4^\circ$. Adding another 7.6° for the effect of ω_{gc} shifted to the right, $\theta_m = 37.4^\circ + 7.6^\circ = 45^\circ$. Using Eq. (6.3), we get

$$\alpha = \frac{(1 - \sin\theta_m)}{(1 + \sin\theta_m)} = \frac{(1 - \sin 45^\circ)}{(1 + \sin 45^\circ)} = 0.172$$

As stated in step 6

$$20\log\sqrt{\alpha} = 20\log\sqrt{0.172} = -7.65 \text{ dB}$$

Corresponding to -7.65 dB, the new gain crossover frequency which is nothing but ω_m is obtained from the $G(j\omega)$ magnitude plot as

$$\begin{aligned} 40\log\frac{\omega_m}{6.32} &= 7.65 \\ \omega_m &= 9.8 \end{aligned}$$

Using Eq. (6.4), we get

$$\begin{aligned} T &= \frac{1}{\omega_m\sqrt{\alpha}} = \frac{1}{9.8\sqrt{0.172}} = 0.246 \\ \frac{1}{\alpha T} &= \frac{1}{0.172 \times 0.246} = 23.63 \\ \alpha T &= 0.0423 \end{aligned}$$

The compensator T.F. is

$$G_c(s) = \frac{\alpha(1 + Ts)}{(1 + \alpha Ts)} = \frac{0.172(1 + 0.246s)}{(1 + 0.0423s)}$$

Taking 0.172 into the over all gain of the system which will ultimately satisfy the given K_v, the T.F. of the compensated system is obtained as

$$G_c(s)G(s) = \frac{20(1 + 0.246s)}{(1 + 0.0423s)s(1 + \frac{s}{2})}$$

The block diagram of the compensated system is shown in Fig. 6.5b

$$\boxed{\begin{aligned} T &= 0.246 \\ \alpha &= 0.172 \end{aligned}}$$

Now we draw the Bode diagram for the compensated system

$$G_c(s)G(s) = \frac{20(1+0.246s)}{(1+0.0423s)s(1+\frac{s}{2})}$$

$$\phi = \tan^{-1} 0.246\omega - 90 - \tan^{-1}\frac{\omega}{2} - \tan^{-1} 0.0423\omega$$

The following table is prepared for the phase angle:

ω	0	1	2	3	5	10	∞
ϕ	$-90°$	$-105°$	$-114°$	$-117°$	$-119°$	$-124°$	$-180°$

The Bode diagram for the compensated system is shown in Fig. 6.5c.

The phase margin is checked as follows.

At $\omega = 9.85$

$$\begin{aligned} \phi &= -90° - \tan^{-1}\frac{9.85}{2} - \tan^{-1}\frac{9.85}{23.63} + \tan^{-1}\frac{9.85}{4.06} \\ &= -90° - 78.5° - 22.6° + 67.5° \\ &= -123.5° \end{aligned}$$

Phase margin, $\phi_m = 180° - 123.5°$

$$\boxed{\phi_m = 56.5°}$$

The required phase margin is 55°. Hence, the required phase margin is met. Since the phase angle of the compensated system is always less than 180°, the gain margin is ∞. The required gain margin is 12 dB only. Thus, the lead compensator is designed.

Example 6.2

Design cascade phase lead compensator for a system whose open loop T.F. is

$$GH(s) = \frac{10}{s(1+s)}$$

to have a phase margin of 45° using Bode plot.

(*Anna University, December, 2003*)

Solution

1. The Bode plot of the uncompensated system is shown in Fig. 6.6a. The gain crossover frequency is obtained as

$$40 \log \omega_{gc} = 20$$
$$\omega_{gc} = 3.16$$

Corresponding to $\omega_{gc} = 3.16$, the phase angle of the uncompensated system is

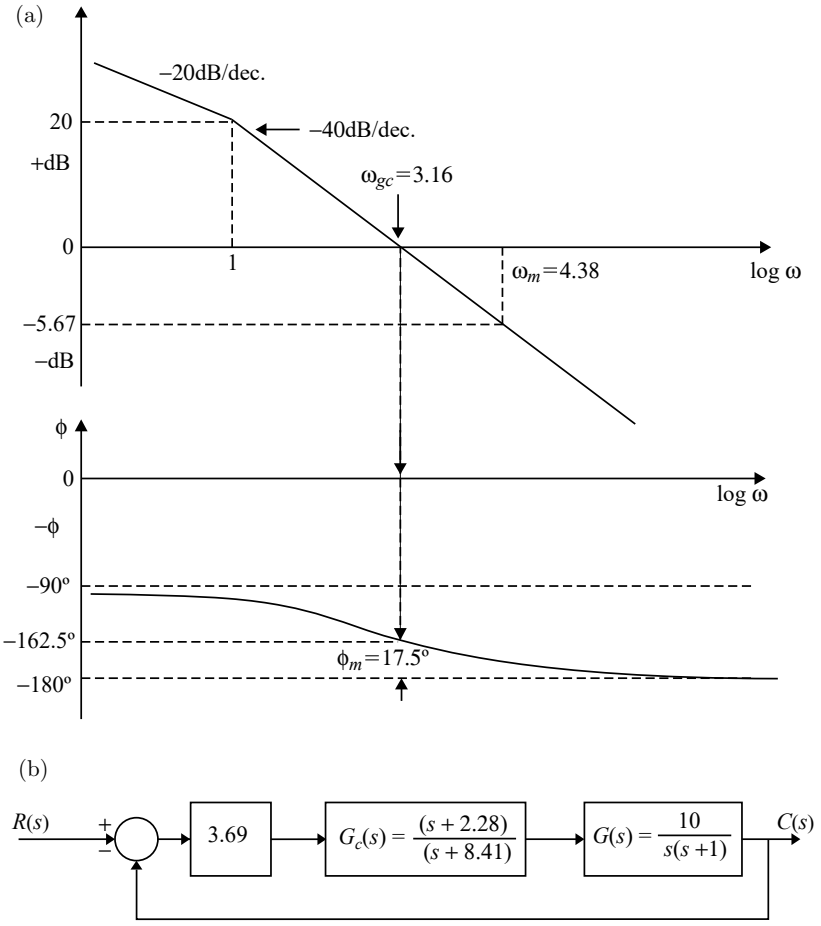


Fig. 6.6 **a** Bode plot of uncompensated system. **b** Block diagram of compensated system. **c** Bode diagram for the compensated system

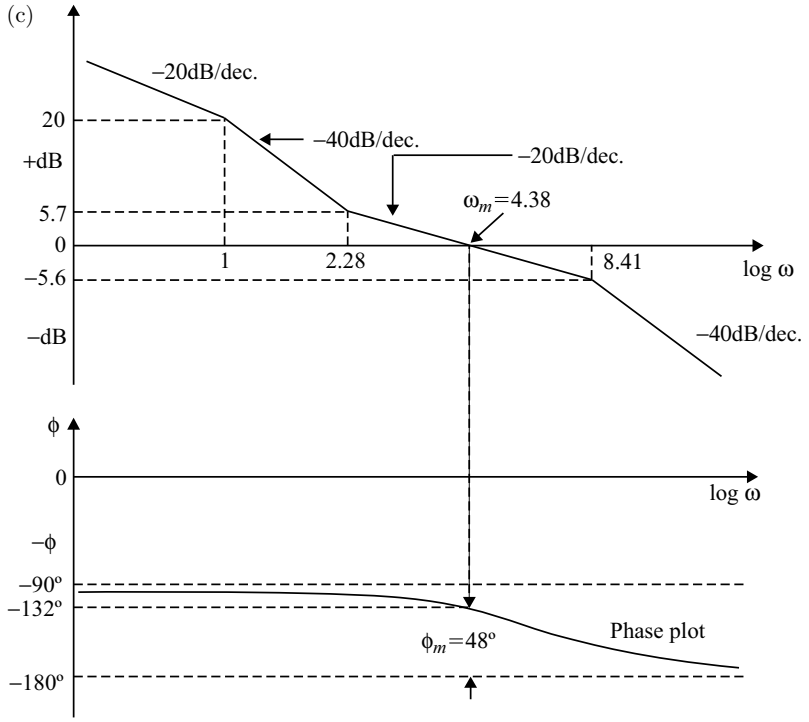


Fig. 6.6 (continued)

$$\begin{aligned}\angle G(j\omega_{gc}) &= -90^\circ - \tan^{-1}\omega_{gc} \\ &= -90^\circ - \tan^{-1} 3.16 = -162.5^\circ\end{aligned}$$

The phase margin of the uncompensated system, therefore, is

$$\phi_m = 180^\circ - 162.5^\circ = 17.5^\circ$$

2. The required phase margin is 45°. The phase angle deficiency is $45^\circ - 17.5^\circ = 27.5^\circ$. An allowance of say 7.5° is given for the shifting of the new gain crossover frequency when the lead network is inserted. Thus, the maximum phase angle contribution of the lead compensator becomes $\theta_m = 27.5 + 7.5^\circ = 35^\circ$. The attenuation factor α is determined using Eq. (6.3)

$$\begin{aligned}\alpha &= \frac{(1 - \sin\theta_m)}{(1 + \sin\theta_m)} \\ &= \frac{(1 - \sin 35^\circ)}{(1 + \sin 35^\circ)}\end{aligned}$$

$$\boxed{\alpha = 0.271}$$

3. The gain crossover frequency ω_{gc} is now shifted to the right by adding a magnitude of $20 \log \sqrt{\alpha}$. The new gain crossover frequency now becomes ω_m.

$$\begin{aligned} 20 \log \sqrt{\alpha} &= 20 \log \sqrt{0.271} \\ &= -5.67 \end{aligned}$$

From Fig. 6.6a, the following equation is written for the magnitude plot

$$\begin{aligned} 40 \log \frac{\omega_m}{3.16} &= 5.67 \\ \omega_m &= 4.38 \end{aligned}$$

4. Using Eq. (6.4), the time constant of the zero of the lead compensator is obtained as

$$\begin{aligned} \frac{1}{T} &= \omega_m \sqrt{\alpha} \\ &= 4.38\sqrt{0.271} \\ &= 2.28 \end{aligned}$$

$$\boxed{T = 0.4386}$$

5. The T.F. of the lead compensator becomes

$$\begin{aligned} G_c(s) &= \frac{(s + \frac{1}{T})}{(s + \frac{1}{\alpha T})} \\ &= \frac{(s + 2.28)}{(s + 8.41)} \end{aligned}$$

6. The forward path T.F. of the compensated system now becomes

$$G_c(s)G(s) = \frac{(s + 2.28)}{s(s + 8.41)} \frac{10}{(s + 1)}$$

The overall system gain is reduced by a factor α. Hence, an amplifier with a gain of

$$\frac{1}{\alpha} = \frac{1}{0.271} = 3.69$$

is to be introduced in the forward path. The block diagram of the compensated system is shown in Fig. 6.6b.

7. The Bode diagram of the compensated system is shown in Fig. 6.6c. At the new gain crossover frequency $\omega_m = 4.38$, the phase angle of the compensated system is obtained as

$$\begin{aligned}\phi &= -90 - \tan^{-1}\omega_m - \tan^{-1}\frac{\omega_m}{8.41} + \tan^{-1}\frac{\omega_m}{2.28} \\ &= -90^\circ - \tan^{-1}4.38 - \tan^{-1}\frac{4.38}{8.41} + \tan^{-1}\frac{4.38}{2.28} \\ &= -90^\circ - 77.1^\circ - 27.5^\circ + 62.5^\circ \\ &= -132^\circ\end{aligned}$$

The phase margin of the compensated system is

$$\begin{aligned}\phi_m &= 180^\circ + \phi \\ &= 180^\circ - 132^\circ\end{aligned}$$

$$\boxed{\phi_m = 48^\circ}$$

The required phase margin is only 45°. Thus, the given specifications are met and the compensator designed.

Example 6.3

The open loop T.F. of a unity feedback control system is

$$G(s) = \frac{K}{s(s+2)}$$

Design a suitable lead compensator to meet the following specifications:

1. Velocity error constant $K_v \geq 12\ \text{s}^{-1}$
2. Phase margin $\phi_m \geq 45^\circ$

Solution

$$K_v = \lim_{s\to 0} sG(s)$$

$$\begin{aligned}K_v &= \lim_{s\to 0}\frac{sK}{s(s+2)} \\ &= 12\end{aligned}$$

$$\boxed{K = 24}$$

$$G(s) = \frac{12}{s(1+\frac{s}{2})}$$

The Bode plot of the uncompensated system is shown in Fig. 6.7a. The sinusoidal T.F. is

$$G(j\omega) = \frac{12}{\omega\sqrt{1+\frac{\omega^2}{4}}} \angle -90^\circ - \tan^{-1}\frac{\omega}{2}$$

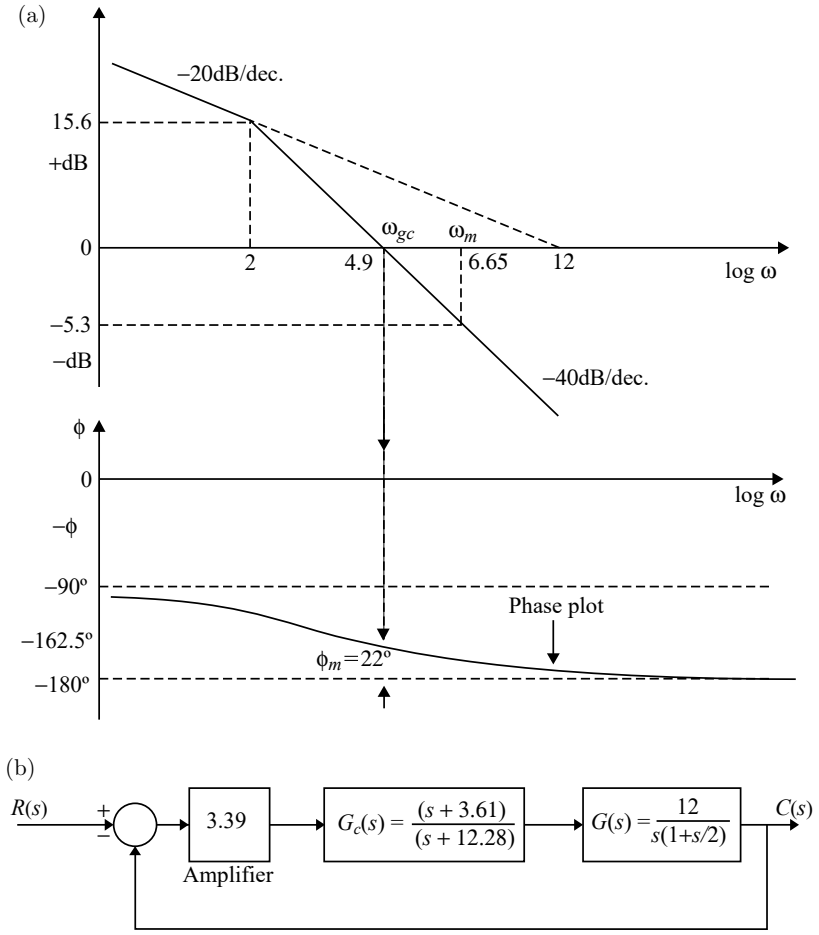


Fig. 6.7 **a** Bode plot for $G(s) = \frac{12}{s(1+\frac{s}{2})}$. **b** Bode diagram for the compensated system. **c** Bode diagram for the compensated system

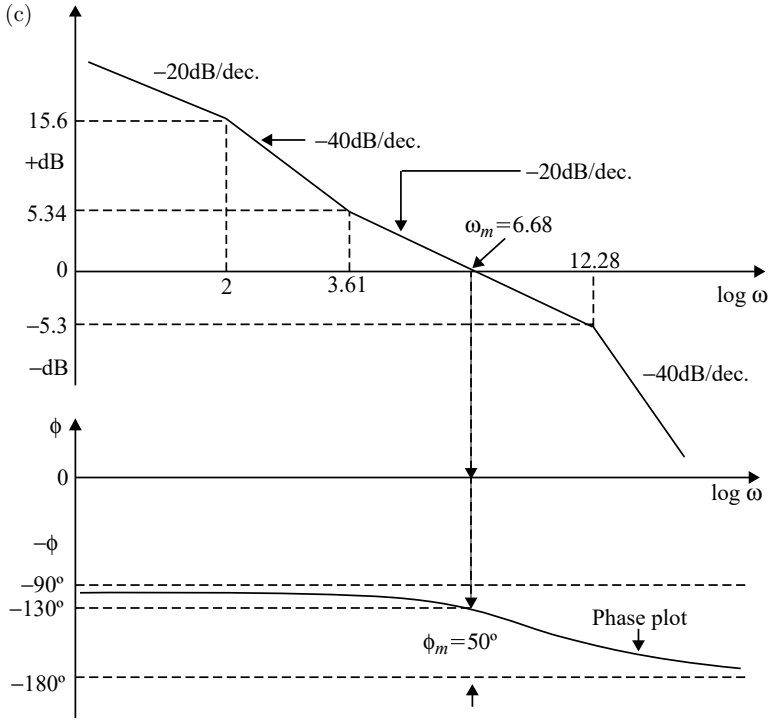


Fig. 6.7 (continued)

The gain crossover frequency is obtained as

$$\begin{aligned} 40 \log \frac{\omega_{gc}}{2} &= 15.6 \\ \omega_{gc} &= 4.9 \end{aligned}$$

The angle contribution when $\omega = \omega_{gc}$ is

$$\begin{aligned} \phi &= -90^\circ - \tan^{-1} \frac{4.9}{2} \\ &= -158^\circ \end{aligned}$$

Hence, the phase margin is obtained as

$$\begin{aligned} \phi_m &= 180^\circ - 158^\circ \\ &= 22^\circ \end{aligned}$$

(Since the phase plot varies from $-90°$ to $-180°$ for $0 \leq \omega \leq \infty$, the general shape of phase plot is known, and hence the phase angle at $\omega = \omega_{gc}$ alone need to be determined. For other values of ω, the phase angle ϕ need not be known).

The required phase margin is 45°. Therefore, the angle contribution from the lead network is $(45° - 22°) = 23°$ and an allowance of 10° is added to compensate for the decrease in phase margin when the lead network is inserted and ω_{gc} is shifted to the right. The angle deficiency now becomes $23° + 10° = 33°$. This is taken as $\theta_m = 33°$. Using Eq. (6.3), we get the attenuation factor as

$$\begin{aligned} \alpha &= \frac{(1 - \sin\theta_m)}{(1 + \sin\theta_m)} \\ &= \frac{(1 - \sin 33°)}{(1 + \sin 33°)} \\ &\boxed{\alpha = 0.295} \end{aligned}$$

When the lead network is inserted, the gain crossover frequency is shifted to the right and becomes $\omega_{gc} = \omega_m$. The amount of attenuation required is

$$\begin{aligned} 20\log\sqrt{\alpha} &= 20\log\sqrt{0.295} \\ &= -5.3 \text{ dB} \end{aligned}$$

Corresponding to -5.3 dB, attenuation ω_m is calculated from Fig. 6.7a as

$$\begin{aligned} 40\log\frac{\omega_m}{4.9} &= 5.3 \\ \omega_m &= 6.65 \end{aligned}$$

Using Eq. (6.4), we get

$$\begin{aligned} \frac{1}{T} &= \omega_m\sqrt{\alpha} \\ &= 6.65\sqrt{0.295} \\ &= 3.61 \\ &\boxed{T = 0.276} \\ \frac{1}{\alpha T} &= \frac{1}{0.295 \times 0.276} \\ &= 12.28 \\ \alpha T &= \frac{1}{12.28} \\ &= 0.0814 \end{aligned}$$

The T.F. of the compensator is

$$G_c(s) = \frac{(s + \frac{1}{T})}{(s + \frac{1}{\alpha T})} = \frac{(s + 3.61)}{(s + 12.28)}$$

To nullify the effect of attenuation factor $\alpha = 0.295$, an amplifier with a gain of

$$\frac{1}{\alpha} = \frac{1}{0.295} = 3.39$$

is connected in cascade with the compensator. The complete block in Fig. 6.7b.

The T.F. of the compensated system is obtained as

$$G_c(s)G(s) = \frac{3.39(s + 3.61)}{s(s + 12.285)} \frac{12}{(1 + \frac{s}{2})} = \frac{12(1 + \frac{s}{3.61})}{s(1 + \frac{s}{12.28})} \frac{1}{(1 + \frac{s}{2})}$$

The velocity error constant of the compensated system is easily verified as 12 from the above equation. The corner frequencies of the above T.F. are 2, 3.61 and 12.28. The Bode magnitude and phase plots are shown in Fig. 6.7c.

From Fig. 6.7c, the phase margin of the given system is checked as follows. From the magnitude plot, the gain crossover frequency $\omega_m = \omega_{gc}$ is obtained as 6.68.

At this frequency, the phase angle of the compensated system is (this alone need to be calculated)

$$\begin{aligned}\phi &= -90^\circ - \tan^{-1}\frac{\omega_m}{2} - \tan^{-1}\frac{\omega_m}{12.28} + \tan^{-1}\frac{\omega_m}{3.61} \\ &= -90^\circ - \tan^{-1}\frac{6.68}{2} - \tan^{-1}\frac{6.68}{12.28} + \tan^{-1}\frac{6.68}{3.61} \\ &= -90^\circ - 73.3^\circ - 28.3^\circ + 61.6^\circ \\ &= -130^\circ\end{aligned}$$

The phase margin is

$$\phi_m = 180^\circ - 130^\circ$$

$$\boxed{\phi_m = 50^\circ}$$

All the given specifications are met with the compensator designed, thus making the completion of the design.

Two Stage Compensator Design

Example 6.4

A certain unity feedback control system has an open loop T.F

$$G(s) = \frac{K}{s^2(1 + 0.2s)}$$

Design a lead compensator to meet the following specifications:

(a) Acceleration error constant $K_a = 10$.
(b) Phase margin $\phi_m = 35°$.

(*Anna University, December, 2006*)

Solution

$$G(s) = \frac{K}{s^2(1 + 0.2s)}$$

$$K_a = \lim_{s \to 0} s^2 G(s) = \lim_{s \to 0} \frac{s^2 10}{s^2(1 + 0.2s)} = 10$$

$$G(s) = \frac{10}{s^2(1 + 0.2s)}$$

The Bode plot of the uncompensated system is shown in Fig. 6.8a. The gain crossover frequency is

$$\omega_{gc} = \sqrt{10} = 3.16$$

At this frequency the phase angle of $G(j\omega)$ is

$$\angle G(j\omega_{gc}) = -180° - \tan^{-1} 3.16 \times 0.2 = -212.3°$$

The phase margin is

$$\phi_m = 180° - 212.3° = -32.3°$$

which is negative and the uncompensated system is unstable. However, the required phase margin is 35°. To achieve this, the total phase lead to be contributed by the lead network is $35° + 32.3° = 67.3°$. When the lead network is inserted, the gain crossover frequency 3.16 is shifted to the right and further phase lead to be contributed as is evident from Fig. 6.8a. An allowance of 22.7° is added and the required phase lead contribution now becomes 90°. If a single phase lead network is used, the attenuation factor α becomes

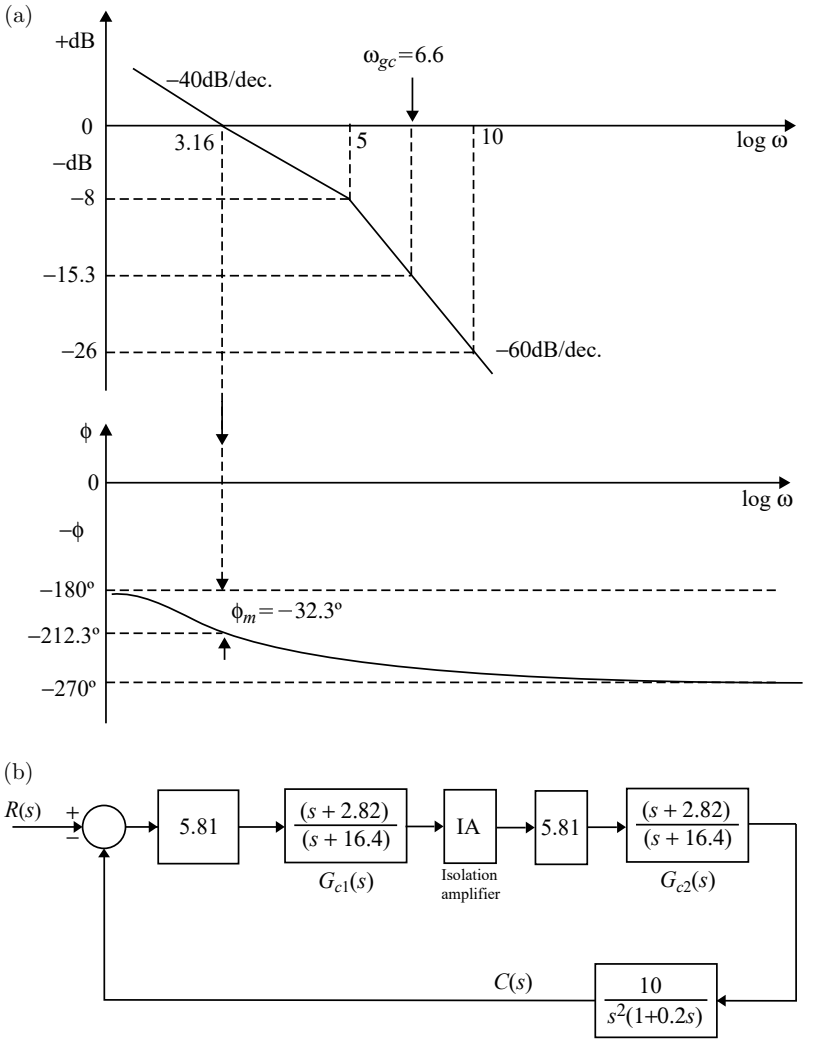


Fig. 6.8 **a** Bode diagram for an uncompensated system $G(s) = \dfrac{10}{s^2(1+0.2s)}$. **b** Two stage compensator system. **c** Bode diagram for the compensated system

$$\alpha = \frac{(1-\sin 90^\circ)}{(1+\sin 90^\circ)} = 0$$

To meet the steady state error requirement, an amplifier with a gain $\frac{1}{\alpha} = \infty$ is needed which is impractical. As stated earlier, the minimum value of α is restricted to 0.07 for a lead network, so that the noise level is within the permissible value. Therefore, a two stage lead network with an isolating amplifier is used to achieve compensation.

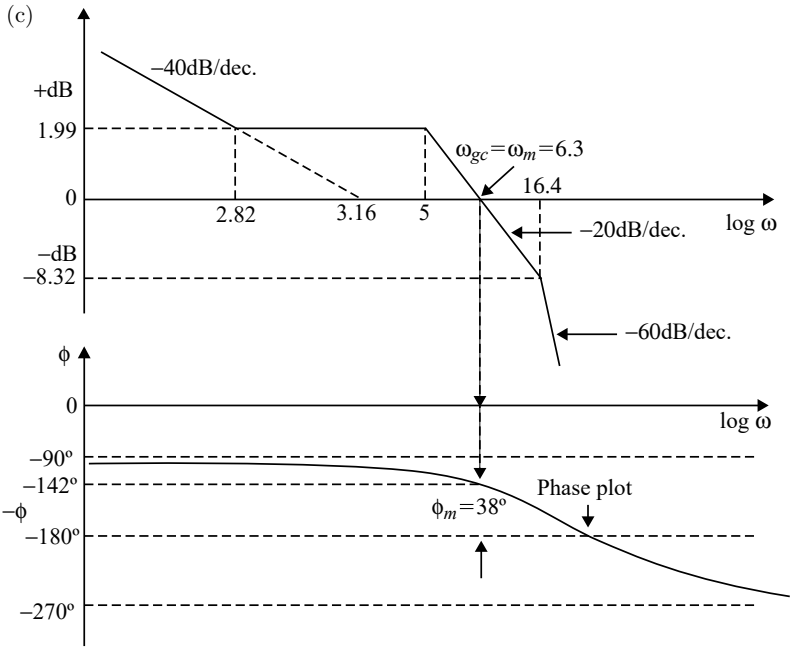


Fig. 6.8 (continued)

The maximum phase θ_m of each phase is chosen as 45°. Thus, α of a single stage network is calculated as

$$\alpha = \frac{(1 - \sin 45^\circ)}{(1 + \sin 45^\circ)}$$

$$\boxed{\alpha = 0.172}$$

To make ω_m as the new gain crossover frequency, when the two stage compensator is introduced the attenuation provided by both the stages are to be considered. Thus, the attenuation in dB is calculated as

$$2 \times 20 \log \sqrt{\alpha} = 2 \times 20 \log \sqrt{0.172} = -15.3 \text{ dB}$$

Corresponding to −15.3 dB, the gain crossover frequency $\omega_{gc} = \omega_m$ is calculated from Fig. 6.8a as

$$60 \log \frac{\omega_{gc}}{5} = 7.3$$
$$\omega_{gc} = \omega_m = 6.6$$
$$\omega_m = \frac{1}{\sqrt{\alpha} T}$$
$$\frac{1}{T} = \sqrt{\alpha}\omega_m = \sqrt{0.172} \times 6.6 = 2.82$$
$$\boxed{T = 0.354}$$
$$\alpha T = 0.172 \times 0.354 = 0.061$$
$$\frac{1}{\alpha T} = 16.4$$

The T.F. of a single stage lead compensator is

$$G_c(s) = \frac{(s + \frac{1}{T})}{(s + \frac{1}{\alpha T})} = \frac{(s + 2.82)}{(s + 16.4)}$$

To maintain the steady state error, an amplifier with a gain of

$$\frac{1}{\alpha} = \frac{1}{0.172} = 5.81$$

is to be connected in cascade with each stage. The block diagram of the compensated system is shown in Fig. 6.8b.

$$G_c(s) = G_{c1}(s)G_{c2}(s) = \frac{(1 + \frac{s}{2.82})^2}{(1 + \frac{s}{16.4})^2}$$
$$G_c(s)G(s) = \frac{(1 + \frac{s}{2.82})^2}{(1 + \frac{s}{16.4})^2} \frac{10}{s^2(1 + 0.2s)}$$

The Bode diagram of the compensated system is shown in Fig. 6.8c. From Fig. 6.8c, the gain crossover frequency $\omega_{gc} = \omega_m$ is obtained as Fig. 6.3. At this frequency, the phase angle is

$$\phi = -180^\circ - \tan^{-1} 0.2 \times 6.3 + 2 \tan^{-1} \frac{6.3}{2.82} - 2 \tan^{-1} \frac{6.3}{16.4}$$
$$= -180^\circ - 51.6^\circ + 131.5^\circ - 42^\circ = -142^\circ$$

The phase margin is

$$\phi_m = 180^\circ - 142^\circ = 38^\circ$$

The required phase margin is 35° only. The given specifications are met by the two stage compensator.

Example 6.5

The T.F. of a phase lead compensator is given by

$$G_c(s) = \frac{(1+3Ts)}{(1+Ts)}$$

where $T > 0$. Determine the maximum phase shift provided by the compensator. At what frequency it occurs?

Solution

$$G_c(s) = \frac{(1+3Ts)}{(1+Ts)}$$

$$G_c(j\omega) = \frac{(1+j3T\omega)}{(1+jT\omega)}$$

$$\angle G_c(j\omega) = \theta = (\tan^{-1} 3T\omega - \tan^{-1} T\omega) = \tan^{-1} \frac{3T\omega T\omega}{1+3T^2\omega^2}$$

$$\tan\theta = \frac{2\omega T}{(1+3T^2\omega^2)}$$

Let $\tan\theta = N$

$$N = \frac{2T\omega}{(1+3T^2\omega^2)}$$

Differentiating N with respect to ω, and equating to zero to get maxima we get

$$\frac{dN}{d\omega} = \frac{2T\omega(T^2\omega)6 - (1+3T^2\omega^2)T2}{(1+3T^2\omega^2)^2} = 0$$

$$12T^2\omega^2 - 2 - 6T^2\omega^2 = 0$$

$$6T^2\omega^2 = 2$$

$$\omega = \omega_m = \frac{1}{\sqrt{3}T}$$

$$\boxed{\omega_m = \frac{1}{\sqrt{3}T}}$$

Substituting this value of ω in N, the maximum phase shift is obtained

$$N = \frac{\frac{2T}{\sqrt{3}T}}{(1+1)} = \frac{1}{\sqrt{3}}$$

$$\tan\theta_m = \frac{1}{\sqrt{3}}$$

$$\boxed{\theta_m = 30^\circ}$$

6.4.3 Lag Compensator

The electric network lag compensator is shown in Fig. 6.9a. From Fig. 6.9

$$Z_1(s) = R_1$$

$$Z_2(s) = R_2 + \frac{1}{Cs} = \frac{(1+R_2Cs)}{Cs}$$

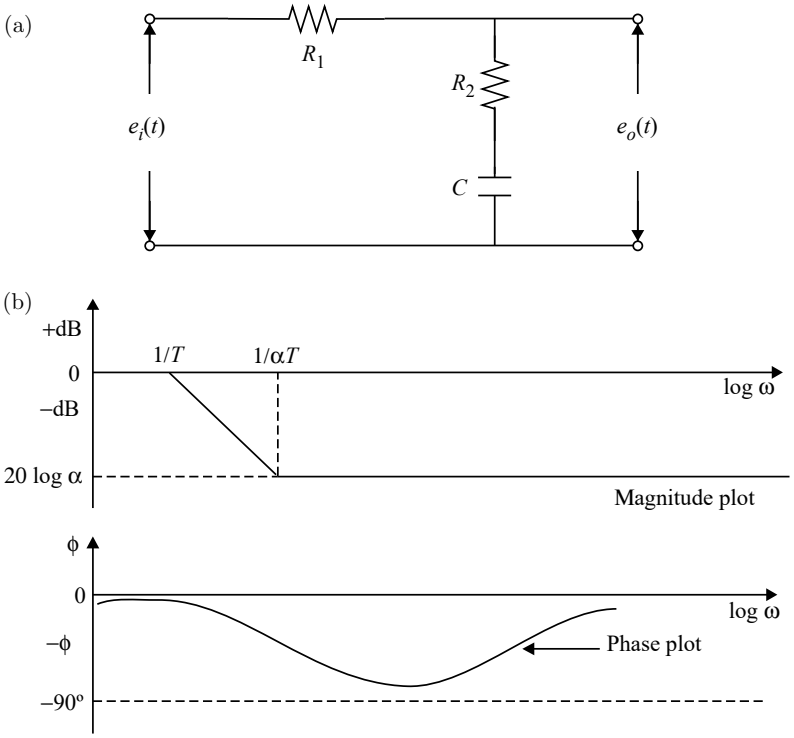


Fig. 6.9 **a** A lag compensator electric network. **b** Bode diagram for lag compensator. **c** Bode diagram with lag compensator

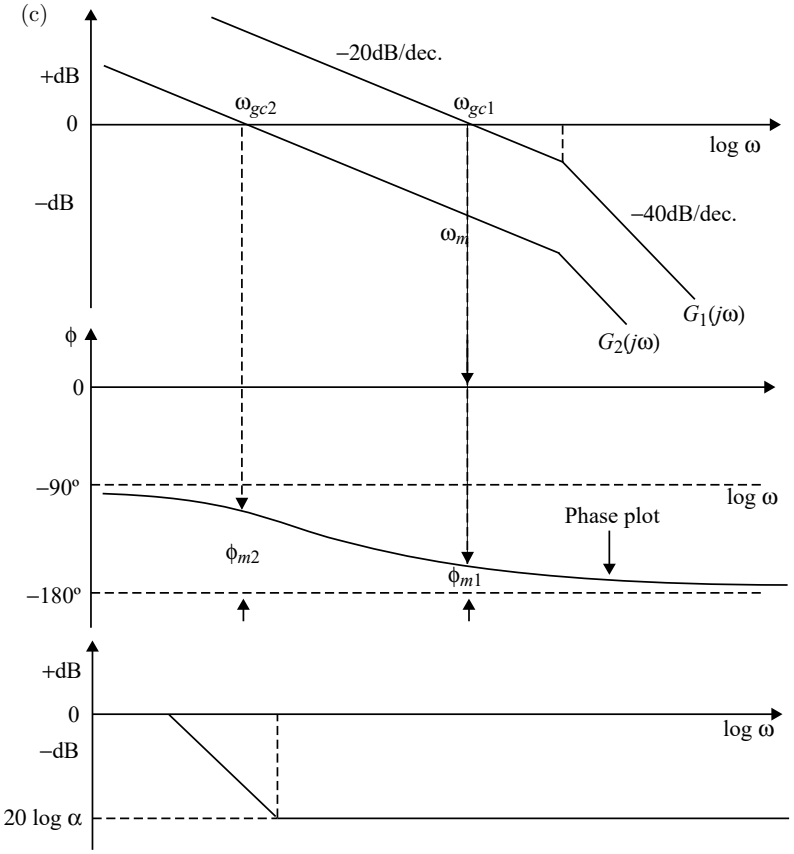


Fig. 6.9 (continued)

The T.F. of the network is

$$\frac{E_o}{E_i} = G_c(s)$$

$$\frac{E_o}{E_i} = \frac{Z_2(s)}{Z_1(s) + Z_2(s)}$$

$$= \frac{\frac{(1+R_2Cs)}{Cs}}{R_1 + \frac{(1+R_2Cs)}{Cs}}$$

$$= \frac{(1 + R_2Cs)}{[1 + (R_1 + R_2)Cs]}$$

$$\boxed{G_c(s) = \frac{(1 + \alpha T s)}{(1 + T s)}} \tag{6.5}$$

where

$$\alpha = \frac{R_2}{(R_1 + R_2)} < 1$$

and

$$T = (R_1 + R_2)C$$

Equation (6.5) which is in time constant form can be expressed in pole–zero form as

$$\boxed{G_c(s) = \frac{\alpha(s + z)}{(s + p)}} \tag{6.6}$$

where

$$z = \frac{1}{R_2 C} \quad \text{and} \quad p = \frac{1}{(R_1 + R_2)C}$$

and

$$\boxed{p = \alpha z} \tag{6.7}$$

The Bode diagram for Eq. (6.5) is shown in Fig. 6.9b. The values of $1/T$ and $1/\alpha T$ are chosen small so that they are closer to the origin. This provides attenuation for a wide range of frequencies.

The Bode diagram of a certain second order system is shown in Fig. 6.9c. $G_1(j\omega)$ plot shows the magnitude plot when the compensator is not inserted. The gain crossover frequency is ω_{gc1} and the phase margin is ϕ_{m1} which may not be sufficient enough to meet the specification requirement. When the lag compensator is introduced, the phase shift is not very much affected. However, the attenuation effect of the lag compensator shifts the $G_1(j\omega)$ plot downwards and is shown as $G_2(j\omega)$ in Fig. 6.9c. The gain crossover frequency is now shifted towards left and denoted as ω_{gc_2}. At this frequency, the phase angle contribution of $G_c(j\omega)$ is negligibly small and the phase margin is estimated from $G_2(j\omega)$ phase plot and is measured as ϕ_{m2}. Therefore, the design problem is approached in the reverse way. For the given phase margin, the phase contribution of the lag compensator, roughly 10° phase angle is added and the gain crossover frequency with this phase margin is determined and the design is completed as described below. The important things to be noted are:

1. In the design of the lead compensator, phase lead is contributed by the lead network at the gain crossover frequency.
2. In the design of the lag compensator, attenuation is provided at the gain crossover frequency.

The following step-by-step procedure is followed to design the lag compensator:

1. Determine the value of K of the uncompensated system which meets the steady state error requirement.
2. Determine the phase margin of the uncompensated system. If the phase margin requirement is not met, go to step 3.
3. To the given phase margin, add (say) 10°. For this phase margin, determine the frequency in the Bode diagram. This frequency becomes the gain crossover frequency.
4. The zero of the lag compensator is arbitrarily fixed at one tenth of the gain crossover frequency determined in step 3. By doing so, the effect of attenuation is shifted towards the left where the phase margin is high.
5. At the gain crossover frequency, the attenuation provided by the lag compensator is $20 \log \alpha$ in dB. The magnitude of the compensator is α. This makes

$$\alpha |G(j\omega_{gc})| = 1 \tag{6.8}$$

   Using Eq. (6.8), α is calculated.
6. To place the zero of the compensator at one tenth of the gain crossover frequency, we get

$$\frac{1}{\alpha T} = \frac{\omega_{gc}}{10} \tag{6.9}$$

   Knowing α and ω_{gc}, T is calculated using Eq. (6.9).
7. With the values of α and T determined in steps 5 and 6, the lag compensator is designed.
8. The Bode diagram for the compensated system is drawn and it is verified that the compensated system satisfies the required specifications. If not, the procedure is repeated from step three by adding more phase angle which is supposed to have been contributed by the lag network at the gain crossover frequency. With the new gain crossover frequency, the procedure is repeated.

Example 6.6

A unity feedback control system has an open loop T.F

$$G(s) = \frac{4}{s(1+2s)}$$

It is desired to obtain a phase margin of 38° without sacrificing K_v. Design a suitable lag network.

(*Anna University, April, 2005*)

Solution The Bode plot of uncompensated system is shown in Fig. 6.10a. The gain crossover of the uncompensated system is obtained from

$$40 \log \frac{\omega_{gc}}{0.5} = 18$$
$$\omega_{gc} = \sqrt{2}$$

At $\omega_{gc} = \sqrt{2}$, the phase angle of the uncompensated system is

$$\begin{aligned}\phi &= -90^\circ - \tan^{-1} 2\omega \\ &= -90^\circ - \tan^{-1} 2\sqrt{2} \\ &= -161^\circ\end{aligned}$$

The phase margin is $(180^\circ - 161^\circ) = 19^\circ$. But the required phase margin is 38°. To account for the phase contribution of the lag network when inserted, 12° is added to the required phase margin. The total phase angle at the new gain crossover frequency is $\phi = 180^\circ - (38^\circ + 12^\circ) = -130^\circ$. This point is identified at A in phase plot. The new gain crossover frequency is calculated from

$$-90^\circ - \tan^{-1} 2\omega = -130^\circ$$
$$\tan^{-1} 2\omega = 40^\circ$$
$$\omega = 0.42$$

$$\boxed{\omega_{gc} = 0.42}$$

$$\alpha |G(j\omega_{gc})| = 1$$
$$\alpha \frac{4}{\omega_{gc}\sqrt{1 + 4\omega_{gc}^2}} = 1$$
$$\alpha = \frac{0.42}{4}\sqrt{1 + 4(0.42)^2}$$

$$\boxed{\alpha = 0.137}$$

Using Eq. (6.9), we get

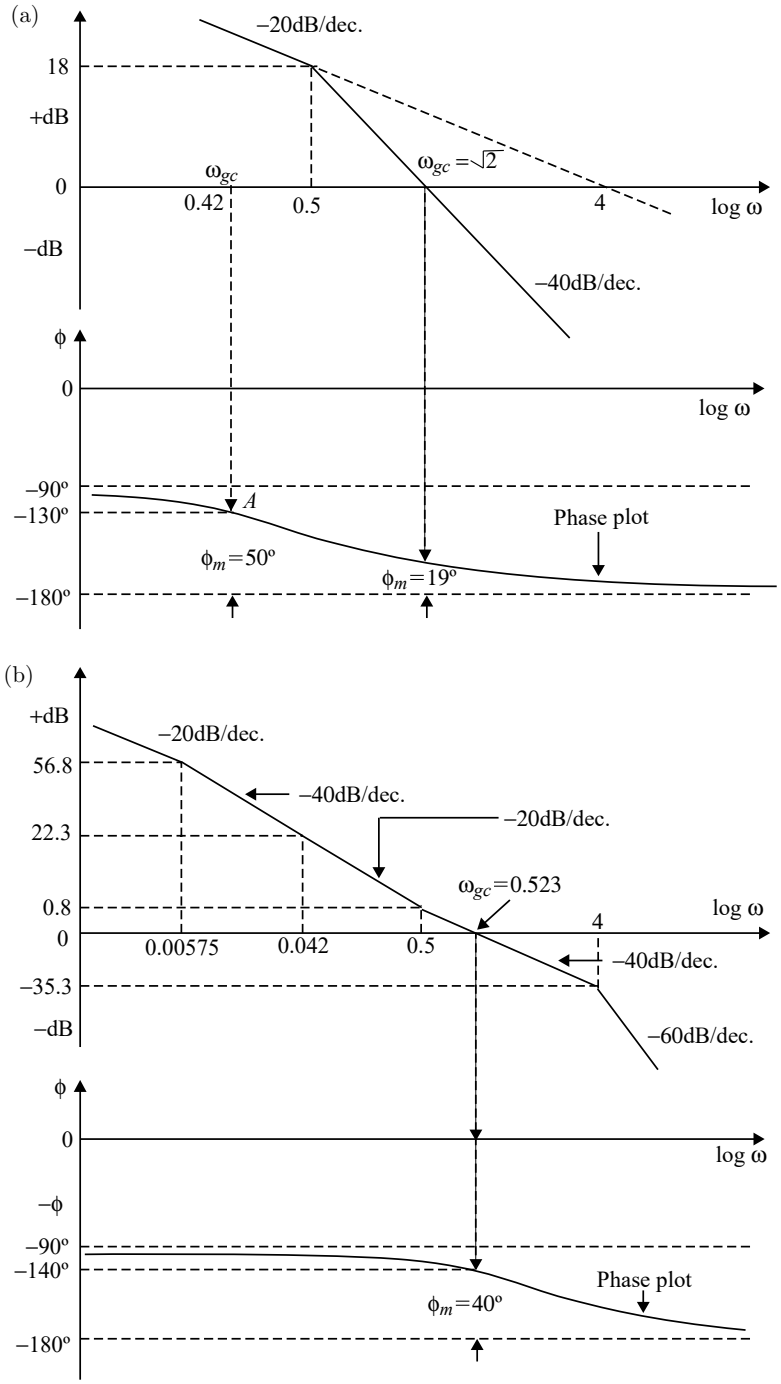


Fig. 6.10 **a** Bode plot for $G(s) = \frac{4}{s(1+2s)}$ **b** Bode diagram for compensated system

$$\frac{1}{\alpha T} = \frac{\omega_{gc}}{10} = 0.042$$
$$\alpha T = 23.8$$
$$T = \frac{23.8}{0.137}$$
$$\boxed{T = 173.8}$$

$$G_c(s) = \frac{(1 + \alpha T s)}{(1 + Ts)}$$
$$\boxed{G_c(s) = \frac{(1 + 23.8s)}{(1 + 173.8s)}}$$

The T.F. of the compensated system is given by

$$G_c(s)G(s) = \frac{(1 + 23.8s)4}{(1 + 173.8s)s(1 + 2s)}$$

The Bode diagram drawn for the above T.F. is shown in Fig. 6.10b. From Fig. 6.10b, the gain crossover frequency is obtained from

$$40 \log \frac{\omega_{gc}}{0.5} = 0.8$$
$$\omega_{gc} = 0.523$$

The phase angle corresponding to this frequency is

$$\phi = -90^\circ + \tan^{-1} 23.8\omega_{gc} - \tan^{-1} 173.8\omega_{gc} - \tan^{-1} 2\omega_{gc}$$
$$= -90^\circ + \tan^{-1} 23.8 \times 0.523 - \tan^{-1} 173.8 \times 0.523 - \tan^{-1} 2 \times 0.523$$

$$\phi = -90^\circ + 85.4^\circ - 89.4^\circ - 46.0^\circ$$
$$= -140^\circ$$

The phase margin is

$$\phi_m = 180^\circ - 140^\circ$$
$$\boxed{\phi_m = 40^\circ}$$

The given specification is satisfied with the designed controller.

Example 6.7

Consider the unity feedback system whose open loop T.F. is

$$G(s) = \frac{K}{s(s+1)(1+0.5s)}$$

Design a lag compensator for the system so that the static velocity error constant $K_v \geq 5\ \text{s}^{-1}$, the phase margin is at least 38° and the gain margin is at least 10 dB.

(*Anna University, December, 2004*)

Solution Let

$$\begin{aligned} G(s) &= \frac{K}{s(1+s)(1+0.5s)} \\ K_v &= \lim_{s\to 0} sG(s) \\ &= \lim_{s\to 0} \frac{sK}{s(1+s)(1+0.5s)} \\ &= K \end{aligned}$$

$$\boxed{K = 5}$$

Hence, an amplifier with a gain of $K = 5$ is chosen for the open loop system to satisfy the given velocity error constant. The T.F. of the uncompensated system is

$$G(s) = \frac{5}{s(1+s)(1+0.5s)}$$

The Bode plot of the uncompensated system is shown in Fig. 6.11a. The gain crossover frequency of the uncompensated system is calculated as

$$\begin{aligned} 60 \log \frac{\omega_{gc}}{2} &= 2 \\ \omega_{gc} &= 2.2 \end{aligned}$$

At this frequency, the phase angle of the uncompensated system is

$$\begin{aligned} \phi &= -90^\circ - \tan^{-1} \omega_{gc} - \tan^{-1} 0.5\omega_{gc} \\ &= -90^\circ - \tan^{-1} 2.2 - \tan^{-1} 0.5 \times 2.2 \\ &= -90^\circ - 66.6^\circ - 47.7^\circ \\ &= -204.3^\circ \\ \phi_m &= 180^\circ - 204.3^\circ \end{aligned}$$

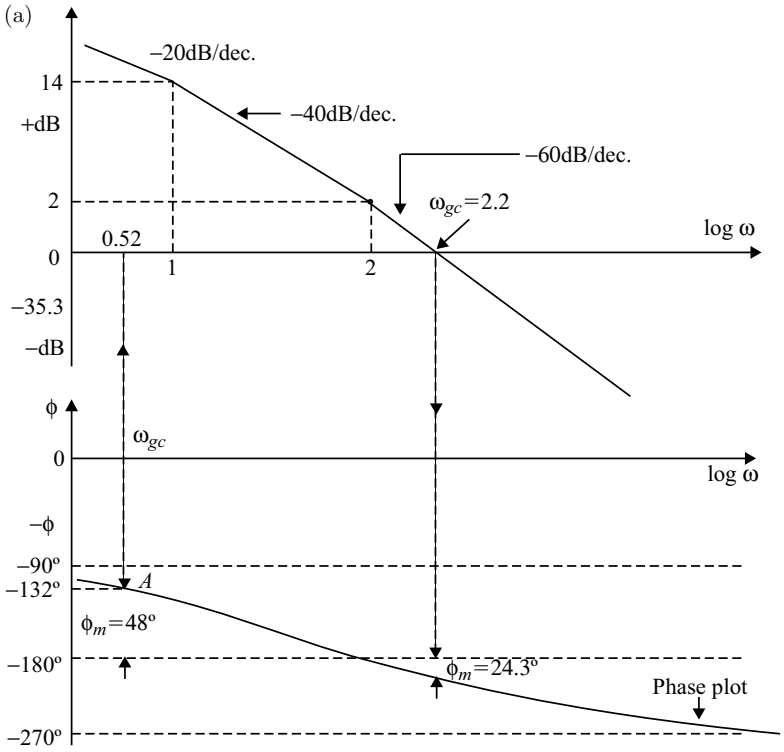


Fig. 6.11 **a** Bode diagram for $G(s) = \frac{5}{s(s+1)(1+0.5s)}$. **b** Bode diagram for the compensated system. **c** Block diagram representation of compensated system

$$= -24.3^\circ$$

If the phase margin is negative, the uncompensated system is unstable. The required phase margin is 38°. Adding an allowance of say 10°, the phase angle of the system is $-180^\circ + (38^\circ + 10^\circ) = -132^\circ$. Corresponding to this phase, point A in the phase plot is identified and the new gain crossover frequency is calculated. Analytically this is calculated as follows:

$$\begin{aligned} -90^\circ - \tan^{-1}\omega - \tan^{-1} 0.5\omega &= -132^\circ \\ \tan^{-1}\omega + \tan^{-1} 0.5\omega &= 42^\circ \\ \tan^{-1} \frac{1.5\omega}{1 - 0.5\omega^2} &= 42^\circ \end{aligned}$$

Taking tan on both sides, we get

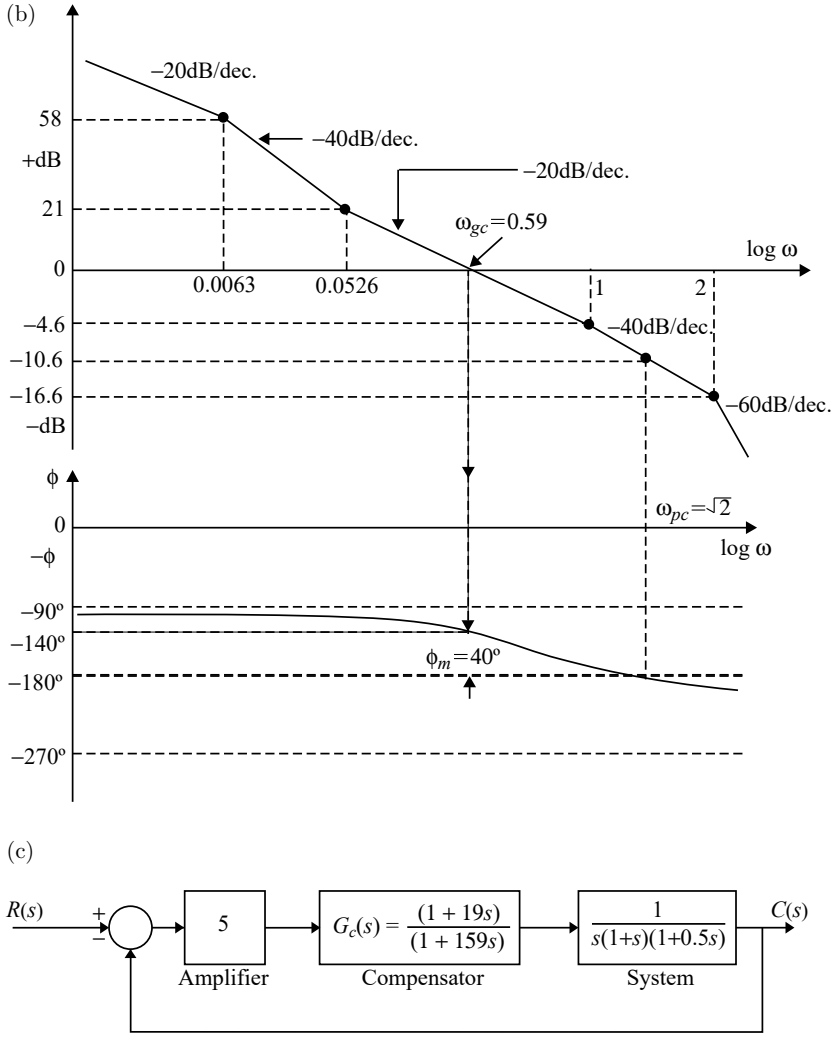


Fig. 6.11 (continued)

$$\frac{1.5\omega}{(1-0.5\omega^2)} = 0.9$$

$$\omega^2 + 3.33\omega - 2 = 0$$

Solving for ω, we get

$$\boxed{\omega = 0.52}$$

This frequency is taken as the new gain crossover frequency. Thus

$$\omega_{gc} = 0.52$$

From Eq. (6.8), we get

$$\alpha|G(j\omega_{gc})| = 1$$

$$\alpha \frac{5}{\omega_{gc}\sqrt{(1+\omega_{gc}^2)(1+0.25\omega_{gc}^2)}} = 1$$

$$\alpha = \frac{0.52}{5}\sqrt{1.27 \times 1.0676}$$

$$\boxed{\alpha = 0.121}$$

The zero of the compensator is placed at one tenth of this gain crossover frequency

$$\frac{1}{\alpha T} = 0.052$$

$$\frac{1}{T} = 0.052 \times 0.121$$

$$\boxed{T = 159.0}$$

$$\alpha T = \frac{1}{0.052}$$

$$= 19.0$$

The T.F. of the lag compensator is

$$G_c(s) = \frac{(1+19s)}{(1+159s)}$$

The T.F. of the compensated system is

$$G_c(s)G(s) = \frac{5(1+19s)}{s(1+159s)(1+s)(1+0.5s)}$$

The Bode diagram for the compensated system is shown in Fig. 6.11b. From Fig. 6.11b, the actual gain crossover frequency is obtained from

$$20\log\frac{\omega_{gc}}{0.0526} = 21$$

$$\boxed{\omega_{gc} = 0.59}$$

Corresponding to this frequency, the phase angle of the compensated system is obtained as

$$\begin{aligned}\phi &= -90^\circ - \tan^{-1}\omega_{gc} - \tan^{-1}0.5\omega_{gc} - \tan^{-1}159\omega_{gc} + \tan^{-1}19\omega_{gc}\\ &= -90^\circ - \tan^{-1}0.59 - \tan^{-1}0.5\times 0.59 - \tan^{-1}159\times 0.59 + \tan^{-1}19\times 0.59\\ &= -90^\circ - 30^\circ - 16^\circ - 89^\circ + 85^\circ\\ &= -140^\circ\end{aligned}$$

The phase margin of the compensated system is

$$\phi_m = 180^\circ - 140^\circ$$

$$\boxed{\phi_m = 40^\circ}$$

The required phase margin is 38°. The actual phase margin is obtained as 40°. The gain margin is calculated as follows. The phase crossover frequency is calculated from

$$-90^\circ - \tan^{-1}\omega_{pc} - \tan^{-1}0.5\omega_{pc} - \tan^{-1}159\omega_{pc} + \tan^{-1}19\omega_{pc} = -180^\circ$$

ω_{pc} is greater than ω_{gc}. Hence, at this frequency, $\tan^{-1}159\omega_p$ is roughly equals $\tan^{-1}19\omega_{pc}$. In other words, the phase angle contribution of the lag network and phase crossover frequency is negligibly small. Hence, we write

$$\begin{aligned}-90^\circ - \tan^{-1}\omega_{pc} - \tan^{-1}0.5\omega_{pc} &= -180^\circ\\ \tan^{-1}\omega_{pc} + \tan^{-1}0.5\omega_{pc} &= 90^\circ\\ \frac{1.5\omega_{pc}}{1-0.5\omega_{pc}^2} &= \tan 90^\circ\\ (1-0.5\omega_{pc}^2) &= 0\end{aligned}$$

$$\boxed{\omega_{pc} = \sqrt{2}}$$

At this frequency, the dB is calculated as

$$\begin{aligned} \text{dB}|_{\omega_{pc}} &= -4.6 - 40 \log \sqrt{2} \\ &= -10.6 \end{aligned}$$

$$\boxed{G_m = 10.6 \text{ dB}}$$

The gain margin obtained with the compensator is more than the required gain margin of 10 dB. Hence, all the prescribed specifications are not with the compensator designed. The block diagram of the compensated system is shown in Fig. 6.11c.

Example 6.8

A unity feedback system has an open loop T.F

$$G(s) = \frac{K}{s(1+2s)}$$

Design a suitable lag compensator so that the phase margin is 40° and the steady state error for ramp input is less than or equal to 0.2

(*Anna University, December, 2009*)

Solution

$$G(s) = \frac{K}{s(1+2s)}$$

The velocity error coefficient K_v is

$$\begin{aligned} K_v &= \lim_{s \to 0} sG(s) \\ &= \lim_{s \to 0} \frac{sK}{s(1+2s)} \\ &= K \end{aligned}$$

The steady state error is

$$\begin{aligned} e_{ss} &= \frac{R}{K_v} \\ &= \frac{1}{K} \\ &= 0.2 \end{aligned}$$

$$\boxed{K = 5}$$

$$G(s) = \frac{5}{s(1+2s)}$$

The Bode diagram is drawn as shown in Fig. 6.12a. The gain crossover frequency of the uncompensated system is given by

$$40 \log \frac{\omega_{gc}}{0.5} = 20$$
$$\omega_{gc} = 1.6$$

Corresponding to this frequency, the phase angle is

$$\phi = -90^\circ - \tan^{-1} 2 \times 1.6$$
$$= -162^\circ$$

The phase margin of the uncompensated system is

$$\phi_m = 180^\circ - 162^\circ$$
$$= 18^\circ$$

The above phase margin does not meet the given specification. The required phase margin is 40°. An additional phase angle of 10° is added to this to compensate the phase contribution of the lag compensator. Thus, to have a phase margin of $(40^\circ + 10^\circ) = 50^\circ$, the new gain crossover frequency is obtained as follows:

$$-90^\circ - \tan^{-1} 2\omega = -130^\circ$$
$$\tan^{-1} 2\omega = 40^\circ$$
$$\omega = 0.42$$
$$\omega_{gc} = 0.42$$

At this gain crossover frequency, the magnitude of the compensated system is

$$\alpha |G(j\omega_{gc})| = 1$$
$$\alpha \frac{5}{\omega_{gc}\sqrt{(1+4\omega_{gc}^2)}} = 1$$
$$\alpha = \frac{0.42}{5}\sqrt{1+4\times(0.42)^2}$$

$$\boxed{\alpha = 0.11}$$

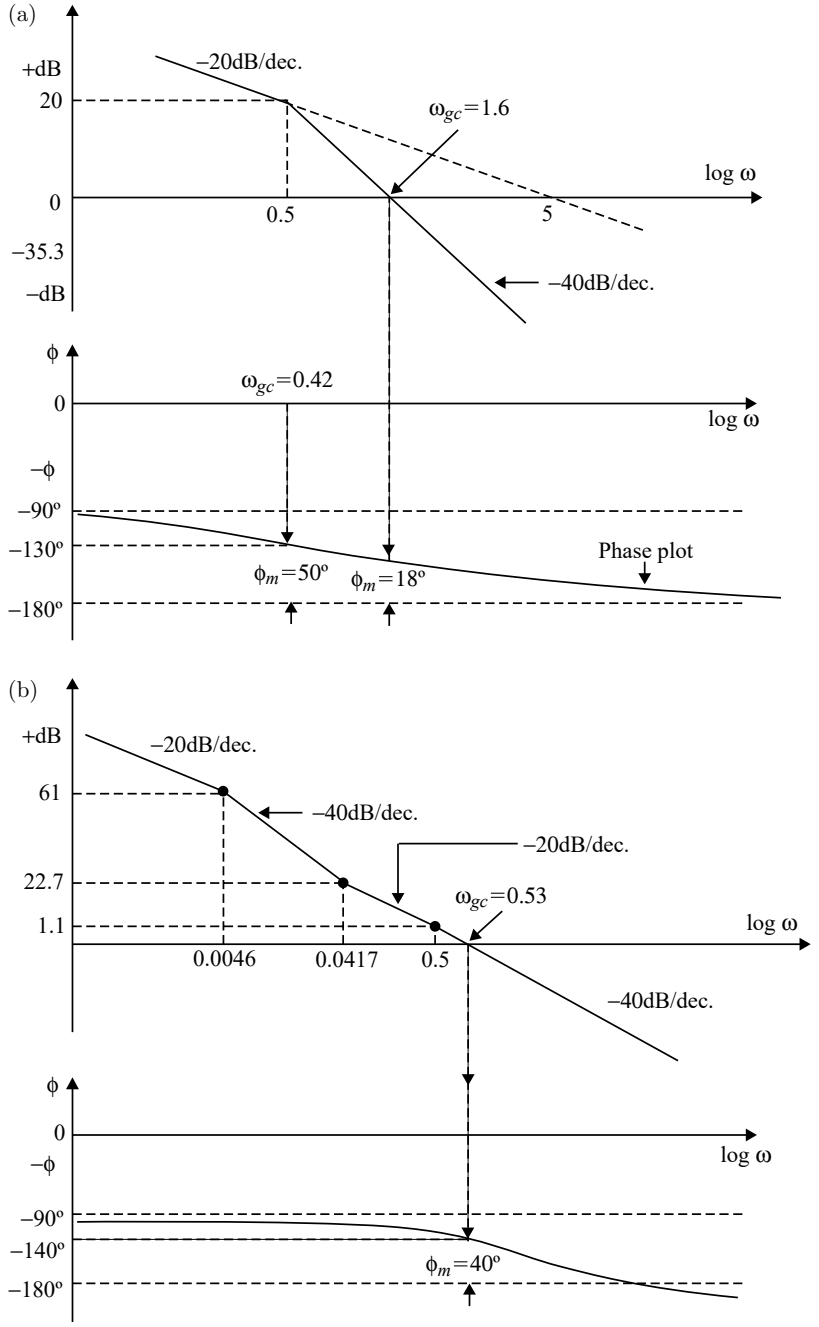


Fig. 6.12 **a** Bode diagram for $G(s) = \frac{5}{s(1+2s)}$. **b** Bode diagram for the compensated system

The zero of the compensator is placed at one tenth of the new gain crossover frequency

$$\begin{aligned}\frac{1}{\alpha T} &= \frac{1}{10}0.42\\ \alpha T &= 24\\ T &= \frac{24}{0.11}\end{aligned}$$

$$\boxed{T = 216}$$

The T.F. of the compensator is

$$G_c(s) = \frac{(1+\alpha T s)}{(1+Ts)}$$

$$\boxed{G_c(s) = \frac{(1+24s)}{(1+216s)}}$$

The T.F. of the compensated system is

$$G_c(s)G(s) = \frac{(1+24s)}{(1+216s)s}\frac{5}{(1+2s)}$$

The Bode diagram for the above T.F. is shown in Fig. 6.12b. From Fig. 6.12b, the gain crossover frequency ω_{gc} is determined as follows:

$$\begin{aligned}40\log\frac{\omega_{gc}}{0.5} &= 1.1\\ \omega_{gc} &= 0.53\end{aligned}$$

At $\omega = 0.53$, the phase angle is calculated as follows:

$$\begin{aligned}\phi &= -90^\circ - \tan^{-1} 2\omega_{gc} - \tan^{-1} 216\omega_{gc} + \tan^{-1} 24\omega_{gc}\\ &= -90^\circ - \tan^{-1} 2 \times 0.53 - \tan^{-1} 216 \times 0.53 + \tan^{-1} 24 \times 0.53\\ &= -90^\circ - 46.2 - 89.3^\circ + 85.5^\circ = -140^\circ\end{aligned}$$

The phase margin of the compensated system is

$$\phi_m = 180^\circ - 140^\circ$$

$$\boxed{\phi_m = 40^\circ}$$

The designed lag compensator satisfies the given specifications.

6.4.4 Lag–Lead Compensator

As discussed in Sects. 6.4.1 and 6.4.2, the lead and lag compensators have their own advantages and disadvantages. A lag–lead compensator is designed which provides the advantages of these two compensators simultaneously. The lag–lead compensating network is shown in Fig. 6.13a. The T.F. of the compensator is determined as described below. Let

$$Z_1 = \text{Parallel combination of } R_1 \text{ and } C_1 = \frac{R_1}{1 + R_1 C_1 s}$$

$$\begin{aligned} Z_2 &= \text{Series combination of } R_2 \text{ and } C_2 \\ &= R_2 + \frac{1}{C_2 s} \\ &= \frac{(1 + R_2 C_2 s)}{C_2 s} \\ \frac{E_o}{E_i}(s) &= \frac{Z_2}{Z_1 + Z_2} \\ &= \frac{(1 + R_2 C_2 s)/C_2 s}{\frac{R_1}{1+R_1 C_1 s} + \frac{(1+R_2 C_2 s)}{C_2 s}} \\ &= \frac{(1 + R_1 C_1 s)(1 + R_2 C_2 s)}{(1 + R_1 C_1 s + R_2 C_2 s + R_1 C_2 s + R_1 C_1 R_2 C_2 s^2)} \end{aligned}$$

$$\boxed{\frac{E_o}{E_i}(s) = G_c(s) = \frac{(1 + \alpha_1 T_1 s)(1 + \alpha_2 T_2 s)}{(1 + T_1 s)(1 + T_2 s)}} \tag{6.10}$$

where

$$\begin{aligned} \alpha_1 T_1 &= R_2 C_2 \\ \alpha_2 T_2 &= R_1 C_1 \\ T_1 + T_2 &= R_1 C_1 + R_2 C_2 + R_1 C_2 \\ \alpha_1 \alpha_2 &= 1 \\ \alpha_1 &< 1 \end{aligned}$$

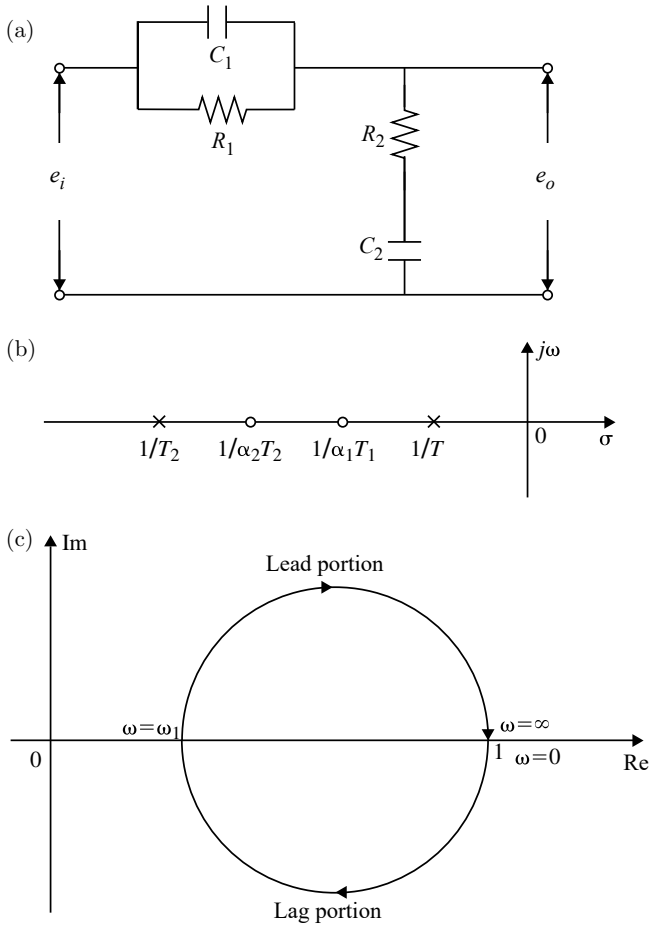


Fig. 6.13 **a** Lag–lead compensating network. **b** Pole–zero locations of $G_c(s)$. **c** Frequency response characteristic of lag–lead compensator in polar plot. **d** Bode diagram of lag–lead compensator

$$\alpha_2 > 1 \tag{6.11}$$

$$\frac{1}{T_1} < \frac{1}{\alpha_1 T_1} < \frac{1}{\alpha_2 T_2} < \frac{1}{T_2} \tag{6.12}$$

The T.F

$$G_{c1}(s) = \frac{(1 + \alpha_1 T_1 s)}{(1 + T_1 s)} \tag{6.13}$$

represents the lag portion of the compensator while

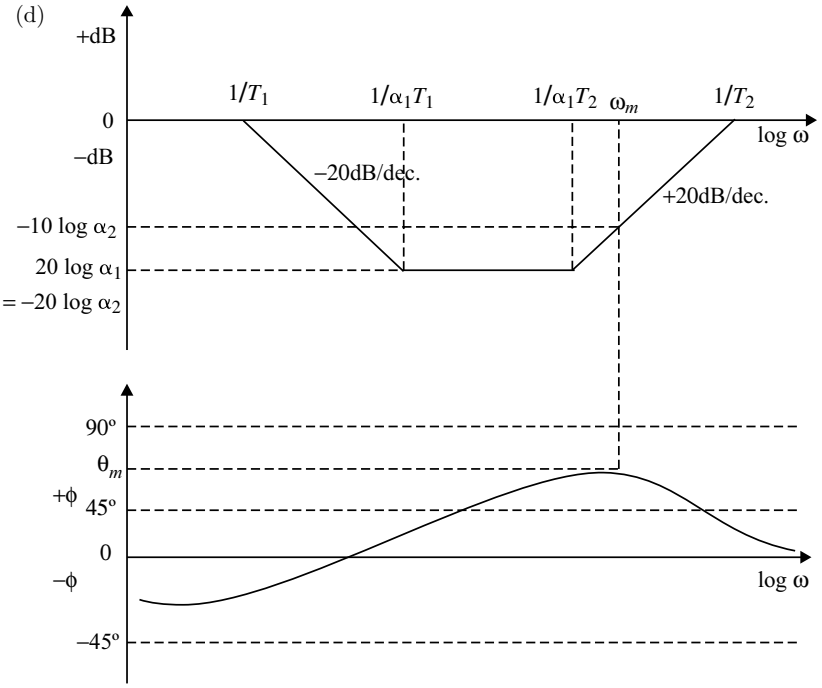


Fig. 6.13 (continued)

$$G_{c2}(s) = \frac{(1 + \alpha_2 T_2 s)}{(1 + T_2 s)} \tag{6.14}$$

represents the lead portion. The sinusoidal T.F. of the lag–lead compensator is given by

$$G_c(j\omega) = \frac{(1 + j\omega\alpha_1 T_1)(1 + j\omega\alpha_2 T_2)}{(1 + j\omega T_1)(1 + j\omega T_2)}$$

$$|G_c(j\omega)| = \frac{\sqrt{(1 + \omega^2\alpha_1^2 T_1^2)(1 + \omega^2\alpha_2^2 T_2^2)}}{\sqrt{(1 + \omega^2 T_1^2)(1 + \omega^2 T_2^2)}} \tag{6.15}$$

$$\angle G_c(j\omega) = \tan^{-1}\alpha_1 T_1\omega + \tan^{-1}\alpha_2 T_2\omega - \tan^{-1} T_1\omega - \tan^{-1} T_2\omega \tag{6.16}$$

The poles and zeros of $G_c(s)$ are shown in Fig. 6.13b. The polar plot of the lag–lead compensator is shown in Fig. 6.13c.

The Bode diagram of Eq. (6.10) is shown in Fig. 6.13d. From Fig. 6.13d, it is evident that the phase angle contribution at high frequency is due to the lead network

and the lag network contributes negligibly small phase. The maximum phase angle contribution ϕ_m occurs at ω_m which is in between

$$\omega = \frac{1}{\alpha_2 T_2} \quad \text{and} \quad \omega = \frac{1}{T_2}$$

Now consider the lead portion of the compensator which is given in Eq. (6.14). The sinusoidal T.F. of G_{c2} is written as

$$G_{c2}(j\omega) = \frac{(1 + j\omega\alpha_2 T_2)}{(1 + j\omega T_2)}$$

$$|G_{c_2}(j\omega)| = \frac{\sqrt{1 + \omega^2\alpha_2^2 T^2}}{\sqrt{1 + \omega^2 T^2}}$$

$$\theta = \angle G_{c_2}(j\omega) = \tan^{-1}\omega\alpha_2 T_2 - \tan^{-1}\omega T_2 \tag{6.17}$$

$$= \tan^{-1}\frac{\omega T_2(\alpha_2 - 1)}{(1 + \omega^2\alpha_2 T_2^2)} \tag{6.18}$$

Taking tan on both sides we get

$$\tan\theta = \frac{\omega T_2(\alpha_2 - 1)}{(1 + \omega^2\alpha_2 T_2^2)}$$

Let us define $\tan\theta = N$

$$N = \frac{\omega T_2(\alpha_2 - 1)}{(1 + \omega^2\alpha_2 T_2^2)}$$

$$\frac{dN}{d\omega} = 0 = \omega T_2(\alpha_2 - 1)(2\omega\alpha_2 T_2^2) - (1 + \omega^2\alpha_2 T_2^2)T_2(\alpha_2 - 1)$$

$$2\omega^2\alpha_2 T_2^2 = 1 + \omega^2\alpha_2 T_2^2$$

$$\boxed{\omega_m = \frac{1}{T_2\sqrt{\alpha_2}}} \tag{6.19}$$

The maximum phase angle at $\omega = \frac{1}{T_2\sqrt{\alpha_2}}$ is obtained from Eq. (6.18) as

$$\theta_m = \tan^{-1}\sqrt{\alpha_2} - \tan^{-1}\frac{1}{\sqrt{\alpha_2}}$$

$$= \tan^{-1}\frac{(\sqrt{\alpha_2} - \frac{1}{\sqrt{\alpha_2}})}{(1 + 1)}$$

Taking tan on both sides, we get

$$\tan\theta_m = \frac{(\alpha_2 - 1)}{2\sqrt{\alpha_2}}$$

The above equation is represented below.
From the triangle, we can write

$$\sin\theta_m = \frac{(\alpha_2 - 1)}{(\alpha_2 + 1)}$$

$$\boxed{\alpha_2 = \frac{(1 + \sin\theta_m)}{(1 - \sin\theta_m)}} \tag{6.20}$$

From the magnitude plot of Fig. 6.13d, at ω_m, the magnitude in dB is obtained as

$$\begin{aligned} \text{dB}|_{\omega=\omega_m} &= 20\log\frac{1}{T_2\omega_m} \\ &= -10\log\alpha_2 \end{aligned}$$

or

$$|G_{c2}(j\omega)| = \frac{1}{\sqrt{\alpha_2}}$$

If ω_m is taken as the new gain crossover frequency, then the following relationship holds good

$$\frac{|G(j\omega_{gc})|}{\sqrt{\alpha_2}} = 1$$

$$\boxed{|G(j\omega_{gc})| = \sqrt{\alpha_2}} \tag{6.21}$$

Using Eqs. (6.19), (6.20) and (6.21), the lag–lead compensator is designed in the frequency domain following the step-by-step procedure given below.

Step-by-Step Procedure to Design Lag–Lead Compensator

1. To satisfy the steady state error requirement, determine the value of open loop gain K. For this value of K, draw the Bode diagram.
2. Determine the phase margin of the uncompensated system and find the angle deficiency that must be contributed by the compensator. Using Eq. (6.20), find α_2.

3. Using Eq. (6.21), determine the gain crossover frequency ω_{gc}. Make $\omega_{gc} = \omega_m$. Using Eq. (6.19), find T_2. Now the lag portion of the compensator is to be designed.
4. Using the relationship $\alpha_1\alpha_2 = 1$ as derived in Eq. (6.11), determine α_1.
5. Arbitrarily place the zero of the lag portion at one tenth of the zero of the lead portion. Thus, we get

$$\frac{1}{\alpha_1 T_1} = 0.1\frac{1}{\alpha_2 T_2} \tag{6.22}$$

   From this, find T_1. Now the lag portion of the compensator is designed.
6. Draw the Bode plot for the compensated system and verify whether all the given specifications are satisfied. If not satisfied, go to step 2 and change θ_m and follow the procedure. If necessary, modify Eq. (6.22) by appropriate factor to get the desired result.

Example 6.9

The open loop T.F. of a unity feedback control system is

$$G(s) = \frac{K}{s(s+2)(s+20)}$$

Design a lag–lead compensator to obtain the following specifications:

(a) Static velocity error = 0.1
(b) Phase margin $\phi_m \geq 50°$
(c) Gain margin $G_m \geq 10$ dB

(*Anna University, December, 2011*)

Solution

$$K_v = \lim_{s\to 0} sG(s)$$
$$= \lim_{s\to 0} \frac{sK}{s(s+2)(s+20)}$$

$$K_v = \frac{K}{40}$$
$$e_{ss} = \frac{1}{K_v} = 0.1 \quad \text{or} \quad K_v = 10$$

$$\boxed{K = 400}$$

$$G(s) = \frac{10}{s(1+\frac{s}{2})(1+\frac{s}{20})}$$

The corner frequencies are 2 and 20. The Bode magnitude and phase plots are shown in Fig. 6.14a. From Fig. 6.14a, the gain crossover frequency of uncompensated system is obtained as

$$40 \log \frac{\omega_{gc}}{2} = 14$$
$$\omega_{gc} = 4.47$$

The phase angle at this frequency is

$$\phi = -90^\circ - \tan^{-1}\frac{4.47}{2} - \tan^{-1}\frac{4.47}{20}$$
$$= -90^\circ - 66^\circ - 13^\circ$$

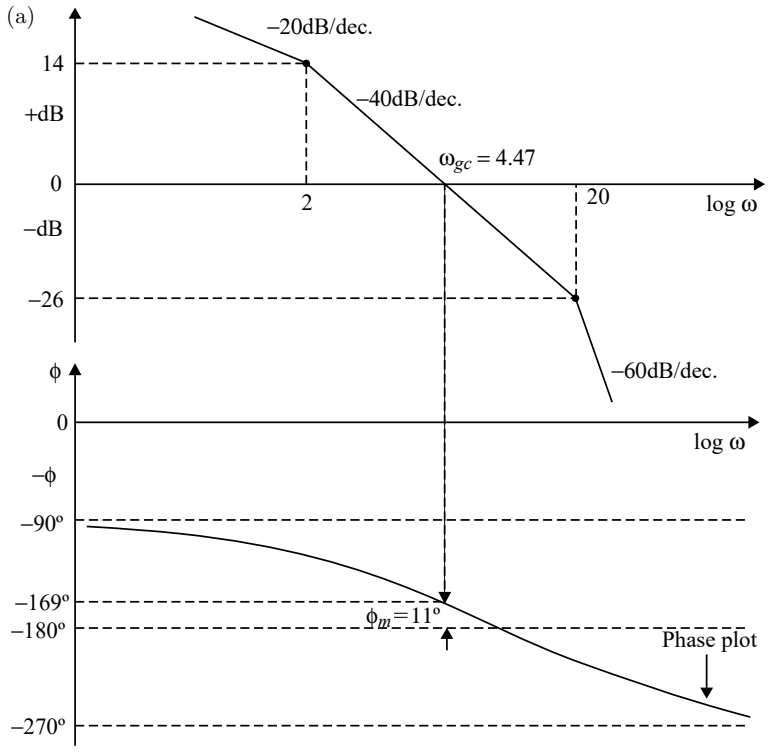


Fig. 6.14 **a** Bode diagram for uncompensated system with $G(s) = \frac{10}{s(1+\frac{s}{2})(1+\frac{s}{20})}$. **b** Bode diagram for the compensated system

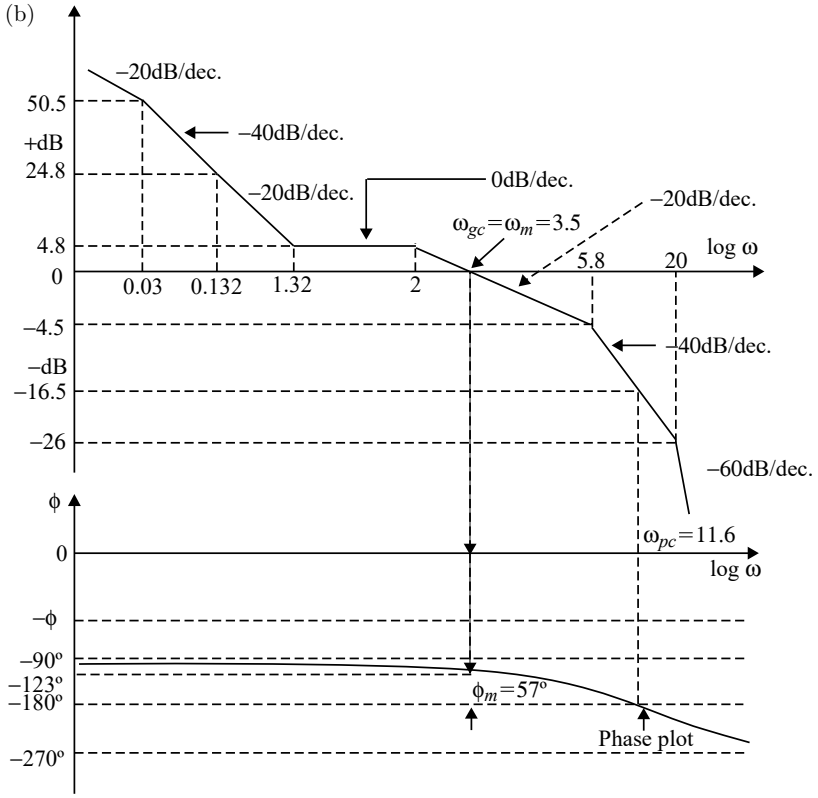


Fig. 6.14 (continued)

$$= -169^\circ$$

The phase margin

$$\phi_m = 180 - 169^\circ = 11^\circ$$

But the required phase margin is 50°. The balance phase that should be contributed by the lag–lead compensator is

$$\theta_m = 50^\circ - 11^\circ = 39^\circ$$

α_2 is calculated using Eq. (6.20)

$$\alpha_2 = \frac{(1 + \sin\theta_m)}{(1 - \sin\theta_m)}$$
$$= \frac{(1 + \sin 39^\circ)}{(1 - \sin 39^\circ)}$$

$$\boxed{\alpha_2 = 4.4}$$

The gain crossover frequency ω_{gc} is calculated using Eq. (6.21)

$$\begin{aligned} |G(j\omega_{gc})| &= \sqrt{\alpha_2} \\ \frac{400}{\omega_{gc}\sqrt{(\omega_{gc}^2+4)(\omega_{gc}^2+400)}} &= \sqrt{4.4} \\ \omega_{gc}^2(\omega_{gc}^2+4)(\omega_{gc}^2+400) &= 36363.6 \end{aligned}$$

Substituting $\omega_{gc}^2 = x$, we get

$$\begin{aligned} x(x+4)(x+400) &= 36363.6 \\ x^3 + 404x + 1600 &= 36363.6 \end{aligned}$$

Solving for x, we get $x = 7.65$

$$\omega_{gc} = \sqrt{7.65}$$

$$\boxed{\omega_{gc} = 2.77}$$

$$\omega_m = \omega_{gc} = 2.77$$

Using Eq. (6.19), we get

$$\begin{aligned} T_2 &= \frac{1}{\omega_m\sqrt{\alpha_2}} \\ &= \frac{1}{2.77\sqrt{4.4}} \end{aligned}$$

$$\boxed{T_2 = 0.172}$$

$$\begin{aligned} \frac{1}{\alpha_2 T_2} &= \frac{1}{4.4 \times 0.172} \\ &= 1.32 \end{aligned}$$

Using Eq. (6.22), we get

$$\begin{aligned}\frac{1}{\alpha_1 T_1} &= \frac{0.1}{\alpha_2 T_2} \\ &= 0.132 \\ \alpha_1 T_1 &= 7.575\end{aligned}$$

Using Eq. (6.11), we get

$$\begin{aligned}\alpha_1 \alpha_2 &= 1 \\ \alpha_1 &= \frac{1}{\alpha_2} \\ &= \frac{1}{4.4}\end{aligned}$$

$$\boxed{\alpha_1 = 0.227}$$

$$\begin{aligned}T_1 &= \frac{7.575}{\alpha_1} \\ &= \frac{7.575}{0.227}\end{aligned}$$

$$\boxed{T_1 = 33.33}$$

Thus, the lag–lead compensator design is completed as

$$\boxed{G_c(s) = \frac{(1 + 7.575s)(1 + 0.757s)}{(1 + 33.33s)(1 + 0.172s)}}$$

The T.F. of the compensated system is given by

$$G_c(s)G(s) = \frac{(1 + 7.575s)(1 + 0.757s)10}{(1 + 33.33s)(1 + 0.172s)s(1 + \frac{s}{2})(1 + \frac{s}{20})}$$

The corner frequencies are 0.03, 0.132, 1.32, 2, 5.8 and 20. The Bode diagram for the compensated system is shown in Fig. 6.14b. From the magnitude plot, the gain crossover frequency of the compensated system is obtained as

$$20 \log \frac{\omega_{gc}}{2} = 4.8$$

$$\boxed{\omega_{gc} = 3.5}$$

The actual gain crossover frequency is slightly shifted to the right when we take into account the lag portion of the compensator. Without lag portion, the gain crossover frequency was obtained as 2.77. Now, with $\omega_{gc} = 3.5$, we calculate below the phase angle of the compensated system

$$\begin{aligned}\phi = \tan^{-1} 7.575\omega_{gc} + \tan^{-1} 0.757\omega_{gc} - 90^\circ - \tan^{-1} 33.33\omega_{gc} \\ - \tan^{-1} 0.172\omega_{gc} - \tan^{-1} 0.5\omega_{gc} - \tan^{-1} 0.05\omega_{gc}.\end{aligned}$$

Substituting $\omega_{gc} = 3.5$, we get

$$\begin{aligned}\phi &= 87.8^\circ + 69.7^\circ - 90^\circ - 89.5^\circ - 31^\circ - 60^\circ - 10^\circ \\ &= -123^\circ\end{aligned}$$

The phase margin of the compensated system is

$$\begin{aligned}\phi_m &= 180^\circ + \phi \\ &= 180^\circ - 123^\circ\end{aligned}$$

$$\boxed{\phi_m = 57^\circ}$$

The required phase margin is at least 50°. However, we get phase margin as 57° which satisfies the phase margin requirement. Now we calculate the gain margin. As seen from Fig. 6.14b, the phase crossover frequency $\omega_{pc} > \omega_{gc}$. As seen from the calculation of phase angle of the system with the compensator, at higher frequency the phase angle contribution is approximately written as

$$\phi = \tan^{-1} 0.757\omega - 90^\circ - \tan^{-1} 0.172\omega - \tan^{-1} 0.5\omega - \tan^{-1} 0.05\omega$$

At $\omega = \omega_{pc}$

$$\begin{aligned}\phi &= 180^\circ \\ 90^\circ &= \tan^{-1} 0.757\omega - \tan^{-1} 0.172\omega \\ &\quad -(\tan^{-1} 0.5\omega + \tan^{-1} 0.05\omega) \\ \tan^{-1} 0.757\omega - \tan^{-1} 0.172\omega &= 90^\circ + (\tan^{-1} 0.5\omega + \tan^{-1} 0.05\omega) \\ \tan^{-1} \frac{0.585\omega}{(1 + 0.13\omega^2)} &= 90^\circ + \tan^{-1} \frac{0.55\omega}{(1 - 0.025\omega^2)}\end{aligned}$$

Taking tan on both sides, we get

$$\frac{(0.585\omega)}{(1+0.13\omega^2)} = \frac{(1-0.025\omega^2)}{0.55\omega}$$

The above equation is simplifier as

$$\omega^4 - 133\omega^2 - 312.5 = 0$$
$$\omega^2 = 135.7$$
$$\omega_{pc} = \sqrt{135.7}$$

$$\boxed{\omega_{pc} = 11.6}$$

At the phase crossover frequency $\omega_{pc} = 11.6$, in Fig. 6.14b, the dB is calculated as

$$\text{dB}|_{\omega_{pc}} = -4.5 - 40\log\frac{11.6}{5.8}$$
$$= -16.5$$

The gain margin is, therefore

$$\boxed{G_m = 16.5}$$

All the given specifications are satisfied and thus the design is validated.

Summary of Performance Characteristic of Lead, Lag and Lag–Lead Compensators

I. Lead compensator

1. The phase lead compensator adds phase lead at the gain crossover frequency which improves the phase margin of the system. This increases the overall damping of the closed loop system. Further, the rise time and setting time are reduced and thus the transient response is faster.
2. As a result, increased phase margin and gain margin are achieved when a lead compensator is inserted, thereby increasing the relative stability.
3. The bandwidth of the closed loop system is increased.
4. Since the phase lead compensator provides lead phase angle at higher frequency, the steady state error is not corrected.
5. One of the limitations of the lead compensator is that high frequency noise enters the system.
6. To meet the steady state requirement, an amplifier with a gain of $1/\alpha$ is to be inserted which is costly.
7. In practical applications, the maximum phase lead provided by the compensator is between 50° and 55°. If the phase lead requirement is more than this, a single stage lead compensator is not enough. In such cases, a two stage lead compensator with an isolation amplifier is designed, which increases the cost.

II. Lag compensator

1. Lag compensator provides attenuation at the new gain crossover frequency which is shifted towards the low frequency region of the Bode diagram of the uncompensated system. This provides the required phase margin.
2. The phase angle curve of the uncompensated system is not appreciably affected when the lag network is introduced. The steady state error is appreciably reduced.
3. The system gain at high frequencies is reduced which reduces the bandwidth of the system. As a result, the transient response is very slow with increased rise time and settling time.

III. Lag–Lead compensator

1. Basically, the lead compensator is used to improve the transient response which has no effect on steady state error. Similarly, the lag compensator is used to improve the steady state error. However, with the lag compensator, the transient response becomes slow. By using the lag–lead compensator, the good effects of lag and lead compensators are achieved. It provides a good transient response and improves the steady state error. However, the additional components used increase the cost of the compensator.

6.5 Compensator Design in the Time Domain

In the control system analysis, transient response specifications are more informative than the frequency response specifications. The time domain specifications include peak overshoot, rise time, settling time and the locations of the dominant roots which are expressed in terms of damping ratio and undamped natural frequency of oscillation. Root locus technique gives a viable solution to the design problem, in that it reshapes the root loci of the uncompensated system so that the desired transient response specifications are met in addition to satisfying the steady state error requirement. We discuss below the design methodologies for lead, lag and lag–lead compensators using root locus method.

6.5.1 Design of Lead Compensator Using Root Locus

From Eq. (6.1), the T.F. of the lead compensator is given by

$$G_c(s) = \frac{\alpha(1 + Ts)}{(1 + \alpha Ts)}$$

The above equation is written in pole–zero form as

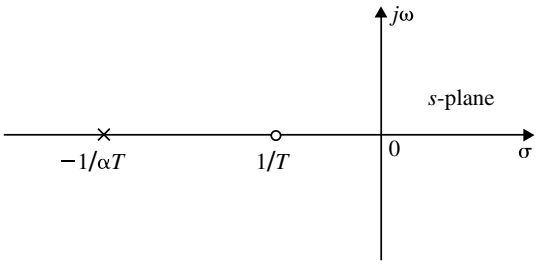


Fig. 6.15 Pole and zero of lead compensator

$$G_c(s) = \frac{s + \frac{1}{T}}{(s + \frac{1}{\alpha T})} \tag{6.23}$$

The pole and zero of Equation (6.23) are represented in Fig. 6.15.

Step-by-Step Procedure to Design Lead Compensator

1. For the given uncompensated system determine the value of K which satisfies the steady state error requirement. Locate the poles and zeros of the uncompensated system in the s-plane as shown in Fig. 6.16.
2. Identify the point P in the s-plane that satisfies the given time domain specification. Draw the line OP. Draw the line PQ which is horizontal to the negative real axis.
3. Draw the line PC which is the bisector of OPQ.
4. Draw the vectors from all the poles and zeros of the uncompensated system to the point P. If P is to be a point on the root locus, then the total angle of all the vectors drawn from the zeros minus the total angle of all the vectors drawn from the poles of the uncompensated system should be an odd multiple of 180°. If this phase angle condition is not satisfied, then determine the phase angle deficiency ϕ which is to be contributed by the lead network.

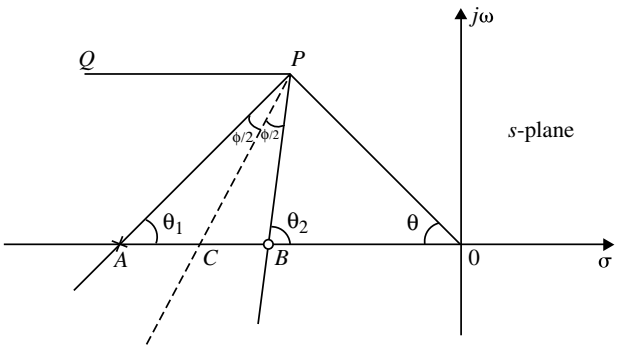


Fig. 6.16 Lead compensator design using root locus

5. Draw on either side of the bisector line PC, straight lines PA and PB which make $\phi/2$ with the bisector line PC as reference. These lines cut the negative real axis at point A and at point B, respectively.
6. It can be shown that by locating the pole of the compensator at point A and the zero at point B, the compensator provides the phase angle deficiency of ϕ when inserted in cascade with the uncompensated system.
7. θ_2 is the phase angle contribution of the zero of the compensator and θ_1 is the phase angle contribution of the pole. It is to be proved that $(\theta_2 - \theta_1) = \phi$. From triangle ABP, in Fig. 6.16

$$\theta_1 + (180^\circ - \theta_2) + \phi = 180^\circ$$
$$(\theta_2 - \theta_1) = \phi$$

8. It is to be noted that the bisector need not be always drawn. One may fix point B where the zero of the lead compensator is placed and making angle $\angle BPA = \phi$, the point A is fixed where the pole of the lead compensator is placed. This may be necessary to satisfy the magnitude criterion of root locus at point P and also static error constants simultaneously.

Example 6.10

The forward path T.F. of a unity feedback control system is

$$G(s) = \frac{K}{s(s+1)}$$

Design a lead compensator to meet the following specifications:

1. Damping ratio $\zeta = 0.7$
2. Settling time $t_s = 1.4$ s
3. Velocity error constant $K_v \geq 2\ \text{s}^{-1}$

(*Anna University, December, 2006*)

Solution

1.

$$G(s) = \frac{K}{s(s+1)}$$
$$K_v = \lim_{s \to 0} sG(s)$$
$$= \lim_{s \to 0} \frac{sK}{s(s+1)}$$
$$2 = K$$

$$\boxed{K = 2}$$

2. For 2% error tolerance in the steady state

$$t_s = 3.91T$$

where

$$\begin{aligned} T &= \frac{1}{\zeta\omega_n} \\ 1.4 &= 3.91T \\ T &= \frac{1.4}{3.91} \\ \zeta\omega_n &= \frac{3.91}{1.4} \\ \omega_n &= \frac{3.91}{1.4 \times 0.7} \end{aligned}$$

$$\boxed{\omega_n = 4}$$

3.

$$\begin{aligned} \omega_d &= \omega_n\sqrt{1-\zeta^2} \\ &= 4\sqrt{1-0.49} \\ &= 2.8 \\ \zeta\omega_n &= 0.7 \times 4 = 2.8 \end{aligned}$$

The point P is located at $s = -2.8 + j2.8$.

4. From point P, the horizontal straight line PQ is drawn. The open loop poles are placed at $s = 0$ and $s = -1$. From Fig. 6.17a, the phase angle contribution from these open loop poles is $(134^\circ + 120^\circ) = 254^\circ$. For the point P to be a point on the root locus, the lead compensator should contribute $+74^\circ(-180^\circ + 254^\circ).\phi = 74^\circ$.
5. The bisector of OPQ is drawn as PC. The lines PA and PB are drawn with angle 37° from PC as shown in Fig. 6.17a. At point A, the pole of the compensator is placed. It is at $s = -7.5$. At point B, the zero of the compensator is placed. It is at $s = -2$.
   Thus

$$\frac{1}{T} = 2$$

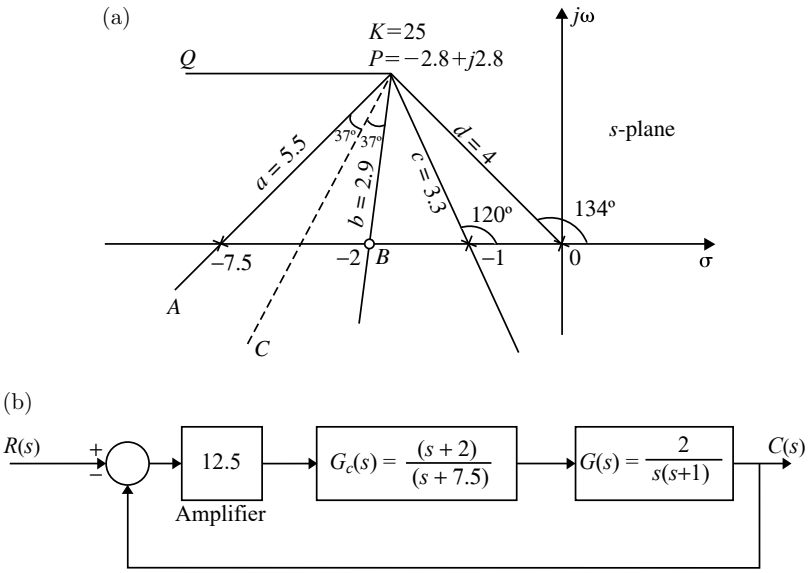


Fig. 6.17 Pole–zero locations for Example 6.10. **b** Block diagram of compensated system

$$\boxed{T = 0.5}$$

$$\frac{1}{\alpha T} = 7.5$$

$$\alpha = \frac{1}{7.5 \times 0.5}$$

$$\boxed{\alpha = 0.267}$$

6. The T.F. of the lead compensator is

$$G_c(s) = \frac{(s+2)}{(s+7.5)}$$

7. From Fig. 6.17a, the gain at point P is calculated as

$$\begin{aligned} K &= \frac{acd}{b} \\ &= \frac{5.5 \times 3.3 \times 4}{2.9} \\ &= 25 \end{aligned}$$

Already the gain available is 2. Hence, an amplifier with a gain of 12.5 is to be provided. Under this condition, the velocity error constant K_v is

$$\begin{aligned} K_v &= \lim_{s \to 0} \frac{s(s+2)}{(s+7.5)} \frac{2 \times 12.5}{s(s+1)} \\ &= \frac{50}{7.5} \\ &= 6.67 \text{ s}^{-1} \end{aligned}$$

This is more than the given specification of $K_v \geq 2 \text{ sec}^{-1}$. The block diagram of the compensated system is shown in Fig. 6.17b.

Example 6.11

Design a suitable lead compensator for a system with unity feedback and having the open loop T.F.

$$G(s) = \frac{K}{s(s+1)(s+4)}$$

to meet the following specifications:

1. Damping ratio $\zeta = 0.5$.
2. Undamped natural frequency of oscillation $\omega_n = 2$.
3. The velocity error constant $K_v \geq 1.5$.

Solution

$$\begin{aligned} G(s) &= \frac{K}{s(s+1)(s+4)} \\ K_v &= 1.5 \\ K_v &= \lim_{s \to 0} sG(s) \\ &= \lim_{s \to 0} \frac{sK}{s(s+1)(s+4)} \\ 1.5 &= \frac{K}{4} \end{aligned}$$

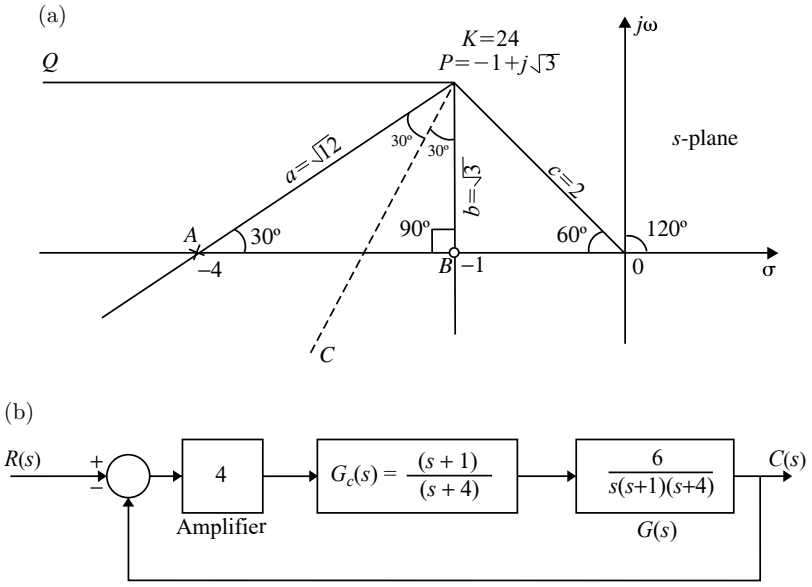


Fig. 6.18 **a** Pole–zero diagram for Example 6.11. **b** Block diagram of the compensated system for Example 6.11

$$\boxed{K = 6}$$

Hence

$$G(s) = \frac{6}{s(s+1)(s+4)}$$

The poles of $G(s)$ are placed as shown in Fig. 6.18a. Since $\omega_n = 2$ and $\zeta = 0.5$

$$\begin{aligned} \omega_d &= \omega_n\sqrt{1-\zeta^2} \\ &= 2\sqrt{0.75} \\ &= \sqrt{3} \\ \zeta\omega_n &= 0.5 \times 2 \\ &= 1 \end{aligned}$$

The point P is fixed at $-1 + j\sqrt{3}$. The horizontal line PQ is drawn. The bisector PC of OPQ is drawn. The phase lead to be contributed by the compensator is $\phi = -180° + 120° + 90° + 30° = 60°$. The lines PA and PB are drawn with $\angle APC = \angle BPC = 30°$. The point A is located at -4 where the pole of the compensator is placed. The point B is located at $s = -1$ where the zero of the compensator is placed. The value of K at point P is calculated as

$$K = \frac{a \times a \times b \times c}{b}$$
$$= 12 \times 2$$

$$\boxed{K = 24}$$

$$G_c(s) = \frac{(s+1)}{(s+4)}$$

The zero of the compensator is $\frac{1}{T}$. Hence

$$\frac{1}{T} = 1$$

$$\boxed{T = 1}$$

The pole of the compensator is $1/\alpha T$. Hence

$$\frac{1}{\alpha T} = 4$$

$$\boxed{\alpha = 0.25}$$

The T.F. of the compensated system is

$$G_c(s)G(s) = \frac{K_a(s+1)6}{(s+4)s(s+1)(s+4)}$$
$$= \frac{6K_a}{s(s+4)^2}$$

Here, pole–zero cancellation takes place. In addition to the system gain of $K = 6$, an amplifier with a gain of $K_a = 4$ is to be connected so that at point P, the magnitude criterion of gain $= 24$ is satisfied. For the compensated system, the velocity error constant is

$$K_v = \lim_{s \to 0} \frac{s24}{s(s+4)^2}$$
$$K_v = 1.5$$

The specification is satisfied. The block diagram of the compensated system is shown in Fig. 6.18b.

Example 6.12

Design a lead compensator for a control system having a T.F. as

$$G(s) = \frac{1}{s(s+1)(s+5)}$$

to meet the following specifications. (a) Over-shoot for step input is 25%. (b) Settling time of $t_s = 5$ s

---

(*Anna University, December, 2003*)

Solution

1. For 25% overshoot due to dominant roots, we have

$$\%M_p = e^{\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}} \times 100$$

$$e^{\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}} \times 100 = 25$$

Solving for ζ, we get

$$\boxed{\zeta = 0.4}$$

The settling time t_s is given (for 2% error tolerance) as

$$t_s = 3.91T$$
$$5 = 3.91T$$
$$T = 1.28$$
$$T = \frac{1}{\zeta\omega_n}$$
$$\omega_n = \frac{1}{T\zeta}$$
$$= \frac{1}{1.28 \times 0.4}$$

$$\boxed{\omega_n = 2 \text{ rad/s}}$$

$$\omega_d = \omega_n\sqrt{1-\zeta^2}$$
$$= 2\sqrt{1-(0.4)^2} = 1.833$$

2. The point P is located at $s = -0.8 + j1.833$. The horizontal line PQ is drawn. Vectors from the poles of $G(s)$ from $s = 0$, $s = -1$ and $s = -5$ are drawn to point P. The angles contributed by these poles are measured. The sum of these angle is $\phi = 110^\circ + 80^\circ + 20^\circ = 210^\circ$. The phase lead to be contributed by the compensator is $210^\circ - 180^\circ = 30^\circ$.
3. The bisector of OPQ is drawn as PC. Lines PA and PB are drawn with $\angle APC = \angle BPC = 15^\circ$. The pole of the compensator is located at point A which is at $s = -3$ and the zero of the compensator is placed at point B which is at $s = -1.5$. The T.F. of the compensator is

$$\boxed{G_c(s) = \frac{(s+1.5)}{(s+3)}}$$

4. The gain of the compensated system at point P is calculated as

$$K = \frac{4.8 \times 2.9 \times 1.9 \times 2}{2}$$

$$\boxed{K = 26.5}$$

The block diagram of the compensated system with an amplifier gain as 26.5 (Since the gain of the original system is 1) is shown in Fig. 6.19b.

5. The velocity error constant of the uncompensated system is

$$K_v = \lim_{s\to 0} sG(s)$$
$$= \lim_{s\to 0} s\frac{1}{s(s+1)(s+5)}$$
$$= 0.2$$

The velocity error constant of the compensated system is

$$K_v = \lim_{s\to 0} sG_c(s)G(s)$$
$$= \lim_{s\to 0} s\frac{(s+1.5)26.5}{(s+3)s(s+1)(s+5)}$$
$$K_v = 2.65$$

The increased velocity error constant reduces the steady state error.

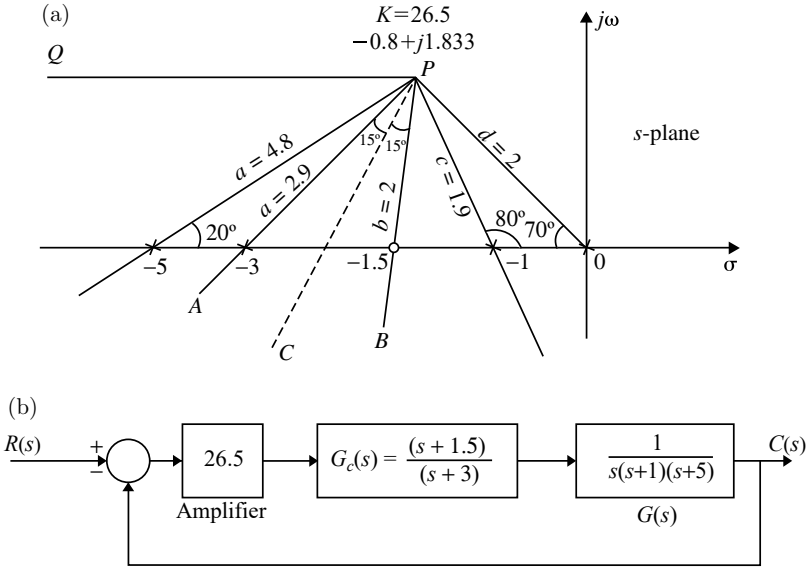


Fig. 6.19 **a** Pole–zero diagram for Example 6.12. **b** Block diagram of the compensated system for Example 6.12

6.5.2 *Design of Lag Compensator Using Root Locus*

To improve the steady state error, the lag network is inserted in cascade with the given system. By this method, for the given time domain specification, the gain of the system is increased as much as possible without sacrificing the transient response characteristics. For this, the pole and zero of the lag network are placed quite nearer to the origin of the s-plane which influence the low frequency response and hence the steady state characteristic of the system. From Eqs. (6.6) and (6.7), we write

$$G_c(s) = \frac{\alpha(s+z)}{(s+p)}$$

where

$$p = \alpha z$$

and

$$\alpha < 1$$

The pole–zero diagram of the lag compensator is shown in Fig. 6.20.

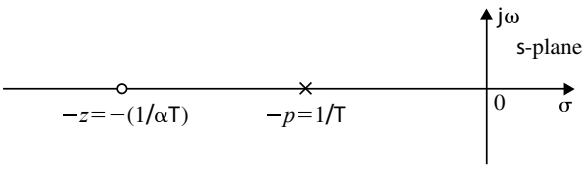


Fig. 6.20 Pole–zero diagram of a lag compensator

Step-by-Step Procedure to Design Lag Compensator

1. Locate the poles and zeros of the uncompensated system in the s-plane. Draw the root locus for the uncompensated system.
2. Identify the point P in the root locus which is chosen to satisfy the given time domain specifications for the dominant roots.
3. Determine the gain of the uncompensated system at the point P using magnitude criterion. Let this gain be K_u.
4. Determine the value of K of the open loop T.F. that satisfies the steady state error requirement.
5. When the compensator is inserted in the forward path with the T.F

$$G_c(s) = \frac{\alpha(s+z)}{(s+p)}$$

the forward path gain together with the gain of the uncompensated system gain becomes $K\alpha$. This should satisfy the magnitude criterion of root locus at point P chosen in step 2. However, the introduction of the compensator slightly modifies this gain roughly by a factor of 1.1. Thus, the gain at point P is $1.1K\alpha$. This should be equal to K_u determined in step 3. Thus, we write

$$1.1K\alpha = K_u$$

$$\boxed{\alpha = \frac{K_u}{1.1K}} \tag{6.24}$$

Knowing K_u and K, calculate α.

6. Place the zero of the compensator on the negative real axis which is roughly about one tenth of the distance of the second pole of the uncompensated system pole from the origin.
7. The pole of the compensator is determined using $p = \alpha z$. Thus, the lag compensator design is completed.
8. Draw the root locus for the compensated system and verify whether the given specifications are satisfied by the compensated system. If not, go to step 5 and repeat the procedure till the required specifications are obtained.

Example 6.13

The forward path T.F. of a certain unity negative feedback control system is given by

$$G(s) = \frac{K}{s(s+2)(s+4)}$$

The system is to have a damping factor $\zeta = 0.6$ for the dominant poles. The steady state error for unit ramp input should be limited to 0.2 rad. Design a suitable lag network.

Solution

1.

$$\begin{aligned} G(s) &= \frac{K}{s(s+2)(s+4)} \\ K_v &= \lim_{s\to 0} sG(s) \\ &= \lim_{s\to 0} \frac{sK}{s(s+2)(s+4)} = \frac{K}{8} \\ e_{ss} &= \frac{1}{K_v} \\ 0.2 &= \frac{8}{K} \end{aligned}$$

$$\boxed{K = 40}$$

2. The root locus plot for $G(s)$ is shown in Fig. 6.21. For the damping ratio, $\zeta = 0.6$

$$\begin{aligned} \cos\theta &= \zeta \\ \theta &= \cos^{-1}\zeta \\ &= \cos^{-1} 0.6 \\ &= 53^\circ \end{aligned}$$

The damping factor line for $\zeta = 0.6$ is drawn. This line makes 53° from −ve real axis and cuts the root locus at point P. The point P is identified as $-0.75 + j1.1$ in the s-plane. The gain at the point P is calculated using magnitude criterion of root locus. Thus, we obtain

$$K_u = 1.3 \times 1.75 \times 3.5$$

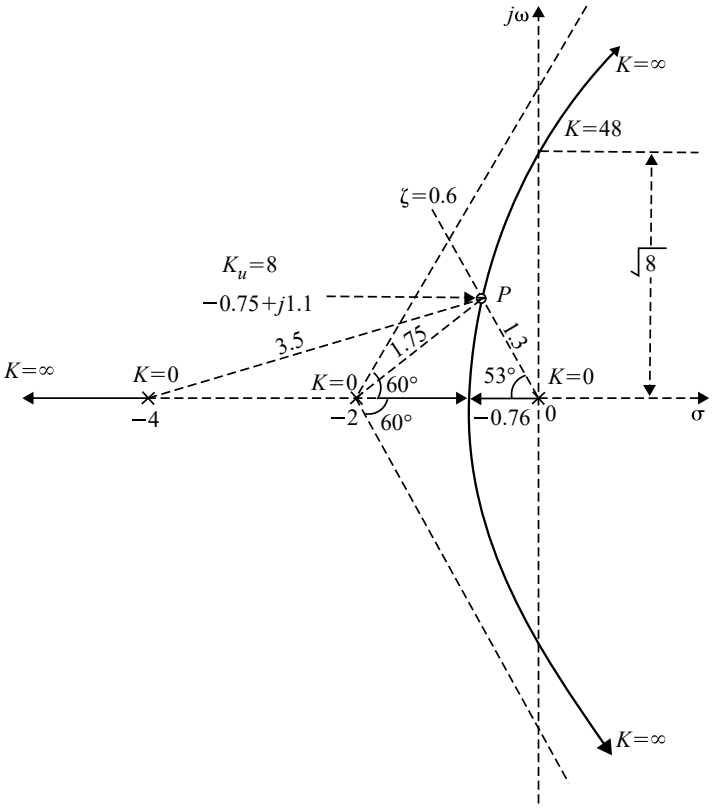


Fig. 6.21 Root locus plot for uncompensated system

$$\boxed{K_u = 8}$$

3. Using Eq. (6.24), the value of α is calculated as

$$\alpha = \frac{K_u}{1.1K} = \frac{8}{1.1 \times 40}$$

$$\boxed{\alpha = 0.182}$$

4. The second pole of $G(s)$ is at $s = -2$. The zero of the compensator is placed at one tenth of this value. Thus

$$\boxed{z = 0.2}$$

5. Using the relation $p = \alpha z$, we get

$$p = 0.18 \times 0.2$$

$$\boxed{p = 0.036}$$

6. Thus, the compensator T.F. is obtained as

$$G_c(s) = \frac{\alpha(s+z)}{(s+p)}$$

$$\boxed{G_c(s) = \frac{0.182(s+0.2)}{(s+0.036)}}$$

7. The T.F. of the compensated system is written as

$$\begin{aligned} G_c(s)G(s) &= \frac{0.182(s+0.2)40}{(s+0.036)s(s+2)(s+4)} \\ K_v &= 5.05 \\ e_{ss} &= \frac{1}{5.05} \\ &= 0.198 \end{aligned}$$

which is less than the specified value.

Example 6.14

Design a lag compensator for a system with open loop T.F

$$G(s) = \frac{K}{s(s+1)(s+4)}$$

The system is to be compensated to meet the following specifications:

(a) Damping ratio $\zeta = 0.4$.
(b) Settling time for 5% tolerance $t_s = 10$ s.
(c) Velocity error constant $K_v \geq 5\ \text{s}^{-1}$.
(d) The specifications in the frequency domain are: Phase margin $\phi_m = 43°$.

(*Anna University, May, 2009*)

Solution The design is first carried out in the time domain using root locus and for the compensated system, the frequency domain specifications are verified.

1.

$$G(s) = \frac{K}{s(s+1)(s+4)}$$

$$K_v = \lim_{s\to 0} sG(s)$$

$$= \lim_{s\to 0} \frac{sK}{s(s+1)(s+4)}$$

$$5 = \frac{K}{4}$$

$$\boxed{K = 20}$$

The settling time for 5% error tolerance is

$$t_s = 3T$$

where

$$T = \frac{1}{\zeta\omega_n}$$

$$\omega_n = \frac{3}{0.4 \times 10}$$

$$\boxed{\omega_n = 0.75}$$

$$\omega_d = \omega_n\sqrt{1-\zeta^2}$$

$$= 0.75\sqrt{1-(0.4)^2}$$

$$= 0.7$$

2. The root locus for the given open loop T.F. is drawn as shown in Fig. 6.22a. The point P corresponds to $-\zeta\omega_n + j\omega_d = -0.3 + j0.7$ or $\zeta = 0.4$ line.
3. The point P should satisfy magnitude criterion of root locus. The gain of the uncompensated system at point P is

$$K_u = \frac{\text{Directed distances from all poles to } P}{\text{Directed distances from all zeros to } P}$$

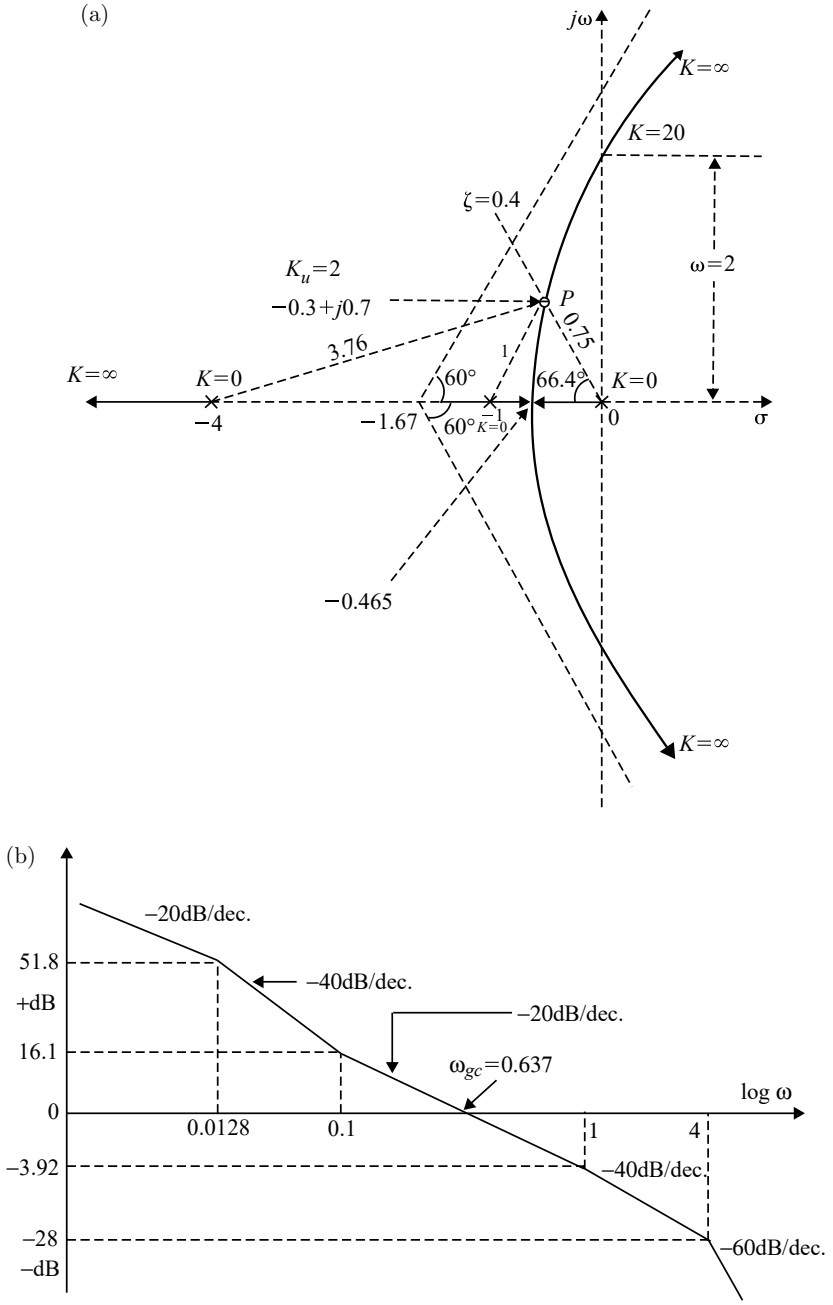


Fig. 6.22 **a** Root locus for $G(s) = \dfrac{K}{s(s+1)(s+4)}$. **b** Bode magnitude plot of the compensated system

$$K_u = 0.75 \times 1.0 \times 3.76$$

$$\boxed{K_u = 2.82}$$

4. Using Eq. (6.23), the value of α is calculated as

$$\alpha = \frac{K_u}{1.1 \times K}$$
$$= \frac{2.82}{1.1 \times 20}$$

$$\boxed{\alpha = 0.128}$$

5. Put the zero of the lag compensator at one tenth value of the second pole nearer to the origin

$$z = 0.1$$

6. The pole of the compensator is given by

$$P = \alpha z$$
$$= 0.128 \times 0.1$$

$$\boxed{P = 0.0128}$$

7. The T.F. of the lag compensator is

$$G_c(s) = \frac{\alpha(s+z)}{(s+p)}$$
$$G_c(s) = \frac{0.128(s+0.1)}{(s+0.0128)}$$

8. The T.F. of the compensated system is

$$G_c(s)G(s) = \frac{0.128(s+0.1)20}{s(s+0.0128)(s+1)(s+4)}$$

In time constant form, the above equation is written as

$$G_c(s)G(s) = \frac{5(1+\frac{s}{0.1})}{s(1+\frac{s}{0.0128})(1+s)(1+\frac{s}{4})}$$

The corner frequencies are 0.0128, 0.1, 1 and 4. The Bode magnitude plot is shown in Fig. 6.22b.

9. From Fig. 6.22b, the gain crossover frequency is obtained as $\omega_{gc} = 0.637$. At this frequency, the phase angle is

$$\begin{aligned}\phi &= -90^\circ - \tan^{-1}\frac{0.637}{0.0128} - \tan^{-1} 0.637 - \tan^{-1}\frac{0.637}{4} + \tan^{-1}\frac{0.637}{0.1} \\ &= -90^\circ - 50^\circ - 32.5^\circ - 9^\circ + 81.5^\circ \\ &= -100^\circ\end{aligned}$$

Phase margin, $\phi_m = 180 + \phi = 180^\circ - 100^\circ$

$$\boxed{\phi_m = 80^\circ}$$

6.5.3 *Design of Lag–Lead Compensator Using Root Locus*

The lag compensator with its pole and zero located closer to the origin in the s-plane improves the steady state response of the system and has little effect on the transient response. On the other hand, the lead network, which contributes phase lead at the gain crossover frequency, improves the transient response and has little effect in improving the steady state error. To accomplish the advantages of both lag and lead compensators, the lag–lead network is designed. The T.F. of the lag compensator as given in Eq. (6.10) is

$$G_c(s) = \frac{(1 + \alpha_1 T_1 s)(1 + \alpha_2 T_2 s)}{(1 + T_1 s)(1 + T_2 s)}$$

The above equation is written in pole–zero form as

$$G_c(s) = \frac{(s + \frac{1}{\alpha_1 T_1})(s + \frac{1}{\alpha_2 T_2})}{(s + \frac{1}{T_1})(s + \frac{1}{T_2})} \tag{6.25}$$

The poles and zeros of Equation (6.25) are as shown in Fig. 6.23. As already stated, the pole and zero of the lag portion of the compensator is located closer to the origin. Further, for any point chosen on the root locus, the magnitude contribution of the lag network is almost one and the phase contribution is legibly small. If these conditions are satisfied, by arbitrarily placing the pole and zero of the lag portion, then the lead portion of the compensator is designed exactly by the method described in Sect. 6.5.1. The step-by-step procedure to design the lag–lead compensator is given below.

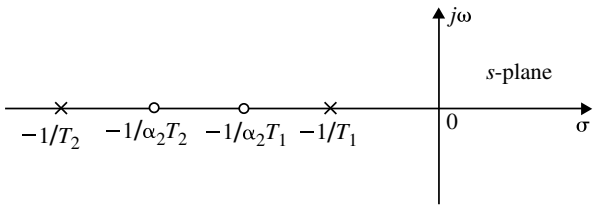


Fig. 6.23 Pole–zero diagram of lag–lead compensator

Step-by-Step Procedure to Design Lag–Lead Compensator

1. Determine the value of K which satisfies the steady state error requirement.
2. Locate the dominant pole location in the s-plane that satisfies the time domain specification. Let this point be P. Join OP. Draw horizontal line PQ.
3. The lead portion $\frac{(s+\frac{1}{\alpha_2 T_2})}{(s+\frac{1}{T_2})}$ contributes maximum phase angle to the point P and the lag portion $\frac{(s+\frac{1}{\alpha_1 T_1})}{(s+\frac{1}{T_1})}$ contributes negligibly small phase. Further, it is assumed that

$$\left|\frac{(s+\frac{1}{\alpha_1 T_1})}{(s+\frac{1}{T_1})}\right|_{s=p} = 1$$

$$\angle\frac{(s+\frac{1}{\alpha_1 T_1})}{(s+\frac{1}{T_1})}\Bigg|_{s=p} = 0^\circ$$

4. Calculate magnitude and phase angle contributions of $G(s)$, to the point P. The angle deficiency should be contributed by the lead network. If ϕ is the angle deficiency draw lines PA and PB such that $\angle APB = \phi$ as shown in Fig. 6.24.
5. Locate the pole of the lead portion at A and zero at point B. The magnitude ratio of $\frac{|PA|}{|PB|}$ should be such that the gain of the system with the lead portion is the same as that calculated in step 1. Otherwise, the locations of A and B are adjusted such that the gain requirement is satisfied noting that the angle requirement $\angle APO$ is always ϕ which satisfies the phase angle requirement of root locus.

Example 6.15

The open loop T.F. of a certain unity feedback control system is given by

$$G(s) = \frac{K}{s(s+5)(s+10)}$$

Design a lag–lead compensator to meet the following specifications:

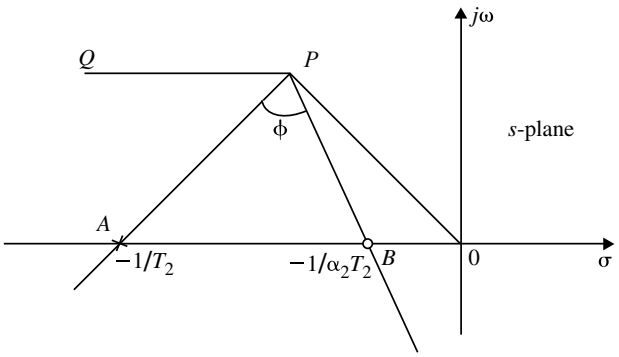


Fig. 6.24 Pole–zero location of the lead portion of the lag–lead compensator

1. The velocity error constant $K_v \geq 4.8\ \text{sec}^{-1}$.
2. The damping ratio for the dominant poles should be 0.6.
3. The undamped natural frequency of oscillation is 5 rad/s.

Solution

1.

$$\begin{aligned} K_v &= \lim_{s\to 0} sG(s) \\ &= \lim_{s\to 0} \frac{sK}{s(s+5)(s+10)} \\ 4.8 &= \frac{K}{50} \end{aligned}$$

$$\boxed{K = 240}$$

2.

$$\begin{aligned} \zeta\omega_n &= 0.6 \times 5 = 3 \\ \omega_d &= \omega_n\sqrt{1-\zeta^2} \\ &= 5\sqrt{1-0.36} \\ &= 4\ \text{rad/s}. \end{aligned}$$

The point P is at $-3 + j4$ and the gain at that point should be 240 to satisfy K_v requirement.

3. The lines OP and PQ are drawn. The value of K at point P is 240. The poles of $G(s)$ are located as shown in Fig. 6.25. The angle contribution from the poles

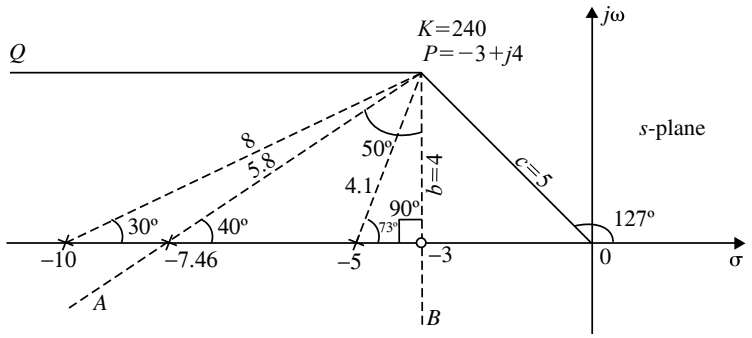


Fig. 6.25 Pole–zero locations of Lag–lead compensator for Example 6.15

of $G(s)$ to point P is

$$\theta = 127^\circ + 73^\circ + 30^\circ = 230^\circ$$

4. For the point P to be the point on the root locus, it should satisfy angle criterion of an odd multiple of 180°. Hence, the phase lead compensator should provide a phase angle lead of $230^\circ - 180^\circ = 50^\circ$.
5. The gain calculated at the point P with the uncompensated system is obtained by the product of directed distances from the poles of $G(s)$ to the point P. As seen from Fig. 6.25 it is $8 \times 4.1 \times 5 = 164$. However, the required gain is 240.
6. Draw any line PA where it cuts the negative real axis at A. Draw the line PB such that $\angle APB = 50^\circ$. Now the phase angle criterion of root locus at point P is satisfied. But the required magnitude at P is 240, while the actual magnitude is 164. By introducing the lead network, the magnitude should be increased by a factor of $\frac{240}{164} = 1.46$. That is $\frac{|PA|}{|PB|} = 1.46$.
7. By trial and error, the points A and B are shifted such that $\frac{|PA|}{|PB|} = 1.46$ and keeping $\angle PAB = 50^\circ$. When these two conditions are satisfied, the pole of the lead portion is fixed at point A and the zero at point B. The point A is at $s = -7.6$ and the point B is at $s = -3$. Thus, the lead portion is designed with $\frac{(s+4)}{(s+7.6)}$. The gain at point P is obtained as

$$K = \frac{8 \times 5.8 \times 4.1 \times 5}{4} = 240$$

8.

$$\frac{1}{T_2} = 7.6$$

$$\boxed{T_2 = 0.1315}$$

$$\frac{1}{\alpha_2 T_2} = 4$$
$$\alpha_2 = \frac{1}{3T_2} = \frac{7.6}{3}$$

$$\boxed{\alpha_2 = 2.53}$$

9.

$$\alpha_1 \alpha_2 = 1$$
$$\alpha_1 = \frac{1}{\alpha_2}$$
$$= \frac{1}{2.53}$$

$$\boxed{\alpha_1 = 0.394}$$

The pole of the lag portion is chosen arbitrarily closer to the origin. Thus, we choose

$$\frac{1}{T_1} = 0.01$$

$$\boxed{T_1 = 100}$$

The zero of the lag portion is then obtained as

$$\frac{1}{\alpha_1 T_1} = \frac{0.01}{0.394}$$
$$= 0.0253$$

10. The T.F. of the lag–lead compensator is

$$\boxed{G_c(s) = \frac{(s + 0.0253)(s + 3)}{(s + 0.01)(s + 7.6)}}$$

The T.F. of the compensated system is

$$G_c(s)G(s) = \frac{(s + 0.0253)(s + 3)240}{(s + 0.01)(s + 7.6)(s)(s + 5)(s + 10)}$$

SUMMARY

1. Compensators and controllers are intentionally inserted into the system for improving the performance deficiency.
2. Series compensators, parallel compensators and series-parallel compensators are normally used. However, the series (cascade) compensators are extensively used. The series compensator design is discussed in this chapter.
3. In control system performance specifications, transient response specifications are more important and they are the ultimate measure of performance indices. Since the root locus method gives direct information about the transient response of the closed loop system, it is preferred for the design of compensators in the time domain.
4. The frequency response approach is a very powerful tool to design compensators even though we get the information about the transient response performance in an indirect way. Bode plot is the choice for the design of the compensators in the frequency domain.
5. The compensators that are normally used are lag, lead and lag–lead compensators. The design methodologies of these compensators in the frequency domain and time domain are explained in detail with illustrative examples.

EXERCISE

Short Answer Type Questions

1. **What do you understand by compensation?**
   The process of adjustment or alteration of a control system to get desired performance specifications is called compensation.
2. **What is a compensator?**
   The additional compensates or circuits that are inserted into the system in order to compensate for a performance deficiency are called compensators.
3. **What are the performance criteria specified for compensators?**
   Transient and steady state errors, good accuracy good speed of response, high relative stability and good damping are the performance criteria that are commonly specified.
4. **What are the performance specifications given in the frequency domain?**
   Phase margin, gain margin, resonant peak, phase crossover frequency, resonant frequency, gain crossover frequency, bandwidth and static error constants are some of the specifications given in the frequency domain.

5. **How frequency domain specifications are correlated to time domain specifications?**
   Phase margin, gain margin and resonant peak in the frequency domain give the rough estimate of the system damping while gain crossover frequency resonant frequency and bandwidth are closely associated with the speed of transient response.
6. **What are the different methods used to connect the compensator in the system?**
   The compensators are connected in
   1. Series or cascade
   2. Parallel
   3. Series-parallel
7. **List out the advantages and disadvantages of the different methods of connecting the compensators?**
   In the series-connected scheme, the signal flows from lower energy level to higher energy level. Therefore, additional amplifiers are required to increase the gain as well as to provide necessary isolation. Thus, it has more number of components and hence it is expensive. In the parallel scheme, the flow of energy is from higher level to lower level, and hence additional components are not required. However, series compensation is simpler than parallel compensation. The advantages of series and parallel scheme are incorporated in the series-parallel scheme.
8. **Why Root locus method is used to design compensators to meet time domain specifications?**
   To meet the time domain specifications root locus method is used in which the poles and zeros of the compensator can reshape the root locus of the system such that the closed loop poles are placed in locations in the s-plane that give desired transient response.
9. **Does the transient response and the stability of a system depend only on pole location?**
   The stability of the system solely depends upon the pole locations of the system. The transient response also, to a great extent, depends upon pole locations. However, the zeros also have their impact on the transient response.
10. **Why Bode plot is preferred to design compensators in the frequency domain?**
    When the compensator is added to the Bode plot of the original system, only the portions that correspond to the poles and zeros of the compensator need to be reshaped. Further, if the open loop gain is varied, the magnitude curve of the Bode plot is shifted up or down without changing the slope of any part of the Bode diagram while retaining the original phase curve. However, when bandwidth requirement is specified, it is preferable to use the Nichols plot.
11. **In the Bode diagram, how the frequency range is classified?**
    In the Bode plot, the frequency region is classified as low frequency region, medium frequency region and high frequency region.

12. **How low frequency region is identified in the Bode plot and what is its implication?**
    In the Bode plot, the frequency range which is far below the gain crossover frequency is termed as low frequency region and it indicates the steady state behaviour of the system.
13. **How medium frequency region is identified in the Bode plot and what is its implication?**
    The region nearer to the gain crossover frequency is called the medium frequency region which indicates the relative stability of the system.
14. **How high frequency region is identified in the Bode plot and what is its implication?**
    The region far above the gain crossover frequency is called the high frequency region and it determines the transient response of the system.
15. **How steady state accuracy is achieved in the low frequency region?**
    By increasing the gain at low frequency, the steady state accuracy is achieved.
16. **How relative stability is achieved in the medium frequency region?**
    The slope of the lag-magnitude curve in the Bode diagram near the gain crossover frequency should be -20 dB/s and to achieve the desired phase margin and high relative stability, this slope is extended over a wide frequency range.
17. **How improved transient response is achieved in the high frequency region?**
    To improve the transient response and to suppress the effects of noise, the gain at high frequency region should be attenuated.
18. **What is the advantage of the frequency response method of compensator design over that of time domain approach using root locus?**
    In the frequency response method, it is easy to implement the design. In the Bode plots, the required gain is easily found from the asymptotic plots, whereas in the root locus, it requires several calculations.
19. **How does a phase lead network perform?**
    The phase lead network provides a phase lead angle which offsets the excessive phase lag which is inherently present in the original system. It is so designed that the maximum phase contribution occurs near the gain crossover frequency, thereby increasing the phase margin.
20. **What is the maximum phase lead that can be practically obtained from a lead compensator?**
    The maximum phase lead that can be obtained from a lead compensator in practice is about 50° to 60°.
21. **If more than** 60° **phase lead is required, how a compensator is designed?**
    If more than 60° phase lead is required, a two stage lead network with an isolation amplifier is designed.
22. **What is the effect of phase lead compensator on the steady state error?**
    The lead compensator will not have much effect on steady state error since it provides attenuation at low frequency region.
23. **What is the effect of lead compensator on the system bandwidth, rise time and settling time?**
    By introducing a lead compensator, the phase margin is increased. This increases

the damping and the bandwidth. The time response is faster and therefore the rise time and settling time are reduced.

24. **What is the limiting value of the attenuation factor α in the lead compensator?**
    The limiting value of the attenuation factor α is normally restricted to 0.07 from the view point of physical realization. Further, if $1/\alpha$ becomes very high, noise is amplified.
25. **How does a lag compensator perform?**
    In the phase lag compensator, attenuation is provided at the gain crossover frequency which shifts the Bode magnitude plot downwards without affecting the phase plot. The new gain crossover frequency is shifted to the left where the enlarged phase margin is obtained which is required in the specifications.
26. **What is the effect of lag compensator on steady state error?**
    Lag compensator, being a low pass filter, provides high gain at low frequencies. This reduces the steady state error.
27. **What is the effect of the lag compensator on system bandwidth, rise time and settling time?**
    The introduction of the lag compensator shifts the gain crossover frequency to the low frequency region of the Bode plot which diminishes the bandwidth of the system. Thus, the rise time and settling time with phase lag compensator are longer.
28. **How does a lag–lead network perform?**
    The lag portion of the compensator provides attenuation near and above the gain crossover frequency which increases the overall gain of the system at low frequency region. This gives improved steady state performance of the system. The phase lead portion provides phase lead angle at the gain crossover frequency which increases the phase margin.
29. **What is the effect of lag–lead network on bandwidth, rise time and settling time?**
    Because of the increased phase margin, the bandwidth is increased. The rise time and settling time are reduced which increases the fastness of response.

Long Answer Type Questions

1. The forward path T.F. of a certain unity feedback control system is given by

$$G(s) = \frac{K}{s(s+10)}$$

It is required to have a velocity error constant $K_v \geq 20$, phase margin is at least 45° and the gain margin is more than 20 dB. Design a suitable lead compensator.
Answer:

$$G_c(s) = \frac{(s + 11.48)}{(s + 17.1)}$$
$$\phi_m = 47^\circ$$
$$G_m = \infty$$

Additional amplifier gain $A = 1.5$.

2. The forward path T.F of a unity feedback control system is given by

$$G(s) = \frac{K}{s(s + 2)(s + 10)}$$

The steady state error for a unit step input is 0.5. The required phase margin is at least 48° and the gain margin is at least 5 dB. Design a phase lead compensator.
Answer:

$$G_c(s) = \frac{(s + 1.463)}{(s + 2.215)}$$
$$\phi_m = 50^\circ$$
$$G_m = 5.86 \text{ dB}$$
$$\omega_{gc} = 5.3$$
$$K = 60.5$$

3. The forward path T.F. of a certain unity feedback system is

$$G(s) = \frac{K(s + 0.2)}{s(s + 1)(s + 2)}$$

The velocity error constant is $K_v \geq 6 \text{ s}^{-1}$. The required phase margin is more than 40° and gain margin more than 8 dB. Design a suitable lag compensator.
Answer:

$$G_c(s) = \frac{(1 + 3s)}{(1 + 13.8s)}$$
$$K = 60$$
$$\alpha = 0.22$$
$$T = 13.8$$
$$\phi_m = 41^\circ$$
$$G_m = \infty$$
$$\omega_{gc} = 3.3$$

4. The forward path T.F. of a certain unity feedback system is

$$G(s) = \frac{K}{s(s+2)(s+3)}$$

It is required that the steady state error is less than 0.2 for unit step input. The phase margin should be more than 40° and gain margin is at least 5 dB. Design a suitable lag compensator.

Answer:

$$\begin{aligned} G_c(s) &= \frac{(1+11.32s)}{(1+66.2s)} \\ K &= 30 \\ \phi_m &= 49^\circ \\ G_m &= 5.5 \text{ dB} \\ \alpha &= 0.171 \\ T &= 66.2 \\ \omega_{gc} &= 0.776 \end{aligned}$$

5. The forward path T.F. of a certain unity feedback system is

$$G(s) = \frac{K}{s(s+2)(s+5)}$$

It is required that the steady state error for unit step input should not exceed 0.2. The phase margin and gain margin should be greater than 45° and 7 dB representatively. Design a suitable lag–lead network.

Answer:

$$\begin{aligned} G_c(s) &= \frac{(s+0.3467)(s+1.2)}{(s+0.1)(s+6.85)} \\ K &= 83 \\ \phi_m &= 53.3^\circ \\ G_m &= 7.5 \text{ dB} \\ \omega_{gc} &= 5.62 \end{aligned}$$

6. The forward path T.F. of a certain unity feedback control system is

$$G(s) = \frac{K}{s(s+2)(s+8)}$$

The velocity error constant $K_v \geq 25$. The required phase margin and gain margin should be greater than 30° and 10 dB, respectively. Design a lag–lead compensator.

Answer:

$$G_c(s) = \frac{(s + 1.216)(s + 3)}{(s + 0.1)(s + 36)}$$
$$K = 400$$
$$\phi_m = 33.4^\circ$$
$$G_m = 21.5 \text{ dB}$$

7. Consider the following open loop T.F. of a certain unity feedback closed loop system

$$G(s) = \frac{K}{s(s + 2)(s + 5)}$$

It is required that the velocity error constant $K_v \geq 4$ sec^{-1}. The phase margin should be greater than 40° and the gain margin is at least 5 dB. The gain crossover frequency $\omega_{gc} > 4$ rad/s Design a lag–lead compensator.
Answer:

$$G_c(s) = \frac{(s + 0.2317)(s + 1.944)}{(s + 0.1)(s + 4.5)}$$
$$K = 40$$
$$\phi_m = 49.1^\circ$$
$$G_m = 5.2 \text{ dB}$$
$$\omega_{gc} = 4.66 \text{ rad/s}.$$

Chapter 7
State Space Modelling and Analysis



Learning Objectives

After completing this chapter, you should be able to

- Represent linear continuous time system by state equation.
- Model mechanical, electromechanical and electrical systems by state equation.
- Convert differential equation to state equation.
- Obtain state equation from transfer function and *vice versa*.
- Find the solution of state equation by the Laplace transform method and by the general form.
- Determine the state transition matrix and the solution of state equation for typical inputs.
- Define and determine state controllability, output controllability and observability of linear time invariant continuous time systems.
- Represent the discrete time system by state equation.
- Model the discrete time system by state equation.
- Find the solution of state equation using the z-transform method.
- Define state controllability, output controllability and observability of discrete time systems.
- Develop a MATLAB program to solve state equations for continuous and discrete time systems.

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S. Palani, *Automatic Control Systems*,
https://doi.org/10.1007/978-3-030-93445-3_7

7.1 Introduction

The transfer function (T.F.) model, for quite a long time, was used for the analysis and design of linear time invariant continuous time systems. However, this model has many limitations in that it is expressed as a ratio of output to input variables, and thus the internal behaviour of the system is hidden. Further, the T.F. method is valid only for linear systems with initial conditions being zero. It is powerless for non-linear, time varying and multi-input and multi-output (MIMO) systems. It is also difficult to handle large-scale complex systems with a transfer function model. Furthermore, system design modelling by T.F. is based on the trial and error procedure which in general will not lead to optimal control systems. All these limitations are overcome by representing the system in the state space model. This is a differential (or difference) equation model which is expressed in n first order equations which are written in a specific format. The model is valid for linear, non-linear and time varying systems and also when initial conditions are not zero. Unlike the T.F. model, it gives a complete description of the internal behaviour of all physical variables in the system. In this chapter, we first develop a state space model of mechanical systems and electrical networks followed by the conversion of the T.F. model to the state space model. The solution of the state equation is derived and illustrated by simple examples. Finally, the discrete time system represented by the difference equation is converted into state equations. Comparison of classical and modern control approaches are given in Table 7.1.

7.2 The State of a System and State Equation of Continuous Time System

For a linear continuous time system, **the state of a system is defined as the minimum number of initial conditions that must be specified at any initial time t_0 so that the complete dynamic behaviour of the system at any time $t > t_0$ is determined when the input $u(t)$ is known.**

When the input $u(t)$ is applied, the future states of the system, for $t > t_0$, also change and we can uniquely determine these states. Since the states of the system vary with respect to time, we call these variables state variables.

The number of state variables depend on the dynamic model selected to describe the physical systems. For a system described by the nth order differential equation, there will be n state variables. If these n state variables form the coordinates of an n dimensional vector space, it is known as state space. For a continuous time system described by nth order differential equation if the variables which represent the states are chosen less than n, then the system is not fully represented and information about the missing states will not be known. Similarly, if the states are chosen more than n, then some of the states chosen are redundant and they can always be expressed in terms of other known states. Hence, for an nth order model of a system it is necessary

Table 7.1 Comparison of classical and modern control theories

S. No.	Classical control theory (Root locus and frequency response methods)	Modern control theory
1.	Generally applicable to linear time invariant system having single-input and single-output It is powerless for time varying, non-linear and multi-input and multi-output systems	Modern control theory can be successfully applied in all these cases
2.	Not applicable to optimal and adaptive control systems	Can be successfully applied in these cases
3.	It is difficult to handle large-scale complex systems	It is not difficult to handle large-scale complex systems with the help of modern computers
4.	This is a complex frequency domain approach	This is essentially a time domain approach
5.	System design is based on trial and error procedure which in general will not lead to optimal control systems	Optimal control system can be designed with ease for the given performance index
6.	Design is carried out for a specific input function such as impulse step and ramp functions	Design can be carried out for a class of inputs
7.	Initial condition is to be zero	Initial condition need not be zero
8.	Information about one output at a time is known	Complete information about all the variables is known at a time

strictly to choose only n appropriate states and they are represented by n first order equations together with the input.

7.3 Vector Matrix Differential Equation of Continuous Time System

State variable equations, whether linear or non-linear, are expressed in the time domain by using compact vector matrix notations. **These equations are called vector matrix differential equations**. The standard form of representing these state equations for a continuous time system is

$$\dot{x}(t) = Ax(t) + Bu(t) \tag{7.1}$$

$$y(t) = Cx(t) + Du(t) \tag{7.2}$$

Equation (7.1) is called the state equation and Eq. (7.2) is called output equation. Equations (7.1) and (7.2) together describe the system dynamics and they are called vector matrix differential equations.

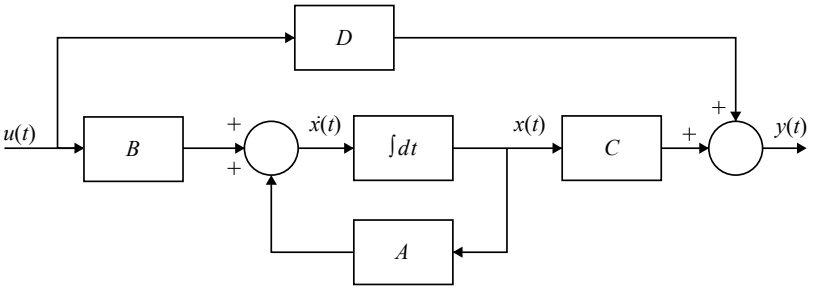


Fig. 7.1 Block diagram representation of state Eqs. (7.1) and (7.2)

In Eqs. (7.1) and (7.2),

$x(t)$ = State vector ($n \times 1$ dimension);
$y(t)$ = Output vector ($p \times 1$ dimension);
$u(t)$ = Input vector ($r \times 1$ dimension);
A = State matrix ($n \times n$ dimension);
B = Input matrix ($n \times r$ dimension);
C = Output matrix ($p \times n$ dimension);
D = Direct transmission matrix ($p \times r$ dimension).

A block diagram representation of Eqs. (7.1) and (7.2) is shown in Fig. 7.1.

Depending upon the dimensions of the vectors x, u and y, the appropriate dimensions of the matrices, A, B and C are chosen. In most practical applications, the direct transmission of input $u(t)$ to the output $y(t)$ is not done and hence $D = 0$. In forming the state Eq. (7.1), it is to be observed that only the first derivative of $x(t)$ appears on the left side of the equation and no derivative of $x(t)$ appears on the right side. The right side of the equation contains only the states and input. The following examples illustrate the method of forming state equations.

7.3.1 State Equations for Mechanical Systems

The dynamic equations of mechanical systems are written from the free body diagram. The physical variables such as displacement and velocity are chosen as the states and for each state variable, the equation for its first derivative is obtained and converted in the format of Eq. (7.1).

Example 7.1

For the mechanical system shown in Fig. 7.2a, form the state equation.

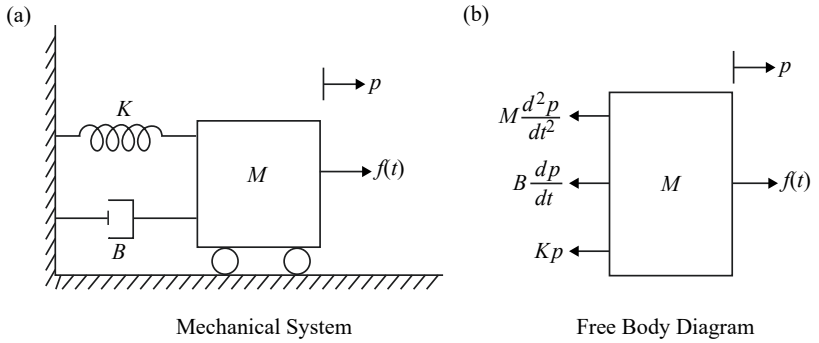


Fig. 7.2 Mechanical system and its free body diagram

Solution

1. The mechanical system is shown in Fig. 7.2a. Its free body diagram is shown in Fig. 7.2b.
2. p is the displacement of mass M and $f(t)$ is the force applied. In the free body diagram, the opposing forces act in the direction opposite to the direction of motion.
3. From Fig. 7.2b, the following dynamic equations for the given mechanical system are written:
$$M\frac{d^2p}{dt^2} + B\frac{dp}{dt} + Kp = f(t) \tag{7.3}$$
4. Let us choose the displacement p, which is the physical variable in the mechanical system as one state variable. Thus,
$$p = x_1(t) \tag{7.4}$$
5. $$\frac{dp}{dt} = \dot{x}_1(t)$$
Let us choose the velocity $\frac{dp}{dt}$ as the second state variable. Thus,
$$\frac{dp}{dt} = \dot{x}_1(t) = x_2(t) \tag{7.5}$$
6. Equation (7.3) gives complete description of the given mechanical system and it is a second order system. Therefore, there should be two states and two state equations. Equation (7.5) represents one state equation. Similar to that, we should obtain an equation for $\dot{x}_2(t)$.
7. Now consider Eq. (7.3). By substituting $p = x_1(t)$, $\frac{dp}{dt} = x_2(t)$ and $f(t) = u(t)$, we get

$$M\dot{x}_2(t) + Bx_2(t) + Kx_1(t) = u(t)$$

Solving for $\dot{x}_2(t)$, we get

$$\dot{x}_2(t) = -\frac{K}{M}x_1(t) - \frac{B}{M}x_2(t) + \frac{1}{M}u(t) \tag{7.6}$$

8. From Eqs. (7.5) and (7.6), the following vector matrix differential equation is formed:

$$\begin{bmatrix}\dot{x}_1(t)\\ \dot{x}_2(t)\end{bmatrix} = [\dot{x}(t)] = \underbrace{\begin{bmatrix}0 & 1\\ -\frac{K}{M} & -\frac{B}{M}\end{bmatrix}}_{A}\begin{bmatrix}x_1(t)\\ x_2(t)\end{bmatrix} + \underbrace{\begin{bmatrix}0\\ \frac{1}{M}\end{bmatrix}}_{B}u(t) \tag{7.7}$$

In Eq. (7.7),

$$A = \begin{bmatrix}0 & 1\\ -\frac{K}{M} & -\frac{B}{M}\end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix}0\\ \frac{1}{M}\end{bmatrix}$$

Note that the A matrix is a square matrix. In the examples to follow, for convenience, the state variables are denoted as x instead of $x(t)$.

Example 7.2

Consider the mechanical system shown in Fig. 7.3a. Form the state equation. The displacement plus velocity of mass M are taken as the output.

Solution

1. The mechanical system is represented in Fig. 7.3a and its free body diagram in Fig. 7.3b. The mass M is given a displacement of p_2. The point A moves with a displacement p_1.
2. From the free body diagram, the following equations are written:

$$M\ddot{p}_2 + B(\dot{p}_2 - \dot{p}_1) + K_2 p_2 = f(t) \tag{7.8}$$

$$B(\dot{p}_1 - \dot{p}_2) + K_1 p_1 = 0 \tag{7.9}$$

The following state variables are chosen:

$$p_2 = x_1$$
$$\dot{p}_2 = x_2$$
$$p_1 = x_3$$

The equations for the first derivatives of x_1, x_2 and x_3 are to be obtained.

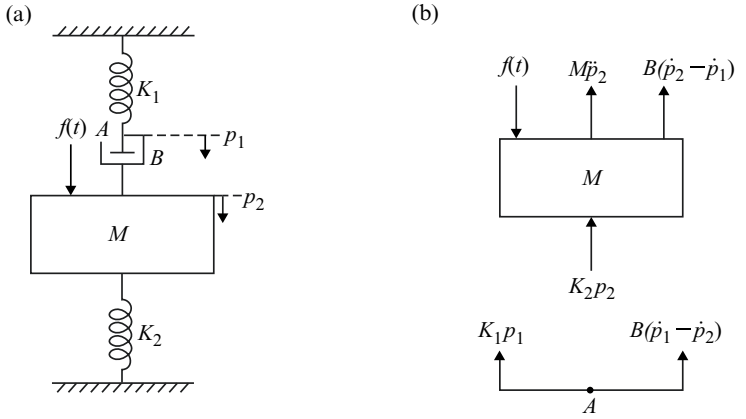


Fig. 7.3 Mechanical system and its free body diagram

$$\dot{x}_1 = \dot{p}_2 = x_2$$
$$M\dot{x}_2 + K_1x_3 + K_2x_1 = u(t)$$

$$\dot{x}_2 = -\frac{K_2}{M}x_1 - \frac{K_1}{M}x_3 + \frac{1}{M}u(t)$$
$$B\dot{x}_3 - Bx_2 + K_1x_3 = 0$$
$$\dot{x}_3 = x_2 - \frac{K_1}{B}x_3$$

Thus

$$\dot{x}(t) = \underbrace{\begin{bmatrix} 0 & 1 & 0 \\ -\frac{K_2}{M} & 0 & -\frac{K_1}{M} \\ 0 & 1 & -\frac{K_1}{B} \end{bmatrix}}_{A} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \underbrace{\begin{bmatrix} 0 \\ \frac{1}{M} \\ 0 \end{bmatrix}}_{B} u(t)$$

The output y is given by

$$y = \underbrace{[1 \quad 1 \quad 0]}_{C} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

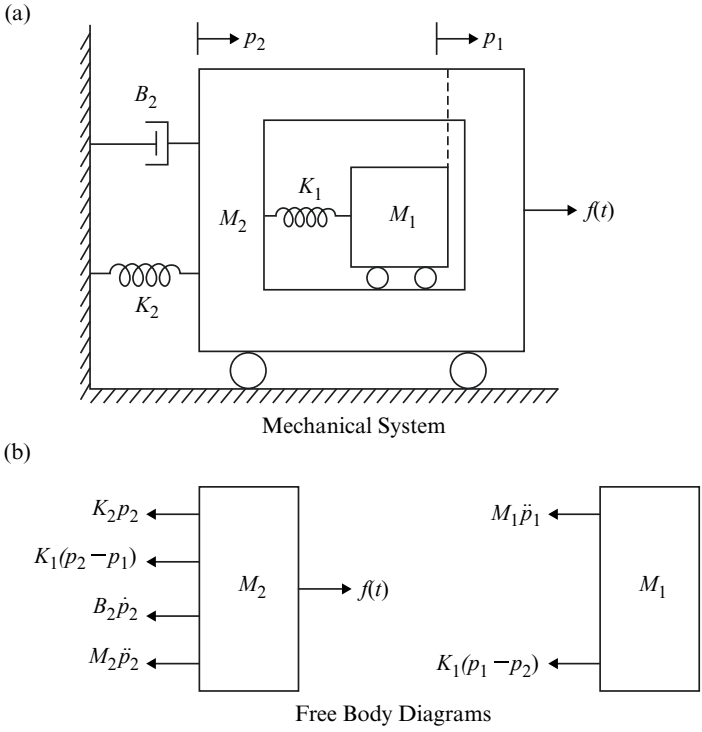


Fig. 7.4 Mechanical system and its free body diagrams

Example 7.3

Consider the mechanical system shown in Fig. 7.4a. Form the state equation.

Solution

1. The mechanical system is shown in Fig. 7.4a and its free body diagram in Fig. 7.4b.
2. From Fig. 7.4b, the following dynamic equations are written:

$$M_2\ddot{p}_2 + B_2\dot{p}_2 + K_2p_2 + K_1(p_2 - p_1) = f(t) \tag{7.10}$$

$$M_1\ddot{p}_1 + K_1(p_1 - p_2) = 0 \tag{7.11}$$

3. The following state variables are chosen:

$$x_1 = p_1$$
$$x_2 = \dot{x}_1 = \dot{p}_1$$
$$x_3 = p_2$$
$$x_4 = \dot{x}_3 = \dot{p}_2$$

4. The first derivatives $\dot{x}_1$ and $\dot{x}_2$ are known. The first derivatives $\dot{x}_3$ and $\dot{x}_4$ are obtained from Eqs. (7.11) and (7.10), respectively, and they are given as follows:

$$\dot{x}_2 = -\frac{K_1}{M_1}x_1 + \frac{K_1}{M_1}x_3 \tag{7.12}$$
$$\dot{x}_4 = \frac{K_1}{M_2}x_1 - \frac{(K_1 + K_2)}{M_2}x_3 - \frac{B_2}{M_2}x_4 + \frac{1}{M_2}f(t) \tag{7.13}$$

5. The vector matrix differential equation is thus obtained as

$$\dot{x}(t) = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -\frac{K_1}{M_1} & 0 & \frac{K_1}{M_1} & 0 \\ 0 & 0 & 0 & 1 \\ \frac{K_1}{M_2} & 0 & -\frac{(K_1 + K_2)}{M_2} & -\frac{B_2}{M_2} \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{1}{M_2} \end{bmatrix} u(t)$$

Example 7.4

For the mechanical system shown in Fig. 7.5, obtain the state space model.

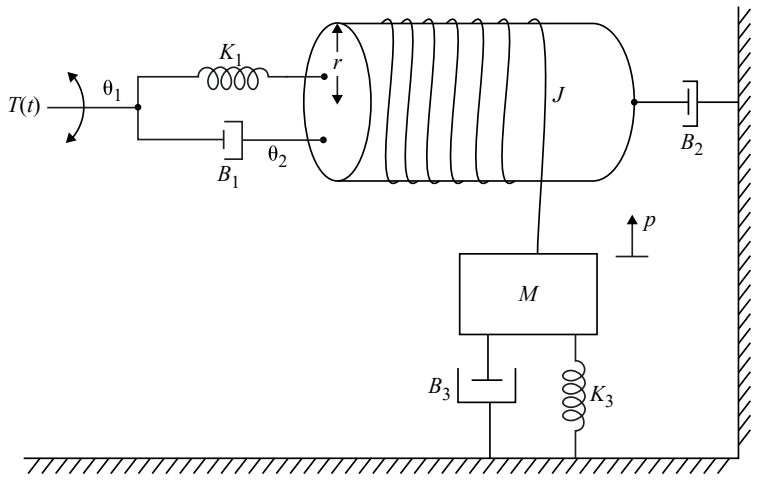


Fig. 7.5 A hybrid mechanical system

Solution

1. A hybrid mechanical system which is a combination of translational and rotational systems is shown in Fig. 7.5. Just by inspection, the following dynamic equations are written.
2. $$T(t) = B_1(\dot{\theta}_1 - \dot{\theta}_2) + K_1(\theta_1 - \theta_2) \tag{7.14}$$

   Also

   $$T(t) = J\ddot{\theta}_2 + B_2\dot{\theta}_2 + r(M\ddot{p} + B_3\dot{p} + K_2 p) \tag{7.15}$$

   Substituting $p = r\theta_2$ in Eq. (7.15), we get

   $$T(t) = (J + r^2M)\ddot{\theta}_2 + (B_2 + r^2B_3)\dot{\theta}_2 + r^2K_3\theta_2 \tag{7.16}$$

3. The following state variables are chosen:

   $$\begin{aligned} x_1 &= \theta_1 \\ x_2 &= \theta_2 \\ x_3 &= \dot{\theta}_2 = \dot{x}_2 \end{aligned}$$

4. The derivatives of x_1 and x_2 are obtained from Eqs. (7.14) and (7.16) and are given as

   $$\begin{aligned} \dot{x}_1 &= -\frac{K_1}{B_1}x_1 + \frac{K_1}{B_1}x_2 + x_3 + \frac{1}{B_1}T(t) \\ \dot{x}_2 &= x_3 \\ \dot{x}_3 &= -\frac{K_{eq}}{J_{eq}}x_2 - \frac{B_{eq}}{J_{eq}}x_3 + \frac{1}{J_{eq}}T(t) \end{aligned}$$

   where $J_{eq} = (J + r^2M)$, $B_{eq} = (B_2 + r^2B_3)$ and $K_{eq} = r^2K_3$.

$$\dot{x}(t) = \begin{bmatrix} -\frac{K_1}{B_1} & \frac{K_1}{B_1} & 1 \\ 0 & 0 & 1 \\ 0 & -\frac{K_{eq}}{J_{eq}} & -\frac{B_{eq}}{J_{eq}} \end{bmatrix} x(t) + \begin{bmatrix} \frac{1}{B_1} \\ 0 \\ \frac{1}{J_{eq}} \end{bmatrix} u(t)$$

7.3.2 State Equations for Electrical Circuits

The number of independent energy-storing elements in the electrical circuit determine the number of state variables. The capacitor and inductor are the two energy-storing elements in electrical circuits: The physical variables, namely the current through the inductor and voltages across the capacitor are chosen as the state variables. The following steps are followed while forming the state space equations for electrical circuits:

1. Choose all independent inductor currents and the capacitor voltages as the state variables.
2. The state variables and their first derivatives are expressed in terms of a set of loop circuits.
3. Write loop equations and eliminate all variables other than the state variables and their first derivatives.

The following examples illustrate the method of forming state equations for electrical networks.

Example 7.5

Write the state equations for the network shown in Fig. 7.6.

Solution

1. Let i be the current passing through the inductor L and v_c be the voltage across the capacitor C. These variables are chosen as state variables.

$$x_1 = i$$
$$x_2 = v_c$$

2. The following loop equation is written:

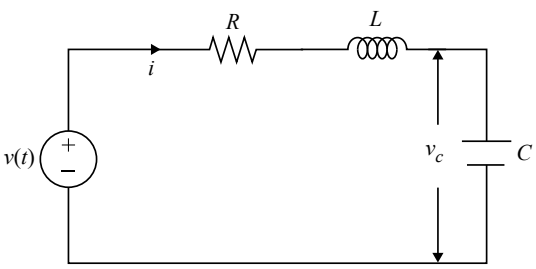


Fig. 7.6 *R-L-C* series circuit

$$L\frac{di}{dt} + Ri + v_c = v(t)$$

$$\dot{x}_1 = -\frac{R}{L}x_1 - \frac{1}{L}x_2 + \frac{1}{L}v(t) \tag{7.17}$$

Also

$$v_c = \frac{1}{C}\int i\,dt$$

Differentiating both sides, we get

$$\dot{v}_c = \frac{1}{C}i$$

$$\dot{x}_2 = \frac{1}{C}x_1 \tag{7.18}$$

3. Equations (7.17) and (7.18) represent the first derivatives of the chosen states. Hence

$$\dot{x}(t) = \begin{bmatrix} -\frac{R}{L} & -\frac{1}{L} \\ \frac{1}{C} & 0 \end{bmatrix} x(t) + \begin{bmatrix} \frac{1}{L} \\ 0 \end{bmatrix} v(t)$$

Example 7.6

For the electrical network shown in Fig. 7.7, form the state equation.

Solution

1. Let i be the current passing through the inductor L and v_c be the voltage across the capacitor C. These variables are taken as the state variables. Thus,

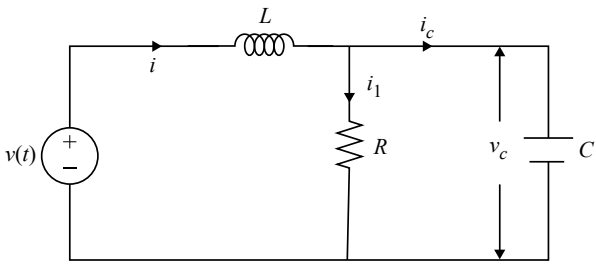


Fig. 7.7 Electrical circuit of Example 7.6

$$x_1 = i$$
$$x_2 = v_c$$

2. The following loop equation connecting L and C is written:

$$L\frac{di}{dt} + v_c = v(t)$$
$$\dot{x}_1 = -\frac{1}{L}x_2 + \frac{1}{L}v(t) \tag{7.19}$$

3. Also

$$i_c = i - i_1$$
$$C\frac{dv_c}{dt} = i - \frac{v_c}{R}$$
$$\dot{x}_2 = \frac{1}{C}x_1 - \frac{1}{RC}x_2 \tag{7.20}$$

4. Combining Eqs. (7.19) and (7.20), we get

$$\dot{x}(t) = \begin{bmatrix} 0 & -\frac{1}{L} \\ \frac{1}{C} & -\frac{1}{RC} \end{bmatrix} x(t) + \begin{bmatrix} \frac{1}{L} \\ 0 \end{bmatrix} v(t)$$

Example 7.7

For the electrical network shown in Fig. 7.8, form the state equations.

Solution

1. There are three energy-storing elements, namely L, C_1 and C_2, and they are independently connected and there should be three state variables. The following

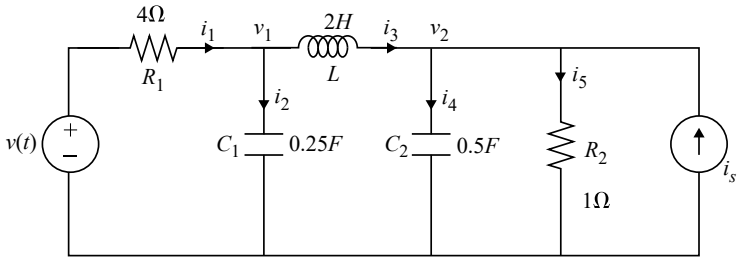


Fig. 7.8 Electrical network of Example 7.7

state variables are chosen:

$$\begin{aligned} x_1 &= v_1 \\ x_2 &= v_2 \\ x_3 &= i_3 \end{aligned}$$

2. The first derivative of these variables is obtained as follows. At v_1 node

$$\begin{aligned} i_1 &= i_2 + i_3 \\ \frac{(v - v_1)}{R_1} &= C_1 \frac{dv_1}{dt} + i_3 \\ \frac{dv_1}{dt} &= -\frac{1}{R_1 C_1} v_1 - \frac{1}{C_1} i_3 + \frac{1}{R_1 C_1} v \\ \dot{x}_1 &= -\frac{1}{R_1 C_1} x_1 - \frac{1}{C_1} x_3 + \frac{1}{R_1 C_1} v \end{aligned} \tag{7.21}$$

3. At note v_2, the following equation is written:

$$\begin{aligned} i_3 + i_s &= i_4 + i_5 \\ &= C_2 \frac{dv_2}{dt} + \frac{v_2}{R_2} \end{aligned}$$

Substituting the state variables, we get

$$\begin{aligned} x_3 + i_s &= C_2 \dot{x}_2 + \frac{x_2}{R_2} \\ \dot{x}_2 &= -\frac{1}{R_2 C_2} x_2 + \frac{1}{C_2} x_3 + \frac{1}{C_2} i_s \end{aligned} \tag{7.22}$$

4. The voltage drop across the inductor L is

$$\begin{aligned} L \frac{di_3}{dt} &= v_1 - v_2 \\ \dot{x}_3 &= \frac{1}{L} x_1 - \frac{1}{L} x_2 \end{aligned} \tag{7.23}$$

5. Combining Eqs. (7.21), (7.22) and (7.23), we get

$$\dot{x}(t) = \begin{bmatrix} -\frac{1}{R_1 C_1} & 0 & -\frac{1}{C_1} \\ 0 & -\frac{1}{R_2 C_2} & \frac{1}{C_2} \\ \frac{1}{L} & -\frac{1}{L} & 0 \end{bmatrix} x(t) + \begin{bmatrix} \frac{1}{R_1 C_1} & 0 \\ 0 & \frac{1}{C_2} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v(t) \\ i_s(t) \end{bmatrix}$$

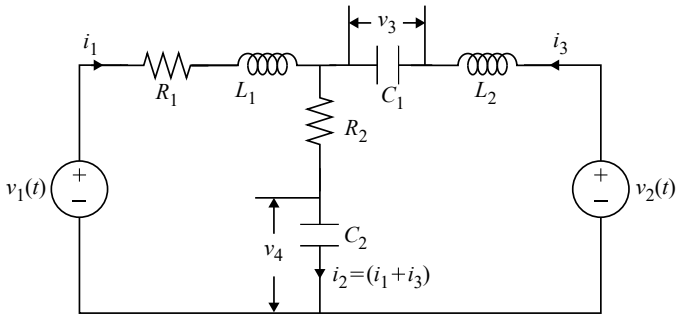


Fig. 7.9 Electrical network of Example 7.8

Substituting the numerical values, we get

$$\dot{x}(t) = \begin{bmatrix} -1 & 0 & -4 \\ 0 & -2 & 2 \\ 0.5 & -0.5 & 0 \end{bmatrix} x(t) + \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v(t) \\ \\ i_s(t) \end{bmatrix}$$

Example 7.8

For the electrical network shown in Fig. 7.9, form the vector matrix differential equation.

Solution

1. There are four energy-storing elements, namely L_1, L_2, C_1 and C_2. Hence, four state variables, the current through the inductors and the voltages across the capacitors are chosen. Thus,

$$\begin{aligned} x_1 &= i_1 \\ x_2 &= i_3 \\ x_3 &= v_3 \\ x_4 &= v_4 \end{aligned}$$

   The dimension of A matrix is 4×4. Since there are two inputs, namely $v_1(t)$ and $v_2(t)$, the dimension of B matrix is 4×2.
2. The following loop equation is written connecting $v_1(t)$, R_1, L_1, R_2 and C_2:

$$\begin{aligned} v_1(t) &= L_1 \frac{di_1}{dt} + i_1 R_1 + (i_1 + i_3) R_2 + v_4 \\ \dot{x}_1 &= -\frac{(R_1 + R_2)}{L_1} x_1 - \frac{R_2}{L_1} x_2 - \frac{1}{L_1} x_4 + \frac{1}{L_1} v_1(t) \end{aligned} \tag{7.24}$$

3. The following loop equation is written connecting $v_2(t)$, L_2, C_1, R_2 and C_2:

$$v_2(t) = L_2\frac{di_3}{dt} + v_3 + (i_1 + i_3)R_2 + v_4$$
$$\dot{x}_2 = -\frac{R_2}{L_2}x_1 - \frac{R_2}{L_2}x_2 - \frac{1}{L_2}x_3 - \frac{1}{L_2}x_4 + \frac{1}{L_2}v(t) \tag{7.25}$$

4. For the capacitance C_1, the following equation is written:

$$C_1\frac{dv_3}{dt} = i_3$$
$$\dot{x}_3 = \frac{1}{C_1}x_2 \tag{7.26}$$

5. For the capacitance C_2, the following equation is written:

$$C_2\frac{dv_4}{dt} = i_1 + i_3$$
$$\dot{x}_4 = \frac{1}{C_2}x_1 + \frac{1}{C_2}x_2 \tag{7.27}$$

6. Combination Eqs. (7.24)–(7.26) and (7.27), the following vector matrix differential equation is obtained:

$$\dot{x}(t) = \begin{bmatrix} -\frac{(R_1+R_2)}{L_1} & -\frac{R_2}{L_1} & 0 & -\frac{1}{L_1} \\ -\frac{R_2}{L_2} & -\frac{R_2}{L_2} & -\frac{1}{L_2} & -\frac{1}{L_2} \\ 0 & \frac{1}{C_1} & 0 & 0 \\ \frac{1}{C_2} & \frac{1}{C_2} & 0 & 0 \end{bmatrix} x(t) + \begin{bmatrix} \frac{1}{L_1} & 0 \\ 0 & \frac{1}{L_2} \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_1(t) \\ v_2(t) \end{bmatrix}$$

Example 7.9

Develop the state model for the electrical network shown in Fig. 7.10.

(*Anna University, April, 2004*)

Solution

1. There are two energy-storing elements L and C. The current i_1 through L_1 and voltage v_c across C are chosen as the state variables. Thus

$$x_1 = i_1$$
$$x_2 = v_c$$

2. The following equation is written connecting R_1, R_2 and C:

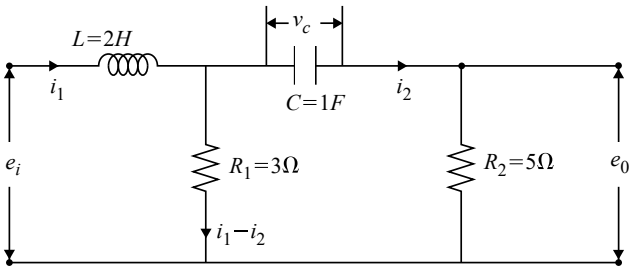


Fig. 7.10 Electrical network of Example 7.9

$$v_c + i_2 R_2 = (i_1 - i_2) R_1$$
$$i_2 = i_1 \frac{R_1}{(R_1 + R_2)} - \frac{v_c}{(R_1 + R_2)} \tag{7.28}$$

The following equation is written connecting e_i, L, C and R_2:

$$e_i = L\frac{di_1}{dt} + v_c + i_2 R_2$$
$$= L\frac{di_1}{dt} + v_c + i_1 \frac{R_1 R_2}{R_1 + R_2} - \frac{v_c R_2}{(R_1 + R_2)}$$
$$\dot{x}_1 = -\frac{R_1 R_2}{L(R_1 + R_2)} x_1 + \frac{1}{L}\left(\frac{R_2}{(R_1 + R_2)} - 1\right) x_2 + \frac{1}{L} e_i \tag{7.29}$$

3. For the capacitor C, the following equation is written:

$$C\frac{dv_c}{dt} = i_2$$
$$= \frac{i_1 R_1}{(R_1 + R_2)} - \frac{v_c}{(R_1 + R_2)}$$
$$\dot{x}_2 = -\frac{R_1}{C(R_1 + R_2)} x_1 - \frac{1}{C(R_1 + R_2)} x_2 \tag{7.30}$$

4. Combining Eqs. (7.29) and (7.30), we get

$$\dot{x}(t) = \begin{bmatrix} -\frac{(R_1 R_2)}{L(R_1+R_2)} & \frac{1}{L}\left(\frac{R_2}{R_1+R_2} - 1\right) \\ \frac{R_1}{C(R_1+R_2)} & -\frac{1}{C(R_1+R_2)} \end{bmatrix} x(t) + \begin{bmatrix} \frac{1}{L} \\ 0 \end{bmatrix} e_i$$

Substituting the numerical value, we get

$$\dot{x}(t) = \begin{bmatrix} -\frac{15}{16} & -\frac{3}{16} \\ \frac{3}{8} & -\frac{1}{8} \end{bmatrix} x(t) + \begin{bmatrix} 0.5 \\ 0 \end{bmatrix} e_i$$

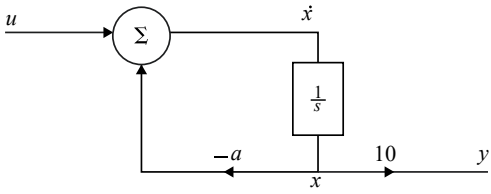


Fig. 7.11 First order T.F. realization

5. The output is

$$\begin{aligned} e_0 &= i_2 R_2 \\ &= \frac{R_1 R_2}{R_1 + R_2} i_1 - \frac{v_c}{R_1 + R_2} \\ &= \frac{1}{8}(15x_1 - x_2) \\ y = e_0 &= \frac{1}{8}[15 \quad -1] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \end{aligned}$$

7.4 State Equations from Transfer Function

State equations can be easily obtained from the transfer function of the system. Consider the first order system with the following transfer function:

$$H(s) = \frac{10}{(s + a)}$$

The system realization is shown in Fig. 7.11.

From Fig. 7.11, the following equations are derived:

$$\begin{aligned} \dot{x} &= -ax + u \\ y &= 10x \end{aligned} \tag{7.31}$$

The output of each integrator ($\frac{1}{s}$) is chosen as one state variable. Thus, for an nth order system, n integrators are required. The following methods of realization are used to determine the state equation:

1. The direct form or controllable canonical form.
2. The cascade form.
3. The parallel form or diagonal form.

7.4.1 General Case of Representation—Phase Variable or Controllable Canonical Form

The state space description can be done in several ways. However, the state variables obtained from the direct form are quite convenient since the state equations can be immediately written just by inspection of the transfer function. Consider the general Nth order transfer function given as follows:

$$\begin{aligned} H(s) &= \frac{b_0 s^N + b_1 s^{N-1} + b_2 s^{N-2} + \cdots + b_N}{s^N + a_1 s^{N-1} + a_2 s^{N-2} + \cdots + a_N} \\ &= \frac{\left(b_0 + \frac{b_1}{s} + \frac{b_2}{s^2} + \cdots + \frac{b_N}{s^N}\right)}{\left(1 + \frac{a_1}{s} + \frac{a_2}{s^2} + \cdots + \frac{a_N}{s^N}\right)} \end{aligned} \tag{7.32}$$

Equation (7.32) is realized in direct form structure and is shown in Fig. 7.12. From Fig. 7.12, the following equations for the state variable are written:

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= x_3 \\ \dot{x}_3 &= x_4 \\ &\vdots \\ \dot{x}_{N-1} &= x_N \\ \dot{x}_N &= -a_N x_1 - a_{N-1} x_2, \ldots, -a_2 x_{N-1} - a_1 x_N + u(t) \end{aligned} \tag{7.33}$$

$$\begin{aligned} y(t) &= b_N x_1 + b_{N-1} x_2 + \cdots + b_1 x_N + b_0 \dot{x}_N \\ &= (b_N - b_0 a_N) x_1 + (b_{N-1} - b_0 a_{N-1}) x_2 + \cdots \\ &\quad + (b_1 - b_0 a_1) x_N + b_0 u(t) \end{aligned} \tag{7.34}$$

Equations (7.33) and (7.34) can be represented in matrix form given as follows:

$$\begin{aligned} \dot{x}(t) &= \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & \cdots & 1 \\ -a_N & -a_{N-1} & -a_{N-2} & \cdots & -a_2 & -a_1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{N-1} \\ x_N \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} u(t) \\ y(t) &= [(b_N - b_0 a_N)(b_{N-1} - b_0 a_{N-1}) \ldots (b_1 - b_0 a_1)] x + b_0 u(t) \\ &= [\bar{b}_N \quad \bar{b}_{N-1} \ldots \bar{b}_1] x + b_0 u(t) \end{aligned} \tag{7.35}$$

where $\bar{b}_N = (b_N - b_0 a_N)$. The A matrix given in Eq. (7.35) is said to be in phase variable canonical form.

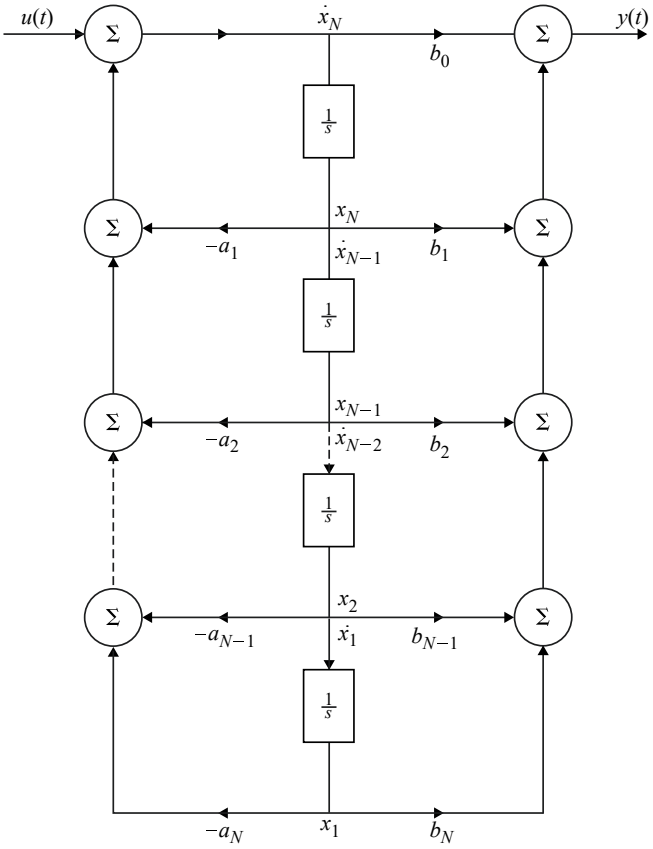


Fig. 7.12 Direct form realization of Nth order LTIC system

Step-by-Step Procedure to Determine A, B and C Matrices

1. If the system is described by linear differential equation, convert that in T.F. form. Find the coefficients of numerator polynomial $b_0, b_1, \ldots, b_N$ and the coefficients of the denominator polynomials $a_0, a_1, \ldots, a_N$.
2. The elements of the A matrix are written in phase variable canonical form. The elements of the last row are written in the reverse order with a negative sign as $-a_N, -a_{N-1}, \ldots, -a_2, a_1$. The elements of the B matrix and identified with 0s in all the rows and 1 by the last row.
3. The elements of C matrix are identified as given in Eq. (7.35). To remember this in an easier way, the elements of the first column are obtained from the product $b_0 a_N$ being subtracted from b_N, the second column from the product $b_0 a_{N-1}$ being subtracted from b_{N-1} and so on. This can be easily viewed from Fig. 7.12. For the state x_1, the right-side branch gain is b_N. The left-side branch gain after being

multiplied by b_0 is $-b_0 a_N$. The sum of these two is $(b_N - b_0 a_N)$. Similarly, the second column of C is obtained which corresponds to the state x_2. This is nothing but $(b_{N-1} - b_0 a_{N-1})$.

Example 7.10

Consider the following differential equation which describes the dynamics of a continuous time system:

$$5\frac{d^4y}{dt^4} + 2\frac{d^3y}{dt^3} + 4\frac{d^2y}{dt^2} + 7\frac{dy}{dt} + 8y = 8\frac{du(t)}{dt} + 7u(t)$$

Form the state space equation.

Solution

$$5\frac{d^4y}{dt^4} + 2\frac{d^3y}{dt^3} + 4\frac{d^2y}{dt^2} + 7\frac{dy}{dt} + 8y = 8\frac{du(t)}{dt} + 7u(t)$$

Taking the Laplace transform on both sides of the above equation, we get

$$\begin{aligned} H(s) &= \frac{Y(s)}{X(s)} \\ &= \frac{(8s+7)}{5\left(s^4 + \frac{2}{5}s^3 + \frac{4}{5}s^2 + \frac{7}{5}s + \frac{8}{5}\right)} \end{aligned}$$

where

$$b_0 = 0,\ b_1 = 0,\ b_2 = 0,\ b_3 = \frac{8}{5},\ b_4 = \frac{7}{5},\ a_1 = \frac{2}{5},\ a_2 = \frac{4}{5},\ a_3 = \frac{7}{5},\ a_4 = \frac{8}{5}$$

The state equation is written in phase variable canonical form given as follows:

$$\dot{x}(t) = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -\frac{8}{5} & -\frac{7}{5} & -\frac{4}{5} & -\frac{2}{5} \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} u(t)$$

$$\bar{b}_4 = b_4 - b_0 a_4 = \frac{7}{5}$$

$$\bar{b}_3 = b_3 - b_0 a_3 = \frac{8}{5}$$

$$\bar{b}_2 = b_2 - b_0 a_2 = 0$$

$$\bar{b}_1 = b_1 - b_0 a_1 = 0$$

$$y = \frac{1}{5}\underbrace{[7 \quad 8 \quad 0 \quad 0]}_{C} x$$

Example 7.11

Consider the following T.F. of a certain continuous time system:

$$H(s) = \frac{7s^3 + 11s^2 + 14s + 10}{s^3 + 8s^2 + 5s + 4}$$

Form the state equations in controllable canonical form.

Solution

$$H(s) = \frac{7s^3 + 11s^2 + 14s + 10}{s^3 + 8s^2 + 5s + 4}$$

where

$$b_0 = 7,\ b_1 = 11,\ b_2 = 14,\ b_3 = 10,\ a_1 = 8,\ a_2 = 5,\ a_3 = 4$$

$$\dot{x}(t) = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -4 & -5 & -8 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u(t)$$

$$\bar{b}_3 = (b_3 - b_0 a_3) = (10 - 28) = -18$$

$$\bar{b}_2 = (b_2 - b_0 a_2) = (14 - 35) = -21$$

$$\bar{b}_1 = (b_1 - b_0 a_1) = (11 - 56) = -45$$

$$y = [-18 \quad -21 \quad -45]x + 7u(t)$$

7.4.2 General Case of Representation-Observable Canonical Form

Phase variable controllable canonical form of state equation is well suited when the controller is designed for the desired pole location. However, when an observer is designed, it is convenient to put the state equation in observable canonical form. Consider the following T.F. of a continuous time system:

$$\frac{Y(s)}{U(s)} = H(s) = \frac{b_0 s^2 + b_1 s + b_2}{s^2 + a_1 s + a_2} = \frac{b_0 + \frac{b_1}{s} + \frac{b_2}{s^2}}{1 + \frac{a_1}{s} + \frac{a_2}{s^2}} \tag{7.36}$$

Cross-multiplying the above equation, we get

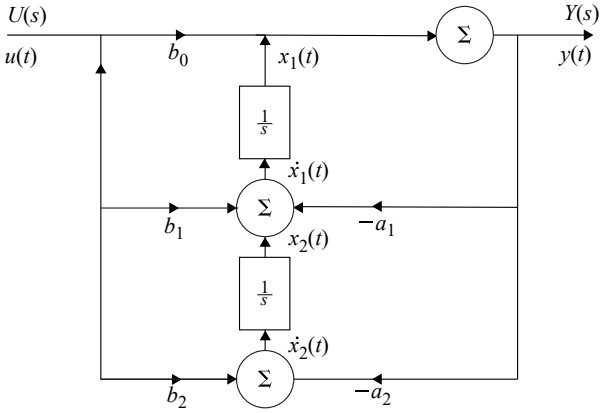


Fig. 7.13 Block diagram to represent Eq. (7.37)

$$\left[1+\frac{a_1}{s}+\frac{a_2}{s^2}\right]Y(s)=\left[b_0+\frac{b_1}{s}+\frac{b_2}{s^2}\right]U(s)$$

$$Y(s)=\left[-\frac{a_1}{s}-\frac{a_2}{s^2}\right]Y(s)+\left[b_0+\frac{b_1}{s}+\frac{b_2}{s^2}\right]U(s) \quad (7.37)$$

Equation (7.37) is represented in Fig. 7.13.

From Fig. 7.13, the following state equations are written:

$$y(t)=x_1(t)+b_0u(t) \quad (7.38)$$

$$\begin{aligned}\dot{x}_1(t)&=-a_1y(t)+x_2(t)+b_1u(t)\\&=-a_1x_1(t)-a_1b_0u(t)+x_2(t)+b_1u(t)\\&=-a_1x_1(t)+x_2(t)+(b_1-b_0a_1)u(t) \quad (7.39)\end{aligned}$$

$$\begin{aligned}\dot{x}_2(t)&=-a_2y(t)+b_2u(t)\\&=-a_2x_1(t)-a_2b_0u(t)+b_2u(t)\\&=-a_2x_1(t)+(b_2-b_0a_2)u(t) \quad (7.40)\end{aligned}$$

Equations (7.39) and (7.40) are combined and expressed in matrix form as

$$x(t)=\begin{bmatrix}-a_1 & 1\\ -a_2 & 0\end{bmatrix}x(t)+\begin{bmatrix}(b_1-b_0a_1)\\ (b_2-b_0a_2)\end{bmatrix}u(t) \quad (7.41)$$

The output equation is written using Eq. (7.38) as

$$y(t) = [1 \quad 0]x(t) + b_0 u(t) \tag{7.42}$$

Now consider the following T.F. in general:

$$\frac{Y(s)}{U(s)} = H(s) = \frac{b_0 s^n + b_1 s^{n-1} + b_2 s^{n-2} + \cdots + b_{n-1}s + b_n}{s^n + a_1 s^{n-1} + a_2 s^{n-2} + \cdots + a_{n-1}s + a_n} \tag{7.43}$$

$$= \frac{b_0 + \frac{b_1}{s} + \frac{b_2}{s^2} + \cdots + \frac{b_{n-1}}{s^{n-1}} + \frac{b_n}{s^n}}{1 + \frac{a_1}{s} + \frac{a_2}{s^2} + \cdots + \frac{a_{n-1}}{s^{n-1}} + \frac{a_n}{s^n}} \tag{7.44}$$

Cross-multiplying the above equation, we get

$$\left[1 + \frac{a_1}{s} + \frac{a_2}{s^2} + \cdots + \frac{a_n}{s^n}\right] Y(s) = \left[b_0 + \frac{b_1}{s} + \frac{b_2}{s^2} + \cdots + \frac{b_n}{s^n}\right] U(s)$$

$$Y(s) = -\left[+\frac{a_1}{s} + \frac{a_2}{s^2} + \cdots + \frac{a_n}{s^n}\right] Y(s)$$

$$+ \left[b_0 + \frac{b_1}{s} + \frac{b_2}{s^2} + \cdots + \frac{b_n}{s^n}\right] U(s) \tag{7.45}$$

Equation (7.45) is represented in Fig. 7.14. From Fig. 7.14, we write the following equation:

$$\begin{aligned} y(t) &= x_1(t) + b_0 u(t) \\ \dot{x}_1(t) &= -a_1 y(t) + x_2(t) + b_1 u(t) \\ &= -a_1 x_1(t) - b_0 a_1 u(t) + x_2(t) + b_1 u(t) \\ &= -a_1 x_1(t) + x_2(t) + (b_1 - b_0 a_1) u(t) \end{aligned} \tag{7.46}$$

$$\begin{aligned} \dot{x}_2(t) &= -a_2 y(t) + b_2 u(t) + x_3(t) \\ &= -a_2 x_1(t) - b_0 a_2 u(t) + b_2 u(t) \\ &= -a_2 x_1(t) + x_3(t) + (b_2 - b_0 a_2) u(t) \end{aligned} \tag{7.47}$$

$$\begin{aligned} \dot{x}_3(t) &= -a_3 y(t) + b_3 u(t) \\ &= -a_3 x_1(t) - b_0 a_3 u(t) + b_3 u(t) \\ &= -a_3 x_1(t) + x_4(t) + (b_3 - b_0 a_3) u(t) \end{aligned} \tag{7.48}$$

In general, we write

$$\dot{x}_n = -a_n x_1(t) + (b_n - b_0 a_n) u(t) \tag{7.49}$$

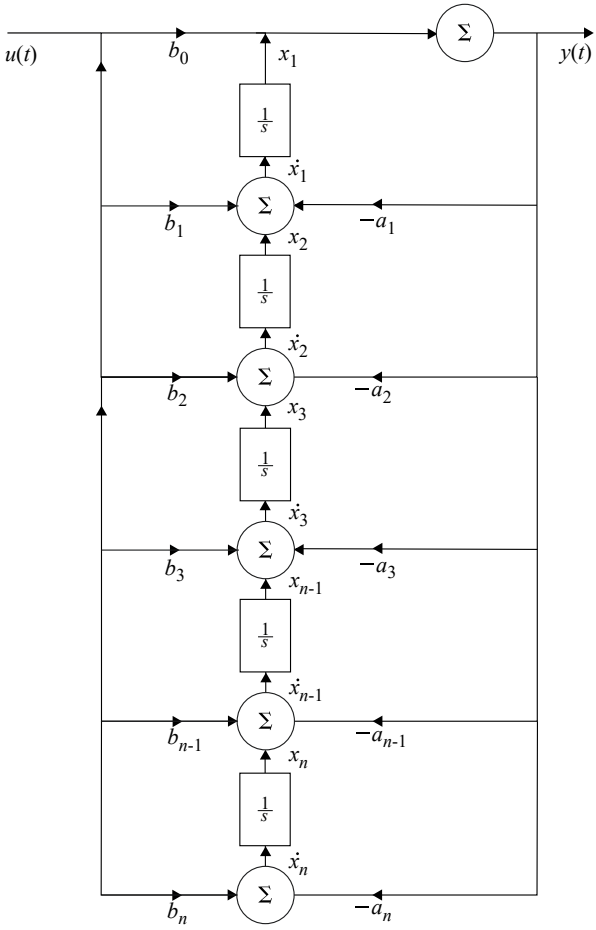


Fig. 7.14 Block diagram to represent Eq. (7.49)

From the above Eqs. (7.46)–(7.49), the following vector matrix differential equation is written:

$$\dot{x}(t) = \begin{bmatrix} -a_1 & 1\,0\,0\,0\cdots 0 \\ -a_2 & 0\,1\,0\,0\cdots 0 \\ -a_3 & 0\,0\,1\,0\cdots 0 \\ \vdots & \vdots\;\vdots\;\vdots\;\vdots\;\vdots\;\vdots \\ -a_{n-1} & 0\,0\,0\,0\;\;0\;\;1 \\ -a_n & 0\,0\,0\,0\;\;0\;\;0 \end{bmatrix} x(t) + \begin{bmatrix} \bar{b}_1 \\ \bar{b}_2 \\ \bar{b}_3 \\ \vdots \\ \bar{b}_{n-1} \\ \bar{b}_n \end{bmatrix} u(t)$$

$$y(t) = [1\;\;0\;\;0\;\;\cdots\;\;0]x(t) + b_0 u(t) \tag{7.50}$$

where

$$\begin{aligned}
\bar{b}_1 &= b_1 - b_0 a_1 \\
\bar{b}_2 &= b_2 - b_0 a_2 \\
&\vdots \\
\bar{b}_{n-1} &= b_{n-1} - b_0 a_{n-1} \\
\bar{b}_n &= b_n - b_0 a_n
\end{aligned} \tag{7.51}$$

Step-by-Step Procedure to Form Observable Canonical Form

1. For the given T.F., from the numerator polynomial determine the coefficients $b_0, b_1, b_2, \ldots, b_n$ and from the denominator polynomial determine the coefficients $a_1, a_2, \ldots, a_n$. The elements of the A matrix are written in observable canonical form. The elements of the first column are written in the ascending order with a negative sign as $-a_1, -a_2, \ldots, -a_n$, where $-a_1$ represents the coefficient of the first column and the first row while $-a_n$ is the coefficient of the first row and the last column. The other elements are represented by $1s$ and $0s$ as shown in Eq. (7.50).
2. The elements of B matrix are determined using Eq. (7.51).
3. The elements of the C matrix are given as

$$C = [1 \ 0 \ 0 \ \cdots \ 0]$$

4. The D matrix has the element b_0

Example 7.12

Consider the following T.F. given in Example 7.11:

$$H(s) = \frac{7s^3 + 11s^2 + 14s + 10}{s^3 + 8s^2 + 5s + 4}$$

Form the state equation in observable canonical form. Also find the output equation.

Solution

$$H(s) = \frac{7s^3 + 11s^2 + 14s + 10}{s^3 + 8s^2 + 5s + 4}$$

$$b_0 = 7; \quad b_1 = 11; \quad b_2 = 14; \quad b_3 = 10;$$
$$a_1 = 8; \quad a_2 = 5; \quad a_3 = 4$$

$$\begin{aligned}\bar{b}_1 &= b_1 - b_0 a_1 \\ &= (11 - 7 \times 8) = -45 \\ \bar{b}_2 &= b_2 - b_0 a_2 \\ &= (14 - 7 \times 5) = -21 \\ \bar{b}_3 &= b_3 - b_0 a_3 \\ &= (10 - 7 \times 4) = -18\end{aligned}$$

$$\begin{aligned}\dot{x}(t) &= \begin{bmatrix} -a_1 & 1 & 0 \\ -a_2 & 0 & 1 \\ -a_3 & 0 & 0 \end{bmatrix} x(t) + \begin{bmatrix} \bar{b}_1 \\ \bar{b}_2 \\ \bar{b}_3 \end{bmatrix} u(t) \\ y(t) &= [1 \ 0 \ 0] x(t) + b_0 u(t)\end{aligned}$$

Substituting the numerical values, we get

$$\begin{aligned}\dot{x}(t) &= \begin{bmatrix} -8 & 1 & 0 \\ -5 & 0 & 1 \\ -4 & 0 & 0 \end{bmatrix} x(t) + \begin{bmatrix} -45 \\ -21 \\ -18 \end{bmatrix} u(t) \\ y(t) &= [1 \ 0 \ 0] x(t) + 7u(t)\end{aligned}$$

Example 7.13

Determine the state space model of a continuous time system whose transfer function is given by

$$H(s) = \frac{3s^2 + 24s + 44}{(s^3 + 12s^2 + 44s + 48)}$$

Use the following methods:

(a) The controllable and observable form.
(b) The cascade form.
(c) The diagonal or Jordan canonical form.

Show that the A matrix is not unique for the given system.

Solution

(a) **The Controllable and Observable Form**

$$H(s) = \frac{3s^2 + 24s + 44}{(s^3 + 12s^2 + 44s + 48)}$$

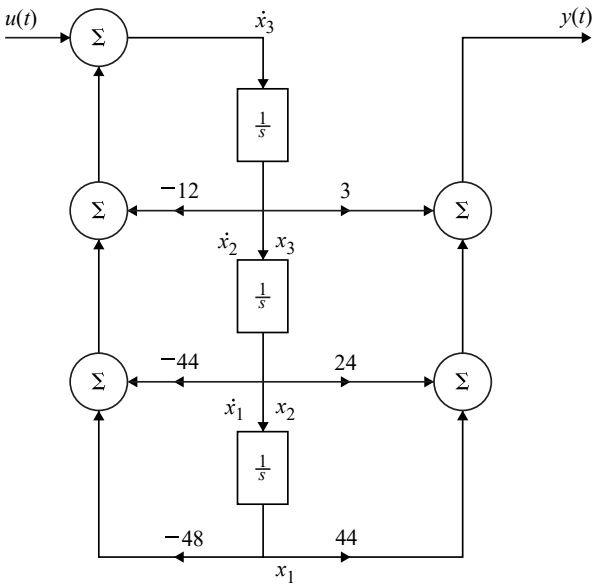


Fig. 7.15 Controllable canonical form realization of $H(s) = \dfrac{3s^2 + 24s + 44}{(s^3 + 12s^2 + 44s + 48)}$

Here, $b_0 = 0; b_1 = 3; b_2 = 24; b_3 = 44; a_1 = 12; a_2 = 44; a_3 = 48$. The direct form realization of $H(s)$ is shown in Fig. 7.15. From Fig. 7.15, the following equations are written:

$$\begin{aligned}
\dot{x}_1 &= x_2 \\
\dot{x}_2 &= x_3 \\
\dot{x}_3 &= -48x_1 - 44x_2 - 12x_3 + u(t)
\end{aligned}$$

The state equation for $H(s)$ is therefore written as

$$\dot{x}(t) = \underbrace{\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -48 & -44 & -12 \end{bmatrix}}_{A} x(t) + \underbrace{\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}}_{B} u(t) \tag{7.52}$$

Also from Fig. 7.15, the output $y(t)$ is obtained as

$$\begin{aligned}
y(t) &= 44x_1 + 24x_2 + 3x_3 \\
y(t) &= \underbrace{[44 \quad 24 \quad 3]}_{C} x(t)
\end{aligned}$$

The Observable Canonical Form

Since $b_0 = 0$,

$$\dot{x}(t) = \begin{bmatrix} -a_1 & 1 & 0 \\ -a_2 & 0 & 1 \\ -a_3 & 0 & 0 \end{bmatrix} x(t) + \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} u(t)$$

$$y(t) = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} x(t)$$

$$\dot{x}(t) = \begin{bmatrix} -12 & 1 & 0 \\ -44 & 0 & 1 \\ -48 & 0 & 0 \end{bmatrix} x(t) + \begin{bmatrix} 3 \\ 24 \\ 44 \end{bmatrix} u(t)$$

$$y(t) = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} x(t)$$

(b) **The Cascade Form**

$$H(s) = \frac{3s^2 + 24s + 44}{(s^3 + 12s^2 + 44s + 48)}$$

The above equation can be written as

$$H(s) = \frac{(s + 5.155)}{(s + 4)} \frac{(s + 2.845)}{(s + 6)} 3 \frac{1}{(s + 2)}$$
$$= H_1(s) H_2(s) H_3(s)$$

The cascade form realization is shown in Fig. 7.16.
From Fig. 7.16, the following equations are written for the first derivatives of the states:

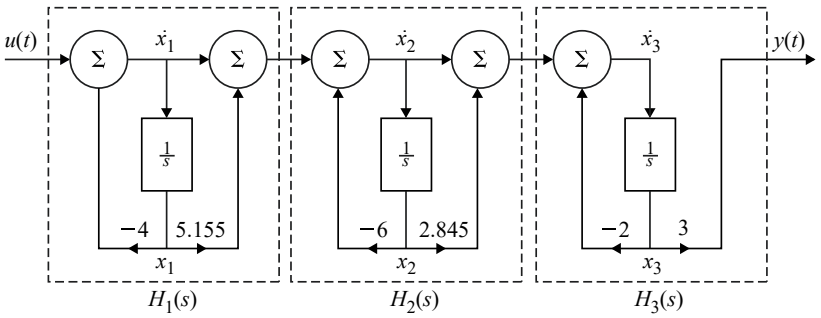


Fig. 7.16 The cascade realization of $H(s) = \dfrac{3(s + 5.155)(s + 2.845)}{(s + 4)(s + 6)(s + 2)}$

$$\begin{aligned}
\dot{x}_1 &= -4x_1 + u(t) \\
\dot{x}_2 &= -6x_2 + \dot{x}_1 + 5.155x_1 \\
&= -6x_2 - 4x_1 + u(t) + 5.155x_1 \\
&= 1.155x_1 - 6x_2 + u(t) \\
\dot{x}_3 &= -2x_3 + \dot{x}_2 + 2.8455x_2 \\
&= -2x_3 + 1.155\dot{x}_1 - 6x_2 + u(t) + 2.845x_2 \\
\dot{x}_3 &= 1.155x_1 - 3.155x_2 - 2x_3 + u(t) \\
y(t) &= 3x_3
\end{aligned}$$

The state equations are given as follows:

$$\begin{aligned}
\dot{x}(t) &= \begin{bmatrix} -4 & 0 & 0 \\ 1.155 & -6 & 0 \\ 1.155 & -3.155 & -2 \end{bmatrix} x(t) + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} u(t) \\
y(t) &= [0 \quad 0 \quad 3]\, x(t)
\end{aligned} \tag{7.53}$$

(c) **The Diagonal or Jordan Canonical Form**

$$\begin{aligned}
H(s) &= \frac{3s^2 + 24s + 44}{(s^3 + 12s^2 + 44s + 48)} \\
&= \frac{3s^2 + 24s + 44}{(s+2)(s+4)(s+6)} \\
&= \frac{1}{(s+2)} + \frac{1}{(s+4)} + \frac{1}{(s+6)} \\
&= H_1(s) + H_2(s) + H_3(s)
\end{aligned}$$

The parallel form realization of $H(s)$ is shown in Fig. 7.17.
From Fig. 7.17, the following equations are written for the first derivatives of the states:

$$\dot{x}_1 = -2x_1 + u(t)$$

$$\begin{aligned}
\dot{x}_2 &= -4x_2 + u(t) \\
\dot{x}_3 &= -6x_3 + u(t) \\
y(t) &= x_1 + x_2 + x_3 \\
\dot{x}(t) &= \begin{bmatrix} -2 & 0 & 0 \\ 0 & -4 & 0 \\ 0 & 0 & -6 \end{bmatrix} x(t) + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} u(t) \\
y(t) &= [1 \quad 1 \quad 1] x(t)
\end{aligned} \tag{7.54}$$

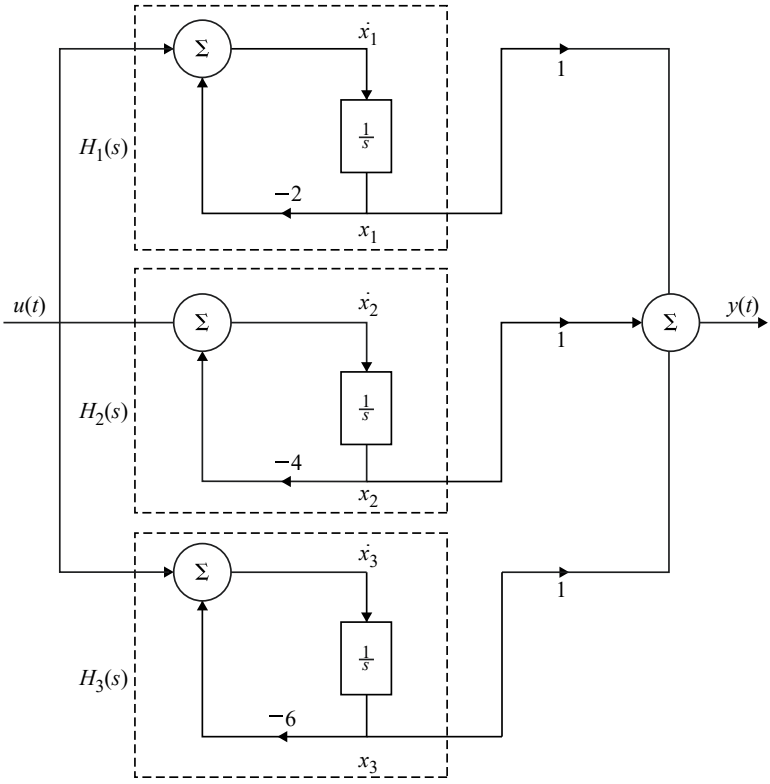


Fig. 7.17 The parallel form realization of $H(s)$

Equations (7.52), (7.53) and (7.54) give the state space description of the system T.F. and **the system A matrices are not unique**. But all of them will give the same characteristics of the system. In parallel form representation, the eigenvalues of the system T.F. form the diagonal elements of the A matrix.

Example 7.14

Consider the following T.F. of a certain system:

$$H(s) = \frac{10(s+2)}{s^2(s+1)^2(s+4)}$$

Determine A, B and C matrices using the Jordan canonical form (Fig. 7.18).

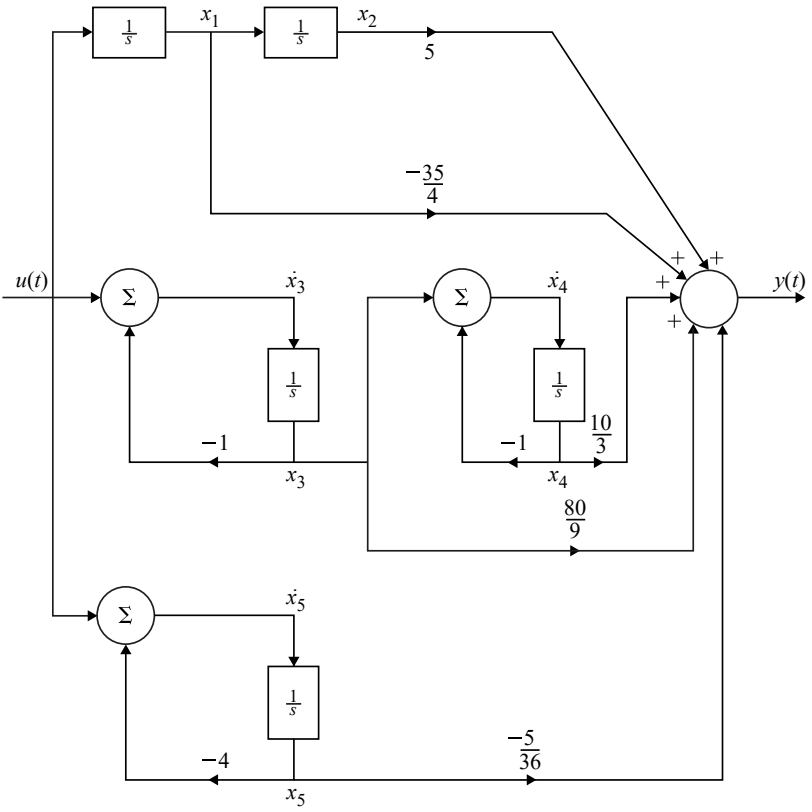


Fig. 7.18 The parallel form realization of $H(s)$ for Example 7.14

Solution

$$H(s) = \frac{10(s+2)}{s^2(s+1)^2(s+4)}$$
$$= \frac{A_1}{s^2} + \frac{A_2}{s} + \frac{A_3}{(s+1)^2} + \frac{A_4}{(s+1)} + \frac{A_5}{(s+4)} \tag{7.55}$$

The residues A_1, A_2, A_3, A_4 and A_5 are determined by any one method discussed in previous chapters and are given as follows:

$$A_1 = 5$$
$$A_2 = -\frac{35}{4}$$
$$A_3 = \frac{10}{3}$$

$$A_4 = \frac{80}{9}$$

$$A_5 = -\frac{5}{36}$$

$$H(s) = -\frac{35}{4}\frac{1}{s} + \frac{5}{s^2} + \frac{80}{9}\frac{1}{(s+1)} + \frac{10}{3}\frac{1}{(s+1)^2} - \frac{5}{36}\frac{1}{(s+1)} \tag{7.56}$$

Equation (7.36) is represented in Fig. 7.15. From Fig. 7.15, the following equations are in terms of state variables:

$$\begin{aligned}
\dot{x}_1 &= u(t) \\
\dot{x}_2 &= x_1 \\
\dot{x}_3 &= -x_3 + u(t) \\
\dot{x}_4 &= x_3 - x_4 \\
\dot{x}_5 &= -4x_5 + u(t) \\
y(t) &= -\frac{35}{4}x_1 + 5x_2 + \frac{80}{9}x_3 + \frac{10}{3}x_4 - \frac{5}{36}x_5
\end{aligned}$$

The above equations are written in vector matrix differential equation form as

$$\dot{x}(t) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & -4 \end{bmatrix} x(t) + \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \\ 1 \end{bmatrix} u(t)$$

$$y(t) = \begin{bmatrix} -\frac{35}{4} & 5 & \frac{80}{9} & \frac{10}{3} & -\frac{5}{36} \end{bmatrix} x(t)$$

7.5 Transfer Function of Continuous Time System from State Equations

Consider the state Eqs. (7.1) and (7.2)

$$\begin{aligned}
\dot{x}(t) &= Ax(t) + Bu(t) \\
y(t) &= Cx(t) + Du(t)
\end{aligned}$$

Let the initial conditions be zero. Taking the Laplace transform on both sides of the above equations, we get

$$[sI - A]X(s) = BU(s)$$

Pre-multiplying both sides by $[sI - A]^{-1}$, we get

$$X(s) = [sI - A]^{-1} BU(s) \tag{7.57}$$

Similarly,

$$\begin{aligned} Y(s) &= CX(s) + DU(s) \\ &= C[sI - A]^{-1} BU(s) + DU(s) \end{aligned}$$

$$\boxed{\frac{Y(s)}{U(s)} = H(s) = C[sI - A]^{-1} B + D} \tag{7.58}$$

$[sI - A]^{-1}$ is called the state transition matrix (STM). In Eq. (7.58),

$$[sI - A]^{-1} = \frac{\text{Adjoint } [sI - A]}{\text{Determinant } [sI - A]}$$

Example 7.15

Consider the following state equations:

$$\begin{aligned} \dot{x} &= \begin{bmatrix} -3 & 1 \\ -2 & 0 \end{bmatrix} x + \begin{bmatrix} 1 \\ 0 \end{bmatrix} x(t) \\ y &= [0 \quad 1] u(t) \end{aligned}$$

Determine the system function.

Solution

$$\begin{aligned} \dot{x} &= \begin{bmatrix} -3 & 1 \\ -2 & 0 \end{bmatrix} x + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u \\ y &= [0 \quad 1] x \\ (sI - A) &= \begin{bmatrix} (s+3) & -1 \\ 2 & s \end{bmatrix} \\ (sI - A)^{-1} &= \frac{1}{(s+1)(s+2)} \begin{bmatrix} s & 1 \\ -2 & (s+3) \end{bmatrix} \\ (sI - A)^{-1} B &= \frac{1}{(s+1)(s+2)} \begin{bmatrix} s & 1 \\ -2 & (s+3) \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ &= \frac{1}{(s+1)(s+2)} \begin{bmatrix} s \\ -2 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} H(s) &= \frac{Y(s)}{U(s)} \\ &= C[sI - A]^{-1}B \\ &= \frac{1}{(s+1)(s+2)}[0 \quad 1]\begin{bmatrix} s \\ -2 \end{bmatrix} \end{aligned}$$

$$\boxed{H(s) = \frac{-2}{(s+1)(s+2)}}$$

7.6 Solution of State Equations

The state equations of a linear time invariant system are solved in both time and frequency domains. In the frequency domain, the Laplace transform method is used. These two methods are discussed ahead.

7.6.1 Laplace Transform Solution of State Equations

Consider the vector matrix differential equation of (7.1)

$$\dot{x} = Ax + Bu(t)$$

Taking the Laplace transform on both sides of the above equation, we get

$$\begin{aligned} sX(s) - x(0) &= AX(s) + BU(s) \\ (sI - A)X(s) &= x(0) + BU(s) \end{aligned} \tag{7.59}$$

Pre-multiplying both sides of Eq. (7.59) by $[sI - A]^{-1}$, we get

$$\begin{aligned} X(s) &= [sI - A]^{-1}[x(0) + BU(s)] \\ &= \phi(s)[x(0) + BU(s)] \end{aligned} \tag{7.60}$$

where

$$\begin{aligned} \phi(s) &= [sI - A]^{-1} \\ X(s) &= \phi(s)x(0) + \phi(s)BU(s) \end{aligned} \tag{7.61}$$

Taking the inverse Laplace transform, we get

$$x(t) = L^{-1}[\phi(s)x(0)] + L^{-1}[\phi(s)BU(s)] \tag{7.62}$$

$\phi(s)$ defined in Eq. (7.61) is the STM. In Eq. (7.62), $L^{-1}[\phi(s)x(0)]$ gives the zero input response and $L^{-1}[\phi(s)BU(s)]$ gives the zero state response.

Example 7.16

A certain system is described by the following state equation:

$$\dot{x} = Ax + Bu(t)$$

where

$$A = \begin{bmatrix} -3 & 1 \\ -2 & 0 \end{bmatrix}; \quad B = \begin{bmatrix} 1 \\ 0 \end{bmatrix}; \quad u(t) = u(t)$$

The initial conditions are $x_1(0) = 1$ and $x_2(0) = -1$. Find STM and hence $x(t)$. Also find $y(t)$ if $C = [0 \quad 1]$.

Solution

$$\dot{x} = \begin{bmatrix} -3 & 1 \\ -2 & 0 \end{bmatrix} x + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u(t)$$

$$(sI - A) = \begin{bmatrix} (s+3) & -1 \\ 2 & s \end{bmatrix}$$

The STM is

$$\phi(s) = [sI - A]^{-1} = \frac{1}{(s+1)(s+2)} \begin{bmatrix} s & 1 \\ -2 & s+3 \end{bmatrix}$$

$$X(s) = [sI - A]^{-1}x(0) + [sI - A]^{-1}BU(s)$$

$$(zI - A)^{-1}x(0) = \frac{1}{(s+1)(s+2)} \begin{bmatrix} s & 1 \\ -2 & s+3 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$= \frac{1}{(s+1)(s+2)} \begin{bmatrix} (s-1) \\ -(s+5) \end{bmatrix}$$

$$= \begin{bmatrix} \dfrac{(s-1)}{(s+1)(s+2)} \\ -\dfrac{(s+5)}{(s+1)(s+2)} \end{bmatrix}$$

Given $u(t) = u(t)$ and $U(s) = \frac{1}{s}$,

$$[sI - A]^{-1}BU(s) = \frac{1}{(s+1)(s+2)} \begin{bmatrix} s & 1 \\ -2 & s+3 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \frac{1}{s}$$

$$= \frac{1}{s(s+1)(s+2)} \begin{bmatrix} s \\ -2 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{(s+1)(s+2)} \\ -\frac{2}{s(s+1)(s+2)} \end{bmatrix}$$

$$x(t) = L^{-1}\{[sI - A]^{-1}x(0) + [sI - A]^{-1}BU(s)\}$$

$$= L^{-1}\begin{bmatrix} \frac{(s-1)}{(s+1)(s+2)} \\ \frac{(s+5)}{(s+1)(s+2)} \end{bmatrix} + L^{-1}\begin{bmatrix} \frac{1}{(s+1)(s+2)} \\ \frac{-2}{s(s+1)(s+2)} \end{bmatrix}$$

$$= L^{-1}\begin{bmatrix} -\frac{2}{(s+1)} + \frac{3}{(s+2)} \\ -\frac{4}{(s+1)} + \frac{3}{(s+2)} \end{bmatrix}$$

$$+ L^{-1}\begin{bmatrix} \frac{1}{(s+1)} - \frac{1}{(s+2)} \\ -\frac{1}{s} + \frac{2}{(s+1)} - \frac{1}{(s+2)} \end{bmatrix}$$

$$x(t) = \begin{bmatrix} 2e^{-2t} - e^{-t} \\ 2e^{-2t} - 2e^{-t} - 1 \end{bmatrix}$$

$$\boxed{\begin{aligned} x_1(t) &= (2e^{-2t} - e^{-t})u(t) \\ x_2(t) &= (2e^{-2t} - 2e^{-t} - 1)u(t) \\ y(t) &= Cx(t) = 2e^{-2t} - 2e^{t} - 1 \end{aligned}}$$

7.6.2 *Time Domain Solution to State Equations*

Consider the state equation

$$\dot{x} = Ax + Bu(t) \tag{7.63}$$

Pre-multiplying Eq. (7.47) by e^{-At} on both sides, we get

$$\begin{aligned} e^{-At}\dot{x} &= e^{-At}Ax + e^{-At}Bu(t) \\ (e^{-At}\dot{x} - e^{-At}Ax) &= e^{-At}Bu(t) \\ \frac{d}{dt}[e^{-At}x] &= e^{-At}Bu(t) \end{aligned} \tag{7.64}$$

By integrating both sides of the above equation from 0 to t, we get

$$e^{-At}x - x(0) = \int_0^t e^{-A\tau}Bu(\tau)d\tau$$

$$\boxed{x(t) = e^{At}x(0) + \int_0^t e^{A(t-\tau)}Bu(\tau)d\tau} \tag{7.65}$$

In Eq. (7.65), e^{At} is the STM; Eq. (7.65) can be generalized to any initial value t_0 and hence it can be modified as

$$\begin{aligned} x(t) &= e^{A(t-t_0)}x(t_0) + \int_{t_0}^t e^{A(t-\tau)}Bu(\tau)d\tau \\ &= \text{Free response} + \text{Forced response} \end{aligned} \tag{7.66}$$

7.6.3 *Determination of e^{At}—The Cayley–Hamilton Theorem*

To determine $x(t)$ which is the solution of the vector matrix differential Eq. (7.65), it is necessary to determine the STM e^{At}. This can be obtained using the Cayley–Hamilton theorem. According to this theorem, an $n \times n$ square matrix A satisfies its own characteristic equation $|\lambda I - A| = 0$ where λ_s are the eigenvalues of system matrix A. Let

$$\begin{aligned} F(A) &= \sum_{k=0}^{n-1} \alpha^k A^k \\ F(\lambda) &= \sum_{k=0}^{n-1} \alpha^k \lambda^k \\ e^{At} &= \sum_{k=0}^{n-1} \alpha^k A^k \end{aligned} \tag{7.67}$$

Equation (7.67) should satisfy for all the eigenvalues of A.

7.6.3.1 Determination of e^{At} for Distinct Eigenvalues of A

If the eigenvalues of A are distinct, then the following procedure is followed to evaluate e^{At}:

1. Determine the eigenvalues of the matrix A from $|\lambda I - A| = 0$.
2. For distinct eigenvalues, the following equations are written:

$$\begin{aligned} e^{\lambda_1 t} &= \alpha_0 + \alpha_1\lambda_1 + \alpha_2\lambda_1^2 + \cdots + \alpha_{n-1}\lambda_1^{n-1} \\ e^{\lambda_2 t} &= \alpha_0 + \alpha_1\lambda_2 + \alpha_2\lambda_2^2 + \cdots + \alpha_{n-1}\lambda_2^{n-1} \\ &\vdots \\ e^{\lambda_n t} &= \alpha_0 + \alpha_1\lambda_n + \alpha_2\lambda_n^2 + \cdots + \alpha_{n-1}\lambda_n^{n-1} \end{aligned} \tag{7.68}$$

Thus, we will have n simultaneous equations if there are n distinct eigenvalues. By solving these simultaneous equations, $\alpha_0,\ \alpha_1,\ \alpha_2, \ldots, \alpha_n$ can be determined.

3. Using the following equation, e^{At} can be evaluated:

$$e^{At} = \alpha_0 I + \alpha_1 A + \alpha_2 A^2 + \cdots + \alpha_{n-1}A^{n-1} \tag{7.69}$$

where $I = n \times n$ identity matrix.

7.6.3.2 Determination of e^{At} for Multiple Eigenvalues of A

Let us assume that $\lambda = \lambda_i$ has multiplicity of m. If all the other eigenvalues are distinct, then the number of distinct eigenvalues are $(n - m + 1)$. Corresponding to these eigenvalues, we will have $(n - m + 1)$ independent equations. For the rest $(m - 1)$, we use Cauchy's residue theorem.

$$\left.\frac{d^i f(\lambda)}{d\lambda^i}\right|_{\lambda=\lambda_i} = \frac{d^i}{d\lambda_i}\left[\sum_{k=0}^{n-1}\alpha_k\lambda^k\right]_{\lambda=\lambda_i} \tag{7.70}$$

where $i = 1, 2, 3, \ldots, (m - 1)$. The following examples illustrate the method of evaluating STM and the solution for $x(t)$.

Example 7.17

Consider the following vector matrix differential equation:

$$\dot{x} = \begin{bmatrix} -3 & 1 \\ -2 & 0 \end{bmatrix} x + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u(t)$$

with the initial conditions

$$x(0) = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Find (a) e^{At} and (b) $x(t)$ for unit step input signal.

Solution

$$\begin{aligned} A &= \begin{bmatrix} -3 & 1 \\ -2 & 0 \end{bmatrix} \\ |\lambda I - A| &= \lambda^2 + 3\lambda + 2 \\ &= (\lambda + 1)(\lambda + 2) \\ \lambda_1 &= -1 \\ \lambda_2 &= -2 \end{aligned}$$

Using Eq. (7.68), we get

$$\begin{aligned} e^{-t} &= \alpha_0 - \alpha_1 \\ e^{-2t} &= \alpha_0 - 2\alpha_1 \end{aligned}$$

Solving the above simultaneous equations for α_0 and α_1, we get

$$\begin{aligned} \alpha_0 &= 2e^{-t} - e^{-2t} \\ \alpha_1 &= e^{-t} - e^{-2t} \\ e^{At} &= \alpha_0 I + \alpha_1 A \\ &= \begin{bmatrix} (2e^{-t} - e^{-2t}) & 0 \\ 0 & (2e^{-t} - e^{-2t}) \end{bmatrix} + (e^{-t} - e^{-2t}) \begin{bmatrix} -3 & 1 \\ -2 & 0 \end{bmatrix} \\ &= \begin{bmatrix} (-e^{-t} + 2e^{-2t}) & (e^{-t} - e^{-2t}) \\ 2(-e^{-t} + e^{-2t}) & (2e^{-t} - e^{-2t}) \end{bmatrix} \\ x(t) &= e^{At}x(0) + \int_0^t e^{A(t-\tau)} Bu(\tau) d\tau \end{aligned}$$

Free Response $x_{FR}(t)$

$$\begin{aligned} x_{FR}(t) = e^{At}x(0) &= \begin{bmatrix} (-e^{-t} + 2e^{-2t}) & (e^{-t} - e^{-2t}) \\ 2(-e^{-t} + e^{-2t}) & (2e^{-t} - e^{-2t}) \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} \\ &= \begin{bmatrix} (3e^{-2t} - 2e^{-t}) \\ (3e^{-2t} - 4e^{-t}) \end{bmatrix} \end{aligned}$$

Forced Response $x_{FO}(t)$

$$\begin{aligned} x_{FO}(t) &= \int_0^t e^{A(t-\tau)} Bu(\tau) d\tau \\ &= \int_0^t \begin{bmatrix} -e^{-(t-\tau)} + 2e^{-2(t-\tau)} & e^{-(t-\tau)} - e^{-2(t-\tau)} \\ 2(-e^{-(t-\tau)} + e^{-2(t-\tau)}) & 2e^{-(t-\tau)} - e^{-2(t-\tau)} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} d\tau \\ &= \int_0^t \left[\begin{Bmatrix} -e^{-(t-\tau)} + 2e^{-2(t-\tau)} \\ 2(-e^{-(t-\tau)} + e^{-2(t-\tau)}) \end{Bmatrix} \right] d\tau \end{aligned}$$

$$= \begin{bmatrix} \left\{ \begin{matrix} -e^{-(t-\tau)} + 2e^{-2(t-\tau)} \\ 2(-e^{-(t-\tau)} + e^{-2(t-\tau)}) \end{matrix} \right\}_0^t \end{bmatrix}$$
$$= \begin{bmatrix} e^{-t} - e^{-2t} \\ -1 + 2e^{-t} - e^{-2t} \end{bmatrix}$$

Hence, the total response is

$$x(t) = x_{FR}(t) + x_{FO}(t)$$
$$= \begin{bmatrix} 3e^{-2t} - 2e^{-t} \\ 3e^{-2t} - 4e^{-t} \end{bmatrix} + \begin{bmatrix} e^{-t} - e^{-2t} \\ 2e^{-t} - e^{-2t} - 1 \end{bmatrix}$$

$$\boxed{x(t) = \begin{bmatrix} (2e^{-2t} - e^{-t}) \\ (2e^{-2t} - 2e^{-t} - 1) \end{bmatrix}}$$

The result is the same as derived in Example 7.16.

Example 7.18

Determine e^{At} for the following A matrix:

$$A = \begin{bmatrix} 4 & 1 & -2 \\ 1 & 0 & 2 \\ 1 & -1 & 3 \end{bmatrix}$$

Solution

$$A = \begin{bmatrix} 4 & 1 & -2 \\ 1 & 0 & 2 \\ 1 & -1 & 3 \end{bmatrix}$$
$$F(\lambda) = |\lambda I - A|$$
$$= \lambda^3 - 7\lambda^2 - 9 + 15\lambda$$
$$= (\lambda - 1)(\lambda - 3)^2$$

This is the case of repeated eigenvalues.

$$\begin{aligned}\lambda_1 &= 1\\ \lambda_2 &= 3\\ \lambda_3 &= 3\\ e^{t} &= \alpha_0 + \alpha_1 + \alpha_2\\ e^{3t} &= \alpha_0 + 3\alpha_1 + 9\alpha_2\\ \frac{d}{d\lambda}e^{\lambda t} &= \frac{d}{d\lambda}(\alpha_0 + \alpha_1\lambda + \alpha_2\lambda^2)\end{aligned}$$

For $\lambda = 3$

$$te^{3t} = \alpha_1 + 6\alpha_2$$

Solving the above three simultaneous equations, we get

$$\begin{aligned}\alpha_0 &= \frac{1}{4}(9e^{t} + 6te^{3t} - 5e^{3t})\\ \alpha_1 &= \frac{1}{4}(-6e^{t} - 8te^{3t} + 6e^{3t})\\ \alpha_2 &= \frac{1}{4}(e^{t} + 2te^{3t} - e^{3t})\\ e^{At} &= \alpha_0 I + \alpha_1 A + \alpha_2 A^2\\ &= \alpha_0\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix} + \alpha_1\begin{bmatrix}4&1&-2\\1&0&2\\1&-1&3\end{bmatrix} + \alpha_2\begin{bmatrix}15&6&-12\\6&-1&4\\6&-2&5\end{bmatrix}\end{aligned}$$

$$\boxed{e^{At} = \begin{bmatrix}(-te^{3t} + e^{3t}) & (te^{3t}) & (-2te^{3t})\\ (te^{3t}) & (2e^{t} + te^{3t} - e^{3t}) & -2(e^{t} + te^{3t} - e^{3t})\\ (te^{3t}) & (e^{t} + te^{3t} - e^{3t}) & (-e^{t} - 2te^{3t} + 2e^{3t})\end{bmatrix}}$$

Example 7.19

A system is represented by the state equation

$$\dot{x}(t) = \begin{bmatrix}0&1\\-4&-5\end{bmatrix}x(t) + \begin{bmatrix}0\\1\end{bmatrix}u(t)$$

Find the STM.

Solution The Laplace transform of STM is given by

$$\phi(s) = [sI - A]^{-1}$$
$$= \frac{\text{Adjoint of } [sI - A]}{|sI - A|}$$
$$[sI - A] = \begin{bmatrix} s & -1 \\ 4 & s+5 \end{bmatrix}$$
$$\text{Adjoint of } [sI - A] = \begin{bmatrix} s+5 & 1 \\ -4 & s \end{bmatrix}$$
$$|sI - A| = s(s+5) + 4$$
$$= (s+1)(s+4)$$
$$\phi(s) = [sI - A]^{-1}$$
$$= \frac{1}{(s+1)(s+4)} \begin{bmatrix} s+5 & 1 \\ -4 & s \end{bmatrix}$$
$$= \begin{bmatrix} \frac{(s+5)}{(s+1)(s+4)} & \frac{1}{(s+1)(s+4)} \\ \frac{-4}{(s+1)(s+4)} & \frac{s}{(s+1)(s+4)} \end{bmatrix}$$
$$= \begin{bmatrix} \frac{A_1}{s+1} + \frac{A_2}{s+4} & \frac{A_3}{s+1} + \frac{A_4}{s+4} \\ \frac{A_5}{s+1} + \frac{A_6}{s+4} & \frac{A_7}{s+1} + \frac{A_8}{s+4} \end{bmatrix}$$

$$A_1(s+4) + A_2(s+1) = s+5$$

For $s = -1$, $A_1 = \frac{4}{3}$

For $s = -4$, $A_2 = -\frac{1}{3}$

$$A_3(s+4) + A_4(s+1) = 1$$

For $s = -1$, $A_3 = \frac{1}{3}$

For $s = -4$, $A_4 = -\frac{1}{3}$

$$A_5(s+4) + A_6(s+1) = -4$$

For $s = -1$, $A_5 = -\frac{4}{3}$

For $s = -4$, $A_6 = \frac{4}{3}$

$$A_7(s+4) + A_8(s+1) = s$$

For $s = -1$, $A_7 = -\frac{1}{3}$

For $s = -4$, $A_8 = \frac{4}{3}$

Substituting these values in $\phi(s)$, we get

$$\phi(s) = \begin{bmatrix} \left(\frac{4}{3}\frac{1}{s+1} - \frac{1}{3}\frac{1}{s+4}\right) & \frac{1}{3}\left(\frac{1}{s+1} - \frac{1}{s+4}\right) \\ -\frac{4}{3}\left(\frac{1}{s+1} - \frac{1}{s+4}\right) & -\frac{1}{3}\left(\frac{1}{s+1} - \frac{4}{s+4}\right) \end{bmatrix}$$

Taking the inverse Laplace transform, the following STM is obtained:

$$\phi(t) = \begin{bmatrix} \left(\frac{4}{3}e^{-t} - \frac{1}{3}e^{-4t}\right) & \frac{1}{3}(e^{-t} - e^{-4t}) \\ -\frac{4}{3}(e^{-t} - e^{-4t}) & -\frac{1}{3}(e^{-t} - 4e^{-4t}) \end{bmatrix}$$

Example 7.20

Determine the transform matrix for the state model given as follows:

$$\dot{x} = \begin{bmatrix} -1 & -1 \\ 3 & -5 \end{bmatrix} x + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u$$

(*Anna University, November, 2010*)

Solution

$$A = \begin{bmatrix} -1 & -1 \\ 3 & -5 \end{bmatrix}$$

$$[sI - A] = \begin{bmatrix} s+1 & 1 \\ -3 & s+5 \end{bmatrix}$$

$$|sI - A| = (s+1)(s+5) + 3$$
$$= s^2 + 6s + 8$$
$$= (s+2)(s+4)$$

The eigenvalues are

$$\lambda_1 = -2; \quad \lambda_2 = -4$$

$$\phi(s) = [sI - A]^{-1}$$
$$= \frac{1}{(s+2)(s+4)} \begin{bmatrix} s+1 & 1 \\ -3 & s+5 \end{bmatrix}$$
$$\phi(s) = \begin{bmatrix} \frac{(s+1)}{(s+2)(s+4)} & \frac{1}{(s+2)(s+4)} \\ \frac{-3}{(s+2)(s+4)} & \frac{(s+5)}{(s+2)(s+4)} \end{bmatrix}$$
$$= \begin{bmatrix} \frac{1}{2}\left(\frac{3}{s+2} - \frac{1}{s+4}\right) & \frac{1}{2}\left(-\frac{1}{s+2} + \frac{1}{s+4}\right) \\ \frac{3}{2}\left(\frac{1}{s+2} - \frac{1}{s+4}\right) & \frac{1}{2}\left(-\frac{1}{s+2} + \frac{3}{s+4}\right) \end{bmatrix}$$
$$\phi(t) = L^{-1}\phi(s)$$

The STM is

$$\phi(t) = e^{At} = \begin{bmatrix} \left(\frac{3}{2}e^{-2t} - \frac{1}{2}e^{-4t}\right) & \left(-\frac{1}{2}e^{-2t} + \frac{1}{2}e^{-4t}\right) \\ \left(\frac{3}{2}e^{-2t} - \frac{3}{2}e^{-4t}\right) & \left(-\frac{1}{2}e^{-2t} + \frac{3}{2}e^{-4t}\right) \end{bmatrix}$$

Example 7.21

Determine the time response for a system given as follows:

$$\begin{aligned} \dot{x}_1 &= -x_1 \\ \dot{x}_2 &= x_1 - x_2 + u(t) \\ x(0) &= \begin{bmatrix} 1 \\ 0 \end{bmatrix} \end{aligned}$$

(*Anna University, November, 2010*)

Solution

$$\begin{aligned} \dot{x} &= \begin{bmatrix} -1 & 0 \\ 1 & -1 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u \\ [sI - A] &= \begin{bmatrix} s+1 & 0 \\ -1 & s+1 \end{bmatrix} \\ |sI - A| &= (s+1)^2 \end{aligned}$$

The eigenvalues are $\lambda_1 = -1$ and $\lambda_2 = -1$.

$$\begin{aligned} e^{\lambda_1 t} &= \alpha_0 + \lambda_1 \alpha_1 \\ e^{-t} &= \alpha_0 - \alpha_1 \\ \frac{d}{d\lambda} e^{\lambda t} &= \alpha_1 \end{aligned}$$

$$\begin{aligned} -te^{-t} &= \alpha_1 \\ \alpha_0 &= \alpha_1 + e^{-t} \\ &= e^{-t} - te^{-t} \\ \phi(t) &= e^{At} \\ &= \alpha_0 I + \alpha_1 A \\ &= \begin{bmatrix} e^{-t} - te^{-t} & 0 \\ 0 & e^{-t} - te^{-t} \end{bmatrix} + (-te^{-t}) \begin{bmatrix} -1 & 0 \\ 1 & -1 \end{bmatrix} \end{aligned}$$

$$= \begin{bmatrix} e^{-t} & 0 \\ -te^{-t} & e^{-t} \end{bmatrix}$$

$$x(t) = e^{At}x(0) + \int_0^t \begin{bmatrix} e^{-(t-\tau)} & 0 \\ -(t-\tau)e^{-(t-\tau)} & e^{-(t-\tau)} \end{bmatrix} Bu(\tau)d\tau$$

$$= \begin{bmatrix} e^{-t} & 0 \\ -te^{-t} & e^{-t} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \int_0^t \begin{bmatrix} e^{-(t-\tau)} & 0 \\ -(t-\tau)e^{-(t-\tau)} & e^{-(t-\tau)} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} d\tau$$

$$= \begin{bmatrix} e^{-t} \\ -te^{-t} \end{bmatrix} + \int_0^t \begin{bmatrix} 0 \\ e^{-(t-\tau)} \end{bmatrix} d\tau$$

$$= \begin{bmatrix} e^{-t} \\ -te^{-t} \end{bmatrix} + \begin{bmatrix} 0 \\ e^{-t}e^{\tau} \end{bmatrix}_0^t$$

$$= \begin{bmatrix} e^{-t} \\ -te^{-t} \end{bmatrix} + \begin{bmatrix} 0 \\ (1-e^{-t}) \end{bmatrix}$$

$$= \begin{bmatrix} e^{-t} \\ 1-e^{-t}(1+t) \end{bmatrix}$$

$$\boxed{\begin{aligned} x_1 &= e^{-t} \\ x_2 &= 1-e^{-t}(1+t) \end{aligned}}$$

Example 7.22

For a system represented by the state equation

$$\dot{x}(t) = Ax(t)$$

$$x(t) = \begin{bmatrix} e^{-t} \\ -e^{-t} \end{bmatrix} \quad \text{for } x(0) = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$x(t) = \begin{bmatrix} e^{-3t} \\ -3e^{-3t} \end{bmatrix} \quad \text{for } x(0) = \begin{bmatrix} 1 \\ -3 \end{bmatrix}$$

Determine the state transition matrix $\phi(t)$ and the system matrix A.

(*Anna University, November, 2010*)

Solution Let the STM e^{At} be expressed as

$$e^{At} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

$$x(t) = e^{At}x(0)$$

$$\begin{bmatrix} e^{-t} \\ -e^{-t} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$
\begin{aligned}
e^{-t} &= a_{11} - a_{12} \\
-e^{-t} &= a_{21} - a_{22} \qquad (7.71)\\
\begin{bmatrix} e^{-3t} \\ -3e^{-3t} \end{bmatrix} &= \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} 1 \\ -3 \end{bmatrix} \\
e^{-3t} &= a_{11} - 3a_{12} \\
-3e^{-3t} &= a_{21} - 3a_{22} \qquad (7.72)
\end{aligned}
$$

From Eqs. (7.71) and (7.72), the following simultaneous equations are formed:

$$
\begin{aligned}
e^{-t} &= a_{11} - a_{12} \\
e^{-3t} &= a_{11} - 3a_{12} \\
\hline
a_{12} &= \frac{1}{2}[e^{-t} - e^{-3t}] \\
a_{11} &= \frac{1}{2}[3e^{-t} - e^{-3t}] \qquad (7.73)\\
-e^{-t} &= a_{21} - a_{22} \\
-3e^{-3t} &= a_{21} - 3a_{22} \\
\hline
a_{22} &= \frac{3}{2}e^{-3t} - \frac{1}{2}e^{-t} \\
a_{21} &= \frac{3}{2}[e^{-3t} - e^{-t}] \qquad (7.74)
\end{aligned}
$$

From Eqs. (7.73) and (7.74), the state transition matrix is written as

$$
e^{At} = \phi(t) = \begin{bmatrix} \left(\frac{3}{2}e^{-t} - \frac{1}{2}e^{-3t}\right) & \frac{1}{2}(e^{-t} - e^{-3t}) \\ \frac{3}{2}(e^{-3t} - e^{-t}) & \frac{1}{2}(3e^{-3t} - e^{-t}) \end{bmatrix} \qquad (7.75)
$$

Taking the Laplace transform, we get

$$
\begin{aligned}
\phi(s) &= \begin{bmatrix} \frac{1}{2}\left(\frac{3}{s+1} - \frac{1}{s+3}\right) & \frac{1}{2}\left(\frac{1}{s+1} - \frac{1}{s+3}\right) \\ \frac{3}{2}\left(\frac{1}{s+3} - \frac{1}{s+1}\right) & \frac{1}{2}\left(\frac{3}{s+3} - \frac{1}{s+1}\right) \end{bmatrix} \\
&= \frac{1}{(s+1)(s+3)} \begin{bmatrix} (s+4) & 1 \\ -3 & s \end{bmatrix} \qquad (7.76)
\end{aligned}
$$

From Eq. (7.76), the determinant is

$$
(s+1)(s+3) = 0
$$

The eigenvalues are $\lambda_1 = -1$ and $\lambda_2 = -3$. But

$$[\phi(s)]^{-1} = \frac{\text{Adjoint of } \phi(s)}{\text{Determinant of } \phi(s)}$$

Since

$$\begin{aligned}
\phi(s) &= [sI - A]^{-1} \\
sI - A &= [\phi(s)]^{-1} \\
\text{Adjoint of } \phi(s) &= \text{Adjoint of} \begin{bmatrix} s+4 & 1 \\ -3 & s \end{bmatrix} \\
&= \begin{bmatrix} s & -1 \\ 3 & s+4 \end{bmatrix}
\end{aligned}$$

Thus

$$\begin{aligned}
[sI - A] &= \begin{bmatrix} s & -1 \\ 3 & s+4 \end{bmatrix} \\
A &= \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} s & -1 \\ 3 & s+4 \end{bmatrix} \\
A &= \begin{bmatrix} 0 & 1 \\ -3 & -4 \end{bmatrix}
\end{aligned}$$

7.7 Controllability of Linear Continuous Time System

The controllability concept was introduced by R. Kalman of the USA in the year 1960. This concept is related to linear algebraic equations whose solution exists iff the rank of the system A matrix is full. However, to achieve this there is no restriction in the choice of the input. Therefore, the concept of controllability is defined as follows.

A linear time invariant continuous time system is said to be completely state controllable at time $t = t_0$ if it is possible to transfer the initial state $x(t_0)$ to any final state $x(t_f)$ by applying any control vector $u(t)$ which has no constraints in its choice in a finite time interval $t_0 \leq t \leq t_f$.

7.7.1 State Controllability Condition

For a given linear time invariant continuous time system, the controllability condition is derived as follows.

Consider the vector matrix differential equation of the linear system given by (7.1)

$$\dot{x}(t) = Ax(t) + Bu(t)$$

$$y(t) = Cx(t) \tag{7.77}$$

where the symbols and notations have the same meaning and dimensions as discussed before. The solution of Eq. (7.77) is given by Eq. (7.65) as

$$x(t) = e^{At}x(0) + \int_0^t e^{A(t-\tau)}Bu(\tau)d\tau \tag{7.78}$$

Without loss of generality, we can choose the origin of the state space as the final state at $t = t_f$. Equation (7.78) is written as

$$\begin{aligned} 0 &= e^{At}x(0) + \int_0^t e^{A(t-\tau)}Bu(\tau)d\tau \\ x(0) &= -\int_0^{t_f} e^{-A\tau}Bu(\tau)d\tau \end{aligned} \tag{7.79}$$

Using the Cayley–Hamilon theorem, $e^{-A\tau}$ can be written as

$$e^{-A\tau} = \sum_{k=0}^{n-1} \alpha_k(\tau)A^k \tag{7.80}$$

Substituting Eq. (7.80) in (7.79), we get

$$x(0) = -\sum_{k=0}^{n-1} A^k B \int_0^{t_f} \alpha_k(\tau)u(\tau)d\tau \tag{7.81}$$

Let

$$\int_0^{t_f} \alpha_k(\tau)u(\tau)d\tau = \beta_k \tag{7.82}$$

Equation (7.82) can be estimated without putting any restriction in the choice of the input $u(\tau)$. Substituting Eq. (7.82) in (7.81), we get

$$\begin{aligned} x(0) &= -\sum_{k=0}^{n-1} A_k B\beta_k \\ &= -[B \quad AB \quad A^2B \cdots A^{n-1}B] \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_{n-1} \end{bmatrix} \end{aligned} \tag{7.83}$$

It is well known from linear algebra that in order to have a solution of Eq. (7.83), it is necessary that the inverse of the following matrix exists:

$$\boxed{Q = [B \quad AB \quad A^2B \cdots A^{n-1}B]} \tag{7.84}$$

Further, we know that the linear algebraic equations with n unknowns, Eq. (7.83) has a unique solution iff the matrix Q has rank n. The matrix Q in Eq. (7.84) is called the controllability matrix. With the controllability matrix having full rank, it gives the sufficient condition for the system to be completely state controllable.

7.7.2 *Output Controllability Condition*

A linear time invariant continuous time system is said to be completely output controllable at time $t = t_0$ if it is possible to transfer the initial output $y(t_0)$ to any final output $y(t_f)$ by applying any control vector $u(t)$ which has no constraints in its choice in a finite time interval $t_0 \leq t \leq t_f$.

In Eq. (7.7), $y(t)$ is the output vector which has dimension $m \times 1$ where m is less than n. Let the final output at $t = t_f$ be chosen as the origin of the state space. The output $y(t)$ can be obtained from Eq. (7.78) and can be written as

$$y(t_f) = 0 = Ce^{At_f}\left[x(0) + \int_0^{t_f} e^{-A\tau}Bu(\tau)d\tau\right] \tag{7.85}$$

By following the procedure for obtaining the condition for complete state controllability, we can obtain the following condition for output controllability. For complete output controllability, the rank of the composite matrix P whose dimension is $m \times nr$ should have rank m. The matrix P is called the output matrix and is given by

$$\boxed{P = [CB \quad CAB \quad CA^2B \cdots CA^{n-1}B]} \tag{7.86}$$

Example 7.23

Consider the following vector matrix differential equation:

$$\dot{x} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & -1 & 10 \end{bmatrix} x + \begin{bmatrix} 0 \\ 0 \\ 10 \end{bmatrix} u$$

Test whether the system is controllable.

Solution For the given system

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & -1 & 10 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 0 \\ 0 \\ 10 \end{bmatrix}$$

$$AB = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & -1 & 10 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 10 \end{bmatrix} = \begin{bmatrix} 0 \\ 10 \\ 100 \end{bmatrix}$$

$$A^2B = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & -1 & 10 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & -1 & 10 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 10 \end{bmatrix} = \begin{bmatrix} 10 \\ 90 \\ 990 \end{bmatrix}$$

The controllability matrix is given by

$$Q = [B \quad AB \quad A^2B] = \begin{bmatrix} 0 & 0 & 10 \\ 0 & 10 & 90 \\ 10 & 100 & 990 \end{bmatrix}$$

The determinant of the controllability matrix is

$$\begin{aligned} |Q| &= \begin{vmatrix} 0 & 0 & 10 \\ 0 & 10 & 90 \\ 10 & 100 & 990 \end{vmatrix} \\ &= 0 \times \begin{bmatrix} 10 & 90 \\ 100 & 990 \end{bmatrix} - 0 \times \begin{bmatrix} 0 & 90 \\ 10 & 990 \end{bmatrix} + 10 \begin{bmatrix} 0 & 10 \\ 10 & 100 \end{bmatrix} \\ &= 0 - 0 + (-1000) = -1000 \end{aligned}$$

The determinant does not vanish. The rank of the controllability matrix is full. Hence, the system is completely state controllable. Further, just by observation of the controllability matrix Q, the three columns are seen to be linearly independent and thus conclude that the rank of Q is 3 and the system is completely state controllable.

Example 7.24

Consider the system described by the following state equation. Determine whether the system is completely state controllable

$$\dot{x} = \begin{bmatrix} 0 & 1 & -3 \\ 2 & -5 & 4 \\ -8 & 6 & 7 \end{bmatrix} x + \begin{bmatrix} 0 & -2 \\ 1 & -4 \\ 5 & -3 \end{bmatrix} u$$

Solution

$$AB = \begin{bmatrix} 0 & 1 & -3 \\ 2 & -5 & 4 \\ -8 & 6 & 7 \end{bmatrix} \begin{bmatrix} 0 & -2 \\ 1 & -4 \\ 5 & -3 \end{bmatrix} = \begin{bmatrix} -14 & 5 \\ 15 & 4 \\ 41 & -29 \end{bmatrix}$$

$$A^2B = \begin{bmatrix} 0 & 1 & -3 \\ 2 & -5 & 4 \\ -8 & 6 & 7 \end{bmatrix} \begin{bmatrix} -14 & 5 \\ 15 & 4 \\ 41 & -29 \end{bmatrix} = \begin{bmatrix} -108 & 91 \\ 61 & -126 \\ 265 & -219 \end{bmatrix}$$

The controllability matrix is

$$Q = [B \quad AB \quad A^2B] = \begin{bmatrix} 0 & -2 & -14 & 5 & -108 & 91 \\ 1 & -4 & 15 & 4 & 61 & -126 \\ 5 & -3 & 41 & -29 & 265 & -219 \end{bmatrix}$$

Considering any 3×3 square matrix, we may choose, for example,

$$Q = \begin{bmatrix} 0 & -2 & -14 \\ 1 & -4 & 15 \\ 5 & -3 & 41 \end{bmatrix}$$

The three columns are linearly independent. In other words, one column cannot be expressed as the linear combination of the other, and hence the rank of Q is 3 and the system is completely state controllable. Thus, in this case it is not necessary to calculate A^2B. Now the above fact is proved by determining the determinant of Q.

$$Q = \begin{bmatrix} 0 & -2 & -14 \\ 1 & -4 & 15 \\ 5 & -3 & 41 \end{bmatrix} = -306$$

The determinant Q does not vanish. Hence, the rank of the controllability matrix is 3. The system is completely state controllable.

Example 7.25

Consider the system described by the following state and output equations:

$$\dot{x} = \begin{bmatrix} -2 & 0 \\ 0 & -1 \end{bmatrix} x + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u$$
$$y = [1 \ \ 1]x$$

Test whether the system is completely state controllable and completely output controllable.

Solution Test for State Controllability

$$A = \begin{bmatrix} -2 & 0 \\ 0 & -1 \end{bmatrix}$$
$$B = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$
$$AB = \begin{bmatrix} -2 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -2 \\ -1 \end{bmatrix}$$

The controllability matrix is

$$Q = [B \quad AB] = \begin{bmatrix} 1 & -2 \\ 1 & -1 \end{bmatrix}$$

The columns of Q are linearly independent and hence the rank of Q is 2. The system is completely state controllable. The determinant of Q is obtained as $|Q| = -1 + 2 = 1$.

Test for Output Controllability

The output controllability matrix

$$P = [CB \quad CAB]$$
$$CB = [1 \ 1] \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 2$$
$$CAB = [1 \ 1] \begin{bmatrix} -2 \\ -1 \end{bmatrix} = -3$$
$$P = [2 \quad -3]$$

Rank of P is 1. The system is output controllable.

Example 7.26

Consider the system described by the following state equation:

$$\dot{x} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

Test the state controllability of the system.

Solution

$$A = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$AB = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$

The controllability matrix Q is

$$Q = [B \quad AB] = \begin{bmatrix} 0 & 0 \\ 1 & -1 \end{bmatrix}$$

The second column of Q can be obtained by multiplying the first column by a factor -1. Hence, the matrix Q is linearly dependent. Hence, it is singular matrix and the system is not completely state controllable. $|Q| = 0$.

7.7.3 *Controllability and Observability Tests in the Frequency Domain*

In the previous sections, the controllability concept was introduced in the time domain for the state space model. The controllability condition can be obtained in the frequency domain as explained below.

Let $H(s)$ be the transfer function of a single-input single-output system.

$$H(s) = C[sI - A]^{-1}B$$

$H(s)$ is the ratio of two polynomials which correspond to the poles and zeros of $H(s)$. A very powerful and simple theorem has been developed that gives the single condition for controllability. According to this theorem,

A single-input single-output system is said to be completely state (output) controllable if no pole–zero cancellation occurs in the transfer function $H(s)$.

Consider the following T.F:

$$H(s) = \frac{(s+2)}{(s+2)(s+3)}$$

By pole–zero cancellation, the above T.F. becomes

$$H(s) = \frac{1}{(s+3)}$$

In the above equation, one degree of freedom is lost. The original second order is reduced to a first order system. Thus, it is not possible to take one of the two states to the desired location by applying any input. The system now becomes not completely

controllable. For the example cited above, the rank of the controllability matrix is obtained as one and not two.

7.8 Observability of Linear Continuous Time System

Consider the following vector matrix differential equation of an autonomous system:

$$\begin{aligned} \dot{x} &= Ax \\ y &= Cx \end{aligned} \tag{7.87}$$

where the symbols have the usual meaning and dimension.

A linear time invariant continuous time system is said to be completely observable if every initial state $x(t_0)$ can be determined by measuring $y(t)$ in a finite interval of time $t_0 \leq t \leq t_f$.

The controllability gives the necessary and sufficient condition for designing a feedback controller so that all the eigenvalues of the given system are placed in the desired locations in the s-plane as one wishes. If any state is not available for measurement, then one has to construct an observer by means of which the unmeasurable states are reconstructed by measuring the known states. For the design of the observer, the system should be observable. Similar to the controllability condition, we establish below the observability condition.

7.8.1 Observability Condition

For a free system, the output as given by Eq. (7.65) is

$$y(t) = Ce^{At}x(0)$$

Using the Cayley–Hamilton theorem, we get

$$e^{At} = \sum_{k=0}^{n-1} \alpha_k A^k \tag{7.88}$$

Substituting Eq. (7.88) in (7.87), we get

$$\begin{aligned} y(t) &= \sum_{k=0}^{n-1} \alpha_k C A^k x(0) \\ &= \alpha_0 Cx(0) + \alpha_1 CAx(0) + \cdots + \alpha_{n-1} CA^{n-1}x(0) \end{aligned} \tag{7.89}$$

By measuring the output $y(t)$, in a finite time interval $t_0 \leq t \leq t_f$, $x(0)$ can be determined. Equation (7.89) can be written as

$$\frac{y(t)}{\sum_{k=0}^{n-1} \alpha^k} = \begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix} x(0) \tag{7.90}$$

For the solution of Eq. (7.90), for $x(0)$, the inverse of the following matrix should exist:

$$S = \begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix} \tag{7.91}$$

For the existence of inverse of S, the rank of the matrix should be n which can be obtained using the theory of linear algebra. Therefore, the observability condition is stated as follows.

For a continuous time system to be completely observable, the necessary and sufficient condition is that the $n \times nm$ matrix

$$S = [C^T \quad A^T C^T \cdots (A^T)^{n-1} C^T] \tag{7.92}$$

should have rank n which implies that the matrix S has n linearly independent column vectors.

Example 7.27

Determine the controllability and observability of the following state model:

$$\dot{x} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & -2 & 1 \\ 0 & 0 & -1 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \\ -2 \end{bmatrix} u$$
$$y = [1 \quad 0 \quad 0]x$$

(*Anna University, December, 2009*)

Solution Controllability Test

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & -2 & 1 \\ 0 & 0 & -1 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 \\ 1 \\ -2 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$$

$$AB = \begin{bmatrix} 1 & 1 & 0 \\ 0 & -2 & 1 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ -2 \end{bmatrix} = \begin{bmatrix} 1 \\ -4 \\ 2 \end{bmatrix}$$

$$A^2B = \begin{bmatrix} 1 & 1 & 0 \\ 0 & -2 & 1 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ -4 \\ 2 \end{bmatrix} = \begin{bmatrix} -3 \\ 10 \\ -2 \end{bmatrix}$$

The controllability matrix Q is given by

$$Q = \begin{bmatrix} 0 & 1 & -3 \\ 1 & -4 & 10 \\ -2 & 2 & -2 \end{bmatrix}$$

By observation of Q, no column of Q can be expressed as the linear combination of the other columns. Hence, Q has full rank and hence the system is completely state controllable. This can also be verified by determining the determinant of Q

$$|Q| = \begin{vmatrix} 0 & 1 & -3 \\ 1 & -4 & 10 \\ -2 & 2 & -2 \end{vmatrix} = -1(-2 + 20) - 3(2 - 8) = 0$$

The determinant is zero and hence the rank of the controllability matrix is not full. The system is not controllable.

Observability Test

$$C^T = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$A^T = \begin{bmatrix} 1 & 0 & 0 \\ 1 & -2 & 0 \\ 0 & 1 & -1 \end{bmatrix}$$

$$(A^T)^2 = \begin{bmatrix} 1 & 0 & 0 \\ 1 & -2 & 0 \\ 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 1 & -2 & 0 \\ 0 & 1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 4 & 0 \\ 1 & -3 & 1 \end{bmatrix}$$

$$A^T C^T = \begin{bmatrix} 1 & 0 & 0 \\ 1 & -2 & 0 \\ 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

$$(A^T)^2 C^T = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 4 & 0 \\ 1 & -3 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

The observability matrix is given by

$$S = \begin{bmatrix} C^T & A^T C^T & (A^T)^2 C^T \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$|S| = 1$$

Observation of the observability matrix S shows that all the columns are linearly independent. Further, the determinant of S is determined as $|S| = 1$. Hence, the rank of S is full (3). The system is observable. The given system is not controllable but observable.

Example 7.28

Determine the observability of the system given as follows:

$$\dot{x} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -2 & -3 \end{bmatrix} x + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 3 & 4 & 1 \end{bmatrix} x$$

(*Anna University, December, 2010*)

Solution

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -2 & -3 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$C = \begin{bmatrix} 3 & 4 & 1 \end{bmatrix}$$

$$C^T = \begin{bmatrix} 3 \\ 4 \\ 1 \end{bmatrix}$$

$$A^T = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & -2 \\ 0 & 1 & -3 \end{bmatrix}$$

$$(A^T)^2 = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & -2 \\ 0 & 1 & -3 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & -2 \\ 0 & 1 & -3 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -2 & 6 \\ 1 & -3 & 7 \end{bmatrix}$$

$$A^T C^T = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & -2 \\ 0 & 1 & -3 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

$$(A^T)^2 C^T = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -2 & 6 \\ 1 & -3 & 7 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -2 \\ -2 \end{bmatrix}$$

The observability matrix is given by

$$S = \begin{bmatrix} C^T & A^T C^T & (A^T)^2 C^T \end{bmatrix} = \begin{bmatrix} 3 & 0 & 0 \\ 4 & 1 & -2 \\ 1 & 1 & -2 \end{bmatrix}$$

By observing the controllability matrix S, it is seen that column 3 can be expressed by multiplying column 2 by -2. Hence, the rank of the matrix S is not full and the system is not observable. The determinant of S is determined as

$$|S| = 3(-2 + 2) = 0$$

7.9 State Equation of Discrete Time System

The discrete time system is described by the difference equation. A continuous time system which is described by an nth order differential equation is converted into n first order equations and the descriptions for n state variables are given. In an analogous manner to continuous time system, the discrete time system which is described by N^{th} order difference equation is converted into N first order difference equations, and the description for N state variables are given in the form of vector matrix difference equation.

A linear time invariant discrete time system can be represented by the following vector matrix difference equation:

$$\begin{aligned} x[k+1] &= Ax[k] + Bu[k] \\ y[k] &= Cx[k] + Du[k] \end{aligned} \tag{7.93}$$

where the symbols of Eq. (7.93) are defined as

$$\begin{aligned} x[k] &= N \times 1 \text{ (State Vector)} \\ y[k] &= p \times 1 \text{ (Output Vector)} \\ u[k] &= r \times 1 \text{ (Input Vector)} \end{aligned}$$

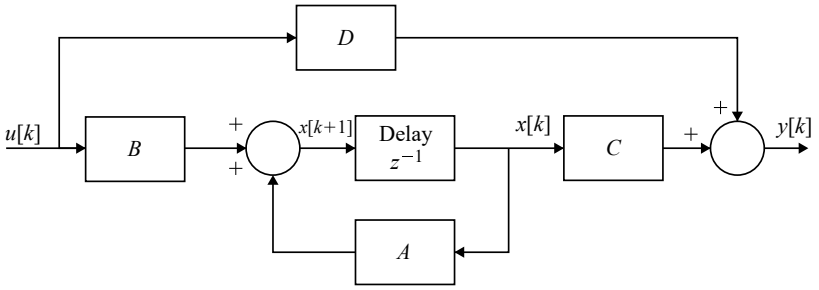


Fig. 7.19 Block diagram representation of state equation of discrete time system

$$
\begin{aligned}
A &= N \times N \text{ (State Matrix)} \\
B &= N \times r \text{ (Input Matrix)} \\
C &= p \times N \text{ (Output Matrix)} \\
D &= p \times r \text{ (Direct Transmission Matrix)}
\end{aligned}
$$

The block diagram representation of Eq. (7.93) is shown in Fig. 7.19.
For example, consider the following second order difference equation:

$$y[k+2] + a_1 y[k+1] + a_2 y[k] = bu[k]$$

Let us choose

$$
\begin{aligned}
y[k] &= x_1[k] \\
y[k+1] &= x_1[k+1] \\
&= x_2[k] \\
y[k+2] &= x_2[k+1]
\end{aligned}
$$

Substituting the above equations in the given second order difference equation, we get

$$
\begin{aligned}
x_2[k+1] + a_1 x_2[k] + a_2 x_1[k] &= bu[k] \\
x_2[k+1] &= -a_2 x_1[k] - a_1 x_2[k] + bu[k]
\end{aligned}
$$

From the above equations, the following vector matrix difference equation is obtained:

$$
\begin{aligned}
x[k+1] &= \begin{bmatrix} 0 & 1 \\ -a_2 & -a_1 \end{bmatrix} x[k] + \begin{bmatrix} 0 \\ b \end{bmatrix} u[k] \\
y[k] &= [1 \quad 0] x[k]
\end{aligned}
$$

7.9.1 Discrete Time State Equation in Controllable Canonical Form

Let $H(z)$ be the discrete time system transfer function which can be expressed as follows:

$$\frac{Y[z]}{U[z]} = H[z] = \frac{b_0 z^N + b_1 z^{N-1} + \cdots + b_{N-1} z + b_N}{z^N + a_1 z^{N-1} + \cdots + a_{N-1} z + a_N} = \frac{b_0 + b_1 z^{-1} + \cdots + b_{N-1} z^{N-1} + b_N z^N}{1 + a_1 z^{-1} + \cdots + a_{N-1} z^{N-1} + a_N z^N} \tag{7.94}$$

Let $H[z] = H_1(z) H_2(z)$, where

$$H_1[z] = (b_0 + b_1 z^{-1} + \cdots + b_{N-1} z^{N-1} + b_N z^N)$$
$$H_2[z] = \frac{1}{(1 + a_1 z^{-1} + \cdots + a_{N-1} z^{N-1} + a_N z^N)}$$

$$Y[z] = H_1[z] W[z] \tag{7.95}$$

where

$$W[z] = H_2[z] U[z] \tag{7.96}$$

Equations (7.95) and (7.96) are represented in Fig. 7.20. From Fig. 7.20, the following equations are written:

$$\begin{aligned} x_1[n+1] &= x_2[n] \\ x_2[n+1] &= x_3[n] \\ &\vdots \\ x_{N-1}[n+1] &= x_N[n] \\ x_N[n+1] &= -a_N x_1[n] - a_{N-1} x_2[n] - \ldots - a_1 x_N[n] + u[n] \\ y[n] &= b_N x_1[n] + b_{N-1} x_2[n] + \ldots + b_1 x_N[n] + b_0 u[n] \end{aligned} \tag{7.97}$$

Substituting for $x_{N+1}[n]$ from Eq. (7.97), we get

$$\begin{aligned} y[n] &= (b_N - b_0 a_N) x_1[n] + (b_{N-1} - b_0 a_{N-1}) x_2[n] + \cdots \\ &\quad + (b_1 - b_0 a_1) x_N[n] + b_0 u[n] \\ &= \bar{b}_N x_1[n] + \bar{b}_{N-1} x_2[n] + \cdots + \bar{b}_1 x_N + b_0 u[n] \end{aligned} \tag{7.98}$$

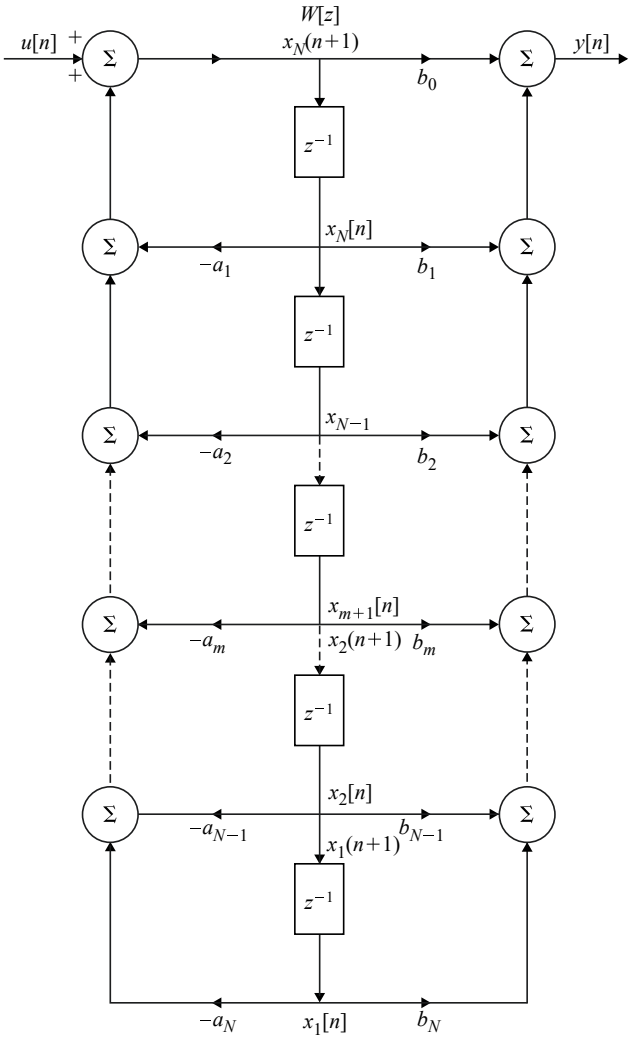


Fig. 7.20 Direct form II realization of Nth order discrete time system

where $\bar{b}_i = (b_i - b_0 a_i)$. Equation (7.97) represents the state equations and Eq. (7.98) represents the output equation. The above equations are written in matrix form as

$$\underbrace{\begin{bmatrix} x_1[n+1] \\ x_2[n+1] \\ x_3[n+1] \\ \vdots \\ x_{N-1}[n] \\ x_N[n] \end{bmatrix}}_{} = \underbrace{\begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & 1 \\ -a_N & -a_{N-1} & -a_{N-2} & -a_2 & -a_1 \end{bmatrix}}_{} \underbrace{\begin{bmatrix} x_1[n] \\ x_2[n] \\ x_3[n] \\ \vdots \\ x_{N-1}[n] \\ x_N[n] \end{bmatrix}}_{} + \underbrace{\begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}}_{}$$

$$x[n+1] = \underbrace{[\bar{b}_N \quad \bar{b}_{N-1} \ldots \bar{b}_1]}_{A} x[n] + \underbrace{b_0}_{B} u[n] \tag{7.99}$$

$$y[n] = \underbrace{[\bar{b}_N \quad \bar{b}_{N-1} \ldots \bar{b}_1]}_{C} + \underbrace{b_0}_{D} u[n] \tag{7.100}$$

The general form of state and output equations is therefore written as

$$\begin{aligned} x[n+1] &= Ax[n] + Bu[n] \\ y[n] &= Cx[n] + Du[n] \end{aligned} \tag{7.101}$$

The state equations for a discrete system can be obtained by several methods.

However, we represent here by controllable canonical form, observable canonical form and diagonal form. The following analogy between continuous and discrete time systems is to be noted:

1. In a continuous time system, the output of each integrator is identified as a state. In a discrete time system, the output of each delay element is identified as the state.
2. In a continuous time system, the input of each integrator is identified as a first derivative from which the first order differential equation is formed. In a discrete time system, the input to each delayed element is identified to form the first order difference equation.

The following examples illustrate the method of forming state equations.

Example 7.29

Form the state equations of canonical form for the following T.F. of a discrete time system:

$$H[z] = \frac{5z^4 + 7z^3 + 8z^2 + 2z + 10}{z^4 + 6z^3 + 7z^2 + 4z + 9}$$

Solution

$$H[z] = \frac{5z^4 + 7z^3 + 8z^2 + 2z + 10}{z^4 + 6z^3 + 7z^2 + 4z + 9} = \frac{5 + 7z^{-1} + 8z^{-2} + 2z^{-3} + 10z^{-4}}{1 + 6z^{-1} + 7z^{-2} + 4z^{-3} + 9z^{-4}}$$

where

$$\begin{aligned} &b_0 = 5;\ b_1 = 7;\ b_2 = 8;\ b_3 = 2;\ b_4 = 10; \\ &a_1 = 6;\ a_2 = 7;\ a_3 = 4;\ a_4 = 9 \end{aligned}$$

$$x[n+1] = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -9 & -4 & -7 & -6 \end{bmatrix} x[n] + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} u[n]$$

$$\bar{b}_4 = b_4 - b_0 a_4 = 10 - 5 \times 9 = -35$$
$$\bar{b}_3 = b_3 - b_0 a_3 = 2 - 5 \times 4 = -18$$
$$\bar{b}_2 = b_2 - b_0 a_2 = 8 - 5 \times 7 = -27$$
$$\bar{b}_1 = b_1 - b_0 a_1 = 7 - 5 \times 6 = -23$$
$$y[n] = [\bar{b}_4 \ \bar{b}_3 \ \bar{b}_2 \ \bar{b}_1] u[n] + b_0 u[n]$$

$$y[n] = [-35 \quad -18 \quad -27 \quad -23] x[n] + 5u[n]$$

Example 7.30

Consider the following difference equation:

$$4y[n-3] + 6y[n-2] - 5y[n-1] + y[n] = u[n] + 5x[n-1]$$

Form state equations.

Solution

$$\frac{Y[z]}{U[z]} = H[z] = \frac{(1 + 5z^{-1})}{(1 - 5z^{-1} + 6z^{-2} + 4z^{-3})}$$

where $b_0 = 1;\ b_1 = 5;\ b_2 = 0;\ b_3 = 0;\ a_1 = -5;\ a_2 = 6;\ a_3 = 4$

$$x[n+1] = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -4 & -6 & 5 \end{bmatrix} x[n] + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u[n]$$

$$\bar{b}_3 = b_3 - b_0 a_3 = -4$$
$$\bar{b}_2 = b_2 - b_0 a_2 = -6$$
$$\bar{b}_1 = b_1 - b_0 a_1 = 5 + 5 = 10$$

$$y[n] = [-4 \quad -6 \quad 10] x[n] + u[n]$$

7.9.2 Observable Canonical Form Model

Consider the following system function:

$$\frac{Y[z]}{U[z]} = H[z] = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2}}{1 + a_1 z^{-1} + a_2 z^{-2}}$$

Cross-multiplying the above equation, we get

$$\begin{aligned}[1 + a_1 z^{-1} + a_2 z^{-2}]Y[z] &= [b_0 + b_1 z^{-1} + b_2 z^{-2}]U[z] \\ Y[z] &= -a_1 z^{-1} Y[z] - a_2 z^{-2} Y[z] + b_0 U[z] \\ &\quad + b_1 z^{-1} U[z] + b_2 z^{-2} U[z] \end{aligned} \tag{7.102}$$

Equation (7.102) is represented in Fig. 7.21, where the states and the delay elements are shown.

From Fig. 7.21, the following equations are written in terms of states:

$$\begin{aligned} y[n] &= x_1[n] + b_0 u[n] \\ x_1[n+1] &= -a_1 y[n] + x_2 y[n] + b_1 u[n] \\ &= -a_1 x_1[n] + x_2[n] + (b_1 - b_0 a_1) u[n] \qquad (7.103) \\ x_2[n+1] &= -a_2 y[n] + b_2 u[n] \\ &= -a_2 x_1[n] + (b_2 - b_0 a_1) u[n] \qquad (7.104) \end{aligned}$$

Equations (7.103) and (7.104) can be combined and expressed in matrix form as

$$\begin{aligned} x[n+1] &= \begin{bmatrix} -a_1 & 1 \\ -a_2 & 0 \end{bmatrix} x[n] + \begin{bmatrix} (b_1 - b_0 a_1) \\ (b_2 - b_0 a_2) \end{bmatrix} u[n] \\ y[n] &= [1 \quad 0] x[n] + b_0 u[n] \end{aligned}$$

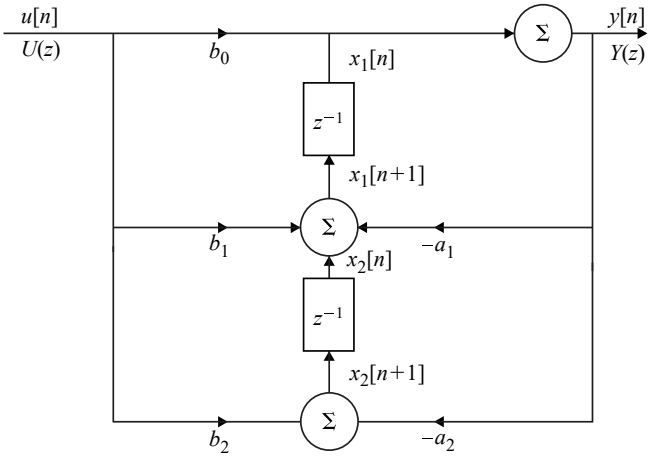


Fig. 7.21 Canonical form structure

In general, for an Nth order difference equation,

$$x[n+1] = \begin{bmatrix} -a_1 & 1\,0\,0\cdots 0 \\ -a_2 & 0\,1\,0\cdots 0 \\ -a_3 & 0\,0\,1\cdots 0 \\ \vdots & \vdots\,\vdots\,\vdots\,\vdots\,\vdots \\ -a_{N-1} & 0\,0\,0\cdots 1 \\ -a_N & 0\,0\,0\cdots 0 \end{bmatrix} x[n] + \begin{bmatrix} (b_1 - b_0 a_1) \\ (b_2 - b_0 a_2) \\ (b_3 - b_0 a_3) \\ \vdots \\ (b_{N-1} - b_0 a_{N-1}) \\ (b_N - b_0 a_N) \end{bmatrix} u[n] \quad (7.105)$$

where $\bar{b}_i$ is defined as $\bar{b}_i = (b_i - b_0 a_i)$

$$y[n] = [1 \quad 0 \quad 0\cdots 0]x[n] + b_0 u[n] \quad (7.106)$$

Example 7.31

Form the state equations of observable canonical form for the following T.F. of a discrete time system:

$$H[z] = \frac{5z^4 + 7z^3 + 8z^2 + 2z + 10}{z^4 + 6z^3 + 7z^2 + 4z + 9}$$

Solution

$$H[z] = \frac{5z^4 + 7z^3 + 8z^2 + 2z + 10}{z^4 + 6z^3 + 7z^2 + 4z + 9} = \frac{5 + 7z^{-1} + 8z^{-2} + 2z^{-3} + 10z^{-4}}{1 + 6z^{-1} + 7z^{-2} + 4z^{-3} + 9z^{-4}}$$

where $b_0 = 5;\ b_1 = 7;\ b_2 = 8;\ b_3 = 2;\ b_4 = 10;\ a_1 = 6;\ a_2 = 7;\ a_3 = 4;\ a_4 = 9$

$$\bar{b}_1 = b_1 - b_0 a_1 = 7 - 5 \times 6 = -23$$
$$\bar{b}_2 = b_2 - b_0 a_2 = 8 - 5 \times 7 = -27$$
$$\bar{b}_3 = b_3 - b_0 a_3 = 2 - 5 \times 4 = -18$$
$$\bar{b}_4 = b_4 - b_0 a_4 = 10 - 5 \times 9 = -35$$

$$x[n+1] = \begin{bmatrix} -6\,1\,0\,0 \\ -7\,0\,1\,0 \\ -4\,0\,0\,1 \\ -9\,0\,0\,0 \end{bmatrix} x[n] + \begin{bmatrix} -23 \\ -27 \\ -18 \\ -35 \end{bmatrix} u[n]$$

$$y[n] = [1 \quad 0 \quad 0 \quad 0]x[n] + 5u[n]$$

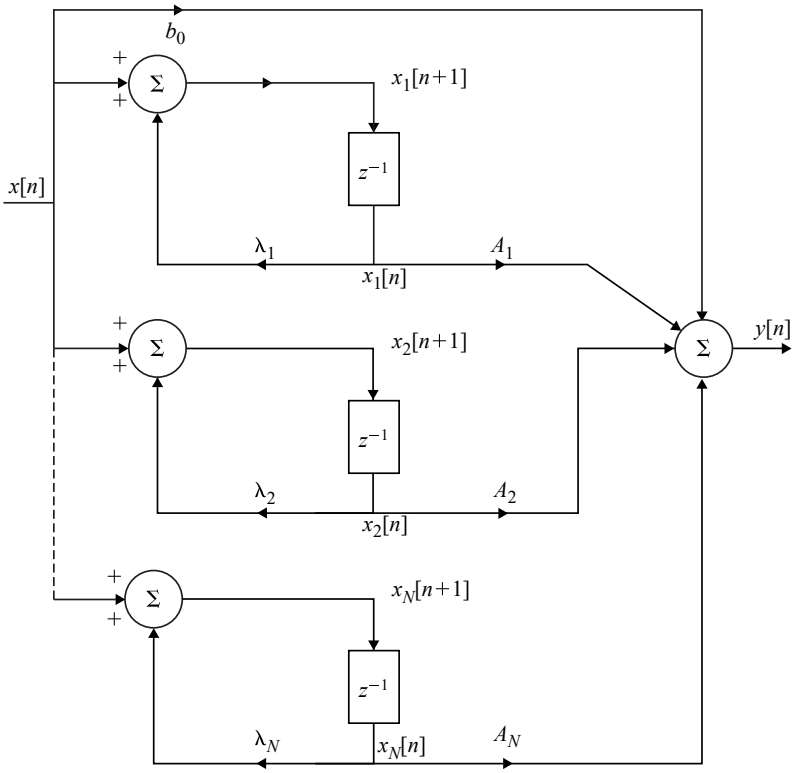


Fig. 7.22 Diagonal form model

7.9.3 Diagonal Form (Parallel Form) Model

Consider the system transfer function given as follows:

$$H[z] = \frac{b_0 z^N + b_1 z^{N+1} + \cdots + b_{N-1} z + b_N}{z^N + a_1 z^{N+1} + \cdots + a_{N-1} z + a_N} \tag{7.107}$$

$$= \frac{b_0 + b_1 z^{-1} + \cdots + b_{N-1} z^{-(N-1)} + b_N z^{-N}}{1 + a_1 z^{-1} + \cdots + a_{N-1} z^{-(N-1)} + a_N z^{-N}}$$

$$= b_0 + \frac{A_1}{(z - \lambda_1)} + \cdots + \frac{A_N}{(z - \lambda_N)} \tag{7.108}$$

where $\lambda_1,\ \lambda_2, \ldots, \lambda_n$ are the distinct eigenvalues of $H[z]$. Equation (7.108) is represented in Fig. 7.22.

From Fig. 7.22, the following state equations are written:

$$x_1[n+1] = \lambda_1 x_1[n] + u[n]$$

$$\begin{aligned} x_2[n+1] &= \lambda_2 x_2[n] + u[n] \\ &\vdots \\ x_N[n+1] &= \lambda_N x_N[n] + u[n] \end{aligned} \tag{7.109}$$

$$y[n] = A_1 x_1[n] + A_2 x_2[n] + \cdots + A_N x_N[n] + b_0 u[n] \tag{7.110}$$

Equations (7.109) and (7.110) can be represented in matrix form as

$$x[n+1] = \begin{bmatrix} \lambda_1 & 0 & 0 & \cdots & 0 \\ 0 & \lambda_2 & 0 & \cdots & 0 \\ 0 & 0 & \lambda_3 & \cdots & 0 \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ 0 & 0 & 0 & \cdots & \lambda_N \end{bmatrix} x[n] + \begin{bmatrix} 1 \\ 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} u[n] \tag{7.111}$$

$$y[n] = [A_1 \quad A_2 \quad A_3 \cdots A_N] x[n] + b_0 u[n] \tag{7.112}$$

Example 7.32

A certain discrete time system has the following T.F.:

$$H[z] = \frac{z^3 + 10z^2 + 32z + 29}{z^3 + 9z^2 + 26z + 24}$$

Form the state equation with its A matrix in diagonal form.

Solution Dividing the numerator polynomial with denominator polynomial, we get

$$\begin{array}{r} 1 \\ z^3 + 9z^2 + 26z + 24 \overline{)\, z^3 + 10z^2 + 32z + 29} \\ \underline{z^3 + \ 9z^2 + 26z + 24} \\ z^2 + 6z \ + \ 5 \end{array}$$

$$H[z] = 1 + \frac{(z^2 + 6z + 5)}{(z^3 + 9z^2 + 26z + 24)}$$

$$(z^2 + 6z + 5) = (z + 1)(z + 5)$$

$$(z^3 + 9z^2 + 26z + 24) = (z + 2)(z + 3)(z + 4)$$

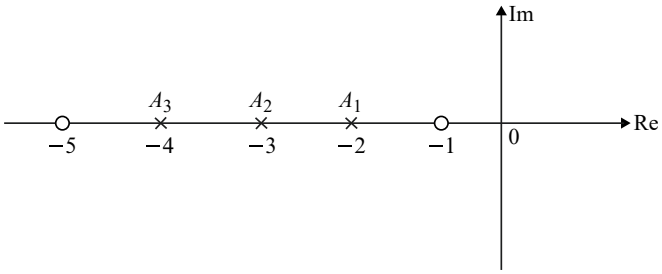


Fig. 7.23 Pole–zero diagram

$$H[z] = 1 + \frac{(z+1)(z+5)}{(z+2)(z+3)(z+4)}$$

$$\frac{(z+1)(z+5)}{(z+2)(z+3)(z+4)} = \frac{A_1}{(z+2)} + \frac{A_2}{(z+3)} + \frac{A_3}{(z+4)}$$

The pole–zero locations are shown in Fig. 7.23.

$$A_1 = \frac{-3 \times 1}{2 \times 1} = -\frac{3}{2}$$

$$A_2 = \frac{2 \times 2}{1 \times 1} = 4$$

$$A_3 = \frac{-3 \times 1}{2 \times 1} = -\frac{3}{2}$$

$$H[z] = 1 - \frac{3}{2}\frac{1}{(z+2)} + \frac{4}{(z+3)} - \frac{3}{2}\frac{1}{(z+4)}$$

The eigenvalues are $\lambda_1 = -2,\ \lambda_2 = -3$ and $\lambda_3 = -4$

$$x[n+1] = \begin{bmatrix} -2 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -4 \end{bmatrix} x[n] + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} u[n]$$

$$y[n] = \begin{bmatrix} -\frac{3}{2} & 4 & -\frac{3}{2} \end{bmatrix} x[n] + u[n]$$

Example 7.33

Find the state variable matrices A, B, C and D for the equation

$$y[n] - 3y[n-1] - 2y[n-2] = u[n] + 5x[n-1] + 6x[n-2]$$

(*Anna University, November, 2007*)

Solution Taking z-transform on both sides of the equation, we get

$$\frac{Y[z]}{U[z]} = H[z] = \frac{(1 + 5z^{-1} + 6z^{-2})}{(1 - 3z^{-1} - 2z^{-2})}$$

where $b_0 = 1;\ b_1 = 5;\ b_2 = 6;\ a_1 = -3;\ a_2 = -2$

$$\bar{b}_2 = b_2 - b_0 a_2 = 6 + 2 = 8$$
$$\bar{b}_1 = b_1 - b_0 a_1 = 5 + 3 = 8$$
$$D = b_0 = 1$$

$$x[n+1] = \underbrace{\begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix}}_{A} x[n] + \underbrace{\begin{bmatrix} 0 \\ 1 \end{bmatrix}}_{B} u[n]$$

$$y[n] = \underbrace{[8 \quad 8]}_{C} x[n] + \underbrace{1}_{D} u[n]$$

Example 7.34

A continuous time system has the state variable description given as follows.

$$A = \begin{bmatrix} 2 & -1 \\ 1 & 0 \end{bmatrix}; \quad B = \begin{bmatrix} 1 \\ 0 \end{bmatrix}; \quad C = [3 \quad 1]; \quad D = [2]$$

Determine the transfer function.

Solution

$$(zI - A) = \begin{bmatrix} (z-2) & 1 \\ -1 & z \end{bmatrix}$$

$$(zI - A)^{-1} = \frac{\text{Adjoint}}{\text{Determinant}}$$

$$= \frac{1}{(z^2 - 2z + 1)} \begin{bmatrix} z & 1 \\ 1 & (2-z) \end{bmatrix}$$

$$(zI - A)^{-1}B = \frac{1}{(z^2 - 2z + 1)}\begin{bmatrix} z & 1 \\ 1 & (2 - z) \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix}$$
$$= \frac{1}{(z^2 - 2z + 1)}\begin{bmatrix} z \\ 1 \end{bmatrix}$$
$$C(zI - A)^{-1}B = \frac{1}{(z^2 - 2z + 1)}[3 \quad 1]\begin{bmatrix} z \\ 1 \end{bmatrix}$$
$$= \frac{(3z + 1)}{(z^2 - 2z + 1)}$$
$$C(zI - A)^{-1}B + D = \frac{(3z + 1)}{(z^2 - 2z + 1)} + 2$$

$$\boxed{H(z) = \frac{(2z^2 - z + 3)}{(z^2 - 2z + 1)}}$$

7.9.4 Solution of State Equation

The solution of state Eq. (7.93) of discrete time system can be obtained by the following methods:

1. By the classical method.
2. The z-transform method.

7.9.4.1 Solution of State Equation by Classical Method—The Recursive Method

Consider the homogeneous equation of (7.93) which is written as

$$x[k + 1] = Ax[k]$$
$$k = 0, 1, 2, \ldots \tag{7.113}$$

The solution of the above equation may be obtained as

$$x[k] = A\phi[k]$$

where $\phi[k]$ is called the fundamental or state transition matrix which is unique for the given system and satisfies the equation

$$\Phi[k + 1] = A\phi[k] \tag{7.114}$$

where

$$\Phi[0] = I$$
$$\phi[k] = n \times n \text{ matrix}$$

The solution of (7.113) may be written as

$$x[k] = \phi[k]x[0] \tag{7.115}$$

For $k = 0$

$$\Phi[1] = A$$

For $k = 1$

$$\Phi[2] = A\Phi[1]$$
$$= A^2$$

In general

$$\phi[k] = A^k \quad \text{for } k = 0, 1, 2, \ldots$$

$$\boxed{x[k] = A^k x[0]} \tag{7.116}$$

For a non-homogenous Eq. (7.113), the solution is obtained as

$$\boxed{x[k] = A^k x[0] + \sum_{j=1}^{k} A^{k-j} Bu(j-1)} \tag{7.117}$$

7.9.4.2 Solution by using z-Transform Method

$$x[k+1] = Ax[k] + Bu[k] \tag{7.118}$$

Taking z-transform on both sides of the above equation, we get

$$Zx[z] - zx[0] = Ax[z] + Bu[z]$$
$$[zI - A]X[z] = zx[0] + BU[z]$$

$$\boxed{X[z] = [zI - A]^{-1} zx[0] + [zI - A]^{-1} BU[z]} \tag{7.119}$$

Taking inverse z-transform for Eq. (7.119), $x[k]$ is obtained.

Example 7.35

Consider the following vector matrix difference equation:

$$x[k+1] = \begin{bmatrix} -3 & 1 \\ -2 & 0 \end{bmatrix} x[k] + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u[k]$$

$$x[0] = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

The input is a unit step sequence. Derive an expression for $x[k]$.

Solution

$$A = \begin{bmatrix} -3 & 1 \\ -2 & 0 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$x[0] = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

From Eq. (7.118), we get

$$X(z) = [zI - A]^{-1} zx[0] + [zI - A]^{-1} BU[z]$$

Let

$$X_0[z] = [zI - A]^{-1} zx[0]$$

$$X_f[z] = [zI - A]^{-1} BU(z)$$

For unit step sequence

$$U(z) = \frac{z}{z-1}$$

$$[zI - A] = \begin{bmatrix} (z+3) & -1 \\ 2 & z \end{bmatrix}$$

$$[zI - A]^{-1} = \begin{bmatrix} z & 1 \\ -2 & z+3 \end{bmatrix} \frac{1}{(z+1)(z+2)}$$

$$
\begin{aligned}
X_0(z) &= \frac{z}{(z+1)(z+2)} \begin{bmatrix} z & 1 \\ -2 & z+3 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} \\
\frac{X_0[z]}{z} &= \frac{1}{(z+1)(z+2)} \begin{bmatrix} (z-1) \\ -(z+5) \end{bmatrix} \\
&= \begin{bmatrix} \frac{3}{(z+2)} - \frac{2}{(z+1)} \\ \frac{3}{(z+2)} - \frac{4}{(z+1)} \end{bmatrix} \\
X_0[z] &= \begin{bmatrix} \frac{3z}{(z+2)} - \frac{2z}{(z+1)} \\ \frac{3z}{(z+2)} - \frac{4z}{(z+1)} \end{bmatrix}
\end{aligned}
$$

Taking the inverse z-transform, we get

$$
\begin{aligned}
x_0[k] &= \begin{bmatrix} 3(-2)^k - 2(-1)^k \\ 3(-2)^k - 4(-1)^k \end{bmatrix} \\
[zI - A]^{-1}B &= \frac{1}{(z+1)(z+2)} \begin{bmatrix} z & 1 \\ -2 & z+3 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\
&= \frac{1}{(z+1)(z+2)} \begin{bmatrix} z \\ -2 \end{bmatrix} \\
X_f[z] &= [zI - A]^{-1}BU[z] \\
&= \frac{z}{(z-1)(z+1)(z+2)} \begin{bmatrix} z \\ -2 \end{bmatrix} \\
\frac{X_f[z]}{z} &= \frac{1}{(z-1)(z+1)(z+2)} \begin{bmatrix} z \\ -2 \end{bmatrix} \\
&= \begin{bmatrix} \frac{1}{2(z+1)} - \frac{2}{3}\frac{1}{(z+2)} + \frac{1}{6(z-1)} \\ \frac{1}{(z+1)} - \frac{2}{3(z+2)} - \frac{1}{3(z-1)} \end{bmatrix} \\
X_f[z] &= \begin{bmatrix} \frac{1}{2}\frac{z}{z+1} - \frac{2}{3}\frac{z}{(z+2)} + \frac{z}{6(z-1)} \\ \frac{z}{(z+1)} - \frac{2}{3}\frac{z}{z+2} - \frac{z}{z(z-1)} \end{bmatrix}
\end{aligned}
$$

Taking the inverse z-transform, we get

$$
x_f[k] = \begin{bmatrix} \frac{1}{2}(-1)^k - \frac{2}{3}(-2)^k + \frac{1}{6} \\ (-1)^k - \frac{2}{3}(-2)^k + \frac{1}{3} \end{bmatrix}
$$

The total response is given by

$$
\begin{aligned}
x[k] &= x_0[k] + x_f[k] \\
&= \begin{bmatrix} 3(-2)^k - 2(-1)^k \\ 3(-2)^k - 4(-1)^k \end{bmatrix} + \begin{bmatrix} \frac{1}{2}(-1)^k - \frac{2}{3}(-2)^k + \frac{1}{6} \\ (-1)^k - \frac{2}{3}(-2)^k - \frac{1}{3} \end{bmatrix} \\
x[k] &= \begin{bmatrix} \frac{7}{2}(-2)^k - \frac{3}{2}(-1)^k + \frac{1}{6} \\ \frac{7}{2}(-2)^k - 3(-1)^k - \frac{1}{3} \end{bmatrix}
\end{aligned}
$$

$$\boxed{\begin{aligned} x_1[k] &= \frac{7}{2}(-2)^k - \frac{3}{2}(-1)^k + \frac{1}{6} \\ x_2[k] &= \frac{7}{2}(-2)^k - 3(-1)^k - \frac{1}{3} \end{aligned}}$$

7.10 Controllability and Observability of Discrete Time System

In order to understand what is going on inside the system just by observation, the given dynamic system should be observable. The dual concepts of controllability and observability are related to linear systems of algebraic equations. The necessary and sufficient conditions for the existence of these two concepts are derived below.

7.10.1 Controllability Condition for Discrete Time System

Consider the vector matrix difference equation which is given as follows:

$$x[k+1] = Ax[k] + BU[k] \tag{7.120}$$

The linear discrete time system Eq. (7.120) is said to be absolutely state controllable if it is possible to transfer any initial state $x[0] = x_0$ to any final state x_f in a finite time. This implies whether it is possible to find a control sequence $u[0], u[1] \cdots u[n-1]$ such that $x[k] = x_f$. Equation (7.120) for various input sequences is given as

$$\begin{aligned} x[1] &= Ax[0] + Bu[0] \\ x[2] &= Ax[1] + Bu[1] = A^2x[0] + ABu[0] + Bu[1] \\ &\vdots \\ x[n] &= A^n x[0] + A^{n-1}Bu[0] + \cdots + Bu[n-1] \end{aligned} \tag{7.121}$$

$$x[n] - A^n x[0] = \begin{bmatrix} B & AB & A^2B \cdots A^{n-1}B \end{bmatrix} \begin{bmatrix} u[n-1] \\ u[n-2] \\ \vdots \\ u[1] \\ u[0] \end{bmatrix} \tag{7.122}$$

$$\begin{bmatrix} u[n-1] \\ u[n-2] \\ \vdots \\ u[1] \\ u[0] \end{bmatrix} = Q^{-1}[x[n] - A^n x[0]] \tag{7.123}$$

where $Q = [B \quad AB \quad A^2B \cdots A^{n-1}B]$ is a $n \times n$ square matrix. For any $x[n] = x_f$, the input sequence transfers the initial state x_0 to any final state x_f in n steps. This requires the existence of $\boldsymbol{Q}$ **which is called the controllability matrix**. In other words, for the discrete system to be completely state controllable, the controllability matrix Q should have full rank.

7.10.2 Observability Condition for Discrete Time System

Consider the following state and output equations of a linear time invariant discrete time autonomous system:

$$\begin{aligned} x[k+1] &= Ax[k] \\ y[k] &= Cx[k] \end{aligned} \tag{7.124}$$

where the symbols have the usual meaning and the matrices A and C have appropriate dimensions. The discrete time system is said to be observable if it is possible to establish the unknown initial state $x[0]$ by measuring the output $y[k]$. Since $x[0]$ is an n dimensional state vector, by taking n measurements it is possible to establish $x[0]$. For $k = 0, 1, 2, \ldots, (n-1)$, the output sequence is expressed as

$$\begin{aligned} y[0] &= Cx[0] \\ y[1] &= Cx[1] = CAx[0] \\ y[2] &= Cx[2] = CA^2x[0] \\ &\vdots \\ y[n-1] &= Cx[n-1] = CA^{n-1}x[0] \end{aligned} \tag{7.125}$$

$$\begin{bmatrix} y[0] \\ y[1] \\ y[2] \\ \vdots \\ y[n-1] \end{bmatrix} = \begin{bmatrix} C \\ CA \\ CA^2 \\ \vdots \\ CA^{n-1} \end{bmatrix} x[0]$$
$$= Sx[0] \tag{7.126}$$

where

$$S = \begin{bmatrix} C \\ CA \\ CA^2 \\ \vdots \\ CA^{n-1} \end{bmatrix} \tag{7.127}$$

$$x[0] = S^{-1}y[k] \tag{7.128}$$

S **is called the observability matrix**. From Eq. (7.128), $x[0]$ can be obtained from $y[k]$ if the inverse of the observability matrix S is non-singular. In other words,

$$S = [C^T \ A^TC^T \ (A^T)^2C^T \cdots (A^T)^{n-1}C^T] \tag{7.129}$$

should have rank n.

Example 7.36

Consider the discrete time system described by the following state equation:

$$x[k+1] = \begin{bmatrix} 0 & 6 \\ -1 & -5 \end{bmatrix} x + \begin{bmatrix} 3 & 6 \\ -1 & -2 \end{bmatrix} u$$

Test whether the system is state controllable.

Solution

$$A = \begin{bmatrix} 0 & 6 \\ -1 & -5 \end{bmatrix}$$
$$B = \begin{bmatrix} 3 & 6 \\ -1 & -2 \end{bmatrix}$$
$$AB = \begin{bmatrix} 0 & 6 \\ -1 & -5 \end{bmatrix} \begin{bmatrix} 3 & 6 \\ -1 & -2 \end{bmatrix} = \begin{bmatrix} -6 & -12 \\ 2 & 4 \end{bmatrix}$$

The controllability matrix Q is

$$Q = \begin{bmatrix} B & AB \end{bmatrix} = \begin{bmatrix} 3 & 6 & -6 & -12 \\ -1 & -2 & 2 & 4 \end{bmatrix}$$

By observing, it is seen column two is obtained by multiplying column one by a factor 2. Similarly, column three is obtained by multiplying column two by -1 and column four is obtained by multiplying column three by 2. Thus, the column vectors are linearly dependent. No square matrix can be formed which will give full rank which is 2. Hence, the given system is not completely state controllable.

Example 7.37

Consider the system described by the following vector matrix difference equation:

$$x[k+1] = \begin{bmatrix} 0 & 1 & 0 \\ 3 & 0 & 2 \\ -12 & -7 & -6 \end{bmatrix} x[k] + \begin{bmatrix} 1 \\ 4 \\ 6 \end{bmatrix} u[k]$$

Test whether the given system is completely state controllable.

Solution

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 3 & 0 & 2 \\ -12 & -7 & -6 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 \\ 4 \\ 6 \end{bmatrix}$$

$$AB = \begin{bmatrix} 0 & 1 & 0 \\ 3 & 0 & 2 \\ -12 & -7 & -6 \end{bmatrix} \begin{bmatrix} 1 \\ 4 \\ 6 \end{bmatrix} = \begin{bmatrix} 4 \\ 15 \\ -76 \end{bmatrix}$$

$$A^2B = \begin{bmatrix} 0 & 1 & 0 \\ 3 & 0 & 2 \\ -12 & -7 & -6 \end{bmatrix} \begin{bmatrix} 4 \\ 15 \\ -76 \end{bmatrix} = \begin{bmatrix} 15 \\ -140 \\ 303 \end{bmatrix}$$

The controllability matrix is given by

$$Q = \begin{bmatrix} B & AB & A^2B \end{bmatrix} = \begin{bmatrix} 1 & 4 & 15 \\ 4 & 15 & -140 \\ 6 & -76 & 303 \end{bmatrix}$$

By observation none of the column vectors of Q can be expressed as a linear combination of other column vectors. The column vectors are linearly independent and hence the matrix Q has full rank. Therefore, the system is completely state controllable.

The rank of the controllability matrix is determined as

$$\begin{aligned}|Q| &= 1(15 \times 303 - 140 \times 76) - 4(4 \times 303 + 6 \times 140) \\ &\quad +15(4 \times (-76) - 6 \times 15) \\ &= -6095 - 8208 - 5910 = -20213\end{aligned}$$

The determinant of Q does not vanish. Hence, the rank of Q is 3. The system is completely state controllable.

Example 7.38

Verify whether the system described by the following state equation of discrete time system is observable:

$$\begin{aligned}x[k+1] &= \begin{bmatrix}-5 & 4 \\ -6 & 5\end{bmatrix} x[k] + \begin{bmatrix}1 \\ 1\end{bmatrix} u[k] \\ y[k] &= [-2 \quad 3]\end{aligned}$$

Solution

$$\begin{aligned}C^T &= \begin{bmatrix}-2 \\ 3\end{bmatrix} \\ A^T &= \begin{bmatrix}-5 & -6 \\ 4 & 5\end{bmatrix} \\ A^T C^T &= \begin{bmatrix}-5 & -6 \\ 4 & 5\end{bmatrix} \begin{bmatrix}-2 \\ 3\end{bmatrix} = \begin{bmatrix}-8 \\ 7\end{bmatrix}\end{aligned}$$

The observability matrix

$$\begin{aligned}S &= \begin{bmatrix}C^T & A^T C^T\end{bmatrix} = \begin{bmatrix}-2 & -8 \\ 3 & 7\end{bmatrix} \\ |S| &= -14 + 24 = 10\end{aligned}$$

The determinant of S is not zero. Hence, the determinant of the observability matrix S exists and the system is observable.

7.11 Sampled Data System

7.11.1 Introduction

Sampled data system has now become very popular with the introduction of inexpensive microprocessors and mini-computers. Sampled data systems or otherwise called digital control systems arise whenever the measurements necessary for control are obtained in an intermittent form. Also when the complexity of the system to be controlled increases and for large scale systems, it is now customary to intentionally introduce a digital computer in the system. The digital computer sends out a control signal to each plant only periodically. In an analog system, the plant or the process to be controlled is described by means of differential equations. On the other hand, in the discrete time system (Digital control system) the signals vary at discrete intervals of time. Such systems are governed by difference equations. In such systems, the components connected respond only to discrete signals and the variables are discrete in nature. If both analog and digital components are connected in a system, then the system is called a sampled data system. Here, the variables appear in discrete form at one point and in analog form at some other point.

7.11.2 Advantages of Sampled Data Control Systems

1. By changing the software, it is easy to modify the control law in digital control systems whereas it is not so in analog systems.
2. Sampled data control systems use control elements which require low energy signals. Thus, large power can be controlled with low power signals.
3. The drift associated with analog components is more and hence the accuracy is less. The drift is minimum in digital control systems and hence the accuracy is more here.
4. The use of differential controllers in both analog and digital control systems has become popular now. The use of a differential controller in continuous systems not only enhances the noise but also produces additional noise. Further, in a sampled data control system, there is no need for amplification of the analog signal, and this reduces the noise in the system.
5. In sampled data control systems, the time intervals between successive signals are used to control many other separate plants and thus the overall cost is reduced.
6. In a sampled data control system, the use of digital computers helps in data acquisition and monitoring of all the variables in the system. The digital computer ensures the safety of the plant. The startup and shutdown procedures are made simple in computer controlled systems.
7. It is easy to implement optimal and adaptive control in sampled data time system.

7.11.3 Disadvantages

1. The analysis and design of a sampled data control system are more complex and tedious as compared to continuous control systems.
2. The use of A/D, D/A converters and digital computer in sampled data time system introduces a delay in sending the signal, and thus it is difficult to achieve the performance objectives as predicted by theoretical calculation which assume no delay.
3. Generally, the sampled data system is less stable compared to the continuous system.
4. In sampled data control system, while reconstructing the continuous signal from a discrete signal, there is a loss of signal information.
5. In sampled data system, it is very difficult to get information about the system behaviour in between sampling intervals.

7.11.4 A Sampled Data Closed Loop Control System

The block diagram of a typical sampled data closed loop control system is shown in Fig. 7.24. The input $r(t)$, output $c(t)$ and error $e(t)$ are continuous signals. The error signal $e(t)$ is converted into a sequence of numbers $m(kT)$ by means of an analog to digital (A/D) converter. For this, it is necessary to sample the analog signal $e(t)$ to $e(kT)$ at instants kT where $k = 0, 1, 2, \ldots$, and T is the time interval between any sampling (T is called the sampling period). The A/D converter samples $e(t)$ into $e(kT)$ and sends a sequence of numbers $m(kT)$ suitable for use by the digital computer. Thus, the A/D converter first converts the continuous signal $e(t)$ into $e(kT)$. This process is called digitization. The discrete signal $e(kT)$ is converted into $m(kT)$. This process is known as quantization. The output of the digital computer appears as $u(kT)$. Since the plant requires continuous signal $u(t)$, it is necessary to convert the discrete signal $u(kT)$ to $u(t)$. This is done by means of a digital to analog (D/A) converter which is followed by a data extrapolator or otherwise called a hold circuit. In the computer hardware, there is a clock which sends a pulse every T seconds to A/D and D/A converters. The converter sends a number to the computer. In a sampled data system, the A/D converter, digital computer, D/A converter and the clock are all represented by a sampling switch followed by a hold circuit.

7.11.5 Sampling Process

The process of converting a continuous signal into a discrete signal is known as sampling. This process may occur in one or more places in the system. In a block diagram, it is represented by a switch.

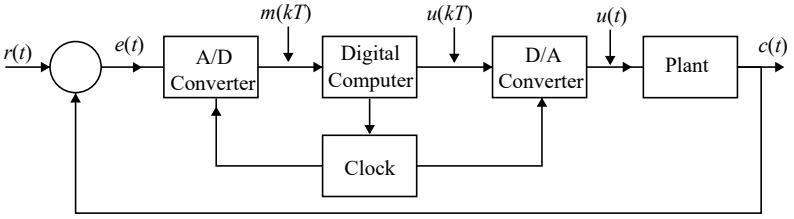


Fig. 7.24 A sampled data closed loop control system

The input signal $x(t)$ is a continuous signal. The sampler output $x^*(t)$ is a train of pulses. The continuous signal $x(t)$ and the output of the sampler $x^*(t)$ may be related by

$$x^*(t) = \delta_T(t)x(t) \tag{7.130}$$

where $\delta_T(t)$ represents a train of unit impulses. The sampler output is equal to the product of continuous signal $x(t)$ and the train of unit impulses. In other words, sampler may be considered as a modulator with the input $x(t)$ as the modulating signal and the train of unit impulses as the carrier. The output $x^*(t)$ is the modulated signal. The sampling process is shown in Fig. 7.25. Now the train of impulses may be represented by the following equation:

$$\delta_T(t) = \sum_{-\infty}^{\infty} \delta_T(t - kT) \tag{7.131}$$

where $\delta_T(t - kT)$ is the unit impulse function occurring at $t = kT$ where $k = 0, 1, 2, 3, \ldots$ and T is the sampling period. If the continuous signal is sampled in a periodic manner, the sampled signal may be represented by

$$x^*(t) = \sum_{-\infty}^{\infty} x(t)\delta_T(t - kT) \tag{7.132}$$

$$= \sum_{-\infty}^{\infty} x(kT)\delta_T(t - kT) \tag{7.133}$$

7.11.6 Sampled Data System Variables

The different variables appearing in a sampled data system can be described in terms of time and amplitude. They are grouped as continuous amplitude and continuous (C-C), continuous amplitude and discrete time (C-D), discrete amplitude and continuous time (D-C) and discrete amplitude and discrete time (D-D). The commonly sampled

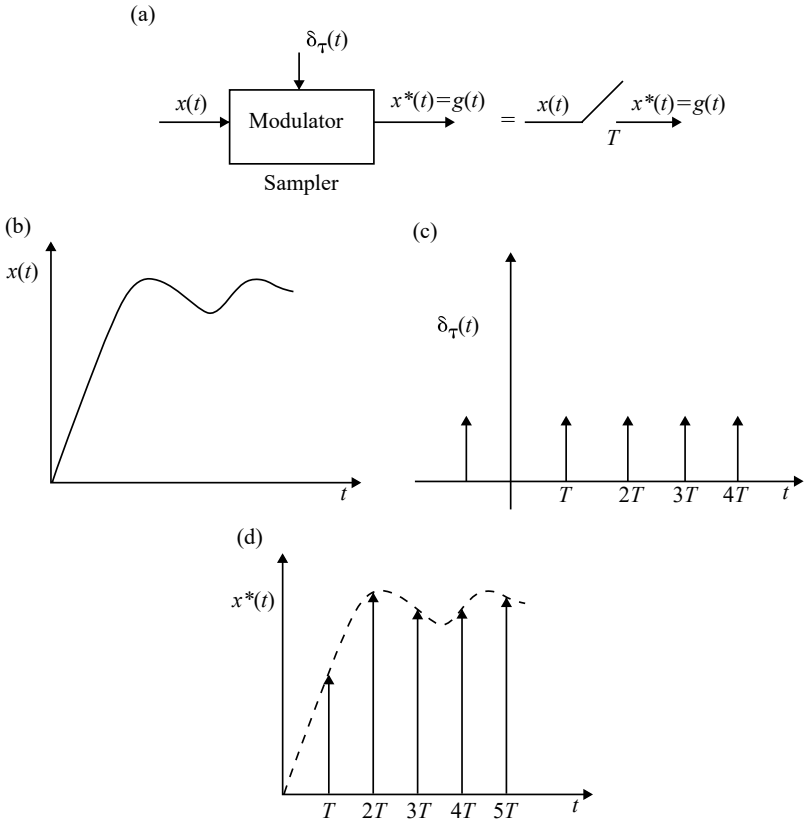


Fig. 7.25 Representation of the sampling process

data system is represented in the block diagram form in Fig. 7.26a, and the signals at different points are presented in Fig. 7.26b. The continuous amplitude, continuous time signal (C-C) (Fig. 7.26b) is applied as the input to the sampler as $e(t)$, and the output of the sampler appears as $e^*(t)$ which has continuous amplitude but in discrete time as shown in Fig. 7.26c. This signal appears as the input to a hold circuit. The output of the hold circuit has continuous amplitude and continuous time (C-C) as shown in Fig. 7.26d. It is to be noted here that $e^*(t)$ can be an infinite number of values within some predetermined range of amplitude.

Figure 7.27a represents the block of an open loop sampled data control system. The continuous signal $e(t)$ (C-C) is sampled as $e^*(t)$ in continuous amplitude and discrete time (C-D). The A/D converter quantizes $e^*(t)$ into $e(kT)$ as a (D-D) signal (Fig. 7.27d) and is applied to the digital computer. It is to be noted here that $e^*(t)$ is quantized such that it can be only one of a finite number of values within a specific value range. The digital computer itself manipulates the quantized value (a base 2 number) into another discrete amplitude–discrete time signal $f(kT)$ (Fig. 7.27e).

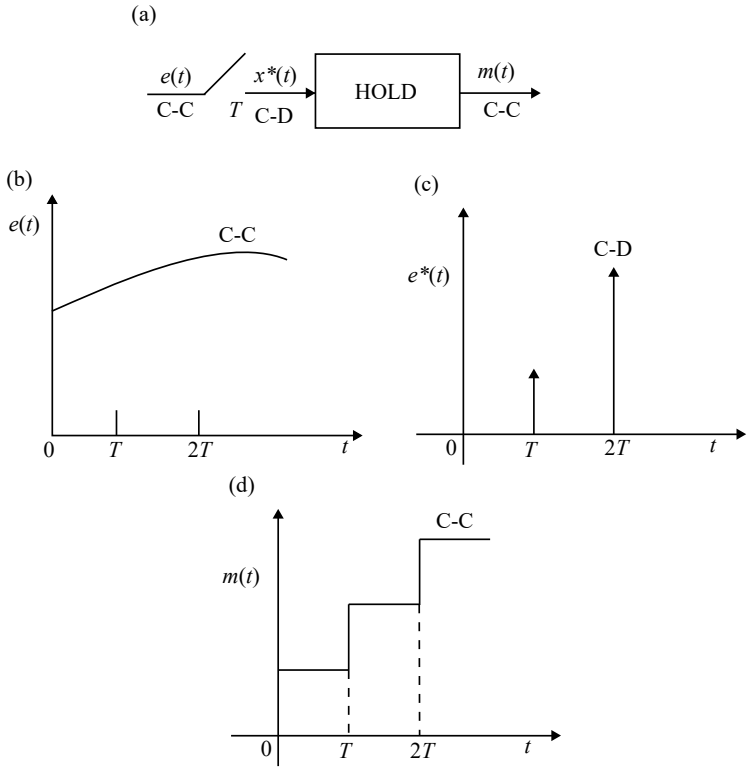


Fig. 7.26 Input–output of a hold circuit

Finally, the D/A converter or the hold device transforms $f(kT)$ into $m(t)$ which has continuous amplitude in continuous time (Fig. 7.27f).

7.11.7 Hold Devices

The essential element of a discrete system is the sampler. It admits the continuous signal T sec. After being quantized, it is applied to the digital computer. The output of the digital computer is a sampled signal. The sampled signal has to be converted into a continuous signal before it is applied to the controller. This is done by a data extrapolator or a data hold circuit. This device reconstructs a sampled signal into a continuous time signal based on the knowledge of the past samples. It is usually a part of the D/A converter. The hold devices are classified according to the number of prior samples utilized for predicting the sampled function during waiting intervals.

It is evident from Fig. 7.28 that the value of the reconstructed function using a zero order hold circuit during any waiting period is equal to the value of the sampled

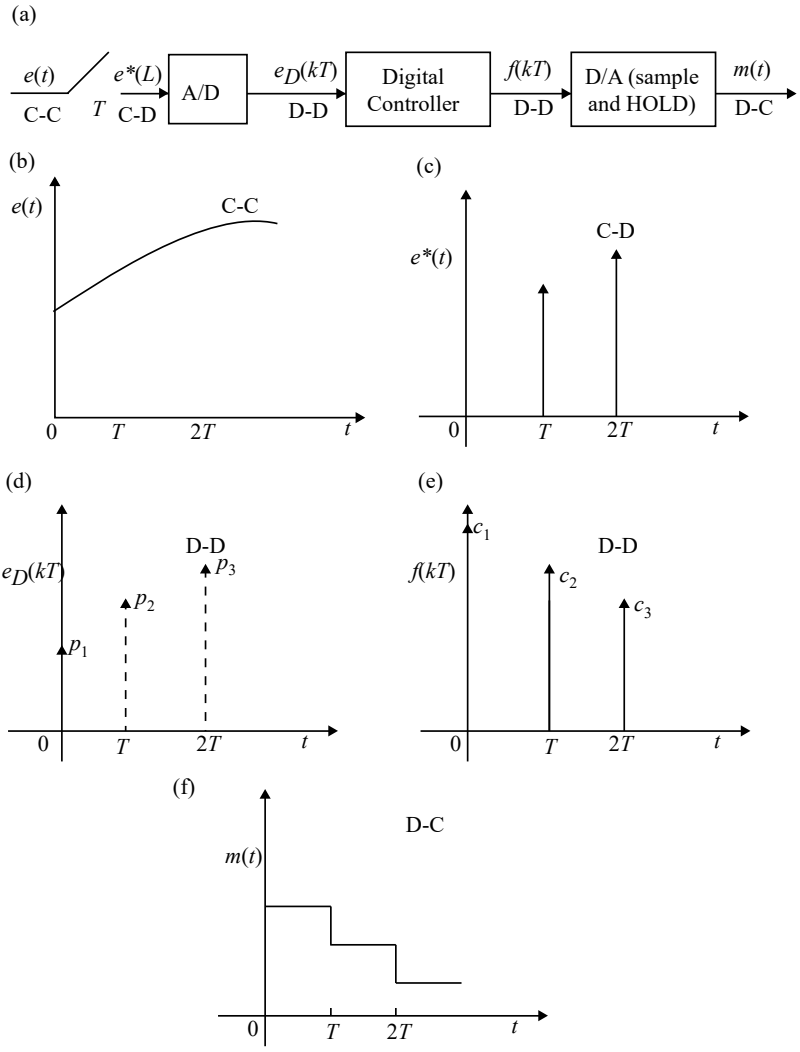


Fig. 7.27 Representation of open loop sampled data system

function at the beginning of the interval. This leads to the staircase approximation of a continuous time function.

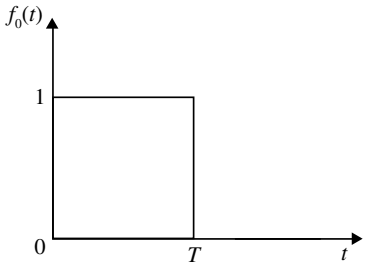


Fig. 7.28 Characteristic of zero prder hold device

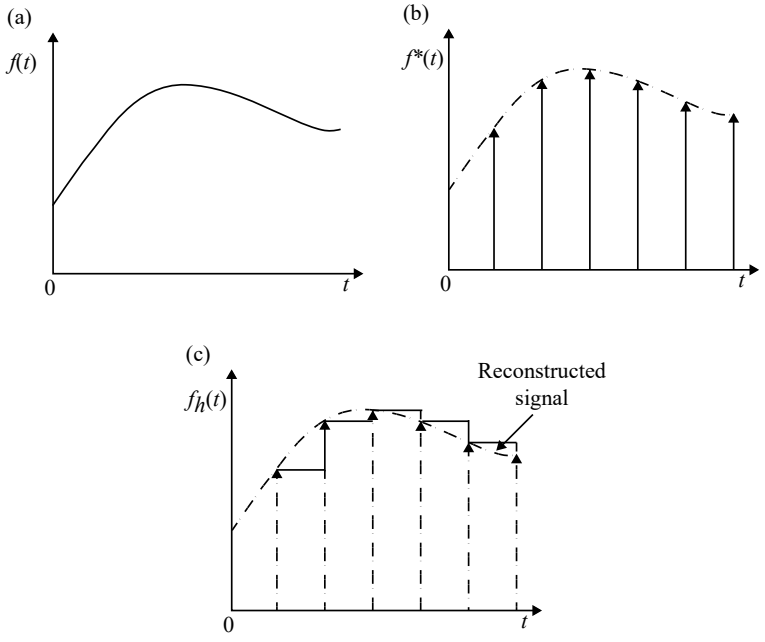


Fig. 7.29 Signal reconstruction using zero order hold circuit

7.11.8 Signal Reconstruction Using Zero Order Hold Device

Figure 7.29 shows the reconstruction of continuous signal using zero order hold (ZOH). Figure 7.29a shows the continuous signal. Figure 7.29b shows the signal after the sampler. Figure 7.29c shows the output of the zero order hold circuit. The zero order hold circuit keeps its output level equal to the magnitude of an input pulse and then resets itself when a new pulse arrives. On the arrival of the new pulse, the signal jumps to the new height and holds the output at the level. Thus, the output of the ZOH leads to the staircase approximation as shown in Fig. 7.29c.

7.11.9 Transfer Function of a ZOH

Figure 7.28 shows the characteristics of a ZOH. It is a composition of positive unit step and a negative unit step of delay T. The transfer function of a unit positive step is $1/s$ at that of a negative unit step with a time delay $-\frac{1}{s}e^{-sT}$. Hence, the transfer function of a ZOH is

$$G_{oh} = \frac{1}{s}[1 - e^{-sT}] \tag{7.134}$$

7.11.10 The Sampling Theorem

The signal $x(t)$, $\delta(t)$ and $g(t)$ shown in Fig. 7.25 are connected by the following equation:

$$g(t) = x(t)\delta_T(t) \tag{7.135}$$

where

$$\delta_T(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s) \tag{7.136}$$

$\delta_{T_s}(t)$ is called sampling function, T_s is the sampling period and $\omega_s = \frac{2\pi}{T_s}$ is the fundamental sampling frequency. This is also called radian frequency and is related to cyclic frequency as

$$\omega_s = \frac{2\pi}{T_s} = 2\pi f_s$$

where $f_s = \frac{1}{T_s}$ is the cyclic frequency. The sampled signal $g(t)$ shown in Fig. 7.25(e) consists of impulses spaced every T_s seconds, which is the sampling interval. $\delta_T(t)$ can be expressed as a trigonometric Fourier series as

$$\delta_T(t) = \frac{1}{T_s}[1 + 2\cos\omega_s t + 2\cos 2\omega_s t + 2\cos 3\omega_s t + \cdots] \tag{7.137}$$

$$\begin{aligned} g(t) &= x(t)\delta_T(t) \\ &= \frac{1}{T_s}[x(t) + 2x(t)\cos\omega_s t + 2x(t)\cos 2\omega_s t + \cdots] \end{aligned} \tag{7.138}$$

From the knowledge of the Fourier transform of continuous time signal, the following equations are written:

$$\begin{aligned} x(t) &\overset{\text{FT}}{\longleftrightarrow} X(\omega) \\ 2x(t)\cos\omega_s t &\overset{\text{FT}}{\longleftrightarrow} X(\omega-\omega_s)+X(\omega+\omega_s) \\ 2x(t)\cos 2\omega_s t &\overset{\text{FT}}{\longleftrightarrow} X(\omega-2\omega_s)+X(\omega+2\omega_s) \\ g(t) &\overset{\text{FT}}{\longleftrightarrow} G(\omega) \end{aligned} \tag{7.139}$$

Substituting Eq. (7.139) in Eq. (7.138), we get

$$\begin{aligned} G(\omega) = \frac{1}{T_s}[&X(\omega)+X(\omega-\omega_s)+X(\omega+\omega_s) \\ &+X(\omega-2\omega_s)+X(\omega-2\omega_s)+\cdots] \end{aligned} \tag{7.140}$$

$$\boxed{G(\omega)=\frac{1}{T_s}\sum_{n=-\infty}^{\infty}X(\omega-n\omega_s)} \tag{7.141}$$

In Eq. (7.140), the first term in the bracket is $X(\omega)$. The second term is $X(\omega-\omega_s)+X(\omega+\omega_s)$. This represents the spectrum of $X(j\omega)$ shifted to ω_s and $-\omega_s$. Similarly, the third term $X(\omega-2\omega_s)+X((\omega+2\omega_s))$ which represents the spectrum $X(\omega)$ is shifted by $2\omega_s$ and $-2\omega_s$ and so on. The frequency spectrum $X(\omega)$ and $\delta(\omega)$ are represented in Fig. 7.30a and b, respectively. In Fig. 7.30a, ω_m is the maximum frequency content of the continuous time signal. Figure 7.30c represents the frequency spectrum $G(\omega)$ of the continuous sampled signal for $\omega_s > 2\omega_m$. Spectrum of the sampled signal $G(\omega)$ for $\omega_s < 2\omega_m$ is shown in Fig. 7.30d.

To reconstruct the continuous time signal $x(t)$ from sampled signal $g(t)$, we should be able to recover $X(\omega)$ from $G(\omega)$. This recovery is possible if there is no overlap between successive $G(\omega)$. From Fig. 7.30c, this is possible if $\omega_s > 2\omega_m$ or

$$\boxed{f_s > 2f_m} \tag{7.142}$$

The minimum sampling rate $f_s = 2f_m$ is called the **Nyquist rate** of $x(t)$ and the corresponding time interval $T_s = \frac{1}{f_s} = \frac{1}{2f_m}$ is called **Nyquist interval** of $x(t)$. Samples of a signal taken at its Nyquist rate are the **Nyquist samples**.

From Eq. (7.142), the Shannon sampling theorem or simply sampling theorem is stated as follows.

A band-limited signal of finite energy which has no frequency component higher than f_m can be completely described and recovered back if the sampling frequency is twice the highest frequency of the given signal.

The proof of the theorem is given in Eq. (7.141) and Fig. 7.30c. If $f_s < 2f_m$, Eq. (7.141) is represented in Fig. 7.30d and here overlapping between successive samples occurs, and therefore it is not possible to recover $x(t)$ from the frequency spectrum $G(\omega)$ when passed through low-pass filter.

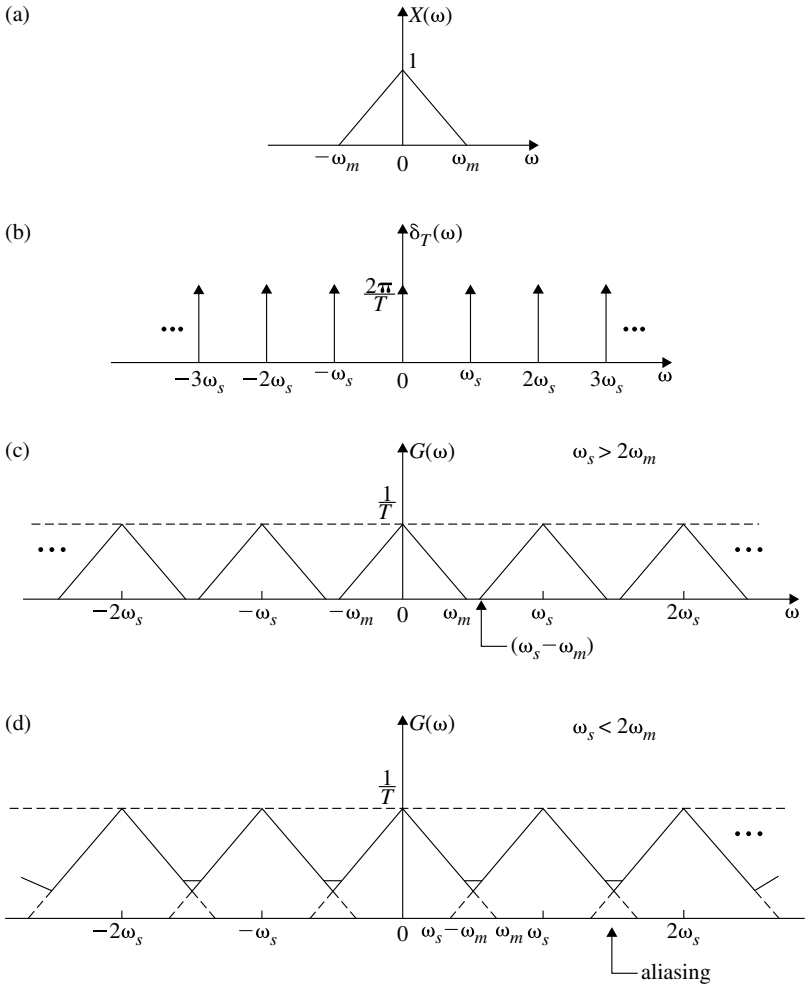


Fig. 7.30 Continuous signal, impulse train sampled signal spectrum

7.12 MATAB Program

7.12.1 Conversion of State Space Model to Transfer Function Model and Vice Versa

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Example 7.39

Determine the transfer function and the eigenvalues of the system represented in state space using MATLAB.

$$\frac{dx(t)}{dt} = \begin{bmatrix} 4 & 1 & -2 \\ 1 & 0 & 2 \\ 1 & -1 & 3 \end{bmatrix} x(t) + \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} u(t)$$
$$y(t) = [2 \quad -6 \quad 5]x(t)$$

PROGRAM

```
clc;
A = [41 − 2; 102; 1 − 13];
B = [1; 2; 3];
C = [2 − 65];
D = 0;
[num,den]=ss2tf(A,B,C,D)
sys=tf(num,den)
EigenValues=roots(den)
```

OUTPUT

```
num = 0   5.0000   − 37.0000   60.0000
den = 1.0000   − 7.0000   15.0000   − 9.0000
Transfer function:
5s^2 − 37s + 60/s^3 − 7s^2 + 15s − 9
Eigen Values = 3.0000
               3.0000
               1.0000
```

Example 7.40

Write a Program to obtain the transfer function of the system defined by the following state space equations:

$$\frac{dx(t)}{dt} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -5 & -10 & -3 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u(t)$$
$$y(t) = [4 \quad 5 \quad 1]x(t)$$

PROGRAM

```
clc;
A = [010; 001; −5 − 10 − 3];
B = [0; 0; 1];
C = [451];
D = 0;
[num,den]=ss2tf(A,B,C,D) sys=tf(num,den)
```

OUPUT

```
num = 0   1.0000   5.0000   4.0000
den = 1.0000   3.0000   10.0000   5.0000
Transfer function:
s^2 + 5s + 4/s^3 + 3s^2 + 10s + 5
```

7.12.2 Conversion of Transfer Function to State Model

Example 7.41

Write a Program to obtain the state space equations for the transfer function given as follows:

$$T(s) = \frac{5s + 2}{s^3 + 7s^2 + 3s + 5}$$

PROGRAM

```
clc;
num=[0 0 5 2];
den=[1 7 3 5];
[A,B,C,D]=tf2ss(num,den)
```

OUTPUT

$$A = \begin{bmatrix} -7 & -3 & -5 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$C = [0 \quad 5 \quad 2]$$
$$D = [0]$$

7.12.3 To Write a Program to Obtain the STM

Example 7.42

Write a Program to find the state transition matrix for the following A matrix:

$$A = \begin{bmatrix} -3 & 1 \\ -2 & 0 \end{bmatrix}$$

PROGRAM

```
clc;
A = [−31; −20];
T=sym('t')
STM=expm(A*t)
```

OUTPUT

```
STM =[2/exp(2*t) - 1/exp(t), 1/exp(t) - 1/exp(2*t)]
[2/exp(2*t) - 2/exp(t), 2/exp(t) - 1/exp(2*t)]
```

7.12.4 To Determine Controllability and Observability of the System

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Example 7.43

A certain control system is described by the following vector matrix differential equation:

$$\frac{dx}{dt} = \begin{bmatrix} 1 & 2 & 1 \\ -1 & -4 & -3 \\ 1 & 2 & 3 \end{bmatrix} x(t) + \begin{bmatrix} 1 \\ 4 \\ 6 \end{bmatrix} u(t)$$
$$y(t) = [1 \quad 1 \quad 2]x(t)$$

Determine whether the above system is completely state controllable, completely output controllable and observable.

PROGRAM

```
clc;
A = [121; −1 − 4 − 3; −123];
B = [1; 4; 6];
C = [112];
D=0;


disp('Rank of the Matrix')
Rankc=rank([B A*B A^2*B]) % To check the controllability
Ranko= rank([C' A'*C' A'^2*C']) % To check the observability
Rankoc= rank([C*B C*A*B C*A^2*B]) % To check the output Controllability
```

OUTPUT

```
Rankc =3
Ranko = 3
Rankoc = 1
```

From the above, the system is completely state controllable and observable. The system is not output controllable since the rank of the matrix is not three.

SUMMARY

1. An Nth order system (continuous as well as discrete) can be described in terms of N variables with N first order equations. These variables are called state variables.
2. State variable representation is not unique. However, the solution of state equation is unique.
3. The state equations are written in a specific format. Such an equation is called vector matrix differential/difference equation.
4. The state variables give internal and external descriptions of the system. Thus, physical variables can be chosen as state variables, and their behaviour can be readily studied.
5. State equations are written from system structure or from block diagrams.
6. State equations are solved by time domain or frequency domain methods.

EXERCISE

Short Answer Type Questions

1. **Define the state of a system.**
   The state of a system is defined as the minimum number of initial conditions that must be specified at any initial time t_0 so that the complete behaviour of the system at any time $t > t_0$ is determined if the input $u(t)$ is known.
2. **What do you understand by state vector?**
   If N state variables are required to completely describe the behaviour of the system, then these N variables are the N components of a vector q. Such a vector is called a state vector.
3. **What is state space?**
   The N dimensional space whose coordinate axes consist of x_1 axis, x_2 axis, $\ldots, x_N$ axis is called state space.
4. **What are state space equations?**
   Input variables, output variables and state variables are involved in the modelling of dynamic systems. The dynamic equations involving these three variables are called state space equations.
5. **What is a vector matrix differential/difference equation?**
   State variable equations are expressed in the time domain by using compact vector matrix notations. These equations when written for CT systems are called vector matrix differential equations and when written for DT systems are called vector matrix difference equations.
6. **What is state transition matrix (STM)?**
   The matrix which is unique for a given system that transforms any initial state $q(t_0)$ to any final state $q(t_f)$ is called the state transition matrix. It contains all the information about the system dynamics at all times. For a continuous time system, the STM $\phi(t) = e^{At}$.
7. **What is the Cayley–Hamilton theorem?**
   According to the Cayley–Hamilton theorem, the matrix A satisfies its own characteristic equation. This property is used to evaluate the STM e^{At}.
8. **What physical variables are chosen as state variables in an electrical circuit and mechanical systems?**
   The current through the inductor and the voltage across the capacitor are chosen as state variables in electrical circuits. The displacement and velocity of energy-storing elements such as mass (inertia) and spring are chosen as state variables in mechanical systems.
9. **What are the advantages of the state space model over that of the transfer function model?**

   (a) The state space model is applicable to linear, non-linear and time varying systems, whereas the T.F. model is applicable only to linear systems.

(b) The T.F. model requires initial conditions to be zero whereas for the state space model, the initial conditions need not be zero.
(c) System design modelled by T.F. is based on trial and error and it will not in general lead to optimal control. The famous optimal control theory which follows a systematic design procedure uses the state space model of the system.

10. **For a particular system, the A matrix is represented in more than one form. What is the nature of characteristic equation?**
In the state space model, even though the system A matrix is not unique, the characteristic equation is the same and is unique.

Long Answer Type Questions

1. **Consider the mechanical system shown in Fig. 7.31. Form the state space equation.**

$$\dot{x}(t) = \begin{bmatrix} -\frac{R}{L} & 0 & 0 \\ 0 & 0 & 1 \\ \frac{K_s}{M_{eq}} & -\frac{K_{eq}}{M_{eq}} & -\frac{B_{eq}}{M_{eq}} \end{bmatrix} x(t) + \begin{bmatrix} \frac{1}{L} \\ 0 \\ 0 \end{bmatrix} e(t)$$

where

$$M_{eq} = \frac{l_1}{l_2} M_1 + \frac{l_2}{l_1} M_2$$
$$B_{eq} = \frac{l_1}{l_2} B_1 + \frac{l_2}{l_1} B_2$$
$$K_{eq} = \frac{l_1}{l_2} K_1 + \frac{l_2}{l_1} K_2$$

2. **Consider the electrical circuit shown in Fig. 7.32. Form the state space equation.**

$$\dot{x}(t) = \begin{bmatrix} -\frac{11}{4} & \frac{1}{4} \\ \frac{3}{8} & -\frac{1}{8} \end{bmatrix} x(t) + \begin{bmatrix} \frac{5}{4} \\ -\frac{1}{8} \end{bmatrix} v(t)$$

3. **Consider the following T.F. of a certain continuous time system. Form the state space equations in canonical form I model.**

$$\boldsymbol{H(z) = \frac{7z^2 + 8z + 4}{z^3 + 5z^2 + 9z + 10}}$$

$$\dot{x}(t) = \begin{bmatrix} -5 & 1 & 0 \\ -9 & 0 & 1 \\ -10 & 0 & 0 \end{bmatrix} x(t) + \begin{bmatrix} 7 \\ 8 \\ 4 \end{bmatrix} u(t)$$

$$y(t) = [1 \quad 0 \quad 0] x(t)$$

4. **Consider the following T.F. of continuous time system. Form the state space equation in canonical form II model.**

$$\boldsymbol{H(z) = \frac{5z^3 + 6z^2 + 2z + 10}{z^3 + 7z^2 + 4z + 5}}$$

$$\dot{x}(t) = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -5 & -4 & -7 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u(t)$$

$$y(t) = [-15 \quad -18 \quad -29] x(t) + 5u(t)$$

5. **Consider the following T.F. of continuous time system. Form the state space equation in diagonal form model.**

$$\boldsymbol{H(z) = \frac{(z^2 + 6z + 8)}{(z^3 + 27z^2 + 230z + 600)}}$$

$$\dot{x}(t) = \begin{bmatrix} -5 & 0 & 0 \\ 0 & -10 & 0 \\ 0 & 0 & -12 \end{bmatrix} x(t) + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} u(t)$$

$$y(t) = \begin{bmatrix} \frac{3}{35} & -\frac{24}{5} & \frac{40}{7} \end{bmatrix} x(t)$$

6. **Consider the following differential equation:**

$$\boldsymbol{\frac{d^3y}{dt^3} - 5\frac{d^2y}{dt^2} + 6\frac{dy}{dt} + 7y = 2\frac{dx}{dt} + 5u(t)}$$

Form the state space equation in canonical form II model.

$$\dot{x}(t) = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -7 & -6 & 5 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u(t)$$

$$y(t) = [5 \quad 2 \quad 0] x(t)$$

7. **Consider the following T.F. of a certain discrete time system:**

$$\boldsymbol{H(z) = \frac{4z^2 - 5z + 10}{z^3 + 2z^2 - 7z + 9}}$$

Form the state variable equation in canonical form II model.

$$x[n+1] = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -9 & 7 & -2 \end{bmatrix} x[n] + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u[n]$$

$$y[n] = [10 \quad -5 \quad 4]\, x[n]$$

8. **Consider the following T.F. of a certain discrete system given below. Form the state space equation of canonical form I model.**

$$\boldsymbol{H(z) = \frac{2z^2 + 6z + 9}{z^3 + 8z^2 + 7z + 16}}$$

$$x[n+1] = \begin{bmatrix} -8 & 1 & 0 \\ -7 & 0 & 1 \\ -16 & 0 & 0 \end{bmatrix} x[n] + \begin{bmatrix} 2 \\ 6 \\ 9 \end{bmatrix} u[n]$$

$$y[n] = [1 \quad 0 \quad 0]\, x[n]$$

9. **A certain discrete time system is described by the following difference equation:**

$$\boldsymbol{y[n] - \frac{1}{12}y[n-1] - \frac{1}{6}y[n-2] + \frac{1}{48}y[n-3] = x[n-1] - \frac{3}{8}x[n-2] + \frac{1}{32}x[n-3]}$$

Determine the system T.F. Form the state space equation in the diagonal form. What are the eigenvalues of the system?

$$x[n+1] = \begin{bmatrix} \frac{1}{2} & 0 & 0 \\ 0 & -\frac{1}{6} & 0 \\ 0 & 0 & -\frac{1}{4} \end{bmatrix} x[n] + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} u[n]$$

$$y[n] = \begin{bmatrix} \frac{3}{16} & -\frac{35}{16} & 3 \end{bmatrix} x[n]$$

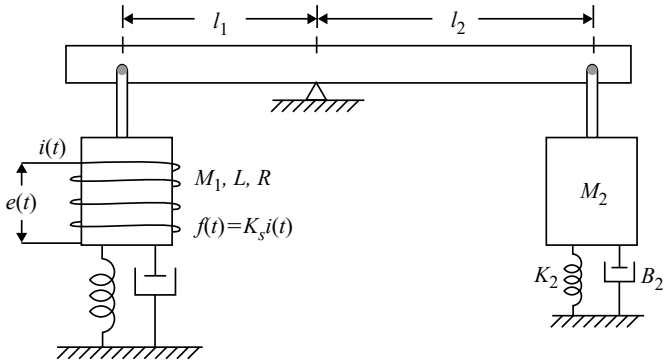


Fig. 7.31 Mechanical system with lower arrangement

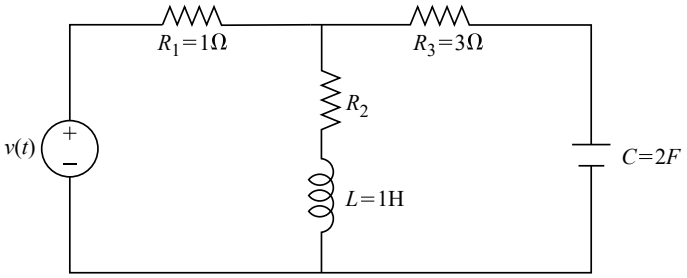


Fig. 7.32 Electrical circuit for Question 2

$$H[z] = \frac{z^{-1} - \frac{3}{8}z^{-2} + \frac{1}{32}z^{-3}}{1 - \frac{1}{12}z^{-1} - \frac{1}{6}z^{-2} - \frac{1}{48}z^{-3}}$$

The eigenvalues are $\lambda_1 = \frac{1}{2},\ \lambda_2 = -\frac{1}{6},\ \lambda_3 = -\frac{1}{4}$.

10. **Find the state equation of a continuous time LTI system described by**

$$\ddot{y}(t) + 3\dot{y}(t) + 2y(t) = u(t)$$

(*Anna University, May, 2007*)

$$\dot{x}(t) = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t)$$

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S. Palani, *Automatic Control Systems*,
https://doi.org/10.1007/978-3-030-93445-3

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